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QUANTITATIVE APTITUDE

QUANTUM CAT

Also Useful For

GAT | XAT | IIFT | SNAP | GMAT | CSAT

Sarvesh K Verma

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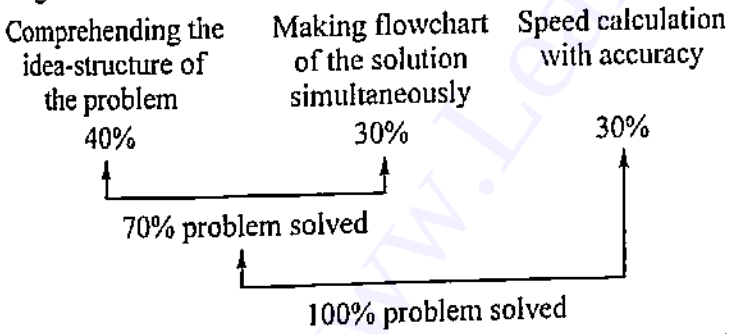
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1 FUNDAMENTALS

First of all I would like to mention that this chapter is deliberately written to promote your interest for CAT preparation in a systematic approach which will lay the strong foundation. Since this chapter subsumes fundamentals of **calculation techniques** along with various **number systems** besides basic tenets of maths, hence it is named as 'Fundamentals'. Again since most of the problems (*i.e.*, 20–40%) in CAT paper (*viz.*, QA section) belong to this chapter alone hence it becomes inevitable to discuss the nuances and subtleties of the various concepts objectively and comprehensively. That's why this chapter has become very bulky too.

In your own interest it would be better to adhere to my advices and stipulations stated at appropriate places in this chapter even throughout the book. As per my own experiences there are basically three stages in the solution of a problem and each of the stage has its own contribution. The following diagrams illustrates the importance of each stage.



If any of the three above mentioned activities is botched up then you are prone to failure. Hence, it is obvious that speed calculation invariably enhances your rate of success in CAT and most importantly it eliminates the stress and anxiety to ensconce you in your own comfort zone. Remember that you can succeed only when you operate in your comfort zone else it turns out to be a challenge. Due to above mentioned reasons it is imperative to become handy in quick calcualtion which is required in DI too.

The latter part, even major section, of the chapter is devoted to numbers, their kinds, operations and behaviour in mathematical milieu. You are supposed to not to skip a bit of it since every bit and every concept is equally important. It is advised that you should make the flow chart of the solution while reading, since going back to the problem several times means irritation and wastage of invaluable time. Also, you should try to solve maximum no. of problems without pencil and paper since it makes you quick respondent and saves a lot of time. Finally, avoid rote method of learning in maths instead be inqusitive and explorer to gain an advantage. The bottomline is that perceive logic and apply logic, since it is logical.

CALCULATION TECHNIQUES

Multiplication : Multiplication is nothing but the shortest method of addition. In day-to-day life when we have to add-up the same quantity many a times, so we multiply the given number by the number of times to which the given number is to be added repeatedly *e.g.*, $3+3+3+3+3+3+3$ can be added seven times but to get quick result we just multiply 3 by 7. Hence, we get $3 \times 7 = 21$. Thus, I am sure that now you must have become aware of that how much important is the multiplication table. So I hope that you will learn the multiplication table.

Now, I would like to suggest you that in the competitive exams like CAT, where speed and accuracy are totally indispensable. So there are some novel and enticing techniques for multiplication given in this chapter. But in some techniques you have to know the square of some relevent numbers, normally the squares of 1 to 100. Last but not the least you don't need to bother about since I have some very innovative techniques to learn and calculate the squares very readily. Even most of the times you need not to use a pencil and paper for this work and that's my purpose.

Basically we divide these techniques into two categories. The first one which I have derived from some algebraic operations and the second one belongs to our ancient maths *i.e.*, Vedic maths.

In this book I have laid emphasis on my own approach of multiplication, because years of research in numeric maths I have found that I myself and my students find it comfortable as it is very convenient and pragmatic to do within a few seconds.

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Case 1. Multiplication by 5, 25 $(=5^2)$, 125 $(=5^3)$, 625 $(=5^4)$ etc.

EXAMPLE 1 *Multiply* 187 *by* 5 *i.e.*, 187×5

SOLUTION **Step 1:** Associate as many zeros at the end of the given number as there is the power to 5. Thus we write 187 as 1870. (because $5 = 5^1$)

Step 2: Divide the resultant number by 2 as many times as there is the power to 5.

Thus, $$\frac{1870}{2} = 935$$

EXAMPLE 2 *Multiply* 369 *by* 125.

SOLUTION **Step 1:** 369000 ($\because 125 = 5^3$) hence 3 zeros will be associated.

Step 2: $$\frac{369000}{2 \times 2 \times 2} = \frac{369000}{8} = 46125$$

(Since, there is 5^3 hence it will be divided by 2^3.)

In general, if you want to multiply any number K by 5^n then you just need to divide that number by 2^n, after writing or (associating) 'n' zeros at the end of the number *i. e.*,

$$K \times 5^n = \frac{K0000\ldots, n \text{ zeros}}{2^n}$$

e.g., $$732 \times 5^6 = \frac{732000000}{2 \times 2 \times 2 \times 2 \times 2 \times 2}$$

$$= \frac{732000000}{64} = 11437500$$

Case 2. Multiplication of a number by 9^n *i.e.*, 9, 9^2 $(=81)$, 9^3 $(=729)$ etc.

EXAMPLE 1 *Multiply* 238 *by* 9.

SOLUTION Multiply the given number by 10 and then subtract the given number from the resultant number. So

$$238 \times 9 = 238 \times (10 - 1)$$
$$= 2380 - 238 = 2142$$

[using distributive law as $a \times (b - c) = ab - ac$]

EXAMPLE 2 *Multiply* 238 *by* 81.

SOLUTION **Step 1:** $238 \times 81 = 238 \times 9 \times 9$

Step 2: $238 \times (10 - 1) \times (10 - 1)$

Step 3: $[2380 - 238] \times (10 - 1)$

Step 4: $2142 \times (10 - 1)$

Step 5: $21420 - 2142 = 19278$

NOTE To multiply a number by the higher powers of 9 *i.e.*, 9 to power 3, 4, 5, ..., etc. it is better to do it conventionally.

Case 3. Multiplication of a number by the numbers whose unit (*i.e.*, last) digit is 9.

EXAMPLE 1 *Multiply* 287 *by* 19.

SOLUTION $287 \times (10 + 9) = 287 \times 10 + 287 \times 9$
$= 2870 + 287 \times (10 - 1)$
$= 2870 + 2870 - 287$
$= 5740 - 287 = 5453$

or $287 \times 19 = 287 \times (20 - 1)$
$= 287 \times 20 - 287$
$= 287 \times 2 \times 10 - 287$
$= 574 \times 10 - 287$
$= 5740 - 287 = 5453$

Case 4. Multiplication of a number by the number which contains all the digits as 9.

EXAMPLE 1 *Multiply* 743 *by* 99.

SOLUTION $743 \times 99 = 743 \times (100 - 1)$
$= 74300 - 743 = 73557$

EXAMPLE 2 *Multiply* 23857 *by* 9999.

SOLUTION $23857 \times 9999 = 23857 \times (10000 - 1)$
$= 238570000 - 23857 = 238546143$

NOTE In general to multiply a number K by a number which is purely consists of n 9s, put n zeros at the end of the number K then subtract the given number from the resultant number.
e.g., $K \times 9999\ldots n$ times $= (K0000\ldots n \text{ zeros}) - K$

Case 5. To multiply those two numbers, the sum of whose unit digit is 10 and the rest digits (left to the unit digit) are same in both the numbers.

EXAMPLE 1 *Multiply* 38 *by* 32.

SOLUTION **Step 1:** Multiply the unit digits as $8 \times 2 = 16$

Step 2: Add 1 to the left digits then multiply this increased number by the original digits *i. e.*,

$3 \times (3 + 1) = 12$

Thus, we get $38 \times 32 = 1216$

EXAMPLE 2 *Multiply* 83 *by* 87.

SOLUTION **Step 1:** $3 \times 7 = 21$

Step 2: $8 \times (8 + 1) = 72$

$\therefore$ $83 \times 87 = 7221$

EXAMPLE 3 *Multiply* 71 *by* 79.

SOLUTION **Step 1:** $9 \times 1 = 09$

Step 2: $7 \times (7 + 1) = 56$

$\therefore$ $71 \times 79 = 5609$

NOTE The last two places are reserved for the product of unit digits as in the above example $1 \times 9 = 9$ but we fill the last two places by writing "09". So don't write 569 since it is wrong.

EXAMPLE 4 *Multiply* 125 *by* 125.

SOLUTION **Step 1:** $5 \times 5 = 25$

Step 2: $12 \times (12 + 1) = 156$

So, $125 \times 125 = 15625$

EXAMPLE 5 *Multiply* 431 *by* 439.

SOLUTION **Step 1:** $1 \times 9 = 09$

Step 2: $43 \times (43 + 1) = 1892$

So, $431 \times 439 = 189209$

EXAMPLE 6 *Multiply* 1203 *by* 1207.

SOLUTION **Step 1:** $3 \times 7 = 21$

Step 2: $120 \times (120 + 1) = 14520$

So, $1203 \times 1207 = 1452021$

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Case 6. To multiply the two numbers whose difference is always an even number.

EXAMPLE 1 *Multiply* 38 *by* 52.

SOLUTION **Step 1:** $52 - 38 = 14$ (here 14 is an even number)

Step 2: $\frac{14}{2} = 7$ **Step 3:** $38 + 7$ or $52 - 7 = 45$

Step 4: $(45)^2 = 2025$ **Step 5:** $7^2 = 49$

Step 6: $2025 - 49 = 1976$

EXAMPLE 2 *Multiply* 76 *by* 96.

SOLUTION **Step 1:** $96 - 76 = 20$ (here 20 is an even number)

Step 2: $\frac{20}{2} = 10$

Step 3: $76 + 10 = 96 - 10 = 86$

Step 4: $(86)^2 = 7396$

Step 5: $(10)^2 = 100$ **Step 6:** $7396 - 100 = 7296$

EXAMPLE 3 *Multiply* 185 *by* 215.

SOLUTION **Step 1:** $215 - 185 = 30$

(here 30 is an even number)

Step 2: $\frac{30}{2} = 15$

Step 3: $185 + 15 = 215 - 15 = 200$

Step 4: $(200)^2 - (15)^2 = 40000 - 225 = 39775$

NOTE 1. In general, if there are any two numbers N_1 and N_2 and the difference between N_1 and N_2 is $2d$ then their product i.e.,

$$N_1 \times N_2 = (N_1 + d)^2 - d^2$$

or $$N_1 \times N_2 = (N_2 - d)^2 - d^2$$

2. Even this method can be applied for the two numbers whose difference is an odd number but since it contains decimal values so some students may feel it slightly difficult.

EXAMPLE 1 *Multiply* 17 *by* 22.

SOLUTION **Step 1:** $22 - 17 = 5$

Step 2: $\frac{5}{2} = 2.5$

Step 3: $17 + 2.5 = 22 - 2.5 = 19.5$

Step 4: $(19.5)^2 - (2.5)^2 = 380.25 - 6.25 = 374.00$

SHORTCUT To multiply any two numbers say, N_1 and N_2 we apply the following methods.

$$N_1 \times N_2 = \left(\frac{N_1 + N_2}{2}\right)^2 - \left(\frac{N_2 - N_1}{2}\right)^2$$

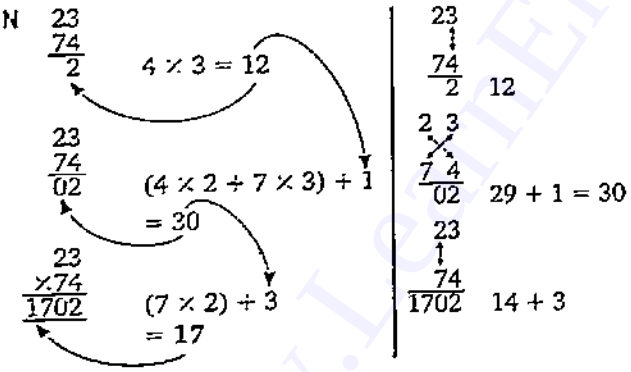
So, if you know the square of the required numbers then using this method you can multiply any two number within 2-3 seconds as

$$56 \times 64 = \left(\frac{56 + 64}{2}\right)^2 - \left(\frac{64 - 56}{2}\right)^2$$
$$= (60)^2 - (4)^2 = 3600 - 16 = 3584$$

VEDIC METHODS OF MULTIPLICATION

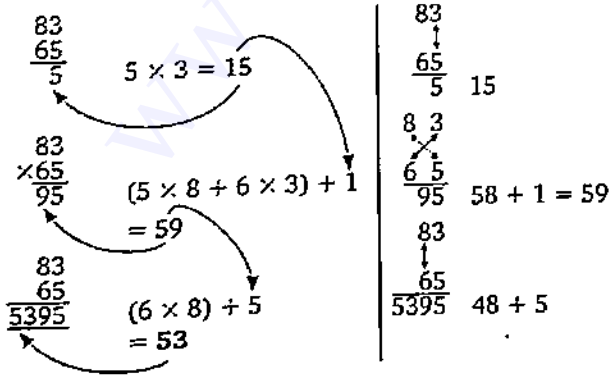
EXAMPLE 1 *Solve* 23×74.

SOLUTION



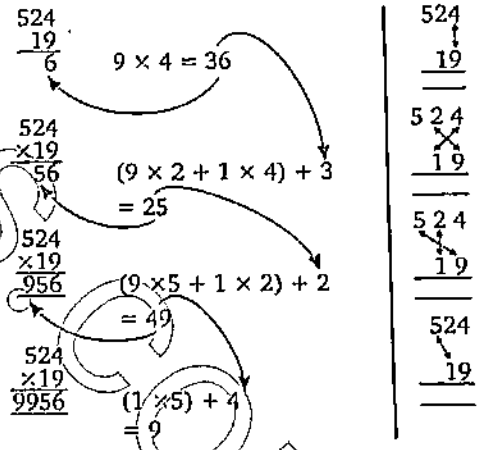
EXAMPLE 2 *Solve* 83×65.

SOLUTION



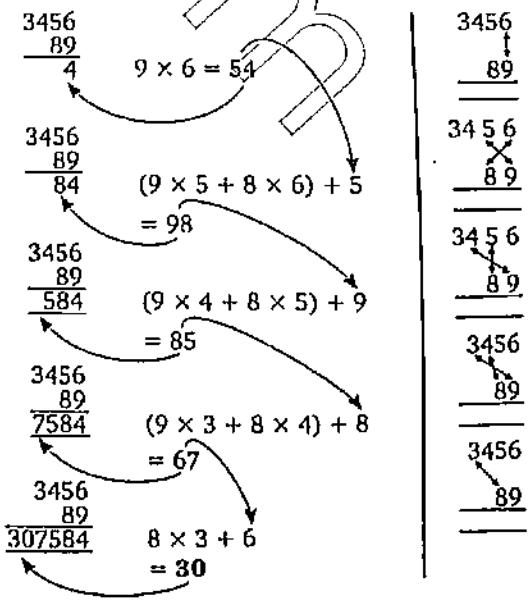
EXAMPLE 3 *Solve* 524×19.

SOLUTION



EXAMPLE 4 *Evaluate* 3456×89.

SOLUTION



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EXAMPLE 5 *Evaluate* 245×367.

SOLUTION

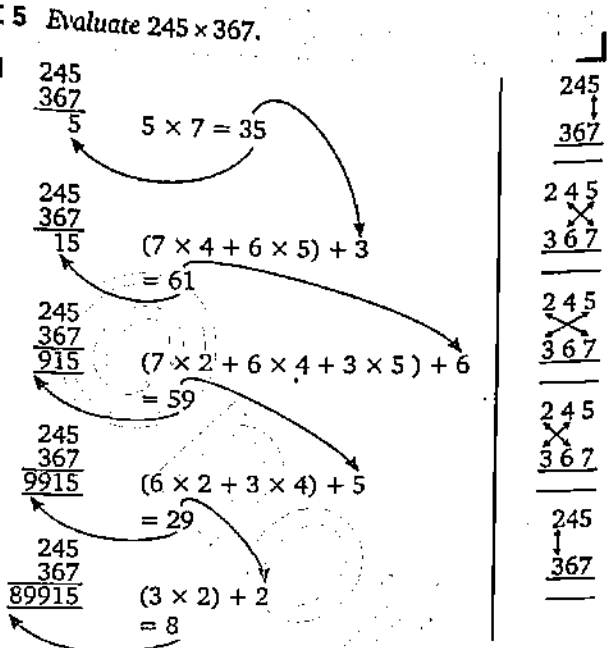
245
367
5 $\quad 5 \times 7 = 35$

245
367
15 $\quad (7 \times 4 + 6 \times 5) + 3 = 61$

245
367
915 $\quad (7 \times 2 + 6 \times 4 + 3 \times 5) + 6 = 59$

245
367
9915 $\quad (6 \times 2 + 3 \times 4) + 5 = 29$

245
367
89915 $\quad (3 \times 2) + 2 = 8$



EXAMPLE 6 *Evaluate* 123456×789.

SOLUTION

123456
789
4 $\quad 9 \times 6 = 54$

123456
789
84 $\quad (9 \times 5 + 8 \times 6) + 5 = 98$

123456
789
784 $\quad (9 \times 4 + 8 \times 5 + 7 \times 6) + 9 = 127$

123456
789
6784 $\quad (9 \times 3 + 8 \times 4 + 7 \times 5) + 12 = 106$

123456
789
06784 $\quad (9 \times 2 + 8 \times 3 + 7 \times 4) + 10 = 80$

123456
789
406784 $\quad (9 \times 1 + 8 \times 2 + 7 \times 3) + 8 = 54$

123456
789
7406784 $\quad (8 \times 1 + 7 \times 2) + 5 = 27$

123456
789
97406784 $\quad (7 \times 1) + 2 = 9$



EXAMPLE 7 *Find the value of* 4567×1289.

SOLUTION

4567
1289
3 $\quad 9 \times 7 = 63$

4567
1289
63 $\quad (9 \times 6 + 8 \times 7) + 6 = 116$

4567
1289
863 $\quad (9 \times 5 + 8 \times 6 + 2 \times 7) + 11 = 118$

4567
1289
6863 $\quad (9 \times 4 + 8 \times 5 + 2 \times 6 + 1 \times 7) + 11 = 10$

4567
1289
86863 $\quad (8 \times 4 + 2 \times 5 + 1 \times 6) + 10 = 58$

4567
×1289
886863 $\quad (2 \times 4 + 1 \times 5) + 5 = 18$

4567
1289
5886863 $\quad (4 \times 1) + 1$

EXAMPLE 8 *Find the value of* 325768×1234.

SOLUTION

325768
1234
2 $\quad 4 \times 8 = 32$

325768
1234
12 $\quad (4 \times 6 + 3 \times 8) + 3 = 51$

325768
1234
712 $\quad (4 \times 7 + 3 \times 6 + 2 \times 8) + 5 = 67$

325768
1234
7712 $\quad (4 \times 5 + 3 \times 7 + 2 \times 6 + 1 \times 8) + 6 = 67$

325768
1234
97712 $\quad (4 \times 2 + 3 \times 5 + 2 \times 7 + 1 \times 6) + 6 = 49$

325768
1234
997772 $\quad (4 \times 3 + 3 \times 2 + 2 \times 5 + 1 \times 7) + 4 = 39$

325768
1234
1997712 $\quad (3 \times 3 + 2 \times 2 + 1 \times 5) + 3 = 21$

325768
1234
01997712 $\quad (2 \times 3 + 1 \times 2) + 2 = 10$

325768
1234
401997712 $\quad (1 \times 3) + 1 = 4$

DIVISIBILITY RULES

While solving the mathematical problems we need to divide one number by another number. In general we use four terms for this process as

Divisor) Dividend (Quotient
xyz
Remainder

or Dividend = (Divisor × Quotient) + Remainder

for example $30 = (7 \times 4) + 2$

where quotient is a whole number *i.e.*, 0, 1, 2, 3, ... etc and remainder is always less than 'divisor' or zero.

So, when the remainder is zero, we can say that the given number is divisible by the particular divisor number.

Now, $238 \div 6 = 39\frac{4}{6} = 39.66$ (decimal notation)

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Since the above number when divided by 6 leaves a remainder of 4 then it is said to be not divisible by 6 where 238 is dividend, 39 is quotient obtained when divided by 6 as a divisor. Thus we get 4 as a remainder.

Further if we subtract the remainder from the dividend then the dividend (the resultant number) becomes divisible by particular divisor. *e.g.*,

$238 - 4 = 234$ (dividend–remainder)

$$6\overline{)234}\,(39$$
$$\underline{18}\;\downarrow$$
$$54$$
$$\underline{54}$$
$$\times$$

Here, the remainder is zero and the number is perfectly divisible by 6.

Another point is that if we add the difference of divisor and remainder to the given number (*i.e.*, dividend) the number becomes divisible by that particular divisor. *e.g.*,

$6 - 4 = 2$ (Divisor–Remainder)

$238 + 2 = 240$

$$6\overline{)240}\,(40$$
$$\underline{240}$$
$$\times$$

Thus $(238+2)$ becomes perfectly divisible by the particular divisor.

For negative integers

e.g., $-37 \div 5$

$$\frac{-37-3+3}{5} = \frac{-40+3}{5} = \frac{-40}{5} + \frac{3}{5}$$

Here 8 is quotient and 3 is remainder.

NOTE The remainder is always a non-negative integer.

Rule 1. If a number N is divisible by D then the product of N with any other integral number K is also divisible by D.

i.e., $\frac{N}{D} \rightarrow$ Remainder $0 \Rightarrow \frac{NK}{D} \rightarrow$ Remainder 0

Rule 2. If N_1 and N_2 two different numbers are divisible by D individually then their sum (*i.e.*, $N_1 + N_2$) and their difference ($N_1 \sim N_2$) is also divisible by D.

Rule 3. If a number N_1 is divisible by N_2 and N_2 is divisible by N_3 then N_1 must be divisible by N_3.

Rule 4. If two numbers N_1 and N_2 are such that they divide mutually each other it means they are same *i.e.*, $N_1 = N_2$ only.

HOW TO CHECK THE DIVISIBILITY

We have certain rules to check the divisibility by certain integral numbers. With the help of the following rules it has become easier to know whether a certain number is divisible or not by a particular number without actually dividing the number.

Divisibility by 2

Any integral number whose last digit (*i.e.*, unit digit)is even or in other words the unit digit is divisible by 2. It means any number whose last digit is either 0, 2, 4, 6 or 8, then this number must be divisible by 2.

e.g., 395672, 132, 790, 377754, etc.

Divisibility by 4

If the number formed by last two digits of the given number is divisible by 4 then the actual number must be divisible by 4. *i.e.*, the last two digits of a number can be 00, 04, 08, 12, 16, 20, 24, 28, 32, ..., 96.

e.g., 33932, 7958956, 2300, 1996, 3819764280 etc.

Divisibility by 8

If the number formed by last three digits of the given number is divisible by 8 then the actual number must be divisible by 8 *i.e.*, the last three digits of the divisible number can be 000, 008, 016, 024, 032, 040, ..., 096, 104, ..., 992.

e.g., 8537000, 9317640, 3945080, 23456008, 12345728, 3152408 etc.

Divisibility by 16

If the number formed by last four digits of the given number is divisible by 16 then the actual number must be divisible by 16 *i.e.*, the last 4 digits of the divisible number can be 0000, 0016, 0032, 0048, 0064, 0080, 0096, 0112, 0128, 0144, 0160, ..., 0960, 0976, ..., 0992, ..., 1600, ..., 9984.

Divisibility by 32, 64, 128, ... can be checked just by checking the last 5, 6, 7, ... digits number formed from the given number as in the above cases.

Divisibility by 3

If the sum of the digits of the given number is divisible by 3 then the actual number will also be divisible by 3.

e.g., 12375 is divisible by 3 since the sum of digits $1+2+3+7+5=18$ is divisible by 3.

Similarly 63089154 is also divisible by 3 since the sum of its digits $6+3+0+8+9+1+5+4=36$ is divisible by 3.

Divisibility by 9

If the sum of the digits of the given number is divisible by 9 then the actual number will also be divisible by 9.

e.g., 7329753 is divisible by 9, since

$7+3+2+9+7+5+3=36$ is divisible by 9.

Similarly the divisibility by 27, 81, 243, ... can be checked.

Divisibility by 5

A number is divisible by 5 if and only if the last (*i.e.*, unit) digit is either 0 or 5.

e.g., 5, 10, 15, 20, 25, 30, 35, ..., 275, 365, ..., 995, 70000, 438915 etc.

Divisibility by some special numbers *e.g.*, 7, 11, 13, 17, 19, 23, etc.

Why are these numbers treated differently for checking the divisibility? Since, if we multiply these numbers by any other number (except 10 or multiples of 10) these numbers can never be divisible by 10. So to make the process easier we bring the given number (*i.e.*, divisor) closer to the multiples of 10 with the difference of 1 (*e.g.*, 9 or 11, 19 or 21, 29 or 31, 39 or 41 etc).

Now if the number is one less than the multiples of 10 (*i.e.*, 9, 19, 29, 39, ...) we need to increase it by 1 to make it

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further multiple of 10. Hence we call it "one more" osculator and the value of multiple of 10 is called the value of "one more" osculator.

Similarly if the number is one more than the multiple of 10 (*i.e.*, 11, 21, 31, 41, ..., etc) we need to decrease it by 1 to make it the multiple of 10. So we call it 'negative' osculator and the value of multiple of 10 is the value of 'negative' osculator.

e.g., The given number is 7 (divisor number) then we bring it closer to the multiple of 10 with the difference of 1, then

$$7 \times 3 = 21 = 20 + 1$$

$$\Rightarrow \quad 20 = 21 - 1 = (2 \times 10) \qquad \ldots(1)$$

Now in the above expression 2 is the multiple of 10 and is known as the negative osculator for 7.

Now let us consider the next example of 13 and then bring it closer to the multiples of 10 with a difference of 1. Thus

$$13 \times 3 = 39$$

and $$39 + 1 = 40 = (4 \times 10) \qquad \ldots(2)$$

In the above expression 4 is the multiple of 10 and 4 is the value of 'one more' osculator.

Similarly for 17 we can find the osculator as

$$17 \times 3 = 51 = 50 + 1$$

So, $$50 = 51 - 1 = (5 \times 10)$$

Here 5 is the negative osculator for 17.

Similarly for 31 the osculator is negative and the value of negative osculator is 3.

$$\because \quad 31 = 30 + 1$$

$$\Rightarrow \quad 30 = 31 - 1 \Rightarrow (3 \times 10)$$

Now we will apply the osculator techniques to check the divisibility by 7, 11, 13, 17, 19, ... etc.

Divisibility by 7

To check the divisibility of a number by 7 we apply the following method. Let the number be 133.

Step 1. $133 \Rightarrow 13 - 3 \times 2 = 13 - 6 = 7$

Since 7 is divisible by 7, so the given number 133 will also be divisible by 7.

In the above process 2 is multiplied with the last digit is the negative osculator for 7, which I have earlier discussed.

EXAMPLE 1 *Check whether 1071 is divisible by 7.*

SOLUTION **Step 1.** $1071 \Rightarrow 107 - 1 \times 2 = 105$

Step 2. $105 \Rightarrow 10 - 5 \times 2 = 0$

Since 0 is divisible by 7 hence the given number 1071 is also divisible by 7.

EXAMPLE 2 *Check whether 939715 is divisible by 7.*

SOLUTION **Step 1.** $939715 \Rightarrow 93971 - 5 \times 2 = 93961$

Step 2. $93961 \Rightarrow 9396 - 1 \times 2 = 9394$

Step 3. $9394 \Rightarrow 939 - 4 \times 2 = 931$

Step 4. $931 \Rightarrow 93 - 1 \times 2 = 91$

Step 5. $91 \Rightarrow 9 - 1 \times 2 = 7$

Hence, it is divisible by 7.

NOTE 1. In all the above examples we have to multiply the last digit by the appropriate osculator and then this value will be subtracted from the number formed by the rest digits of the number and this process is continued till you know that the resultant value is divisible by 7. Even you can stop the process in midway when you guess that the obtained value is divisible by 7. For example in the latest problem (example-3) we can stop at step 4 if we know that the 91 is divisible by 7 also we can stop even at step 3 if we have any idea that 931 is divisible by 7.

2. Remember that if the operating osculator is "one more" osculator then we add the product of last digit and one more osculator in the number formed by the rest digit else. We subtract if the operating osculator is 'negative' osculator.

Divisibility by 13

EXAMPLE 1. *Check whether 2366 is divisible by 13.*

SOLUTION **Step 1.** $2366 \Rightarrow 236 + 6 \times 4 = 260$

[Since, the osculator for 13 is 4 and it is 'one more' osculator. So we use addition]

Step 2. $260 \Rightarrow 26 + 0 \times 4 = 26$

Since 26 (or 260) is divisible by 13 hence 2366 is also divisible by 13.

EXAMPLE 2. *Check whether 377910 is divisible by 13.*

SOLUTION **Step 1.** $377910 \Rightarrow 37791 + 0 \times 4 = 37791$

Step 2. $37791 \Rightarrow 3779 + 1 \times 4 = 3783$

Step 3. $3783 \Rightarrow 378 + 3 \times 4 = 390$

Step 4. $390 \Rightarrow 39 + 0 \times 4 = 39$

Since 39 is divisible by 39. So 377910 is also divisible by 13.

Divisibility by 17

EXAMPLE 1 *Find out whether 323 is divisible by 17.*

SOLUTION **Step 1.** $32\,3 \Rightarrow 32 - 3 \times 5 = 17$

[$\because$ 5 is the negative osculator of 17]

Therefore 323 is divisible by 17.

EXAMPLE 2 *Checkout that 12716 is divisible by 17 or not.*

SOLUTION **Step 1.** $1271\,6 \Rightarrow 1271 - 6 \times 5 = 1241$

$$1241\,1 \Rightarrow 124 - 1 \times 5 = 119$$

$$119 \Rightarrow 11 - 9 \times 5 = -34$$

so we can conclude that 12716 is divisible by 17. Since -34 and 119 both are simply visible that these two numbers are divisible by 17.

As I have already mentioned that you can stop your checking process as soon as you can get a number which is easy to know that the particular number is divisible by the given divisor or not. Further you should know that every resultant value in the right hand side might be divisible by the divisor whether you readily recognise it or not.

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Divisibility by 19

EXAMPLE 1 *Find out whether 21793 is divisible by 19.*

SOLUTION **Step 1.** $\underline{2179}\,\underline{3} \Rightarrow 2179 + 3 \times 2 = 2185$

[∵ 2 is the "one more" osculator of 19]

Step 2. $\underline{218}\,\underline{5} \Rightarrow 218 + 5 \times 2 = 228$

Step 3. $\underline{22}\,\underline{8} \Rightarrow 22 + 8 \times 2 = 38$

Hence 21793 is divisible by 19.

Shortcut rule for the divisibility by 7, 11 and 13 : A number can be divisible by 7, 11 or 13 if and only if the difference of the number formed by the last three digits and the number formed by the rest digits is divisible by 7, 11 or 13 respectively.

For example we have to check that 139125 is divisible by 7 or not.

So we take the difference as given below

$$139 - 125 = 14$$

Since, the difference is divisible by 7. Hence the given number is also divisible by 7.

EXAMPLE 1 *Check whether 12478375 is divisible by 13 or not.*

SOLUTION **Step 1.** $12478 - 375 = 12103$

Step 2. $12 - 103 = -91$

Since 91 is divisible by 13 hence 12478375 is also divisible by 13.

Divisibility by 11

A number is divisible by 11 if the difference between the sum of the digits at odd places and sum of the digits at even places is equal to zero or multiple of 11 (*i.e.*, 11, 22, 33 etc.)

For example : 57945822

Here sum of the digits at odd places $= 2 + 8 + 4 + 7 = 21$

and sum of the digits at even places $= 2 + 5 + 9 + 5 = 21$

and thus the difference $= 0\ (= 21 - 21)$

Hence 57945822 is divisible by 11.

Divisibility by Composite Numbers *e.g.*, 4, 6, 8, 10, 12, 14, 15 etc.

Divisibility by 6

A number is divisible by 6 only when it is divisible by 2 and 3 both. So first of all we see that the number is even or not then we check for the divisibility by 3.

Divisibility by 10

A number is divisible by 10 if and only if when it is divisible by both 2 and 5. So it can be easily observed that a number is divisible by 10 must ends up with zero(s) at the right end (*i.e.*, last digits) of itself.

NOTE Thus we can say that if there are '*n*' zeros at the end of the given number the number can be divided by 10^n means 10000... *n* zeros.

Divisibility by 12

A number is divisible by 12 only when it is divisible by 4 and 3 both at the same time. So first of all check the divisibility by 4 then by 3

Divisibility by 15

A number is divisible by 15 only when it is divisible by 3 and 5 both simultaneously. So first of all check the number by 5 then by 3.

Thus we can conclude that any number which is divisible by a composite number, as mentioned above, must be divisible by all its factors whose L.C.M. is the given divisor.

INTRODUCTORY EXERCISE-1.1

1. What is the largest possible two digit number by which 2179782 can be divided ?
   (a) 88 (b) 50
   (c) 66 (d) 99
2. At least which number must be subtracted from 9999999 so that it will become the multiple of 125?
   (a) 124 (b) 4
   (c) 24 (d) none of these
3. A number of the form $10^n - 1$ is always divisible by 11 for every *n* is a natural number, when :
   (a) *n* is odd (b) *n* is prime
   (c) *n* is even (d) can't say
4. Out of the following numbers which is divisible by 132?
   (a) 31218 (b) 78520
   (c) 38148 (d) 52020
5. If 653*xy* is divisible by 80 then the value of $x + y$ is :
   (a) 2 (b) 3
   (c) 4 (d) 6
6. The value of *k* if *k*35624 is divisible by 11 :
   (a) 2 (b) 5
   (c) 7 (d) 6
7. If 42573*k* is divisible by 72 then the value of *k* is :
   (a) 4 (b) 5
   (c) 6 (d) 7
8. How many numbers between 1 and 1000 are divisible by 7?
   (a) 777 (b) 142
   (c) 143 (d) none of these
9. How many numbers between 55 and 555 including both the extreme values are divisible by 5?
   (a) 100 (b) 111
   (c) 101 (d) none of these
10. How many numbers are there from 100 to 200 ?
    (a) 100 (b) 101
    (c) 99 (d) none of these
11. How many numbers are divisible by 3 in the set of numbers 300, 301, 302, ..., 499, 500?
    (a) 200 (b) 66
    (c) 67 (d) none of these
12. How many numbers are there between 200 and 800 which are divisible by both 5 and 7?

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(a) 35 (b) 16
(c) 17 (d) can't be determined

13. In the above question total numbers in the set of numbers $S = \{200, 201, ..., 800\}$ which are either divisible by 5 or by 7 is :
(a) 210 (b) 190
(c) 199 (d) can't be determined

14. How many numbers are there in the set $S = \{200, 201, 202, ..., 800\}$ which are divisible by neither of 5 or 7?
(a) 411 (b) 412
(c) 410 (d) none of these

15. Total number of numbers lying in the range of 1331 and 3113 which are neither divisible by 2, 3 or 5 is :
(a) 477 (b) 594
(c) 653 (d) none of these

16. Atleast what number must be subtracted from 434079 so that it becomes divisible by 137 ?
(a) 173 (b) 63
(c) 97 (d) can't be determined

17. In the above question, at least what number be added to 434079, so that it will become divisible by (or multiple of) 137 ?
(a) 97 (b) 74
(c) 75 (d) none of these

18. Which one number is closest to 193 which is divisible by 18 is :
(a) 180 (b) 195
(c) 198 (d) 108

19. The product of two numbers $ab7$ and $cd5$ could be, where $ab7$ and $cd5$ are individually three digit numbers :
(a) 8135 (b) 79236
(c) 8735255 (d) none of these

20. When a 3 digit number 984 is added to another 3 digit number $4p3$, we get a four digit number $13q7$, which is divisible by 11. The value of $p + q$ is :
(a) 10 (b) 11
(c) 12 (d) 13

21. When a number divided by 9235, we get the quotient 888 and the remainder 222, such a least possible number is :
(a) 820090 (b) 8200920
(c) 8200680 (d) none of these

22. The number which when divided by 33 is perfectly divisible and closer to 1000 is :
(a) 990 (b) 999
(c) 1023 (d) can't be determined

23. A number which when divided by 32 leaves a remainder of 29. If this number is divided by 8 the remainder will be :
(a) 0 (b) 1
(c) 5 (d) 3

24. A number when divided by 5 leaves a remainder of 4, when the double (*i.e.*, twice) of that number is divided by 5 the remainder will be :
(a) 0 (b) 1
(c) 3 (d) can't be determined

25. When a number 'N' is divided by a proper divisor 'D' then it leaves a remainder of 14 and if the thrice of that number *i.e.*, $3N$ is divided by the same divisor D, the remainder comes out to be 8. Again if the 4 times of the same number *i.e.*, '$4N$' is divided by D the remainder will be :
(a) 35 (b) 22
(c) 5 (d) can't be determined

26. A number when divided by 5 gives a number which is 8 more than the remainder obtained on dividing the same number by 34. Such a least possible number is :
(a) 175 (b) 75
(c) 680 (d) does not exist

27. When a natural number divided by a certain divisor, we get 15 as a remainder. But when the 10 times of the same number is divided by the same divisor we get 6 as a remainder. The maximum possible number of such divisors is :
(a) 6 (b) 7
(c) 15 (d) can't be determined

28. A certain number 'C' when divided by N_1 it leaves a remainder of 13 and when it is divided by N_2 it leaves a remainder of 1, where N_1 and N_2 are the positive integers. Then the value of $N_1 + N_2$ is, if $\frac{N_1}{N_2} = \frac{5}{4}$:
(a) 36
(b) 27
(c) 54
(d) can't be determined uniquely

29. In the above problem the value of c is :
(a) $50 < c < 100$ (b) any multiplied 11
(c) $20 < c < 50$ (d) can't be determined

30. In how many parts a rod of length 19.5 m can be broken of equal length of 65 cm?
(a) 20 (b) 30
(c) 3 (d) 130

31. A six digit number which is consisting of only one digits either 1, 2, 3, 4, 5, 6, 7, 8 or 9, e.g., 111111, 222222... etc. This number is always divisible by :
(a) 7 (b) 11
(c) 13 (d) all of these

32. The maximum possible difference between the 4 digit numbers formed by using the 4 different digits 1, 2, 3, 5 is :
(a) 4086 (b) 5076
(c) 4386 (d) 3242

33. The sum of all digits except the unity that can be substituted at the place of k inorder to be divisible by 8 in the number $23487k2$:
(a) 5 (b) 14
(c) 9 (d) none of these

34. A certain number N when multiplied by 13, the resultant values consists entirely of sevens. The value of N is :
(a) 123459 (b) 58829
(c) 59829 (d) none of these

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SQUARE TABLE

1	1	21	441	41	1681	61	3721	81	6561
2	4	22	484	42	1764	62	3844	82	6724
3	9	23	529	43	1849	63	3969	83	6889
4	16	24	576	44	1936	64	4096	84	7056
5	25	25	625	45	2025	65	4225	85	7225
6	36	26	676	46	2116	66	4356	86	7396
7	49	27	729	47	2209	67	4489	87	7569
8	64	28	784	48	2304	68	4624	88	7744
9	81	29	841	49	2401	69	4761	89	7921
10	100	30	900	50	2500	70	4900	90	8100
11	121	31	961	51	2601	71	5041	91	8281
12	144	32	1024	52	2704	72	5184	92	8464
13	169	33	1089	53	2809	73	5329	93	8649
14	196	34	1156	54	2916	74	5476	94	8836
15	225	35	1225	55	3025	75	5625	95	9025
16	256	36	1296	56	3136	76	5776	96	9216
17	289	37	1369	57	3249	77	5929	97	9409
18	324	38	1444	58	3364	78	6084	98	9604
19	361	39	1521	59	3481	79	6241	99	9801
20	400	40	1600	60	3600	80	6400	100	10,000

SQUARING

Dear students the importance of squaring of numbers is somewhat crucial particularly in the context of CAT. I think that initially you may not consider the importance of squaring but later on you have to delve deep into the basics of the squaring because most of the calculative problems can be solved easily if you learn the squares just upto 100. Get surprised! No there will be no any difficulty in learning these squares, since I have some breakthrough techniques to find the square of any number faster than any one and I can consider that you will find no where these methods and their efficacy as in this book. Now we will proceed step by step toward our learning goal. Last but not the least you will consciously observe the "square table" you will find some peculiarities,as these peculiarities of the numbers (*i.e.*, their distinct nature) help us in remembering for long time and making a network with other numbers.

1. Square of the numbers whose unit (*i.e.*, last) digit is zero (0) : Any number whose last digit is zero, we double the number of zeros at the right side of the number and in the left of there zeros we write the square of the non-zero numbers (or rest digits) which is in the left of the zero(s)

e.g., square of 40 :

$(40)^2 \Rightarrow$ **Step 1.** 00

Step 2. $4^2 = 16$

Step 3. 1600

Similarly square of 700 :

$(700)^2 \Rightarrow$ **Step 1.** 0000

Step 2. $(7)^2 = 49$

Step 3. 490000

Square of 13000 :

$(13000)^2 \Rightarrow$ **Step 1.** 000000

Step 2. $(13)^2 = 169$

Step 3. 169000000

Square of 21000 :

$(21000)^2 \Rightarrow$ **Step 1.** 000000

Step 2. $(21)^2 = 441$

Step 3. 441000000

2. Square of the numbers whose unit digit is 5 : Any number whose unit digit is 5. We write 25 in the rightmost in place of 5, then we multiply the rest (left digits to the 5 of original number) number with its successive number (i,e, one more than the previous one)

For example : (i) The square of 35 :

$(35)^2 \Rightarrow$ **Step 1.** 25

Step 2. $3 \times (3+1) = 12$

Step 3. 1225

(ii) The square of 65 :

$(65)^2 \Rightarrow$ **Step 1.** 25

Step 2. $6 \times (6+1) = 42$

Step 3. 4225

(iii) The square of 75 :

$(75)^2 \Rightarrow$ **Step 1.** 25

Step 2. $7 \times (7+1) = 56$

Step 3. 5625

(iv) The square of 125 :

$(125)^2 \Rightarrow$ **Step 1.** 25

Step 2. $12 \times (12+1) = 156$

Step 3. 15625

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(v) The square of 1405

Step 1. 25

Step 2. $140 \times (140+1) = 19740$

Step 3. 1974025

(vi) The square of 139005

$(139005)^2 \Rightarrow$ **Step 1.** 25

Step 2. $13900 \times (13900+1) = 193223900$

Step 3. 19322390025

(vii) The square of 155 :

$(155)^2 \Rightarrow$ **Step 1.** 25

Step 2. $15 \times (15+1) = 240$

Step 3. 24025

3. (i) Square of the numbers ending with the digit 1 : These numbers are very easy to solve. To get the square of any such number just write the square of the previous number and then add the previous number and the number whose square is being asked.

For example the square of 21

$$(21)^2 = (20)^2 + (20+21) = 441$$

Similarly, $(31)^2 = (30)^2 + (30+31) = 961$

and $(41)^2 = (40)^2 + (40+41) = 1681$

and $(81)^2 = (80)^2 + (80+81) = 6561$

and $(131)^2 = (130)^2 + (130+131)$

$= 16900 + 261 = 17161$

and $(151)^2 = (150)^2 + (150+151)$

$= 22500 + 301 = 22801$

and $(1111)^2 = (1110)^2 + (1110+1111)$

$= (1110)^2 + 2221$

$= 1232100 + 2221 = 1234321$

(ii) Square of the numbers whose unit digit is 9 : It is very similar to the previous case. The difference is that here we have to subtract instead of addition. For example the square of 39

$$(39)^2 = (40)^2 - (39+40)$$

$$= 1600 - 79 = 1521$$

Similarly, $(49)^2 = (50)^2 - (49+50)$

$= 2500 - 99 = 2401$

and $(89)^2 = (90)^2 - (89+90)$

$= 8100 - 179 = 7921$

and $(139)^2 = (140)^2 - (139+140)$

$= 19600 - 279 = 19321$

and $(249)^2 = (250)^2 - (249+250)$

$= 62500 - 499 = 62001$

NOTE In the case of unit digit of 9, we subtract from the base square because base is higher than the number whose square is to be calculated.

(iii) Square of the numbers whose last digit is 2 or 8:

For example $(52)^2 = (50)^2 + 2(50+52)$

$= 2500 + 204 = 2704$

Similarly, $(112)^2 = (110)^2 + 2(110+112)$

$= 12100 + 444 = 12544$

and $(38)^2 = (40)^2 - 2(38+40)$

$= 1600 - 156 = 1444$

and $(88)^2 = (90)^2 - 2(88+90)$

$= 8100 - 356 = 7744$

(iv) Square of the numbers whose unit digit is 3 or 7:

For example : $(23)^2 = (20)^2 + 3(20+23)$

$= 400 + 129 = 529$

Similarly, $(53)^2 = (50)^2 + 3(50+53)$

$= 2500 + 309 = 2809$

and $(123)^2 = (120)^2 + 3(120+123)$

$= 14400 + 729 = 15129$

and $(47)^2 = (50)^2 - 3(47+50)$

$= 2500 - 291 = 2209$

and $(137)^2 = (140)^2 - 3(137+140)$

$= 19600 - 831 = 18769$

and $(2347)^2 = (2350)^2 - 3(2347+2350)$

$= 5522500 - 4691 = 5517809$

(v) Square of the numbers which ends with 4 or 6 :

For example : $(34)^2 = (30)^2 + 4(30+34)$

$= 900 + 256 = 1156$

or $(34)^2 = (35)^2 - (34+35)$

$= 1225 - 69 = 1156$

Similarly, $(36)^2 = (35)^2 + (35+36)$

$= 1225 + 71 = 1296$

and $(126)^2 = (125)^2 + (125+126)$

$= 15625 + 251 = 15876$

In general square of any number can be found by using the square of any convenient number as base square *e.g.*, the numbers whose unit digit is either 0 or 5, since the square of these numbers is easy to find and learn. However we can calculate the square by considering any base square. For example if

$$(32)^2 = (30)^2 + 2(30+32) = 1024$$

or $(32)^2 = (31)^2 + (30+31)$

$= 961 + 61 = 1024$

or $(32)^2 = (33)^2 - (32+33)$

$= 1089 - 65 = 1024$

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or $(32)^2 = (35)^2 - 3(32+35)$

$= 1225 - 201 = 1024$

Now, it can be generalized as :

If N is the number whose square is to be calculated and B is the base and d is difference between N and B then

$(N)^2 = (B)^2 + d(B+N)$ when $B < N$

or $(N)^2 = B^2 - d(B+N)$ when $B > N$

Properties of Squares

(i) If the unit digit of the number is zero then the unit digit of the square of this number will also be zero and the number of zeros will be double in the square than that of its root.

e.g., $(60)^2 = 3600$, $(130)^2 = 16900$

(ii) If the unit digit of the number is 5 then the unit digit of its square is also 5 and the number formed by last two digits is 25.

e.g., $(35)^2 = 1225$, $(45)^2 = 2025$, $(55)^2 = 3025$ etc.

(iii) If the unit digit of any number is 1 or 9 then the unit digit of the square of its number is always 1.

e.g., $(71)^2 = 5041$, $(31)^2 = 961$, $(19)^2 = 361$

(iv) If the unit digit of any number is 2 or 8 then the unit digit of the square of its number is always 4.

(v) If the unit digit of any number is 3 or 7 then the unit digit of its square is always 9.

e.g., $(23)^2 = 529$, $(27)^2 = 729$

(vi) If the unit digit of any number is 4 or 6 then the unit digit of its square is always 6.

e.g., $(26)^2 = 676$, $(24)^2 = 576$,

$(14)^2 = 196$, $(16)^2 = 256$ etc.

(vii) The square of any number is always positive irrespective of the nature of the given number.

Remember :

(a) The numbers with unit digit 0, 1, 5 and 6 always give the same unit digits respectively, on squaring.

(b) 2, 3, 7 and 8 never appear as unit digit in the square of a number.

Perfect Square : The square of any natural number is known as perfect square *e.g.*, 1,4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, ... etc. are the squares of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, ... etc.

SQUARE ROOT

A number n is called the square root of a number $n^2 = (n \times n)$ where n^2 is obtained by multiplying the number n with itself only once. The symbol for square root is '$\sqrt{\ }$'.

For example the square root of 4 *i.e.*, $\sqrt{4} = 2$, and $\sqrt{9} = 3$, $\sqrt{25} = 5$ etc.

Generally there are two methods for finding the square root :

(i) Prime Factorisation method

(ii) Division method

(i) Factorisation Method : In this method first we find out the prime factors and then we paired them as given below.

EXAMPLE 1 *Find the square root of* 3600.

SOLUTION The factors of 3600

$= \underline{2 \times 2} \times \underline{2 \times 2} \times \underline{3 \times 3} \times \underline{5 \times 5}$

So the square root of 3600 *i.e.*,

$\sqrt{3600} = 2 \times 2 \times 3 \times 5$

$= 60$

2	3600
2	1800
2	900
2	450
5	225
5	45
3	9
3	3
	1

EXAMPLE 2 *Find the square root of* 144.

SOLUTION Factors of $144 = \underline{2 \times 2} \times \underline{2 \times 2} \times \underline{3 \times 3}$

So $\sqrt{144} = 2 \times 2 \times 3$

$\sqrt{144} = 12$

2	144
2	72
2	36
2	18
3	9
3	3
	1

EXAMPLE 3 *Find the square root of* 10404.

SOLUTION The factors of 10404

$= \underline{2 \times 2} \times \underline{3 \times 3} \times \underline{17 \times 17}$

$\therefore \sqrt{10404} = 2 \times 3 \times 17$

or $\sqrt{10404} = 102$

2	10404
2	5202
3	2601
3	867
17	289
17	17
	1

(ii) Division Method : In this method first of all we make pairs from the right side towards left and then solve as given below.

EXAMPLE 1 *Find the square root of* 3600.

SOLUTION

$\therefore \sqrt{3600} = 60$

	60
6	36 00
6	36
120	00 00
0	00 00

EXAMPLE 2 *Find the square root of* 10404.

SOLUTION

$\therefore \sqrt{10404} = 102$

	102
1	1 04 04
1	1
202	0 04 04
02	4 04
	× ×

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EXAMPLE 3 *Find the square root of 15876.*

SOLUTION

	126
1	1 58 76
1	1
22	58
2	44
246	1476
6	1476
	×

$\therefore \quad \sqrt{15876} = 126$

EXAMPLE 4. *Find the square root of 120409.*

SOLUTION

	347
3	12 04 09
3	9
64	304
4	256
687	4809
7	4809
	×

$\therefore \quad \sqrt{120409} = 347$

EXAMPLE 5 *Find the square root of 5793649.*

SOLUTION

	2407
2	5 79 36 49
2	4
44	179
4	176
4807	33649
07	33649
	×

$\therefore \quad \sqrt{5793649} = 2407$

Did you know If $(11)^2 = 121$ then $(101)^2 = 10201$
$(21)^2 = 441$ then $(201)^2 = 40401$ and so on...
If $(11)^2 = 121$ then $(1001)^2 = 1002001$
$(31)^2 = 961$ then $(3001)^2 = 9006001$
$(12)^2 = 144$ then $(1002)^2 = 1004004$ and so on
Go in depth and find the limitation of this logic ...

Did you know

$1^2 = 1$
$11^2 = 121$
$111^2 = 12321$
$1111^2 = 1234321$
$11111^2 = 123454321$
$111111^2 = 12345654321$
... ...
... ...
... ...

INTRODUCTORY EXERCISE-1.2

1. The square root of 6280036 is :
   (a) 1308 (b) 2903
   (c) 2506 (d) none of these
2. The square root of 1296 is :
   (a) 33 (b) 44
   (c) 34 (d) 36
3. The square root of 7744 is :
   (a) 94 (b) 88
   (c) 77 (d) none of these
4. The square root of 56169 is :
   (a) 359 (b) 323
   (c) 227 (d) none of these
5. The square root of 1238578 is :
   (a) 3254 (b) 3724
   (c) 3258 (d) none of these
6. The least number by which we multiply to the 11760, so that we can get a perfect square number :
   (a) 2 (b) 3
   (c) 5 (d) none of these
7. In the above question, by which least possible number we divide to the 11760 so the resultant number becomes a perfect square :
   (a) 3 (b) 15
   (c) 7 (d) can't be determined
8. The least possible positive number which should be added to 575 to make a perfect square number is :
   (a) 0 (b) 1
   (c) 4 (d) none of these
9. The least possible number which must be subtracted from 575 to make a perfect square number is :
   (a) 5 (b) 50
   (c) 46 (d) 37
10. The square of a number 'A' is the sum of the square of other two numbers 'B' and 'C'. Where $5B = 12C$ and B, C are positive numbers. The least possible positive value of A is :
    (a) 10 (b) 12
    (c) 13 (d) 16
11. Lieutenant Kalia when arranged all his 1500 soldiers in such a way that the number of soldiers in a line were the same as there were the number of lines. So he was left with 56 soldiers, who were not a part of this arrangement. The number of lines in this arrangement is :
    (a) 44 (b) 36
    (c) 38 (d) none of these
12. $\sqrt{289} \div \sqrt{x} = \frac{1}{5}$, then the value of x is :
    (a) $\frac{17}{25}$ (b) $\frac{34}{35}$
    (c) 235 (d) 7225
13. Out of the following statements which one is incorrect?
    (a) $\sqrt{5184} = 72$ (b) $\sqrt{15625} = 125$
    (c) $\sqrt{1444} = 38$ (d) $\sqrt{1296} = 34$
14. If $2 * 3 = \sqrt{13}$ and $3 * 4 = 5$, then the value of $5 * 12$ is :
    (a) 17 (b) $\sqrt{29}$
    (c) 21 (d) 13
15. If $a * b * c = \sqrt{\frac{(a+2)(b+3)}{(c+1)}}$, then the value of $(6 * 15 * 3)$ is :
    (a) 6
    (b) 3
    (c) 4
    (d) can't be determined

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CUBE

If a number is multiplied by itself, twice, then the resultant value is called as the cube of that number. For example $n \times n \times n = n^3$, read as n cube. As most of you are aware of the ice cubes, dice and some boxes, which are the examples of cubes, so there is no problem in getting it.

If we say that what is the cube of 4 then it means

$$4^3 = 4 \times 4 \times 4 = 64$$

Similarly $7^3 = 7 \times 7 \times 7 = 343$

and $8^3 = 8 \times 8 \times 8 = 512$

But for the greater (or larger) number sometimes it becomes tedious to calculate the cube very quickly. So use a new approach to calculate the cube of larger numbers from 11 to 99. Since you can easily calculate the cubes of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30, 40, 50, 60, 70, 80, 90 and 100 etc. So you can use it for further calculation.

The most important point which you should know that a CAT aspirant must use his brain with his own genius, because it is a test of virtual intelligence in real time. So you have to delve deep into your own mind and evolve the new techniques, and solutions as an effective manager. So think now.

Method to find cube of two digit numbers.

Step 1. Write down the cube of tens digit.

Step 2. Write down the resultant value in the same ratio as the ratio of tens and unit digit in four terms as a G.P.

Step 3. Write down twice the value obtained in the second and third term below the second and third term respectively.

Step 4. Now you can add all the numbers in the prescribed manner as given in the following examples :

NOTE While doing the addition (in step 4) subscripts should be treated as a "carryover" for the next term.

EXAMPLE 1 $(11)^3 \Rightarrow$ **Step 1.** $1^3 = 1$

Step 2. 1 1 1 1

Step 3. 1 1 1 1
+ 2 2

Step 4. 1 3 3 1

$\therefore$ $(11)^3 = 1331$

EXAMPLE 2 $(12)^3 \Rightarrow$ **Step 1.** $1^3 = 1$

Step 2. 1 2 4 8

Step 3. 1 2 4 8
4 8

Step 4. 1 7 2 8

$\therefore$ $(12)^3 = 1728$

EXAMPLE 3 $(13)^3 \Rightarrow$ **Step 1.** $1^3 = 1$

Step 2. 1 3 9 27

Step 3. 1 3 9 27
6 18

Step 4. 2 1 $_2$9 $_2$7

$\therefore$ $(13)^3 = 2197$

EXAMPLE 4 $(25)^3 \Rightarrow$ **Step 1.** $2^3 = 8$

Step 2. 8 20 50 125

Step 3. 8 20 50 125
40 100

Step 4. 15 $_7$6 $_{16}$2 $_{12}$5

$\therefore$ $(25)^3 = 15625$

EXAMPLE 5 $(26)^3 \Rightarrow$ **Step 1.** $2^3 = 8$

Step 2. 8 24 72 216

Step 3. 8 24 72 216
+ 48 144

Step 4. 17 $_9$5 $_{23}$7 $_{21}$6

$\therefore$ $(26)^3 = 17576$

EXAMPLE 6 $(27)^3 \Rightarrow$ **Step 1.** $2^3 = 8$

Step 2. 8 28 98 343

Step 3. 8 28 98 343
+ 56 196

Step 4. 19 $_{11}$6 $_{32}$8 $_{34}$3

$\therefore$ $(27)^3 = 19683$

EXAMPLE 7 $(53)^3 \Rightarrow$ **Step 1.** $5^3 = 125$

Step 2. 125 75 45 27

Step 3. 125 75 45 27
+ 150 90

Step 4. 148 $_{23}$8 $_{13}$7 $_2$7

$\therefore$ $(53)^3 = 148877$

EXAMPLE 8 $(67)^3 \Rightarrow$ **Step 1.** $(6)^3 = 216$

Step 2. 216 252 294 343

Step 3. 216 252 294 343
+ 504 588

Step 4. 300 $_{84}$7 $_{91}$6 $_{34}$3

$\therefore$ $(67)^3 = 300763$

EXAMPLE 9 $(94)^3 \Rightarrow$ **Step 1.** $(9)^3 = 729$

Step 2. 729 324 144 64

Step 3. 729 324 144 64
+ 648 288

Step 4. 830 $_{101}$5 $_{43}$8 $_6$4

$\therefore$ $(94)^3 = 830584$

EXAMPLE 10 $(88)^3 \Rightarrow$ **Step 1.** $(8)^3 = 512$

Step 2. 512 512 512 512

Step 3. 512 512 512 512
+ 1024 1024

Step 4. 681 $_{169}$4 $_{158}$7 $_{51}$2

$\therefore$ $(88)^3 = 681472$

EXAMPLE 11 $(55)^3 \Rightarrow$ **Step 1.** $(5)^3 = 125$

Step 2. 125 125 125 125

Step 3. 125 125 125 125
+ 250 250

Step 4. 166 $_{41}$3 $_{38}$7 $_{12}$5

$\therefore$ $(55)^3 = 166375$

EXAMPLE 12 $(72)^3 \Rightarrow$ **Step 1.** $(7)^3 = 343$

Step 2. 343 98 28 8

Step 3. 343 98 28 8
+ 196 56

Step 4. 373 $_{30}$2 $_8$4 8

$\therefore$ $(72)^3 = 373248$

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CUBE ROOT

The cube root of a number n is expressed as $\sqrt[3]{n}$ or $(n)^{1/3}$ or in other words $n = (n)^{1/3} \times (n^{1/3}) \times (n^{1/3})$

For example cube root of 27 *i.e.*, $\sqrt[3]{27} = 3$

and $\sqrt[3]{343} = 7$ and $\sqrt[3]{64} = 4$

EXAMPLE 1 *Find the cube root of* 125.

SOLUTION $\because \quad 125 = 5 \times 5 \times 5$

$\therefore \quad \sqrt[3]{125} = 5$

5	125
5	25
5	5
	1

EXAMPLE 2 *Find the cube root of* 1728.

SOLUTION $1728 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3$

$\sqrt[3]{1728} = 2 \times 2 \times 3$

$\sqrt[3]{1728} = 12$

2	1728
2	864
2	432
2	216
2	108
2	54
3	27
3	9
3	3
	1

NOTE In this method first we factorize the given numbers then we make the triplet of the factors and thus we multiply these triplets to get the cube root of the given number.

EXAMPLE 3 *On 14th November, in my school, each child received as many packs of chocolates as there were total number of the students in the school. Further, each pack of chocolates contains as many chocolates as there were the total number of packs which a child had. Total how many chocolates have been distributed among all the children of school :*

(a) 729 (b) 196
(c) 961 (d) can't be determined

SOLUTION (d) is the appropriate answer since we don't have sufficient data to calculate.

EXAMPLE 4 *In the above question if there were 8 packs of chocolates with every child. Then the total number of chocolates distributed among them was :*

(a) 512 (b) 64
(c) 256 (d) can't be determined

SOLUTION Let there are n children it means each child has n packs of chocolates and since every pack contains n chocolates. Therefore total number of chocolates

$= n \times n \times n = n^3$

Hence total number of chocolates $= (8)^3 = 512$

Thus option (a) is correct.

PRACTICE EXERCISE

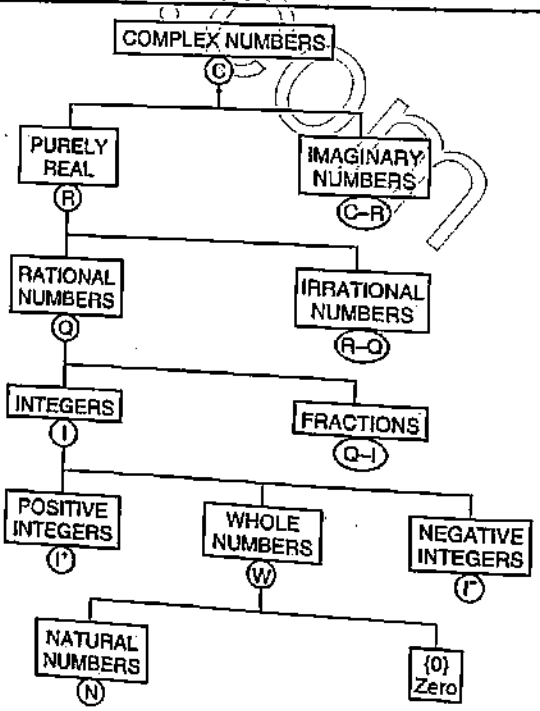
1. The least possible number which we should add to 1720 to make a perfect cube number is :
(a) 0 (b) 1
(c) 8 (d) 7
2. The least positive number which is subtracted from 1369 to make it a perfect cube is :
(a) 17 (b) 38
(c) 34 (d) none of these
3. The least possible natural number by which if we multiply to the 1372, we get a perfect cube number is :
(a) 2 (b) 3
(c) 5 (d) can't be determined
4. The least possible number by which if we divide 1372, it will become a perfect cube number is :
(a) 2 (b) 7
(c) 3 (d) 4

ANSWERS

1. (c) 2. (b) 3. (a) 4. (d)

NUMBER SYSTEMS

It has been said by a great mathematician that "Mathematics is the queen of Science and Arithmetic is the queen of Mathematics." Arithmetic is the maths of numerals and digits and so the other branches of Maths are dependent upon Arithmetic. Either the counting of the currency or counting of the stars in all the ways it has been a very popular subject especially among Indians. One thing which is very well known that the great Indian Mathematician Aryabhatta has introduced '0' (zero) in the counting numbers and the whole world has adopted it. None other than computer is dead if this figure does not exist. So one can say zero has revolutionised the whole world. Here we will discuss all sorts of numbers, their properties, application besides the different number systems used widely. The following chart briefly illustrates the different kinds of numbers with their relationship.



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Natural Numbers : The counting numbers 1, 2, 3, 4, 5, ... are called the natural numbers and are denoted by N. As you can recall that a child at his very young stage try to count the things *e.g.,* chocolates, toys, sweets etc. He just know the numbers *viz.*, 1, 2, 3, 4, 5, ... and has no idea about zero. Here we are discussing the numbers which are represented in international symbols. So $N = \{1, 2, 3, 4, 5, 6, 7...\}$

PROPERTIES OF NATURAL NUMBERS

1. **Successor :** The next natural number just after any natural number n is called its successor 'n^+' where $n^+ = n+1$ for example the successor of 2 is 3, successor of 6 is 7 etc.
2. **Closure law :** For any two natural numbers a and b
   $(a+b) \in N$ and $(a \times b) \in N$
   e.g., $3+4=7 \in N$ **and** $3 \times 4 = 12 \in N$
3. **Commutative law :** For any two natural numbers a and b
   $a+b=b+a$
   and $a \times b = b \times a$
   e.g., $5+6=6+5=11$
   and $5 \times 6 = 6 \times 5 = 30$
4. **Associative law :** For any three natural numbers
   $(a+b)+c=a+(b+c)$
   $(a \times b) \times c = a \times (b \times c)$
   e.g., $(7+8)+9=7+(8+9)=24$
   $(7 \times 8) \times 9 = 7 \times (8 \times 9) = 504$
5. **Multiplicative Identity :** 1 is the multiplicative identity of every natural number as
   $4 \times 1 = 4$
   $5 \times 1 = 5$
   $10 \times 1 = 10$
   $17 \times 1 = 17$
6. **Cancellation law :** For any three natural numbers a, b, c
   $a+b=c+b \Rightarrow a=c$
   and $a \times b = c \times b \Rightarrow a=c$
7. **Distributive law :** For any three natural numbers a, b, c
   $a \times (b+c) = a \times b + a \times c$
   but $a+(b \times c) \neq (a+b) \times (a+c)$
   e.g., $3 \times (4+5) = 3 \times 4 + 3 \times 5 = 27$

Even Numbers : All the natural numbers which are divisible by 2 are known as even numbers *e.g.*, 2, 4, 6, 8, 10, ...

Odd Numbers : All the natural numbers which are not divisible by 2 are known as 'Odd numbers' *e.g.*, 1, 3, 5, 7, ...

IMPORTANT NOTE

- Even + Even = Even
- Even − Even = Even
- Even × Even = Even
- Even + Even = Even or odd

- Odd + Odd = Even
- Odd − Odd = Even
- Odd × Odd = Odd
- Odd ÷ Odd = Odd

- Even + Odd = Odd
- Even − Odd = Odd
- Even × Odd = Even
- Even ÷ Odd = Even

- Odd + Even = Odd
- Odd − Even = Odd
- Odd × Even = Even
- Odd ÷ Even = (never divisible)

- $(\text{even})^{\text{even/odd}} = \text{Even}$
- $(\text{odd})^{\text{odd/even}} = \text{Odd}$

Prime Numbers : Except 1 each natural number which is divisible by only 1 and itself is called as prime number *e.g.*, 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, ... etc.

- There are total 25 prime numbers upto 100
- There are total 46 prime numbers upto 200
- 2 is the only even prime number and the least prime number.
- 1 is neither prime nor composite number.
- There are infinite prime numbers.
- A list of all the prime numbers upto 100 is given below.

Table of Prime Numbers (1 – 100) :

2	11	23	31	41	53	61	71	83	97
3	13	29	37	43	59	67	73	89	
5	17			47			79		
7	19								

How to test whether a number is prime or not : To test a number n take the square root of n and consider as it is , if it is a natural number otherwise just increase the square root of it to the next natural number. Then divide the given number by all the prime numbers below the square root obtained. If the number is divisible by any of these prime numbers then it is not a prime number else it is a prime number.

EXAMPLE *Check that whether 241 is prime.*

SOLUTION When we take the square root of 241 it is approximate 15, so we consider it 16. Now we divide 241 by all the prime numbers below 16 *viz.*, 2, 3, 5, 7, 11, 13

Since 241 is not divisible by any one of the prime numbers below 16. So it is a **prime number.**

NOTE Any digit if it is written continuously 3 times, 6 times, 9 times ... etc. then it is divisible by 3
e.g., 111; 555555; 777777; 222222222; 888, 222 etc.

Co-prime Numbers : Two natural numbers are called co-prime (or relatively prime) numbers if they have no common factor other than 1 or in other words. The highest common factor *i.e.*, H.C.F. between co-prime numbers is 1. *e.g.*, (15, 16), (14, 25), (8, 9), (13, 15) etc.

NOTE It is not necessary that the numbers involved in the pair of co-primes will be prime even they can be composite numbers as seen in the above examples.

Composite Numbers : A number other than one which is not a prime number is called a composite number. It means it

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is divisible by some other number(s) other than 1 and the number itself. *e.g.*, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26...

- Every natural number except 1 is either prime or composite.

Trichotomy law : If there are any two natural numbers a and b then there exists one and only one relation necessarily is

(i) $a > b$ (ii) $a = b$ (iii) $a < b$

Whole Numbers : The extended set of natural numbers in which '0' is also included is called as the set of whole numbers and is denoted by $W = \{0, 1, 2, 3, 4, 5, ...\}$

- '0' (zero) is an even number

Consecutive Numbers : A series of numbers in which the next number is 1 more than the previous number or the predecessor number is 1 less than the successor or just they can be differed by 1 *e.g.*, 10, 11, 12, or 17, 18, 19, 20, 21 or 717, 718, 719, 720, 721... etc.

Perfect Number : When the sum of all the factors (including 1 but excluding the number itself) of the given number is the same number then this number is called as Perfect Number. For example 6, 28, 496, 8128, ...etc

So far only 27 perfect numbers are known.

As the factors of 28 are 1, 2, 4, 7, 14, 28

Now, $1 + 2 + 4 + 7 + 14 = 28$

Hence 28 is a perfect Number.

Triangular Number : A triangular number is obtained by adding the previous number to the nth position in the sequence of triangular numbers, where the first triangular number is 1. The sequence of triangular numbers is given as follows

1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, ...etc.

Integers : The extended set of whole numbers in which negative integers are also included is known as the set of integers and is denoted by

Z or $I = \{..., -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, ...\}$

(a) **Positive integers :** The set of integers $\{1, 2, 3, ...\}$ is known as positive integers.

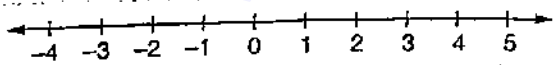
(b) **Negative integers :** The set of integers $\{-1, -2, -3, -4, ...\}$ is known as negative integers.

(c) **Non-negative integers :** The set of integers $\{0, 1, 2, 3, 3....\}$ is called as non-negative integers.

(d) **Non-positive integers :** The set of integers $\{0, -1, -2, -3, ...\}$ is called as non-positive integers.

- '0' is neither positive nor negative integer.

REPRESENTATION OF THE INTEGERS ON A NUMBER LINE



All the integers (whole number, natural number etc.) can be shown on this number line, where every integer is represented by some point on the line.

It can be said that :

(a) There is no any largest or smallest integer.

(b) Every integer has a predecessor and a successor.

(c) An integer is smaller than all those integers which are on the right side of it and is greater than all those integers which are on the left side of it on the number line.

e.g., $-4 > -5, 3 > 1, 10 > -4$ etc.

or $-6 < 0, -2 < 1$ etc.

EXAMPLE 1 *Arrange the following integers in ascending order*
$-3, -7, 8, 5, 0, 3, 17, -23$

SOLUTION $-23, -7, -3, 0, 3, 5, 8, 17$

EXAMPLE 2 *Arrange the following integers in descending order*
$-17, 18, 15, 32, 81, -5, 87.$

SOLUTION $87, 81, 32, 18, 15, -5, -17$

Properties of Integers

1. Closure law is followed by all the integers
2. Commutative law and Associative law is not followed by all the integers for the subtraction.

e.g., $4 - 6 \neq 6 - 4$

and $(4 - 6) - 2 \neq 4 - (6 - 2)$

but it is valid for the addition and multiplication as

$4 + 6 = 6 + 4$

or $(-3) + (-2) = (-2) + (-3)$

and $4 \times 6 = 6 \times 4$

or $-3 \times -2 = -2 \times -3$

and $(-2) + [(-3) + (-7)] = [(-2) + (-3)] + (-7)$ etc.

3. Additive identity of all the integers is zero (0) and Multiplicative identity of all the integers is 1.

e.g., $-3 + 0 = -3, \quad 8 + 0 = 8$

and $-5 \times 1 = -5, \quad 7 \times 1 = 7$ etc.

4. Additive inverse of an integer a is $-a$.

e.g. the additive inverse of 7, 8, 9, 15, $-3, -5, -6$ etc. are $-7, -8, -9, -15, 3, 5$ and 6 respectively.

5. Distributive law of multiplication over addition or subtraction

i.e., $a \times (b \pm c) = a \times b \pm a \times c$

For example: $3 \times (4 \pm 6) = 3 \times 4 \pm 3 \times 6$

NOTE
1. Division by zero is not defined in Mathematics.
2. Division by 1 is actually unification (not division) so it is an improper divisor.

Some important rules regarding the sign convention in mathematical operations :

(i) $(a) + (b) = +(a + b)$

$(-a) + (b) = b - a$

$(a) + (-b) = a - b$

$(-a) + (-b) = -(a + b)$

i.e., $(+) + (+) = +$

$(-) + (+) = +$ if the numerical value of $+$ is greater

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$(-)+(+)=-$ if the numerical value of $-$ is greater

$(-)+(-)=-$

For example $(4)+(7)=11, (-3)+(8)=5$

$(-8)+(3)=-5$

$(-5)+(-3)=-8$ etc.

Similarly, $(5)-(2)=3, (2)-(5)=-3$

$(-5)-(2)=-7, (5)-(-2)=7$

$(-5)-(-2)=-3, (-2)-(-5)=3$

(ii) $a\times b=ab$ *i.e.,* $(+)\times(+)=+$

$-a\times b=-ab$ *i.e.,* $(-)\times(+)=-$

$a\times -b=-ab$ *i.e.,* $(+)\times(-)=-$

$-a\times -b=ab$ *i.e.,* $(-)\times(-)=+$

For example, $(2)\times(3)=6, (-2)\times(3)=-6, 2\times(-3)=-6$ and $(-2)\times(-3)=6$

Similarly, $\frac{8}{2}=4, \frac{-8}{2}=-4, \frac{8}{-2}=-4, \frac{-8}{-2}=4$

i.e., $\frac{(+)}{(+)}=(+), \frac{(-)}{(+)}=(-), \frac{(+)}{(-)}=(-), \frac{(-)}{(-)}=(+)$

PRACTICE EXERCISE

1. Evaluate the following

(a) $4+(-5)$ (b) $-4-(-2)$

(c) $8-(-5)$ (d) $-13+(56)$

(e) $(-94)+(-239)$ (f) $(-526)-(-217)$

(g) $7+(-5)+(-2)$ (h) $-6+(-2)-(-3)+1$

(i) 6×9 (j) $(-9)\times(-13)$

(k) -7×8 (l) $13\times(-15)$

(m) $(-3)\times(8)\times(-5)$ (n) $(-8)\times(-19)\times(-15)$

(o) $[(-6)\times(-8)]\times5$ (p) $(-18)\div6$

(q) $126\div(-14)$ (r) $-13\times(7-8)$

(s) $(-32)\div(-4)$ (t) $(-31)+(31)$

ANSWERS

(a) -1	(b) -2	(c) 13	(d) 43	(e) -333
(f) -309	(g) 0	(h) -4	(i) 54	(j) 117
(k) -56	(l) -195	(m) 120	(n) -2280	(o) 240
(p) -3	(q) -9	(r) 13	(s) 8	(t) 0

Numerical Expression : Collection of numbers connected by one or more operations of addition, subtraction, multiplication and division is called a numerical expression. It can also involve some brackets.

e.g., $7+18\div3\div1\times6$

$84-6\div2-3\times(-7)$

and $9+(-2)\times\{72\div8\}+21$ etc.

RULE OF SIMPLIFICATION OR CALCULATION

The order in which various mathematical operations must be done can be remembered with the word 'BODMAS'

Where B → Brackets

O → Off

D → Division

M → Multiplication

A → Addition

S → Subtraction

So first of all we solve the inner most brackets moving outwards. Then we perform 'of' which means multiplication then, Division, Addition and Subtraction.

- Addition and Subtraction can be done together or separately as required.
- Between any two brackets if there is no any sign of '+' or '–' it means we have to do multiplication *e.g.,*

$(7)(2)=7\times2=14$

$[3(5)+7]=15+7=22$

Brackets : They are used for the grouping of things or entities. The various kind of brackets are :

(i) '–' is known as line (or bar) bracket or vinculum

(ii) ( ) is known as parenthesis or common bracket

(iii) { } is known as curly bracket or brace.

(iv) [ ] is known as rectangular (or big) bracket.

The order of eliminating brackets is :

(i) line bracket

(ii) common bracket

(iii) curly bracket

(iv) rectangular bracket

EXAMPLE 1 *Solve the following expressions :*

(a) $3+2-1\times4+2$ *(b)* $7\times3+8-2$

(c) $53\times2-1\times6$ *(d)* $12+(-3)+5-(-2)$

SOLUTION (a) $3+2-1\times4+2=3+2-1\times2=3+2-2=3$

(b) $7\times3+8-2=21+8-2=27$

(c) $53\times2-1\times6=106-6=100$

(d) $12+(-3)+5-(-2)=12-3+5+2=9+7=16$

EXAMPLE 2 *Evaluate or simplify the following expression*

(a) $9-\{7-24\div(8+6\times2-16)\}$

(b) $(-3)\times(-12)\div(-4)+3\times6$

(c) $17-\{8\div(2\times3-4)\}$

(d) $5\times2-[3-\{5-(7+2\text{ of }4-19)\}]$

SOLUTION (a) $9-\{7-24\div(8+6\times2-16)\}$

$=9-\{7-24\div(8+12-16)\}$

$=9-\{7-24\div4\}$

$=9-\{7-6\}=9-\{1\}=8$

(b) $(-3)\times(-12)\div(-4)+3\times6=36\div(-4)+18$

$=-9+18=9$

(c) $17-\{8\div(2\times3-4)\}=17-\{8\div(6-4)\}$

$=17-\{8\div2\}=17-\{4\}=13$

(d) $5\times2-[3-\{5-(7+2\text{ of }4-19)\}]$

$=5\times2-[3-\{5-(7+8-19)\}]$

$=10-[3-\{5-(-4)\}]$

$=10-[3-\{9\}]$

$=10-[-6]=10+6=16$

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EXAMPLE 3 *Which one of the following options is correct in the given expressions?*

(i) $45-[28-\{37-(15-k)\}]=58$, *the value of k is :*

(a) 19 (b) -19

(c) -39 (d) *none of these*

(ii) $1+[1\div\{5\div4-1\div(13\div3-1+3)\}]$ *equal to :*

(a) 2/5 (b) 2

(c) 3/2 (d) *none of these*

(iii) $2-[3-\{6-(5\div\overline{4-3})\}]$:

(a) 0 (b) 1

(c) -3 (d) *none of these*

ANS. (i) (a), (ii) (b), (iii) (a)

EXAMPLE 4 *A student gets 4 marks for a correct answer and 1 mark is deducted for a wrong answer. If she has done 80 questions at all and she has gotten only 240 marks. The correct answers she has attempted is :*

(a) 40 (b) 60

(c) 64 (d) *can't be determined*

SOLUTION If she has done all the 80 questions correctly she must have gotten 320 $(=80\times4)$ marks, but if she does one wrong answer it means she is liable to lose 5 marks $(4+1)$. Thus for every wrong answer she loses 5 marks.

Now since she has lost total 80 marks $(320-240=80)$.

This implies that she has done 16 questions wrong $\left(\because \frac{80}{5}=16\right)$. It means she has done only 64 questions $(80-16=64)$ correctly. Hence (c) is the correct option.

Alternatively: Since she has obtained less marks than the maximum possible marks it means she must has done some questions wrong. Now, we can check through options. Let us consider option (c).

Correct answer	Wrong answers	Total questions
64	16	80
Marks for correct answers	Marks for wrong answers	Net marks
$64\times4=256$	$16\times-1=-16$	$256-16=240$

Hence the assumed option (c) is correct.

Alternatively : If she has done x question correctly it means she has done $(80-x)$ questions wrong.

So $4\times(x)-1\times(80-x)=240$

$\Rightarrow 5x-80=240$

$\Rightarrow 5x=320$

$\Rightarrow x=64$

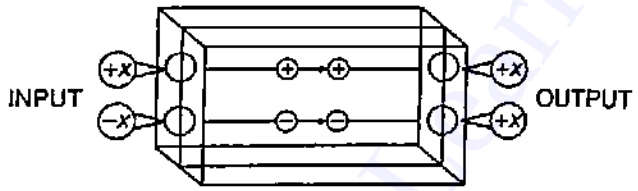
Hence she has done 64 correct answers.

Absolute value of an integer or Modulus : The absolute value of an integer is its numerical value irrespective of its sign (or nature)

The absolute value of an integer x is written as $|x|$ and is defined as

$$|x|=\begin{cases}x \text{ if } x\geq 0\\ -x \text{ if } x<0\end{cases}$$

MODULUS (MACHINE MODEL)



It means any integer whether it is positive or negative if it is operated upon modulus, it always gives a positive integer.

for example $|-7|=7, |-4|=4$

$|3|=3$ etc.

Also, $\text{Max}\{x,-x\}=|x|$

and $-\text{Min}\{x,-x\}=|x|$

and $\sqrt{(x)^2}=|x|$

Properties of a Modulus or Mod

1. $|a|=|-a|$ 2. $|ab|=|a||b|$
3. $\left|\frac{a}{b}\right|=\frac{|a|}{|b|}$
4. $|a+b|\leq|a|+|b|$

(The sign of equality holds only when the sign of a and b are same)

5. If $|a|\leq k \Rightarrow -k\leq a\leq k$
6. If $|a-b|\leq k \Rightarrow -k\leq a-b\leq k \Rightarrow b-k\leq a\leq b+k$

EXAMPLE 1 *Solution set of the equation* $|x-2|=5$ *is :*

(a) $\{3,-7\}$ (b) $\{-3,7\}$

(c) $\{3,6\}$ (d) *none of these*

SOLUTION $|x-2|=5 \Rightarrow x-2=5$ or $x-2=-5$

$\Rightarrow x=7$ or $x=-3$

Hence $x=\{-3,7\}$, (b) is the correct option.

NOTE
- { } consists only the mentioned value particularly,
- ( ) does not include the extreme values
- [ ] includes the values of extremity
- ( ] includes only the highest value of extreme but not the lowest extreme value
- [ ) includes only lowest extreme value but not includes the highest extreme value.

EXAMPLE 2 *The maximum value of the expression* $27-|9x-8|$ *is:*

(a) 27 (b) 17 (c) 44 (d) 26

SOLUTION The maximum value of the expression $27-|9x-8|$ will be maximum only when the value of $|9x-8|$ is minimum, but the minimum possible value of any $|k|$ is zero. Hence the maximum value of

$$27-|9x-8|=27-(0)=27$$

Hence option (a) is correct.

EXAMPLE 3 *The minimum value of the expression* $|17x-8|-9$ *is:*

(a) 0 (b) -9

(c) $\frac{8}{17}$ (d) *none of these*

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SOLUTION The minimum value of expression $|17x-8|-9$ is minimum only when $|17x-8|$ is minimum. But the minimum value of $|k|$ is zero.

Hence minimum value of $|17x-8|-9 = 0-9 = -9$

EXAMPLE 4 *If* $2a-9=b+a$, *then the value of* $(|a-b|+|b-a|)$ *is :*

(a) 18 *(b) 11* *(c) 1* *(d) 0*

SOLUTION $\because \quad 2a-9=b+a$

$\Rightarrow \quad 2a-a=b+9$

$\Rightarrow \quad a=b+9$

$\Rightarrow \quad a-b=9 \text{ or } b-a=-9$

Hence $|a-b|+|b-a| = |9|+|-9|$

$= 9+9 = 18$

Thus (a) is the correct option.

EXAMPLE 5 *The value of x for which the value of* $|3x+15|$ *is minimum :*

(a) 3 *(b) 5*

(c) −5 *(d) none of these*

SOLUTION The minimum value of $|3x+15|=0$

$\Rightarrow \quad 3x+15=0$

$\Rightarrow \quad x=-5$

Hence (c) is correct option.

FACTORS AND MULTIPLES

When two or more integers are multiplied together then the resultant value is called their product :

For example

$3\times4=12, \quad 2\times7\times5=70, \quad 3\times5\times11=165$

where 12, 70 and 165 are called the products.

But we see that $12=4\times3$ it means 4 and 3 are the **factors** of 12.

and $70=2\times5\times7$

where 2, 5, 7 are known as **factors** of 70.

Again 12 is called as the **multiple of** 3 or **multiple** of 4.

Similarly 70 is called as the **multiple of** 2 or 5 or 7 or 10 or 35 or 14.

Thus a number which divides a given number exactly is called factor (or divisor) of that given number and the given number is called a multiple of that number.

Now, 15 is exactly divisible by 1, 3, 5, 15 so 1, 3, 5, 15 are called as the factors of 15 while 15 is called as the multiple of these factors. Where 1, 15 are **improper factors** and 3, 5 are called **proper factors of** 15.

Therefore 1 and itself (the number) are called the improper factors of the given number.

So the factors of $24 = 1, 2, 3, 4, 6, 8, 12, 24$

Factors of $35 = 1, 5, 7, 35$

Factors of $36 = 1, 2, 3, 4, 6, 9, 12, 18, 36$

Similarly the multiples of 2 are 2, 4, 6, 8, 10, 12, ...

multiples of 7 are 7, 14, 21, 28, 35, 42, 49....

multiples of 10 are 10, 20, 30, 40, 50...

- 1 is a factor of every number.
- Every number is a factor of itself.
- Every number, except 1 has atleast two factors *viz.*, 1 and itself.
- Every factor of a number is less than or equal to that number.
- Every multiple of a number is greater than or equal to itself.
- Every number has infinite number of its multiples.
- Every number is a multiple of itself.

Prime Factorisation : If a natural number is expressed as the product of prime numbers (factors) then the factorisation of the number is called its prime factorisation.

For example :

(i) $72 = 2\times2\times2\times3\times3$

2	72
2	36
2	18
3	9
3	3
	1

(ii) $420 = 2\times2\times3\times5\times7$

2	420
2	210
3	105
5	35
7	7
	1

(iii) $3432 = 2\times2\times2\times3\times11\times13$

2	3432
2	1716
2	858
3	429
11	143
13	13
	1

PRACTICE EXERCISE

1. Find the prime factors of the following :
   (i) 210 (ii) 120 (iii) 3315 (iv) 3465
   (v) 1197 (vi) 157573
2. Find the least number by which 22932 must be multiplied or divided so as to make it a perfect square :
   (a) 10 (b) 4 (c) 11 (d) 13
3. How many natural numbers upto 150 are divisible by 7?
   (a) 21 (b) 14 (c) 22 (d) 17
4. How many numbers between 333 and 666 are divisible by 5?
   (a) 67 (b) 70 (c) 75 (d) 55
5. How many numbers between 11 and 111 are the multiples of both 2 and 5?
   (a) 10 (b) 12 (c) 11 (d) 70
6. Find the total number of prime factors in $2^{17}\times6^{31}\times7^{5}\times10^{11}\times11^{10}\times(323)^{23}$:
   (a) 162 (b) 161 (c) 346 (d) 97

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ANSWERS & SOLUTIONS

1. (i) $2\times3\times5\times7$ (ii) $2\times2\times2\times3\times5$ (ii) $3\times5\times13\times17$
(iv) $3\times3\times5\times7\times11$ (v) $3\times3\times7\times19$ (vi) $13\times17\times23\times31$

2. (d) **3.** (a) **4.** (a) **5.** (a)

6. (a) $2^{17}\times6^{31}\times7^5\times10^{11}\times11^{10}\times(323)^{23}$
$=2^{17}\times2^{31}\times3^{31}\times7^5\times2^{11}\times5^{11}\times11^{10}\times17^{23}\times19^{23}$
$=2^{59}\times3^{31}\times5^{11}\times7^5\times11^{10}\times17^{23}\times19^{23}$
$\therefore$ Total number of prime factors =
$59+31+11+5+10+23+23=162$

NUMBER OF FACTORS (OR DIVISORS) OF A GIVEN NUMBER (COMPOSITE NUMBER)

Let us assume a composite number say 24 then find the no. of factors.

$24=1\times24$
2×12
3×8
4×6

We see that there are total 8 factors namely,
1, 2, 3, 4, 6, 8, 12 and 24.
But for the larger numbers it becomes difficult to find the total number of factors. So we use the following formula which can be derived with the help of theory of permutation and combination.

Let there be a composite number N and its prime factors be $a, b, c, d, \ldots$ etc. and $p, q, r, s, \ldots$ etc. be the indices (or powers) of the $a, b, c, d, \ldots$ etc respectively *i.e.*, if N can be expressed as

$$N=a^p.b^q.c^r.d^s\ldots$$

then the number of total divisors or factors of N is

$$(p+1).(q+1)(r+1)(s+1)\ldots$$

EXAMPLE 1 *Find the total number of factors of 24.*

SOLUTION $24=2^3\times3^1$ [$\because 24=2\times2\times2\times3$]
$\therefore$ Number of factors $=(3+1).(1+1)=8$

EXAMPLE 2 *Find the total number of factors of 540 :*
(a) 24 *(b)* 20
(c) 30 *(d) none of these*

SOLUTION $540=2\times2\times3\times3\times3\times5$
or $540=2^2\times3^3\times5^1$
Therefore total number of factors of 540 is
$(2+1)(3+1)(1+1)=24$

EXAMPLE 3 *The total number of divisors of 10500 except 1 and it self is :*
(a) 48 *(b)* 50 *(c)* 46 *(d)* 56

SOLUTION $10500=2\times2\times3\times5\times5\times5\times7$
$10500=2^2\times3^1\times5^3\times7^1$
$\therefore$ Total number of divisors of 10500 is
$(2+1)(1+1)(3+1)(1+1)=48$
But we have to exclude 1 and 10500
So there are only $48-2=46$ factors of 10500 except 1 and 10500. So (c) is the correct option.

EXAMPLE 4 *Total number of factors of 36 is :*
(a) 4 *(b)* 9
(c) 6 *(d) none of these*

SOLUTION $36=2^2\times3^2$
$\therefore$ Number of factors $=(2+1)(2+1)=3\times3=9$
Hence (b) is the correct option.

NOTE A perfect square number always contains odd no. of factors.

PRACTICE EXERCISE

Find the number of factors of the following :
1. 1008 **2.** 101 **3.** 111 **4.** 7056 **5.** 18522
6. 7744 **7.** 3875 **8.** 1458 **9.** 1339 **10.** 512

ANSWERS

1. 30 **2.** 2 **3.** 4 **4.** 45 **5.** 32
6. 21 **7.** 8 **8.** 14 **9.** 4 **10.** 10

SUM OF FACTORS OF A GIVEN NUMBER

Once again if you want to find the sum of smaller composite numbers, then you can do it manually, but for larger numbers its a problem.

e.g., Sum of factors of 24
$=1+2+3+4+6+8+12+24=60$

Let N be the composite number and $a, b, c, d..$ be its prime factors and p, q, r, s be the indices (or powers) of a, b, c, d *i.e.*, if N can be expressed as

$$N=a^p.b^q.c^r.d^s\ldots$$

then the sum of all the divisors (or factors) of N

$$=\frac{(a^{p+1}-1)(b^{q+1}-1)(c^{r+1}-1)(d^{s+1}-1)}{(a-1)(b-1)(c-1)(d-1)}\ldots$$

EXAMPLE 1 *Find the sum of factors of 24.*

SOLUTION $\because$ $24=2^3\times3^1$
$\therefore$ Sum of factors of $24=\frac{(2^4-1)(3^2-1)}{(2-1)(3-1)}$
$=\frac{15\times8}{1\times2}=60$

EXAMPLE 2 *Find the sum of factors of 270.*

SOLUTION $\because$ $270=2\times3^3\times5$
$\therefore$ Sum of factors of $270=\frac{(2^{1+1}-1)(3^{3+1}-1)(5^{1+1}-1)}{(2-1)(3-1)(5-1)}$
$=\frac{3\times80\times24}{1\times2\times4}=720$

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EXAMPLE 3 *The sum of factors of 1520 except the unity is :*
(a) 3720 (b) 2730
(c) 2370 (d) *none of these*

SOLUTION Since $1520 = 2^4 \times 5 \times 19$

$\therefore$ Sum of all the factors of $1520 = \dfrac{(2^5-1)(5^2-1)(19^2-1)}{(2-1)(5-1))(19-1)}$

$= \dfrac{31 \times 24 \times 360}{1 \times 4 \times 18} = 3720$

But since unity is to be excluded.
$\therefore$ The net sum of the factors = 3719
$\therefore$ (d) is the correct option.

EXAMPLE 4 *The sum of factors of 19600 is :*
(a) 54777 (b) 33667
(c) 5428 (d) *none of these*

SOLUTION $19600 = 2^4 \times 5^2 \times 7^2$

$\therefore$ Sum of factors of $19600 = \dfrac{(2^{4+1}-1)(5^{2+1}-1)(7^{2+1}-1)}{(2-1)(5-1)(7-1)}$

$= \dfrac{31 \times 124 \times 342}{1 \times 4 \times 6}$

$= 54777$

Hence (a) is the correct option.

PRODUCT OF FACTORS

Let us assume a very small number 24 and see the factors

$24 = 1 \times 24$
$= 2 \times 12$
$= 3 \times 8$
$= 4 \times 6$

Now, it is obvious from the above explanation that the product of factors of 24

$= (1 \times 24) \times (2 \times 12) \times (3 \times 8) \times (4 \times 6)$
$= 24 \times 24 \times 24 \times 24 = (24)^4$

Thus, the product of factors of composite number $N = N^{n/2}$, where n is the total number of factors of N.

EXAMPLE 1 *Product of divisors of 7056 is :*
(a) $(84)^{48}$ (b) $(84)^{44}$
(c) $(84)^{45}$ (d) *none of these*

SOLUTION $\because$ $7056 = 2^4 \times 3^2 \times 7^2$

$\therefore$ Number of factors/divisors of 7056
$= (4+1)(2+1)(2+1) = 45$

$\therefore$ Product of factors $= (7056)^{45/2} = (84)^{45}$

Hence (c) is the correct option.

EXAMPLE 2 *Product of factors of 360 is :*
(a) $(360)^{12}$ (b) $(36)^{120}$ (c) $(360)^{22}$ (d) $6^{24} \times 10^{10}$

SOLUTION $\because$ $360 = 2^3 \times 3^2 \times 5^1$

$\therefore$ Number of factors of $360 = (3+1)(2+1)(1+1) = 24$

Thus the product of factors $= (360)^{24/2} = (360)^{12}$

Hence (a) is the correct option.

NOTE As we know the number of divisors of any perfect number is an odd number so we take the square root of the perfect square to eliminate the $\frac{1}{2}$ from the power of the number as it can be seen in the above example no. 1.

NUMBER OF ODD FACTORS OF A GIVEN NUMBER

Let us assume a smaller number *e.g.*, 24

$24 = 1 \times 24$ also $24 = 2^3 \times 3^1$
$= 2 \times 12$
$= 3 \times 8$

We can see that there are total 2 odd factors namely 1 and 3.
Further assume another number say 36

$36 = 1 \times 36$ also $36 = 2^2 \times 3^2$
$= 2 \times 18$
$= 3 \times 12$
$= 4 \times 9$
$= 6 \times 6$

So, we can see that there are only 3 odd factors *viz.*, 1, 3 and 9.

Once again we assume another number say 90

then $90 = 1 \times 90$ also $90 = 2 \times 3^2 \times 5$
$= 2 \times 45$
$= 3 \times 30$
$= 5 \times 18$
$= 6 \times 15$
$= 9 \times 10$

Thus there are only 6 odd factors namely 1, 3, 5, 9, 15, 45.

To get the number of odd factors of a number N first of all express the number N as

$$N = (p_1^{\,a} \times p_2^{\,b} \times p_3^{\,c} \times \ldots) \times (e^x)$$

(where, $p_1, p_2, p_3, \ldots$ are the odd prime factors and e is the even prime factor.)

Then the total number of odd factors

$= (a+1)(b+1)(c+1)\ldots$

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EXAMPLE 1 *The number of odd factors (or divisors) of 24 is*

SOLUTION ∵ $24 = 2^3 \times 3^1$

Here 3 is the odd prime factor

So, total number of odd factors $= (1 + 1) = 2$

EXAMPLE 2 *The number of odd factors of 36 is ...*

SOLUTION ∵ $36 = 2^2 \times 3^2$

∴ Number of odd factors $= (2 + 1)\ 3$

EXAMPLE 3 *The number of odd factors of 90 is ....*

SOLUTION ∵ $90 = 2^1 \times 3^2 \times 5^1$

∴ Total number of odd factors of $90 = (2 + 1)(1 + 1) = 6$

NUMBER OF EVEN FACTORS OF A COMPOSITE NUMBER

Number of even factors of a number
= (Total number of factors of the given number
− Total number of odd factors)

Thus in the above
Example (1) number of even factors $= 8 - 2 = 6$
Example (2) number of even factors $= 9 - 3 = 6$
Example (3) number of even factors $= 12 - 6 = 6$

NUMBER OF WAYS OF EXPRESSING A COMPOSITE NUMBER AS A PRODUCT OF TWO FACTORS

It has long been discussed in the previous examples of composite numbers but you might have not noticed it.

Let us consider an example of small composite number say, 24

then $24 = 1 \times 24$
2×12
3×8
4×6

So it is clear that the number of ways of expressing a composite no. as a product of two factors

$= \frac{1}{2} \times$ the no. of total factors

EXAMPLE 1 *Find the number of ways of expressing 180 as a product of two factors.*

SOLUTION $180 = 2^2 \times 3^2 \times 5^1$

Number of factors $= (2 + 1)(2 + 1)(1 + 1) = 18$

Hence, there are total $\frac{18}{2} = 9$ ways in which 180 can be expressed as a product of two factors.

NOTE As you know when you express any perfect square number 'N' as a product of two factors namely $\sqrt{N}$ and $\sqrt{N}$, and you also know that since in this case $\sqrt{N}$ appears two times but it is considered only once while calculating the no.of factors so we get always an odd number as number of factors so we can not divide the odd number exactly by 2 as in the above formula. So if we have to consider these two same factors then we find the number of ways of expressing N as a product of two factors $= \frac{(\text{Number of factors} + 1)}{2}$.

Again if it is asked that find the no. of ways of expressing N as a product of Two distinct factors then we do not consider 1 way (*i.e.*, $N = \sqrt{N} \times \sqrt{N}$) then no. of ways $= \frac{(\text{Number of factors} - 1)}{2}$.

EXAMPLE 2 *Find the number of ways expressing 36 as a product of two factors.*

SOLUTION $36 = 2^2 \times 3^2$

Number of factors $= (2 + 1)(2 + 1) = 9$

Hence the no. of ways of expressing 36 as a product of two factors $= \frac{(9 + 1)}{2} = 5$.

as $36 = 1 \times 36,\ 2 \times 18,\ 3 \times 12,\ 4 \times 9$ and 6×6

EXAMPLE 3 *In how many ways can 576 be expressed as the product of two distinct factors?*

SOLUTION ∵ $576 = 2^6 \times 3^2$

∴ Total number of factors $= (6 + 1)(2 + 1) = 21$

So the number of ways of expressing 576 as a product of two distinct factors $= \frac{(21 - 1)}{2} = 10$.

NOTE Since the word 'distinct' has been used therefore we do not include $576 = 26 \times 26$.

LCM AND HCF

In the previous articles I have discussed in detail about the factors. Now I will move towards common factor, Highest common factors (HCF) and least common multiple (LCM).

Common Factors : When any factor which is the factor of two or more given numbers then it is said that this particular factor is common.

For example $6 = 2 \times 3$
$15 = 3 \times 5$

We see that 3 is a common factor in both 6 and 15.

Similarly, $6 = 2 \times 3$
$8 = 2 \times 2 \times 2$
$30 = 2 \times 3 \times 5$

So, we can see that only 2 is common in all the three numbers 6, 8 and 30. No other factor is common in all.

Highest Common factor (HCF) or Greatest Common divisor (GCD) : HCF of two or more than two numbers is the greatest possible number that can divide all these numbers exactly, without leaving any remainder. For example

Find the HCF of 84, 126.

∴ $84 = 2 \times 3 \times 7 \times 2 = 42 \times 2$
$126 = 2 \times 3 \times 7 \times 3 = 42 \times 3$

So, the HCF of 84 and $126 = 2 \times 3 \times 7 = 42$

Thus we can see that 42 is the greatest common divisor since it can divide exactly both 84 and 126.

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Basically there are two methods of finding the HCF.
(i) Factor Method (ii) Division Method

(i) Factor Method : In this method first we break (or resolve) the numbers into prime factors then take the product of all the common factors. This obtained resultant product is known as the HCF of the given numbers.

EXAMPLE 1 *Find the HCF of 1680 and 3600.*

SOLUTION $\because 1680 = 2\times2\times2\times2\times3\times5\times7$

$3600 = 2\times2\times2\times2\times3\times5\times3\times5$

So the product of common factors $= 2\times2\times2\times2\times3\times5 = 240$

Hence, the HCF of 1680 and 3600 is 240.

EXAMPLE 2 *Find the HCF of 750, 6300, 18900.*

SOLUTION $750 = 2\times3\times5\times5\times5$

$6300 = 2\times3\times5\times5\times3\times2\times7$

$18900 = 2\times3\times5\times5\times3\times2\times3\times7$

So the product of common factors $= 2\times3\times5\times5 = 150$

Hence 150 is the HCF of 750, 6300, 18900.

(ii) HCF by Division Method : Consider two smallest numbers, then divide the larger one of them by the smaller one and then divide this divisor by the remainder and again divide this remainder by the next remainder and so on until the remainder is zero.

If there are more than two numbers of which HCF is to be found, we continue this process as we divide the third lowest number by the last divisor obtained in the above process.

EXAMPLE 1 *Find the HCF of 120 and 180.*

SOLUTION

```
120)180(1
    120
    ---
    60)120(2
       120
       ---
         x
```

Hence, 60 is the HCF of 120 and 180.

EXAMPLE 2 *Find the HCF of 420 and 1782.*

SOLUTION

```
420)1782(4
   1680
   ----
    102)420(4
        408
        ---
         12)102(8
            96
            --
             6)12(2
               12
               --
                x
```

Hence 6 is the HCF of 420 and 1782.

EXAMPLE 3 *Find the HCF of 210, 495 and 980.*

SOLUTION First we consider the two smallest number *i.e.*, 210 and 495.

```
210)495(2
   420
   ---
    75)210(2
       150
       ---
        60)75(1
           60
           --
           15)60(4
              60
              --
               x

15)980(65
   975
   ---
    5)15(3
       15
       --
        x
```

Hence the HCF is 5.

Did you notice something?

If you watch carefully the above discussion you will find that the HCF is the factor of difference of the given numbers. So there is short cut for you to find the HCF.

You can divide the given numbers by their lowest possible difference if these numbers are divisible by this difference then this difference itself is the HCF of the given numbers. If this difference is not the HCF then any factor of this difference must be the HCF of the given numbers.

EXAMPLE 1 *Find the HCF of 63 and 84.*

SOLUTION The difference $= 84 - 63 = 21$

Now divide 63 and 84 by 21, since the given numbers 63 and 84 are divisible by their difference 21, thus 21 itself is the HCF of 63 and 84.

EXAMPLE 2 *Find the HCF of 42 and 105.*

SOLUTION The difference $= 105 - 42 = 63$

Now divide 42 and 105 by 63 but none of these is divisible by their difference 63. So we factorize 63 and then divide 42 and 105 by the factor of 63. The greatest factor of 63 which can divide 42 and 105 that will be the HCF of the given numbers.

So $63 = 1\times63,\ 3\times21,\ 7\times9$

$\therefore$ The factor of 63 are 1, 3, 7, 9, 21 and 63.

Obviously we divide 42 and 105 by 21 (since the division by 63 is not possible) and we see that the given numbers are divisible hence 21 is the HCF of 42 and 105.

EXAMPLE 3 *Find the HCF of 30, 42 and 135.*

SOLUTION The difference between 30 and 42 = 12

The difference between 42 and 135 = 93

Now divide, 30, 42 and 135 by 12 but it can not divide all the numbers.

So take the factors of 12.

The factors of 12 are 1, 2, 3, 4, 6, 12

Now we divide 30, 42 and 135 by 6 but it can not divide all these numbers. So we divide now 30, 42 and 135 by 4, but 4 also can not divide all the numbers, Now we divide the given numbers by 3. We will find that all the numbers are divisible by 3 hence 3 is the HCF of the given numbers.

1. Euclidean Algorithm : If 'H' be the HCF of any two positive integers a and b then there exists unique integers p and q such that $H = ap + bq$.

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INTRODUCTORY EXERCISE-1.3

1. The product of two numbers is 84 and their HCF is 2. Find the numbers of such pairs.
2. The product of two numbers is 15120 and their HCF is 6, find the number of such pairs.
3. The number of pairs of two numbers whose product is 300 and their HCF is 5 :
   (a) 2 (b) 3
   (c) 4 (d) can't be determined

HCF WITH REMAINDERS

Case 1. Find the greatest possible number with which when we divide 37 and 58, it leaves the respective remainder of 2 and 3.

SOLUTION Since when we divide 37 and 58 by the same number then we get remainders 2 and 3 respectively.

So $(37-2)$ and $(58-3)$ must be divisible hence leaving the remainders zero.

It means 35 and 55 both are divisible by that number so the HCF of 35 and 55 is 5.

Hence the greatest possible number is 5.

Case 2. Find the largest possible number with which when 60 and 98 are divided it leaves the remainders 3 in each case.

SOLUTION Since 60 and 98 both leave the remainders 3 when divided by such a number.

Thus $57 = (60-3)$ and $95 = (98-3)$ will be divisible by the same number without leaving any remainder. So the HCF of 57 and 95 is 19. Hence 19 is the highest possible number.

Case 3. Find the largest possible number with which when 38, 66 and 80 are divided the remainders remains the same.

SOLUTION In this case (since we donot know the value of remainder) we take the HCF of the differences of the given numbers.

So the HCF of $(66-38), (80-66), (80-38)$

$= \text{HCF of } 28, 14, 42 = 14$

Hence 14 is the largest possible number which leaves same remainder $(= 10)$ when it divides either 38, 66 or 80.

CAT CONCISE

In case 1, the remainders are different
In case 2, the remainders are same in each case.
In case 3, the remainders are same in each case, but the value of remainder is unknown.

INTRODUCTORY EXERCISE-1.4

1. The largest possible number with which when 76, 132 and 160 are divided the remainders obtained are the same is :
   (a) 6 (b) 14
   (c) 18 (d) none of these
2. Find the number of pairs of two numbers whose HCF is 5 and their sum is 50.
3. The largest possible length of a tape which can measure 525 cm, 1050 cm and 1155 cm length of cloths in a minimum number of attempts without measuring the length of a cloth in a fraction of the tape's length
   (a) 25 (b) 105
   (c) 75 (d) none of these
4. In the above question minimum how many attempts are required to measure whole length of cloths?
5. Minimum how many similar tiles of square shape are required to furnish the floor of a room with the length of 462 cm and breadth of 360 cm?
6. The ratio of two numbers is 15 : 11. If their HCF be 13 then these numbers will be :
   (a) 15 : 11 (b) 75 : 55
   (c) 105 : 77 (d) 195 : 143
7. The three numbers are in the ratio 1 : 2 : 3 and their HCF is 12. These numbers are :
   (a) 4, 8, 12 (b) 5, 10, 15
   (c) 24, 48, 72 (d) 12, 24, 36
8. There are three drums with 1653 litre 2261 litre and 2527 litre of petrol. The greatest possible size of the measuring vessel with which we can measure up the petrol of any drum while every time the vessel must be completely filled :
   (a) 31 (b) 27
   (c) 19 (d) 41
9. Two pencils are of 24 cm and 42 cm. If we want to make them of equal size then minimum no. of similar pencils is :
   (a) 6 (b) 11
   (c) 12 (d) none of these
10. The HCF of two numbers is 27 and their sum is 216. These numbers are :
    (a) 27, 189 (b) 81, 189
    (c) 108, 108 (d) 154, 162
11. Mr. Baghwan wants to plant 36 mango trees, 144 orange trees and 234 apple trees in his garden. If he wants to plant the equal no. of trees in every row, but the rows of mango, orange and apple trees will be separate, then the minimum number of rows in his garden is :
    (a) 18 (b) 23
    (c) 36 (d) can't be determined

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LCM (LEAST COMMON MULTIPLE)

In this chapter I have already discussed the multiples of a number for example integral multiples of 7 are 7, 14, 21, 28, 35, 42, ...

and the integral multiples of 8 are 8, 16, 24, 32, 40, 48, 56, ...etc. in short we can write the positive integral multiples of any number N as NK, where $K = 1, 2, 3, 4, 5, 6, ...$

Now if we consider altcast two numbers say 2 and 3 then we write the multiples of each as 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30... and 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, ...

Thus it is clear from the above illustrations that the some of the multiples of 2 and 3 are common *viz.*, 6, 12, 18, 24, 30...

Again if we consider any 3 numbers say, 3, 5 and 6 then the multiples of each of the 3, 5, 6 are

3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48, 51, 54, 57, 60

5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60...

6, 12, 18, 24, 30, 36, 42, 48, 54, 60...

So, we can see that the common multiples of 3, 5 and 6 are 30, 60, 90, 120, ...etc.

But out of these common multiples, 30 is the least common multiple. Similarly in the previous example 6 is the least common multiple.

NOTE Generally there is no any greatest common multiple unless otherwise the condition is stated.

Basically, there are two methods to find the LCM.

(i) Factor Method (ii) Division Method

(i) Factor Method : Resolve the given numbers into their prime factors, then take the product of all the prime factors of the first number with those prime factors of second number which are not common to the prime factors of the first number. Now this resultant product can be multiplied with those prime factors of the third number which are not common to the factors of the previous product and this process can be continued for further numbers if any.

EXAMPLE 1 *Find the LCM of* 48, 72, 140.

SOLUTION We can write the given numbers as

$48 = \boxed{2} \times \boxed{2} \times 2 \times 2 \times 3 = \boxed{2} \times \boxed{2} \times 2 \times 3 \times 2$

$72 = \boxed{2} \times \boxed{2} \times 2 \times 3 \times 3 = \boxed{2} \times \boxed{2} \times 2 \times 3 \times 3$

$140 = \boxed{2} \times \boxed{2} \times 5 \times 7 = \boxed{2} \times \boxed{2} \times 5 \times 7$

So the LCM of 48, 72 and 140

$= 2 \times 2 \times 2 \times 3 \times 2 \times 3 \times 5 \times 7 = 5040$

EXAMPLE 2 *Find the LCM of* 42, 63 *and* 231.

SOLUTION $42 = 2 \times 3 \times 7 = \boxed{3} \times \boxed{7} \times 2$

$63 = 3 \times 3 \times 7 = \boxed{3} \times \boxed{7} \times 3$

$231 = 3 \times 7 \times 11 = \boxed{3} \times \boxed{7} \times 11$

$\therefore$ The LCM of 42, 63 and $231 = 3 \times 7 \times 2 \times 3 \times 11 = 1386$

(ii) Division Method : First of all write down all the given numbers in a line separating by the comma (,) then divide these numbers by the least common prime factors say 2, 3, 5, 7, 11... to the given numbers, then write the quotients just below the actual numbers separating by comma (,). If any number is not divisible by such a prime factor then write this number as it is just below itself, then continue this process of division by considering higher prime factors, if the division is complete by lower prime factor, till the quotient in the last line is 1. Then take the product of all the prime factors by which you have divided the numbers (or quotient) in different lines (or steps). This product will be the LCM of the given numbers.

EXAMPLE 3 *Find the LCM of* 108, 135 *and* 162.

SOLUTION

2	108, 135, 162
2	54, 135, 81
3	27, 135, 81
3	9, 45, 27
3	3, 15, 9
3	1, 5, 3
5	1, 5, 1
	1, 1, 1

Thus the required LCM $= 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 5 = 1620$

EXAMPLE 4 *Find the LCM of* 420, 9009, 6270.

SOLUTION

2	420, 9009, 6270
2	210, 9009, 3135
3	105, 9009, 3135
3	35, 3003, 1045
5	35, 1001, 1045
7	7, 1001, 209
11	1, 143, 209
13	1, 13 19
19	1, 1, 19
	1, 1, 1

Thus the required LCM $= 2 \times 2 \times 3 \times 3 \times 5 \times 7 \times 11 \times 13 \times 19$

$= 3423420$

EXAMPLE 5 *Find the least possible number which can be divided by 32, 36 and 40.*

SOLUTION The number which is divisible by 32, 36 and 40 it must be the common multiple of all the given numbers. Since we need such a least number then we have to find out just the LCM of 32, 36 and 40.

The LCM of 32, 36 and 40 = 1440

Hence, the least possible no. is 1440, which is divisible by all the given numbers.

EXAMPLE 6 *In the above problem what is the least possible number of* 5 *digits which is divisible by all the numbers* 32, 36 *and* 40.

SOLUTION Since the least possible no. is 1440, but it is a four digit number. So we can take the integral multiples of 1440 which must be divisible by the given numbers.

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Now since the least possible 5 digit number is 10000, so the required number must be equal to or greater than 10,000. So when we multiply 1440 by 7. We get the required result *i.e.*, 10,080.

Thus 10080 is the least possible 5 digit number which is divisible by 32, 36 and 40.

Alternatively : Divide the least possible 5 digit no.by 1440 and then add the difference of the divisor and remainder to the least possible 5 digit number. This will be the required number.

So,

$$1440\overline{)10000}\,(6$$
$$\underline{8640}$$
$$1360$$

Now, $1440 - 1360 = 80$

Thus the required number = 10000 + 80 = 10080

EXAMPLE 7 *According to above problem find the largest possible number of 4 digits which is exactly divisible by 32, 36 and 40.*

SOLUTION Since the largest 4 digit number is 9999. So the required number can not exceed 9999 any how, now we take the appropriate multiple of 1440. Since 1440 is divisible by the given numbers, so the multiples of 1440 must be divisible by the given numbers.

Hence $1440 \times 6 = 8640$ is the largest possible number since $1440 \times 7 = 10080$ is greater than 9999 which is not admissible. Hence, 8640 is the largest possible 4 digit no. which is divisible by all the given numbers.

Alternatively : Divide the greatest 4 digit number by 1440 (the LCM of the given no.) and then subtract the remainder from the greatest 4 digit number. So,

$$1440\overline{)9999}\,(6$$
$$\underline{8640}$$
$$1359$$

Hence, the required number = 9999 – 1359 = 8640

EXAMPLE 8 *Find the number of numbers lying between 1 and 1000 which are divisible by each of 6, 7 and 15.*

SOLUTION The least possible number which is divisible by 6, 7, and 15 = LCM of 6, 7, 15 = 210

So, the first such number is 210 and the other numbers are the multiples of 210 *i.e.*, 210, 420, 630, 840.

Thus there are total 4 numbers lying between 1 and 1000 which are divisible by 6, 7 and 15.

EXAMPLE 9 *Find the number of numbers lying between 1000 and 1,00,000 which are divisible by 15, 35 and 77 and are even also.*

SOLUTION The required numbers must be the multiples of the LCM of 15, 35 and 77.

Now the LCM of 15, 35 and 77 = 1155.

So the other numbers are

1155, 2310, 3465, 4620, 5775, ..., 99330

Thus there are total 86 numbers which are divisible by 15, 35 and 77 but only 43 numbers are even *i.e.*, every alternate number is even.

Thus there are total 43 required numbers.

EXAMPLE 10 *Find the least possible perfect square number which is exactly divisible by 6, 40, 49 and 75.*

SOLUTION The required number must be divisible by the given numbers so it can be the LCM or its multiple number.

Now the LCM of 6, 40, 49 and 75

$$= 2 \times 2 \times 2 \times 3 \times 5 \times 5 \times 7 \times 7$$

But the required number is a perfect square

Thus the LCM must be multiplied by $2 \times 3 = 6$.

Thus the required number

$$= (2 \times 2 \times 2 \times 3 \times 5 \times 5 \times 7 \times 7) \times (2 \times 3)$$
$$= 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 5 \times 7 \times 7$$
$$= 176400$$

EXAMPLE 11 *Three bells in the bhootnath temple toll at the interval of 48, 72 and 108 second individually. If they have tolled all together at 6 : 00 AM then at what time will they toll together after 6 : 00 AM?*

SOLUTION The three bells toll together only at the LCM of the times they toll individually.

Thus the LCM of 48, 72 and 108 is 432 seconds.

Therefore all the bells will toll together at 6 : 07 : 12 AM

(∵ 432 seconds = 7 minuts 12 seconds)

EXAMPLE 12 *In the above problem how many times these bells will toll together till the 6 : 00 PM on the same day.*

SOLUTION The total time since 6 : 00 AM till 6 : 00 PM

$$= 12 \times 60 \times 60 \text{ seconds.}$$

Now since all these bells toll together at the interval of 432 seconds.

So the number of times when they will toll together

$$= \frac{\text{Total time}}{\text{least duration of tolling together}}$$
$$= \frac{12 \times 60 \times 60}{432} = 100$$

but since the bells toll together at 6 : 00 PM also. Hence total 101 (= 100 + 1) times these bells will toll together in the given duration of time.

LCM WITH REMAINDERS

Case 1. When the remainders are same for all the divisors.

Case 2. When the remainders are different for different divisors but the respective difference between the divisors and the remainders remain constant.

Case 3. When neither the divisors are same nor the respective difference between divisors and the remainders remain constant.

CASE 1

EXAMPLE 13 *What is the least possible number which when divided by 24, 32 or 42 in each case it leaves the remainder 5?*

SOLUTION Since we know that the LCM of 24, 32 and 42 is divisible by the given numbers. So the required number is

= (LCM of 24, 32, 42) + (5) = 672 + 5 = 677

Hence such a least possible number is 677.

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EXAMPLE 14 *In the above question how many numbers are possible between 666 and 8888?*

SOLUTION Since the form of such a number is $672m + 5$, where $m = 1, 2, 3, \ldots$

So, the first no. $= 672 \times 1 + 5 = 677$ and the highest possible number in the given range $= 672 \times 13 + 5$

$= 8736 + 5 = 8741$

Thus the total numbers between 666 and 8888 are 13.

EXAMPLE 15 *What is the least possible number which when divided by 21, 25, 27 and 35 it leaves the remainder 2 in each case?*

SOLUTION The least possible no.

$=$ (LCM of 21, 25, 27 and 35) $m + 2$

$= 4725m + 2$

$= 4727$ (for the least possible value we take $m = 1$)

EXAMPLE 16 *What is the least possible no. which must be added to 4722 so that it becomes divisible by 21, 25, 27 and 35?*

SOLUTION The number which is divisible by 21, 25, 27 and 35 is the LCM of 21, 25, 27 and $35 = 4725$

So the required number $= 4725 - 4722 = 3$

EXAMPLE 17 *What is the least number which when divided by 8, 12 and 16 leaves 3 as the remainder in each case, but when divided by 7 leaves no any remainder?*

SOLUTION The possible value $= m$ (LCM of 8, 12 and 16) $+ 3$

$= m\,(48) + 3$

Now put the value of m such that "$m\,(48) + 3$" becomes divisible by 7. So $3 \times 48 + 3 = 147$ which is the least possible required number.

CASE 2

EXAMPLE 18 *What is the least possible number which when divided by 18, 35 or 42 it leaves the 2, 19, 26 as the remainders respectively?*

SOLUTION Since the difference between the divisors and the respective remainders is same.

Hence the least possible number

$=$ (LCM of 18, 35 and 42) $- 16$

$= 630 - 16$ $[\because (18 - 2) = (35 - 19) = (42 - 26) = 16]$

$= 614$

EXAMPLE 19 *What is the least possible number which when divided by 2, 3, 4, 5, 6 it leaves the remainders 1, 2, 3, 4, 5 respectively?*

SOLUTION Since the difference is same as

$(2 - 1) = (3 - 2) = (4 - 3) = (5 - 4)\,(6 - 5) = 1$

Hence the required number $=$ (LCM of 2, 3, 4, 5, 6) $- 1$

$= 60 - 1 = 59$

EXAMPLE 20 *In the above problem what is the least possible 3 digit number which is divisible by 11?*

SOLUTION Since the form of the number is $60m - 1$, where $m = 1, 2, 3, \ldots$.

but the number $(60m - 1)$ should also be divisible by 11 hence at $m = 9$ the number becomes 539 which is also divisible by 11. Thus the required number $= 539$.

EXAMPLE 21 *Find the least possible 5 digit number which when divided by 2, 4, 6 and 8, it leaves the remainders 1, 3, 5 and 7 respectively.*

SOLUTION The possible value $=$ (LCM of 2, 4, 6, 8) $m - 1$

$= 24m - 1$

Since, the least 5 digit number is 10000. So the required number must be atleast 10000. So putting the value of $m = 417$, we get $(10008 - 1) = 10007$, which is the required number.

CASE 3

EXAMPLE 22 *What is the least possible number which when divided by 13 leaves the remainder 3 and when it is divided by 5 it leaves the remainder 2.*

SOLUTION Let the required number be N then it can be expressed as follows

$$N = 13k + 3 \quad \ldots(1)$$

and

$$N = 5l + 2 \quad \ldots(2)$$

where k and l are the quotients belong to the set of integers.

Thus $5l + 2 = 13k + 3$

$\Rightarrow$ $5l - 13k = 1$

$\Rightarrow$ $5l = 13k + 1$

$\Rightarrow$ $l = \dfrac{13k + 1}{5}$

now we put the value of k such that numerator will be divisible by 5 or l must be integer so considering $k = 1, 2, 3, \ldots$ we find that $k = 3$, l becomes 8. So the number $N = 5 \times 8 + 2 = 42$

Thus the least possible number $= 42$

To get the higher numbers which satisfy the given conditions in the above problem we just add the multiples of the given divisors (*i.e.*, 13 and 5) to the least possible number (*i.e.*, 42)

NOTE The next higher number $=$ (Multiple of LCM of 13 and 5) $+ 42$

$= 65m + 42$

EXAMPLE 23 *In the above problem what is the greatest possible number of 4 digits?*

SOLUTION Since the general form of this number is $65m + 42$. So by putting $m = 153$ we get $(9945 + 42 =) 9987$, which is required number.

HINT To get the value of 9945 we can simply divide 9999 (which is the greatest 4 digit number) by 65 and then subtract the remainder from 9999.

EXAMPLE 24 *How many numbers lie between 11 and 1111 which when divided by 9 leave a remainder of 6 and when divide by 21 leave a remainder of 12?*

SOLUTION Let the possible number be N then it can be expressed as

$$N = 9k + 6$$

and $$N = 21l + 12$$

$\therefore$ $9k + 6 = 21l + 12$

$\Rightarrow$ $9k - 21l = 6$

or $3\,(3k - 7l) = 6$

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or $3k = 7l + 2$ or $k = \frac{7l+2}{3}$

So put the min. possible value of l such that the value of k is an integer or in other words numerator (*i.e.*, $7l + 2$) will be divisible by 3.

Thus at $l = 1$, we get $k = 3$ (an integer). So the least possible number $N = 9 \times 3 + 6 = 21 \times 1 + 12 = 33$.

Now the higher possible values can be obtained by adding 33 in the multiples of LCM of 9 and 21. *i.e.*, The general form of the number is $63m + 33$. So the other number in the given range including 33 are 96, 159, 222, 285, 348, ..., 1104. Hence there are total 18 numbers which satisfy the given condition.

Important Formula : Product of two numbers = Product of their HCF and their LCM.

EXAMPLE 25 *The HCF and LCM of the two numbers is 12 and 600 respectively. If one of the number is 24, then the other number will be*
(a) 300 *(b)* 400
(c) 1500 *(d)* *none of these*

SOLUTION The answer is (a)

since product of numbers = HCF × LCM

$\therefore \quad N \times 24 = 12 \times 600 \Rightarrow N = 300$

Successive Division : If the quotient in a division is further used as a dividend for the next divisor and again the latest obtained divisor is used as a dividend for another divisor and so on, then it is called the "successive division" *i.e.*, if we divide 625 by 3, we get 208 as quotient and 1 as a remainder then if 208 is divided by another divisor say 4 then we get 52 as a quotient and '0' (zero) as remainder and again, if we divide 52 by another divisor say 6 we get 8 as quotient and 4 as a remainder *i.e.*, we can represent it as following

3 | 625
4 | 208 → 1 (Remainder)
6 | 52 → 0 (Remainder)
8 → 4 (Remainder)

Now you can see that the quotient obtained in the first division behaves as a dividend for another divisor 4. Once again the quotient 52 is treated as a dividend for the next divisor 4. Thus it is clear from the above discussion as

Dividend	Divisor	Quotient	Remainder
625	3	208	1
208	4	52	0
52	6	8	4

So the 625 is successively divided by 3, 4 and 6 and the corresponding remainders are 1, 0 and 4.

EXAMPLE 1 *The least possible number of 3 digits when successively divided by 2, 5, 4, 3 gives respective remainders of 1, 1, 3, 1 is :*
(a) 372 *(b)* 275 *(c)* 273 *(d)* 193

SOLUTION The problem can be expressed as

2 | A
5 | B → 1
4 | C → 1
3 | D → 3
E → 1

} Remainder

So it can be solved as

$$((((E \times 3) + 1)4 + 3)5 + 1)2 + 1) = A$$

(where A is the required number)

So for the least possible number E

= 1 (the least positive integer)

then $A = ((((1 \times 3) + 1) \times 4 + 3)5 + 1)2 + 1)$

[Since at E = 0, we get a two digit number]

Alternatively : We use the following convention while solving this type of problem. First we write all the divisors as given below then their respective remainders just below them.

2 ↖ 5 ↖ 4 ↖ 3] Divisors
↓1 ↓1 ↓3 1] Remainders

Where the arrow downwards means to add up and the arrow slightly upward (at an angle of 45°) means multiplication.

So we start from the right side remainder and move towards left since while writing down the divisors and remainders we write the first divisor first (*i.e.*, leftmost), second divisor at second position (from the left) and so on.

Now solve it as : **Step 1.** $(1 \times 4) + 3 = 7$

Step 2. $7 \times 5 + 1 = 36$

Step 3. $36 \times 2 + 1 = 73$

or $((((1 \times 4) + 3)5 + 1)2 + 1) = 73$

but we are required to find a three digit number so the next higher numbers can be obtained just by taking the multiples of the product of the divisors and then adding to it the least such number.

The next higher number = $(2 \times 5 \times 4 \times 3)m + 73 = 120m + 73$

So by putting $m = 1, 2, 3 \ldots$ we can get the higher possible numbers.

Now we just need a least possible 3 digit number so we can get it by putting $m = 1$

Hence the required number = $120 \times 1 + 73 = 193$

Hence (d) is the correct answer.

EXAMPLE 2 *A number when successively divided by 2, 3, 5 it leaves the respective remainders 1, 2 and 3. What will be the remainder if this number will be divided by 7?*

SOLUTION Write the divisors and remainders as given below

2 ↖ 3 ↖ 5
↓1 ↓2 3

then solve it as follows

$$((3 \times 3) + 2)2 + 1) = 23$$

or

$3 \times 3 = 9$
$9 + 2 = 11$
$11 \times 2 = 22$
$22 + 1 = 23$

So, the least possible number is 23 and the higher numbers can be obtained as $(2 \times 3 \times 5)m + 23 = 30m + 23$.

So the higher numbers are 53, 83, 113, 143, 173, 213, 333, ...etc.

But when we divide all these possible numbers by 7 we get the different remainders. So we can not conclude the single figure as a remainder.

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EXAMPLE 3 *A number when divided successively by 6, 7 and 8, it leaves the respective remainders of 3, 5 and 4, what will be the last remainder when such a least possible number is divided successively by 8, 7 and 6?*

SOLUTION First we find the actual least possible no. then we move further

$$\downarrow 6 \nwarrow 3 \quad \downarrow 7 \nwarrow 5 \quad 8 \nwarrow 4$$

$$((((4\times7)+5)\times6)+3) = (((28+5)\times6)+3)$$
$$= 33\times6+3 = 198+3 = 201$$

Now we divide 201 successively by 8, 7 and 6.

8	201	
7	25	→ 1 (Remainder)
6	3	→ 4 (Remainder)
	0	→ 3 (Remainder)

So 3 is the last remainder.

EXAMPLE 4 *How many numbers lie between 100 and 10000 which when successively divided by 7, 11 and 13 leaves the respective remainders of 5, 6 and 7?*

SOLUTION The least possible number can be obtained as

$$\downarrow 7 \nwarrow 5 \quad \downarrow 11 \nwarrow 6 \quad 13 \nwarrow 7$$

$$(((7\times11)+6)7+5) = ((77+6)7+5)$$
$$= (83\times7+5) = (581+5) = 586$$

The general form for the higher numbers is

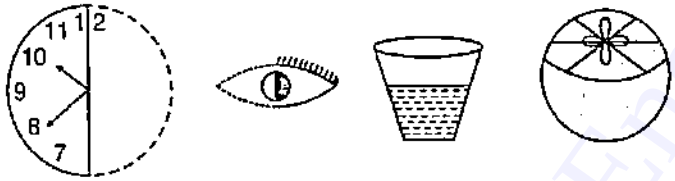
$$(7\times11\times13)m + 586 = (1001)m + 586$$

So, the numbers can be obtained by considering $m = 0, 1, 2, 3, \ldots$ so the first number is 586 and the last number is 9595 which can be attained at $m = 9$. So there are total 10 such numbers lying between 100 and 10000.

FRACTIONS

Suppose you have borrowed Rs. 20 from your friend in the last month. Now he has asked you to return his money. But you are paying him only Rs. 10 and the rest amount you want to return in the next month. It means you are paying the total amount not in a one lot but in a fraction that means "in parts".

From figures you can see that these things are in fraction.



So, we can say that when any unit of a thing is divided into equal parts and some parts are considered, then it is called a fraction. For example $\frac{1}{2}, \frac{3}{4}, \frac{2}{5}, \frac{3}{7}$ etc.

$\frac{1}{2}$ is read as one half

$\frac{3}{4}$ is read as three fourth



Denominator : The lower value indicates the number of parts into which the whole thing or quantity is being equally divided and it is known as Denominator.

Numerator : The upper value indicates the number of parts taken into consideration (or for use) out of the total parts in known as Numerator.

So in the fractions $\frac{2}{3}, \frac{5}{7}, \frac{1}{4}$ etc. 2, 5, 1 are the numerators and 3, 7, 4 are the denominators.

The numerators and denominators of a fraction are also called the 'terms' of a fraction.

Proper Fraction : A fraction whose numerator is less then its denominator,but not equal to zero) is called a proper fraction. For example $\frac{1}{2}, \frac{3}{4}, \frac{2}{7}, \frac{11}{20}$, etc.

Improper Fraction : A fraction whose numerator is equal to or greater than its denominator is called an improper fraction . For example $\frac{7}{2}, \frac{8}{5}, \frac{215}{15}, \frac{63}{15}$ etc.

NOTE
1. Every natural number can be expressed as a fraction, for example $7 = \frac{7}{1}, 8 = \frac{8}{1}, 11 = \frac{11}{1}$ etc. which are improper fraction.
2. When the numerator and denominator of a fraction are equal, then it is equal to unity.
3. The denominator of a fraction can never be equal to zero.

Mixed Fraction : A number which consists of two parts (i) a natural number (ii) a proper fraction is called a **mixed fraction.** For example $2\frac{5}{13}, 1\frac{7}{8}, 18\frac{9}{17}$ etc. where 2, 1, 18 are natural numbers and $\frac{5}{13}, \frac{7}{8}$ and $\frac{9}{17}$ are proper fractions.

and $2\frac{5}{13} = 2+\frac{5}{13}, 1\frac{7}{8} = 1+\frac{7}{8}$ and $18\frac{9}{17} = 18+\frac{9}{17}$

NOTE
1. Every mixed fraction can be written as an improper fraction and every improper fraction can be written as a mixed fraction as $\frac{45}{7} = 6\frac{3}{7}$
2. The numbers for example $-\frac{2}{3}, \frac{-7}{8}, \frac{-9}{5}$ etc are not the fraction numbers, but these are called as fraction like numbers.

Like Fractions : The fractions whose denominators are same are called like fractions, for example $\frac{3}{11}, \frac{4}{11}, \frac{9}{11}, \frac{8}{11}, \frac{17}{11}$ etc.

Equivalent Fractions : The fractions whose values are same *i.e.,* the ratio is same are called equivalent fractions for example,

$$\frac{2}{3} = \frac{4}{6} = \frac{6}{9} = \frac{8}{12} = \frac{10}{15} = \frac{12}{18} = \frac{20}{30} = \frac{26}{39}$$

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It implies that :

1. If we multiply the numerator and the denominator by the same non-zero number, the value of the fraction remains unchanged.
2. If we divide the numerator and the denominator by the same non zero number, the value of the fractions remains unchanged.

Cancellation : Division of the numerator and denominator by the same (non-zero) number is called the cancellation when the numerator and denominator of a fraction have no any common factors between them, then it is said the Reduced or Simplest form of fraction or in lowest term.

For example $\frac{2}{3}, \frac{3}{4}, \frac{7}{8}$ etc are the reduced fraction in its lowest term.

Reduction of a fraction to its lowest terms or simplest form :

1. HCF Method : Divide the numerator and denominator both by their HCF. *e.g.,* to reduce $\frac{21}{35}$ to its lowest term just divide 21 and 35 by their HCF.

So, $$\frac{21 \div 7}{35 \div 7} = \frac{3}{5}$$

2. Prime Factorisation Method : Just cancel out the common prime factors of both the numerator and denominator. For example to reduce 42 and 140 we cancel out the common prime factors as

$$\frac{42}{140} = \frac{2 \times 3 \times 7}{2 \times 2 \times 5 \times 7} = \frac{3}{10}$$

Reduction of the given fractions into like fractions : Obtain the LCM of the denominators of the given fractions and then make all the denominators equal to the LCM obtained, in such a way that the value of every fraction remain unchanged.

For example $\frac{6}{35}, \frac{10}{21}$.

Since the LCM of the 35 and 21 is 105 so the new fraction will be as

$$\frac{6 \times 3}{35 \times 3}, \frac{10 \times 5}{21 \times 5} \Rightarrow \frac{18}{105}, \frac{50}{105}$$

Relationship among fractions : If the denominators of all the fractions are same then the fraction whose numerator is greater, the fraction will be greater.

For example $\frac{8}{19} > \frac{7}{19} > \frac{5}{19} > \frac{4}{19}$

NOTE

1. To equalize the denominator of the fractions, take the LCM of the denominators.
2. (A) If $\frac{x}{y}$ be proper positive fraction and '*a*' be a positive integers when $y \neq 0$

   1. $\frac{x+a}{y+a} > \frac{x}{y}$ 2. $\frac{x-a}{y-a} < \frac{x}{y}$

   If '*b*' being a positive integer such that $a > b$, then

   3. $\frac{x}{y} \times \frac{a}{b} > \frac{x}{y}$ 4. $\frac{x}{y} \times \frac{b}{a} < \frac{x}{y}$

   (B) If $\frac{x}{y}$ be an improper positive fraction, then

   5. $\frac{x+a}{y+a} < \frac{x}{y}$ 6. $\frac{x-a}{y-a} > \frac{x}{y}$
   7. $\frac{x}{y} \times \frac{a}{b} > \frac{x}{y}$ 8. $\frac{x}{y} \times \frac{b}{a} < \frac{x}{y}$

   (C) For any fraction $\frac{x}{y}$

   9. $\frac{x+nx}{y+ny} = \frac{x}{y}$ 10. $\frac{x-nx}{y-ny} = \frac{x}{y}$
   11. $\frac{x \pm m}{y \pm n} \neq \frac{x}{y}$ if $\frac{m}{n} \neq \frac{x}{y}$

PRACTICE EXERCISE

1. State (a) if the fraction is proper (b) if the fraction is improper (c) if the fraction is mixed (d) none of these
   (i) $\frac{3}{7}$ (ii) $\frac{7}{3}$
   (iii) $\frac{35}{12}$ (iv) $4\frac{3}{11}$
2. State (a) if the fractions are equivalent and (b) if the fractions are like fractions else (c)
   (i) $\frac{7}{16}, \frac{8}{16}, \frac{9}{16}, \frac{31}{16}$ (ii) $\frac{-5}{-6}, \frac{15}{18}, \frac{-45}{-54}$
   (iii) $\frac{3}{5}, \frac{7}{10}, \frac{14}{20}$ (iv) $\frac{9}{25}, \frac{4}{16}, \frac{13}{29}$
3. State (a) if the fractions are in ascending order, (b) if the fractions are in descending order, else (c)
   (i) $\frac{3}{5}, \frac{7}{9}, \frac{13}{8}$ (ii) $\frac{9}{25}, \frac{11}{26}, \frac{13}{28}$
   (iii) $\frac{17}{16}, \frac{35}{52}, \frac{65}{75}$ (iv) $\frac{30}{45}, \frac{30}{50}, \frac{38}{80}$

ANSWERS

1. (i) a, (ii) b, (iii) b, (iv) c
2. (i) b, (ii) a, (iii) c, (iv) c
3. (i) a, (ii) a, (iii) a, (iv) b

Reciprocal Fraction : The reciprocal fraction of a number k be $1/k$ *i.e.,* the product of a reciprocal number with the number itself is unity (*i.e.,* 1) hence the reciprocal of 6 is $\frac{1}{6}$, the reciprocal of $\frac{1}{8}$ is 8, the reciprocal of $\frac{3}{5}$ is $\frac{5}{3}$.

Compound Fraction : The fraction of a fraction is called its compound fraction for examples $\frac{1}{3}$ of $\frac{1}{2}$ $\left(i.e., \frac{1}{3} \times \frac{1}{2} = \frac{1}{6} \right)$ is the compound fraction.

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Complex Fraction : If the numerator or denominator or both of a fraction are fraction then it is called as complex fraction. For example $\frac{2/3}{7/8}, \frac{7/5}{2}, \frac{3/7}{8/9}, \frac{2}{3/7}$ etc.

Remember :

$$\frac{a/b}{c/d} = \frac{a}{b} \times \frac{d}{c}$$

$$\frac{a/b}{c} = \frac{a}{b \times c}$$

$$\frac{a}{b/c} = \frac{a \times c}{b}$$

Continued Fraction : A continued fraction consists of the fractional denominators

e.g., (i) $1+\cfrac{1}{2+\cfrac{1}{3+\cfrac{1}{4+\frac{1}{5}}}}$ (ii) $2+\cfrac{3}{5+\cfrac{2}{7+\cfrac{6}{5+\frac{2}{3}}}}$

NOTE These fractions are solved starting from the bottom towards upside.

SIMPLIFICATION OF FRACTIONS

EXAMPLE 1 *Solve the following expressions :*

(i) $\frac{3}{5}+\frac{2}{7}$ (ii) $\frac{7}{3}+\frac{9}{2}$ (iii) $8\frac{3}{7}+6\frac{1}{5}$ (iv) $\frac{8}{3}-\frac{13}{6}$

(v) $\frac{6}{55}-\frac{7}{88}$ (vi) $9\frac{3}{5}-5\frac{3}{9}$ (vii) $\frac{16}{5}-\frac{3}{8}+\frac{11}{10}$

(viii) $\frac{193}{75}+\frac{37}{81}-\frac{18}{65}$

(v) $\frac{1}{2}$ *of* $\left(\frac{16}{7}+\frac{5}{3}\right)$ (vi) $\frac{2}{3}$ *of* $21\frac{4}{9}-9\frac{4}{11}$

(vii) $\frac{6}{15}\div\frac{3}{2}$ (viii) $\frac{14}{9}\div\frac{3}{7}$

(ix) $\frac{3}{7}\div\frac{2}{7}\times\frac{21}{8}$ (x) $\frac{9}{5}$ *of* $\frac{37}{8}\times\frac{36}{74}$

(xi) $5\frac{1}{3}\div\frac{4}{15}$ (xii) $36\div 4$ *of* $\frac{1}{2}+\frac{3}{4}\times\frac{3}{2}$

SOLUTION (i) $\frac{3}{5}+\frac{2}{7}=\frac{3\times7}{5\times7}+\frac{2\times5}{7\times5}=\frac{21+10}{35}=\frac{31}{35}$

(ii) $\frac{7}{3}+\frac{9}{2}=\frac{7\times2}{3\times2}+\frac{9\times3}{2\times3}=\frac{14+27}{6}=\frac{41}{6}=6\frac{5}{6}$

(iii) $8\frac{3}{7}+6\frac{1}{5}=\frac{59}{7}+\frac{31}{5}=\frac{59\times5}{7\times5}+\frac{31\times7}{5\times7}$

$=\frac{295+217}{35}=\frac{512}{35}=14\frac{22}{35}$

(iv) $\frac{8}{3}-\frac{13}{6}=\frac{8\times2}{3\times2}-\frac{13}{6}=\frac{16}{6}-\frac{13}{6}=\frac{3}{6}=\frac{1}{2}$

(v) $\frac{6}{55}-\frac{7}{88}=\frac{6\times8}{55\times8}-\frac{7\times5}{88\times5}=\frac{48}{440}-\frac{35}{440}=\frac{13}{440}$

(vi) $9\frac{3}{5}-5\frac{3}{9}=\frac{48}{5}-\frac{48}{9}=\frac{48\times9}{5\times9}-\frac{48\times5}{9\times5}$

$=\frac{432}{45}-\frac{240}{45}=\frac{192}{45}=\frac{64}{15}=4\frac{4}{15}$

(vii) $\frac{16}{5}-\frac{3}{8}+\frac{11}{10}=\frac{16\times8}{5\times8}-\frac{3\times5}{8\times5}+\frac{11\times4}{10\times4}$

$=\frac{128-15+44}{40}=\frac{157}{40}=3\frac{37}{40}$

(viii) $\frac{193}{75}+\frac{37}{81}+\frac{18}{65}=\frac{193\times27\times13}{75\times27\times13}+\frac{37\times25\times13}{81\times25\times13}+\frac{18\times81\times5}{65\times81\times5}$

$=\frac{67743+12025+7290}{26325}$

$=\frac{87058}{26325}=3\frac{8083}{26325}$

EXAMPLE 2 *Simplify the following expressions :*

(i) $\frac{3}{7}\times\frac{5}{4}$ (ii) $\frac{3}{8}\times\frac{9}{16}\times\frac{7}{5}$

(iii) $\frac{9}{7}\times\frac{21}{6}\times\frac{5}{8}\times3\frac{5}{6}$ (iv) $\frac{4}{9}$ *of* $\frac{729}{48}$

SOLUTION (i) $\frac{3}{7}\times\frac{5}{4}=\frac{15}{28}$

(ii) $\frac{3}{8}\times\frac{9}{16}\times\frac{7}{5}=\frac{189}{640}$

(iii) $\frac{9}{7}\times\frac{21}{6}\times\frac{5}{8}\times3\frac{5}{6}=\frac{9}{7}\times\frac{21}{6}\times\frac{5}{8}\times\frac{23}{6}=\frac{345}{32}=10\frac{25}{32}$

(iv) $\frac{4}{9}$ of $\frac{729}{48}=\frac{4}{9}\times\frac{729}{48}=\frac{1}{9}\times\frac{729}{12}$

$=\frac{1}{1}\times\frac{81}{12}=\frac{1}{1}\times\frac{27}{4}=\frac{27}{4}=6\frac{3}{4}$

(v) $\frac{1}{2}$ of $\left(\frac{16}{7}+\frac{5}{3}\right)=\frac{1}{2}\times\left(\frac{16}{7}+\frac{5}{3}\right)$

$=\frac{1}{2}\times\left(\frac{48}{21}+\frac{35}{21}\right)=\frac{1}{2}\left(\frac{83}{21}\right)=\frac{83}{42}=1\frac{41}{42}$

(vi) $\frac{2}{3}$ of $21\frac{4}{9}-9\frac{4}{11}=\frac{2}{3}\times21\frac{4}{9}-9\frac{4}{11}$

$=\frac{2}{3}\times\frac{193}{9}-\frac{103}{11}=\frac{386}{27}-\frac{103}{11}$

$=\frac{386\times11}{27\times11}-\frac{103\times27}{11\times27}=\frac{4246-2781}{297}$

$=\frac{1465}{297}=4\frac{277}{297}$

(vii) $\frac{6}{15}\div\frac{3}{2}=\frac{6/15}{3/2}=\frac{6}{15}\times\frac{2}{3}=\frac{4}{15}$

(viii) $\frac{14}{9}\div\frac{3}{7}=\frac{14/9}{3/7}=\frac{14}{9}\times\frac{7}{3}=\frac{98}{27}=3\frac{17}{27}$

(ix) $\frac{3}{7}\div\frac{2}{7}\times\frac{21}{8}=\frac{3}{7}\times\frac{7}{2}\times\frac{21}{8}=\frac{63}{16}=3\frac{15}{16}$

(x) $\frac{9}{5}$ of $\frac{37}{8}\times\frac{36}{74}=\frac{9}{5}\times\frac{37}{8}\times\frac{36}{74}=\frac{81}{20}=4\frac{1}{20}$

(xi) $5\frac{1}{3}\div\frac{4}{15}=\frac{16}{3}\div\frac{4}{15}=\frac{16}{3}\times\frac{15}{4}=\frac{20}{1}=20$

(xii) $36\div4$ of $\frac{1}{2}+\frac{3}{4}\times\frac{3}{2}=36\div4\times\frac{1}{2}+\frac{3}{4}\times\frac{3}{2}$

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$$= 36 \div 2 + \frac{9}{8} = 18 + \frac{9}{8}$$

$$= \frac{153}{8} = 19\frac{1}{8}$$

EXAMPLE 3 *Simplify the following expressions :*

(a) $1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{6}}}$

(b) $1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1}}}}}$

(c) $2 - \cfrac{1}{3 + \cfrac{1}{4 - \cfrac{1}{5 + \cfrac{1}{6 - \cfrac{1}{7}}}}}$

SOLUTION (a) $1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{6}}} = 1 + \cfrac{1}{1 + \cfrac{1}{7/6}}$

$$= 1 + \frac{1}{1 + \frac{6}{7}} = 1 + \frac{1}{13/7}$$

$$= 1 + \frac{7}{13} = \frac{20}{13} = 1\frac{7}{13}$$

(b) $1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1}}}}} = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \frac{1}{2}}}}$

$$= 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{3/2}}} = 1 + \frac{1}{1 + \frac{1}{1 + \frac{2}{3}}}$$

$$= 1 + \frac{1}{1 + \frac{1}{5/3}} = 1 + \frac{1}{1 + \frac{3}{5}} = 1 + \frac{1}{8/5}$$

$$= 1 + \frac{5}{8} = \frac{13}{8} = 1\frac{5}{8}$$

(c) $2 - \cfrac{1}{3 + \cfrac{1}{4 - \cfrac{1}{5 + \cfrac{1}{6 - \frac{1}{7}}}}} = 2 - \cfrac{1}{3 + \cfrac{1}{4 - \cfrac{1}{5 + \cfrac{1}{41/7}}}}$

$$= 2 - \cfrac{1}{3 + \cfrac{1}{4 - \cfrac{1}{5 + \frac{7}{41}}}} = 2 - \cfrac{1}{3 + \cfrac{1}{4 - \frac{1}{212/41}}}$$

$$= 2 - \cfrac{1}{3 + \cfrac{1}{4 - \frac{41}{212}}} = 2 - \cfrac{1}{3 + \frac{1}{807/212}}$$

$$= 2 - \frac{1}{3 + \frac{212}{807}} = 2 - \frac{1}{2633/807} = 2 - \frac{807}{2633}$$

$$= \frac{4459}{2633} = 1\frac{1826}{2633}$$

HCF AND LCM OF FRACTIONS

HCF of fractions : The greatest common fraction is called the HCF of the given fractions.

$$\text{HCF of fractions} = \frac{\text{HCF of Numerator}}{\text{LCM of denominator}}$$

For example, The HCF of $\frac{4}{3}, \frac{4}{9}, \frac{2}{15}, \frac{36}{21}$

$$= \frac{\text{HCF of } 4,\ 4,\ 2,\ 36}{\text{LCM of } 3,\ 9,\ 15,\ 21} = \frac{2}{315}$$

LCM of fractions : The least possible number of fraction which is exactly divisible by all the given fractions is called the LCM of the fractions.

$$\text{LCM of fractions} = \frac{\text{LCM of Numerator}}{\text{HCF of Denominator}}$$

For example The LCM of $\frac{4}{3}, \frac{4}{9}, \frac{2}{15}, \frac{36}{21}$

$$= \frac{\text{LCM of } 4,\ 4,\ 2,\ 36}{\text{HCF of } 3,\ 9,\ 15,\ 21} = \frac{36}{3} = 12$$

EXAMPLE 1 *Find the smallest positive number which is exactly divisible by* $\frac{1}{3}, \frac{1}{2}, \frac{3}{7}$ *and* $\frac{4}{11}$.

SOLUTION The LCM of the given numbers will be the required number. So, the LCM of $\frac{1}{3}, \frac{1}{2}, \frac{3}{7}, \frac{4}{11}$

$$= \frac{\text{LCM of } 1,\ 1,\ 3,\ 4}{\text{HCF of } 3,\ 2,\ 7,\ 11} = \frac{12}{1} = 12$$

Thus 12 is the smallest positive numbers, which is required.

EXAMPLE 2 *Four runners started running the race in the same direction around a circular path of 7 km. Their speeds are 4, 3, 9 and 3.5 km/hr individually. If they have started their race at 6 o'clock in the morning, then at what time they will be at the starting point?*

SOLUTION Time required by everyone to complete one revolution individually is $\frac{7}{4}, \frac{7}{3}, \frac{7}{9}, \frac{7}{3.5}$ hours.

Therefore everyone must reach at the starting point after the time of the LCM of the individual time period for one revolution.

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So the LCM of $\frac{7}{4}, \frac{7}{3}, \frac{7}{9}, \frac{7}{3.5} = \frac{7}{4}, \frac{7}{3}, \frac{7}{9}, \frac{2}{1}$

$= \frac{\text{LCM of } 7, 7, 7, 2}{\text{HCF of } 4, 3, 9, 1} = \frac{14}{1} = 14$ hours.

Hence after 14 hours i.e., at 8 o'clock in the evening of the same day they will meet at the starting point.

Square, square root, cube and cube root of fractions

Remember : $\sqrt{ab} = \sqrt{a}\sqrt{b}$ and $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$

$(ab)^2 = a^2 b^2$ and $\left(\frac{a}{b}\right)^2 = \frac{a^2}{b^2}$

Square of a fraction $= \frac{\text{Square of numerators}}{\text{Square of denominators}}$

Similarly, Cube of a fraction $= \frac{\text{Cube of numerator}}{\text{Cube of denominator}}$

EXAMPLE 3 *Evaluate the following :*

(a) $\left(\frac{3}{4}\right)^2$ (b) $\left(\frac{17}{25}\right)^2$ (c) $\left(\frac{19}{-31}\right)^2$ (d) $\left(\frac{78}{36}\right)^2$

(e) $\left(\frac{9}{7}\right)^3$ (f) $\left(-\frac{13}{12}\right)^3$ (g) $\left(\frac{20}{7}\right)^3$ (h) $\left(6\frac{1}{2}\right)^3$

SOLUTION (a) $\frac{9}{16}$ (b) $\frac{289}{625}$ (c) $\frac{361}{961}$ (d) $\frac{1521}{324}$

(e) $\frac{729}{343}$ (f) $-\frac{2197}{1728}$ (g) $\frac{8000}{343}$ (h) $\frac{2197}{8}$

EXAMPLE 4 *Evaluate the following :*

(a) $\sqrt{\frac{961}{625}}$ (b) $\frac{\sqrt{7744}}{20}$ (c) $\frac{31}{\sqrt{1681}}$ (d) $\sqrt{\frac{1156}{529}}$

(e) $\sqrt{50} \times \sqrt{18}$ (f) $\sqrt{39} \times \sqrt{\frac{75}{2197}}$

SOLUTION (a) $\sqrt{\frac{961}{625}} = \frac{31}{25}$

(b) $\frac{\sqrt{7744}}{20} = \frac{88}{20} = 4\frac{8}{20} = 4\frac{2}{5}$

(c) $\frac{31}{\sqrt{1681}} = \frac{31}{41}$

(d) $\frac{\sqrt{1156}}{\sqrt{529}} = \frac{34}{23}$

(e) $\sqrt{50} \times \sqrt{18} = \sqrt{50 \times 18} = \sqrt{900} = 30$

(f) $\sqrt{39} \times \sqrt{\frac{75}{2197}} = \sqrt{\frac{39 \times 75}{2197}}$

$= \sqrt{\frac{13 \times 3 \times 3 \times 25}{13 \times 13 \times 13}} = \frac{15}{13} = 1\frac{2}{13}$

EXAMPLE 5 *The value of* $\left\{\frac{1}{(\sqrt{2}-2)} + \frac{1}{(\sqrt{2}+2)} + (\sqrt{2}+2)\right\}$ *is :*

(a) 2 (b) 4

(c) $2\sqrt{2}$ (d) *none of these*

SOLUTION $\left\{\frac{(\sqrt{2}+2)+(\sqrt{2}-2)}{(\sqrt{2}-2)(\sqrt{2}+2)} + (\sqrt{2}+2)\right\} = 2$

EXAMPLE 6 *The value of* $\left\{\frac{1}{(\sqrt{6}-\sqrt{5})} - \frac{1}{(\sqrt{5}-\sqrt{4})} + \frac{1}{(\sqrt{4}-\sqrt{3})} - \frac{1}{(\sqrt{3}-\sqrt{2})} + \frac{1}{(\sqrt{2}-1)}\right\}$ *is :*

(a) 5 (b) 7

(c) $\sqrt{12}$ (d) *none of these*

SOLUTION $\frac{1}{(\sqrt{6}-\sqrt{5})} - \frac{1}{(\sqrt{5}-\sqrt{4})} + \frac{1}{(\sqrt{4}-\sqrt{3})} - \frac{1}{(\sqrt{3}-\sqrt{2})} + \frac{1}{(\sqrt{2}-1)}$

$= (\sqrt{6}+\sqrt{5}) - (\sqrt{5}+\sqrt{4}) + (\sqrt{4}+\sqrt{3}) - (\sqrt{3}+\sqrt{2}) + (\sqrt{2}+1)$

$= \sqrt{6} + 1$

$\left[\because \frac{1}{\sqrt{a}-\sqrt{b}} = \frac{(\sqrt{a}+\sqrt{b})}{(\sqrt{a}-\sqrt{b})(\sqrt{a}+\sqrt{b})} = \frac{\sqrt{a}+\sqrt{b}}{(a-b)}\right]$

Hence (d) is the correct option.

PRACTICE EXERCISE

1. Find the value of $\frac{1}{\sqrt{9}-\sqrt{4}}$.
2. Find the value of x if $\sqrt{1+\frac{x}{169}} = \frac{14}{13}$.
3. Find the value of x if $140\sqrt{x} + 315 = 1015$.
4. If $\sqrt{\left(1+\frac{27}{169}\right)} = \left(1+\frac{x}{13}\right)$ then find the value of x.
5. Find the value of $\left[6\frac{1}{4} \div \left\{2\frac{1}{4} - \frac{1}{2}\left(2\frac{1}{2} - \overline{\frac{1}{4} - \frac{1}{6}}\right)\right\}\right]$.
6. Find the value $\frac{3}{2}$ of $\left(\frac{4}{3} \div \frac{5}{7}\right) + \frac{1}{2} \div \left[3\frac{4}{5} - \left\{\frac{2}{5} \div \left(\frac{1}{3} + \frac{1}{2} + \overline{\frac{1}{5} - \frac{1}{6}}\right)\right\}\right]$.
7. Find the value of $1 + \cfrac{1}{2 + \cfrac{1}{4 + \cfrac{1}{8 + \cfrac{1}{16}}}}$.
8. Find the value of $1 - \cfrac{1}{2 + \cfrac{1}{2 - \cfrac{1}{2 + \frac{1}{2}}}}$.
9. Calculate the value of $\frac{2}{5} + 2\frac{4}{9} \div \left[\left(7\frac{5}{12} - 5\frac{3}{4}\right) \div 22\frac{1}{2} + 10 \times \frac{5}{18}\right] - \frac{4}{5}$.
10. Calculate the value of $\cfrac{1}{2 + \cfrac{1}{2 + \cfrac{1}{2 - \frac{1}{2}}}}$.

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11. Calculate the value of $\left(1\frac{29}{36}+1\frac{7}{11}\times 4\frac{1}{8}\right)\div\left(5\frac{1}{9}-\frac{63}{8}\div 9\frac{9}{20}\right)$.

12. Calculate the value of $\frac{4}{2\frac{1}{3}}+\frac{3}{1\frac{3}{4}}-\frac{5}{3\frac{1}{2}}$.

ANSWERS

1. 1 **2.** 27 **3.** 25 **4.** 1 **5.** 6
6. $2\frac{2061}{2170}$ **7.** $1\frac{532}{1193}$ **8.** $\frac{13}{21}$ **9.** $\frac{16}{35}$ **10.** $\frac{8}{19}$
11. 2 **12.** 2

EXAMPLE 1 *From a rope of length* $38\frac{3}{5}$ *m, a piece of length* $5\frac{3}{38}$ *m is cut off. The length of the remaining rope is :*

(a) $190\frac{99}{33}$ *(b)* $33\frac{90}{190}$
(c) $33\frac{99}{190}$ *(d) none of these*

SOLUTION Total length of the rope $= 38\frac{3}{5}=\frac{193}{5}$

The length of the rope which has been removed $= 5\frac{3}{38}=\frac{193}{38}$

$\therefore$ The length of the remaining part of the rope

$$=\frac{193}{5}-\frac{193}{38}=33\frac{99}{199}$$

Hence (c) is the correct option.

EXAMPLE 2 *The cost of 1 metre cloth is Rs.* $21\frac{1}{2}$ *then the cost of* $\frac{42}{43}$ *metre :*

(a) Rs. 21 *(b) Rs. 42*
(c) Rs. $\frac{43}{2}$ *(d) none of these*

SOLUTION $\because$ The price of 1m cloth $= 21\frac{1}{2}=\frac{43}{2}$

$\therefore$ the price of $\frac{42}{43}$ m cloth $=\frac{42}{43}\times\frac{43}{2}=21$

Hence option (a) is correct.

EXAMPLE 3 *A drum of kerosene oil is* $\frac{3}{4}$ *full. When 15 litres of oil is drawn from it, it is* $\frac{7}{12}$ *full. The capacity of the drum is :*

(a) 45 *(b) 90*
(c) 60 *(d) can't be determined*

SOLUTION Let the capacity of drum be x

then $\frac{3}{4}x-15=\frac{7}{12}x$

$\Rightarrow \frac{9x-7x}{12}=15$

$\Rightarrow x=90$

Thus (b) is the correct option.

Alternatively : If you consider option (b) then

$$90\times\frac{3}{4}-15=90\times\frac{7}{12}$$

Alternatively: The decrease in amount $=\frac{x}{6}=15$

$\Rightarrow x=90$ litre.

EXAMPLE 4 *A sum of Rs. 11200 is shared among Mr. Khare, Mr. Patel and Mr. Verma. Mr. Khare gets* $\frac{1}{4}$*th of it while Mr. Patel gets* $\frac{1}{5}$*th of it. The amount of Mr. Verma is :*

(a) 6006 *(b) 6160*
(c) 3080 *(d) data insufficient*

SOLUTION The share of Mr. Khare $=\frac{1}{4}$

The share of Mr. Patel $=\frac{1}{5}$

$\therefore$ The share of Mr. Verma $=1-\left(\frac{1}{4}+\frac{1}{5}\right)=\frac{11}{20}$

So, the amount of Mr. Verma $=\frac{11}{20}\times 11200=6160$

Thus option (b) is correct.

EXAMPLE 5 *When Sarvesh travelled 33 km, he found that* $\frac{2}{3}$*rd of the entire journey was still left. The length of the total journey is :*

(a) 66 *(b) 132* *(c) 99* *(d) 100*

SOLUTION Since $\frac{2}{3}$rd journey is left, it means Sarvesh has travelled only $\frac{1}{3}$rd of his journey which is equal to 33. Thus *i.e.*,

$$\frac{1}{3}x=33 \Rightarrow x=99$$

Hence (b) is the right choice.

EXAMPLE 6 *Mrs. Verma earns Rs. 18000 per month. She spends* $\frac{7}{12}$ *on house hold items and* $\frac{1}{8}$ *on rest of the things. The amount she saves is :*

(a) 7120 *(b) 5250*
(c) 5520 *(d) none of these*

SOLUTION Her total expenditure $=\frac{7}{12}+\frac{1}{8}=\frac{17}{24}$

$\therefore$ Her savings $=1-\frac{17}{24}=\frac{7}{24}$

$\therefore$ Her saving in Rs. $=\frac{7}{24}\times 18000$

$=5250$

Hence (b) is the correct option.

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EXAMPLE 7 *Neha, a working lady, earns Rs. x per month. If she spends $\frac{2}{5}$th of her earning for personal uses and $\frac{3}{4}$th of the personal uses, she spends in entertainment while $\frac{7}{20}$th of the expenditure in entertainment she spends in movies only. Her salary would be, if her expenditure in movies is in integers.*

(a) 4225 *(b)* 2175 *(c)* 200 *(d)* 3465

SOLUTION Since her salary is Rs. x then her expenditure in movies is

$$\frac{7}{20}\text{ of }\frac{3}{4}\text{ of }\frac{2}{5}\text{ of }x=\frac{7}{20}\times\frac{3}{4}\times\frac{2}{5}\times x=\frac{21x}{200}$$

Now, In order to $\frac{21x}{200}$ be an integer so x must be equal to 200.

Thus (c) is the correct option.

DECIMAL FRACTIONS

In the previous topics I have discussed the integral and fractional numbers. Now in the ongoing discussion. I am extenting the theory of numbers as "decimal numbers".

In the number 8637, the place value of 7 is $7\times1=7$, the place value of 3 is $10\times3=30$, the place value of 6 is $6\times100=600$ and the place vale of 8 is $8\times1000=8000$.

Thus, as we move from right to left each next digit is multiplied by 10, 100, 1000, 10000, 100000 etc.

It means moving from left to right the place value of each digit is divided by 10, 100, 1000, etc.....

In the above number 7 is known as unit digit (the right most digit), 3 is known as tens digit, 6 as hundreds digit and 8 as thousands digit.

Now if we move towards the right of unit place, the place value of the digit next to the right of unit digit (is the tenths digit) is being divided by 10, while the place value of the next digit is the hundredth digit is being divided by 100 and so on.

So, for our convenience we put a small dot just right to the unit digit and it is known as decimal point.

Hence, the place value of a digit in a number depends on the place or position with respect to the unit digit position while the **face value** of a digit is always fixed as the face value of 1 is 1, 2 is 2, 3 is 3, 4 is 4 and so on.

Place Value Chart

Places	Ten thousands	Thousands	Hundreds	Tens	Unit	Decimal point	Tenths	Hundredths	Thousandths
Value	10000	1000	100	10	1	.	$\frac{1}{10}$	$\frac{1}{100}$	$\frac{1}{1000}$

Thus a decimal number say 4637.938 can be expressed as

$$4\times1000+6\times100+3\times10+7\times1+\frac{9}{10}+\frac{3}{100}+\frac{8}{1000}.$$

NOTE The digits written in the right side of decimal point read separately as we read 2.359 as two point three five nine.

Thus the fraction whose denominators are 10, 100, 1000 etc. are called decimal fractions.

The number left to the decimal point is a whole or integral number, while the number to right to the decimal point is a **decimal number.** In the number 735.82, 735 is a whole number and .82 is the **decimal number.**

Conversion of decimal numbers into decimal fractions : Write the number of zeros as the number of digits in the decimal number preceded by 1, below the actual number without decimal point as

$$23.6=\frac{236}{10},\quad 4.579=\frac{4579}{1000}$$

$$75.89=\frac{7589}{100},\quad 0.18870=\frac{18870}{100000}=\frac{1887}{10000}$$

Conversion of decimal fractions into decimal numbers : Count the number of zeros in the denominator and then count the same number of digits in the numerator starting from the unit digit of the numerator moving to the left and then place the decimal point as :

$$\frac{2375}{10}=237.5,\quad \frac{2375}{100}=23.75,\quad \frac{2375}{1000}=2.375,$$

$$\frac{2375}{100000}=0.02375,\quad \frac{2375}{10000000}=0.0002375\text{ etc.}$$

NOTE 0.35 = 0.350 = 0.3500 = 0.35000 etc. because increasing the number of zeros in the rightmost of a decimal number is inconsiderable.

For example $0.35=\frac{35}{100}$

$$0.350=\frac{350}{1000}=\frac{35}{100}$$

and $0.3500=\frac{3500}{10000}=\frac{35}{100}$

Some mathematical operations on decimal numbers :

(i) Addition and Subtraction : The decimal numbers are written in such a manner that decimal points of all the numbers fall in one column or fall below one another. For example

$$\begin{array}{r} 2.358 \\ 709.21 \\ 0.35 \\ +\ 2.0067 \\ \hline 713.9247 \end{array}\qquad\text{and}\qquad \begin{array}{r} 635.888 \\ -\ 28.0125 \\ \hline 607.8755 \end{array}$$

(ii) Multiplication : Suppose there is no decimal point as natural numbers and then multiply them.

Then put the decimal point in the product as the sum of the decimal places in the multiplicands. For example

	Decimal Places
(i) $23\times1.1=25.3$	$0+1=1$
(ii) $3.8\times4.6=17.48$	$1+1=2$
(iii) $2.456\times7.8=19.1568$	$3+1=4$
(iv) $8.125\times100=812.500$	$(3+0)=3$
(v) $75\times0.3=22.5$	$(0+1)=1$

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(iii) Division : Make the divisor as a natural number by shifting the decimal in the right hand side equally in dividend and divisor. Now divide the resultant dividend by this resultant divisor as usual.

For Example :

$$65 \div 1.3 = \frac{65}{1.3} = \frac{650}{13} = 50$$

$$235 \div 4.7 = \frac{235}{4.7} = \frac{2350}{47} = 50$$

$$1.17 \div 13 = \frac{1{\cdot}17}{13} = 0.09 \quad \left[\because \frac{1{\cdot}17}{13} = \frac{117}{13} \times \frac{1}{100} = 9 \times \frac{1}{100}\right]$$

$$172.8 \div 0.12 = \frac{172.8}{0.12} = \frac{17280}{12} = 1440$$

$$1{\cdot}068 \div 8.9 = \frac{1.068}{8.9} = \frac{10.68}{89} = 0.12$$

$$78 \div 0.0039 = \frac{78}{0.0039} = \frac{780000}{0039}$$

$$= \frac{780000}{39} = 20000$$

Recurring Decimal : A decimal number in which a digit or a set of digits repeats regularly, over a constant period, is called a recurring decimal or periodic decimal. For example

2.33333..., 7.5555, ..., 108.232323...

9.142857142857142857.....

where $2.3333\ldots = 2.\overline{3}$ or $2.\dot{3}$

$7.5555\ldots = 7.\overline{5}$ or $7.\dot{5}$

$108.232323\ldots = 108.\overline{23}$

and $9.142857142857142857\ldots = 9.\overline{142857} = 9.\dot{1}4285\dot{7}$

So the recurring of the decimal number is expressed by putting the bar or dot over the period of a recurring decimal.

(i) Pure Recurring decimal : A decimal fraction in which all the figures (or digits) occur repeatedly, is called a pure recurring decimal as 7.4444, ..., 2.666, ..., 9.454545, ... etc.

Mixed Recurring decimal : A decimal, number in which some of the digits do not recur is called a mixed recurring decimal for example 327.63454545...

Non-recurring decimals : A decimal number in which there is no any regular pattern of repetition of digits after decimal point is called a non-recurring decimal.

e.g., 3.2466267628....

CONVERSION OF RECURRING DECIMAL INTO VULGAR FRACTION

(a) Pure Recurring Decimal

Write down as many 9's in the denominator as the number of digits in the period of decimal number, below the given decimal no., for example,

(i) $0.77777\ldots = 0.\overline{7} = \frac{7}{9}$

(ii) $0.545454\ldots = 0.\overline{54} = \frac{54}{99}$

(iii) $2.357357357357\ldots = 2.\overline{357} = 2 + 0.\overline{357}$

$$= 2 + \frac{357}{999} = \frac{2355}{999}$$

(iv) $0.768397683976839\ldots = 0.\overline{76839} = \frac{76839}{99999}$

(b) Mixed Recurring Decimal

Step 1. Write down the number and then subtract from it the no. formed by the non recurring digits after decimal point.

Step 2. Write down as many 9's as there are digits in the period of the recurring decimal followed by as many zeros as there is number of digits which are not recurring after the decimal point.

Step 3. Divide the value of step 1 by the value obtained in step 2.

For example :

1. $0.17777\ldots = 0.1\overline{7} = \frac{17-1}{90} = \frac{16}{90} = \frac{8}{45}$

2. $0.8737373\ldots = 0.\overline{873} = \frac{873-8}{990} = \frac{865}{990} = \frac{173}{198}$

3. $0.78943943943\ldots = 0.7\overline{8943} = \frac{78943-78}{99900}$

$$= \frac{78865}{99900} = \frac{15773}{19980}$$

4. $0.250121212\ldots = 0.250\overline{12} = \frac{25012-250}{99000}$

$$= \frac{24762}{99000} = \frac{12381}{49500}$$

5. $5.00\overline{72} = 5 + 0.00\overline{72} = 5 + \left(\frac{72-00}{9900}\right)$

$$= 5 + \frac{72}{9900} = 5 + \frac{2}{275} = 5\frac{2}{275}$$

ADDITION AND SUBTRACTION OF RECURRING DECIMALS

EXAMPLE 1 $5.7\overline{32} + 8.\overline{613}$

SOLUTION $5.7\overline{32} + 8.\overline{613} = 5 + 0.7\overline{32} + 8 + 0.\overline{613}$

$$= 5 + \frac{732-7}{990} + 8 + \frac{613}{999}$$

$$= 5 + \frac{725}{990} + 8 + \frac{613}{999}$$

$$= 13 + \frac{725}{990} + \frac{613}{999} = 13 + \frac{145}{198} + \frac{613}{999}$$

$$= 13 + \frac{145 \times 3 \times 37 + 613 \times 2 \times 11}{2 \times 3 \times 9 \times 11 \times 37}$$

$$= \frac{13 \times 2 \times 3 \times 9 \times 11 \times 37 + 145 \times 3 \times 37 + 613 \times 2 \times 11}{2 \times 3 \times 9 \times 11 \times 37}$$

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$$= \frac{285714 + 16095 + 13486}{21978} = \frac{315295}{21978} = 14.3459368$$

Alternatively : Step 1. Express the numbers without bar as

$5.732323232 + 8.613613613613$

Step 2. Write the numbers as one above other *i.e.*,

5.732323232
8.613613613

Step 3. Divide this number into two parts. In the first part *i.e.*, left side write as many digits as there will be integral value with non recurring decimal. In the right side write as many digits as the LCM of the number of recurring digits in the given decimal number *e.g.*,

$5.7 \mid 323232$ (Since 5.7 is the integral + non-recurring part]
$8.6 \mid 136136$ (The LCM of 2 and 3 is 6]

Step 4. Now add or subtract as usual.

5.7	323232
8.6	136136
14.3	459368

Step 5. Put the bar over the digits which are on the right side in the resultant value.

$14.3\overline{459368}$

Thus $5.7\overline{32} + 8.\overline{613} = 14.3\overline{459368}$

EXAMPLE 2 *Solve the following :* $19.\overline{368421} - 16.20\overline{53}$

SOLUTION $19.3684216842168421 - 16.2053535353...$

$= 19.3684216842168421$
$= 16.205353535353535353$

19 . 36	8421684216
− 16.20	5353535353
3.16	3068148863

(Since 16.20 is integral part with non recurring digits]
[The LCM of 5 and 2 is 10]
In the first no. there are 5 recurring digits and in the second no. there are 2 recurring digits]

$19.\overline{368421} - 16.20\overline{53} = 3.16\overline{3068148863}$

MULTIPLICATION AND DIVISION OF RECURRING DECIMALS

It can be done as usual. Just convert the decimals into vulgar fractions and then operate as required.

Example 1. $97.2\overline{81} \times 100 = 97.2818181 \times 100$

$= 9728.181818 = 9728.\overline{181}$

(Remember that the set of recurring digits is not altered as in the above problem the set of recurring digit will remain $\overline{81}$ but not $\overline{18}$ since initially it was $\overline{81}$.)

Example 2. $25.6\overline{32} \times 55 = (25 + 0.6\overline{32}) \times 55$

$$= 25\frac{632-6}{990} \times 55 = 25\frac{626}{990} \times 55$$

$$= 25\frac{313}{495} \times 55 = \frac{12688}{495} \times 55$$

$$= \frac{12688}{9} = 1409.7777 = 1409.\overline{7}$$

Example 3. $13.00\overline{5} \times 20 \times 8.4\overline{23} = 13\frac{5}{900} \times 20 \times 8\frac{423-4}{990}$

$$= 13\frac{1}{180} \times 20 \times 8\frac{419}{990}$$

$$= \frac{2341}{180} \times 20 \times \frac{8339}{990} = \frac{2341 \times 8339}{9 \times 990}$$

$$= \frac{19521599}{9 \times 990} = 2190.9763187$$

Example 4. $0.08\overline{9} \div 100 = 0.089999... \div 100$

$= 0.00089999... = 0.00089$

Example 5. $53.08\overline{53} \div 6 = \frac{53 + 0.08\overline{53}}{6} = \frac{53 + \frac{0853-08}{9900}}{6}$

$$= \frac{53 + \frac{845}{9900}}{6} = \frac{53 + \frac{169}{1980}}{6}$$

$$= \frac{\left(\frac{105109}{1980}\right)}{6} = \frac{105109}{1980 \times 6} = \frac{105109}{11880}$$

$= 8.847558922558922558922$
$= 8.847558922$

Alternatively : $53.08\overline{53} \div 6 = 53.08535353... \div 6$

$6 \overline{)} 53.08535353... (8.847558922558922$

48
50
48
28
24
45
42
33
30
35
30
53
48
55
54
13
12
15
12
33
30
35
30
53
48
55
54
1 etc.

Square of Decimals : First we assume that there is no decimal point and then square up the given decimal number. But at last we put the dot (as decimal point), counting the digits from the rightmost digit. The no. of places will be double as that of the given number.

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For example :

1. $(2.3)^2 = 5.29$ $[(23)^2 = 529]$
2. $(1.07)^2 = 1.1449$ $[(107)^2 = 11449]$
3. $(0.11)^2 = 0.0121$ $[(11)^2 = 121]$
4. $(1.352)^2 = 1.827904$ $[\because (1352)^2 = 1827904]$
5. $0.1 = 0.01$ $[\because (1)^2 = 1]$
6. $(9.99)^2 = 99.8001$ $[(999)^2 = 998001]$

Square root of Decimals : The process of finding the square root is same as that of integers. Here we put the decimal point as in the division of decimal numbers.

For example :

1. Find the square root of 5.29

	2.3
2	5.29
2	4
43	129
3	129
	×

$\therefore \sqrt{5.29} = 2.3$

2. Find the square root of 0.0121

	0.11
0	0.0121
0	0
01	01
1	01
21	21
1	21
	×

$\therefore \sqrt{0.0121} = 0.11$

3. Find the square root of 536.85374

	23.1701
2	536.85374
2	4
43	136
3	129
461	785
1	461
4627	32437
7	32389
463401	484000
01	463401

The process can be continued if required. But generally we stop our calculation after 2-3 places of decimal point.

NOTE To find the nth power of a decimal number in which the decimal point is placed at m places before the rightmost digit, we simply solve it considering as an integer then we put the decimal point in the resultant value before the $m.n$ places from the right most digit, where m, n are positive integers.

For example : $(2.13)^4 = 20.58346161$ [The no. of places of decimal point $= 2 \times 4 = 8$]

HCF AND LCM OF DECIMALS

HCF

Step 1. First of all equate the number of places in all the numbers by using zeros, wherever required.

Step 2. Then considering these numbers as integers find the HCF of these numbers.

Step 3. Put the decimal point in the resultant value as many places before the right most digit as that of in the every equated number.

EXAMPLE 1 *Find the HCF of 0.0005, 0.005, 0.15, 0.175, 0.5 and 3.5.*

SOLUTION.

$0.0005 \Rightarrow 5$
$0.0050 \Rightarrow 50$
$0.1500 \Rightarrow 1500$
$0.1750 \Rightarrow 1750$
$0.5000 \Rightarrow 5000$
$3.5000 \Rightarrow 35000$

Then the HCF of 5, 50, 1500, 1750, 5000 and 35000 is 5.

So the HCF of the given number is 0.0005 (since there are four digits in all the adjusted (or equated) decimal places.

EXAMPLE 2 *Find the HCF of 0.9, 0.36 and 1.08.*

SOLUTION

$0.9, 0.36, 1.08 \rightarrow 0.90, 0.36, 1.08 \rightarrow 90, 36, 108$

Now the HCF of 90, 36 and 108 is 18.

So the required HCF of the given numbers is 0.18.

(Since there are two digits after decimal places in every number in second step)

LCM

Step 1. First of all equate the no. of places in all the given numbers by putting the minimum possible no. of zeros at the end of the decimal numbers, wherever ever required.

Step 2. Now consider the equated numbers as integers and then find the LCM of these numbers.

Step 3. Put the decimal point in the LCM of the numbers as many places as that of in the equated numbers.

EXAMPLE 1 *Find the LCM of 1.8, 0.54 and 7.2*

SOLUTION

$1.8, 0.54, 7.2 \rightarrow 1.80, 0.54, 7.20 \rightarrow 180, 54, 720$

Now the LCM of 180, 54 and 720 is 2160

Therefore the required LCM is 21.60.

EXAMPLE 2 *Find the LCM of 4.44, 37 and 55.5*

SOLUTION

$4.44, 37.0, 55.5 \rightarrow 4.44, 37.00, 55.50 \rightarrow 444, 3700, 5550$

Now the LCM of 444, 3700, 5550 is 11100.

Hence the required LCM is 111.00 = 111.

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INDICES AND SURDS

In the previous articles we have studied the multiplication, as $2\times2=4, 2\times2\times2=8, 2\times2\times2\times2=16, \ldots$

Now, here we extend our discussion to the higher power or index of a number.

as
$$2\times2=2^2=4$$
$$2\times2\times2=2^3=8$$
$$2\times2\times2\times2=2^4=16$$
$$2\times2\times2\times2\times2=2^5=32$$

Similarly, $2\times2\times2\times\ldots..$ upto n times $=2^n$

Thus $\underbrace{a\times a\times a\times a\times a\ldots a}_{n \text{ factors}}=a^n$

Where a is any number and n is a natural number. *i.e.*, a^n is continued product of n equal quantities, each equal to a and is called the nth power of a. 'n' is called the **index or exponent** and 'a' is called the **base** of a^n.

Therefore a^n is the exponential expression. a^n is read as 'a raised to the power n' or 'a to the power n'

Laws of Indices

If m and n are positive integers, then

1. $a^m \times a^n = a^{m+n}$
2. $\frac{a^m}{a^n} = a^{m-n}; (a\neq0, m\geq n)$
3. $(ab)^n = a^n . b^n$
4. $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}; (b\neq0)$
5. $(a^m)^n = a^{m.n}$
6. $(a^0)=1; (a\neq0)$

Some Important Results

1. If $a^x = k$, then $a=(k)^{1/x}$
2. If $a^{1/x}=k$, then $a=k^x$
3. If $a^x=b^y$, then $a=(b)^{y/x}$ and $b=(a)^{x/y}$
4. If $a^x=a^y$, then $x=y$, where $a\neq0,1$
5. $a^{-n}=\frac{1}{a^n}$ and $a^n=\frac{1}{a^{-n}}$
6. $a^{b^c}\neq(a^b)^c; b\neq c$

EXAMPLE 1 *Solve the following :*
(i) $(5)^3$ *(ii)* $(-6)^4$ *(iii)* $(-2)^5$ *(iv)* $(4)^3$ *(v)* $(-4)^3$

SOLUTION (i) $5^3=5\times5\times5=125$

(ii) $(-6)^4=(-6)\times(-6)\times(-6)\times(-6)=1296$

(iii) $(-2)^5=(-2)\times(-2)\times(-2)\times(-2)\times(-2)=-32$

(iv) $(4)^3=4\times4\times4=64$

(v) $(-4)^3=(-4)\times(-4)\times(-4)=-64$

EXAMPLE 2 *Solve the following expressions.*
(i) $(-2)^3\times5^2$ *(ii)* $(-4)^3\times5^2\times7^0$
(iii) $(32)^{-1/5}$ *(iv)* $(243)^{4/5}$
(v) $(36)^{1/6}$ *(vi)* $\left(-\frac{1}{343}\right)^{-2/3}$
(vii) $3^{-3}+(-3)^3$ *(viii)* $(2^2+2^3+2^{-2}+2^{-3})$
(ix) $2^{2x-1}=\frac{1}{8^{(x-3)}}$, *then* $x=?$
(x) $4^{2x+1}=8^{x+3}$ *then* $x=?$
(xi) $3^{x-1}+3^{x+1}=90$, *then* $x=?$

SOLUTION (i) $(-2)^3\times5^2=-8\times25=-200$

(ii) $(-4)^3\times5^2\times7^0=-64\times25\times1=-1600$

(iii) $(32)^{-1/5}=(2^5)^{-1/5}=2^{5\times-\frac{1}{5}}=2^{-1}=\frac{1}{2}$

(iv) $(243)^{4/5}=(3^5)^{4/5}=3^{5\times\frac{4}{5}}=3^4=81$

(v) $(36)^{1/6}=(6^2)^{1/6}=6^{2\times\frac{1}{6}}=6^{\frac{1}{3}}$

(vi) $\left(-\frac{1}{343}\right)^{-2/3}=\left(-\frac{1}{7^3}\right)^{-2/3}=(-7^{-3})^{-2/3}$
$=(-7)^{-3\times-2/3}=(-7)^2=49$

(vii) $3^{-3}+(-3)^3=\frac{1}{3^3}+(-3)^3=\frac{1}{27}-27$
$=\frac{1-729}{27}=-\frac{728}{27}$

(viii) $(2^2+2^3+2^{-2}+2^{-3})=\left(4+8+\frac{1}{4}+\frac{1}{8}\right)$
$=12+\frac{3}{8}=\frac{99}{8}$

(ix) $2^{2x-1}=\frac{1}{8^{x-3}} \Rightarrow 2^{2x-1}=\frac{1}{2^{3(x-3)}}$

$\Rightarrow 2^{2x-1}=\frac{1}{2^{3x-9}}$

$\Rightarrow (2^{2x-1})(2^{3x-9})=1$

$\Rightarrow 2^{(2x-1)+(3x-9)}=1$

$\Rightarrow 2^{5x-10}=1 \Rightarrow 2^{5(x-2)}=1$

$\Rightarrow 2^{5(x-2)}=2^0$

$\Rightarrow 5(x-2)=0$

$\Rightarrow x-2=0 \Rightarrow x=2$

(x) $4^{2x+1}=8^{x+3}$

$\Rightarrow (2^2)^{(2x+1)}=(2^3)^{(x+3)}$

$\Rightarrow 2^{4x+2}=2^{3x+9}$

$\Rightarrow 4x+2=3x+9$ (∵ base in both side is same)

$\Rightarrow x=7$

(xi) $3^{x-1}+3^{x+1}=90$

$\Rightarrow \frac{3^x}{3}+3.3^x=90$

$\Rightarrow 3^x+3^2.3^x=90\times3$

$\Rightarrow 3^x(1+3^2)=270$

$\Rightarrow 3^x(10)=270$

$\Rightarrow 3^x=27$

$\Rightarrow 3^x=3^3$

$\Rightarrow x=3$

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EXAMPLE 3 *Solve the followings :*

(i) $\left[\left(x+\frac{1}{y}\right)^a\left(x-\frac{1}{y}\right)^b\right]\div\left[\left(y+\frac{1}{x}\right)^a\left(y-\frac{1}{x}\right)^b\right]$ *is equal to :*

(a) $\left(\frac{x}{y}\right)^{a+b}$ (b) $\left(\frac{y}{x}\right)^{(a+b)}$ (c) $\frac{x^a}{y^b}$ (d) $(xy)^{a+b}$

(ii) $\left(\frac{x^b}{x^c}\right)^a\times\left(\frac{x^c}{x^a}\right)^b\times\left(\frac{x^a}{x^b}\right)^c$ *is equal to :*

(a) 0 (b) 1
(c) abc (d) none of these

(iii) *If* $x^{x^{3/2}}=(x^{3/2})^x$, *then the value of x is :*

(a) $\frac{3}{2}$ (b) $\frac{9}{4}$ (c) $\frac{16}{25}$ (d) $\frac{8}{27}$

(iv) *If* $x^a=y^b=z^c$ *and* $y^2=zx$ *then the value of* $\frac{1}{a}+\frac{1}{c}$ *is :*

(a) $\frac{b}{2}$ (b) $\frac{c}{2}$ (c) $\frac{2}{b}$ (d) 2a

(v) $(a^{m-n})^l\times(a^{n-l})^m\times(a^{l-m})^n$

(a) 1 (b) 0 (c) 2 (d) a^{lmn}

(vi) *If* $a^x=b$, $b^y=c$ *and* $c^z=a$, *then the value of xyz is:*

(a) 0 (b) 1 (c) $x+y+z$ (d) abc

(vii) $\left(\frac{x^a}{x^b}\right)^{a+b}\times\left(\frac{x^b}{x^c}\right)^{b+c}\times\left(\frac{x^c}{x^a}\right)^{c+a}$ *is equal to :*

(a) 1 (b) 0
(c) 2 (d) none of these

(viii) *The value of*

$$\frac{1}{1+x^{b-a}+x^{c-a}}+\frac{1}{1+x^{c-b}+x^{a-b}}+\frac{1}{1+x^{a-c}+x^{b-c}}$$ *is :*

(a) 0 (b) x^{abc} (c) 1 (d) $x^{(a+b+c)}$

(ix) $2^{2^x}=16^{2^{3x}}$, *then x is equal to*

(a) −1 (b) 0
(c) 1 (d) none of these

(x) *The value of the expression*

$$\frac{4^n\times20^{n-1}\times12^{m-n}\times15^{m+n-2}}{16^m\times5^{2m+n}\times9^{m-1}}$$ *is :*

(a) 500 (b) 1 (c) 200 (d) $\frac{1}{500}$

(xi) *The value of the expression*

$$\frac{(0.3)^{1/3}\cdot\left(\frac{1}{27}\right)^{1/4}\cdot(9)^{1/6}\cdot(0.81)^{2/3}}{(0.9)^{2/3}\cdot(3)^{-1/2}\cdot\left(\frac{1}{3}\right)^{-2}\cdot(243)^{-1/4}}$$ *is :*

(a) 0.3 (b) 0.9 (c) 1.27 (d) 0.09

(xii) *The value of expression* $\frac{(0.6)^0-(0.1)^{-1}}{\left(\frac{3}{2^3}\right)^{-1}\cdot\left(\frac{3}{2}\right)^3+\left(-\frac{1}{3}\right)^{-1}}$ *is :*

(a) $-\frac{3}{2}$ (b) $\frac{2}{3}$ (c) $\frac{3}{2}$ (d) $\frac{9}{4}$

(xiii) *If* $\frac{9^n\times3^2\times(3^{-n/2})^{-2}-(27)^n}{3^{3m}\times2^3}=\frac{1}{27}$ *then the value of* $(m-n)$ *is :*

(a) −1 (b) 1 (c) 2 (d) −2

SOLUTION (i) $\left[\left(x+\frac{1}{y}\right)^a\cdot\left(x-\frac{1}{y}\right)^b\right]\div\left[\left(y+\frac{1}{x}\right)^a\left(y-\frac{1}{x}\right)^b\right]$

$$=\frac{\left(x+\frac{1}{y}\right)^a\left(x-\frac{1}{y}\right)^b}{\left(y+\frac{1}{x}\right)^a\left(y-\frac{1}{x}\right)^b}=\frac{\left(\frac{xy+1}{y}\right)^a\left(\frac{xy-1}{y}\right)^b}{\left(\frac{xy+1}{x}\right)^a\left(\frac{xy-1}{x}\right)^b}$$

$$=\frac{\frac{(xy+1)^a}{y^a}\cdot\frac{(xy-1)^b}{y^b}}{\frac{(xy+1)^a}{x^a}\cdot\frac{(xy-1)^b}{x^b}}$$

$$=\frac{(xy+1)^a(xy-1)^b}{y^{(a+b)}}\times\frac{x^{(a+b)}}{(xy+1)^a(xy-1)^b}$$

$$=\frac{x^{a+b}}{y^{a+b}}=\left(\frac{x}{y}\right)^{a+b}$$

Hence (a) is the correct option.

(ii) $\left(\frac{x^b}{x^c}\right)^a\times\left(\frac{x^c}{x^a}\right)^b\times\left(\frac{x^a}{x^b}\right)^c=(x^{b-c})^a\times(x^{c-a})^b\times(x^{a-b})^c$

$$=x^{(ab-ac)}\times x^{(bc-ab)}\times x^{(ac-bc)}$$
$$=x^{(ab-ac)+(bc-ab)+(ac-bc)}$$
$$=x^0=1$$

Hence (b) is the correct option.

(iii) $x^{x^{3/2}}=(x^{3/2})^x$

$\Rightarrow$ $x^{x^{3/2}}=x^{(3/2)x}$

$\Rightarrow$ $x^{3/2}=\frac{3}{2}x$

$\Rightarrow$ $x^{1/2}=\frac{3}{2}\Rightarrow x=\frac{9}{4}$

Hence (b) is the correct option.

(iv) If $x^a=y^b=z^c$ and $y^2=zx$

Let $x^a=y^b=z^c=k$

$\Rightarrow$ $x=k^{1/a}$, $y=k^{1/b}$, $z=k^{1/c}$

Now, $\because$ $y^2=zx$

$\therefore$ $(k^{1/b})^2=(k^{1/c}).(k^{1/a})$

$\Rightarrow$ $k^{2/b}=k^{\frac{1}{c}+\frac{1}{a}}$

$\Rightarrow$ $\frac{1}{a}+\frac{1}{c}=\frac{2}{b}$

Hence (c) is the correct option.

(v) $(a^{m-n})^l\times(a^{n-l})^m\times(a^{l-m})^n=a^{ml-nl}\times a^{nm-lm}\times a^{ln-mn}$

$$=a^{ml-nl+nm-lm+ln-mn}$$
$$=a^0=1$$

Thus (a) is correct option.

(vi) If $a^x=b$, $b^y=c$, $c^z=a$

$\because$ $a^x=b$

$\therefore$ $(c^z)^x=b$

$\Rightarrow$ $c^{zx}=b\Rightarrow(b^y)^{zx}=b$

$\Rightarrow$ $b^{yzx}=b\Rightarrow xyz=1$

Hence (b) is the correct option.

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(vii) $\left(\frac{x^a}{x^b}\right)^{a+b} \times \left(\frac{x^b}{x^c}\right)^{b+c} \times \left(\frac{x^c}{x^a}\right)^{c+a}$

$= (x^{a-b})^{(a+b)} \times (x^{b-c})^{(b+c)} \times (x^{c-a})^{(c+a)}$

$= x^{(a-b)(a+b)} \times x^{(b-c)(b+c)} \times x^{(c-a)(c+a)}$

$= x^{a^2-b^2} \times x^{b^2-c^2} \times x^{c^2-a^2}$

$= x^{a^2-b^2+b^2-c^2+c^2-a^2} = x^0 = 1$

Thus (a) is the correct option.

(viii) $\frac{1}{1+x^{b-a}+x^{c-a}} + \frac{1}{1+x^{c-b}+x^{a-b}} + \frac{1}{1+x^{a-c}+x^{b-c}}$

$= \frac{1}{1+\frac{x^b}{x^a}+\frac{x^c}{x^a}} + \frac{1}{1+\frac{x^c}{x^b}+\frac{x^a}{x^b}} + \frac{1}{1+\frac{x^a}{x^c}+\frac{x^b}{x^c}}$

$= \frac{x^a}{x^a+x^b+x^c} + \frac{x^b}{x^b+x^c+x^a} + \frac{x^c}{x^c+x^a+x^b}$

$= \frac{x^a+x^b+x^c}{x^a+x^b+x^c} = 1$

Hence (c) is correct option.

(ix) $2^{2^x} = 16^{2^{3x}}$

$\Rightarrow 2^{2^x} = (2^4)^{2^{3x}}$

$\Rightarrow 2^{2^x} = 2^{4 \cdot 2^{3x}} = 2^{2^{3x+2}}$

$\Rightarrow 2^{2^x} = 2^{2^{3x+2}}$ [Since base in both sides is equal]

$\Rightarrow 2^x = 2^{3x+2}$

$\Rightarrow x = 3x + 2$

$\Rightarrow 2x = -2$

$\Rightarrow x = -1$

Hence (a) is correct option.

(x) $\frac{4^n \times 20^{m-1} \times 12^{m-n} \times 15^{m+n-2}}{16^m \times 5^{2m+n} \times 9^{m-1}}$

$= \frac{2^{2n} \times 2^{2m-2} \times 5^{m-1} \times 2^{2m-2n} \times 3^{m-n} \times 3^{m+n-2} \times 5^{m+n-2}}{2^{4m} \times 5^{2m+n} \times 3^{2m-2}}$

$= 2^{2n+2m-2+2m-2n-4m} \times 3^{m-n+m+n-2-2m+2} \times 5^{m-1+m+n-2-2m-n}$

$= 2^{-2} \times 3^0 \times 5^{-3}$

$= \frac{1}{4} \times \frac{1}{125} = \frac{1}{500}$

Hence (d) is the correct option.

(xi) $\frac{(0.3)^{1/3} \cdot \left(\frac{1}{27}\right)^{1/4} \cdot (9)^{1/6} \cdot (0.81)^{2/3}}{(0.9)^{2/3} \cdot (3)^{-1/2} \cdot \left(\frac{1}{3}\right)^{-2} \cdot (243)^{-1/4}}$

$= \frac{\left(\frac{3}{10}\right)^{1/3} \left(\frac{1}{3}\right)^{3/4} (3)^{1/3} \cdot \left(\frac{81}{100}\right)^{2/3}}{\left(\frac{9}{10}\right)^{2/3} \left(\frac{1}{3}\right)^{1/2} \cdot (3)^2 \, (3)^{-5/4}}$

$= \frac{3^{\frac{1}{3}+\frac{1}{3}-\frac{3}{4}+\frac{8}{3}} \times 10^{-\frac{1}{3}-\frac{4}{3}}}{3^{\frac{4}{3}+2-\frac{1}{2}-\frac{5}{4}} \times 10^{-\frac{2}{3}}}$

$= \frac{3^{31/12} \times 10^{-\frac{5}{3}}}{3^{\frac{19}{12}} \times 10^{-\frac{2}{3}}}$

$= 3 \times 10^{-1} = \frac{3}{10} = 0.3$

(xii) $\frac{(0.6)^0 - (0.1)^{-1}}{\left(\frac{3}{2^3}\right)^{-1} \cdot \left(\frac{3}{2}\right)^3 + \left(-\frac{1}{3}\right)^{-1}}$

$= \frac{1 - \left(\frac{1}{10}\right)^{-1}}{(3^{-1})(2^3) \times 3^3 \times 2^{-3} + (-3)^1}$

$= \frac{1-10}{3^{3-1} \cdot 2^0 - 3} = \frac{-9}{9-3} = \frac{-9}{6} = \frac{-3}{2}$

Hence (a) is the correct option.

(xiii) $\frac{9^n \times 3^2 \times (3^{-n/2})^{-2} - (27)^n}{3^{3m} \times 2^3} = \frac{1}{27}$

$\Rightarrow \frac{3^{2n} \times 3^2 \times 3^n - 3^{3n}}{3^{3m} \times 2^3} = \frac{1}{3^3}$

$\Rightarrow \frac{3^{2n+2+n} - 3^{3n}}{8 \times 3^{3m}} = \frac{1}{3^3}$

$\Rightarrow \frac{3^{3n+2} - 3^{3n}}{3^{3m} \times 8} = 3^{-3}$

$\Rightarrow \frac{3^{3n}(3^2-1)}{3^{3m} \times 8} = 3^{-3}$

$\Rightarrow 3^{3n-3m} = 3^{-3}$

$\Rightarrow 3n - 3m = -3$

$\Rightarrow m - n = 1$

Hence (b) is the correct option.

HCF : The HCF of $(a^m - 1)$ and $(a^n - 1)$ is equal to the $(a^{\text{HCF of } m, n} - 1)$

EXAMPLE 4 *Find the HCF of* $(2^{315} - 1)$ *and* $(2^{25} - 1)$ *is*

(a) 5 *(b)* 2

(c) 31 *(d) none of these*

SOLUTION The required HCF $= (2)^{\text{HCF of } 315, 25} - 1$

$= 2^5 - 1 = 31$

Hence (c) is the correct option.

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INTRODUCTORY EXERCISE-1.5

1. Simplify : $(x^{2^{n-1}} + y^{2^{n-1}})(x^{2^{n-1}} - y^{2^{n-1}})$
(a) $x^{2^n} - y^{2^n}$ (b) $x^{2n} + y^{2n}$
(c) $(x^n - y^n)^2$ (d) $(x^2 - y^2)^{n-1}$

2. If $x = 2^{1/3} + 2^{-1/3}$, then the value of $2x^3$ is :
(a) $6x + 5$ (b) $5x + 6$
(c) $6x - 5$ (d) $5x - 6$

3. If $x = 3 + 3^{2/3} + 3^{1/3}$, then the value of $x^3 - 9x^2 + 18x - 12$ is :
(a) 1 (b) 0
(c) -1 (d) $\sqrt{2}$

4. If $x^{1/3} + y^{1/3} + z^{1/3} = 0$, then the value of $(x + y + z)^3$ is :
(a) 27 (b) $27xyz$
(c) 81 (d) $(xyz)^3$

5. If $x^{x(\sqrt{x})} = (x\sqrt{x})^x$, then the value of x is :
(a) $\frac{9}{4}$ (b) $\frac{4}{9}$
(c) 8 (d) 4

6. The value of (x, y) if $5^x + 3^y = 8, 5^{x-1} + 3^{y-1} = 2$ is :
(a) 2, 3 (b) 3, 2
(c) 1, 1 (d) 0, 1

7. The value of (x, y) if $x^y = y^x$ and $x^2 = y^3$ is :
(a) $\frac{27}{8}, \frac{9}{4}$ (b) $\frac{9}{8}, \frac{27}{4}$
(c) $\frac{8}{27}, \frac{4}{9}$ (d) $\frac{8}{9}, \frac{4}{27}$

8. The value of $(ax^{1/4} + 3a^{1/2}x^{1/2} + 4x^{3/4})(a - 3a^{1/2}x^{1/4} + 4x^{1/2})$ is :
(a) $a^2x^{1/4} - ax^{3/4} + 16x^{5/4}$
(b) $ax^2 + x^{3/4} + 9x^{1/4}$
(c) 0
(d) 1

9. The value of $\frac{1}{x^b + x^{-c} + 1} + \frac{1}{x^c + x^{-a} + 1} + \frac{1}{x^a + x^{-b} + 1}$ given that $a + b + c = 0$ is :
(a) 1 (b) 0
(c) abc (d) x

10. The value of $\frac{1}{1 + p + q^{-1}} + \frac{1}{1 + q + r^{-1}} + \frac{1}{1 + r + p^{-1}}$ given that $pqr = 1$ is :
(a) 1 (b) 0
(c) pqr (d) $p + q + r$

SURDS

In the previous articles we have studied about the square roots and cube roots etc. There we have studied those numbers whose roots can be found in the form of an integer, but here we study only those numbers whose roots are not the integers.

Surds : When a root of a rational number (*i.e.*, quantities of the type $\sqrt[n]{a}$, a being a rational number) can not be exactly obtained, then this root is called a **surd.**

For Example $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}, \sqrt{10}$ etc.

$\sqrt[3]{2}, \sqrt[3]{3}, \sqrt[3]{4}, \sqrt[3]{5}, \sqrt[3]{6}, \sqrt[3]{7}, \sqrt[3]{9}, \sqrt[3]{10}$ ...etc

$\sqrt[4]{2}, \sqrt[4]{3}, \sqrt[4]{4}, \sqrt[4]{5}, \sqrt[4]{6}, \sqrt[4]{7}$... etc.

But $\sqrt{4} = 2, \sqrt[3]{27}, \sqrt[4]{16}, \sqrt[5]{32}, \sqrt[6]{729}$... etc are not the surds.

Thus it can be said that "all surds are irrational numbers, but all irrational numbers are not the surds".

For example π, e are irrational, but not surds.

Order of Surds : In a surd $\sqrt[n]{a}$ $[= (a)^{1/n}]$, the value of n is called the order of the surd and a is called the **radicand.**

The surds of second order are called quadratic surds *e.g.*, $\sqrt{5}, \sqrt{6}, (7)^{3/2}$... The surds of third order are called cubic surds *e.g.*, $\sqrt[3]{3}, \sqrt[3]{7}, (6)^{5/3}, (9)^{2/3}$.

The surds of fourth order are called biquadratic surds *e.g.*, $\sqrt[4]{2}, \sqrt[4]{3}, \sqrt[4]{7}, (5)^{7/4}$.

Mixed Surds : The product of a rational number with a surd is known as a **mixed surd.**

For example, $3\sqrt{2}, 4\sqrt{7}, 5\sqrt[3]{6}, \sqrt{18}\ (=3\sqrt{2})$

NOTE A surd having no rational factor is known as pure surd. *e.g.*, $\sqrt[4]{2}, \sqrt[3]{7}, \sqrt[5]{8}$ etc.

Similar Surds : If the radicands of two or more rationalised surds are same, then these surds are called the **similar surds.** For example, $\sqrt{5}, 3\sqrt{5}, \sqrt{20}\ (=2\sqrt{5})$ and $\sqrt{80}\ (=4\sqrt{5})$ etc.

Conjugate or Complementary Surds : For every surd of the form $\sqrt{a} + \sqrt{b}$ there exists another surd $\sqrt{a} - \sqrt{b}$, which is called the conjugate surd of previous surd and *vice-versa.* For example

$(2\sqrt{3} + 5\sqrt{3})$ and $(2\sqrt{3} - 5\sqrt{3})$, $(\sqrt{7} + \sqrt{8})$ and $(\sqrt{7} - \sqrt{8})$, ... etc are the pair of conjugate surds.

Properties of Surds

1. A quadratic surd cannot be equal to the sum or difference of a rational number and a quadratic surd *e.g.*, $\sqrt{a} + b \neq \sqrt{c}$ or $\sqrt{a} - b \neq \sqrt{c}$, where $\sqrt{a}, \sqrt{c}$ are quadratic surds and b is a rational number.
2. The product and quotient of two dissimilar quadratic surds can not be rational *e.g.*, $\sqrt{2} \cdot \sqrt{5} = \sqrt{10}$ or $\frac{\sqrt{6}}{\sqrt{2}} = \sqrt{3}$
3. If $a + \sqrt{b} = c + \sqrt{d}$ or $a - \sqrt{b} = c - \sqrt{d}$, then $a = c$ and $b = d$.
4. If $a \pm \sqrt{b} = 0$ then $a = b = 0$
5. If $a + \sqrt{b} = c + \sqrt{d}$, then $a - \sqrt{b} = c - \sqrt{d}$ and its *vice-versa.*
6. If $\sqrt{a + \sqrt{b}} = \sqrt{c} + \sqrt{d}$, then $\sqrt{a - \sqrt{b}} = \sqrt{c} - \sqrt{d}$ and its *vice-versa.*

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Addition : Only similar surds can be simplified

e.g., $2\sqrt{3}+7\sqrt{3}=9\sqrt{3}$.

$8\sqrt{5}+3\sqrt{5}+\sqrt{20}=8\sqrt{5}+3\sqrt{5}+2\sqrt{5}=13\sqrt{5}$

Subtraction : Only similar surds can be simplified

e.g., $7\sqrt{3}-2\sqrt{3}=5\sqrt{3}$

$8\sqrt{5}-3\sqrt{5}=5\sqrt{5}$ etc.

NOTE $8\sqrt{5}\pm 2\sqrt{6}$ can not be simplified, because radicands are different.

Multiplication : $3^{1/4}\times 6^{1/2}\times 9^{1/3}$

$=3^{3/12}\times 6^{6/12}\times 9^{4/12}$

$=(3^3\times 6^6\times 9^4)^{1/12}$

$=(27\times 46656\times 6561)^{1/12}$

$=(8264970432)^{1/12}$

Division : $25\div\sqrt{5}=\dfrac{25}{\sqrt{5}}=\dfrac{(5)^2}{(5)^{1/2}}=5^{2-\frac{1}{2}}$

$=5^{3/2}=\sqrt{125}$

Rationalisation : Making two or more surds a rational number by multiplication, is called the rationalisation

e.g., $3\sqrt{2}\times 4\sqrt{2}=24$

$7\sqrt{5}\times\sqrt{5}=35$

$(9\sqrt{3}-5\sqrt{3})(9\sqrt{3}+5\sqrt{3})=(9\sqrt{3})^2-(5\sqrt{3})^2=168$

LAWS OF SURDS

(i) $\sqrt[n]{a.b}=\sqrt[n]{a}.\sqrt[n]{b}$ (ii) $\sqrt[n]{\dfrac{a}{b}}=\dfrac{\sqrt[n]{a}}{\sqrt[n]{b}}$

(iii) $(\sqrt[n]{a})^m=\sqrt[n]{a^m}$ (iv) $(\sqrt[n]{a})^n=(a^{1/n})^n=a$

(v) $\sqrt[n]{\sqrt[m]{a}}=\sqrt[mn]{a}$

EXAMPLE 1 *Which one is smaller out of $\sqrt[3]{2}$ and $\sqrt[4]{3}$?*

SOLUTION $\sqrt[3]{2}=(2)^{1/3}=2^{4/12}=(16)^{1/12}$

and $\sqrt[4]{3}=(3)^{1/4}=3^{3/12}=(27)^{1/12}$

(Taking the LCM of surd)

Hence $\sqrt[3]{2}<\sqrt[4]{3}$

EXAMPLE 2 *Which one is greatest out of $\sqrt[3]{5}$, $\sqrt[4]{3}$, $\sqrt[5]{4}$?*

SOLUTION $\sqrt[3]{5}=(5)^{1/3}=(5)^{20/60}=(5^{20})^{1/60}=\sqrt[60]{5^{20}}$

$\sqrt[4]{3}=(3)^{1/4}=(3)^{15/60}=(3^{15})^{1/60}=\sqrt[60]{3^{15}}$

$\sqrt[5]{4}=(4)^{1/5}=(4)^{12/60}=(4^{12})^{1/60}=\sqrt[60]{4^{12}}$

Obviously $\sqrt[3]{5}$ is the greatest.

EXAMPLE 3 *If $p=\sqrt[3]{5}$, $q=\sqrt[4]{3}$, $r=\sqrt[5]{4}$ then the correct relationship is :*

(a) $p>q>r$ *(b)* $q<r<p$

(c) $r>p>q$ *(d) can't be determined*

SOLUTION In the previous question we have found that p is the greatest.

Again $q=\sqrt[60]{3^{15}}$ and $r=\sqrt[60]{4^{12}}$

Now $3^{15}\ \text{⋔}\ 4^{12}$ [Here the sign ⋔ signifies the trichotomy-relation *i.e.*, if x ⋔ y, then either $x<y$ or $x=y$ or $x>y$]

$\Rightarrow\quad 3^{12}.3^3\ \text{⋔}\ 4^{12}$

$\Rightarrow\quad 3^3\ \text{⋔}\ \dfrac{4^{12}}{3^{12}}$

$\Rightarrow\quad 3^3\ \text{⋔}\ \left(\dfrac{4}{3}\right)^{12}$

$\Rightarrow\quad 3^3\ \text{⋔}\ (1.33)^{12}$

$\Rightarrow\quad 3^3\ \text{⋔}\ [(1.33)^4]^3$

$\Rightarrow\quad 3^3<(3.16)^3$

Thus $q<r$

Hence $p>r>q$, so (b) is the correct option.

EXAMPLE 4 *If $5\sqrt{5}\times 5^3\div 5^{-3/2}=5^{a+2}$ then the value of a is :*

(a) 4 *(b)* 5 *(c)* 9 *(d)* 16

SOLUTION $\dfrac{5\sqrt{5}\times 5^3}{5^{-3/2}}=5^{a+2}$

$\Rightarrow\quad \dfrac{5^{4+\frac{1}{2}}}{5^{-\frac{3}{2}}}=5^{a+2}\quad\Rightarrow\quad 5^{4+\frac{1}{2}+\frac{3}{2}}=5^{a+2}$

$\Rightarrow\quad 5^6=5^{a+2}$

$\Rightarrow\quad 6=a+2\quad\Rightarrow\quad a=4$

Hence (a) is the correct option.

EXAMPLE 5 *Find the value of $\dfrac{2+\sqrt{3}}{2-\sqrt{3}}$.*

SOLUTION $\dfrac{2+\sqrt{3}}{2-\sqrt{3}}=\dfrac{2+\sqrt{3}}{2-\sqrt{3}}\times\dfrac{2+\sqrt{3}}{2+\sqrt{3}}$

(Multiplying numerator and denominator by conjugate)

$=\dfrac{(2+\sqrt{3})^2}{4-3}=\dfrac{4+3+4\sqrt{3}}{1}=7+4\sqrt{3}$

EXAMPLE 6 *The value of $\dfrac{\sqrt{5}-2}{\sqrt{5}+2}$ is :*

(a) $4-9\sqrt{5}$ *(b)* $9+4\sqrt{5}$

(c) $9-4\sqrt{5}$ *(d)* $7+4\sqrt{5}$

SOLUTION $\dfrac{\sqrt{5}-2}{\sqrt{5}+2}=\dfrac{(\sqrt{5}-2)}{(\sqrt{5}+2)}\times\dfrac{(\sqrt{5}-2)}{(\sqrt{5}-2)}$

$=\dfrac{(\sqrt{5}-2)^2}{5-4}=\dfrac{5+4-4\sqrt{5}}{1}=9-4\sqrt{5}$

Thus (c) is the correct option.

EXAMPLE 7 *Find the value of $\dfrac{5}{6\sqrt{6}}\times\dfrac{12\sqrt{30}}{25\sqrt{5}}$.*

SOLUTION $\dfrac{5}{6\sqrt{6}}\times\dfrac{12\sqrt{30}}{25\sqrt{5}}=\dfrac{5}{6\sqrt{6}}\times\dfrac{12\sqrt{6}\ \sqrt{5}}{25\sqrt{5}}=\dfrac{2}{5}$

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EXAMPLE 8 *Arrange the following in descending order*
$\sqrt{3}-\sqrt{2}, \sqrt{4}-\sqrt{3}, \sqrt{5}-\sqrt{4}, \sqrt{2}-1.$

SOLUTION $\sqrt{3}-\sqrt{2}=\dfrac{\sqrt{3}-\sqrt{2}}{1}\times\dfrac{\sqrt{3}+\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

$=\dfrac{3-2}{\sqrt{3}+\sqrt{2}}=\dfrac{1}{\sqrt{3}+\sqrt{2}}$

Similarly, $\sqrt{4}-\sqrt{3}=\dfrac{1}{\sqrt{4}+\sqrt{3}}, \sqrt{5}-\sqrt{4}=\dfrac{1}{\sqrt{5}+\sqrt{4}}$

and $\sqrt{2}-1=\dfrac{1}{\sqrt{2}+1}$

As we know, if the numerator is same then the fraction whose denominator is larger the fraction will be lower. Hence the correct order of descending is

$$(\sqrt{2}-1)>(\sqrt{3}-\sqrt{2})>(\sqrt{4}-\sqrt{3})>(\sqrt{5}-\sqrt{4})$$

EXAMPLE 9 $\sqrt[r]{\dfrac{9^{\left(r+\frac{1}{4}\right)}\sqrt{3.3^{-r}}}{3.\sqrt{3^{-r}}}}=k$, *then the value of* k *is*

(a) 3 (b) 3^2 (c) 3^3 (d) $\sqrt[r]{3}$

SOLUTION $\sqrt[r]{\dfrac{3^{2r+\frac{1}{2}}3^{\left(\frac{1-r}{2}\right)}}{3^{1-\frac{r}{2}}}}=\sqrt[r]{\dfrac{3^{\frac{4r+1+1-r}{2}}}{3^{\frac{2-r}{2}}}}$

$=\sqrt[r]{3^{\left(\frac{3r+2}{2}\right)-\left(\frac{2-r}{2}\right)}}=\sqrt[r]{3^{2r}}=3^{\frac{2r}{r}}=3^2$

Hence (b) is the correct option.

EXAMPLE 10 *If* m, n *are the positive integers* $(n>1)$ *such that* $m^n=121$, *then value of* $(m-1)^{n+1}$ *is :*

(a) 12321 (b) 1 (c) 1000 (d) 11

SOLUTION $\because$ $m^n=121$

and $n>1$

$\therefore$ $(11)^2=121$

$\Rightarrow$ $m=11$ and $n=2$

Hence $(m-1)^{n+1}=(10)^3=1000$

Thus (c) is correct option.

EXAMPLE 11 *Simplify :* $\sqrt{192}-\dfrac{1}{2}\sqrt{48}-\sqrt{75}$

SOLUTION $\sqrt{192}-\dfrac{1}{2}\sqrt{48}-\sqrt{75}=8\sqrt{3}-\dfrac{4}{2}\sqrt{3}-5\sqrt{3}$

$=8\sqrt{3}-2\sqrt{3}-5\sqrt{3}=\sqrt{3}$

EXAMPLE 12 *Find the square root of* $7-2\sqrt{10}$.

SOLUTION $7-2\sqrt{10}=5+2-2\sqrt{5\times2}$

$\Rightarrow$ $7-2\sqrt{10}=(\sqrt{5})^2+(\sqrt{2})^2-2\sqrt{5}.\sqrt{2}$

$\Rightarrow$ $7-2\sqrt{10}=(\sqrt{5}-\sqrt{2})^2$

Thus the $\sqrt{7-2\sqrt{10}}=\pm(\sqrt{5}-\sqrt{2})$

EXAMPLE 13 *Find the value of* $\left(\dfrac{1}{\sqrt{5}-2}+\dfrac{1}{\sqrt{5}+2}\right)^2$.

SOLUTION $\left(\dfrac{1}{\sqrt{5}-2}+\dfrac{1}{\sqrt{5}+2}\right)^2=\left[\dfrac{\sqrt{5}+2+\sqrt{5}-2}{(\sqrt{5}-2)(\sqrt{5}+2)}\right]^2$

$=\left[\dfrac{2\sqrt{5}}{(\sqrt{5})^2-(2)^2}\right]^2$

$=\dfrac{4\times5}{5-4}=\dfrac{20}{1}=20$

INTRODUCTORY EXERCISE-1.6

1. If $x=\dfrac{1}{\sqrt{2}-1}$, then the value of $x^2-6+\dfrac{1}{x^2}$ is :
   (a) 0 (b) 1
   (c) $\sqrt{2}$ (d) can't be determined

2. The value of $\dfrac{3}{5\sqrt[4]{27}}$, when its denominator is a rational number :
   (a) $\dfrac{\sqrt[4]{3}}{6}$ (b) $\dfrac{\sqrt[4]{3}}{5}$
   (c) $\dfrac{\sqrt[3]{4}}{5}$ (d) none of these

3. The value of denominator when $\dfrac{7\sqrt{3}-5\sqrt{2}}{\sqrt{18}+\sqrt{48}}$ is rationalised :
   (a) 15 (b) 25
   (c) 30 (d) 50

4. If $\sqrt{5+\sqrt[3]{x}}=3$, then the value of x is :
   (a) 9 (b) 27
   (c) 64 (d) 343

5. If $A=\sqrt{2}$, $B=\sqrt[3]{3}$ and $C=\sqrt[4]{4}$ then which of the relation is correct?
   (a) $A>B>C$ (b) $A>B=C$
   (c) $A=C<B$ (d) none of these

Factorial : The product of n consecutive natural numbers (or positive integers) starting from 1 to n is called as the factorial 'n'.

i.e., $n!=1\times2\times3\times4\times5\times6....(n-2)(n-1)n$

So, the $4!=1\times2\times3\times4=4\times3\times2\times1$

$5!=1\times2\times3\times4\times5=5\times4\times3\times2\times1$

$6!=1\times2\times3\times4\times5\times6=6\times5\times4\times3\times2\times1$

NOTE 1. Factorial n is written as "n!".
2. $0!=1$ and $1!=1$

Properties

1. $n!$ is always an even number if $n\geq2$
2. $n!$ always ends with zero if $n\geq5$

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EXAMPLE 1 *If* $m^n - m = (m-n)!$ *where* $m > n > 1$ *and* $m = n^2$ *then the value of* $m^2 + n^2$ *is :*

(a) 272 *(b)* 90
(c) 20 *(d) none of (a), (b), (c)*

SOLUTION Let us consider $n = 2$ $(\because n > 1)$

Then $m^n - m = 4^2 - 4 = 12$ $(\because m = n^2)$

and $(m-n)! = (4-2)! = 2! = 2$

Hence it is impossible.

Again consider $n = 3$, then

$$m^n - m = 9^3 - 9 = 720$$

and $(m-n)! = (9-3)! = 6! = 720$

Thus we get $n = 3$ and $m = 9$ the probable values

Now $m^2 + n^2 = 9^2 + 3^2 = 81 + 9 = 90$

Hence (b) is the correct option.

EXAMPLE 2 *If* $P + P! = P^3$, *then the value of* P *is :*

(a) 4 *(b)* 6 *(c)* 0 *(d)* 5

SOLUTION Consider $P = 5$, then

$$5 + 5! = 5^3$$
$$5 + 120 = 125$$
$$125 = 125$$

Thus (d) is correct option.

EXAMPLE 3 *If* $P + P! = P^2$ *then the value of* P *is :*

(a) 5 *(b)* 7 *(c)* 3 *(d)* 0

SOLUTION Consider $P = 3$ then

$$3 + 3! = 3^2$$
$$3 + 6 = 9$$
$$9 = 9$$

Hence (c) is the correct option.

EXAMPLE 4 *If* $n! - n = n$, *then the value of* n *is :*

(a) 4 *(b)* 5 *(c)* 6 *(d)* 3

SOLUTION Consider $n = 3$ then

$$3! - 3 = 3$$
$$6 - 3 = 3$$
$$3 = 3$$

Hence (d) is the correct answer.

EXAMPLE 5 *The appropriate value of* P *for the relation* $(P! + 1) = (P+1)^2$ *is :*

(a) 3 *(b)* 4
(c) 5 *(d) none of these*

SOLUTION Let us consider $P = 4$ then

$$4! + 1 = (4+1)^2$$
$$24 + 1 = 5^2$$
$$25 = 25$$

Hence (b) is the correct option.

EXAMPLE 6 *The value of* $8! \div 4!$ *is :*

(a) 4 *(b)* 16 *(c)* 2 *(d)* 1680

SOLUTION $$\frac{8!}{4!} = \frac{8.7.6.5.4.3.2.1}{4.3.2.1}$$

$$= 8.7.6.5 = 1680$$

Hence (d) is the correct option.

EXAMPLE 7 *If* $n! = \frac{(n+4)!}{(n+1)!}$, *then the value of* n *is :*

(a) 5 *(b)* 18 *(c)* 6 *(d)* 9

SOLUTION $$n! = \frac{(n+4)!}{(n+1)!},$$

$$= \frac{1.2.3\ldots n.(n+1)(n+2)(n+3)(n+4)}{1.2.3\ldots n.(n+1)}$$

$$= (n+2)(n+3)(n+4)$$

$$n! = (n+2)(n+3)(n+4)$$

now $5! \neq 7 \times 8 \times 9$

and $18! \neq 20 \times 21 \times 22$

and $9! \neq 11 \times 12 \times 13$

but $6! = 8 \times 9 \times 10$

$720 = 720$

Hence $n = 6$ is the correct answer, thus (c) is the right choice.

EXAMPLE 8 *The value of*

$(1.2.3\ldots9).(11.12.13\ldots19).(21.22.23\ldots29).(31.32.33\ldots39).$
$(41.42.43\ldots49)\ (51.52.53\ldots59)\ldots(91.92\ldots99)$

(a) $\frac{100!}{36288 \times 10^{11}}$ *(b)* $\frac{99!}{388 \times 10^{11}}$

(c) $\frac{99!}{36288 \times 10^{10}}$ *(d) can't be determined*

SOLUTION $(1.2.3\ldots9).(11.12.13\ldots19).(21.22.23\ldots29)\ldots$
$\ldots.(91.92.93\ldots99)$

$$= (1.2.3\ldots9)\frac{10}{10}(11.12.13\ldots19)\frac{20}{20}$$

$$(21.23.23\ldots29)\frac{30}{30}\ldots(91.92.93\ldots99)$$

$$= \frac{1.2.3\ldots99}{10.20.30\ldots90} = \frac{99!}{362880 \times 10^9} = \frac{99!}{36288 \times 10^{10}}$$

Hence (c) is the correct option.

EXAMPLE 9 *The expression* $1! + 2! + 3! + 4! + \ldots + n!$ *(where* $n \geq 5$*) is not a/an :*

(a) composite number *(b) odd number*
(c) perfect square *(d) multiple of 3*

SOLUTION $1! + 2! + 3! + 4! + 5! = 1 + 2 + 6 + 24 + 120 = 153$

Since, the option (a), (b) and (d) are ruled out so the rest option (c) is the correct one.

EXAMPLE 10 *The HCF and LCM of* 13! *and* 31! *are respectively :*

(a) 12! *and* 32! *(b)* 13! *and* 31!
(c) 26 *and* 403 *(d) can't be determined*

SOLUTION Since 13! is contained in 31! so the LCM is 31! and 13! is common in 13! and 31!, so the HCF is 13!

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Number of zeros at the end of the product of numbers : We know that $10=5\times2$, $100=5^2\times2^2$, $1000=5^3\times2^3$ etc. So we can say that for n number of zeros at the end of the product we need exactly n combinations of "5×2". For example.

$2\times3\times5\times7=210$ there is only one zero at the end of the product (or resultant value)

Again, $2\times3\times5\times6\times7\times15=2\times3\times5\times2\times3\times7\times3\times5$

$=2\times2\times5\times5\times3\times3\times3\times7$

$=100\times189=18900$

Thus there are two zeros because there are two combinations of "5×2".

Now, $1\times2\times3\times4\times5\times6\times7\times8\times9$

$=1\times2\times3\times2\times2\times5\times2\times3\times7\times2\times2\times2\times3\times3$

$=1\times2\times2\times2\times2\times2\times2\times2\times5\times3\times3\times3\times3\times7$

$=1\times2^7\times5\times3^4\times7$

$=1\times2^6\times2\times5\times3^4\times7$

$=10\times2^6\times3^4\times7$

$=10\times64\times81\times7$

$=10\times36288=362880$

Thus there is only one zero at the end of the product since there is only one combination of 5×2..

Again $4\times125\times3=2\times2\times5\times5\times5\times3$

$=2^2\times5^3\times3$

$=2^2\times5^2\times5\times3$

$=(2\times5)^2\times15$

$=100\times15=1500$

Thus there are only two zeros at the end of the product since there are only two combinations of "5×2".

NOTE The number of zeros at the end of the product depends upon 2×5, but the condition is that
(i) $2^k\times5^l$ gives k number of zeros if $k<l$
(ii) $2^k\times5^l$ gives l number of zeros if $l<k$

EXAMPLE 1 *Find the number of zeros in the product of* 10!.

SOLUTION $10!=1\times2\times3\times4\times5\times6\times7\times8\times9\times10$

It is obvious from the above expression that there are only two 5's and eight 2's.

Since the number of 5's are less (than the number of 2's) so the number of 5's will be effective to form the combination of '5×2. Thus there are only 2 zeros at the end of the product of 10!.

EXAMPLE 2 *Find the number of zeros at the end of the product of* $2^{222}\times5^{555}$.

SOLUTION Since the number of 2's are less than the number of 5's hence the restriction is imposed by the number of 2's.

Thus there can be only 222 pairs of (5×2). Hence the number of zeros at the end of the product of the given expression will be 222.

EXAMPLE 3 *Find the number of zeros at the end of the product of the expression* $2^1\times5^2\times2^3\times5^4\times2^5\times5^6\times2^7\times5^8\times2^9\times5^{10}$

SOLUTION Since the number of 2's are less than the number of 5's hence number of 2's will be effective. Thus

$2^1\times2^3\times2^5\times2^7\times2^9\times5^2\times5^4\times5^6\times5^8\times5^{10}$

$=2^{25}\times5^{30}=2^{25}\times5^{25}\times5^5=(2\times5)^{25}\times5^5$

$=10^{25}\times5^5=5^5\times10^{25}$

Thus there will be 25 zeros at the end of the product of the given expression.

EXAMPLE 4 *The number of zeros at the end of the expression* $10+100+1000+\ldots+10000000000$ *is :*

(a) 1 *(b)* 10
(c) 55 *(d)* none of these

SOLUTION
10
100
1000
.....
.....
10000000000
11111111110

Thus there is only one zero at the end of resut.

Hence (a) is the correct option.

EXAMPLE 5 *The number of zeros at the end of the product of the expression* $10\times100\times1000\times10000\times\ldots10000000000$ *is :*

(a) 10 *(b)* 100 *(c)* 50 *(d)* 55

SOLUTION $10\times100\times1000\times\ldots10000000000$

$=10^1\times10^2\times10^3\times\ldots\times10^{10}$

$=10^{(1+2+3+\ldots+10)}=10^{55}$

Hence (d) is the correct option.

EXAMPLE 6 *Number of zeros at the end of the following expression* $(5!)^{5!}+(10!)^{10!}+(50!)^{50!}+(100!)^{100!}$ *is :*

(a) 165 *(b) 120*
(c) 125 *(d) none of these*

SOLUTION The number of zeros at the end of $(5!)^{5!}=120$

[$\because$ 5! = 120 and thus $(120)^{120}$ will give 120 zeros]

and the number of zeros at the end of the $(10!)^{10!}$, $(50!)^{50!}$ and $(100!)^{100!}$ will be greater than 120.

Now since the number of zeros at the end of the whole expression will depend on the number which has least number of zeros at the end of the number among other given numbers.

So, the number of zeros at the end of the given expression is 120.

Divisibility of a factorial number by the largest power of any number : Let us start with very simple example. As if we consider 1!, 2!, 3! or 4!, none of them is divisible by 5, because 5 is not the factor, involved in the given factorial number.

Now, if we consider any factorial number greater than 4!, every number consists 5 or the higher powers of 5 in the factorial numbers. For example starting from 5! every number has 5 as its factor

i.e., $5!=1\times2\times3\times4\times5$

$6!=1\times2\times3\times4\times5\times6$

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$7! = 1\times2\times3\times4\times5\times6\times7$

... ...

$10! = 1\times2\times3\times4\times5\times6\times7\times8\times9\times10$ etc.

Thus it is obvious that every number greater than 4! is divisible by 5. But as we move towards higher number we will find that the frequency of the occurrence of 5 (or any particular number) will increase. As 5! contains only one 5, 10! contains two 5's, 15! contains three 5's, 20! contains four 5's but 25! contains suddenly six 5's instead of five 5's. So it becomes a very tedious work to calculate the occurrence of any particular number involved as a factor of any higher factorial number.

Similarly if we have to know that how many times 3 is contained in the 30! or in other words what is the largest power of 3 that can divide 30!

Now, if we try to do it manually, we see that there are 10 multiples of 3 *viz.*, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. But there are some multiples which contains 3 at least two times for example 9, 18 and 27. Again there are some multiples which contain 3 three times as in 27. Thus the factor 3 totally occurs 14 (=10+3+1) times in 30!. Hence you can understand that how tedious can be the calculation if we have to find the highest power of 3 that can exactly divide to 9235!

Therefore to make the calculation easier we can use the following formula.

Suppose we have to find the highest power of k that can exactly divided n!, we divide n by k, n by k^2, n by k^3 ... and so on till we get $\left[\frac{n}{k^x}\right]$ equal to 1 (where, $[P]$ means the greatest integer less than or equal to P) and then add up as $\left[\frac{n}{k}\right]+\left[\frac{n}{k^2}\right]+\left[\frac{n}{k^3}\right]+\left[\frac{n}{k^4}\right]+\ldots+\left[\frac{n}{k^x}\right]$.

NOTE For more information about greatest integer number (or function) *i.e.*, GIF refer the chapter "FUNCTIONS".

EXAMPLE 1 *Find the largest power of 5 contained in 124!*

SOLUTION $\left[\frac{124}{5}\right]+\left[\frac{124}{5^2}\right] = 24 + 4 = 28$

[We cannot do it further since 124 is not divisible by 5^3]
Hence, there are 28 times 5 involved as a factor in 124!

EXAMPLE 2 *Find the largest power of 2 that can divide 268!*

SOLUTION $\left[\frac{268}{2}\right]+\left[\frac{268}{2^2}\right]+\left[\frac{268}{2^3}\right]+\left[\frac{268}{2^4}\right]+\left[\frac{268}{2^5}\right]+\left[\frac{268}{2^6}\right]+\left[\frac{268}{2^7}\right]+\left[\frac{268}{2^8}\right]$

$= 134 + 67 + 33 + 16 + 8 + 4 + 2 + 1 = 265$

Thus the greatest power of 2 is 265 that can divide exactly 268!

Dear students you might have notice that if we further increase the value of n !and consider a very typical number say 7,13, 17, 19 etc. for the divisibility then we feel the real pain in solving the problem since we don't know the values of higher powers of these divisors say $17^4, 17^5, 19^8, 19^9, \ldots$ etc. So this formula in this present form is not appropriate.

Now you must have noticed that we consider only integral values of the quotient q as $[q]$ which gives an integer just less than or equal to q. So we can do this calculation slightly in a easier manner. Here we just divide the given number and then the succeeding quotients will be divided by the same remainder as in the case of successive division, so for better understanding we solve the previous problem in this manner as follows.

2	268	[Divide successive quotients till we get 0 as the last quotient]
2	134 →	
2	67 →	
2	33 →	
2	16 →	
2	8 →	265 (Add up all the quotients)
2	4 →	
2	2 →	
2	1 →	
	0	

So we can frankly say that this method of calculation is easy since we need not to know the values of $2^1, 2^2, 2^3, 2^4, \ldots, 2^8$ etc. Also the division by 2 is easier than the division by 2^x, where x is any large integer.

EXAMPLE 3 *Find the largest power of 7 that can exactly divide 777!*

SOLUTION

7	777	
7	111 →	
7	15 →	128
7	2 →	
	0	

Thus the highest power of 7 is 128 by which 777! can be divided.

EXAMPLE 4 *Find the largest value of n in the 10^n which can exactly divide 1000!*

SOLUTION Since 10 is made up of 5 and 2 *i.e.*, $10 = 2\times5$

So you can see that before 10! there is the presence of 10 in 5!, 6!, 7!, 8! or 9!.

So $10^n = (5\times2)^n = 5^n \times 2^n$

Thus we find the powers of 2 and 5 individually

2	1000	
2	500 →	
2	250 →	
2	125 →	
2	62 →	
2	31 →	994
2	15 →	
2	7 →	
2	3 →	
	1 →	

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and

5	1000
5	200 →
5	40 → 249
5	8 →
	1 →

So there is 2^{994} and 5^{249} but we can make only 249 combinations due to the restriction imposed by the lower power of 5!.

Thus the highest power of 10 is 249 which can divide 1000! $\left(\because 2^{994} \times 5^{249} \Rightarrow 2^{745} \times 2^{249} \times 5^{249} \right.$ $\left. \Rightarrow 2^{745} \times (10)^{249}\right)$

EXAMPLE 5 *Find the number of zeros at the end of* 1000!

SOLUTION Since we know that the zeros at the end of any product are due to the presence of 10 as the factor of the product and the number of zeros depends upon the number of times 10 is involved. For example if there are seven 10's (*i.e.*, 7 combinations of 5×2) in the $n!$ then the number of zeros at the end of $n!$ will be 7.

But since, we have solved the same problem previous to this problem so we can conclude that the number of 10's in 1000! is 249. Hence the number of zeros at the end of the 1000! is 249.

EXAMPLE 6 *The number of zeros at the end of* 100! *is :*

(a) 25 *(b)* 50 *(c)* 24 *(d)* 100

SOLUTION

2	100
2	50 →
2	25 →
2	12 → 97
2	6 →
2	3 →
	1 →

and

5	100
5	20 → 24
	4 →

So, there will be only 24 combinations of 5×2 it means there will be 24 zeros at the end of 100!.

Hence (c) is the correct answer.

EXAMPLE 7 *Find the highest power of* 63 *which can exactly divide* 6336!.

SOLUTION $63 = 3^2 \times 7$

Since this technique is applicable only for the prime factors. So we solve it by breaking 63 in its prime factors.

3	6336
3	2112 →
3	704 →
3	234 →
3	78 → 3164
3	26 →
3	8 →
	2 →

and

7	6336
7	905 →
7	129 → 1054
7	18 →
	2 →

$\because \quad 63 = 9 \times 7 = 3^2 \times 7$

$\therefore$ Since to make a 9 we need 3×3 *i.e,* two times. So we divide 3164 by 2 and get 1582.

So $3^{3164} \times 7^{1054} \Rightarrow 9^{1582} \times 7^{1054}$

Thus the largest power of 63 is 1054. That can exactly divide 6336!

EXAMPLE 8 *Find the highest power of* 40 *which can exactly divide* 4000!

SOLUTION $40 = 8 \times 5 = 2^3 \times 5$

So,

2	4000
2	2000 →
2	1000 →
2	500 →
2	250 →
2	125 →
2	62 → 3994
2	31 →
2	15 →
2	7 →
2	3 →
	1 →

and

5	4000
5	800 →
5	160 →
5	32 → 999
5	6 →
	1 →

Now since $2^{3m} \times 5^n = 8^m \times 5^n$

$\therefore \quad 2^{3994} \times 5^{999} = (2^3)^{1331} \times 2 \times 5^{999}$

$\Rightarrow \quad 2 \times (8^{1331} \times 5^{999}) \Rightarrow 2 \times 8^{332} \times (8 \times 5)^{999}$

Thus the highest power of 40 is 999 that can exactly divide 4000!.

EXAMPLE 9 *Find the highest power of* 81 *that can divide* 1800!.

SOLUTION $81 = 3^4$

So,

3	1800
3	600 →
3	200 →
3	66 → 897
3	22 →
3	7 →
	2 →

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Thus we get 3^{897}. But we need 81.

Now since $81 = 3^4$

Therefore $3^{897} = (3^4)^{224} \times 3^1 = (81)^{224} \times 3$.

Therefore the highest power of 81 is 224 which can divide exactly 1800! so $(81)^{224}$ can divide 1800!

NOTE Sometimes for your convenience you can skip some unnecessary calculation.
As if you are asked to find the highest power of 30 which can exactly divide 1357!
Since $30 = 2 \times 3 \times 5$. Now if you have been serious while doing the previous examples so you can conclude that you need not to solve to find the highest powers of 2 and 3. What you need is just to find the highest power of 5 (which is the greatest factor therefore it occurs least frequently) and as you know that the least frequent factor impose the restriction and hence it is the only effective value. Hence we just calculate the highest power of 5 and get the required result. Still there are some special points which are really difficult to explain on the paper. These points can be effectively explained only in the classroom by teacher and last but not the least if you are intelligent enough so you can pick up these subtle but the important points by yourself as you are learning the concept religiously.
Please note that there can be some shortcuts which I have not mentioned because it is difficult to explain exactly on paper and even some students don't fathom the logic of the shortcut. So it becomes dangerous and hence it is better to find the shortcuts by themselves just by intensive practice or seeking the help of an expert.

CONCEPT OF UNIT DIGIT

Look at the following :

$$1 \times 5 = 5$$
$$3 \times 5 = 15$$
$$5 \times 5 = 25$$
$$7 \times 5 = 35$$
$$9 \times 5 = 45$$
$$11 \times 5 = 55$$
$$\ldots \ldots \ldots \ldots$$
$$\ldots \ldots \ldots \ldots$$

i.e., if the number whose last digit is 5, is multiplied by any odd number, the unit digit of the product will always be 5.

For example

$$13 \times 15 = 195, 19 \times 35 = 665 \text{ etc.}$$

Now,
$$2 \times 5 = 10$$
$$4 \times 5 = 20$$
$$6 \times 5 = 30$$
$$8 \times 5 = 40$$
$$10 \times 5 = 50$$
$$12 \times 5 = 60$$
$$\ldots \ldots \ldots \ldots$$
$$\ldots \ldots \ldots \ldots$$

i.e., if the number whose last is 5, is multiplied by any even number (including zero), the unit digit of the product is always zero. For example

$$82 \times 15 = 1230, 156 \times 45 = 7020$$
$$62 \times 13 \times 65 = 52390 \text{ etc.}$$

Now we will discuss that the unit digit of the resultant value depends upon the unit digits of all the participating numbers *i.e.*,

$$12 + 17 + 13 + 47 = 89$$

Thus it is clear that the unit digit of the resultant value 89 depends upon the unit digits 2, 7, 3, 7.

Similarly, $6 \times 7 \times 9 = 378$
$$3 \times 7 \times 8 = 168$$

So we can find out the unit digit of the resultant value only by solving the unit digits of the given expression.

EXAMPLE 1 *Find the unit digit of* $123 + 345 + 780 + 65 + 44$.

SOLUTION We can find the unit digit just by adding the unit digits 3, 5, 0, 5, 4 as

$$3 + 5 + 0 + 5 + 4 = 17$$

So the unit digit (or the last digit) of the resultant value of the expression $123 + 345 + 780 + 65 + 44$ will be 7. (you can verify it by doing the whole sum)

EXAMPLE 2 *Find the unit digit of* $676 \times 543 \times 19$.

SOLUTION We can find the unit digit of the product of the given expression just by multiplying the unit digits (6, 3, 9) instead of doing the whole sum.

Thus $6 \times 3 \times 9 = 162$

Hence, the unit digit of the product of the given expression will be 2. (you can verify it by doing the complete sum)

EXAMPLE 3 *Find the unit digit of* $135 \times 361 \times 970$.

SOLUTION The unit digit can be obtained by multiplying the unit digits 5, 1, 0. then $5 \times 1 \times 0 = 0$ thus the unit digit will be zero.

EXAMPLE 4 *Find the unit digit of the product of all the odd prime numbers.*

SOLUTION The prime numbers are 3, 5, 7, 11, 13, 17, 19...etc.
Now we know that if 5 is multiplied by any odd number it always gives the last digit 5. So the required unit digit will be 5.

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Now look at the following

$1 \times 1 = 1$

$1 \times 1 \times 1 = 1$

$1 \times 1 \times 1 \times 1 = 1$

$1 \times 1 \times 1 \times 1 \times 1 = 1$ etc.

$2^1 = 2$

$2^2 = 2 \times 2 = 4$

$2^3 = 2 \times 2 \times 2 = 8$

$2^4 = 2 \times 2 \times 2 \times 2 = 16$

$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$

$2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$

$2^7 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 128$

$2^8 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 256$

$2^9 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 512$

$2^{10} = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 1024$

$2^{11} = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2048$

$2^{12} = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 4096$

Similarly, $3^1 = 3$

$3^2 = 9$

$3^3 = 27$

$3^4 = 81$

$3^5 = 243$

$3^6 = 729$

$3^7 = 2187$

$3^8 = 6561$

$3^9 = 19683$ etc.

Similarly, $4^1 = 4$

$4^2 = 16$

$4^3 = 64$

$4^4 = 256$

$4^5 = 1024$

$4^6 = 4096$

Thus we can say that the unit digit follows a periodic pattern that is after a particular period it repeats in a cyclic form.

the unit digit of $2^1, 2^5, 2^9, 2^{13}, \ldots$ is the same which is 2.

Similarly $2^2, 2^6, 2^{10}, 2^{14}, \ldots$ etc. has the same unit digit, which is 4.

Again the last digit of 3^1 is 3.

and the last digit of 3^2 is 9.

and the last digit of 3^3 is 7.

and the last digit of 3^4 is 1.

and the last digit of 3^5 is 3.

and the last digit of 3^6 is 9.

and the last digit of 3^7 is 7.

and the last digit of 3^8 is 1.

Thus the last digit must follow a pattern. It can be seen that the last digit of 2 repeats after every four steps and the last digit of 4 repeats after every 2 steps.

NOTE

1. The last digit (or unit digits) of 0, 1, 5 and 6 is always the same irrespective of their powers raised on them.
2. The last digit of 4 and 9 follows the pattern of odd-even *i.e.*, their period is 2.
3. The last digit of 2, 3, 7, 8 repeats after every 4 steps *i.e.*, their cyclic period is 4.

EXAMPLE 1 *Find the last digit of* 2^{35}.

SOLUTION The last digit of $2^1 \Rightarrow 2$

$2^2 \Rightarrow 4$

$2^3 \Rightarrow 8$

$2^4 \Rightarrow 6$

$2^5 \Rightarrow 2$

Since its cyclic period is four, it means the unit digit of 2 will repeat after every 4 steps. Hence we can say that the last digit of $2^1, 2^5, 2^9, 2^{13}, 2^{17}, 2^{21}, 2^{25}, 2^{29}, 2^{33}, 2^{37}$ is the same and the last digit of $2^2, 2^6, 2^{10}, 2^{14}, 2^{18}, \ldots, 2^{30}, 2^{34}, 2^{38}$ is same and the last digit of $2^3, 2^7, 2^{11}, \ldots, 2^{31}, 2^{35}, 2^{39}$ is same and the last digit of $2^4, 2^8, 2^{12}, 2^{16}, \ldots, 2^{32}, 2^{36}, 2^{40}$ etc. is same.

Hence the last digit of 2^{35} is 8.

Alternatively : You can see that the powers which are divisible by 4 (*i.e.*, cyclic period) give the same unit digit as 2^4. As $2^8, 2^{12}, 2^{16}, \ldots$ etc.

Again the powers which leave the remainder 1 when divided by 4 (which is the cyclic period) give the same unit digit as 2^1. As $2^5, 2^9, 2^{13}, \ldots$ etc.

Similarly the powers which leave the remainder 2 when divided by 4, give the same unit digit as 2^2. As $2^6, 2^{10}, 2^{14}, 2^{18}, \ldots$etc.

Similarly the powers which leave the remainder 3 when divided by 4 give the same unit digit as 2^3. As $2^7, 2^{11}, 2^{15}, \ldots$etc.

Since when 35 is divided by 4, the remainder is 3. Hence, the unit digit of 2^{35} is equal to the unit digit of 2^3, which is equal to 8.

EXAMPLE 2 *Find the unit digit of* $(12)^{78}$.

SOLUTION The unit digit of $(12)^{78}$ will be same as $(2)^{78}$. Now since we know that the cyclic period of unit digit of 2 is 4. The remainder when 78 is divided by 4 is 2. Hence the unit digit of 2^{78} will be same as 2^2 which is 4. Thus the unit digit of 12^{78} is 4.

EXAMPLE 3 *Find the unit digit of* $(33)^{123}$.

SOLUTION Since we know that the unit digit of $(33)^{123}$ will be same as $(3)^{123}$. Now the unit-digit of 3^{123} will be 7 since it will be equal to the unit digit of 3^3.

Thus the unit digit of $(3)^{123}$ is 7.

EXAMPLE 4 *Find the unit digit of* $3^{47} + 7^{52}$.

SOLUTION The unit digit of the given expression will be equal to the unit digit of the sum of the unit digits of both the terms individually.

Now, unit digit of 3^{47} is 7 (since it will be equal to 3^3) and the unit digit of 7^{52} is 1 (since it will be equal to 7^4)

Thus the unit digit of $3^{47} + 7^{52}$ is $7 + 1 = 8$.

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EXAMPLE 5 *Find the unit digit of* $3^6 \times 4^7 \times 6^3 \times 7^4 \times 8^2 \times 9^5$.

SOLUTION The unit digit of 3^6 is 9
The unit digit of 4^7 is 4
The unit digit of 6^3 is 6
The unit digit of 7^4 is 1
The unit digit of 8^2 is 4
The unit digit of 9^5 is 9
Therefore the unit digit of the given expression is 6, (since $9 \times 4 \times 6 \times 1 \times 4 \times 9 = 7776$).

EXAMPLE 6 *Find the unit digit of* 111! *(factorial* 111).

SOLUTION $111! = 1 \times 2 \times 3 \times 4 \times 5 \times \ldots \times 110 \times 111$
Since there is a product of 5 and 2 hence it will give zero as the unit digit.
Hence the unit digit of 111! is 0 (zero).

EXAMPLE 7 *Find the unit digit of the product of all the prime number between* 1 *and* $(11)^{11}$.

SOLUTION The set of prime number $S = \{2, 3, 5, 7, 11, 13, \ldots\}$
Since there is one 5 and one 2 which gives 10 after multiplying mutually, it means the unit digit will be zero.

EXAMPLE 8 *Find the unit digit of the product of all the elements of the set which consists all the prime numbers greater than 2 but less than* 222.

SOLUTION The set of required prime number = $\{3, 5, 7, 11, \ldots\}$
Since there is no any even number in the set so when 5 will multiply with any odd number, it will always give 5 as the last digit.
Hence the unit digit will be 5.

EXAMPLE 9 *Find the last digit of* $222^{888} + 888^{222}$.

SOLUTION The last digit of the expression will be same as the last digit of $2^{888} + 8^{222}$.
Now the last digit of 2^{888} is 6 and the last digit of the 8^{222} is 4.
Thus the last digit of $2^{888} + 8^{222}$ is 0 (zero), since $6 + 4 = 10$.

EXAMPLE 10 *Find the last digit of* $32^{32^{32}}$.

SOLUTION The last digit of $32^{32^{32}}$ is same as $2^{32^{32}}$.
But $2^{32^{32}} = 2^{32 \times 32 \times 32 \times \ldots \times 32 \text{ times}}$
$\Rightarrow$ $2^{32^{32}} = 2^{4 \times 8 \times (32 \times 32 \times \ldots \times 31 \text{ times})}$
$\Rightarrow$ $2^{32^{32}} = 2^{4n}$,
where $n = 8 \times (32 \times 32 \times \ldots \times 31 \text{ times})$
Again $2^{4n} = (16)^n \Rightarrow$ unit digit is 6, for every $n \in N$
Hence, the required unit digit = 6.

EXAMPLE 11 *Find the last digit of the expression :*
$1^2 + 2^2 + 3^2 + 4^2 + \ldots + 100^2$.

SOLUTION The unit digit of the whole expression will be the equal to the unit digit of the sum of the unit digits of the expression.
Now adding the unit digits of $1^2 + 2^2 + 3^2 + \ldots + 10^2$
we get $1 + 4 + 9 + 6 + 5 + 6 + 9 + 4 + 1 + 0 = 45$
Hence, the unit digit of $1^2 + 2^2 + 2^2 + \ldots + 10^2$ is 5.
Now since there are 10 similar columns of numbers which will yield the same unit digit 5. Hence the sum of unit digits of all the 10 columns is 50 $(= 5 + 5 + 5 + \ldots + 5)$.
Hence, the unit digit of the given expression is 0 (zero).

EXAMPLE 12 *Find the unit digit of* $1^1 + 2^2 + 3^3 + \ldots 10^{10}$.

SOLUTION
The unit digit of $1^1 = 1$
The unit digit of $2^2 = 4$
The unit digit of $3^3 = 7$
The unit digit of $4^4 = 6$
The unit digit of $5^5 = 5$
The unit digit of $6^6 = 6$
The unit digit of $7^7 = 3$
The unit digit of $8^8 = 6$
The unit digit of $9^9 = 9$
The unit digit of $10^{10} = 0$
Thus the unit digit of the given expression will be 7. $(\because 1 + 4 + 7 + 6 + 5 + 6 + 3 + 6 + 9 = 47)$

EXAMPLE 13 *Find the unit digit of*
$13^{24} \times 68^{57} + 24^{13} \times 57^{68} + 1234 + 5678$.

SOLUTION The unit digit of 3^{24} is 1
The unit digit of 8^{57} is 8
The unit digit of 4^{13} is 4
The unit digit of 7^{68} is 1
So the resultant value of the unit digits
$= 1 \times 8 + 4 \times 1 + 4 + 8$
$= 8 + 4 + 4 + 8 = 24$
Thus the unit digit of the whole expression is 4.

EXAMPLE 14 *The unit digit of the expression*
$\dfrac{1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9 \times 10}{100}$ *is :*

(a) 7 *(b)* 9
(c) 8 *(d) none of these*

SOLUTION Since in the numerator of the product of the expression there will be 2 zeros at the and these two zeros will be cancelled by 2 zeros of the denominator. Hence finally we get a non-zero unit digit in the expression.

Now, $\dfrac{1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9 \times 10}{100}$

$= \dfrac{1 \times 2^8 \times 3^4 \times 5^2 \times 7^1}{5^2 \times 2^2}$

$= 1 \times 2^6 \times 3^4 \times 7$

Therefore, the unit digit of the given expression will be same as that of $1 \times 2^6 \times 3^4 \times 7$.
Now, the unit digit of $1 \times 2^6 \times 3^4 \times 7$ is 8.
$(\because$ the product of unit digits of $1, 2^6, 3^4, 7$ is $1 \times 4 \times 1 \times 7 = 28)$
Hence, the unit digit of $\dfrac{10!}{100}$ is 8.

EXAMPLE 15 *Find the unit digit of the expression*
$888^{92351} + 222^{92351} + 666^{23591} + 999^{99991}$.

SOLUTION First of all we find the unit digit individually of all the four terms,
So, the unit digit of 888^{92351} is equal to the unit digit of 8^{92351}
Now, the unit digit of 8^{92351} is equal to the unit of 8^4 (since 92351 is divisible by 4), which is 6.

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Again the unit digit $222^{9235!}$ is equal to the unit digit of $2^{9235!}$ and the unit digit of $2^{9235!}$ is same as that of 2^4 which is 6, since 9235! is divisible by 4. Further the unit digit of 666^{23591} is 6, which always constant for all the powers except zero and the unit digit of $999^{9999!}$ is 1 since the value of 9999! is even.

Thus the unit digit of the expression is 9. ($\because 6 + 6 + 6 + 1 = 19$)

EXAMPLE 16 *The last digit of the following expression is :*

$(1!)^1 + (2!)^2 + (3!)^3 + (4!)^4 + \ldots + (10!)^{10}$

(a) 4 (b) 5 (c) 6 (d) 7

SOLUTION The unit digit of the given expression will be equal to the unit digit of the sum of the unit digits of every term of the expression.

Now,

The unit digit of $(1!)^2 = 1$
The unit digit of $(2!)^2 = 4$
The unit digit of $(3!)^3 = 6$
The unit digit of $(4!)^4 = 6$
The unit digit of $(5!)^5 = 0$
The unit digit of $(6!)^6 = 0$

Thus the last digit of the $(7!)^7$, $(8!)^8$, $(9!)^9$, $(10!)^{10}$ will be zero.

So, the unit digit of the given expression = 7

($\because 1 + 4 + 6 + 6 + 0 + 0 + 0 + 0 + 0 + 0 = 17$)

EXAMPLE 17 *The last 5 digits of the following expression will be*

$(1!)^5 + (2!)^4 + (3!)^3 + (4!)^2 + (5!)^1 + (10!)^5 + (100!)^4 + (1000!)^3 + (10000!)^2 + (100000!)$

(a) 45939 (b) 00929
(c) 20929 (d) can't be determined

SOLUTION The last digit of $(1!)^5 = 1$
The last digit of $(2!)^4 = 16$
The last digit of $(3!)^3 = 216$
The last digit of $(4!)^2 = 576$
The last digit of $(5!)^1 = 120$
The last 5 digit of $(10!)^5 = 00000$
The last 5 digit of $(100!)^4 = 00000$
$(1000!)^3 = 00000$
$(10000!)^2 = 00000$
$(100000!)^1 = 00000$

Thus the last 5 digits of the given expression = 00929

[$\because 1 + 16 + 216 + 576 + 120 + 00000 + 00000 + 00000 + 00000 + 00000 = 00929$]

INVOLUTION

As in the previous articles we have been studied that $a.a.a..(n \text{ times}) = a^n$. So the process of multiplication of a number several times by itself is known as INVOLUTION.

Similarly,we use the same method in some algebraic expressions as follows.

1. $(a+b)^2 = a^2 + b^2 + 2ab = (a+b)(a+b)$
2. $(a-b)^2 = a^2 + b^2 - 2ab = (a-b)(a-b)$
3. $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$
   $= (a+b+c)(a+b+c)$
4. $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$
5. $(a-b)^3 = a^3 - b^3 - 3ab(a-b)$
6. $(a+b+c)^3 = a^3 + b^3 + c^3 + 3(a+b)(b+c)(c+a)$
7. $(a+b)^2 = (a-b)^2 + 4ab$
8. $(a-b)^2 = (a+b)^2 - 4ab$
9. $a^2 - b^2 = (a+b)(a-b)$
10. $a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$
11. $a^3 - b^3 = (a-b)(a^2 + b^2 + ab)$
12. $a^3 + b^3 + c^3 - 3abc = (a+b+c)$
    $(a^2 + b^2 + c^2 - ab - bc - ac)$

The above rules are used widely. But these are important in finding the square, cube, etc. of a number besides help in finding the remainder or divisibility of a number.

1. $a^n + b^n$ is divisible by $(a+b)$, when n is odd.
2. $a^n + b^n$ is never divisible by $(a-b)$
3. $a^n + b^n$ is not divisible by $(a+b)$, when n is even.
4. $a^n - b^n$ is always divisible by $(a-b)$
5. $a^n - b^n$ is divisible by $(a+b)$ only when n is even.
6. $a^n - b^n$ is not divisible by $(a+b)$ when n is odd.

EXAMPLE 1 *Find the value of* $107 \times 107 - 93 \times 93$

SOLUTION $(107)^2 - (93)^2$ [$\because a^2 - b^2 = (a+b)(a-b)$]
$= (107 + 93)(107 - 93)$
$= 200 \times 14 = 2800$

EXAMPLE 2 *Find the value of* 734856×9999..

SOLUTION $734856 \times 9999 = 734856 \times (10000 - 1)$
$= 7348560000 - 734856$
$= 7347825144$

EXAMPLE 3 $6798 \times 223 + 6798 \times 77 = k$, *then the value of k will be :*

(a) 2034900 (b) 3029400
(c) 2039400 (d) none of these

SOLUTION $6798 \times 223 + 6798 \times 77 = 6798(223 + 77)$
$= 6798(300) = 2039400$

EXAMPLE 4 $123 \times 123 + 77 \times 77 + 2 \times 123 \times 77 = ?$

SOLUTION $123 \times 123 + 77 \times 77 + 2 \times 123 \times 77$
$= (123)^2 + (77)^2 + 2 \times 123 \times 77$
$= (123 + 77)^2$ [$(a+b)^2 = a^2 + b^2 + 2ab$]
$= (200)^2 = 40000$

EXAMPLE 5 $\dfrac{(941 + 149)^2 + (941 - 149)^2}{(941 \times 941 + 149 \times 149)} = ?$

SOLUTION $\dfrac{(941 + 149)^2 + (941 - 149)^2}{(941 \times 941 + 149 \times 149)}$

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$$\because \quad \frac{(a+b)^2+(a-b)^2}{a^2+b^2}=\frac{(a^2+b^2+2ab)+(a^2+b^2-2ab)}{(a^2+b^2)}$$

$$=\frac{2(a^2+b^2)}{(a^2+b^2)}=2$$

Hence, the value of the given expression is 2. So you need not to calculate the given numbers in the expression, where $a=941, b=149$

EXAMPLE 6 $\dfrac{888\times888\times888-222\times222\times222}{888\times888+888\times222+222\times222}=?$

SOLUTION $\dfrac{888\times888\times888-222\times222\times222}{888\times888+888\times222+222\times222}$

$$=\frac{(888)^3-(222)^3}{(888)^2+888\times222+(222)^2}$$

$$=\frac{(888)^3-(222)^3}{(888)^2+888\times222+(222)^2}$$

$$=888-222 \qquad \left[\because \frac{a^3-b^3}{(a^2+b^2+ab)}=\frac{(a-b)(a^2+b^2+ab)}{(a^2+b^2+ab)}=(a-b)\right]$$

$$=666$$

EXAMPLE 7 *If* $(64)^2-(36)^2=20k$ *then the value of* k *is :*

(a) 140 (b) 120
(c) 80 (d) none of these

SOLUTION

$$(64)^2-(36)^2=20k$$
$$\Rightarrow \quad (64+36)(64-36)=20k$$
$$\Rightarrow \quad 100\,.\,28=20k$$
$$\Rightarrow \quad 20\times5\times28=20k$$
$$\Rightarrow \quad k=28\times5=140$$

Hence (a) is the correct answer.

EXAMPLE 8 *If* $a+\dfrac{1}{a}=3$, *then the value of* $a^2+\dfrac{1}{a^2}$ *is :*

(a) 6 (b) 7
(c) 9 (d) can't be determined

SOLUTION $\because \quad \left(a+\dfrac{1}{a}\right)=3$

$$\therefore \quad \left(a+\frac{1}{a}\right)^2=a^2+\frac{1}{a^2}+2a\times\frac{1}{a}=9$$

$$\Rightarrow \quad a^2+\frac{1}{a^2}+2=9 \Rightarrow a^2+\frac{1}{a^2}=7$$

Hence (b) is the correct option.

EXAMPLE 9 *If* $a+\dfrac{1}{a}=3$, *then the value of* $a^3+\dfrac{1}{a^3}$ *is :*

(a) 15 (b) 18
(c) 27 (d) none of these

SOLUTION $\left(a+\dfrac{1}{a}\right)^3=a^3+\dfrac{1}{a^3}+3a.\dfrac{1}{a}\left(a+\dfrac{1}{a}\right)$

$$\Rightarrow \quad (3)^3=a^3+\frac{1}{a^3}+3(3) \qquad \left[\because \ a\times\frac{1}{a}=1 \text{ and } (a+b)^3=a^3+b^3+3ab(a+b)\right]$$

$$\Rightarrow \quad 27=a^3+\frac{1}{a^3}+9 \Rightarrow a^3+\frac{1}{a^3}=18$$

Hence (b) is the correct option.

EXAMPLE 10 *What is the value of* $\dfrac{2.75\times2.75\times2.75-2.25\times2.25\times2.25}{2.75\times2.75+2.75\times2.25+2.25\times2.25}$?

SOLUTION $\because \quad a^3-b^3=(a-b)(a^2+b^2+ab)$

$$\therefore \quad \frac{a^3-b^3}{(a^2+b^2+ab)}=\frac{(a-b)(a^2+b^2+ab)}{(a^2+b^2+ab)}=(a-b)$$

$\therefore$ The value of the expression $= 2.75-2.25=0.50$

EXAMPLE 11 *If* $(\sqrt{a}+\sqrt{b})=17$ *and* $(\sqrt{a}-\sqrt{b})=1$, *then the value of* $\sqrt{ab}$ *is :*

(a) 72 (b) 27
(c) 35 (d) none of these

SOLUTION $(x+y)^2-(x-y)^2=(x^2+y^2+2xy)-(x^2+y^2-2xy)$

$$(x+y)^2-(x-y)^2=4xy$$
$$xy=\frac{(x+y)^2-(x-y)^2}{4}$$

Thus
$$\sqrt{ab}=\frac{(\sqrt{a}+\sqrt{b})^2-(\sqrt{a}-\sqrt{b})^2}{4}$$
$$=\frac{(17)^2-(1)^2}{4}=\frac{288}{4}=72$$

Hence (a) is the correct option.

EXAMPLE 12 *Find the value of* $a^3+b^3+c^3-3abc$ *if* $a+b+c=12$ *and* $ab+bc+ca=47$.

SOLUTION $\because \quad a+b+c=12$

$$\therefore (a+b+c)^2=a^2+b^2+c^2+2(ab+bc+ac)=144$$
$$\Rightarrow \quad a^2+b^2+c^2+2\times47=144$$
$$\Rightarrow \quad a^2+b^2+c^2=50$$

Now, since $a^3+b^3+c^3-3abc$

$$=(a+b+c)(a^2+b^2+c^2-ab-bc-ac)$$

or $a^3+b^3+c^3-3abc=12(50-47)=12\times3=36$

EXAMPLE 13 *If* $a+b+c=0$, *then the value of* $a^3+b^3+c^3$ *is :*

(a) 0 (b) abc
(c) 3abc (d) none of these

SOLUTION $\because \quad a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-ab-bc-ac)$

or $\quad a^3+b^3+c^3-3abc=0$
$$\Rightarrow \quad a^3+b^3+c^3=3abc$$

Hence (c) is the correct option.

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EXAMPLE 14 *If* $x + y + z = 0$ *then the value of* $\frac{x^2y^2 + y^2z^2 + z^2x^2}{x^4 + y^4 + z^4}$ *is :*

(a) 0 (b) 1/2 (c) 1 (d) 2

SOLUTION $\because \quad (x + y + z) = 0$

$\Rightarrow \quad x^2 + y^2 + z^2 + 2(xy + yz + zx) = 0$

$\Rightarrow \quad x^2 + y^2 + z^2 = -2(xy + yz + zx)$

$\Rightarrow \quad (x^2 + y^2 + z^2)^2 = 4(xy + yz + zx)^2$

$\Rightarrow \quad x^4 + y^4 + z^4 + 2(x^2y^2 + y^2z^2 + z^2x^2)$
$= 4[x^2y^2 + y^2z^2 + z^2x^2 + 2xyz(x + y + z)]$

$\therefore \quad x^4 + y^4 + z^4 = 2(x^2y^2 + y^2z^2 + z^2x^2)$
$[\because (x + y + z) = 0]$

$\therefore \quad \frac{x^2y^2 + y^2z^2 + z^2x^2}{x^4 + y^4 + z^4} = \frac{1}{2}$

Hence (b) is the correct option.

EXAMPLE 15 *The value of* $a^3 + b^3 + c^3 - 3abc$, *when* $a = 87$, $b = -126$ *and* $c = 39$ *is :*

(a) 0 (b) 1259
(c) −48 (d) none of these

SOLUTION $\because \quad a + b + c = 0$

$\therefore \quad a^3 + b^3 + c^3 - 3abc = 0$

Thus (a) is the correct option.

EXAMPLE 16 *The greatest divisor of*
$(a - b)(a + b)(a^2 + b^2)(a^4 + b^4)(a^8 + b^8)(a^{16} + b^{16})$ *is :*

(a) $a^{16} - b^{16}$ (b) $a^{16} + b^{16}$
(c) $a^{31} - b^{31}$ (d) $a^{32} - b^{32}$

SOLUTION $(a - b)(a + b)(a^2 + b^2)(a^4 + b^4)(a^8 + b^8)(a^{16} + b^{16})$
$= a^{32} - b^{32}$

Hence, the greatest divisor is $a^{32} - b^{32}$. Hence (d) is the correct option.

EXAMPLE 17 *In the above question which of the following can not divide the given expression :*

(a) $a^4 - b^4$ (b) $a^{12} - b^{12}$
(c) $a^{16} - b^{16}$ (d) *can't be determined*

SOLUTION Since $a^{12} - b^{12}$ is not the factor of the given expression.

Hence, it can not divide the given expression. Thus (b) is the appropriate option.

EXAMPLE 18 *The expression* $31^n + 17^n$ *is divisible by, for every odd positive integer n :*

(a) 48 (b) 14
(c) 16 (d) both (a) and (c)

SOLUTION $a^n + b^n$ is divisible by $(a + b)$ for every odd n, $n \in N$

Hence $31^n + 17^n$ is divisible by 48 (= 31 + 17) and since 16 is the factor of 48. So the given expression is divisible by both (a) and (c). Hence (d) is the most appropriate answer.

EXAMPLE 19 *Which is not the factor of* $4^{6n} - 6^{4n}$ *for any positive integer n?*

(a) 5 (b) 25
(c) 7 (d) none of these

SOLUTION $\because 4^{6n} - 6^{4n} = (64)^{2n} - (36)^{2n}$
$= (64^n + 36^n)(64^n - 36^n)$

For $n = 1, 3, 5, \ldots$ etc. $(64^n + 36^n)$ is divisible by 100 and all its factors. Also $(64^n - 36^n)$ is divisible by 28 and all its factors.

Again for $n = 2, 4, 6, \ldots$ etc. $(64^n - 36^n)$ is always divisible by 100 and all its factors. Also it is divisible by 28 and all its factors. Hence (d).

EXAMPLE 20 $19^n - 1$ *is :*

(a) *always divisible by* 9 (b) *always divisible by* 20
(c) *is never divisible by* 19 (d) *only (a) and (c) are true*

SOLUTION $19^n - 1$ is divisible by 18 (= 19 − 1) when n is even or odd. So (a) is correct.

$19^n - 1$ is divisible by 20 only when n is even so (b) is wrong.

$19^n - 1$ is never divisible by 19 which is correct. Thus (d) is the most appropriate answer.

CONCEPTS OF REMAINDER

Some Important Results

1. When $a_1, a_2, a_3, \ldots$ are divided by 'd' individually the respective remainders obtained are $R_1, R_2, R_3, \ldots$ etc. and when $a_1 + a_2 + a_3, \ldots)$ is divided by d the remainders can be obtained by dividing $(R_1 + R_2 + R_3, \ldots)$ by d.

EXAMPLE 1 *Find the remainder when* 38 + 71 + 85 *is divided by* 16.

SOLUTION When we divide 38, 71 and 85 we obtain the respective remainders as 6, 7 and 5. Now we can obtain the required remainder by dividing the sum of remainders (*i.e.*, 6 + 7 + 5 = 18) by 16, which is 2 instead of dividing the sum of actual given numbers (*i.e.*, 38 + 71 + 85 = 194) by 16, which also gives us the same remainder 2.

EXAMPLE 2 *Find the remainder when* 1661 + 1551 + 1441 + 1331 + 1221 *is divided by* 20.

SOLUTION The remainders when 1661, 1551, 1441, 1331, 1221 are divided by 20 are 1, 11, 1, 11, 1. So the required remainder can be obtained just by dividing 1 + 11 + 1 + 11 + 1 (= 25) by 20. Hence the required remainder is 5.

EXAMPLE 3 *What is the remainder when* 678 + 687 + 6879 + 6890 *is divided by 17?*

SOLUTION The individual remainders when 678, 687, 6879, 6890 are divided by 17 are 15, 7, 11, 5 respectively. Hence the required remainder can be obtained by dividing 15 + 7 + 11 + 5 (= 38) by 17. Thus the required remainder is 4.

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EXAMPLE 4 *What is the remainder when* $(10+10^2+10^3+10^4+10^5)$ *is divided by 6?*

SOLUTION The remainder when 10 is divided by 6 is 4
The remainder when 100 is divided by 6 is 4
The remainder when 1000 is divided by 6 is 4
The remainder when 10000 is divided by 6 is 4
The remainder when 100000 is divided by 6 is 4
So the required remainder can be obtained by dividing $4+4+4+4+4\ (=20)$ by 6. Thus the required remainder is 2.

2. When 'a' is divided by 'd' the remainder is R and 'a_1' and 'a_2' are divided by 'd' the remainders are R_1 and R_2, then the remainder R will be equal to the difference of remainders R_1 and R_2 if $a_1 - a_2 = a$.

For example the remainder when 63 is divided by 35 is 28 and this can be obtained by taking the 63 as difference of any two numbers say 100 and 37 then the remainders when 100 and 37 are divided by 35 are 30 and 2 then the required remainder is $28\ (=30-2)$

[*i.e.*, Remainder of $\frac{63}{35}$

$= \text{Remainder of } \frac{100}{35} - \text{Remainder of } \frac{37}{35} = 30-2 = 28$]

NOTE If the obtained remainder in a division has negative sign, then the actual remainder can be obtained by adding the divisor with initially obtained remainder containing negative sign.

For example, the remainder of $\frac{63}{35}$

$= \text{Remainder of } \frac{120}{35} - \text{Remainder of } \frac{57}{35} = 15-22 = -7$

So, the required remainder $= 35-7 = 28$

3. When $a_1, a_2, a_3, \ldots$ are divided by a divisor d the respective remainders obtained are $R_1, R_2, R_3, \ldots$ then the remainder when $(a_1 \times a_2 \times a_3 \times \ldots)$ is divided by 'd' can be obtained by dividing $(R_1 \times R_2 \times R_3 \times \ldots)$ by d.

For example when $365 \times 375 \times 389$ is divided by 35 the remainders can be obtained by dividing the product of the remainders which are obtained by dividing 365, 375, 389 individually.

Now since the remainders are 15, 25, 4 so the required remainder = Remainder of $(15 \times 25 \times 4)$ when divided by 35.

Thus the required remainder $= \frac{15 \times 25 \times 4}{35} = \frac{60 \times 25}{35}$

$= \frac{25 \times 25}{35} = \frac{625}{35} = 30$

Thus 30 is the remainder.

NOTE The required remainder can be obtained by considering the remainders as the dividends.

Thus the required remainder $= R_1 \times R_2 \times R_3 \times R_4$
$= R_5 \times R_3 \times R_4$
$= R_6 \times R_4 = R_7$

where R_7 is the required remainder and $R_1, R_2, \ldots, R_6$ are the remainders which are obtained by dividing the remainders perse.

EXAMPLE 1 *Find the remainder when* 123×1234 *is divided by* 15.

SOLUTION The remainder of $\frac{123 \times 1234}{15}$ = Remainder of $\frac{3 \times 4}{15}$

(3 and 4 are the remainders of 123 and 1234 when divided by 15)

$= \text{Remainder of } \frac{12}{15} = 12$

EXAMPLE 2 *Find the remainder when* $1719 \times 1721 \times 1723 \times 1725 \times 1727$ *is divided by* 18.

SOLUTION Since you can understand that applying the traditional method of getting the remainder is very difficult since first of all you have to take the product of all the numbers as given in the expression, then this a very large product has to be divided by 18, which is very lengthy and tedious process.

Thus we find the remainder of the remainders.

Hence, the remainder of $\frac{1719 \times 1721 \times 1723 \times 1725 \times 1727}{18}$

$= \frac{9 \times 11 \times 13 \times 15 \times 17}{18}$

(9, 11, 13, 15, 17 are the remainder when the given numbers are divided by 18.)

$= \frac{99 \times 13 \times 15 \times 17}{18}$

$= \frac{9 \times 13 \times 255}{18} = \frac{117 \times 255}{18}$

$= \frac{9 \times 3}{18} = \frac{27}{18} = 9$

EXAMPLE 3 *Find the remainder when* 7^7 *is divided by* 4.

SOLUTION $\frac{7^7}{4} = \frac{7 \times 7 \times 7 \times 7 \times 7 \times 7 \times 7}{4}$

$= \frac{3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3}{4}$

$= \frac{9 \times 9 \times 9 \times 3}{4}$

$= \frac{1 \times 1 \times 1 \times 3}{4} = \frac{3}{4}$

Thus the required remainder is 3.

EXAMPLE 4 *Find the remainder when* 11^8 *is divided by* 7.

SOLUTION $\frac{11^8}{7} = \frac{11 \times 11 \times 11 \times 11 \times 11 \times 11 \times 11 \times 11}{7}$

$= \frac{4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4}{7}$

$= \frac{16 \times 16 \times 16 \times 16}{7}$

$= \frac{2 \times 2 \times 2 \times 2}{7} = \frac{16}{7}$

Thus the required remainder = 2

Now you can see that, if in the above problem it is to be asked that find the remainder when 11^{75} is divided by 7, then it would be very difficult to solve it within a short span of time because it would involve a very large calculations if done in the above shown manner.

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So follow the steps in which I will show the appropriate process, which is generally known as "pattern method to find the remainder."

The remainder when 11^1 is divided by 7 is 4
The remainder when 11^2 is divided by 7 is 2
The remainder when 11^3 is divided by 7 is 1
The remainder when 11^4 is divided by 7 is 4
The remainder when 11^5 is divided by 7 is 2
The remainder when 11^6 is divided by 7 is 1
The remainder when 11^7 is divided by 7 is 4

So you can see that you are getting a pattern, if you follow it you can get the remainder for higher powers of 11.

Now you are seeing that after every 3 steps the cycle of remainders is repeating its course as after every 7 days a particular day repeats itself.

Now if we assume that the month of March starts with Monday, then which day will fall on 24th March of the same year? So you will quickly respond that it would be Wednesday.

Let us see how?

M	T	W	Th	F	Sat	Sun
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

As we can see that after every 7 days Monday comes on 1st, 8th, 15th, 22nd etc. So what we do is just divide the given date by 7. If it is divisible by 7 then the last day of the week (*i.e.*, last day of the cycle) will be the required day. If it is not divisible by 7 then it must leaves some remainder then we calculate the day according to the remainder. For instance in the above example when we divide 24 by 7, then we get the remainder 3. So we calculate the 3rd day starting from the first day of the week (or cycle). So the required day will be Wednesday.

Now, If I ask that which day will fall on 7th, 14th, 21st or 28th of this month then your response will be Sunday since all the dates given above are divisible by 7 which is the period of the cycle.

If the dates were 1, 8, 15, ... etc then the required day would have been Monday since when we divide these given dates by 7 the remainder is 1 so the dates which are corresponding to 1 *i.e.*, 8, 15, 22, 29 etc. will fall on Monday.

So I hope this discussion will be very helpful to you while solving the concerned problems.

Thus we can say that the remainder when 11^{75} is divided by 7 is 1, since 75 is divisible by 3.

EXAMPLE 5 *Find the remainder when 5^{123} is divided by 7.*

SOLUTION The remainder when 5^1 is divided by 7 is 5
The remainder when 5^2 is divided by 7 is 4
The remainder when 5^3 is divided by 7 is 6
The remainder when 5^4 is divided by 7 is 2
The remainder when 5^5 is divided by 7 is 3
The remainder when 5^6 is divided by 7 is 1
The remainder when 5^7 is divided by 7 is 5
The remainder when 5^8 is divided by 7 is 4 etc.

So we see that the cyclic period of remainder is 6,since after 6 steps the remainders start repeating. Now we divide the power of 5 *i.e.*, 123 by 6, then it leaves the remainder 3. It means the required remainder will be equal to the corresponding remainder when 5^3 is divided by 7, which is 6. So the remainder when 5^{123} is divided by 7 is 6.

Explanation : $\text{Remainder } \frac{5^{123}}{7} = \text{Remainder } \frac{5^{120} \times 5^3}{7}$

$= \text{Remainder } \frac{(5^6)^{20} \times 5^3}{7}$

$= \text{Remainder} = \frac{5^3}{7}$

= Remainder is 6.

NOTE Remember that when you get the remainder 1 while obtaining the cyclic period, you can stop the further calculation because at this step your cyclic period ends and in the next step the remainder starts repeating as in the above examples at 5^6 we get the remainder 1. So we need not to calculate further for $5^7, 5^8, 5^9$ etc., because all these numbers will give the same remainders as $5^1, 5^2, 5^3$.. etc. respectively.

EXAMPLE 6 *Find the remainder when 123^{321} is divided by 5.*

SOLUTION The remainder when 123 is divided by 5 is 3
So, the remainder when 123^{321} is divided by 5 is same as when 3^{321} is divided by 5.
Now, the remainder when 3^1 is divided by 5 is 3
The remainder when 3^2 is divided by 5 is 4
The remainder when 3^3 is divided by 5 is 2
The remainder when 3^4 is divided by 5 is 1
The remainder when 3^5 is divided by 5 is 3
So the cyclic period is 4 since at 3^4, we get the remainder 1 (after which the cyclic starts repeating)
Thus the remainder when 3^{321} is divided by 5 is 3 since we get the remainder 1 when 321 is divided by 4 (the cyclic period).

or $\text{Rem. } \frac{123^{321}}{5} = \text{Rem. } \frac{3^{321}}{5} = \text{Rem. } \frac{3^{320} \times 3^1}{5}$

$= \text{Rem } . \frac{(3^4)^{80} \times 3^1}{5} = \text{Rem. } \frac{3^1}{5}$

= Remainder is 3.

EXAMPLE 7 *Find the remainder when $923^{888} + 235^{222}$ is divided by 4.*

SOLUTION $\text{Rem } \frac{923^{888} + 235^{222}}{4} = \text{Rem. } \frac{3^{888} + 3^{222}}{4}$

$= \text{Rem. } \frac{1+1}{4} = \frac{2}{4}$

Thus the remainder is 2.

EXAMPLE 8 *Find the remainder of $\frac{3^{9415}}{80}$.*

SOLUTION $\text{Rem } \frac{3^{9415}}{80} = \text{Rem } \frac{3^{9412} \times 3^3}{80}$

$= \text{Rem } \frac{(3^4)^{2353} \times 3^3}{80}$ [Since the cyclic period is 4]

$= \text{Rem } \frac{(81)^{2353} \times 3^3}{80}$

$= \text{Rem } \frac{1 \times 27}{80} = \text{Rem } \frac{27}{80}$

Thus the remainder is 27.

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EXAMPLE 9 *Find the remainder when* $10^1 + 10^2 + 10^3 + 10^4 + 10^5 + \ldots + 10^{99}$ *is divided by 6.*

SOLUTION The remainder when 10^1 is divided by 6 is 4
The remainder when 10^2 is divided by 6 is 4
The remainder when 10^3 is divided by 6 is 4
The remainder when 10^4 is divided by 6 is 4
The remainder when 10^5 is divided by 6 is 4
Thus the remainder is always 4.

So, the required remainder $= \frac{4 + 4 + 4 + \ldots \text{ 99 times}}{6} = \frac{396}{6}$

Thus the remainder is zero.

EXAMPLE 10 *The remainder of* $\frac{888^{222} + 222^{888}}{3}$ *is :*

(a) 0 *(b)* 1 *(c)* 2 *(d)* 3

SOLUTION Remainder $\frac{888^{222} + 222^{888}}{3}$ = Remainder is zero,

since 888 and 222 both (bases) are divisible by 3.

EXAMPLE 11 *The remainder of* $\frac{888^{222} + 222^{888}}{5}$ *is :*

(a) 0 *(b)* 1 *(c)* 3 *(d)* 4

SOLUTION Rem. $\frac{888^{222} + 222^{888}}{5}$

$= \text{Rem.}\frac{888^{222}}{5} + \text{Rem.}\frac{222^{888}}{5}$

$= \text{Rem.}\frac{3^{222}}{5} + \text{Rem.}\frac{2^{888}}{5}$

$= \text{Rem.}\frac{(3^4)^{55} \times 3^2}{5} + \text{Rem.}\frac{(2^4)^{222}}{5}$

$= \text{Rem.}\frac{1 \times 9}{5} + \text{Rem. } 1$

$= \text{Rem.}\frac{4}{5} + \text{Rem.}\frac{1}{5} = \text{Rem.}\frac{4+1}{5} \Rightarrow \frac{5}{5}$

Thus the remainder is zero.

Alternatively: [To check the divisibility by 5 just see the sum of the unit digits which is 10 (= 4 +6)

$\because$ $8^{222} \rightarrow 4$ (unit digit)

and $2^{888} \rightarrow 6$ (unit digit)

Hence it is divisible. So there is no remainder]

EXAMPLE 12 *The remainder of* $\frac{32^{32^{32}}}{7}$ *:*

(a) 2 *(b)* 1
(c) 4 *(d) none of these*

SOLUTION $32^{32^{32}}$ means $32^{(32.\,32.\,32.\,32 \ldots 32 \text{ times})}$

Now the remainder of $\frac{32}{7}$ is 4.

Again $\frac{4^{32.\,32.\,32.\,32 \ldots 32 \text{ times}}}{7} \Rightarrow \frac{4^{2.\,2.\,2 \ldots 32 \text{ times}}}{7}$

$\left[\because \frac{4^1}{7} \rightarrow 4, \frac{4^2}{7} \rightarrow 2, \frac{4^3}{7} \rightarrow 1\right]$,

Remainder = 4

Since remainder of $\frac{4^2}{7} = \frac{4^8}{7} = \frac{4^{32}}{7} = \frac{4^{128}}{7} = \ldots$ is 2

and remainder of $\frac{4^4}{7} = \frac{4^{16}}{7} = \frac{4^{64}}{7} = \frac{4^{256}}{7} = \ldots$ is 4

NOTE 1. $\frac{(a+1)^n}{a}$ gives always remainder 1.

2. $\frac{a^n}{(a+1)}$ gives remainder 1 when n is even and gives the remainder a itself when n is odd, where a is any integer and n being the positive integer.

EXAMPLE 13 *The remainder when* 8^{1785} *is divided by 7 is :*

(a) 5 *(b)* 1
(c) 6 *(d) can't be determined*

SOLUTION $\because \frac{(a+1)^n}{a}$ leaves always remainder 1.

So $\frac{8^{1785}}{7} = \frac{(7+1)^{1785}}{7}$ gives the remainder 1.

EXAMPLE 14 *The remainder when* $(16)^{3500}$ *is divided by 17 is :*

(a) 1 *(b) 0*
(c) 16 *(d) none of these*

SOLUTION $\because \frac{a^n}{a+1}$ gives remainder 1 when n is even

So the remainder of $\frac{16^{3500}}{17} = \frac{16^{3500}}{(16+1)}$ is 1.

Hence (a) is the correct option.

EXAMPLE 15 *The remainder when* $(3)^{81}$ *is divided by 28 is :*

(a) 1 *(b)* 8 *(c)* 18 *(d)* 26

SOLUTION The remainder of $\frac{3^{81}}{28}\left[= \frac{(3^3)^{27}}{28} = \frac{(27)^{27}}{28}\right]$ is 27.

Since $\frac{a^n}{(a+1)}$ gives remainder a when n is odd.

EXAMPLE 16 *The remainder of* $\frac{2^{243}}{3^2}$ *is :*

(a) 8 *(b) 10*
(c) 4 *(d) none of these*

SOLUTION $\frac{2^{243}}{3^2} = \frac{(2^3)^{81}}{9} = \frac{8^{81}}{9} = \frac{8^{81}}{(8+1)}$

Hence the remainder is 8 since the power of 8 is odd.

EXAMPLE 17 *The remainder when* $(3)^{67!}$ *is divided by 80 :*

(a) 0 *(b)* 1
(c) 2 *(d) can't be determined*

SOLUTION $\because 3^4 = 81$ so $\frac{3^{4n}}{80}$ gives remainder 1.

Thus $\frac{3^{67!}}{80}$ will also give the remainder 1.

Since $67! = 4n$ for a positive integer n.

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EXAMPLE 18 *The remainder of* $\frac{39^{931}}{40}$ *is :*

(a) 0 (b) 1 (c) 39 (d) 13

SOLUTION $\because \frac{a^n}{a+1}$ gives remainder 1, when n is even, Now since 931 is an even number, hence the remainder is 1. Thus (b) is the correct choice.

EXAMPLE 19 *The remainder of* $\frac{2^{591}}{255}$ *is :*

(a) 0 (b) 1 (c) 55 (d) 5

SOLUTION 2^{591} can be expressed as 2^{8n}

So, $\text{Rem.}\frac{2^{591}}{255} = \text{Rem.}\frac{2^{8n}}{255} = \text{Rem.}\frac{(256)^n}{255}$

Hence the remainder is 1.

Rational Numbers : The numbers which can be expressed as the ratio of any two integers *i.e,* in the form of $\frac{p}{q}$ where p, q are integers, prime to each other (*i.e,* p and q must be coprime) and $q \neq 0$ are called the rational numbers. These numbers are denoted by 'Q'.

Thus $3, -4, \frac{1}{7}, -\frac{3}{8}, \sqrt{16}, \sqrt{25}$ etc. are rational numbers.

All integers and all fractions are rational numbers which are also called commensurable quantities.

FACTS ABOUT RATIONAL NUMBERS

1. The denominator of any rational number can never be zero since the division by zero is undefined.
2. Every integer *e.g.*, $0, \pm1, \pm2, \pm3$.. etc. is a rational number since these can be expressed as $7 = \frac{7}{1}, -3 = -\frac{3}{1}, 0 = \frac{0}{1}$ etc.
3. All the decimal numbers which are terminating are rational numbers *e.g.*,
$0.14 = = \frac{14}{100}, -8.5 = \frac{-85}{10}, 0.006 = \frac{6}{1000}$ etc.
4. All the recurring decimals, which are non terminating, are the rational numbers.
e.g., 0.4444..., 1.4141...., 6.785785..., 1.42857142857 ... $\frac{22}{7}$ etc.

NOTE π is not a rational number, since the exact value of π is not $\frac{22}{7}$.

PROPERTIES OF RATIONAL NUMBERS

1. **Closure law :** Addition and multiplication of any two rational number is also a rational number.
2. **Commutative law :** For any two rational numbers $\frac{a}{b}, \frac{c}{d}$.
$\frac{a}{b} + \frac{c}{d} = \frac{c}{d} + \frac{a}{b}$ and $\frac{a}{b} \times \frac{c}{d} = \frac{c}{d} \times \frac{a}{b}$
3. **Associative law :** For any three rational numbers $\frac{a}{b}, \frac{c}{d}$ and $\frac{e}{f}$.
$\left(\frac{a}{b} + \frac{c}{d}\right) + \frac{e}{f} = \frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f}\right)$
and $\left(\frac{a}{b} \cdot \frac{c}{d}\right) \cdot \frac{e}{f} = \frac{a}{b} \cdot \left(\frac{c}{d} \cdot \frac{e}{f}\right)$
4. **Additive and Multiplicative Identity :**
$\frac{a}{b} + 0 = \frac{a}{b}$ [$\because 0 = \frac{0}{1}$ is an additive identity]
and $\frac{a}{b} \times 1 = \frac{a}{b}$ [$\because 1 = \frac{1}{1}$ is a multiplicative identity]
5. **Additive and Multiplicative Inverse :**
$\frac{a}{b} + \left(-\frac{a}{b}\right) = \frac{0}{1}$ [$-\frac{a}{b}$ is additive inverse of $\frac{a}{b}$]
$\frac{a}{b} \times \left(\frac{b}{a}\right) = \frac{1}{1}$ [$\frac{b}{a}$ is multiplicative inverse of $\frac{a}{b}$]
where $a \neq 0, b \neq 0$
6. **Rational Numbers :** Do not follow the Commutative law and Associative law for the subtraction and division.
7. **Distribution law :** For any three rational numbers a, b, c
$a.(b+c) = a.b + a.c$
and $a.(b-c) = a.b - a.c$
8. If x and y are any two rational numbers such that $x < y$ then $\frac{1}{2}(x+y)$ is also a rational number lying between x and y which shows that **there are infinite number of rational numbers between any two rational numbers.**

Irrational Numbers : The numbers which can't be expressed in the form $\frac{p}{q}$, where p, q are two integers prime to each other and $q \neq 0$ are called irrational numbers. Thus $\sqrt{2}, \sqrt{3}, -\sqrt{3}, \sqrt[3]{4}, \sqrt[3]{6}, \sqrt{5}, \pi$, etc. are irrational numbers.

The decimal representation of these numbers is non-repeating and non-terminating.

e.g., 7.2030030003 ..., $\sqrt{2} = 1.41421...., \sqrt{3} = 1.732....$

Properties of Irrational Numbers

(i) The set 'P' of all irrational numbers is not closed for addition since the sum of two irrationals need not be irrational *e.g.*, $(3+\sqrt{5}) \in P$, $(3-\sqrt{5}) \in P$ but $(3+\sqrt{5}) + (3-\sqrt{5}) = 6 \notin P$

(ii) The set P of all irrational number is not closed for multiplication since the product of two irrationals need not be irrational.

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$\therefore$ *e.g.*, $\sqrt{3} \in P, -\sqrt{3} \in P$

but $\sqrt{3} \times -\sqrt{3} = -3 \notin P$

Real Numbers : All the rational and all irrational numbers are called as real numbers *i.e.* the set of real numbers is the union of entire rational and irrational numbers.

e.g., $-3, 2, 0, \frac{5}{7}, \sqrt{4}, \sqrt{2}, \sqrt{3}, \pi, e, \frac{22}{7}, 0$, etc.

NOTE For any real number a and b

1. $a > b \Rightarrow -a < -b$
2. $a > b \Rightarrow \frac{1}{a} < \frac{1}{b}$
3. $a > b \Rightarrow a + k > b + k$
4. $a > b \Rightarrow ak > bk$ when $k > 0$
5. $a.b = 0 \Rightarrow$ either a is zero or b is zero or both a and b are zero.

Imaginary Numbers : If the square of a number is negative then this number is called as an imaginary number. *e.g.*, $\sqrt{-1}, \sqrt{-2}, \sqrt{-3}$ etc.

An imaginary number is denoted by 'i', where $i = \sqrt{-1}$

FACTS ABOUT IMAGINARY NUMBERS

1. $i^0 = 1, i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1$ etc.
2. $i^n = i^{4p+r} = (i^4)^p \times i^r = 1 \times i^r = i^r$
3. $\sqrt{a} \times \sqrt{b} \neq \sqrt{ab}$ if both a, b are negative *i.e.*, imaginary otherwise atleast one of a, b must be whole number *e.g.*,

$$\sqrt{-5} \times \sqrt{-7} = \sqrt{5}i \times \sqrt{7}i = \sqrt{35}i^2 = -\sqrt{35}$$

but $\sqrt{-5} \times \sqrt{-7} = \sqrt{-5 \times -7} \neq \sqrt{35}$

EXAMPLE 1 *Find the value of* i^{63}.

SOLUTION $i^{63} = (i^4)^{15} \times i^3 = i^3 = -i$

EXAMPLE 2 *Find the value of* i^{170}.

SOLUTION $i^{170} = (i^4)^{42} \times i^2 = i^2 = -1$

EXAMPLE 3 *Find the value of* $\frac{1}{i^{123}}$

SOLUTION $\frac{1}{i^{123}} = \frac{1}{i^{120} \times i^3} = \frac{1}{1 \times i^3} = \frac{1}{i^3} = \frac{1}{i^3} \times \frac{i}{i} = \frac{i}{i^4} = i$

EXAMPLE 4 *Find the value of* $\sqrt{-16} \times \sqrt{-25}$.

SOLUTION $\sqrt{-16} \times \sqrt{-25} = 4i \times 5i = 20i^2 = -20$

EXAMPLE 5 *Find the value of* $(\sqrt{-1})^{4n+1}$.

SOLUTION $(\sqrt{-1})^{4n+1} = i^{4n+1} = i^{4n} \times i = i$

EXAMPLE 6 *Find the value of* i^{-343}.

SOLUTION $i^{-343} = \frac{1}{i^{343}} = \frac{1}{(i^4)^{85} \times i^3} = \frac{1}{1 \times i^3} = \frac{1 \times i}{i^3 \times i} = i$

EXAMPLE 7 *Find the value of* $i^{248} + i^{341} + i^{442} + i^{543}$.

SOLUTION $i^{248} + i^{341} + i^{442} + i^{543} = 1 + i + (-1) + (-i) = 0$

EXAMPLE 8 *The value of* $\frac{1}{i^n} + \frac{1}{i^{(n+3)}} + \frac{1}{i^{(n+2)}} + \frac{1}{i^{(n+1)}}$ *is :*

(a) -1 (b) 1
(c) 0 (d) *can't be determined*

SOLUTION $\frac{1}{i^n} + \frac{1}{i^{(n+3)}} + \frac{1}{i^{(n+2)}} + \frac{1}{i^{(n+1)}}$ is :

$$= \frac{1}{i^n} + \frac{1}{i^n \times i^3} + \frac{1}{i^n \times i^2} + \frac{1}{i^n \times i}$$

$$= \frac{1}{i^n} + \frac{1}{i^n} \times \frac{i}{i^4} + \frac{1}{i^n} \times \frac{i^2}{i^4} + \frac{1}{i^n} \times \frac{i^3}{i^4}$$

$$= \frac{1}{i^n} + \frac{i}{i^n} + \frac{i^2}{i^n} + \frac{i^3}{i^n} = \frac{1}{i^n} + \frac{i}{i^n} + \frac{-1}{i^n} - \frac{i}{i^n} = 0$$

Hence (c) is the correct option.

COMPLEX NUMBERS

The combination of real number and imaginary number is known as complex number *i.e.*, the number of the form $a + ib$ where a and b are purely real numbers and $i = \sqrt{-1}$, is called as a complex number. It is denoted by $z = a + ib$ where real $z = a$ and imaginary $z = b$.

1. **Property of order :** For any two complex numbers $(a + ib)$ and $(c + id)$ $(a + ib) <$ (or $>$) $c + id$ is not defined *e.g.*, $9 + 4i < 3 - i$ makes no sense.
2. A complex number is said to be purely real if $\text{Im}(z) = 0$ is and said to be purely imaginary if $Re(z) = 0$. The complex number $0 = 0 + i.0$ is both purely real and purely imaginary.
3. The sum of four consecutive powers of i is zero.

i.e., $i^k + i^{k+1} + i^{k+2} + i^{k+3} = 0$

OPERATIONS ON COMPLEX NUMBERS

Let z_1, z_2 be two complex numbers such that $z_1 = (a_1 + ib_1)$ and $z_2 = (a_2 + ib_2)$ then :

1. $z_1 \pm z_2 = (a_1 \pm a_2) + i(b_1 \pm b_2)$
2. $z_1 . z_2 = (a_1 a_2 - b_1 b_2) + i(a_1 b_2 + b_1 a_2)$
3. $z_1 = z_2 \Leftrightarrow a_1 = a_2$ and $b_1 = b_2$

Conjugate Complex Number : The complex number $z = a + ib$ and $\bar{z} = (a - ib)$ are called the complex conjugate of each other, where $i = \sqrt{-1}$, $b \neq 0$ and a, b are real numbers.

Properties of Conjugate

For any z there exists the mirror image of z along the real axis denoted as $\bar{z}$, then

(i) $z = \bar{z} \Leftrightarrow z$ is purely real

(ii) $z = -\bar{z} \Leftrightarrow z$ is purely imaginary

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(iii) $\overline{(\bar{z})} = z$

(iv) $\text{Re}(z) = \text{Re}(\bar{z}) = \dfrac{z+\bar{z}}{2}$

(v) $\text{Im}(z) = \dfrac{z-\bar{z}}{2i}$

(vi) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$

(vii) $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$

(viii) $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$

(ix) $\overline{\left(\dfrac{z_1}{z_2}\right)} = \dfrac{\bar{z}_1}{\bar{z}_2}$

(x) $z_1\bar{z}_2 + \bar{z}_1 z_2 = 2\,\text{Re}(\bar{z}_1 z_2) = 2\,\text{Re}(z_1\bar{z}_2)$

(xi) $\overline{z^n} = (\bar{z})^n$

(xii) If $z = f(z_1)$, then $\bar{z} = f(\bar{z}_1)$

Modulus of a Complex Number : The modulus of a complex number $z = (a + ib)$ is represented as $|z| = \sqrt{a^2 + b^2}$.

Properties of Modulus

(i) $|z| \geq 0 \Rightarrow |z| = 0$ iff $z = 0$ and $|z| > 0$ iff $z \neq 0$

(ii) $-|z| \leq \text{Re}(z) \leq |z|$ and $-|z| \leq \text{Im}(z) \leq |z|$

(iii) $|z| = |\bar{z}| = |-z| = |-\bar{z}|$

(iv) $z\bar{z} = |z|^2$

(v) $|z_1 \pm z_2| \leq |z_1| + |z_2|$
In general $|z_1 \pm z_2 \pm z_3 \pm \ldots z_n| \leq |z_1| \pm |z_2| \pm |z_3| \pm \ldots |z_n|$

(vi) $|z_1 \pm z_2| \geq ||z_1| - |z_2||$

(vii) $|z_1 z_2| = |z_1||z_2|$

(viii) $\left|\dfrac{z_1}{z_2}\right| = \dfrac{|z_1|}{|z_2|}$

(ix) $|z^n| = |z|^n$

(x) $||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$

(xi) $|z_1 \pm z_2|^2 = (z_1 \pm z_2)(\bar{z}_1 \pm \bar{z}_2)$
$= |z_1|^2 + |z_2|^2 \pm (z_1\bar{z}_2 + \bar{z}_1 z_2)$

(xii) $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$

nth ROOT OF UNITY

1. Unity has n roots namely $1, \omega, \omega^2, \omega^3, \ldots, \omega^{n-1}$ which are in geometric progression and the sum of these n roots is zero.
2. Product of n roots is $(-1)^{n-1}$.

Square Root

The square root of $z = a + ib$ are

$$\pm\left(\sqrt{\frac{|z|+a}{2}} + i\sqrt{\frac{|z|-a}{2}}\right) \text{ for } b > 0$$

and $$\pm\left(\sqrt{\frac{|z|+a}{2}} - i\sqrt{\frac{|z|-a}{2}}\right) \text{ for } b < 0$$

Remember

1. The square root of i are $\pm\left(\dfrac{1+i}{\sqrt{2}}\right)$ when $b = 1$
2. The square root of $-i$ are $\left(\dfrac{1-i}{\sqrt{2}}\right)$ when $b = -1$

or $$\sqrt{a+ib} = \pm(x+iy)$$

and $$\sqrt{a-ib} = \pm(x-iy)$$

where $$x = \left(\frac{a+\sqrt{a^2+b^2}}{2}\right)^{1/2}$$

and $$y = \left(\frac{\sqrt{a^2+b^2}-a}{2}\right)^{1/2}$$

CUBE ROOT OF UNITY

Cube root of unity are $1, \omega, \omega^2$ where

(a) $1 + \omega + \omega^2 = 0$

(b) $\omega^3 = 1$

(c) $\bar{\omega} = \omega^2$ and $(\bar{\omega})^2 = \omega$

(d) $\omega^{3n} = 1, \omega^{3n+1} = \omega, \omega^{3n+2} = \omega^2$

NOTE Iota (i) is neither 0, nor greater than 0, or nor less than 0.

EXAMPLE 1 *Find the modulus of* $5 + \sqrt{-11}$.

SOLUTION $\because \quad z = 5 + \sqrt{-11} = 5 + \sqrt{11}\,i$

$\therefore \quad |z| = \sqrt{25 + 11} = \sqrt{36} = 6$

EXAMPLE 2 *If the multiplicative inverse of a complex number is* $\dfrac{\sqrt{5}+6i}{41}$ *then the complex number itself is:*

(a) $\sqrt{5} - 6i$ (b) $\sqrt{5} + 6i$

(c) $6 + \sqrt{7}i$ (d) *none of these*

SOLUTION Since $z \cdot \dfrac{(\sqrt{5}+6i)}{41} = 1$

$$\Rightarrow \quad z = \frac{41}{(\sqrt{5}+6i)} \times \frac{\sqrt{5}-6i}{\sqrt{5}-6i} = \sqrt{5} - 6i$$

Hence (a) is the correct option.

EXAMPLE 3 *Which of the following is correct?*

(a) $7 + i > 5 - i$ (b) $8 + i > 9 - i$

(c) $13 + i < 25 - 3i$ (d) *none of these*

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SOLUTION Since the relation is not defined i.e., $(a+ib) \gtrless (c+id)$ is indeterminable.

Hence (d) is the correct option.

EXAMPLE 4 *The smallest positive integer n for which* $\left(\frac{1+i}{1-i}\right)^n = -1$ *is :*

(a) 5 (b) 1
(c) 2 (d) none of these

SOLUTION Given that $\left(\frac{1+i}{1-i}\right)^n = -1$

$$\Rightarrow \left[\frac{(1+i)(1+i)}{(1-i)(1+i)}\right]^n = -1$$

$$\Rightarrow i^n = -1$$

$$\Rightarrow (-1)^{\frac{n}{2}} = (-1) \Rightarrow \frac{n}{2} = 1$$

$$\therefore n = 2$$

Hence (c) is the correct option.

EXAMPLE 5 *The smallest positive integer n for which* $\left(\frac{1+i}{1-i}\right)^n = 1$ *is:*

(a) 2 (b) 4 (c) 6 (d) 8

SOLUTION $\left(\frac{1+i}{1-i}\right)^n = \left(\frac{(i+i)(1+i)}{(1-i)(1+i)}\right)^n$

$$= \left(\frac{(1+i)^2}{2}\right)^n = i^n = 1$$

$$\Rightarrow i^n = 1 = i^4 \Rightarrow n = 4$$

Hence (b) is the correct option.

EXAMPLE 6 $(3+2i)^3 = ?$

(a) $9-46i$ (b) $9+46i$
(c) $-9+46i$ (d) none of these

SOLUTION $(3+2i)^3 = 3^3 + (2i)^3 + 3.3.2i\,(3+2i)$

$= 27 + 8i^3 + 18i\,(3+2i)$

$= 27 + 8i^3 + 54i + 36i^2$

$= 27 - 8i + 54i - 36 = (-9+46i)$

Hence (c) is the correct option.

EXAMPLE 7 *The multiplicative inverse of* $(3+2i)^2$ *is :*

(a) $\frac{12}{169} - \frac{5i}{169}$ (b) $\frac{5}{169} - \frac{12i}{169}$
(c) $\frac{5}{13} - \frac{12i}{13}$ (d) none of these

SOLUTION $(3+2i)^2 = 9 + 4i^2 + 12i$

$= 9 - 4 + 12i = 5 + 12i$

then $z^{-1} = \frac{1}{5+12i} = \frac{1}{5+12i} \times \frac{5-12i}{5-12i}$

$$= \frac{5-12i}{25-144i^2} = \frac{5-12i}{169} = \left(\frac{5}{169} - \frac{12i}{169}\right)$$

Thus (b) is the correct option.

EXAMPLE 8 *If* $|z| = 1$ *then* $\left(\frac{1+z}{1+\bar{z}}\right)$ *equals :*

(a) $\bar{z}$ (b) z
(c) z^{-1} (d) none of these

SOLUTION $\frac{1+z}{1+\bar{z}} = \left(\frac{1+z}{1+\bar{z}}\right)\frac{z}{z} = \frac{z(1+z)}{(z+1)} = z$ $(\because |z| = 1)$

Hence (b) is the correct option.

EXAMPLE 9 *The value of* $\sqrt{i} + \sqrt{-i}$ *is :*

(a) 1 (b) $\sqrt{2}$
(c) $-i$ (d) none of these

SOLUTION $\sqrt{i} + \sqrt{-i} = \sqrt{(\sqrt{i} + \sqrt{-i})^2}$

$$= \sqrt{i - i + 2\sqrt{i}\sqrt{-i}} = \sqrt{2\sqrt{-i^2}} = \sqrt{2}$$

Hence (b) is the correct option.

EXAMPLE 10 *The value of* $\log_e(-1)$ *is :*

(a) $i\pi$ (b) 0
(c) i^{-1} (d) does not exist

SOLUTION $\log_e(-1) = \log_e(e^{i\pi}) = i\pi$

Hence (a) is the correct option.

EXAMPLE 11 *The number of solutions of the equation* $z^2 + |z|^2 = 0$, *where z is a complex number is :*

(a) 1 (b) 2
(c) 4 (d) infinitely many

SOLUTION $z^2 + |z|^2 = 0 \Rightarrow z^2 + z\bar{z} = 0$

$\Rightarrow z(z+\bar{z}) = 0$

or $z \cdot 2\text{Re}(z) = 0$

$\therefore z = 0$ and $\text{Re}(z) = 0$

If $z = x + iy \Rightarrow 0 = x + iy \Rightarrow (x, y) = (0, 0)$

and if $z = 0 + ib \Rightarrow (a, b) = (0, b)$ but $b \in R$

So, there are infinitely many solutions. Since $b \in R$ Hence (d) is the correct option.

EXAMPLE 12 *The value of* $\left(\frac{i+\sqrt{3}}{2}\right)^{100} + \left(\frac{i-\sqrt{3}}{2}\right)^{100}$ *is :*

(a) -1 (b) 1 (c) 0 (d) i

SOLUTION $\frac{i+\sqrt{3}}{2} = \left(\frac{i+\sqrt{3}}{2}\right)\frac{i}{i} = \left(\frac{-1+\sqrt{3}i}{2i}\right) = \frac{\omega}{i}$

($\because$ A complex number $a + ib$ for which $|a : b| = 1 : \sqrt{3}$ or $\sqrt{3} : 1$ can always be expressed in terms of i, ω or ω^2)

Similarly, $\frac{i-\sqrt{3}}{2} = \frac{\omega^2}{i}$

$$\therefore \left(\frac{i+\sqrt{3}}{2}\right)^{100} + \left(\frac{i-\sqrt{3}}{2}\right)^{100} = \left(\frac{\omega}{i}\right)^{100} + \left(\frac{\omega^2}{i}\right)^{100}$$

$= \omega^{99} \cdot \omega + \omega^{198} \cdot \omega^2$ $[\because \omega^{99} = \omega^{198} = 1]$

$= \omega + \omega^2 = -1$

Hence (a) is the correct option.

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EXAMPLE 13 *If 1, ω, ω² be the cube roots of unity, then the value of* $(1-\omega+\omega^2)^5+(1+\omega-\omega^2)^5$ *is :*

(a) 0 *(b)* 16 *(c)* 32 *(d)* 64

SOLUTION ∵ $1+\omega+\omega^2=0$

$\therefore \quad 1+\omega=-\omega^2$ and $1+\omega^2=-\omega$

$\therefore (1-\omega+\omega^2)^5+(1+\omega-\omega^2)^5=(-2\omega)^5+(-2\omega^2)^5$

$=-32(\omega^5+\omega^{10})$ [∵ $\omega^5=\omega^2$ and $\omega^{10}=\omega$ and $\omega+\omega^2=-1$]

$=32$

Hence (c) is the correct option.

EXAMPLE 14 *If 1, ω, ω², ω³, …, ω^{n−1} be the n, nth roots of unity, the value of* $(9-\omega)(9-\omega^2)\ldots(9-\omega^{n-1})$ *will be :*

(a) 0 *(b)* $\frac{9^n-1}{8}$

(c) $\frac{8^n-1}{3}$ *(d) none of these*

SOLUTION Let $x^n=1 \Rightarrow x=(1)^{1/n}$

$\Rightarrow \quad x^n-1=0$

or $x^n-1=(x-1)(x-\omega)(x-\omega^2)\ldots(x-\omega^{n-1})$

$\Rightarrow \quad \frac{x^n-1}{x-1}=(x-\omega)(x-\omega^2)\ldots(x-\omega^{n-1})$

Putting $x = 9$ in both sides, we have

$(9-\omega)(9-\omega^2)(9-\omega^3)\ldots(9-\omega^{n-1})=\frac{9^n-1}{8}$

Thus (b) is the correct choice.

EXAMPLE 15 *If* $\left(\frac{1-i}{1+i}\right)^{100}=x+iy$ *then the value of (x, y) is :*

(a) (0, 1) *(b)* (0, 0)
(c) (1, 0) *(d)* (−1, 0)

SOLUTION $\frac{1-i}{1+i}=\frac{(1-i)(1-i)}{(1+i)(1-i)}=\frac{-2i}{2}=-i$

Hence $\left(\frac{1-i}{1+i}\right)^{100}=(-i)^{100}=1$

Thus $x+iy=1 \Rightarrow x=1, y=0$

So the value of $(x, y) = (1, 0)$

Hence (c) is the correct option.

EXAMPLE 16 *The inequality a + ib < c + id holds if*

(a) $a<c, b<d$ *(b)* $a<c, b>d$
(c) $a>c, b<d$ *(d) none of these*

SOLUTION Option (d) is correct.

EXAMPLE 17 *The points z_1, z_2, z_3, z_4 in the complex plane form the vertices of a parallelogram iff :*

(a) $z_1+z_2=z_3+z_4$ *(b)* $z_1+z_3=z_2+z_4$
(c) $z_1+z_3>z_2+z_4$ *(d) none of these*

SOLUTION If the mid point of z_1z_3 is the same as that of z_2z_4.

i.e., $\frac{1}{2}(z_1+z_3)=\frac{1}{2}(z_2+z_4)$

$\Rightarrow z_1+z_3=z_2+z_4$, only then these points can form the parallelogram.

EXAMPLE 18 *If* $a+ib=\sqrt{\frac{(u+iv)}{(x+iy)}}$ *then the value of* a^2+b^2 *is :*

(a) $\sqrt{\frac{u^2+v^2}{x^2+y^2}}$ *(b)* $\frac{u^2-v^2}{x^2-y^2}$

(c) can't be determined *(d) none of these*

SOLUTION $a^2+b^2=|a+ib|^2$

$=\left|\frac{u+iv}{x+iy}\right|$ (Putting the value of $a+ib$)

$=\frac{\sqrt{u^2+v^2}}{\sqrt{x^2+y^2}}$

Hence (a) is the correct option.

EXAMPLE 19 *The maximum value of |z| when z satisfies the condition* $\left|z+\frac{2}{z}\right|=2$ *is :*

(a) $\sqrt{3}-1$ *(b)* $\sqrt{3}$ *(c)* $\sqrt{3}+1$ *(d)* 1

SOLUTION ∵ $|z|=\left|z+\frac{2}{z}-\frac{2}{z}\right|$

$|z| \le \left|z+\frac{2}{z}\right|+\left|\frac{2}{z}\right|$

It means $|z| \le 2+\frac{2}{|z|}$

$\Rightarrow \quad |z|^2 \le 2|z|+2$

$\Rightarrow \quad |z|^2-2|z|+1 \le 2+1$

$\Rightarrow \quad (|z|-1)^2 \le 3$

$\Rightarrow \quad -\sqrt{3} \le |z|-1 \le \sqrt{3}$

$\Rightarrow \quad 1-\sqrt{3} \le |z| \le 1+\sqrt{3}$

Therefore the maximum value of $|z|$ is $1+\sqrt{3}$

EXAMPLE 20 *The modulus of the complex number* $\frac{(2-i\sqrt{5})}{(1+2\sqrt{2}i)}$ *is:*

(a) −1 *(b)* 1 *(c)* $\sqrt{3}$ *(d)* $-\sqrt{3}$

SOLUTION The modulus of $\left(\frac{2-i\sqrt{5}}{1+2\sqrt{2}i}\right)=\left|\frac{2-i\sqrt{5}}{1+2\sqrt{2}i}\right|$

$=\frac{\sqrt{4+5}}{\sqrt{1+8}}=\frac{3}{3}=1$

Hence (b) is the correct answer.

EXCLUSIVE DISCUSSION ABOUT NUMBERS

Generally we use decimal system in our day to day mathematical applications *e.g.*, counting of the things, accounting in offices etc. Why we call it decimal system?

Since there we use 10 symbols *viz.*, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 to represent the data, so we said the representation is in decimal system. The sequence of decimal numbers goes on as 0, 1, 2, 3 4, 5, 6, 7, 8, 9, 10, 11,12, 13, 14, 15, 16, 17, 18, 19, 20, 21.. etc. So after 9, each successive number is a combination of 2 (or more) unique symbols of this system.

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The decimal system is a positional-value system in which the value of a digit in a number depends on its position. For example 879. Here 8 represents 800, 7 represents 70 and 9 represents 9 units. In essence, 8 carries the most weight of three digits, it is referred as the most significant digit (MSD). The 9 carries the least weight and is called as the least significant digit (LSD)

i.e., $879 = 8\times10^2 + 7\times10^1 + 9\times10^0$

Similarly $63.78 = 6\times10^1 + 3\times10^0 + 7\times10^{-1} + 8\times10^{-2}$

So, this number is actually equal to 6 tens plus 3 units plus 7 tenths plus 8 hundredths.

So $3859276 = 3\times10^6 + 8\times10^5 + 5\times10^4 + 9\times10^3 + 2\times10^2 + 7\times10^1 + 6\times10^0$

and $9253.827 = 9\times10^3 + 2\times10^2 + 5\times10^1 + 3\times10^0 + 8\times10^{-1} + 2\times10^{-2} + 7\times10^{-3}$

In general, any number is simply the sum of the products of each digit value and its positional (or place) value.

Positional values (weights) →	10^3	10^2	10^1	10^0	10^{-1}	10^{-2}	10^{-3}
	9	2	5	3	8	2	7
	MSD			Decimal point			LSD

Thus a two digit numbers can be expressed as $x\times10^1 + y\times10^0$, where x is the tens and y is the unit digit.

Thus the two digit number xy is expressed as $10x + y$

Similarly a three digit number xyz is expressed as $10^2x + 10^1y + 10^0z$

or xyz is expressed as $100x + 10y + z$ where x, y, z are the hundreds tens and unit digit respectively.

In general an n digit number in decimal representation, can be expressed as

$$k_1\times10^{n-1} + k_2\times10^{n-2} + k_3\times10^{n-3} + \ldots + k_n\times10^0$$

Some Facts about Decimal Numbers

1. The digits which are always same *i,e*, when they are seen upside down, they appears the same. *e.g.*, 0, 1 and 8.
2. The digits which create confusion when they are written upside down they change their face values. *e.g.*, 6 and 9.
3. When the digits of a two digits number are reversed *i.e.*, when their position is interchanged, then the difference of the original number and this resultant number is always divisible by 9. Also the difference of these two numbers is exactly the product of the difference of the two digits (*i.e.*, difference of their face values) with 9.

For example,
$$\begin{array}{r} xy \\ -\ yx \\ \hline 9(x-y) \end{array}$$

$[\because (10x + y) - (10y + x) = 9(x - y)]$

e.g., $53 - 35 = 9\times2 = 18$ $(\because 5 - 3 = 2)$

e.g., $82 - 28 = 9\times6 = 54$ $(\because 8 - 2 = 6)$

NOTE The sum of these two numbers is always divisible by 11.

4. When the digits of a three digit number are reversed *i.e.*, the unit digit becomes hundreds digit and vice versa, then the difference between these two numbers is always divisible by 99. Also the difference of these two numbers (*i.e.*, original and resultant number) is exactly equal to the product of the difference of unit and hundreds digit, with 99, *i.e.*, $xyz - zyx = 99(x - z)$

$[\because (100x + 10y + z) - (100z + 10y + x) = 99(x - z)]$

For example (i) $852 - 258 = 99\times6$ $[\because 8 - 2 = 6]$

$= 594$

(ii) $703 - 307 = 99\times4$ $[\because 7 - 3 = 4]$

$= 396$

5. When the digits of a four digit number are reversed *i.e*, unit digit with thousands digit and vice versa and tens digit with hundreds digit and vice versa, then the difference between the original number and the resultant number is always divisible by 9.

EXAMPLE 1 *The difference between the highest and lowest two digit numbers is :*

(a) 88 (b) 89 (c) 22 (d) 99

SOLUTION Simply 99 – 10 = 89

Hence (b) is the correct option.

EXAMPLE 2 *When a two digit number is reversed, then the new number becomes $\frac{5}{6}$th of the original number. The original number is :*

(a) 56 (b) 45 (c) 48 (d) 54

SOLUTION The best way is to go through options.

Now, we consider those options which are divisible by 6. Since, the number is an integer. Again we pick up only those values which on being reversed (their digits) decreases. So only option (d) is suitable. Check it out.

Alternatively: Let the original number be $(10x + y)$ then the new number will be $(10y + x)$, then

$$(10y + x) = \frac{5}{6}(10x + y)$$

$$\Rightarrow \quad 55y = 44x$$

$$\Rightarrow \quad \frac{x}{y} = \frac{5}{4}$$

Thus the only possible values of x, y are 5, 4 respectively. Hence the original number is 54.

Alternatively : When you multiply the given (original) number by $\frac{5}{6}$ then the digit of the resultant value gets interchanged.

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EXAMPLE 3 *The number of two digit numbers which on being reversed (i.e., their digits exchanged the position) gives out perfect square two digit numbers :*

(a) 1 *(b)* 4 *(c)* 6 *(d)* 10

SOLUTION These numbers are 61, 52, 63, 94, 46 and 18. Since these numbers on being reversed give out 16, 25, 36, 49, 64, 81 the two digit perfect square numbers.

Hence (c) is the correct answer.

EXAMPLE 4 *The number of two digit numbers which are prime :*

(a) 25 *(b)* 23

(c) 21 *(d) can't be determined*

SOLUTION Prime numbers between 1 and 100 are 25

Prime numbers of 1 digit are 4 (*viz*, 2, 3, 5, 7)

So, only two digit prime numbers = 25 – 4 = 21

Thus (c) is the correct option.

EXAMPLE 5 *The number of two digit prime numbers, with distinct digits, on being reversed they give again the prime number is :*

(a) 3 *(b)* 5 *(c)* 8 *(d)* 11

SOLUTION See, the total number of prime numbers in this case (when there are distinct digits *i.e.*, there are no same digits) the number of prime numbers must be an even number hence (c) is the correct option.

Alternatively : These numbers are 13, 31, 17, 71, 37, 73 and 79, 97.

Thus there are total 8 such numbers.

EXAMPLE 6 *The total number of two digit numbers whose unit digit is either, same, double, triple or quadruple of the tens digit :*

(a) 13 *(b)* 9 *(c)* 18 *(d)* 90

SOLUTION 11, 22, 33, 44, 55, 66, 77, 88, 99

12, 24, 36, 48

13, 26, 39

14, 28

Thus there are total 18 such numbers. Hence (c) is the correct answer.

EXAMPLE 7 *The number of two digit numbers which are perfect square and perfect cube both is :*

(a) 0 *(b)* 2

(c) can't be determined *(d) none of these*

SOLUTION Two digit perfect cubes are 27, 64

Two digit perfect squares are 16, 25, 36, 49, 64, 81

Thus there is only one value *i.e.*, 64 which is both perfect square and perfect cube.

Thus (d) is the most appropriate choice.

EXAMPLE 8 *Aprajita multiplies a number by 72 instead of 27, then her new answer will increase by, than the actual result :*

(a) $\frac{7}{2}$ *(b)* $\frac{8}{3}$

(c) $\frac{5}{3}$ *(d) none of these*

SOLUTION Let the number be k then the original product will be $k \times 27$ and the new product will be $k \times 72$

So, the difference in the product value = $72k - 27k = 45k$

Thus, new product will increase by $\frac{45k}{27k} = \frac{5}{3}$

Hence (c) is the correct answer.

Alternatively : $\frac{72-27}{27} = \frac{5}{3}$

EXAMPLE 9 *If the numerator and denominator of a fraction are exchanged then the product of these two fractions becomes equal to 1. The total number of such fractions is :*

(a) 1 *(b)* 31

(c) 13 *(d) infinitely many*

SOLUTION Let $\frac{x}{y}$ be the fraction where $x, y \neq 0$, then

$$\frac{x}{y} \times \frac{y}{x} = 1 \quad \text{(always)}$$

Now since x and y can assume infinite values so (d) is the best answer.

EXAMPLE 10 $\frac{3}{4}$*th of a number is 20 more than half of the same number. The required number is :*

(a) 50 *(b)* 180 *(c)* 90 *(d)* 80

SOLUTION Since the number is an integer so it must be divisible by 4. Hence option (a) and (c) are ruled out.

Now, if we check option (b) we find it is wrong so (d) is the correct.

As $80 \times \frac{3}{4} = 80 \times \frac{1}{2} + 20$

$60 = 60$

Alternatively : $x \times \frac{3}{4} = x \times \frac{1}{2} + 20 \Rightarrow x = 80$

EXAMPLE 11 *When 50 is added to the 50% of a number, then the number becomes itself. The required number is :*

(a) 375 *(b)* 100 *(c)* 150 *(d)* 500

SOLUTION

$$50 + \frac{x}{2} = x$$

$$\Rightarrow \quad \frac{x}{2} = 50 \Rightarrow x = 100$$

Hence (b) is the correct option.

EXAMPLE 12 *If we reverse the digits of a two digit number then the difference between the original number and new number is 27, the difference between the digits is :*

(a) 9 *(b)* 3

(c) can't be determined *(d) none of these*

SOLUTION Since $xy - yx = 9(x - y)$

Here $9(x - y) = 27$

Thus $x - y = 3$

So (b) is the correct answer.

EXAMPLE 13 *A two digit number is 4 times the sum of its digits and the unit digit is 3 more than the tens digit. The number is :*

(a) 52 *(b)* 61 *(c)* 63 *(d)* 36

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SOLUTION Go through option

$$36 = 4(3 + 6)$$

and $$6 - 3 = 3$$

Hence (d) is the correct answer.

Alternatively : Let x, y be the tens and unit digits respectively, then

$$(10x + y) = 4(x + y) \quad ...(1)$$

$$\Rightarrow \quad 6x = 3y$$

$$\Rightarrow \quad \frac{x}{y} = \frac{1}{2} \quad ...(2)$$

and $$y - x = 3 \quad ...(3)$$

from equation (2) and (3)

$$x = 3 \quad \text{and} \quad y = 6$$

Thus the required number is 36.

EXAMPLE 14 *When a two digit number is subtracted from the other two digit number which consists of the same digits but in reverse order, then the difference comes out to be a two digit perfect square. The number is :*

(a) 59 *(b)* 73
(c) 36 *(d) not unique*

SOLUTION Since the difference between number is a perfect square. So this difference can be only 36, because 36 is the only two digit perfect square contains 9 as a factor.

But there are total 5 numbers possible *viz.*, 15, 26, 37, 48, 59. Since the only condition is that

$$(10x + y) - (10y + x) = 9(x - y) = 36$$

$$\Rightarrow \quad (x - y) = 4$$

Thus (d) is the most appropriate answer.

EXAMPLE 15 *There is a three digit number such that the sum of its end digits (unit digit and hundredth place digit) is always a single digit number. Another three digit number is obtained by reversing the position of end digits of the original number. Then what can be the possible sum of the tens digits of both these numbers?*

(a) 5 *(b)* 12 *(c)* 15 *(d)* 26

SOLUTION Since in both the numbers the middle digits *i.e.*, the tens digits are same it means the sum of these tens digits is always an even number ($\because x + x = 2x$)

So the option (a) and (c) are ruled out.

Now, the largest possible digit is 9 so the maximum possible sum of these two digits can be 18 (= 9 + 9). So the option (d) is also ruled out.

Thus the possible answer is (b).

EXAMPLE 16 *A three digit number which on being subtracted from another three digit number consisting of the same digits in reverse order gives 594. The minimum possible sum of all the three digits of this number is :*

(a) 8 *(b)* 12
(c) 6 *(d) can't be determined*

SOLUTION Let x, y, z be the hundred, tens and unit digits of the original number then

$$(100z + 10y + x) - (100x + 10y + z) = 594$$

$$\Rightarrow \quad 99(z - x) = 594 \quad \Rightarrow \quad (z - x) = 6$$

So the possible values of (x, z) are (1, 7), (2, 8) and (3, 9). Again the tens digit can have the values *viz.*, 0, 1, 2, 3, ... 9.

So the minimum possible value of $x + y + z = 1 + 0 + 7 = 8$.

Hence (a) is the correct option.

NOTE x and z can never be zero since if the left most digit becomes zero, then it means this number is only two digit number.

DIGITAL NUMBER SYSTEMS

There are various number systems used in digital representation. The most common number systems are binary octal, decimal, hexadecimal systems.

Base : The number that determines the positional value of every digit in a number is called the base of that number. For example the base of binary system is 2, octal is 8, decimal is 10 and hexadecimal is 16. It is also known as **"Radix."**

Also the base tells us that the number of distinct (or unique) symbols used in that system. For example we use only 2 distinct symbols *viz.*, 0 and 1 in binary system. Also in decimal system there are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 unique symbols.

In the binary system (base is 2) there are only two symbols or possible digits *viz.*, 0 and 1. Even it can represent a very-very large number that can be represented in any other number system.

CONVERSION OF DECIMAL NUMBERS INTO ANOTHER

To convert any decimal number in any base 'n'. We divide the number successively by 'n' *i.e.*, first of all divide the given number, then divide the quotients obtained one by one succevily, till the last quotient becomes zero, then write down all the remainders in proper order. As the last remainder is MSD (most significant digit) and the first remainder is least significant digit. It means the first remainder will be the right most digit and the last remainder will be left most digit.

EXAMPLE 1 *Convert* $(17)_{10}$ *to binary system (or in base 2)*

SOLUTION

2 | 17		
2 | 8	→ 1	→ Remainders
2 | 4	→ 0	
2 | 2	→ 0	
2 | 1	→ 0	
0	→ 1	

So $(17)_{10} = (10001)_2$

EXAMPLE 2 *Convert* $(35)_{10}$ *into base 3.*

SOLUTION

3 | 35		
3 | 11	→ 2	→ Remainders
3 | 3	→ 2	
3 | 1	→ 0	
0	→ 1	

So $(35)_{10} = (1022)_3$

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EXAMPLE 3 *Convert* $(169)_{10}$ *in base 7.*

SOLUTION

7	169	
7	24	→ 1
7	3	→ 3
	0	→ 3

→ Remainders

$\therefore \quad (169)_{10} = (331)_7$

EXAMPLE 4 *Convert* $(8976)_{10}$ *in base 9.*

SOLUTION

9	8976	
9	997	→ 3
9	110	→ 7
9	12	→ 2
9	1	→ 3
	0	→ 1

→ Remainders

So $(8976)_{10} = (13273)_9$

EXAMPLE 5 *Convert* $(128)_{10}$ *into binary system.*

SOLUTION

2	128	
2	64	→ 0
2	32	→ 0
2	16	→ 0
2	8	→ 0
2	4	→ 0
2	2	→ 0
2	1	→ 0
	0	→ 1

Thus $(128)_{10} = (10000000)_2$

EXAMPLE 6 *Convert* $(127)_{10}$ *into binary system.*

SOLUTION

2	127	
2	63	→ 1
2	31	→ 1
2	15	→ 1
2	7	→ 1
2	3	→ 1
2	1	→ 1
	0	→ 1

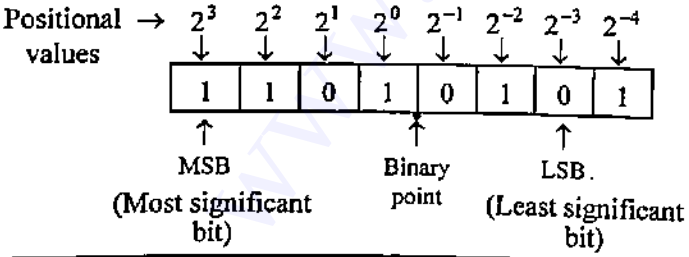
Thus $(127)_{10} = (1111111)_2$

REPRESENTATION OF BINARY NUMBER SYSTEM

The binary system is also a positional value system, wherein each binary digit has its own value or weight expressed as a power of 2.

NOTE We have studied the representation of the Decimal number system in the previous articles.

The number 1101.0101 is shown represented in the figure



Here, places to the left of the binary point (*i.e.*, counter part of the decimal point) are positive powers of 2 and places to the right are negative powers of 2.

CONVERSION OF BINARY INTO DECIMAL.

To find the decimal equivalent of binary number, we take the sum of the products of each digit value (0 or 1) with its positional value.

EXAMPLE 1 *Convert* $(100101)_2$ *into decimal representation.*

SOLUTION $1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

$= 32 + 0 + 0 + 4 + 0 + 1 = 37$

EXAMPLE 2 *Convert* $(1101.0101)_2$ *in base 10 (i.e., decimal system)*

SOLUTION $(1101.0101) = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} + 1 \times 2^{-4}$

$= 8 + 4 + 0 + 1 + 0 + 0.25 + 0 + 0.0625 = (13.3125)_{10}$

EXAMPLE 3 *The decimal number corresponding to the binary number* $(111000.0101)_2$ *is :*

(a) $(5.6312)_{10}$ (b) $(56.3125)_{10}$

(c) $(563.125)_{10}$ (d) $(5631.2)_{10}$

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SOLUTION
$$1\times2^5+1\times2^4+1\times2^3+0\times2^2+0\times2^1+0\times2^0+0\times2^{-1}+1\times2^{-2}+0\times2^{-3}+1\times2^{-4}$$
$$=32+16+8+0+0+0+0+0.25+0+0.0625$$
$$=(56.3125)_{10}$$

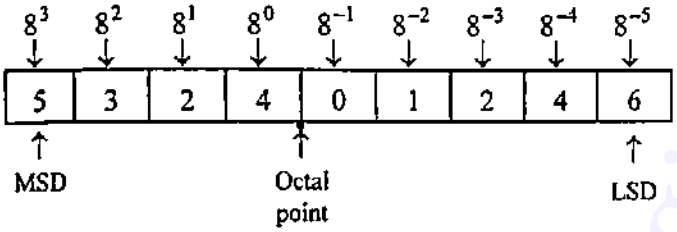
NOTE We use the subscripts to denote the base number so as to avoid confusion. For example $(10010)_2$ means the base of the number is 2 and $(201021)_3$ means the base of the number is 3 etc.
In the binary system the term "binary digit" is often abbriviated as "bit". The leftmost bit carries the largest weight and hence is called the most significant bit (MSB). The rightmost bit carries the smallest weight and hence called least significant bit (LSB). So the sequence of binary numbers goes as
00, 01, 10, 11, 100, 101, 110, 111, 1000, 1001, ... etc.
See the peculiarity in the pattern of binary numbers.

00
01
10
11
100
101
110
111
1000
1001
1010
1011
1100
1101
1110
1111
10000
10001

Now, you can see that the unit digit goes as 0, 1, 0, 1, 0, 1... and tens digit goes as 0, 0, 1, 1, 0,0, 1, 1, 0, 0, 1, 1 and Hundreds digit goes as 0000, 1111, 0000, 1111, 0000, 1111, ... etc. and thousands digit goes as 00000000, 11111111, 00000000, 11111111, ... etc.

REPRESENTATION OF OCTAL NUMBERS

This system has base of 8. So there are 8 unique symbols *viz.*,0, 1, 2, 3, 4, 5, 6, 7 are used for representation. For example consider a number 5324.01246. Then



The decimal equivalent of $(5324.012)_8$ is given below :
$$(5324.012)_8=5\times8^3+3\times8^2+2\times8^1+4\times8^0+0\times8^{-1}+1\times8^{-2}+2\times8^{-3}$$
$$=2560+192+16+4+0.015625+2\times0.001953125$$
$$=(2772.01953125)_{10}$$

The sequence of octal number goes as :
0, 1, 2, 3, 4, 5, 6, 7, 10, 11, 12, 13, 14, 15, 16, 17, 20, 21, 22, 23, 24, 25, 26, 27, 30, 31, ...

HEXADECIMAL NUMBER SYSTEM

It has base 16. So it uses 16 unique symbols *viz.*, 1, 2, 3, ...9 and *A*, *B*, *C*, *D*, *E*, and *F*.

Thus $(1)_{10}=(1)_{16}$, $(2)_{10}=(2)_{16}$,
$(3)_{10}=(3)_{16}$ .... $(9)_{10}=(9)_{16}$
and $(10)_{10}=(A)_{16}$
$(11)_{10}=(B)_{16}$, ... $(15)_{10}=(F)_{16}$.

CONVERSION OF DECIMAL FRACTIONS INTO BINARY

To convert a decimal fraction into binary, we multiply successively the decimal fraction by the radix i.e., base 2 (in case of binary) and write down all the integral values (i.e., the figures left of the decimal point) in order. As the first integral figure (0 or 1) will be just right to the binary point and the last integral will be the right most figure in the binary representation.

EXAMPLE 1 *Convert* $(0.875)_{10}$ *to binary.*

SOLUTION

	Integer part
$2\times0.875=1.750$	1
$2\times0.75=1.50$	1
$2\times0.50=1.00$	1

$\therefore\ (0.875)_{10}=(0.111)_2$

EXAMPLE 2 *Convert* $(0.3125)_{10}$ *to binary equivalent.*

SOLUTION

	Integer part
$2\times0.3125=0.625$	0
$2\times0.625=1.25$	1
$2\times0.25=0.50$	0
$2\times0.50=1.00$	1

Thus $(0.3125)_{10}=(0.1010)_2$

EXAMPLE 3 *Convert* 0.454545 *to binary equivalent.*

SOLUTION

	Integer part
$2\times0.454545=0.909090$	0
$2\times0.90909=1.81816$	1
$2\times0.81818=1.63638$	1
$2\times0.63636=1.27272$	1
$2\times0.27272=0.54544$	0
$2\times0.54544=1.09088$	1
$2\times0.09088=0.18176$	0
$2\times0.18176=0.36352$	0
$2\times0.36352=0.72704$	0
$2\times0.72704=1.45408$	1

Since, the sequence is continuous so can never be expressed as exactly in binary. Therefore writing down the integral parts.
$$(0.454545)_{10}=(0.1000101110)_2$$

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EXAMPLE 4 *Convert $(38.21)_{10}$ to its binary equivalent.*

SOLUTION First we convert the integer part *i.e.*,38.

2	38		
2	19	→ 0	Remainder
2	9	→ 1	
2	4	→ 1	
2	2	→ 0	
2	1	→ 0	
	0	→ 1	

$(38)_{10} = (100110)_2$

Now we convert the fractional part *i.e*, 0.21

	Integer part
$2 \times 0.21 = 0.42$	0
$2 \times 0.42 = 0.84$	0
$2 \times 0.84 = 1.68$	1
$2 \times 0.68 = 1.36$	1
$2 \times 0.36 = 0.72$	0
$2 \times 0.72 = 1.44$	1
$2 \times 0.44 = 0.88$	0
$2 \times 0.88 = 1.76$	1
$2 \times 0.76 = 1.52$	1
$2 \times 0.52 = 1.04$	1

Thus $(0.21)_{10} = (0.0011010111)_2$

$\therefore$ $(38.21)_{10} = (100110.0011010111)_2$

EXAMPLE 5 *Convert $(0.825)_{10}$ to octal.*

SOLUTION

	Integer part
$8 \times 0.825 = 6.600$	6
$8 \times 0.6 = 4.8$	4
$8 \times 0.8 = 6.4$	6
$8 \times 0.4 = 3.2$	3
$8 \times 0.2 = 1.6$	1
$8 \times 0.6 = 4.8$	4

Since its recurring number so it will not be stop.
Thus we write only 5 figures.
Hence, $(0.825)_{10} = (.64631)_8$

OCTAL TO BINARY CONVERSION

The following table helps in quick conversion.

Octal Digit	0	1	2	3	4	5	6	7
Binary Equivalent	000	001	010	011	100	101	110	111

EXAMPLE 6 *Convert $(273)_8$ to binary.*

SOLUTION

2	7	3
↓	↓	↓
010	111	011

Hence $(273)_8 = (010111011)_2$

EXAMPLE 7 *Convert $(16504)_8$ to binary.*

SOLUTION

1	6	5	0	4
↓	↓	↓	↓	↓
001	110	101	000	100

Hence $(16504)_8 = (001110101000100)_2$

EXAMPLE 8 *Convert $(37.42)_8$ to binary.*

SOLUTION

3	7	4	2
↓	↓	↓	↓
011	111	100	010

$\therefore$ $(37.42)_8 = (0111\ 111.100010)_2$

BINARY TO OCTAL CONVERSION

This is the inverse process of the previous conversion. In this process we make the triplets starting from the right (*i.e.*, LSB). If the number of digits in the binary representation are not in the group of 3 (*i.e.*, 3, 6, 9, 12, ... etc) then we associate either 1 or 2 zeros at the left of the MSB, then we write the octal equivalent of each triplet.

EXAMPLE 9 *Convert $(100011010110)_2$ to octal.*

SOLUTION

100	011	010	110	(see the table)
↓	↓	↓	↓	
4	3	2	6	

Hence $(100011010110)_2 = (4326)_8$

EXAMPLE 10 *Convert $(1001011001)_2$ to octal.*

SOLUTION

001	001	011	001	[two more zeros are added in the left to make the triplet]
↓	↓	↓	↓	
1	1	3	1	

Thus $(1001011001)_2 = (1131)_8$

EXAMPLE 11 *Convert $(10110.0101)_2$ to octal.*

SOLUTION

010	110	010	100
↓	↓	↓	↓
4	6	2	4

Thus $(10110.0101)_2 = (46.24)_8$

Note that, after binary point, the groups of 3 bits are made starting from left to right. That's why we added two zeros to make a group of three bits as the, last group had only 1.

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DECIMAL TO HEX CONVERSION

EXAMPLE 12 *Convert $(725)_{10}$ to hexadecimal.*

SOLUTION

16	725		$[\because (13)_{10} = (D)_{16}]$
16	45	→ 5	
16	2	→ D	
	0	→ 2	

$\therefore \quad (725)_{10} = (2D5)_{16}$

EXAMPLE 13 *Convert $(0.03125)_{10}$ to base 16.*

SOLUTION

	Integer part
$16 \times 0.03125 = 0.5$	0
$16 \times 0.5 = 8.0$	8

So $(0.03125)_{10} = (0.08)_{16}$

HEX TO DECIMAL CONVERSION

EXAMPLE 14 *Convert $(765)_{16}$ to decimal.*

SOLUTION $(765)_{16} = 7 \times 16^2 + 6 \times 16^1 + 5 \times 16^0$

$= 1792 + 96 + 5 = 1893$

Thus $(765)_{16} = (1893)_{10}$

EXAMPLE 15 *Convert $(3C8.08)_{16}$ to decimal.*

SOLUTION $(3C8.08)_{16} = 3 \times 16^2 + C \times 16^1 + 8 \times 16^0 + 0 \times 16^{-1} + 8 \times 16^{-2}$

$= 3 \times 16^2 + 12 \times 16^1 + 8 \times 16^0 + 0 \times 16^{-1} + 8 \times 16^{-2}$

$= 768 + 192 + 8 + 0 + 0.03125 = (968.03125)_{10}$

So $(3C8.08)_{16} = (968.03125)_{10}$

INTRODUCTORY EXERCISE-1.7

1. The value of x when $(25)_x = (85)_{10}$ is :
   (a) 2 (b) 8
   (c) 40 (d) can't be determined
2. Which one is an invalid number?
   (a) $(325)_7$ (b) $(345)_5$
   (c) $(543)_6$ (d) none of these
3. $(35)_n$, $(37)_n$, $(45)_n$ and $(51)_n$ all are written in base n then the value of n, such that all these four numbers when written in decimal system, they must be the consecutive prime numbers :
   (a) $n = 8$ (b) $n = 2$
   (c) $n = 12$ (d) $n = 6$
4. The last three digits in the binary representation of $(365247728)_{10}$ is :
   (a) 000 (b) 100
   (c) 110 (d) 010
5. The last three digits, if $(12345956)_{10}$ is expressed in binary system :
   (a) 210 (b) 100
   (c) 110 (d) 010
6. The last four digits in the binary representation of $(8009)_{10}$ is :
   (a) 1001 (b) 0001
   (c) 0101 (d) none of these
7. A number of decimal system when represented in binary system then its first and last digits are same and the rest digits are also of another kind. Further this number appears to be palindrome of numerals. The number is :
   (a) 57 (b) 19
   (c) 8 (d) 9
8. When a two digit number is reversed, the new number is reduced by 63 and the sum of digits of these individual numbers is 9. Now, if these same numbers are converted into base 'x' the larger number becomes 5 times that of smaller one the value of x is :
   (a) 36 (b) 13
   (c) 15 (d) 18
9. If $(25)_n \times (31)_n = (1015)_n$ then the value of $(13)_n \times (52)_n$ is when $n > 0$:
   (a) $(626)_4$ (b) $(462)_n$
   (c) $(716)_n$ (d) $(676)_n$
10. What is the decimal equivalent of the hexadecimal number $(\underbrace{100\ldots001}_{23\text{ zeros}})_{16}$?
    (a) $2^{23} + 1$ (b) $2^{24} + 1$
    (c) $2^{92} + 1$ (d) $2^{96} + 1$
11. If the decimal number 2^{111} is written in the octal system, then what is its unit place digit?
    (a) 0 (b) 1
    (c) 2 (d) 3
12. Match list I (Binary) with list II (Octal) and select the correct answer using the codes given below the lists :

List I (Binary)	List II (Octal)
I - 101110	(A) 135
II - 1101110	(B) 56
III - 1011101	(C) 176
IV - 1111110	(D) 156

Codes :
(a) I-A, II-C, III-B, IV-D (b) I-B, II-D, III-A, IV-C
(c) I-A, II-D, III-B, IV-C (d) I-B, II-C, III-A, IV-D

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PROGRESSIONS

This topic is directly related to our real life. For example whenever the things (or activities) occur in a definite pattern, we say there is a progression.

Suppose A has 2 books, B has 4 books, C has 6 books, D has 8 books etc., then we can say that E must has 10 books, if all these form a progression.

So if the same quantity is added in the next term or value, then it is said to be in Arithmetic Progression *e.g.*, $a, (a+d), (a+2d), (a+3d), (a+4d), \ldots$

Here you can see that each next term is 'd' more than its previous term.

Some more examples of **Arithmetic Progression** are as follows

1, 2, 3, 4, 5, 6, 7, ...
3, 5, 7, 9, 11, ...
7, 11, 15, 19, 23, ...
$-9, -4, 1, 6, 11, 16, \ldots$
$13, 9, 5, 1, -3, -7, -11, \ldots$

The first term of the series is denoted by 'a' and the common difference by 'd' and nth term by t_n and sum of the n terms of the series is denoted by 'S_n'.

Also the last term is denoted by 'l'

The common difference is the difference between any two consecutive terms of the series.

Let a_1 be the first term and a_2 be the second term and a_3 be the third term then $d = a_3 - a_2 = a_2 - a_1$ etc.

The nth term of a series is $T_n = a + (n-1)d$

and the sum of the n terms of a series is

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

and the arithmetic mean A.M. $= \left(\frac{a_1 + a_2 + a_3 + \ldots + a_n}{n}\right)$

and the last term $= l = a + (n-1)\,d.$

EXAMPLE 1 *Find the 10th term of the following series 7, 10, 13, 16, ... etc.*

SOLUTION $T_{10} = 7 + (10-1)\,3$ [Here $a = 7$, $d = 3$]

$T_{10} = 34$

EXAMPLE 2 *Find the sum of the 10 terms of the following series $-11, -8, -5, -2, \ldots$*

SOLUTION $S_n = \frac{n}{2}[2a + (n-1)\,d]$

$S_{10} = \frac{10}{2}[2 \times (-11) + (9)\,3]$ [Here $a = -11$, $d = 3$]

$= 5 \times (-22 + 27) = 25$

EXAMPLE 3 *Find the arithmetic mean of the following series $-17, -13, -9, -5, -1, \ldots.$*

SOLUTION A.M. $= \frac{(-17) + (-13) + (-9) + (-5) + (-1)}{5}$

$= -\frac{45}{5} = -9$

NOTE
1. The arithmetic mean (A.M.) of odd numbers of consecutive terms is the middle most term itself.
2. If a, b, c be three consecutive terms of an A.P. then the A.M. $= b = \frac{a+c}{2}$
3. The consecutive odd number of terms of an A.P. are written as $(a-2d)\,(a-d), a, (a+d), (a+2d), \ldots$ etc.
4. The consecutive even number of terms on an A.P. are written as $(a-5d)\,(a-3d)\,(a-d)\,(a+d)\,(a+3d)\,(a+5d)\ldots$ etc.

GEOMETRIC SERIES (OR PROGRESSION)

In the series each successive (or next) term is multiplied by a fix number.

For example: 1, 3, 9, 27, 81, ...
2, 8, 32, 128, 512, ...
$-3, 6, -12, 24, -48, \ldots$

If a is the first term, l is the last term, r is the common ratio, T_n is the nth and S_n is the sum of n term.

then, $l = ar^{(n-1)}$

$T_n = ar^{(n-1)}$

$S_n = \frac{a(r^n - 1)}{r - 1}$ if $r > 1$

and $S_n = \frac{a(1 - r^n)}{1 - r}$ if $r < 1$

also the geometric mean G.M. $= (a_1 \,.\, a_2 \,.\, a_3 \,.\, a_4 \ldots a_n)^{1/n}$

EXAMPLE 1 *Find the 6th term of 2, 4, 8, ...*

SOLUTION $T_6 = 2(2)^5 = 2^6 = 64$ [$\because T_n = a(r)^{n-1}$]

EXAMPLE 2 *Find the sum of 7 terms of the following series 1, 3, 9, 27, 81, ...*

SOLUTION $\because S_n = \frac{a(r^n - 1)}{r - 1}$

$\therefore \quad S_7 = \frac{1(3^7 - 1)}{3 - 1} = \frac{(2187 - 1)}{2} = 1093$

EXAMPLE 3 *Find the Geometric mean of 3, 9, 27.*

SOLUTION G.M. $= (a_1 \,.\, a_2 \,.\, a_3)^{1/3}$

$= (3 \,.\, 9 \,.\, 27)^{1/3} = (3^6)^{1/3} = 3^2 = 9$

NOTE
1. The G.M. of odd number of consecutive terms of a G.P. is the middlemost term itself.
2. If a, b, c be three consecutive terms of a G.P. then G.M. $= b = \sqrt{ac} = (a \,.\, c)^{1/2}$
3. The consecutive odd number of terms of a G.P. is written as ... $\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2 \ldots$ etc.
4. The consecutive even number of terms of a G.P. is written as ..., $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3, \ldots$ etc.

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SOME IMPORTANT FORMULAE

1. The sum of first n terms of natural numbers $= \frac{n(n+1)}{2}$
2. The sum of first n even natural numbers $= n(n+1)$
3. The sum of first n odd natural numbers $= n^2$
4. The sum of first n perfect squares (numbers) $= \frac{n(n+1)(2n+1)}{6}$
5. The sum of first n perfect cubes (numbers) $= \left[\frac{n(n+1)}{2}\right]^2$
6. The sum of the first n terms of the following series : 1, 3, 6, 10, 15, 21, ... is $\frac{n(n+1)(n+2)}{6}$.

FUNCTIONS/OPERATIONS/CODING-DECODING

In this article we will learn to operate the numbers with specially defined *i.e.*, user defined operations. Basically this is a kind of coding-decoding.

Directions : *Solve the following questions (1–3) on the basis of information given below :*

If $a \oplus b = a^3 + b^2 + 1$

$a \copyright b = (a-b)^2 + 1$

$a * b = a^3 - b^2 + 1$

EXAMPLE 1 *If* $P = (2 \oplus 3) \copyright (3 * 5)$, *then the value of P is :*

(a) 225 (b) 226

(c) 256 (d) none of these

SOLUTION $P = (2 \oplus 3) \copyright (3 * 5)$

$= (2^3 + 3^2 + 1) \copyright (3^3 - 5^2 + 1)$

$= 18 \copyright 3 = (18-3)^2 + 1$

$P = 226$

EXAMPLE 2 *The value of* $P = (((1 \oplus 2) * 2) \copyright 1)$ *is :*

(a) 40001 (b) 10804

(c) 40401 (d) none of these

SOLUTION $(((1 \oplus 2) * 2) \copyright 1) = ((6 * 2) \copyright 1)$

$= (213 \copyright 1)$

$= (212)^2 + 1 = 44945$

EXAMPLE 3 *The value of* $P = (((2 \oplus 2) \copyright 2) \copyright 2)$ *is :*

(a) 14441 (b) 14401

(c) 144001 (d) none of these

SOLUTION $P = (((2 \oplus 2) \copyright 2) \copyright 2)$

$= (((13) \copyright 2) \copyright 2)$

$= (122 \copyright 2) = 14401$

Directions for example number 4, 5, 6: *In the following division problem some digits are replaced by letters a, b, c.... etc. Every missing digit is uniquely determinable by a, b, c, ... etc.*

$$\begin{array}{r} ab3\overline{)c61de1}\,(4cfd \\ \underline{eg2} \\ 6gd \\ \underline{fa5} \\ 8be \\ \underline{d3j} \\ 8f1 \\ \underline{8f1} \\ \times \end{array}$$

EXAMPLE 4 *The value of J is :*

(a) 1 (b) 6

(c) 8 (d) none of these

EXAMPLE 5 *The value of f is :*

(a) 4 (b) 6

(c) 0 (d) can't be determined

EXAMPLE 6 *The value of a + g is :*

(a) 7 (b) 8

(c) 10 (d) none of these

SOLUTION (4–6) From the second step, $g = 9$, since $11 - 2 = 9$

again from the second step, $c = 5$, since $ab3 \times c = fa5$ (using unit digit property)

Now, from the last step $ab3 \times d = 8f1$

$\Rightarrow$ $d = 7$ (only)

so, the value of $a = 1$ (only)

therefore $f = 6$ (only) $\Rightarrow$ $b = 2$

Therefore, $e = 4$, from first step (since $a = 1$, $b = 2$)

Thus $a = 1$, $b = 2$, $c = 5$, $d = 7$, $e = 4$, $f = 6$, $g = 9$, $j = 8$

EXAMPLE 7 *If* $F = a.(|b|).c.(|d|).e.(|f|).g(|h|).i.(|j|)$ *and* $G = a.b.c.d.e.f.g.h.i.j$

Again, if $a = (-1)^1$, $b = (2)^{-2}$, $c = (-3)^3$, $d = (4)^{-4}$, ... $j = (10)^{-10}$ *then the correct relation is :*

(a) $F = -G$ (b) $F = G$

(c) $F \cdot G = 1$ (d) none of these

SOLUTION For F, the modulus of b, d, f, h, j is ineffective since for the given values of b, d, f, h, j the value of $(|b|) = b, |d| = d$.... etc.

Therefore $F = G$

EXAMPLE 8 *If* $2f(x) = f(2x)$, $3f(x) = f(3x)$, $4f(x) = f(4x)$.... *etc, then* $f(1) + f(2) + f(3) + \ldots + f(n)$ *equals to where* $f(1) = 1$ *:*

(a) $n(n+1)$ (b) 0

(c) $\frac{n(n+1)}{2}$ (d) $n!$

SOLUTION

$f(1) = 1$

$f(2) = 2$

$f(3) = 3$

$f(4) = 4$

... ...

$f(n) = n$

$\therefore$ $f(1) + f(2) + f(3) + \ldots f(n) = \frac{n(n+1)}{2}$.

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EXAMPLE 9 *If* $f\left(x-\frac{1}{x}\right)=x^2+\frac{1}{x^2}$, *then* $f(t)$ *equals to :*

(a) $t^2-\frac{1}{t^2}$ (b) $t^2+\frac{1}{t^2}$ (c) t^2+2 (d) $(t-1)^2$

SOLUTION Since $f\left(x-\frac{1}{x}\right)=x^2+\frac{1}{x^2}$

$$\Rightarrow \quad f\left(x-\frac{1}{x}\right)=\left(x-\frac{1}{x}\right)^2+2$$

Hence $f(t)=t^2+2$

✍ ***Directions for example number 10, 11 and 12 :***

$p * q = p^2 - q^2$

$p \$ q = p^2 + q^2$

$p @ q = pq + p + q$

$p \Delta q =$ Remainder of $\frac{p}{q}$

$p © q =$ greatest integer less than or equal to $\frac{p}{q}$

EXAMPLE 10 *If* $p=11$ *and* $q=7$, *then the value of* $(p * q) @ (p \$ q)$ *is :*

(a) 14641 (b) 12482
(c) 12243 (d) none of these

SOLUTION $(11 * 7) @ (11 \$ 7) = 72 @ 170$

$= 72 \times 170 + 72 + 170 = 12482$

EXAMPLE 11 *If* $p=8$ *and* $q=10$, *then the value of* $[(p \$ q) \Delta (p @ q)] * [(q * p) @ (q © p)]$ *is :*

(a) 5329 (b) 5239 (c) 12100 (d) −973

SOLUTION $[(8 \$ 10) \Delta (8 @ 10)] * [(10 * 8) @ (10 © 8)]$

$= (164 \Delta 98) * (36 @ 1)$

$= 66 * 73 = -973$

EXAMPLE 12 *If* $p=15$ *and* $q=25$, *then the value of the expression* $[(q © p) @ (p \$ q)] © [(q * p) \Delta (p \$ q)]$ *is :*

(a) 4 (b) 5 (c) 101 (d) 6

SOLUTION $[(25 © 15) @ (15 \$ 25)] © [(25 * 15) \Delta (15 \$ 25)]$

$= (1 @ 850) © (400 \Delta 850)$

$= (1701) © (400) = 4$

✍ ***Directions for question number 13 :*** *Follow the steps to solve the given problem :*

Step 1. *Put* $x=2$ *and* $y=3$

Step 2. *Multiply* x *by 2*

Step 3. *Multiply* y *by 3*

Step 4. *If* y = *greatest possible 3 digit perfect square, then stop, else follow step 5.*

Step 5. *Go to step 2.*

EXAMPLE 13 *What is the maximum possible value of* x?

(a) 32 (b) 64
(c) less than 50 (d) none of these

SOLUTION $x = 2, 4, 8, 16, 32, 64$

$y = 3, 9, 27, 81, 243, 729$

Hence, the required maximum value of $x = 64$.

EXAMPLE 14 $p @ q = p^2 - 20$ *if* $p > q$, $p @ q = p^2 + 20$ *if* $p \le q$, *then for* $a=5$ *and* $b=4$ *the value of* $[(a @ b) @ (a @ b)] - [(b @ a) @ (b @ a)]$ *is :*

(a) 416 (b) −461
(c) −1271 (d) none of these

SOLUTION $[(5 @ 4) @ (5 @ 4)] - [(4 @ 5) @ (4 @ 5)]$

$= [5 @ 5) - (36 @ 36)$

$= 45 - 1316 = -1271$

EXAMPLE 15 *If* $x \# y = \frac{x+y}{xy}$, *then the value of* $p \# (q \# r)$ *for every* $p, q, \in N$ *:*

(a) $\frac{pq+qr+r^2}{p(p+q+r)}$ (b) $\frac{pqr+q+r}{p(q+r)}$
(c) $\frac{p(qr+q+r)}{pqr}$ (d) none of these

SOLUTION $p \# (q \# r) = p \# \left(\frac{q+r}{qr}\right)$

$$= \frac{p+\left(\frac{q+r}{qr}\right)}{p\left(\frac{q+r}{qr}\right)} = \frac{\frac{pqr+q+r}{qr}}{\frac{p(q+r)}{qr}}$$

$$= \frac{pqr+q+r}{p(q+r)}$$

EXAMPLE 16 *If* $f(x, y) = \frac{(x+y)^2}{xy}$ *for every* $x, y > 0$ *then* $f(x, y)$ *must be :*

(a) ≥ 2 (b) < 5
(c) ≥ 4 (d) none of these

SOLUTION $f(x, y) = \frac{x^2+y^2+2xy}{xy} = \frac{x}{y} + \frac{y}{x} + 2$

The minimum value of $\frac{x}{y} + \frac{y}{x} = 2$ at $x = y$

Therefore $f(x, y) = \left(\frac{x+y}{xy}\right)^2 \ge 4$

EXAMPLE 17 *The function* f_1 *and* f_2 *are defined as follows :* $f_1 = a! + b!$ *and* $f_2 = a^a + b^b$ *for* $a, b > 1$ *and* $a, b \in N$ *then the correct relation is :*

(a) $f_1 < f_2$ (b) $f_1 = f_2$
(c) $f_1 > f_2$ (d) can't say

SOLUTION Check for any value of $(a, b) > 1$ and $a, b \in N$

since $a! = a.(a-1)(a-2)\ldots 1$

but $a^a = a.a.a.\ldots.a$ times

Therefore, $f_1 < f_2$

✍ ***Directions :*** *For any natural number* p *and* q

(i) $p \# q = p^3 + q^3 + 3$ *and* $p * q = p^2 + q^2 + 2$ *and* $p \$ q = |p - q|$

(ii) $Max\,(p, q)$ = *Maximum of* (p, q) *and* $Min\,(p$ *and* $q) =$ *Minimum of* (p, q)

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EXAMPLE 18 *The value of* $[(4 \# 5) \$ (14 * 15)]$ *is :*
(a) -196 *(b)* 231 *(c)* -225 *(d)* 229

SOLUTION $[(4 \# 5) \$ (14 * 15)]$
$= (64 + 125 + 3) \$ (196 + 225 + 2)$
$= (192 \$ 423) = 231$

EXAMPLE 19 *The value of* $[(1 \$ 2) \# (3 \$ 4)] * [(5 \$ 6) \# (7 \$ 8)]$ *is :*
(a) 25 *(b)* 52
(c) 36 *(d) none of these*

SOLUTION $[(1 \$ 2) \# (3 \$ 4)] * [(5 \$ 6) \# (7 \$ 8)]$
$= (1 \# 1) * (1 \# 1) = 5 * 5 = 52$

EXAMPLE 20 *The value of :*
$[\text{Max}\,(2, 4) \$ \text{Min}\,(6, 8)] \# [\text{Min}\,(10, 12) \$ \text{Max}\,(14, 16)]$
(a) 227 *(b)* 225
(c) 224 *(d) none of these*

SOLUTION $[\text{Max}\,(2, 4) \$ \text{Min}\,(6, 8)] \# [\text{Min}\,(10, 12) \$ \text{Max}\,(14, 16)]$
$= (4 \$ 6) \# (10 \$ 16)$
$= 2 \# 6 = 227$

ROMAN NUMBER SYSTEM

In this system there are basically seven symbols used to represent the whole number system. The symbols and their respective values are given below.

$I = 1,\quad V = 5,\quad X = 10,\quad L = 50,$
$C = 100,\quad D = 500 \text{ and } M = 1000$

In general, the symbols in the numeral system are read from left to right, starting with the symbol representing the largest value; the same symbol cannot occur contiguously more than three times; the value of the numeral is the sum of the values of the symbols.

For example LX VII = 50 + 10 + 5 + 1 + 1 = 67.

An exception to the left to the right reading occurs when a symbol is followed immediately by a symbol of greater value, then the smaller value is subtracted from the larger. For example,

CDXLVIII $= (500 - 100) + (50 - 10) + 5 + 1 + 1 + 1 = 448$.

IMPORTANT CONVERSION

1 trillion $= 10^{12} = 1000000000000$
1 billion $= 10^{9} = 1000000000$
1 million $= 10^{6} = 1000000$
1 crore $= 10^{7} = 100$ lakh
10 lakh $= 10^{6} = 1$ million
1 lakh $= 10^{5} = 100000 = 100$ thousand
1 thousand $= 10^{3} = 1000$

EXAMPLE 1 *The value of the numeral MCDLXIV is :*
(a) 1666 (b) 664 (c) 1464 (d) 656

EXAMPLE 2 *Which of the following can represent the numeral for 2005 ?*
(a) DDDDV (b) MMV
(c) MDDXV (d) MDVX

EXAMPLE 3 *The value of the numeral MDCCLXXXIX is :*
(a) 1789 (b) 1987 (c) 2989 (d) 2311

EXAMPLE 4 *Which of the following represents the numeral for 2949?*
(a) MMMDXL (b) MMXMIX
(c) MMCMIL (d) MMCMXLIX

SOLUTION

1. MCDLXIV $= 1000 + (500 - 100) + 50 + 10 + (5 - 1)$
$= 1464$
2. $2005 = 2000 + 5 = 1000 + 1000 + 5 = \text{MMV}$
3. MDCCLXXXIX $= 1000 + 500 + 100 + 100 + 50 + 10 + 10 + 10 + (10 - 1)$
4. $2949 = 2000 + 900 + 40 + 9$
$= (1000 + 1000) + (1000 - 100) + (50 - 10) + (10 - 1)$
$= \text{MMCMXLIX}$

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EXERCISE

MISCELLANEOUS

1. $\frac{-2}{3}, \frac{5}{2}, \frac{-1}{6}$ in descending order can be arranged as :

(a) $\frac{5}{2}, \frac{-1}{6}, \frac{-2}{3}$ (b) $\frac{-2}{3}, \frac{5}{2}, \frac{-1}{6}$

(c) $\frac{-1}{6}, \frac{5}{2}, \frac{-2}{3}$ (d) $\frac{5}{2}, \frac{-2}{3}, \frac{-1}{6}$

2. The sum of a number and its reciprocal is thrice the difference of the number and its reciprocal. The number is :

(a) $\pm\sqrt{2}$ (b) $\pm\frac{1}{\sqrt{2}}$

(c) $\pm\sqrt{3}$ (d) $\pm\frac{1}{\sqrt{3}}$

3. If x and y are positive real number, then :

(a) $x > y \Rightarrow -x > -y$ (b) $x > y \Rightarrow -x < -y$

(c) $x > y \Rightarrow \frac{1}{x} > \frac{1}{y}$ (d) $x > y \Rightarrow \frac{-1}{x} < \frac{-1}{y}$

4. If x be a rational number and y be an irrational number, then :

(a) both $x + y$ and xy are necessarily irrational

(b) both $x + y$ and xy are necessarily rational

(c) xy is necessarily irrational, but $x + y$ can be either rational or irrational

(d) $x + y$ necessarily irrational, but xy can be either rational or irrational.

5. The smallest number must be added to 1780 to make it a perfect square is :

(a) 69 (b) 156

(c) 49 (d) 59

6. If p be a prime, $p > 3$ and let x be the product of positive integers 1, 2, 3, ..., $(p-1)$, then consider the following statements :

1. x is a composite number divisible by p.
2. x is a composite number not divisible by p, but some prime greater than p may divide x.
3. x is not divisible by any prime $(p-2)$.
4. All primes less than $(p-1)$ divide x.

Of these statements :

(a) 1 and 2 are correct (b) 2 and 3 are correct

(c) 3 and 4 are correct (d) 4 alone is correct

7. Consider the following statements :

1. If $p > 2$ is a prime, then it can be written as $4n + 1$ or $4n + 3$ for suitable n.
2. If $p > 2$ is a prime, then $(p-1)(p+1)$ is always divisible by 4.

Of these statements :

(a) 1 and 2 are false (b) 1 and 2 are true

(c) 1 is true but 2 is false (d) 1 is false but 2 is true

8. 111111111111 is divisible by :

(a) 3 and 37 only (b) 3, 37 and 11 only

(c) 3, 11, 37 and 111 only (d) 3, 11, 37, 111 and 1001

9. An integer is divisible by 16 if and only if its last X digits are divisible by 16. The value of X would be :

(a) three (b) four

(c) five (d) six

10. The number of pairs lying between 40 and 100 such that HCF is 15, is :

(a) 3 (b) 4

(c) 5 (d) 6

11. If the product of the HCF and the LCM of 3 natural numbers p, q, r equals pqr, then p, q, r must be :

(a) such that $(p, q, r) = 1$

(b) prime number

(c) odd number

(d) such that $(p, q) = (p, r) = (q, r) = 1$

12. If x and y be integer such that $3x + 2y = 1$ then consider the following statements regarding x and y :

1. x and y can be found using euclidean algorithm.
2. x and y are positive.
3. x and y are uniquely determined.

Of these statements :

(a) 1 and 2 are correct (b) 1 alone is correct

(c) 1 and 3 are correct (d) 1, 2 and 3 are correct

13. In the equation $4^{x+2} = 2^{x+3} + 48$ the value of x will be :

(a) $-\frac{3}{2}$ (b) -2

(c) -3 (d) 1

14. A lad was asked his age by his friend. The lad said, "The number you get when you subtract 25 times my age from twice the square of my age will be thrice your age". If the friend's age is 14, then the age of the lad is :

(a) 21 (b) 28

(c) 14 (d) 25

15. Consider the following statements :

Assertion (A): $a^0 = 1, a \neq 0$

Reason (R): $a^m \div a^n = a^{m-n}$, m, n being integers.

Of these statements :

(a) both A and R are true and R is the correct explanation of A

(b) both A and R are true and R is not the correct explanation of A

(c) A is true, but R is false

(d) A is false, but R is true

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16. If the sum of two numbers added to the sum of their squares is 42 and the product of these numbers is 15, then the numbers are :
(a) 15, 1 (b) $\frac{15}{6}$, 6
(c) $2\frac{1}{2}$, 6 (d) 5, 3

17. If the equality $\frac{1}{x-1} = \frac{2}{x-2}$ is satisfied by x then the value of x will be :
(a) 2 (b) 1
(c) 1/2 (d) 0

18. If $x + \frac{1}{x} = 2$, then the value of $x^2 + \frac{1}{x^2}$ is :
(a) 6 (b) 4
(c) 2 (d) 0

19. If $a = 24$, $b = 26$, $c = 28$, then the value of $a^2 + b^2 + c^2 - ab - bc - ac$ will be :
(a) 0 (b) 4
(c) 8 (d) 12

20. A number n is said to be 'perfect' if the sum of all its divisors excluding n 'itself' is equal to n. An example of perfect number is :
(a) 9 (b) 15
(c) 21 (d) 6

21. The product $\frac{a}{b^2} \times \frac{b}{a^2}$ expressed as the sum of two identical terms, is :
(a) $\frac{1}{a+b} + \frac{1}{a+b}$ (b) $\frac{a}{b} + \frac{a}{b}$
(c) $\frac{b}{a} + \frac{b}{a}$ (d) $\frac{1}{2ab} + \frac{1}{2ab}$

22. If $x = 6$ and $y = 3$ then the value of $[x + y]^{x/y}$ is :
(a) 30 (b) 36
(c) 81 (d) 18

23. If the arithmetic mean of the numbers $x_1, x_2, x_3, \ldots, x_n$ is $\bar{x}$, then the arithmetic mean of the numbers $ax_1 + b, ax_2 + b, ax_3 + b, \ldots, ax_n + b$, where a and b are two constants, would be :
(a) $\bar{x}$ (b) $na\bar{x} + nb$
(c) $a\bar{x}$ (d) $a\bar{x} + b$

24. The arithmetic mean of first 50 odd natural numbers is :
(a) 50 (b) 625
(c) 175 (d) none of these

25. A boy draws n squares with sides 1, 2, 3, 4, 5, ... in inches. The average area covered by these n squares will be :
(a) $\left(\frac{n+1}{2}\right)$ (b) $\left(\frac{n+1}{2}\right)\left(\frac{2n+1}{3}\right)$
(c) $\left(\frac{n+1}{2}\right)\left(\frac{2n+1}{3}\right)^{-1}$ (d) $\left(\frac{n+1}{2}\right) - 1\left(\frac{2n+1}{3}\right)$

26. Which one of the following sets of surds is in correct sequence of ascending order of their values?
(a) $\sqrt[4]{10}, \sqrt[3]{6}, \sqrt{3}$ (b) $\sqrt{3}, \sqrt[4]{10}, \sqrt[3]{6}$
(c) $\sqrt{3}, \sqrt[3]{6}, \sqrt[4]{10}$ (d) $\sqrt[4]{10}, \sqrt{3}, \sqrt[3]{6}$

27. Which one of the following statements is not correct :
(a) every integer is a rational number
(b) every natural number is an integer
(c) every natural number is a real number
(d) every real number is a rational number

28. If $(a^2 + b^2)^3 = (a^3 + b^3)^2$ and $ab \neq 0$ then $\left(\frac{a}{b} + \frac{b}{a}\right)^6$ is equal to :
(a) $\frac{a^6 + b^6}{a^3 b^3}$ (b) $\frac{64}{729}$
(c) 1 (d) $\frac{a^6 + a^3b^3 + b^6}{a^2b^4 + a^4b^2}$

29. A page contains 60 lines. A chapter contains 125 pages. A book contains 5 chapters. 20 such books form a bound. If there are totally 30 lakhs lines in an x number of bounds then the value of x is :
(a) 4 (b) 2
(c) 5 (d) 6

30. If x is a natural number which is a perfect square, then the number $x + \sqrt{x}$ must end in :
(a) 0 or 5 (b) 0 or 1 or 9
(c) 0 or 2 or 6 (d) 0 or 4 or 8

31. If $2^p + 1$ is a prime number, then p must be power of :
(a) 2 (b) 3
(c) 5 (d) 6

32. The number of composite numbers between 101 and 120 is :
(a) 11 (b) 12
(c) 13 (d) 14

33. The number $10^N - 1$ is divisible by 11 for :
(a) even values of N
(b) all values of N
(c) odd values of N
(d) N must be a multiple of 11

34. A merchant has 140 litres, 260 litres, 320 litres of three kinds of oil. He wants to sell the oil by filling the three kinds separately in tins of equal volume. The volume of such a tin is :
(a) 20 litres (b) 13 litres
(c) 16 litres (d) 70 litres

35. If d is the HCF of a and b, then $d = \lambda a + \mu b$ where :
(a) λ and μ are uniquely determined
(b) λ and μ are both positive
(c) λ and μ are both negative
(d) one of the λ and μ is negative and the other is positive

36. The expression $\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{4}\right)\left(1 + \frac{1}{5}\right)\ldots\left(1 + \frac{1}{n}\right)$ simplifies to :
(a) $\frac{n+1}{3}$ (b) $\frac{n}{n+1}$
(c) $\frac{3}{n}$ (d) $1 + \frac{1}{3}\cdot\frac{1}{4}\cdot\frac{1}{5}\ldots\frac{1}{n}$

37. If $x = 2^{1/3} + 2^{-1/3}$ then the value of $2x^3 - 6x$ will be :
(a) 5 (b) −5
(c) 1 (d) 0

38. On the set of integers I, if a binary operation 'o' be defined as $aob = a - b$ for every $a, b \in I$, then :
(a) association law holds
(b) commutative law holds
(c) I is not closed under this operation
(d) I is closed under this operation

39. If $x - y = 1$, then the value of $x^3 - y^3 - 3xy$ will be :
(a) 1 (b) −1
(c) 3 (d) −3

40. Among the expression $(1 - 3P)$, $[1 - (3P)^2]$, $[1 - (3P)^3]$ and $[1 - (3P)^4]$ the number of factors of $(1 - 81P^4)$ is :
(a) one (b) two
(c) four (d) three

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41. If the sum of three consecutive integers is 21, then the sum of the two smaller integers is :
(a) 11 (b) 5
(c) 12 (d) 13

42. If $2s = a + b + c$, then the value of $(s-a)^2 + (s-b)^2 + (s-c)^2 + s^2 - a^2 - b^2 - c^2$ will be :
(a) -1 (b) 1
(c) 2 (d) 0

43. If $ab + bc + ca = 0$ then the value of $\frac{1}{a^2 - bc} + \frac{1}{b^2 - ca} + \frac{1}{c^2 - ab}$ will be :
(a) -1 (b) $a + b + c$
(c) abc (d) 0

44. If $4^{x+3} \times 2^{x-3} - 128 = 0$ then the value of x is :
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $\frac{3}{3}$ (d) $\frac{4}{3}$

45. If $a^m \cdot a^n = a^{mn}$, then $m(n-2) + n(m-2)$ is :
(a) 0 (b) 1
(c) -1 (d) $\frac{1}{2}$

46. $\frac{6^6 + 6^6 + 6^6 + 6^6 + 6^6 + 6^6}{3^6 + 3^6 + 3^6} \div \frac{4^6 + 4^6 + 4^6 + 4^6}{2^6 + 2^6} = 2^n$, then the value of n is :
(a) -1 (b) 0
(c) $\frac{1}{2}$ (d) 1

47. If $a^{1/m} = b^{1/n} = c^{1/p}$ and $abc = 1$ then $m + n + p$ is equal to :
(a) 0 (b) 2
(c) 1 (d) -2

48. Which one of the following among the numbers $-\frac{3}{7}, \frac{2}{3}$ and $\frac{-1}{3}$ is in descending order?
(a) $\frac{-1}{3}, \frac{-3}{7}, \frac{2}{3}$ (b) $\frac{2}{3}, \frac{-1}{3}, \frac{-3}{7}$
(c) $\frac{-3}{7}, \frac{2}{3}, \frac{-1}{3}$ (d) $\frac{2}{3}, \frac{-3}{7}, \frac{-1}{3}$

49. A rational equivalent to $\frac{-24}{20}$ with denominator 25 is :
(a) $\frac{-30}{25}$ (b) $\frac{30}{25}$
(c) $\frac{-29}{25}$ (d) $\frac{-19}{25}$

50. $\frac{3}{4} \times \left(\frac{-2}{3} + \frac{3}{5}\right)$ is equal to :
(a) $\frac{-3}{20}$ (b) $\frac{-19}{20}$
(c) $\frac{-1}{20}$ (d) $\frac{1}{20}$

51. In a problem involving division, the divisor is eight times the quotient and four times the remainder. If the remainder is 12, then the dividend is :
(a) 300 (b) 288
(c) 512 (d) 524

52. Which one of the following is not correct?
(a) $\sqrt{0.4096} = 0.64$ (b) $\sqrt{40.96} = 6.4$
(c) $\sqrt{0.04096} = 0.064$ (d) $\sqrt{4096} = 64$

53. Let $\frac{a}{b} = \frac{c}{d}$, (where a and b are odd prime numbers). If $c > a$ and $d > b$, then :
(a) c is not a multiple of a (b) d is not a multiple of b
(c) $c = ka$, $d = kb$ with $k > 1$ (d) $c = ka$, $d = lb$ with $k \neq l$

54. Which one of the following numbers belong to the given series 18, 26, 34, 42, 50?
(a) 438 (b) 338
(c) 232 (d) 132

55. The number of prime factors in the expression $6^4 \times 8^6 \times 10^8 \times 12^{10}$ is :
(a) 48 (b) 64
(c) 72 (d) 80

56. x is a five digit number. The digit in ten thousands place is 1. The number formed by its digits in units and tens places is divisible by 4. The sum of all the digits is divisible by 3. If 5 and 7 also divide x, then x will be :
(a) 14020 (b) 12060
(c) 10020 (d) 10080

57. The sum of all possible factors of 500 (including 1 and 500 themselves) equals :
(a) 784 (b) 980
(c) 1092 (d) 1350

58. HCF of 1007 and 1273 is :
(a) 1 (b) 17
(c) 23 (d) 19

59. If $2s = 9$ then the value of $s^2 + (s-1)^2 + (s-3)^2 + (s-5)^2$ is :
(a) 9 (b) 25
(c) 45 (d) 35

60. The sum of the digits of a two digit number is 8. The number obtained by reversing the digits is 18 less than the original number. The original number is :
(a) 62 (b) 17
(c) 53 (d) 35

61. If $x^y = y^x$ and $y = 2x$ then x is equal to :
(a) 2 (b) -2
(c) 1 (d) -1

62. Value of $[(x^l)^{l-\frac{1}{l}}]^{\frac{1}{l-1}}$ is :
(a) x (b) 1
(c) x^{-1} (d) x^{-l}

63. If $x^a \times x^b \times x^c = 1$, then $a^3 + b^3 + c^3$ is equal to :
(a) 9 (b) abc
(c) $a + b + c$ (d) $3abc$

64. Which one of the following is a rational number?
(a) $(\sqrt{2})^2$ (b) $2\sqrt{2}$
(c) $2 + \sqrt{2}$ (d) $\frac{\sqrt{2}}{2}$

65. Rational number $\frac{-18}{5}$ lies between consecutive integers :
(a) -2 and -3 (b) -3 and -4
(c) -4 and -5 (d) -5 and -6

66. $1 \div \frac{1}{1 \div \frac{1}{1 \div \frac{1}{3}}}$ is equal to :
(a) $\frac{1}{3}$ (b) 1
(c) 3 (d) $1\frac{1}{3}$

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67. The divisor when the quotient, dividend and the remainder are respectively 547, 171282 and 71 is equal to :
(a) 333 (b) 323
(c) 313 (d) 303

68. The value of $\sqrt{900} + \sqrt{0.09} - \sqrt{0.000009}$ is :
(a) 30.297 (b) 30.197
(c) 30.097 (d) 30.397

69. The greatest integer that divides 358, 376, 232 leaving the same remainder in each case is :
(a) 6 (b) 7
(c) 8 (d) 9

70. The units digit of every prime number (other than 2 and 5) must be necessarily :
(a) 1, 3 or 5 (b) 1, 3, 7 or 9
(c) 7 or 9 (d) 1 or 7

71. If the number 243 x 51 is divisible by 9 then the value of the digit marked as x would be :
(a) 3 (b) 1
(c) 2 (d) 4

72. The GCD of two whole numbers is 5 and their LCM is 60. If one of the number is 20, then other number would be :
(a) 25 (b) 13
(c) 16 (d) 15

73. The number of possible pairs of numbers, whose product is 5400 and HCF is 30 :
(a) 1 (b) 2
(c) 3 (d) 4

74. The least number which when divided by 2, 3, 4, 5 and 6 leaves the remainder 1 in each case. If the same number is divided by 7 it leaves no remainder. The number is :
(a) 231 (b) 301
(c) 371 (d) 441

75. Three bells, toll at interval of 36 sec, 40 sec and 48 sec respectively. They start ringing together at particular time. They next toll together after :
(a) 6 minutes (b) 12 minutes
(c) 18 minutes (d) 24 minutes

76. A has certain amount in his account. He gives half of this to his eldest son and one third of the remaining to his youngest son. The amount with him now is :
(a) 1/3 of the original (b) 2/3 of the original
(c) 3/4 of the original (d) 1/6 of the original

77. If $x + y + z = 0$, then $x^3 + y^3 + z^3$ is equal to :
(a) 0 (b) $3xyz$
(c) $\dfrac{xy + yz + zx}{xyz}$ (d) $xyz\,(xy + yz + zx)$

78. The remainder when $x^4 - y^4$ is divided by $x - y$ is :
(a) 0 (b) $x + y$
(c) $x^2 - y^2$ (d) $2y^4$

79. If $x - \dfrac{1}{x} = 2$, then the value of $x^4 + \dfrac{1}{x^4}$ is :
(a) 4 (b) 8
(c) 12 (d) 34

80. If the sum and the product of two numbers are 25 and 144 respectively then the difference of the numbers must be :
(a) 3 (b) 5
(c) 7 (d) 11

81. The sum of two numbers is 9 and the sum of their squares is 41. The numbers are :
(a) 4 & 5 (b) 1 & 8
(c) 3 & 6 (d) 2 & 7

82. If $3^n = 27$ then 3^{n-2} is :
(a) 3 (b) 1/3
(c) 1/9 (d) 9

83. If $a = b^x$, $b = c^y$, $c = a^z$, then xyz is :
(a) -1 (b) 0
(c) 1 (d) abc

84. If $p = x^{1/3} + x^{-1/3}$, then $p^3 - 3p$ is equal to :
(a) 3 (b) $\dfrac{1}{2}(x + x^{-1})$
(c) $x + x^{-1}$ (d) $2(x + x^{-3})$

85. The sum of squares of first ten natural numbers is :
(a) 375 (b) 385
(c) 475 (d) 485

86. If a language of natural numbers has binary vocabulary of 0 and 1, then which one of the following strings represents the natural number 7?
(a) 11 (b) 101
(c) 110 (d) 111

87. The least number which must be subtracted from 6708 to make it exactly divisible by 9 is :
(a) 1 (b) 2
(c) 3 (d) 4

88. Which one of the following statements is not correct?
(a) If 'a' is a rational number and b is irrational, then $a + b$ is irrational.
(b) The product of a non-zero rational number with an irrational number is always irrational.
(c) Addition of any two irrational numbers can be an integer.
(d) Division of any two integers is an integer.

89. The value of $\dfrac{10\sqrt{6.25}}{\sqrt{6.25} - 0.5}$ is :
(a) 125 (b) 0.125
(c) 1.25 (d) 12.5

90. The fundamental arithmetical operations on 2 recurring decimals can be performed directly without converting them to vulgar fractions :
(a) only in addition and subtraction
(b) only in addition and multiplication
(c) only in addition, subtraction and multiplication
(d) in all the four arithmetical operations

91. The greatest number that will divide 398, 436 and 542 leaving 7, 11 and 15 as remainders, respectively is :
(a) 16 (b) 17
(c) 18 (d) 19

92. Which one of the following statements is correct?
(a) If $x^6 + 1$ is divided by $x + 1$, then the remainder is -2
(b) If $x^6 + 1$ is divided by $x - 1$, then the remainder is 2
(c) If $x^6 + 1$ is divided by $x + 1$, then the remainder is 1
(d) If $x^6 + 1$ is divided by $x - 1$, then the remainder is -1

93. The value of
$$\frac{(1.5)^3 + (4.7)^3 + (3.8)^3 - 3 \times 1.5 \times 4.7 \times 3.8}{(1.5)^2 + (4.7)^2 + (3.8)^2 - (1.5 \times 4.7) - (4.7 \times 3.8) - (1.5 \times 3.8)}$$ is :
(a) 8 (b) 9
(c) 10 (d) 11

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94. The smallest number which must be added to 803642 in order to obtain a multiple of 9 is :
(a) 1 (b) 2
(c) 3 (d) 4

95. Simplified value of $\frac{8.73\times8.73\times8.73+4.27\times4.27\times4.27}{8.73\times8.73-8.73\times4.27+4.27\times4.27}$ is :
(a) 11 (b) 12
(c) 13 (d) 14

96. If $\left(x-\frac{1}{x}\right)=5$, then $x^3-\frac{1}{x^3}$ equals :
(a) 125 (b) 130
(c) 135 (d) 140

97. Mohan gets 3 marks for each correct sum and loses 2 marks for each wrong sum. He attempts 30 sums and obtains 40 marks. The number of sums solved correctly is :
(a) 10 (b) 15
(c) 20 (d) 25

98. If x satisfies $|x-1|+|x-2|+|x-3|\geq 6$, then :
(a) $0\leq x\leq 4$ (b) $x\leq 0$ or $x\geq 4$
(c) $x\leq -2$ or $x\geq 4$ (d) $x\geq -2$ or $x\leq 4$

99. It costs Rs. 10 a kilometer to fly and Rs. 2 a km to drive. If one travels 200 km covering x km of the distance by flying and the rest by driving, then the cost of the trip is :
(a) Rs. 2000 (b) Rs. 24000
(c) Rs. $(8x+400)$ (d) Rs. $(12x+400)$

100. If $2^{x+3}\cdot 4^{2x-5}=2^{3x+7}$, then the value of x is :
(a) 3 (b) 4
(c) 6 (d) 7

101. The value of $\left(\frac{x^a}{x^b}\right)^{(a^2+ab+b^2)}\left(\frac{x^b}{x^c}\right)^{(b^2+bc+c^2)}\left(\frac{x^c}{x^a}\right)^{(c^2+ca+a^2)}$ is :
(a) -1 (b) x^{abc}
(c) 1 (d) $x^{(a+b+c)}$

102. The value of $4\times100+3\times10+\frac{9}{1000}$ is :
(a) 430.09 (b) 430.0009
(c) 430.009 (d) 430.900

103. Which one of the following statements is correct?
(a) There can be a real number which is both rational and irrational
(b) The sum of two irrational number is always irrational
(c) For any real numbers x and y, $x<y\Rightarrow x^2<y^2$
(d) Every integer is a rational number

104. The least number which is a perfect square and has 540 as a factor is :
(a) 8100 (b) 6400
(c) 4900 (d) 3600

105. The set of natural numbers is closed under the binary operations of :
(a) addition, subtraction, multiplication and division
(b) addition, subtraction, multiplication but not division
(c) addition and multiplication but not subtraction and division
(d) addition and subtraction but not multiplication and division

106. If $\frac{100\sqrt{25}}{\sqrt{25}+x}=50$ then the value of x is :
(a) 25 (b) $\frac{1}{\sqrt{25}}$
(c) $\sqrt{25}$ (d) $\frac{1}{25}$

107. The total number of prime numbers between 120 and 140 is :
(a) 7 (b) 6
(c) 5 (d) 4

108. The number 12375 is divisible by :
(a) 3, 11 and 9 (b) 3 and 11 only
(c) 11 and 9 only (d) 3 and 9 only

109. $\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{5}\right)\ldots\left(1-\frac{1}{n}\right)$ equals :
(a) $\frac{1}{n}$ (b) $\frac{2}{n}$
(c) $\frac{3}{n}$ (d) $\frac{4}{n}$

110. If $\frac{x}{2}=\frac{y}{3}$, then $\left[\frac{4}{5}+\frac{y-x}{y+x}\right]$ equals :
(a) $\frac{3}{5}$ (b) $\frac{4}{5}$
(c) 1 (d) $\frac{6}{5}$

111. The value of $\frac{(119)^2+(119)(111)+(111)^2}{(119)^3-(111)^3}$ is :
(a) 8 (b) $\frac{1}{8}$
(c) 230 (d) $\frac{1}{230}$

112. A number consists of two digits. The sum of the digits is 11, reversing the digits, the number decreases by 45, the number is :
(a) 38 (b) 65
(c) 74 (d) 83

113. For a journey the cost of a child ticket is 1/3rd of the cost of an adult ticket. If the cost of the tickets for 4 adults and 5 children is Rs. 85, the cost of a child ticket is :
(a) Rs. 5 (b) Rs. 6
(c) Rs. 10 (d) Rs. 15

114. $9^{3/2}\div(243)^{-2/3}$ simplifies to :
(a) $3^{10/3}$ (b) $3^{19/3}$
(c) $3^{1/3}$ (d) 3^{19}

115. The solution of $(25)^{x-2}=(125)^{2x-4}$:
(a) 3/4 (b) 0
(c) 2 (d) -2

116. If $a^x=b^y=c^z$ and $\frac{b}{a}=\frac{c}{b}$ then $\frac{2z}{x+z}$ is equal to :
(a) $\frac{y}{x}$ (b) $\frac{x}{y}$
(c) $\frac{x}{2}$ (d) $\frac{z}{x}$

117. If $x<0<y$, then :
(a) $\frac{1}{x^2}<\frac{1}{xy}<\frac{1}{y^2}$ (b) $\frac{1}{x^2}>\frac{1}{xy}>\frac{1}{y^2}$
(c) $\frac{1}{x}<\frac{1}{y}$ (d) $\frac{1}{x}>1$

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118. The expression $33.33 \div 1.1$ simplifies as :
(a) 33.3 (b) 303
(c) 33.0 (d) 30.3

119. If $|x-2| < 3$ then :
(a) $1 < x < 5$ (b) $-1 < x < 2$
(c) $-1 < x < 5$ (d) $0 < x < 6$

120. A fraction becomes 4 when 1 is added to both the numerator and denominator and it becomes 7 when 1 is subtracted from both the numerator and denominator. The numerator of the given fraction is :
(a) 2 (b) 3
(c) 7 (d) 15

121. The expression $(a-b)^3 + (b-c)^3 + (c-a)^3 = 0$ if :
(a) $a < b < c$ (b) $a > b > c$
(c) $a = b = c$ (d) $a \neq b = c$

122. If $a+b+c=11$, $a^2+b^2+c^2=51$, what is the value of $ab+bc+ac$?
(a) 24 (b) 28
(c) 32 (d) 35

123. If $x+y+z=0$, then the value of $\frac{x^2}{yz}+\frac{y^2}{zx}+\frac{z^2}{xy}$ is :
(a) 3 (b) -3
(c) 0 (d) $3xyz$

124. The value of $\left(\frac{e^x+e^{-x}}{2}\right)^2 - \left(\frac{e^x-e^{-x}}{2}\right)^2$ is :
(a) 0 (b) 1
(c) 4 (d) 16

125. The continued product of $(1+x), (1+x^2), (1+x^4), (1+x^8)$ and $(1-x)$ is :
(a) $(1-x^8+x^{16})$ (b) (x^8+x^{16})
(c) $(1-x^{16})$ (d) $(x^{16}-1)$

126. Find the value of x and y in the given equation $5\frac{7}{x} \times y\frac{1}{13} = 12$:
(a) $-2, 9$ (b) 9, 2
(c) 4, 3 (d) 9, 4

127. The square root of $32\frac{137}{484}$ is :
(a) $5\frac{15}{22}$ (b) $15\frac{5}{32}$
(c) $5\frac{15}{28}$ (d) none of these

128. Simplify $\frac{\sqrt{25.4016}-\sqrt{1.0609}}{\sqrt{25.4016}+\sqrt{1.0609}}$:
(a) $\frac{401}{607}$ (b) $\frac{104}{706}$
(c) 41/76 (d) none of these

129. The square root of $\frac{\left(3\frac{1}{4}\right)^4 - \left(4\frac{1}{3}\right)^4}{\left(3\frac{1}{4}\right)^2 - \left(4\frac{1}{3}\right)^2}$:
(a) $5\frac{7}{12}$ (b) $5\frac{5}{12}$
(c) $5\frac{12}{13}$ (d) $5\frac{1}{6}$

130. Two numbers 34041 and 32506 when divided by certain number of three digits, leave the same remainder. The number is :
(a) 307 (b) 211
(c) 247 (d) 299

131. Mr. Black has three kinds of wine, of the first kind 403 litres, of the second 434 litres of the third 465 litres what is the least number of full carks of equal size in which this can be stored without mixing?
(a) 31 (b) 39
(c) 42 (d) 51

132. The largest sum of money which is contained in both Rs. 49.56 and Rs. 38.94 an exact number of time is :
(a) 2.7 (b) 3.54
(c) 4.28 (d) none of these

133. Tanya gives away to each of four girls $\frac{1}{12}, \frac{5}{18}, \frac{7}{30}, \frac{7}{48}$ of the apples in a basket and has only just enough apples to be able to do so without dividing an apple. The minimum number of apple had she?
(a) 250 (b) 720
(c) 750 (d) none of these

134. Abhishek, Bobby and Charlie start from the same point and travel in the same direction round an Island 6 km in circumference. Abhishek travels at the rate of 3, Bobby at the rate of $2\frac{1}{2}$ and Charlie at the rate of $1\frac{1}{4}$ km/hour. In how many hours will they come together again?
(a) 6 hrs (b) 12 hrs
(c) 24 hrs (d) 15 hrs

135. The value of $\frac{4\frac{1}{7}-2\frac{1}{4}}{3\frac{1}{2}+1\frac{1}{7}} \div \frac{1}{2+\frac{1}{2+\frac{1}{5-\frac{1}{5}}}}$:
(a) 7/29 (b) 5/6
(c) 1 (d) 9/13

136. Mallika travelled $\frac{5}{16}$ of his journey by coach and $\frac{7}{20}$ by rail, she walked the remaining 10 km, how far did she go altogether?
(a) $27\frac{17}{29}$ (b) $33\frac{7}{27}$
(c) $19\frac{7}{27}$ (d) $29\frac{17}{27}$

137. $2^{\sqrt{3}}$ is :
(a) a natural number (b) an integer
(c) a rational number (d) an irrational number

138. If $p > 0, q < 0$ then which of the following is correct?
(a) $p+q > 0$ (b) $p-q > 0$
(c) $p+q < 0$ (d) $p-q < 0$

139. Let n be a positive integer. If $\frac{1}{n}$ has a terminating decimal expansion, then which one of the following is true?
(a) n is of the form 5^x, where x is a positive integer
(b) n is of the form 2^y, where y is non-positive integer
(c) n is of the form $2^x \cdot 5^y$ for some non-negative x and y
(d) n is of the form 10^z for some positive integer z

140. If a number of two digit is 'k' times the sum of its digits then the number formed by interchanging the digits is the sum of the digits multiplied by :
(a) $(9+k)$ (b) $(10-k)$
(c) $(11-k)$ (d) $(k-1)$

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LEVEL 1

1. When the natural numbers 1, 2, 3, ..., 500 are written, then the digit 3 is used n times in this way. The value of n is :
(a) 100 (b) 200
(c) 300 (d) 280

2. In how many ways can 1146600 be written as the product of two factors?
(a) 100 (b) 108
(c) 216 (d) 273

3. How many natural numbers upto 990 are divisible by 5 & 9 both but not by 7?
(a) 18 (b) 19
(c) 22 (d) none of a, b, c

4. The number of solution of $|x| + |y| \le 0$, for $(x, y) \in R$, is :
(a) 0 (b) 1
(c) 2 (d) infinitely many

5. The remainder when 2^{39} is divided by 39 is :
(a) 0 (b) 2
(c) 8 (d) 1

6. The unit digit of the following expression
$(11)^{99} + (21)^{98} + (31)^{97} + (41)^{96} + \ldots (991)^1$ is :
(a) 1 (b) 3
(c) 7 (d) 6

7. The sum of all four digit numbers which are divisible by 7 is :
(a) 7071071 (b) 7^7
(c) 7107073 (d) 10019996

8. When the numerator of a positive fraction is increased by 2 and the denominator of the same fraction is multiplied by 2, the new fraction can be reduced to $\frac{1}{2}$ to its lowest term. The sum of the numerator and denominator of the original fraction can be :
(a) 13
(b) 45
(c) 16
(d) any even integer greater than 3

9. In the given expression $pq = p - q + 9$; q is a fraction and p is any positive integer. The value of p, which is inadmissible is :
(a) 5 (b) 4
(c) 8 (d) 7

10. The digits of a three digit number are in G.P. when the digits of this number are reversed and this resultant number is subtracted from the original number the difference comes out to be 792. The actual number is :
(a) 842 (b) 961
(c) 421 (d) 931

11. How many even integers n; $13 \le n \le 313$ are of the form of $3k + 4$, where k is any natural number?
(a) 101 (b) 51
(c) 50 (d) none of these

12. In the above question the number of values of n which are odd :
(a) 10 (b) 51
(c) 32 (d) none of a, b, c

13. If a and b are two odd distinct prime numbers and if $a > b$ then $a^2 - b^2$ can never be divided by :
(a) 13 (b) 11
(c) 17 (d) none of these

14. If $P = (101)^{100}$ and $Q = (100)^{101}$, then the correct relation is :
(a) $P > Q$ (b) $P < Q$
(c) $P = Q$ (d) $P = \frac{11}{10}Q$

15. If $k^2 - 25$ is an odd integer then which one of the following values gives the even number?
(a) $k^{(2k-1)}$ (b) $(3k - 3)(2k - 2)$
(c) $(k + 5)(k - 5)$ (d) any two of these

16. $(a + 1)(b - 1) = 625$; $(a \ne b) \in I^+$ then the value of $(a + b)$ is :
(a) $a + b \ge 25$ (b) $a + b \ge 50$
(c) $a + b = 24$ (d) $a + b = 26$

17. If $p + \frac{1}{p} = q$, then for $p > 0$:
(a) $q = 0$ (b) $-2 < q < 2$
(c) $q \ge 2$ (d) none of a, b, c

18. If $a^b = b^a$; $(a \ne b) > 1$, then the value of $(a + b)$ is :
(a) 5 (b) 6
(c) 7 (d) does not exist

19. If $m^n - n^m = (m + n)$; $(m, n) \in$ prime numbers, then what can be said about m and n :
(a) m, n are only even integers
(b) m, n are only odd integers
(c) m is even and n is odd
(d) none of these

20. There is unique 3 digit number which is cube of a natural number, if we shift the position of the digits of this number. The new number also becomes the cube of another number. The number is :
(a) 343 (b) 729
(c) 125 (d) does not exist

21. The given expression $n^4 - n^2$ is divisible by for $n \in I^+$ and $n > 2$:
(a) 4 (b) 8
(c) 12 (d) all of these

22. If a, b represents two distinct positive integers and thus $(aa)^b = abba$ is a valid relation. Then the value of $(a^b . b^a + b^a . a^b)$ is :
(a) 4 (b) 6
(c) 13 (d) none of these

23. At our training institute we have p-1, p-2, p-3 and p-4 processors in the ratio of $\frac{1}{6}, \frac{1}{5}, \frac{1}{3}$ and $\frac{1}{2}$ respectively. Minimum number of processors in our institute is :
(a) 16 (b) 30
(c) 32 (d) 36

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24. $(392)^n - (392)^{n-1}$ is not divisible by :
(a) 56 (b) 23
(c) 13 (d) 17

25. Mr. Chaalu while travelling by Ferry Queen has travelled the distance one kilometre more, than the fare he paid per km. Initially he had total amount of Rs. 350 in his wallet. Now he is only left with the minimum sum of (if all the distance travelled by him is in integers) :
(a) Rs. 26 (b) Rs. 8
(c) Rs. 19 (d) can't be determined

26. A person starts typing the numbers from 1 to 1999. He press the keys total 'n' number of times. The value of n is :
(a) 6889 (b) 1000
(c) 2888 (d) none of these

27. The remainder when $(20)^{23}$ is divided by 17 is :
(a) 11 (b) 3
(c) 6 (d) can't be determined

28. Let p be a prime number such that $3 < p < 50$, then $p^2 - 1$ is :
(a) always divisible by 8 (b) always divisible by 24
(c) always divisible by 12 (d) all of a, b, c

29. If p be a prime number, then $p^2 + 1$ can not have its unit digit is :
(a) 3 (b) 9
(c) 7 (d) all of these

30. The number of numbers from 1 to 200 which are divisible by neither 3 nor 7 is :
(a) 115 (b) 106
(c) 103 (d) less than 100

31. p is a prime number and $(p^2 + 3)$ is also a prime no. The no. of numbers that p can assume is :
(a) 3 (b) 2
(c) 1 (d) can't say

32. In a bangle shop, the shopkeeper displays the bangles in the form of a square then he is left with 38 bangles with him. If he wanted to increase the size of square by one unit each side of the square he found that 25 bangles fall short of in completing the square. The actual number of bangles which he had with him in the shop was :
(a) 1690 (b) 999
(c) 538 (d) can't be determined

33. If a and b are two integers which are multiples of 5 which of the following is not necessarily true :
(a) $a - b$ is divisible by 5
(b) $a + b$ is divisible by 10
(c) $a^2 - b^2$ is divisible by 5
(d) none of the above

34. What are the values of the digits a and b, respectively, in the number a 5523879 b, if it is divisible by both a and b is :
(a) (8, 6) (b) (7, 2)
(c) (8, 1) (d) not unique

35. A six digit number $abcabc$ such that $a, b, c \in N$, then which is the most correct statement is :
(a) It is divisible by 91
(b) It can be divided by 143
(c) It is divisible by 6
(d) Only a and b are correct

36. Two numbers a and b are such that one is odd and the other is even. Which statement is necessarily true?
(A) a^{2b} is even (b) $(a + b)$ is even
(C) $a^b \times b^a$ is even (d) $a^2 - b^2$ is even

37. A gardener plants his garden with 5550 trees and arranged them so that there is one plant more per row as there are rows then number of trees in a row is :
(a) 56 (b) 74
(c) 76 (d) 75

38. The value of 'a' when $3^a = 9^b$ and $4^{(a+b+2)b} = 16^{ab}$ is :
(a) 2 (b) 1
(c) 4 (d) none of these

39. The number of two digit prime numbers which remain prime even on inverting the position of its digits is :
(a) 4 (b) 5
(c) 9 (d) 10

40. Half way through the journey from Delhi to Lahore Atalji began to look out of the window of the Samjhauta Express and continued it until the distance which was remained to cover was half of what he has covered. Now at this time how much distance he has to cover?
(a) $\frac{2}{2}$ (b) $\frac{1}{4}$
(c) $\frac{1}{3}$ (d) $\frac{1}{6}$

41. Some times ago the height of my son was 90 cm at the end of 1996. When I have observed that his height is increasing equally every year. After seven years in the last month of 2003. I have found that his height was $\frac{1}{9}$th more than that at the end of 2002. Can you find the height of my son at the end of 2008?
(a) 360 cm (b) 450 cm
(c) 250 cm (d) 270 cm

42. Chris widener had a servant who was determined to be paid \$ 250, a wrist watch and a ration for whole year? But after 9 months Widener migrated to India and he has just paid him \$ 270 and the ration for the 9 months. What is the cost of the wrist watch?
(a) \$ 20 (b) \$ 120
(c) \$ 110 (d) data insufficient

43. The sum of the squares of a two digit number is 10. If we add 18 to this number we get another number consisting of the same digits written in reverse order. The original number is :
(a) 10 (b) 46
(c) 13 (d) none of (a), (b), (c)

44. A two digit number ab is added to another number ba, which is obtained by reversing the digits then we get a three digit number. Thus $a + b$ equals to :
(a) at least 18 (b) $2ab$
(c) $2(a + b)$ (d) $(a + b) \geq 10$

45. At Wharton School every student is awarded with the grades A, B or C only. 57.1428571428 ...% students obtained 'A' grade while 26.4444...% students obtained 'B' grade. If there are less than 3500 students then the maximum number of students obtained the grade 'C' while no one is declared fail.

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(a) 517 (b) 533
(c) 428 (d) can't be determined

46. Which one of the following is wrong?
(a) The sum of two even numbers, each raised to an odd power is even.
(b) The sum of two odd numbers, each raised to an odd power is even
(c) The remainder when dividing an even number by an odd number is even or zero.
(d) The remainder when dividing an odd number by an even number is always odd.

47. The sum of the expression $551 + 552 + 553 + \ldots + 560$ is :
(a) 3450 (b) 5555
(c) 555 (d) 6060

48. Find x, if $x^2 + x = x^3 - x$:
(a) 1 (b) 2
(c) 3 (d) 4

49. In the above problem the number of values of x is :
(a) 1 (b) 2
(c) 3 (d) 4

50. How many 3's you have to write down while writing the numbers from 3301 to 3401?
(a) 220 (b) 218
(c) 198 (d) none of these

51. $(2^{19} + 1)$ is divisible by :
(a) 3 (b) 4
(c) 6 (d) 3 & 6 both

52. Which of these is a prime number 3123, 219, 573, 467 :
(a) 219 (b) 467
(c) 573 (d) 3123

53. Which is not a prime number?
(a) 97 (b) 1001
(c) 127 (d) 101

54. What is the least number which must be multiplied to 5400 to get a perfect square?
(a) 2 (b) 3
(c) 6 (d) 10

55. For every p, q positive integers at $x = 0$ or $x = 1$, the valid relation can be :
(a) $p^x q^{(1-x)} = qx + p(1 - x)$
(b) $p^x q^{(1-x)} = px + q(1 - x)$
(c) $p^x q^{(1-x)} = p^{(1-x)} qx$
(d) either (b) or (c)

56. How many times does the sum of 3780 and 2835 contains their difference?
(a) 4 (b) 5
(c) 6 (d) 7

57. The expression $(x + y)^{-1} \cdot (x^{-1} + y^{-1})$ is equivalent to :
(a) 1 (b) $(xy)^{-1}$
(c) x^y (d) $xy^{-1} + x^{-1}y$

58. For any odd prime number p there exists a positive integer k where $1 < k < p$, such that the remainder of $\frac{k^2}{p}$ is 1. Then the number of positive integers k is :
(a) 0 (b) 1
(c) $p - 1$ (d) can't be determined

59. At Lucknow Public School $\frac{1}{9}$ students were absent in an exam and only $\frac{19}{24}$ of those who appeared for the exam passed it. Now we know that 500 students failed in the exam. Total number of students registered for the exam :
(a) 2000 (b) 2400
(c) 2700 (d) 3000

60. If, $0 < m < n < 1$. Then the expression $km < kn$ is true if :
(a) $k < 0$ (b) $k > 0$
(c) $k = 1$ only (d) all of a, b, c

61. If $a = 0.1818181818 \ldots$ and $b = 0.3030030003 \ldots$ then $(a + b)$ is :
(a) a rational no. (b) perfect number
(c) an irrational no. (d) both (b) and (c)

62. Half life of a substance is defined as the time period in which a substance becomes just half of it. If it is known that the half life of a substance "DECAY" is 1122 years, then after 4488 years, 80 gm of "DECAY" becomes.
(a) 4 gm (b) 20 gm
(c) 5 gm (d) none of these

63. In the examination of CBSE, a candidate must get $\frac{2}{5}$th marks to pass, out of total marks. Vinod appeared in the same exam and got 210 marks and still failed by 40 marks. The maximum marks which a candidate can get is :
(a) 500 (b) 625
(c) 390 (d) can't be determined

64. Sunny gets $3\frac{1}{2}$ times as many marks in 'QA' as in 'English'. If his total combined marks in both the papers is 90. His marks in 'QA' is :
(a) 50 (b) 60
(c) 70 (d) none of these

65. I know a two digit number, but when its digits swap their places and we get another two digit number But when these two digit numbers are added, it amounts to 99. Further if I just consider the difference between these numbers, it comes out to be 45. What is the number which I know?
(a) 27 (b) 38
(c) 72 (d) data insufficient

66. Which of the following is/are correct?
(i) $a^{x+y} = ax + ay$ (ii) $(a^x)^y = ya^x$
(iii) $a(x \cdot y) = ax \cdot ay$ (iv) $\frac{a^x}{a^y} = a^{x-y}$
(a) (i) only (b) (ii) & (iii) only
(c) (iv) only (d) none of these

67. In the MOCK CAT at 'i4 IIM' a student while doing the multiplication he reversed the digits (*i.e.*, tens to unit and unit to tens) of both the numbers respectively, still his answer was correct. Such a pair of numbers is :
(a) 16, 32 (b) 28, 42
(c) 31, 23 (d) 12, 63

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Fundamentals

68. The relation $\frac{p}{(p+1)} > 1$ is valid when :

(a) $p \geq -1$ (b) $p > -1$
(c) $p < -1$ (d) $-1 < p < 0$

69. A number when divided by 14 leaves a remainder of 8, but when the same number is divided by 7, it will leave the remainder :

(a) 3 (b) 2
(c) 1 (d) can't be determined

70. The unit digit of $(316)^{3^{4n}} + 1$ is :

(a) 4 (b) 5
(c) 1 (d) 7

71. The sum of two numbers is 18. The greatest product of these two numbers can be :

(a) 17 (b) 81
(c) 80 (d) can't be determined

72. Largest four digit number which when divided by 15 leaves a remainder of 12 and if the same number is divided by 8 it leaves the remainder 5. Such a greatest possible number is :

(a) 9963 (b) 9957
(c) 9945 (d) 9999

73. In a mobileshop $\frac{7}{12}$ mobiles are imported and rest are manufactured in India. Further $\frac{1}{5}$th Indian mobiles are coloured while $\frac{5}{7}$th imported mobile are black and white. If there are total 150 coloured mobile in his shop, then total number of mobile phones in his shop is :

(a) 500 (b) 600
(c) 800 (d) data insufficient

74. In a call centre at New Delhi, it is observed that it gets a call at a interval of every 10 minutes from California, at every 12 minutes from Texas, at the interval of 20 minutes from Washington DC and after every 25 minutes it gets the call from London. If in the early morning at 5 : 00 a.m. it has recieved the calls simultaneously from all the four destinations, then at which time it will receive the calls at a time from all the places on the same day?

(a) 10 : 00 a.m. (b) 3 : 00 a.m.
(c) 5 : 00 p.m. (d) both (a) and (b)

75. The H.C.F. and L.C.M. of 2^4, 8^2, 16^2, 20^3 are :

(a) 2^3; 32000 (b) 2^4; 32000
(c) 2^4; 25600 (d) 2^2; 3200

76. When we divide 15192327 by 99 the remainder will be :

(a) 98 (b) 84
(c) 30 (d) none of these

77. The number of numbers lying between 1 and 200 which are divisible by either of 2, 3 or 5 is :

(a) 146 (b) 145
(c) 158 (d) none of these

78. If xAy means $x + y$, xSy means $x - y$, xMy means $x \times y$ and xDy means $x \div y$. Then the value of $4D2S3M6A12$ is :

(a) -4 (b) 18
(c) $\frac{-47}{4}$ (d) none of these

79. The L.C.M. of two numbers is 1020 and their H.C.F. is 34 the possible pair of numbers is :

(a) 255, 34 (b) 102, 204
(c) 204, 170 (d) none of these

80. The sum of 100 terms of the series

$1 - 3 + 5 - 7 + 9 - 11 + 13 - 15 + \ldots$ is :

(a) 100 (b) 50
(c) 200 (d) none of these

81. The value of $\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{5}\right)\ldots\left(1-\frac{1}{n}\right)$ is :

(a) 1 (b) $\left(1-\frac{1}{n}\right)^n$
(c) $\frac{1}{n}$ (d) can't be determined

82. The minimum and maximum possible values of $\frac{x}{y}$, where $2 \leq x \leq 8$ and $16 \leq y \leq 32$ respectively are :

(a) $\frac{1}{8}, \frac{1}{4}$ (b) $\frac{1}{16}, \frac{1}{2}$
(c) 2, 16 (d) not unique

83. A rectangular floor in my office has its area equal to 56 m^2. The minimum number of tiles required, if all the tiles are in square shape is :

(a) 15 (b) 9
(c) 14 (d) can't be determined

84. A string of length 221 metre is cut into two parts such that one part is $\frac{9}{4}$th as long as the rest of the string, then the difference between the larger piece and the shorter piece is :

(a) 58 m (b) 53 m
(c) 85 m (d) none of these

85. Total number of prime numbers between 1 and 200 is :

(a) 34 (b) 46
(c) 56 (d) 71

86. What is the remainder of $\frac{6^{36}}{215}$?

(a) 0 (b) 1
(c) 2 (d) none of these

87. The remainder when $(12^{13} + 23^{13})$ is divided by 11 :

(a) 0 (b) 1
(c) 2 (d) none of these

88. The four digit smallest positive number which when divided by 4, 5, 6 or 7, it leaves always the remainder as 3 :

(a) 1000 (b) 1257
(c) 1263 (d) 1683

89. Which one of the following is correct?

(i) $13^{31} > 31^{13}$ (ii) $10^{100} < 100^{10}$
(iii) $2^{32} < 32^2$

(a) (i) and (ii) (b) (i) and (iii)
(c) (ii) and (iii) (d) (i) only

90. If $n_1, n_2, n_3 \ldots, n_k$ are such that, out of these k elements $\frac{k}{2}$ elements are even and rest are odd numbers. Which is necessarily even?

(a) $(n_1 + 2n_2 + 3n_3 + 4n_4 + \ldots + k.n_k)$

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(b) $n_1 \cdot n_2 \cdot n_3 \ldots n_{\left(\frac{k}{2}+1\right)}$
(c) $(n_1 + n_3 + n_5 + n_7 + \ldots + n_{k-1})$
(d) $n_2 + n_4 + n_6 + \ldots + n_k$

91. The H.C.F. of two numbers is 43 and their sum is 430. Total number of distinct pairs of two such numbers is :
(a) 5 (b) 2
(c) 6 (d) data insufficient

92. $2^{73} - 2^{72} - 2^{71}$ is same as :
(a) 2^{72} (b) 2^{71}
(c) 2^{70} (d) none of these

93. $N = 55^3 + 17^3 - 72^3$, then N is divisible by :
(a) 3 & 17 (b) 40 & 11
(c) 11 & 15 (d) all of these

94. $abcde$ is a five digit number when multiplied by 13 it gives a number, purely formed by the digit 9. Then the value of $a + b + c + d + e$ is :
(a) divisible by 8 (b) equal to 27
(c) divisible by 11 (d) all of these

95. The sum of 3 consecutive even numbers is always divisible by :
(a) 24 (b) 48
(c) 10 (d) none of these

96. The remainder obtained when $23^3 + 31^3$ is divided by 54 :
(a) 0 (b) 1
(c) 3 (d) can't be determined

97. The largest possible number by which the product of any five consecutive natural numbers can be divided :
(a) 120 (b) 160
(c) 100 (d) none of these

98. If $x^2 + y^2 = 25$ and $xy = 12$, then the value of $x^{-1} + y^{-1}$ is :
(a) $\frac{12}{5}$ (b) $\frac{7}{12}$
(c) $\frac{-7}{12}$ (d) both (b) and (c)

99. The remainder when $75^{75^{75}}$ is divided by 37 :
(a) 0 (b) 1
(c) 5 (d) 7

100. Let p be a prime number strictly greater than 3. Then $p^2 + 17$ will leave remainder k, when divided by 12, the value of k is :
(a) 1 (b) 5
(c) 6 (d) none of these

101. The unit digit of $(12345k)^{72}$ is 6. The value of k is :
(a) 6 (b) 8
(c) 2 (d) all of these

102. What is the least number which must be subtracted from 369 to make it a perfect cube?
(a) 8 (b) 26
(c) 2 (d) 25

103. The product of two consecutive even numbers is 624, then one of the numbers is :
(a) 13 (b) 25
(c) 26 (d) 28

104. What is the remainder when $1719 \times 1715 \times 1713$ is divided by 17?
(a) 1 (b) 16
(c) 3 (d) none of these

105. The quotient when 12.5 is divided by $\frac{1}{6}$ is :
(a) 75 (b) 25
(c) 2.48 (d) none of these

106. Kavita had to do a multiplication. Instead of taking 23 as a multiplicant she took 32 and got the result increased by 315. The original product was :
(a) 135 (b) 805
(c) 775 (d) data insufficient

107. A fraction in its lowest form is such that when it is squared and then its numerator is reduced by $\frac{1}{3}$rd and denominator is reduced to $\frac{1}{5}$th, it results as twice of the original fraction the sum of numerator and denominator can be :
(a) 7 (b) 8
(c) 9 (d) 17

108. The largest number which will divide the numbers 104, 221 and 377 leaving the same remainder in each case is :
(a) 56 (b) 13
(c) 39 (d) it does not exist

109. If $(ab^2)^{1/3} = 125$ where $a > b > 1$ and $(a, b^2) \in N$ then the correct relation is :
(a) $a = b$ (b) $a = b^4$
(c) $a^{3/2} = b^{2/3}$ (d) none of these

110. If a five digit number 'm21n2" is divisible by 24 then the maximum number of possible combinations of m and n :
(a) 4 (b) 9
(c) 16 (d) 10

111. If a fraction is divided by its reciprocal and then multiplied by itself then the fraction becomes $\frac{512}{125}$. The difference between the actual fraction and its reciprocal fraction is :
(a) $\frac{29}{30}$ (b) $\frac{39}{40}$
(c) data insufficient (d) none of these

112. A perfect square less than 500 is such that when a prime number p_1 is subtracted or another prime number p_2 is added to it in both cases it becomes the perfect square when $p_2 = 2 + p_1$. Then the $p_1 + p_2$ is necessarily divisible by :
(a) 8 (b) 11
(c) 4 (d) all of these

113. If $(11)^a = (19)^b = (209)^c$ then the correct relation is :
(a) $ab = (c)^{ab}$ (b) $a^b = c$
(c) $ab = c(a + b)$ (d) none of these

114. The remainder when $(2^{54} - 1)$ is divided by 9 is :
(a) 0 (b) 8
(c) 7 (d) none of these

115. A number $1 < N < 100$ is such that it is a perfect square and perfect cube both, then the sum of digits of N is :
(a) 1 (b) 10
(c) 20 (d) can't be determined

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116. The product of the digits of a three digit number which is a perfect square and perfect cube both is :
(a) 126 (b) 256
(c) 18 (d) none of these

117. A natural number N is such that $N = a^2 = b^4 = c^8$, where a, b, c are distinct positive integers, then the least possible value of N is :
(a) 729 (b) 1000
(c) 256 (d) none of these

118. The possible value of $|x| + |x-1| = 2$ is :
(a) $\left\{\frac{3}{2}, -\frac{1}{2}\right\}$ (b) $\left\{\frac{3}{2}, \frac{1}{2}\right\}$
(c) $\left\{\frac{2}{3}, -\frac{1}{2}\right\}$ (d) none of these

119. If $(pqr)^2 = (ijk\ pqr)$, where $i, j, k, p, q, r \in W$, and pqr and $ijkpqr$ are three digits and 6 digits numbers respectively. Then the value of $i \times j \times k \times p \times q \times r$ is :
(a) 0 (b) 720
(c) can't say (d) none of these

120. $3^{6n} + 4^{6n}$ is necessarily divisible by 25 when :
(a) n is an even integer (b) n is an odd integer
(c) only n is a prime (d) $n \geq 6; n \in I^+$

121. The number of zeros at the end of 100! is :
(a) $(100)^{100}$ (b) 100×10
(c) data insufficient (d) none of these

122. The greatest power of 3, when 41! is expressed in prime factors :
(a) 13 (b) 18
(c) can't be determined (d) none of these

123. A two digit number is such that it is the product of the two distinct perfect squares. The tens digit, unit digit and the sum of these two digits are in A.P. Further if we reverse the digits mutually, it increases by 27. The original number is :
(a) 72 (b) 19
(c) 36 (d) none of these

124. The value of the expression
$$7777 + 7777 \times 7777 \times (5 \div 77) \times (11 \div 35) :$$
(a) 1234321 (b) 12344321
(c) 7^{7777} (d) none of these

125. The sum and difference of a number with its reciprocal are $\frac{113}{56}$ and $\frac{15}{56}$ respectively, the number is :
(a) $\frac{11}{4}$ (b) $\frac{13}{6}$
(c) $\frac{14}{8}$ (d) $\frac{7}{8}$

126. If 4 is added to the numerator of a fraction, it becomes 1/3 and if 3 is added to the denominator of the same fraction it becomes 1/6 then the sum of the numerator and denominator is :
(a) 32 (b) 7
(c) 4 (d) 3

127. In an opera HOUSE, there are 7777 chairs to be placed, but the organiser of the event arranged all the chairs in such a way that there were as many columns as there were rows. So he had to remove minimum 'n' chairs from the total 7777 chairs. The minimum value of n is :
(a) 121 (b) 44
(c) 33 (d) 25

128. If $p^r \cdot p^{-1} \cdot p^s = (\sqrt{p^3})^2$ and $p^{3/2} \cdot p^r = p^s \cdot p^{-1/2}$ then the value of $(r+s)^{(r+s)}$ is :
(a) 27 (b) 64
(c) 4 (d) none of these

129. If $5^{-k} = \frac{1}{l}$, then the value of 5^{3k} is equal to :
(a) $\frac{1}{3l}$ (b) $3l$
(c) l^3 (d) data insufficient

130. $5^{x-1} + 5^x + 5^{x+1} = 775$ then the value of x for every positive integer x, is :
(a) 1 (b) 3
(c) 2 (d) can't be determined

131. If a, b are two perfect square digits and ab is a two digit perfect square number, such that $(a \times b) + (a + b) = ab$ then the value of $[ba - (b + a)]$ is :
(a) 27 (b) b^2
(c) 81 (d) either of b and c

132. The greatest possible number which can always divide the sum of the cubes of any three consecutive integer is :
(a) 3 (b) 4
(c) 9 (d) 11

133. The least possible divisor of 25930800 by which we divide this number, we get the quotient as a perfect square :
(a) 2 (b) 3
(c) 5 (d) can't be determined

134. Arun and Prabhat have some books with them. Once Prabhat said to Arun that if Arun gives 3 books to Prabhat then Arun will have only $\frac{1}{2}$ of the books that Prabhat will have with him. Then Arun asked frankly that if Prabhat gives him only two books (to Arun), then Prabhat will have as many books as Arun will have. The total number of books that Arun and Prabhat have with them is :
(a) 25 (b) 56
(c) 30 (d) can't be determined

135. The number of co-primes of 200 lying between 1 to 100 is :
(a) 100 (b) 40
(c) 50 (d) none of these

136. A soda water bottle is exactly filled with coke weighs 1600 gm but when it is $\frac{1}{3}$rd filled it weighs only 900 gm. The weight of the empty bottle is :
(a) 650 (b) 1100
(c) 550 (d) data insufficient

137. If ab, cd, ba and dc are two digit numbers then the maximum value of $(ab \times cd) - (ba \times dc)$ is, where a, b, c, d are distinct non-zero integers :
(a) 7938 (b) 7128
(c) 6930 (d) none of these

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138. In how many ways 12600 can be expressed as a product of two factors which are relatively prime is :
(a) 12 (b) 4
(c) 8 (d) 72

139. The solution set of the expression $\frac{1}{(1+p)^p} = 1$ is :
(a) $[-1, 2]$ (b) $(0, 1)$
(c) $\{-2, 0\}$ (d) $\{-1, 1\}$

140. The number of zeros at the end of $(2^{123} - 2^{122} - 2^{121}) \times (3^{234} - 3^{233} - 3^{232})$:
(a) 0 (b) 1
(c) 121 (d) none of these

LEVEL 2

1. In Mahabalipuram Temple there are some magical bells which tolls 18 times in a day, simulateneously. But every bell tolls at a different interval of time, but not in fraction of minutes. The maximum number of bells in the temple can be :
(a) 18 (b) 10
(c) 24 (d) 6

2. Three numbers p, q, r are such that $p^q = q^r$, where $p, q, r > 1$ then the correct relation between q and r is :
(a) $\frac{q}{r} = 1$ (b) $q < r$
(c) $q > r$ (d) indeterminable

3. $p^q - q^r = (p+q)^{r-q}$, where $1 < q < r < p < 10$ then the value of $p + q + r$ is :
(a) 31 (b) 21
(c) 15 (d) 12

4. If $a^2 + b^2 + c^2 + d^2 = 1$, then the maximum value of $a.b.c.d$ is :
(a) 1 (b) 2
(c) $\frac{1}{16}$ (d) data insufficient

5. The value of n in the expression $n^2 - 2(n!) + n = 0$ for every $n \in N$ is :
(a) 6 (b) 1
(c) 3 (d) both b, c are true

6. If m, n, p are in A.P. and $m^n = p^m = n! + p$; $m, n, p \in N$, then the value of $m.n.p$ is, where $2 < m, n < p < 10$:
(a) 136 (b) 72
(c) 162 (d) none of these

7. The value of $(n!)^n$ if $n + (n-1) + (n-2) = n(n-1)(n-2)$, where $n^3 > 9$, a positive number :
(a) 27 (b) 216
(c) 256 (d) 331776

8. If $(n)^3 - (n)^2 - n = n$, then the number of values of n that satisfy the given relation is :
(a) 1 (b) 2
(c) 3 (d) can't be determined

9. If $k = (k_1, k_2, k_3, \ldots, k_n) \in I$ and
$k_1k_2k_3 + k_2k_3k_4 + k_3k_4k_5 + \ldots + k_{(n-2)}\,k_{(n-1)}\,k_n = 0$
Then minimum how many entities *i.e.* k_i $(i = 1, 2, 3, \ldots)$ must be zero? If there are total 12 terms in the above expression :
(a) 3 (b) 4
(c) 6 (d) ${}^nC_3$

10. The given relation $n^{(n-1)} + n^{(n+1)} = (n^2+1)^2 - (n^2+1)$ is valid for every $n \in N$ if n equals to :
(a) 3 (b) 5
(c) 1 (d) both (a) and (c)

11. The smallest possible number that can be expressed as the sum of cube of two natural numbers in two different combinations.
(a) 1000 (b) 1728
(c) 1729 (d) none of these

12. $8^6 - 5^6$ is divisible by individually :
(a) 91 (b) 49
(c) 129 (d) all of these

13. Total number of factors of a number is 24 and the sum of its 3 prime factors out of four, is 25. The product of all 4 prime factors of this number is 1365. Then such a greatest possible number can be :
(a) 17745 (b) 28561
(c) 4095 (d) can't be determined

14. If p be any odd natural number, greater than 3 then which digit will never appear as the last digit in the product of $(p^2-1)(p^2+1)$ is :
(a) 9 and 7 (b) 5 and 3
(c) 1 and 5 (d) all of these

Directions for question number 15 to 18 : The relation $R(m, n)$ can be defined for every positive integer m, n as $R(m, n) = m \times (m+1) \times (m+2) \times (m+3) \times \ldots (m+n)$ and the relation $R(1, n)$ is equal to $n!$ or can be written as $R(n)$

15. The value of $\frac{R(135)}{R(100, 35)}$ is :
(a) 99! (b) 100!
(c) 270 (d) none of these

16. The value of $R(17) \cdot R(19, 62)$ is :
(a) $\frac{81!}{18}$ (b) $(81!) \times 18$
(c) 36! (d) $17 \times (19 + 62)$

17. The L.C.M. of $R(2, 995)$ and $R(996, 1)$ is :
(a) 1994 (b) 996!
(c) 997! (d) can't be determined

18. The H.C.F. of $R(139, 2)$ and $R(141)$:
(a) 141 (b) 2743860
(c) 32,16,839 (d) 19599

19. A six digit number of the form $abcabc$ is written where $a, b, c \in I^+$, then which statement is true about this number :
(a) it is always divisible by 7 and 11
(b) it is divisible by 143
(c) it is divisible by 1001
(d) all of (a), (b) and (c) are correct

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20. How many natural numbers upto 1155 are divisible by either 5 or 7 but not by 11?
(a) 105 (b) 330
(c) 333 (d) none of these

21. The number of solution set (x, y) for the given equation $4x + 7y = 3$, such that $-99 \le x \le 99$ and $-100 \le y \le 100$, where $x, y \in I$:
(a) 14 (b) 29
(c) 15 (d) 30

22. When any two natural numbers N_1 and N_2, such that $N_2 = N_1 + 2$, are multiplied with each other, then which digit appears least time as a unit digit if $N_2 \le 1000$?
(a) 0 (b) 9
(c) 4 (d) both (a) and (c)

23. In the above problem (no. 22), if all such unit digits will be added the maximum sum can be :
(a) 4491 (b) 4500
(c) 3609 (d) 5400

24. A diamond expert cuts a huge cubical diamond into 960 identical diamond pieces in minimum number of 'n' cuts. If he wants to maximize the number of identical diamond pieces making same number of n cuts to it. So the maximum number of such diamond pieces are :
(a) 1000 (b) 1331
(c) 1200 (d) none of (a), (b), (c)

25. The sum of all the factors of 45000 which are exactly the multiples of 10 is :
(a) 152295 (b) 141960
(c) 600 (d) none of these

26. The unit digit of the expression
$(1!)^{1!} + (2!)^{2!} + (3!)^{3!} + \ldots + (100!)^{100!}$:
(a) 0 (b) 1
(c) 2 (d) 7

27. When $(1!)^{1!} + (2!)^{2!} + (3!)^{3!} + \ldots + (100!)^{100!}$ is divided by 5, the remainder obtained is :
(a) 2 (b) 0
(c) 4 (d) none of these

28. The digit at the tens place in the sum of the expression :
$(1!) + (2!)^2 + (3!)^3 + (4!)^4 + (5!)^5 + \ldots (111!)^{111}$ is :
(a) 0 (b) 1
(c) 8 (d) 9

29. A number is divided strictly into two unequal parts such that the difference of the squares of the two parts equals 50 times the difference between the two parts. The number is :
(a) 100 (b) 250
(c) 50 (d) can't be determined

30. A positive number p is such that $(p + 4)$ is divisible by 7. 'N' being a smallest possible number larger than first prime number, which can make $(p + N^2)$ divisible by 7. The value of N is :
(a) 3 (b) 9
(c) 5 (d) 7

31. Anjali and Bhagwat fired 45 shots each. Total 66 bullets hit the target and the remaining bullets missed it. How many times does the Anjali hit the target if it is known that the number of hits per one miss shown by the Anjali is twice that of Bhagwat?
(a) 30 (b) 36
(c) 40 (d) 35

32. The remainder when n is divided by 3 is 1 and the remainder when $(n + 1)$ is divided by 2 is 1. The remainder when $(n - 1)$ is divided by 6 is :
(a) 2 (b) 3
(c) 5 (d) none of (a), (b), (c)

33. The expression $2222^{7777} + 7777^{2222}$ is divisible by :
(a) 99 (b) 101
(c) 13 (d) any two of these

34. In a Business school we have found that every two student share the book of 'Organisational Behaviour' between them. Every 3 student shared among them the book of Financial Management and every four students used to share Operation Research's book from the library. But it is well known that there were only 91 books of the above mentioned topics. The number of students in the school was :
(a) 81 (b) 84
(c) 48 (d) can't be determined

35. After two successive rises, the salary of Eckhart Tolle, working in a BPO company became equal to $\frac{15}{8}$ times of his initial salary. By how much percent was the salary raised second time if the first time rise in salary was half of the rise in second time in percent?
(a) 82.5 % (b) 60 %
(c) 50 % (d) can't be determined

36. The number of digits in the product of $5^{72} \times 8^{27}$ is :
(a) 77 (b) 75
(c) 99 (d) none of (a), (b), (c)

37. A student of 5th standard started writing down the counting numbers as 1, 2, 3, 4, ... and then he added all those numbers and got the result 500. But when I checked the result I have found that he had missed a number. What is the missing number?
(a) 25 (b) 32
(c) 30 (d) 28

38. A stairway of 20 ft height is such that each step account for half a foot upward and one foot forward. What distance will an ant travel if it starts from ground level to reach the top of the stairway?
(a) 59 (b) 60
(c) 58 (d) none of these

39. Paltry and Sundry, the two bird hunters went to woods. Paltry fires 5 shots when Sundry fires 7 shots. But Paltry kills 2 out of 5 while Sundry kills 3 out of 7. When Sundry has missed 32 shots, then how many birds has Paltry killed?
(a) 25 (b) 24
(c) 16 (d) 12

40. A cigarette pack is $\frac{5}{6}$th full of its capacity, then 5 cigarettes were taken out and 2 another cigarettes were put inside the

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pack. Now it is $\frac{4}{5}$ full. How many cigarettes can this pack contain when it is full?

(a) 90 (b) 80
(c) 72 (d) can't be determined

41. Around a square table chairs are arranged in a sequence starting from one corner, numbered as 1, 2, 3, ... etc. The chair number 2 is opposite to chair number 14. So how many chairs are there in all?

(a) 10 (b) 20
(c) 14 (d) can't be determined

42. The value of y for which the expression $p = \frac{1}{(|y-1|-3)}$ becomes undefined :

(a) {2, 8} (b) {– 2, 4}
(c) {– 1, 3} (d) {1, 2}

43. What is the sum of the following series?

$$\frac{1}{1\times2}+\frac{1}{2\times3}+\frac{1}{3\times4}+\ldots+\frac{1}{100\times101}$$

(a) $\frac{100}{101}$ (b) $\frac{1}{101}$
(c) $\frac{101}{100}$ (d) 101!

44. If Rupert has 4 more coins than Laxmi, Laxmi has 1 more coin than Bill and Bill has 1 more coin than Hawkins. Finally Hawkins has 4 more coins than Ajim. Then minimum number of coins that must be transfered if all of them wish to have an equal number of coins :

(a) 5 (b) 6
(c) 7 (d) none of these

45. If $f(x) = \frac{7}{x^7}+\frac{5}{x^5}+\frac{3}{x^3}+1+3x^3+5x^5+7x^7$. Now, if the value of $f(2)$ is 1081.58 then the value of $f\left(\frac{1}{2}\right)$ is :

(a) 540.79 (b) $\frac{1}{1081.58}$
(c) 1081.58 (d) 367.42

46. The last digit of the expression

$4\times9^2\times4^3\times9^4\times4^5\times9^6\times\ldots\times4^{99}\times9^{100}$ is :

(a) 4 (b) 6
(c) 9 (d) 1

47. The last digit of the expression

$4+9^2+4^3+9^4+4^5+9^6+\ldots+4^{99}+9^{100}$ is :

(a) 0 (b) 3
(c) 5 (d) none of these

48. p, q, r are the decimal numbers (*e.g.*, 5.8) and $[x]$ means the greatest integer less than or equal to x and $A = [p+q+r]$ and $B = [p]+[q]+[r]$, then the maximum value of $A - B$ is :

(a) 0 (b) 2
(c) 2.99 (d) none of these

49. If p, q, r be integers such that $p^2 = q^2.r$ then :

(a) p is an even number (b) q is an even number
(c) r is an even number (d) r is a perfect square

50. The sum of n positive integers $k_1, k_2, k_3, \ldots, k_n$ is an even number, then number of odd integers involve in the expression is :

(a) odd (b) even
(c) $(n-1)$ (d) none of these

51. If $1+2+3+\ldots+k = N^2$ and N is less than 100 then the value of k can be, where $N \in$ Natural Numbers :

(a) 8 (b) 1 and 49
(c) 8 and 36 (d) both (a) and (b)

52. If a and b be two co-prime numbers, then $(a+b)$ and $(a-b)$:

(a) are always co-primes
(b) have atleast one common factor other than 1
(c) if $(a+b)$ or $(a-b)$ is not a prime number, then their HCF is 2
(d) none of the above

53. The G.M. of two positive numbers is 35 and the A.M. of the same number is $43\frac{3}{4}$, then the greater of these numbers is :

(a) 28 (b) 30
(c) 70 (d) 35

54. When a number is divided by 1, 2, 3, 4, 5, ..., $(n-1)$, n individually it leaves 0, 1, 2, 3, 4, ..., $(n-2)$, $(n-1)$ respective remainders, then this number can be :

(i) $n!$ (ii) $(n!-1)$
(iii) [(L.C.M. of 1, 2, 3, ..., n) – 1]

(a) both (i) and (ii) (b) both (ii) and (iii)
(c) only (iii) (d) only (ii)

55. Four consecutive even numbers are such that 3 times of the first number is equal to twice of the third number. The sum of all the four numbers is :

(a) 20 (b) 40
(c) 44 (d) can't be determined

56. In the morning batch at "i4IIM" we have observed that when five-five students took seat on a bench, 4 students remained unseated. But when eleven students took seat per bench, 4 benches remained vaccant. The number of students in our morning batch were?

(a) 55 (b) 48
(c) 26 (d) none of these

57. When $\left(\frac{m}{n}\right) = 25\left(\frac{n}{m}\right)$, then the value of $m : n$ is :

(a) 1/25 (b) 5
(c) 1/5 (d) 2.5

58. Which one of the following is greatest one?

(a) 3^{3322} (b) 33^{322}
(c) 333^{22} (d) 22^{333}

59. The product of any two integers is 25, then the minimum possible sum is :

(a) 5 (b) 10
(c) 26 (d) none of these

60. If $4 \le p \le 5$ and $-10 \le q \le -9$, then the least value is given by the expression :

(a) $p.q$ (b) pq^6
(c) p^6q (d) $(pq)^5$

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61. The remainder obtained when $1! + 2! + 3! + \ldots + 77!$ is divided by 7 is :
(a) 0 (b) 5
(c) 4 (d) can't be determined

62. If $(a, n) \in I^+$ and $(a, n) > 1$, then the remainder when $[(a+1)^{2n-1} - 1]$ is divided by $(a-1)$ is :
(a) 1 (b) $a-1$
(c) n (d) none of these

63. A typist starts to type the serial numbers of candidates in a list, upto 500. Minimum how many times he has to press the keys of numerals only?
(a) 1389 (b) less than 1000
(c) 1392 (d) can't say

64. If a, b, c, d, e, f are sequentially the terms of an A.P. belong to set $\{1, 2, 3, \ldots, 9\}$ where all the terms $a, b, c, \ldots$are in increasing order, then the last digit of $a^b \times c^d \times e^f$ is :
(a) 5 (b) 2
(c) 7 (d) either of (a) and (b)

65. Total number of factors of a greatest possible number which when divides 1313 and 621, the respective remainders obtained are 17 and 9 :
(a) 9 (b) 10
(c) 11 (d) can't be determined

66. The set of values of x for which $|(x-5)x| > 0$ is :
(a) all real numbers (b) $R - \{0\}$
(c) $R - \{0, 5\}$ (d) $R - (0, 5)$

67. The sum of four prime numbers of two digits is 204. The sum of the first and last *i.e.* $(p_1 + p_4)$ is same as the sum of second number p_2 and third number p_3 (*i.e.* $p_2 + p_3$). Again $p_3 - p_2 = 2(p_2 - p_1) = 2(p_4 - p_3)$. The average number of all the four numbers is not a prime number, but a product of two prime numbers one of these prime numbers (out of four) is :
(a) 23 (b) 89
(c) 71 (d) can't be determined

68. Jai Bhan wanted to sell his mobilephone consists of the instrument and a SIM card, but Praveen who intended to buy it, asked the price of SIM card only? Jai Bhan told him that the price of the SIM card is Rs. 4000 less than the price of the instrument but if he wished to buy the complete set he had to pay Rs. 5000 only. The price of the instrument was :
(a) 5500 (b) 2500
(c) 4500 (d) can't be determined

69. If $|x + y| = |x - y|$ then the number of ordered pairs of (x, y) which satisfy the given condition is :
(a) 1 (b) 4
(c) infinite (d) none of these

70. If $292k + 7 = 23l$, where $(k, l) \in I$, then the value of l is :
(a) 23 (b) 31
(c) does not exist (d) none of these

71. If $ab + 4 = cd$ and $ba + 40 = dc$, where ab, cd, ba and dc are the two digit prime numbers. Further b and d are the prime number digits and a, c are neither prime nor composite. The value of $\frac{(ab + ba)}{(cd + dc)}$ is :
(a) 1 (b) $\frac{1}{2}$
(c) 2 (d) can't be determined

72. A man sells chocolates which are in the boxes. Only either full box or half a box of chocolates can be purchased from him. A customer comes and buys half the number of boxes which the seller had plus half a box more. A second customer comes and purchases half the remaining number of boxes plus half a box. After this the seller is left with no chocolate boxes. How many chocolate boxes the seller had initially?
(a) 2 (b) 3
(c) 4 (d) 3.5

73. If $n^2 = 123454321$, then the value of n is :
(a) 1001 (b) not a natural number
(c) 111, 111 (d) none of these

74. $1^2 - 2^2 + 3^2 - 4^2 + \ldots - 198^2 + 199^2$:
(a) 19900 (b) 12321
(c) 19998 (d) none of these

75. The quotient when L.C.M. is divided by the H.C.F. of a G.P. with first term 'a' and common ratio 'r' is :
(a) r^{n-1} (b) r^n
(c) $a^{-1}r^{n-2}$ (d) $(r^n - 1)$

76. Once I met two persons of the same parents namely Ashmit and Amisha. Mean while Ashmit told me that he has twice the number of sisters as the number of brothers. Further Amisha told me that she has twice the number of brothers as the number of sisters. Actually it was very confusing for me, so do you know that how many brothers and sisters are in their family?
(a) 4 (b) 5
(c) 6 (d) can't be determined

77. If $p < q$, then $p @ q = p \# q$ else $p @ q = q \# p$ where $a \# b = \frac{a}{b}$. Then the value of $(4 @ 5) @ (6 @ 5)$ is :
(a) $\frac{24}{25}$ (b) $\frac{2}{3}$
(c) 3/4 (d) none of these

78. A six digit number is such that every alternate digit is a prime digit and the three leftmost digit forms a G.P. while last three digits (*i.e.* Hundreds, tens and unit) form an A.P. If it is expressed as $pqrstu$, where $p + q + r = u$, $q + r = t$, $p + r = s$, $\frac{r}{t} = \frac{2}{3}$ and $p \neq q \neq r \neq s \neq t \neq u$. Then the sum of all the digits must be :
(a) 25 (b) 16
(c) 21 (d) can't be determined

79. Total number of digits in the product of $(4)^{1111} \times (5)^{2222}$ is :
(a) 3333 (b) 2223
(c) 2222 (d) can't be determined

80. If $p = N + 5$ when N is the product of any three consecutive positive integer. Then :
(a) p is prime (b) p is odd
(c) p is divisible by 6 (d) either of (b), (c)

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81. If $\frac{p}{q}$ and $\frac{r}{s}$ are two rational numbers then the relation $\left|\frac{p}{q}\right| < \left|\frac{r}{s}\right|$ is :
(a) always true (b) always false
(c) never true (d) none of these

82. If $u^v + v^w + w^x = 0$ for every negative integer u, v, w, x the value of $u \times v \times w \times x$ is necessarily be :
(a) 0 (b) less than zero
(c) even (d) odd

83. The unit digit of $2^{3^4} \times 3^{4^5} \times 4^{5^6} \times 5^{6^7} \times 6^{7^8} \times 7^{8^9}$ is :
(a) 0 (b) 5
(c) can't be determined (d) none of these

84. If $(a-7)(b-10)(c-12) = 1000$, then the least possible value of $(a+b+c)$ equals :
(a) 59
(b) 29
(c) 14
(d) any integer less than 1000

85. If the number $23^{32} - 9$ is divided by 16, then the remainder is :
(a) 8 (b) 0
(c) 6 (d) none of these

86. If $(x-5)(y+6)(z-8) = 1331$, then the minimum value of $x+y+z$ is :
(a) 40 (b) 33
(c) 19 (d) not unique

87. If $x+y+z = 21$, then the maximum value of $(x-6)(y+7)(z-4)$ is :
(a) 343 (b) 216
(c) 125 (d) not unique

88. The remainder 'R' when $3^{37} + 4^{37}$ is divided by 7 is :
(a) 0 (b) 1
(c) $2 < R < 6$ (d) none of these

89. $7^{74} - 4^{74}$ is divisible by :
(a) 3 (b) 11
(c) 7 (d) both (a) and (b)

90. The factorial of a number n is exactly divisible by $(2^{11} \times 11^2)$ then the least possible value of n is :
(a) 22 (b) 25
(c) does not exist (d) none of these

91. The number of zeros at the end of the product of :
$2^3 \times 3^4 \times 4^5 \times 5^6 + 3^5 \times 5^7 \times 7^9 \times 8^{10} + 4^5 \times 5^6 \times 6^7 \times 7^8 - 10^2 \times 15^3 \times 20^4$ is :
(a) 5 (b) 6
(c) 28 (d) none of these

92. A nine digit number *abcdefghi* is such that *a* is divisible by 1, *ab* is divisible by 2, *abc* is divisible by 3 and *abcd* is divisible by 4 and so on where none of *a, b, c, d, ...* is same and every digit is a non-zero digit such a number is :
(a) 123456789
(b) 381654729
(c) 126453789
(d) 826435791

Directions for question numbers 93, 94 and 95 :
If Minimum (x, y, z) = Minimum of (xy, yz, zx)
Maximum (x, y, z) = Maximum of (x^y, y^z, z^x)
Labh (x, y, z) = Average of (x, y, z)
Hani (x, y, z) = Modulus of $(x - y - z)$ i.e., $|x - y - z|$

93. The value of Minimum (1, 2, 3) + Maximum (1, 2, 3) is :
(a) 10 (b) 8
(c) 12 (d) 4

94. Labh [Hani (1, 2, 3), Hani (2, 3, 4), Hani (3, 4, 5)] is equal to :
(a) 4 (b) 5
(c) 6 (d) can't be determined

95. Minimum [Hani (1, 2, 3), Hani (2, 3, 4), Hani (3, 4, 5)] + Maximum [Labh (1, 2, 3), Labh (2, 3, 4), Labh (3, 4, 5)] is :
(a) 101 (b) 111
(c) 60 (d) none of these

96. When $(55)_{10}$ is represented in base 25 then the expression is :
(a) $(25)_{25}$ (b) $(35)_{25}$
(c) $(55)_{25}$ (d) none of these

97. For $(P, Q) > 0$ the number 340 PQQ 0 is divisible by both 3 and 8 then the total possible values of (P, Q) is :
(a) 2 (b) 3
(c) 6 (d) none of these

98. A typist while typing the numbers from 600 to 799 mistakenly he typed 8 every time in place of 6. So the total number of times he has typed 8 is :
(a) 300 (b) 230
(c) 180 (d) none of these

99. If p and q are two distinct integers such that $p^2 - pq = 0$ then we can deduce that :
(a) $p = q$
(b) q is only negative integer
(c) $p = 0$
(d) either (a) or (c)

100. In a survey it was found that YTC sells the cigarettes of Rs. 15990 per day. If the cost of a pack is not less than Rs. 100 then what can be the price of each pack which it sells per day is :
(a) 150 (b) 420
(c) 78 (d) 205

101. If $5n + 12 < 50$; where $n \in N$, then the value of n which satisfies the given inequality is :
(a) $5 < n < 12$ (b) $3n^2 \leq 150$
(c) $4n + 5 \leq 33$ (d) $4n^2 + 3 > 200$

102. If A and B are divided by D, then the remainder obtained are 13 and 31 respectively where A, B, D are natural numbers. Further $A + B$ is divided by the same divisor D the remainder is 4, then the divisor D is :
(a) 14 (b) 23
(c) 40 (d) 18

103. The unit digit of N^{2p} and $(N)^{(2p+1)}$ are 6 and 4 respectively. The greatest two digit number that follows the afore-said condition is :
(a) 49 (b) 84
(c) 94 (d) 98

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104. If $f(x) = x^3 + 3x^2 - 4x$ then $f(x+1)$ is :
(a) $x^3 + 7x^2 - 5$ (b) $x^3 + 6x^2 + 5x$
(c) $(x+1)^3$ (d) none of these

105. If $a + b = n$, $n \in I$ and $ab = 60$ then n cannot be equal to :
(a) 33 (b) – 19
(c) 17 (d) 32

Direction for questions 106 to 108 : If a sequence is as given below :
1, 1, 2, 3, 5, 8, 13, 21, 34, ...

106. The unit digit of the 75th term of this sequence will be :
(a) 0 (b) 5
(c) 7 (d) none of these

107. In the above sequence the 55th term will be :
(a) an even number (b) an odd number
(c) either even or odd (d) can't be determined

108. The unit digit of the sum of the 88th and 89th term of the same sequence will be :
(a) 5 (b) 2
(c) 0 (d) none of these

109. The unit digit of the product of n consecutive numbers of the form $2k + 1$, where $k \in I^+$ and n must be at least 5.
(a) zero (b) any odd number
(c) 5 (d) both (b) and (c)

110. In the above question the sum of the n terms of the sequence is :
(a) $n^2 - 1$ (b) $n(2n - 1)$
(c) $[(n-1)!]\, n$ (d) none of these

111. If $n_1, n_2, n_3, \ldots, n_k \in I$ such that $-5 \le n_k \le 25$, then the least possible product of any 3 numbers can be :
(a) – 60 (b) 0
(c) – 3000 (d) none of these

112. If p, q, r are such that $q - p = r - q = 2$ for every $p, q, r \in I^+$, then the least possible value of $(p+q)(q+r)(r+p)$ is :
(a) 192 (b) 720
(c) 129 (d) data insufficient

113. The H.C.F. of 38024 and 16296 is :
(a) 1234 (b) 6431
(c) 5432 (d) none of these

114. For any positive real number R, $[R]$ is the greatest integer less than or equal to R, then the value of $\left[\frac{R}{2}\right] + \left[\frac{R+1}{2}\right]$ is :
(a) $[R]$ (b) $[2R]$
(c) $\left[R + \frac{1}{3}\right]$ (d) can't be determined

115. For the natural numbers n_1 and n_2 if $2n_1 + 3n_2 = n_1 \times n_2$, then the least possible value of the $2n_1 + 3n_2$ is :
(a) 6 (b) 12
(c) 24 (d) can't be determined

116. A company offers total 150 pens to its customers. As per the scheme one pen will be offered on the purchase of a "Quantitative Aptitude" book. Out of 150 pens the cost of some pens is Rs. 3 and the cost of the rest pens is Rs. 5. Maximum how many customers can avail a pen of Rs. 5 as an offer from the company if the total cost of the pens cannot exceed Rs. 745.
(a) 45 (b) 120
(c) can't be determined (d) none of these

Directions for question numbers 117 to 118 : A number of 109 digits is written as follows :
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 ...

117. What is the least possible a positive number is required to add up in the given number to make it divisible by 2 and 5 both?
(a) 0 (b) 1
(c) 3 (d) can't be determined

118. What is the least possible whole number which must be subtracted from the given number so that it will become divisible by 3 and 11 both?
(a) 8 (b) 3
(c) 0 (d) none of these

119. Each family in Gyanpur village has atmost two adults and the total number of boys in this village is less than the number of girls. Similarly the number of girls is less than the number of adults in the village. Raghubir Singh, the chief of this village is the only adult in his family. The minimum number of families in his village is :
(a) 2 (b) 3
(c) 4 (d) none of these

120. Which of the following is correct if $A = 3^{3^3}$, $B = 3^{33^3}$, $C = 3^{3^{33}}$ and $D = 3^{333}$?
(a) $A > B = C > D$ (b) $C > A > B > D$
(c) $A > C > D > B$ (d) $C > B > D > A$

121. When two positive integers A and B are divided by another number D, then the remainders obtained were 23 and 3 respectively. Further when $(A + B)$ is divided by D, the remainder is 26, then the value of D is :
(a) 62 (b) 69
(c) 26
(d) Any value greater than 26

122. In how many ways can 729 be expressed as a difference of the square of whole numbers?
(a) 4 (b) 6
(c) 8 (d) none of these

123. In how many ways can 2310 be expressed as a product of 3 factors?
(a) 41 (b) 23
(c) 56 (d) 46

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Final Round

1. Sania wanted to cut a cubical cake into 120 identical pieces applying minimum number of 'n' cuts. Later on she realised that she had to cut this cake into maximum number of identical pieces *i.e.*, 125 pieces. Now she applies the number of cuts is :
(a) n (b) $n + 5$
(c) $n + 1$ (d) can't be determined

Directions for questions 2 to 4 :
If $s_1 = (1)$; $s_2 = (2), (3)$;
$s_3 = (4), (5), (6)$;
$s_4 = (7), (8), (9), (10)$;
$s_5 = (11), (12), (13), (14), (15)$ ...*etc.*
where $s_1, s_2, s_3, \ldots$*etc. are the first, second and third terms ... of the given sequence.*

2. The last digit and tens digit of the sum of all the elements of the first 21 terms of this sequence respectively are :
(a) 7, 0 (b) 6, 9
(c) 9, 6 (d) none of (a), (b), (c)

3. The number of zeros at the end of the product of the elements of s_{19}.
(a) 4 (b) 5
(c) 1 (d) none of these

4. The largest power of 10 that can exactly divide the product of all the elements of s_{19} and s_{20} is :
(a) 10 (b) 9
(c) 19 (d) can't be determined

5. A set 'S' contains first 50 elements of the form $2n$; $n \in N$. Further a subset 'P' of set 'S' is formed such that the product of any 3 elements of 'P' is not divisible by 16. Then maximum number of elements that 'p' can have is :
(a) 12 (b) 13
(c) 25 (d) none of (a), (b), (c)

6. A cuboid of dimensions 51, 85 and 102 cm is first painted by red colour then it is cut into minimum possible identical cubes. Now the total surface area of all those faces of cubes which are not red is :
(a) 119646 cm^2 (b) 52020 cm^2
(c) 18514 cm^2 (d) 36414 cm^2

7. A number 'p' is such that it is divisible by 7 but not by 2. Another number 'q' is divisible by 6 but not by 5, then the following expression which necessarily be an integer is :
(a) $\frac{7p + 6q}{42}$ (b) $\frac{5p + 6q}{71}$
(c) $\frac{6p + 7q}{42}$ (d) none of these

8. If $p^q - q^r = (p + q)^{r-q}$; $p > r > q \in$ Prime numbers less than 11 then $p + q$ is equal to :
(a) $r(r - q)$ (b) $r(q - p)$
(c) $r(p + q)$ (d) pq

9. To visit the Republic Day Parade on 26th January 2005, the people from every nook and corner including intellectuals, artists, farmers and mathematicians thronged in New Delhi. There were 100 seats in front row numbered 1, 2, 3, ..., 100. But the smart mathematician chose not to sit on those seats which are the multiples of any number greater than unity. How many mathematician could sit on these front row chairs?
(a) 25 (b) 26
(c) 33 (d) none of these

Direction for questions 10 to 12 : *In South-Asia the New Desh follows a septarian calender in which every month starts with Monday and a week has 7 days. There are only four months, the first 3 months consists of 98 days each and the last month has 70 days only.*

10. Which day falls on the 88th day of the first month?
(a) Sunday (b) Monday
(c) Thursday (d) Saturday

11. Which date cannot fall on the Thursday of the first month?
(a) 46 (b) 18
(c) 81 (d) 64

12. Which day occurs maximum number of time in a year?
(a) Monday (b) Saturday
(c) Sunday (d) none of these

13. The last two digit in the expansion of $(1989)^{91}$ are :
(a) 9, 1 (b) 8, 1
(c) 6, 9 (d) 8, 9

14. Earlier when I have created my e-mail-ID, the password was consisting of first 4 prime numbers. Recently when I tried to check my emails I got dumbfounded since I could not remember my password exactly. So when I have written 2735, my computer indicated me that no digit is correctly placed. Again I tried 5273, I got the same response. So once again I have written only 3 as the left most digit for my password it again indicated me that it was wrong. Finally I have taken one more attempt and got the account open. The code of my password is :
(a) 2537 (b) 7352
(c) 7325 (d) none of these

15. The remainder when $(8881)^{9999}$ is divided by 77 is :
(a) 1 (b) 2
(c) 3 (d) none of these

16. We publish a monthly magazine of 84 pages. Once I found that in a magazine 4 pages were missing. One out of them was page number 29 it is known that the page number of the last page of the magazine is 84, (including the coverpages). The numbers printed on the missing pages were :
(a) 29, 52, 53 (b) 30, 55, 56
(c) 28, 52, 53 (d) can't be determined

17. There are six locks exactly with one key for each lock. All the keys are mixed to each other. The maximum number of attempts needed to get the correct combination is :
(a) 21 (b) 15
(c) 6 (d) can't be determined

18. If n is an integer, how many values of n will give an integral value of $51n^2 + 17n + 6$?
(a) 4 (b) 3
(c) 2 (d) none of these

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19. Sania always beats Plexur in tennis, but loses to Venus. Lindse usually beats Plexur and sometimes Sania, but cannot win against Venus. The worst player can be :
(a) Venus (b) Plexur
(c) Sania (d) can't say

20. Winner and Loser two brothers are playing a game in recreational room at Amausi Airport, Lucknow. In this game each one in turn has to pick up a number m such that $1 \le m \le n$; $m, n \in I^+$. The game stops as soon as the sum of all the numbers picked so far attains the value of $2n + 1$ or exceeds and thus the player who picked last number was loser. With which number winner starts if he were to pick up first to defeat necessarily to loser is?
(a) $(n + 1)$ (b) n
(c) $(n - 1)$ (d) $2n$

21. The sum of the last 10 digits of the sum of the expression :
$(1^1 \times 2^2 \times 3^3 \times 4^4 \times 5^5) + (1^6 \times 2^7 \times 3^8 \times 4^9 \times 5^{10}) + (1^{11} \times 2^{12} \times 3^{13} \times 4^{14} \times 5^{15}) + \ldots + (1^{96} \times 2^{97} \times 3^{98} \times 4^{99} \times 5^{100})$ is :
(a) 16 (b) 18
(c) 20 (d) none of these

22. One day very early morning Ravishankar went to temple to offer some flowers as a part of *Puja*. He purchased some flowers but the seller offered him that if he would give him all his Rs. 2, he could get all the remaining 6 flowers and thus could gain 60 paise per dozen. If each time the transaction is possible only in rupees then how many flowers did Ravishankar purchase initially?
(a) 6 (b) 3
(c) 4 (d) 12

23. Maximum number of squares possible that can be constructed using 31 pencils of equal length on the table :
(a) 30 (b) 20
(c) 15 (d) 29

Directions for question number 24 and 25 : Kavita, a student of i4 IIM, told me that she did everyday 3 more passages of English than that of previous day and thus she completed all the passages in 10 days. Later on she told me that the number of passages she did on the last but one day were four times that she did on the second day.

24. Number of passages she has done on the last day :
(a) 30 (b) 41
(c) 32 (d) none of these

25. Total number of passages that she has completed in those 10 days :
(a) 84 (b) 180
(c) 175 (d) can't be determined

26. Recently, a small village, in Tamilnadu where only male shephered reside with four sheep each, was devastated by Tsunami waves. Therefore 8 persons and 47 sheep were found to be dead and the person who luckily survived, left the village with one sheep each since 21 sheep were too injured to move so have been left on their own luck, in the village. The number of sheep which were earlier in the village is :
(a) 84 (b) 120
(c) can't be determined (d) none of these

27. The number of 3-digit numbers which consists of the digits in A.P., strictly in increasing order using the non-zero digits of the decimal system is :
(a) 14 (b) 16
(c) 15 (d) none of these

28. The sum of :
$(2^2 + 4^2 + 6^2 + \ldots + 100^2) - (1^2 + 3^2 + 5^2 + \ldots + 99^2)$ is :
(a) 5555 (b) 5050
(c) 888 (d) 222

29. If an integer p is such that $(8p + 1)$ is prime, where $p > 2$, then $(8p - 1)$ is :
(a) divisible by 7 (b) divisible by 3
(c) a prime number (d) none of these

30. The remainder when $3^0 + 3^1 + 3^2 + \ldots + 3^{200}$ is divided by 13 is :
(a) 0 (b) 12
(c) 3 (d) none of these

31. The remainder when $4^0 + 4^1 + 4^2 + 4^3 + \ldots + 4^{40}$ is divided by 17 is :
(a) 0 (b) 16
(c) 4 (d) none of these

32. In the problem number 31, if the divisor is 7 then the remainder is :
(a) 4 (b) 5
(c) 6 (d) 1

33. A monkey wanted to climb on the smooth vertical pole of height of 35 metre. In the first one minute he climbs up 5 metre in the next one minute he slips down by 2 metre. Further he repeated the same process till he has reached on the top of the pole. How many times it has to go upward to reach the apex of the pole?
(a) 35 (b) 12
(c) 11 (d) can't say

34. In the above question the minimum time required for this job is :
(a) 21 minute (b) 22 minute
(c) 24 minute (d) none of these

35. In the same problem, if the height of the pole is 36 metre then the time taken by monkey to reach at the top of the pole is :
(a) 22 min. 36 sec. (b) 22 min. 24 sec.
(c) 23 min. 12 sec. (d) none of these

36. The remainder when $1^3 + 2^3 + 3^3 + \ldots + 999^3 + 1000^3$ is divided by 13 is :
(a) 7 (b) 11
(c) 12 (d) none of these

37. If $22^3 + 23^3 + 24^3 + \ldots + 87^3 + 88^3$ is divided by 110 then the remainder wil be :
(a) 55 (b) 1
(c) 0 (d) 44

38. The sum of the n terms of a series is $n! + n^2$ then the 6th term is, if $n \in N$:
(a) 756 (b) 611
(c) data insufficient (d) none of these

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39. A smallest possible number which is divisible by either 3, 5 or 7 when represented by only two digits either 0 or 1, then the minimum number of digits required to represent it :
(a) 6 (b) 5
(c) 7 (d) can't be determined

40. The sum of first n odd numbers (*i.e.*, $1 + 3 + 5 + 7 + \ldots + 2n - 1$) *is divisible by 11111 then the value of* n is :
(a) 12345 (b) 11111
(c) can't be determined (d) none of these

41. Anjuli bought some chocolates from Nestle's exclusive shop, she gave to Amit one less than half of what she had initially. Then she had given 3 chocolates to Babli and then half of the chocolates which she had then given to Charles. Thus finally she gave one chocolate to Deepak and the remaining one she ate herself. The number of chocolates she had purchased.
(a) 9 (b) 12
(c) 10 (d) 15

42. If $(a_1b_1c_1)^{100} + (a_2b_2c_2)^{100} + (a_3b_3c_3)^{100} + \ldots (a_{100}b_{100}c_{100})^{100}$, where $a_ib_ic_i$ is a three digit positive number and in the expression all the 100 numbers are any consecutive 3 digit numbers. The last digit is :
(a) 0 (b) 1
(c) 3 (d) none of these

Directions for question number 43 to 45 : If [x] read as the greatest integer less than or equal to x, {x} is the least integer greater than or equal to x. Further $f(x, y) = [x] + \{y\}$
and $g(x, y) = [x] - \{y\}$
and $P(x, y) = f(x, y) + g(x, y)$
and $Q(x, y) = f(x, y) - g(x, y)$

43. If $x = 16$ and $y = 25$ then the value of $P(x, y) + Q(x, y)$ is :
(a) 90 (b) 200
(c) can't be determined (d) none of these

44. If $x^2 = 16$ and $y^2 = 25$ then $P(x, y)Q(x, y)$ is :
(a) 80 (b) – 80
(c) 72 (d) none of these

45. If $x, y \in I^+$ then $P(x, y) + Q(x, y)$ is always :
(a) an even number (b) an odd number
(c) can't say (d) none of these

46. Which of the following is/are true?
(i) $43^3 - 1$ is divisible by 11
(ii) $56^2 + 1$ is divisible by 19
(iii) $50^2 - 1$ is divisible by 17
(iv) $(729)^5 - 729$ is divisible by 5
(a) (i) and (ii) (b) (iii) and (iv)
(c) (ii), (iii) and (iv) (d) (ii) and (iii)

47. Capt. Manoj Pandey once decided to distribute 180 bullets among his 36 soldiers. But he gave n bullets to a soldier of nth row and there were same number of soldiers in each row. Thus he distributed all his 180 bullets among his soldiers. The number of soldiers in $(n - 1)$th row was :
(a) 3 (b) 8
(c) 9 (d) none of these

48. If $(n - 5)$ is divisible by 17 for every $n \in I^+$ then the greatest integer which will necessarily divide $(n + 12)(n + 29)$ is :
(a) 578
(b) 289
(c) such a number does not exist
(d) none of these

49. A certain number 'n' can exactly divide $(3^{24} - 1)$ then this number can also divide the number :
(a) $(3^{16} + 1)$ (b) $(3^8 - 1)$
(c) $(3^{70} - 1)$ (d) $(3^{96} - 1)$

50. If a number 'n' can exactly divide $(5^{14} - 1)$ then 'n' can divide necessarily :
(a) $(5^{28} - 1)$ (b) $(5^{42} - 1)$
(c) $(5^{21} + 1)$ (d) both (a) and (b)

51. The nth term of a series of which all the terms are positive is defined as $T_n = n^2 + n$ then the sum of n terms of the series is :
(a) $\frac{n(n+1)(n+2)}{6}$ (b) $\frac{n^3 + 4n}{n}$
(c) $\frac{5n}{9}[n^2 + n]$ (d) $\frac{n(n+1)(n+2)}{3}$

52. The number of zeros at the end of the product of $222^{111} \times 35^{53} + (7!)^{61} \times (10!)^{51} + 42^{42} \times 25^{25}$ is :
(a) 42 (b) 53
(c) 1055 (d) none of these

53. $\frac{12345}{12346} + \frac{12346}{12347} + \frac{12347}{12345}$ is equal to :
(a) 2.67 (b) 6.27
(c) 3 (d) 5

Directions for question numbers 54 to 56 : The set $S_1 = \{1\}$, $S_2 = \{3,5\}$, $S_3 = \{7,9,11\}$, *etc. forms a sequence.*

54. Sum of all the elements of S_{10} is :
(a) 55 (b) 300
(c) 3375 (d) none of these

55. The 11th element of the set S_{21} is :
(a) 21 (b) 121
(c) 221 (d) 441

56. The sum of the first and last element of the set S_{51} is :
(a) 5202 (b) 5151
(c) 5152 (d) 5102

57. During my studies once I brought a book from library which was written in early days, when there were only 9 digits *i.e.*, the digit 0 did not exist. There was a sum in that book as follows :

$$k = 13 + 17 + 31 + 2$$

Then the value of k if $9 + 1 = 11$, $19 + 4 = 24$ etc.
(a) 70 (b) 71
(c) 63 (d) 64

58. The sum of $4 + 16 - 5 + 12$ is according to that book.
(a) 27 (b) 26
(c) 29 (d) none of these

59. The value of x for which the unit digits of $(2357)^{\log_{10} x}$ and $(5723)^x$ is same for $x > 1$.
(a) 10 (b) 100
(c) 1000 (d) none of these

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60. The value of x for which the unit digits of the following two expressions $(1+2+3+4+7)^{x^2+x}$ and $(11\times11\times13)^x$ are same for $x>0$:
(a) 1 (b) 2
(c) 3 (d) none of these

61. When any odd number greater than unity multiplied by even times by itself then dividing this product by 8, we get the remainder as :
(a) 1 (b) 7
(c) not unique (d) none of these

62. Stephen's birthday, this year falls on 2nd April, Wednesday. But coincidently his marriage anniversary is 2 days before the 23rd of the same month. On which day he will celebrate his marriage anniversary?
(a) Monday (b) Wednesday
(c) Friday (d) can't be determined

63. In the above problem if there are only 6 days in a week *i.e.*, there is no Sunday and the week starts with Monday and ends with Saturday then his marriage anniversary will fall on :
(a) Wednesday (b) Thursday
(c) Friday (d) data insufficient

64. An N.G.O. (non-government organisation) STRANGE working for the relief of Tsunami Victims in Srilanka consisting of 7 members S, T, R, A, N, G, E of the same family.

The eldest one '*S*' spoke to me "I have deployed equal number of brothers and sisters for medical relief and psychological counselling under the supervision of myself". Later on the youngest member *E* spoke to me as "we have been working for rehabilitation and food supply as twice the number of sisters as the number of brothers have been deployed there, but I did not actually do any thing due to a severe injury in my leg". Then we can conclude that :
(a) youngest person is a lady
(b) eldest person's wife is youngest
(c) the brother of *S* is youngest
(d) nothing can be said

65. When the sum of n digits of an n digit number is subtracted from the number itself, where the number must be atleast two digit number, then the correct statement is :
(a) the difference is a prime number
(b) the resultant value is a perfect square
(c) the resultant value is an odd number
(d) the resultant value is a multiple of 9

66. For every natural number x and y the value of $x-y$ when $y+\frac{7}{x}+\frac{y}{13}=6\frac{3}{143}$, is :
(a) 5 (b) 6
(c) 2 (d) not unique

Directions for question 67 and 68 : The $S=\{(1,3,5,7,9,\ldots,99)\ (102,104,106,\ldots,200)\}$ *i.e., in the first part there are odd integers less than 100 and in the second part there are even integers greater than 100, but upto 200.*

67. The highest power of 3 in the product of the element of the set is :
(a) 52 (b) 51
(c) 97 (d) can't be determined

68. The highest power of 5 that can exactly divide the product is :
(a) 25 (b) 24
(c) 30 (d) 26

69. The number of zeros at the end of the following expression :
$P=\{(2\times4\times6\times8\times10\times\ldots50)\times(55\times60\times65\times70\times75\times\ldots100)\}$
(a) less than 20 (b) 57
(c) 20 (d) 36

70. 1, 2, 3, 4, 10, 11, 12, 13, 14, 20, 21, 22, 23, 24, ... are the consecutive numbers written in base 5. The twenty fifth number in the above sequence would be :
(a) 52 (b) 100
(c) 25 (d) none of these

71. Watch India Corporation made a wrist watch in which the minute hand makes one complete round of dial in 12 minute and accordingly the hour hand too. When I have set this watch at 12 : 00 noon on Ist February this year. Which time will be shown by this watch at 3 O'clock on the same day.
(a) 2 : 30 P.M. (b) 6 : 15 A.M.
(c) 3 O'clock (d) can't be determined

72. The sum of first n numbers of the form $(5k+1)$, where $k\in I^+$ is :
(a) $\frac{n}{2}[5n^2-3]$ (b) $n(20-3n)$
(c) $\frac{n}{2}(5n+7)$ (d) none of these

73. A series is given as : 1, 4, 9, 16, 25, 36, ...
Then the value of $T_{n+1}-T_n$ is, where T_n is the nth term of the series is :
(a) n^2-1 (b) $2n+1$
(c) n^2+1 (d) none of these

74. The area of paper can be divided into 144 squares, but if the dimensions of each square were reduced by 2 cm each, then the number of squares so formed are 400. The area of the paper initially was :
(a) 544 cm² (b) 1444 cm²
(c) 3600 cm² (d) none of these

75. If $(ab)^2=bcb$ and $(dd)^2=cqff$, where a, f, d are strictly in increasing order of G.P. and b, c, d are in increasing order of A.P. Then the value of f will be, where ab and bcb etc. are the two digit and 3 digit numbers etc.
(a) 9 (b) 8
(c) can't be determined (d) none of these

76. If $A=555!$ and $B=(278)^{555}$ then which one of the following relation is appropriate?
(a) $A>B$ (b) $A=B$
(c) $A<B$ (d) can't say

Directions for question number 77 to 81 : *For any natural number n the sets* $S_1, S_2, \ldots$ *are defined as below :*
$S_1=\{1\}$, $S_2=\{2,3\}$, $S_3=\{4,5,6\}$
$S_4=\{7,8,9,10\}$, $S_5=\{11,12,13,14,15\}$ ... *etc.*

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77. The last element in the S_{24} is :
(a) 576 (b) 600
(c) 300 (d) 625

78. The middlemost element of the set S_{15} is :
(a) 196 (b) 169
(c) 131 (d) none of these

79. The sum of the elements of set S_{25} is :
(a) 7825 (b) 3125
(c) 3250 (d) none of these

80. In which set, there are maximum number of prime number elements among $S_1, S_2, S_3, \ldots, S_{13}$ is :
(a) S_{12} (b) S_{13}
(c) S_{12} and S_{13} (d) S_9, S_{12}, S_{13}

81. Of which set the sum of all the elements of the set is even :
(a) S_{39} (b) S_{50}
(c) S_{72} (d) S_{94}

Directions for question numbers 82 to 85 : *The sequence of sets* $S_1, S_2, S_3, S_4, \ldots$ *is defined as* $S_1 = \{1\}$, $S_2 = \{3, 5\}$, $S_3 = \{7, 9, 11\}$, $S_4 = \{13, 15, 17, 19\}$ ... *etc.*

82. The first element of the nth set S_n is :
(a) n (b) $n^2 - 1$
(c) $2n + 1$ (d) $n^2 - n + 1$

83. The last element of the set S_{100} is :
(a) 10099 (b) 9899
(c) 9900 (d) none of these

84. The middlemost element of an odd numbered set S_{125} is :
(a) 12500 (b) 15625
(c) 3125 (d) none of these

85. The sum of all the elements of S_{101} :
(a) 1531441 (b) 1189811
(c) 1030301 (d) none of these

86. The sum of the series :
$$S = \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \frac{1}{4.5} + \ldots + \frac{1}{99.100} \text{ is :}$$
(a) $\frac{98}{99}$ (b) $\frac{99}{100}$
(c) $S > 1$ (d) none of these

87. A number P when divided by D it leaves the remainder 18 and if another number Q is divided by the same divisor D it leaves the remainder 11. Further if we divide $P + Q$ by D then we obtain the remainder 4. Then the common divisor D is :
(a) 22 (b) 15
(c) 25 (d) can't be determined

88. If the product of $1 \times 2 \times 3 \times 4 \times \ldots n$ contains 68 zeros in the end of the number. Then the maximum possible number of values of n is :
(a) 1 (b) 3
(c) 5 (d) 6

89. The remainder when $6^{6^{6^{6^{\ldots}}}}$ (∞ times) is divided by 10 is :
(a) 3 (b) 6
(c) 0 (d) can't be determined

90. $53^3 - 46^3 - 7^3$ is divisible by :
(a) 6 and 9 (b) 2 and 21
(c) 21 and 23 (d) both (b) and (c)

91. A number of decimal system when written in base n, $2 < n < 10$, we get a two digit number. Further if we reverse the digits of the obtained number in base 'n' we get a number which is twice of the original number in decimal system. The sum of original and resultant number both in decimal system for the largest possible value of n is :
(a) 45 (b) 63
(c) 77 (d) can't be determined

92. In the above question how many values of n are possible?
(a) 0 (b) 2
(c) 4 (d) 7

93. Pandavas won a hen in the war of the Mahabharat. They brought it on the Ist January, 2002. This hen gave birth to the 7 new hens on the very first day. After it every new hen irrespective of its age everyday gave birth (only once in a lifetime) to 7 new hens. This process continued throughout the year, but no any hen had been died so far. On the 365th day all the Pandavas shared equally all the hens among all the five brothers. The remaining (if these can not be shared equally) hens were donated to Krishna. The number of hens which the Krishna had received is :
(a) 3 (b) 2
(c) can't be determined (d) none of these

94. Total number of natural numbers being the perfect square whose square root is equal to the sum of the digits of the perfect square is :
(a) 0 (b) 1
(c) 2 (d) 12

95. At our training Institute the number of boys is same as that of the girls. Last week, except $\frac{2}{3}$rd of the girls all the students went to picnic, where they bought some samosas but later on they found exactly one dozen samosas were not fresh so those 12 samosas have been thrown away. After it half-half the samosas were divided between boys and girls. Further when boys dealtout the samosas equally among themselves 39 samosas left undistributed, but when the girls dealtout the same number of samosas equally among themselves 12 samosas were still left undistributed. The number of students at out training institute is :
(a) 60 (b) 156
(c) 162 (d) can't be determined

96. Darwin Miya has 6 kinds of fruits in large amount and has sufficient number of identical boxes to store the fruits. He can put at least 10 and atmost 15 fruits in any box and he put only one kind of fruits in a box. Further not more than 5 boxes can contain same number of fruits. Maximum number of fruits that he can put in the boxes is :
(a) 325 (b) 375
(c) 75 (d) can't be determined

97. In the above question if he is allowed to put the equal number of fruits in atmost 7 boxes and he has only 33 boxes and now he can put any kind of fruit with any other kind of fruits. At least how many boxes are there in which the number of fruits are same if he fills every box to its maximum capacity.
(a) 5 (b) 6
(c) 3 (d) can't be determined

98. If $n \in 1, 3, 5, 7, \ldots$ etc., then the value of $19^n - 23^n - 43^n + 47^n$ is necessarily divisible by :
(a) 264 (b) 246
(c) 76 (d) 129

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99. The sum of the following series :

$$1.1^2\left(1-\frac{0}{1}\right)+2.2^2\left(1-\frac{1}{2}\right)+3.3^2\left(1-\frac{2}{3}\right)+4.4^2\left(1-\frac{3}{4}\right)+\ldots \text{upto } n \text{ terms is :}$$

(a) $n.n^2\left(1-\frac{1}{n}\right)$ (b) $\frac{n(n+1)(2n+1)}{6}$

(c) $\frac{n(n+1)^2}{4}$ (d) none of these

100. The distance between the houses of Sarvesh and Ravi is 900 km and the house of former is at 100th milestone where as the house of Ravi's is at 1000th milestone. There are total 901 milestones at a regular interval of 1 km each. When you go to Ravi's house from the house of Sarvesh which are on the same highway, you will find that if the last digit (*i.e.*, unit digit) of the 3 digit number on every milestone is same as the first (*i.e.*, hundreds digit) of the number on the next mile stone is same, then these milestones must be red colour and rest will be of black. Total number of red colour milestones is :

(a) 179 (b) 90

(c) can't be determined (d) none of these

101. The highest power of 17 which can divide exactly the following expression :

$(18^2-1)(18^4-1)(18^6-1)(18^8-1)(18^{10}-1)\times\ldots(18^{16}-1)(18^{18}-1)$ is :

(a) 1 (b) 17

(c) 9 (d) can't be determined

102. "Help India Foundation" and "People for People Organisation" decided to distribute the blankets among 22 men and 28 women who are Tsunami victims. When HIF and PPO distributed their respective blankets evenly among 28 women they were left with 24 and 16 blankets respectively. If they distributed their blankets evenly among 22 men they were left with 12 blankets each. So finally they decided to combine all their blankets and then distributed among 22 men and 28 women altogether then no any blanket remained undistributed. Minimum total blankets distributed by them were :

(a) 960 (b) 700

(c) 1300 (d) none of these

103. In the above problem the ratio of blankets between HIF and PPO is :

(a) $\frac{43}{157}$ (b) $\frac{147}{179}$

(c) $\frac{59}{101}$ (d) can't be determined

104. The total number of 3 digit numbers which have two or more consecutive digits identical is :

(a) 171 (b) 170

(c) 90 (d) 180

105. The expression, for $p \neq 1$, $(1+p^{256})\times(1+p^{128})\times(1+p^{64})\times(1+p^{32})\times(1+p^{16})\times(1+p^8)\times(1+p^4)\times(1+p^2)\times(1+p)$ is equivalent to :

(a) $\frac{1+p^{256}}{1+p^{128}}$ (b) $\frac{1-p^{512}}{1-p}$

(c) $\frac{1-p^{256}}{1-p}$ (d) p^{255}

106. For the given fixed perimeter of 50 cm, the total number of rectangles which must have its sides in integers (cm) is :

(a) 50 (b) 25

(c) 12 (d) infinite

107. The total number of factors of a number is 24 and the product of the prime factors of the same number is 30. The square root of the ratio of such a smallest and the greatest possible numbers is :

(a) 5/6 (b) 6/9

(c) 9/25 (d) 4/25

108. In the above problem, maximum how many numbers are possible?

(a) 9 (b) 8

(c) 7 (d) data insufficient

109. Mr. Oberai appeared in CAT for four consecutive years, but coincidentlly each time his net score was 75. He told me that there was $\frac{1}{3}$rd negative marking for every wrong answer and 1 mark was alloted for every correct answer. He has attempted all the questions every year, but certainly some answers have been wrong due to stress and conceptual problems. Which is not the total number of questions asked for CAT in any year, in that period?

(a) 231 (b) 163

(c) 150 (d) 123

110. A thief somehow managed to steal some golden coins from a bank's cash but while coming out of it at the first door he was caught by the watch man and he successfully dealt him by paying 1 coin plus half of the rest coins. Further he had to pay 2 coins, then half of the rest to the second watchman. Once again at the third gate (outermost) he gave 3 coins and then half of the rest. After it he was left with only one coin. How many coins had he stolen?

(a) 32 (b) 36

(c) 25 (d) none of these

111. The number $\log_2 7$ is :

(a) an integer (b) a rational number

(c) an irrational number (d) a prime number

112. The product of n positive numbers is unity. Then their sum is :

(a) a positive integer (b) divisible by n

(c) equal to $\left(n+\frac{1}{n}\right)$ (d) never less than n

113. Let $n > 1$, be a positive integer. Then the largest integer m, such that (n^m+1) divides $(1+n+n^2+n^3+\ldots+n^{127})$ is :

(a) 127 (b) 63

(c) 64 (d) 32

114. Number of divisors of the form $4n+2$; $n \geq 0$ which can divide 240 is :

(a) 4 (b) 8

(c) 10 (d) 3

115. If the integers m and n are chosen at random between 1 and 100, then atmost distinct numbers of the form 7^m+7^n is divisible by 5 equals to :

(a) 1250 (b) 10000

(c) 2500 (d) none of these

116. If a, b, c, d are positive real numbers such that $a+b+c+d=2$, then $M=(a+b)(c+d)$ satisfies the relation is :

(a) $0 \leq M \leq 1$ (b) $1 \leq M \leq 2$

(c) $2 \leq M \leq 3$ (d) $3 \leq M \leq 4$

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Test of Your Learning-1

1. A monkey wants to climb up a pole of 50 metre height. He first climbs up 1 metre but he falls back by the same height. Again he climbs up 2 metre but he falls back by 1 metre. Once again he climbs up 3 metre but he falls back by 1 metre. Again he climbs up 4 metre but he falls back by 1 metre. In this way he reaches at the top of the pole. If it is known that the monkey needs 10 seconds for 1 metre in upward direction and 5 seconds for 1 metre in downward direction, then the total time required by monkey to reach at the top of the pole. For your kind information it must be clear to you that when he will reach on the top, he will not be slip back. The total time required by this monkey to reach on the top of the poll is :

(a) 9 minutes 10 seconds (b) 10 minutes
(c) 8 minutes 20 seconds (d) none of these

2. Which one of the following is correct?

(i) $(123)^{369} > (369)^{123}$ (ii) $(246)^{642} < (642)^{246}$
(iii) $(50)^{501} = (501)^{50}$ (iv) $3^{4^5} > 5^{4^3}$

(a) (i) and (ii) (b) (ii) and (iii)
(c) (i) only (d) (i) and (iv)

3. The tens digit of the 2^{248} is :

(a) 1 (b) 7
(c) 5 (d) none of these

4. For a community marriage 300 boys and 300 girls gathered for their marriage. But out of 600 people 200 were from America and 200 from France and rest 200 were from India itself. The rule was that no any person can marry with a person of different ethnicity or nationality. Minimum how many married couples will be formed, after marriage?

(a) 100 (b) 150
(c) 200 (d) none of these

5. The 444th term of the following sequence 1, 2, 3, 10, 11, 12, 13, 20, 21, 22, 23, 30, 31, ... is :

(a) 1331 (b) 17776
(c) 12330 (d) 13330

6. If $P = 3[x] + 5$ and $Q = 4[x + 3] + 6$ and $P : Q = 1 : 2$, where $[x]$ is the greatest integer less than or equal to x then the value of $\left[P + \frac{Q}{2}\right]$ is :

(a) 34 (b) 51
(c) 40 (d) data insufficient

7. The maximum possible value of $a + b + c + d$ will be where $(a + b) = c, (b + c) = d, (c + d) = a$, for every $a, b, c, d \in I^+$:

(a) – 4 (b) 0
(c) 16 (d) can't be determined

8. The number of divisors of 720 which are multiples of perfect square of first prime number is :

(a) 12 (b) 14
(c) 18 (d) can't be determined

9. The number of zeros which are not possible at the end of the $n!$ is :

(a) 82 (b) 73
(c) 156 (d) none of these

Directions for questions number 10 and 11 : The following sequence is given below :

1, 3, 3, 3, 5, 5, 5, 5, 5, 7, 7, 7, 7, 7, 7, 9, 9, 9, 9, 9, 9, 9, 9, 9, ...

10. The 111th term of this sequence will be :

(a) 111 (b) 19
(c) 21 (d) none of these

11. The sum of the 1st, 3rd, 8th, 15th, 24th, 35th, ..., 99th term will be :

(a) 100 (b) 101
(c) 121 (d) 81

Directions for questions number 12 to 14 : The following sequence is given below :

1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6, 6, ... etc.

12. The sum of the first 100 terms of this sequence is :

(a) 945 (b) 358350
(c) 1001 (d) none of these

13. The 77th element is :

(a) 10 (b) 12
(c) 15 (d) 13

14. The sum of the 1st, 2nd, 4th, 7th, 11th, 16th, ..., 211th terms is :

(a) 221 (b) 400
(c) 231 (d) 211

15. Which one of the following is true?

(I) The least positive value of n for which $n!$ can be divided by $(n + 1)$ is 5

(ii) The square root of 123454321 is 1111

(iii) The sum of the squares of the first 24 natural numbers is a perfect square

(a) (i) and (ii) (b) (ii) and (iii)
(c) (i) and (iii) (d) all of these

16. The number of zeros at the end of $1^{99} \times 2^{98} \times 3^{97} \times ... 99^1$ is :

(a) 1100 (b) 1111
(c) 1099 (d) none of these

Directions for questions number 17 to 20 : Mewalal is selling fruits in Charbagh, Lucknow, which were imported from Kesharbagh and Dayalbagh only. He has 13 apples, 25 bananas, 17 cherries, 29 goose berries, 35 mangoes and 42 pine apples. When I wanted to purchase some fruits from his shop, he told me that he can sell only in lots (i.e., either total apples or total bananas or total cherries etc., but anyhow he can't sell say 12 bananas or 10 apples) and he told me that he himself can import any type of fruits without any lot, say 20 mangoes from Kesharbagh and 15 mangoes from Dayalbagh and so on if he wished.

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17. Mewalal sold some fruits and then he told me that he is now left with twice the fruits of Kesharbagh than that of Dayalbagh. The number of fruits of Dayalbagh left after this selling is :
(a) 35 (b) 43
(c) 46 (d) can't be determined

18. In the above question (no. 17) how many lots or category of fruits from A, B, C, G, M, P he has sold?
(a) 1 (b) 2
(c) 3 (d) can't be determined

19. As per the given constraints if Mewalal has sold some fruits in such a way that he is now left with $\frac{3}{4}$ fruits of Dayalbagh than that of Kesarbagh. Maximum how many lots (*i.e.*, categories) of fruits he can sell out in such a way?
(a) 1 (b) 2
(c) 3 (d) can't be determined

20. If all the fruits were bought by a single person and were distributed equally among maximum possible number of children and each child has must received more than one fruit, then minimum how many children had received different kind of fruits?
(a) 2 (b) 3
(c) 4 (d) can't be determined

21. Total number of factors of the expression $62^3 - 54^3 - 8^3$ is :
(a) 60 (b) 62
(c) 46 (d) can't be determined

22. The least possible number with which (180)! should be multiplied so that it can be divided by $(45)^{44}$ is :
(a) 44 (b) 25
(c) 9 (d) none of these

23. If $P = abc$ and $Q = uv$ are three digits and 2 digits two natural numbers respectively, such that u and v must be distinct integers. How many pairs of P and Q are there in total which gives the same result when we multiply abc with uv as the product of cba with vu (*i.e.*, the position of digits is inter changed) :
(a) 2 (b) 8
(c) 5 (d) can't be determined

24. There is a three digits number 'abc' and a two digit number 'xy' such that the product of 'abc' with 'xy' is same as the product of cba with 'yx', if x and y can be similar digits, then the number of such pairs of P and Q is :
(a) 99 (b) 900
(c) 810 (d) can't be determined

25. If $(N)_{10} = (ab)_n$ and $(3N)_{10} = (ba)_n$; where $3 < n < 10$, then the least possible value of n is :
(a) 4 (b) 6
(c) 7 (d) can't be determined

26. At his birthday party Pravesh invited his 100 friends and they took their seats numbered as 1, 2, 3, ..., 100. Starting from seat number 1 every third guest was served with Chapatis, every fourth guest was served with Poori and every fifth guest was served with Dahibara and the remaining guests were served with coca-cola only. How many guests enjoyed coca-cola in that party?
(a) 33 (b) 13
(c) 41 (d) can't be determined

27. In the above question (no. 26) if Pravesh served every guest who were sitting on the chair number 3, 6, 9, 12, 15, ... etc. with chapatis and to those who were, sitting on the chair number 5, 10, 15, 20, ... etc. with poori and to the rest with Idli. The number of guests who has not received the Idlis is :
(a) 38 (b) 47
(c) 41 (d) none of these

28. The number of ways in which 5^7 can be expressed as a product of three factors :
(a) 7 (b) 8
(c) 9 (d) none of these

29. If $[x]$ means the greatest integer less than or equal to x and $\{x\}$ means the least integer greater than or equal to x and $|x|$ means the absolute value of x, then $\{x\} - [x]$ equals to :
(a) -1 (b) 0
(c) 1 (d) any one of (a), (b), (c)

30. In the previous question if x is not an integer then the value of $|\{x\} - [x]|$ is :
(a) 0 (b) -1
(c) 1 (d) either (a) or (c)

31. If mnp is a three digit number such that :
$m^3 + n^3 + p^3 = mnp = \Sigma(n!)$, then the value of $m \times n \times p$ is :
(a) 30 (b) 21
(c) 15 (d) none of these

32. A square field is fenced by fixing the polls all around it in such a way that in each side of the field the farmer fixed 19 polls. The total number of polls required to fence the field is :
(a) 76 (b) 75
(c) 72 (d) 80

33. Swarn Jayanti Park is a regular hexagonal park, around which there are 'n' trees on each side. Which of the following can't be the total number of trees around it?
(a) 74070 (b) 81474
(c) 591 (d) 325960

34. Pentabhai purchased the ticket of a movie for all his 200 friends. Luckily he was the first person to buy the ticket for the evening show. So he got the tickets number 1, 2, 3, 4, 6, 7, ... etc. Since he denied to have the tickets on which the digit '5' is printed as the part of ticket number. The last ticket number which he has purchased if no one else has bought the tickets until he has purchased for all his friends.
(a) 244 (b) 242
(c) 222 (d) can't be determined

35. The expression
$$N = \frac{1}{1+1\times1} + \frac{1}{2+2\times2} + \frac{1}{3+3\times3} + \frac{1}{4+4\times4} + \ldots \frac{1}{n+n\times n},$$ where $n = 100$, then value of N is :
(a) $\frac{100}{101}$ (b) $\frac{99}{100}$
(c) 9900 (d) none of these

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36. Anjuli has written all the letters (or characters) of English alphabets (*i.e.*, *A, B, C, D, ... Y, Z*) horizontally in a single line. She started counting as *A*-1, *B*-2, *C*-3, *D*-4, ... *Y*-25, *Z*-26 and reversed back as *Y*-27, *X*-28, *W*-29, *V*-30, ... *B*-50, *A*-51 and further reversed back as *B*-52, *C*-53, *D*-54, ... etc. She continued counting till she reached the 777th character. At which letter or character she stopped counting?
(a) *Y* (b) *B*
(c) *R* (d) none of these

37. The reading style of Sunny is quite unusual. He reads one page on the first day, 2 pages on the second day, 3 pages on the third day etc. How many pages Sunny can read in 24 days?
(a) 242 (b) 300
(c) 276 (d) none of these

38. All the natural numbers less than 1000 but not containing the digit '5' are arranged horizontally in descending order. What is the 555th number on the list?
(a) 111 (b) 212
(c) 213 (d) none of these

39. A shopkeeper told me when I have asked him the total number of cellphones in his shop, that he had all Kyocera except three, all Nokia except three, all samsung except three and all L.G. except three. The number of mobile (*i.e.*, cell phones) he had is :
(a) 24 (b) 256
(c) can't be determined (d) none of these

40. The number of zeros at the end of the product of all the prime numbers between 1 and 1111 is :
(a) 222 (b) 21
(c) can't be determined (d) none of these

Test of Your Learning-2

1. The remainder when 888222888222888222 ... upto 9235 digits is divided by 5^3 is :
(a) 103 (b) 38
(c) can't be determined (d) none of these

2. Shakuntala asked Aryabhatta to assume any two values of three digits say *P* and *Q* then she told him to multiply *P* by *R* and *Q* by *S* where the values of *R* and *S* were given by Shakuntala herself. Aryabhatta exactly told her the values of $PR \div QS = 888222$. Then Shakuntala told him the value of *P* by *Q* (*i.e.*, *P*/*Q*) is :
(a) 1/4 (b) 4
(c) can't be determined (d) none of these

3. In the above question the value of $P + Q$ is :
(a) 1001 (b) 1110
(c) 3108 (d) none of these

4. The sum of the following series
$(1^2 + 1) + (2^2 + 2) + (3^2 + 3) + (4^2 + 4) + ... + (n^2 + n)$ is :
(a) n^3 (b) $\frac{(n^2 + n)5}{3}$
(c) $\frac{n(n+1)(n+2)}{3}$ (d) can't be determined

Directions for questions number 5 to 8 : Earlier when I attended a close door meeting to discuss the outsourcing of CAT papers. Including the chairman there were n people viz. A, B, C, D, E, F, ... etc.
As per the convention everyone has gotten some chocolates in the following manner. As A, B, C, D, E, ... etc. have received 1, 2, 3, 4, 5, ... etc. chocolates respectively. Before anyone has eaten a bit of chocolate, due to some urgent call, the chairman left the meeting with his chocolates. Later on the rest attendants recollected their chocolates in a box and then redistributed all the chocolates evenly among themselves and thus everyone has received 13 chocolates.

5. Who is the Chairman of the meeting?
(a) A (b) Q
(c) M (d) can't be determined

6. Total number of people, initially who have attended the meeting is :
(a) 18 (b) 25
(c) 30 (d) can't be determined

7. Minimum number of person who attended the party can be:
(a) 18 (b) 15
(c) 24 (d) can't be determined

8. As per the question number 7, maximum how many chocolates were there to be received by all of them, initially?
(a) 300 (b) 676
(c) 351 (d) none of these

Directions for questions number 9 to 10 : Under the scheme of Kisan Vikas, the Govt. of U.P. purchased 't' number of tractors and these were allocated equally among 7 districts of U.P. and thus 4 tractors remained unallocated. Similarly every district allocated these tractors equally to 10 blocks in each district and still 3 tractors remained without allocation, in every district. Further every block assigned these tractors equally to every 16 village in each block and thus 2 tractors per block remained unallocated.
According to the scheme every village must receive at least one tractor.

9. The number of tractors allocated to each block is :
(a) 18 (b) 30
(c) 34 (d) can't be determined

10. If the state govt. of U.P. purchased '*k*' tractors to provide exactly 1 tractor to each village as mentioned in the scheme but later on it has sent one tractor to each village of U.P. (as mentioned in the scheme) directly without any involvement of a block or a district. The number of tractors which the govt. of U.P. has saved more in comparison to the previous scheme.
(a) 161 (b) 181
(C) 24 (d) can't be determined

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11. If 2^n can exactly divide $p!$ Such that the quotient is an odd positive integer, then the value of n which is not possible is :
(a) 43 (b) 44
(c) 45 (d) all of these

12. Each of the numbers $x_1, x_2, x_3, x_4, x_5, \ldots, x_n, x_n \geq 4$ is equal to -1 or 1. If $x_1x_2x_3 + x_2x_3x_4 + x_3x_4x_5 + \ldots x_{n-3}x_{n-2}x_{n-1} + x_{n-2}x_{n-1}x_n = 0$, then the value of n is :
(a) odd (b) even
(c) prime (d) multiple of 3

13. In the above question number 12, the minimum number of x_i (*i.e.*, $x_1, x_2, x_3, \ldots, x_n$) are equal to -1, is :
(a) $\frac{n}{3}$ (b) $\frac{n}{2}$
(c) $\frac{(n-2)}{2}$ (d) can't be determined

14. If in the above question (no. 13) the value of x_i is either 0 or 1 only (*i.e.*, each of $x_1, x_2, x_3, \ldots, x_n$ is either zero or 1) then the minimum number of x_i, such that $x_i = 0$ is required, when $n = 6, 12, 18, 24, \ldots$ etc.
(a) $\frac{n}{3}$ (b) $\frac{(n+2)}{3}$
(c) $\frac{(n-2)}{2}$ (d) none of these

15. If $m + n = mn - 5$, then the maximum number of ordered pairs of (m, n) for every m, n is a natural number.
(a) 4 (b) 5
(c) 6 (d) none of these

16. At the eve of marriage anniversary of Tristan and Iseult some special couples were invited. Out of them only 23 couples were there having one child per couple with them. When I observed it, I found that 13 men were without any child and 20 women were also without any child. It is known that there were only couples allowed to attend the ceremony. For your kind information we will not consider the couple (*i.e.*, Tristan and Iseult) for any calculation, since they are hosts. The total number of couples who attended the party (*i.e.*, anniversary).
(a) 28 (b) 30
(c) 33 (d) can't be determined

17. The number of numbers less than or equal to 666 which are the products of exactly 4 distinct positive prime numbers is :
(a) 4 (b) 8
(c) 7 (d) can't be determined

18. Tata, Hutch and Idea started of with a game. The rule is that the loser of the game distributes $\frac{1}{3}$rd of his money amongst the other two players in the ratio of the amount as they are holding with them. After playing three round game Tata, Hutch and Idea endup with Rs. 450, 150 and 300 respectively. The total amount of money what they were initially holding is :
(a) 575 (b) 750
(c) $\frac{5}{4}(6!)$ (d) can't be determined

19. Remainder when $4^{4^n} + 3$ is divided by 7 is :
(a) 0 (b) 1
(c) 5 (d) none of these

20. All the soldiers are arranged in the form of an equilateral triangle *i.e.*, one soldier in the front and 2 soldiers in the second row and 3 soldiers in the third row, 4 soldiers in the fourth row and so on. If 669 more soldiers of another company are added in such a way that all the soldiers now are in the form of an square and each of the sides then contain 8 soldiers less than each side of equilateral triangle. Initially, how many soldiers were there?
(a) 2056 (b) 1540
(c) 1400 (d) 1220

21. A natural number when divided by the quotient, which in turn obtained by dividing the leading digit by the unit digit of the same number, we get another number consisting of the same digits, but in reverse order. The original number is :
(a) 98658823 (b) 91703
(c) 87912 (d) 27918

22. One day my friend Dorsey told me from LA, that he gets the same salary in each month but when he adds up his six figure salary of two months, three months, four months, five months or six months each time the figures (or digits) of his salary remains the same, but not in a fix order. Later, he told me that when he add up his salary of 7 months his total salary become $(10^6 - 1)$. What is the salary of Dorsey?
(a) 123456 (b) 285614
(c) 142857 (d) can't be determined

23. A leading chocolate producing company produces '*abc*' chocolates per hour (*abc* is a three digit positive number). In how many hours it will produce '*abcabc*' chocolates?
(a) *abc* (b) 101
(c) 1001 (d) can't be determined

24. A number D236D0 can be divided by 36 if D is :
(a) 8
(b) 6
(c) 1
(d) more than one values are possible

25. In the christmas eve of his 7th birthday anniversary, Martin, the eldest son of his parents, went to picnic with all his family members. There his family purchased the tickets for everyone. The entire family consists of 3 children, 1 grand father, 1 grand mother, two fathers and two mothers 1 brother and 1 sister. Also there is one daughter-in-law, one father in law, one mother in law. Minimum how many tickets his family has purchased for the entire family?
(a) 5 (b) 6
(c) 8 (d) 14

26. In the above question (no. 25), if the rate of a ticket for a child (considered to be unmarried and below 18 years) is Rs. 6, while the price of a ticket for an adult (*i.e.*, married and above 18) is Rs. 10. So what is the minimum price has to pay for the tickets?
(a) Rs. 52 (b) Rs. 58
(c) Rs. 90 (d) data insufficient

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27. Which one of the following is not the correct relation?
(a) $(25)^3 + (38)^3 + (87)^3 = (90)^3$
(b) $(3)^3 + (4)^3 + (5)^3 = (6)^3$
(c) $(17)^3 + (29)^3 = (57)^3$
(d) $1^3 + 6^3 + 8^3 = 9^3$

28. In a college of 300 students, every student reads 5 newspapers and every newspapers is read by 60 students. The number of newspapers is :
(a) at least 30
(b) at most 20
(c) 25
(d) none of (a), (b) and (c)

29. At East End Mall, burgers can be bought in quantities of either 6, 9 or 20 only. For example one can purchase 26 (20 + 6), 15 (6 + 9) and 24 (9 × 2 + 6) etc. but can't purchase 10 burgers *i.e.*, without any lot. What is largest number of burgers that can not be purchased?
(a) 83 (b) 37
(c) can't be determined (d) none of (a), (b), (c)

30. Let there be a fraction whose denominator is one less than the square of its numerator. If we add 2 to both the numerator and denominator, the fraction will exceed to $\frac{1}{3}$ and if we subtract 3 from numerator and denominator the fraction will lie between 0 and $\frac{1}{10}$. The actual fraction is :
(a) $\frac{3}{8}$ (b) $\frac{7}{48}$
(c) $\frac{4}{17}$ (d) none of these

31. When a two digit number is subtracted from another two digits number consisting of the same digits in reverse order, then the resultant value we get is equal to the sum of the digits of that number. The square of sum of digits of that number can be :
(a) 49 (b) 64
(c) 81 (d) 36

32. If $A = [k^3 + (k+2)^3 + (k+4)^3 + (k+6)^3 + (k+8)^3 + \ldots (k+38)^3]$ and
$B = [k + (k+2) + (k+4) + (k+6) + (k+8) + \ldots (k+38)]$
Now, if 'A' is divided by 'B', then the remainder will be (for every positive integer k) :
(a) 0 (b) 1
(c) k (d) none of these

33. If $1^3 = 1$, $2^3 = 3 + 5$, $3^3 = 7 + 9 + 11$, $4^3 = 13 + 15 + 17 + 19$, $5^3 = 21 + 23 + 25 + \ldots + 29$, etc. Then the value of $(100)^3$ is equal to :
(a) $9901 + 9903 + \ldots + 10099$
(b) $9999 + 10001 + 10003 + \ldots + 10199$
(c) $9989 + 9991 + \ldots + 10089$
(d) Any two of (a), (b), (c)

34. If $\{x\}$ denotes the fractional part of x, then $\left\{\frac{5^{200}}{8}\right\}$ is :
(a) 1/8 (b) 1/4
(c) 5/8 (d) 0.625

35. Which one of the following is correct?
(a) $25^{26} > 26^{25}$ (b) $100^{101} < 101^{100}$
(c) $199^{198} > 198^{199}$ (d) $399^{400} < 400^{399}$

36. If x and y are positive prime number and if $x^2 - 2y^2 = 1$, then the value of $x + y$ is :
(a) 7 (b) 24
(c) 13 (d) none of these

37. The greatest possible divisor of $3^{2n+3} - 24n - 27$ for every $n \in N$, which necessarily divides is :
(a) 64 (b) 24
(c) 96 (d) none of these

38. If $(\sqrt{3} + 1)^5 = I + F$, where I is an integer and F is a proper fraction, then the value of $F.(I + F)$ is :
(a) $(\sqrt{3})^5$ (b) 32
(c) $(1.99)^5$ (d) 31.232

39. When the product of 'r' consecutive positive integers is divided by 'r!', then the quotient is :
(a) any natural number (b) a perfect square
(c) a proper fraction (d) either (b) or (a)

40. Two digits p, q, where $q > p > 1$ such that $p + q < pq$ and $(p+q)$, $(p+q)^n$, $(p+q)^{n+1}$ have the same unit digits and $(p.q)$, $(p.q)^n$, $(pq)^{n+1}$ have the same unit digits. The value of p is :
(a) 2 (b) 3
(c) 4 (d) 5

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Answers

INTRODUCTORY EXERCISE-1.1

1. (d)	2. (a)	3. (c)	4. (c)	5. (d)	6. (d)	7. (c)	8. (b)	9. (c)	10. (b)
11. (c)	12. (c)	13. (b)	14. (a)	15. (a)	16. (b)	17. (b)	18. (c)	19. (d)	20. (a)
21. (d)	22. (a)	23. (c)	24. (c)	25. (d)	26. (b)	27. (b)	28. (d)	29. (d)	30. (b)
31. (d)	32. (a)	33. (b)	34. (c)						

INTRODUCTORY EXERCISE-1.2

1. (c)	2. (d)	3. (b)	4. (d)	5. (d)	6. (d)	7. (b)	8. (b)	9. (c)	10. (c)
11. (c)	12. (d)	13. (d)	14. (d)	15. (a)					

INTRODUCTORY EXERCISE-1.3

1. (d)	2. (d)	3. (a)							

INTRODUCTORY EXERCISE-1.4

1. (d)	2. (d)	3. (b)	4. (d)	5. (d)	6. (d)	7. (d)	8. (c)	9. (b)	10. (a)
11. (b)									

INTRODUCTORY EXERCISE-1.5

1. (a)	2. (a)	3. (b)	4. (b)	5. (a)	6. (c)	7. (a)	8. (a)	9. (a)	10. (a)

INTRODUCTORY EXERCISE-1.6

1. (a)	2. (b)	3. (c)	4. (c)	5. (c)					

INTRODUCTORY EXERCISE-1.7

1. (c)	2. (b)	3. (a)	4. (a)	5. (b)	6. (a)	7. (d)	8. (b)	9. (c)	10. (d)
11. (a)	12. (b)								

MISCELLANEOUS

1. (a)	2. (a)	3. (b)	4. (a)	5. (a)	6. (d)	7. (b)	8. (d)	9. (b)	10. (b)
11. (b)	12. (c)	13. (d)	14. (c)	15. (a)	16. (d)	17. (d)	18. (c)	19. (d)	20. (d)
21. (d)	22. (c)	23. (d)	24. (a)	25. (b)	26. (b)	27. (d)	28. (b)	29. (a)	30. (c)
31. (a)	32. (d)	33. (a)	34. (a)	35. (a)	36. (a)	37. (a)	38. (d)	39. (a)	40. (d)
41. (d)	42. (d)	43. (d)	44. (d)	45. (a)	46. (b)	47. (a)	48. (b)	49. (a)	50. (c)
51. (a)	52. (c)	53. (c)	54. (b)	55. (c)	56. (d)	57. (c)	58. (d)	59. (d)	60. (c)
61. (a)	62. (a)	63. (d)	64. (a)	65. (b)	66. (c)	67. (c)	68. (a)	69. (d)	70. (b)
71. (a)	72. (d)	73. (b)	74. (b)	75. (b)	76. (a)	77. (b)	78. (a)	79. (d)	80. (c)
81. (a)	82. (a)	83. (c)	84. (c)	85. (b)	86. (d)	87. (c)	88. (d)	89. (d)	90. (d)
91. (b)	92. (b)	93. (c)	94. (d)	95. (c)	96. (d)	97. (c)	98. (b)	99. (c)	100. (d)
101. (c)	102. (c)	103. (d)	104. (a)	105. (c)	106. (c)	107. (d)	108. (a)	109. (b)	110. (c)
111. (b)	112. (d)	113. (a)	114. (b)	115. (c)	116. (a)	117. (c)	118. (d)	119. (c)	120. (d)
121. (c)	122. (d)	123. (a)	124. (b)	125. (c)	126. (b)	127. (a)	128. (a)	129. (b)	130. (a)
131. (c)	132. (b)	133. (b)	134. (c)	135. (c)	136. (d)	137. (d)	138. (b)	139. (c)	140. (c)

LEVEL-1

1. (b)	2. (b)	3. (b)	4. (b)	5. (c)	6. (c)	7. (a)	8. (d)	9. (d)	10. (d)
11. (c)	12. (b)	13. (d)	14. (b)	15. (d)	16. (b)	17. (c)	18. (b)	19. (c)	20. (c)
21. (d)	22. (b)	23. (d)	24. (c)	25. (b)	26. (a)	27. (a)	28. (d)	29. (d)	30. (a)
31. (c)	32. (b)	33. (b)	34. (d)	35. (d)	36. (c)	37. (d)	38. (c)	39. (c)	40. (c)

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41. (b)	42. (c)	43. (c)	44. (d)	45. (a)	46. (c)	47. (b)	48. (b)	49. (c)	50. (a)
51. (a)	52. (b)	53. (b)	54. (c)	55. (b)	56. (d)	57. (b)	58. (b)	59. (c)	60. (d)
61. (c)	62. (c)	63. (b)	64. (c)	65. (d)	66. (c)	67. (d)	68. (c)	69. (c)	70. (d)
71. (b)	72. (b)	73. (b)	74. (d)	75. (b)	76. (b)	77. (b)	78. (a)	79. (c)	80. (d)
81. (c)	82. (b)	83. (c)	84. (c)	85. (b)	86. (b)	87. (c)	88. (c)	89. (d)	90. (b)
91. (b)	92. (b)	93. (d)	94. (b)	95. (d)	96. (a)	97. (a)	98. (d)	99. (b)	100. (c)
101. (d)	102. (b)	103. (c)	104. (b)	105. (a)	106. (b)	107. (b)	108. (c)	109. (b)	110. (b)
111. (b)	112. (c)	113. (c)	114. (a)	115. (b)	116. (a)	117. (c)	118. (a)	119. (a)	120. (b)
121. (d)	122. (b)	123. (c)	124. (d)	125. (d)	126. (a)	127. (c)	128. (d)	129. (c)	130. (b)
131. (d)	132. (c)	133. (b)	134. (c)	135. (b)	136. (c)	137. (c)	138. (c)	139. (c)	140. (b)

LEVEL-2

1. (b)	2. (d)	3. (c)	4. (c)	5. (d)	6. (c)	7. (b)	8. (c)	9. (b)	10. (d)
11. (c)	12. (d)	13. (a)	14. (d)	15. (a)	16. (a)	17. (c)	18. (b)	19. (d)	20. (b)
21. (b)	22. (b)	23. (a)	24. (a)	25. (b)	26. (d)	27. (a)	28. (d)	29. (c)	30. (c)
31. (b)	32. (b)	33. (b)	34. (b)	35. (c)	36. (b)	37. (d)	38. (a)	39. (c)	40. (a)
41. (b)	42. (b)	43. (a)	44. (b)	45. (c)	46. (b)	47. (a)	48. (b)	49. (d)	50. (b)
51. (d)	52. (c)	53. (c)	54. (b)	55. (c)	56. (d)	57. (b)	58. (a)	59. (d)	60. (d)
61. (b)	62. (d)	63. (c)	64. (d)	65. (a)	66. (c)	67. (c)	68. (c)	69. (c)	70. (d)
71. (b)	72. (b)	73. (d)	74. (a)	75. (a)	76. (a)	77. (a)	78. (a)	79. (b)	80. (b)
81. (d)	82. (c)	83. (a)	84. (a)	85. (a)	86. (a)	87. (b)	88. (a)	89. (d)	90. (a)
91. (b)	92. (b)	93. (a)	94. (b)	95. (d)	96. (a)	97. (c)	98. (c)	99. (c)	100. (d)
101. (c)	102. (c)	103. (c)	104. (a)	105. (a)	106. (a)	107. (b)	108. (c)	109. (c)	110. (d)
111. (c)	112. (a)	113. (c)	114. (a)	115. (c)	116. (d)	117. (b)	118. (b)	119. (c)	120. (b)
121. (d)	122. (a)	123. (a)							

FINAL ROUND

1. (a)	2. (b)	3. (b)	4. (a)	5. (c)	6. (a)	7. (c)	8. (a)	9. (b)	10. (c)
11. (d)	12. (d)	13. (d)	14. (b)	15. (d)	16. (b)	17. (b)	18. (a)	19. (b)	20. (c)
21. (b)	22. (c)	23. (b)	24. (d)	25. (c)	26. (d)	27. (b)	28. (b)	29. (b)	30. (a)
31. (d)	32. (b)	33. (c)	34. (a)	35. (a)	36. (d)	37. (a)	38. (b)	39. (c)	40. (b)
41. (b)	42. (a)	43. (d)	44. (d)	45. (a)	46. (b)	47. (d)	48. (d)	49. (d)	50. (d)
51. (d)	52. (a)	53. (c)	54. (d)	55. (d)	56. (a)	57. (d)	58. (a)	59. (d)	60. (b)
61. (a)	62. (a)	63. (b)	64. (c)	65. (d)	66. (c)	67. (a)	68. (b)	69. (a)	70. (b)
71. (c)	72. (c)	73. (b)	74. (c)	75. (d)	76. (c)	77. (c)	78. (d)	79. (a)	80. (d)
81. (c)	82. (d)	83. (a)	84. (b)	85. (c)	86. (b)	87. (c)	88. (c)	89. (b)	90. (d)
91. (b)	92. (b)	93. (a)	94. (c)	95. (c)	96. (b)	97. (a)	98. (a)	99. (b)	100. (a)
101. (c)	102. (c)	103. (d)	104. (a)	105. (b)	106. (c)	107. (d)	108. (a)	109. (c)	110. (c)
111. (c)	112. (d)	113. (c)	114. (a)	115. (a)	116. (a)				

TEST OF YOUR LEARNING-1

1. (a)	2. (d)	3. (c)	4. (a)	5. (c)	6. (a)	7. (d)	8. (c)	9. (b)	10. (c)
11. (a)	12. (a)	13. (b)	14. (c)	15. (c)	16. (a)	17. (d)	18. (d)	19. (b)	20. (a)
21. (a)	22. (d)	23. (b)	24. (c)	25. (c)	26. (a)	27. (b)	28. (b)	29. (d)	30. (c)
31. (c)	32. (c)	33. (d)	34. (b)	35. (a)	36. (a)	37. (b)	38. (c)	39. (d)	40. (d)

TEST OF YOUR LEARNING-2

1. (a)	2. (c)	3. (b)	4. (c)	5. (d)	6. (d)	7. (c)	8. (c)	9. (d)	10. (a)
11. (d)	12. (b)	13. (c)	14. (a)	15. (a)	16. (a)	17. (c)	18. (c)	19. (a)	20. (b)
21. (c)	22. (c)	23. (c)	24. (a)	25. (b)	26. (a)	27. (c)	28. (c)	29. (d)	30. (d)
31. (c)	32. (a)	33. (a)	34. (a)	35. (a)	36. (d)	37. (d)	38. (b)	39. (a)	40. (a)

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Hints & Solutions

INTRODUCTORY EXERCISE 10.1

1. As per the given choices first of all we check it by 99 since in the question largest possible number of two digits is being asked.
Now, since $(2 + 7 + 7 + 2) - (8 + 9 + 1) = 18 - 18 = 0$
Hence it is divisible by 11
again $2 + 1 + 7 + 9 + 7 + 8 + 2 = 36$
Hence it is also divisible by 9.
Thus we can say that the given number 2179782 is divisible by 99 $(= 11 \times 9)$
So, the option (d) is correct answer.

2. A number which is multiple of 125 must be divisible by 125. So the number formed by last three digits must be divisible by 125 hence it can be 000, 125, 250, 375, 500, 675, 750, 875.
So the number which is divisible by 125 must be 9999875 since we have to subtract minimum possible number. Thus we see that we have to subtract 124.
[$\because$ 9999999 – 9999875 = 124]
Alternatively : Divide 9999999 by 125 and then the obtained remainder is the required result.

3. As per the divisibility rule of 11, the difference of sum of the digits at odd places and sum of the digits at even places be either 0, 11, 22, 33, ... etc. So n must be an even number.
Correct choice is (c).
Alternatively : Go through the options. Now when you write prime of any odd value of n you will find odd prime or odd number of nines then you will also observe that these numbers are not divisible by 11.
Further using option (c) you will find that it is always correct for any even number. Thus option (c) is correct.

4. If we look at the options we see that option (a) is not correct since in the first instance it is not divisible by 4. Again if we look at option (b) we see that it is not divisible by 3. Similarly option (d) is wrong because it is not divisible by 11 and at last when we check the option (c) then we find it is correct. Remember a number is divisible by 132 if the number is divisible by its factors as 3, 4, and 11 simultaneously as $132 = 3 \times 4 \times 11$
Thus option (c) is correct.

5. Since $80 = 8 \times 10$ or $80 = 16 \times 5$
Thus y (*i.e.*, unit digit) must be zero.
$\therefore$ $653xy = 653x0$, where $653x0$ must be divisible by 16 or $653x$ is divisible by 8.
Thus the last 3-digit number $53x$ will be divisible by 8.
Hence, at $x = 6$, we get the required result.
$\therefore$ $x + y = 6 + 0 = 6$

6. The best way to use the options
Alternatively : $(4 + 6 + 3) - (2 + 5 + k) = 0 / 11 / 22 \ldots$
$(13) - (7 + k) = 0$ [$\because$ 11/22 ... are not possible]
$\Rightarrow$ $k = 6$
Hence option (d) is correct.

7. The quickest method is to use appropriate option. As the divisor is 72 so the given number must be divisible 9 and 8 both. Therefore, it is clear that option (b) and (d) are useless, since unit digit can't be odd inorder to be divisible by 8. Now option (a) and (c) can be tried out. Thus we can get the value of $k = 6$. Hence option (c) is correct.

8. The first number which is divisible by 7 is 7 and the last number is 994. It can be easily calculated just by dividing 1000 by 7 and then subtract the remainder from 1000. So total numbers between 1 and 1000 which are divisible by 7 are 142.The sequence is as follows 7, 14, 21, 28, 35, ... 987, 994.

9. The sequence is as follows 55, 60, 65, 70, ..., 550, 555. So the number of number divisible by 5 is $\left(\frac{555 - 55}{5}\right) + 1 = 101$
Note we have added 1 from outside because here both the extremes (i.e., 55 and 555) are included.
SHORTCUT If D be the divisor of numbers lying between N_1 and N_2, $N_2 > N_1$ then the number of numbers which are divisible by D (or multiple of D) is
(a) When no one extreme is divisible by such a divisor D :
$$\left(\frac{N_2 - N_1}{D}\right) - 1$$
(b) When any one of the extreme is divisible by D :
$$\left(\frac{N_2 - N_1}{D}\right)$$
(c) When both the extremes are divisible by D is :
$$\left(\frac{N_2 - N_1}{D}\right) + 1$$

10. $(200 - 100) + 1 = 101$
Alternatively : Since you know that you are counting the numbers from 100, so only 99 numbers (1, 2, 3, ..., 99) are not included
i.e., 1, 2, ... 99, 100, 101, 102, 103, ..., 200.
Hence total numbers = 200 – 99 = 101.

11. The first number which is divisible by 3 is 300 and the last number which is divisible by 3 is 498.
So, the total numbers = $\left(\frac{498 - 300}{3}\right) + 1 = 66 + 1 = 67$
Hence (c) is the correct option.
Alternatively : When we form the sequence of numbers which are divisible by 3, the number 300 is lying at the 100th position and 498 at the 166th position (i.e., 3, 6, 9, 12, ... 297, 300, 303, 306, ..., 498) so there are total 166 numbers upto 498 but 99 numbers are not included in this sequence. Thus the required numbers are 166 – 99 = 67 numbers, which are divisible by 3.

12. In the given range, the least number which is divisible by both 5 and 7. i.e., 35 is 210 and the highest number is 770. So the total number of numbers between 200 and 800 which are divisible by both 5 and 7 is $\left(\frac{770 - 210}{35}\right) + 1 = 17$
Hence option (c) is correct.

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13. Total numbers in the set are

$$(800 - 200) + 1 = 601$$

Total number of numbers which are divisible by 5

$$= \frac{(800 - 200)}{5} + 1 = 121$$

Total number of numbers which are divisible by 7

$$= \frac{(798 - 203)}{7} + 1 = 86$$

Total number of numbers which are divisible by both 5 and 7

$$= \frac{770 - 210}{35} + 1 = 17$$

So the total number of numbers which are either divisible by 5 or 7 or both = (121 + 86) – 17 = 190
Hence option (b) is correct.

14. Total numbers in the set $= (800 - 200) + 1 = 601$
Number of numbers which are divisible by 5

$$= \frac{(800 - 200)}{5} + 1 = 121$$

Number of numbers which are divisible by 7

$$= \frac{(798 - 203)}{7} + 1 = 86$$

Number of numbers which are divisible by both 5 & 7

$$= \left(\frac{770 - 210}{35}\right) + 1 = 17$$

∴ Number of numbers which are either divisible by 5 or 7 or both

$$= (121 + 86) - 17 = 190$$

Thus the number of numbers in the given set which are neither divisible by 5 nor by 7 = 601 – 190 = 411.
Hence (a) is correct option.

15. To find the answer of this problem first of all we need to calculate total number of numbers in the given set which are divisible by any of 2, 3 or 5 then these numbers will be subtracted from the total number of the original set of numbers. Thus we get the total number of numbers which are neither divisible by 2, 3 or 5.
Total numbers in the set = (3113 – 1331) + 1

$$= 1782 + 1 = 1783$$

Total number of numbers in the set which are divisible by 2

$$= \left(\frac{3112 - 1332}{2}\right) + 1 = 891$$

Total numbers divisible by 3 $= \left(\frac{3111 - 1332}{3}\right) + 1 = 594$

Total numbers divisible by 5 $= \left(\frac{3110 - 1335}{5}\right) + 1 = 356$

Total numbers which are divisible by 2 and 3 both

$$= \left(\frac{3108 - 1332}{6}\right) + 1 = 297$$

Total numbers which are divisible by 3 and 5 both

$$= \left(\frac{3105 - 1335}{15}\right) + 1 = 119$$

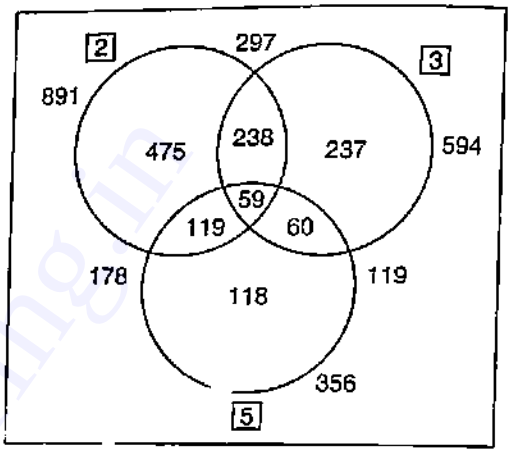
Total numbers which are divisible by 2 and 5 both

$$= \left(\frac{3110 - 1340}{10}\right) + 1 = 178$$

Total numbers which are divisible by 2, 3 and 5

$$= \left(\frac{3090 - 1350}{30}\right) + 1 = 59$$

So, the total number of numbers in the given set which are neither divisible by any of the 2, 3 or 5 = 477.



$\because\ n(A \cup B \cup C) = [n(A) + n(B) + n(C)]$
$- [n(A \cap B) + n(B \cap C) + n(A \cap C)] + n(A \cap B \cap C)$
So the divisible numbers
= (891 + 594 + 356) – (297 + 119 + 178) + 59
= 1306
∴ Number of numbers which are not divisible by 2, 3 or 5.
= 1783 – 1306 = 477
Hence (a) is the correct option.
Alternatively : Number of numbers in the given set which are only divisible by 2 = 891
the number of numbers which are only divisible by 3, but not by 2

$$= \frac{(3111 - 1335)}{6} + 1 = 297$$

The number of numbers which are only divisible by 5 but not by 2or 3

$$= \left\{\frac{(3105 - 1335)}{10} + 1\right\} - \left\{\frac{(3105 - 1335)}{30} + 1\right\}$$
$$= 178 - 60 = 118$$

So the total number of numbers which are divisible by any of the 2, 3 or 5 = 891 + 297 + 118 = 1306
Thus the total number of numbers which are neither divisible by 2, 3 or 5 = 1783 – 1306 = 477.

16. Just divide 434079 by 137 and then the obtained remainder is your correct answer which is 63, so (b) is the right choice.

17. To make 434079 divisible by 137 we have to add just the difference of remainder and the divisor. So the choice (b) is correct and the answer is 74.

18. Obviously (c) is the right choice because there can be maximum two numbers closest to any particular number. So out of 180 and 198 the number 198 is closer than the number 180 and 195 is not divisible by 18 also 108 is the farthest from 193.

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19. Since when we multiply the least three digit numbers ($100 \times 100 = 10000$) we get a 5 digit number and when we multiply the greatest possible 3 digit numbers ($999 \times 999 = 998001$) we get a 6 digit number. So anyhow the product of a 3 digit number with another 3 digit number can never be a number of less than 5 digit or greater than 6 digit number. So the option (a) and (c) are ruled out because these are 4 digit and 7 digit numbers respectively.
Further whenever we multiply any two numbers whose unit digits are 7 and 5 individually then the unit digit of the resultant number is always 5 ($\because 5\times1=5,\ 5\times3=15,\ 5\times5=25,\ 5\times7=35,\ 5\times9=45$ and $5\times2=10,\ 5\times4=20,\ 5\times6=30,\ 5\times8=40$, and $5\times10=50$)
Thus the option (b) is also wrong since the required unit digit can never be 6 (i.e., an even number).
Hence the appropriate option is (d).

20.
$$\begin{array}{r} 9\ 8\ 4 \\ +\ 4\ p\ 3 \\ \hline 13\ q\ 7 \end{array}$$

It is clear from the above expression that there is no any carry over obtained in the addition of unit digits. Further it is very simple to see that there is no carry over value taken while adding hundredths digit of both the numbers. So we get $8 + p = q$
Now if $13q7$ is divisible by 11 then

$$(7+3)-(1+q)=11k \qquad \text{where, } k=0,1,2,\ldots\ldots$$
$$10-(1+q)=0$$
$$\Rightarrow \quad q=9$$

here $k = 0$ is the only possible value because if
$k = 1$ then q must be either -2 or 20.
but neither -2 nor 20 is possible because q is a single positive digit i.e., $q \in 1, 2, 3, 4, 5, 6, 7, 8, 9, 0$.
Thus the value of $p + q = 1 + 9 = 10$ $\quad [\because p = q - 8,\ \therefore p = 1]$

Hence, (a) is the correct option.

21. Since Dividend = Divisor × Quotient + Remainder
$\therefore$ Dividend $= 9235 \times 888 + 222$
Thus the number = 8200902
Hence (d) is the correct choice.

22. Option (a) is correct. There are only two relevant options (a) and (c). Option (b) is redundant since 999 is not divisible by 33. Now 990 and 1023 both are close to 1000 and both are divisible by 33 but it is obvious that 990 is much closer to 1000 than 1023.

23. Let this number be N then

$$N = 32 \times Q_1 + 29 \qquad \ldots(1)$$
Again $$N = 8 \times Q_2 + R \qquad \ldots(2)$$
From (1) and (2)
$$32Q_1 + 29 = 8Q_2 + R \quad (\text{where } R \text{ is the remainder})$$
$$8Q_2 - 32Q_1 = 29 - R$$
$$8(Q_2 - 4Q_1) = 29 - R$$
or $$(Q_2 - 4Q_1) = \frac{29-R}{8}$$

Since Q_1, Q_2, R are integers also $Q_2 - 4Q_1$ is an integer. Therefore $29 - R$ must be divisible by 8.
So the probable values of R are 29, 21, 13 and 5.
But since the remainder can never be greater than the divisor so the only possible value of R is 5. Hence (c) is the correct option.
Alternatively : $N = 32Q + 29 = 8 \times 4Q + 8 \times 3 + 5$ (where Q is Quotient)

$$\Rightarrow \quad N = 8(4Q + 3) + 5$$
$$\Rightarrow \quad N = 8P + 5$$

(where p is any positive integer)
Thus the remainder is 5.
Shortcut : In this kind of question where the same number is divided by another divisor (proper factor of the original divisor), we just need to divide the original remainder by the new divisor to get the value of new remainder e.g. when we divide 29 by the new divisor '8' then we get new remainder 5.

24. Let the number be N and the quotient be Q then $N = 5Q + 4$

Again $$2N = 2(5Q + 4) = 10Q + 8$$
or $$2N = 5 \times 2Q + 5 + 3$$
or $$2N = 5(2Q + 1) + 3$$

Hence, the remainder will be 3.
Alternatively : Since the number becomes double then the remainder will also be double but when the remainder will become twice it will exceed the divisor which is not possible so to get the correct remainder we further divide the twice of the remainder by the divisor and in this way we obtain the required remainder which is admissible. For example in the given problem the divisor is 5 and the original remainder is 4. So this remainder will become twice i.e., 8 (since the number i.e.,dividend becomes twice) but since $8 > 5$ which is not possible so we divide 8 by 5 then we get the correct remainder 3.

25. In the first case $N = DQ + 14$
In the second case $3N = 3(DQ + 14)$

$$= 3DQ + 42$$
$$= 3DQ + 34 + 8$$
$$3N = (3DQ + 34) + 8$$

It shows that 34 must be divisible by 34 or its factor i.e., 1,2, 17,34. But 1 and 2 can't be values of D. Since the divisor D must be greater than the remainder (*viz.* 8, 14) Hence we are now left with two possible values of D *viz.* $D = 17$ and $D = 34$
Now, if remainder $D = 17$, then

$$N = 17Q + 14$$
$$4N = 4(17Q + 14) = 4 \times 17Q + 56$$
$$= 17 \times 4Q + 17 \times 3 + 5$$
$$4N = 17(4Q + 3) + 5$$

Thus the remainder will be 5.
But if we consider $D = 34$, then

$$N = 34Q + 14$$
$$4N = 4(34Q + 14)$$
$$= 34 \times 4Q + 56$$
$$= 34 \times 4Q + 34 + 22$$
$$4N = 34(4Q + 1) + 22$$

Thus the remainder will be 22.
Hence we can't say exactly whether the remainder will be 5 or 22. So (d) is the correct option.

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26. Let the number be N then

$$N = 34Q + R, \quad \ldots(1)$$

where Q is any quotient

Again $N = 5D$ and D is also a quotient

but $D = R + 8$

so $N = 5(R + 8)$ $\quad \ldots(2)$

$\therefore \quad 5(R + 8) = 34Q + R$

$5R + 40 = 34Q + R$

$\Rightarrow \quad 34Q - 40 = 4R$

$\Rightarrow \quad 17Q - 2R = 20$

So the minimum possible value of $Q = 2$ and the corresponding value of $R = 7$

So $N = 34 \times 2 + 7$

$N = 75,$

Hence (b) is correct choice.

27. Let N be the number and d be the divisor then $N = dQ + 15$

Again, $10N = 10(dQ + 15)$

$\Rightarrow \quad 10N = 10dQ + 150$

$\Rightarrow \quad 10N = d\,10Q + 144 + 6$

$\Rightarrow \quad 10N = (d \times 10Q + 144) + 6$

$\Rightarrow \quad 10N = d\left(10Q + \frac{144}{d}\right) + 6$

$\therefore$ 144 must be divisible by d. Thus the probable values of d are 1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 36, 48, 72, 144 (as these numbers are the factors of 144).

But the value of divisor d must be greater than the remainder (here 6 and 15) so 1, 2, 3, 4, 6, 8, 9, 12 can not be the values of d. Hence only 7 values (greater than 15) are possible.

Thus the option (b) is correct.

28. $C = N_1Q_1 + 13$ and $C = N_2Q_2 + 1$

where Q_1 and Q_2 are the quotients and hence Q_1 and Q_2 must be integers

So, $N_1Q_1 + 13 = N_2Q_2 + 1$

$N_2Q_2 - N_1Q_1 = 12$

$\frac{4}{5}N_1Q_2 - N_1Q_1 = 12$

$N_1(4Q_2 - 5Q_1) = 60$

where N_1 and $(4Q_2 - 5Q_1)$ both will be integers so that N_1 can be one of the values of factors of 60 *i.e.*, 1, 2, 3, 4, 15, 20, 30 and 60. But N_1 can not be less than the remainder 13. So the possible values are 15, 20, 30 and 60. Thus we see that the corresponding values of N_2 are 12, 16, 24 and 48.

So there are more than one possible values of N_1 and N_2. Thus there is no unique value of $N_1 + N_2$. Hence (d) is correct option.

29. Since we know that the values of N_1 and N_2 are not unique so the value of C is also not unique hence it is indeterminable.

30. Number of parts $= \dfrac{\text{Total length}}{\text{length of each piece}}$

$= \dfrac{1950}{65} = 30$ $\quad (\because 19.5 \text{ m} = 1950 \text{ cm})$

Thus option (b) is correct.

31. Since the 7, 11 and 13 all are the factors of such a number so (d) is the correct answer.

32. (a) is the correct option. Since 5321 is the greatest and 1235 is the least number then $5321 - 1235 = 4086$.

33. The possible values of k are 5 and 9. So the sum of $5 + 9 = 14$.

34. When we multiply 59829 with 13 we get the required number 777777. So (c) is the correct option.

INTRODUCTORY EXERCISE 1.2

1. Option (a) and (b) are ruled out due to unit digit property. Now to check whether (c) is correct or not. Now it would be better to square 2506 instead of finding the square root of 6280036.

Let us find the square of 2506.

Step 1. $(2505)^2 + (2505 + 2506)$

Step 2. $(2500)^2 + 5(2500 + 2505) + 5011$

Step 3. $6250000 + 25025 + 5011$

Step 4. $6275025 + 5011 = 6280036$

Hence option (c) is correct.

2. From your basic knowledge you know that $(30)^2 = 900$ again $(40)^2 = 1600$. This shows that the square root lies between 30 and 40.

Further you know that $(35)^2 = 1225$. From this it is clear that only option (d) is correct.

3. Since you know that $(80)^2 = 6400$ and $(90)^2 = 8100$. It shows that the square root of 7744 must lie between 80 and 90. So now there is only one option to check i.e.,,88. So we try to square 88 instead of finding the square root of 7744.

So $(88)^2 = (11 \times 8)^2 = 11^2 \times 8^2$

$= 121 \times 64 = 7744$

Hence option (b) is correct.

4. First of all option (a) is ruled out due to unit digit property. Again if we do some rough calculation we see that $(300)^2 = 90000$ then the value of $(323)^2$ will be very large even more than 90000 so this option is also ruled out. Now, if we find the square of 225 we will get $(225)^2 = 50625$ and the value of $(230)^2 = 52900$ since the value of 227 lies between 225 and 230, but 56169 does not lie between these values so this value is also not correct. Hence option (d) is correct.

5. (d) is right option since 8 can't be the unit digit of the square of any natural number.

6. Since the factors of 11760 are $2 \times 2 \times 2 \times 2 \times 3 \times 5 \times 7 \times 7$ so we need to multiply it with 3×5 because all the factors are paired but 3 and 5 are unpaired, hence (d) is the correct choice.

7. Since we know that there are two factors viz. 3 and 5 which are unpaired so if we remove them the number will become a perfect square. Hence we have to divide11760 by 15 $(= 3 \times 5)$ in order to make a perfect square number.

8. This problem can't be solved by factorisation because we need not factor. So we have to solve it by division method as follows

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	23.9
2	5 75
2	4
43	175
3	129
469	4600
9	4221

(If the number is not a perfect square then by putting decimal we can increase the zeros in pairs for further calculation.)
The result obtained is ≈ 23.9.
So by adding some number we can make it the perfect square of 24. Now since we know that $(24)^2 = 576$. So we need to add 1 (∵ $576 - 575 = 1$)
Thus (b) is the correct option.
Alternatively : Using options we can solve this problem as if we consider option (a) then 575 itself be a perfect square but its not a perfect square. Again if we add 1 (i.e.,, using option (b) we get the number 576 and then check it, we find that 576 is a perfect square. Hence (b) is correct.
Alternatively : Since we know that $(20)^2 = 400$ and $(25)^2 = 625$. It means the value of perfect square must lies in the range of 400 and 625. So we can try it manually and get that $(23)^2 = 529$ and $(24)^2 = 576$. So simply we need to add 1 to make a perfect square number.

9. From the discussion done in the previous question we know that we have to subtract 46 to make the perfect square number. Since the closest perfect square numbers are 529 and 576. Since we have to subtract the number therefore we choose 529 and then we get (575 – 529 =) 46.
Hence (c) is the correct option.
Alternatively : If we subtract 5 then 570 is not a perfect square. Again if we subtract 38 then 538 is not a perfect square (simply due to unit digit concept of 2, 3, 7 and 8). Further if we subtract 46 we get 529 which is found to be a perfect square.

NOTE In the checking through option we have to follow the basic constraints imposed by the problem and here we have to find the least possible number so we start from the least option towards higher valued option *i.e.*, 5 then 37 then 46 and then 50.

10. It is given that $A^2 = B^2 + C^2$
But $\frac{B}{C} = \frac{12}{5}$ and B, C are integers.
So the least possible values of B and C are 12 and 5 respectively. Hence

$$A^2 = 12^2 + 5^2$$
$$A^2 = 169$$
$$A = 13$$

11. Since Lieutenant Kalia can arrange only 1444 soldiers in the square form. Now if he had arranged n soldiers in each of the n rows then

$$n \times n = 1444 \Rightarrow n = 38$$

Hence option (c) is correct.

12. $$\frac{\sqrt{289}}{\sqrt{x}} = \frac{1}{5} \Rightarrow \frac{289}{x} = \frac{1}{25}$$
$$\Rightarrow x = 7225$$

13. Option (d) is the required answer.

14. By observation we get to know that

$$2 * 3 = \sqrt{2^2 + 3^2} = \sqrt{13}$$
and $$3 * 4 = \sqrt{3^2 + 4^2} = 5$$
then $$5 * 12 = \sqrt{5^2 + 12^2} = 13$$

15. $$6 * 15 * 3 = \sqrt{\frac{(6+2)(15+3)}{(3+1)}} = \sqrt{\frac{8 \times 18}{4}} = 6$$

INTRODUCTORY EXERCISE 1.3

1. Let the numbers be $2a$ and $2b$ since 2 is the HCF of the numbers.
Therefore $2a \times 2b = 84$
$\Rightarrow$ $ab = 21$
Now, since $21 = 1 \times 21 = 3 \times 7$
$\therefore$ $(a, b) = (1, 21)$ and $(3, 7)$
Hence the values of pairs = (2, 42), (6, 14)
Thus there are only 2 pairs of numbers whose HCF is 2 and the product is 84.

2. Let the numbers be $6a$ and $6b$, since 6 is the HCF of the required numbers.
then $6a \times 6b = 15120$
$\Rightarrow$ $ab = 420$
Now $420 = 1 \times 420$
$= 2 \times 210 \rightarrow \times$
$= 3 \times 140$
$= 4 \times 105$
$= 5 \times 84$
$= 6 \times 70 \rightarrow \times$
$= 7 \times 60$
$= 10 \times 42 \rightarrow \times$
$= 12 \times 35$
$= 14 \times 30 \rightarrow \times$
$= 15 \times 28$
$= 20 \times 21$
Thus out of 12 pairs only 8 pairs are possible since the rest 4 pairs are not the coprimes.
Since 6 is the highest common factor of the two numbers (*i.e.*,, pair of numbers)
So further there must not be any number (or factor) common in both the numbers otherwise the HCF will be changed so the only possible values of (a, b) are
(1, 420), (3, 140), (4, 105), (5, 84), (7, 60), (12, 35), (15, 28) and (20, 21).
Thus the possible numbers can be
(6, 520), (18, 840), (24, 630), (30, 504), (42, 360), (72, 210), (90, 168), (120, 126).
Hence, there are 8 pairs of such numbers whose HCF is 6 and the product is 15120.

3. (a) Only two pair of numbers.

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INTRODUCTORY EXERCISE 1.4

1. 28 is the largest possible number by which we can divide the given numbers in order to obtain the same remainders and thus dividing by 28 we get the remainder 20 in each division.

2. Let the two numbers of a pair be $5a$ and $5b$ then

$$5a + 5b = 50 \Rightarrow 5(a + b) = 50$$

$$\Rightarrow (a + b) = 10$$

So the possible values of (a, b) can be $(1, 9), (2, 8), (3, 7), (4, 6), (5, 5)$ but $(2, 8), (4, 6)$ and $(5, 5)$ are inadmissible since these pairs are not the coprimes, hence they can change the value of HCF. Thus only two pairs [$(1, 9)$ & $(3, 7)$] are possible. Thus the actual pairs of such numbers are $(5, 45)$ and $(15, 35)$.

3. The largest possible length of the tape = HCF of 525, 1050, 1155 = 105

Hence (b) is the correct answer.

4. Minimum number of attempts $= \dfrac{\text{Total length of cloths}}{\text{Max. possible length of tape}}$

$$= \frac{2730}{105} = 26$$

Hence in 26 attempts the whole length of cloth can be measured using the tape of 105 cm length.

5. In order to have minimum number of tiles we need the largest possible tiles of square shape (in area).
So the maximum possible dimension (length and breadth) of a tile is the HCF of the dimension of the room.
Thus the HCF of 462 and 360 is 6 cm.
Hence the area of a square tile be $6 \times 6 = 36\ \text{cm}^2$.
Now the number of tiles required

$$= \frac{\text{Area of floor of the room}}{\text{Area of one tile}}$$

$$= \frac{462 \times 360}{6 \times 6} = 77 \times 60 = 4620$$

Thus total 4620 tiles are required.

6. (d) is correct answer. The numbers will be 195 and 143.

7. (d) is the correct option. In the options (a), (b) and (c) the HCF is not 12.

8. The maximum capacity of the vessel = HCF of 1653, 2261 and 2527 = 19
Hence (c) is the correct option.

9. To be the minimum number of pencils the size of all the pencils must be maximum. So the length of each pencil = HCF of 24 and 42 = 6

Thus the number of pencils $= \dfrac{\text{Total length of pencils}}{\text{length of one pencil}}$

$$= \frac{24 + 42}{6} = 11$$

Thus (b) is the correct choice since there can be min. 11 pencils of same size.

10. (a) The correct pair of numbers is (27, 189).

11. Minimum number of rows means max. number of trees per row, also equal number of trees per row is required so we need to find the HCF of 36, 144 and 234 to find the maximum number of trees in a row.
Thus HCF of 36, 144 and 234 = 18

Thus the number of rows $= \dfrac{\text{Total no. of trees}}{\text{No. of trees in a row}}$

$$= \frac{36 + 144 + 234}{18} = 23$$

Hence (b) is correct answer.

INTRODUCTORY EXERCISE 1.7

1. Going through options :
Option (a) is invalid since $5 > 2$
Try option (c) $2 \times 40^1 + 5 \times 40^0 = 80 + 5 = 85$
Hence (c) is the correct option.

Alternatively : $(25)_x = (85)_{10}$

$$\Rightarrow 2 \times x^1 + 5 \times x^0 = (85)_{10}$$

$$\Rightarrow 2x + 5 = 85$$

$$\Rightarrow 2x = 80$$

$$\Rightarrow x = 40$$

Hence (c) is the correct option.

2. Obviously option (b) is the required answer.
Since base must be greater than any integer of the number.

3. Going through option (b) is the invalid option.
Option (d) is also wrong since $(53)_6 = 33$ which is not a prime number and option (c) is also wrong since $(53)_{12} = 60 + 3 = 63$ which is not a prime number.
Hence option (a) is the correct. You can verify by assuming the value of n.

4. Since the last three digits in the given number are divisible by 2^3 (= 8). Hence the equivalent binary digits will be 000.
Alternatively : Divide the given number successively by 2 and get the last three digits as

2	365247728	
2	182623864 →	0
2	91311932 →	0
	45655966 →	0

Remainders

Thus the last three digits in the binary representation will be 000.

5. Since last three digits (*i.e.*, 956) is not divisible by 2^3 (= 8)
Hence all the last three digits will not be zero. Now since 956 is divisible by 4. Hence the last two digits will be 00. Thus only option (b) is correct.

NOTE Option (a) is invalid.

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6.

2	8009	
2	4000 →	1
2	2000 →	0
2	1000 →	0
	500 →	0

Remainders

Hence the last four digits in the binary representation of the given number are 0001.
Hence (a) is the correct option.

7. Going through option :
Let us assume option (c) ,then

2	8	
2	4 →	0
2	2 →	0
2	1 →	0
	0 →	1

Since $(1000)_2$ does not satisfy the given condition hence presumed option is wrong.
Now, consider option (d).

2	9	
2	4 →	1
2	2 →	0
2	1 →	0
	0 →	1

Thus $(9)_{10} = (1001)$, which satisfies all the given condition hence correct option.

8. Let the tens digit and unit digit of the original number be a, b then the original number $= 10a + b$
Again if the digits are reversed then the new number $= 10b + a$
So $(10a + b) - (10b + a) = 9(a - b)$
Now since it is given that $a + b = 9$ and '9 . $(a - b) = 63$
So solving for $(a + b)$ and $(a - b)$, we get $a = 8$ and $b = 1$
It means the given two digit number is 81 and its reverse is 18.
Further $(81)_x = 5(18)_x$

$$\Rightarrow \quad 8x + 1 = 5[x + 8]$$
$$\Rightarrow \quad 8x + 1 = 5x + 40$$
$$\Rightarrow \quad 3x = 39$$
$$x = 13$$

Thus the value of $x = 13$ and hence option (b) is correct.

9.
$$(25)_n \times (31)_n = (1015)_n$$
$$\Rightarrow \quad (2n + 5)(3n + 1) = n^3 + n + 5$$
$$\Rightarrow \quad 6n^2 + 17n + 5 = n^3 + n + 5$$
$$\Rightarrow \quad n^3 - 6n^2 - 16n = 0$$
$$\Rightarrow \quad n(n^2 - 6n - 16) = 0$$
$$\Rightarrow \quad n = 0 \quad \text{or} \quad n^2 - 6n - 16 = 0$$
$$\text{or} \quad n^2 - 6n - 16 = 0$$
$$\Rightarrow \quad (n - 8)(n + 2) = 0$$
$$\Rightarrow \quad n = 8 \quad \text{or} \quad n = -2$$

So the only admissible value of $n = 8$
Now, $(13)_8 \times (52)_8 = (11)_{10} \times (42)_{10}$
$= (462)_{10} = (716)_8$
Hence option (c) is correct.

10. $(100000\ldots001)_{16} = 1 \times (16)^{24} + \underbrace{0 + 0 + \ldots}_{23 \text{ zeros}} + 1 \times 16^0$

$$= 16^{24} + 1$$
$$= (2^4)^{24} + 1 = 2^{96} + 1$$

11. Since 2^{111} is divisible by 2^3 (*i.e.*,,8) hence the unit digit in octal system will be 0.

12. Option (b) is correct.

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MISCELLANEOUS

1. $\because \frac{-2}{3} = -0.66$ and $\frac{5}{2} = 2.5$ and $\frac{-1}{6} = -0.16$

$\therefore$ The correct required order is

$$\frac{5}{2} > \frac{-1}{6} > \frac{-2}{3}$$

Thus (a) is the correct answer.

2. $x + \frac{1}{x} = 3\left(x - \frac{1}{x}\right) \Rightarrow \frac{x^2+1}{x} = 3\left(\frac{x^2-1}{x}\right)$

$\Rightarrow \quad 2x^2 = 4 \Rightarrow x^2 = 2 \Rightarrow x = \pm\sqrt{2}$

Hence, (a) is correct.

3. Let us assume some values as 4 and 2, then

$$4 > 2 \Rightarrow -4 < -2$$

Hence, (b) is correct.

NOTE Other three option are wrong as

(a) $4 > 2 \Rightarrow -4 < -2$

(c) $4 > 2 \Rightarrow \frac{1}{4} < \frac{1}{2}$

(d) $4 > 2 \Rightarrow -\frac{1}{4} > -\frac{1}{2}$

4. Go back to the Basics.

5. $1780 + x = 1849$, which is nearest perfect square.

So, $x = 1849 - 1780$

$x = 69$

Hence, (a) is the least possible number.

Remark : If you have any problem in finding the value of nearest perfect square so try the following method.

$(40)^2 = 1600$, $(50)^2 = 2000$ it means your required value is lying somewhere between 40^2 & 50^2. Now $45^2 = 2025$, so it is clear that the value is lying between $(40)^2$ & $(45)^2$. Now in this way you can attain the required value.

6. Statement (1) and (2) are wrong since when p is a prime number so it does not have any factor so when all the factors (or numbers) before 'p' do not involve in the product so it is not divisible by p or any prime number greater than p. Statement (3) is wrong, since $1 \times 2 \times 3 \times 4 \times 5 \times 6$ is divisible by 5.

7. Statement (1) is true since all the prime numbers are either at a difference of 2 or 4. So for a suitable value of n it can be expressed. Statement (2) is correct since $(p-1)$ and $(p+1)$ both must be even numbers so $2k \times 2l = 4kl$. Thus it must be divisible by 4.

8. Since we know that $111 = 3 \times 37$, so the whole number is divisible by 3, 37 and 111. Now since the digit 1 is being repeated even number of times so it is divisible by 11. Again since the given number is such that the digits are being repeated and can be broken into triplets, so it is also divisible by 1001.

9. Since $16 = 2^4$ so the last four digits, thus $x = 4$ or go back to the Basics.

10. The pair will be of the form $(15k, 15l)$, where k and l must be coprimes. The values of k and l be such that $15k$ and $15l$ will remain between 40 and 100. So, the pairs are (45, 60), (45, 75), (60 75), (75, 90).

11. It is possible only when there is no common factor among p, q and r. Hence, the HCF of $p, q, r = 1$

and the LCM of $p, q, r = p \cdot q \cdot r$

$\therefore$ (HCF $\times$ LCM) of $p, q, r = 1 \times p \cdot q \cdot r = pqr$

12. Statement (2) is wrong since $3 \times (1) + 2 \times (-1) = 1$, here y is negative. Again since x and y depends on each other. Hence they are uniquely determined.

13. Go through options and put $x = 1$.

Alternatively: Solve with the help of quadratic equations.

14. **Go through options:** Then consider option (c). As

$$2 \times (14)^2 - 25 \times 14 = 3 \times 14$$

Alternatively: Let the age of the lad be x then

$$2x^2 - 25x = 42$$

$\Rightarrow \quad 2x^2 - 25x - 42 = 0$

Now, either solve this quadratic equation to get the answer or put the appropriate value from the option to get the required answer.

15. Go back to the Basics.

16. Check through options: As

$$(5+3) + (5^2 + 3^2) = 42$$

and $5 \times 3 = 15$

17. Check through alternatives: As

$$\frac{1}{0-1} = \frac{2}{0-2} \Rightarrow \frac{1}{-1} = \frac{2}{-2} = \frac{1}{-1}$$

Option (a) and (b) are clearly inadmissible.

Alternatively: $\frac{1}{x-1} = \frac{2}{x-2}$

$\Rightarrow \quad x - 2 = 2x - 2 \Rightarrow x = 0$

18. $\because \quad \left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 \cdot \left(x \cdot \frac{1}{x}\right)$

$\Rightarrow \quad (2)^2 = x^2 + \frac{1}{x^2} + 2(1)$

$\left(\text{Substituting the value of } x + \frac{1}{x}\right)$

$\Rightarrow \quad 4 = x^2 + \frac{1}{x^2} + 2$

$\Rightarrow \quad x^2 + \frac{1}{x^2} = 2$

Hence, (c) is the right choice.

19. $\because \quad a^2 + b^2 + c^2 - ab - bc - ac$

$$= \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$$

$$= \frac{1}{2}[(2)^2 + (-2)^2 + (4)^2] = 12$$

Alternatively: Substitute the values of a, b, c into given expression.

20. Go through options or Go back to the Basics.

The factors of 6 = 1, 2, 3, 6

Now, since $1 + 2 + 3 = 6$

Hence, (d) is correct choice.

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21. $\because \quad \frac{a}{b^2} \times \frac{b}{a^2} = \frac{1}{ab}$

Now, consider option (d).

$$\frac{1}{2ab} + \frac{1}{2ab} = 2\left(\frac{1}{2ab}\right) = \frac{1}{ab}$$

Hence, (d) is the appropriate option.

22. Substitute the value of x and y in the given expression

$$(6+3)^{6/3} = 9^2 = 81$$

23. AM of $ax_1 + b, ax_2 + b, ax_3 + b, \ldots, ax_n + b$ is

$$= \frac{(ax_1 + b) + (ax_2 + b) + (ax_3 + b) + \ldots + (ax_n + b)}{n}$$

$$= \frac{(ax_1 + ax_2 + ax_3 + \ldots + ax_n) + (b + b + b + \ldots + b)}{n}$$

$$= \frac{a(x_1 + x_2 + \ldots + x_n) + nb}{n}$$

$$= \frac{a(x_1 + x_2 + x_3 + \ldots + x_n)}{n} + \frac{nb}{n} = a\bar{x} + b$$

24. The arithmetic mean of first 'n' odd natural number $= \frac{n^2}{n} = n$

Thus, the required answer is 50. Hence, (a) is correct.

25. Average area $= \frac{1^2 + 2^2 + 3^2 + \ldots + n^2}{n} = \frac{\left(\frac{n(n+1)(2n+1)}{6}\right)}{n}$

$$= \frac{(n+1)(2n+1)}{6} = \frac{(n+1)}{2} \cdot \frac{(2n+1)}{3}$$

Hence, (b) is the correct option.

26. $\sqrt[4]{10} = (10)^{1/4} = (10)^{3/12} = (1000)^{1/12}$

$\sqrt[3]{6} = (6)^{1/3} = (6)^{4/12} = (1296)^{1/12}$

$\sqrt{3} = (3)^{1/2} = (3)^{6/12} = (729)^{1/12}$

$\therefore \sqrt{3} < \sqrt[4]{10} < \sqrt[3]{6}$ is the correct order and hence (b) is correct.

27. Go back to the Basic. As $\sqrt[3]{2}$ is a real number but not a rational number.

28. $\because \quad (a^2 + b^2)^3 = (a^3 + b^3)^2$

$\Rightarrow \quad a^6 + b^6 + 3a^2b^2(a^2 + b^2) = a^6 + b^6 + 2a^3b^3$

$\Rightarrow \quad 3a^2b^2(a^2 + b^2) = 2a^3b^3$

$\Rightarrow \quad 3(a^2 + b^2) = 2ab$

$\Rightarrow \quad \frac{a^2 + b^2}{ab} = \frac{2}{3}$

Now, $\left(\frac{a}{b} + \frac{b}{a}\right)^6 = \left(\frac{a^2 + b^2}{ab}\right)^6 = \left(\frac{2}{3}\right)^6 = \frac{64}{729}$

Hence, (b) is the correct choice.

29. Total number of lines (in one bound)

= [Number of lines per page
× Number of page per chapter
× Number of chapter per book
× Number of books per bound]

$= 60 \times 125 \times 5 \times 20 = 750000$

$\therefore$ Number of bounds $= \frac{3000000}{750000} = 4$

Hence, there are total 4 bounds.

30.

$$1 + \sqrt{1} = 2$$
$$4 + \sqrt{4} = 6$$
$$9 + \sqrt{9} = 12$$
$$16 + \sqrt{16} = 20$$
$$25 + \sqrt{25} = 30$$
$$36 + \sqrt{36} = 42$$

Thus, we get the unit digits as 0, 2 and 6 only. Hence (c) is the correct option.

31. Since $2^p + 1$ is a prime number so

$$2^0 + 1 = 2$$
$$2^1 + 1 = 3$$
$$2^2 + 1 = 5$$
$$2^{(2)^2} = 2^4 + 1 = 17$$
$$2^{(2)^3} = 2^8 + 1 = 257$$

Hence, the value of p is 2 or the power of 2.

32. Since, the prime number between 101 and 120 is 4 *viz.*, 103, 107, 109, 113.

Hence, the number of composite numbers between 101 and 120 = 18 − 4 = 14.

33. Since $(10^N - 1) = (10^{2n} - 1)$

$= (10^2)^n - 1 = (100)^n - 1$

So, $= 100^n - 1^n = (100 - 1)[X] = 99[X]$

Thus, for any even power it is divisible by 11. *i. e.*, $N = 2n$.
Hence, (a) is the correct option.

34. The volume of tin is = HCF of 140, 260 and 320 = 20
Hence, (a) is the correct answer.

35. Go back to the Basics.

36. $\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{4}\right)\left(1 + \frac{1}{5}\right)\ldots\left(1 + \frac{1}{n}\right)$

$$= \frac{4}{3} \times \frac{5}{4} \times \frac{6}{5} \times \ldots \frac{n}{(n-1)} \times \left(\frac{n+1}{n}\right) = \frac{n+1}{3}$$

Hence, (a) is the correct option.

37. $(x)^3 = (2^{1/3} + 2^{-1/3})^3 = 2 + 2^{-1} + 3 \times 2^{1/3} \times 2^{-1/3}(x)$

$$= 2 + \frac{1}{2} + 3x$$

$\Rightarrow \quad x^3 - 3x = \frac{5}{2}$

$\Rightarrow \quad 2(x^3 - 3x) = 5$

$\Rightarrow \quad 2x^3 - 6x = 5$

Hence, (a) is the correct answer.

38. Go back to Basics.

39. $\because \quad (x - y)^3 = x^3 - y^3 - 3xy(x - y)$

$\therefore \quad (1)^3 = x^3 - y^3 - 3xy(1)$

[Substituting the value of $(x - y)$]

$\Rightarrow \quad x^3 - y^3 - 3xy = 1$

40. $\because (1 - 81P^4) = (1 + 9P^2)(1 - 9P^2)$

$= (1 + 9P^2)(1 + 3P)(1 - 3P)$

Thus, $(1 - 3P)$, $[1 - (3P)^2]$ and $[1 - (3P)^4]$ are the 3 factors possible.

41. Let the three consecutive integers be $(n - 1)$, n and $(n + 1)$ then

$$(n-1) + n + (n+1) = 21$$

$\Rightarrow \quad 3n = 21 \Rightarrow n = 7$

$\therefore \quad (n-1) + n = (7-1) + 7 = 13$

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42. $\because (s-a)^2+(s-b)^2+(s-c)^2+s^2-a^2-b^2-c^2$
$= s^2+a^2-2as+s^2+b^2-2bs+s^2+c^2-2cs$
$\quad +s^2-a^2-b^2-c^2$
$= 4s^2+(a^2+b^2+c^2)-2s(a+b+c)-(a^2+b^2+c^2)$
$= 4s^2-2s(a+b+c)$
$= 4s^2-4s^2=0 \qquad [\because 2s=(a+b+c)]$
Hence, (d) is the correct option.

43. Given that $ab+bc+ca=0$
Now $a^2-bc=a^2+ab+ca \qquad [\because -bc=ab+ca]$
$\Rightarrow a^2-bc=a(a+b+c)$
Similarly $b^2-ca=b(a+b+c)$
and $c^2-ab=c(a+b+c)$
$\therefore \frac{1}{a^2-bc}+\frac{1}{b^2-ca}+\frac{1}{c^2-ab}$
$=\frac{1}{a(a+b+c)}+\frac{1}{b(a+b+c)}+\frac{1}{c(a+b+c)}$
$=\frac{1}{a+b+c}\left[\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right]$
$=\frac{1}{a+b+c}\left[\frac{bc+ac+ab}{abc}\right]=0 \qquad [\because ab+bc+ac=0]$

44. $4^{x+3}\times 2^{x-3}-128=0$
$\Rightarrow 2^{2x+6}\times 2^{x-3}-2^7=0$
$\Rightarrow 2^{3x+3}-2^7=0$
$\Rightarrow 3x+3=7$
$\Rightarrow 3x=4 \Rightarrow x=\frac{4}{3}$
Hence, (d) is the correct option.

45. $\because a^m\cdot a^n=a^{mn}$
$\Rightarrow a^{m+n}=a^{mn}$
$\Rightarrow m+n=mn$
$\therefore m(n-2)+n(m-2)=mn-2m+nm-2n$
$=(m+n)-2m+(m+n)-2n$
$=2m+2n-2m-2n=0$
Hence, (a) is the correct option.

46. $\frac{6^6+6^6+6^6+6^6+6^6+6^6}{3^6+3^6+3^6}+\frac{4^6+4^6+4^6+4^6}{2^6+2^6}=2^n$
$\Rightarrow \frac{6(6)^6}{3(3^6)}+\frac{4(4^6)}{2(2^6)}=2^n$
$\Rightarrow 1=2^n \Rightarrow 2^0$
thus, $n=0$
Hence, (b) is the correct option.

47. $a^{1/m}=b^{1/n}=c^{1/p}=k$
$\Rightarrow a=k^m,\ b=k^n,\ c=k^p$
$\because a\cdot b\cdot c=1$
$\Rightarrow k^m\cdot k^n\cdot k^p=1$
$\Rightarrow k^{m+n+p}=1=k^0$
$\Rightarrow m+n+p=0$
Hence, (a) is the correct option.

48. $\frac{-3}{7}=-0.42$ and $\frac{-1}{3}=-0.33$
So, the value in descending order will be as follows:
$\frac{2}{3}>\frac{-1}{3}>\frac{-3}{7}$
Hence, (b) is the correct option.

49. $\frac{-24}{20}=\frac{x}{25} \Rightarrow x=-30$
Hence, the equivalent rational number is $\frac{-30}{25}$.

50. $\frac{3}{4}\times\left(\frac{-2}{3}+\frac{3}{5}\right)=\frac{3}{4}\times\left(\frac{-1}{15}\right)=\frac{-1}{20}$

51. If the dividend is X, divisor be y and quotient be q, then
$X=yq+r$
Now, since $y=8q$
and $y=4r=4\times 12=48$ (r is remainder)
$\Rightarrow q=6$
$\therefore X=48\times 6+12$
$X=300$

52. Go back to the Basics.

53. Options (a) and (b) are wrong since when $c>a$ and $d>b$ still $\frac{a}{b}=\frac{c}{d}$ it means $\frac{a}{b}$ and $\frac{c}{d}$ are equivalent rational number and $\frac{a}{b}$ is in standard form.
Thus, $\frac{c}{d}=\frac{ka}{kb};\ k>1$
Hence, (c) is the correct answer.

54. Since all the numbers in the given series are of the form $4n+2; n\in N$.
It means the number after division by 4 leaves remainder 2.
So, the possible value is 338. Hence, (b) is the correct option.

55. $6^4\times 8^6\times 10^8\times 12^{10}=(2\times 3)^4\times(2^3)^6\times(2\times 5)^8\times(2^2\times 3)^{10}$
$=2^4\times 3^4\times 2^{18}\times 2^8\times 5^8\times 2^{20}\times 3^{10}$
$=2^{50}\times 3^{14}\times 5^8$
Thus, there are total 72 (= 50 + 14 + 8) prime factors.

56. Check the options for the given condition. 10080 fulfills all the requirements.

57. $500=2^2\times 5^3$
The sum of all the factors $=\frac{(2^{2+1}-1)\times(5^{3+1}-1)}{(2-1)(5-1)}$
$=\frac{7\times 624}{1\times 4}=1092$
Hence, (c) is the correct option.

58. Go through option. It would be better to try to divide the given numbers. Then 19 divides 1007 and 1273 both. Hence, (d) is the correct option.

59. $s^2+(s-1)^2+(s-3)^2+(s-5)^2$
$=\left(\frac{9}{2}\right)^2+\left(\frac{7}{2}\right)^2+\left(\frac{3}{2}\right)^2+\left(\frac{-1}{2}\right)^2 \qquad \left[\because 2s=9\Rightarrow s=\frac{9}{2}\right]$
$=\frac{81}{4}+\frac{49}{4}+\frac{9}{4}+\frac{1}{4}=35$
Hence, (d) is the correct option.

60. Go through option $5+3=8$
$53-35=18$
Hence, option (c) is correct.

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61. $x^y = y^x \Rightarrow y = x^{y/x}$

$\Rightarrow \quad 2x = x^{y/x} = x^{2x/x} = x^2$

$\Rightarrow \quad 2x = x^2 \Rightarrow x = 2$

Hence, (a) is the correct option.

Alternatively: Go through options and put $x = 2$ in both the values if they satisfy, then the presumed option is correct. As

$y = 2x = 4$

then $2^4 = 4^2$

$16 = 16$

Hence, it is correct.

62. $[(x^l)^{1-\frac{1}{l}}]^{\frac{1}{l-1}} = (x^{l-1})^{\frac{1}{l-1}} = x$

Hence, (a) is the correct choice.

63. $\because x^a \times x^b \times x^c = 1 \Rightarrow x^{a+b+c} = 1 = x^0$

$\Rightarrow \quad a + b + c = 0$

Now since $a^3 + b^3 + c^3 - 3abc$

$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$

$\Rightarrow \quad a^3 + b^3 + c^3 - 3abc = 0$

$\therefore \quad a^3 + b^3 + c^3 = 3abc$

Hence, (d) is the right choice.

64. $(\sqrt{2})^2 = (2^{1/2})^2 = 2$ is a rational number.

65. $\frac{-18}{5} = -3.6$, which lies between -3 and -4.

Hence, (b) is the correct option.

66. $1 \div \frac{1}{1 \div \frac{1}{1 \div \frac{1}{3}}} = 1 \div \frac{1}{1 \div \frac{1}{3}} = 1 \div \frac{1}{3} = 3$

Hence, (c) is the correct answer.

67. $171282 = 547 \times \text{divisor} + 71$

$\Rightarrow \quad 171211 = 547 \times \text{divisor}$

$\Rightarrow \quad \text{divisor} = 313$

Hence, (c) is the correct answer.

68. $\sqrt{900} + \sqrt{0.09} - \sqrt{0.000009}$

$= 30 + 0.3 - 0.003 = 30.297$

Hence, (a) is the correct option.

69. $376 - 358 = 18, \quad 358 - 232 = 126, \quad 376 - 232 = 144$

Now, the HCF of 18, 126 and 144 = 18

Hence, 18 and all its factor can fulfill the required condition. But 9 is the greatest available such number. Hence, (d) is the correct choice.

Alternatively: According to the given condition go through options starting with the greatest value from the choices available and see the remainders must be same.

71. $2 + 4 + 3 + x + 5 + 1 = 18$ (the only possible sum)

$\Rightarrow \quad x = 3$

Hence, (a) is the correct choice.

72. $5 \times 60 = 20 \times N$

$\Rightarrow$ 15, Hence (d).

73. $30k \times 30l = 5400$, where k and l are coprimes.

$k \times l = 6$

Now, since $6 = 1 \times 6$ and 2×3

Hence, there are two pairs *viz.*, (30, 1800) and (60 and 90). Thus (b).

74. The required number = (LCM of 2, 3, 4, 5, 6) $K + 1 = 7l$

$= 60k + 1 = 7l$

$\Rightarrow \quad \frac{60k + 1}{7} = l$

Now put the least possible value of k such that l must be a positive integer. Hence at $k = 5$, l is an integer. Thus, the required value is $60 \times 5 + 1 = 301$.

75. The required time = LCM of 36, 40 and 48

$= 720$ second $= 12$ minutes

Hence, (b) is the right choice.

76. $\left(1 - \frac{1}{2}\right) = \frac{1}{2}$

Now $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$

Thus $1 - \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$

Thus, (a)is the correct answer.

77. See question number 63 or see the list of formulae and then solve it.

78. Since $x^4 - y^4 = (x^2)^2 - (y^2)^2$

$= (x^2 + y^2)(x + y)(x - y)$

Thus, $(x^4 - y^4)$ is exactly divisible by $(x - y)$ as one of the factors. Hence, no remainder will leave.

79. $\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 \cdot x \cdot \frac{1}{x}$

$\Rightarrow \quad 4 = \left(x^2 + \frac{1}{x^2}\right) + 2$

$\Rightarrow \quad x^2 + \frac{1}{x^2} = 6$

Now $\left(x^2 + \frac{1}{x^2}\right)^2 = x^4 + \frac{1}{x^4} + 2x^2 \cdot \frac{1}{x^2}$

$\Rightarrow \quad 36 = x^4 + \frac{1}{x^4} + 2$

$\Rightarrow \quad x^4 + \frac{1}{x^4} = 34$

Hence, (d) is the correct answer.

80. $(x + y) = 25$ and $xy = 144$

$(x - y)^2 = (x + y)^2 - 4xy$

$(x - y)^2 = 625 - 576$

$(x - y)^2 = 49$

$\Rightarrow \quad x - y = 7$

Hence, (c) is correct.

81. $x + y = 9$ and $x^2 + y^2 = 41$

$(x + y)^2 = x^2 + y^2 + 2xy$

$81 = 41 + 2xy$

$\Rightarrow \quad 40 = 2xy \Rightarrow xy = 20$

Now $(x - y)^2 = (x + y)^2 - 4xy$

$(x - y)^2 = 81 - 80$

$(x - y)^2 = 1 \Rightarrow x - y$

Thus $x + y = 9$ and $x - y = 1$

Hence, $x = 5$ and $y = 4$

Therefore (a) is the correct answer.

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Alternatively: Go through options:

$$4 + 5 = 9$$

and $$16 + 25 = 41$$

82. $3^n = 27 \Rightarrow 3^n = 3^3 \Rightarrow n = 3$

$\therefore$ $3^{n-2} = 3^1 = 3$

Thus, (a) is the right choice.

83. $a = b^x = (c^y)^x = (c)^{xy} = (a^z)^{xy} = a^{xyz}$

Thus $a = a^{xyz} \Rightarrow xyz = 1$

Hence, (c).

84. See the question number 37. Hence (c).

85. $1^2 + 2^2 + \ldots + 10^2 = \dfrac{10 \times 11 \times 21}{6} = 385$

$$\left[\because 1^2 + 2^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}\right]$$

86. $\because$

2 | 7	
2 | 3	$\rightarrow 1$
2 | 1	$\rightarrow 1$
0	$\rightarrow 1$

Thus $(7)_{10} = (111)_2$

87. The remainder will be the required answer.

Since $6 + 7 + 5 = 18$

$\therefore$ 6705 will be divisible by 5 and 3 will be remainder. Hence, (c).

88. For example $\dfrac{3}{2} = 1.5$ which is not an integer, while 3, 2 are integers. Hence, (d).

89. $\dfrac{10\sqrt{6.25}}{\sqrt{6.25} - 0.5} = \dfrac{10 \times 2.5}{2.5 - 0.5} = \dfrac{25}{2} = 12.5$

90. Go back to the Basics.

91. The required number = HCF of $(398 - 7)$, $(436 - 11)$ and $(542 - 15)$

= HCF of (391, 425 and 527) = 17

Hence, (b) is the right choice.

Alternatively: If your speed of calculation is very fast then pickup the suitable option and then divide the given numbers and see the required result.

92. Option (a) and (d) are invalid since remainder is always a positive value. Again if you know the remainder theorem, then apply it. Otherwise consider any appropriate value of x and then perform the division.

93. $\because$ $\dfrac{a^3 + b^3 + c^3 - 3abc}{a^2 + b^2 + c^2 - ab - bc - ac}$

$$= \frac{(a+b+c)(a^2 + b^2 + c^2 - ab - bc - ac)}{(a^2 + b^2 + c^2 - ab - bc - ac)}$$

$$= a + b + c$$

$\therefore$ The required answer = 10 = (1.5 + 4.7 + 3.8)

94. Go through options

$\because$ $8 + 0 + 3 + 6 + 4 + 2 = 23$

So, the required number = 27 – 23 = 4

Alternatively: Divide 803642 and find the remainder. Now the required number = (9 – Remainder)

95. $\because$ $\dfrac{a^3 + b^3}{a^2 + b^2 - ab} = \dfrac{(a+b)(a^2 + b^2 - ab)}{(a^2 + b^2 - ab)} = (a + b)$

Hence, the required number = 8.73 + 4.27 = 13

96. $\left(x - \dfrac{1}{x}\right)^3 = x^3 - \dfrac{1}{x^3} - 3x \cdot \dfrac{1}{x}\left(x - \dfrac{1}{x}\right)$

$$125 = x^3 - \frac{1}{x^3} - 3(5)$$

$\Rightarrow$ $x^3 - \dfrac{1}{x^3} = 140$ Hence, (d).

97. Best way is to go through option. Consider option (c)

Correct answer = 20, marks for correct answer = 60

So wrong answer = 10, marks for wrong answer = –20

$\therefore$ Net marks = 60 – 20 = 40

Hence, presumed option is correct.

Alternatively: $30 \times 3 - x \times 5 = 40$

$$90 - 5x = 40$$

$$x = 10$$

Hence, the wrong answer = 10

Thus, the correct answer = 20

(for more detailed discussion see the example)

98. Since $|x| = x$ or $-x$

$\therefore$ $|x - 1| + |x - 2| + |x - 3| \geq 6$

$\Rightarrow$ $3x - 6 \geq 6$ or $6 - 3x \geq 6$

$\Rightarrow$ $x \geq 4$ or $x \leq 0$

Hence, (b).

99. Total cost = Distance covered × Rate (or charge/km)

$$= x \times 10 + (200 - x) \times 2$$

$$= 10x + 400 - 2x = 8x + 400$$

Hence, (c).

100. $2^{x+3} \cdot 4^{2x-5} = 2^{3x+7}$

$\Rightarrow$ $2^{x+3} \cdot 2^{4x-10} = 2^{3x+7}$

$\Rightarrow$ $2^{5x-7} = 2^{3x+7}$

$\Rightarrow$ $5x - 7 = 3x + 7$

$\Rightarrow$ $2x = 14$

$x = 7$ Hence, (d).

101. $\left(\dfrac{x^a}{x^b}\right)^{(a^2+b^2+ab)} \cdot \left(\dfrac{x^b}{x^c}\right)^{(b^2+c^2+bc)} \cdot \left(\dfrac{x^c}{x^a}\right)^{(c^2+a^2+ac)}$

$= x^{(a-b)(a^2+b^2+ab)} \cdot x^{(b-c)(b^2+c^2+bc)} \cdot x^{(c-a)(c^2+a^2+ac)}$

$= x^{(a^3 + ab^2 + a^2b - a^2b - b^3 - ab^2 + b^3 + bc^2 + b^2c - b^2c - c^3 - bc^2 + c^3 + a^2c + ac^2 - ac^2 - a^3 - a^2c)}$

$= x^0 = 1$ Thus, (c).

102. $4 \times 100 + 3 \times 10 + \dfrac{9}{1000} = 400 + 30 + 0.009$

$= 430.009$ Hence, (c).

103. The same number can not be rational and irrational both, simultaneously. So (a) is wrong.

Again $(2 + \sqrt{3}) + (2 - \sqrt{3}) = 4$

So, (b) is wrong, as irrational + irrational = Rational.

Once again, $-3 < 2 \Rightarrow 9 > 4$

For more information go back to the basics.

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104. $\because \quad 540 = 2^2 \times 3^3 \times 5$

So, the least perfect square

$= 2^2 \times 3^4 \times 5^2 = [(2^2 \times 3^3 \times 5) \times (3 \times 5)]$

$= 8100$ Hence, (a).

105. Go back to the basics.

Since $2 - 2 = 0$

[Natural number – Natural number] = Not a natural number

$\frac{3}{2} = 1.5$

$\frac{\text{Natural number}}{\text{Natural number}}$ = Not a natural number

Hence, it is not closed for subtraction and division.

106. $\frac{100\sqrt{25}}{\sqrt{25}+x} = 50 \Rightarrow \frac{500}{5+x} = 50 \Rightarrow x = 5$

$\because \quad 5 = \sqrt{25}$ Hence (c).

107. 127, 131, 137, 139.

108. $1 + 2 + 3 + 7 + 5 = 18$, hence, divisible by 3 and 9 both.

Again $(5 + 3 + 1) - (7 + 2) = 0$

Hence, divisible by 11 also. Thus (a).

109. $\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{5}\right)\ldots\left(1-\frac{1}{n}\right) = \frac{2}{3}\times\frac{3}{4}\times\frac{4}{5}\times\ldots\times\frac{n-1}{n}$

$= \frac{2}{n}$

Thus, (b) is correct.

110. $\frac{4}{5} + \frac{y-x}{y+x} = \frac{4}{5} + \frac{3-2}{3+2}$ $\left(\because \frac{x}{2} = \frac{y}{3} \Rightarrow \frac{x}{y} = \frac{2}{3} \Rightarrow x = 2 \text{ and } y = 3\right)$

$= \frac{4}{5} + \frac{1}{5} = \frac{5}{5} = 1$ Hence, (c).

111. $\frac{a^2+ab+b^2}{a^3-b^3} = \frac{a^2+ab+b^2}{(a-b)(a^2+ab+b^2)} = \frac{1}{(a-b)}$

$= \frac{1}{119-111} = \frac{1}{8}$ Thus, (b).

112. Go through options : $8 + 3 = 11$

and $83 - 38 = 45$ Hence, (d).

Alternatively: $x + y = 11$

and $9(x - y) = 45$

$\Rightarrow \quad x - y = 5$

Thus, $8 - 3 = 5$, hence 83.

113. $4A + \frac{5A}{3} = 85$

$\Rightarrow \quad A = 15 \Rightarrow \frac{A}{3} = 5$

Hence, the cost of child ticket = Rs. 5/-. Thus (a) is the correct option.

114. $9^{3/2} \div (243)^{-2/3} = (3^2)^{3/2} \div \left(\frac{1}{3^5}\right)^{2/3}$

$= 3^3 \div 3^{-10/3} = 3^{3+\frac{10}{3}} = 3^{19/3}$

115. At $x = 2$, both sides are equal.

116. $\frac{b}{a} = \frac{c}{b} = b^2 = ca$

Again, $a^x = b^y = c^z = k$

$\Rightarrow \quad a = k^{1/x}, b = k^{1/y}, c = k^{1/z}$

$\therefore \quad b^2 = ac$

$\Rightarrow \quad k^{2/y} = k^{1/x} \cdot k^{1/z}$

$\Rightarrow \quad \frac{2}{y} = \frac{1}{x} + \frac{1}{z} = \frac{z+x}{zx}$

$\Rightarrow \quad \frac{2z}{x+z} = \frac{y}{x}$

117. $-\frac{1}{2} < 1$, hence (d) is wrong.

$-\frac{1}{2} < \frac{1}{2}$, hence, it is correct ($x \in I^-$ and $y \in I^+$)

$\because \quad x < 0 \Rightarrow \frac{1}{x} < 0$

and $y > 0 \Rightarrow \frac{1}{y} > 0$

$\therefore \quad \frac{1}{x} < \frac{1}{y}$ Hence, (c).

118. $33.33 \div 1.1 = \frac{33.33}{1.1} = \frac{33.33}{1.10}$

$= \frac{3333}{110} = 30.3$

119. $|x - 2| < 3$

$\Rightarrow \quad (x - 2) < 3 \text{ or } (2 - x) < 3$

$\Rightarrow \quad x < 5 \text{ or } x > -1$

$\Rightarrow \quad -1 < x < 5$

120. Let the fraction be $\frac{x}{y}$, then

$\frac{x+1}{y+1} = 4$ ...(1)

and $\frac{x-1}{y-1} = 7$ ...(2)

Solving equations (1) and (2),

$x = 15$ and $y = 3$

Thus, (d) is correct.

121. $\because \quad (a-b)^3 + (b-c)^3 + (c-a)^3 = 0$

$\Rightarrow \quad (a - b) = 0 \Rightarrow a = b$

$(b - c) = 0 \Rightarrow b = c$

$(c - a) = 0 \Rightarrow c = a$

Thus, $a = b = c$ Hence, (c)

122. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ac)$

$121 = 51 + 2(ab + bc + ac)$

$70 = 2(ab + bc + ca)$

$\Rightarrow \quad (ab + bc + ca) = 35$

123. $\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy} = \frac{x^3}{xyz} + \frac{y^3}{xyz} + \frac{z^3}{xyz}$

$= \frac{x^3+y^3+z^3}{xyz} = \frac{3xyz}{xyz} = 3$

($\because a^3 + b^3 + c^3 = 3abc$, when $a + b + c = 0$)

124. $\left(\frac{e^x+e^{-x}}{2}\right)^2 - \left(\frac{e^x-e^{-x}}{2}\right)^2 = \frac{e^{2x}+e^{-2x}+2}{4} - \frac{e^{2x}+e^{-2x}-2}{4}$

$= 1$

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125. $(1-x)(1+x)(1+x^2)(1+x^4)(1+x^8)$

$=(1-x^2)(1+x^2)(1+x^4)(1+x^8)$

$[\because (a-b)(a+b)=a^2-b^2]$

$=(1-x^4)(1+x^4)(1+x^8)$

$=(1-x^8)(1+x^8)=(1-x^{16})$ Hence, (c)

126. Go through option : Consider the value of x such that $(5x+7)$ must be divisible by 13. Since, the product is an integer and the value of y must be such that $(13y+1)$ will be divisible by x.

Thus option (b) is correct.

127. Option (b) is clearly wrong. Again since $(28)^2=784$.

So, option (c) is also wrong. Now check for option (a)

$$\left(5\frac{15}{22}\right)^2=\left(\frac{125}{22}\right)^2=\frac{15625}{484}=32\frac{137}{484}$$

Thus, (a) is the correct option.

128. $\because$ $\sqrt{254016}=504$

and $\sqrt{10609}=103$

$\therefore$ $\dfrac{\sqrt{25.4016}-\sqrt{1.0609}}{\sqrt{25.4016}+\sqrt{1.0609}}=\dfrac{5.04-1.03}{5.04+1.03}$

$=\dfrac{4.01}{6.07}=\dfrac{401}{607}$

129. $\sqrt{\dfrac{\left(3\frac{1}{4}\right)^4-\left(4\frac{1}{3}\right)^4}{\left(3\frac{1}{4}\right)^2-\left(4\frac{1}{3}\right)^2}}=\sqrt{\dfrac{\left(\frac{13}{4}\right)^4-\left(\frac{13}{3}\right)^4}{\left(\frac{13}{4}\right)^2-\left(\frac{13}{3}\right)^2}}$

$=\sqrt{\left(\frac{13}{4}\right)^2+\left(\frac{13}{3}\right)^2}$

$=\sqrt{169\left(\frac{25}{144}\right)}=13\times\frac{5}{12}$

$\left[\because \dfrac{a^2-b^2}{a-b}=\dfrac{(a+b)(a-b)}{(a-b)}=a+b;\right.$

$\left.a=\left(\frac{13}{2}\right)^2 \text{ and } b=\left(\frac{13}{3}\right)^2\right]$

$=\dfrac{65}{12}=5\dfrac{5}{12}$

130. The required number must be a factor of

$=34041-32506=1535$

Now since $1535=5\times307$

So, clearly 307 is the required number because it has 3 digits and is also a prime number.

131. Minimum number of corks $=\dfrac{403+434+465}{\text{HCF of }(403, 434, 465)}$

$=\dfrac{1302}{31}=42$

Hence, (c). For detailed discussion go back to examples.

132. The required value = HCF of 49.56 and 38.94 = 3.54

Hence, the sum = Rs. 3.54.

Thus, (b) is correct.

133. The required value = LCM of 12, 30, 18, 48 = 720

Hence, (b).

134. Time taken for each of three persons A, B and C is respectively $\frac{6}{3}$, $\frac{6}{2\frac{1}{2}}$ and $\frac{6}{1\frac{1}{4}}$ hrs

i.e., $\frac{2}{1}$, $\frac{12}{5}$ and $\frac{24}{5}$ hrs.

So, it is required to find the LCM of $\frac{2}{1}, \frac{12}{5}, \frac{24}{5}=\frac{24}{1}=24$ hr

Hence, (c).

135. $\because$ $\dfrac{4\frac{1}{7}-2\frac{1}{4}}{3\frac{1}{2}+1\frac{1}{7}}=\dfrac{53}{130}$ and $\dfrac{1}{2+\dfrac{1}{2+\dfrac{1}{5-\frac{1}{5}}}}=\dfrac{53}{130}$

$\therefore$ $\dfrac{53}{130}\div\dfrac{53}{130}=1$ Hence, (c).

136. Let the total journey be x kms then,

$$\frac{5x}{16}+\frac{7x}{20}+10=x$$

$\Rightarrow$ $x=29\dfrac{17}{27}$

Hence, (d).

137. Since $\sqrt{3}$ is an irrational number.

138. $p>0$ and $q<0$

Now, let $p=3$ and $q=-4$

therefore $p-q=3-(-4)=7>0$

Hence, (b) is correct.

140. $(10x+y)=k(x+y)$ ...(1)

Again $(10y-x)=l(x+y)$ ...(2)

Now, adding equation (1) and (2)

$11(x+y)=(k+l)(x+y)$

$\Rightarrow$ $k+l=11 \Rightarrow l=(11-k)$

Hence, (c).

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1. $5\times(3, 13, 23, 33, 43, 53, 63, 73, 83, 93)$
$+ 5\times(30, 31, 32, 33, 34, 35, 36, 37, 38, 39)$
$+ (301, 302, 303, 304, \ldots, 399)$
$= 5\times10 + 5\times10 + 100 = 200$

Explanation : In every set of 100 numbers there are 10 numbers whose unit digit is 3. Similarly in every set of 100 numbers there are 10 numbers whose tens digit is 3 and there are total 100 numbers whose hundreds digit is 3.

2. $1146600 = 2^3\times3^2\times5^2\times7^2\times13$
Thus total number of factors of
$1146600 = (3+1)(2+1)(2+1)(2+1)(1+1)$
$= 4\times3\times3\times3\times2 = 216$
So the number of ways in which 1146600 can be expressed as a product of two factors $= \frac{216}{2} = 108$
(for more discussion go back to the Basics)

3. The numbers which are divisible by 5 & 9 both means the number be divisible by 45. So the total number which are divisible by $45 = 22$ [45, 90, 135, ..., 990]
[i.e. all the numbers of the form $45k$, where $k = 1, 2, 3, \ldots, 22$]
Now since these obtained numbers will not be divisible by 7 so, the numbers of the form $45\times7l$ must be subtracted from these 22 numbers.
Thus the required number $= 22 - 3 = 19$
($\because k = 7l$). Thus the possible values of $l = 1, 2, 3$ for $k = 1, 2, \ldots, 22$

4. $\because |x| + |y| \le 0$ for $x, y \in R$
$\Rightarrow \quad |x| = 0$ and $|y| = 0$
$\Rightarrow \quad x = 0$ and $y = 0$
So there is only one solution for the expression i.e. $(x, y) = (0, 0)$

5. $\frac{2^{39}}{39} \to \frac{(2^6)^6\times2^3}{39} \to \frac{64^6\times8}{39} \to \frac{25^6\times8}{39} \to \frac{(625)^3\times8}{39} \to \frac{(1)^3\times8}{39} \to \frac{8}{39}$

Thus the remainder is 8.
(Since the final remainder can be obtained by the dividing the factors of the given number) for more information GO BACK TO THE BASICS.

6. The unit digit of $(1!)^{99}$ is 1
The unit digit of $(2!)^{98}$ is 4
The unit digit of $(3!)^{97}$ is 6
The unit digit of $(4!)^{96}$ is 6
The unit digit of $(5!)^{95}$ is 0
Now since we know that all the factorial numbers starting with 5! has its unit digits 0. So we need not to calculate it.
Thus the required unit digit
= unit digit of the sum of the unit digits
Thus the unit digit = 7 ($\because 1 + 4 + 6 + 6 + 0 = 17$)

7. The first four digit number which is divisible by 7 is 1001 and the last digit is 9996 and such total numbers are
$= \frac{8995}{7} + 1 = 1286$
So the sum of all these numbers $= \left(\frac{9946 + 1001}{2}\right)\times1286$
$= 7071071$
[Since all these numbers are in A.P. with common difference 7]

8. Let the fraction be $\frac{x}{y}$ then,
$$\frac{x+2}{2y} = \frac{1}{2}$$
$$\Rightarrow \quad \frac{x+2}{y} = 1$$
$$\Rightarrow \quad x + 2 = y$$
$$\Rightarrow \quad y - x = 2$$
Thus any even number greater than 3 can be the sum of x and y. Where x and y are some integers where $y - x = 2$
Further we know that the even – even = even
or odd – odd = even
Again even + even = even
and odd + odd = even
Thus in all the cases $x + y$ must be an even number.

9.
$$pq = p - q + 9$$
$$\Rightarrow \quad pq + q = p + 9$$
$$\Rightarrow \quad q(p+1) = p + 9$$
$$\Rightarrow \quad q = \left(\frac{p+9}{p+1}\right)$$
Now since p is an integer such that q is a fraction.
So go through options and you will find that at $p = 7$ q is not a fraction. Hence (d) is the required answer
$\because q = \frac{7+9}{7+1} = 2$, which is inadmissible.

10. Option (b) is clearly invalid. Now since the difference between these numbers $= 792 = 99\times8$
Hence the difference between unit digit and hundreds digit must be 8. Thus option (d) is the right choice.
As $931 - 139 = 792$ and 1, 3, 9 are in G.P. also.
Alternatively :
$$(100a + 10ar + ar^2) - (100ar^2 + 10ar + a) = 792$$
$$\Rightarrow \quad 100a(1 - r^2) + a(r^2 - 1) = 792$$
$$\Rightarrow \quad 99a(1 - r^2) = 792$$
$$\Rightarrow \quad a(1 - r^2) = 8$$
But since $8 = 1\times8, 2\times4, 4\times2, 8\times1$
So at $a = 1$, $(1 - r^2) = 8$
$$\Rightarrow \quad r^2 = 9$$
$$\Rightarrow \quad r = \pm3$$
Therefore the digits are 1, 3 and 9.
(Note at $a = 2, 4$ and 8, r^2 is not a perfect square, thus inadmissible)

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11. The possible values can be obtained by substituting

$$k = 4, 6, 8, 10, \ldots, 102$$

Thus the number of required values $= \left(\frac{102-4}{2}\right) + 1 = 50$

12. Here $k = 3, 5, 7, \ldots, 103$

Thus the number of required values $= \left(\frac{103-3}{2}\right) + 1 = 51$

13. $\because a^2 - b^2 = (a+b)(a-b)$

Let us consider two odd prime number as 3 and 29.

Then $(29)^2 - 3^2 = (29+3)(29-3)$

$= 32 \times 26,$

which is divisible by 13.

Again consider 7 and 29,

Then $(29)^2 - 7^2 = (29+7)(29-7)$

$= 36 \times 22$, which is divisible by 11.

Further consider 3 and 37, then $(37)^2 - (3)^2 = 40 \times 34$ which is divisible by 17.

Hence (d) is correct.

NOTE As we know odd + odd = even = odd – odd. Thus consider such two number whose difference must be even multiple of the required divisor. As

$29 - 3 = 26 = 13 \times 2$

$29 - 7 = 22 = 11 \times 2$

$37 - 3 = 34 = 17 \times 2$

14. $P = (101)^{100}$, $Q = (100)^{101}$

$= (101)^{100}$ $= (100)^{100} \times (100)$

$= \frac{(101)^{100}}{(100)^{100}}$ $= \frac{(100)^{100}}{(100)^{100}} \times 100$ [Dividing both numbers by $(100)^{100}$]

$= (1.01)^{100}$ $<$ 100

Thus $P < Q$ hence (b).

15. $k^2 - 25 =$ An odd number

$\Rightarrow k^2$ must be even

$\Rightarrow k$ must be even

Thus both (a) and (b) are even numbers so (d) is correct as $(\text{even})^{\text{odd/even}} \rightarrow$ always even number

and odd $\times$ even $\rightarrow$ even

16. $(a+1)(b-1) = 625$

$\Rightarrow$ But $625 = 1 \times 625 = (a+1) \times (b-1) \Rightarrow a = 0, b = 626$

$5 \times 125 = (a+1) \times (b-1) \Rightarrow a = 4, b = 126$

$25 \times 25 = (a+1) \times (b-1) \Rightarrow a = 24, b = 26$

$125 \times 5 = (a+1) \times (b-1) \Rightarrow a = 124, b = 6$

$625 \times 1 = (a+1) \times (b-1) \Rightarrow a = 624, b = 2$

Thus $(a+b)$ is always equal to or greater than 50. Since the min $(a+b) = 50 = (24+26)$

Alternatively : $(a+1)(b-1) = 625$

$ab + b - a - 1 = 625$

$\Rightarrow$ $b(a+1) - a = 626$

$\Rightarrow$ $b(a+1) = a + 626$

$\Rightarrow$ $b = \frac{(a+626)}{(a+1)} = \frac{(a+1)+625}{(a+1)} = \frac{625}{(a+1)} + 1$

Let us consider $a = 4$ then $b = 126$

Thus the only suitable option is (b) since option (c) and (d) are wrong and at $a = 24$, $b = 26 \Rightarrow a + b = 50$, which is least possible answer because in this case when the difference between a and b is least then $(a+b)$ is also least.

17. Since $p > 0$ then at $p = 1$, we get $q = 2$ and for the rest values $q > 2$

as when $0 < p < 1$ $\quad p + \frac{1}{p} > 2$

and also at $p > 1$ $\quad p + \frac{1}{p} > 2$

Hence $\quad p + \frac{1}{p} = q \geq 2$

18. $a^b = b^a; (a \neq b) > 1$

Let us consider $a = 2$ and $b = 4$ then

$2^4 = 4^2$

$16 = 16$ Hence (b)

NOTE $a + b = 5$ is not admissible since

$2^3 \neq 3^2$

and $a + b = 7$ is also not possible since

$2^5 \neq 5^2$ and $3^4 \neq 4^3$; where a and b are greater than 1.

19. $m^n - n^m = m + n$

Consider $m = 2$ and $n = 5$, then

$2^5 - 5^2 = 5 + 2$

$7 = 7$

Thus option (a) and (b) are wrong and option (c) is correct.

20. Since 334 or 433 are not the perfect cube hence (a) is wrong

Again 792, 279, 297, 972 or 927 are not the perfect cubes hence (b) is also wrong.

Further 512 is also a perfect cube hence (c) is correct answer.

Alternatively : See the perfect cubes of three digits as

$5^3 = 125$, $6^3 = 216$, $7^3 = 343$, $8^3 = 512$, $9^3 = 729$

So the digits of $5^3 = 125$ and $8^3 = 512$ are same. Hence (c) is the correct answer.

21. $n^4 - n^2 = n^2(n^2 - 1) = n^2(n+1)(n-1)$

Thus at $n = 3$ the given expression is divisible by 8, 4 and 12.

Hence it is divisible by all of a, b, c.

22. Since $(11)^4 = 14641$, hence b will be less than 4 always because 11 is the least two digit appropriate number but 14641 is a 5 digit number. Also $b \neq 1$, hence b lies between 1 and 4 *i.e.* b is equal to either 2 or 3.

Now try for suitable values of (aa) and b, we find that the only possible relation is

$(11)^3 = 1331$

Thus $1^3.3^1 + 3^1.1^3 = 6$

Hence (b) is the correct option.

23. $\frac{1}{6} : \frac{1}{5} : \frac{1}{3} : \frac{1}{2} = \frac{30}{6} : \frac{30}{5} : \frac{30}{3} : \frac{30}{2} = 5 : 6 : 10 : 15$

Therefore min. number of processers $= (5 + 6 + 10 + 15) = 36$

Hence (d) is correct.

24. $(392)^n - (392)^{n-1} = (392)^{n-1}[392 - 1]$

$= 392^{(n-1)}(391)$

Now since 13 is not the factor of either 391 or 392.

Hence (c) is the required answer.

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25. Let the fair (*i.e.*, charge) per km be Rs. x, then the total distance travelled by Mr. Chaalu is $(x + 1)$.
Hence the total expenses $= x.(x + 1)$
but $x(x + 1) \le 350$
Thus the greatest possible value of $x = 18$
hence $18 \times 19 = 342$
Thus he is left with Rs. 8 only.

26. The number of times he pressed the keys for one digit number $= 9$
The number of times he pressed the keys for two digit number $= 90 \times 2 = 180$
The number of times he pressed the keys for three digit number $= 900 \times 3 = 2700$
The number of times he pressed the keys for the number 1000 to 1999 is $1000 \times 4 = 4000$
Then total number of times the keys are pressed = 6889

27. $\frac{(20)^{23}}{17} \to \frac{(3)^{23}}{17} \to \frac{(3^3)^7 \times 3^2}{17} \to \frac{(27)^7 \times 9}{17} \to \frac{(10)^7 \times 9}{17}$
$\to \frac{(10^2)^3 \times 90}{17} \to \frac{(15)^3 \times 5}{17} \to \frac{225 \times 75}{17} \to \frac{4 \times 7}{17} \to 11$
Hence the required remainder is 11.

28. Check for all the prime numbers between 3 and 50 you will find that option (d) is correct since all such numbers are always divisible by 8, 12 and 24.
Because $p^2 - 1 = (p + 1)(p - 1)$.

29. Since the unit digit of primes number = 1, 2, 3, 5, 7, 9
So the unit digit of the squares of prime number
$= 1, 4, 9, 5, 9, 1$
Thus $1 + 1 = 2$, $4 + 1 = 5$, $9 + 1 = 10$, $5 + 1 = 6$, $9 + 1 = 10$ and $1 + 1 = 2$,
Hence the possible unit digits are 2, 5, 6, 0.
Thus 3, 7 or 9 are not possible hence (d) is correct.

30. The required number = Number of number, which are (divisible by 3 + divisible by 7 − divisible by 21)
Number of numbers which are divisible by 3
$= \left(\frac{198 - 3}{3}\right) + 1 = 66$
Number of numbers which are divisible by 7
$= \left(\frac{196 - 7}{7}\right) + 1 = 28$
Number of numbers which are divisible by 21
$= \left(\frac{189 - 21}{21}\right) + 1 = 9$
Thus the divisible value $= (66 + 28 - 9) = 85$
Hence the number of numbers which are not divisible
$= 200 - 85 = 115$
Hence (a) is the correct answer.

31. Only one value P can assume, which is 2 at $P = 2$, $P^2 + 3 = 7$, which is also prime.
Again at $P = 3, 5, 7, 11, \ldots$
$P^2 + 3$ = an even number which can not be a prime number.

32. Let there be x bangles each side, then the total number of bangles he had $= x^2 + 38$
If he increases the size of the square by one unit each side, then the total number of bangles
$= (x + 1)^2 - 25$
Thus $x^2 + 38 = (x + 1)^2 - 25$
$x^2 + 38 = x^2 + 1 + 2x - 25$
$\Rightarrow 2x = 62$
$\Rightarrow x = 31$
Thus total number of bangles $= x^2 + 38$
$= 961 + 38 = 999$

Alternatively : Go through options, consider a suitable value from the options and check that in each case it must produce a perfect square number.
As $999 - 38 = 961$ is a perfect square
and $999 + 25 = 1024$ is also a perfect square.

33. Let $a = 20$ and $b = 15$
$a + b = 35$, which is not divisible by 10.

34. At $a = 7$ and $b = 2$, the following number becomes 755238792, which is divisible by both 7 and 2.
But there are other values of (a, b) are possible consider $a = 4$ and $b = 2$, $a = 8$, $b = 2$ etc.
So there is no any unique value of a and b. Hence (d).

35. Any number of the form "*abcabc*" must be divisible by 1001 or its factors *i.e.* $1001 = 7 \times 13 \times 11$
Hence (a) and (b) are correct, thus (d) is the most appropriate statement.

36. Let suppose $a \to$ odd and $b \to$ even
then $\text{odd}^{\text{even}} \times \text{even}^{\text{odd}} = \text{odd} \times \text{even} = \text{even}$
even if a is even and b is odd, the result is an even number.

37. Let there be n rows, then the number of trees in each row $= (n + 1)$
Thus total number of trees $= n(n + 1)$
$\therefore n(n + 1) = 5550$
Now, at this moment this problem can be solved in two ways. First by finding the roots of quadratic equation. Second by using the values from options.
Again since the value of $(n + 1)$ is given in the options so consider option (d)
$74(75) = 5550.$
Hence (d) is correct.

38. $3^a = 9^b \Rightarrow a = 2b$
and $4^{(a+b+2)b} = 16^{ab}$
$\Rightarrow (a + b + 2)b = 2ab$
$\Rightarrow a + b + 2 = 2a$
$\Rightarrow a - b = 2$
$\Rightarrow b = 2$ $(\because a = 2b)$
$\therefore a = 4$

39. These numbers are 11, 13, 31, 17, 71, 37, 73, 79, 97.

40. Since he has covered twice the distance which yet he has to cover. It means he has covered $\frac{2}{3}$ of the whole journey and remaining journey is $\frac{1}{3}$rd.

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41. Let the x cm height increases per year then the height at the end of $2002 = 6x + 90$

and the height at the end of $2003 = 7x + 90$

but $$7x + 90 = (6x + 90) + (6x + 90)\frac{1}{9}$$

$$\Rightarrow \quad (7x + 90) = (6x + 90) \times \frac{10}{9}$$

$$\Rightarrow \quad 63x + 810 = 60x + 900$$

$$\Rightarrow \quad 3x = 90$$

$$x = 30 \text{ cm}$$

Thus the height at the end of $2008 = 90 + 12x$

$$= 90 + 12 \times 30 = 450 \text{ cm}$$

42. Let the cost of watch be Rs. x

Then the total amount for 12 months except ration $= 250 + x$

Therefore the amount for 9 months $= (250 + x) \times \frac{3}{4}$

Thus $$(250 + x)\frac{3}{4} = 270$$

$$\Rightarrow \quad 750 + 3x = 1080$$

$$\Rightarrow \quad 3x = 330$$

$$\Rightarrow \quad x = 110$$

Therefore the cost of wrist watch $= 110$

(It can also be solved using options.)

43. $$1^2 + 3^2 = 10$$

$$13 + 18 = 31$$

44. When two "two digit" numbers are added and the resultant value is a 'three digit number". It means there must be carry over (*i.e.* the sum of the unit digits be greater than 9. Similarly the sum of the tens digit is also greater than 9.)

Hence (d) is correct as $\begin{array}{r} 72 \\ 27 \\ \hline 99 \end{array}$ inadmissible but $\begin{array}{r} 64 \\ 46 \\ \hline 110 \end{array}$ is admissible value.

45. '*A*' grade student $= 57.\overline{142857}\% = \left[57 + \frac{1}{7}\right]\%$

$$= \frac{400}{7}\% = \frac{4}{7} \text{ of the total student}$$

'*B*' grade student $= 26.444\%$

$$= \left[26 + \frac{4}{9}\right]\% = \frac{238}{9}\%$$

$$= \frac{238}{900} \text{ of the total student}$$

Now the total student of '*A*' and '*B*' grade

$$= \frac{4}{7} + \frac{238}{900}$$

$$= \frac{3600 + 1666}{6300} = \frac{5266}{6300}$$

$$= \frac{2633}{3150} \text{ of the total student}$$

$\therefore$ The number of students who achieved the grade (c)

$$= 1 - \frac{2633}{3150} = 517 \text{ students}$$

Since the maximum number of students are below 3500. So there can not be the number of students in the multiple of 3150.

46. $3\overline{)16}\,(5$ Remainder is odd.
$\quad\underline{15}$
$\quad\quad 1$

47. $551 + 552 + 553 + \ldots + 559 + 560$

$= (500 + 500 + 500 \ldots 10 \text{ times}) + (51 + 52 + 53 + \ldots + 60)$

$= 5000 + (50 + 50 + 50 \ldots 10 \text{ times}) + (1 + 2 + 3 + \ldots + 10)$

$= 5000 + 500 + 55 = 5555$

48. $$x^2 + x = x^3 - x$$

$$\Rightarrow \quad x^3 - x^2 - 2x = 0$$

$$\Rightarrow \quad x(x^2 - x - 2) = 0$$

$$\Rightarrow \quad x = 0 \quad \text{or} \quad x^2 - x - 2 = 0$$

If $$x^2 - x - 2 = 0$$

$$\Rightarrow \quad (x - 2)(x + 1) = 0$$

$$\Rightarrow \quad x = 2 \quad \text{or} \quad x = -1$$

Thus $x = 2$ is the possible value of x.

Hence (b).

Alternatively : Go through options

$$2^2 + 2 = 2^3 - 2$$

$$6 = 6$$

49. As we have seen in above problem that the equation is of 3 degree equation. Hence there are 3 solution for x *viz.* $x = 0, 2$ and -1.

50. The required number of digits written as the thousands digit $= 101$

The required number of digits written as the hundreds digit $= 99$

The required number of digits written as the tens digit $= 10$

The required number of digits written as the unit digit $= 10$

Thus total number of 3-digits $= 220$

51. $(2^{19} + 1) = (2^{19} + 1^{19})$

Now, $2^{19} + 1^{19}$ is divisible by $3 (= 2 + 1)$

Since $(a^n + b^n)$ is always divisible by $(a + b)$, when 'n' is odd.

Again since $(2^{19} + 1)$ is an odd number, hence no any even number can divide it. Thus (a) is the correct option.

52. Except to 467 all the rest numbers are divisible by 3.

53. $\because \quad 1001 = 7 \times 11 \times 13$

Hence (b).

54. $\because \quad 5400 = 2^3 \times 3^3 \times 5^2$

To get a perfect square there must be even number of powers of each prime factors.

So the least such perfect square $= 2^3 \times 3^3 \times 5^2 \times 2 \times 3$

$$= 32400$$

Thus we have to multiply it by 6 inorder to get a perfect square.

55. At $x = 0$ and 1 the valid relation is

$$p^x q^{(1-x)} = px + q(1 - x)$$

Hence (b).

56. $$3780 - 2835 = 945$$

Now $$\frac{3780 + 2835}{945} = \frac{6615}{945} = 7$$

57. $(x + y)^{-1}.(x^{-1} + y^{-1}) = \frac{1}{(x + y)}\left(\frac{1}{x} + \frac{1}{y}\right)$

$$= \frac{1}{(x + y)}\left(\frac{x + y}{xy}\right) = \frac{1}{xy} = (xy)^{-1}$$

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58. In every set there is only one such relation exists as if $p = 3$, $k = 2$

Then the remainder is 1 when $\frac{k^2}{p} = \frac{4}{3}$

Similarly if $p = 7$ and $k = 6$

Then the remainder is 1 when $\frac{k^2}{p} = \frac{36}{7}$ etc.

Hence k is always $(p - 1)$.

59. Go through option consider the correct option (c).

Absent students $= 2700 \times \frac{1}{9} = 300$

$\therefore$ Number of students appeared in exam = 2400

Number of students who passed $= 2400 \times \frac{19}{24} = 1900$

Thus number of failed students $= 2400 - 1900 = 500$

Hence presumed option is correct.

Alternatively : Failed students $= 500 = \frac{5}{24}$

$\therefore$ Passed students $= \frac{19}{24} = 1900$

Thus the number of students appeared in exam $= 2400 = \frac{8}{9}$

$\therefore$ Total number of students registered $= \frac{2400 \times 9}{8} = 2700$

Alternatively : Let the x students registered for the exam then the number of students appeared in exam $= x - \frac{x}{9} = \frac{8x}{9}$

Again the number of passed students $= \frac{8x}{9} \times \frac{19}{24} = \frac{19x}{27}$

Thus the number of failed students $= \frac{5x}{27} = 500$

$\Rightarrow$ $x = 2700$

60. $km < kn \Rightarrow \frac{km}{kn} < 1$

Since for any value of k the above relation is true thus we can't specify that whether for $k < 0$, or $k > 0$ this is True.

Hence (d).

61. Rational + Irrational = Irrational Number

62. In first 1122 years 'Decay' becomes 40 gm
In next 1122 years 'Decay' becomes 20 gm
In next 1122 years 'Decay' becomes 10 gm
and In last 1122 years 'Decay' becomes 5 gm

63. Vinod gets 210 marks and failed by 40 marks

It means the min. marks required $= 250 = \frac{2}{5}$th of total.

Therefore maximum (or total) marks $= 250 \times \frac{5}{2} = 625$

64. Marks in QA : Marks in English $= \frac{3\frac{1}{2}}{1} = \frac{\frac{7}{2}}{1} = \frac{7}{2}$

Thus his marks in QA $= 70 \left(= 90 \times \frac{7}{9}\right)$

65. $\begin{array}{r} 72 \\ +27 \\ \hline 99 \end{array}$ and $\begin{array}{r} 72 \\ -27 \\ \hline 45 \end{array}$

But we are not sure whether it is larger or smaller number so (d) is the most appropriate answer.

66. Go back to the Basics. (Refer Indices)

67. $12 \times 63 = 756$ and $21 \times 36 = 756$

68. The best way to consider the values from given range in options.

Thus option (a) is obviously ruled out since at $p = -1$ the relation is undefined. Similarly if we consider option (d) then at $p = -\frac{1}{2}$.

$\frac{p}{p-1} = -1$, which is wrong. Thus (b) is also wrong.

Hence (c) is correct.

69. The form of the required number is $14k + 8$

but when $14k + 8$ is divided by 7, then

$$\frac{14k + 8}{7} = \frac{14k + (7 + 1)}{7} = \frac{(14k + 7) + 1}{7}$$

Thus the remainder will be 1.

NOTE $14k$ is already divisible by 7 and thus $14k + 7$ will also be divisible by 7.

Remember : In this type of problem just divide the existing remainder and get the required answer. As $\frac{8}{7} \rightarrow$ Remainder is 1.

70. The unit digit of $(316)^{3^{4n}}$ is always 6.

So the unit digit of $(316)^{3^{4n}} + 1$ will be 7.

71. $a + b = 18$

So maximum of $(a \times b)$ will be only when $a = b$

$\therefore$ $a = b = 9$

$\therefore$ Maximum of $a \times b = 9 \times 9 = 81$

72. Let the smallest possible number be x, then

$$x = 15k + 12$$

and $$x = 8l + 5$$

$\Rightarrow$ $15k + 12 = 8l + 5$

$\Rightarrow$ $15k + 7 = 8l$

$\Rightarrow$ $l = \frac{15k + 7}{8}$,

l must be an integer putting $k = 1, 2, 3, \ldots$ etc.

But at $k = 7$, we get a number which on being divided by 8, gives 'l' as an integer.

So the $x = 15 \times 7 + 12$

$x = 117$

The next higher numbers = (L.C.M. of divisors) $m + 117$
= (L.C.M. of 15 and 8) $m + 117$
= 120 m + 117

So consider the highest possible value of m such that $120m + 117 \leq 9999$ (largest possible number of four digit)
Thus at m = 82, the value of 120m + 117 = 9957, which is the required number.

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Alternatively : Go through option.
It is very obvious from the option that the option (a), (c) and (d) are wrong because when you multiply any number by 15, it will give the unit digit either 0 or 5 and when you add 12 to this product the unit digit will be either 2 or 7.
Or in other way divide the values in the given options by 8 and it must leave the remainder 5.

73. Go through alternatives. Consider option (b)

Total mobiles = 600

$$\text{Indian mobiles} = 600 \times \frac{5}{12} = 250$$

and $$\text{imported mobiles} = 600 \times \frac{7}{12} = 350$$

$$\text{The coloured mobiles of Indian origin} = 250 \times \frac{1}{5} = 50$$

The coloured mobiles of which are imported

$$= 350 \times \frac{2}{7} = 100$$

Thus total coloured mobiles = 50 + 100 = 150, which is same as given in the problem. Hence the presumed option is correct.

Alternatively : Solve through equation.

74. The minimum time interval, at which time from all the destinations, the call centre received the calls

= the L.C.M. of 10, 12, 20 and 25
= 300 minutes
= 5 hours

Thus 5 + 5 = 10 A.M. and 10 + 5 = 15 = 3 : 00 P.M.
Hence (d).

75. H.C.F. of $2^4, 8^2, 16^2, 20^3 = 2^4$
L.C.M. of $2^4, 8^2, 16^2, 20^3 = 2^8 \times 125 = 32{,}000$

76. Go through options : Consider option (b)

$$15192327 - 84 = 15192243$$

Now, $(3 + 2 + 9 + 5) - (4 + 2 + 1 + 1) = 11$
and $1 + 5 + 1 + 9 + 2 + 2 + 4 + 3 = 27$
Thus 15192243 is divisible by both 11 & 9 hence by 99.
Thus the presumed remainder 84, is correct.

77. The total numbers (integral) between 1 and 200 = 198
The number of numbers divisible by 2 = 99 (2, 4, 6, ..., 198)
The number of numbers which are purely divisible by 3 but not by 2.
(Since the division by 2 is taken into account, already)

$$= \frac{(195 - 3)}{6} + 1 = 33 \qquad (3, 9, 15, 21, \ldots, 195)$$

The number of numbers which are purely divisible by 5 but not by 2 and 3

$$= \left\{\frac{(195 - 5)}{10} + 1\right\} - \left\{\frac{195 - 15}{30} + 1\right\}$$

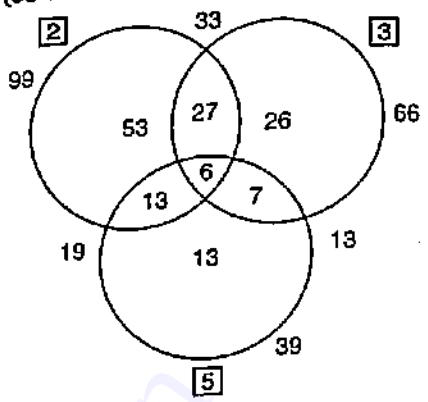
$$= 20 - 7 = 13$$

Thus the total number of numbers lying between 1 and 200, which are divisible by 2, 3 or 5 = 99 + 33 + 13 = 145.

NOTE In ill case we did not consider 10, 20, 30, ... etc. for the division of 5 because these numbers have been divided already by 2.

Alternatively : So the required number

= (53 + 13 + 26) + (27 + 13 + 7) + 6 = 145



(for more clarification of the solution see the chapter of set theory or the example discussed in the theory of this same chapter.)

78.

$$xAy \to x + y$$
$$xSy \to x - y$$
$$xMy \to x \times y$$
$$xDy \to x \div y$$

$$4D2S3M6A12 = 4 \div 2 - 3 \times 6 + 12$$
$$= 2 - 3 \times 6 + 12$$
$$= 2 - 18 + 12 = -4$$

79. Go through options
L.C.M. of 204 and 170 = 1020.

80. $1 - 3 + 5 - 7 + 9 - 11 + \ldots + 197 - 199$
$= (-2) + (-2) + (-2) + \ldots + (-2)$ [50 times]
$= 50 \times (-2) = -100,$ Hence (d).

81. $$\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{5}\right)\ldots\left(1 - \frac{1}{n}\right)$$
$$= \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \ldots \times \frac{(n-2)}{(n-1)} \times \frac{(n-1)}{n} = \frac{1}{n}$$

82. $$\text{Minimum of } \frac{x}{y} = \frac{\text{Max. of } x}{\text{Min. of } y} = \frac{2}{32} = \frac{1}{16}$$

$$\text{and Maximum of } \frac{x}{y} = \frac{\text{Max. of } x}{\text{Min. of } y} = \frac{8}{16} = \frac{1}{2}$$

83. $56 = 1 \times 56 \Rightarrow$ H.C.F. is 1, hence number of tiles = 56
$2 \times 28 \Rightarrow$ H.C.F. is 2, hence number of tiles = $1 \times 14 = 14$
$4 \times 14 \Rightarrow$ H.C.F. is 2, therefore number of tiles = $2 \times 7 = 14$
$7 \times 8 \Rightarrow$ H.C.F. is 1, therefore number of tiles = $7 \times 8 = 56$
hence the minimum number of tiles can be 14. Thus (c) is correct option.

84. $$x + \frac{9x}{4} = 221 \Rightarrow x = 68 \text{ metre}$$

and $$\frac{9}{4}x = 153 \text{ metre}$$

Thus the difference between two parts = 153 − 68 = 85 metre

85. Total prime numbers between 1 and 100 = 25
and total prime numbers between 101 and 200 = 21 } 46

86. $$\frac{6^{36}}{215} \to \frac{(6^3)^{12}}{215} \to \frac{(216)^{12}}{215} \to \frac{1^{12}}{215} \to 1 \text{ is the remainder.}$$

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87. $\frac{12^{13}}{11} \rightarrow$ Remainder is 1.

$\frac{23^{13}}{11} \rightarrow$ Remainder is 1.

$\therefore$ The remainder of $\frac{(12^{13} + 23^{13})}{11}$ is $(1 + 1) = 2$

88. The least possible number = (L.C.M. of 4, 5, 6, and 7) + 3

$= 420 + 3 = 423$

The next higher number $= (420m) + 3$

Now put the least possible value of m such that

$(420m + 3) \geq 1000$

So at $m = 3$, the required value $= 1260 + 3 = 1263$

Alternatively : Option (a) and (b) are ruled out since when we divide these numbers by 5 they do not leave the remainder 3.

Again check the option (c) instead of (d) because we are required to find out the smallest possible number.

89. $13^{31} > 31^{13}$

$\frac{13^{31}}{13^{13}} > \frac{31^{13}}{13^{13}}$

$13^{18} > (2.4)^{13}$

Hence it is correct.

Again $10^{100} < 100^{10}$

$10^{100} < (10^2)^{10}$

$10^{100} < 10^{20}$ which is wrong.

Again, $2^{32} < 32^2$

$2^{32} < (2^5)^2$

$2^{32} < 2^{10}$ which is also wrong.

Hence (d) is the correct option.

90. Since it is not clear that which particular numbers are odd and which are even.

So we can't say about option (a), (c) and (d). Again in option (b) there are 1 more than half of the total numbers, so we can infer that there must be atleast one even number. Further we know that the product with even number is always even

i.e. even × odd = even

and even × even = even

Thus option (b) is confirm.

91. (43 and 387) and (129, 301)

$43k + 43l = 430$

$\Rightarrow \quad 43(k + l) = 430$

$\Rightarrow \quad k + l = 10$

but k, l must be coprimes.

So $(k, l) = (1, 9)$ and $(3, 7)$

92. $2^{73} - 2^{72} - 2^{71} = 2^{71}(2^2 - 2^1 - 1)$

$= 2^{71}(4 - 2 - 1) = 2^{71}$

93. Since we know that if $a + b + c = 0$, then

$a^3 + b^3 + c^3 = 3abc$

Hence $55^3 + 17^3 - 72^3 = 3 \times 55 \times 17 \times (-72)$

$= -(3 \times 5 \times 11 \times 17 \times 2^3 \times 3^2)$

Thus the given number is divisible by 3, 17, 40, 11 and 15.

Hence (d) is the most appropriate answer.

94.

$$
\begin{array}{r)l}
13 & 999999 \; (76923 \\
 & \underline{91} \\
 & 89 \\
 & \underline{78} \\
 & 119 \\
 & \underline{117} \\
 & 29 \\
 & \underline{26} \\
 & 39 \\
 & \underline{39} \\
 & \times
\end{array}
$$

Thus $abcde = 76923$

$\therefore \quad a + b + c + d + e = 7 + 6 + 9 + 2 + 3 = 27$

95. $n + (n + 2) + (n + 4) = 3n + 6 = 3(n + 2)$,

where $n = 2, 4, 6, \ldots$

So it is always divisible by 12.

Hence (d).

96. Since we know that $(a^3 + b^3) = (a + b)(a^2 + ab + b^2)$ which is divisible by $(a + b)$ hence leaves no any remainder.

97. $N = n(n + 1)(n + 2)(n + 3)(n + 4)$

at $n = 1$, $N = 120 = 1 \times 2 \times 3 \times 4 \times 5$

$n = 2$, $N = 720 = 2 \times 3 \times 4 \times 5 \times 6$

$n = 3$, $N = 2520 = 3 \times 4 \times 5 \times 6 \times 7$

$n = 4$, $N = 6720 = 4 \times 5 \times 6 \times 7 \times 8$ etc.

Hence the possible number is 120.

98. $(x - y)^2 = x^2 + y^2 - 2xy = 25 - 24$

$\Rightarrow \quad (x - y) = \pm 1$

Similarly $(x + y) = \pm 7$

So, solving for x and y, we get

$x = -3, -4, 3$ and 4

and $y = -4, -3, 4$ and 3

Thus $x^{-1} + y^{-1} = \frac{1}{x} + \frac{1}{y} = \frac{7}{12}$ and $\frac{-7}{12}$

Hence (d).

99. $\frac{75^{75^{75}}}{37} = \frac{1^{75^{75}}}{37} \rightarrow$ Remainder is 1.

100. $\frac{p^2 + 17}{12} \rightarrow \frac{(p^2 + 5)}{12} \rightarrow$ Now since $p^2 = 12m + 1$.

Therefore $\frac{p^2 + 5}{12} \rightarrow \frac{(12m + 1) + 5}{12} \rightarrow \frac{12m + 6}{12}$

Hence the remainder is 6.

101. The unit digit of $(12345k)^{72}$ is 6.

Means the unit digit of k^{72} is 6.

Now, go through options and verify

$6^{72} \rightarrow$ unit digit is 6

$8^{72} \rightarrow$ unit digit is 6

$2^{72} \rightarrow$ unit digit is 6

Hence (d).

102. **Go through option.** If you subtract 2, 8 or 25 from 369, then you will not obtain a perfect cube number. So (b) is the correct answer.

Alternatively : If you write down the nearest perfect cube such that $k \leq 369$, where k is a perfect cube, then

$7^3 = 343$, which is just below 369.

So $369 - 343 = 26$

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103. $n(n+2) = 624$

$\therefore$ from options $24 \times 26 = 624$

or find the roots of quadratic equation $n^2 + 2n - 624 = 0$.

104. $\frac{1719 \times 1715 \times 1713}{17} \to \frac{2 \times 15 \times 13}{17} \to \frac{30 \times 13}{17} \to \frac{13 \times 13}{17}$

$\to \frac{169}{17} \to 16$, is the required remainder.

105. $\frac{12.5}{1/6} = \frac{12.5 \times 6}{1} = 75$

106. $32k - 23k = 315$

$\Rightarrow \quad 9k = 315 \Rightarrow k = 35$

$\therefore$ The original product $= 23 \times 35 = 805$

107. Let the fraction be $\frac{x}{y}$, then

$$\left(\frac{x}{y}\right)^2 = \frac{x^2}{y^2}$$

then $$\frac{\frac{2}{3}x^2}{\frac{1}{5}y^2} = \frac{10x^2}{3y^2}$$

Thus $$\frac{10x^2}{3y^2} = 2\frac{x}{y}$$

$\Rightarrow$ $$\frac{x}{y} = \frac{3}{5}$$

Hence $x + y = 3 + 5 = 8$

108. The required number

= H.C.F. of (221–104), (377–221) and (377–104)

= H.C.F. of 117, 156, 273 = 39

109. $(ab^2)^{1/3} = 125 = 5^3$

$\Rightarrow \quad ab^2 = (5^3)^3$

$ab^2 = 5^9$

Now since $a \neq b \neq 1$ and $(a, b) > 1$.

Therefore a and b must be equal to the some power of 5.

Again since $a > b$, so check the option (b).

Consider $a = b^4$

$\therefore \quad ab^2 = 5^9$

$b^6 = 5^9$

$\Rightarrow \quad b^2 = 5^3$

then $a = b^4 = 5^6$

$\therefore \quad ab^2 = 5^6 \times 5^3 = 5^9$, which is correct.

Thus the presumed option (b) is correct.

Option (a) is clearly wrong and option (c) is also wrong.

Because $a^{3/2} = b^{2/3} \Rightarrow a^{9/6} = b^{4/6} \Rightarrow a < b$, which is inadmissible.

110. $24 = 8 \times 3$, so the given number must be divisible by both 8 & 3.

Now for 8, $n = 1, 5, 9$

Again for 3, when $n = 1$ then $m = 3, 6, 9$

and when $n = 5$ then $m = 2, 5, 8$

and when $n = 9$ then $m = 1, 4, 7$

Thus there are total 9 combinations of m and n.

111. Let the fraction be $\frac{x}{y}$, then its reciprocal be $\frac{y}{x}$,

$\therefore \quad \frac{\frac{x}{y}}{\frac{y}{x}} = \frac{x^2}{y^2}$

Now $\frac{x^2}{y^2} \times \frac{x}{y} = \frac{x^3}{y^3} = \frac{512}{125} = \frac{8^3}{5^3}$

$\Rightarrow \quad \frac{x}{y} = \frac{8}{5}$

$\therefore \quad \frac{y}{x} = \frac{5}{8}$

$\therefore \quad \frac{x}{y} - \frac{y}{x} = \frac{8}{5} - \frac{5}{8} = \frac{64 - 25}{40} = \frac{39}{40}$

112. Consider the values as $441 - 41 = 400$

$441 + 43 = 484$

So, $41 + 43 = 84$, which is divisible by 4.

113. $(11)^a = (19)^b = (209)^c = z$

$\Rightarrow \quad 11 = z^{1/a}, 19 = z^{1/b}, 209 = z^{1/c}$

Therefore $11 \times 19 = 209$

$z^{1/a} \times z^{1/b} = z^{1/c} = z^{(1/a + 1/b)} = z^{1/c}$

$\Rightarrow \quad \frac{1}{a} + \frac{1}{b} = \frac{1}{c}$

$\Rightarrow \quad c(a + b) = ab$

114. $\frac{2^{54} - 1}{9} = \frac{(2^6)^9 - 1}{9} = \frac{(64)^9 - 1}{9}$

$\Rightarrow \quad \frac{(1)^9 - 1}{9} \Rightarrow 0$

Hence there will be no any remainder.

115. 64 is the only number which is both a perfect square and a perfect cube.

So $6 + 4 = 10$.

116. The only possible number is 729 which is the cube of 9 and square of 27.

So $7 \times 2 \times 9 = 126$

117. In order to N be a least possible number $c < b < a$.

But $c \neq 1$, since $a \neq b \neq c$.

So the least possible value of $c = 2$, then

$c^8 = 2^8 = 256 = 4^4 = 16^2$

Thus $a = 16$, $b = 4$ and $c = 2$ the required number is 256.

118. Go through option.

119. There are only 4 digits whose square also gives the same unit digit which are 0, 1, 5 and 6.

But the number whose unit digit is 0 (zero), this number after squaring makes the tense digit zero also. Again it means the tens digit of the original number be zero. Now if we square such a number whose tens and unit digits are zero, this number gives 4 zeros (at least) on the end of the square which is not possible since 000 can not be considered a relevent number. Thus zero cannot be the unit digit.

Now, if we consider unit digit 1, and any digit except zero.

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NOTE Zero shows the same behaviour as in the first case), then unit digit remains the same on being squared the given number but tens digit gets changed so, 1 as a unit digit is also impossible. Now if we consider 6 as probable unit digit similar behaviour is seen with 6 also.
Thus, if we consider 5 as a unit we always get 5 as a unit digit after squaring any number whose unit digit is 5. Also we always get the 2 as the tens digit when we square any number whose unit digit is 5. Now to maintain the tens digit '2' we consider a number whose tens and unit digits be respectively 2, 5. Now if we square to 25, we get a three digit number 625. It means if it is possible, then the three digit given number must be 625.

Now, $(625)^2 = 390625$,

which satisfies the given conditions.

Now $3 \times 9 \times 0 \times 6 \times 2 \times 5 = 0$

Hence (a) is correct.

120. $3^{6n} + 4^{6n} \to 9^{3n} + 16^{3n}$

Now $9^{3n} + 16^{3n}$ is divisible by $25\ (= 9 + 16)$

Since $(a^n + b^n)$ is divisible by $(a + b)$ only when 'n' is odd.

Hence n must be odd.

121.

5	100
5	20 →
	4 →

Number of zeros = 24

122.

3	41
3	13 →
3	4 →
	1 →

$41! = 3^{18} \times k^a \times l^b$

123. Go through options

$$36 = 4 \times 9 = (2)^2 \times (3)^2$$

3, 6, 9 are in G.P.; $9 = (3 + 6)$

$$36 + 27 = 63$$

124. $7777 + 7777 \times 7777 \times (5 \div 77) \times (11 \div 35)$

$$= 7777 + 7777 \times 7777 \times \frac{5}{77} \times \frac{11}{35}$$

$$= 7777 + 1111 \times 1111$$

$$= 7777 + 1234321 = 1242098$$

Hence (d) is the answer.

125. Go through options

$$\frac{7}{8} + \frac{8}{7} = \frac{113}{56}$$

and $\frac{8}{7} - \frac{7}{8} = \frac{15}{56}$

Alternatively $a + \frac{1}{a} = \frac{113}{56}$

and $a - \frac{1}{a} = \frac{15}{56}$

Solve these equations.

126. Let $\frac{x}{y}$ be the fraction, then

$$\frac{x+4}{y} = \frac{1}{3}$$

and $\frac{x}{y+3} = \frac{1}{6}$

$\Rightarrow$ $3x + 12 = y$

and $6x = y + 3$

$\Rightarrow$ $x = 5$ and $y = 27$

Thus $x + y = 32$

127. Go through options and subtract the reasonable value from 7777 and see that whether the resultant number is a perfect square.

Alternatively : $n^2 \le 777 \Rightarrow n^2 = 7744$

Hence 33 (= 7777 – 7744) chairs are required to remove from that place.

128. $p^r \cdot p^{-1} \cdot p^s = (\sqrt{p^3})^2$

$\Rightarrow$ $p^{r+s-1} = p^3$

$\Rightarrow$ $r + s - 1 = 3$ ...(1)

$\Rightarrow$ $r + s = 4$

Again $p^{3/2} \cdot p^r = p^s \cdot p^{-1/2}$

$\Rightarrow$ $p^{r+3/2} = p^{s-1/2}$

$\Rightarrow$ $r + \frac{3}{2} = s - \frac{1}{2}$

$\Rightarrow$ $r - s = -2$ ...(2)

$\Rightarrow$ $r = 1$ and $s = 3$

Thus $(r+s)^{r+s} = (4)^4 = 256$

Hence (d).

129. $5^{-k} = \frac{1}{l} \Rightarrow l = 5^k$

$5^{3k} = (5^k)^3 = (l)^3$

Hence (c) is correct.

130. $5^{x-1} + 5^x + 5^{x+1} = 775$

$\Rightarrow$ $5^{x-1}(1 + 5 + 5^2) = 775$

$\Rightarrow$ $5^{x-1}(31) = 775$

$\Rightarrow$ $5^{x-1} = 25 \Rightarrow 5^2$

$\Rightarrow$ $x - 1 = 2 \Rightarrow x = 3$

131. The perfect square digits are 1, 4 and 9.
So the number formed by using any of these two digits which is also a perfect square is 49.

where $(4 \times 9) + (4 + 9) = 49$

$36 + 13 = 49$

$49 = 49$

So, $94 - (9 + 4) = 81$

132. Consider some consecutive values.

133. $25930800 = 2^4 \times 3^3 \times 5^2 \times 7^2$

So, if this number is divided by 3, then we get the quotient as a perfect square.

134. Let the Arun has x books and Prabhat has y books with them, then

$2(x - 3) = (y + 3)$

$2x - 6 = y + 3$

$\Rightarrow$ $2x - y = 9$ ...(1)

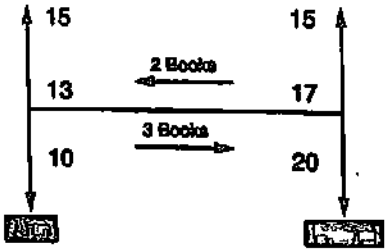
and $(x + 2) = (y - 2)$

$\Rightarrow$ $x - y = -4$ ...(2)

Solving equations (1) and (2), we get $x = 13$ and $y = 17$ thus $x + y = 30$.

Alternatively : Go through option.

Consider Option (c) :



(Solve in reverse)

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136. Weight of Bottle + Total volume of water = 1600 gm

Weight of Bottle + $\frac{1}{3}$ volume of water = 900 gm

$\Rightarrow \frac{2}{3}$ volume of water = 700 gm

$\Rightarrow$ total volume of water = $700 \times \frac{3}{2} = 1050$ gm

$\therefore$ The weight of empty bottle = 550 gm

$= (1600 - 1050)$

137. $(10a + b) \times (10c + d) - (10b + a) \times (10d + c)$

$= (100a.c + 10b.c + 10a.d + b.d)$

$- (100b.d + 10b.c + 10a.d + a.c)$

$= 99\,(a.c - b.d)$

Now, inorder to the difference be maximum so $a \times c$ will be maximum and $b.d$ will be minimum, thus $a \times c = 9 \times 8 = 72$

and $b \times d = 1 \times 2 = 2$

Hence $99\,(ac - bd) = 99\,(72 - 2) = 99 \times 70 = 6930$

138. $12600 = 2^3 \times 3^2 \times 5^2 \times 7$

Required value $= 2^{(n-1)} = 2^{(4-1)} = 2^3 = 8$

where $n \rightarrow$ number of prime factors.

Here 2, 3, 5, 7 are total 4 prime factors.

139. Go through options.

140. $(2^{123} - 2^{122} - 2^{121}) \times (3^{234} - 3^{233} - 3^{232})$

$= 2^{121}\,(2^2 - 2 - 1) \times 3^{232}\,(3^2 - 3 - 1)$

$= 2^{121}\,(1) \times 3^{232} \times 5$

$= 2^{121} \times 3^{232} \times 5^1$

Therefore there will be only one zero at the end of the product.

LEVEL 2

1. Since these bells tolls 18 times in 24 hrs

So the min. time interval when they toll together $= \frac{24}{18}$

$= 80$ minutes

So the required number of bells

= total number of different factors of 80

Now since $80 = 2^4 \times 5$

$\therefore$ Total number of factors $= (4 + 1)(1 + 1) = 10$

Thus the maximum number of bells = 10

NOTE The time interval of all the bells individually will be the factor of 80.

2. Consider the following relation and compare with $p^q = q^r$

$2^4 = 4^2 \Rightarrow p = 2,\ q = 4,\ r = 2$

$\Rightarrow q > r$

Again $4^2 = 2^4 \Rightarrow p = 4,\ q = 2,\ r = 4$

$\Rightarrow q < r$

So we can't say exactly whether $q < r$ or $q > r$.

NOTE If we know the relation between p and q then only we can say about q and r.

3. By Hit and trial plus your intelligence and familiarity with the numbers will help you in finding the following relation.

Now follow the given directions :

$7^3 - 3^5 = (7 + 3)^{5-3}$

$343 - 243 = 10^2$

$100 = 100$

Hence $p + q + r = 7 + 5 + 3 = 15$.

Thus (c) is correct.

4. Given that $a^2 + b^2 + c^2 + d^2 = 1$

Now, the maximum value of $a.b.c.d$ will be only when

$a = b = c = d = k$

Then $a^2 + b^2 + c^2 + d^2 = 4k^2 = 1$

$\Rightarrow k = \pm \frac{1}{2}$

Now $a.b.c.d = k^4 = \left(\pm \frac{1}{2}\right)^4 = \frac{1}{16}$

Hence (c) is correct.

5. $n^2 - 2\,(n!) + n = 0$

$\Rightarrow n^2 + n = 2\,(n!)$

$\Rightarrow n\,(n + 1) = 2\,(n!)$

Now, going through options, we see that 1 and 3 are the two possible values hence (d).

6. Try and find the following suitable values for the given relation

$3^6 = 9^3 = (6!) + 9$

$729 = 729 = 720 + 9$

$729 = 729 = 729$

where $m = 3,\ n = 6$ and $p = 9$

$\therefore m.n.p = 162$

7. $n + (n - 1) + (n - 2) = n\,(n - 1)(n - 2)$

$\Rightarrow 3n - 3 = n\,(n - 1)(n - 2)$

$\Rightarrow 3\,(n - 1) = n\,(n - 1)(n - 2)$

$\Rightarrow n\,(n - 2) = 3$

$\Rightarrow n = 3, -1$

But the only possible values is $n = 3$.

Hence $(n!)^n = (3!)^3 = (6)^3 = 216$

8. $n^3 - n^2 - n = n$

$\Rightarrow n^3 - n^2 - 2n = 0$

$\Rightarrow n\,(n^2 - n - 2) = 0$

$\Rightarrow n = 0$ or $(n^2 - n - 2) = 0$

Now, if $(n^2 - n - 2) = 0$

$\Rightarrow n = -1, 2$

$\therefore n = -1, 0, 2$ Hence (c).

9. $\frac{12}{3} = 4$

First of all you should know that if $n_1 + n_2 + n_3 + \ldots = 0$

$\Rightarrow n_1 = n_2 = n_3 \ldots = 0$

Again, each term is the product of three quantities (or numbers). So only one number which is equal to zero is enough to make the product zero. Thus a particular single number, which is equal to zero can make three terms zero, because it present in three consecutive terms.

For example if $k_3 = 0$, then $k_1k_2k_3 = k_2k_3k_4 = k_3k_4k_5 = 0$

and if $k_6 = 0$, then $k_4k_5k_6 = k_5k_6k_7 = k_6k_7k_8 = 0$

and if $k_9 = 0$, then $k_7k_8k_9 = k_8k_9k_{10} = k_9k_{10}k_{11} = 0$

and if $k_{12} = 0$, then $k_{10}k_{11}k_{12} = k_{11}k_{12}k_{13} = k_{12}k_{13}k_{14} = 0$ etc.

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10. $3^2 + 3^4 = (3^2 + 1)^2 - (3^2 + 1)$

$9 + 81 = 100 - 10$

$90 = 90$

and $1^0 + 1^2 = (1^2 + 1)^2 - (1^2 + 1)$

$1 + 1 = 4 - 2$

$2 = 2$

Alternatively : $n^{(n-1)} + n^{(n+1)} = (n^2 + 1)^2 - (n^2 + 1)$

$\Rightarrow \quad n^{(n-1)}(1 + n^2) = (n^2 + 1)[(n^2 + 1) - 1]$

$n^{(n-1)} = [n^2]$

$\Rightarrow \quad n - 1 = 2 \Rightarrow n = 3$

and if $n = 1$, then $1^0 = 1^2 = 1$

Thus $n = 1, 3.$

11. $1^3 + (12)^3 = (10)^3 + (9)^3$

$1 + 1728 = 1000 + 729$

$1729 = 1729$

12. $8^6 - 5^6 = (8^2)^3 - (5^2)^3 = (64)^3 - (25)^3$

$= (64 - 25)(64^2 + 25^2 + 64 \times 25) = 39 \times 6321$

$= 13 \times 3 \times 3 \times 7 \times 7 \times 43$

So it can be divided by $13 \times 7 = 91, 7 \times 7 = 49, 3 \times 43 = 129$

Hence, all of these are possible.

13. Since the product of 4 prime factors = 1365

$= 3 \times 5 \times 7 \times 13$

and the sum of the 3 prime factors = $25 = (5 + 7 + 13)$

Now, total number of factors of the required number $N = 24$

$= 2^3 \times 3 \Rightarrow (1 + 1)(1 + 1)(1 + 1)(2 + 1)$

Let N can be expressed as $N = 3^p \times 5^q \times 7^r \times 13^s$

Thus, in order to N will be greatest possible number in the above expressed form, the power of the greatest prime factors will be greater.

So $N = 3 \times 5 \times 7 \times 13^2$

$= 105 \times 169 = 17745$

14. Obviously (as per the given options) no any odd digit will appear as the last (or unit digit).

Hence (d) since even × even = even.

15. $\dfrac{R(135)}{R(100, 35)} = \dfrac{(135)!}{100 \times 101 \times 102 \times 103 \times \ldots \times 135}$

$= \dfrac{1 \times 2 \times 3 \times \ldots \times 99 \times 100 \times 101 \times 102 \times \ldots \times 135}{100 \times 101 \times 102 \times \ldots \times 135}$

$= 1 \times 2 \times 3 \times \ldots \times 99 = (99)!$

16. $R(17) . R(19, 62)$

$= (1 \times 2 \times 3 \times 4 \times 5 \times 6 \times \ldots \times 17) \times (19 \times 20 \times 21 \times 22 \times \ldots \times 81)$

$= \dfrac{(1 \times 2 \times 3 \times \ldots \times 81)}{18} = \dfrac{81!}{18}$

17. $R(2, 995) = 2 \times 3 \times 4 \times \ldots \times \ldots \times 995 \times 996 \times 997$

$R(996, 1) = 996 \times 997$

∴ L.C.M. of $R(2, 995)$ and $R(996, 1)$

$= 1 \times 2 \times 3 \times \ldots \times 997 = 997!$

18. $R(139, 2) = 139 \times 140 \times 141$

$R(141) = 1 \times 2 \times 3 \times \ldots \times 141$

∴ H.C.F. of $R(139, 2)$ and $R(141) = 139 \times 140 \times 141$

$= 2743860$

19. Since $abc \times 1001 = abcabc$

Thus the number $abcabc$ is always divisible by 1001 and its all factors. So (d) is the best option. Since $1001 = 7 \times 11 \times 13$.

20. Numbers divisible by 5 upto 1155 = 231

Numbers divisible by 7 upto 1155 = 165

Numbers which are divisible by both 5 and 7 (*i.e.*, 35) upto 1155 = 33

∴ Numbers which are exactly divisible by 5 or $7 = (231 + 165) - 33 = 363$

Numbers which are divisible by 5 and 11 both (*i.e.*, 55) = 21

Numbers which are divisible by 7 and 11 both (*i.e.*, 77) = 15

Numbers which are divisible by 5, 7 and 11 simultaneously = 3

∴ Numbers which are only divisible by either 5 and 11 or 7 and 11

$= 21 + 15 - 3 = 33$

Hence, the total number of numbers which are divisible by 5 or 7 but not by 11 = 363 − 33 = 330.

21. $4x + 7y = 3$

$\Rightarrow \quad y = \dfrac{3 - 4x}{7}; \quad x, y \in I$

Now, in the given range of x and y, the least possible value of x is − 99 and then $y = 57$.

Again the largest possible value of x is 97 and the corresponding value of y is −55. So in this range there are total 29 solution sets are possible.

(Since the possible values of x are −99, −92, −85, ... 99 and the possible values of y are 57, 53, 49, ... −51, −55)

22. See the following pattern :

$N_1 \times N_2 \rightarrow$ Unit digit	$N_1 \times N_2 \rightarrow$ Unit digit
$1 \times 3 \rightarrow 3$	$10 \times 12 \rightarrow 0$
$2 \times 4 \rightarrow 8$	$11 \times 13 \rightarrow 3$
$3 \times 5 \rightarrow 5$	$12 \times 14 \rightarrow 8$
$4 \times 6 \rightarrow 4$	$13 \times 15 \rightarrow 5$
$5 \times 7 \rightarrow 5$	$14 \times 16 \rightarrow 4$
$6 \times 8 \rightarrow 8$	$15 \times 17 \rightarrow 5$
$7 \times 9 \rightarrow 3$	$16 \times 18 \rightarrow 8$
$8 \times 10 \rightarrow 0$	$17 \times 19 \rightarrow 3$
$9 \times 11 \rightarrow 9$	$18 \times 20 \rightarrow 0$ etc.

Except to first column in all the rest columns the least occuring unit digits are 9 and 4, and they occur same number of times. But in the first column 9 is not occuring. Thus finally '9' as the unit digit occurs least number of times *i.e.*, one times less as '4' occurs.

23. Except the first column the sum of all the unit digits

$= 99 (9 + 0 + 3 + 8 + 5 + 4 + 5 + 8 + 3 + 0)$

$= 99 (45) = 4455$

and the sum of all the unit digits of first column

$= (3 + 8 + 5 + 4 + 5 + 8 + 3 + 0) = 36$

Thus the sum of all the unit digits = 4455 + 36 = 4491

Hence (a).

24. Since a cube has to be cut from three sides (*i.e.*, along 3 dimensions). Hence the number of cuts will be equal to the sum of the 3 factors of 960, when 960 is expressed as the product of 3 factors.

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Again since the number of cuts to be applied are minimum. Hence the sum of these 3 factors of 960 must be minimum. So, in order to be the sum of 3 factors of 960 be minimum we have to have the minimum possible difference between the 3 factors of 960.

Thus $960 = 8 \times 10 \times 12$

Hence the minimum possible number of cuts

$$= (8 + 10 + 12) = 30$$

Now, if we want to maximize the product of any 3 factors whose sum is constant *i.e.*, 30, then it is possible only when all the 3 factors be equal. Thus a, b, c be the three factors such that :

$$a + b + c = 30$$

Then Maximum $(a \times b \times c) = 10 \times 10 \times 10 = 1000$

So, the maximum number of 1000 identical pieces can be formed by applying the same number of n cuts.

25. Since $45000 = 10 \times 4500$

So, the sum of all the factors of 45000 which are the multiples of 10 will be same as sum of all the factors of 4500 multiplied by 10.

Now, $4500 = 2^2 \times 3^2 \times 5^3$

Therefore the sum of all the factors of

$$4500 = \frac{(2^3 - 1) \times (3^3 - 1)(5^4 - 1)}{(2 - 1)(3 - 1)(5 - 1)}$$

$$= \frac{7 \times 26 \times 624}{1 \times 2 \times 4} = 14196$$

Hence the sum of all the products of 45000 which are the multiples of $10 = 10 \times 14196 = 141960$.

26. The unit digit of $(1!)^{1!} = 1$

The unit digit of $(2!)^{2!} = 4$

The unit digit of $(3!)^{3!} = 6$

The unit digit of $(4!)^{4!} = 6$

The unit digit of $(5!)^{5!} = 0$

Now since we know that 5!, 6!, 7!, 8!, ... all have their unit digits zero.

Thus the sum of all the unit digits = 7

(Since $1 + 4 + 6 + 6 + 0 = 17$)

27. To know the remainder, when any number is divided by 5, we just need to know the unit digit of the dividend. Further from the previous question, we know that the unit digit of the sum of the whole expression is 7.

So divide 7 by 5 and get 2 as the remainder.

28. The last two digits of $(1!) = 01$

The last two digits of $(2!)^2 = 04$

The last two digits of $(3!)^3 = 16$

The last two digits of $(4!)^4 = 76$

The last two digits of $(5!)^5 = 00$

Now, we can conclude that the last two digits of the higher numbers *e.g.*, $(6!)^6$, $(7!)^7$, $(8!)^8$ ... etc. are "00". So the last two digits of the whole expression = 9

(Since $01 + 04 + 16 + 76 + 00 = 97$)

29. Let the two parts be x and y then

$$x^2 - y^2 = 50\,(x - y)$$

$$\Rightarrow \quad (x + y) = 50$$

Hence the number is 50.

30. If $p = 3$, then $n = 5$ (Since $n > 2$)

If $p = 10$, then $n = 5$ also

If $p = 17$, then $n = 5$ also etc.

Hence $n = 5$.

31. Go through options :

	Anjali	Bhagwat	
Total Shots →	45	45	= 90
Total Hits →	36	30	= 66
Total miss shots →	9	15	

Since, $\frac{36}{9} = 4$ and $\frac{30}{15} = 2$, $\frac{4}{2} = 2$,

Hence, the presumed option is correct.

32. This is possible only when $n = 4, 10, 16, 22, 28, 34, \ldots$ hence the remainder will be 3 when $(n - 1)$ will be divided by 6.

33. Option (b) is obvious, as 2222 and 7777 both numbers are divisible by 101.

Now we know that if the sum of the remainders of two or more numbers are divisible by the given divisor then the required expression is also divisible by the divisor. The remainder when $(2222)^{7777}$ is divided by 13 is 12 and the remainder when $(7777)^{2222}$ is divided by 13 is 9.

Hence the given expression is not divisible by 13.

(Since $(12 + 9) \neq 13m$ for any positive integer m)

Again the remainder when $(2222)^{7777}$ is divided by 99 is 44 and the remainder when $(7777)^{2222}$ is divided by 99 is 77.

Hence the whole expression cannot be divided by 99.

(Since $44 + 77 = 121$ which is not divisible by 99.)

34. Let there be x students, then

$$\frac{x}{2} + \frac{x}{3} + \frac{x}{4} = 91$$

$$\Rightarrow \quad \frac{6x + 4x + 3x}{12} = 91$$

$$\Rightarrow \quad \frac{13x}{12} = 91$$

$$\Rightarrow \quad x = 84 \text{ students}$$

35. Best way is to go through convenient options.

$$1 + \frac{1}{4} \times 1 = \frac{5}{4} \qquad \left(\because \frac{1}{4} \to 25\%\right)$$

$$\frac{5}{4} + \frac{1}{2} \times \frac{5}{4} = \frac{15}{8} \qquad \left(\frac{1}{2} \to 50\%\right)$$

Hence option (c) is correct.

Alternatively : Let the first time rise in salary be x%, and the original salary be k then

$$k + \frac{kx}{100} = \frac{k\,(100 + x)}{100}$$

Again $\frac{k\,(100 + x)}{100} + \frac{k\,(100 + x)}{100} \times \frac{2x}{100} = \frac{k \times 15}{8}$

$$\frac{k\,(100 + x)}{100}\left[\frac{(100 + 2x)}{100}\right] = \frac{15k}{8}$$

$$\Rightarrow \quad \frac{(100 + x)}{100}\,\frac{(100 + 2x)}{100} = \frac{15}{8}$$

So simplify and solve the quadratic equation and get the values of x, then twice the value of x will be the required value.

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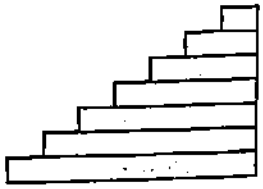
36. $5^{72}\times 8^{27} = 5^{72}\times 2^{81} = (5^{72}\times 2^{72})\times 2^9$
$= 512\times 10^{72}$

Therefore there are total 75 (= 3 + 72) digits in the above expression.

37. Find the sum of n natural numbers which is just greater than 500. So if we consider that he had to be added 32 natural numbers then the ideal answer would have been 528. It means he had missed 28 during the process.

NOTE For clarification of the concept just write the natural numbers in sequence and then add up them one by one but not as the child did, then you will must get the lucid explanation of this problem.

38. Total distance to be covered
= total horizontal movement
+ total upward movement
$= (20\times 2 - 1) + (20)$
$= 39 + 20 = 59$ feet



39.

	Paltry	Sundry
Fired shots	5	7
Hit shots	2	3
Missed shots	3	4

When Sundry missed 32 shots, it means Paltry missed 24 shots.
When Paltry missed 24 shots, it means Paltry hit 16 shots.

40. It can be solved easily by options :
As consider option (a), then

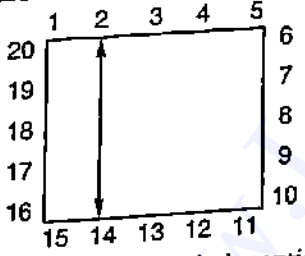
$$90\times\frac{5}{6} = 75$$

$$75 - 5 + 2 = 72$$

Again $$90\times\frac{4}{5} = 72$$

Hence, the assumed option is correct.

41. See the figure and visualise the actual situation.



42. Go through options and check the option (b).

Thus at $y = -2$ and $y = 4$, $p = \frac{1}{0}$, which is undefined.

43. $\frac{1}{1\times 2}+\frac{1}{2\times 3}+\frac{1}{3\times 4}+\frac{1}{4\times 5}+\ldots+\frac{1}{100\times 101}$

$=\left(\frac{1}{1}-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+\left(\frac{1}{4}-\frac{1}{5}\right)+\ldots+\left(\frac{1}{100}-\frac{1}{101}\right)$

$=1-\frac{1}{101}=\frac{100}{101}$

44.

$$\begin{matrix} R\to x+10 \\ L\to x+6 \\ B\to x+5 \\ H\to x+4 \\ A\to x \end{matrix}\begin{matrix}x\\x\\x\\x\\x\end{matrix} + \left(25\right)\begin{matrix}x+5\\x+5\\x+5\\x+5\\x+5\end{matrix}\begin{matrix}(5)^+\\(1)^+\\(1)^-\\(5)^-\end{matrix}$$

Thus total 6 coins have to be transferred.

45. As per the given expression the value of $f(x) = f\left(\frac{1}{x}\right)$

Try to put $\left(\frac{1}{x}\right)$ in place of x.

Thus the value of $f\left(\frac{1}{2}\right) = f(2) = 1081.58$

46. The unit digit of each pair is 4, and there are 50 such pairs which are mutually multiplied together. Thus finally we get 6 as unit digit. As

$$\underset{\text{Unit digit}}{\underbrace{4\times 9^2}_{4}\times\underbrace{4^3\times 9^4}_{4}\times\underbrace{4^5\times 9^6}_{4}\times\ldots\times\underbrace{4^{99}\times 9^{100}}_{4}}$$

Again $4\times 4\times 4\times 4\ldots 4$ (upto 50 times)
i.e., the unit digit of 4^{50}, which is 6
[Since unit digit of 4^{2n} is 6 for $n = 1, 2, 3, \ldots$ etc.]

47. The unit digit of each pair is 5 and there are total 50 such pairs.

$$\underset{\text{Unit digit}}{\underbrace{4+9^2}_{5}+\underbrace{4^3+9^4}_{5}+\underbrace{4^5+9^6}_{5}+\ldots+\underbrace{4^{99}+9^{100}}_{5}}$$

Thus $5 + 5 + 5 + \ldots 5$ (50 times)
Hence the unit digit = 0
[Since $5\times 50 = 250 \to$ unit digit is zero.]

48. Consider some appropriate values :
As $p = 3.99,\quad q = 4.99,\quad r = 6.99$
$A = [p + q + r] = [3.99 + 4.99 + 6.99] = [15.97] = 15$
$B = [p] + [q] + [r] = [3.99] + [4.99] + [6.99]$
$= 3 + 4 + 6 = 13$
Hence $A - B = 2$

49. Options (a), (b) and (c) are irrelevent.
as $(25)^2 = (5)^2\times 25$
or $(27)^2 = (9)^2\times 9$
or $(6)^2 = (4)^2\times 9$
or $(6)^2 = (3)^2\times 4$ etc.
Hence (d) is correct.

NOTE If $a = b.c$ then $a^2 = b^2.c^2$
$a^2 = b^2.k$
Thus k must be a perfect square.

50. Since odd × even = even
even × even = even
even × even = even
It means when the odd numbers are added even number of times the resultant is always an even number for example :
$3 + 5 = 8 \to$ even
$3 + 5 + 9 = 17 \to$ odd
$3 + 5 + 9 + 13 = 30 \to$ even etc.
Again, even + even = even
and odd + even = odd,
so the sum of odd numbers cannot be odd in order to be the resultant sum be an even number.

51. Go through options

$$1 + 2 + 3 + \ldots + k = N^2 \qquad \text{(a perfect square)}$$

$$\frac{k(k+1)}{2} = N^2$$

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$\frac{1\times 2}{2} = (1)^2 \rightarrow$ a perfect square number,

$\frac{8\times 9}{2} = 4\times 9 = (2\times 3)^2 \rightarrow$ a perfect square number.

and $\frac{49\times 50}{2} = 49\times 25 = (7\times 5)^2 \rightarrow$ a perfect square number.

52. Check through options consider some appropriate values.

$(17 + 15)$ and $(17 - 15)$

32 and 2, have a common factor 2.

$(7 + 4)$ and $(7 - 4)$

11 and 3, have no any common factor.

Hence options (a) and (b) are wrong and option (c) is correct.

53. If a and b be two such numbers, then

$$(ab)^{1/2} = 35 \Rightarrow ab = 1225$$

and $$\left(\frac{a+b}{2}\right) = 43\frac{3}{4} \Rightarrow (a+b) = \frac{175}{2}$$

$$\therefore \quad (a-b)^2 = (a+b)^2 - 4ab$$

$$(a-b)^2 = \frac{30625}{4} - 4900$$

$$(a-b) = \sqrt{\frac{30625 - 19600}{4}}$$

$$(a-b) = \frac{105}{2}$$

Now solving $(a+b) = \frac{175}{2}$

and $(a-b) = \frac{105}{2}$

we get $a = 70$ and $b = \frac{35}{2}$

Hence (c) is correct.

Alternatively : From option

$$\sqrt{70\times x} = 35$$

$$\Rightarrow \quad x = \frac{35}{2}$$

and $$\frac{70 + \frac{35}{2}}{2} = 43\frac{3}{4}$$

Hence (c) is the right answer.

54. Such a least possible number = [L.C.M. of 1, 2, 3, ..., n] $- 1$

Now if the L.C.M. of $(1, 2, 3, ..., n) = L$

then the next higher number $= (m.L) - 1$

and $m.L = n!$ for a suitable value of m; $m \in N$

Hence option (b) is most appropriate.

55. Let $n + (n+2) + (n+4) + (n+6) = k$

$$\therefore \quad 3n = 2(n+4)$$

$$\Rightarrow \quad n = 8$$

$$\therefore \quad k = 8 + 10 + 12 + 14$$

$$k = 44$$

56. Go through option and match the conditions.

Alternatively : Let there be k number of benches.

$$5k + 4 = 11(k - 4)$$

$$\Rightarrow \quad 6k = 48$$

$$\Rightarrow \quad k = 8$$

Hence $5k + 4 = 44$

Thus there are 8 benches and 44 students.

57. $$\frac{m}{n} = 25\left(\frac{n}{m}\right)$$

$$\Rightarrow \quad \left(\frac{m}{n}\right)^2 = 25 \Rightarrow \frac{m}{n} = \frac{5}{1} = 5$$

58. 3^{3322} and 33^{332}

$$\Rightarrow \quad (3^4)^{830}\times 3^2 \text{ and } 33^{332}$$

Thus $(81)^{830}\times 3^2 > 33^{332}$

Again 3^{3322} and 333^{22}

$$\Rightarrow \quad (3^6)^{553}\times 3^4 \text{ and } (333)^{22}$$

Thus $(729)^{553}\times 3^4 > (333)^{22}$

Again 3^{3322} and 22^{333}

$$\Rightarrow \quad (3^3)^{1107}\times 3 \text{ and } 22^{333}$$

Thus $(27)^{1107}\times 3 > 22^{333}$

Hence (a).

59. $$a\times b = 25$$

$$(-1)\times(-25) = 25$$

Thus the minimum required value of $a + b = (-1) + (-25)$

$$= -26$$

Hence (d).

60. The least value of $p.q = 5\times(-10) = -50$

The least value of $p.q^6 = 4\times(-9)^6 = 4\times(729)^2$

The least value of $p^6.q = 5^6\times(-10) = (-156250)$

The least value of $(pq)^5 = [5\times(-10)]^5 = (-312500000)$

Hence (d) is the correct option.

61. Please remember that all the factorial numbers greater than or equal to n are divisible by n. Hence 7!, 8!, 9!, 10!, ... etc. are divisible by 7. So we have to check only $1! + 2! + 3! + 4! + 5! + 6!$

Hence $1! + 2! + 3! + 4! + 5! + 6!$

$$= 1 + 2 + 6 + 24 + 120 + 720 = 873$$

Thus 873 leaves a remainder of 5 when divided by 7.

Therefore $1! + 2! + 3! + 4! + 5! + ... + 77!$ leaves remainder 5.

62. It does not satisfy options a, b, c. Therefore option (d) is correct.

63. For one digit number $1\times 9 = 9$

for two digit numbers $2\times 90 = 180$

for three digit numbers upto $500 = 3\times 401 = 1203$

Thus the required value $= 9 + 180 + 1203 = 1392$

64. If a, b, c, d, e, f all the six forms of an A.P. can have only 9 values hence the common difference of the A.P. must be 1. (Since common difference cannot be greater than 1.)

$\therefore$ If $a = 1$, then

$1^2\times 3^4\times 5^6 \rightarrow$ unit digit is 5

If $a = 2$, then

$2^3\times 4^5\times 6^7 \rightarrow$ unit digit is 2

If $a = 3$, then

$3^4\times 5^6\times 7^8 \rightarrow$ unit digit is 5

If $a = 4$, then

$4^5\times 6^7\times 8^9 \rightarrow$ unit digit is 2

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65. The greatest possible number
= H.C.F. of (1313 – 17) and (621 – 9)
= H.C.F. of 1296 and 612 = 36
Now the total number of factors of 36 = (2 + 1)(2 + 1) = 9
(Since $36 = 2^2 \times 3^2$)

66. Except to $x = 0$ and $x = 5$ all the values are acceptible, hence the range of x is $R - \{0, 5\}$ where R represents all the real numbers.

67.
$$p_1 + p_2 + p_3 + p_4 = 204 \quad \ldots(1)$$
$$p_1 + p_4 = p_2 + p_3 \quad \ldots(2)$$
$$\Rightarrow \quad p_2 - p_1 = p_4 - p_3 \quad \ldots(3)$$
and
$$p_3 - p_2 = 2(p_2 - p_1)$$
$$= 2(p_4 - p_3)$$
$$\Rightarrow \quad p_1 + x = p_2$$
$$p_2 + 2x = p_3 \quad \ldots(4)$$
$$p_3 + x = p_4$$
Also $\left(\frac{p_1 + p_2 + p_3 + p_4}{4}\right) = 51 = 17 \times 3$

Thus from the relation 2, we can deduce that
$$p_1 + p_4 = p_2 + p_3 = 102$$
Thus by hit and trial we will find the only possible values of p_1, p_2, p_3 and p_4 = 31, 41, 61 and 71 respectively.

NOTE The unit digits of (p_1 and p_4) or (p_2 and p_3) can be only (1 and 1), (3 and 9), but not necessarily in the order. So consider only those two digit prime number which satisfy all the given conditions.

68. Let the price of Sim card be x and instrument be y
then $x + y = 5000$
and $x = y - 4000$
$\Rightarrow \quad (y - 4000) + y = 5000$
$\Rightarrow \quad 2y = 9000 \Rightarrow y = 4500$
Alternatively : Go through option

Instrument	Sim card
4500	500 = 5000
4500	– 500 = 4000

69. At $x = 0$ or $y = 0$, we get infinite number of ordered solutions set.

70.
$$l = \frac{292k + 7}{23}$$
Therefore at $k = 1$, $l = 13$
Hence (d).

71. Since a, c are neither prime nor composite it means $a = c = 1$
and $b = 2, 3, 5, 7$ and $d = 2, 3, 5, 7$ but $b \neq d$.
Thus from $ab + 4 = cd$
$\Rightarrow$ $ab = 13$ and $cd = 17$, the only possible combination.
$\therefore \quad ba = 40 = dc$
$\Rightarrow \quad 31 + 40 = 71$
Therefore $\frac{ab + ba}{cd + dc} = \frac{13 + 31}{17 + 71} = \frac{44}{88} = \frac{1}{2}$

72. The best way to go through options.
Let there are initially 3 boxes then,
1st customer gets = $\frac{3}{2} + \frac{1}{2} = 2$
Remaining boxes = 3 – 2 = 1
2nd customer gets = $\frac{1}{2} + \frac{1}{2} = 1$
Remaining boxes = 0
Hence presumed option (b) is correct.

73. $123454321 = (11111)^2$
Hence (d).

74. $1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + \ldots - 198^2 + 199^2$
$= (1^2 - 2^2) + (3^2 - 4^2) + (5^2 - 6^2) + \ldots + (197^2 - 198^2) + 199^2$
$= (-3) + (-7) + (-11) + (-15) + \ldots + (-395) + (199)^2$
$= \left[-\left(\frac{3 + 395}{2}\right) \times 99\right] + 39601$
$= (-199 \times 99) + 39601$
$= (-19701) + 39601 = 19900$

75. The G.P. is as following :
$a, ar, ar^2, ar^3, \ldots$ etc.
then LCM of this G.P. = ar^{n-1}
and HCF of this G.P. = a
$\therefore \quad \frac{\text{LCM}}{\text{HCF}} = \frac{ar^{n-1}}{a} = r^{n-1}$

76. From option (a)
Total number of brothers and sisters = 4
Also there are two brothers and 2 sisters.

77.
$$(4 @ 5) = (4 \# 5) = \frac{4}{5} \quad \because 4 < 5$$
and
$$(6 @ 5) = (5 \# 6) = \frac{5}{6} \quad \because 6 > 5$$
and
$$\left(\frac{4}{5}\right) @ \frac{5}{6} = \frac{\frac{4}{5}}{\frac{5}{6}} \quad \because \frac{4}{5} < \frac{5}{6}$$
$$\therefore \quad (4 @ 5) @ (6 @ 5) = \frac{\frac{4}{5}}{\frac{5}{6}} = \frac{24}{25}$$

78. The only possible values of p, q, r are 1, 2, 4
So $s = 5$, $t = 6$, $u = 7$.
Hence the number is 124567 therefore
$$p + q + r + s + t + u = 25$$

79. $4^{1111} \times 5^{2222} = 2^{2222} \times 5^{2222} = 1 \times 10^{2222}$
Hence there will be 2223 number of digits.

80. Since N is always even. Thus p is always odd
[As even + odd = odd]

81. Let $\frac{p}{q} = -7$ and $\frac{r}{s} = 3$
then $\left|\frac{p}{q}\right| = 7$ and $\left|\frac{r}{s}\right| = 3$
Thus $\left|\frac{p}{q}\right| > \left|\frac{r}{s}\right|$.
Again if $\frac{p}{q} = 4$ and $\frac{r}{s} = 7$ then $\left|\frac{p}{q}\right| > \left|\frac{r}{s}\right|$, hence (d).

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82. Since all the numbers u, v, w, x are negative, but when $u^v + v^w + w^x = 0$, then u^v it means there must be some terms whose value will be positive and thus it adding up with negative value makes the expression zero.

e.g., $(k) + (-k) = 0$

Hence we can say that there must be some even integers which converts a negative number into a positive number. Further we know that an even number when multiplied with any other (even or odd) number it finally makes an even number.

Explanation : $u^v + v^w + w^x = 0$

$\Rightarrow$ $(-u)^{-v} + (-v)^{-w} + (-w)^{-x} = 0$ $\{\because u, v, w, x \in I^-\}$

Again, if $k + l + m = 0$

$\Rightarrow$ $k + l = m$ or $k + m = l$ or $l + m = k$

i.e., half of the numerical value will be positive and half of the numerical value will be negative since there are every non zero integer.

Now, if $(-k)^{\text{even}} \rightarrow$ positive value

So, if there exists some even integer n, then

$k \times l \times m \times n \rightarrow$ even (though k, l, m can be odd).

83. The unit digit of 2^{3^4} is 2.

The unit digit of 3^{4^5} is 1.

The unit digit of 4^{5^6} is 4.

The unit digit of 5^{6^7} is 5.

The unit digit of 6^{7^8} is 6.

The unit digit of 7^{8^9} is 1.

Therefore the unit digit of :

$$2^{3^4} \times 3^{4^5} \times 4^{5^6} \times 5^{6^7} \times 6^{7^8} \times 7^{8^9} = 0$$

(Since $2 \times 1 \times 4 \times 5 \times 6 \times 1 = 240$)

Alternatively : 5 always gives unit digit 5 irrespective of its (positive) power and similarly 6 also gives 6 as a unit digit irrespective of its power.

So, $5 \times 6 = 30$. Hence you will get the unit digit zero without solving the complete expression.

84. In order to be a, b, c be least, then

$$(a-7) = (b-10) = (c-12)$$

Again $(a-7)(b-10)(c-12) = 1000$

$\Rightarrow$ $10 \times 10 \times 10 = 1000$

$\Rightarrow$ $a = 17, \quad b = 20$ and $c = 22$

Therefore $(a + b + c) = 59$.

85. $\dfrac{23^{32} - 9}{16} \rightarrow \left(\dfrac{23^{32}}{16} - \dfrac{9}{16}\right) \rightarrow \left(\dfrac{7^{32}}{16} - \dfrac{9}{16}\right)$

$$\rightarrow \left[\frac{(7^2)^{16}}{16} - \frac{9}{16}\right] \rightarrow \left[\frac{1^{16}}{16} - \frac{9}{16}\right]$$

$$\rightarrow \frac{1-9}{16} \rightarrow \frac{-8}{16} \rightarrow \frac{16-8}{16} \rightarrow \frac{8}{16} \rightarrow 8$$

86. Very similar to the problem number 84.

Hence $(x-5) = (y+6) = (z-8) = 11$

$\Rightarrow$ $x = 16, y = 5, z = 19$

Thus $(x + y + z) = 40$.

87. $(x-6)(y+7)(z-4)$ will be maximum only when $(x-6) = (y+7) = (z-4) = k$ (say) for a given value of $(x + y + z) = 21$

$\Rightarrow$ $x = k + 6$

$y = k - 7$

$z = k + 4$

$\therefore$ $x + y + z = 21$

$\Rightarrow$ $(k+6) + (k-7) + (k+4) = 21$

$\Rightarrow$ $3k + 3 = 21$

$\Rightarrow$ $k = 6$

Therefore, $(x-6).(y+7)(z-4) = k^3 = (6)^3 = 216$

88. $\dfrac{3^{37} + 4^{37}}{7} \rightarrow \left(\dfrac{3^{37}}{7} + \dfrac{4^{37}}{7}\right) \rightarrow \left[\dfrac{(3^2)^{18} \times 3}{7} + \dfrac{(4^2)^{18} \times 4}{7}\right]$

$$\rightarrow \left(\frac{2^{18} \times 3}{7} + \frac{2^{18} \times 4}{7}\right) \rightarrow \left(\frac{(2^3)^6 \times 3}{7} + \frac{(2^3)^6 \times 4}{7}\right)$$

$$\rightarrow \left(\frac{3}{7} + \frac{4}{7}\right) = \left(\frac{7}{7}\right) \rightarrow \text{Remainder zero.}$$

Alternatively : $a^n + b^n$ is divisible by $(a + b)$ if n is odd. Hence $(3^{37} + 4^{37})$ is divisible by 7 $(= 3 + 4)$. Hence leaves no remainder.

89. $7^{74} - 4^{74}$ is divisible by 3

since $a^n - b^n$ is divisible by $(a - b)$ and

Again $7^{74} - 4^{74} = (7^2)^{37} - (4^2)^{37} = 49^{37} - 16^{37}$

Now, since $a^n - b^n$ is divisible by $(a - b)$ if n is odd, then $7^{74} - 4^{74}$ (*i.e.*, $49^{37} - 16^{37}$) is divisible by 11.

Hence (d).

90. Try to find out 11^2 in less than 22!.

You can't find. So (a) is the correct choice.

91. The number of zeros at the end of $2^3 \times 3^4 \times 4^5 \times 5^6 = 6$

The number of zeros at the end of $3^5 \times 5^7 \times 7^9 \times 8^{10} = 7$

The number of zeros at the end of $4^5 \times 5^6 \times 6^7 \times 7^8 = 6$

The number of zeros at the end of $10^2 \times 15^3 \times 20^4 = 7$

So the number of zeros at the end of the whole expression = 6.

93. Min. (1, 2, 3) + Max. (1, 2, 3)

= Min. (2, 6, 3) + Max. (1, 8, 3) = 2 + 8 = 10

94. Labh [Hani (1, 2, 3), Hani (2, 3, 4), Hani (3, 4, 5)]

$$= \text{Labh } [4, 5, 6] = \frac{4+5+6}{3} = 5$$

95. Min. [Hani (1, 2, 3), Hani (2, 3, 4), Hani (3, 4, 5)]

+ Max. [Labh (1, 2, 3), Labh (2, 3, 4), Labh (3, 4, 5)]

= Min. [4, 5, 6] + Max. [2, 3, 4]

= Min. (20, 30, 24) + Max. (8, 81, 16)

= 20 + 81 = 101

96. $\therefore$ $(55)_{10} = (25)_{25}$

25	55
25	2 → 5
	0 → 2

97. $Q = 4, 8$

$P = 3, 6, 9, 1, 4, 7;$ since $P, Q > 0$

So there are total 6 values of (P, Q).

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98. Total number of 6's = Number of actual 6's + Number of wrong 6's

Number of Actual 6's = 20 (as unit digits) + 20 (at tens digit) + 100 (as hundreds digit) = 140 6's

Number of wrong 6's (which are at the place of 8's) = 20 (at unit's place) + 20 (at tens place) = 40

Hence total number of 6's = 180

99. If $p^2 - pq = 0 \Rightarrow p(p-q) = 0 \Rightarrow p = 0$ or $p = q$, but since $p \neq q$, then $p = 0$ is only possible solution.

100. $15990 = 2 \times 3 \times 5 \times 13 \times 41$

$= 2 \times 3 \times 13 \times 5 \times 41 = 78 \times 205$

Hence the only possible option is (d).

101. $n = 1, 2, 3, \ldots, 7$.

102. If you go through option, then you will find option (c) is correct. Since divisor is always greater than remainder.

Alternatively : $\frac{A}{D} \rightarrow$ Remainder 13

$\frac{B}{D} \rightarrow$ Remainder 31

$\frac{A+B}{D} \rightarrow$ Remainder should be 44

But it is given that the remainder is 4.
Hence, the divisor must be 40 (= 44 – 4).

103. Since $(N)^{2p} \rightarrow$ unit digit 6

and $(N)^{(2p+1)} \rightarrow$ unit digit 4

It means every even power of N has unit digit 6 and every odd power has unit digit 4. It means the unit digit of N is 4. Hence the largest two digit number is 94, whose unit digit is 4.

104. Just replace x by $(x+1)$ and simplify or consider some numerical value of x then check through options.

105. $$60 = \pm \begin{bmatrix} 1 \times 60 \\ 2 \times 30 \\ 3 \times 20 \\ 4 \times 15 \\ 5 \times 12 \\ 6 \times 10 \end{bmatrix}$$

$$\Rightarrow \quad a + b = \begin{Bmatrix} = (-4) + (-15) & = -19 \\ = 5 + 12 & = 17 \\ = 2 + 30 & = 32 \end{Bmatrix}$$

106. To find the required unit digit just add up the single unit digits of the two previous consecutive numbers successively and you will find that every 15th term of this sequence has unit digit zero. Hence the unit digit of the 75th term of this sequence is 0.

107. Every nth term has its unit digit 5 if n is divisible by 5 but it has unit digit zero (0) if n is divisible by 15. Since we have to find out the unit digit of 55th term. So here 55 is divisible by 5 but not by 15, hence the unit digit will be 5.

108. The unit digit of the sum of the 88th term plus 89th term will be equal to the unit digit of 90th term. Hence it will be zero. (See the question number 106).

109. Always 5. Since when we multiply a number whose unit digit is 5 with any odd number the unit digit of the resultant number is 5.

111. The least possible product of any 3 numbers can be found by multiplying numerically greatest possible number, but at least one number must be negative.

Thus the required product $= (-5) \times (24) \times (25) = (-3000)$

Hence (c).

112. Since $q - p = r - q = 2$
Therefore p, q, r are in A.P.

But the least possible value of p, q, r can be 1. So if $p = 1$, then $q = 3$ and $r = 5$.

Hence $(p+q)(q+r)(r+p) = 4 \times 8 \times 6 = 192$

114. Check for any value of R.

116. Option (a) and (b) are wrong since the maximum cost of total pens is nearly Rs. 745. Now, to maximize the number of pens of Rs. 5, we have to minimize the number of pens of Rs. 3 and the total cost cannot exceed Rs. 745.6
So by hit and trial get the required result. As :

Number of pens of Rs. 3 each	Number of pens of Rs. 5 each	Total Cost
$1 \times 3 = 3$	$149 \times 5 = 745$	748
$2 \times 3 = 6$	$148 \times 5 = 740$	746
$3 \times 3 = 9$	$147 \times 5 = 735$	744

Hence the maximum number of pens of Rs. 5 is 147.

117. The last digit of this number is 9.

Since $(1 \times 9) + (2 \times 50) = 109$

i.e., there are 9 digits of one digit number and 100 digits are of 50 numbers of 2 digit.
Again the 50th two digit number is 59, hence the last digit of the given number is 9.
Now, if we add 1 to this number it will become divisible by both 2 and 5.

118. If the sum of all the 109 digits of the given number is divisible by 3, then the given number itself is divisible by 3.
Now, the sum of all the 109 digits

= [sum of all the unit digits of all the 59 numbers + sum of all the tens digit of all the 2 digit numbers upto 59]

$= 6 \times (45) + 5 \times (15)$

$= 270 + 75 = 345$

Now since the sum of all the digits is 345. Hence it is divisible by 3.
Again a number is divisible by 11 if the difference of sum of all the digits at even places and sum of all the digits at odd places is divisible by 11 or it is zero.
Now, 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 ... 575859.
The sum of all the digits at odd places

$= 5 \times (45) + (9 + 7 + 5 + 3 + 1) = 250$

[Counted from right side]

The sum of all the digits at even places

$= 10 \times (1 + 2 + 3 + 4 + 5) + (8 + 6 + 4 + 2)$

$= 150 + 20 = 170$

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Thus the required difference = 250 – 170 = 80

But if we subtract the least possible number *i.e.*, 3 from the given number (1 2 3 4 5 ... 5859) the new number will become (1 2 3 4 5 ... 5856)

Now the required difference = 247 – 170 = 77

Since 77 is divisible by 11 hence the given number is also divisible by 11.

Again, the number (12345 ... 5856) is also divisible by 3. Since if a number is divisible by 3 and we subtract a multiple of 3 from this number then the resultant number is also divisible by 3.

119. Since $F < B < G < A$

Now, if $F = 1$, then $1 < 2 < 3 < 4$, adultas cannot be 4 in one family.

if $F = 2$, then $2 < 3 < 4 < 5$, adultas cannot be 5 in 2 families.

if $F = 3$, then $3 < 4 < 5 < 6$, not possible since one family has only one adult.

Again, if $F = 4$ then $4 < 5 < 6 < 7$, possible since

$$(3 \times 2) + (1 \times 1) = 7$$

120. $$A = 3^{3^{3^3}} = 3^{3^{27}}$$

and $$C = 3^{3^{33}} = 3^{3^{33}}$$

Hence $C > A$.

Hence either (b) or (d) option is correct.

Now $$A = 3^{3^{3^3}} = 3^{3^{27}}$$

and $$D = 3^{333}$$

Hence $$A > D \qquad (\text{Since } 3^{27} > 333)$$

Thus the correct relation is $C > A > B > D$.

Hence, option (b) is correct.

121. $$\frac{A}{D} \rightarrow \text{Remainder } 23$$

and $$\frac{B}{D} \rightarrow \text{Remainder } 3$$

$$\frac{A+B}{D} \rightarrow \text{Remainder } 26$$

Therefore it is clear that the divisor D cannot less than or equal to 26. Thus the possible values of D are greater than 26.

122. $a^2 - b^2 = (a+b)(a-b) = m \times n$

$$\Rightarrow \qquad a = \frac{m+n}{2} \text{ and } b = \frac{m-n}{2}$$

Now, $729 = 1 \times 729 \Rightarrow a = 365, b = 364$

$3 \times 243 \Rightarrow a = 123, b = 120$

$9 \times 81 \Rightarrow a = 45, b = 36$

$27 \times 27 \Rightarrow a = 27, b = 0$

Thus we can express it in 4 ways, Hence (a).

Alternatively : The required answer is always equal to the number of ways in which a given product can be expressed as the product of two factors.

123. $2310 = 2 \times 3 \times 5 \times 7 \times 11$, Now since 2310 has to be expressed as a product of 3 factors, as $2310 = a \times b \times c$. Now, we exclude one prime factor say a then we find the rest part as a product of 2 factors.

Thus when $a = 1$, then $b \times c$ can be expressed as

$$\frac{(1+1)(1+1)(1+1)(1+1)(1+1)}{2} = 16 \text{ ways}$$

Now, if $a = 2$, then

$$b \times c = \frac{(1+1)(1+1)(1+1)(1+1)}{2} = 8 \text{ ways}$$

when $a = 3$, then

$$b \times c = \frac{(1+1)(1+1)(1+1)(1+1)}{2} = 8 \text{ ways}$$

similarly when $a = 5, 7, 11$ the value of $b \times c = 8$ ways

But when $a = 2$, then the $(b \times c) = 8 - 1 = 7$ since one way of expression has been included in the case of $a = 1$.

Similarly when $a = 3, 5, 7$ and 11 then the number of ways will be 6 (= 8 – 2), 5 (= 8 – 3), 4 (= 8 – 4), 3 (= 8 – 5) respectively.

Thus the total number of ways in which 729 can be expressed as the product of 3 factors = 16 + 7 + 6 + 5 + 4 + 3 = 41 ways.

Final Round

1. $120 = 4 \times 5 \times 6$ (Since minimum number of cuts are possible only when all the three factors are very close)

Again $$125 = 5 \times 5 \times 5$$

But since 4 + 5 + 6 = 5 + 5 + 5 = 15, hence the same number of cuts are required as previously. Hence '*n*' number of cuts are required to get 125 identical pieces.

2. Note that the first element of *n*th term of the given sequence is $\left(\frac{n^2 - n + 2}{2}\right)$ and the last element of the *n*th element is $\frac{n(n+1)}{2}$.

So the last element of the 21st term of the sequence

$$= \frac{21 \times 22}{2} = 231$$

Now to find the sum of first 21 term we have to find the sum of first 231 natural numbers.

Hence $$\frac{231 \times 232}{2} = 231 \times 116 = 26796$$

Hence the last digit is 6 and tens digit is 9.

3. The elements of the set

$$S_{19} = \{172, 173, 174, 175, \ldots, 190\}$$

Now $175 = 5^2 \times 7$

$180 = 5 \times 36$

$185 = 5 \times 37$

$190 = 5 \times 38$

Therefore the number of 5's contained in the product of the element of the set is 5. Note that the number of 2's are more than the number of 5's. So the total number of zeros at the end of the product of the element is 5.

4. Note the number of zeros in the product of S_{19} and S_{20} will depend upon the number of 5's contained in S_{19} and S_{20}.

Now, the $S_{20} = \{191, 192, 193, 194, 195, \ldots, 210\}$

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Thus $195 = 5 \times 39$
$200 = 5 \times 5 \times 8$
$205 = 5 \times 41$
$210 = 5 \times 42$

Therefore the total number of 5's contained in $S_{20} = 5$ hence the number of zeros at the end of the product is 5. Thus the number of zeros at the end of the product of S_{19} and $S_{20} = 10$. Thus the largest power of 10 is 10, which can exactly divide the product of S_{19} and S_{20}.

5. $S = \{2, 4, 6, 8, 10, 12, 14, 16, 18, \ldots, 96, 98, 100\}$
If the product of any 3 elements contains 4 or more than 4 two's then it will be divisible by 16 (since $16 = 2^4$) otherwise it will not be divisible. Hence remove those elements which are multiples of either 4 or 8 or 16 or 32 or 64 in the given set.
Since $(2k \times 2l \times 2m)$ is not divisible by 16 if k, l, m are odd numbers.
Thus there are only 25 elements in p.
As $P = \{2, 6, 10, 14, 18, 22, 26, 30, \ldots, 94, 98\}$

6. Since $51 = 17 \times 3$
$85 = 17 \times 5$
$102 = 17 \times 6$
Thus the minimum possible number of cubes $= 3 \times 5 \times 6 = 90$
The total surface area of non-red faces
= (Total surface area of all the cubes
– total surface area of cuboid)
$= (90 \times 6 \times 17 \times 17) - 2 \times (51 \times 85 + 85 \times 102 + 102 \times 51)$
$= 90 \times 6 \times 289 - 2 \times 289\ (15 + 30 + 18)$
$= 90 \times 6 \times 289 - 2 \times 289\ (63)$
$= 289 \times 18\ (5 \times 6 - 7)$
$= 289 \times 18 \times 23 = 119646\ \text{cm}^2$

7. Go through option
$$\frac{6p + 7q}{42} \rightarrow \frac{6p}{42} + \frac{7q}{42}$$
Since p is divisible by 7 and q is divisible by 6 thus
$$\frac{6p}{42} + \frac{7q}{42} \rightarrow \frac{6 \times 7m + 7 \times 6n}{42}$$
$$= \frac{42(m + n)}{42} = (m + n)$$

8. $p^q - q^r = (p + q)^{r-q}$; $11 > p > r > q \in$ Prime numbers.
$7^3 - 3^5 = (7 + 3)^{5-3}$
$100 = 100$
$\therefore \quad (p + q) = 7 + 3 = 10$
$= 5(5 - 3) = r(r - q)$

9. Since there are 25 prime numbers upto 100 also 1 is not divisible by any number.
Hence there are only 26 seats available on the front row for the mathematicians.

10.

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
1	2	3	4	5	6	7
8	9	10	11	12	13	14

$\frac{88}{7}$ = Remainder is 4.
So the 4th day in the week is Thursday.

11. Since Thursday will fall on those dates which can be expressed as $7k + 4$; $k = 0, 1, 2, 3, \ldots$
Now since 64 can't be expressed as $7k + 4$.
Hence (d) is the required answer.

12. According to the septarion calendar there are exactly 364 days and each day occurs equal number of times in whole year.

13. Find the pattern
hence, $(89)^{91} \rightarrow (89^{10})^9 \times 89^1$
$(89)^{91} \rightarrow 01 \times 89$
$(89)^{91} \rightarrow 89$
Thus the last two digits of $(1989)^{91}$ are 8 and 9.

Since		
$(89)^1 \rightarrow 89$	$(89)^{11} \rightarrow 89$	
$(89)^2 \rightarrow 21$	$(89)^{12} \rightarrow 21$	
$(89)^3 \rightarrow 69$	$(89)^{13} \rightarrow 69$	
... ...	... ...	
... ...	... ...	
$89^{10} \rightarrow 01$	$(89)^{20} \rightarrow 01$	

NOTE To find the last 'n' digits we find the pattern for last 'n' digits.

14. Step 1. | 2 | 7 | 3 | 5 | → No any digit is placed at the right place.
Step 2. | 5 | 2 | 7 | 3 | → No any digit is correctly placed.
Step 3. | 3 | | | | → This digit is not acceptable at this place.

It means the first digit will be 7, since, 2, 5 and 3 are inadmissible.

7			

Again since 5 and 3 are not acceptable at unit digit place and 7 is correctly filled up as the first digit, hence only one digit *i.e.*, '2' is left for the unit digit place.

7			2

Now, since 3 and 7 are not acceptable and 2 has been used, hence only one digit *i.e.*, '5' is left for the tens place.

7		5	2

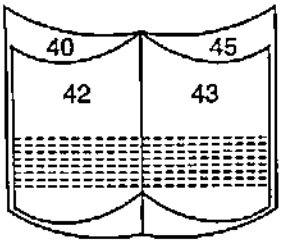
Thus, we can complete the password as

7	3	5	2

Alternatively : Go through options.

15. $(888!)^{999} = (1 \times 2 \times 3 \times 4 \times \ldots \times 76 \times 77 \times 78 \times \ldots \times 888)^{999}$
Since it has a factor as 77. Thus it has no remainder.

16. The middle most pages of the magazines are shown in the figure numbered as 42 and 43.
Did you notice that the front pages on the left side of the magazine (as shown in figure) are divisible by 2 and the odd numbers are on the back side of these numbers ? Note that I am talking about the left side of the magazine when it is open as shown in figure.



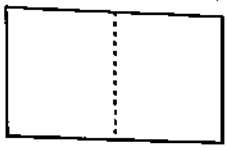
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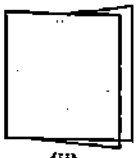
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So the missing page numbers with 29 will be 30 and the corresponding number in the right side will be 51 and 52.

Alternatively : Since the magazine has 84 pages it means there are 21 sheets of paper, which are folded in middle. Now see the pattern of page numbers :

Left side	Right side
1, 2	83, 84
3, 4	81, 82
5, 6	79, 80
7, 8	77, 78
9, 10	75, 76
⋮ ⋮	⋮ ⋮
29, 30	55,56
⋮ ⋮	⋮ ⋮
41, 42	43, 44


(i)


(ii)

Thus we can say that the sum of the page numbers on a sheet in one side is always 85. Hence the page numbers printed on the sheet of paper is 29, 30 and 55, 56.

17. Let there be six locks *A, B, C, D, E* and *F*.
Then we need maximum 5 attempts to know the right key for 'A'. Again we need maximum 4 attempts to know the right key for 'B'. Similarly we need 3, 2 and 1 attempts to know the proper key for the locks *C, D* and *E*. Now 1 key is automatically left for the lock 'F'. Thus the total number attempts required

$$= 5 + 4 + 3 + 2 + 1 = 15$$

18. $\dfrac{51n^2 + 17n + 6}{n} = 51n + 17 + \dfrac{6}{n}$

Here the first two terms are divisible for any value of n, but the last term is divisible by only 1, 2, 3 and 6 *i.e.*, only 4 values, which are actually the factors of 6.

19. $V >> S >> P$

$V >> L >> P$

$V >> L > S$

Therefore, $V >> L > S >> P$

Thus the worst player can be Plexur.

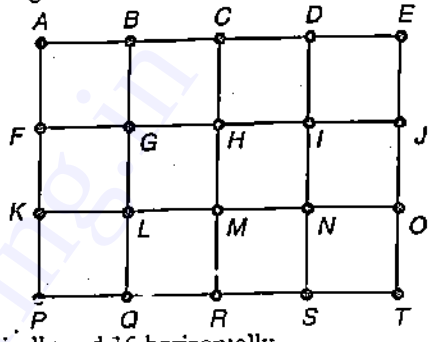
20. Obviously $(n-1)$. Since if the Winner first picks up $(n-1)$ sticks then there will be only one stick left, which must be picked up by the Loser and hence the last player will lose it.

21. The last ten digits of $1^1 \times 2^2 \times 3^3 \times 4^4 \times 5^5$ is (00,86,40,00,00)
The last ten digits of $1^6 \times 2^7 \times 3^8 \times 4^9 \times 5^{10}$ is (00,00,00,00,00)
The last ten digits of $1^{11} \times 2^{12} \times 3^{13} \times 4^{14} \times 5^{15}$ is (00,00,00,00,00)
The last ten digits of $1^{96} \times 2^{97} \times 3^{98} \times 4^{99} \times 5^{100}$ is (00,00,00,00,00)
Thus the sum of the last tens digits of the expression is (00,86,40,00,00)
Hence the sum of the last ten digits of the sum of the expression = 18. (∵ 8 + 6 + 4 = 18)

22. Initially Ravi Shankar has only Rs. 2 with him. Since we know that the Ravi Shankar can pay the sum in rupees only it means he cannot pay in paise. Therefore he must have purchased initially the flowers of exactly Re. 1.
Now from the options : Consider option (c).
The number of flowers he has purchased for Re. 1 = 4.
Later on he could purchased the number of flowers for total Rs. 2 = 4 + 6 = 10.
Thus the initial cost of one dozen flowers = Rs. 3
and the changed cost of one dozen flowers = Rs. 2.40
Thus he could gain 60 paise per dozen. Hence the presumed option (c) is correct.

NOTE The initial number of flowers must be less than 6, for Re. 1 only then he can get the benefit by purchasing 6 flowers for the next Re. 1.

23. See the figure there are 31 pencils of equal length.



15 vertically and 16 horizontally.
Thus there are 12 small squares of the dimension of equivalent to 1 pencil and 6 squares of dimension of equivalent to 2 pencil and 2 squares of dimensions of equivalent to 3 pencils. Therefore total number of squares = 12 + 6 + 2 = 20.

24. Let she did x number of pages on the first day. It means second, third, fourth, ..., nineth and tenth day she did $(x+3)(x+6)(x+9)\ldots(x+24)$ and $(x+27)$ passages.

Now since $(x + 24) = 4(x + 3)$

$\Rightarrow \quad 3x = 12$

$\Rightarrow \quad x = 4$

Therefore the number of passages she did on the last day

$$= x + 27 = 31.$$

25. Total number of passages

$$= x + (x+3) + (x+5) + \ldots + (x+27)$$
$$= (4 \times 10) + (3 + 6 + 9 + \ldots + 27)$$
$$= 40 + 3(1 + 2 + 3 + \ldots + 9)$$
$$= 40 + 135 = 175$$

26. Let there be x people in the village then there must be $4x$ sheep.
Now the number of persons live = $(x - 8)$
and the number of sheep to be carried with the survived people = $4x - (47 + 21) = 4x - 68$
Since we know that each person has carried only one sheep with him. Therefore

$$(x - 8) = 4x - 68$$

$\Rightarrow \quad 3x = 60$

$\Rightarrow \quad x = 20$

$\Rightarrow \quad 4x = 80$

Hence there were total 80 sheep.

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27. (123) (234) (345) (456) (567) (678) (789)
(135) (246) (357) (468) (579)
(147) (258) (369)
(159)
Thus there are total 16 such numbers.

28. $(2^2 + 4^2 + 6^2 + 8^2 + \ldots + 100^2) - (1^2 + 3^2 + 5^2 + \ldots + 99^2)$
$= (2^2 + 4^2 + 6^2 + \ldots + 100^2)$
$- [(1^2 + 2^2 + 3^2 + 4^2 + \ldots + 99^2 + 100^2)$
$- (2^2 + 4^2 + 6^2 + \ldots + 100^2)]$
$= 2(2^2 + 4^2 + \ldots + 100^2) - (1^2 + 2^2 + 3^2 + \ldots + 100^2)$
$= 8(1^2 + 2^2 + 3^2 + \ldots + 50^2)$
$- (1^2 + 2^2 + 3^2 + 4^2 + \ldots + 100^2)$
$= 8\left(\frac{50 \times 51 \times 101}{6}\right) - \left(\frac{100 \times 101 \times 201}{6}\right)$
$= \frac{101 \times 100}{6}(51 \times 4 - 201) = \frac{10100}{6} \times (3) = 5050$, hence (b).

Alternatively :
$(2^2 + 4^2 + 6^2 + 8^2 + \ldots + 100^2) - (1^2 + 3^2 + 5^2 + \ldots + 99^2)$
$= (4 + 16 + 36 + 64 + \ldots + 10000)$
$- (1 + 9 + 25 + \ldots + 9801)$
$= [(4 - 1) + (16 - 9) + (36 - 25) + \ldots + (10000 - 9801)]$
$= 3 + 7 + 11 + 15 + \ldots + 199 = \left(\frac{3 + 199}{2}\right) \times 50$
$= \frac{202 \times 50}{2} = 101 \times 50 = 5050$

29. Consider some proper values.

30. $S_n = 3^0 + 3^1 + 3^2 + \ldots + 3^{200}$
$S_n = \frac{3^0 (3^{201} - 1)}{(3 - 1)} = \frac{(3^{201} - 1)}{2}$
Now, $\frac{(3^{201} - 1)}{2} = \frac{(3^3)^{67} - 1^3}{2} = \frac{[(27)^{67} - 1]}{2}$
Now since $a^n - b^n$ is divisible by $(a - b)$ therefore $[(27)^{67} - 1]$ is divisible by $(27 - 1) = 26$.
Hence it is also divisible by 13.
Thus the remainder is zero since the given expression is divisible by 13.

31. Let the sum of the expression be S_n then
$S_n = 4^0 + 4^1 + 4^2 + 4^3 + \ldots + 4^{40}$
$\Rightarrow \quad S_n = (1 + 4 + 16 + 64) + 4^4 (1 + 4 + 16 + 64) + \ldots$
$+ 4^{36} (1 + 4 + 16 + 64) + 4^{40}$
Since $(1 + 4 + 16 + 64) = 85$ is divisible by 17, hence except 4^{40} remaining expression is divisible by 17.
$\therefore \quad \frac{4^{40}}{17} \to \frac{(4^4)^{10}}{17} \to \frac{(1)^{10}}{17}$,
hence the required remainder is 1.

32. Let $S_n = 4^0 + 4^1 + 4^2 + 4^3 + 4^4 + 4^5 + \ldots + 4^{40}$
$\Rightarrow \quad S_n = (1 + 4 + 16) + 4^3 (1 + 4 + 16)$
$+ 4^6 (1 + 4 + 16) + \ldots + 4^{40}$
$\Rightarrow \quad S_n = 21 + 4^3 (21) + 4^6 (21) + 4^9 (21) + \ldots$
$+ 4^{36} (21) + (4^{39} + 4^{40})$
Here we can see that except $(4^{39} + 4^{40})$ remaining expression is divisible by 7.

Now $\frac{(4^{39} + 4^{40})}{7} \to \frac{4^{39}(5)}{7} \to \frac{(4^3)^{13} \times 5}{7} \to \frac{5}{7}$
Hence the required remainder is 5.

33. The total distance which a monkey has to overcome is 35 metre. Since the monkey first climbs up 5 metre and then he slips down by 2 metre, it means he climbs up only 3 metre in one round of 2 (= 1 + 1) minutes.
Now, note that if the monkey has to just climbup 5 metre then he will not be slipped back. Thus he will cover the 30 metre height in 10 rounds and the rest 5 metre height of the top will covere in only one attempt. Since when he will touch (or reach) the top of the pole he will not be slipped back.
Thus total number of attempts required = 11 (= 10 + 1).

NOTE In this type of question subtract the height which he/she climbs up in first attempt (*i.e.*, 5 metre in this question) then assuming the resultant height and solve it as usual. Now the length of the top will be covered in only one attempt of upward direction and then addup all the results obtained.

34. Since first he covers 30 metres in 20 minutes and the rest height of 5 metre (at the top) covers in only one minute.
Hence total time = 20 + 1 = 21 minutes.

35. Since the actual height is 36 metre.
Therefore 36 – 5 = 31, which is not divisible by 3.
Hence he will climb 33 (= 3 × 11) metre in 11 attempts taking 22 (= 11 × 2) minutes.
Now the rest height of 3 metre will be covered in $\frac{3}{5}$ minute
i.e., 36 seconds.
Hence total time taken to cover 36 metre height is 22 minutes 36 seconds.

36. $1^3 + 2^3 + 3^3 + 4^3 + \ldots + 1000^3$
$= \left[\frac{1000\,(1001)}{2}\right]^2 \left\{\because 1^3 + 2^3 + \ldots + n^3 = \left[\frac{n\,(n+1)}{2}\right]^2\right\}$
$= (500 \times 1001)^2 = 250000 \times 1001 \times 1001$,
which is divisible by 13. Hence the remainder is zero.

37. $(22)^3 + (23)^3 + (24)^3 + \ldots + (87)^3 + (88)^3$
$= (22^3 + 88^3) + (23^3 + 87^3) + (24^3 + 86^3)$
$+ (25^3 + 85^3) + \ldots + (54^3 + 56^3) + 55^3$
Now since we know that $a^n + b^n$ is divisible by $(a + b)$ when n is an odd number. Therefore all the terms, except $(55)^3$, is divisible by 110.
Now the remainder when $(55)^3$ is divided by 110 is 55.
Hence the required remainder of the whole expression is 55.

38. The value of nth term = Sum of n terms – Sum of $(n - 1)$ term
Therefore value of 6th term = Sum of 6 terms
– Sum of 5 terms
$= (6^1 + 6^2) - (5^1 + 5^2) = (720 + 36) - (120 + 25) = 611$

39. The smallest possible number which is divisible by either 3, 5 or 7 is 105. Now we express it in binary representation, then

2	105		
2	52	→ 1	→ LSB
2	26	→ 0	
2	13	→ 0	
2	6	→ 1	
2	3	→ 0	
2	1	→ 1	
	0	→ 1	→ MSB

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Thus $(105)_{10} = (1101001)_2$.

Hence the minimum 7 digits are required to represent it.

40. Go through option

$$S_n = 1 + 3 + 5 + 7 + \ldots + 22221$$

$$S_{11111} = (11111)^2 \qquad [\because S_n = n^2]$$

Hence it is divisible by 11111. Thus option (b) is correct.

41. Go through options

Anjuli had chocolates = 12

She gave to Amit $= \frac{12}{2} - 1 = 5$

Now she is left with 7 chocolates
then she gave 3 chocolates to Bablu
Now she has 4 chocolates
then she gave 2 chocolates to Charles
Now she has only two chocolates.
Then she gave 1 chocolate to Deepak
and she is finally left with one chocolate.

Alternatively : Solve the problem in reverse.

Anjuli's chocolate = 1 (Finally)
Before she has given to Deepak = 2 chocolates
Before she has given to Charles = 4 chocolates
Before she has given to Bablu = 7 chocolates
Before she has given to Amit = 12 chocolates.

Alternatively : By forming the equation assuming any variable x as the original number of chocolates which Anjuli had.

42. Since $a_1b_1c_1$, $a_2b_2c_2$, $a_3b_3c_3$, ... etc. are consecutive 3 digit numbers. Hence $c_1, c_2, c_3, \ldots$ etc. must be the consecutive unit digits. Hence there will be all the 10 digits (*viz.*, 0, 1, 2, ..., 9) 10 times as unit digits.

It means the required unit digit is equal to the sum of all the unit digits.

Thus $[(1)^{100} + (2)^{100} + (3)^{100} + \ldots (9)^{100} + (10)^{100}] \times 10$

$= [1 + 6 + 1 + 6 + 5 + 6 + 1 + 6 + 1 + 0] \times 10$

$= [3] \times 10 = 0$

Thus the unit digit of the whole expression is 0 (zero).

43. $P(x, y) + Q(x, y)$

$= (f(x, y) + g(x, y)) + (f(x, y) - g(x, y))$

$= 2f(x, y) = 2 \times ([x] + \{y\})$

$= 2 \times ([16] + \{25\}) = 2 \times (16 + 25)$

$= 2 \times 41 = 82$

44. $P(x, y) . Q(x, y) = (f(x, y) + g(x, y)) . (f(x, y) - g(x, y))$

Now since $f(x, y) = [x] + \{y\} = [\pm 4] + \{\pm 5\}$

$f(\pm 4, \pm 5) = \pm 9$ or ± 1

and $g(x, y) = [x] - \{y\} = [\pm 4] - \{\pm 5\}$

$g(\pm 4, \pm 5) = \pm 9$ or ± 1

Now since there are too many combinations of $f(x, y)$ and $g(x, y)$.

So there is no unique value of $P(x, y) . Q(x, y)$.

45. Since $P(x, y) + Q(x, y) = 2f(x, y)$

So the required value is always an even number.

46. $(50^2 - 1) = (50 + 1)(50 - 1) = (17 \times 3) \times (7 \times 7)$

hence divisible by 17.

and $(729)^5 - 729 = 729(729^4 - 1)$

$= 729(729^2 - 1)(729^2 + 1)$

$= (729)(729 - 1)(729 + 1)(729^2 + 1)$

$= 729 \times 728 \times 730 \times (729^2 + 1)$

Hence it is divisible by 5.

Alternatively : $(a^p - a)$ is divisible by p if p is a prime number.

47. Let there be n rows, then the number of soldier in each row $= \frac{36}{n}$

(Since the number of soldier in each row is same)

Now the number of bullets he gave to the first row

= 1 bullet × number of soldiers

and the number of bullets he gave to the second row,

= 2 bullets × number of soldiers

and the number of bullets he gave to the third row

= 3 bullets × number of soldiers

Hence total number of bullets

$$= 1 \times \frac{36}{n} + 2 \times \frac{36}{n} + 3 \times \frac{36}{n} + \ldots + n \times \frac{36}{n}$$

$$180 = \frac{36}{n}(1 + 2 + 3 + \ldots + n)$$

or $$180 = \frac{36}{n} \times \frac{n(n+1)}{2}$$

or $$180 = \frac{36(n+1)}{2}$$

$\Rightarrow$ $$(n + 1) = 10$$

$\Rightarrow$ $$n = 9$$

Thus the number of soldiers in each row $= \frac{36}{9} = 4$

Hence, there are 4 soldiers in 8th row.

48. Given that $(n - 5)$ is divisible by 17 and let $(n - 5) = k = 17m$

So $(n + 12) \times (n + 29) = (n - 5 + 17) \times (n - 5 + 34)$

$= [(n - 5) + 17] \times [(n - 5) + 34]$

$= (k + 17) \times (k + 34)$

$= k^2 + 51k + 2(17)^2$

Now since k is divisible by 17 so,

$k^2 + 51k + 2.(17)^2 = (17m)^2 + 51 \times 17m + 2.(17)^2$

$= 17^2 (m^2 + 3m + 2)$

Now the least possible value of $m = 1$, therefore $(n + 12) \times (n + 29)$ is divisible by $(17)^2 \times 6 = 1734$.

49. Go through option :

$(3^{96} - 1) = (3^{48} + 1)(3^{48} - 1)$

$= (3^{48} + 1)(3^{24} + 1)(3^{24} - 1)$

Thus if any number 'n' can divide $(3^{24} - 1)$ it will certainly divide $(3^{96} - 1)$.

50. $(5^{28} - 1) = (5^{14} + 1)(5^{14} - 1)$

and $(5^{42} - 1) = (5^{14})^3 - 1$

$= (5^{14} - 1)(5^{28} + 1 + 5^{14}.1)$

Hence both (a) and (b). Thus the appropriate answer is (d).

51. Check through some values.

Alternatively : $T_1 = 1^2 + 1$

$T_2 = 2^2 + 2$

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$$T_3 = 3^3 + 3$$
.....
.....
$$T_n = n^2 + n$$

$$S_n = T_1 + T_2 + T_3 + \ldots + T_n$$
$$= (1^2 + 1)(2^2 + 2)(3^2 + 3) + \ldots + (n^2 + n)$$
$$= (1^2 + 2^2 + 3^2 + \ldots + n^2) + (1 + 2 + 3 + \ldots + n)$$
$$= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}$$
$$= \frac{n(n+1)}{2}\left[\frac{2n+1}{3} + 1\right] = \frac{n(n+1)}{2}\left[\frac{2n+4}{3}\right]$$
$$S_n = \frac{n(n+1)(n+2)}{3}$$

52. The number of zeros at the end of $222^{111} \times 35^{53}$ is 53.
The number of zeros at the end of $(71)^{61} \times (101)^{51}$ is 960.
The number of zeros at the end of $42^{42} \times 25^{25}$ is 42.
Thus the number of zeros at the end of the whole expression is 42.

53. $\frac{12345}{12346} + \frac{12346}{12347} + \frac{12347}{12345}$
$$= 1 - \frac{1}{12346} + 1 - \frac{1}{12347} + 1 + \frac{2}{12345}$$
$$= 3 + \frac{2}{12345} - \frac{1}{12346} - \frac{1}{12347} = 3$$

54. The sum of all the elements of $S_n = n^3$
Therefore the sum of all the element of $S_{10} = (10)^3 = 1000$

55. The middle most element of any odd number of set
$$S_n = n^2$$
Therefore the 11th (*i.e.*, the mid/most) element of
$$S_{21} = (21)^2 = 441$$

56. The sum of the first and last elements of the odd number of set $S_n = 2n^2$.
Therefore the required answer $= 2 \times (51)^2$
$$= 2 \times 2601 = 5202$$

57. $k = 13 + 17 + 31 + 2$
$$= (9 + 3) + (9 + 7) + (9 + 9 + 9 + 1) + 2$$
$$= 9 + 9 + 9 + 9 + 9 + 7 + 2 + 3 + 1$$
$$= 9 + 9 + 9 + 9 + 9 + 9 + 4 = 64$$
[$\because 9 + 9 = 19, 19 + 9 = 29, 29 + 9 = 39, 39 + 9 = 49,$
$49 + 9 = 59, 59 + 4 = 64$]

58. $4 + 16 - 5 + 12 = 12 + 4 + 16 - 5$
$$= 16 + 11 = (9 + 6) + (9 + 1)$$
$$= 9 + 9 + 7 = 27$$

59. Check for options (a), (b) and (c) then you will find distinct unit digits. Now consider $x = 10^4$.
The unit digit of $(7)^4 = 1$ [$\because \log_{10}(10^4) = 4$]
and the unit digit of $(3)^{10000}$ is 1.
Hence (d).

60. $(1 + 2 + 3 + 4 + 7)^{x^2 + x}$ and $(11 \times 11 \times 13)^x$
$$(17)^{x^2 + x} \text{ and } (1573)^x$$
at $x = 2$, $x^2 + x = 6$
$\therefore$ The unit digit of $(17)^6$ is 9 and the unit digit of $(1573)^2$ is 9
Hence (b) is correct.

61. For example

	Remainder
$\frac{3^2}{8}$	$\to 1$
$\frac{3^4}{8}$	$\to 1$
$\frac{5^6}{8}$	$\to 1$
$\frac{7^2}{8}$	$\to 1$
$\frac{11^2}{8}$	$\to 1$ etc.

62.

	Wednesday
	2
	9
Thus $(23 - 2) = 21$, will	16
fall on Monday.	23
	30

63.

	Wednesday
	2
	8
21 will fall on Thursday.	14
	20
	26

64. Eldest said $\to$ Equal number of brothers and sisters (excluding eldest one)
$\Rightarrow$ 3 brothers and 3 sisters (excluding eldest one)
Youngest said $\to$ Twice the number of sisters as the number of brothers
$\Rightarrow$ 4 sisters and 2 brothers (excluding youngest)
It means there are 3 male and 4 female persons.
Now, as the youngest said that there are 4 sisters and 2 brothers engaged in work. Therefore it is obvious that the youngest person itself is a male. Hence it can be inferred that the brother of *S* is youngest.

	F	*F*	*F*	*F*	*M*	*M*	*M*
As :	*S*	*T*	*R*	*A*	*N*	*G*	*E*
	↑						↑
	Eldest						Youngest

65. Since $(10x + y) - (x + y) = 9x$
$(100x + 10y + z) - (x + y + z) = 9(11x + y)$
$(1000w + 100x + 10y + z) - (w + x + y + z)$
$= 9(111w + 11x + y)$ etc.
Hence all these numbers are divisible by 9.

66. $$y + \frac{7}{x} + \frac{y}{13} = 6\frac{3}{143}$$
$\Rightarrow$ $$\frac{14y}{13} + \frac{7}{x} = \frac{861}{143}$$
Now, the value of x can be only 11 or 143 or its multiples, inorder to satisfy the relation.
Let $x = 11$, then
$$\frac{14y}{13} = \frac{861}{143} - \frac{7}{11}$$
$\Rightarrow$ $$\frac{14y}{13} = \frac{770}{143}$$
$\Rightarrow$ $$y = 5$$

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Therefore $x - y = 7 - 5 = 2$

Again, $x = 143$, then

$$\frac{14y}{13} = \frac{861}{143} - \frac{7}{143} \Rightarrow \frac{14y}{13} = \frac{854}{143}$$

$\Rightarrow y = \frac{61}{11}$, which is not a natural number.

Therefore it is clear that when $x \neq 143$, then the multiples of 143 are also not acceptable.

Thus the only acceptable value is $x = 7$ and $y = 5$.

Hence $x - y = 2$, is the correct answer.

67. In the first subset, the number of elements

Divisible by 3 → (3, 9, 15, 21, ..., 99) ⇒ 17
Divisible by 9 → (9, 27, 45, ... 99) ⇒ 6
Divisible by 27 → (27, 81) ⇒ 2
Divisible by 81 is 162 only ⇒ 1
} 26

In the second subset, the number of elements

divisible by 3 → {102, 108, ..., 198} ⇒ 17
divisible by 9 → {108, 126, ..., 198} ⇒ 6
divisible by 27 → (108, 162) ⇒ 2
divisible by 81 is 162 only ⇒ 1
} 26

Hence the greatest power of 3 is 52 which can divide the product of all the elements of the given set.

68. In the first subset, the number of elements

divisible by 5 → {5, 15, 25, ..., 95} ⇒ 10
divisible by 25 → (25, 75) ⇒ 2
} 12

In the second subset, the number of elements

divisible by 5 → {110, 120, 130, ..., 200} ⇒ 10
divisible by 25 → {150, 200} ⇒ 2
} 12

Thus the highest power of 5 is 24 that can divide the product of all the elements of the set.

69. In the first subset, the number of elements

divisible by 5 → (10, 20, 30, 40, 50) → 5
divisible by 25 → (50) → 1

In the second subset, the number of elements

divisible by 5 → (55, 60, 65, 70, ...) → 10
divisible by 25 → (75, 100) → 2

(Note, that in the whole set the number of 2's are greater than the number of 5's)

Hence the number of zeros at the end of the product of the element is 18. Hence (a).

70. Since the 5th number is 10,
10th number is 20
15th number is 30
20th number is 40
25th number will be 100

71. Since in normal one hour this watch completes 5 hours. So in normal 3 hours this watch will completes 15 hours hence it will show 3 o'clock As, 15 = 12 + 3.

72.
$$T_1 = 5 \times 1 + 1$$
$$T_2 = 5 \times 2 + 1$$
$$T_3 = 5 \times 3 + 1$$
$$\ldots\ldots\ldots\ldots$$
$$\ldots\ldots\ldots\ldots$$
$$T_n = 5 \times n + 1$$

$$\therefore \quad S_n = 5(1 + 2 + 3 + \ldots + n) + (1 + 1 + \ldots + n \text{ times})$$
$$= 5\left[\frac{n(n+1)}{2}\right] + n = \frac{5n^2 + 7n}{2} = \frac{n}{2}(5n + 7)$$

73. Go through options or solve similarly as Q. No. 72.

74. Let the original dimension of a square be x unit then the area will be x^2, hence total area of paper = $144x^2$

Again the area of reduced square = $(x - 2)(x - 2) = (x - 2)^2$

therefore the total area of paper = $400\,(x - 2)^2$

Thus $144x^2 = 400\,(x - 2)^2$

$\Rightarrow \quad 9x^2 = 25\,(x - 2)^2$

$\Rightarrow \quad 9x^2 = 25x^2 + 100 - 100x$

$\Rightarrow \quad 16x^2 - 100x + 100 = 0$

$\Rightarrow \quad x = 5$ cm

$\therefore$ Area of paper $= 5 \times 5 \times 144 = 3600$

Alternatively : $\frac{3600}{114} \rightarrow 25 \rightarrow$ side of square = 5 cm

$\frac{3600}{400} \rightarrow 9 \rightarrow$ side of new square = 3 cm

Hence option (c) is correct.

75. $(ab)^2 = bcb$, where ab is a two digit number and bcb is a three digit number.

Now 'b' can be equal to 0, 1, 5 or 6, which are probable values of b as a unit digit.

at $b = 0$, bcb is not a three digit number.

at $b = 1$ either $a = b$ or $b = c$ or $b \neq b$, which is not possible.

at $b = 5$, c is always 2 and $b \neq b$, hence not possible.

at $b = 6$ and $a = 2$, $(ab)^2 = bcb$ *i.e.*, $(26)^2 = 676$, the only possible value.

Thus $a = 2, b = 6, c = 7$.

Since b, c, d are in AP hence $d = 8$

Therefore $(dd)^2 = (88)^2 = 7744 = ccff$

$\Rightarrow \quad f = 4$

76. $A = 555! = 1 \times 2 \times 3 \times 4 \times \ldots \times 277 \times 278 \times 279 \times \ldots \times 554 \times 555$
$= (1 \times 555) \times (2 \times 554) \times (3 \times 553) \times (4 \times 552) \times \ldots$
$\times (277 \times 279) \times (278)$

and $B = (278)^{555} = (278 \times 278) \times (278 \times 278)$
$\times (278 \times 278) \ldots (278)$

Hence $A < B$.

77. The last element of nth set $= \frac{n(n+1)}{2}$

Therefore last element of $S_{24} = \frac{24 \times 25}{2} = 300$

78. The middlemost element of any odd number set is $\frac{n^2 + 1}{2}$

Therefore the middlemost element of the set S_{15} is

$$\frac{15^2 + 1}{2} = 113$$

79. The sum of all the elements of an odd numbered set

$S_n = n \times$ (middlemost term)

Therefore, the sum of all the elements of set

$$S_{25} = 25 \times \left(\frac{25^2 + 1}{2}\right) = 25 \times 313 = 7825$$

80. Write the elements and then find out.

S_9, S_{12}, S_{13} have maximum number of prime number elements equally.

81. The sum of all the elements of the set S_n is even if n is divisible by 4. Hence (c) is the required answer.

Thus sum of all the elements of S_{72} is even since 72 is divisible by 4.

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82. Check the options.

83. The last element of the set $S_n = n^2 + n - 1$
Therefore the last element of the set $S_{100} = 10099$

84. The middlemost element of an odd number set $S_n = n^2$
Therefore the middlemost element of $S_{125} = (125)^2 = 15625$

85. The sum of all the elements of any set $S_n = n^3$
Therefore the required sum of the set

$$S_{101} = (101)^3 = 1030301$$

86. $S = \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \ldots + \frac{1}{99.100}$

$$= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \ldots + \left(\frac{1}{99} - \frac{1}{100}\right)$$

$$= 1 - \frac{1}{100} = \frac{99}{100}$$

87. $\frac{P}{D} \rightarrow$ Remainder 18

$\frac{Q}{D} \rightarrow$ Remainder 11

$\frac{P+Q}{D} \rightarrow$ Remainder 29, but the remainder is 4.

Hence, the divisor = (29 – 4) = 25.

88. Note $1 \times 2 \times 3 \times 4 \times \ldots \times (n-1).\, n = n!$
Now if n = 240, 241, 242, 243, 244 (5 values)
then the number of zeros in the end of the product of $n!$ is 68.
Alternatively : The number of zeros in the given expression will depend on number of 5's (but do not depend on number of 2's). Now, since you know that the number of 5's changed at the interval of 5 consecutive numbers. Then there will be no any change in the number of zeros (since number of 5's) for the five consecutive values of n.

89. Since $\frac{6}{10} \rightarrow$ Remainder is 6.

$\frac{6^6}{10} \rightarrow$ Remainder is 6

$\frac{6^{6^6}}{10} \rightarrow$ Remainder is 6

90. Since we know that if $(a + b + c) = 0$, then

$$a^3 + b^3 + c^3 = 3abc$$

Therefore $53^3 - 46^3 - 7^3 = 3 \times 53 \times 46 \times 7$

91. Let the original number in base 'n' be $(xy)_n$ then its reverse number be $(yx)_n$

Again $2(nx + y) = (ny + x)$

$$\Rightarrow \quad \frac{x}{y} = \frac{n-2}{2n-1}$$

So at $n = 3$, $\frac{x}{y} = \frac{1}{5}$

at $n = 4$, $\frac{x}{y} = \frac{2}{7}$

and at $n = 5$ $\frac{x}{y} = \frac{3}{9} = \frac{1}{3}$

and at $n = 8$ $\frac{x}{y} = \frac{6}{15} = \frac{2}{5}$

but the values of x, y cannot be equal to or greater than their respective bases. Hence there are only two values of n are possible *viz.*, $n = 5$ and $n = 8$.

Now since the greatest possible value of n is 8. Hence the numbers in base n = 8 are 25 and 52. Again when we convert them into base 10, then the numbers are 21 and 42.
Thus the sum of these numbers in decimal system is 63.

92. $n = 5, 8$. Hence (b).

93. $7^0 + 7^1 + 7^2 + 7^3 + \ldots + 7^{365} = \frac{1(7^{366} - 1)}{(7-1)} = \frac{(7^{366} - 1)}{6}$,

its divisible since $(a^n - b^n)$ is divisible by $(a - b)$.

Now $\frac{(7^{366} - 1)}{5} \rightarrow \frac{(2^{366} - 1)}{5} \rightarrow \frac{(4)^{183} - 1}{5} \rightarrow \frac{4-1}{5} \rightarrow \frac{3}{5}$

Since the remainder is 3.
Hence Krishna will receive 3 hens.

94. These numbers are 1 and 81.

95. $\frac{\text{Actual number of boys}}{\text{Actual number of girls}} = \frac{1}{1}$

Since $\frac{2}{3}$rd girls didnot go to picnic.

Hence the ratio of boys to the girls at picnic = 3 : 1.
Now, let there be k samosas to be distributed among boys or girls equally then

$\frac{k}{3x} \rightarrow$ Remainder 39

and $\frac{k}{x} \rightarrow$ Remainder 12

Now, since $39 - 12 = 27$

Thus x = 27, or 3, or 9 [$\because 27 = 1 \times 27$ or $27 = 3 \times 9$]

but since x cannot be less than 12

hence $x = 27$

Thus the number of boys = $3x = 81$
and the number of girls = $3x = 81$
Hence total number of students = 162

96. In any box the maximum number of fruits can be packed is 15 and since not more than 5 boxes contain same number of fruits. So the number of fruits in this case = $5 \times 15 = 75$
Again we can put 14 fruits in other 5 boxes = $5 \times 14 = 70$
Similarly 13 fruits in each of the 5 boxes can be contained

$$= 5 \times 13 = 65$$

Since the minimum number of fruits we have to put in a box = 10
So the total number of fruits which can be put

$$= 5 \times 15 + 5 \times 14 + 5 \times 13 + 5 \times 12 + 5 \times 11 + 5 \times 10$$
$$= 5(10 + 11 + 12 + 13 + 14 + 15) = 375$$

97. To maximize the contents in the boxes first we fill up 15–15 fruits in maximum possible number (*i.e.*, 7) of boxes.

7	15	15	15	15	15	15	15
7	14	14	14	14	14	14	14
7	13	13	13	13	13	13	13
7	12	12	12	12	12	12	12
5	11	11	11	11	11		

Then we fill up 14, 13, 12, in 7–7 boxes. Thus we filled up 28 boxes containing maximum number of fruits. Now in the rest 5 boxes we fill up 11 fruits in each box to its maximum possible capacity. Now we see that there are at least 5 boxes in which the number of fruits are same.

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98. $(19)^n - (23)^n - (43)^n + (47)^n$

$= [(47)^n + (19)^n] - [(43)^n + (23)^n]$

$= (47 + 19)\,k - (43 + 23)\,l$

[since $a^n + b^n$ is divisible by $(a + b)$ if n is odd]

$= 66\,[k - l]$

Hence, it is divisible by 2, 3, 6, 11, 22, 33 and 66.

Again $(47)^n - (23)^n - (43)^n + (19)^n$

$= (47)^n - (23)^n - [(43)^n - (19)^n]$

$= (47 - 23)\,(k) - (43 - 19)\,(l) = 24\,(k - l)$

[$\because$ since $(a^n - b^n)$ is divisible by $(a - b)$]

Therefore it is divisible by 2, 12, 3, 8, 4, 6.

Thus the whole expression is divisible by 264.

Since $264 = 11 \times 24$.

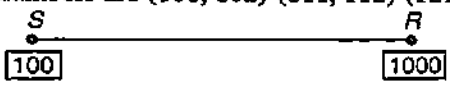
99. $1\cdot(1)^2\left(1-\frac{0}{1}\right) + 2\cdot 2^2\left(1-\frac{1}{2}\right) + 3\cdot 3^2\left(1-\frac{2}{3}\right) + 4\cdot 4^2\left(1-\frac{3}{4}\right)$

$= 1\cdot(1)^2\cdot 1 + 2\cdot(2)^2\cdot\frac{1}{2} + 3\cdot(3)^2\cdot\frac{1}{3} + 4\cdot(4)^2\cdot\frac{1}{4} + \ldots$

$= 1^2 + 2^2 + 3^2 + 4^2 + \ldots + n^2$

$= \frac{n\,(n+1)(2n+1)}{6}$

100. These numbers are (101, 102) (111, 112) (121, 122)



(131, 132) (141, 142) ... (202, 203) (212, 213) ... (303, 304) (313, 314) ... (808, 809) ... (898, 899, 900) ... (909, 910) (919, 920) ... (989, 990).

Thus there are total $20 \times 7 + 21 + 18 = 179$, mile stones which are red, because the hundreds digit on next mile stone is same as the unit of previous mile stone.

101. $(18^2 - 1) = (17)(18 + 1)$

$(18^4 - 1) = (18^2 + 1)(18^2 - 1)$

$= (18^2 + 1)(18 + 1)(18 - 1)$

$(18^6 - 1) = (18^3)^2 - 1$

$= (18^3 + 1)(18^3 - 1) = 17 \times k$ etc.

Hence there will only 9 times 17 in the whole expression.

102. $\frac{M}{28} \to$ Remainder 24 $\Rightarrow$ $M = 28k + 24$

and $\frac{M}{22} \to$ Remainder 12 $\Rightarrow$ $M = 22l + 12$

$\Rightarrow$ $M = 28k + 24 = 22l + 12$

$\Rightarrow$ at $k = 9$, $l\,(= 12)$, is an integer, for the least solution.

Hence, $M = 276$ ...(1)

and higher such numbers can be obtained as

$M' = 308p_1 + 276$ ...(2)

Again $\frac{N}{28} \to$ Remainder 16 $\Rightarrow$ $N = 28m + 16$

and $\frac{N}{22} \to$ Remainder 12 $\Rightarrow$ $N = 22n + 12$

$\Rightarrow$ $N = 28m + 16 = 22n + 12$

at $m = 3$, $n\,(= 4)$ is an integer, for the least solution.

Hence $N = 100$

and higher such numbers can be obtained as

$N' = 308p_2 + 100$

Now, $28 + 22 = 50$, hence $(M' + N')$ must be divisible by 5.

So $M' + N' = 308\,(p_1 + p_2) + 376$

Inorder that $M' + N'$ must be divisible by 50, either $p_1 = 1$ and $p_2 = 2$ or $p_1 = 2$ and $p_2 = 1$, for the least possible solution.

Hence, $M' + N' = 308 \times 3 + 376 = 1300$.

103. Since $M' + N' = 308\,(p_1 + p_2) + 376$

As we can observe that the values of p_1 and p_2 are not constant so we cannot determine uniquely the ratio of $M' : N'$.

104. In each set of 100 numbers, there are 10 numbers whose tens digit and unit digit are same. Again in the same set there are 10 numbers whose hundreds and tens digits are same. But there is one number in each set of 100 numbers whose Hundreds, Tens and Unit digit are same as 111, 222 etc. Hence there are exactly $(10 + 10 - 1) = 19$ numbers in each set of 100 numbers. Further there are 9 such sets of number. Hence total required number = 171.

Alternatively :

H	T	U
9	10	10

$-$

H	T	U
9	9	9

$900 - 729 = 171$

105. $(1 + p)^{256} \times (1 + p^{128}) \times \ldots \times (1 + p^2)(1 + p) \times \frac{(1 - p)}{(1 - p)}$

$= \frac{(1 + p^{256})(1 - p^{256})}{1 - p} = \frac{(1 - p^{512})}{1 - p}$

106. Let the length and breadth of the given perimeter be l and b respectively. Then

$2\,(l + b) = 50$

$\Rightarrow$ $l + b = 25$

Now $(b, l) = (1, 24), (2, 23), (3, 22), (4, 21), (5, 20), (6, 19), (7, 18), (8, 17), \ldots, (12, 13)$

Therefore the required number of rectangles = 12.

107. The product of prime factors $= 30 = 2 \times 3 \times 5$

Now the total number of prime factors = 24

$= 2 \times 2 \times 6$ $\Rightarrow$ $(1 + 1)(1 + 1)(5 + 1)$

So the least possible number $= 2^5 \times 3^1 + 5^1 = 480$

and the greatest possible number $= 2^1 \times 3^1 \times 5^5 = 18750$

So the ratio of $= \frac{480}{18750} = \frac{2^4}{5^4}$

$\therefore$ The square root $= \frac{2^2}{5^2} = \frac{4}{25}$

108. Let the powers of 2, 3, 5 be a, b, c respectively then the total number of factors = 24

$= (a + 1)(b + 1)(c + 1) = 24$

So if $(a + 1) = k$, $(b + 1) = l$, $(c + 1) = m$, then

$k \times l \times m = 24$, where $k, l, m \geq 2$

$2 \times 2 \times 6$
$2 \times 3 \times 4$
$2 \times 6 \times 2$
$2 \times 4 \times 3$
$3 \times 2 \times 4$
$3 \times 4 \times 2$
$4 \times 2 \times 3$
$4 \times 3 \times 2$
$6 \times 2 \times 2$

Hence there are total 9 numbers possible.

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109. The best way is to go through options. You will find that when you consider 150 question you can not get 75 marks any how.

Alternatively : Since 75 is an integer, hence he must have to do $(75 + 4x)$ problems, where x is a whole number. Now put the values of x and satisfy the given options.

Explanation : Since he gets the net marks in integers it means when he does $3n$ questions wrong , he loses actually $4n$ marks. For more classification see the table below :

Total Questions	Correct	Marks	Wrong	Marks	Net marks
75	75	75	0	0	75
79	76	76	3	–1	75
83	77	77	6	–2	75
87	78	78	9	–3	75

110. Solve it through options

	Total given coins	Rest coins
$(25-1)\times\frac{1}{2}=24\times\frac{1}{2}=12$	13	12
$(12-2)\times\frac{1}{2}=10\times\frac{1}{2}=5$	7	5
$(5-3)\times\frac{1}{2}=2\times\frac{1}{2}=1$	4	1

Hence he must have been stolen 25 coins.

111. Let $y=\log_2 7 \Rightarrow 2^y=7$ $\quad(\because x^y=a\Rightarrow \log_x a=y, x\neq 1)$

[Clearly y neither can be an integer nor a fraction]

Now if y is a rational number, we can write as

$$y=\frac{p}{q};\ q\neq 0$$

Thus $\quad 2^{p/q}=7 \ \Rightarrow\ 2^p=7^q.$

Since in L.H.S. the quantity is even while in the right hand side the given quantity is odd. Thus, y can't be a rational number.

112. Let us consider $2\times\frac{1}{2}=1$,

then $\quad 2+\frac{1}{2}=2\cdot 5$

Thus the option (a) and (b) are ruled out.

Again, let us take some different values

viz., $\quad \frac{1}{3}\times\frac{3}{2}\times 2=1$

then, $\quad \frac{1}{3}+\frac{3}{2}+2=\frac{23}{6}\neq 3+\frac{1}{3}\left(=\frac{10}{3}=\frac{20}{6}\right)$

hence option (c) is ruled out.

Finally, option (d) is correct, as can be seen from the above illustrations.

Alternatively : Let $a_1, a_2, a_3, \ldots, a_n$ be n positive real numbers such that $a_1 . a_2 \ldots a_n = 1$.

Now, we know that A.M. ≥ G.M. for some given quantities.

So $\quad \frac{(a_1+a_2+a_3+\ldots+a_n)}{n}\geq (a_1 . a_2 . a_3 \ldots a_n)^{1/n}=1$

$\Rightarrow \quad (a_1+a_2+a_3+\ldots+a_n)\geq n$

Hence option (d) is correct choice.

113. Let $S=1+n+n^2+\ldots+n^{127}$

$$=\frac{1(n^{128}-1)}{(n-1)} \qquad \left[\because S_n=\frac{a(r^n-1)}{r-1};\ r>1\right]$$

$$=\frac{n^{128}-1}{(n-1)}=\frac{(n^{64}-1)(n^{64}+1)}{(n-1)}$$

Thus at $m = 64$ the given expression is divisible by $(n^m + 1)$.

Hence (c).

114. We have $240 = 2^4 \times 3 \times 5$

$\therefore$ Number of divisors

$= 5 \times 2 \times 2 = 20$, which are given below :

1	2	4	8	16
3	6	12	24	48
5	10	20	40	80
15	30	60	120	240

Thus only 4 divisors are of the form $4n + 2$

Hence option (a) is correct.

115. We know that if a number is divisible by 5, then its unit digit must be either 0 or 5.

Again, the unit digits of $7^1 = 7$

$$7^2 = 9$$

$$7^3 = 3 \quad \text{and} \quad 7^4 = 1$$

i.e., we can get only 7, 9, 3, 1 as unit digits.

Now, the combinations of 7, 9, 3, 1 never gives unit digit 5, but $7 + 3 = 9 + 1 = 10$, these two combinations gives us unit digit zero.

So $7^m + 7^n$ is divisible where

case (i) $m = 1, 5, 9, \ldots, 97$ correspondingly $n = 3, 7, 11, \ldots, 99$

case (ii) $m = 2, 6, 10, \ldots, 98$ correspondingly

$n = 4, 8, 12, \ldots, 100$

or the values of m and n can be reversed in each case mutually but then we get the same values.

Hence total distinct numbers so formed, divisible by 5, are

$$1250 = 25 \times 25 + 25 \times 25$$

Hence option (a) is correct.

116. Let us assume the values of a, b, c, d such that

$$a + b + c + d = 2$$

i.e., $\quad \frac{1}{3}+\frac{2}{3}+\frac{1}{4}+\frac{3}{4}=2$

then $\quad M=(a+b)(c+d)=\left(\frac{1}{3}+\frac{2}{3}\right)\left(\frac{1}{4}+\frac{3}{4}\right)=1$

Hence option (c) and (d) are ruled out.

now, if a, b, c, d be $\frac{1}{3}, \frac{1}{4}, \frac{2}{3}, \frac{3}{4}$, then

$$M=(a+b)(c+d)=\left(\frac{1}{3}+\frac{1}{4}\right)\left(\frac{2}{3}+\frac{3}{4}\right)=\frac{119}{144}<1$$

Hence option (b) is also ruled out. Thus only option (a) is correct.

Alternatively : Let $a + b = P$ and $c + d = Q$,

$$P + Q = 2 \quad \text{and} \quad PQ > 0$$

(Since P and Q are positive real numbers)

Further we know that

$$\text{AM} \geq \text{GM} \ \Rightarrow\ \frac{P+Q}{2}\geq\sqrt{PQ}$$

$$\Rightarrow \quad 1\geq\sqrt{PQ} \ \Rightarrow\ 1\geq PQ$$

$$\therefore \quad PQ = M \leq 1$$

Hence option (a) is correct.

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2 AVERAGES

In MAT, CET, FMS and other entrance exams the questions are directly asked from this chapter and they are very moderate, but in CAT this chapter constitutes a very low proportion of questions *i.e.*, they are rarely asked based on direct formula. As in last 10-12 years on an average 1 question per year has appeared in CAT and each question was almost of applied nature and blended with the logical reasoning.

So, when we said that on an average there was 1 question was asked per year in the last decade (*i.e.*, 10 years), it means we can conclude that almost 10 questions were asked by CAT in the last 10 years.

In general average is the central value of the given data. For example if the heights of three persons A, B and C be 90 cm, 110 cm and 115 cm respectively, then the average height of A, B and C together will be $\frac{90+110+115}{3} = 105$ cm.

So we can say that the height of each person *viz.* A, B and C is near about 105 cm. Thus in layman's language it can be said that every one is almost 105 cm tall.

Basically the average is the arithmetic mean of the given data. For example if the $x_1, x_2, x_3, x_4 \ldots x_n$ be any 'n' quantities (*i.e.*, data), then the average (or arithmetic mean) of these 'n' quantities

$$= \frac{x_1 + x_2 + x_3 + \ldots x_n}{n}$$

PROPERTIES OF AVERAGES

1. The average of any two or more quantities (or data) necessarily lies between the lowest and highest values of the given data. *i.e.*, if x_l and x_h be the lowest and highest (or greatest) values of the given data $(x_1, x_2, \ldots x_l, \ldots x_h, \ldots x_n)$ then $x_l <$ Average $< x_h$; $x_l \neq x_h$

$$i.e., \quad x_l < \frac{(x_1 + x_2 + x_3 + x_l \ldots + x_h \ldots + x_n)}{n} < x_h$$

EXAMPLE 1 *The average of 8, 9, 12, 13, 15, 9 is :*
(a) 11 *(b) 6* *(c) 16* *(d) 18*

SOLUTION From the above mentioned property (1) we can say that options (b), (c) and (d) are invalid since 6, 16, and 18 are out of range *i. e.*, either below 8 (which is the least value) or above 15 (which is the highest value of the data).
So, option (a) can be acceptable only.
Alternatively :

$$\text{Average} = \frac{8+9+12+13+15+9}{6} = \frac{66}{6} = 11$$

EXAMPLE 2 *A has 8 pencils, B has 10 pencils and C has 15 pencils, then the average number of pencils with them :*
(a) 8 *(b) 10*
(c) 15 *(d) lies between 9 and 15*

SOLUTION Average number of pencils $= \frac{8+10+15}{3} = 11$

So, option (d) is correct.

2. If each quantity is increased by a certain value 'K' then the new average is increased by K.

EXAMPLE 3 *A, B, C, D, E are the five electronic shops in the Nazaa market, which have 20, 30, 60, 80 and 50. T.V. sets with them respectively, then the average number of T.V. sets in each shop is :*
(a) 24 *(b) 48* *(c) 50* *(d) 60*

SOLUTION Average number of T.V. sets

$$= \frac{20+30+60+80+50}{5}$$

$$= 48$$

So option (b) is correct.

EXAMPLE 4 *In the previous example, if each of A, B, C, D and E have imported 12 new T.V. sets, then the average number of T.V. sets is:*
(a) 50 *(b) 56* *(c) 60* *(d) 144*

SOLUTION Average number of T.V. sets

$$= \frac{(20+12)+(30+12)+(60+12)+(80+12)+(50+12)}{5}$$

$$= \frac{300}{5} = 60$$

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Alternatively :

$$\frac{(20+30+60+80+50)+(12+12+12+12+12)}{5}$$

$$=\frac{(240)+(5\times 12)}{5}=48+12=60$$

Thus, the new average will be 60.

EXAMPLE 5 *The average number of shirts with Salman, Ambani and Dalmiya is 60, if all of them reached a shopping mall (AMBI) in Gurgaon and purchased 6 shirts each of them then the average number of shirts each of them now has :*

(a) 66 *(b) 63*
(c) 62 *(d) can't be determined*

SOLUTION Required average = Old average + New average
$= 60 + 6 = 66$

3. If each quantity is decreased by a certain value K, then the new average is also decreased by K.

EXAMPLE 6 *A, B, C, D, E, F are the only six families in Indira nagar. A, B, C, D, E and F has 7, 8, 10, 13, 6, and 10 members in their families respectively.*
If 1 member from all the six families left their respective families to accomodate themselves in the hostel of IIM Lucknow, then the average number of members now in each family of Indira nagar is :

(a) 8 *(b) 9* *(c) 10* *(d) 13*

SOLUTION Required average =

$$\frac{(7-1)+(8-1)+(10-1)+(13-1)+(6-1)+(10-1)}{6}$$

$$=\frac{(7+8+10+13+6+10)}{6}-\frac{(6\times 1)}{6}$$

$$=9-1=8$$

EXAMPLE 7 *Last year Sahara, Tata, Singhania and Birla each has 250 industries. This year everyone has sold out 10 factories due to the recession of the economy and poor turnout. The average number of industries which each of them now has :*

(a) 260 *(b) 240*
(c) 25 *(d) none of these*

SOLUTION Required average = Old average – New average
$= (250) - (10) = 240$

4. If each quantity is multiplied by a certain value K, then the new average is the product of old average with K.

EXAMPLE 8 *In a flower shop there were 6 flowers in each bouquet. If the seller has doubled the number of flowers in each bouquet then the new average of flowers in each bouque is :*

(a) 12 *(b) 8*
(c) 3 *(d) none of these*

SOLUTION Now required average = Old average × Multiplicand
$= 6 \times 2 = 12$

5. If each quantity is divided by a certain quantity 'K', then the new average becomes $\frac{1}{K}$ times of the initial average, where $K \neq 0$.

EXAMPLE 9 *The average of 100, 200, 300, 400, ..., 1000 is 550. If each number is divided by 5, then the new average will be equal to :*

(a) 450 *(b) 45*
(c) 55 *(d) none of these*

SOLUTION Actual average

$$=\frac{(100+200+300+\ldots+1000)}{10}=550$$

the new average $=\frac{550}{5}=110$

Hence, (d) is correct.

6. If 'A' be the average of $x_1, x_2, x_m, \ldots y_1, y_2, \ldots y_n$, where $x_1, x_2, \ldots, x_m$ be the below A and $y_1, y_2, y_3, \ldots y_n$ be the above A, then

$$(A-x_1)+(A-x_2)+\ldots(A-x_m)$$
$$=(y_1-A)+(y_2-A)+\ldots(y_n-A)$$

i.e., the surplus above the average is always equal the net deficit below average.

EXAMPLE 10 *The salary of A, B, C, D, E is Rs. 8000. Rs. 5000. Rs. 11000, Rs. 7000, Rs. 9000 per month respectively, then the average salary of A, B, C, D and E per month is :*

(a) Rs. 7000 *(b) Rs. 8000*
(c) Rs. 8500 *(d) Rs. 9000*

SOLUTION Average salary

$$=\frac{8000+5000+11000+7000+9000}{5}$$

$= \text{Rs. } 8000$

NOTE The average salary = Rs. 8000
and the sum of salaries above Rs. 8000

Therefore, the average surplus =

again, the sum of salaries below Rs. 8000

Therefore, the average deficit = 8000 –

Hence the surplus above the average
= deficit below the average

Alternatively : Let the average be 10,000
Then,
10,000 – 8,000 = 2,000
10,000 – 5,000 = 5,000
10,000 – 7,000 = 3,000 and 11,000 – 10,000 = 1,000
10,000 – 9,000 = 1,000 Surplus = 1,000
Deficit → 11000

$\therefore$ Net difference = Surplus – deficit
$= 1{,}000 - 11{,}000 = -10{,}000$

$\therefore$ Average variation $=\frac{-10{,}000}{5}=-2000$

Hence, the actual average = assumed average + average variation
$= 10{,}000 - 2{,}000 = 8{,}000$

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EXAMPLE 11 *The average of 6 numbers 13, 17, 25, 11, 26, 10 is :*
(a) 20 *(b) 17* *(c) 15* *(d) 25*

SOLUTION Let the average be 20.

then
$20 - 13 = 7$
$20 - 17 = 3$
$20 - 11 = 9$
$20 - 10 = 10$
$\overline{29}$ (deficit)

and

$20 - 25 = 5$
$20 - 26 = 6$
$\overline{11}$
(Surplus)

The net difference (surplus − deficit) = $(11 - 29) = -18$

Therefore average variation = $\frac{-18}{6} = -3$

Hence, the actual-average = $20 + (-3) = 17$

NOTE This method is very helpful in Data Interpretation section where there are very large values are to be calculated and most of the time we need just the lumpsum value.

7. Weighted average : When the average of groups or sets, instead of individuals, having different number of elements is being calculated, then it is called the weighted average. Since in this case the number of elements is different for the different sets thus they carry different weightage. If the number of elements in n different groups be $K_1, K_2, K_3, K_4, \ldots, K_n$ and the averages of the respective groups be $A_1, A_2, A_3, A_4 \ldots A_n$ then the

$$\text{weighted average} = \frac{K_1A_1 + K_2A_2 + K_3A_3 + \ldots K_nA_n}{K_1 + K_2 + K_3 + \ldots K_n}$$

EXAMPLE 12 *The average salary of 12 employees of STAR plus is Rs. 18,000 per month and 15 employees of NDTV is Rs. 16,000 per month. The average salary of all the 27 employees is :*
(a) Rs. 17,000 *(b) Rs. 16,500*
(c) Rs. 16,888.88 *(d) none of these*

SOLUTION Required average salary =

$$\frac{12 \times 18{,}000 + 15 \times 16{,}000}{12 + 15} = \frac{4{,}56{,}000}{27} = \text{Rs. } 16{,}888.88$$

EXAMPLE 13 *The number of students at i4 IIM in morning batch, evening batch and weekends batch is 30, 40 and 60 respectively and their respective averages ages (in years) are 22, 21 and 25, then the average age of all the students (combined) is :*
(a) $22\frac{1}{13}$ *years* *(b)* $23\frac{1}{13}$ *years*
(c) 24.5 years *(d) none of these*

SOLUTION $\frac{30 \times 22 + 40 \times 21 + 60 \times 25}{130} = \frac{3000}{130} = 23\frac{1}{13}$ years

8. Problems based on ages :

(i) If the average age of 'n' members of a family is x years then K years back, the average age of the family is $(x - K)$ years, provided that no any person is expired or born in K years period.

(ii) If the present age of 'n' members of a family is x years then K years later the average age of the same family will be $(x + K)$ years.

EXAMPLE 14 *The average age of 7 members of Patel's family is 25 years. The average age of the same family 3 years ago was :*
(a) 21 years *(b) 22 years*
(c) 25 years *(d) none of these*

SOLUTION Present average age of family = 25 years
3 years ago average age of family = 25 − 3 = 22 years

EXAMPLE 15 *The average age of all the 16 professors of Lucknow University is 52 years. Four years later the average age of all the 16 professors will be, if there is no any retirement or recruitment :*
(a) 68 *(b) 64*
(c) 56 *(d) none of these*

SOLUTION Present average age = 52 years
4 years later the average age = 52 + 4 = 56 years

EXAMPLE 16 *6 months ago the present age of the student of class 10th was 14 years. 6 months hence, the age of the same students will be :*
(a) 15 years *(b)* $15\frac{1}{2}$ *years*
(c) 20 years *(d) none of these*

SOLUTION Since the time difference between two dates is 1 year, hence the average age will be increased by 1 year.
Thus the average age of the class 6 months hence will be 15 years.

EXAMPLE 17 *The average age of Priyambada's family consisting of 5 members 3 years ago was 35 years. One year ago a new baby was born in this family. Three years hence the average age of the family will be :*
(a) 36 years *(b)* $34\frac{5}{6}$ *years*
(c) $35\frac{4}{5}$ *years* *(d) none of these*

SOLUTION 3 years ago total age of 5 members
$= 5 \times 35 = 175$ years
At the time of birth of new baby the total age of family
$= 175 + (2 \times 5)$
$= 185$ years
The present age of family $= 185 + (1 \times 6) = 191$ years
3 years hence, the average age of family
$= \frac{191 + (3 \times 6)}{6} = 34\frac{5}{6}$ years

EXAMPLE 18 *10 years ago the average age of all the 25 teachers of the Girls college was 45 years. 4 years ago, the principal has retired from her post at the age of 60 year. So after one year a new principal whose age was 54 years recruited from outside. The present average age of all the teachers is, if principal is also considered as a teacher :*
(a) $54\frac{18}{25}$ *years* *(b)* $55\frac{17}{25}$ *years*
(c) $49\frac{1}{2}$ *years* *(d) none of these*

SOLUTION 10 years ago average age of 25 teachers = 45 years
4 years ago (just before the reteriment of principal) average age of 25 teachers = 45 + 6 = 51 years
and the same time total age of 25 teachers = 51 × 25
= 1275 years

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and the total age of remaining 24 teachers when just the principal has retired

$$= 1275 - 60 = 1215 \text{ years}$$

1 year later (*i.e.*, 3 years ago from present) total age of 24 teachers (just before the recruitment of new principal)

$$= 1215 + (1 \times 24) = 1239 \text{ years}$$

and the total age of 25 teachers including new principal just after the recruitment

$$= 1239 + 54 = 1293 \text{ years}$$

Thus the present age of all the 25 teachers

$$= 1293 + (3 \times 25) = 1368 \text{ years}$$

Hence, the present average age of the 25 teachers

$$= \frac{1368}{25} = 54\frac{18}{25} \text{ years}$$

Alternatively : 10 years ago, the average age of 25 teachers = 45 years

Let us assume that the principal has not retired from her post then the present average age of all the 25 teachers,

$$= 45 + 10 = 55 \text{ years}$$

Thus the total age of all the 25 teachers

$$= 55 \times 25 = 1375 \text{ years}$$

Now assume that the new principal has replaced the old principal 4 years ago instead of 3 years ago, when the age of new principal would have been 53 years. Thus the age of new principal was 7 years less than the age of old principal, which results in the reduction of total age of the group of 25 teachers by 7 years.

Thus the actual total age of the 25 teachers (presently)

$$= 1375 - 7 = 1368 \text{ years}$$

Hence, the present average age of the 25 teachers

$$= \frac{1368}{25} = 54\frac{18}{25} \text{ years}$$

EXAMPLE 19 *The ratio of the ages of the father and the daughter at present is 3 : 1. Four years ago the ratio was 4 : 1. The average age of the father and daughter 2 years hence will be :*

(a) 24 *(b) 26* *(c) 25* *(d) 36*

SOLUTION Let the present ages of father be $3x$ and daughter be x.

So the 4 years ago father's age and daughter's age was $(3x - 4)$ and $(x - 4)$

Therefore $\left(\frac{3x-4}{x-4}\right) = \frac{4}{1}$

$\Rightarrow$ $x = 12$ years and $3x = 36$ years

Hence, the present average age of father and daughter = 24 years and the average age 2 years hence will be 26 years.

9. Problems based on income/salary

Income = Expenditure + Savings

EXAMPLE 20 *The average salary of Rajesh, Bahadur and Amir is Rs. 8000 per month. The average expenditure of the Rajesh, Bahadur and Amir per month is Rs. 5000. The average savings of all the 3 persons per month is :*

(a) Rs. 3000 *(b) Rs. 5000*
(c) Rs. 2500 *(d) Rs. 9000*

SOLUTION Average saving

= Average Income − Average Expenditure

= 8000 − 5000 = 3000

EXAMPLE 21 *The average salary of A, B and C is Rs. 10,000 and average expenditure of A is Rs. 6000 then the average savings of B and C is :*

(a) Rs. 5500 *(b) Rs. 4500*
(c) Rs. 4000 *(d) can't be determined*

SOLUTION Total income = Total Expenditure + Total savings

We can not find the average savings of B and C, since data is insufficient.

EXAMPLE 22 *The average salary of A, B is Rs. 6000 and that of C, D and E is Rs. 8000. The average salary of all the 5 people is :*

(a) Rs. 7200 *(b) Rs. 7000*
(c) Rs. 7500 *(d) can't be determined*

SOLUTION Required average salary =

$$\frac{6000 \times 2 + 8000 \times 3}{(2+3)} = \frac{36000}{5} = \text{Rs. } 7200$$

EXAMPLE 23 *The average salary of all the 60 employees in an office is Rs. 12,000 per month. If the number of executives is twice the number of non executive employees, then the average salary of all the non executive employees is :*

(a) Rs. 9000 *(b) Rs. 8000*
(c) Rs. 6000 *(d) can't be determined*

SOLUTION $\frac{\text{No. of executives}}{\text{No. of non-executives}} = \frac{2}{1}$

Therefore number of executives = 40

and number of non-executive employees = 20

Now, go through the options

Total salary = 40 × salary of executive + 20 × salary of non-executive

$60 \times 12000 = 40 \times k + 20 \times l$, here k, l are unknowns

So we can't determine the required average salary.

EXAMPLE 24 *In the above problem, if the average salary of non-executives be $\frac{2}{5}$th of the average salary of executives, then the average salary of non-executive employees is :*

(a) Rs. 9000 *(b) Rs. 8000*
(c) Rs. 6000 *(d) data insufficient*

SOLUTION By options : $60 \times 12000 = 20 \times 6000 + 40 \times 15000$

Hence option (c) is correct.

Alternatively : $60 \times 12000 = 20 \times x + 40 \times \frac{5}{2}x$

$\Rightarrow$ $x = 6000$

10. Problems based on speed, time and distance :

Case 1: When the distance travelled in different time slots or parts is same *i.e.*, if a person or vehicle moves x km at a speed of u km/hr and further he goes or comes back the same distance x km at a speed of v km/hr. Then the average speed $= \frac{2uv}{(u+v)}$.

If there are 3 parts of distance x km travelled with 3 different speeds *i.e.*, if a person goes first x km @ speed of u km/hr and next x km @ v km/hr and the last x km @ w km/hr. Then the average speed $= \frac{3uvw}{(uv + vw + wu)}$.

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NOTE In this type of questions the average speed is independent of the distance travelled.

General Formula : Average speed $= \dfrac{\text{Total distance}}{\text{Total time}}$

The above two formulae are derived with the help of the general formula of average speed.

Proportion Method : If the half of the distance is covered at u km/hr and the rest half of the journey is covered at v km/hr then the average speed can be found as follows :

Step 1. Divide the difference of u and v in the ratio of $u : v$ (where $u < v$)

Step 2. $u + \dfrac{(u \sim v) \times u}{(u + v)}$

or $v - \dfrac{(u \sim v)\, v}{u + v}$

Case 2. When the distances travelled at different speeds are different then we calculate the average speed with the help of general formula of average speed. e.g., A person first goes x_1 km at the speed of u km/hr and x_2 km at the speed of v km/hr and x_3 km at the speed of w km/hr and so on, then the

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{x_1 + x_2 + x_3 + \ldots}{t_1 + t_2 + t_3 + \ldots} = \frac{x_1 + x_2 + x_3 + \ldots}{\frac{x_1}{u} + \frac{x_2}{v} + \frac{x_3}{w} + \ldots}$$

EXAMPLE 25 *Einstien goes from Ahmedabad to Lucknow at the speed of 40 km/hr and returned at the speed of 60 km/hr. The average speed of Einstien during the whole journey is :*

(a) *48 km/hr* (b) *24 km/hr*
(c) *50 km/hr* (d) *none of these*

SOLUTION $\dfrac{2 \times 40 \times 60}{(40 + 60)} = \dfrac{4800}{100} = 48$ km/hr

Alternatively: $40 \xleftarrow{(20)} 60$ (2 : 3)

So the average speed $= 40 + \dfrac{2}{5}(20) = 48$ km/hr

or Average Speed $= 60 - \dfrac{3}{5}(20) = 48$ km/hr

EXAMPLE 26 *Sri Krishna took the chariot and started his journey from Mathura to Gokul by his chariot at the speed of 40 km/hr and then, the same distance he travelled on his foot at the speed of 10 km/hr from Gokul to Brindaban. Then he returned from Brindaban to Mathura via Gokul at the speed of 24 km/hr riding on the horse. The average speed of the whole trip is :*

(a) *20 km/hr* (b) *25 km/hr*
(c) *19.2 km/hr* (d) *18.5 km/hr*

SOLUTION Since the distance from Mathura to Gokul is same as that of Gokul to Brindaban.

So the average speed from Mathura to Brindaban $= \dfrac{2 \times 40 \times 10}{(40 + 10)} = 16$ km/hr

Again since he returned on the same path, so the distance from Mathura to Brindaban is same in both the directions. Thus the required average speed $= \dfrac{2 \times 16 \times 24}{(16 + 24)} = 19.2$ km/hr

EXAMPLE 27 *Anoop travels first $\frac{1}{3}$rd of the total distance at the speed of 10 km/hr and the next $\frac{1}{3}$rd distance at the speed of 20 km/hr and the last $\frac{1}{3}$rd distance at the speed of 60 km/hr. The average speed of Anoop is :*

(a) *15 km/hr* (b) *18 km/hr*
(c) *25 km/hr* (d) *30 km/hr*

SOLUTION Since all the three distances are same, hence the average speed

$$= \frac{3 \times 10 \times 20 \times 60}{(200 + 1200 + 600)} \quad \left(= \frac{3uvw}{uv + vw + wu}\right)$$

$$= \frac{36000}{2000} = 18 \text{ km/hr}$$

EXAMPLE 28 *Columbus started his journey from Lucknow to Kolkata, which is 200 km, at the speed of 40 km/hr then he went to Bangalore which is 300 km, at the speed of 20 km/hr. Further he went to Ahmedabad which is 500 km, at the speed of 10 km/hr. The average speed of Columbus is :*

(a) $14\frac{2}{7}$ *km/hr* (b) $14\frac{5}{7}$ *km/hr*
(c) *15.6 km/hr* (d) *none of these*

SOLUTION Average speed $= \dfrac{\text{Total distance}}{\text{Total time}}$

$$= \frac{(200 + 300 + 500)}{\left(\frac{200}{40} + \frac{300}{20} + \frac{500}{10}\right)} = \frac{1000}{70} = \frac{100}{7} = 14\frac{2}{7} \text{ km/hr}$$

$= 14.2857$ km/hr

11. Some Important Concepts

1. $a + b = k$ and $b + c = l$ and $a + b + c = m$
then $[(a + b) + (b + c)] - (a + b + c) = (k + l) - m$
or $b = k + l - m$

2. $a + b = k$, $d + e = l$ and $a + b + c + d + e = m$
then $c = (a + b + c + d + e) - [(a + b) + (d + e)]$
$= m - [k + l]$

EXAMPLE 29 *The average weight of all the 11 players of Indian cricket team is 50 kg. If the average of first six lightest weight players is 49 kg and that of the six heaviest players is 52 kg. The average weight of the player which lies in the sixth position in the list of players when all the 11 players are arranged in the order of increasing or decreasing weights :*

(a) *56 kg* (b) *52 kg*
(c) *51 kg* (d) *none of these*

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SOLUTION Let $A, B, C, D, E, F, G, H, I, J, K$ be the 11 players in the order of increasing weight then,

$$A + B + C + D + E + F = 49 \times 6 = 294$$
$$F + G + H + I + J + K = 52 \times 6 = 312$$
$$\text{and } A + B + C + D + E + F + G + H + I + J + K = 50 \times 11 = 550$$

Therefore $F = (A + B + C + D + E + F) + (F + G + H + I + G + K) - (A + B + C + D + \ldots J + K)$
$= 294 + 312 - 550 = 56$

Hence the average weight of $F = 56$ kg.

EXAMPLE 30 *The average presence of students in a class on Monday, Tuesday and Wednesday is 30 and on the Wednesday, Thursday, Friday and Saturday is 28 then number of students who attended the class on Wednesday is, if the average number of students on all the six days is 27 :*

(a) 24 (b) 25 (c) 20 (d) 40

SOLUTION Since $W = (M + T + W) + (W + Th + F + S) - (M + T + W + Th + F + S)$
$= (30 \times 3) + (28 \times 4) - (27 \times 6)$
$= 202 - 162 = 40$

EXAMPLE 31 *The average age of A, B, C, D and E is 40 years. The average age of A and B is 35 years and the average age of C and D is 42 years. The average age of E is :*

(a) 46 (b) 48
(c) 32 (d) none of these

SOLUTION $A + B + C + D + E = 40 \times 5 = 200$
$A + B = 35 \times 2 = 70$
$C + D = 42 \times 2 = 84$
Therefore $E = (A + B + C + D + E) - [(A + B) + (C + D)]$
$= 200 - (70 + 84) = 46$
Thus the average age of $E = 46$ years.

EXAMPLE 32 *The average temperature on Monday, Tuesday and Wednesday is 38°C. The average temperature on Tuesday, Wednesday and Thursday is 43°C. If the average temperature on Monday and Thursday is 18.5°C. The average temperature on Monday is :*

(a) 11°C (b) 21°C (c) 35°C (d) 27°C

SOLUTION $(M + T + W) = 38 \times 3 = 114$ ...(i)
$(T + W + Th) = 43 \times 3 = 129$ ...(ii)
Therefore $Th - M = 15$ [(ii) – (i)]
and $Th + M = 37$
Thus the temperature on Monday is 11°C and on Thursday is 26°C.

12. Average of Some Important Series of Numbers :

(i) Average of first 'n' natural numbers $= \frac{n+1}{2}$

(ii) Average of first 'n' even numbers $= (n + 1)$

(iii) Average of first 'n' odd numbers $= n$

(iv) If there are $(p + q)$ elements in a set or group but the average of p elements is r and the average of q elements is s, then the average of all the elements of the set (or group) is $\frac{(pr + sq)}{(p + q)}$.

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EXERCISE

LEVEL 1

1. The average weight of a class of 20 students is 45 kgs. A new student whose weight is 40 kgs replaces an old student of this class. Hence, the average weight of the whole class decreases by 1 kg. The weight of the replaced student is :
(a) 55 kgs (b) 50 kgs
(c) 60 kgs (d) none of these

2. The average length of first 3 fingers is 3 inches and the average length of the other 2 fingers *i. e*, thumb and the index finger is 2.8 inches. If the length of the index fingers is 3 inches then the length of the thumb is :
(a) 2 inches (b) 2.6 inches
(c) 3 inches (d) none of these

3. Three types of rice whose rates are Rs. 38, Rs. 43 and Rs. 49 per kg are blended together to make a 15 kg of new blend of rice in which there are 8 kgs, 4 kgs 3 kgs of the respective types of rice. The average price of the new blend of rice is :
(a) Rs. 41.53 (b) Rs. 43
(c) Rs. 40 (d) Rs. 43.3

4. Pankaj went to the post-office at the speed of 60 km/hr while returning for his home he covered the half of the distance at the speed of 10 km/hr, but suddenly he realized that he was getting late so he increased the speed and reached the home by covering rest half of the distance at the speed of 30 km/hr. The average speed of the Pankaj in the whole length of journey is :
(a) 5.67 km/hr (b) 24 km/hr
(c) 22.88 km/hr (d) 5.45 km/hr

5. 123 typists typed 984 papers in 1/15 hour. The number of papers typed per minute by an average typist is :
(a) 1 (b) 2
(c) 3 (d) 5

6. The cost of the Red, Green and Blue colours per kg is Rs. 20, Rs. 15 and Rs. 18 respectively. Rang Mahal is a renowned building in which these three colours are being used in the ratio of 3 : 2 : 4. The average cost of all the three colours used per kg is :
(a) 18 (b) 20
(c) 17.66 (d) can't be determined

7. Rajeev earns 3/2 times in January, April, July and October than his average earning of Rs. 600 per month in the rest of the months. So his savings in the January, April, July and October goes to 5/4 times that of the rest months saving of Rs. 400 per month in the year. The average expenditure of per month is :
(a) Rs. 266.66 (b) Rs. 250
(c) Rs. 233.33 (d) Rs. 433.33

8. The average of 3 prime numbers lying between 47 and 74 is 191/3. The greatest possible difference between any two out of the 3 prime numbers is :
(a) 12 (b) 14
(c) 18 (d) can't be determined

9. The average of 9 numbers is 11. If each of these 9 numbers is multiplied by 5 and then 5 is added to each of these resultant numbers, then the new average is :
(a) 20 (b) 30
(c) 60 (d) 50

10. The average score of Dhoni after 48 innings is 48 and in the 49th innings Dhoni scores 97 runs. In the 50th innings the minimum number of runs required to increase his average score by 2 than it was before the 50th innings :
(a) 99 (b) 149
(c) 151 (d) can't be determined

11. The average age of Sachin and Ganguli is 35 years. If Kaif replaces Sachin, the average age becomes 32 years and if Kaif replaces Ganguli, then the average age becomes 38 years. If the average age of Dhoni and Irfan be half of the average age of Sachin, Ganguli and Kaif, then the average age of all the five people is :
(a) 28 years (b) 32 years
(c) 25 years (d) none of these

12. Out of these five people (in question no. 11) whose age is the greatest?
(a) Sachin (b) Ganguli
(c) Kaif (d) can't be determined

13. In a village the average age of *n* people is 42 years. But after the verification it was found that the age of a person had been considered 20 years less than the actual age, so the new average, after the correction, increased by 1. The value of *n* is :
(a) 21 (b) 20
(c) 22 (d) none of these

14. The average rainfall in the months of January and February is 6 cm and in the months of March to June is 5 cm and July to October is 10 cm and in the November and December, it is 6 cm. The average rainfall for the whole year is :
(a) 7 (b) 5.5
(c) 7.5 (d) none of these

15. On an average 300 people watch the movie in Sahu Cinema hall on Monday, Tuesday and Wednesday and the average number of visitors on Thursday and Friday is 250. If the average number of visitors per day in the week be 400, then the average number of people who watch the movie in weekends (*i. e*, on Saturday and Sunday) is :

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(a) 500 (b) 600
(c) 700 (d) none of these

16. The average weight of 11 players of Indian cricket team is increased by 1 kg, when one player of the team weighing 55 kg replaced by a new player. The weight of the new player is :
(a) 55 kg (b) 64 kg
(c) 66 kg (d) none of these

17. The average age of a family of 6 members 4 year ago was 25 years. Mean while a child was born in this family and still the average age of the whole family is same today. The present age of the child is :
(a) 2 years (b) $1\frac{1}{2}$ years
(c) 1 year (d) data insufficient

18. Amitabh's average expenditure for the January to June is Rs. 4200 and he spends Rs. 1200 in January and Rs 1500 in July. The average expenditure for the months of February to July is :
(a) Rs. 4250 (b) Rs. 4520
(c) Rs. 4060 (d) none of these

19. The average of a, 11, 23, 17 is 15 and the average of a, b, 12, 25 is 16. The value of $\frac{a}{b}$ is :
(a) 1/3 (b) 1/2
(c) 2/3 (d) 3/4

20. The average salary is being paid to all its employees by the Biotech corporation is Rs. 15,500. The average salary of the senior employees is Rs. 18000 per month and the average salary of the junior employees is Rs. 12,000 per month. If there are only two levels of employees viz junior and senior level, then what fraction of the total employees is the junior level employees are :
(a) $\frac{7}{10}$ (b) $\frac{5}{12}$
(c) 5/10 (d) none of these

21. The average of any 5 consecutive odd numbers a, b, c, d, e is :
(a) $\frac{(abcde)}{5}$ (b) $\frac{bd}{3}$
(c) $\left(\frac{a+c+e}{5}\right)$ (d) none of these

22. The average age of 30 students of a class is 30 years. When the average age of class teacher is also included, the average age of the whole class increases by 1 year. The age of the class teacher is :
(a) 31 years (b) 60 years
(c) 61 years (d) none of these

23. There were five sections in MAT paper. The average score of Pooja in first 3 sections was 83 and the average in the last 3 sections was 97 and the average of all the sections (i.e., whole paper) was 92, then her score in the third section was :
(a) 85 (b) 92
(c) 88 (d) none of these

24. The average age of 18 pupils of Dronacharya was 25 years. If the age of Dronacharya was also included, the average age of 19 people becomes 26 years. The average age of the Dronacharya at that time was :
(a) 33 (b) 44
(c) 50 (d) 51

25. The average of 7 consecutive odd numbers if the smallest of those numbers is denoted by k :
(a) $k+4$ (b) $k+7$
(c) $k+6$ (d) $7k$

26. If the average marks of $\frac{1}{4}$th class is 85% and that of $\frac{1}{3}$rd class is 70% and the average marks of the rest class is 56%, then the average of the whole class is (for the given subjects) :
(a) 67.916% (b) 72.33%
(c) 69.165% (d) can't be determined

27. The average length of any four fingers of my left hand is 600 mm. Then the average length of all the five fingers of my left hand is :
(a) 800 mm (b) 750 mm
(c) 480 mm (d) can't be determined

28. The average of $4\frac{3}{5}, 2\frac{2}{3}, 6\frac{8}{9}, 7\frac{7}{15}, 3\frac{5}{9}$ is :
(a) $5\frac{3}{225}$ (b) $5\frac{8}{225}$
(c) $6\frac{3}{45}$ (d) $25\frac{8}{45}$

29. The average of 1000.0001, 100.001, 10.01, 1.1 is :
(a) 277.777 (b) 322.222
(c) 11.11 (d) 233.333

30. The average of 7 consecutive numbers which are positive integers is 10. The average of lowest and highest such numbers is :
(a) 7 (b) 10
(c) 15 (d) data insufficient

31. The average of first 100 natural number is :
(a) 100 (b) 50
(c) 50.50 (d) 55

32. The average of first 50 odd natural numbers is :
(a) 50 (b) 55
(c) 51 (d) 101

33. The average of first 99 even numbers is :
(a) 9999 (b) 100
(c) 9801 (d) 9009

34. The average of a, b and c is 79 and the average of a and c is also 79. Then the value of b is :
(a) 0 (b) 79
(c) −79 (d) none of these

35. The average value of property of Mittal, Ambani and Singhania is Rs. 11111 crore. The property of Singhania is as less as the property of Mittal is greater than the average property of both the Singhania and Mittal. The value of property of Ambani is :
(a) Rs. 111 crore (b) Rs. 11111 crore
(c) Rs. 3703.7 crore (d) can't be determined

36. I went to Delhi @ speed of 200 km/hr but suddenly I returned to the same place @ speed of 600 km/hr. What is my average speed :
(a) 300 km/hr (b) 400 km/hr
(c) 366.66 km/hr (d) none of these

37. The average of A and B is 400 and the average of C and D is 600 the average of A, B, C and D is :
(a) 500 (b) 450
(c) 525 (d) 625

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38. The average weight of liquid in 100 bottles is 500 gm. The total weight of all the bottles is 20 kg. The average weight of a bottle with liquid is :
(a) 0.65 kg (b) 0.7 kg
(c) 70 ml (d) none of these

39. The average score of Sehwag in 10 innings was 77 runs. In the 11th innings he had scored zero runs. The overall average score of Sehwag in all the 11 innings was :
(a) 77 (b) 7.7
(c) 11 (d) none of these

40. The average age of 3 children of Arihant Singh is 12 years and their ratio of ages is 3 : 4 : 5. The average age of the youngest and eldest child is if he had only 3 children :
(a) 12 (b) 21
(c) 8 (d) 9

41. The average income of all the Infosys employees is Rs. 20,000 per month. Recently the company announced the increment of Rs. 2,000 per month for all the employees. The new average of all the employees is :
(a) Rs. 22,000 (b) Rs. 40,000
(c) Rs. 22,00 (d) data insufficient

42. The average age of 10 students in a class is 20 years, if a new student is also included, then the new average age of all the students increases by 1 year. The age of the new student is :
(a) 21 years (b) 30 years
(c) 31 years (d) none of these

43. The average weight of 20, four wheelers is 180 kg. If an old car is removed from this group of four wheelers, the new average weight decreases by 2 kg. The weight of the removed car is :
(a) 220 (b) 218
(c) 182 (d) none of these

44. The average price of 3 diamonds of same weights is Rs. 5 crore, where the average price of the two costliest diamonds is double the price of the cheapest diamond. The price of the cheapest diamond is :
(a) 3 crore (b) 5 crore
(c) 1.66 crore (d) can't be determined

45. In the previous question, the price of the costliest diamond is :
(a) 5 crore (b) 6 crore
(c) 8 crore (d) can't be determined

46. Praveen gets 40 marks out of 50 in Computer Science, 70 out of 100 in Manufacturing science and 142 out of 150 in proffessional communication. The average marks of Praveen (in percent) in all the three subjects is :
(a) 84% (b) 76%
(c) 71% (d) 60%

47. The average of all the prime and composite numbers upto 100 is :
(a) 51 (b) 49.50
(c) 50.50 (d) 55

48. The average of all the perfect squares upto 100 is :
(a) 38.5 (b) 1000
(c) 100 (d) 385

49. The average of all the non-negative integers upto 99 is :
(a) 50.49 (b) 49.50
(c) 50.50 (d) 99

50. The average of 7, 14, 21, 28 ... 77 is :
(a) 7 (b) 11
(c) 42 (d) 66

51. The average weight of A, B, C and D is 40 kg. A new person E is also included in the group, then the average weight of the group is increased by 1 kg. Again a new person F replaces A, then the new average of 5 persons becomes 42. The average weight of B, C, D, F is :
(a) 42 (b) 41.25
(c) 42.5 (d) none of these

52. The average of 3 consecutive natural numbers (which are in increasing order) is k. If two more consecutive number, just next the first set of numbers, is added, then the new average becomes :
(a) $k + 2$ (b) $k + 1$
(c) $\frac{2k+1}{2}$ (d) $2k - 1$

53. The average of any 5 consecutive odd natural numbers is k. If two more such numbers, just next to the previous 5 numbers are added, the new average becomes :
(a) $\frac{2}{7}(k + 1)$ (b) $2k - 3$
(c) $2k + 1$ (d) $k + 2$

54. The average weight of the 5 officers of a regiment is 42 kg. If a senior officer was replaced by a new officer and thus the average increased by 500 gm, the weight of the new officer is :
(a) 44.5 kg (b) 45 kg
(c) 42.5 kg (d) none of these

55. The average age of 6 servants in my farm house is 28 years. A new and young servant replaces an old servant, then the new average reduces by 1 year, the age of the new servant is :
(a) 26 years (b) 22 years
(c) 35 years (d) can't be determined

56. In the above question (no. 55) if the age of the replaced servant was 31 years, then the age of the new servant is :
(a) 25 years (b) 35 years
(c) 24 years (d) none of these

57. The average income of A, B and C is Rs. 12,000 per month and the average income of B, C and D is Rs. 15,000 per month. If the average salary of D be twice that of A, then the average salary of B and C is (in Rs.) :
(a) 8,000 (b) 18,000
(c) 13,500 (d) 9,000

58. The average price of 80 computers in an electronic shop is Rs. 30,000. If the highest and lowest price computers are sold out then the average price of the remaining 78 computers is Rs. 29,500. The cost of the highest price computer is Rs. 80,000. The cost of lowest price computer is :
(a) Rs. 19,000 (b) Rs. 20,000
(c) Rs. 29,000 (d) can't be determined

59. A has 50 coins of 10 paise denominations. While B has 10 coins of 50 paise denominations. C has 20 coins of 25 paise denominations while D has 25 coins of 20 paise denominations. The average number of paise per person is :
(a) 450 paise (b) 500 paise
(c) 600 paise (d) can't be determined

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60. A travel agency has three types of vehicles viz. four seater, autorickshaw, 10 seater maxi cab and 20 seater minibus. The rate of each passanger (irrespective of its age or weight or seniority) for the auto rickshaw is Rs. 12 and for the maxicab is Rs. 15 and for the minibus is Rs. 8 for the one round. The average occupancy of the seats is 100%, 80% and 75% respectively. If he has only one vehicle of each kind, then the average earning for one round of each vehicle is :
(a) Rs. 96 (b) Rs. 90
(c) Rs. 86 (d) Rs. 70

LEVEL 2

1. A train normally covers a certain distance at a speed of 60 km/hr. However, if it were to halt for a fixed time interval in each hour its average speed reduced to 50 km/hr. What is the time interval for which the train halt in each hour?
(a) 10 minutes (b) 20 minutes
(c) 6 minutes (d) 12 minutes

2. If p, q, r be three positive numbers such that $p > q > r$ when the smallest number is added to the difference of the rest two numbers, then the average of the resultant number and the original numbers except to the smallest number is 21 more than the average of all the three original numbers. The value of $(p - q)$ is :
(a) 7 (b) 14
(c) 63 (d) 42

3. Progressive express left for New Delhi, increasing its speed in each hour. It started its journey from Lucknow, but after four hours of its journey it met with accident. Its speed in the fourth hour was $\frac{7}{5}$ times that of the third hour and the speed in the third hour was $\frac{10}{7}$ times that of the second hour and in the second hour it was $\frac{7}{5}$ times that of the first hour. If it would have been travelled with the half of the speed that of the third hour, then it would have gone 160 km less in the same time (i. e , in four hours). The average speed of the train during the journey of 4 hours was :
(a) 50 km/hr (b) 90 km/hr
(c) 80 km/hr (d) can't be determined

4. In a Mock CAT 123 students appeared and the average score obtained was 120. But later it was found that the top three students were repeaters, so their score has been eliminated and then the new average score was found to be decreased by 1.5. Also, it is known that all the students obtained the marks in integers and the scores of the toppers were distinct. If the second highest topper has scored more than 185 marks, then the highest possible score of the third highest topper is :
(a) 166 (b) 167
(c) 168 (d) 170

5. The average age of all the 100 employees in an office is 29 years, where $\frac{2}{5}$ employees are ladies and the ratio of average age of men to women is 5 : 7. The average age of female employees is :
(a) 18 years (b) 35 years
(c) 25 years (d) none of these

6. There are two houses in Parliament. One is Lok Sabha and the other one is Rajya Sabha and the member of Parliaments (MPs) in both the houses is 300 and 200 respectively. The average age of the members of Lok Sabha and Rajya Sabha is 40 years and 50 years respectively. A member of the Rajya Sabha when elected for the Lok Sabha also, he left the Rajya Sabha and becomes the member of the Lok Sabha. Thus the average age of both the houses increases. Which one of the following statements is true?
(a) The age of this member is greater than 50 years
(b) The age of this member is less than 40 years
(c) The age of this member is greater than 40 but less than 50 years
(d) none of these

Directions for questions 7, 8 and 9 : Eight years ago there were 5 members in the Arthur's family and then the average age of the family was 36 years. Mean while Arthur got married and gave birth to a child. Still the average age of his family is same now.

7. The present age of his wife is :
(a) 25 years (b) 26 years
(c) 32 years (d) data insufficient

8. The age of his wife at the time of his child's birth was. If the difference between the age of her child and herself was 26 years :
(a) 25 years (b) 26 years
(c) 20 years (d) can't be determined

9. The age of Arthur at the time of his marriage was :
(a) 22 years (b) 23 years
(c) 26 years (d) can't be determined

10. Eleven years earlier the average age of a family of 4 members was 28 years. Now the age of the same family with six members is yet the same, even when 2 children were born in this period. If they belong to the same parents and the age of the first child at the time of the birth of the younger child was same as there were total family members just after the birth of the youngest members of this family, then the present age of the youngest member of the family is :
(a) 3 years (b) 5 years
(c) 6 years (d) none of these

11. Mr. Patel walked 6 km to reach the station from his house, then he boarded a train whose average speed was 60 km/hr and thus he reached his destination. In this way he took total of 3 hours. If the average speed of the entire journey was 32 km/hr then the average speed of walking is :
(a) 3 km/hr (b) 4.5 km/hr
(c) 4 km/hr (d) none of these

Directions for question numbers 12 and 13: During the winter session all the women which constitutes $\frac{1}{5}$th strength of the Lok Sabha, left the house (i.e., parliament) due to the

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rejection of their demand. Actually they were asking for the 50% reservation of seats for the women in the Lok Sabha. Thus the average age of the remaining members of the house (i.e., the Lok Sabha) increases by $\frac{1}{4}$th than it was earlier when all the members (i.e., men and women) were present.

12. The total strength of the Lok Sabha is :
(a) 300 MPs (b) 475 MPs
(c) 526 MPs (d) none of these

13. If the average age of women be 30 years then the average age of male members in the Lok Sabha will be :
(a) 26 (b) 34
(c) 40 (d) none of these

14. An artist in a circus moves along the hexagonal path of each side 20 metres in such a way that for the first 20 metres he goes with a speed of 20 m/s and the next 20 metres with a speed of 10 m/s. Similarly he continues for the rest of the hexagonal path with the same alternating speeds *i. e.*, 20 m/s and 10 m/s. The average speed of the artist per round of the circus is :
(a) 13.33 m/s (b) 23.66 m/s
(c) $39\frac{1}{3}$ m/s (d) 15 m/s

15. The average of 26, 29, n, 35 and 43 lies between 25 and 35. If n is always an integer and greater than the average of the given integers then the value of n is :
(a) $33 < n < 47$ (b) $34 < n < 43$
(c) $33 < n < 42$ (d) none of these

16. The average of 4 distinct prime numbers a, b, c, d is 35, where $a < b < c < d$. a and d are equidistant from 36 and b and c are equidistant from 34 and a, b are equidistant from 30 and c and d are equidistant from 40. The difference between a and d is :
(a) 30 (b) 14
(c) 21 (d) can't be determined

17. There are four numbers whose product is 9261000 and each of these four numbers is formed by 3 distinct prime numbers. The average of all the four numbers is :
(a) 61.75 (b) 67.25
(c) 82.33 (d) data insufficient

18. A number is such that when it is multiplied by 8, it gives another number which is as much above from 270 as the original number (itself) is below 270. The average of the original number and the resultant number is :
(a) 33.75 (b) 190
(c) can't be determined (d) none of these

19. In the above question (no. 18) the average of the original no. and 270 is :
(a) 165 (b) 185
(c) 135 (d) none of these

20. In a particular week the average number of people who visited the Tajmahal is 40. If we exclude the holidays then the average is increased by 16. Further if we exclude also the day on which the maximum number of 112 people visited the Tajmahal, then the average becomes 42. The number of holidays in the week is :
(a) 1 (b) 2
(c) 3 (d) data insufficient

21. The average age of all the 20 students of a class is 24. The minimum age of a student is 18 and the maximum age of another student in the same class is 30 years. When the two students whose average age was 26 years resticated from the class but later on one of the resticated student was readmitted. Now the average age of the class is :
(a) 23.89 years (b) 28.39 years
(c) 25 years (d) can't be determined

22. In a set of prime and composite numbers, the composite numbers are twice the number of prime numbers and the average of all the numbers of the set is 9. If the number of prime numbers and composite numbers are exchanged then the average of the set of numbers is increased by 2. If during the exchange of the numbers the average of the prime numbers and composite numbers individually remained constant, then the ratio of the average of composite numbers to the average of prime numbers (initially) was :
(a) $\frac{7}{13}$ (b) $\frac{13}{7}$
(c) 9/11 (d) none of these

23. The total age of all the guests in the party was 540 years. If a South Indian couple (guests) left the party, then the average of the remaining guests still remained unchanged, where the age of both the husband and wife (the South Indian couple) was same, then the average age of this couple and the total number of guests in the party, respectively, can be :
(a) 18, 27 (b) 20, 27
(c) 15, 38 (d) can't be determined

24. In the command hospital Lucknow the sum of the ages of all the 29 people *i. e.*, physicians, surgeons and nurses is 696. If the age of each physician, each surgeon and each nurse be 1 year, 6 years and 3 years more, then the average age of the whole staff would have been 3 years more. If the number of surgeon is a square root of a two digit number which is also a perfect cube, then the number of nurses in the hospital is :
(a) 12 (b) 15
(c) 16 (d) none of these

25. The average expenditure of Sarvesh for the January to June is Rs. 4200 and he spents Rs. 1200 in January and Rs. 1500 in July. The average expenditure for the months of February to July is :
(a) 4250 (b) 4520
(c) 4060 (d) none of these

26. The average marks of Sameer decreased by 1, when he replaced the subject in which he has scored 40 marks by the other two subjects in which he has just scored 23 and 25 marks respectively. later he has also included 57 marks of Computer Science, then the average marks increased by 2. How many subjects were there initially?
(a) 6 (b) 12
(c) 15 (d) can't be determined

27. In a combined family the average age of 4 males and 7 females is 42 and 20 years respectively. If two persons whose average age is 13 years have left the family and other three

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people joined the family whose respective ages are 11, 15 and 28 years, then the average age of the new family is increased by :
(a) 4 years (b) 1 year
(c) 3 years (d) none of these

28. A teacher gave sum to his class to find the average of n numbers viz. 1, 2, 3, 4, 5, 6 ... etc. But when the teacher checked the solution, he has found that during the calculation a student just missed a number for the addition thus his average of the n numbers was 15. The value of n is :
(a) 30 (b) 26
(c) 31 (d) not unique

29. The average earning of a group of persons is Rs. 50 per day. The difference between the highest earning and lowest earning of any two persons of the group is Rs. 45. If these two people are excluded the average earning of the group decreases by Re. 1. If the minimum earning of the person in the group lies between 42 and 47 and the number of persons initially in the group was equal to a prime number, with both its digits prime. The number of persons in the group initially was :
(a) 29 (b) 53
(c) 31 (d) none of these

30. There are three categories of jobs A, B and C. The average salary of the students who got the job of A and B categories is 26 lakh per annum. The average salary of the students who got the job of B and C category is 44 lakh per annum and the average salary of those students who got the job of A and C categories is 34 lakh per annum. The most appropriate (or closest) range of average salary of all the three categories (if it is known that each student gets only one category of jobs *i. e.*, A, B and C) :
(a) lies between 30 and 44 (b) lies between 28 and 34
(c) lies between 34 and 43 (d) lies between 29 and 48

31. Out of the five integral numbers C is the average of A and D. B is greater than C and less than D. Also B is the average of A and E. The middle most number in the sequence is :
(a) A (b) B
(c) C (d) D

32. The average age of Donald, his wife and their two children is 23 years. His wife is just 4 year younger than Donald himself and his wife was 24 years old when his daughter was born. He was 32 years old when his son was born. The average age of Donald and his daughter is :
(a) 25 years (b) 22.5 years
(c) 26 years (d) can't be determined

33. There are only five people in the Aman Verma's family. Aman, his wife, a son and two daughters. The younger daughter's age is $\frac{4}{5}$th of the elder daughter's age. The age of eldest daughter is $\frac{3}{8}$ times that of her father Aman and the age of the son is $\frac{1}{5}$th that of his father Aman. 4 years ago the age of her wife was 8 times that of his son and now the sum of the ages of the younger daughter and wife is same as the sum of the ages of Aman and his son. The average age of the family is :
(a) 22.22 years (b) 25.4 years
(c) 21.2 years (d) none of these

34. The average weight of a political party is decreased by 1, when some new politicians joined the party, whose strength is $\frac{1}{4}$th of the existing (or old) politicians and the total weight of the new politicians is 209 kgs. What is the new average weight of all the politicians if it is known that the average weight of the politicians always must be greater than 50 kgs and number of politicians always must be greater than 100 :
(a) 15 kgs (b) 16 kgs
(c) 18 kgs (d) 19 kgs

35. Ravi went to Kanpur from Lucknow by his four wheeler. During the journey he had to use the spare wheel (*i.e.* stepney). Thus he finished his 160 km journey. The average distance covered by the wheels of his car is :
(a) 40 km (b) 120 km
(c) 128 km (d) 48 km

36. There are 6 consecutive odd numbers in increasing order. The difference between the average of the squares of the first 4 numbers and the last four numbers is 64. If the sum of the squares of the first and the last element (*i. e.*, odd numbers) is 178, then the average of all the six numbers is :
(a) 7 (b) 8
(c) 9 (d) 10

37. The average age of board of directors of a company, having 10 directors was 48 years. Coincindentally when a director aged 53 resigned from the board of directors, another director died on the same day. So a new director joined the board of directors aged 34. Next year in the same month the average age of all the 9 directors was found to be 46 years. The age of the late (*i. e.*, dead) director at the time of his death was :
(a) 56 years (b) 53 years
(c) 57 years (d) 61 years

38. In an office the average weight of 24 employees is 60 kg. If n employees were included whose average weight was 54 kg, then the total number of employees in the office, (given that the new average weight of all the $(24 + n)$ employees is a whole number) :
(a) 36 (b) 30
(c) 34 (d) 25

39. The average age of 100 nurses in a nursing home in 1982 was 50 years. In 1984, 20 nurses retired from their job, whose average age was 60 years. After a huge gap in 1987, 40 nurses were employed whose average age was 38 years. The average age of all the nurses in 1990 was :
(a) 53 years (b) 51 years
(c) 48.5 years (d) data insufficent

✍ ***Directions for question numbers 40, 41, 42, 43 and 44:*** *Bhartiya Idol is a talent search programme launched and run by TV Tarana. In this programme each participating candidate has to appear for the audition in such a manner that a candidate will play the Antakchhary with its aspiring opponent until he fails, otherwise he can continue and become winner. If a candidate fails on his part, he will have to leave*

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the contest and in place of it another candidate will start off with the same existing opponent. The number of points a candidate scores is equal to the number of times he responds correctly to his opponents. The candidate who finishes the game (or who wins over the last opponent) will be declared as a winner, even if he/she has scored less points and started as the last participant. The points scored individually by all the 10 candidates are as shown below. Besides if a person loses the contest only if he/she responds incorrectly.

	Participant	Points
1.	Rajesh	8
2.	Radhe	10
3.	Harish	7
4.	Prajakta	12
5.	Aditya	25
6.	Rahul	24
7.	Shekhar	18
8.	Amit Tandon	23
9.	Amit Sana	29
10.	Abhijeet	30

40. The average score of all the 10 participants is :
(a) 21.5 (b) 18.6
(c) 18.3 (d) none of these

41. A candidate can face maximum '*n*' opponents. The value of '*n*' is :
(a) 3 (b) 4
(c) 1 (d) can't be determined

42. The average score of five participants who started earlier :
(a) 23 (b) 12
(c) 19.5 (d) cant't be determined

43. If Abhijeet has not started off earlier and he is the winner then the minimum number of participants who has lost the game before he has started off :
(a) 4 (b) 6
(c) 5 (d) none of these

44. For the above question the maximum possible average of the opponents of Abhijeet could be :
(a) 13.5 (b) 9.25
(c) 13.0 (d) 17.5

45. In an NGO, the daily average wages of 20 illiterate employees is decreased from Rs. 25 to Rs. 10, thus the average salary of all the literate (educated) and illiterate employees is decreased by Rs. 10 per day. The number of educated employees working in the NGO is :
(a) 15 (b) 20
(c) 10 (d) data insufficient

46. Mr. Tyagi while going from Meerut to Saharanpur covered half the distance by train at the speed of 96 km/hr then he covered half of the rest distance by his scooter at the speed of 60 km/hr and finally he covered the rest distance at the speeed of 40 km/hr by car. The average speed at which Mr Tyagi completed his journey is :
(a) 64 km/hr (b) 56 km/hr
(c) 60 km/hr (d) 36 km/hr

47. There are four types of candidates in our coaching preparing for the CAT. The number of students of Engineering, Science, Commerce and Humanities is 40, 60, 50 and 30 respectively and the respective percentage of students who qualified the CAT is 80%, 75%, 60% and 50% respectively the overall percentage of successful candidates in our institute is :
(a) 67.77% (b) 66.66%
(c) 68.5% (d) none of these

48. Mr. Manmohan calculated the average of 10, 'three digit numbers'. But due to mistake he reversed the digits of a number and thus his average increased by 19.8. The difference between the unit digit and hundreds digit of that number is :
(a) 8 (b) 4
(c) 2 (d) can't be determined

49. Once my peon went to the office of SKYLINE COURIER with 4 different envelopes. The clerk in the office measured the weights in all possible pairs. The weights obtained are 59 gm, 61 gm, 62 gm, 63 gm, 64 gm and 66 gm. The weight of the heaviest envelope is :
(a) 35 gm (b) 36 gm
(c) 34 gm (d) can't be determined

50. The average expenditure of the hotel when there are 10 guests is Rs. 60 per guests and the average expenditure is Rs. 40 when there are 20 guests. If it is known that there are some fixed expenses irrespective of the number of guests then the average expenditure per guest when there are 40 guests in the hotel :
(a) Rs. 30 (b) Rs. 25
(c) Rs. 20 (d) can't be determined

Final Round

✍ ***Directions for question numbers 1 to 15:*** *There are 3 sets of natural numbers 1 to 100. Set A contains all the natural numbers which are prime, upto 100. Set B contains all the non-prime even natural numbers upto 100. Set C contains all the non-prime odd natural numbers upto 100 i.e.,*
$A = \{2, 3, 5, 7, 11, \ldots 89, 97\}$
$B = \{4, 6, 8, 10, 12, \ldots 98, 100\}$
$C = \{1, 9, 15, 21, 25, 27, 33 \ldots 95, 99\}$

1. The average of all the elements of *A*, *B* and *C* is :
(a) 49.50 (b) 50.50
(c) 55 (d) none of these

2. The average of all the elements of *B* is :
(a) 52 (b) 48
(c) 49 (d) none of these

3. If the average (factually correct) of the set *A* is 42.46, then the average of the Set *C* is :
(a) 52 (b) 49.87
(c) 55.46 (d) can't be determined

4. The average of the elements of the Set *A* and *C* combined is :
(a) 49.0588 (b) 49.0372
(c) 50 (d) none of these

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5. If an element less than 50 belongs to Set A is transferred to Set B, then the average of Set B :
(a) increases (b) decreases
(c) remains constant (d) can't say

6. If any two elements, greater than 50, belong to Set A are transferred to Set C, then the average of Set C :
(a) remains constant (b) decreases
(c) increases (d) can't say

7. Any 10 elements of the Set A are transferred to the Set B, then the average of Set B :
(a) increases (b) decreases
(c) remains constant (d) can't say

8. If a least and a greatest element of Set C are transferred from Set C to Set A then the average of Set A :
(a) increases (b) decreases
(c) remains constant (d) can't be determined

9. If a smallest and a greatest element of the Set B is transferred to Set A, then the averages of *A*, *B*, *C* respectively :
(a) decreases, decreases, increases
(b) decreases, constant and increased
(c) increases, constant, constant
(d) can't be determined

10. If an element 2 is also included in the Set B, then the average of B :
(a) decreases by 3 (b) increases by 2
(c) decreases by 1 (d) can't be determined

11. The average of all the perfect squares of the Set C is :
(a) 35 (b) 33
(c) 30.5 (d) none of these

Directions for question number 12, 13, 14: 5 elements below 25 from the Set A are transferred to Set B, and 10 elements lying between 25 and 50 from the Set B are transferred to Set C and 15 elements above 50 from the Set C are transferred to Set A.

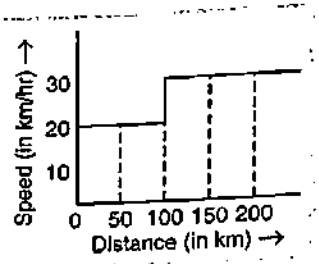
12. The overall average of all the elements of *A*, *B* and *C* is :
(a) 39.8 (b) 50.50
(c) 71.2 (d) can't be determined

13. The maximum increase in the average of set :
(a) A (b) B
(c) C (d) can't say

14. When we intended to minimise the loss in average of Set B then the new average of Set B is :
(a) 54.4772 (b) 45.74
(c) 51.84 (d) can't be determined

15. If 10-10 elements are transferred from Set A to Set B, then Set B to Set C and then from Set C to Set A but the received elements can not be transferred to the next set. e.g. the elements obtained from Set A can not be transferred to C through Set B. The average of which Set is maximum :
(a) A (b) B
(c) C (d) can't say

Directions for question numbers 16 and 17: In the following graph the relation between speed and distance is given :



16. The average speed for the first 200 kms. (in km/hr)
(a) 24 (b) 25
(c) 26 (d) 40

17. The average speed for the first 150 km :
(a) 15 (b) 18
(c) 25 (d) none of these

18. Abhay working in Tele Bharti as a salesperson. His monthly salary is just Rs. 200. But he gets the bonus as per the given rule. If he sells the simcards of Rs. x then his bonus will be Rs. $\left[\left(\frac{x}{100}\right)^2 + 10\right]$. In the first quarter of the year his average sale was Rs. 3000 per month. In the next 5 months his average sale was Rs. 5000 per month. In the last four months his average sale was Rs. 8000 per month. What is the average earning per month for the whole year?
(a) Rs. 3350 (b) Rs. 3610
(c) Rs. 3560 (d) none of these

19. The price of Shirts at Sahara Ganj is defined as Rs. $(100 + 10x^2)$, where x is the number of shirts. Mallika purchased 5 shirts from the above shop. The average price of a shirt is :
(a) Rs. 70 (b) Rs. 50
(c) Rs. 75 (d) none of these

20. In the above question if the sister of Mallika purchased k shirts but the average price was same as that of Mallika. The value of k could be :
(a) 6 (b) 4
(c) 2 (d) none of these

21. There are 10 compartments in passenger train which carries on an average 20 passengers per compartment. If atleast 12 passengers were sitting in each compartment and no any compartment has equal number of passengers then maximum how many passengers can be accomodated in any compartment :
(a) 64 (b) 45
(c) 56 (d) none of these

22. There are twice the number of two wheelers as there are three wheelers and the number of 4 wheelers are equal to the number of two wheelers. The average number of wheel per vehicle is :
(a) 3 (b) 4
(c) 5 (d) none of these

23. Sone lal has 'n' magical eggs whose average weight is 'k' gm. Each of the 'n' eggs produces 'n' eggs next day such that the average weight of 'n' eggs produced is same as that of the parental (previous generation) egg for each 'n' groups individually i.e., each egg produces 'n' eggs in its next generation and the average weight of all the 'n' eggs of next generation is same as the weight of the mother egg. This process is continued without any change in pattern. What is the total weight of all the eggs of rth generation, where the initial number of eggs with Sone Lal are considered as the eggs of first generation :
(a) rnk (b) $n^r k$
(c) nk^r (d) $n^{r+1}k$

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Directions for question numbers 24 to 27 : *A CAT training institute was established on January 1, 2004 with 3, 4, 5 and 6 faculty members in the Logical Reasoning (LR), Data Interpretation (DI), English Language and Quantitative Analysis (QA) areas respectively, to start with. No faculty member retired or joined the institute in the first three months, of the year 2004. In the next four years, the institute recruited one faculty member in each of the four areas. All these new faculty members, who joined the institute subsequently over the years were 25 years old at the time of their joining the institute. All of them joined the institute on April 1. During these four years, one of the faculty members retired at the age of 60. The following diagram gives the area-wise average age (in terms of number of completed years) of faculty members as on April 1 of 2004, 2005, 2006 and 2007.*

Faculty	2004	2005	2006	2007
LR	49.33	44	45	46
DI	50.5	51.5	52.5	47.8
English	50.2	49	45	46
Quants	45	43	44	45

24. In which year did the new faculty member join as the faculty of English?
(a) 2004 (b) 2005
(c) 2006 (d) 2007

25. What was the age of the new faculty member, who joined the faculty of QA, as on April 1, 2007?
(a) 25 (b) 26
(c) 27 (d) 28

26. From which area did the faculty member retire?
(a) English (b) LR
(c) DI (d) Quants

27. Professors Sarvesh and Manish, two faculty members in the LR area, who have been with the Institute since its inception, share a birthday, which falls on 30th November. One was born in 1951 and the other one in 1954. On April 1, 2009, what was the age of the third faculty member, who has been in the same area since inception?
(a) 47 (b) 50
(c) 51 (d) 52

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Answers

LEVEL-1

1 (c)	2. (b)	3. (a)	4. (b)	5. (b)	6. (a)	7. (a)	8. (c)	9. (c)	10. (b)
11. (a)	12. (a)	13. (b)	14. (a)	15. (c)	16. (c)	17. (c)	18. (a)	19. (b)	20. (b)
21. (d)	22. (c)	23. (d)	24. (b)	25. (c)	26. (a)	27. (b)	28. (b)	29. (a)	30. (b)
31. (c)	32. (a)	33. (b)	34. (b)	35. (b)	36. (a)	37. (a)	38. (b)	39. (d)	40. (a)
41. (a)	42. (c)	43. (b)	44. (a)	45. (d)	46. (a)	47. (a)	48. (a)	49. (b)	50. (c)
51. (b)	52. (b)	53. (d)	54. (d)	55. (d)	56. (a)	57. (c)	58. (a)	59. (b)	60. (a)

LEVEL-2

1 (a)	2. (c)	3. (b)	4. (b)	5. (b)	6. (c)	7. (d)	8. (b)	9. (d)	10. (a)
11. (c)	12. (d)	13. (d)	14. (a)	15. (c)	16. (b)	17. (a)	18. (d)	19. (a)	20. (b)
21. (d)	22. (a)	23. (b)	24. (d)	25. (a)	26. (c)	27. (d)	28. (d)	29. (d)	30. (a)
31. (b)	32. (a)	33. (a)	34. (a)	35. (c)	36. (b)	37. (a)	38. (a)	39. (b)	40. (b)
41. (b)	42. (d)	43. (c)	44. (a)	45. (c)	46. (a)	47. (a)	48. (c)	49. (c)	50. (a)

FINAL ROUND

1 (b)	2. (a)	3. (c)	4. (a)	5. (b)	6. (c)	7. (d)	8. (a)	9. (c)	10. (c)
11. (b)	12. (b)	13. (a)	14. (c)	15. (d)	16. (a)	17. (d)	18. (b)	19. (a)	20. (c)
21. (c)	22. (a)	23. (b)	24. (c)	25. (c)	26. (a)	27. (d)			

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Hints & Solutions

LEVEL 1

1. Initially the total weight = $20 \times 45 = 900$
Now, the total weight when a student has been replaced $= 20 \times 44 = 880$
It means the weight of the new student is 20 kgs less than the replaced student. Hence the weight of the replaced student $= 40 + 20 = 60$ kgs
Alternatively: Since you know that there are total 20 students and when their average weight decreases by 1, it means on an average 1 kg weight is reduced from each of the students. Thus there is 20 kg weight loss, in total. Again this happens due to the student whose weight is 40 kg replaces an old student. Thus we can say that the weight of the old (or existing) student of the class was 60 kg, which is reduced by a 40 kg student.

2. The only useful data is that the average length of the thumb and index finger is 2.8 and the length of index finger is 3 inches.
Now the total length of index finger and thumb
$$= 2.8 \times 2 = 5.6 \text{ inches}$$
Thus the length of thumb is $(5.6 - 3) = 2.6$ inches.

3. The average price of the new mixture
$$= \frac{(38 \times 8) + (43 \times 4) + (49 \times 3)}{15} = 41.53$$

4. Average speed when Pankaj was returning
$$= \frac{2 \times 10 \times 30}{40} = 15 \text{ km/hr}$$
Now the average speed of the whole journey
$$= \frac{2 \times 15 \times 60}{75} = 24 \text{ km/hr}$$

5. Use unitary method :
In $\frac{1}{15}$ hours, 123 typists can type 984 papers
In 1 minute, 123 typists can type $\frac{984}{4} = 246$ papers
In 1 minute, 1 typist can type $\frac{246}{123} = 2$ papers
(**Hint :** $\frac{1}{15}$ hours = 4 minutes)

6. Average cost $= \frac{20 \times 3x + 15 \times 2x + 18 \times 4x}{9x} = 18$

7. Earning in the 8 months $= 600 \times 8 = 4800$
Earning in the 4 months $= \left(600 \times \frac{3}{2}\right) \times 4 = 3600$
Total earning = Rs. 8400
Saving in 8 months $= 400 \times 8 = 3200$
Saving in 4 months $= \left(400 \times \frac{5}{4}\right) \times 4 = 2000$
Total savings = 5200
Total expenditure for 12 months
$$= 8400 - 5200 = 3200$$
Therefore average saving per month $= \frac{3200}{12} = 266.66$

8. There are 6 prime numbers between 47 and 74 as given below:
53, 59, 61, 67, 71, 73
There are only two combinations whose average is $\frac{191}{3}$ or whose total sum is 191.
These are, {53, 67, 71} and {59, 61, 71}
We observe that in the first set the difference between greatest and smallest prime number is 18.
Hence choice (c) is correct.

9. Go back to the fundamentals and see the property number 4 and 2 of averages.
Hence required average $= 11 \times 5 + 5 = 60$

10. The average score after 48th innings = 48
The average score after 49th innings = 49
Since $\frac{(48 \times 48 + 97)}{49} = 49$
Now the requirement of runs $= 49 + (50 \times 2) = 149$
or $(50 \times 51) - (49 \times 49) = 149$

11.

		Average	Total
$S + G$	→	35	70
$K + G$	→	32	64
$S + K$	→	38	76
$S + K + G$	→	35	105
$D + I$	→	$\frac{35}{2}$	35

$$\therefore \quad \frac{S + K + G + D + I}{5} = \frac{105 + 35}{5} = 28$$

12. Kaif → 35
Sachin → 41
Ganguli → 29
Dhoni < 35 and Irfan < 35
Hence Sachin is the senior person

13. It is the same as a person with 20 years more age replaces an existing person of the group (or village)
Since the total age of the village having n persons, is being increased by 20 years and the average age of village is being increased by 1 year, hence there are total 20 people in the village.
Alternatively : $(n \times 42) + 20 = (n \times 43)$
$\Rightarrow \quad n = 20$

14. Average rainfall $= \frac{2 \times 6 + 4 \times 5 + 4 \times 10 + 2 \times 6}{12} = 7$

15. $400 \times 7 = (300 \times 3) + (250 \times 2) + (n \times 2) \Rightarrow 700$

16. The new player must be 11 kg heavier than the replaced one.
Hence the weight of the new person = 55 + 11 = 66 kg.

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17.

	No. of family members	Average age	Total age
4 years ago	6	25	150
Presently	6	29	174

But the no. of family members (presently) = 7
and average age (presently) = 25
Therefore the total age = $25 \times 7 = 175$
Hence, the age of child = $175 - 174 = 1$ year

18. Amitabh's total expenditure for Jan-June = $4200 \times 6 = 25200$
Expenditure for Febuary-June = $25200 - 1200 = 24000$
Expenditure for the months of Febuary-July
$= 24,000 + 1500 = 25,500$
The average expenditure = $\frac{25,500}{6} = 4250$

19. Total value of $a + 11 + 23 + 17 = 60 \Rightarrow a = 9$
again $a + b + 12 + 25 = 64 \Rightarrow b = 18$
therefore $\frac{a}{b} = \frac{1}{2}$

20. By the method of Alligation

12,000 — 18,000 — 15,500 — 2,500 — 3,500

Therefore the ratio of number of Jun. level employee to the senior level employee = 5 : 7
Hence the required fraction = $\frac{5}{(5+7)} = \frac{5}{12}$

Alternatively: Go through options

$$\frac{J}{J+S} = \frac{5x}{12x}$$

$$\Rightarrow \quad \frac{J}{S} = \frac{5x}{7x}$$

Therefore $15,500 \times 12x = 12,000 \times 5x + 18,000 \times 7x$
L.H.S. = R.H.S., Hence correct

21. None of a, b, c is correct.
The correct answer can be $\frac{a+b+c+d+e}{5}$
or $\frac{a+e}{2} = \frac{b+d}{2} = c$
where a, b, c, d, e are consecutive odd numbers.

22. $31 \times 31 - 30 \times 30 = 61$ years

23. $a + b + c + d + e = 5 \times 92 = 460$
$a + b + c = 3 \times 83 = 249$
$c + d + e = 3 \times 97 = 291$
$\therefore \quad c = (a + b + c) + (c + d + e) - (a + b + c + d + e)$
or $\quad c = 540 - 460 \quad$ or $\quad c = 80$

24. $19 \times 26 - 18 \times 25 = 44$ years

25. $\frac{k + k + 2 + k + 4 + k + 6 + k + 8 + k + 10 + k + 12}{7} = k + 6$

26. $\frac{\frac{x}{4} \times \frac{85}{100} + \frac{x}{3} \times \frac{70}{100} + \frac{5x}{12} \times \frac{56}{100}}{x} \times 100 = 67.916$

Here, x is the total number of students in the whole class.

27. Since there are five possibilities or combinations (as ${}^5C_4 = 5$)
therefore average = $\frac{5 \times 600}{4} = 750$ mm

28. $\frac{\frac{23}{5} + \frac{8}{3} + \frac{62}{9} + \frac{112}{15} + \frac{32}{9}}{5} = 5\frac{8}{225}$

29. $\frac{1000.0001 + 100.001 + 10.01 + 1.1}{4} = \frac{1111.1111}{4} = 277.777$

30. $\frac{7 + 8 + 9 + 10 + 11 + 12 + 13}{7} = 10$

$\Rightarrow \quad \frac{7 + 13}{2} = 10$

31. $\left(\frac{1 + 2 + 3 + \ldots 100}{100}\right) = \frac{\left(\frac{100 \times 101}{2}\right)}{100} = 50.50$

32. $\left(\frac{1 + 3 + 5 + 7 + \ldots 99}{50}\right) = \left(\frac{50 \times 50}{50}\right) = 50$

33. $\frac{2 + 4 + 6 + \ldots 198}{99} = \frac{99 \times 100}{99} = 100$

34. $a + b + c = 237$
$a + c = 158$
$\Rightarrow \quad b = 79$

35. $\frac{M + A + S}{3} = 11111$
also $\quad \frac{S + M}{2} = 11111 \quad (\because M - 11111 = 11111 - S)$
$\Rightarrow \quad A = 11111$

36. $\frac{2 \times 200 \times 600}{800} = 300$ km/hr

37. $\frac{A + B + C + D}{4} = \frac{400 \times 2 + 600 \times 2}{4} = 500$

38. Average weight = $\frac{20 + 50}{100} = 0.7$ kg ($\because$ 1 kg = 1000 gm)

39. Average = $\frac{770}{11} = 70$

40. $\frac{3x + 4x + 5x}{3} = 12 \Rightarrow x = 3$
$\therefore$ The ages are 9, 12 and 15 years
and the required average = $\frac{9 + 15}{2} = 12$

41. Since the salary of each employee is being increased therefore the average salary will also increased by Rs. 2000. Thus the required average = 22,000.

42. $11 \times 21 - 10 \times 20 = 31$

43. $20 \times 180 - 19 \times 178 = 218$

44. Let the price of $A > B > C$
then $\frac{A + B}{2} = 2C$
$A + B = 4C$
Now, $A + B + C = 5 \times 3 = 15$ crore
$\Rightarrow \quad 5C = 15$ crore
$\Rightarrow \quad C = 3$ crore

45. Since we don't know about the price of B.

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46. $\left(\frac{40+70+142}{50+100+150}\right)\times 100 = 84\%$

47. Since 1 is neither prime nor composite number. Thus there are only 99 number viz. 2, 3, 4, 5, 6, ... 99, 100.

Hence $\left(\frac{2+3+4+5+6+\ldots 100}{99}\right)$

$$=\left(\frac{(1+2+3+4+\ldots 100)-1}{99}\right)$$

$$=\frac{5050-1}{99}=\frac{5049}{99}=51$$

48. $\frac{1+4+9+16+\ldots 100}{10}$

$$=\left(\frac{10\times 11\times 21}{6\times 10}\right)$$

$$\left[\because 1^2+2^2+\ldots n^2=\frac{n(n+1)(2n+1)}{6}\right]$$

$= 38.5$

49. The non-negative integers upto 99 are 0, 1, 2, 3, ... 99.

Therefore average $=\left(\frac{0+1+2+3+\ldots 99}{100}\right)$

$$=\frac{\left(99\times\frac{100}{2}\right)}{100} \qquad \left[\because 1+2+3+\ldots n=\frac{n(n+1)}{2}\right]$$

$= 49.5$

50. $\frac{7+14+21+\ldots 77}{11}=\frac{7(1+2+\ldots 11)}{11}$

$$=\frac{7\times 11\times 12}{11\times 2}=42 \qquad \left[\because 1+2+3\ldots n=\frac{n(n+1)}{2}\right]$$

Alternatively: Since all the numbers are in A.P. Further there are odd number of numbers (*i. e.*, 11) in the sequence. Thus the middle most term is the average of the sequence, which is 42.

51.

	Average	Total
A, B, C, D	40	160
A, B, C, D, E	41	205
F, B, C, D, E	42	210

Thus $E = 205 - 160 = 45$

Hence $F, B, C, D = 210 - 45 = 165$

Therefore average of $F, B, C, D = 41.25$

52. Three consecuntive natural numbers whose average is k are $(k-1), (k), (k+1)$

The next two numbers will be $(k+2)$ and $(k+3)$

Therefore average of $(k-1), (k), (k+1)(k+2)$ and $(k+3)$ is $(k+1)$

Alternatively: Consider any 3 conseculitive natural number *eg.* 4, 5, 6 the average is 5

Again 4, 5, 6, 7, 8 the average is 6.

To verify it consider some different numbers *eg.* 10, 11, 12 average = 11 and 10, 11, 12, 13, 14 average = 12

Hence, proved that the average is increased by 1.

53. The 5 consevtive odd numbers whose average is k are $(k-4)(k-2), k, (k+2)(k+4)$

Again the average of $(k-4), (k-2), (k), (k+2), (k+4), (k+6), (k+8)$ is $(k+2)$

Alternatively: Consider some appropriate numbers.

54. The increase in weight $= (5\times 42.5) - (5\times 42) = 2.5$ kg

But we don't know the weight of the replaced officer. So we can't determine.

55. The decrease in age $= 6\times(28-27) = 6$ years

But we don't know the age of the old servant which is being replaced. So we cannot determine the average age of new servant.

56. The age of new servant $= 31 - 6 = 25$ years

57.
$$A+B+C = 12{,}000\times 3$$
$$B+C+D = 15{,}000\times 3$$
$$\Rightarrow \quad D-A = 3000\times 3$$
$$D-A = 9000$$
also $$D = 2A$$
$$\Rightarrow \quad D = 18{,}000 \quad \text{and} \quad A = 9000$$

Therefore average salary of B and $C = \frac{(45{,}000-18{,}000)}{2}$

$= 13{,}500$

58. The price of the costliest and cheapest computer

$= (80\times 30{,}000) - (78\times 29{,}500) = 99{,}000$

Therefore the price of the cheapest computer

$= 99{,}000 - 80{,}000 = 19{,}000$

59. $\frac{10\times 50+50\times 10+20\times 25+25\times 20}{4} = 500$

60.

	Auto Rickshaw	Maxi. cab	Mini. Bus
No. of seats	4	10	20
No. of seats occupied	4	8	15
Rate per seat	12	15	8
Total amount (in Rs.)	48	120	120

Therefore average earning $= \frac{48+120+120}{3} = 96$

LEVEL 2

1. For this type of questions take the LCM of speeds and assume the LCM as the distance

then the time taken @ speed of 60 km/hr $= \frac{300}{60} = 5$ hrs

again the time taken @ speed of 50 km/hr $= \frac{300}{50} = 6$ hrs

Thus we see that in place of 5 hrs train takes 6 hours. It means the train takes 1 hour extra and this one hour is stopping period in the total time of 6 hours. Thus in 6 hour train halts for 1 hour. So in 1 hour train will stop for $\frac{1}{6}$ hours or 10 minutes.

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Alternatively : (short cut)

Halting (or stopping) time $= \left(1 - \frac{\text{slower speed}}{\text{faster speed}}\right)$ hours

$= 1 - \frac{50}{60} = \frac{10}{60} = \frac{1}{6}$ hours

$= 10$ minutes

($\because$ 1 hour = 60 minutes)

2. $\frac{[r + (p - q)] + p + q}{3} = 21 + \frac{p + q + r}{3}$

$\Rightarrow \quad \frac{2p + r}{3} - 21 = \frac{p + q + r}{3}$

$\Rightarrow \quad \frac{p - q}{3} = 21$

$\Rightarrow \quad p - q = 63$

3. Let the speed for the first hour be x km/hr

then the speed for the second hour be $\frac{7}{5}x$ km/hr

then the speed for the third hour be $\frac{10}{7} \times \frac{7}{5}x = 2x$ km/hr

then the speed for the fourth hour be $2x \times \frac{7}{5} = \frac{14x}{5}$ km/hr

Therefore total distance in four hours $= x + \frac{7}{5}x + 2x + \frac{14x}{5}$

$= \frac{36x}{5}$ km

$\therefore$ Average speed $= \frac{\text{Total Distance}}{\text{Total Time}} = \frac{\left(\frac{36x}{5}\right)}{4} = \frac{9x}{5}$ km/hr

Again the distance in 4 hours @ speed of x km/hr which is half of the third hour's speed is $4x$ km

Hence $\frac{36x}{5} - 4x = 160$ km

$\Rightarrow \quad x = 50$

Hence, the average speed $= \frac{9 \times 50}{5} = 90$ km/hr

4. The total score of 3 toppers $= 123 \times 120 - 120 \times 118.5 = 540$

The highest possible score of the third highest topper is possible when the score of other two toppers was minimum

So,

1st rankers score = 187 (minimum)
2nd rankers score = 186 (minimum) } 540
3rd rankers score = 167 (maximum)

5. Go through options : $40 \times 35 + 60 \times 25 = 29 \times 100$

Since there are 40 ladies and 60 gents

Alternatively: $40 \times 7x + 60 \times 5x = 29 \times 100$

$\Rightarrow \quad x = 5 \quad \therefore \quad 7x = 35$

6.

	Lok Sabha	Rajya Sabha
No. of MPs	300	200
Ave. Age	40	50

Since when a member of Rajya Sabha joins the Lok Sabha and the average age of both the houses increases, it means the average age of this member must lie between 40 and 50. When the age of this member is greater than 40, then the average age of the Lok Sabha increases.

Again when the age of this member is less than 50, then after leaving it, the average age of the Rajya Sabha increases.

Solutions for question numbers 7, 8 and 9:

	No. of family members	Average	Total
8 years ago →	5	36	180
Presently →	(if) 5	(36 + 8) = 44	220
	7	36	252

7. From the above explanation we have no any clue about his wife's age.

8. Since we know that the difference between the age of any two persons remains always constant, while the ratio of their ages gets changed as the time changes.

So, if the age of his child be x (presently)

Then the age of wife be $x + 26$ (presently)

Thus, the total age $= x + (x + 26) = 32 \quad [\because 252 - 220 = 32]$

$\Rightarrow \quad x = 3$

Therefore the age of her child is 3 years and her self is 29 years. Hence her age at the time of the birth of her child was 26 years.

Alternatively: As we have mentioned above that the age difference remains always constant. Therefore her age at the time of her child's birth was 26 years.

9. Since there is no clue. So we can't determine.

10.

	No. of family members	Average	Total
Eleven years earlier	4	28	112
Presently	if 4	39	156
	6	28	168

Since it is obvious that just after the birth of the youngest member (*i.e.*, child) was 6 family members in the family. Therefore at the time of the birth of the youngest child the elder child's age was 6 years.

Now the sum of their ages $= x + (x + 6) = 12 = (168 - 156)$

$\Rightarrow \quad x = 3 \quad \text{and} \quad (x + 3) = 9$

11. Go through options

$\frac{6}{4} = 1.5$ hour, when he was walking

Therefore time of journey by train $= 3 - 1.5 = 1.5$

Now, the distance travelled by train $= 1.5 \times 60 = 96 - 6 = 90$

Hence correct.

Alternatively: Total distance $= 32 \times 3 = 6 + 60 \times x$

$\Rightarrow \quad x = 1.5$ hours

Thus, the speed of walking $= \frac{6}{1.5} = 4$ km/hr

12. Let the number of total Mps $= n$ and their average age be x then

$$n \times x = \frac{4}{5}n \times \frac{5}{4}x + \frac{n}{5} \times y$$

$\Rightarrow \quad$ either $n = 0$ or $y = 0$

[Since there are only 80% MPs remained in the house which

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is equal to $\frac{4}{5}n$ and the increase in average age = 20% = $\frac{5}{4}x$]

Thus, there cannot be any possible value of n.

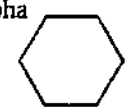
13. $nx = \frac{4}{5}n \times \frac{5}{4}x + \frac{n}{5} \times 30$

$\Rightarrow \quad n = 0$ which is impossible

So, there is no any woman MP in the Lok Sabha

14. Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

$= \frac{6 \times 20}{9} = 13.33 \text{ m/s}$



Hexagonal path has six sides

15. Average of 26, 29, 35 and 43 is 33.25. Also the average of 26, 29, n, 35 and 43 lies between 25 and 35 *i. e.*,

$$25 < \frac{26 + 29 + n + 35 + 43}{5} < 35$$

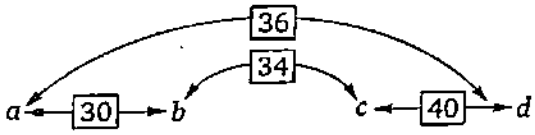
$\Rightarrow \quad 125 < 26 + 29 + n + 35 + 43 < 175$

$\Rightarrow \quad 125 < 133 + n < 175$

$\Rightarrow \quad n < 42$

Since the value of n is an integer and greater than 33.25 then $33 < n < 42$, for every integer n.

16. Given that $a < b < c < d$



The only possible prime number between 30 and 34 is 31

Hence $b = 31$, therefore $a = 29$

Similarly $c = 37$ and $d = 43$

Therefore $d - a = 43 - 29 = 14$

17. $9261000 = 2^3 \times 3^3 \times 5^3 \times 7^3$

$= (2 \times 3 \times 5) \times (2 \times 3 \times 7) \times (2 \times 5 \times 7) \times (3 \times 5 \times 7)$

$= 30 \times 42 \times 70 \times 105$

Therefore the average of 30, 42, 70 and 105 is 61.75.

18. Since the number is as below 270 as its multiple is as above 270. It means these two numbers are equidistant from 270. Hence their average is 270.

Alternatively : Let the number be x then.

$270 - x = 8x - 270$

$\Rightarrow \quad x = 60$ and $8x = 480$

Therefore the average of 60 and 480 is 270.

19. The average of 60 and 270 is 165

20. Number of days in a week = 7

Average number of visitors = 40

Total visitors = 280 (= 7 × 40)

Now, if n be the number of holidays in a week then

$(7 - n) \times 56 = 280 \quad (40 + 16 = 56)$

$\Rightarrow \quad n = 2$

The rest data is redundant or useless, since our problem is solved here without using it.

Alternatively : Go through options

$40 \times 7 = 280 = 56 \times 5$ hence proved.

21. Since we don't know their ages individually so we cannot calculate the average of the class when a student of unknown age readmitted in the class.

22. Let the average of prime numbers be P and average of composite numbers be C.

Again the number of prime numbers be x, then the number of composite numbers be $2x$.

Then $\frac{Px + 2Cx}{3x} = 9 \Rightarrow P + 2C = 27$ ...(1)

and $\frac{2Px + Cx}{3x} = 11$

$\Rightarrow \quad 2P + C = 33$ ...(2)

on adding eq. (1) and (2)

we get $P + C = 20$

and on subtracting eq. (1) from (2)

we get $P - C = 6$

Therefore $P = 13$ and $C = 7$

Thus $\frac{C}{P} = \frac{7}{13}$

23. Solve through option $\frac{540}{27} = 20$

$\frac{540 - (2 \times 20)}{25} = \frac{500}{25} = 20$

Hence option (b) is correct.

24. Number of members in the staff × average age = Total age

$29 \times 24 = 696$

$29 \times 27 = 783$

hence change in total age = 87

	Physician	Surgeon	Nurse
No. of members	$(21 - x)$	8	x
Increase in average age	1	6	3
Increase in total age	$(21 - x)$	48	$3x$

Since $P + S + N = 29$

therefore if there would be x nurses then there must be $(21 - x)$ physicians.

Again total change (or increase) in age

$= (21 - x) + 48 + 3x = 87$

$\Rightarrow \quad x = 9$

25. Total exp. Jan-June = 4200 × 6 = Rs. 25200

Total exp. Feb-June = 25200 − 1200 = Rs. 24000

Total exp. Feb-July = 24000 + 1500 = Rs. 25500

The average expenditure Feb-July = $\frac{25500}{6} = 4250$

26. Let the number of subjects be n and average marks be x, then, total marks = nx

Again $(n + 1)(x - 1) = (nx - 40) + (23 + 25)$

$\Rightarrow \quad x - n = 9$ ...(1)

Further $(n + 2)(x + 1) = (nx - 40) + (23 + 25) + 57$

$\Rightarrow \quad nx + 2x + n + 2 = nx + 65$

$\Rightarrow \quad 2x + n = 63$ ...(2)

Solving equations (1) and (2), we get

$n = 15$ and $x = 24$

27. Initially, the total age of family = $4 \times 42 + 7 \times 20 = 308$

and the average age = $\frac{308}{11} = 28$

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Now, the total age of family = $308 - (2 \times 13) + (11 + 15 + 28)$

$= 308 + 28 = 336$

Now, the new average of the family = $\frac{336}{12} = 28$

Since the average age of the original family and that of new family is same (i.e., 28)

Hence, the average age of the new family is increased by 0 year.

28. Let there be n number and he missed a number k, then the average (which he has calculated) = $\frac{\left(\frac{n(n+1)}{2} - k\right)}{n} = 15$

$\Rightarrow \quad n^2 - 29n = 2k$

$\Rightarrow \quad n(n - 29) = 2k$

Thus at $n = 29$ or $n < 29$, the expression is invalid since the value of k is neither zero nor negative, which is actually a natural number. So for the least possible valued of $n = 30$

$k = 15$

and for $n = 31, k = 31$

Again for $n > 31$, k is beyound the range *i. e.*, greater than n. Since k can not be greater than n. Hence there are only two values of k. So there is no unique value of n.

29. Let there be n people (initially) in the group, then the total earning of the group $= n \times 50$

Again $n \times 50 = (n - 2) \times 49 + (2x + 45)$

$\Rightarrow n = 2x - 53$; where x is the lowest earning of any person.

Now, since $42 < x < 47$ and $n \in$ prime numbers

Then the only possible value of $n = 37$ for $x = 45$

30. Let the number of students who got the jobs of A, B and C categories is a, b and c respectively,

then the total salary = $\frac{26(a + b) + 44(b + c) + 34(c + a)}{2(a + b + c)}$

$= \frac{60a + 70b + 78c}{2(a + b + c)}$

$= \frac{30(a + b + c) + (5b + 9c)}{a + b + c}$

$= 30 +$ some positive value

So the minimum salary must be Rs. 30 lakh and the maximum salary can not exceed 44, which is the highest of the three.

31. $D - C - A$ ...(1)

$D > B > C$ ...(2)

from (1) and (2)

$D > B > C > A$ ...(3)

Again $E - B - A$

but $B > A$, from (3)

So $E > D > B > C > A$

[Since B is the average of E and A so it is equidistant from both E and A.]

32. Let Donald be denoted by H (Husband)
His wife be denoted by W (Wife)
His daughter be denoted by D (Daughter)
His son be denoted by S (Son)

The average age of 4 persons = $\frac{(H + W + D + S)}{4} = 23$

$\Rightarrow \quad H + W + D + S = 92$

Again $H = W + 4$

So

	H		W	D	S
At the time when daughter is born	28	$\xleftarrow{(+4)}$	24	0	×
	↓				
At the time when son is born	32	$\xrightarrow{(-4)}$	28	4	0

So at the time of birth of his son, total age of his family = 64 years $(32 + 28 + 4 + 0 = 64)$

and presently the total age of his family = 92 years

It means total increase in age of the whole family = 28 years

Thus the average increase in age = $\frac{28}{4} = 7$ years

It means the age of Donald = 39 years

and age of his daughter = 11 years

Therefore the average age of Donald and his daughter is 25 years.

33.

Aman	Wife	Son	El. D	Yg. D
$5x$	$\longleftrightarrow$	x	$5z$	$\longleftrightarrow 4z$
$8y$	$\longleftrightarrow$		$3y$	
$\Rightarrow 40K$	(W)	$8K$	$15K$	$12K$

Again since Yg. $D + W = A + S$ $(K = x.y)$

$\Rightarrow \quad 12K + W = 40K + 8K$ $W \rightarrow$ Age of wife

$\Rightarrow \quad W = 36K$

Thus 4 years ago

$(36K - 4) = 8(8K - 4)$

$\Rightarrow \quad 28K = 28 \Rightarrow K = 1$

Therefore, the age of Aman = 40

Wife = 36

Son = 8

Elder daughter = 15

Younger daughter = 12

Hence, the average age of the family = $\frac{111}{5} = 22.22$ years

34. Let there be n politicians (initially) in the party and their average weight be x kg, then

$nx + 209 = \frac{5}{4} n(x - 1)$

$\frac{n}{4}(x - 5) = 209$

$\Rightarrow \quad x = \frac{209 \times 4}{n} + 5$

$x = \frac{4 \times 11 \times 19}{n} + 5$

So the possible value of n is 76 $(= 19 \times 4)$

Thus $x = 16$

Therefore the average weight of all the politicians is 15 kg.

35. Total distance covered by all the wheels = 4×160

Number of wheels used = 5

Therefore average distance covered by each wheel

$= \frac{4 \times 160}{5} = 128$ km

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36. Let the numbers be $(a-5), (a-3), (a-1), (a+1), (a+3), (a+5)$
then their average
$$= \frac{(a-5)+(a-3)+(a-1)(a+1)(a+3)(a+5)}{6} = a$$
Again the value of 'a' can be found by using the last statement
i.e., $(a-5)^2 + (a+5)^2 = 178$
$\Rightarrow$ $a^2 = 64 \Rightarrow a = 8$

37.

	No. of Directors	Average Age	Total Age
Just before death and resignation	10	48	480
Just after death and resignation	9		{480 – (53 + x) + 34}
One year later	9	46	414

So one year later, after the incident
total age = {480 – (53 + x) + 34} + 9 × 1 = 414
$\Rightarrow$ $x = 56$ years
where x is the age of the dead person at the time of his death.

38. Go through options $= \frac{24 \times 60 + 12 \times 54}{(24+12)}$
$= 58$, which is a whole number.

39.

	Year/ Time	No. of Nurses	Average Age	Total Age
	1982	100	50	5000
Just before retirement	1984	100	52	5200
Just after retirement	1984	80	50	(5200 – 20 × 60) = 4000
Just before recruitment	1987	80	53	4240
Just after recruitment	1987	(80 + 40) = 120	48	(4240 + 38 × 40) = 5760
	1990	120	51	6120

40. Average score
$$= \frac{8+10+7+12+25+24+18+23+29+30}{10} = \frac{186}{10} = 18.6$$

41. The candidate who has scored maximum (means continuing for long time and number of responses were maximum) can be such a required person.
Now, consider Abhijeet, whose score was 30.
Again consider Rajesh, Radhe, Harish and one more candidate
since $8 + 10 + 7 < 30 \Rightarrow 25 < 30$
So there are exactly 3 persons which can be full time opponents. Further there are 5 (= 30 – 25) more chances. So these chances can be utilised by any other candidate but he or she must be in the beginning or in the ending of the Abhijeet, since the score of all the candidates is greater than 5. Hence there can be maximum 3 + 1 = 4 opponents of Abhijeet.

42. There is no such an information.

43. The minimum number of participants who have lost the game before Abhijeet started will be possible only when the number of participant with Abhijeet be maximum, which is 4. Hence excluding these 4 players (opponents of Abhijeet) and Abhijeet himself there are 5 people left. So minimum 5 players have lost the game (or contest).

44. The opponents of Abhijeet (as per the requirement) are Rajesh, Radhe, Harish and Amit Sana
So, the average score $= \frac{8+10+7+29}{4} = 13.5$

45. Go through options
or Total employs $= \frac{(25-10)\times 20}{10} = 30$
Hence number of educated employees = 30 – 20 = 10

46. Average speed of the later half journey
$$= \frac{2 \times 40 \times 60}{100} = 48 \text{ km/hr}$$
Now the average speed of the whole journey
$$= \frac{2 \times 48 \times 96}{144} = 64 \text{ km/hr}$$

47. $\frac{40 \times 0.8 + 60 \times 0.75 + 50 \times 0.6 + 30 \times 0.5}{180} \times 100$
$= \frac{122}{180} \times 100 = 67.777 \ldots \%$

48. Remember $\frac{abc - cba}{99(a-c)}$
where abc and cba are the three digit numbers and $(a, c) \neq 0$
Again since the difference in average = 19.8
Therefore the difference in total = 19.8 × 10 = 198
Thus, $99 \times (a-c) = 198$
$\Rightarrow$ $(a-c) = 2$

49. If the highest weight be 35 gm, then the second highest weight will be 31 gm. Again if the second highest will be 31, then the third highest will be 33 which is inadmissible, since then 35 + 33 = 68 which is not the greatest possible combination. Hence wrong.
Similarly 36 (*i.e.*, option *b*) is also invalid

Highest	Sec. Highest	Third Highest	
36	30	34	

Thus 36 + 34 = 70 > 66, hence wrong.
The greatest possible combination can not be greater than 66.
Now, consider option (c)

Highest	Sec. Highest	Third Highest	
34	32	32 ✗	(since weights are different)
	32	31 ✗	(since 65 is not a combination)
	32	30 ✓	

So, the highest weight 34
Sec. highest weight 32
Third highest weight 30
Lowest weight 29
Since, all the weights obtained give all the 6 different combinations, hence 34 is the highest possible weight of an envelope.

50. Let the fixed expenditure of the hotel be Rs. x and the variable expenditure (which is dependent on the guest) is Rs. y, then
$$x + 10y = 600 \quad \ldots(1)$$
$$x + 20y = 800 \quad \ldots(2)$$
$\Rightarrow$ $10y = 200$
y = Rs. 20 and $x = 400$
hence the total expenditure when there are 40 guests
$= 400 + 40 \times 20 = 1200$
Therefore average expenditure $= \frac{1200}{40}$ = Rs. 30

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1. Since all the total 100 elements of Sets A, B, C are the natural numbers upto. Thus the average of these first 100 natural numbers is the required average.

$$\therefore \text{ Average} = \frac{1+2+3+4\ldots 100}{100} = \frac{100\times 101}{2\times 100} = 50.50$$

2. Except to 2 there are all the even numbers upto 100

So, the required average $= \frac{(2+4+6+\ldots 100)-2}{49}$

$$= \frac{50\times 51-2}{49} = \frac{2548}{49} = 52$$

NOTE There are only 49 elements in the Set B. Apply the formula of sum of first even numbers. Also use the property of AP.

3. The total value of all the 25 elements of the Set A

$$= 25\times 42.4 = 1060$$

Since, there are 25 prime numbers upto 100 in the Set A Again in the Set A and C there are 50 odd numbers and one even number. So the sum of all the elements of A and C

$$= (1+3+5+7+\ldots 99)+2$$
$$= (50)^2+2 = 2502$$

Therefore the sum of all the elements of Set C

$$= 2502-1060 = 1442$$

Hence, the average of the Set C $= \frac{1442}{26} = 55.4615$

4. The average of all the elements of the Set A and C

$$= \frac{2502}{51} = 49.0588$$

Solutions for question number 5 to 15 :

Set	No. of elements	Average	Least element	Greatest element
A	25	42.4	2	97
B	49	52	4	100
C	26	55.46	1	99

5. Since the value of element which is transferred to Set B is less than 50, which in turn less than the average of Set B, hence the average of Set B decreases.

NOTE If a quantity which is less than the average of the group introduces from outside then the new average of the group decreases.

6. The least possible numbers of Set A which are greater than 50 are 53 and 59 whose average is always greater than the average of C. Hence the average of C will necessarily increases.

7. Can't say, since we don't know which 10 numbers are being transferred. Whether their average is greater, less or equal to the average of B.

8. Definitily increases, since the average of those numbers (*viz.* 1 and 99) is 50 which is greater than the average of Set A.

9. The average of those numbers (viz. 4 and 100) is 52. Hence average of A will increase and average of B will remain constant and the average of C remains unaffected because Set C is not involved.

NOTE If an element or average of some elements is equal to the average of the group then this element (or subset of elements) does not change the average of the group, when it joins the group or leaves the group.

10. After the insertion of new element *viz.* 2 in the Set B the new average $= \frac{2+4+6+\ldots 100}{50} = 51$

Hence, the new average of Set B decreases by 1.

11. The perfect square number of the Set C are 1, 9, 25, 49, 81

Hence, the average of these number $= \frac{165}{5} = 33$

12. Since there is no net change (*i. e.*, all the elements even after being transferred are same). Hence their average is also same as in question no. 1.

13. Obviously A. Since the average of all those 15 elements which are joining the Set A is greater than the average of all those 5 elements which are leaving the Set A and this difference in average is largest in companision to Set B or Set C. Even in Set C there is decrease in average.

14. To minimize the loss in average of Set B, we have to transfer the least possible values of the given range and have to bring the highest possible values from the Set A to the Set B.

Thus the absolute decrease in Set B

$$= (26+28+30+32\ldots 44)-(23+19+17+13+11)$$
$$= 350-83 = 267$$

Hence, the decrease in total value of Set B

$$= 2548-267 = 2281$$

Therefore, new average $= \frac{2281}{44} = 51.84$

NOTE Now there are only 44 elements in Set B.

15. There is no relevent information regarding the numbers which are being transferred from one set to another set.

16. Average speed $= \frac{\text{Total distance}}{\text{Total time}}$

$$= \frac{200}{\left(5+\frac{10}{3}\right)} = \frac{200\times 3}{25} = 24 \text{ km/hr}$$

Since for the first 100 km time required is $\frac{100}{20} = 5$ hrs and for the last 100 km time required $= \frac{100}{30} = \frac{10}{3}$ hours

17. The average speed $= \frac{150}{5+\frac{5}{3}} = \frac{150\times 3}{20} = 22.5$ km/hr

18. Average bonus for the first 3 months $= \left(\frac{3000}{100}\right)^2 + 10 = 910$

Average bonus for the next 5 months $= \left(\frac{5000}{100}\right)^2 + 10 = 2510$

Average bonus for the last 4 months $= \left(\frac{8000}{100}\right)^2 + 10 = 6410$

His average bonus for the whole year

$$= \frac{910\times 3+2510\times 5+6410\times 4}{12} = \text{Rs. } 3410$$

Hence his average earning per month

$$= 3410+200 = \text{Rs. } 3610$$

19. Total price of 5 shirts = Rs $[100+10\times(5)^2]$ = Rs. 350

Hence, the average price $= \frac{350}{5} = \text{Rs. } 70$

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20. Check the option (c).

$$\text{Total price} = 100 + 10 \times (2)^2 = \text{Rs. } 140$$

$$\text{Average price} = \frac{140}{2} = \text{Rs. } 70$$

Hence, the average price is same as that of Mallika.

21. Total number of passengers = $10 \times 20 = 200$

In the 9 compartments the total number of passangers

$= 144 (= 12 + 13 + 14 + 15 + 16 + 17 + 18 + 19 + 20)$

So the no. of passengers in the 10th coach = $200 - 144 = 56$

22.

	No. of 2 wheelers	No. of 3 wheelers	No. of 4 wheelers
No. of wheels	$2x$ $2 \times 2x = 4x$	x $3 \times x = 3x$	$2x$ $2x \times 4 = 8x$

Therefore average number of wheels = $\frac{4x + 3x + 8x}{5x} = 3$

23. The average weight of eggs of first generation is k gm and the no. of eggs is 'n'.

Let $a_1, a_2, a_3, \ldots a_n$ be the weights of n eggs of the first generation

$$\therefore \quad k = \frac{a_1 + a_2 + a_3 + \ldots a_n}{n}$$

$$\therefore \quad nk = a_1 + a_2 + a_3 + \ldots a_n \qquad \ldots(1)$$

where a_1 is the average weight of its 'n' child eggs, a_2 is the average weight of its own 'n' child eggs and so on. child egg is referred to the egg of next generation produced by its mother egg.

$$\therefore \quad a_1 = \frac{a_1 + b_1 + c_1 + \ldots n_1}{n}$$

$$a_2 = \frac{a_2 + b_2 + c_2 + \ldots n_2}{n}$$

$$a_3 = \frac{a_3 + b_3 + c_3 + \ldots n_3}{n} \text{ etc.}$$

Substituting the values of $a_1, a_2, a_3 \ldots$ in equation ...(1)

$$nk = \left(\frac{a_1 + b_1 + c_1 + \ldots}{n}\right) + \left(\frac{a_2 + b_2 + c_2 + \ldots}{n}\right) + \left(\frac{a_3 + b_3 + c_3 + \ldots}{n}\right) + \ldots \left(\frac{a_n + b_n + c_n + \ldots}{n}\right)$$

$$\therefore nk = \frac{(a_1 + b_1 + c_1) + (a_2 + b_2 + c_2 + \ldots) + (a_3 + b_3 + c_3 + \ldots)}{n}$$

$\Rightarrow$ Therefore n^2k is the total weight of all the eggs of second generation.

Similarly, each of $a_1, b_1, c_1, a_2, b_2, c_2 \ldots$ are the average weights of n eggs of their child eggs. Hence in the third generation total weight will be n^3k. Thus the total weight of all the eggs of rth generation is n^rk.

✍ ***Solutions for question number 24 to 27:*** *Before going for the final solution we need to look for the fundamental concept of averages i.e., if a person of higher age than the average age of the group leaves the group, then the average age of the group decreases. Also if a person of less age than the average age of the existing group joins the group, then the average age of the group decreases.*

Besides it we also know that the average age of the same group after k years increases by K years.

Faculty of LR :

Year	No. of faculty	Average age	Total age
2004	3	49.33	148
2005	4	44	$176 = 148 + 3 + 25$
2006	4	45	$180 = 176 + 4$
2007	4	46	$184 = 180 + 4$

$176 = 148 + 3 + 25$, implies that due to 3 existing professors their total age will be increased by 3 years after one year time period and 25 years age will be added due to a new entrant in the faculty of LR.

Faculty of DI :

Year	No. of faculty	Average age	Total age
2004	4	50.5	202
2005	4	51.5	$206 = 202 + 4$
2006	4	52.5	$210 = 206 + 4$
2007	5	47.8	$239 = 210 + 4 + 25$

Faculty of English :

Year	No. of faculty	Average age	Total age
2004	5	50.2	251
2005	4	49	$196 = 251 + 5 - 60$
2006	5	45	$225 = 196 + 4 + 25$
2007	5	46	$230 = 225 + 5$

Faculty of Quants :

Year	No. of faculty	Average age	Total age
2004	6	45	270
2005	7	43	$301 = 270 + 6 + 25$
2006	7	44	$308 = 301 + 7$
2007	7	45	$315 = 308 + 7$

24. In the year 2006, a new faculty member joined the English faculty.

25. The new faculty member who joined on April 1, 2005 became 27 years old on April 1, 2007.

26. From the faculty of English a professor retired on April 1, 2005.

27. Age of Sarvesh on April 1, 2004 = 52years + 4 months $\approx$ 52years

Similarly age of Manish on April 1, 2004

$\approx$ 49 years + 4 months $\approx$ 49 years

$\therefore$ Age of the third professor on April 1, 2004

$= 148 - (52 + 49) = 47$ years

Hence the age of the third professor on April 1, 2009

$= 47 + 5 = 52$ years

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3 ALLIGATIONS

As we have studied the normal average in the previous chapter, so here we study particularly weighted average. In this chapter we will learn to find the average of two different groups with different number of elements which is known as weighted average. So alligation method is the simplified technique to solve the complex average problems. In this chapter, various kinds of problems will be discussed, since this chapter works as a means not as an end for.

Basically in the entrance tests or competitive exams like CAT, the problems are not directly asked from this chapter, rather the techniques used in this chapter are applied for the higher level problems of CAT. As the alligation plays a vital role as a time saving method in solving the problems of Ratio Proportion, S.I., C.I. Profit Loss and some other topics, so we need to emphasize on the concept of Alligation and its applications extensively applied.

Remember, in some exams like MAT, IRMA, CET etc., the problems of this chapter are directly asked which you need to solve within seconds, even orally.

EXAMPLE 1 *The average weight of a class of 40 students is 30 and the average weight of a class of 20 students is 15. Find the average weight of both the combined classes :*

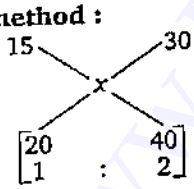
(a) 20 (b) 25 (c) 17.5 (d) 15

SOLUTION To find the solution of this problem, we can use two widely used techniques :

1. Weighted average method

$$\text{The required average} = \frac{40 \times 30 + 20 \times 15}{(40 + 20)} = 25$$

2. Alligation method :



In this graphical representation of the solution, x is the weighted average where 15 and 30 are the averages of different class. So remember that the weighted average is always written in middle and the individual averages are written at the top, the smaller one in the left and the greater one in the right just for convenience and the number of elements (or the fixed quantities) are written below correspondingly.

Now, the difference between the two averages is divided in the inverse ratio of the quantities written below. As in the above problem, the difference 15 (= 30 - 15) will be divided in the ratio of 2:1 (as 40 : 20) but not in the ratio of 1 : 2, which must be clear from the arrows indicating cross proportion.

So, the value of $x = 15 + \frac{2}{3} \times (30 - 15) = 25$

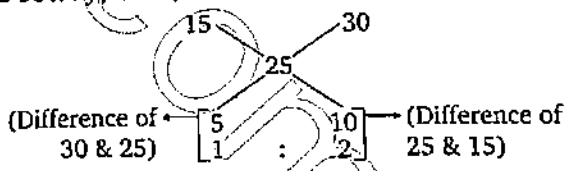
or $x = 30 - \frac{1}{3} \times (30 - 15) = 25$

Therefore, the average weight of both the classes is 25.

EXAMPLE 2 *If the average weight of a class of students is 15 and the average weight of another class of students is 30, then find the ratio of the students of the first class to the another class of 30 students when the average weight of both the classes is 25 :*

(a) 1 : 2 (b) 2 : 1 (c) 1 : 3 (d) 3 : 4

SOLUTION Let the ratio of the students with 15 students to 30 students be $x : y$, then



Here we know that the averages of individual classes is 15 and 30. Again we know that the averages of both the combined classes is 25. So the difference between 25 and 15 *i. e.*, 10 and 30 & 25, *i.e.*, 5 is written diagonally opposite.

Thus the required ratio $= \frac{5}{10} = \frac{1}{2}$

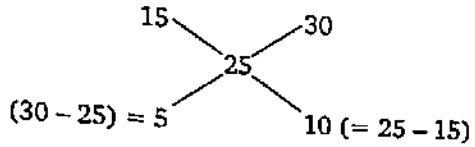
EXAMPLE 3 *The average weight of girls is 15 and the average weight of boys is 30 and the average weight of boys and girls both is 25. If the number of boys are 12, then the number of girls is :*

(a) 4 (b) 6 (c) 10 (d) 18

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SOLUTION



Since the ratio of girls to boys is $\frac{5}{10} = \frac{1}{2}$

Hence if there are 12 boys, so there will be 6 girls.

EXAMPLE 4 *The ratio of number of girls to number of boys is 1 : 2. If the average weight of the boys is 30 kg and the average weight of both the boys and girls be 25 kg, then the average weight of the girls is :*

(a) 15 kg (b) 20 kg (c) 35 kg (d) 40 kg

SOLUTION



$$\Rightarrow \quad \left(\frac{30-25}{25-G} = \frac{x}{2x}\right)$$

So $$\frac{(30-25)}{(25-G)} = \frac{1}{2} \Rightarrow G = 15$$

Therefore, the average weight of the girls is 15 kg. In general, if the average of group 1 be A_1 and the number of the elements be n_1 and the average of the group 2 be A_2 & the number of element be n_2 then the weighted average

$$A_w = \frac{n_1A_1 + n_2A_2}{n_1 + n_2}$$

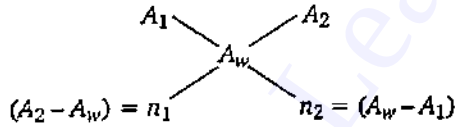
$$\Rightarrow \quad n_1A_w + n_2A_w = n_1A_1 + n_2A_2$$

$$\Rightarrow \quad n_1(A_w - A_1) = n_2(A_2 - A_w)$$

$$\Rightarrow \quad \frac{n_1}{n_2} = \frac{(A_2 - A_w)}{(A_w - A_1)}$$

(known as Alligation Equation)

So this can be represented in the graphical form as given below

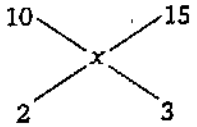


So all these problems concerned to the topic can be solved either by the formula of weighted average or Alligation equation or graphical representation method, as discussed above.

EXAMPLE 5 *Two varieties of soda water with different prices is mixed in the ratio of 2 : 3. The price of first soda water is Rs. 10 per litre while the price of second soda water is Rs. 15 per litre, respectively. The average price of the mixture (per litre) is :*

(a) Rs. 12 (b) Rs. 13 (c) Rs. 14 (d) Rs. 15

SOLUTION



$$\Rightarrow \quad \frac{(x-10)}{(15-x)} = \frac{3}{2}$$

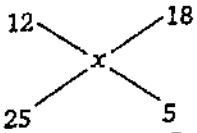
$$\Rightarrow \quad 2(x-10) = 3(15-x)$$

$$\Rightarrow \quad x = 13$$

EXAMPLE 6 *5 kg of superior quality of sugar is mixed with 25 kg of inferior quality sugar. The price of superior quality and inferior quality sugar is Rs. 18 and Rs. 12 respectively. The average price per kg of the mixture is :*

(a) Rs. 13 (b) Rs. 15 (c) Rs. 18 (d) Rs. 21

SOLUTION



$$\frac{(x-12)}{(18-x)} = \frac{5}{25}$$

$$\Rightarrow \quad x = 13$$

SHORTCUT The corresponding ratio is 25 : 5 = 5 : 1

Now reverse the ratio, which is 1 : 5

Now divide the differences of 12 and 18 in the ratio of 1 : 5 *i.e.*, divide 6 into two parts in the ratio of 1 : 5

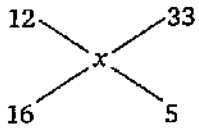
then the average price of mixture is 12 + 1 = 13

or 18 – 5 = 13

EXAMPLE 7 *16 litres of kerosene is mixed with 5 litres of petrol. The price of kerosene is Rs. 12 per litre and the price of petrol is Rs 33 per litre. The average price of the mixture per litre is :*

(a) Rs. 15 (b) Rs. 17 (c) Rs. 23 (d) Rs. 27

SOLUTION



Now divide the difference of 12 and 33 in the ratio of 5 : 16 (not in the ratio of 16 : 5)

Here, the difference of 33 and 12 = 21

Again on dividing 21 in the ratio of 5 : 16, we get the first part = 5 and second part = 16

So, the average price = 12 + 5 = 17

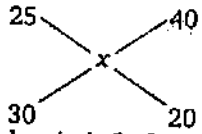
or = 33 – 16 = 17

NOTE This cross among the numbers written above shows that the difference between x and 12 is 5 and the difference between 33 and x is 16.

EXAMPLE 8 *Amit travels 30 minutes at the speed of 25 km/hr. Further he travels 20 minutes at the speed of 40 km/hr. Find his average speed.*

(a) 25 km/hr (b) 30 km/hr
(c) 31 km/hr (d) none of these

SOLUTION



Since the actual ratio is 3 : 2, so reverse it, *i.e.*, the required ratio is 2 : 3.

Now divide the difference of both the speeds *i.e.*, 25 and 40 in the ratio of 2 : 3.

i.e., $(40-25)\times\frac{2}{5} = 6$ and $(40-25)\times\frac{3}{5} = 9$

So the average speed is 25 + 6 = 31

or 40 – 9 = 31

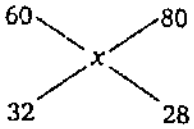
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EXAMPLE 9 *A milkman has two types of milk. In the first container the percentage of milk is 80% and in the second container the percentage of milk is 60%. If he mixes 28 litres of milk of the first container to the 32 litres of milk of the second container, then the percentage of milk in the mixture is :*

(a) 63.99 (b) 69.33 (c) 72.5 (d) 75.2

SOLUTION



8 : 7

7 : 8 (reversed ratio)

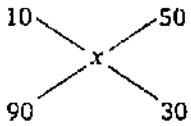
So $(80 - 60) \times \frac{7}{7+8} = 20 \times \frac{7}{15} = 9.33$

Thus the required percentage = 60 + 9.33 = 69.33

EXAMPLE 10 *Modern electronic shop sold the 30% hardware at the profit of 50% and 90% software at the profit of 10%. The average profit percent of the Modern electronic shop is, if it sells only these two kinds of things :*

(a) 15 (b) 20 (c) 25 (d) 45

SOLUTION



The actual ratio is 3 : 1.

So the reversed ratio is 1 : 3.

Now divide the difference of 10 and 50 in the ratio of 1 : 3 then add the first part to the 10 or subtract the second part from the 50.

i.e., Required average $= 10 + (50 - 10) \times \frac{1}{4}$

$= 10 + 10 = 20$

or $50 - (50 - 10) \times \frac{3}{4} = 20$

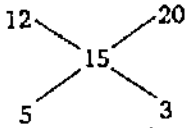
NOTE All the problems discussed above can be solved in any way, *i.e.*, either by using the formula of weighted average or Alligation Equation. But we have emphasised on the graphical (cross proportion) method. Since this technique is quite handy, *i.e.*, some times you need not to calculate on paper.

EXAMPLE 11 *Avinash covered 150 km distance in 10 hours. The first part of his journey he covered by car, then he hired a rickshaw. The speed of car and rickshaw is 20 km/hr and 12 km/hr respectively. The ratio of distances covered by car and the rickshaw respectively are :*

(a) 2 : 3 (b) 4 : 5

(c) 1 : 1 (d) none of these

SOLUTION The average speed of Avinash $= \frac{150}{10} = 15$ km/hr



It means the rickshaw took 5/8 and car took 3/8 of the total time

i. e., the ratio of time taken by rickshaw to car is 5 : 3

So the ratio of distances covered by rickshaw to car is

$5 \times 12 : 3 \times 20 \Rightarrow 1 : 1$

NOTE

1. In this calculation (*i.e.*, In alligation method) distance never involves directly. Only time and speeds are involved.
2. Since we have to find generally the average speed (not the average time) so speeds are written on the top and corresponding time taken is written below.

EXAMPLE 12 *A mixture of rice is sold at Rs. 3.00 per kg. This mixture is formed by mixing the rice of Rs. 2.10 and Rs. 2.52 per kg. What is the ratio of price of cheaper to the costlier quality in the mixture if the profit of 25% is being earned.*

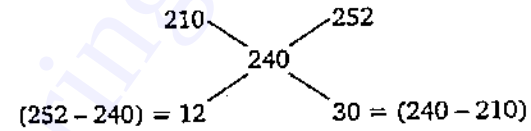
(a) 5 : 2 (b) 2 : 7 (c) 2 : 5 (d) 15 : 8

SOLUTION Let the cost price of the mixture be Rs. x per kg, then selling price $= x + \frac{25 \times x}{100} = 3$

$\Rightarrow \quad 1.25x = 3$

$\Rightarrow \quad x = 2.4$

Now the average cost price of mixture = Rs. 2.4



Therefore, the ratio of cheaper to costlier rice is 12 : 30, *i.e.*, 2 : 5.

NOTE Re 1 = 100 paise. There is no change in the ratio when we change the rupees into paise, just for our convenience in calculation.

EXAMPLE 13 *A milkman has 20 litres of milk. If he mixes 5 litres of water, which is freely available, in 20 litres of pure milk. If the cost of pure milk is Rs. 18 per litre, then the profit of the milkman, when he sells all the mixture at cost price, is :*

(a) 20% (b) 25% (c) 33.33% (d) 18%

SOLUTION When the water is freely available and all the water is sold at the price of the milk, then the water gives the profit on the cost of 20 litres of milk.

Therefore, profit percentage $= \frac{5}{20} \times 100 = 25\%$

Since, the profit % $= \frac{\text{profit}}{\text{cost price}} \times 100$

NOTE Here the milkman cheats his customer by false practise of creating illusion that instead of 20 litres milk, there is 25 litres of milk. So this extra 5 litres of milk (actually water) is the part of profit.

EXAMPLE 14 *In what ratio should water and wine be mixed so that after selling the mixture at the cost price a profit of 33.33% is made?*

(a) 1 : 4 (b) 1 : 3 (c) 2 : 3 (d) 3 : 4

SOLUTION 33.33% profit means there is one part water and 3 part is pure wine. So the required ratio of water and wine in the mixture is 1 : 3.

NOTE The above problem (No. 14) and the previous problem are basically related to the topics of ratio and profit-loss.

EXAMPLE 15 *In what ratio should freely available water be mixed with the wine worth Rs. 60 per litre so that after selling the mixture at Rs. 50 per litre, the profit will be 25%?*

(a) 1 : 2 (b) 2 : 3 (c) 3 : 4 (d) 4 : 5

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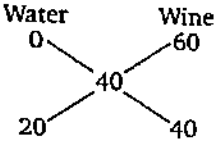
SOLUTION Selling price = Rs. 50

Therefore, the cost price

= Rs. 40, which is the average price.

It means the wine, worth Rs. 60 becomes worth Rs. 40 when the water was mixed in it. So we can conclude that in the mixture of Rs. 60, there is wine worth Rs. 40 and the rest is water. Therefore, the ratio of water and wine is 20 : 40 i.e., 1 : 2 :

Alternatively :



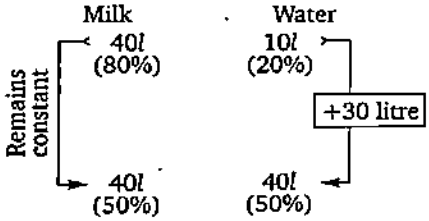
⇒ 1 : 2

EXAMPLE 16 *A mixture of water and milk contains 80% milk. In 50 litres of such a mixture, how many litres of water is required to increase the percentage of water to, 50%?*

(a) 20 *(b) 15*

(c) 30 *(d) none of these*

SOLUTION Total amount of mixture is 50 litre



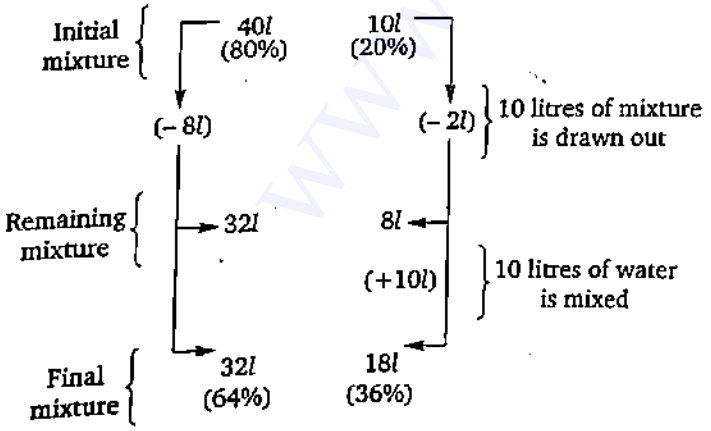
Note that, in this process the quantity of milk remains constant but its percentage decreases as the quantity (and percentage) of water increases.

Initially we have 40 litre milk and 10 litre water. Now we are required to have 50% water and remaining 50% milk in the new mixture. So we have to have 40 litre water, equal to the amount of pure milk (which is constant) available in the mixture. Thus we have to add up 30 litre (= 40 – 10) water in the original mixture.

EXAMPLE 17 *In a 50 litre mixture of water and milk, water is only 20%. The milkman gives 10 litres of this mixture to a customer and then he adds up 10 litres of pure water in the remaining mixture. The percentage of water in the final mixture is :*

(a) 84% *(b) 74%* *(c) 26%* *(d) 36%*

SOLUTION Total quantity of mixture = 50 litre



See the chart for the solution.

Initially there is 40 litres (80%) milk in the 50 litres mixture. When 10 litres of mixture is drawn out, it means 8 litres of milk is drawn out. Now when the 10 litres of water is added, it means in the 50 litres of mixture, there is only 32 litres of milk, i.e., 64%. Hence the percentage of water is 36%.

EXAMPLE 18 *There are three types of milk, Parag, Amul and Nestle. The ratio of fat to the non-fat contents in milk is 4 : 5, 5 : 6, 6 : 7 respectively. If all these three types of milk is mixed in equal quantity, the ratio of fat to the non-fat contents in the mixture will be :*

(a) 1751 : 2110 *(b) 175 : 543*

(c) 3 : 5 *(d) 10 : 18*

SOLUTION In this type of questions, we consider only one item (*viz.*, either fat or non-fat) as the fraction of the total quantity. Again we equate the denominators.

Let us consider fraction of 'fat'.

	Parag	Amul	Nestle
Fraction of fat →	$\frac{4}{9}$	$\frac{5}{11}$	$\frac{6}{13}$
	$\frac{4\times11\times13}{9\times11\times13}$	$\frac{5\times9\times13}{11\times9\times13}$	$\frac{6\times9\times11}{13\times9\times11}$
	$\frac{572}{1287}$	$\frac{585}{1287}$	$\frac{594}{1287}$

So the total fat in the total mixture

$$= \frac{572 + 585 + 594}{1287 + 1287 + 1287} = \frac{1751}{3861}$$

Hence, the ratio of fat to the non-fat contents in the mixture

$= 1751 : 2110$ $\because (3861 - 1751 = 2110)$

EXAMPLE 19 *Sharabi Chand purchased two different kinds of alcohol. In the first mixture the ratio of alcohol to water is 3 : 4 and in the second mixture it is 5 : 6. If he mixes the two given mixtures and makes a third mixture of 18 litres in which the ratio of alcohol to water is 4 : 5, the quantity of first mixture (whose ratio is 3 : 4) is required to make the 18 litres of the third kind of mixture is :*

(a) 6 *(b) 7* *(c) 8* *(d) 9*

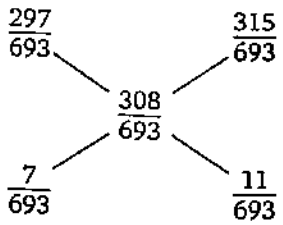
SOLUTION The fraction of alcohol in the different mixtures is as follows :

$$\text{First mixture } \frac{3}{7} = \frac{297}{693}$$

$$\text{Second mixture } \frac{5}{11} = \frac{315}{693}$$

$$\text{Final mixture } \frac{4}{9} = \frac{308}{693}$$

Therefore



Hence the ratio of first mixture is to second mixture is 7 : 11. Thus he has to mix 7 litres of first type of alcohol to make 18 litres of required mixture.

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EXAMPLE 20 *Some amount out of Rs. 6000 was lent out at 10% per annum and the rest amount @ at 20% per annum and thus in 4 years the total interest from both the amounts collected was Rs. 3400. What is the amount which was lent out @ 10% per annum?*

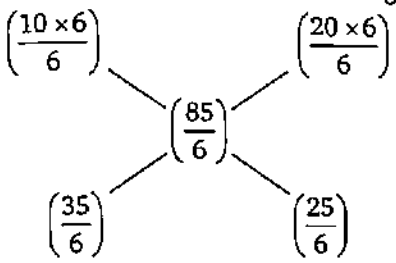
(a) Rs. 2500 *(b) Rs. 2800*
(c) Rs. 3200 *(d) Rs. 3500*

SOLUTION The total interest of one year

$$= \text{Rs. } 850 = \left(\frac{3400}{4}\right)$$

Therefore, the average rate of interest $= 14\frac{1}{6}\% = \frac{85}{6}\%$

Thus,



Hence the ratio of amount which is lent @ 10% per annum to the amount lent @ 20% per annum is 35 : 25, *i.e.*, 7 : 5.

Therefore, the amount which is lent out @ 10% per annum is Rs. 3500.

EXAMPLE 21 *From the 50 litres of milk, 5 litres of milk is taken out and after it 5 litres of water is added to the rest amount of milk. Again 5 litres of mixture of milk and water is drawn out and it was replaced by 5 litres of water. If this process is continued similarly for the third time, the amount of milk left after the third replacement :*

(a) 45 L *(b) 36.45 L* *(c) 40.5 L* *(d) 42.5 L*

SOLUTION

Milk	Water	
50 L	0 L	50 litre (initially)
− 5 L	− 0 L	withdrawn amount
45 L	0 L	50 litre (after first replacement)
+ 0 L	+ 5 L	
− 4.5 L	− 0.5 L	withdrawn amount
40.5 L	4.5 L	50 litre (after second replacement)
+ 0 L	+ 5 L	
− 4.05 L	− 0.95 L	withdrawn amount
36.45 L	8.55 L	50 litre (after third replacement)
	+ 5 L	

SHORTCUT $50 \times \left(1 - \frac{5}{50}\right) \times \left(1 - \frac{5}{50}\right) \times \left(1 - \frac{5}{50}\right)$

$$= 50 \times \frac{45}{50} \times \frac{45}{50} \times \frac{45}{50} = 50 \times \left(\frac{45}{50}\right)^3 = 50 \times \left(\frac{9}{10}\right)^3$$

$$= 36.45 \text{ L}$$

GENERAL FORMULA

Final or reduced concentration = Initial concentration $\left(1 - \frac{\text{amount being replaced in each operation}}{\text{total amount}}\right)^n$

where n is the number of times the same operation is being repeated. The "amount being replaced" could be pure or mixture as per the case. Similarly, "total amount" could also be either pure or mixture. Here amount being replaced denotes the quantity which is to be withdrawn in each time.

EXAMPLE 22 *From a tank of petrol, which contains 200 litres of petrol, the seller replaces each time with kerosene when he sells 40 litres of petrol (or its mixture). Everytime he sells out only 40 litres of petrol (pure or impure). After replacing the petrol with kerosene 4th time, the total amount of kerosene in the mixture is :*

(a) 81.92 l *(b) 96 l*
(c) 118.08 l *(d) none of these*

SOLUTION The amount of petrol left after 4 operations

$$= 200 \times \left(1 - \frac{40}{200}\right)^4 = 200 \times \left(\frac{4}{5}\right)^4$$

$$= 200 \times \frac{256}{625} = 81.92 \text{ litres}$$

Hence the amount of kerosene = 200 − 81.92 = 118.08 litres

EXAMPLE 23 *From a container of wine, a thief has stolen 15 litres of wine and replaced it with same quantity of water. He again repeated the same process. Thus in three attempts the ratio of wine and water became 343 : 169. The initial amount of wine in the container was :*

(a) 75 litres *(b) 100 litres*
(c) 150 litres *(d) 120 litres*

SOLUTION $\frac{\text{wine (left)}}{\text{water (added)}} = \frac{343}{169}$

It means $\frac{\text{wine (left)}}{\text{wine (initial amount)}} = \frac{343}{512}$ $(\because 343 + 169 = 512)$

Thus, $343x = 512x\left(1 - \frac{15}{K}\right)^3$

$\Rightarrow$ $\frac{343}{512} = \left(\frac{7}{8}\right)^3 = \left(1 - \frac{15}{K}\right)^3$

$\Rightarrow$ $\left(1 - \frac{15}{K}\right) = \frac{7}{8} = \left(1 - \frac{1}{8}\right)$

$\Rightarrow$ $K = 120$

Thus the initial amount of wine was 120 litres.

EXAMPLE 24 *A jar was full with honey. A person used to draw out 20% of the honey from the jar and replaced it with sugar solution. He has repeated the same process 4 times and thus there was only 512 gm of honey left in the jar, the rest part of the jar was filled with the sugar solution. The initial amount of honey in the jar was :*

(a) 1.25 kg *(b) 1 kg*
(c) 1.5 kg *(d) none of these*

SOLUTION Let the initial amount of honey in the jar was K, then

$$512 = K\left(1 - \frac{1}{5}\right)^4 \quad \left(\because 20\% = \frac{20}{100} = \frac{1}{5}\right)$$

or $512 = K\left(\frac{4}{5}\right)^4$

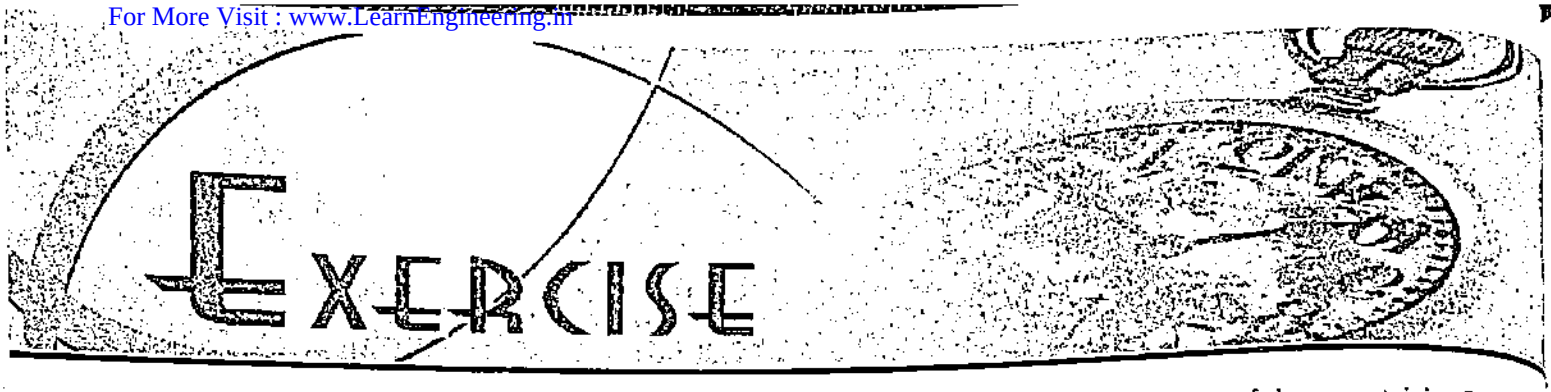
$\Rightarrow$ $K = \frac{512 \times 625}{256}$

$\therefore$ $K = 1250$

Hence initially the honey in the jar = 1.25 kg.

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1. How much Pepsi at Rs. 6 a litre is added to 15 litre of 'dew' at Rs. 10 a litre so that the price of the mixture be Rs. 9 a litre?
(a) 5 (b) 8
(c) 10 (d) none of these

2. In a municipal parking there are some two wheelers and rest are 4 wheelers. If wheels are counted, there are total 520 wheels but the incharge of the parking told me that there are only 175 vehicles. If no vehicle has a stepney, then the no. of two wheelers is :
(a) 75 (b) 100
(c) 90 (d) 85

3. In my pocket there are Rs. 25 consisting of only the denominations of 20 paise and 50 paise. Thus there are total 80 coins in my pocket. The no. of coins of the denomination of 50 paise is :
(a) 30 (b) 70
(c) 50 (d) 25

4. There are some shepherds and their sheep in a grazing field. The no. of total heads are 60 and total legs are 168 including both men and sheep. The no. of sheep is :
(a) 18 (b) 26
(c) 24 (d) 36

5. In the 75 litres of mixture of milk and water, the ratio of milk and water is 4 : 1. The quantity of water required to make the ratio of milk and water 3 : 1 is :
(a) 1 litre (b) 3 litres
(c) 4 litres (d) 5 litres

6. In my office the average age of all the female employees is 21 years and that of male employees is 32 years, where the average age of all the (male and female) employees is 28 years. The total no. of employees in my office could be :
(a) 35 (b) 78
(c) 231 (d) 90

7. A car agency has 108 cars. He sold some cars at 9% profit and rest at 36% profit. Thus he gains 17% on the sale of all his cars. The no. of cars sold at 36% profit is :
(a) 25 (b) 32
(c) 35 (d) 75

8. Rs. 69 were divided among 115 students so that each girl gets 50 paise less than a boy. Thus each boy received twice the paise as each girl received. The no. of girls in the class is :
(a) 92 (b) 42
(c) 33 (d) 23

9. In what proportion water be mixed with spirit to gain 12.5% by selling it at cost price?
(a) 3 : 5 (b) 1 : 8
(c) 2 : 7 (d) 1 : 9

10. A butler stole wine from a butt of sherry containing 50% of spirit, then he replenished it by different wine containing 20% spirit. Thus there was only 30% strength (spirit) in the new mixture. How much of the original wine did he steal?
(a) 1/3 (b) 2/3
(c) 1/2 (d) 1/4

11. Mr. Mittal purchased two steel factories, one in India and other one in Malaysia for total Rs. 72 crores. Later on he sold the Indian factory at 16% profit and Malasian factory at 24% profit. Thus he gained a total profit of 19%. The selling price of Indian factory is :
(a) 45 crore (b) 52.2 crore
(c) 8.55 crore (d) can not be determined

12. In a 25 litre mixture of milk and water, the water is only 20%. How many litres of water is required to increase the percentage of water to 90%?
(a) 45 litre (b) 70 litre
(c) 115 litre (d) 175 litre

13. A milkman sells the milk at the cost price but he mixes the water (freely available) in it and thus he gains 9.09%. The quantity of water in the mixture of 1 litre is :
(a) 83.33 mL (b) 90.90 mL
(c) 99.09 mL (d) can't be determined

14. The price of petrol is Rs. 60 per litre and the price of spirit is Rs. 40 per litre. In what ratio the petrol and spirit be mixed such that the profit after selling the mixture at Rs. 75 per litre be 25%?
(a) 1 : 1
(b) 3 : 2
(c) 5 : 1
(d) such a mixture is not possible

15. A trader sells total 315 TV sets. He sells black and white TV sets at a loss of 6% and colour TV sets at a profit of 15%. Thus he gains 9% on the whole. The no. of B/W TV sets, which he has sold, is :
(a) 126 (b) 216
(c) 135 (d) 90

16. In a class of 30 students, the average weight of boys is 20 kg and the average weight of the girls is 25 kg. The fraction of boys out of the total students of the class is :
(a) $\frac{4}{5}$ (b) $\frac{5}{6}$
(c) $\frac{3}{4}$ (d) data insufficient

17. Baniya sells two types of tea *viz.* Desi Chai and Videshi Chai. He sells Desi Chai at Rs. 18 per kg and incurs a loss of 10% whereas on selling the Videshi Chai at Rs. 30 per kg, he gains

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20%. In what proportion should the Desi Chai and Videshi Chai be mixed such that he can gain a profit of 25% by selling the mixture at Rs. 27.5 per kg?
(a) 3 : 2 (b) 2 : 3
(c) 2 : 5 (d) 3 : 5

18. The average age of boys in class is 16.66, while the average age of girls is 18.75. Thus the average age of all the 40 students of the class is 17.5. If the difference between the no. of boys and girls is 8, then the no. of girls in the class is :
(a) 12 (b) 16
(c) 18 (d) data insufficient

19. The ratio of water and alcohol in two different containers is 2 : 3 and 4 : 5. In what ratio we are required to mix the mixtures of two containers in order to get the new mixture in which the ratio of alcohol and water be 7 : 5?
(a) 7 : 3 (b) 5 : 3
(c) 8 : 5 (d) 2 : 7

20. The average marks of the students in four sections *A*, *B*, *C* and *D* together is 60%. The average marks of the students of *A*, *B*, *C* and *D* individually are 45%, 50%, 72% and 80% respectively. If the average marks of the students of sections *A* and *B* together is 48% and that of the students of *B* and *C* together is 60%. What is the ratio of number of students in sections *A* and *D*?
(a) 2 : 3 (b) 4 : 3
(c) 5 : 3 (d) 3 : 5

21. The diluted wine contains only 8 litres of wine and the rest is water. A new mixture whose concentration is 30%, is to be formed by replacing wine. How many litres of mixture shall be replaced with pure wine if there was initially 32 litres of water in the mixture ?
(a) 4 (b) 5
(c) 8 (d) none of these

22. The average weight of boys in a class is 30 kg and the average weight of girls in the same class is 20 kg. If the average weight of the whole class is 23.25 kg, what could be the possible strength of boys and girls respectively in the same class ?
(a) 14 and 26 (b) 13 and 27
(c) 17 and 27 (d) none of these

23. The shopkeeper mixed 40 kg refined oil with vegetable oil worth Rs. 60 per kg. Thus he gains Rs. 10 after selling the mixture of the two oils. The price of the first oil is :
(a) 20 (b) 25
(c) 45 (d) can't be determined

24. In a mixture of milk and water, there is only 26% water. After replacing the mixture with 7 litres of pure milk, the percentage of milk in the mixture become 76%. The quantity of mixture is :
(a) 65 litre (b) 91 litre
(c) 38 litre (d) none of these

25. The ratio of expenditure and savings is 3 : 2. If the income increases by 15% and the savings increases by 6%, then by how much per cent should his expenditure increases?
(a) 25 (b) 21
(c) 12 (d) 24

26. 4 kg of a metal contains $\frac{1}{5}$ copper and rest is zinc. Another 5 kg of metal contains $\frac{1}{6}$ copper and rest is zinc. The ratio of copper and zinc into the mixture of these two metals :
(a) 49 : 221 (b) 39 : 231
(c) 94 : 181 (d) none of these

27. 450 litres of a mixture of milk and water contain the milk and water in the ratio 9 : 1. How much water should be added to get a new mixture containing milk and water in the ratio 3 : 1?
(a) 54 (b) 90
(c) 45 (d) 63

28. The ratio of petrol and kerosene in the container is 3 : 2 when 10 litres of the mixture is taken out and is replaced by the kerosene, the ratio becomes 2 : 3. The total quantity of the mixture in the container is :
(a) 25 (b) 30
(c) 45 (d) cannot be determined

29. From a container, 6 litres milk was drawn out and was replaced by water. Again 6 litres of mixture was drawn out & was replaced by the water. Thus the quantity of milk and water in the container after these two operations is 9 : 16. The quantity of mixture is :
(a) 15 (b) 16
(c) 25 (d) 31

Answers

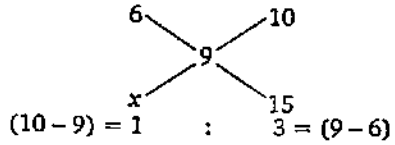
1. (a)	2. (c)	3. (a)	4. (c)	5. (d)	6. (c)	7. (b)	8. (a)	9. (b)	10. (b)
11. (b)	12. (d)	13. (a)	14. (d)	15. (d)	16. (d)	17. (a)	18. (b)	19. (b)	20. (b)
21. (b)	22. (b)	23. (d)	24. (b)	25. (b)	26. (a)	27. (b)	28. (b)	29. (a)	

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Hints & Solutions

1. Let x litre pepsi is required.



Therefore $\frac{x}{15} = \frac{1}{3}$

$\Rightarrow$ $x = 5$ litre

Alternatively : Go through options.

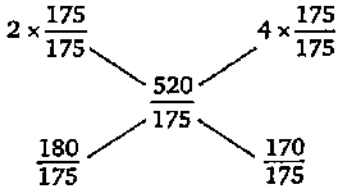
2. Go through options :

$$90 \times 2 + 85 \times 4 = 520$$

If 2 wheelers be 90 then the four wheelers will be

$$85 = (175 - 90)$$

Alternatively : The average number of wheels per vehicle $= \frac{520}{175}$



$\Rightarrow$ 180 : 170

$\Rightarrow$ 90 : 85

$\Rightarrow$ 18 : 17

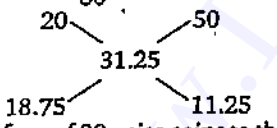
Therefore the ratio of two wheelers to four wheelers is 18 : 17.
Hence there are 90 "two wheelers".

3. Go through options :

$$30 \times 50 + 50 \times 20 = 2500 \text{ paise}$$

Alternatively : Since the average price of a coin

$$= \frac{2500}{80} = 31.25 \text{ paise}$$



So the ratio of no. of 20 paise coins to the no. of 50 paise coins

$$= 18.75 : 11.25$$

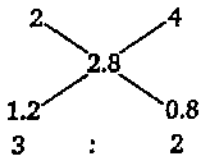
$$= 75 : 45 = 5 : 3$$

Therefore, the no. of coins of the denominations of 50 paise is 30.

4. Go through option :

$$24 \times 4 + 36 \times 2 = 168$$

Alternatively :



$\Rightarrow$ 3 : 2

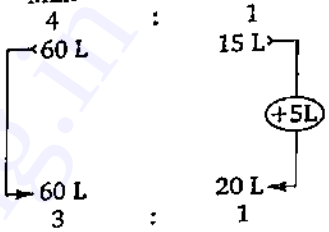
Therefore, the ratio of men and sheep is 3 : 2

Alternatively : Suppose there are only men, then the no. of legs $= 60 \times 2 = 120$.
Now since there are $48 = (168 - 120)$ legs extra, it means there are $24 = \left(\frac{48}{2}\right)$ sheep, since a sheep has 2 extra legs than a man has.

5. Total quantity of mixture = 75 litre

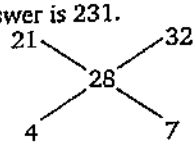
Therefore Milk : Water

4 : 1

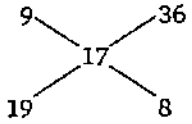


3 : 1

6. Since the ratio of no. of female and male employees is 4 : 7, so the total no. of employees must be the multiples of 11. Hence the possible answer is 231.

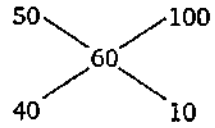


7. Since the ratio of cars sold at profit of 9% to the 36% is 19 : 8, hence the no. of cars sold at 36% profit is 32.



8. Here each girl receives 50 paise and each boy receives 100 paise and the average receiving of each student

$$= \frac{6900}{115} = 60 \text{ paise}$$



$\Rightarrow$ (G) 4 : (B) 1

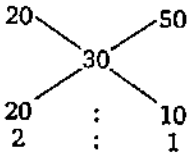
Thus the no. of girls = 92. [Number of boys = 23]

9. Profit = 12.5% = $\frac{1}{8}$

Hence the ratio of water to spirit is 1 : 8

since profit % = $\frac{\text{profit}}{\text{cost}} \times 100$.

10.



Since the ratio of 20% wine to 50% wine is 2 : 1, it means there is $\frac{2}{3}$ wine which is replaced with wine in which the concentration of spirit is 20%.

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11. 16, 24, 19, 5, 3

Thus the cost price of Indian factory is Rs. 45 crore.
Therefore, the selling price of Indian factory is

$$= 45 + \frac{45 \times 16}{100} = 52.2 \text{ crore}$$

12.

Milk		Water
80%		20%
4	:	1
20 L		5 L
20 L		180 L (+175 L)
10%		90%
1	:	9

13. Profit (%) = 9.09% = $\frac{1}{11}$

Since the ratio of water and milk is 1 : 11,
Therefore the ratio of water is to mixture = 1 : 12
Thus the quantity of water in mixture of 1 litre

$$= 1000 \times \frac{1}{12} = 83.33 \text{ mL}$$

14. The selling price of mixture = Rs. 75

∴ The cost price of mixture = Rs. 60

Now we know that if he mixes the spirit (worth Rs. 40) with petrol (worth Rs. 60), the cost price of mixture must be less than Rs. 60, which is impossible. Hence there is no spirit with the petrol.

15. −6, 15, 9, 6, 15

$\Rightarrow$ 2 : 5

Thus the ratio of B/W TV sets to the no. of colour TV sets

$$= 2 : 5$$

Therefore, no. of B/W TV sets = 90

16. Since we do not know either the average weight of the whole class or the ratio of no. of boys to girls.

17. The S.P. of Desi Chai = Rs. 18
The S.P. of Videshi Chai = Rs. 30
The C.P. of Desi Chai = Rs. 20
The C.P. of Videshi Chai = Rs. 25
The S.P. of mixture = Rs. 27.5
The C.P. of mixture = Rs. 22

20, 25, 22, 3, 2

Therefore, the ratio of Desi Chai is to Videshi Chai is 3 : 2

18. 16.66, 18.75, 17.5, (Boys) x, y (Girls)

$\Rightarrow$ $\frac{50}{3} \times \frac{4}{4}$, $\frac{75}{4} \times \frac{3}{3}$, $\frac{35}{2} \times \frac{6}{6}$, x, y

$\Rightarrow$ $\frac{200}{12}$, $\frac{225}{12}$, $\frac{210}{12}$, $\frac{15}{12}$, $\frac{10}{12}$

$\Rightarrow$ 3 : 2

Thus the no. of girls = 16 and no. of boys = 24

19.

$W_1 : A_1$	$W_2 : A_2$	$W_N : A_N$
2 : 3	4 : 5	5 : 7
$\frac{W_1}{W_1 + A_1} = \frac{2}{5}$	$\frac{W_2}{W_2 + A_2} = \frac{4}{9}$	$\frac{W_N}{W_N + A_N} = \frac{5}{12}$
$= \frac{72}{180}$	$= \frac{80}{180}$	$= \frac{75}{180}$

$\frac{72}{180}$, $\frac{80}{180}$, $\frac{75}{180}$, $\frac{5}{180}$, $\frac{3}{180}$

$\Rightarrow$ 5 : 3

Therefore, the ratio is 5 : 3

20. Since the average marks of sections B and C together are equal the average marks of all the four sections (i.e., A, B, C and D), therefore the average marks of the remaining two sections A and D together will also be equal i.e., 60%.

45, 80, 60, 20, 15

$\Rightarrow$ 4 : 3

Hence, the required ratio is 4 : 3.

21.

Wine		Water
8 L		32 L
1	:	4
20%		80% (original ratio)
30%		70% (required ratio)

In this case, the percentage of water being reduced when the mixture is being replaced with wine.

So the ratio of **left quantity** to the initial quantity is 7 : 8.

Therefore $\frac{7}{8} = \left[1 - \frac{K}{40}\right] \Rightarrow \frac{7}{8} = \left[\frac{40 - K}{40}\right]$

$\Rightarrow$ $K = 5$ litre

Alternatively : Go through options.

22. Therefore no. of boys : Number of girls = 13 : 27

20, 30, 23.25, 6.75, 3.25

27 : 13

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23. Since there is insufficient data.

24. Milk Water

74% 26% (Initially)

76% 24% (after replacement)

$$\text{Left amount} = \text{Initial amount}\left(1 - \frac{\text{replaced amount}}{\text{total amount}}\right)$$

$$24 = 26\left(1 - \frac{7}{K}\right)$$

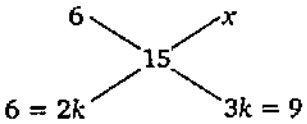
$$\Rightarrow \quad \frac{12}{13} = \left(1 - \frac{7}{K}\right)$$

$$\Rightarrow \quad \frac{1}{13} = \frac{7}{K}$$

$$\Rightarrow \quad K = 91 \text{ litre}$$

NOTE In case of replacement, the initial (pure) amount is equal to the amount of final mixture.

25.



Therefore $x = 21\%$

26. Copper in 4 kg $= \frac{4}{5}$ kg

and Zinc in 4 kg $= 4 \times \frac{4}{5} = \frac{16}{5}$ kg

Copper in 5 kg $= 5 \times \frac{1}{6} = \frac{5}{6}$ kg

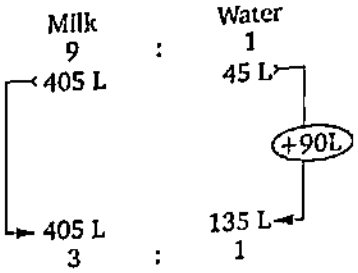
Zinc in 5 kg $= 5 \times \frac{5}{6} = \frac{25}{6}$ kg

Therefore, Copper in mixture $= \frac{4}{5} + \frac{5}{6} = \frac{49}{30}$ kg

and Zinc in the mixture $= \frac{16}{5} + \frac{25}{6} = \frac{221}{30}$ kg

Therefore, the required ratio = 49 : 221.

27.



28. Petrol : Kerosene

3 : 2 (initially)

2 : 3 (after replacement)

$$\frac{\text{Remaining (or left) quantity}}{\text{Initial quantity}} = \left(1 - \frac{\text{replaced quantity}}{\text{total quantity}}\right)$$

$$\text{(for petrol)} \quad \frac{2}{3} = \left(1 - \frac{10}{K}\right)$$

$$\Rightarrow \quad \frac{1}{3} = \frac{10}{K}$$

$$\Rightarrow \quad K = 30 \text{ litre}$$

Therefore, the total quantity of the mixture in the container is 30 litre.

29.

$$\frac{9}{25} = \left(1 - \frac{6}{K}\right)^2$$

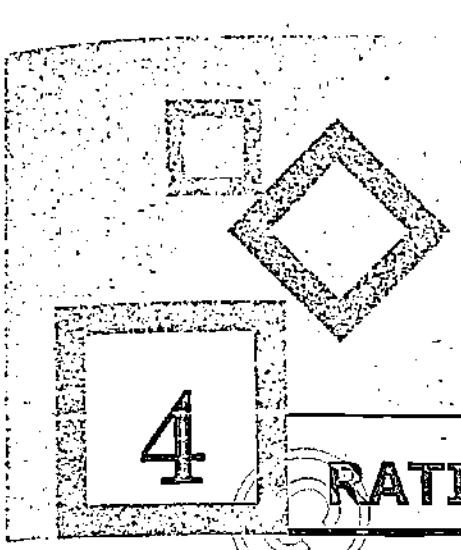
$$\Rightarrow \quad \frac{3}{5} = \left(1 - \frac{6}{K}\right)$$

$$\Rightarrow \quad K = 15 \text{ litre}$$

□□□

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4 RATIO, PROPORTION & VARIATION

The basic applications of the concepts involved in this chapter are comparisons of two or more quantities and changes in their magnitudes, *e.g.*, comparison of the ages, weights income, savings, heights, volume, density, temperature etc.

So this chapter is very useful in solving the problems of Data interpretation. Also each and every year one or two problems from this chapter is/are asked in CAT, either directly or application based for QA section. Last but not the least, the concepts of ratio, proportion and variations are very useful in solving the maximum arithmetic problems. That's why these problems are usually asked in most of the competitive exams like FMS, IIFT, MAT, SYMBIOSIS etc.

RATIO

The comparison between two quantities in terms of magnitude is called the ratio, *i.e.*, it tells us that the one quantity is how many times the other quantity.

For example, Amit has 5 pens and Sarita has 3 pens. It means the ratio of number of pens between Amit and Sarita is 5 is to 3. It can be expressed as '5 : 3'.

NOTE It should be noted that in a ratio, the order of the terms is very important. For example, in the above illustration the required ratio is 5 : 3 while 3 : 5 is wrong.

So the ratio of any two quantities is expressed as $\frac{a}{b}$ or $a : b$.

The numerator 'a' is called the **antecedent** and denominator 'b' is called as **consequent**.

Rule of Ratio

The comparison of two quantities is meaningless if they are not of the same kind or in the same units (of length, volume or currency etc). We do not compare 8 boys and 6 cows or 15 litres and 5 toys or 5 metres and 25 centimetres. Therefore, to find the ratio of two quantities (of the same kind), it is necessary to express them in same units.

NOTE

1. We do not compare 8 boys and 6 cows, but we can compare the number (8) of boys and number (6) of cows. Similarly, we cannot compare the number (15) of litres and the number (5) of toys etc.
2. Ratio has no units.

PROPERTIES OF RATIOS

1. The value of a ratio does not change when the numerator and denominator both are multiplied by same quantities *i.e.*,

$$\frac{a}{b} = \frac{ka}{kb} = \frac{la}{lb} = \frac{ma}{mb} \text{ etc.}$$

e.g., $\frac{3}{4} = \frac{6}{8} = \frac{9}{12}$ ... etc. have the same ratio.

2. The value of a ratio does not alter (or change) when the numerator and denominator both are divided by same quantities *i.e.*,

$$\frac{a}{b} = \frac{a/k}{b/k} = \frac{a/l}{b/l} = \frac{a/m}{b/m} \text{ etc.}$$

e.g., $\frac{3}{4} = \frac{3/2}{4/2} = \frac{3/3}{4/3} = \frac{3/4}{4/4}$ ... etc are same ratio.

3. The ratio of two fractions can be expressed in ratio of integers. *e.g.*,

$$\frac{3/4}{5/4} = \frac{3}{4} \times \frac{4}{5} = \frac{3}{5}$$

Since $\frac{a/b}{c/d} = \frac{a}{b} \times \frac{d}{c}$ (Refer to the **fractions** in Fundamentals)

4. When two or more than two ratios are multiplied with each other, then it is called as compounded ratio *e.g.*, $\frac{2}{3} \times \frac{4}{5} \times \frac{6}{7} = \frac{16}{35}$ is the compounded ratio of $\frac{2}{3}, \frac{4}{5}, \frac{6}{7}$

So, $\frac{a}{b} \times \frac{c}{d} \times \frac{e}{f} \ldots = \frac{k}{l}$ (compounded ratio)

5. When the ratio is compounded with itself, it is called as duplicate, triplicate ratios etc. *e.g.*,

$\frac{a}{b} \times \frac{a}{b} = \frac{a^2}{b^2} = \left(\frac{a}{b}\right)^2$ is called as duplicate ratio of $\frac{a}{b}$

and $\frac{a}{b} \times \frac{a}{b} \times \frac{a}{b} = \left(\frac{a}{b}\right)^3$ is called as triplicate ratio of $\frac{a}{b}$.

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Similarly, $\sqrt{\left(\frac{a}{b}\right)}=\left(\frac{a}{b}\right)^{1/2}$ is called as sub-duplicate ratio and $\sqrt[3]{\left(\frac{a}{b}\right)}=\left(\frac{a}{b}\right)^{1/3}$ is called as sub-triplicate ratio of $\frac{a}{b}$.

6. $\frac{a}{b}=\frac{c+am}{d+bm}$ if and only if $\frac{c}{d}=\frac{a}{b}$

i.e., $\frac{30}{40}=\frac{30+3m}{40+4m}=\frac{33}{44}=\frac{36}{48}$ ... etc at $m=1,2,\ldots$

This property is very useful when we compare two fractions

e.g., to compare between $\frac{10}{20}$ and $\frac{12}{22}$

We see that $\frac{10}{20}=\frac{1}{2}$

Now if $\frac{10}{20}$ and $\frac{12}{22}$ are equal then $\frac{12}{22}=\frac{10+m}{22+2m}$

Now putting $m=2$, we don't get the required fraction.

So $\frac{12}{22}\neq\frac{10}{20}$ for any value of m, $\frac{12}{22}\neq\frac{10}{20}$ (in terms of ratio)

7. $\frac{a+k}{b+k}<\frac{a}{b}$ if for every positive k, $\frac{a}{b}>1$

and $\frac{a-k}{b-k}>\frac{a}{b}$

8. $\frac{a+k}{b+k}>\frac{a}{b}$ if for every positive k, $\frac{a}{b}<1$ and $\frac{a-k}{b-k}<\frac{a}{b}$

9. $\frac{a+c}{b+d}>\frac{a}{b}$ if $\frac{c}{d}>\frac{a}{b}$

10. $\frac{a+c}{b+d}<\frac{a}{b}$ if $\frac{c}{d}<\frac{a}{b}$

11. If $\frac{a}{b}=\frac{c}{d}=\frac{e}{f}=\frac{g}{h}=\ldots=k$

then $\frac{a+c+e+g+\ldots}{b+d+f+h+\ldots}=k$

12. Let $\frac{a}{b},\frac{c}{d},\frac{e}{f},\frac{g}{h}\ldots$ be some different ratios, then the value of $\left(\frac{a+c+e+g+\ldots}{b+d+f+h+\ldots}\right)$ must lies between the lowest and highest ratios.

13. If $a:b$ and $b:c$ are given, then

$$a:b:c=\overset{a}{\underset{b}{\downarrow}}\swarrow\overset{b}{\underset{c}{\downarrow}}=(a\cdot b):(b\cdot b):(b\cdot c)$$

14. If the ratios between $a:b$, $b:c$, $c:d$, $d:e$ are given individually, then the combined ratio of $a:b:c:d:e$

$$\begin{matrix}a:b\\b:c\\c:d\\d:e\end{matrix}\quad\begin{matrix}a\\b\\c\\d\end{matrix}\Big|\;:\;\begin{matrix}b\\b\\c\\d\end{matrix}\Big|\;:\;\begin{matrix}b\\c\\c\\d\end{matrix}\Big|\;:\;\begin{matrix}b\\c\\d\\d\end{matrix}\Big|\;:\;\begin{matrix}b\\c\\d\\e\end{matrix}$$

i.e., $a:b:c:d:e=(a.b.c.d):(b.b.c.d)$
$:(b.c.c.d):(b.c.d.d):(b.c.d.e)$

EXAMPLE 1 *Find the ratio of 25 to 40.*

SOLUTION $\frac{25}{40}=\frac{5\times5}{5\times8}=\frac{5}{8}$

NOTE To get the ratio, we rationalize the fractions by cancelling out the common factors of numerator and denominator.

EXAMPLE 2 *Find the ratio of 90 cm to 1.5 m.*

SOLUTION 1.5 m = 150 cm (units must be same)

So, the required ratio $=\frac{90}{150}=\frac{3}{5}$

EXAMPLE 3 *The number of boys and girls in a school are 576 and 480 respectively. Express the ratio of the number of boys to that of girls in the simplest form.*

SOLUTION Required ratio $=\frac{576}{480}=\frac{6}{5}$ $\left(\because\frac{576}{480}=\frac{96\times6}{96\times5}\right)$

EXAMPLE 4 *Shukla earns Rs. 14,000 per month and Mishra earns Rs. 18,000 per month. Find the ratio of Shukla's salary to Mishra's salary.*

SOLUTION Required ratio $=\frac{14,000}{18,000}=\frac{7}{9}$

EXAMPLE 5 *Out of 144 persons working in an office, 56 are men and the remaining are women. Find the ratio of number of women to number of men.*

SOLUTION Ratio $=\frac{88}{56}=\frac{11}{7}$ ($\because$ number of women = 144 − 56)

EXAMPLE 6 *In a club having 100 members, 20 play carrom, 24 play table-tennis and 16 play cricket and the remaining members do not play any game. No member plays more than one game. Find the ratio of the number of members who play.*

(a) *Carrom to the number of those who play table-tennis.*
(b) *Cricket to the number of those who play carrom.*
(c) *Cricket to the number of those who do not play any game.*
(d) *Table-tennis to the number of those who do not play any game.*
(e) *Some game to the number of those who do not play any game.*

SOLUTION Total members = 100
Carrom = 20
Table-tennis = 24
Cricket = 16
No any game = 40

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(a) $\frac{20}{24}=\frac{5}{6}$ (b) $\frac{16}{20}=\frac{4}{5}$

(c) $\frac{16}{40}=\frac{2}{5}$ (d) $\frac{24}{40}=\frac{3}{5}$

(e) $\frac{60}{40}=\frac{3}{2}$

EXAMPLE 7 *A person earns Rs. 1200 per day and spends Rs. 800. Find the ratio of his savings to expenditure.*

SOLUTION $\frac{400}{800}=\frac{1}{2}$ (Savings = Income – Expenditure)

$= 1 : 2$

EXAMPLE 8 *Simplify the following ratios :*

(a) $\frac{1}{6}:\frac{1}{8}$ (b) $2\frac{1}{3}:3\frac{1}{2}$ (c) $\frac{5}{6}:\frac{3}{8}:3\frac{3}{4}$

SOLUTION (a) $\frac{1}{6}:\frac{1}{8}=\frac{1/6}{1/8}=\frac{1}{6}\times\frac{8}{1}=\frac{4}{3}$ or $4:3$

Alternatively: $\frac{1}{6}:\frac{1}{8}=\frac{48}{6}:\frac{48}{8}$ $\left(\text{or } \frac{24}{6}:\frac{24}{8}\right)$

$= 8 : 6 = 4 : 3$

NOTE In case of fractions, convert them to whole numbers by multiplying each term by the L.C.M. of their denominators.

(b) $2\frac{1}{3}:3\frac{1}{2}=\frac{7}{3}:\frac{7}{2}=\frac{7/3}{7/2}=\frac{2}{3}$ or $2:3$

NOTE In case of two fractions if numerators are same, then the required ratio is the inverse ratio of the fractions.

(c) $\frac{5}{6}:\frac{3}{8}:3\frac{3}{4}=\frac{5}{6}:\frac{3}{8}:\frac{15}{4}=\frac{5}{6}\times24:\frac{3}{8}\times24:\frac{15}{4}\times24$

$= 20 : 9 : 90$ {∵ LCM of 6, 8 and 4 is 24}

EXAMPLE 9 *Divide 14 toffees among Ankita and Anshul in the ratio 5 : 2*

SOLUTION According to the question, if there are 7 toffees then Ankita will have 5 and Anshul will have 2 toffees but since there are 14 toffees, which is twice of 7. So Ankita will have 10 toffees and Anshul will have 4 toffees.

SHORTCUT

Ankita	:	Anshul
5	:	2
$14\times\frac{5}{(5+2)}$	:	$14\times\frac{2}{(5+2)}$
10	:	4

EXAMPLE 10 *Three boys are aged 2 years, 4 years and 8 years. They want to divide seventy rupees in the ratio of their ages. How much money would each get?*

SOLUTION The ratio of their ages $= A : B : C = 2 : 4 : 8 = 1 : 2 : 4$

A	:	B	:	C
1	:	2	:	4
$70\times\frac{1}{1+2+4}$	:	$70\times\frac{2}{1+2+4}$	:	$70\times\frac{4}{1+2+4}$
10	:	20	:	40

So A gets Rs. 10, B gets Rs. 20 and C gets Rs. 40 if the ratio of their ages is $2 : 4 : 8$.

EXAMPLE 11 *An amount of Rs. 100 is being divided among two persons in the ratio $\frac{1}{10}:\frac{1}{15}$. How much money does each get?*

SOLUTION $\frac{1}{10}:\frac{1}{15}=\frac{1}{10}\times30:\frac{1}{15}\times30=3:2$ (here 30 is the LCM of 10 and 15)

So, the ratio of amount of money $= \frac{3}{5}\times100:\frac{2}{5}\times100$

$= 60$ and 40

EXAMPLE 12 *The lengths of sides of a triangle are in the ratio 2 : 3 : 4. If the perimeter of the triangle is 63 cm, find the lengths of the sides of the triangle.*

SOLUTION Let the sides of triangle be $2x$, $3x$ and $4x$, then

$2x + 3x + 4x = 63$

$\Rightarrow \quad 9x = 63 \Rightarrow x = 7$

$\therefore$ The sides of triangle $= 2x, 3x, 4x = 14, 21, 28$

Alternatively:

$63\times\frac{2}{(2+3+4)}$ and $63\times\frac{3}{(2+3+4)}$ and $63\times\frac{4}{(2+3+4)}$

$= 14, 21$ and 28.

EXAMPLE 13 *Divide 1224 into three parts such that first part be double that of second part and second part be $\frac{1}{3}$ of the third part.*

SOLUTION Let A, B, C be three parts respectively, then

$A : B = 2 : 1$

and $B : C = 1 : 3$

$\therefore \quad A : B : C = 2 : 1 : 3$

$A = 1224\times\frac{2}{6}=408$

$B = 1224\times\frac{1}{6}=204$

$C = 1224\times\frac{3}{6}=612$

EXAMPLE 14 *If $A : B = 3 : 4$, $B : C = 5 : 2$ then find the value of $A : B : C$.*

SOLUTION $A : B = 3 : 4$

and $B : C = 5 : 2$

$A : B : C = (3\times5) : (4\times5) : (4\times2)$

or $A : B : C = 15 : 20 : 8$



EXAMPLE 15 *The ratio of $A : B = 1 : 3$, $B : C = 2 : 5$, $C : D = 2 : 3$. Find the value of $A : B : C : D$.*

SOLUTION $A : B = 1 : 3$

$B : C = 2 : 5$

$C : D = 2 : 3$

$A : B : C : D = (1\times2\times2) : (3\times2\times2) : (3\times5\times2) : (3\times5\times3)$

$A : B : C : D = 4 : 12 : 30 : 45$

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EXAMPLE 16 *There are two types of mixtures of milk and water. In the first mixture, out of 12 litres of mixture, 5 litre is milk only and in the second mixture, 6 litre is milk and 12 litre is water. Which one mixture is better in terms of milk's strength?*

SOLUTION

First Mixture	Second Mixture	
$\frac{5}{12}$	$\frac{6}{18}$	$(6 + 12 = 18)$
$\frac{5}{12} \times 36$	$\frac{6}{18} \times 36$	(milk + water = mixture)
15	12	

So the first mixture has more milk in comparison to water.

EXAMPLE 17 *The ratio of salary of $A : B = 1 : 2$, $B : C = 3 : 4$, $C : D = 5 : 6$ and $D : E = 7 : 8$. What is the ratio of salary of A and E?*

SOLUTION

$A : B \quad 1 : 2$
$B : C \quad 3 : 4$
$C : D \quad 5 : 6$
$D : E \quad 7 : 8$

$A : B : C : D : E = (1 \times 3 \times 5 \times 7) : (2 \times 3 \times 5 \times 7) : (2 \times 4 \times 5 \times 7) : (2 \times 4 \times 6 \times 7) : (2 \times 4 \times 6 \times 8)$

or $A : B : C : D : E = 105 : 210 : 280 : 336 : 384$

So the ratio of salary of $A : E = 105 : 384 = 35 : 128$

NOTE In every next step, we leave the left term and adopt right term.

EXAMPLE 18 *If $\frac{a}{b} = \frac{3}{4}$, then find the value of $7a - 4b : 3a + b$*

(a) 7 : 1 (b) 5 : 13
(c) 12 : 1 (d) *none of these*

SOLUTION Simply substitute the value of a and b as 3 and 4 in the given algebraic ratio

as $(7 \times 3 - 4 \times 4) : (3 \times 3 + 4) = 5 : 13$

EXAMPLE 19 *If $a : b = 3 : 2$ and $b : c = 6 : 5$ then $a : b : c$ is equal to*

(a) 9 : 6 : 5 (b) 9 : 6 : 10
(c) 3 : 3 : 5 (d) 3 : 6 : 5

SOLUTION

$a : b = 3 : 2$
$b : c = 6 : 5$

$a : b : c = (3 \times 6) : (2 \times 6) : (2 \times 5)$
$= 18 : 12 : 10 = 9 : 6 : 5$

EXAMPLE 20 *The sum of two natural numbers is 64. Which of the following can not be the ratio of these two numbers?*

(a) 3 : 5 (b) 1 : 3 (c) 7 : 9 (d) 3 : 4

SOLUTION Let the numbers be $3x$ and $5x$, then

$3x + 5x = 64 \Rightarrow 8x = 64$

$\Rightarrow x = 8$, which is possible

The numbers are 24, 40.

Take another option:

$x + 3x = 64 \Rightarrow x = 16$

The numbers are 16 and 48.

Check for option (c) :

$7x + 9x = 64 \Rightarrow 16x = 64$

$\Rightarrow x = 4$

The numbers are 28 and 36.

Check for option (d)

$3x + 4x = 64 \Rightarrow 7x = 64$

$x = \frac{64}{7}$

Numbers are $\frac{3 \times 64}{7}$ and $\frac{4 \times 64}{7}$ or $\frac{192}{7}$ and $\frac{256}{7}$

which are not the natural numbers.

Hence option (d) is the required answer.

EXAMPLE 21 *Monthly incomes of A and B are in the ratio of 4 : 3 and their savings are in the ratio of 3 : 2. If the expenditure of each will be Rs. 600, then the monthly incomes of each are :*

(a) 1800, 2400 (b) 2400, 1600
(c) 2400, 1800 (d) 1600, 1200

SOLUTION Income = Exp. + Savings

$A \rightarrow 4x = 3y + 600$

$B \rightarrow 3x = 2y + 600$

Therefore, $4x - 3y = 600$ and $3x - 2y = 600$

$\Rightarrow 4x - 3y = 3x - 2y \Rightarrow x = y$

$\therefore 4x - 3x = 600 \Rightarrow x = 600$

Then, the income of $A = 4 \times 600 = 2400$

and income of $B = 3 \times 600 = 1800$

Alternatively: Check the options. Consider (c)

Income	A		B	
	2400		1800	
	(4)	:	(3)	$\Rightarrow$ correct
Again,	2400		1800	
	− 600		− 600	
Savings	1800		1200	
	(3)	:	(2)	$\Rightarrow$ correct

Hence, option (c) is correct. If you check other options, the ratio will not match.

EXAMPLE 22 *A, B and C have 40, x and y balls with them respectively. If B gives 20 balls to A, he is left with half as many balls as C. If together they had 60 more balls, each of them would have had 100 balls on an average. What is value of $x : y$?*

(a) 3 : 2 (b) 4 : 6 (c) 2 : 1 (d) 3 : 4

SOLUTION From the last statement

$\frac{100 + x + y}{3} = 100$

$\Rightarrow x + y = 200$ ...(1)

Again from first statement, $\frac{x - 20}{y} = \frac{1}{2}$

$\Rightarrow 2x - y = 40$ ...(2)

Solving equations (1) and (2), we get

$x = 80$ and $y = 120$

Therefore, required ratio of $x : y = 2 : 3$

Hence, option (b) is correct.

Alternatively: After forming the equation (1), we can go through the options. Let us assume option (b).

$x + y = 200$

$\because x : y = 4 : 6$

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$\Rightarrow \quad x = 80$ and $y = 120$

Now from the first statement, $\frac{x-20}{y} = \frac{1}{2}$

So $\frac{80-20}{120} = \frac{1}{2}$ (verified)

hence option (b) is correct.

EXAMPLE 23 *The incomes of A, B, C are in the ratio of 12 : 9 : 7 and their spendings are in the ratio 15 : 9 : 8. If A saves 25% of his income. What is the ratio of the savings of A, B and C?*

SOLUTION Income = Expenditure + Saving

$A \rightarrow \quad 12x = 15y + 3x \qquad (3x = 25\% \text{ of } 12x)$

$B \rightarrow \quad 9x = 9y + (9x - 9y)$

$C \rightarrow \quad 7x = 8y + (7x - 8y)$

Therefore, $12x - 3x = 15y$

$\Rightarrow \quad \frac{x}{y} = \frac{5}{3}$

$\Rightarrow \quad y = \frac{3x}{5}$

Therefore, savings = (income − expenditure)

$A = 12x - 9x = 3x$

$B = 9x - \frac{27}{5}x = \frac{18}{5}x$

$C = 7x - \frac{24}{5}x = \frac{11}{5}x$

i.e., the ratio of savings of $A : B : C = 3x : \frac{18}{5}x : \frac{11}{5}x$

$= 15x : 18x : 11x$

$= 15 : 18 : 11$

EXAMPLE 24 *There are total 100 coins consisting of 20 paise, 50 paise and Re 1 in the ratio of 7 : 8 : 5. What is the no. of coins of 50 paise if the difference between the amount yielded by 20 paise and Re. 1 coin is 18?*

(a) 32 *(b) 40* *(c) 26* *(d) 56*

SOLUTION $(20 \times 7x) \sim (100 \times 5x) = 1800$ (Rs. 18 = 1800 paise)

$(140x) \sim (500x) = 1800$

$\Rightarrow \quad 360x = 1800$

$\Rightarrow \quad x = 5$

Therefore, number of coins of 50 paise $= 8 \times 5 = 40$

Alternatively: Let there be 40 coins of 50 paise denomination, then

The no. of coins of $A, B, C = 7x : 8x : 5x = 35 : 40 : 25$

Therefore, the amount from 20 paise coins = 35×20 = Rs. 7

and amount from Re 1 coins = 1×25 = Rs. 25

Hence difference = Rs. 18 (25 − 7)

Thus the presumed option is correct.

EXAMPLE 25 *There are 43800 students in 4 schools of a city. If half of the first, two-third of the second, three-fourth of the third and four-fifth of the fourth are the same number of students, then find the ratio of number of students of A and D if A, B, C and D be the first, second, third and fourth schools respectively.*

SOLUTION $\frac{A}{2} = \frac{2B}{3} = \frac{3C}{4} = \frac{4D}{5}$

Therefore, $\frac{A}{B} = \frac{4}{3}$ and $\frac{B}{C} = \frac{9}{8}$ and $\frac{C}{D} = \frac{16}{15}$

$\therefore \quad A : B = 4 : 3$

$B : C = 9 : 8$

$C : D = 16 : 15$

$\therefore \quad A : B : C : D = (4 \times 9 \times 16) : (3 \times 9 \times 16) : (3 \times 8 \times 16) : (3 \times 8 \times 15)$

$A : B : C : D = 576 : 432 : 384 : 360$

Therefore, the ratio of number of students of A and $D = 576 : 360 = 8 : 5$

Alternatively: $\frac{A}{2} = \frac{2}{3}B = \frac{3}{4}C = \frac{4}{5}D = k$

then $A : B : C : D = 2k : \frac{3}{2}k : \frac{4}{3}k : \frac{5}{4}k$

$= 24k : 18k : 16k : 15k$

$\therefore \quad A : B : C : D = 24 : 18 : 16 : 15$

$\therefore \quad A : D = 24 : 15 = 8 : 5$

PROPORTION

An equality of two ratios is called a proportion and we say that the four numbers are in proportion.

i.e., if $\frac{a}{b} = \frac{c}{d}$, or $a : b = c : d$, then we say that a, b, c and d are in proportions and written as $a : b :: c : d$, where the symbol '::' indicates proportion and it is read as 'a is to b as c is to d'.

Here a and d are called **"extremes"** (or extreme terms) and b and c are called as **"means"** (or middle terms). Thus four numbers are said to be in proportion, if the ratio of the first to the second number is equal to the ratio of the third to the fourth number. *e.g.*, 2 : 3 :: 4 : 6. Some important results are derived from the discussion of proportion, which are very important for clear understanding.

Properties of Proportion

1. If four numbers (quantities) are in proportion then product of the extremes is equal to the product of the means and if these are not in proportion, then product of extremes is not equal to the product of the means.

   i.e., if $a : b :: c : d$,

   then $a \times d = b \times c$

   Thus it is clear that if three out of four terms of a proportion are given, then we can find the fourth term by dividing the product of means (or extremes) by the remaining term.
2. If a, b and c are three numbers such that $a : b = b : c$ then these numbers a, b, c are said to be in **continued proportion** or simply in proportion.

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i.e., $a : b = b : c \Rightarrow b^2 = ac$

Here b is said to be the **mean proportional** between a and c, and c is said to be the **third proportional** to a and b. e.g., $3 : 9 :: 9 : 27$

Here $9 \times 9 = 3 \times 27$ which are in continued proportion.

NOTE Sometimes the above idea is also expressed by saying that the three numbers are in the ratio 3 : 9 : 27. Thus if three quantities are proportionals, the first is to the third as the duplicate ratio of the first to the second.

i.e., if $a : b :: b : c$
then $a : c = a^2 : b^2$

$$\left[\because \frac{a}{b} = \frac{b}{c} \Rightarrow \frac{a^2}{b^2} = \frac{ab}{bc} \Rightarrow \frac{a^2}{b^2} = \frac{a}{c} \Rightarrow a : c = a^2 : b^2\right]$$

Some Facts about Proportion

(a) Invertendo: If $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{b}{a} = \frac{d}{c}$

(b) Alternando: If $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{c} = \frac{b}{d}$

NOTE Each of the two results (a) and (b) can be obtained by cross product.

(c) Componendo: If $\frac{a}{b} = \frac{c}{d} \Rightarrow \left(\frac{a+b}{b}\right) = \left(\frac{c+d}{d}\right)$

(d) Dividendo: If $\frac{a}{b} = \frac{c}{d} \Rightarrow \left(\frac{a-b}{b}\right) = \left(\frac{c-d}{d}\right)$

(e) Componendo and Dividendo: If $\frac{a}{b} = \frac{c}{d}$

$$\Rightarrow \left(\frac{a+b}{a-b}\right) = \left(\frac{c+d}{c-d}\right)$$

NOTE The result (e) can be obtained by dividing result (c) by result (d).

EXAMPLE 1 *The first, second and fourth terms of a proportion are 5, 15 and 90 respectively. Find the third term.*

SOLUTION Let the third term be x, then 5, 15, x, 90 are in proportion i. e., $5 : 15 :: x : 90$

$\Rightarrow$ $5 \times 90 = 15 \times x$ (by the I[st] property)

$\Rightarrow$ $x = 30$

EXAMPLE 2 *The ratio of length to width of a rectangular sheet of paper is 5 : 3. If the width of the sheet is 18 cm, find its length.*

SOLUTION Let the length of sheet of paper be x cm. Then the ratio of length to width $= x : 18$

Thus $x : 18 = 5 : 3$

$\Rightarrow$ $x \times 3 = 18 \times 5$

$\Rightarrow$ $x = 30$

Hence the length of paper = 30 cm.

EXAMPLE 3 *If 81, x, x, 256 are in proportion, find x.*

SOLUTION $81 : x :: x : 256$

$\therefore \Rightarrow$ $81 \times 256 = x \times x \Rightarrow x = 144$

EXAMPLE 4 *The ratio between the number of men and women in an office is 5 : 7. If the number of women working in the office is 56, find the number of men working in the office.*

SOLUTION $5 : 7 = x : 56$ (suppose number of men = x)

$\Rightarrow$ $x = 40$ (by the first property)

Therefore, number of men in the office = 40

EXAMPLE 5 *The age of Chandi and Radhika are in the ratio 5 : 3. If Chandi's age is 20 years, find the age of Radhika.*

SOLUTION $5 : 3 = 20 : x$

$\Rightarrow$ $x = \frac{3 \times 20}{5}$ (by the first property)

$\Rightarrow$ $x = 12$

Hence the age of Radhika is 12 years.

EXAMPLE 6 *The ratio of the number of boys to that of girls in a school is 9 : 11. If the number of girls in the school is 2035, find :*
(a) number of boys in school
(b) number of students in school.

SOLUTION (a) $9 : 11 = x : 2035$

$\Rightarrow$ $x = \frac{9 \times 2035}{11}$

$x = 1665$ (number of boys)

(b) number of students
= number of boys + number of girls = 3700

EXAMPLE 7 *What is the least possible number which must be subtracted from 16, 19 and 23 so that the resulting numbers are in continued proportion?*
(a) 2 *(b) 4* *(c) 6* *(d) 7*

SOLUTION Going through options, we find option (d) is correct.

Alternatively:

$(16 - x) : (19 - x) :: (19 - x) : (23 - x)$

$\Rightarrow$ $(19 - x)^2 = (16 - x)(23 - x)$

By solving the above equation, we get $x = 7$

EXAMPLE 8 *If $(a + b) : (a - b) = 15 : 1$, then the value of $a^2 - b^2$ is:*
(a) 56 *(b) 15* *(c) 112* *(d) 8*

SOLUTION $\frac{(a+b)}{(a-b)} = \frac{15}{1}$

$\Rightarrow$ $\frac{a}{b} = \frac{8}{7}$ (by componendo and dividendo)

therefore $a^2 - b^2 = 64 - 49 = 15$

EXAMPLE 9 *The mean proportional between 8 and 98 is :*
(a) 16 *(b) 53* *(c) 112* *(d) 28*

SOLUTION $8 : x :: x : 98$

$x^2 = 8 \times 98$

$\Rightarrow$ $x = 28$

Hence (d) is correct.

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EXAMPLE 10 *The students in three classes are in the ratio of 2 : 3 : 4. If 40 students are added in each class, the ratio becomes 4 : 5 : 6. Find the total number of students in all the three classes is :*

(a) 270 (b) 180 (c) 126 (d) 135

SOLUTION

$$2x + 40 = 4y \quad \ldots(i)$$

$$3x + 40 = 5y \quad \ldots(ii)$$

$$4x + 40 = 6y \quad \ldots(iii)$$

Therefore $2x = 40$

$\Rightarrow$ $x = 20$

Hence, total number of students $= 2x + 3x + 4x = 9x$

$= 9 \times 20 = 180$

Alternatively: It can be solved through options also.

EXAMPLE 11 *The dimensions of a photograph are 4 and 1.8 cms. If the breadth of the enlarged photo is 4.5 cm and it was enlarged proportionally then what is the new length of new photograph?*

(a) 6 (b) 5.4 (c) 10 (d) 9

SOLUTION

$$4 : 1.8 = x : 4.5$$

$\Rightarrow$ $x = 10$

Thus, the length of new photograph is 10 cm.

EXAMPLE 12 *Two equal containers are filled with the mixture of milk and water. The concentration of milk in each of the containers is 20% and 25% respectively. What is the ratio of water in both the containers respectively?*

(a) 15 : 16 (b) 16 : 15 (c) 4 : 5 (d) 5 : 4

SOLUTION

Milk	20%	25%
Water	80%	75%

Thereofre, required ratio $= \frac{80}{75} = \frac{16}{15}$ or 16 : 15

EXAMPLE 13 *A cat takes 7 steps for every 5 steps of a dog, but 5 steps of a dog are equal to 6 steps of cat. What is the ratio of speed of cat to that of dog?*

(a) 24 : 25 (b) 42 : 25 (c) 24 : 19 (d) 25 : 42

SOLUTION

	CAT	**DOG**
Given speed	7 steps	5 steps

but the length of 5 steps of dog = length of 6 steps of cat

it means the ratio of length covered by dog is to cat $= \frac{6}{5}$

Therefore, in each step a dog will cover $\frac{6}{5}$ times distance than that of a cat.

Thus the ratio of actual speed of cat is to dog

$$= 7 : 5 \times \frac{6}{5} = 7 : 6$$

SHORTCUT Actual speed of $A : B$

$$= \frac{\text{Given speed of } A}{\text{no. of steps of } A \text{ in terms of length}} : \frac{\text{Given speed of } B}{\text{no. of steps of } B \text{ in terms of length}}$$

$$= \text{CAT} : \text{DOG}$$

$$= \frac{7}{6} : \frac{5}{5} = 7 : 6$$

EXAMPLE 14 *A camel pursue an elephant and takes 5 leaps for every 7 leaps of the elephant, but 5 leaps of elephant are equal to 3 leaps of camel. What is the ratio of speeds of camel and elephant?*

SOLUTION Ratio of speed of camel and elephant

$$= \frac{5}{3} : \frac{7}{5} = \frac{5}{3} \times 15 : \frac{7}{5} \times 15$$

$$= 25 : 21$$

UNITARY METHOD

A method in which the value of a quantity is first obtained to find the value of any required quantity is called unitary method.

In solving problems based on unitary method.

Direct Proportion

Two quantities are said to be directly proportional if the increase (or decrease) in one quantity causes the increase (or decrease) in the other quantity by same proportion. *e.g.,*

(i) The cost of articles varies directly to the number of articles. More articles more cost, less articles less cost.

(ii) The work done varies directly to the number of men (work force) at work. More men at work, more work done in same time. Less men, less work done in same time.

Inverse Proportion

Two quantities are said to vary inversely if the increase (or decrease) in one quantity causes the decrease (or increase) in the other quantity by same proportion *e.g.,* The time taken to finish a work varies inversely to the number of men at work.

More men at work, less time taken to finish the same work. Less men at work, more time taken to finish the same work.

EXAMPLE 1 *If 6 note books cost Rs. 45, how much would 8 notebooks cost?*

SOLUTION More note books more cost; less note books, less cost

Note books	Cost
6	45
1	45/6
8	$\frac{8 \times 45}{6} = 60$

Hence 8 note books cost Rs. 60.

EXAMPLE 2 *If 45 students can consume a stock of food in 2 months, find for how many days the same stock of food will last for 27 students?*

SOLUTION More students, less days; less students, more days

Students	Days
45	60
1	60×45
27	$\frac{60 \times 45}{27} = 100$ days

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EXAMPLE 3 *A man working 8 hours a day takes 5 days to complete a project. How many hours a day must he work to complete it in 4 days?*

SOLUTION 'More days, less hours; less days, more hours'

Days	Hours
5	8
1	5×8
4	$\frac{5 \times 8}{4} = 10$ hours a day

VARIATION

When two or more quantities are dependent upon each other and then if any one of them is changed, the other (dependent) quantity is also changed.

For example :

(i) When the salary of a person increases, then its savings/expenditure increases.

(ii) When the number of guests in a hotel/number of students in a hostel/number of employees changes, their respective expenses increases.

Basically, as it happens in direct proportion and inverse proportion, there are two types of variation :

(i) Direct variation (ii) Inverse variation

Direct Variation

A quantity A is said to vary directly if the increase (or decrease) in B yields increase (or decrease) in A but not in proportion. It is expressed as

$$A \propto B \Rightarrow A = KB,$$

where K is called proportionality constant

$$\Rightarrow \quad K = \frac{A}{B}$$

Inverse Variation

A quantity A is said to vary inversely if the increase (or decrease) in B yields decrease (or increase) in A but not in same proportion. It is expressed as

$$A \propto \frac{1}{B} \Rightarrow A = \frac{K}{B}$$

or $K = AB$, K is called as proportionality constant.

NOTE

1. If it is not mentioned that a particular quantity is inversely variable, then it means the given quantity is directly variable.
2. A quantity sometimes vary jointly *i.e.*, directly on any quantity and inversely on another quantity.

e.g., $A \propto B$ and $A \propto \frac{1}{C}$

It means $A \propto \frac{B}{C} \Rightarrow A = \frac{KB}{C}$

Here A varies directly as B but inversely as C.

Also it can vary as only directly or inversely as more than one quantities.

e.g., $A \propto BC \Rightarrow A = KBC$

and $A \propto \frac{1}{BC} \Rightarrow A = \frac{K}{BC}$

EXAMPLE 1 *A varies directly as B and inversely as C. A is 12 when B is 6 and C is 2. What is the value of A when B is 12 and C is 3?*

SOLUTION $A \propto B$ and $A \propto \frac{1}{C}$

$$\Rightarrow \quad A \propto \frac{B}{C} \Rightarrow A = K\frac{B}{C}$$

when $A = 12, B = 6, C = 2$, then

$$12 = K\frac{6}{2} \Rightarrow K = 4$$

Again $A = K\frac{B}{C} = 4 \times \frac{12}{3} = 16$

$$\Rightarrow \quad A = 16$$

EXAMPLE 2 *The value of a coin varies directly to the square of its radius, when its thickness is constant. The radius of a coin is 1.5 cm and its value is Rs. 2. What will be the radius of a coin if its value is Rs. 5?*

SOLUTION $V \propto r^2$

$$\Rightarrow \quad V = Kr^2$$

$$2 = K \times (1.5)^2$$

$$\Rightarrow \quad K = \frac{2}{2.25}$$

$$K = \frac{8}{9}$$

Again $5 = \frac{8}{9} \times r^2$

$$\Rightarrow \quad r^2 = \frac{5 \times 9}{8}$$

$$\therefore \quad r = \frac{3}{2} \times \sqrt{\frac{5}{2}}$$

$$= 1.5 \times \sqrt{2.5} = 1.5 \times 1.6$$

$$r \approx 2.4 \text{ cm}$$

Hence, required radius = 2.4 cm

PROBLEMS BASED ON AGES

This article is very suitable as an appendix of Ratio-proportion, since most of the questions based on ages involve the concept of ratio-proportion. *e.g.*, the age of Ravi is 16 years and the age of her mother is twice etc. or the ratio of the ages of the father and son at present is 3 : 1. 4 years, earlier the ratio was 4 : 1. What is the present ages of the father and son etc.

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EXAMPLE 1 *The ratio of ages of Krishna and Balram is 3 : 4. Four years earlier the ratio was 5 : 7. Find the present ages of Krishna and Balram :*

(a) 15 years, 20 years *(b) 24 years, 32 years*
(c) 16 years, 20 years *(d) 32 years, 24 years*

SOLUTION Let the present age of Krishna and Balram be $3x$ and $4x$, then

four years ago their ages be $(3x - 4)$ and $(4x - 4)$

So $$\frac{(3x-4)}{(4x-4)} = \frac{5}{7}$$

$$\Rightarrow \quad 7(3x-4) = 5(4x-4)$$

$$\Rightarrow \quad 21x - 28 = 20x - 20$$

$$\Rightarrow \quad x = 8$$

$\therefore$ Present age of Krishna $= 3x = 24$ years

and age of Balram $= 4x = 32$ years

EXAMPLE 2 *The ratio of age of Aman and her mother is 3 : 11. The difference of their ages is 24 years. What will be the ratio of their ages after 3 years?*

SOLUTION Let the present age of Aman and her mother be $3x$ and $11x$ then 3 years later their ages will be $(3x + 3)$ and $(11x + 3)$ respectively.

Again $11x - 3x = 24$

$$\Rightarrow \quad 8x = 24 \Rightarrow x = 3$$

Therefore the ratio of their ages 3 years after

$$= \frac{3x+3}{11x+3} = \frac{12}{36} = \frac{1}{3}$$

EXAMPLE 3 *The age of Sachin is 4 times that of his son. Five years ago Sachin was nine times as old as his son was at that time. The present age of the Sachin is :*

(a) 25 years *(b) 36 years*
(c) 32 years *(d) 48 years*

SOLUTION Let the age of son is x years, then the age of Sachin will be $4x$ years.

$$\therefore \quad (4x - 5) = 9(x - 5) \Leftrightarrow x = 8$$

$\therefore$ Age of Sachin is 32 years.

EXAMPLE 4 *The ratio of Varun's age and his mother's age is 5 : 11. The difference of their ages is 18 years. The ratio of their ages after 5 years will be :*

(a) 19 : 59 *(b) 2 : 3* *(c) 37 : 75* *(d) 10 : 19*

SOLUTION Let their ages be $5x$ and $11x$.

$$11x - 5x = 18 \Leftrightarrow x = 3$$

So their present ages are 15 and 33 years. Therefore, ratio of their ages after 5 years = 20 : 38 = 10 : 19.

PARTNERSHIP

When two or more than two people run a business jointly by investing their money/resources, then it is called a joint venture or the business in partnership.

All these people, who have invested their resources, are called as Partners.

Partners are basically of two types

(i) **Working partner:** A partner who is directly involved with day-to-day activities of business is called as working partner.

(ii) **Sleeping partner:** A partner who just invests his or her money is called as sleeping partner.

General rules of partnership :

(i) If the partners invest different amounts for the same period of time, then the profits of all the partners are shared in the ratio of their investments.

(ii) If the partners invest same amount for the different time periods, then the profits of all the partners are shared in the ratio of time periods for which their amounts were invested.

(iii) If the partners invest different amounts for different time periods, then their profits are shared in the ratio of products of respective investments with the time period for each partner, individually.

Thus gain or loss is divided in the ratio of 'money-time' capitals.

NOTE Sometimes different problems are solved on the basis of partnership to find the expenses.

EXAMPLE 1 *Bhanu and Shafeeq started a business by investing Rs. 36,000 and Rs. 63,000. Find the share of each, out of an annual profit of Rs. 5,500.*

SOLUTION Ratio of shares of Bhanu and Shafeeq

$= 36000 : 63,000 = 4 : 7$

$\therefore$ Share of Bhanu $= 5500 \times \frac{4}{11} =$ Rs. 2000

and Share of Shafeeq $= 5500 \times \frac{7}{11} = 3500$

EXAMPLE 2 *A starts some business with Rs. 50,000. After 3 months B joins him with Rs. 70,000. At the end of the year, in what ratio should they share the profits?*

SOLUTION Ratio of amount of A and $B = 50,000 : 70,000$

Ratio of time periods for A and $B = 12 : 9$

$\therefore$ Ratio of their money-time capital

Investments $= 50000 \times 12 : 70000 \times 9 = 20 : 21$

EXAMPLE 3 *Harsh Vardhan started a business by investing Rs. 36,000. After 4 months Gyan Vardhan joined him with some investment. At the end of the year, the total profit was divided between them in the ratio of 9 : 7. How much capital was invested by Gyan Vardhan in the business?*

SOLUTION $$\frac{36,000 \times 12}{x \times 8} = \frac{9}{7}$$

$$\Rightarrow \quad x = 42,000$$

$\therefore$ Gyan Vardhan invested Rs. 42,000 for 8 months only.

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EXAMPLE 4 *A started some business with Rs. 26,000. After 3 months B joined him with Rs. 16,000. After some more time C joined them with Rs. 25,000. At the end of the year, out of a total profit of Rs. 15,453, C gets Rs. 3825 as his share. How many months after B joined the business did C join?*

SOLUTION Ratio (of share) of profits

$= 26,000 \times 12 : 16000 \times 9 : 25000 \times C$

$= 312 : 144 : 25C$

Now C's share $= \dfrac{25C}{456 + 25C} = \dfrac{3825}{15453}$

$\Rightarrow \quad C = 6$

$\therefore$ Therefore C joined 3 months later than B joined.

EXAMPLE 5 *A, B and C started a business with their investments in the ratio 1 : 2 : 4. After 6 months A invested the half amount more as before and B invested twice the amount as before while C withdrew $\frac{1}{4}$th of the their investments. Find the ratio of their profits at the end of the year.*

SOLUTION Let us assume their initial investments were x, $2x$ and $4x$ respectively.

Therefore, ratio of their investments during the whole year

$= (x \times 6 + \dfrac{3x}{2} \times 6) : (2x \times 6 + 4x \times 6) : (4x \times 6 + 3x \times 6)$

$= 15x : 36x : 42x$

$= 5x : 12x : 14x$

$= 5 : 12 : 14$

$\therefore$ Ratio of their profits $= 5 : 12 : 14$

EXAMPLE 6 *A started a business with Rs. 52,000 and after 4 months B joined him with Rs. 39,000. At the end of the year, out of the total profits B received total Rs. 20,000 including 25% of the profits as commission for managing the business. What amount did A receive?*

SOLUTION Profit's share of A and B

$= 52,000 \times 12 : 39,000 \times 8 = 2 : 1$

Let the profit be Rs. x, then B receives 25% as commission for managing business, the remaining 75% of the total profit x is shared between A and B in the ratio 2 : 1. Hence B will get $\frac{1}{3}$rd part of this in addition to his commission. Hence his total earning

$= 0.25x + \dfrac{1}{3} \times 0.75x$

$= 0.5x = 20,000$

$\Rightarrow \quad x = 40,000$

So, the remaining profit goes to A, hence the profit of A is Rs. 20,000.

EXAMPLE 7 *A working partner gets 20% as his commission of the profit after his commission is paid. If the working partner's commission is Rs. 8000, then what is the total profit in the business?*

SOLUTION Let the total profit be Rs. x. The remaining profit after paying 20% working partner's commission $= (x - 8000)$. Again since 20% of this is working partner's commission,

therefore $\dfrac{20}{100} \times (x - 8000) = 8000$

$\Rightarrow \quad x = 48,000$

$\therefore$ The total profit in the business is Rs. 48,000.

UNITARY METHOD

INTRODUCTORY EXERCISE-4.1

1. If 40 articles cost Rs. 180, the cost of 18 articles is :
   (a) Rs. 18 (b) Rs. 36
   (c) Rs. 81 (d) Rs. 40.5
2. If 20 persons can do a piece of work in 7 days, then the number of persons required to complete the work in 28 days :
   (a) 4 (b) 5
   (c) 14 (d) 10
3. If 20 men can reap a field in 38 days, in how many days will 19 men reap the field?
   (a) 21 days (b) 19 days
   (c) 76 days (d) 40 days
4. A rope makes 260 rounds of a cylinder with base radius 20 cm. How many times can it go round a cylinder with base radius 26 cm?
   (a) 130 (b) 300
   (c) 200 (d) 150
5. A garrison of 750 men has provision for 20 weeks. If at the end of 4 weeks they are reinforced by 450 men, how long will the remaining provision last?
   (a) 10 weeks (b) 11 weeks
   (c) 15 weeks (d) 16 weeks
6. 40 men can build a wall 20 m high in 15 days. The number of men required to build a similar wall 25 m high in 6 days will be :
   (a) 100 (b) 125
   (c) 150 (d) 200
7. Cost of erecting a fence round a square field of 625 hectares at 15 paise per metre is :
   (a) Rs. 500 (b) Rs. 750
   (c) Rs. 100 (d) Rs. 1500
8. 56 workers can reap a field in 8 days. If the work is to be completed in 7 days, the extra workers needed are :
   (a) 7 (b) 8
   (c) 14 (d) 16
9. If 5 persons weave 180 shawls in 12 days, how many shawls will 6 persons weave in 15 days?
   (a) 260 (b) 370
   (c) 270 (d) 360

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10. 15 men take 42 days of 4 hours each to do a piece of work. How many days of 6 hours each would 21 women take if 3 women do as much work as 2 men?
(a) 15 (b) 22
(c) 25 (d) 30

11. 10 engines consume 80 litres diesel when each is running 9 hours a day., How much diesel will be required for 10 engines, each running 15 hours a day, whereas 6 engines of the former type consume as much as 5 engines of latter type?
(a) 160 litre (b) 75 litre
(c) 80 litre (d) $111\frac{1}{9}$ litre

12. If $\frac{3}{5}$th of a cistern is filled in 30 minutes, how much more time will be required to fill the rest of it?
(a) 50 minutes (b) 20 minutes
(c) 15 minutes (d) 45 minutes

AGE RELATED PROBLEMS

INTRODUCTORY EXERCISE-4.2

1. Amit is as much younger to Barkha as he is older to Chaman. If the sum of the ages of Barkha and Chaman is 48 years, what is the present age of Amit?
(a) 18 years (b) 36 years
(c) 24 years (d) 28 years

2. Bipin is 6 times old as Alok. Bipin's age will be twice of Chandan's age after 10 years. If Chandan's 7th birthday was celebrated 3 years ago, what is Alok's present age?
(a) 15 years (b) 12 years
(c) 5 years (d) none of these

3. Renuka got married 8 years ago. Today her age is $1\frac{1}{3}$ times her age at the time of marriage. Her daughter's age is 1/8 times her age. Her daughter's age is :
(a) 3 years (b) 4 years
(c) 6 years (d) 8 years

4. Ten years ago B was twice of A in age. If the ratio of their present ages is 4 : 3, what is the sum of their present ages?
(a) 25 years (b) 30 years
(c) 40 years (d) 35 years

5. The sum of the ages of Aryabhatta and Shridhar is 45 years. Five years ago the product of their ages was 4 times the Aryabhatta's age at that time. The present ages of Aryabhatta and Shridhar respectively are :
(a) 25 and 20 (b) 35 and 10
(c) 36 and 9 (d) 40 and 5

PARTNERSHIP

INTRODUCTORY EXERCISE-4.3

1. A company make a profit of Rs. 9,00,000, 20% of which is paid as taxes. If the rest is divided among the partners P, Q and R in the ratio of $1 : 1\frac{1}{2} : 2$, then the shares of P, Q and R are respectively :
(a) 2,40,000; 3,20,000; 1,60,000
(b) 3,20,000; 2,40,000; 1,60,000
(c) 1,60,000; 3,20,000; 2,40,000
(d) 1,60,000; 2,40,000; 3,20,000

2. We have to divide a sum of Rs. 13,950 among three persons A, B and C. B must get the double of A's share and C must get Rs. 50 less than the double of B's share. The share of A will be :
(a) Rs. 1950 (b) Rs. 1981.25
(c) Rs. 2000 (d) Rs. 2007.75

3. A started business with Rs. 45,000 and B joined afterward with Rs. 30,000. If the profits at the end of one year were divided in the ratio 2 : 1 respectively, then B would have joined A for business after :
(a) 1 month (b) 2 months
(c) 3 months (d) 4 months

4. A and B are partners in a business. They invest in the ratio 5 : 6, at the end of 8 months A withdraws. If they receive profits in the ratio of 5 : 9, find how long B's investment was used?
(a) 12 months (b) 10 months
(c) 15 months (d) 14 months

5. Four milkmen rented a pasture. A put to graze 16 cows for 3 months, B 20 cows for 4 months, C 18 cows for 6 months and D 42 cows for 2 months. If A's share of rent be Rs. 2400, the rent paid by C is :
(a) Rs. 3200 (b) Rs. 4200
(c) Rs. 4000 (d) Rs. 5400

6. A, B and C subscribe Rs. 47000 for a business. If a subscribes Rs. 7000 more than B and B Rs. 5000 more than C, then out of total profit of Rs. 4700, C receives :
(a) Rs. 1200 (b) Rs. 4500
(c) Rs. 1000 (d) none of these

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RATIO AND PROPORTION

INTRODUCTORY EXERCISE-4.4

1. If $A : B = 4 : 5$, $B : C = 3 : 4$, $C : D = 7 : 11$, then $A : D$ is :
(a) 3 : 4 (b) 21 : 55
(c) 21 : 44 (d) 7 : 5

2. Mean proportional between 17 and 68 is :
(a) 51 (b) 24
(c) 4 (d) 34

3. Third proportional between 16 and 36 is :
(a) 64 (b) 144
(c) 81 (d) 49

4. If $a : b = 2 : 3$, then $(5a + b) : (3a + 2b)$ is :
(a) 13 : 12 (b) 15 : 17
(c) 12 : 13 (d) 13 : 11

5. $a = 2b = 3c = 4d$, then $a : b : c : d$ is :
(a) 12 : 3 : 6 : 4 (b) 3 : 4 : 6 : 12
(c) 6 : 12 : 4 : 3 (d) 12 : 6 : 4 : 3

6. The fourth proportional to 4, 7 and 20 is :
(a) 28 (b) 21
(c) 18 (d) 35

7. If $\sqrt{2} : (1 + \sqrt{3}) :: \sqrt{6} : x$, then x is equal to :
(a) $1 + \sqrt{3}$ (b) $\sqrt{3} - 1$
(c) $\sqrt{3} + 3$ (d) $2\sqrt{3}$

8. If $\frac{a}{3} = \frac{b}{4} = \frac{c}{5}$ then $\frac{a + b + c}{b} = ?$
(a) 2 (b) 3
(c) 4 (d) 5

9. If $\frac{a}{b} = \frac{c}{d}$, then :
(a) $\frac{a + b}{a - b} = \frac{c + d}{c - d}$ (b) $\frac{a + b}{a^2} = \frac{c + d}{d^2}$
(c) $\frac{a + b}{a^2} = \frac{c + d}{c^2}$ (d) $ac = bd$

10. If $a : b = b : c = c : d$ then $\frac{a}{b}, \frac{b}{c}, \frac{c}{d}$ are :
(a) in AP
(b) in continued proportion
(c) in GP
(d) both (b) and (c)

11. If $(a + b) : (a - b) = 3 : 2$, the $(a^2 - b^2) : (a^2 + b^2)$ equals :
(a) 5 : 13 (b) 12 : 13
(c) 9 : 4 (d) none of these

12. Two whole numbers, whose sum is 64, can not be in the ratio :
(a) 1 : 7 (b) 3 : 5
(c) 5 : 11 (d) 1 : 2

13. Two numbers are in the ratio 3 : 4. The difference between their squares is 28. Find the greater number :
(a) 12 (b) 8
(c) 24 (d) 16

14. If a, b, c, d, e, f, g are in continued proportion, then the value of $a.b.c.e.f.g$ is :
(a) d^3 (b) d^7
(c) d^6 (d) none of these

15. If A and B shared Rs. 1300 in the ratio 1 : 12, how much did A get?
(a) 120 (b) 1200
(c) 100 (d) 1000

16. Rs. 3960 are divided among A, B and C so that half of A's part, one-third of B's part and one-sixth of C's part are equal. Then C's part is :
(a) 720 (b) 2160
(c) 1080 (d) 810

17. A sum of Rs. 21000 is divided among A, B, C such that shares of A and B are in the ratio of 2 : 3 and those of B and C are in the ratio 4 : 5. The amount received by A is :
(a) Rs. 6000 (b) Rs. 4500
(c) Rs. 4800 (d) Rs. 8400

18. A certain amount was divided between A and B in the ratio 7 : 9. If B's share was Rs. 7200, the total amount was :
(a) Rs. 1280 (b) Rs. 6300
(c) Rs. 5600 (d) Rs. 12800

19. Rs. 11250 are divided among A, B and C so that A may receive one-half as much as B and C together receive and B receives one-fourth of what A and C together receive. The share of A is more than that of B by :
(a) Rs. 2500 (b) Rs. 1500
(c) Rs. 1800 (d) Rs. 650

20. A girl 1.2 metre tall casts a shadow 1.1 m at the time when a building casts a shadow 6.6 m long. The height of the building is :
(a) 2.7 m (b) 7.2 m
(c) 6.0 m (d) 5.5 m

21. The prices of Bajaj Scooter and Bajaj Pulser are in the ratio of 4 : 9. If the Bajaj Pulser costs Rs. 30,000 more than a Bajaj Scooter, the price of Bajaj Pulser is :
(a) Rs. 63,000 (b) Rs. 45,000
(c) Rs. 54,000 (d) Rs. 60,000

22. What is the ratio whose terms differ by 40 and the measure of which is 2/7?
(a) 16 : 56 (b) 14 : 49
(c) 15 : 36 (d) 16 : 72

23. Two numbers are in the ratio 3 : 5. If 9 be subtracted from each, then they are in the ratio of 12 : 23. The second number is :
(a) 53 (b) 54
(c) 55 (d) 52

24. Consider the following statements :
(1) If both the terms of a ratio are multiplied or divided by the same natural number, then the ratio remains unaltered.

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(2) A statement which states that two ratios are equivalent is called proportion.

(3) If 4 quantities are in proportion, the product of extremes is not equal to the means.

(4) The mean proportion between any two numbers is equal to the square root of their product. The wrong one statements is/are :

(a) 1 (b) 3
(c) 3 and 4 (d) 1 and 4

25. In a mixture of 120 litres, the ratio of milk and water is 2 : 1. If the ratio of milk and water is 1 : 2, then the amount of water (in litres) is required to be added is :

(a) 20 (b) 40
(c) 80 (d) 120

26. A quantity x varies inversely as the square of y. Given that $x = 4$, when $y = 3$, the value of x when $y = 6$ is :

(a) 1 (b) 2
(c) 3 (d) 4

27. Suppose y varies as the sum of two quantities of which one varies directly as x and the other inversely as x. If $y = 6$ when $x = 4$ and $y = 3\frac{1}{3}$ when $x = 3$, then the relation between x and y is :

(a) $x = y + 4$ (b) $y = 2x + \frac{8}{x}$
(c) $y = 2x - \frac{8}{x}$ (d) $y = 2x - \frac{4}{x}$

28. The time period of a pendulum is proportional to the square root of the length of the pendulum. Consider the following statements :

(1) If the length of the pendulum is doubled, then the time period is also doubled.

(2) If the length is halved, then time period becomes one-fourth of the original time period.

The correct assertions are :

(a) 1 (b) 2
(c) neither 1 nor 2 (d) both 1 and 2

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EXERCISE

LEVEL 1

1. Four numbers are in proportion. The sum of the squares of the four numbers is 50 and the sum of the means is 5. The ratio of first two terms is 1 : 3. What is the average of the four numbers?
(a) 2 (b) 3
(c) 5 (d) 6

2. A naughty student breaks the pencil in such a way that the ratio of two broken parts is same as that of the original length of the pencil to one of the larger part of the pencil. The ratio of the other part to the original length of pencil is :
(a) $1 : 2\sqrt{5}$ (b) $2 : (3 + \sqrt{5})$
(c) $2 : \sqrt{5}$ (d) can't be determined

3. If $\frac{x+y}{x-y} = \frac{5}{3}$ and $\frac{x}{(y+2)} = 2$, the value of (x, y) is :
(a) (4, 1) (b) (2, 8)
(c) (1, 4) (d) (8, 2)

4. If $a^3 + b^3 : a^3 - b^3 = 133 : 117$; find $a : b$:
(a) 2 : 3 (b) 5 : 4
(c) 5 : 2 (d) none of these

5. A student obtained equal marks in History and Sociology. The ratio of marks in Sociology and Geography is 2 : 3 and the ratio of marks in History and Philosophy is 1 : 2. If he has scored an aggregate of 55% marks. The maximum marks in each subject is same. In how many subjects did he score equal to or greater than 60% marks?
(a) 1 (b) 2
(c) 3 (d) none of these

6. The ratio of income of Anil and Mukesh is 2 : 3. The sum of their expenditure is Rs. 8000 and the amount of savings of Anil is equal to the amount of expenditure of Mukesh. What is the sum of their savings?
(a) 22,000 (b) 4,000
(c) 16,000 (d) 12,000

7. Hutch and Essar entered into a partnership just 5 months ago. The ratio of profit claimed by Hutch and Essar is 6 : 17. If Essar had just started his business 12 months ago with Rs. 1275, what is the amount contributed by Hutch?
(a) Rs. 980 (b) Rs. 1080
(c) Rs. 1200 (d) Rs. 998

8. A child has three different kinds of chocolates costing Rs. 2, Rs. 5 and Rs. 10. He spends total Rs. 120 on the chocolates. What is the minimum possible number of chocolates, he can buy, if there must be atleast one chocolate of each kind?
(a) 22 (b) 19
(c) 17 (d) 15

9. In the previous problem (no. 8) what is the maximum possible no. of chocolates?
(a) 52 (b) 53
(c) 55 (d) 60

10. Mr. Teremere and Mr. Meretere have 5 chocolates and 3 chocolates with them respectively. Meanwhile Mr. Khabbu Singh joined them and all 8 chocolates were distributed equally among all these three people. In turn Khabbu Singh gave Rs. 16 to Mr. Teremere and Mr. Meretere, since Khabbu Singh did not has any chocolate. What is the difference of amounts received by Teremere and Meretere? Given that the amount was shared in proportion of chocolates received by Khabbu Singh.
(a) Rs. 8 (b) Rs. 12
(c) Rs. 14 (d) Rs. 15

11. Rs. 4536 is divided among 4 men, 5 women and 2 boys. The ratio of share of a man, a woman and a boy is 7 : 4 : 3. What is the share of a woman?
(a) Rs. 336 (b) Rs. 498
(c) Rs. 166 (d) Rs. 256

12. The concentration of petrol in three different mixtures (petrol and kerosene) is $\frac{1}{2}$, $\frac{3}{5}$ and $\frac{4}{5}$ respectively. If 2 litres, 3 litres and 1 litre are taken from these three different vessels and mixed. What is the ratio of petrol and Kerosene in the new mixture?
(a) 4 : 5 (b) 3 : 2
(c) 3 : 5 (d) 2 : 3

13. Time period (T) of pendulum is directly proportional to the square root of length of string by which bob is attached to a fixed point and inversely proportional to the square root of gravitational constant 'g'. Time period of a bob is 3 seconds when the gravitational constant g is 4 m/sec^2 and length of string is 9 metre, what is the time period of a bob having a string of length 64 metre and gravitational constant 16 m/sec^2?
(a) 4 seconds (b) 12 seconds
(c) 16 seconds (d) 10 seconds

14. In a milk shoppe there are three varieties of milk, 'Pure', 'Cure' and 'Lure'. The 'Pure' milk has 100% concentration of milk. The ratio of milk is to water in the 'Cure' is 2 : 5 and in the Lure it is 3 : 8 respectively. Sonali purchased 14 litres of Cure and 22 litres of Lure milk and mixed them. If she wanted to make the concentration of milk in the mixture of purchased milk to 50%. How many litres of 'Pure' milk she is needed?
(a) 6 litres (b) 8 litres
(c) 16 litres (d) 18 litres

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15. In the squadron of Indian Air Force the ratio of Sukhoi is to Mig and Jaguar together is 5 : 7 and the ratio of Jaguar is to Sukhoi and Mig together is 1 : 2 Find the ratio of Sukhoi and Mig :
(a) 2 : 7 (b) 3 : 5
(c) 3 : 1 (d) 5 : 3

16. During our campaign against child labour we have found that in three glass making factories A, B and C there were total 33 children aged below 18 were involved. The ratio of male to female in A, B and C was 4 : 3, 3 : 2 and 5 : 4 respectively. If the no. of female children working in the factories B and C be equal then find the no. of female children working in factory A :
(a) 5 (b) 2
(c) 8 (d) 6

17. The value of a diamond is directly proportional to the square of its weight. A diamond unfortunately breaks into three pieces with weights in the ratio of 3 : 4 : 5 thus a loss of Rs. 9.4 lakh is incurred. What is the actual value of diamond :
(a) 28.8 lakh (b) 13.5 lakh
(c) 14.4 lakh (d) 18.8 lakh

18. In the Ruchika's wallet there are only Rs. 16, consisting of 10 paise, 20 paise and Re. 1 coins. The ratio of no. of coins of 10 paise and 20 paise is 6 : 1. The minimum no. of Re 1 coin is :
(a) 5 (b) 12
(c) 4 (d) 8

19. There are two vessels containing the mixture of milk and water. In the first vessel the water is 2/3 of the milk and in the second vessel water is just 40% of the milk. In what ratio these are required to mix to make 24 litres mixture in which the ratio of water is to milk is 1 : 2?
(a) 4 : 3 (b) 5 : 7
(c) 5 : 2 (d) 7 : 5

20. Nehru Ji had 'n' chocolates. He distributed them among 4 children in the ratio of $\frac{1}{2} : \frac{1}{3} : \frac{1}{5} : \frac{1}{8}$. If he gave them each one a complete chocolate, the minimum no. of chocolates that he had :
(a) 139 (b) 240
(c) 278 (d) none of these

21. The ratio of working efficiency of A and B is 5 : 3 and the ratio of efficiency of B and C is 5 : 8. Who is the most efficient ?
(a) A (b) B
(c) C (d) can't be determined

22. If $4A = 5B$ and $3A = 2C$, the ratio of $B : C$ is :
(a) 4 : 3 (b) 5 : 8
(c) 8 : 15 (d) 10 : 15

23. Equal quantities of three mixtures of milk and water are mixed in the ratio of 1 : 2, 2 : 3 and 3 : 4. The ratio of water and milk in the mixture is :
(a) 193 : 122 (b) 122 : 193
(c) 61 : 97 (d) 137 : 178

24. The ratio of age of A and B is 8 : 9 and the age of B is 2/3 of C's age and age of C is $\frac{9}{13}$ times the age of D. If the age of B is 18 years then the age of C is :
(a) 36 years (b) 39 years
(c) 27 years (d) 54 years

25. A milk man has a mixture of milk in which ratio of milk and water is 5 : 3. He sells 40 litres of milk *i. e.*, mixture then he adds up 15 litres of pure water. Now the ratio of milk and water is 5 : 4. What is the new quantity of mixture?
(a) 72 litres (b) 270 litres
(c) 135 litres (d) data insufficient

26. A and B are two alloys of copper and tin prepared by mixing the respective metals in the ratio of 5 : 3 and 5 : 11 respectively. If the alloys A and B are mixed to form a third alloy C with an equal proportion of copper and tin, what is the ratio of alloys A and B in the new alloy C?
(a) 3 : 5 (b) 4 : 5
(c) 3 : 2 (d) 2 : 3

27. A hotel incurs two types of expenses, one which is fixed and others depend upon no. of guests. When there are 10 guests, total expenses of hotel are Rs. 6000. Also when there are 25 guests average expenses per guests are Rs. 360? What is the total expenses of hotel when there are 40 guests?
(a) Rs. 8,000 (b) Rs. 12,000
(c) Rs. 15,500 (d) none of these

28. The ratio of third proportional to 21 and 42 and mean proportional to 16 and 49 is :
(a) 3 : 1 (b) 2 : 3
(c) 4 : 3 (d) 1 : 3

29. The period of the pendulum is directly proportional to the square root of the length of the string. The period of such a pendulum with string of length 16 cm is 52 seconds. Find the length of the string if the period is 65 seconds :
(a) 4.5 cm (b) 5 cm
(c) 6 cm (d) none of these

30. For any two numbers m, n; $(m + n) : (m - n) : mn = 7 : 1 : 60$ Find the value of $\frac{1}{m} : \frac{1}{n}$
(a) 4 : 3 (b) 8 : 6
(c) 3 : 4 (d) 7 : 8

31. Rs. 960 were distributed among A, B, C and D in such a way that C and D together gets half of what A and B together gets and C gets one-third amount of B. Also D gets $\frac{5}{3}$ times as much as C. What is the amount of A?
(a) Rs. 240 (b) Rs. 280
(c) Rs. 320 (d) data insufficient

32. Find the value of $\frac{p^2 + q^2}{r^2 + s^2}$, if $p : q :: r : s$.
(a) $\frac{1}{4}$ (b) $\frac{1}{9}$
(c) $\frac{ps}{rq}$ (d) $\left(\frac{p - q}{r - s}\right)^2$

33. Find the value of $\frac{p^2 + q^2}{r^2 + s^2}$, if $p : q :: r : s$.
(a) $\frac{1}{4}$ (b) $\frac{pq}{rs}$
(c) $\left(\frac{p + q}{r + s}\right)^2$ (d) both (b) and (c)

34. The speeds of rickshaw, car and scooter are in the ratio of 3 : 5 : 6. What is the ratio of time taken by each one of them for the same distance?
(a) 6 : 5 : 3 (b) 10 : 6 : 5
(c) 12 : 7 : 6 (d) data insufficient

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35. Divide Rs. 6940 in such a way that A gets $\frac{2}{3}$rd of what B gets and B gets $\frac{3}{5}$th of what C gets? What is the share of A and B together?
(a) Rs. 1982 (b) Rs. 1388
(c) Rs. 3470 (d) none of these

36. The ratio of age of A and B is $x : y$. If A's age is increased by 3 years and B's age is increased by 2 years then new ratio of their ages becomes 24 : 25. Given that the sum of their actual ages is 93 years. Find the actual ratio of their ages.
(a) 21 : 22 (b) 42 : 45
(c) 45 : 48 (d) can't be determined

37. $a : b = 4 : 9$ if 4 is added to both of the numbers then the new ratio becomes 21 : 46. What is the difference between a and b?
(a) 80 (b) 100
(c) 125 (d) 130

38. The ratio of ages of Rahul and Deepesh is 3 : 5. 10 years later this ratio becomes 5 : 7. What is the present age of Deepesh?
(a) 20 years (b) 50 years
(c) 25 years (d) 40 years

39. When 5 is added to the numerator and denominator both of a (positive) fraction, then the new ratio of numerator to denominator becomes 11 : 15. What is the original ratio?
(a) 17 : 25 (b) 3 : 5
(c) 28 : 40 (d) none of these

40. Five numbers a, b, c, d and e are in the ratio of 2 : 3 : 5 : 8 : 9 and their sum is 162. Find the average of all these numbers :
(a) 27 (b) 30
(c) 32.4 (d) can't be determined

41. 6 pumps of Kirlosker can fill a tank in 7 days and 2 similar pumps of USHA can fill the same tank in 18 days. What is the ratio of the efficiency of a Kirlosker pump and a USHA pump?
(a) 6 : 7 (b) 7 : 6
(c) 7 : 54 (d) can't be determined

42. 16 persons can reap $\frac{1}{5}$th field in 6 days. How many persons (with same efficiency) are required to reap rest of the field in 8 days?
(a) 27 (b) 54
(c) 48 (d) 64

43. The LCM of two numbers is 210 and their ratio is 2 : 3. The sum of these numbers is :
(a) 210 (b) 175
(c) 315 (d) can't be determined

44. What number must be subtracted from each of the numbers 53, 21, 41, 17 so that the remainders are in proportion?
(a) 1 (b) 3
(c) 5 (d) none of these

45. The angles of a triangle are in the ratio of 2 : 3 : 4. Find the measurement of greatest angle :
(a) 30° (b) 60°
(c) 100° (d) 80°

46. In a wallet the ratio of 25 paise, 50 paise and Re 1 coins are in the ratio of 12 : 4 : 3, which amounts to Rs. 600. Find the no. of coins of 25 paise :
(a) 200 (b) 225
(c) 275 (d) none of these

47. $(x-a):(x-b):(x-c) = 11:9:5$, where $x = \frac{a+b+c}{2}$. What is the ratio of a, b, c?
(a) 10 : 8 : 7 (b) 3 : 5 : 9
(c) 7 : 8 : 10 (d) 6 : 5 : 3

48. Petrol is 7 times heavy than Kerosene and Castrol mobil is 18 times as heavy as Kerosene. What should be the ratio of petrol and mobil in the new mixture to get the mixture which must be 11 times as heavy as kerosene?
(a) 3 : 4 (b) 7 : 4
(c) 9 : 19 (d) 9 : 10

49. A girl buys 2 pigeons for Rs. 182. She sells one at a loss of 5% and another at a profit if 8%. But she neither gains nor loses on the whole. Find the price of pigeon which has sold at a profit :
(a) Rs. 112 (b) Rs. 85
(c) Rs. 70 (d) can't be determined

50. The ratio of prices of Cello and Rotomac pens in 2000 were in the ratio of 3 : 5. In 2005 the price of Cello pen trebles itself and the price of Rotomac pen is increased by Rs. 100, then the new ratio of prices of the same pens becomes 4 : 5. What was the original price of the Rotomac pen in 2000?
(a) Rs. 60 (b) Rs. 80
(c) Rs. 100 (d) Rs. 120

51. A goldsmith has 361 rings of gold. He sells some of them at a loss of 4% and rest at a profit of 15% making overall profit of 8%. Find the no. of rings sold at a profit of 15% :
(a) 171 (b) 133
(c) 218 (d) 228

52. Michel travelled from New York to New Jersey covering total distance of 250 miles in 8 hr. partly by car at 30 miles/hr. and rest by train at 35 miles/hr. The distance travelled by car is :
(a) 150 miles (b) 80 miles
(c) 220 miles (d) 180 miles

53. A rabit takes 22 leaps for every 17 leaps of cat and 22 leaps of a rabit are equal to 17 leaps of the cat. What is the ratio of the speeds of rabit and cat?
(a) 1 : 1 (b) 484 : 289
(c) 17 : 22 (d) none of these

54. Rs. 171 are divided among four friends in the ratio of $\frac{1}{3}:\frac{1}{4}:\frac{1}{5}:\frac{1}{6}$. What is the amount of the person who got the greatest share?
(a) 14 (b) 40
(c) 36 (d) 60

55. 10 years ago the age of Karishma was $\frac{1}{3}$rd of the age of Babita. 14 years hence the ratio of ages of Karishma and Babita will be 5 : 9. Find the ratio of their present ages :
(a) 13 : 29 (b) 11 : 27
(c) 29 : 17 (d) 13 : 25

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56. The ratio of numerator to a denominator of a fraction is $\frac{1}{5}$ when x and $5x$ are added to the numerator and denominator respectively to the given fraction then the ratio of the new fraction will be :
(a) 1 : 1 (b) 1 : 25
(c) 1 : 5 (d) 2 : 7

57. x varies directly as y and x varies inversely as the square of z. When $y = 75$ and $x = 6$, then $z = 5$. Find the value of x when $y = 24$ and $z = 4$:
(a) 1 (b) 2
(c) 3 (d) 4

58. x varies directly as $(y^2 + z^2)$. At $y = 1$ and $z = 2$, the value of x is 15. Find the value of z, when $x = 39$ and $y = 2$:
(a) 2 (b) 3
(c) 4 (d) 6

59. Weight of a sumo is jointly varies as his height and his age. When height is 1.2 m and age is 20 years his weight is 48 kg. Find the weight of the sumo when his height is 1.5 metre and age is 30 years :
(a) 60 kg (b) 72 kg
(c) 90 kg (d) 58 kg

60. If $(a + b) : (b + c) : (c + a) = 5 : 6 : 9$ and $a + b + c = 10$. What is the value of c :
(a) 2 (b) 3
(c) 5 (d) 7

61. A, B, and C have amounts in the ratio of 3 : 4 : 5. First B gives $\frac{1}{4}$th to A and $\frac{1}{4}$th to C then C gives $\frac{1}{6}$th to A . Find the final ratio of amount of A, B and C, respectively :
(a) 4 : 3 : 5 (b) 5 : 4 : 3
(c) 6 : 4 : 2 (d) 5 : 2 : 5

LEVEL 2

1. A tin contains a mixture of Dew and Sprite in the ratio of 7 : 3 and another tin contains the Dew and Sprite in the ratio of 5 : 4. In what proportion should the solution of two tins be mixed to achieve a perfect proportion of 2 : 1 (in which Dew is 2 times that of sprite).
(a) 10 : 3 (b) 4 : 1
(c) 3 : 10 (d) 3 : 1

2. There are two containers A and B filled with mentha oil with different prices and their volumes are 140 litres and 60 litres respectively. Equal quantities are drawn from both A and B in such a manner that the oil drawn from A is poured into B and the oil drawn from B is poured into A. The price per litre becomes equal in both A and B. How much oil is drawn from each of A and B :
(a) 40 litre (b) 21 litre
(c) 42 litre (d) can't be determined

3. A three digit number is such that this number itself is divisible by the sum of its digits. The sum of hundreds and unit digit is 6 while the sum of the tens and unit digit is 5. What is the ratio of unit and tens digit :
(a) 1 : 2 (b) 2 : 3
(c) 3 : 4 (d) 2 : 7

4. Sachin bought 1.5 kg fresh grapes. The ratio of water is to pulp was 4 : 1. When his naughty child crushed these grapes, then some water get wasted. Now the ratio of water is to pulp is 3 : 2. What is the total amount of the crushed grapes?
(a) 0.5 kg (b) 1 kg
(c) 0.75 kg (d) none of these

5. A man bought 9 mangoes for a rupee and sold them at 6 mangoes for a rupee. What is the ratio of profit to the cost price?
(a) $\frac{3}{10}$ (b) $\frac{3}{2}$
(c) 1/2 (d) none of these

6. Half of the volume of Petrol and kerosene mixture of ratio 7 : 5 is converted into a mixture of ratio 3 : 1 by the substitution (or replacement) method. While the mixture of ratio 7 : 5 was formed from the mixture of 7 : 3 by adding the Kerosene in it. If 240 litres petrol is required in the replacement method, what is the total amount of Kerosene was added to prepare the mixture of 7 : 5?
(a) 100 litres (b) 400 litres
(c) 50 litres (d) 200 litres

7. Sometimes ago in a Cinema hall a blockbuster movie was being shown. Due to excessive demand of the show, manager of the Cinema hall increased the prices of tickets in all the three categories. 25% in first class, 12.5% in special class and 40% in balcony. If the collection of one show was Rs. 218925, then find the collection from the balcony only given that the ratio of price of ticket (increased price) for first class, special class and balcony is 5 : 9 : 14 :
(a) 109462.5 (b) 58800
(c) 14400 (d) data insufficient

8. In a mixture of petrol and Kerosene petrol is only 99 litres. if this same quantity of petrol would be presented in another mixture of petrol and Kerosene where total volume would be 198 litres less than the actual mixture then the concentration of petrol in the actual mixture would have been 13.33% point less than that of the new mixture. What is the concentration of petrol in actual mixture?
(a) 20% (b) 16.66%
(c) 26.66% (d) 8.33%

9. A drum of 20 litres is filled with milk. A milkman has only two measuring vessels of 3 litres and 5 litres without any calibration. He has to measure four litres of milk for a customer without using anyother vessel. Minimum how many operations are required for this work, where an operation is counted if the milk is transferred from one vessel to another vessel?
(a) 5 (b) 6
(c) 8 (d) 11

10. The ages of Vinay, Varsha, Veera and Vikram are in arithmetic progression, but not in order. The ratio of ages of Vinay and Varsha is 6 : 5 and Veera is to Vikram is 7 ; 8. Two years later

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the age of Varsha and Vikram will be 2 : 3. Find the ratio of ages of Vinay and Veera :

(a) 7 : 6 (b) 5 : 8
(c) 6 : 7 (d) 8 : 9

11. A container is filled with the mixture of milk and water. The ratio of milk and water is same. Bobby and Sunny increases the concentration to 60%. Bobby makes it by adding the milk and Sunny makes it by replacing the mixture with milk. What is the percentage of milk added by Bobby to that of milk replaced by Sunny :

(a) 100% (b) 120%
(c) 133.33% (d) none of these

12. There are two vessels A and B containing 25 litres each of pure milk and pure water respectively. 5 litres of milk from A is taken and poured into B, then 6 litres of mixture from B is taken and poured in A. What is the ratio of water in A and B respectively :

(a) 4 : 5 (b) 1 : 4
(c) 5 : 4 (d) 2 : 3

13. The ratio of age between A and B is 6 : 5 and the age of each C and D is $\frac{9}{10}$ times that of B. Age of F is less than A but greater than B. The ratio of ages between B and E is 2 : 3 also age of A is 3 years less than E. What is the ratio of ages of A and F if all the ages are in integers?

(a) 12 : 11 (b) 9 : 7
(c) 24 : 19 (d) 12 : 13

14. The ratio of students in a coaching preparing for B.Tech and MBA is 4 : 5. the ratio of fees collected from each of B. Tech and MBA student is 25 : 16. If the total amount collected from all the students is 1.62 lakh, what is the total amount collected from only MBA aspirants?

(a) Rs. 62,000 (b) Rs. 72,000
(c) Rs. 80,000 (d) none of these

15. The cost of the marble varies directly with square of its weight. Marble is broken into 3 parts whose weights are in the ratio 3 : 4 : 5. If marble had been broken into three equal parts by weight then there would have been a further loss of Rs. 1800. What is the actual cost of the original (or unbroken) marble?

(a) Rs. 3600 (b) Rs. 10,800
(c) Rs. 2160 (d) none of these

Directions for 16 and 17: Four friends A, B, C and D have some money among them one day they decided to equate the money, so first A gave B what B had initially, then B gave C what C had initially. Again C gave D what D had initailly and finally D had doubled the money of A. Thus each of them had equal sum of Rs. 48.

16. What was the initial amount of B?

(a) Rs. 36 (b) Rs. 54
(c) Rs. 45 (d) Rs. 42

17. What was the amount with C after second transaction?

(a) Rs. 45 (b) Rs. 69
(c) Rs. 72 (d) Rs. 84

18. Hari and Murli have 24 cows and 30 cows respectively. Both of them together hired a grazing field for the whole month of November. Initially Hari's cows used for grazing then for the remaining days of the month Murli's cows grazed it. If Hari has paid Rs. 3500 and Murli has paid Rs. 5000 for grazing then for how many days Hari used the grazing field :

(a) 14 (b) 16
(c) 21 (d) 20

19. The speeds of scooter, car and train are in the ratio of 1 : 4 : 16. If all of them covers equal distance then the ratio of time taken/velocity for the each of the vehicle is :

(a) 256 : 16 : 1 (b) 1 : 4 : 16
(c) 16 : 4 : 1 (d) 16 : 1 : 4

20. Radhika purchased one dozen bangles. One day she slipped on the floor fell down. What can not be the ratio of broken to unbroken bangles :

(a) 1 : 2 (b) 1 : 3
(c) 2 : 3 (d) 1 : 5

21. If $\frac{a}{b}=\frac{c}{d}=\frac{e}{f}$, then the value of $\left(\frac{a^n p+c^n q+e^n r}{b^n p+d^n q+f^n r}\right)^{1/n}$:

(a) $\frac{ad}{bc}$ (b) $\frac{af}{be}$
(c) $\frac{ck}{dk}$ (d) none of these

22. At a casino in Mumbai, there are 3 tables A, B and C. The pay-offs at A is 10 : 1, at B is 20 : 1 and at C is 30 : 1. If a man bets Rs. 200 at each table and win at two of the tables, what is the maximum and minimum difference between his earnings can be?

(a) Rs. 2500 (b) Rs. 2000
(c) Rs. 4000 (d) none of these

23. If $m=\frac{4pq}{p+q}$, then the value of $\frac{m+2p}{m-2p}+\frac{m+2q}{m-2q}$:

(a) 2 (b) 4
(c) $\frac{2mpq}{(p+q)}$ (d) none of these

24. The ratio of volumes of two cubes is 8 : 27. What is the ratio of surface area of these cubes respectively?

(a) 2 : 3 (b) 4 : 9
(c) 8 : 19 (d) 9 : 4

25. Dudheri Lal has two jars. Jar A is completely filled with milk and another jar B is totally empty. Before selling the milk in a town he transferred some milk in to the empty jar B then he fill the jar A with water. Once again he transferred the mixture of milk from A to B so that B is completely filled. Which one of the following is correct?

(a) Concentration of milk in B cannot be less than 75%
(b) Concentration of milk in B cannot be greater than 75%
(c) Concentration of milk is always 75%
(d) none of the above

26. In ABC corporation there are some management trainees. These trainees are divided into 3 groups A, B and C for 3 different projects in the ratio of 3 : 4 : 5 respectively, where P, Q, R are the projects-in-charge of A, B, C respectively. The difference between the no. of trainees in A and C is not greater than 3. Also P, Q, R belongs to the group of trainees. The no. of assistant of Q is less than the no. of assistance of R by :

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(a) 33.33% (b) 20%
(c) 25% (d) 16.66%

27. If $\frac{a}{b}=\frac{b}{c}=\frac{c}{d}=\frac{d}{e}=\frac{e}{f}=\frac{1}{3}$, find the value of $\left(\frac{a+b+c+d+e}{b+c+d+e+f}\right)$:

(a) $\frac{1}{81}$ (b) $\frac{1}{27}$
(c) $\frac{1}{3}$ (d) 1

28. An engine can move at the speed of $\frac{20}{3}$ m/s without any wagon attached. Reduction in the speed of the train is directly proportional to the square root of the no. of wagons attached to the engine. When there are only four wagons attached its speed is $\left(\frac{50}{9}\right)$ m/s. The greatest no. of wagons with which the engine can move is :

(a) 144 (b) 143
(c) 12 (d) none of these

29. Mrs. Annapurna per day sells exactly four quintal sugar at Rs. 2000 per quintal getting the profit of Rs. 25%. Since she mixes two varities of sugar, one costs Rs. 14 per kg and another costs Rs. 22 per kg. One day due to huge demand in market she had only 3 quintal of the required mixture so she purchased the sugar costs Rs. 17 per kg at Rs. 18 per kg from the wholeseller on that day and then she mixed 300 kg mixture with 100 kg sugar costing Rs. 18 per kg to fullfill the demand of the market selling at the same price. How much percent less does she gain that she would have gained, if she had sufficient quantity of usual mixture of sugar?

(a) 12.5% (b) 18.18%
(c) 62.5% (d) can't be determined

30. Two vessels P and Q contain 'a' litres of petrol and 'b' litres of Kerosene respectively. 'c' litres of petrol and same quantity of Kerosene is taken out and then transferred to Q and P respectively. This process is repeated several times. If after the first operation the quantity of petrol or Kerosene in either of P and Q does not change. What is the value of 'c'?

(a) $\frac{ab}{(a-b)}$ (b) $\frac{2ab}{(a+b)}$
(c) $\frac{ab}{(a+b)}$ (d) $\left(\frac{a}{b}\right)^2$

31. Arvind Singh purchased a 40 seater bus. He started his services on route no. 2 (from Terhipuliya to Charbagh with route length of 50 km). His profit (P) from the bus depends upon the no. of passangers over a certain minimum number of passangers 'n' and upon the distance travelled by bus. His profit is Rs. 3600 with 29 passangers in the bus for a journey of 36 km and Rs. 6300 when there are 36 passengers. travelled for 42 km. What is the minimum no. of passangers are required so that he will not suffer any loss?

(a) 12 (b) 20
(c) 18 (d) 15

32. Three cats are roaming in a zoo in such a way that when cat A takes 5 steps, B takes 6 steps and cat C takes 7 steps. But the 6 steps of A are equal to the 7 steps of B and 8 steps of C. What is the ratio of their speeds :

(a) 140 : 144 : 147 (b) 40 : 44 : 47
(c) 15 : 21 : 28 (d) 252 : 245 : 240

33. In a family there were n people. The expenditure of rice per month in this family is directly proportional to the 5 times the square of no. of people in the family. If the elder son left the family to study in USA there was decrease in consumption of 95 kg rice per month. What is the value of n?

(a) 5 (b) 12
(c) 9 (d) 10

34. One day in summer I wanted to chill me out, I went to a cool corner. I gave him a note of Rs. 10 and asked for a coke costing Rs. 5 per jar and he did so, but he returned me Rs. 5, in the denomination of Re. 1, 50 paise and 25 paise. What could be the ratio of no. of coins of Re. 1, 50 paise and 25 paise respectively :

(a) 2 : 3 : 1 (b) 1 : 7 : 2
(c) 6 : 1 : 3 (d) 2 : 1 : 2

35. If $\frac{a}{b+c}=\frac{b}{c+a}=\frac{c}{a+b}$ and $a+b+c\neq 0$ then the value of $\frac{b}{a+b+c}$ is :

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{1}{4}$ (d) 1

36. The ratio of the density of 3 kinds of petrol P_1, P_2 and P_3 is 9 : 7 : 5. The density of P_1 is 18 gm/cc and P_1, P_2, P_3 are mixed in the ratio of 6 : 5 : 4 by weight. If a litre of P_3 cost Rs. 40, then find the cost of P_3 in 450 kg of mixture of P_1, P_2 and P_3 :

(a) Rs. 380 (b) Rs. 480
(c) Rs. 355 (d) Rs. 448

37. Three persons Amar, Akbar and Anthony agree to pay their hotel bills in the ratio of 3 : 4 : 5. Amar pays the first day's bill which amounts to Rs. 26.65, Akbar pays the second days bill which amounts to Rs. 42.75 and C pays the third day's bill which amounts to Rs. 53.00. When they settle their accounts, which of the following happens?

(a) Amar gives Rs. 3 to C
(b) Akbar gives Rs. 2 to Amar
(c) Amar gives Akbar Rs. 1.95 and Rs. 2 to Anthony
(d) none of the above

38. Find the value of x if $(14x-4):(8x-1)=(3x+8):(9x+5)$:

(a) 1 (b) 1/2
(c) 3/4 (d) none of these

39. Pooja, Shipra and Monika are three sisters. Pooja and Shipra are twins. The ratio of sum of the ages of Pooja and Shipra is same as that of Monika alone. Three years earlier the ratio of age of Pooja and Monika was 2 : 7. What will be the age of Shipra 3 years hence?

(a) 21 years (b) 16 years
(c) 8 years (d) 12 years

40. A couple got married 9 years ago when the age of wife was 20% less than her husband. 6 years from now the age of wife will be only 12.5% less than her husband. Now they have six

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children including single, twins and triplets and the ratio of their ages is 2 : 3 : 4 respectively. What can be the maximum possible value for the present age of this family?
(a) 110 years (b) 103 years
(c) 105 years (d) 83 years

41. The price of a necklace varies directly as the no. of pearls in it. Also, it varies directly as the square root of radius of a pearl. The price of a necklace was Rs. 150. When it had 75 pearls each of radius 1 cm. Find the radius of the pearl of a necklace having 100 pearls whose cost is Rs. 600.
(a) 2 (b) 9
(c) 3 (d) 4

42. The price of a book varies directly as the no. of pages in it and inversely as the time periods in years that have elapsed since the date of purchasing. Two books cost the same, however, the no. of pages in the first book is triple of the second book. if the first book is sold on 18 years ago, how long ago was the second book sold?
(a) 54 years (b) 9 years
(c) 6 years (d) 3 years

43. Akbar and Birbal who purchased the shares for the cost of their basic salaries which are in the ratio of 5 : 6. later on company gave them 40 additional shares to each, due to which the ratio changed to 7 : 8. If the worth of each share is Rs. 75, what is the basic salary of the person who got less shares?
(a) Rs. 10500 (b) Rs. 7500
(c) Rs. 8800 (d) Rs. 9000

44. Distance covered by a train is directly proportional to the time taken and also it varies directly as the square root of fuel used and varies inversely as the no. of wagons attached to it. A train coveres 192 km journey in 20 hours when there are 10 wagons attached to it and total fuel consumption was 256 litre of diesel. Find the consumption of fuel per km when a train goes 200 km in 25 hours with 15 wagons attached to it :
(a) 1.5 *l*/km (b) 2 *l*/km
(c) 2.8 *l*/km (d) 20 *l*/km

45. At Sahara shopping centre, a person can purchase as much articles at a time as his or her age that is a person of n years age can purchase only n similar articles at a time. Amisha is younger to her elder brother who has just entered into his twenties. One day Amisha went to the Sahara shopping centre, she purchased same toffees at a particular rate on the ground floor. But when she reached on third floor she found that she could purchase double the no. of toffees with the same amount as she had spent on the ground floor. Also to purchase the same no. of toffees on the third floor she had to spend Rs. 2 less than that of on the first floor. How many toffees did she buy?
(a) 6 (b) 12
(c) 18 (d) 15

46. A contractor deployed some men to plant 1800 trees in a certain no. of days. But in $\frac{1}{3}$rd of the planned time 120 plants could be less planted so to ful fill the target for the rest of the days every day 20 more plants were planted. Thus it saved one day out of the initially planned no. of days. How many plants he planned to plant each day initially?
(a) 180 (b) 100
(c) 120 (d) 160

47. A and B have to write 810 and 900 pages respectively in the same time period. But A completes his work 3 days ahead of time and B completes 6 days ahead of time. How many pages did A write per hour if B wrote 21 pages more in each hour?
(a) 45 (b) 72
(c) 54 (d) 100

48. Three friends A, B and C decided to share the soda water with D, who had no soda water. A contributed 2 tumbler more than that of B and B contributed 1 tumbler more than that of C and then all of them had equal amount of soda water. In turn D paid money, which was divided among A, B and C in the ratio of their contribution to D. Thus A had gotten thrice as much money as B had gotten. The price of each tumbler of soda water was Rs. 15 and each transaction was integral in numbers either the sharing of money or contribution of soda water. What was the sum of money that B had gotten?
(a) Rs. 15 (b) Rs. 18
(c) Rs. 22.5 (d) none of these

49. In Maa Yatri Temple every devotee offers fruits to the orphans. Thus every orphan receives bananas, oranges and grapes in the ratio of 3 : 2 : 7 in terms of dozen. But the weight of a grape is 24 gm and weight of a banana and an orange are in the ratio of 4 : 5, while the weight of an orange is 150 gm. Find the ratio of all the three fruits in terms of weight, that an orphan gets :
(a) 90 : 75 : 42 (b) 180 : 150 : 82
(c) 75 : 42 : 90 (d) none of these

Final Round

1. In two alloys the ratio of Iron and copper is 4 : 3 and 6 : 1 respectively. If 14 kg of the first alloy and 42 kg of the second alloy are mixed together to form a new alloy, then what will be the ratio of copper to iron in the new alloy :
(a) 11 : 3 (b) 11 : 8
(c) 8 : 11 (d) none of these

2. In a zoo, there are rabbits and pigeons. If heads are counted, there are 340 heads and if legs are counted there are 1060 legs. How many pigeons are there?
(a) 120 (b) 150
(c) 180 (d) 170

3. 6 litre is taken out from a vessel full of Kerosene and substituted by pure petrol. This process is repeated two more times. Finally the ratio of petrol and Kerosene in the mixture becomes 1701 : 27. Find the volume of the original solution :
(a) 14 litre (b) 16 litre
(c) 8 litre (d) 42 litre

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4. In three vessels, each of 25 litres capacity, mixture of milk and water is filled. The ratio of milk and water are 3 : 1, 2 : 3, 4 : 3 in the respective vessels. If all the three vessels are emptied into a single large vessel, then what will be the ratio of water to milk in the resultant mixture?
(a) 179 : 241 (b) 197 : 214
(c) 219 : 117 (d) 179 : 234

5. Two liquids are mixed in the ratio 4 : 3 and the mixture is sold at Rs. 20 with a profit of $33\frac{1}{3}\%$. If the first liquid is costlier than the second by Rs. 7. Find the sum of costs of both the liquids :
(a) Rs. 11 (b) Rs. 29
(c) Rs. 35 (d) Rs. 70

6. Two alloys made up of copper and tin. The ratio of copper and tin in the first alloy is 1 : 3 and in the second alloy it is 2 : 5. In what ratio the two alloys should be mixed to obtain a new alloy in which the ratio of tin and copper be 8 : 3?
(a) 3 : 5 (b) 4 : 7
(c) 3 : 8 (d) 5 : 11

7. Three vessels having volumes in the ratio of 2 : 3 : 5 are full of a mixture of water and milk. In the first vessel ratio of water and milk is 1 : 3, in second 2 : 3 and in third vessel, 2 : 5. If all the three vessels were poured out in a large container, what is the resulting ratio of milk and water?
(a) 43 : 96 (b) 438 : 962
(c) 348 : 962 (d) 962 : 438

8. The number of oranges in three baskets are in the ratio of 3 : 4 : 5. In which ratio the no. of oranges in first two baskets must be increased so that the new ratio becomes 5 : 4 : 3?
(a) 1 : 3 (b) 2 : 1
(c) 3 : 4 (d) 2 : 3

9. A vessel of capacity 2 litre has 25% alcohol and another vessel of capacity 6 litre had 40% alcohol. The total liquid of 8 litre was poured out in a vessel of capacity 10 litre and thus the rest part of the vessel was filled with the water. What is the new concentration of mixture?
(a) 31% (b) 71%
(c) 49% (d) 29%

10. Alloy A contains 40% gold and 60% silver. Alloy B contains 35% gold and 40% silver and 25% copper. Alloys A and B are mixed in the ratio of 1 : 4. What is the ratio of gold and silver in the newly formed alloy is?
(a) 20% and 30% (b) 36% and 44%
(c) 25% and 35% (d) 49% and 36%

11. Dia and Urea are two chemical fertilizers. Dia is consists of N, P and K and Urea conists of only N and P. A mixture of Dia and Urea is prepared in which the ratio of N, P and K is 26%, 68% and 6% respectively. The ratio of N, P and K in Dia is 20%, 70% and 10% respectively. What is the ratio of N and P in the Urea?
(a) 27% and 63% (b) 33% and 67%
(c) 35% and 65% (d) 70% and 30%

12. The ratio of copper and nickel by weight in the two alloys X and Y are 2 : 7 and 5 : 4. How many kilogram of the alloy X and Y are required to make 42 kg of new alloy Z in which the ratio of copper and nickel is same?
(a) 6 kg and 36 kg (b) 10 kg and 32 kg
(c) 7 kg and 35 kg (d) none of these

13. There are two alloys made up of copper and aluminium. In the first alloy copper is half of the aluminium and in the second alloy copper is thrice as much as aluminium. How many times the second alloy must be mixed with first alloy to get the new alloy in which copper is twice as that of aluminium?
(a) 2 (b) 3
(c) 4 (d) 5

14. There are 90 litres castrol and 150 litres CRB mobil oils. The price of castrol is Rs. 80 per litre and price of CRB is Rs. 75 per litre. Equal amount of castrol and CRB is taken out and then CRB is poured out in the vessel of castrol and castrol is poured out in the vessel of CRB. Now the rate of both the mixtures is same. What is the amount of mobil oil taken out from each of the vessel?
(a) 45 litres (b) 56.25 litres
(c) 24.5 litres (d) 36 litres

15. There are two containers, the first contain, 1 litre pure water and the second contains 1 litre of pure milk. Now 5 cups of water from the first container is taken out is mixed well in the second container. Then, 5 cups of this mixture is taken out and is mixed in the first container. Let A denote the proportion of milk in the first container and B denote the proportion of water in the second container then :
(a) $A < B$ (b) $A = B$
(c) $A > B$ (d) can't be determined

16. Mitthu Bhai sells rasgulla (a favorite indian sweets) at Rs. 15 per kg. A rasgulla is madeup of flour and sugar in the ratio of 5 : 3. The ratio of price of sugar and flour is 7 : 3 (per kg). Thus he earns $66\frac{2}{3}\%$ profit. What is the cost price of sugar?
(a) Rs. 10/kg (b) Rs. 9/kg
(c) Rs. 18/kg (d) Rs. 14/kg

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Answers

INTRODUCTORY EXERCISE-4.1

1 (c)	2. (b)	3. (d)	4. (c)	5. (a)	6. (b)	7. (d)	8. (b)	9. (c)	10. (d)
11. (a)	12. (b)								

INTRODUCTORY EXERCISE-4.2

1 (c)	2. (c)	3. (b)	4. (d)	5. (c)					

INTRODUCTORY EXERCISE-4.3

1 (d)	2. (c)	3. (c)	4. (a)	5. (d)	6. (c)				

INTRODUCTORY EXERCISE-4.4

1 (b)	2. (d)	3. (c)	4. (a)	5. (d)	6. (d)	7. (c)	8. (b)	9. (a)	10. (d)
11. (b)	12. (d)	13. (b)	14. (c)	15. (c)	16. (b)	17. (c)	18. (d)	19. (b)	20. (b)
21. (c)	22. (a)	23. (c)	24. (b)	25. (d)	26. (a)	27. (c)	28. (c)		

LEVEL-1

1 (b)	2. (b)	3. (d)	4. (c)	5. (b)	6. (d)	7. (b)	8. (c)	9. (b)	10. (b)
11. (a)	12. (b)	13. (a)	14. (c)	15. (d)	16. (d)	17. (c)	18. (c)	19. (b)	20. (a)
21. (a)	22. (c)	23. (a)	24. (c)	25. (c)	26. (c)	27. (b)	28. (a)	29. (d)	30. (c)
31. (b)	32. (d)	33. (d)	34. (b)	35. (c)	36. (c)	37. (b)	38. (c)	39. (d)	40. (c)
41. (a)	42. (c)	43. (b)	44. (c)	45. (d)	46. (d)	47. (c)	48. (b)	49. (c)	50. (b)
51. (d)	52. (d)	53. (a)	54. (d)	55. (a)	56. (c)	57. (c)	58. (b)	59. (c)	60. (c)
61. (d)									

LEVEL-2

1 (a)	2. (c)	3. (b)	4. (c)	5. (c)	6. (d)	7. (d)	8. (a)	9. (c)	10. (c)
11. (d)	12. (b)	13. (a)	14. (b)	15. (d)	16. (c)	17. (d)	18. (a)	19. (a)	20. (c)
21. (c)	22. (c)	23. (a)	24. (b)	25. (a)	26. (c)	27. (c)	28. (b)	29. (a)	30. (c)
31. (d)	32. (a)	33. (d)	34. (b)	35. (b)	36. (b)	37. (c)	38. (b)	39. (c)	40. (b)
41. (b)	42. (c)	43. (b)	44. (b)	45. (b)	46. (b)	47. (c)	48. (a)	49. (a)	

FINAL ROUND

1 (d)	2. (b)	3. (c)	4. (a)	5. (b)	6. (b)	7. (d)	8. (b)	9. (d)	10. (b)
11. (c)	12. (c)	13. (c)	14. (b)	15. (b)	16. (d)				

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Hints & Solutions

LEVEL 1

1. $a : b :: c : d$

$a^2 + b^2 + c^2 + d^2 = 50$

$b + c = 5$

and $a : b = 1 : 3$

If consider $a : b = 1 : 3$ as it is, then

$c = 2$ $(5 - 3 = 2)$

and $d = 6$ $(\because a : b :: c : d)$

$a^2 + b^2 + c^2 + d^2 = 1^2 + 3^2 + 2^2 + 6^2 = 50$

Hence, the presumed values are correct.

Thus, the average of a, b, c and $d = \dfrac{a+b+c+d}{4}$

$= \dfrac{1+3+2+6}{4} = 3$

Hence (b) is correct.

Alternatively: Assume option (b)

$\because \quad \dfrac{a+b+c+d}{4} = 3$

$\Rightarrow \quad a + b + c + d = 12$

Now $\because \quad b + c = 5$

$\Rightarrow \quad a + d = 7$

Again $a : b = 1 : 3$

$\Rightarrow \quad c : d = 2 : 6$

Now verify that $a^2 + b^2 + c^2 + d^2 = 50$. Since it is correct.

Hence option (b) is correct.

NOTE It can also be solved by forming quadratic equations.

2. $\dfrac{a}{b} = \dfrac{a+b}{a}$

$\Rightarrow \quad a^2 = ab + b^2$ $\quad \overset{a}{\vdash}\!\!\!\!\!\!\dashv\!\!\!\!\!\!\vdash \overset{b}{}$

$\Rightarrow \quad a^2 - b^2 - ab = 0$

Let $b = 1$, then $a : b = a : 1$ (by putting $b = 1$)

$\therefore \quad a^2 - a - 1 = 0$

$\Rightarrow \quad a = \dfrac{1 \pm \sqrt{5}}{2}$ (Solving quadratic equation by Sridharacharya's formula)

$\therefore a = \dfrac{1+\sqrt{5}}{2}$ (negative value can't be considered)

$a : b = \dfrac{1+\sqrt{5}}{2} : 1$

or $a : b = (1 + \sqrt{5}) : 2$

Therefore, $\dfrac{b}{a+b} = \dfrac{2}{1+\sqrt{5}+2} = \dfrac{2}{3+\sqrt{5}}$

Hence option (b) is correct.

3. The best way is to go through options

$\dfrac{x+y}{x-y} = \dfrac{8+2}{8-2} = \dfrac{10}{6} = \dfrac{5}{3}$ (verified)

Again $\dfrac{x}{y+2} = \dfrac{8}{2+2} = 2$ (verified)

Hence option (d) is correct.

4. Best way is to go through options

and $\dfrac{a^3+b^3}{a^3-b^3} = \dfrac{5^3+2^3}{5^3-2^3} = \dfrac{125+8}{125-8} = \dfrac{133}{117}$

Hence option (c) is correct.

5. $H : S = 1 : 1$

and $S : G = 2 : 3$

$H : P = 1 : 2$

$\therefore \quad H : S : G : P = 2 : 2 : 3 : 4$

$= 2x : 2x : 3x : 4x$

Therefore $\dfrac{2x + 2x + 3x + 4x}{4} = \dfrac{11x}{4} = 55$

$\Rightarrow \quad x = 20$

Marks in History = 40

Sociology = 40

Geography = 60

Philosophy = 80

Hence, only in two subjects he scored 60% or above.

Hence option (b) is correct.

6. Let the incomes of A and M is $2x$ and $3x$

Let the savings of A be K, then the expenditure of M be K

Also expenditure of $A = 2x - K$

Given $(2x - K) + K = 8000 \Rightarrow x = 4000$

$\therefore$ Total income of

A and $B = 2x + 3x = 5x = 5 \times 4000 = 20000$

$\therefore$ Total savings of A and $B = 20{,}000 - 8000 =$ Rs. 12,000

7. $\dfrac{\text{Profit of Hutch}}{\text{Profit of Essar}} = \dfrac{\text{time period} \times \text{amount of Hutch invested}}{\text{time period} \times \text{amount of Essar invested}}$

$\dfrac{6}{17} = \dfrac{5 \times K}{12 \times 1275}$

$\Rightarrow \quad K = \dfrac{6 \times 12 \times 1275}{17 \times 5} = 1080$

8. Minimum number of chocolates are possible when he purchases maximum number of costliest chocolates.

Thus $2 \times 5 + 5 \times 2 =$ Rs. 20

Now Rs. 100 must be spend on 10 chocolates as $100 = 10 \times 10$.

Thus minimum number of chocolates $= 5 + 2 + 10 = 17$

9. Maximum number of chocolates are possible only when he purchases minimum number of costlier chocolates and maximum number of cheaper chocolates

$\therefore \quad 2 \times 5 + 1 \times 10 =$ Rs. 20

Now Rs. 100 must be spend on 50 chocolates as $100 = 2 \times 50$.

Thus maximum number of possible chocolates

$= 2 + 1 + 50 = 53$

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10. Meretere → M, Teremere → T and Khabbu Singh → K

$\frac{5+3}{3} = 2.66$ (since all the 8 chocolates were shared by 3)

It means M has given $5 - 2.66 = 2.33$ chocolates to K and T has given $3 - 2.66 = 0.33$ chocolates to K

Thus M and T will receive the amount in the ratio of donations (*i.e.*, share of chocolates)

M	:	T
2.33	:	0.33
$2\frac{1}{3}$	:	$\frac{1}{3}$
$\frac{7}{3}$	:	$\frac{1}{3}$
7	:	1

So the M receives Rs. 14 and T receives Rs. 2

Thus the difference = Rs. 12

11. Share of a man, a woman and a boy = $7x$, $4x$ and $3x$

then the share of 4 men = $4 \times 7x = 28x$

then the share of 5 women = $5 \times 4x = 20x$

then the share of 2 boys = $2 \times 3x = 6x$

Now, the share of all women = $\frac{20x}{(28x + 20x + 6x)} \times 4536$

$= \frac{20}{54} \times 4536 =$ Rs. 1680

Hence, the share of one woman = $\frac{1680}{5} = 336$

12. Concentration of petrol in

A	B	C
$\frac{1}{2}$	$\frac{3}{5}$	$\frac{4}{5}$

Quantity of petrol taken from A = 1 litre out of 2 litre

Quantity of petrol taken from B = 1.8 litre out of 3 litre

Quantity of petrol taken from C = 0.8 litre out of 1 litre

Therefore total petrol taken out from

A, B and $C = 1 + 1.8 + 0.8 = 3.6$ litre

So, the quantity of Kerosene = $(2 + 3 + 1) - (3.6) = 2.4$ litre

Thus, the ratio of petrol to Kerosene = $\frac{3.6}{2.4} = \frac{3}{2}$

13. $T \propto \frac{\sqrt{l}}{\sqrt{g}} \Rightarrow T = k\sqrt{\frac{l}{g}}$

Therefore, $3 = K\sqrt{\frac{9}{4}}$ or $3 = K\frac{3}{2} \Rightarrow K = 2$

Again $T = K\sqrt{\frac{l}{g}} = 2 \times \sqrt{\frac{64}{16}} = 4$

$T = 4$ seconds

14.

Pure	Cure	Lure
100%	40%	37.5%
$\frac{1}{1}$	$\frac{2}{5}$	$\frac{3}{8}$

	Milk	Water	Mixture
Cure	$4l$	$10l$	$14l$
Lure	$6l$	$16l$	$22l$
New mixture	Milk	Water	
	$10l$	$26l$	
	$16l$		
	$26l$	$26l$	

Required mixture 1 : 1

Since in the required mixture the ratio of milk and water is 1 : 1 so she has to add up 16 litre of more milk (pure) to get it, for the fixed quantity of water.

15. $S : (M + J) = 5 : 7 \Rightarrow 7S = 5M + 5J$ ...(1)

$J : (S + M) = 1 : 2 \Rightarrow 2J = S + M$ ...(2)

By solving equations (1) and (2), we get

$S : M : J = 5 : 3 : 4$

so $S : M = 5 : 3$

16.

	Male		Female
A	$4x$	:	$3x$
B	$3y$	:	$2y$
C	$5z$	:	$4z$

but $2y = 4z$

$\therefore$

	Male		Female
A	$4x$	:	$3x$
B	$6z$	:	$4z$
C	$5z$	:	$4z$

Therefore, $7x + 19z = 33$

but z can assume only one value *i.e.*, $z = 1$

Hence, $7x + 19 = 33 \Rightarrow x = 2$

Thus the no. of female children in factory $A = 3x = 6$.

17. The ratio of broken parts (by weight) = $3x : 4x : 5x$

Therefore value of broken parts of diamond

$= (3x)^2 + (4x)^2 + (5x)^2 = 50x^2$

The value of original diamond = $(3x + 4x + 5x)^2 = 144x^2$

Therefore, loss in value = $144x^2 - 50x^2 = 9.4$ lakh

$\Rightarrow 94x^2 = 9.4$ lakh

$\Rightarrow 94x^2 = 940000$

$\Rightarrow x^2 = 10000$

Hence, the actual value of the diamond = $144x^2$

$= 144 \times 10000$

$= 14.4$ lakh

18. $10x + 20y + 100z = 1600$

Again since $x : y = 6 : 1$

$\therefore 60y + 20y + 100z = 1600$

$\Rightarrow 80y + 100z = 1600$

$\Rightarrow 4y + 5z = 80$

Putting $z = 1, 2, 3, 4, 5 \ldots$, we get at $z = 4$, $y = 15$ (an integer)

Hence, min. 4 coins of Re. 1 will be there

19. Concentration of water in first vessel = $\frac{2}{5} = 40\%$

concentration of water in second vessel = $\frac{2}{7} = 28.57\%$

By alligation

$\frac{2}{7}$ $\quad$ $\frac{2}{5}$

$\frac{1}{3}$

$\left(\frac{2}{5} - \frac{1}{3}\right)$ $\quad$ $\left(\frac{1}{3} - \frac{2}{7}\right)$

$\frac{1}{15}$	:	$\frac{1}{21}$
21	:	15
7	:	5

Therefore, ratio of first mixture to the second mixture = 5:7

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20. LCM of 2, 3, 5, 8 = 120

Therefore, $\frac{1}{2} \times 120 = 60$

$\frac{1}{3} \times 120 = 40$

$\frac{1}{5} \times 120 = 24$

$\frac{1}{8} \times 120 = 15$

Therefore, minimum number of chocolates

$= 60 + 40 + 24 + 15 = 139$

21. $A : B = 5 : 3$

$B : C = 5 : 8$

$\therefore \quad A : B : C = 25 : 15 : 24$

So, A is the most efficient.

22. $A : B = 5 : 4 \Rightarrow 10 : 8$

$A : C = 2 : 3 \Rightarrow 10 : 15$

$\therefore \quad A : B : C = 10 : 8 : 15$

$\therefore \quad B : C = 8 : 15$

23.

	A		B		C
(Proportion of milk in mixture)	$\frac{1}{3}$	:	$\frac{2}{5}$	:	$\frac{3}{7}$
or	$\frac{35}{105}$	:	$\frac{42}{105}$	:	$\frac{45}{105}$

quantity of milk in new mixture $= 35 + 42 + 45 = 122l$

quantity of water in new mixture $= (105 \times 3) - 122 = 193l$

Therefore, ratio of water is to milk = 193 : 122

24. $A : B = 8 : 9$

$B : C = 2 : 3$

$C : D = 9 : 13$

$A : B : C : D = 144x : 162x : 243x : 351x$

But we need not to solve this, since we already know that

$B : C = 2x : 3x$

$\therefore \quad 2x : 3x :: 18 : k \Rightarrow k = 27$ years

25.

Milk		Water	
$5x$		$3x$	(initially)
$-25l$		$-15l$	
$5x - 25$		$3x - 15$	(after selling 40l mixture)
		$+15l$	
$\frac{5x-25}{5}$	:	$\frac{3x}{4}$	(after adding 15l water)

$\therefore \quad \frac{5x-25}{3x} = \frac{5}{4} \Rightarrow x = 20$

$\therefore$ New quantity of mixture $= (5x - 25) + 3x = 135$ litre

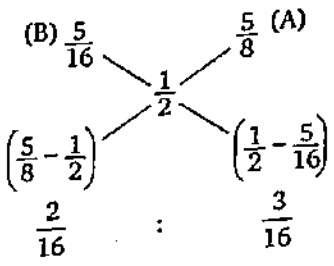
26.

A	B
$C : T$	$C : T$
5 : 3	5 : 11

Concentration of copper in $A = \frac{5}{8}$

Concentration of copper in $B = \frac{5}{16}$

By alligation



$\frac{2}{16} \quad : \quad \frac{3}{16}$

$2 \quad : \quad 3$

So, the required ratio of $A : B = 3 : 2$ (Since $B : A = 2 : 3$)

27. $k + 10x = 6000$

$k + 25x = 9000$ ($\because 25 \times 360 = 9000$)

$\Rightarrow \quad 15x = 3000$

$\Rightarrow \quad x = 200$ and $k = 4000$

$\therefore \quad k + 40x = 4000 + 40 \times 200 = 12{,}000$

where k is the fixed expenditure.

28. Third proportional of 21 and 42 is 84 and mean proportional to 16 and 49 is 28.

Therefore required ratio $= \frac{84}{28} = \frac{3}{1}$ or 3 : 1

29. $P \propto \sqrt{l} \Rightarrow P = k\sqrt{l}$

$\therefore \quad 52 = k\sqrt{16}$

$\Rightarrow \quad k = 13$

Again, $P = k\sqrt{l}$

$65 = 13\sqrt{l}$

$\Rightarrow \quad l = 25$ cm

30. $\frac{m+n}{m-n} = \frac{7x}{x} \Rightarrow \frac{m}{n} = \frac{4x}{3x}$

Again $mn = 4x \times 3x = 12x^2$

and $mn = 60x$

So $60x = 12x^2$

$\Rightarrow \quad x = 5$

$\therefore \quad m = 20$ and $n = 15$

Hence, $\frac{1}{m} : \frac{1}{n} = \frac{1}{20} : \frac{1}{15} = 3 : 4$

Alternatively: $\frac{m}{n} = \frac{4x}{3x}$

$\frac{1}{m} : \frac{1}{n} = 3 : 4$

31. $\underbrace{\frac{A+B}{2}} : \underbrace{\frac{C+D}{1}}$

and $\underbrace{\frac{B}{3} : \frac{C}{1}}$ and $\underbrace{\frac{C}{3} : \frac{D}{5}}$

$\Rightarrow \quad B : C : D$

$9 : 3 : 5$

Again $A + B : C + D$

$\underbrace{7+9}_{16} \quad \underbrace{3+5}_{8}$

Thus $A + B = 16 \Rightarrow A = 7$ when $B = 9$

$\therefore$ Therefore share of $A = \frac{7}{24} \times 9600 =$ Rs. 280

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32. The best way is to consider some values in the proportion and verify the option, but care should be taken that in this type of question you are required to verify a possible correct option atleast with 2-3 times with different values.
Since, if you consider

$$p:q::r:s \Rightarrow 1:2::3:6$$

you will find answer (b) is also correct, but for only particular values. So it is not the general solution. Finally you will realize that option (d) is the most appropriate answer.

33. Solve by componendo and dividendo or as mentioned in the previous solution.

34. Since for the constant distance time is inversely proportional to the speed. So, the required ratio of time taken by each of the rickshaw, car and scooter is

$$\frac{1}{3}:\frac{1}{5}:\frac{1}{6} \Rightarrow 10:6:5$$

$$\left[\frac{1}{3}\times\frac{10}{10}:\frac{1}{5}\times\frac{6}{6}:\frac{1}{6}\times\frac{5}{5}=\frac{10}{30}:\frac{6}{30}:\frac{5}{30}=10:6:5\right]$$

35. $A:B=2:3$ and $B:C=3:5$

$\Rightarrow A:B:C=2:3:5$

$\therefore (A+B):C=5:5=1:1$

Hence, Share of $(A+B)=\frac{1}{2}\times 6940=3470$

36. $x+y=93$ ...(1)

and $\frac{x+3}{y+2}=\frac{24}{25}$

$\Rightarrow 25x-24y=(-27)$ ...(2)

From equations number (1) and (2)

$x=45$ and $y=48$

Alternatively: Go through options.

37. $\frac{a}{b}=\frac{4x}{9x}$ and $\frac{a+4}{b+4}=\frac{21}{46}$

$\Rightarrow a=80$ and $b=180$

$\therefore b-a=100$

38. $\frac{R}{D}=\frac{3x}{5x}$ and $\frac{R}{D}\Rightarrow\frac{3x+10}{5x+10}=\frac{5}{7}$

$\Rightarrow x=5$

Hence, $\frac{R}{D}=\frac{15}{25}$

Alternatively: Go through options.

39. Let the fraction be $\frac{x}{y}$ then

$$\frac{x+5}{y+5}=\frac{11}{15} \Rightarrow 11y-15x=20$$

Since, we have only one equation in two variables, so we cannot find the solution.

40. $\frac{2x+3x+5x+8x+9x}{5}=\frac{162}{5}=32.4$

41. The number of days required by a single Kirlosker pump to fill the tank $=6\times 7=42$ days and the number of days required by a single USHA pump to fill the same tanks $=2\times 18=36$ days.
Now, since efficiency is inversely proportional to the number of days. Hence,

$$\frac{\text{Efficiency of one K-pump}}{\text{Efficiency of U-pump}}=\frac{36}{42}=\frac{6}{7}$$

42. $\frac{1}{5}$th field can be reaped by 96 man days

$\therefore \frac{4}{5}$th field can be reaped by $96\times 4=384$ man days

Now, since there are only 8 days so reqired

number of men $=\frac{384}{8}=48$ men

43. $210=2\times 3\times 5\times 7$

Therefore, $\frac{N_1}{N_2}=\frac{2k\times 5\times 7}{3k\times 5\times 7}$

since, $N_1:N_2=2:3$

Therefore, $N_1=70$ and $N_2=105$

$\therefore N_1+N_2=70+105=175$

44. $\frac{(53-x)}{(21-x)}=\frac{(41-x)}{(17-x)} \Rightarrow x=5$

Alternatively : Go through options.

45. $2x+3x+4x=180$

$\Rightarrow x=20 \quad \therefore \quad 4x=80$

46. $(25\times 12x)+(50\times 4x)+(100\times 3x)=800x=60000$

$\Rightarrow x=75$

$\therefore$ number of coins of 25 paise $=12x=12\times 75=900$

Alternatively : Go through options, choices (a), (b) and (c) are eliminated since neither of 200, 225, 275 is divisible by 12. Hence choice (d) is correct.

47. Go through options.

Alternatively : $\frac{(x-a)}{11}=\frac{(x-b)}{9}=\frac{(x-c)}{5}=k$

$\Rightarrow a=x-11k$

$b=x-9k$

$c=x-5k$

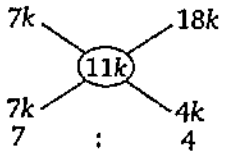
$$\therefore x=\frac{(x-11k)+(x-9k)+(x-5k)}{2}$$

$2x=3x-25k \Rightarrow x=25k$

$a=14k \Rightarrow b=16k$ and $c=20k$

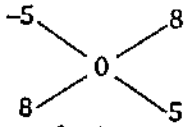
$\therefore a:b:c=14:16:20=7:8:10.$

48. If the weight of kerosene be k kg/unit volume, then
weight of petrol $=7k$/unit volume
weight of castrol $=18k$/unit volume
required weight of the mixture $=11k$/unit volume
By Alligation



The required ratio is 7 : 4.

49. **By Alligation**



The ratio of price of pigeons sold at loss and profit respectively are in the ratio of 8 : 5

$\therefore$ Cost price of profitable pigeon $=\frac{5}{8+5}\times 182=$ Rs. 70.

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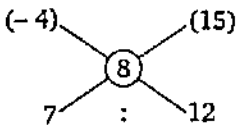
50. $\frac{C}{R} = \frac{3x}{5x}$

Again $\frac{C}{R} = \frac{(3x)3}{(5x+100)} = \frac{4}{5}$

$\Rightarrow \quad x = 16$

$\therefore$ Price of Rotomac pen in 2000 was Rs. 80.

51. By Alligation



Now, we get the ratio of no. of rings sold at a loss and profit is 7 : 12

$\therefore$ Number of rings sold at profit $= \frac{12}{19} \times 361 = 228$

Alternatively : Go through options.

52. Best way is to go through options.
consider option (d)

$$\frac{180}{30} + \frac{70}{35} = 6 + 2 = 8 \text{ hrs}$$

Hence, (d) is correct.

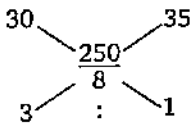
Alternatively : $\frac{x}{30} + \frac{250 - x}{35} = 8$

$\Rightarrow \quad x = 180$ miles

Alternatively : Since the Average Speed

$$= \frac{\text{Total Distance}}{\text{Total Time}} = \frac{250}{8} \text{ miles/hr}$$

$\therefore$**By Alligation**



Therefore, the ratio of time taken @ 30 m/hr and @ 35 m/hr is in the ratio of 3 : 1. It means he has travelled @ 30 m/hr for 6 hr.
Therefore, the distance travelled by car is 180.
($\because$ Distance = Speed $\times$ Time).

53. $\frac{22}{22} : \frac{17}{17} = 1 : 1$

54. $\frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{1}{6}$

$\Rightarrow \quad \frac{20}{60} : \frac{15}{16} : \frac{12}{60} : \frac{10}{60}$ (by taking LCM)

$\Rightarrow \quad 20 : 15 : 12 : 10$

$\therefore \quad$ Largest share $= \frac{20}{(20+15+12+10)} \times 171$

$= \frac{20}{57} \times 171 = 60.$

55. Let the present age of Karishma and Babita be x and y then,

$$\frac{x-10}{y-10} = \frac{1}{3} \quad \ldots(1)$$

Again $\frac{x+14}{y+14} = \frac{5}{9} \quad \ldots(2)$

By solving (1) and (2) we get $x = 26$ and $y = 58$

56. Let the fraction be $\frac{k}{5k}$

then, $\frac{k+x}{5k+5x} = \frac{(k+x)}{5(k+x)} = \frac{1}{5}$

57. $x \propto y$ and $x \propto \frac{1}{z^2}$

$$x \propto \frac{y}{z^2} \Rightarrow x = k\frac{y}{z^2}$$

$$6 = k \times \frac{75}{(5)^2} \Rightarrow k = 2$$

Again $x = 2 \times \frac{24}{(4)^2} \Rightarrow x = 3$

58. $x \propto (y^2 + z^2) \Rightarrow x = k(y^2 + z^2)$

$15 = k(1^2 + 2^2) \Rightarrow k = 3$

again $39 = 3 \times (2^2 + z^2)$

$\Rightarrow \quad 13 = (4 + z^2) \Rightarrow z = 3$

59. $W \propto HA \Rightarrow W = K \times H \times A$

Now, $48 = K \times 1.2 \times 20 \Rightarrow K = 2$

Again $W = 2 \times 1.5 \times 30$

$\therefore \quad W = 90$

60. $(a+b) : (b+c) : (c+a) = 5x : 6x : 9x$

and $a + b + c = 10$

$\Rightarrow \quad (a+b) + (b+c) + (c+a) = 20x$

$\Rightarrow \quad a + b + c = 10x$

$\Rightarrow \quad a + b + c = 10 \quad (\because x = 1)$

$\therefore \quad c = (a+b+c) - (a+b)$

or $c = 10 - 5 = 5$

61.

A	B	C	
$3x$	$4x$	$5x$	
$(3x + x)$	$2x$	$(5x + x)$	(B gives $\frac{1}{4}$ to A and $\frac{1}{4}$ to C)
$= 4x$	$2x$	$6x$	
$(4x + x)$	$2x$	$(5x)$	(C gives $\frac{1}{6}$ to A)
$5x$	$2x$	$5x$	
5	: 2	: 5	

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LEVEL 2

1.

M_1	M_2
$D : S$	$D : S$
7 : 3	5 : 4

New Mixture

D		S
2	:	1

proportion of Dew in $M_1 = \frac{7}{10}$

proportion of Dew in $M_2 = \frac{5}{9}$

proportion of new mixture $= \frac{2}{3}$ (required)

By alligation rule

$\frac{7}{10}$ $\quad$ $\frac{5}{9}$

$\frac{2}{3}$

$\left(\frac{2}{3} - \frac{5}{9}\right)$ $\quad$ $\left(\frac{7}{10} - \frac{2}{3}\right)$

$\frac{3}{27}$ $\quad$ $\frac{1}{30}$

or $\quad \frac{1}{9} \quad : \quad \frac{1}{30}$

or $\quad 10 \quad : \quad 3$

2. The ratio of quantities = 140 : 60 = 7 : 3

$\therefore$ The quantity to be exchanged $= \frac{(140 \times 3) + (60 \times 7)}{2(7 + 3)}$

$= 42$ litre

NOTE In this question individual prices are not required. You can solve it by forming equation.

Alternatively : Go through options by assuming some prices.

3. Let the number be $100x + 10y + z$, then

$x + z = 6$ and $y + z = 5$

Also, from the given options only option (b) is suitable

i.e., $\quad y + z = 5$

or $\quad 3 + 2 = 5$

4.

Water	Pulp
$4x$	x
$3y$	$2y$

but $x = 2y$ since pulp remains contant

	Water	Pulp
$5y$ = loss in water →	$8y$	$2y$
	$3y$	$2y$

but $\quad 10y\ (= 8y + 2y) = 1.5$ kg

$\therefore \quad 5y\ (= 3y + 2y) = 0.75$ kg

5.

Cost price	:	Selling price
$\frac{1}{9}$	:	$\frac{1}{6}$
2	:	3

or CP : SP = 2 : 3

since profit = SP − CP

$\therefore$ Profit = 3 − 2 = 1

$\therefore$ Profit : CP = 1 : 2

6. Flow Chart :

$P : K$

$7x : 3x$ → + Kerosene by addition

Step 1 → $7x : 5x$ ← by replacement

Step 2 → $3x : x$ ←

By the replacement formula from alligation chapter

$$\frac{1}{4} = \frac{5}{12}\left(1 - \frac{240}{12x}\right)$$

$$\Rightarrow \quad \frac{3}{5} = \left(1 - \frac{240}{12x}\right)$$

$$\Rightarrow \quad \frac{2}{5} = \frac{20}{x}$$

$$\Rightarrow \quad x = 50$$

Therefore, half of the initial amount = (350 + 150) litres then, the required amount of kerosene

$= 5x - 3x = 2x = 100$ litres

But for the whole amount required kerosene to be added = 200 litres

7. Data insufficient. Since we don't know that how many persons bought ticket of all the categories individually

i.e., $\quad 5x + 9y + 14z = 213925$

but we don't know anything about x, y and z

8.

Petrol	Kerosene	Total Mixture
99	x	$99 + x$
99	$(x - 198)$	$(x - 99)$

Again $\frac{99}{(x - 99)} \times 100 - \frac{99}{(x + 99)} \times 100 = 13.33$

or $\quad 9900\left(\frac{x + 99 - x + 99}{x^2 - 99^2}\right) = 13.33$

or $\quad \frac{9900\ (198)}{x^2 - 99^2} = \frac{40}{3}$

$\Rightarrow \quad x^2 - 99^2 = 99^2 \times 15$

$\Rightarrow \quad x^2 = (99)^2 \times (16)$

$\Rightarrow \quad x = 99 \times 4 = 396$ litres

Therefore the actual concentration of petrol

$= \frac{99}{(99 + 396)} = 20\%$

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9. Do it yourself.

10. Varsha : Vinay = 5 : 6 = 5x : 6x

Veera : Vikram = 7 : 8 = 7y : 8y

But their ages are in A.P.

Therefore, $6x - 5x = 8y - 7y$

$\Rightarrow \quad x = y$

Again, $\frac{5x+2}{8y+2} = \frac{2}{3}$

$\Rightarrow \quad \frac{5x+2}{8x+2} = \frac{2}{3}$

$\Rightarrow \quad x = 2$

Therefore, the ages of Varsha, Vinay, Veera and Vikram are 10, 12, 14 and 16 years respectively.

Therefore, the ratio of ages of Vinay and Veera = 6 : 7

11.

Bobby			Sunny		
Milk	:	Water	Milk	:	Water
1		1	1		1
50%		50%	50%		50%
②		②			
60%		40%			
③	:	②			

It means bobby will add up 1 litre of milk, in 4 litre of initial mixture, to prepare 5 litres mixture in the ratio of 3 : 2

By replacement method

$\frac{2}{5} = \frac{2}{4}\left(1 - \frac{k}{4}\right)$

$\Rightarrow \quad \frac{4}{5} = \left(1 - \frac{k}{4}\right)$

$\Rightarrow \quad \frac{1}{5} = \frac{k}{4}$

$\Rightarrow \quad k = \frac{4}{5}$

It means Sunny will replace $\frac{4}{5}$ litre of initial mixture by the same quantity of pure milk.

Hence, the percentage of milk added by Bobby to that of replaced by Sunny $= \frac{1}{4/5} \times 100 = 125\%$

12.

	A			B	
	Milk	Water		Milk	Water
After first operation	25l	0		0	25l
	20l	0	→	5l	25l
			1 : 5		
After second operation	21l	5l	←	4l	20l

Therefore, the ratio of water in A and B is 1 : 4

13. $B = \frac{5}{6}A$ ...(i)

and $C = D = \frac{9}{10}B$ ...(ii)

also $B = \frac{2}{3}E$ ...(iii)

and $E - A = 3$ ...(iv)

From (i) and (iii) $\frac{A}{E} = \frac{4}{5}$ or $E = \frac{5}{4}A$ ...(v)

$\therefore \quad E - A = \frac{5A}{4} - A = 3$ from (iv) and (v)

$\Rightarrow \quad A = 12$ and $E = 15$ and $B = 10$

Also $C = D = 9$ and $F = 11$, since $B < F < A$ and F is integer

$\therefore \quad A : F = 12 : 11$

14. The ratio of fees collected from B.Tech : MBA

$= 4x \times 25y : 5x \times 16y$

$= 100xy : 80xy$

$= 5xy : 4xy = 5k : 4k$

The amount collected only from MBA students

$= \frac{4}{9} \times 1.62 \text{ lakh}$

= Rs. 72,000

15. $C \propto (W)^2$

$W_1 : W_2 : W_3 = 3 : 4 : 5$

cost $= (3x)^2 + (4x)^2 + (5x)^2 = 50\,(x)^2$

Again $W_1 : W_2 : W_3 = 4 : 4 : 4$ (when weights are equal)

cost $= (4x)^2 + (4x)^2 + (4x)^2 = 48x^2$

loss $= 50x^2 - 48x^2 = 2x^2$

$1800 = 2x^2$

$x = 30$

$\therefore$ Actual cost of unbroken marble $= (4x + 4x + 4x)^2$

$= (12x)^2 = 144x^2$

$= 144 \times x^2$

$= 144 \times 900 = 129600$

Solutions for 16 and 17 :

	A	B	C	D
Initially	69	45	42	36
$A \to B$	24	90	42	36
$B \to C$	24	48	84	36
$C \to D$	24	48	48	72
$D \to A$	48	48	48	48

Solve it in reverse order (*i.e.*, from the result side)

18. Ratio of price paid by Hari and Murli = 7 : 10

or $(24 \times x) : 30 \times (30 - x) = 7 : 10$

$\Rightarrow \quad x = 14$

or go through options.

19. Since distance is constant. Therefore ratio of speeds of scooter, car and train = 1 : 4 : 16

and therefore, ratio of time taken = 16 : 4 : 1

Therefore, required ratio $= \frac{16}{1} : \frac{4}{4} : \frac{1}{16}$

$= 16 : 1 : \frac{1}{16} = 256 : 16 : 1$

20. Since there are 12 bangles, then the no. of broken to unbroken bangles can not be 2 : 3, since $5x = (2x + 3x)$ can not divide 12 for any integral value of x *i.e.*, all the sum of ratios which are the factors of 12 can possibly be the ratio of broken to unbroken.

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21. $$\frac{a}{b}=\frac{c}{d}=\frac{e}{f}=k$$

$$\Rightarrow \quad \frac{a^n}{b^n}=\frac{c^n}{d^n}=\frac{e^n}{f^n}=k^n$$

$$\Rightarrow \quad \frac{a^n p}{b^n p}=\frac{c^n q}{d^n q}=\frac{e^n r}{f^n r}=k^n$$

$$\therefore \quad \frac{a^n p+c^n q+e^n r}{b^n p+d^n q+f^n r}=(k^n)$$

$$\therefore \quad \left(\frac{a^n p+c^n q+e^n r}{b^n p+c^n q+f^n r}\right)^n=(k^n)^{1/n}$$

$$\therefore \quad \left(\frac{a^n p+c^n q+e^n r}{b^n p+d^n q+f^n r}\right)^{1/n}=k=\frac{a}{b}=\frac{c}{d}=\frac{e}{f}$$

Hence (c) is the possible answer.

22. Maximum earning will be only when he will won on the maximum yielding table.

$A \rightarrow 10:1$

$B \rightarrow 20:1$

$C \rightarrow 30:1$

i.e., he won on B and C but lost on A

$$20\times200+30\times200-1\times200=9800$$

minimum earning will be when he won on table A and B and lose on that table 3.

$$\therefore \quad 10\times200+20\times200-1\times200$$

$$6000-200=5800$$

Therefore, difference $= 9800 - 5800 = 4000$

Alternatively the difference

$$=[(30+20-1)-(10+20-1)]\times200$$

$$=20\times200=\text{Rs. } 4000.$$

23. $$\frac{m+2p}{m-2p}+\frac{m+2q}{m-2q}$$

$$=\frac{m}{2p}+\frac{m}{2q} \text{ (by componendo and dividendo)}$$

$$=m\left(\frac{p+q}{2pq}\right)$$

$$=\left(\frac{4pq}{p+q}\right)\left(\frac{p+q}{2pq}\right) \text{ substituting the value of } m=\frac{4pq}{p+q}=2$$

24. Volume of a cube $= (\text{side})^3 = (a)^3$

surface area of a cube $= 6(a)^2$ ($a \rightarrow$ side)

$$a_1^3 : a_2^3 = 8:27$$

$$\Rightarrow \quad a_1 : a_2 = 2:3$$

$$\Rightarrow \quad a_1^2 : a_2^2 = 4:9$$

$$\Rightarrow \quad 6a_1^2 : 6a_2^2 = 4:9$$

25. Clearly it can not be less than 75%. For clarification of concept consider some values and then verify it.

26. $$A:B:C=3x:4x:5x$$

but $$C-A=5x-3x\le3$$

or $$C-A=2x\le3 \Rightarrow x=1$$

$$\Rightarrow \quad C-A=2$$

$$\therefore \quad A:B:C=3:4:5$$

Therefore, number of assistant trainee (except project in charge) of $Q = 3$ and assistant trainee of $R = 4$

Therefore, required percentage $= \frac{4-3}{4}\times100$

$$=\frac{1}{4}\times100=25\%$$

27. $b = 3a$

$c = 3b = 9a$

$d = 3c = 9b = 27a$

$e = 3d = 9c = 27b = 81a$ and $f = 3e = 243a$

put the values and simplify.

or we know that $\frac{a}{b}=\frac{c}{d}=\frac{e}{f}\ldots=\frac{a+c+e}{b+d+f}$

28. Let w be the number of wagons and s be the speed of engine

without wagon $= \frac{20}{3}$ m/s $= 24$ km/hr.

then speed of the train $= s - k\sqrt{w}$

$$\therefore \quad 20=24-k\sqrt{4} \qquad [\because \tfrac{50}{9}\text{ m/s} = 20\text{ km/hr}]$$

$$\therefore \quad k=2$$

when train will stop its speed becomes zero

$$0=24-2\sqrt{w} \qquad (\because k=2)$$

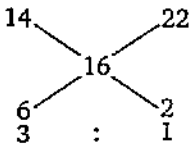
$$\Rightarrow \quad w=144$$

since at 144 wagons train will stop, so at 143 wagons train just can move with its least possible speed having maximum possible wagons.

29. Cost of 1 kg (mixture) of sugar = Rs. 16/kg

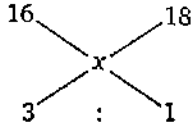
since selling price is Rs. 20/kg

Therefore, ratio of quantity of sugar costing Rs. 14 and Rs. 22 per kg = 3 : 1



So, the available stock of mixture costing Rs. 16 = 3 quintal and the required stock of sugar costing Rs. 18 = 1 quintal

Therefore, the new price of mixture costing 3 quintal of sugar @ Rs. 16/kg and 1 quintal of sugar @ Rs. 18 per kg



$$\therefore \quad x = \text{Rs. } 16.5 \text{ per kg}$$

New price = Rs. 16.5 per kg

Now, original profit = 20 − 16 = Rs. 4 per kg

New profit = 20 − 16.5 = Rs. 3.5 per kg

$\therefore$ Percentage reduction in profit $= \frac{4-3.5}{4}\times100 = 12.5\%$

30.

	P	Q
Initial amounts	a (petrol)	b (kerosene)
After one operation	$a-c$ (petrol) c (kerosene)	$b-c$ (kerosene) c (petrol)

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Since there is no change in concentration in the second, third... etc operations.

In the vessel P;

The fraction of kerosene in P is $\frac{c}{a}$

In the vessel Q;

The fraction of kerosene in Q is $\left(\frac{b-c}{b}\right)$

$$\therefore \quad \left(\frac{b-c}{b}\right) = \frac{c}{a}$$

$$\Rightarrow \quad c(a+b) = ab$$

$$\Rightarrow \quad c = \frac{ab}{a+b}, \qquad \text{Hence (c)}$$

Since at this moment the concentration in both vessels is same.

31. The minimum number of passengers n, at which there is no loss and number of passengers travelling $= m$ and let the distance travelled is d, then

$$P \propto (m-n)d$$

or $\quad p = k(m-n)d$; k is a constant.

when $P = 3600$, $m = 29$ and $d = 36$, then

$$3600 = k(29-n) \times 36 \qquad \ldots(1)$$

Again, when $p = 6300$, $m = 36$, $d = 42$ km

$$6300 = k(36-n) \times 42 \qquad \ldots(2)$$

Dividing equation (2) by (1)

$$\frac{6300}{3600} = \frac{k}{k}\frac{(36-n)\times 42}{(29-n)\times 36}$$

$$\Rightarrow \quad \frac{(36-n)}{(29-n)} = \frac{9}{6}$$

$$\Rightarrow \quad 3n = 45 \quad \Rightarrow \quad n = 15$$

Hence to avoid loss, minimum number of 15 passengers are required.

32. Frequency of step of $A : B : C = 5 : 6 : 7$

but it terms of size of step, $6A = 7B = 8C$

$\therefore$ Ratio of speeds of A, B and $C = \frac{5}{6} : \frac{6}{7} : \frac{7}{8}$

$$= 280 : 288 : 294$$

$$= 140 : 144 : 147$$

33. Expenditure $= 5$ (no. of family members)^2

$$\Rightarrow \quad E_1 = 5(n)^2 \qquad \ldots(1)$$

again $\quad E_2 = 5(n-1)^2 \qquad \ldots(2)$

but $\quad E_1 - E_2 = 95$

$$\therefore \quad 5[n^2 - (n-1)^2] = 95$$

$$5[n^2 - (n^2 + 1 - 2n)] = 95$$

$$n^2 - n^2 - 1 + 2n = 19$$

$$\Rightarrow \quad 2n = 20 \quad \Rightarrow \quad n = 10$$

34. Finally I have Rs. 5

Consider option (a)

$$2 \times 100 + 3 \times 50 + 1 \times 25 = 375 \text{ paise}$$

Its integer multiple can not give the value equals to 500. So can not be true.

Consider option (b)

$1 \times 100 + 7 \times 50 + 2 \times 25 = 500$ paise, can be true

but we also check the option (c) and (d) now when you check the choices (c) and (d) you will find wrong as choice (a). So, only choice be could be the best answer.

35. By componendo and dividendo :

$$\frac{a}{b+c} = \frac{b}{c+a}$$

$$\frac{a+b+c}{a-b-c} = \frac{b+c+a}{b-c-a}$$

$$\Rightarrow \quad a-b-c = b-c-a$$

$$\Rightarrow \quad a = b$$

similarily $\quad a = b = c$

$$\therefore \quad \frac{b}{a+b+c} = \frac{1}{3}$$

36. Density of P_1, P_2 and P_3 are 18, 14 and 10 gm/cc

$$\text{again since volume} = \frac{\text{weight}}{\text{density}}$$

now the weight of P_3 in 450 kg mixture $= \frac{450 \times 4}{15} = 120$ kg

now the volume of $P_3 = \frac{120}{10} = 12$ litre

$\therefore$ The cost of 12 litre P_3 petrol $= 12 \times 40 =$ Rs. 480

37. $26.65 + 42.75 + 53 = 122.40$

Proportionate amount of Amar, Akbar and Anthony in the ratio of 3 : 4 : 5 is Rs. 30.60, Rs. 40.80 and Rs. 51 respectively.

Now	Amar	Akbar	Anthony
	30.60	40.80	51.00
	26.65	42.75	53.00

So, Amar pays Rs. 1.95 to Akbar and Rs. 2 to Anthony.

38. The best way is to go through options

Alternatively: Solve by cross product rule and componendo.

39. Since Pooja and Shipra are twins so their ages be same. Let their ages be x and age of Monika be y, then,

$$x + x = y \qquad \ldots(1)$$

and $\quad \frac{(x-3)}{(y-3)} = \frac{2}{7}$

$$\Rightarrow \quad 7x - 2y = 15$$

Now, from equation (1),

$$7x - 4x = 15 \quad \Rightarrow \quad x = 5$$

So the age of Shipra 3 years hence will be $5 + 3 = 8$ years.

40. $\frac{H-9}{W-9} = \frac{5}{4}$ and $\frac{H+6}{W+6} = \frac{8}{7}$

$\therefore$ Thus the present age of Husband is 34 and present age of his wife is 29 years.

Now, the maximum age of any child must be less than 9 years. Hence their ages can be 2, 3 and 4 years or 4, 6 and 8 years. So the max. possible sum of age of this family

$$= 34 + 29 + (1 \times 4 + 2 \times 6 + 3 \times 8)$$

$$= 103 \text{ years}$$

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41. $P \propto n\sqrt{r} \Rightarrow P = kn\sqrt{r}$

where P is the price of necklace, n is number of pearls and r is the radius of a pearl.

Now $150 = k \times 75 \times \sqrt{1}$

$\Rightarrow k = 2$

again $600 = 2 \times 100 \times \sqrt{r}$

$\Rightarrow \sqrt{r} = 3 \Rightarrow r = 9$ cm

42. $P \propto \frac{N}{T} \Rightarrow P = K\frac{N}{T}$

$P \rightarrow$ Price of a book, $N \rightarrow$ Number of Pages, $T \rightarrow$ Time period

$P_1 = P_2$

$\therefore K\frac{N_1}{T_1} = K\frac{N_2}{T_2}$

$\Rightarrow \frac{3N}{18} = \frac{N}{T}$

$T = 6$ years

43. $\frac{5x + 40}{6x + 40} = \frac{7}{8} \Rightarrow \frac{\text{Akbar}}{\text{Birbal}}$

$\Rightarrow x = 20$

$\therefore$ The actual number of shares of less salaried person

$= 100 \quad \because (5 \times 20 = 100)$

$\therefore$ The salary of Akbar $= 100 \times 75 = 7500$

44. $D \propto \frac{\sqrt{F} \times T}{W}$, $D \rightarrow$ Distance, $F \rightarrow$ Fuel, $T \rightarrow$ Time, $W \rightarrow$ No. of wagons

$D = k\frac{\sqrt{F} \times T}{W}$

$192 = k\frac{\sqrt{256} \times 20}{10}$

$\Rightarrow k = 6$

again $200 = \frac{6 \times \sqrt{F} \times 25}{15}$

$\Rightarrow \sqrt{F} = 20 \Rightarrow F = 400$ litre

$\therefore$ fuel used per km $= \frac{400}{200} = 2$ l/km

45. Just go through option and factorize the product into two factors such that the given conditions must satisfy.

e.g., $6 = 1 \times 6$

2×3

3×2

6×1

It is not true.

Again consider option (b)

$12 = 1 \times 12$

2×6

$3 \times 4 \rightarrow 3 \times 4 = 12$

4×3

$6 \times 2 \rightarrow 6 \times 2 = 12$

12×1

Now you can see that the rate is being half from 4 to 2 so she can purchase double number of toffees as she was already purchasing on the ground floor. Again to purchase the same number of toffees she had to spend Rs. 2 less than the spending on the ground floor :

Rate		Number of toffee/Re		Total Number of toffee
4	×	3	=	12
2	×	6	=	12

and if you check other options (c) and (d) they will not satisfy the given conditions.

46. Go through options.

	Total no. of plants		No. of plants	Days
Previous scheme $\Rightarrow$	1800	=	100	× 18

Again after $\frac{1}{3}$rd days, remaining plants

$= \frac{1800 \times 2}{3} + 120 = 1320$

$1320 = (100 + 20) \times 11$

This shows that in the second case 1 day was saved than the planned no. of day.

47. Go through options.

$\frac{810}{54} - \frac{900}{75} = 3 = (6 - 3 = 3)$

$15 - 12 = 3$

$3 = 3$

Hence choice (c) is correct.

i.e., if A writes 54 page, then B writes 75 (= 54 + 21) pages per hour.

48.

	A	B	C	D
Initial amount of soda water	$(x + 3)$	$(x + 1)$	(x)	0
final amount of soda water	$\left(\frac{3x}{4} + 1\right)$	$\left(\frac{3x}{4} + 1\right)$	$\left(\frac{3x}{4} + 1\right)$	$\left(\frac{3x}{4} + 1\right)$

Therefore to be the integral value x must be $4m$; $m = 1, 2, 3...$

So at $m = 1$

Hence option (a) is correct since B had contributed 1 tumbler.

	A	B	C	D
Initial amount	7	5	4	0
Final amount	4	4	4	4
	3	1	0	
		2	1	

49. Ratio of fruits (by dozen) = 3 : 2 : 7

Ratio of fruits by weight = 120 : 150 : 24

$\therefore$ Ratio of fruits (combined) by weight

$= 3 \times 120 : 2 \times 150 : 7 \times 24$

$= 30 : 25 : 14$

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Final Round

1.

	1st Alloy		2nd Alloy	
	Iron	Copper	Iron	Copper
	4	3	6	1
Proportion of iron in the alloys	$\frac{4}{7}$ ⇩ $\frac{8}{14}$ (×2)		$\frac{6}{7}$ ⇩ $\frac{36}{42}$ (×6)	

So in the new alloy total iron will be 44 kg and copper will be 12 kg.

$\because$ Iron = 8 + 36 = 44

and Copper = (14 + 42) − (8 + 36) = 12

$\therefore$ Ratio of copper to iron = 12 : 44 = 3 : 11, hence (d)

2. Suppose there are all the pigeons then total no. of heads are 340 and total no. of legs are 680. Now since 380 (= 1060 − 680) legs are extra, it means there will be $190\left(=\frac{380}{2}\right)$ rabbits. As we know a rabbit has 2 extra legs than that of a pigeon who has only two legs.

Therefore, number of rabbits = 190

and number of pigeons = 340 − 190 = 150

Alternatively : go through options and consider choice (b)

	Pigeons	Rabbits
Heads (340)	150 (×2)	190] 340 (×4)
Legs (1060)	300	760] 1060

Alternatively : $P + R = 340$ ...(1)

and $2P + 4R = 1060$ ...(2)

Solve these two equations and you will get the answer.

Alternatively : It can be solved through alligation rule.

3. By the replacement formula

Decreased amount

$$= \text{Original amount}\left(1 - \frac{\text{replacing amount}}{\text{original amount}}\right)^{\text{no. of times } (n)}$$

Now, since the ratio of petrol and kerosene is 1701 and 27 it means initially there was (1701 + 27) = 1728 unit of kerosene and the decreased amount of kerosene is 27 unit.

$$\therefore \quad 27 = 1728\left(1 - \frac{6}{k}\right)^3 \quad \Rightarrow \quad k = 8 \text{ litre}$$

NOTE Here 27 : 1728 just show the ratio not the exact amount.

4. The ratio of milk in 3 vessels

$$= \frac{3}{4} \times \frac{5\times7}{5\times7} : \frac{2}{5} \times \frac{4\times7}{4\times7} : \frac{4}{7} \times \frac{4\times5}{4\times5}$$

$$= \frac{105}{140} : \frac{56}{140} : \frac{80}{140}$$

Remember, The value of 25 litre does not matter, the basic thing is that the amount of mixture in all the three quantities is same.

So the total quantity of milk in mixture = 105 + 56 + 80 = 241

So the total amount of water in mixture

$$= [(3\times140) - 241] = 179 \text{ litre}$$

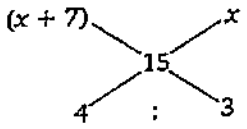
Therefore ratio of water to milk in the new mixture

$$= 179 : 241$$

5. $\because$ Profit = $33\frac{1}{3}$%, it means cost price = Rs. 15

Now, by alligation

$$\frac{(x+7)-15}{(15-x)} = \frac{3}{4}$$



$$\Rightarrow \quad x = 11$$

$$\therefore \quad x = 11 \text{ and } (x+7) = 18$$

Thus the total value of both the prices = 11 + 18 = 29

6.

A_1		A_2	
C	T	C	T
1	3	2	5
Copper → $\frac{1}{4}$		Copper → $\frac{2}{7}$	

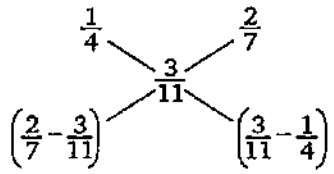
Required copper → $= \frac{3}{11}$

So, the required ratio is 4 : 7

Since it is clear from the above values

(1 + 2 → 3 and 4 + 7 → 11)

Alternatively : By Alligation



$$\frac{1}{77} : \frac{1}{44}$$

$$\Rightarrow \quad \frac{1}{7} : \frac{1}{4}$$

$$4 : 7$$

7. Ratio of W_1 M_1 W_2 M_2 W_3 M_3

1 : 3 2 : 3 2 : 5

Proportion of water $\frac{1}{4} : \frac{2}{5} : \frac{2}{7}$

$$\Rightarrow \quad \frac{1}{4} \times \frac{5\times7}{5\times7} : \frac{2}{5} \times \frac{4\times7}{4\times7} : \frac{2}{7} \times \frac{4\times5}{4\times5}$$

$$\Rightarrow \quad \frac{35}{140} : \frac{56}{140} : \frac{40}{140}$$

Now since all these three mixtures are mixed in the ratio of 2 : 3 : 5

Therefore new ratio $= \frac{35}{140} \times \frac{2}{2} : \frac{56\times3}{140\times3} : \frac{40\times5}{140\times5}$

$$= \frac{70}{280}, \frac{168}{420}, \frac{200}{700}$$

Now, the amount of water = 70 + 168 + 200 = 438

$\therefore$ The amount of milk = (280 + 420 + 700) − 438 = 962

$\therefore$ Ratio of milk to water = 962 : 438

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8. $B_1 : B_2 : B_3 = 3x : 4x : 5x$

again $B_1 : B_2 : B_3 = 5y : 4y : 3y$

Since there is increase in no. of oranges in first two basket only, it means the no. of oranges remains constant in the third basket

$\therefore \quad 5x = 3y$

Hence $3x : 4x : 5x$

$\Rightarrow \quad \frac{9y}{5} : \frac{12y}{5} : \frac{15y}{5} = 9y : 12y : 15y$

and $5y : 4y : 3y \Rightarrow 25y : 20y : 15y$

Therefore, increase in first basket = 16

and increase in second basket = 8

$\therefore$ the required ratio = 2 : 1

9. Amount of alcohol in first vessel = $0.25 \times 2 = 0.5$ litre

amount of alcohol in second vessel = $0.4 \times 6 = 2.4$ litre

Total amount of alcohol out of 10 litres of mixture is $0.5 + 2.4 = 2.9$ litre

Hence, the concetration of the mixture is 29% $\left(= \frac{2.9}{10} \times 100\right)$

10. Assume the weight of Alloy A is 100 kg

$\therefore$ The weight of Alloy B is 400 kg

		Gold	Silver	Copper
$\therefore$	A	40 kg	60 kg	0 kg
	B	140 kg	160 kg	100 kg
$\therefore$	total →	180 kg	220 kg	100 kg

$\therefore$ Ratio of Gold and Silver in new alloy $= \frac{180}{500} : \frac{200}{500}$

$= 36\% : 44\%$

11.

Urea			Dia		
N	P	K	N	P	K
x	y	0	20%	70%	10%

Mixture		
N	P	K
26%	68%	6%

This 6% of K is obtained only from Dia.

$\therefore$

Urea			Dia		
N	P	K	N	P	K
x	y	0	120	420	60

Mixture		
N	P	K
260	680	60

$N_U + N_D = N_M \Rightarrow N_U + 120 = 260$

$N \rightarrow$ Nitrogen, $P \rightarrow$ Phosphorus

and $P_U + P_D = P_M \Rightarrow P_U + 420 = 680$

$U, D, M \rightarrow$ Urea, Dia and mixture

$\therefore$ Amount of Nitrogen in Urea = 140

and amount of Phosphorus in Dia = 260

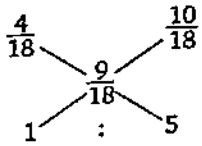
$\therefore$ Ratio of $N : P = 7 : 13 \Rightarrow 35 : 65$

12.

X		Y	
C	N	C	N
2	7	5	4

Copper → $\frac{2}{9} = \frac{4}{18}$ Copper → $\frac{5}{9} = \frac{10}{18}$

$\therefore$ **By alligation**



$\therefore$ amount of $X = \frac{1}{6} \times 42 = 7$ kg

and amount of $Y = \frac{5}{6} \times 42 = 35$ kg

13.

First	Alloy	Second	Alloy
C	Al	C	Al
1	2	3	1

$\therefore$

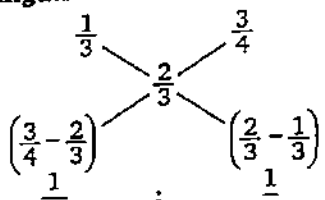
Required	alloy
C	Al
2	1

$\therefore$ Copper in first alloy $= \frac{1}{3}$

copper in second alloy $= \frac{3}{4}$

copper in required alloy (mixture) $= \frac{2}{3}$

Now, **by alligation**



$\Rightarrow \quad 1 : 4$

Therefore, second alloy be mixed 4 times the first alloy.

14. Note in this type of question individual prices does not matter. To prove this solve it algebrically.

$$\text{Exchanged amount} = \frac{3 \times 150 + 5 \times 90}{2(3+5)}$$

$$= \frac{450}{8} = 56.25 \text{ litre}$$

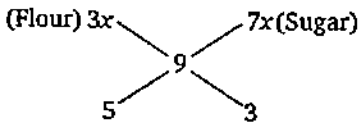
Here 3 and 5 are obtained from the ratio of amounts i.e., from 90 and 150.

15. Here the ratio of mixtures (i.e., milk, water) does not matter. But the important point is that whether the total amount (either pure or mixture) being transferred is equal or not. Since the total amount (i.e., 5 cups) being transferred from each one to another, hence $A = B$.

HINT You can verify it very easly by considering a simple example.

16. CP. of rasgulla = Rs. 9 (since profit is 66.66%)

Now by alligation



$\frac{(9-3x)}{(7x-9)} = \frac{3}{5} \Rightarrow x = 2$

$\therefore$ Price of sugar = $7x$ = Rs. 14 per kg.

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5 PERCENTAGES

It is one of the most important chapters which is the backbone of calculations either involved in commercial arithmetic or in real life. Personally I do maximum arithmetical calculation using percentage and others too. So in the context of calculation it is necessary to know the clear concepts of percentage which plays a very vital role in Data Interpretation besides quantitative Aptitude section. On an average two problems *i.e.*, nearly 4–5 % problems in QA only, are being asked in CAT every year.

In other entrance/competitive exams like MAT, XAT and UPMCAT, etc there are too many questions asked from this chapter.

PERCENTAGE AND ITS APPLICATION

A fraction with denominator 100 is called a per cent. Per cent is an abbreviation for the latin word "percentum" meaning "per hundred " or "hundredhs" and is denoted by symbol %.

NOTE A fraction with denominator 10 is called as decimal.

Since per cent is a form of fraction, we can express per cent as fractions (or decimals) and *vice-versa*.

CONVERSION OF A FRACTION INTO PERCENTAGE

To convert a fraction into a percentage, multiply the fraction by 100 and put "%" sign.

EXAMPLE 1 *Convert the following fractions into percentages:*

(i) $\frac{1}{2}$ *(ii)* $\frac{3}{4}$ *(iii)* $\frac{4}{5}$ *(iv)* $\frac{7}{8}$

SOLUTION (i) $\frac{1}{2} \to \frac{1}{2} \times 100 = 50\%$

(ii) $\frac{3}{4} \to \frac{3}{4} \times 100 = 75\%$

(iii) $\frac{4}{5} \to \frac{4}{5} \times 100 = 80\%$

(iv) $\frac{7}{8} \to \frac{7}{8} \times 100 = 87.5\%$

CONVERSION OF A PERCENTAGE INTO A FRACTION

To convert a percentage into a fraction, replace the % sign with $\frac{1}{100}$ and reduce the fraction to simplest form.

EXAMPLE 2 *Express the following percentage as fraction*

(i) 20% *(ii)* 30% *(iii)* 45% *(iv)* $5\frac{1}{8}$ % *(v)* 155%

SOLUTION (i) $20\% = \frac{20}{100} = \frac{1}{5}$

(ii) $30\% = \frac{30}{100} = \frac{3}{10}$

(iii) $45\% = \frac{45}{100} = \frac{9}{20}$

(iv) $5\frac{1}{8}\% = \frac{41}{8 \times 100} = \frac{41}{800}$

(v) $155\% = \frac{155}{100} = \frac{31}{20} = 1\frac{11}{20}$

CONVERSION OF A PERCENTAGE INTO A RATIO

To convert a percentage into a ratio, first convert the given percentage into a fraction in simplest form and then to a ratio.

EXAMPLE 3 *Solve the following:*

(i) 38% (ii) 25% (iii) 66.66%

SOLUTION (i) $38\% = \frac{38}{100} = \frac{19}{50} = 19:50$

(ii) $25\% = \frac{25}{100} = \frac{1}{4} = 1:4$

(iii) $66.66\% = 66\frac{2}{3}\% = \frac{200}{3 \times 100} = \frac{2}{3} = 2:3$

CONVERSION OF A RATIO INTO A PERCENTAGE

To convert a ratio into a percentage, first convert the given ratio into a fraction then to a percentage.

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EXAMPLE 4 *Express the following ratios as percentage :*
(i) 1 : 5 (ii) 2 : 3 (iii) 4 : 9

SOLUTION (i) $1:5=\frac{1}{5}=\frac{1}{5}\times100=20\%$

(ii) $2:3=\frac{2}{3}=\frac{2}{3}\times100=66.66\%$

(iii) $4:9=\frac{4}{9}=\frac{4}{9}\times100=44.44\%$

CONVERSION OF A PERCENTAGE INTO A DECIMAL

To convert a percentage into a decimal remove the % sign and move the decimal point two places to the left.

EXAMPLE 5 *Convert the following percentages into decimals :*
(i) 36% (ii) 250% (iii) 57.5% (iv) $17\frac{1}{5}\%$ (v) 7%

SOLUTION (i) $36\%=0.36$

(ii) $250\%=2.50=2.5$

(iii) $57.5\%=0.575$

(iv) $17\frac{1}{5}\%=17.2\%=0.172$

(v) $7\%=0.07$

CONVERSION OF A DECIMAL INTO A PERCENTAGE

To convert a decimal into a percentage, move the decimal point two place to the right (adding zeros if necessary) and put % sign.

EXAMPLE 6 *Convert the following decimals into percentages :*
(i) 0.35 (ii) 8.12 (iii) 0.018

SOLUTION (i) $0.35=35\%$

(ii) $8.12=812\%$

(iii) $0.018=1.8\%$

- Work out some more examples so that all these thing rest on your fingure tips.

Remember $\frac{1}{2}=\frac{2}{4}=\frac{3}{6}=\frac{4}{8}=\ldots=50\%$ etc.

Learn and practice all the values given below.

CONVERSION OF FRACTION INTO PERCENTAGE

NUMERATORS

DENOMINATORS	1	2	3	4	5	6	7	8	9	10	11	12
1	100	200	300	400	500	600	700	800	900	1000	1100	1200
2	50	100	150	200	250	300	350	400	450	500	550	600
3	33.33	66.66	100	133.33	166.66	200	233.33	266.66	300	333.33	366.60	400
4	25	50	75	100	125	150	175	200	225	250	275	300
5	20	40	60	80	100	120	140	160	180	200	220	240
6	16.66	33.33	50	66.66	83.33	100	116.66	133.33	150	166.66	183.33	200
7	14.28	28.56	42.85	57.13	71.42	85.71	100	114.28	128.56	142.85	157.13	171.42
8	12.5	25	37.5	50	62.5	75	87.5	100	112.5	125	137.5	150
9	11.11	22.22	33.33	44.44	55.55	66.66	77.77	88.88	100	111.11	122.22	133.33
10	10	20	30	40	50	60	70	80	90	100	110	120
11	9.09	18.18	27.27	36.36	45.45	54.54	63.63	72.72	81.81	90.9	100	109.09
12	8.33	16.66	25	33.33	41.66	50	58.33	66.66	75	83.33	91.66	100
15	6.66	13.33	20	26.66	33.33	40	46.66	53.33	60	66.66	73.33	80

(PERCENTAGE-FRACTION CONVERSION TABLE)

Remember : $\frac{1}{7}=14.28\%$ and $\frac{1}{14}=7.14\%$

$\frac{1}{6}=16.66\%$ and $\frac{1}{15}=6.66\%$

$\frac{1}{9}=11.11\%$ and $\frac{1}{11}=9.09\%$

$\frac{1}{15}=6.66\%$ and $\frac{1}{16}=6.25\%$

$\frac{1}{3}=33.33\%$ and $\frac{3}{10}=30\%$

99.99% is equivalent to 100% (in calculation)

SOME OTHER TYPICAL VALUES

$\frac{1}{13}=7.69\%\approx7.7\%$, $\frac{1}{17}=5.88\%$, $\frac{1}{19}=5.26\%$,

$\frac{1}{21}=4.76\%$, $\frac{1}{23}=4.35\%$, $\frac{1}{24}=4.166\%$

Learnings from the table :

(i) This table is a first hand support as a percentage values of some frequently used fractions.

(ii) All the percentage values whose decimal part is 0.33, 0.66, 0.00 contain the denominator 3 in the fraction.

(iii) Similarly if there is 0.16, 0.33, 0.50, 0.66, 0.83, 0.00 it means there is a 6 in the denominator of the fraction.

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(iv) Similarly if there is 0.28, 0.56, 0.85, 0.13, 0.42, 0.71 it means there is a 7 as denominator.

(v) If there is 0.11, 0.22, 0.33, 0.44 ... 0.99 etc. It means there is 9 as the denominator of the fraction.

(vi) If there is 0.09, 0.18, 0.27 ... it means there is 11 as the denominator of the fraction.

(viii) If two percentage values have different decimal values (there must be different denominators) then their addition or subtraction results always in decimal *i.e.*, never as an integer.

PERCENTAGE OF A QUANTITY

EXAMPLE 1 *Find the no. of male students (i.e., boys), if there are 42% male students in the school and the total no. of students in the school is 1000.*

SOLUTION Required number of male students

$$= 42\% \text{ of } 1000$$

$$= \frac{42}{100} \times 1000 = 420$$

EXAMPLE 2 *A student scored 85% marks. Total marks were 400. How much did he score?*

SOLUTION Marks scored = 85% of 400

$$= \frac{85}{100} \times 400$$

$$= 340$$

EXAMPLE 3 *In an orchard $16\frac{2}{3}\%$ of the trees are mango trees. If the total number of trees in the orchard is 360, find the number of other types of trees in the orchard.*

SOLUTION Total number of trees = 360

Number of mango trees = $16\frac{2}{3}\%$ of 360

$$= \frac{50}{3 \times 100} \times 360 = 60$$

Therefore, the number of other trees = 360 − 60 = 300

Alternatively: Number of mango trees = $16\frac{2}{3}\%$

It means no. of other types of trees = $\left(100 - 16\frac{2}{3}\right)\%$

$$= 83\frac{1}{3}\%$$

Thus number of other types of trees = $83\frac{1}{3}\%$ of 360 = 300

INTRODUCTORY EXERCISE-5.1

1. Find the value of :
   (i) 25% of 200 (ii) 30% of 180
   (iii) 37.5% of 300 (iv) 83.33% of 480
   (v) 100% of 2 quintal (vi) 165% of 330 litre
   (vii) $5\frac{1}{2}\%$ of Rs. 1600 (viii) 10% of 1 hour
   (ix) 66.66% of 300
   (x) 20% of 1 million rupees.
2. Mr. Arvind Vidyarthi spends 30% of his money on education and he has total Rs, 15,000. How many rupees he spends on education?
3. Sonia purchased 80 metres of cloth, out of which 35% was used for making trousers. How much cloth was used by her for making trousers?
4. The total no. of students in the school are 1250. 40% of the students are girls. Find the number of boys.
5. William's monthly salary was Rs. 1140 . His salary is increased by 16.66%. How much increase has he gotten? Also find the salary if his salary increased by 33.33%.
6. The population of Vatican city is 700. If it increases by 7.14% per annum (*i.e.*, every year). Find the population of the Vatican city after one year.
7. The speed of a car is 85 km/hr. It is increased by 20%. Find the increased speed of the car?
8. A shopkeeper annonces a reduction of 8.33% on all its prices after new year. If a wrist watch was earlier for Rs. 2400. How much would it costs now?
9. 44% of the students in a class are females and the number of male students is 42. Find the total no. of students in the class.
10. 30% of a number is 225. Find the number.
11. A horse costing Rs 80,000 one year ago now costs 25% less. Find the changed price.
12. 1700 students took an exam 85% students passed it. Find the number of students who failed in the exam.

CAT Tips Now I would like to suggest you that all the problems mentioned in the exercise must be done using fractions instead of percentage, in order to make the calculation simple and handy.

For example see the solution of question 2 :

$15000 \times 0.3 = 4500$ $\left[\text{Recall } \frac{x \times 30}{100} = 0.3x\right]$

Solution for question 5 : Since $16.66\% = \frac{1}{6}$

So the new salary will increase by $\frac{1}{6}$ it means the absolute value of new salary will be $\frac{7}{6}\left(= 1 + \frac{1}{6}\right)$ times of the original salary.

So the increase in salary $= 1140 \times \frac{1}{6} =$ Rs. 190

and increased salary = original salary + increase in salary

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$$= 1 + \frac{1}{6} = \frac{7}{6} = \frac{7}{6} \times 1140 = \text{Rs.}1330$$

Similarly if the salary is increased by 33. ›3% it means

$$33.33\% = \frac{1}{3}$$

Therefore new salary $= 1 + \frac{1}{3} = \frac{4}{3} = \frac{4}{3} \times 1140 = 1520$

Solution for question 6: Since $7.14\% = \frac{1}{14}$

So the new population of the city $= 700 \times \left(1 + \frac{1}{14}\right)$

$$= 700 \times \frac{15}{14} = 750$$

Solution for question 8: The reduced price of the watch

$$= 2400\left(1 - \frac{1}{12}\right) \qquad \left\{\because 8.33\% = \frac{1}{12}\right\}$$

$$= 2400 \times \frac{11}{12} = 2200$$

EXPRESSING ONE QUANTITY AS A PERCENTAGE OF ANOTHER QUANTITY

EXAMPLE 1 *What per cent is number 3 of number 20 ?*

SOLUTION As per cent means out of 100. Then by unitary method

out of $20 \rightarrow 3$

out of $1 \rightarrow \frac{3}{20}$

out of $100 \rightarrow \frac{3}{20} \times 100 = 15\%$

Hence to find what per cent the first number is of second number, we divide the first number by the second number and multiply the result by 100.

EXAMPLE 2 *Ravi obtained 325 marks out of a maximum of 400 marks. Find the percentage of marks obtained by him.*

SOLUTION Required percentage of marks $= \frac{325}{400} \times 100 = 81.25\%$

EXAMPLE 3 *In a factory of 150 workers, 18 were absent in a day what percentage were present?*

SOLUTION Present $= 150 - 18 = 132$

$$\text{Percentage presence} = \frac{132}{150} \times 100 = 88\%$$

EXAMPLE 4 *Kurla obtained 480 marks out of 600 and Birla obtained 560 marks out of 800. Whose performance is better?*

SOLUTION % marks of Kurla $= \frac{480}{600} \times 100 = 80\%$

$$\% \text{ marks of Birla} = \frac{560}{800} \times 100 = 70\%$$

So, obviously Kurla's performance is better than that of Birla even though getting less absolute marks.

INTRODUCTORY EXERCISE-5.2

1. What per cent is :
   (i) 30 out of 600 ?
   (ii) 25 out of 160 ?
   (iii) 75 out of 225 ?
   (iv) 36 kg of 150 kg ?
   (v) 90 cm of 4.5 metre.
   (vi) 60 litres of 40 litres
   (vii) 800 shirts out of 1200 shirts?
   (viii) 875 m of 2 km. ?
2. Express:
   (i) 20 as a percentage of 500.
   (ii) 60 kg as a per cent age of 80 kg .
   (iii) 350 ml as a percentage of 5.6 litre.
   (iv) Rs. 13 as a percentage of Rs. 39.
   (v) 15 seconds as a percentage of 1 hour.
   (vi) 27° as a percentage of 360°.
3. Manu scored 384 marks out of 450. What per cent marks did she get?
4. In an election, out of 60,000 eligible voters 42000 cast their vote. Calculate the percentage of voters casting their votes.
5. A tin contains 24 litres of milk. Due to leakage, 720 ml is lost. What per cent of milk is still present in the tin?
6. Price of an item increased from 16.50 to Rs. 41. 25. Find the percentage increase in price.
7. The excise duty on a certain item has been reduced to Rs. 3480 from Rs. 5220 . Find the percentage reduction in the excise duty for that item.
8. Out of total production of 6450 tonnes of a coalmine a quantity of 645 tonnes was lost during extraction. What per cent of the total production was the net coal extracted?
9. A cricket team played 24 matches. The team won 9 matches and lost 3 matches. 12 matches ended in draw. What per cent of the total matches did the team lose?
10. In a particular month, Rs. 10,000 were allocated for the food items in a hostel out of total Rs. 50,000 budget. Further Rs. 2000 is allocated for the fruits out of Rs.10,000 (allocated for food items) what per cent of total budget is spend on fruits only?

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SOME MOST IMPORTANT VALUES

$\frac{1}{2} \to 50\%$, $\frac{1}{3} = 33.33\%$,

$\frac{2}{3} = 66.66\%$, $\frac{1}{4} \to 25\%$,

$\frac{3}{4} \to 75\%$, $\frac{1}{5} \to 20\%$,

$\frac{2}{5} \to 40\%$, $\frac{3}{5} \to 60\%$,

$\frac{1}{6} \to 16.66\%$, $\frac{5}{6} \to 83.33\%$

NOTE The application of percentage is very diverse in nature *e.g.* profit and loss, Simple and compound interest etc. All these are exactly based on the percentage increase/decrease of the original (or actual) value.

Percentage increase/decrease in a quantity

$$= \left(\frac{\text{change in quantity}}{\text{original quantity}} \times 100\right)\%$$

EXAMPLE 1 *The height of Abhimanyu some times ago was 110 cm. Now his height is 120 cm. Find the percentage change in his height.*

SOLUTION $\frac{120-110}{110} \times 100 = 9.09\%$

SHORTCUT $\frac{10}{110} \to \frac{1}{11} \to 9.09\%$

Alternatively: $\frac{12}{11} \to 109.09\%$, so increase = 9.09%

EXAMPLE 2 *The total expenses of a hostel were Rs. 8000 per month. Some students left the hostel due to which the new expenses come down by Rs. 1000. Find the percentage decrease in expenses of the hostel.*

SOLUTION $\frac{1000}{8000} \to \frac{1}{8} \to 12.5\%$

EXAMPLE 3 *Salary of Raja is Rs. 9000 per month and salary of Rani is Rs. 10,000 per month.*

(i) What per cent is the salary of Rani to that of Raja ?

(ii) What per cent is the salary of Raja to that of Rani?

SOLUTION (i) $\frac{10,000}{9,000} \times 100 = 111.11\%$

Alternatively:

$\frac{10}{9} \to \frac{9}{9} + \frac{1}{9} \to 100\% + 11.11\% \to 111.11\%$

(ii) $\frac{9}{10} \to 90\%$ $\left(\because \frac{1}{10} \to 10\%\right)$

Hence, salary of Raja is 90% to the salary of Rani.

PERCENTAGE CHANGE AND PERCENTAGE POINT CHANGE

Last year Abhijeet's salary was Rs. 10,000 and Sonu's salary was Rs. 8,000. This year Abhijeet's salary is Rs. 12,000 while Sonu's salary is Rs. 10,000.

1. (i) What is the percentage increase of Abhijeet's salary?

(ii) What is the percentage increase of Sonu's salary?

(iii) Percentage increase in Sonu's salary is how much percent greater than the percentage increase in Abhijeet's salary?

(iv) What is the percentage point change in the salary of Sonu and Abhijeet?

SOLUTION (i) $\frac{12,000-10,000}{10,000} \times 100 = 20\%$

or $\frac{2}{10} \to \frac{1}{5} \to 20\%$

(ii) $\frac{2}{8} \to \frac{1}{4} \to 25\%$

(iii) Percentage increase of Sonu's salary = 25

Percentage increase of Abhijeet's salary = 20

So the required percentage $= \frac{25-20}{20} \times 100 = 25\%$

It means percentage increase of Sonu's salary is 25% greater than the percentage increase of Abhijeet's salary.

(iv) Percentage point change

= (Percentage increase in Sonu's salary

– Percentage increase in Abhijeet's salary)

= 25 – 20 = 5 percentage point

Infact percentage point change is the difference between two percentage values.

ADVANCED CONCEPT OF PERCENTAGE CHANGE

(A) If a value p is increased by $x\%$, then we have to decrease the resultant value by $\left(\frac{x}{x+100} \times 100\right)\%$ to get back to the original value p.

Original value p $\xrightarrow{\text{increasing value}}$ $\frac{p \times x}{100}$ (increasing value) $\longrightarrow$ Increased value $\left(p + \frac{px}{100}\right)$

$$= p\left(\frac{100+x}{100}\right)$$

Now the percentage decrease

$$= \frac{\left[p\left(\frac{100+x}{100}\right) - p\right]}{p\left(\frac{100+x}{100}\right)} \times 100 = \left(\frac{x}{100+x} \times 100\right)\%$$

In other words (*i.e.*, in terms of fraction) if a value is increased by $\frac{n}{d}$ then to get back the same number p from the resultant value, we have to decrease the increased value by $\left(\frac{n}{d+n}\right)$.

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EXAMPLE 1 *Salary of Rajeev Ratan in 2001 was \$100 per day and his salary in 2002 was \$ 125 per day. Again in 2003 his salary was \$ 100 per day.*

(i) What is the percentage increase in his salary in 2002?

(ii) What is the percentage decrease in his salary in 2003 over 2002?

SOLUTION (i) $\frac{125-100}{100}\times 100 = 25\%$

(ii) $\frac{125-100}{125}\times 100 = 20\%$

In fact there is same absolute change but percentage change is different due to different denominator (*i.e.*, base change). So the base change is as much important as the numerator or absolute change in quantity.

EXAMPLE 2 *Age of Ravi is 20 years. If the Shyam's age is 25% greater than that of Ravi then how much per cent Ravi's age is less than Shyam's age ?*

SOLUTION Age of Ravi Age of Shyam

$$20 \underset{(-20\%)}{\overset{(+25\%)}{\rightleftarrows}} 25$$

Solution using percentage change formula

Here $x = 25$, then

percentage decrease (or less) $= \frac{25}{125}\times 100 = 20\%$

Solution using fractional change formula

Since increase $= \frac{1}{4}$ $\quad\because \left(25\% = \frac{1}{4}\right)$

Therefore decrease $= \frac{1}{4+1} = \frac{1}{5} \to 20\%$

EXAMPLE 3 *Gautam has 33.33% more pencils than Rahul has. By how much per cent less pencils Rahul has than that of Gautam?*

SOLUTION

$$33.33\% \to \frac{1}{3}$$

Rahul ⇄ Gautam

$$-25\% = \frac{1}{4} = \frac{1}{3+1}$$

So Rahul has 25% less pencils than Gautam has.

EXAMPLE 4 *Height of Amitabh is 50% greater than the height of Amir Khan. Height of Amir Khan is how much per cent less than that of Amitabh?*

SOLUTION

$$+50\% \to \frac{1}{2}$$

Amir ⇄ Amitabh

$$-33.33\% = \frac{1}{3} = \frac{1}{2+1}$$

Thus the height of Amir is 33.33% less than Amitabh.

Explanation of Concept : Assume A has 10 apples and B has 20 apples, then we say that B has 100% more apples than A has. Again we can say that A has 50% less apples than B has.

To calculate the percentage change :

$$100\% \to \frac{1}{1}$$

$A \rightleftarrows B$

$$-50\% = \frac{1}{2} = \frac{1}{1+1}$$

Consider another example : Let us assume A has 6 chocolates and B has 10 chocolates then we can say that B has 66.66% more chocolates than A. So again we can also say that A has 40% less chocolates than B.

$$66.66\% \to \frac{2}{3}$$

$A \rightleftarrows B$

$$-40\% = \frac{2}{5} = \frac{2}{3+2}$$

Remember the percentage change graphic is almost 70–80% better in giving quick and simple results. So instead of using the percentage formula, use the percentage change graphic method. For this you are just required to know the percentage fraction conversion table.

(B) If a value P is first decreased by x% then to get back the original value P, we have to increase the decreased (resultant) value by $\left(\frac{x}{100-x}\times 100\right)$

$$\underset{P}{\text{original value}} \xrightarrow[\frac{Px}{100}]{\text{(decreasing value)}} \underset{\left(P-\frac{Px}{100}\right)=P\left(1-\frac{x}{100}\right)}{\text{decreased value}}$$

Now, the per cent increase $= \frac{\left[P-P\left(1-\frac{x}{100}\right)\right]}{P\left(1-\frac{x}{100}\right)}\times 100$

$$= \frac{x}{100-x}\times 100$$

In terms of fraction, if a value P is first decreased by n/d then to get back the original number P. We have to increase the decreased (or resultant) value by $\frac{n}{d-n}$.

EXAMPLE 5 *Due to irregular working habits of Ramachandran, his salary was reduced by 20% but after some months his salary was increased to the original salary. What is the percentage increase in salary of Ramachandran?*

SOLUTION Let initial salary be Rs. 100

$$100 \underset{+25\%}{\overset{-20\%}{\rightleftarrows}} 80$$

Alternatively : Here $x = 20$

then % increase $= \frac{20}{(100-20)}\times 100 = 25\%$

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Alternatively:

$$-20\% \to \frac{1}{5}$$

(Initial salary) P ⟶ Q (Decreased salary)

$$+25\% = \frac{1}{4} = \frac{1}{(5-1)} \qquad \left[\frac{n}{d-n}\right]$$

Hence his salary will be incresed by 25%

EXAMPLE 6 *Kajol usually wears saree, which is 16.66% less than the actual length of the saree. By how much per cent the actual length of the saree is greater than the length of saree which kajol usually wears?*

SOLUTION Actual length ⟶ A; Usual length ⟶ U

$$-16.66\% = \frac{1}{6}$$

$$+20\% = \frac{1}{5} = \left(\frac{1}{6-1}\right) \quad \because \left[\frac{n}{d-n}\right]$$

So the actual length of the saree is 20% greater than the usually used saree.

NOTE
(1) If a value X is first increased by p% to Y then Y is again decreased to X by q% then p is always greater than q (for positive values)
(2) If a value X is first decreased by p% to Y and then Y is increased by q% to X, then p is always less than q.
(3) If a value A is increased by p% then again by q% once again it is increased by r%, then the final value will be same as if you change the order of p, q, r i.e., A can be first increased by r% then by q% and then by p% still the result will be same.
(4) The rule 3 is also applicable for the decreasing of the values.
A value 'A' is first decreased by p%, then by q% and then by r% and so on, the resultant value will be same as when A is first decreased by q% then by p% and then by r% etc.
Note : In case 3 and 4 we are discussing the successive increase or decrease in the value.
(5) A value 'A' is first increased by p% then by q% and then it is reduced by r% will give the same results as when A is first decreased by r%, then increased by q% and then by p% etc.

EXAMPLE 7 *Initially Ms. Rakhi Sawant has Rs. 200 in her wallet then she increased it by 20%. Once again she increased her amount by 25%. The final value of money in her wallet will be how much per cent greater than the initial amount.*

SOLUTION $200 \xrightarrow{+20\%} 240 \xrightarrow{+25\%} 300$

So the required % increase $= \frac{300-200}{200} \times 100 = 50\%$

EXAMPLE 8 *The age of B is 50% greater than the age of A. The age of C is 20% less than the age of B. By how much percentage the age of C is greater than the age of A.*

SOLUTION $A \underset{100}{} \xrightarrow{+50\%} B \underset{150}{} \xrightarrow{-20\%} C \underset{120}{}$

So the required percentage change $= \frac{20}{100} \times 100$

$= 20\%$

CONCEPT OF PRODUCT CONSTANCY

It is the same as we know the inverse proportion in the chapter of Ratio, Proportion and Variation.

e.g., When the rate of a pencil is Rs. 1.25 then we can purchase 16 pencils by paying Rs. 20. If the rate of a pencil is decreased by Rs. 0.25 then we can purchase 20 pencils by paying Rs. 20.

Explanation : Rate × No. of pencils = Price

$$1.25 \times 16 = 20$$

$$1.00 \times 20 = 20$$

So you can see that here the product (20) is constant in both the cases. Thus it is clear that if we reduce the price of a pencil to Rs. 0.50, then we can purchase 40 pencils in Rs. 20.

Some more examples of product constancy :
(i) speed × time = distance
(ii) rate × time = cost
(iii) efficiency × time = work
(iv) length × breadth = area
(v) average × no of elements = total value
(vi) rate × quantity = price (or expenditure)

e.g., The price of sugar is increased by 25% then by how much per cent should a customer reduce the consumption (*i.e.,* quantity used) of sugar so that he has not to increase his expenses on sugar.

price × quantity = expenditure

$$100 \times 100 = 10{,}000$$

$$125 \times x = 10{,}000$$

$\Rightarrow x = \frac{10{,}000}{125} = 80$. Therefore % reduction $= 20\%$

or $1 \times 1 = 1$

$1.25 \times k = 1 \Rightarrow k = 0.8$

thus there will be 20% decrease in the consumption of sugar in order to maintain the same expenditure on sugar.

PRODUCT CONSTANCY CONDITIONS

- When one factor of a product is increased by p% then the other factor will be decreased by $\left(\frac{p}{100+p} \times 100\right)$%

  It means when one factor of a product is increased by $\frac{n}{d}$ then the other factor is decreased by $\frac{n}{(d+n)}$

- When one factor of a product is decreased by p% then the other factor will be increased by $\left(\frac{p}{100-p} \times 100\right)$%

  It means when one factor of a product is decreased by $\frac{n}{d}$ then the other factor will must be increased by $\frac{n}{(d-n)}$.

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EXAMPLE 1 *If the price of a commodity be raised by 20% then by how much per cent a house holder reduce his consumption of the same commodity so that his expenditure does not increase.*

SOLUTION Since here product (*i.e.*, expenditure) is constant
rate × consumption = expenditure

$$\text{initially} \rightarrow 1 \times 1 = 1$$

After change $1.2 \times x = 1$

$\Rightarrow x = 0.833 \therefore$ decrease in value = 16.66%

Alternatively:

Increase in rate $\boxed{+20\% = +\frac{1}{5}}$ $\longrightarrow$ Decrease in consumption $\boxed{-16.66\% = -\frac{1}{6}}$

EXAMPLE 2 *If the price of petrol falls down by 20% by how much per cent must a person increase its consumption, so as not to decrease the expenditure on this item?*

SOLUTION Since product is constant

decrease by $20\% = \frac{1}{5}$ $\longrightarrow$ increase by $\frac{1}{4} = 25\%$ $\left[\because \frac{n}{d} \rightarrow \frac{n}{(d-n)}\right]$

EXAMPLE 3 *Due to 50% increase in the price of rice . We purchased 5 kg less rice with the same amount of Rs. 60. What is the new price of rice?*

(a) Rs. 4.66 (b) Rs. 5
(c) Rs. 4 (d) Rs. 6

SOLUTION Increase in price $50\% = \frac{1}{2}$ $\longrightarrow$ Decrease in amount $\frac{1}{3} = 33.33\%$

Since the new quantity of rice decreases by 33.33% which is equal to 5 kg it means initially there was 15 kg rice to be used.

So, the initial price = Rs. 4 $\because \left(\frac{60}{15} = 4\right)$

and final price = Rs. 6 $\because \left(\frac{60}{10} = 6\right)$

Alternatively: From the options.
Let us consider choice (d).
Therefore [6 × 10 = 60] (finally)
Hence [4 × 15 = 60] (initialy)

EXAMPLE 4 *The length of a plot is decreased by 33.33%. By how much % the breadth of the plot will be increased so that the area remains constant?*

SOLUTION (Decrease) (Increase)

$33.33 = \frac{1}{3} \longrightarrow \frac{1}{2} = 50\%$ $\left[\frac{1}{2} = \frac{1}{3-1}\right]$

CONCEPT OF 'BY' AND 'TO'

Please note that there is a clear difference between "by" and "to". *e.g.*, the income is reduced by 40% it means the new income is 60% of the original and the income is reduced to 40% means the new income is 40% of the original value. Thus "by" represents difference and "to" represents final value.

e.g., The income of Sarika is increased by 20% means its new income is 100 + 20 = 120 % of the original income. The income of Sarika is increased to 120% means the new income of Sarika is 120% of the original income.

EXAMPLE 1 *In an election between two candidates, the candidate who got 57% valid votes won by a majority of 420 votes. Find the total no. of valid votes.*

SOLUTION Winner Loser

$0.57x$ $0.43x$ [100 − 57 = 43]

$0.14x$ [57 − 43 = 14]

$0.14x = 420$

$x = 3000$

Hence total valid votes = 3000

EXAMPLE 2 *Due to fall in manpower, the production in the factory decreases by 60% By what per cent should the working hour be increased to restore the original production in the factory?*

SOLUTION Manpower × Working hours = Production

$(-60\%) = \frac{3}{5} \longrightarrow \frac{3}{2} = (+150\%)$ $\left[\frac{3}{5} \rightarrow \frac{3}{5-3} = \frac{3}{2}\right]$

Hence by 150% working hours will be increased. It means the new working hours will be 2.5 times (not 1.5 times) of the original time.

EXAMPLE 3 *Two numbers are respectively 25% and 40% less than a third number. What per cent is the second of the first ?*

SOLUTION Consider *A*, *B*, *C* three numbers and assume *C* = 100 (as a base)

A B C

75 60 ← (−40%) 100

(−25%)

Now $\frac{60}{75} \times 100 = 80\%$

EXAMPLE 4 *A person gives 10% to his wife 10% of the remaining to a hospital (as a donation) again 10% of the remaining to prime minister's relief Fund. Then he has only 7290 Rs. with him. What was the initial sum of money with that person?*

SOLUTION Since he gives 10% so he is left with 90% of the original sum and since he does the same with the remaining (or left) amount. So it forms a chain.

$\therefore$ Remaining amount = $(x) \times 0.9 \times 0.9 \times 0.9$

$= 0.729x = 7290$

$\Rightarrow x = 10{,}000$, where x is supposed to be initial amount.

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EXAMPLE 5 *Initially a shopkeeper had n chocolates. A customer bought 10% chocolate from n then another customer bought 20% of the remaining chocolates, after that one more customer purchased 25% of the remaining chocolates. Finally shopkeeper is left with 270 chocolates in his shop. How many chocolates were there initially in his shop?*

(a) 300 *(b) 450* *(c) 500* *(d) 600*

SOLUTION $n \times 0.9 \times 0.8 \times 0.75 = 270$

$$n = \frac{270 \times 10{,}000}{9 \times 8 \times 75}$$

$$n = 500$$

NOTE These type of problems (see example no. 4 and 5) can be solved with a great convenience if we solve in reverse order, with the aid of given choices.
Let us consider option (c)

$500 \times 0.9 = 450$
$450 \times 0.8 = 360$
$360 \times 0.75 = 270$

Hence presumed option is correct.

POPULATION RELATED MATHEMATICAL PROBLEMS

If the original population of a locality (*i.e.*, region) be P and the annual growth rate be $r\%$. The population after n years

$$= P\left(1 + \frac{r}{100}\right)^n$$

change (or increase) in the population

$$= P\left[\left(1 + \frac{r}{100}\right)^n - 1\right]$$

If there is decrease in population be $r\%$ then,

total population after n years $= P\left(1 - \frac{r}{100}\right)^n$

and decrease in population $= P\left[1 - \left(1 - \frac{r}{100}\right)^n\right]$

EXAMPLE 1 *If the annual increase in the population be 20% and the present population be 10,000. What will be the population after 3 years hence?*

(a) 16,000 *(b) 17,280*
(c) 14,400 *(d) 1,728*

SOLUTION $10{,}000\left(1 + \frac{20}{100}\right)^3 = 10{,}000\left(\frac{6}{5}\right)^3$

$$= 10{,}000 \times (1.2)^3 = 17{,}280$$

Hence (b) is correct.

EXAMPLE 2 *The population of a town in the first year increases by 10% in the next year it decreases by 10%. Once again in the third year it increase by 10% and in the fourth year it decrease by 10%. If the present population be 20,000 then the population after four years will be :*

(a) 16,902 *(b) 19,602*
(c) 20,000 *(d) none of these*

SOLUTION $20{,}000\left(1 + \frac{10}{100}\right)\left(1 - \frac{10}{100}\right)\left(1 + \frac{10}{100}\right)\left(1 - \frac{10}{100}\right)$

$$= 20{,}000\,(1.1)\,(0.9)\,(1.1)\,(0.9)$$
$$= 20{,}000 \times 1.21 \times 0.81 = 19{,}602$$

Thus option (b) is correct.

NOTE If the value of a number is first increased by $x\%$ and then decreased by $x\%$, the net change is always a decreases (or loss) in original value.

Hence, $\%\ \text{loss} = \left(\frac{x}{10}\right)^2\%$

EXAMPLE 1 *Shweta is a very expert in bargaining. Once she went to a nearby shop. When Shweta asked the price of Shampoo Sachet the shopkeeper told her the price by increasing 15% of the original cost. But Shweta insisted to decrease the price by 15% so the shopkeeper sold it by decreasing the price by 15% What is the loss or profit of shopkeeper and by how much percent?*

(a) no loss *(b) profit of 1.5%*
(c) loss of 2.25% *(d) none of these*

SOLUTION Let the actual price be 100

$$100 \xrightarrow{+15\%} 115 \xrightarrow{-15\%} 97.75 \quad \text{(loss of 2.25\%)}$$

Alternatively: $\text{loss}\% = \left(\frac{15}{10}\right)^2 = 2.25\%$

There is always a loss.

EXAMPLE 2 *If the length and breadth of a rectangle are changed by +20% and −10% What is the percentage change in area of rectangle?*

(a) 8% *(b) 10.8%*
(c) 20% *(d) data insufficient*

SOLUTION $l \times b = \text{area}$

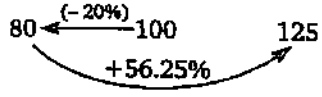
$1 \times 1 = 1$

$1.2 \times 0.9 = 1.08$

So there is 8% increase in the area of rectangle.

EXAMPLE 3 *The salary of A is 20% lower than B's salary and the salary of C is 56.25% greater than A's salary. By how much percent the salary of B is less than the salary of C.*

SOLUTION



(Consider the salary of $B = 100$)

The required value $= \frac{25}{125} \times 100 = 20\%$

Remember: Percentage change

$$= \left(\frac{\text{difference between original value and new value}}{\text{original value}}\right) \times 100$$

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$$\text{Percentage increase} = \frac{\text{increased value} - \text{original value}}{\text{original value}} \times 100$$

$$\text{Percentage decrease} = \frac{\text{original value} - \text{decreased value}}{\text{original value}} \times 100$$

NOTE (1) There is a huge difference between "decreased value" and "decrease in value" and between "increased value" and "increase in value".

For example : Initial value = 70
Final value = 90
it means increased value is = 90
but increase in value is 20
and % increase $= \frac{90-70}{70} \times 100 = 28.57\%$

(2) If there is increase of $\frac{x}{y}$ in any value P then the increased value will be $P\left(1+\frac{x}{y}\right)$.

(3) If there is decrease of $\frac{x}{y}$ in any value, then the decreased value will be $P\left(1-\frac{x}{y}\right)$.

EXAMPLE 1 *Nishith is now 20 years old. Some years later his age will increase by 50% of himself. What will be the new age at that time?*

SOLUTION $20\left(1+\frac{1}{2}\right) = 30$ years $\left[\because 20 + 20\times\frac{1}{2} = 20\left(1+\frac{1}{2}\right)\right]$

EXAMPLE 2 *The average salary of Purushottam in Infosys is 20% less than that was in Microsoft. If the salary of Purushottam in Microsoft be $ 80,000 per month then what is the salary of Purushottam in Infosys?*

SOLUTION $20\% = \frac{1}{5}$

So new salary $= 80{,}000\left(1-\frac{1}{5}\right)$

$= 80{,}000 \times \frac{4}{5} = \$\,64{,}000$

NOTE Remember this type of problems can also be solved exactly through percentages.

As $80{,}000 \times 0.8$ $(100 - 20 = 80\%)$
$= 64{,}000$

but the technique given above is not less important. Some where it becomes very necessary to solve through fractions.
So, keep your eyes open mind focussed and use your wits to solve the problems intelligently as per the situation.

EXAMPLE 3 *Which one of the following is greatest* $\frac{3}{4}, \frac{7}{8}, \frac{16}{19}, \frac{13}{15}$?

SOLUTION

$\frac{3}{4} = 75\%$

$\frac{7}{8} = 87.5\%$

$\frac{16}{19} = 84.21\%$

$\frac{13}{15} = 86.66\%$

So, $\frac{7}{8}$ is the greatest fraction (or rational number)

INTRODUCTORY EXERCISE-5.3

1. What per cent of $\frac{3}{7}$ is $\frac{1}{105}$?
   (a) 10% (b) 2.22%
   (c) 45% (d) 450%
2. What per cent is 3% of 15%?
   (a) 15% (b) 20%
   (c) 40% (d) 66.66%
3. If the cost of a calculator worth Rs 250 is increased by Rs. 100, the rate of increase is
   (a) 100% (b) 40%
   (c) 25% (d) none of these
4. A number increased by 37.5% gives 99 the number is
   (a) 140 (b) 61.5
   (c) 72 (d) 48
5. When 40% of a number is added to 42, the result is the number itself. The number is:
   (a) 105 (b) 72
   (c) 70 (d) 82
6. In an examination 52% of the candidates failed in Science 42% in Mathematics and 17% in both. The no. of those who passed in both the subjects, is :
   (a) 83% (b) 64%
   (c) 23% (d) 55.55%
7. The price of an item is increased by 20% and then decreased by 20%. The final price as compared to original price
   (a) 4% more (b) 20% more
   (c) 20% less (d) 4% less
8. Two candidates fought an election. One got 65% of the votes and won by 600 votes. The total no. of votes polled is:
   (a) 12,000 (b) 10,000
   (c) 2,000 (d) 8,000
9. Out of a no. of electronic items, a person purchases 60% coloured TVs 5% of these were found to be defective. The percentage of defective TVs in all is
   (a) 3% (b) 6%
   (c) 12% (d) can't be determined
10. A's salary is half that of B. If A got a 50% rise in his salary and B got a 25% rise in his salary, then the percentage increase in combined salaries of both is
    (a) 30% (b) 33.33%
    (c) 55% (d) 28%

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EXERCISE

LEVEL 1

1. In our Singapore office there are 60% female employees. 50% of all the male employees are computer literate. If there are total 62% employees computer literate out of the total 1600 employees, then the no. of female employees who are computer literate :
(a) 690 (b) 672
(c) 960 (d) can't be determined

2. The price of a car depreciates in the first year by 25% in the second year by 20% in the third year by 15% and so on. The final price of the car after 3 years, if the present cost of the car is Rs. 10,00,000 :
(a) 7,80,000 (b) 5,10,000
(c) 6,90,000 (d) 1,70,000

3. A shopkeeper charges sales tax of x% up to Rs. 2,000 and above it he charges y%. A customer pays total tax of Rs 320, when he purchases the goods worth Rs. 6,000 and he pay's the total tax of Rs. 680 for the goods worth Rs. 12,000. The value of $(x - y)$ is :
(a) 0 (b) – 2
(c) – 4 (d) 5

4. 40% of a number when added to the square of the same number, then it is increased to 4040% of itself the actual number is :
(a) 175 (b) 400
(c) 40 (d) 120

5. In my office there are 30% female employees and 30% of these earn greater than Rs. 8,000 per month and 80% of male employees earn less than Rs. 8,000 per month. What is the percentage of employees who earn more than Rs. 8,000 per month?
(a) 30% (b) 23%
(c) 60% (d) can't be determined

6. 600 students took the test on physics and chemistry. 35% students failed in Physics and 45% students failed in chemistry and 40% of those who passed in chemistry also passed in Physics, then how many students failed in both :
(a) 162 (b) 138
(c) 60 (d) none of these

7. In an examination 70% of the candidates passed in History and 50% in Geography and 20% students failed in both the subjects. If 500 students passed in both the subjects, then how many candidates appeared for the exam ?
(a) 1,000 (b) 1,500
(c) 2,500 (d) none of these

8. My salary is Rs. 12,345 per month. The salary of my brother is 10% greater than that of mine. The salary of my only sister is 9.09% greater than my only brother. The salary of my wife is $56\frac{12}{23}$% less than the total salary of my brother and sister together, then the salary of my wife is :
(a) greater than my sister's salary
(b) $33\frac{11}{23}$% less than my sister's salary
(c) equal to my salary
(d) $44\frac{11}{23}$% greater than my own salary

9. MDTV is a very popular TV channel. It telecasts the programmes from 8 : 00 a.m. to 12 : 00 p.m. It telecasts 60 advertisements each of 8 seconds and 16 advertisements each of 30 seconds. What is the percentage of time devoted in a day for the advertisements?
(a) 1.5% (b) 1.66%
(c) 2% (d) 2.5%

10. Lagaan is levied on the 60% of the cultivated land. The revenue department collected total Rs. 3,84,000 through the lagaan from the village of Sukhiya. Sukhiya, a very rich farmer, paid only Rs. 480 as lagaan. The percentage of total land of Sukhiya over the total taxable land of the village is :
(a) 0.15% (b) 15%
(c) 0.125% (d) none of these

11. The actual area of a rectangle is 60 cm^2, but while measuring its length a student decreases it by 20% and the breadth increases by 25%. The percentage error in area, calculated by the student is :
(a) 5% (b) 25%
(c) can't be determined (d) none of these

12. The cost of packaging of the mangoes is 40% the cost of fresh mangoes themselves. The cost of mangoes increased by 30% but the cost of packaging decreases by 50%, then the percentage change of the cost of packed mangoes, if the cost of packed mangoes is equal to the sum of the cost of fresh mangoes and cost of packaging :
(a) 14.17% (b) 7.14%
(c) 6.66% (d) none of these

13. Abhinav scores 80% in Physics and 66% in chemistry and the maximum marks of both the papers are 100. What per cent does he score in maths which is of 200 marks, if he scores 80% marks in all the three subjects :
(a) 74% (b) 84%
(c) 87% (d) 83%

14. In the New York Stock Exchange there are 45% female employees and thus the number of male employees is exceeded by 72. Hence the total no. of employees in the New York Stock Exchange is :
(a) 540 (b) 720
(c) 7200 (d) 550

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15. Three candidates *A*, *B* and *C* contested an election. Out of the total votes on a voter list 25% did not vote and 6.66% votes polled were invalid. *C* got 2450 valid votes, which were 40% more than that of *B*. If *A* got only 40% of the total votes, then who is the winner?
(a) A (b) B
(c) C (d) can't be determined

16. The cost of a car is 400% greater than the cost of a bike. If there is an increase in the cost of the car is 15% and that of bike is 20%. Then the total increase in the cost of the 5 cars and 10 bikes is :
(a) 17.5% (b) $16\frac{3}{7}\%$
(c) 18.5% (d) 18.25%

17. The square of a positive number is 2,000% greater than the number itself, then the square of that number is :
(a) 1762 (b) 1635
(c) 441 (d) 139

18. The monthly salary of Shahid and Kareena together is $ 28,000. The salary of Shahid and Kareena is increased by 25% and 12.5% respectively then the new salary of Kareena becomes 120% of the new salary of Shahid. The new (or increased) salary of Shahid is :
(a) $ 15,000 (b) $ 18,000
(c) $ 14,000 (d) $ 16,000

19. 80% of a smaller number is 4 less than 40% of a larger number. The larger number is 85 greater than the smaller one. The sum of these two numbers is :
(a) 325 (b) 425
(c) 235 (d) 500

20. 220% of a number '*X*' is 44. What is 44% of '*X*'?
(a) 88 (b) 8.8
(c) 66 (d) data insufficient

21. The shopkeeper increased the price of a product by 25% so that customer finds it difficult to purchase the required amount. But somehow the customer managed to purchase only 70% of the required amount. What is the net difference in the expenditure on that product ?
(a) 10% more (b) 5% more
(c) 12.5% less (d) 17.5% less

22. In the previous government, party *Q* was in the opposition. Now increasing the seats by 33.33% Q is the ruling party and thus party Q enjoys twice the majority than that of party *P* in the previous government. If there were only two parties *P* and *Q* and the fix no. of seats be 500 in the parliament of Hum-Tum, then the no. of seats of the Q in the new government is :
(a) 225 (b) 200
(c) 275 (d) 300

23. In an examination a candidate got 30% marks and failed by 30 marks. If the passing marks are 60% of the total marks, then the maximum marks will be :
(a) 450 (b) 600
(c) 300 (d) 100

24. In a school there are 1800 students. Last day except 4% of the boys all the students were present in the school. Today except 5% of the girls all the students are present in the school, but in both the days no. of students present in the school, were same. The no. of girls in the school is :
(a) 1200 (b) 800
(c) 1000 (d) 600

25. In a library 60% of the books are in Hindi, 50% of the remaining are in English and rest of the books are in Urdu. If there are 3600 books in English, then the total no. of books in Urdu are :
(a) 2400 (b) 2500
(c) 3000 (d) none of these

26. In a test there are total *n* questions. Bhanu answers 20 out of 25 questions correctly in the first section. In the second section he answers 60% question correct and thus his total score is 66.66% in the test. Given that all the questions carry equal marks, without any negative marking. The total no. of question in the test is :
(a) 50 (b) 60
(c) 75 (d) 100

27. In a class of MBA students 16.66% students are from Science background and 12.5% students are from commerce background and 6.66% students from arts background and rest are from Engineering background. The minimum possible students of engineeering background are :
(a) 45 (b) 77
(c) 100 (d) 120

28. An alloy contains the copper and aluminium in the ratio of 7 : 4. While making the weapons from this alloy, 12% of the alloy got destroyed. If there is 12 kg of aluminium in the weapon, then the weight of the alloy required is :
(a) 48 kg (b) 40 kg
(c) 37.5 kg (d) 14.4 kg

29. Hariharan goes to a shop to buy an FM radio costing Rs. 1404 including sales tax at 8%. He asks the shopkeeper to reduce the price of radio so that he can save the amount equal to the sales tax. The reduction of the price of the radio is :
(a) Rs. 108 (b) Rs.104
(c) Rs. 112.32 (d) none of these

30. The average weight of a class of students is 67.5 kg. The weight of the class teacher is 25% more than the average weight of the class. The average weight of the class is less than the class teacher by *x*%. The value of *x* is :
(a) 33.33% (b) 25%
(c) 20% (d) can't be determined

31. Last year in CAT, each section of the question paper had different weightage. The weightage of QA, DI and VA/RC sections was 8, 9 and 10 respectively. The maximum marks in all the three sections together were 810. Wrong answer did not carry negative marks as a penalty. If Padma had gotten 20% more marks in QA and 8% more marks in DI and 7.14...% more marks in VA/RC, then she must had gotten 100% marks in all the three sections. The total marks that Padma had scored :
(a) 730 (b) 700
(c) 750 (d) 775

32. A salesman gets commission on total sales at 9%. If the sale is exceeded Rs. 10,000 he gets an additional commission as bonus of 3% on the excess of sales over Rs. 10,000. If he gets total commission of Rs. 1380, then the bonus he received is :
(a) Rs. 180 (b) Rs. 120
(c) Rs. 480 (d) data insufficient

33. In Veeru Bhai Pvt. limited company 60% of the employees are men and 48% of the employees are Engineer and 66.6% of these are men. The percentage of women who are not engineers :
(a) 33.33% (b) 60%
(c) 52% (d) 46.66%

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34. Initially Veer had 60% more love letters than that of Zara. In the last month the no. of love letters of Veer increased by 25% and that of Zara decreased by 25%. Again in the present month the no. of love letters of Veer decreased to 60% and that of Zara increased by 60%. Then which of the following statements is correct regarding the present no. of love letters :
(a) Veer has 40% more letters than that of Zara
(b) Zara has 20% less letters than that of Veer
(c) Veer and Zara have equal no of letters
(d) Zara has 37.5% less letters than that of Veer.

35. The charges per hour of internet surfing is increased by 25% then find the percentage decrease in the time period of surfing of a user (a net savy) who can afford only a 10% increase in expenditure :
(a) 22% (b) 12%
(c) 15% (d) 9.09%

36. The average earning of each member of the Ambani family is 20% less than the average earning of each member of the Sahara family and the total earning of Ambani's family is 20% more than the total earning of Saharas's family. The no. of family members in the Sahara is what per cent of the no. of family members of Ambani :
(a) 25% (b) 20%
(c) 66.66% (d) none of these

37. From 2000 onwards, till 2003 the price of computers increased every year by 10%. After that due to government subsidy the price of computers decreases every year by 10%. The price of a computer in 2006 will be approx. how much per cent less than the price in 2000 if the same pattern of price is continued :
(a) 2 (b) 3
(c) 4 (d) none of these

38. A book consists of 30 pages, 25 lines on each page and 35 characters on each line. If this content is written in another note book consisting of 30 lines and 28 characters per line, then the required no. of pages will how much per cent greater than the previous pages?
(a) 4.16% (b) 5%
(c) 6.66% (d) none of these

39. The rate of increase of the price of sugar is observed to be two per cent more than the inflation rate expressed in percentage. The price of sugar on January 1, 2004 is Rs. 20 per kg. The inflation rates of the years 2004 and 2005 are expected to be 8% each. The expected price of sugar on January 1, 2006 would be :
(a) Rs. 23.60 (b) Rs. 24.00
(c) Rs. 24.20 (d) Rs. 24.60

40. A club has raised 75% of the amount it needs for a new building by receiving an average donation of Rs. 600 from the people already solicited. The people already solicited represents 60% of the people the club will ask for donations. If the club is to raise exactly the amount needed for the new building, what should be the average donation from the remaining people to be solicited?
(a) 250 (b) 300
(c) 400 (d) 600

41. A number x is mistakenly divided by 10 instead of being multiplied by 10. What is the percentage error in the result?
(a) – 99% (b) + 99%
(c) – 100% (d) + 100%

42. What is the percentage change in the result when we add 50 to a certain number x, instead of subtracting 50 from the same number x ?
(a) 50% (b) 100%
(c) 300% (d) can't be determined

43. In the Regional Science Centre, Lucknow the rate of ticket is increased by 50% to increase the revenue, but simultaneously 20% of the visitors decreased. What is percentage change in the revenue of Regional Science Centre. If it is known that the centre collects the revenue only from the visitors and it has no other financial supports :
(a) + 20% (b) – 25%
(c) + 30% (d) can't be determined

44. Recently when I visited a show room of shoes shopkeeper told me that he could reduce the price of Bata shoes by 49% and if I were to purchase woodland shoes he could reduce the price to 51% of the original price. If the marked price (*i.e.*, printed price) of both the shoes was same, then which shoes was cheap to buy :
(a) Wood land (b) Bata
(c) both (d) can't say

45. Selling price of a shirt and a coat is Rs. 4000. The cost price of a shirt is 58.33% of the cost price of a coat and so amount of profit on both the shirt and coat is same, then the price of the shirt could be :
(a) Rs 2100 (b) Rs. 2525
(c) Rs. 2499 (d) Rs. 1120

46. On the April 1, 2005 my salary increased from Rs. 10,000 to Rs. 16,000. Simultaneously the rate of income tax decreased by 37.5% , So the amount of income tax paid by me remains constant what is the value of income tax paid by me :
(a) Rs. 3000 (b) Rs. 6000
(c) Rs. 1600 (d) can't be determined

47. In the previous question, if the difference in the rate of income tax be 9 (in percent) then the income tax paid by me :
(a) Rs. 2000 (b) Rs. 2400
(c) Rs. 1600 (d) none of these

48. The average of a set of whole numbers is 27.2. When the 20% of the elements (ie numbers) are eliminated from the set of numbers then the average becomes 34. The number of elements in the new set of numbers can be :
(a) 27 (b) 35
(c) 52 (d) 63

49. In a class, the no. of boys is more than the no. of girls by 12% of the total strength. The ratio of boys to girls is :
(a) 15 :11 (b) 11 : 14
(c) 14 :11 (d) 8 : 11

50. The population of a village is 5000 and it increases at the rate of 2% every year. After 2 years, the population will be :
(a) 5116 (b) 5202
(c) 5200 (d) 5204

51. A customer asks for the production of x number of goods. The company produces y number of goods daily. Out of which z% are unfit for sale. The order will be completed in :
(a) $\frac{x}{100\,y\,(1-z)}$ days (b) $\frac{100\,yz}{x}$ days
(c) $\frac{100\,x}{y\,(100-z)}$ days (d) $\frac{100}{y\,(z-1)}$ days

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LEVEL (2)

1. In the Awadh school Gomti Nagar, there are 500 students. 60% of the students are boys, 40% of whom play hockey and the girls don't play hockey, 75% of girls play badminton. There are only two games to be played. The number of students who don't play any game is :
(a) 10% (b) 36%
(c) 46% (d) can't be determined

2. A fraction in reduced form is such that when it is squared and then its numerater is increased by 25% and the denominator is reduced to 80% it results in 5/8 of the original fraction. The product of the numerator and denominator is :
(a) 6 (b) 12
(c) 10 (d) 7

3. In the Chidambaram's family the ratio of expenses to the savings is 5 : 3. But his expenses is increased by 60% and income increases by only 25% thus there is a deficit of Rs. 3500 in the savings. The increased income of Mr. Chidambaram's family is :
(a) Rs. 35,000 (b) Rs. 28,000
(c) Rs. 25,000 (d) Rs. 18,500

4. In the Presidency College two candidates contested a presidential election. 15% of the voters did not vote and 41 votes were invalid. The elected contestant got 314 votes more than the other candidate. If the elected candidate got 45% of the total eligible votes, which is equal to the no. of all the students of the college. The individual votes of each candidate are :
(a) 2250 and 1936 (b) 3568 and 3254
(c) 2442 and 2128 (d) 2457 and 2143

5. The annual earning of Mr. Sikkawala is Rs. 4 lakhs per annum for the first year of his job and his expenditure was 50%. Later on for the next 3 years his average income increases by Rs. 40,000 per annum and the saving was 40%, 30% and 20% of the income. What is the percentage of his total savings over the total expenditure if there is no any interest is applied on the savings for these four years :
(a) $49\frac{37}{87}\%$ (b) $41\frac{73}{83}\%$
(c) 53% (d) none of these

6. In an election only two candidates contested 20% of the voters did not vote and 120 votes were declared as invalid. The winner got 200 votes more than his opponents thus he secured 41% votes of the total voters on the voter list. Percentage votes of the defeated candidate out of the total votes casted is :
(a) 47.5% (b) 41%
(c) 38% (d) 45%

Directions for questions 7, 8 and 9: Pujari ji, the chief of a temple's trust, has a beautiful daughter Nirjala and a son in law, Radhey. Pujarin, the wife of Pujari ji, lives her own life by receiving the alms from the devotees and receives 9.09% earning of her husband and the daughter together. The earning of Nirjala in each month is Rs. 8000 less than her husband Radhey. The earning of Pujari ji and Radhey together is Rs. 30,000 per month. The earning of Radhey and Nirjala together is Rs. 133.33% greater than that of Pujari ji.

7. The average earning of each Pujari ji. Nirjala and Radhey is:
(a) Rs. 13333.33 (b) Rs. 888.88
(c) Rs. 15,000 (d) none of these

8. What is the earning of Pujarin from the alms?
(a) Rs. 1800 (b) Rs. 2000
(c) Rs. 3600 (d) can't be determined

9. The earning of Radhey is how much per cent greater than that of his wife?
(a) 50% (b) 80%
(c) $47\frac{11}{13}\%$ (d) none of these

10. A sales executive gets 20% bonus of the total sales value and 10% commission besides the bonus on the net profit after charging such commission. If the total sales value be Rs. 10 lakh per annum and the total profit of the company be Rs. 1.32 lakh, then his total earning per annum will be, given that he is not entitled to receive any fixed salary from the company :
(a) 2.3 lakh (b) 3.2 lakh
(c) 2.32 lakh (d) 2.12 lakh

11. Mr Scindia after selling 5.5% stock at Rs. 92 realizes Rs. 32200. Then he invested $\frac{1}{3}$ of the amount in 4.5% stock at Rs. 92, $\frac{2}{5}$ of the amount at Rs. 115 in 5% stock and the remaining in 6% stock at Rs. 56. The change in his income is :
(a) Rs. 56 loss (b) Rs. 78 profit
(c) Rs. 80 profit (d) Rs. 70 loss

12. Each edge of a cube is increased by 20% then the percentage increase in surface area of the cube is :
(a) 144% (b) 40%
(c) 44% (d) 72.8%

Directions for question number 13 and 14: Pati, Patni and Woh (the three persons) were playing a game. At the begining of the game Pati and Patni together had 100% more money than Woh. Patni and Woh together had 300% more than Pati. By the end of the game Pati and Patni together had 100% more money than Woh had and Pati had 12.5% less money than Patni and Woh together had. Finally Pati gained Rs. 800 by the end of the game.

13. Who has suffered the loss?
(a) Patni (b) Woh
(c) Patni and Woh both (d) can't be determined

14. The percentage change of money of Patni is :
(a) 40% (b) 30%
(c) 57.1428% (d) 42.857%

15. The raw material and manufacturing cost formed individually 70% and 30% of the total cost and the profit percentage is 14.28% of the raw material. If the cost of raw material increase by 20% and the cost of manufacturing is increased by 40% and the selling price is increased by 80%, then the new profit percentage is :
(a) 57% (b) 65.8%
(c) 60% (d) can't determined

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16. A, B, C and D purchased a cine-multiplex for Rs. 56 lakhs. The contribution of B, C and D together is 460% that of A, alone. The contribution of A, C and D together is 366.66% that of B's contribution and the contribution of C is 40% that of A, B and D together. The amount contributed by D is :
(a) 10 lakh (b) 12 lakh
(c) 16 lakh (d) 18 lakh

17. In a village three people contested for the post of village Pradhan. Due to their own interest, all the voters voted and no one vote was invalid. The losing candidate got 30% votes. What could be the minimum absolute margin of votes by which the winning candidate led by the nearest rival, if each candidate got an integral per cent of votes?
(a) 4 (b) 2
(c) 1 (d) none of these

18. Every day a mango seller sells half his stock, 10% of the stock overnight gets spoiled. If 1983 mangoes rotted over 3 nights then how many did he start with on the first day?
(a) 25,000 (b) 24,000
(c) 30,000 (d) 32,000

19. A man lost half of his initial amount in the gambling after playing 3 rounds. The rule of gambling is that if he wins he will receive Rs. 100, but he has to give 50% of the total amount after each round. Luckily he won all the three rounds. The initial amount with which he had started the gambling was :
(a) $\frac{500}{3}$ (b) $\frac{700}{3}$
(c) 300 (d) 600

20. In a factory there are three types of Machines M_1, M_2 and M_3 which produces 25% , 35%, and 40% of the total products respectively. M_1, M_2 and M_3 produces 2%, 4% and 5% defective products, respectively. What is the percentage of non-defective products?
(a) 89% (b) 97.1%
(c) 96.1% (d) 86.1%

21. A company has 12 machines of equal efficiency in its factory. The annual manufacturing expenses are Rs. 24,000 and the establishment charges are Rs. 10,000. The annual output of the company is Rs. 48,000. The annual output and manufacturing costs are directly proportional to the no. of machines while the share holders get the 10% profit, which is directly proportional to the annual output of the company. If 8.33% machines remained close throughout the year. Then the percentage decrease in the amount of Share holders is :
(a) 16.66% (b) 14.28%
(c) 8.33% (d) none of these

22. In every month Ravindra consumes 25 kg rice and 9 kg wheat. The price of rice is 20% of the price of wheat and thus he spends total Rs. 350 on the rice and wheat per month. If the price of wheat is increased by 20% then what is the percentage reduction of rice consumption for the same expenditure of Rs. 350? Given that the price of rice and consumption of wheat is constant :
(a) 36% (b) 40%
(c) 25% (d) 24%

23. My friend Siddhartha Ghosh is working in the life insurance Corporation of India (LIC). He was hired on the basis of commission and he got the bonus only on the first years commission. He got the policies of Rs. 2 lakh having maturity period of 10 year. His commission in the first second, third, fourth and for the rest of the years is 20%, 16% 12% 10% and 4% respectively. The bonus is 25% of the commission. If the annual premium is Rs. 20,000 then what is his total commission if the completion of the maturity of all the policies is mandatory :
(a) Rs. 174,00 (b) RS. 23, 600
(c) Rs. 15,000 (d) Rs. 15,500

Directions for question number 24 and 25: DELL Computer has two branches : One in Ohio and second in Texas : The total no. of employees in Ohio office grew this year by 25% to 750 but the ratio of male to female employees is same as in the previous year. The no. of employees in the Texas office grew this year by 9.09% to 1200. The ratio of male to female employees last year in the Texas office was 5 : 6 and the no. of male employees in the Ohio office was 20% less than that of Texas office.

24. The total no. of female employees this year in both the offices is :
(a) 654 (b) 546
(c) 950 (d) can't be determined

25. The total no. of employees in both the offices last year was :
(a) 1500 (b) 1700
(c) 1650 (d) can't be deremined

26. A shepherd had n goats in the year 2000. In 2001 the no. of goats increased by 40%. In 2002 the no. of goats declined to 70%. In 2003 the no. of goats grew by 30%. In 2004, he sold 10% goats, then he had only 34,398 goats. The percentage increase of the no. of goats in this duration was :
(a) 14.66% (b) 16.66%
(c) 20% (d) 33.33%

27. In the above question in which year the no. of goats was minimum?
(a) 2000 (b) 2001
(c) 2002 (d) 2004

Directions for questions number 28, 29 and 30: In the IGNOU (Indira Gandhi National Open University) there are total 16,000 students pursuing MBA, which offers the specialization only in Finance, HR and Marketing. IGNOU accepts only Science, Commerce and Engineering students for the two years course of MBA. The number of science. students is 166.66% of the Commerce students. Number of Engineering students is equal to the number of Science and Commerce students together. Each student can specialize in only one of the marketing, HR and Finance. 20% of Science students opted the Finance, which is 16.66% less than the no. of Commerce students who opted Finance. The total Finance students is equal to 18% the total strength of the MBA students. 32% of Science students opted HR. Commerce students who opted HR is equal to 25% of total students specializing in Finance and Engineering students equal to 6.5% of the total strength of the MBA students opted HR.

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28. The number of Engineering students who opted marketing is :
(a) 7850 (b) 7500
(c) 8850 (d) none of these

29. The percentage of Commerce students who opted HR over the total strength of the MBA students is :
(a) 6.6% (b) 42.5%
(c) 4.5% (d) 62.5%

30. The most preferable course among the MBA students is :
(a) Finance (b) HR
(c) Marketing (d) I. T and systems

31. P% of the students of a class passed the exam. g% of the passed students are girls and b% of the fail students are boys. The percentage of passed boys over the failed girls is :
(a) $\left(\frac{bg}{p} \times 100\right)$ (b) $\frac{100(100-g)p}{(100-p)(100-b)}$
(c) $\frac{(100-g)(100-b)}{(100-p)}$ (d) none of these

32. In the Polo hospital some patients who were suffering from the Hepatites-B were admitted for the treatment , but 9% of the patients were died within half an hour. After treatment, the percentage of patients cured out of the remaining was only 80%. Out of these patients only 70% are completely cured out and the remaining were partially cured out which were equal to 153 patients. The no. of patients (approx.) who were admitted for the treatment for the same was :
(a) 400 (b) 678
(c) 560 (d) 700

33. The total cost of setting up a sugar cane factory is Rs. 1 crore, which produces 5000 ton per annum. Sugar is being sold at Rs. 18 per kg. Manufacturing cost including raw material is Rs. 3.2 per kg, labour and packing charges are Rs. 1.8 per kg. Maintenance and utilities expenses are Rs. 2 per kg and the general expenses (ie all the rest charges) are Rs. 3 per kg, 20% taxes are being paid of the gross annual earnings, then the net profit of the production of the factory per annum is :
(a) Rs. 4.2 crore (b) Rs. 3.2 crore
(c) 5.4 crore (d) none of these

34. A student appeard in the Mock CAT. The test paper contained 3 sections namely QA, DI and VA. The percentage marks in VA was equal to the average of the percentage marks in all the 3 sections. Coincidentally, if we reverse the digits of the percentage marks of QA we get the percentage marks of DI. The percentage marks in VA scored by the student could be :
(a) 48 (b) 66
(c) 69 (d) 81

35. The pressure of a definite mass of a gas is directly proportional to the temperature and inversely proportional to the volume under the given conditions. If temperature is increased by 40% and the volume is decreased by 20% then the new pressure will:
(a) be increased by 75% (b) reduce to 25%
(c) be increase by 20% (d) increase by 28%

36. A computer typist types a page with 20 lines in 10 minutes but he leaves 8% margin on the left side of the page. Now he has to type 23 pages with 40 lines on each page which he leaves 25% more margin than before. How much time is now required to type these 23 pages
(a) $7\frac{1}{2}$ hrs (b) $7\frac{2}{3}$ hrs
(c) $23\frac{1}{2}$ hrs (d) 3.916 hrs

37. A company made a cuboidal box of size $16 \times 12 \times 5$ to sell the ice cream, but later on it was found that the capacity of the box was 14.28% less than the required capacity while the height of the box was correct, which is 5 inches. As per the requirement he had to increase the length and breadth of the box in equal amount then the percentage increase in the area of the base of the box is
(a) 12.5% (b) 6.66%
(c) 16.66% (d) none of these

38. In Sabarmati Express, there are as many wagons as there are the no. of seats in each wagon and not more than one passanger can have the same berth (seat). If the middlemost compartment carrying 25 passangers is filled with 71.428% of its capacity, then find the maximum no. of passangers in the train that can be accommodated if it has minimum 20% seats always vacant
(a) 500 (b) 786
(c) 980 (d) can't be determined

39. The prepaid card of Reliance Infocom gives 19% less talktime than the prepaid card of Tata Indicom, having same price. Again the post-paid card of same price of Tata Indicom gives 10% less talktime than its prepaid card.
Similarly the post paid card of same price of Reliance gives 11.11% less talk time than its prepaid card. How much percent less talk time we get from the Reliance post paid card than the post paid card of Tata Indicom?
(a) 21.11% (b) 20%
(c) 30.11% (d) none of these

40. In the half yearly exam only 70% of the students were passed. Out of these (passed in half yearly) only 60% student are passed in annual exam. Out of those who did not pass the half yearly exam, 80% passed in annual exam. What per cent of the students passed the annual exam ?
(a) 42% (b) 56%
(c) 66% (d) none of these

41. The marks obtained by the students of a school is given below : maximum marks are 50.

Marks	Percentage of Students
< 10	15%
< 20	32%
< 30	40%
< 40	70%
< 50	100%

The ratio of no. of boys to no. of girls who passed the exam is 7 : 6. It is known that a student can pass the exam only when he obtained at least 20 marks in the exam. The total no. of students in the school if the no. of girls who are passed is 408 :
(a) 1100 (b) 1200
(c) 1300 (d) 1430

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Direction for question number 42 to 45: *After defeating Ravana, Ram and his family won a lot of valuable assets in the war. It consists of horses, chariots and some land of Ravan's kingdom. The cost of each horse and chariot was Rs. 20,000 and Rs. 8,000 respectively while the cost of 1 acre land was Rs. 5000. All the property was shared among the four persons in such a way that Ram and Sita got together the same wealth as Laxman and Urmila got together. Ram got more than Sita and Laxman got more than Urmila. Ram got $\frac{1}{3}$rd horses and 20% chariots while Laxman received 50%chariots as the 50% of his total wealth. The no. of horses that Ram and Sita got together was 50% more than that of Laxman and Urmila together had. Sita got 8 horses and Urmila got 7 horses but the Ram and Sita got equal no. of Chariots and Urmila got 20 chariots less than that of Laxman. Urmila got twice the land than that of Sita but 20% less than Laxman.*

42. What is the difference between the wealth of Ram and wealth of Urmila?
(a) 1.2 lakh (b) 1 lakh
(c) 1.4 lakh (d) can't be determined

43. If Laxman wanted to exchange all his chariots with the horses, then who can exchange with his/her horses in terms of wealth :
(a) Ram (b) Sita
(c) Urmila (d) can't be determined

44. The wealth of Urmila is how many per cent less than that of Laxman :
(a) 42% (b) 45%
(c) 35% (d) none of these

45. The wealth due to land and chariot together is how much greater, in per cent, than the wealth due to horses.
(a) 25% (b) 20%
(c) 33.33% (d) none of these

46. A big cube is formed by rearranging the 160 coloured and 56 non-coloured similar cubes in such a way that the exposure of the coloured cubes to the outside is minimum. The percentage of exposed area that is coloured is:
(a) 25.9% (b) 44.44%
(c) 35% (d) none of these

47. Selection into IIMs (Indian Institutes of Management) is quite simple. In our coaching institute some students qualified CAT (The first stage of entrance into IIMs) but coincidentally the no. of boys who qualified the CAT was equal to the no. of girls. Besides these boys and girls got the calls from only IIM Ahamedabad and IIM Bangalore, but each of these from both the IIMs. 60% of the boys failed in the group discussion (the second phase of the selection process) and thus equal no of boys (but distinct) appeared for the personnel interview of IIM-A and IIM-B (interview is the third and final stage of selection of a candidate) but 20% of the boys who appeared for the interview of IIM-A and 60% of the boys who appeared for the interview of IIM-B failed. If it is possible that a candidate can receive the calls from more than one IIMs but he/ she can face the interview of only one IIM. Given that only 24 boys from our coaching institute were selected by the IIM-A and IIM-B also a candidate can appear for the next stage only if he/she qualifies the previous stage of the exam, then find the no. of girls who qualified the CAT (Common Admission Test).
(a) 100 (b) 250
(c) 300 (d) 600

48. In an office there were initially n employees. The HR manager first hired P% employees then after a month q% employees left the office, then there were finally n employees remained in the office, the value of $p-q$ is :
(a) pq (b) $\frac{pq}{100}$
(c) $\frac{p}{q}$ (d) none of these

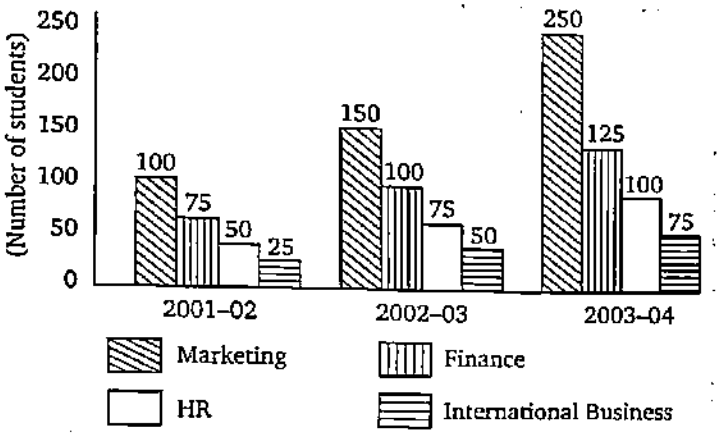
49. In the Garbar Jhala, Aminabad a shopkeeper first raises the price of a Jewellery by x% then he decreases the new price by x%. After one such up down cycle, the price of a Jewellery decreased by Rs. 21025.After a second updown cycle the jewellery was sold for Rs. 484416. What was the original price of the jewellery.
(a) Rs. 5,00,000 (b) Rs. 6,00,625
(c) Rs. 525625 (d) RS. 5,26,000

50. The amount of work in a leather factory is increased by 50%. By what per cent is it necessary to increase the number of workers to complete the new amount of work in previously planned time, if the productivity of the new labour is 25% more.
(a) 60% (b) 66.66%
(c) 40% (d) 33.33%

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Directions for question number 1, 2 and 3: *The bar graph shown below reveals the data about the no. of students in different disciplines of MBA at a prominent Business School in India.*



1. The growth rate of Finance compared to that of marketing is:
(a) less (b) equal
(c) greater (d) none of these

2. Total no. of MBA students in the session of 2004-05 if the no. of students in 2004-2005 is increasing by 9.09% over the previous year :
(a) 555 (b) 600
(c) 777 (d) none of these

3. In the session of 2002-03 what percentage of MBA students are studying International Business?
(a) 13.33% (b) 25%
(c) 6.66% (d) 24%

Directions for question number 4 to 7: *The following table gives the sales details of the books for CAT written by Sarvesh.*

Year	Quant. Aptitude	Data Interpretation	Verbal Ability	Log. Reasoning
2000	4000	3750	4140	4350
2001	4200	3870	4260	4400
2002	4370	3990	4255	4500
2003	4268	3868	5371	4690
2004	4750	4900	5476	4710
2005	4800	5000	5500	4800

4. What is the growth rate of sales of Quantitative Aptitude from 2000 to 2005?
(a) 8% (b) 25%
(c) 20% (d) 40%

5. Which of the categories shows the lowest growth rate from 2000 to 2005?
(a) Q.A (b) DI
(c) VA (d) LR

6. Which category had the highest growth rate in the period?
(a) QA (b) DI
(c) VA (d) LR

7. Which of the categories had either a consistent growth or consistent decline in the period shown?
(a) QA (b) DI
(c) VA (d) LR

Directions for question number 8 to 10: *In India there were only three bicycle making companies in the given period. The following table shows the production of units (in 000)*

Year	HERO	ATLAS	AVON
1990	2.97	1.75	3.77
91	4.22	2.48	4.55
92	5.95	3.14	4.5
93	6.28	3.01	4.76
94	6.33	4.12	4.74
95	8.50	4.21	4.26

UPTREND IN BICYCLES PRODUCTION IN INDIA

8. The simple average rate of growth of production of Hero cycles from 1990-95 :
(a) 37% (b) 42%
(c) 27% (d) 31.5%

9. Which of the following statements is/are correct ?
(i) Atlas cycles have recorded the fastest growth rate.
(ii) Total bicycles production was the highest in 1993
(iii) Hero cycles on an average account for 48% of total bicycles production
(a) I only (b) II only
(c) III only (d) none of these

10. Atlas cycles on an average account for what percentage of the total bicycles production :
(a) 15% (b) 23.4%
(c) 34.5% (d) 29%

Directions for question number 11 to 14 : *Answer the questions based on the following information, which gives data about certain butter producer companies in India.*

Name of the Company	Production ('000) tonnes	Capacity Utilisation (%)	Sales ('000) tonnes	Total sales value (in crore)
Amul	1.54	59.35	1.47	17.45
Nestle	1.64	64.80	1.26	15.25
Parag	2.48	71.20	2.03	26.75
Amrit	2.97	76.50	2.55	31.15
Total (including others)	11.60	61.30	10.67	132.80

11. What is the maximum production capacity (in '000 tonnes) of Nestle for Butter?
(a) 2.53 (b) 2.07
(c) 2.84 (d) 2.97

12. Which company out of the four companies mentioned above has the maximum unutilised capacity in ('000 tonnes)?
(a) Amul (b) Nestle
(c) Parag (d) Amrit

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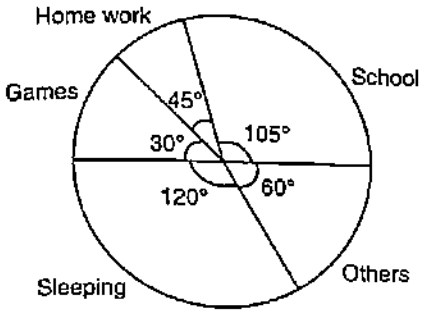
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13. What is the approximate total production capacity (in '000 tonnes) for butter in India?
(a) 7.8 (b) 18.9
(c) 11.60 (d) data insufficient

14. What per cent of the total market share (by sales value) is controlled by others?
(a) 32% (b) 83%
(c) 67% (d) data insufficient

Directions for (Q. no. 15 to 19): The following pie chart shows the hourly distribution of all the major activities of a student.



15. The percentage of time which he spends in school is :
(a) 38% (b) 30%
(c) 40% (d) 25%

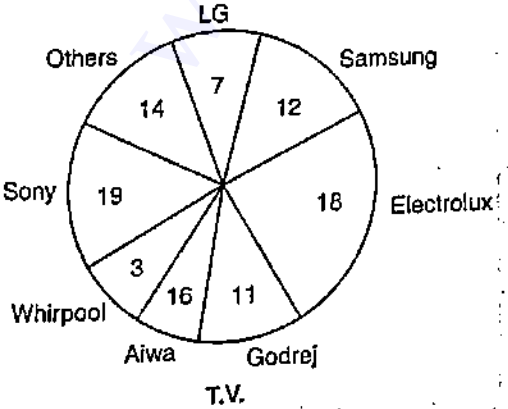
16. How much time (in per cent) he spends in games in comparison to sleeping?
(a) 30% (b) 40%
(c) 25% (d) none of these

17. If he spends the time in games equal to the home work and remains constant in other activities, then the percentage decrease in time of sleeping :
(a) 15% (b) 12.5%
(c) 20% (d) none of these

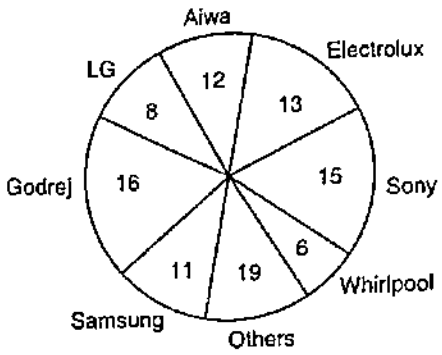
18. What is the difference in time (in hours) spends in school and in home work :
(a) 2 (b) 3
(c) 4 (d) 8

19. If he spends $\frac{1}{3}$rd time of home work in Mathematics then the no. of hours spends in rest of the subjects in homework :
(a) 1 (b) 2
(c) 3 (d) 4

Directions for question number 20 to 23 : The two pie charts show the market share of different companies which produces TV and Refrigerator (both) in the first quarter of 2005-06.



T.V.



REFRIGERATOR

NOTE The graph is not shown to scale.

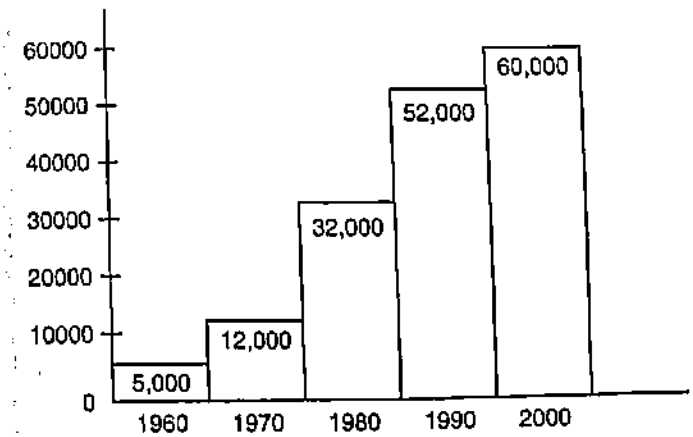
20. The difference in the angle subtended by Sony TV. and refrigerator is :
(a) 7.2° (b) 14.4°
(c) 21.6° (d) none of these

21. If the turn over of Samsung TV is Rs. 31 crores and of Samsung refrigerator is Rs. 9 crore, then what is the overall market share of Electrolux TV and refrigerater combined?
(a) 33.33% (b) 28%
(c) 16.7% (d) 65%

22. If the turnover in the first quarter of 2005-06 of Electrolux TV and Refrigerator is Rs. 42 crores and Rs. 6 crores respectively, then what is the average annual turnover of Godrej and Aiwa in both the product categories together?
(a) Rs. 304 crores
(b) Rs. 284 crores
(c) RS. 178 crores
(d) none of the above

23. Which of the following cann't be deduced from the given data :
(a) The angular difference between Samsung TV and Electrolux refrigerator is 3.6°.
(b) Sony is the market leader in the TV and Refrigerator segment combined.
(c) For every Rs. 100 turnover of Whirlpool Refrigerator, the difference in the turnover of Electrolux and Samsung Refrigerator is Rs. 33.34 crore
(d) none of the above

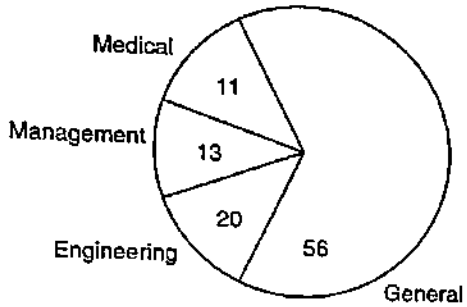
Directions for question number 24 to 28 : Number of different colleges in India in different years is given in the graph below. Distribution of different colleges in the years 1980 and 1990 shown in the pie charts below :



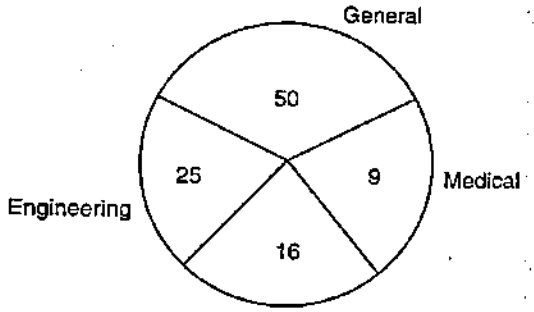
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(1990)



(1990)

NOTE General courses include all the courses except Medical, Engineering and management

24. What is percentage increase in the number of colleges from 1960 to 1990?
(a) 1000% (b) 940%
(c) 1040% (d) 470%

25. The growth rate in no. of medical colleges in 1980 to 1990 is :
(a) 33% (b) 52%
(c) 36% (d) 39%

26. By what percentage did the no. of Engineering colleges go up from 1960 to 1970?
(a) 38% (b) 28%
(c) 42% (d) cann't of be determind

27. Till 1990 what was the highest increase in the number of colleges in any decade?
(a) 150% (b) 166.66%
(c) 200% (d) 140%

28. If the projected increase in the number of colleges in 2000 over 1990 is to be the same in all the categories of colleges, the percentage of Medical colleges in 2000 will be :
(a) 9% (b) 11%
(c) 7% (d) 13%

Directions for question number 29 to 33 : *Answer the questions based on the following information. The following table gives the tariff (in paise per kilo-litre) levied by the Lucknow jal Nigam in 2003-04 in four sectors and the region within them. (Each sector is divided in 5 regions). The table gives the percentage change in the tariff as compared to 2000-01.*

	Region 1		Region 2		Region 3		Region 4		Region 5	
	P/kl	%	P/kl	%	P/kl	%	P/kl	%	P/kl	%
Sector 1	1000	+ 25	400	+ 14.28	250	– 16.66	625	+ 4.166	720	+ 20%
Sector 2	800	+ 33.33	375	+ 7.14	350	– 12.5	750	+ 7.14	360	– 10
Sector 3	625	– 16.66	525	– 12.5	400	+ 14.28	240	– 4%	320	– 20
Sector 4	575	– 4.166	800	– 20%	500	– 16.66	360	+ 20%	400	– 11.11

29. If the amount of water consumed by the various regions in sector 1 is the same, then as compared to 2000-01 the net tariff in 2003-04 :
(a) increased by 20% (b) increased by 13%
(c) decreased by 12% (d) decreased by 20%

30. What was the approximate average tariff in region 2 in 2000-01?
(a) 450 (b) 675
(c) 575 (d) 525

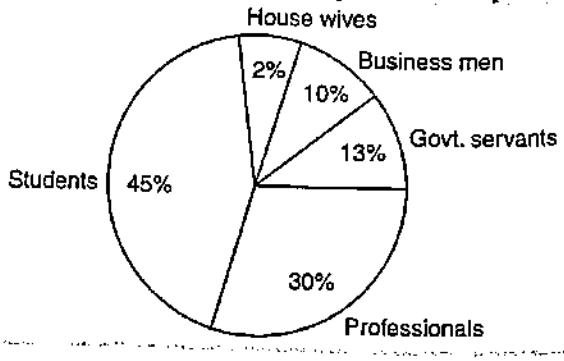
Additional direction for question 31-33 : *The Lucknow Jal Nigam supplies the water under four categories : Urban (25%) rural (15%) domestic (40%) industrial (20%). In 2003-04, the total water supplied by the Lucknow Jal Nigam was 20,000 kilo-litres.*

31. If in 2003-4, there was a 20% decrease in the domestic consumption of water as compared to that in 2000-01 what was the consumption of water in the industrial sector in 2000-01?
(a) 5,000 kl (b) 7500 kl
(c) 10,000 kl (d) 6000 kl

32. In the given two years, what is the total tariff paid by the Urban sector?
(a) Rs. 16,000 (b) Rs. 48,000
(c) Rs. 23,000 (d) can't be determined

33. Which of the following statements is true?
(a) The average tariff in region 2 is 625
(b) The average tariff in region 4 is greater than the average tariff in region 5
(c) In 2000-01 the industrial sector contributed about 30% of the total revenue from water supply
(d) none of the above

Directions for question 34-37 : *A marketing company conducted a survey among 10,000 person in Kanpur :*

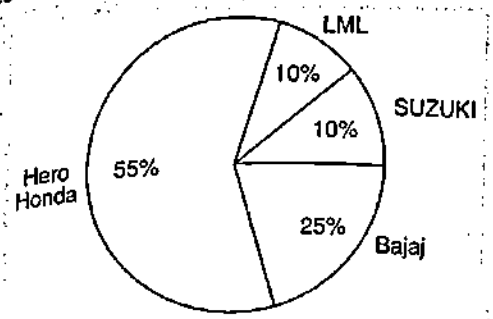


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Percentages



It was observed that some people have more than one bike but from only one company i.e., a particular person can have more than one bike of HeroHonda, but not from bajaj etc. and vice versa. Thus there were total 12,000 bikes with 10,000 persons used for survey. There are only four companies operating in this market.

34. If each of Students, Govt. Servants and housewives use a Hero Honda bike (motorcycle), what per cent of the remaining people drive Hero Honda bike?
(a) 15% (b) 25%
(c) 20% (d) none of these

35. If the number of people who drive one, two and three bikes are in the ratio 15 : 3 : 1 what is the number of people in the survey who do not drive even a single bike?
(a) 750 (b) 400
(c) 600 (d) 500

36. If all the persons driving more than one bike drive only Hero Honda what is the number of people who drive single Hero Honda bike (the data can be used from previous question if necessary)?
(a) 2400 (b) 2100
(c) 4200 (d) 2600

37. If 20% of the persons who drive Bajaj's bike also drive another bike. What is the number of people who drive only Bajaj' bike ?
(a) 2400 (b) 2500
(c) 2660 (d) none of these

Directions for question 38-42 : A table below shows the production and imports of crude oil (in '000 tonnes). Domestic production of crude oil is total of on-shore and off-shore production, which is supplemented by imports to meet the total demand of crude oil in the country.

Year	2001	2002	2003
On shore	12,000	11,500	11,000
Off shore	11,000	19,000	16,000
Imports	21,000	24,000	30,000

(in '000 tonnes)

38. What was the percentage of domestic production of crude oil over imports in 2001?
(a) 80% (b) 140%
(c) 109.52% (d) none of these

39. What was the percentage change in domestic production of crude oil from 2001-2003 ?
(a) 14% (b) 27%
(c) 17.4% (d) – 10%

40. What is the average of total demand of crude oil over the period?
(a) 185 million tonnes (b) 52 million tonnes
(c) 18.5 million tonnes (d) 35 million tonnes

41. What was the approx-percentage increase in imports of crude oil from 2001 to 2003 :
(a) 49% (b) 65%
(c) 43% (d) none of these

42. If in the year 2004, off-shore production declines by 12.5% production on-shore remains the same and total demand increases by 2% what will be the imports of crude oil in 2004 ?
(a) 33.14 million tonnes (b) 63 million tonnes
(c) 39 million tonnes (d) 25 million tonnes.

Directions : (Q. no. 43 to 50) Solve the following question on the basis of given data in the following table :

Production of Rice in India

Year	Quantity (in tonnes)	Percentage change over the previous year
1920-21	1,34,300	+ 06.25%
1930-31	10,97,172	+ 12.50%
1940-41	2,64,280	+ 11.11%
1950-51	1,27,890	– 09.09%
1960-61	2,01,924	+ 20.00%
1970-71	1,12,325	– 16.66%
1980-81	2,13,465	– 25.00%
1990-91	1,69,368	+ 33.33%
2000-01	100,956	+ 50.00%
2010-11	23,800	– 83.33%

43. What is the production of rice in 1959-60?
(a) 1,84,250 (b) 1,68,270
(c) 242308.8 (d) none of these

44. What is the production of rice in 1949-50?
(a) 116263.6 3 (b) 1,23,460
(c) 1,40,679 (d) none of these

45. What is the total production of rice in 1919-20and 1929-30 and 1939-40 ?
(a) 13,26,400 (b) 13,39,516
(c) 1142693.75 (d) can't be determined

46. The production of rice in 2000-01 forms what percentage of total production out of the given years?
(a) 3.12% (b) 3.23%
(c) 4.128% (d) 6.45%

47. What is the difference in production of rice in 1969-70 and 1979-80 ?
(a) 149830 (b) 175752
(c) 53,890 (d) none of these

48. The percentage decrease in production of rice from 1929-30 to 1949-50
(a) 88.3% (b) 85.57%
(c) 66.66% (d) none of these

49. The production of rice in 1959-60 is what percentage of rice in 1960-61.
(a) 86.66% (b) 75.6%
(c) 83.33% (d) 16.66%

50. Out of the given years which year has shown least increase (in amount) over the previous year :
(a) 1940-41 (b) 1960-61
(c) 1920-21 (d) 2000-01

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Answers

INTRODUCTORY EXERCISE-5.1

1. (i) 50, (ii) 54, (iii) 112.5, (iv) 400, (v) 2 quintal, (vi) 544.5, (vii) 88, (viii) 6 minutes, (ix) 200, (x) 2 lakh rupees
2. Rs. 4,500 **3.** 28 metres **4.** 750 **5.** 190 and 1520 **6.** 750 **7.** 102 km/hr **8.** Rs. 2200 **9.** 75 **10.** 750
11. 60000 **12.** 255

INTRODUCTORY EXERCISE-5.2

1. (i) 5%, (ii) 15.625, (iii) 33.33%, (iv) 24%, (v) 20%, (vi) 150%, (vii) 66.66%, (viii) 43.75%
2. (i) 4%, (ii) 75%, (iii) 6.25%, (iv) 33.33%, (v) 0.4166%, (vi) 7.5%
3. 85.33% **4.** 70% **5.** 97% **6.** 150% **7.** $33\frac{1}{3}\%$ **8.** 90% **9.** 12.5% **10.** 4%

INTRODUCTORY EXERCISE-5.3

1 (b)	2. (b)	3. (b)	4. (c)	5. (c)	6. (c)	7. (d)	8. (c)	9. (a)	10. (b)

LEVEL-1

1 (b)	2. (b)	3. (b)	4. (c)	5. (d)	6. (d)	7. (d)	8. (c)	9. (b)	10. (d)
11. (d)	12. (b)	13. (c)	14. (b)	15. (a)	16. (b)	17. (c)	18. (a)	19. (c)	20. (b)
21. (c)	22. (d)	23. (d)	24. (b)	25. (a)	26. (c)	27. (b)	28. (c)	29. (b)	30. (c)
31. (a)	32. (b)	33. (b)	34. (c)	35. (b)	36. (c)	37. (b)	38. (c)	39. (c)	40. (b)
41. (a)	42. (d)	43. (a)	44. (c)	45. (d)	46. (d)	47. (b)	48. (c)	49. (c)	50. (b)
51. (c)									

LEVEL-2

1 (d)	2. (c)	3. (a)	4. (d)	5. (d)	6. (d)	7. (a)	8. (d)	9. (b)	10. (d)
11. (c)	12. (c)	13. (a)	14. (c)	15. (a)	16. (d)	17. (c)	18. (b)	19. (b)	20. (c)
21. (b)	22. (a)	23. (a)	24. (d)	25. (b)	26. (a)	27. (c)	28. (d)	29. (c)	30. (c)
31. (b)	32. (d)	33. (b)	34. (b)	35. (a)	36. (a)	37. (c)	38. (c)	39. (b)	40. (c)
41. (c)	42. (a)	43. (a)	44. (c)	45. (b)	46. (b)	47. (a)	48. (b)	49. (c)	50. (c)

FINAL ROUND

1 (a)	2. (b)	3. (a)	4. (c)	5. (d)	6. (b)	7. (d)	8. (a)	9. (d)	10. (b)
11. (a)	12. (a)	13. (b)	14. (a)	15. (b)	16. (c)	17. (b)	18. (c)	19. (b)	20. (b)
21. (c)	22. (a)	23. (b)	24. (b)	25. (a)	26. (d)	27. (b)	28. (b)	29. (b)	30. (c)
31. (a)	32. (d)	33. (b)	34. (a)	35. (d)	36. (b)	37. (a)	38. (c)	39. (c)	40. (b)
41. (c)	42. (a)	43. (b)	44. (c)	45. (b)	46. (c)	47. (a)	48. (b)	49. (c)	50. (c)

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Hints & Solutions

LEVEL 1

1.

F	M	
$60x$	$40x$	
$42x$	$20x$	(comp. lit.)

$\leftarrow 62x \rightarrow$

$\therefore$ Female comp. literate $1600 \times \frac{42}{100} = 672$

2. $(10,00,000) \times 0.75 \times 0.80 \times 0.85 =$ Rs.5,10,000

3. $2000 \times \frac{x}{100} + 4000 \times \frac{y}{100} = 320$

and $2000 \times \frac{x}{100} + 10,000 \times \frac{y}{100} = 680$

$\therefore \quad x = 4 \quad$ and $\quad y = 6$

$\therefore \quad x - y = -2$

4. Go through option $(40 \times 0.4) + (40)^2 = 1616$

$\left(100 \times \frac{1616}{40} = 4040\right)$

Alternatively: $(x \times 0.4) - x^2 = \frac{x \cdot 4040}{100}$

$\Rightarrow \quad x^2 = 40x \quad \Rightarrow \quad x = 40$

5. Can't be determined. We don't that whether there are some male employees who have exactly Rs. 8000 per month as their salary or not.

6.

	Physics	Chemistry
Failed	35%	45%
Passed	65%	55%

Passed in both = 22% of total student

Percentage of students who are passed in any of the Physics or Chemistry or both = (65 + 55) − 22 = 98%

So, the percentage of students who are failed in both = 2%

Therefore total failed (in both the subject) students = 12

7.

	History	Geography
Pass	$70x$	$50x$
Fail	$30x$	$50x$

both $20x$

$\therefore$ Total failed candidates $= (30x + 50x) - 20x = 60x$

Passed in both $= (100x - 60x) = 40x = 500$

Therefore total students $= 100x = 1250 \quad \because x = \frac{25}{2}$

8. My salary = 100

Salary of my brother = 110

Salary of my sister = 120

Salary of my wife $= 230 - \left(230 \times \frac{1300}{23 \times 100}\right) = 100$

$\because \quad 56\frac{12}{23} = \frac{1300}{23}$

9. $\frac{60 \times 8 + 16 \times 30}{16 \times 60 \times 60} \times 100 = 1.66\%$

10. Total land of Sukhiya $= \frac{480x}{0.6} = 800x$

$\therefore$ Cultivated land of village $= 384000x$

$\therefore$ Required percentage $= \frac{800x}{384000} \times 100 = 0.20833$

11. Area $= l \times b$

$1 = 1 \times 1$

$1 = 0.8 \times 1.25$

So the area remained constant

12. Cost of fresh Mangoes − Packaging cost = Total cost

1	+	0.4	= 1.4
1.3	+	0.2	= 1.5

Percentage increase in cost $= \frac{0.1}{1.4} \times 100 = 7.14\%$

13. $\frac{80}{100} - \frac{66}{100} - \frac{x}{200} = \frac{320}{400}$

$\Rightarrow \quad x = 174 \quad \Rightarrow \quad 87\%$

14.

Male	Female
$55x$	$45x$

$\Rightarrow \quad 10x = 72$

$x = 7.2$

$100x = 7.2 \times 100$

15. Let the total votes be x,

Polled votes $= 0.75x$

valid votes $= 0.70x$

$A + B + C = 0.70x$

2450

1750

$0.4x$

It means $B + C = 0.3x$ since given that $A = 0.4x$

Hence A is the winner.

16.

Bike	Car
x	$5x$
$1.2x$	$5.75x$

Initially total cost $= 25x + 10x = 35x$

Changed cost $= 28.75x + 12x = 40.75x$

Percentage change $= \frac{5.75}{35} \times 100 = 16\frac{3}{7}\%$

18. Go through option

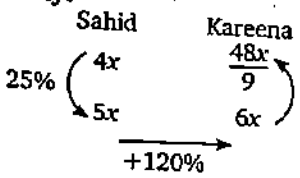
S	K	
15,000	18,000	Going in the reverse direction.
12,000	16,000	

Hence, the presumed option is correct.

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Alternatively:



$\Rightarrow$ Initially $S : K = 4x : \frac{48x}{9} = 3 : 4$

$\Rightarrow$ Sahid's initial salary = 12,000
Sahid's changed salary = 15000

19. Let the smaller number be x and larger number be y

$$0.8x + 4 = 0.4y$$
$\Rightarrow$ $$4y - 8x = 40$$
and $$y - x = 85$$
$\Rightarrow$ $$x = 75 \text{ and } y = 160$$
$\therefore$ $$x + y = 235$$

20. $$\frac{x \times 220}{100} = 44$$

$\Rightarrow$ $$x = 20 \text{ So } \frac{20 \times 44}{100} = 8.8$$

21. Quantity × Rate = Price

$$1 \times 1 = 1$$
$$0.7 \times 1.25 = 0.875$$

$\therefore$ Decrease in price $= \frac{0.125}{1} \times 100 = 12.5\%$

22. Go through options

	P		Q	
In the Ist government	275	←	225	Difference = 50
In the 2nd government	200	←	300	Difference = 100

Hence, the presumed option (d) is correct

NOTE It can also be solved through the equation and variables

23. Passing marks are $0.6x$

So $0.3x + 30 = 0.6x \Rightarrow x = 100$

24. Go through options

Girls	Boys
800	1000

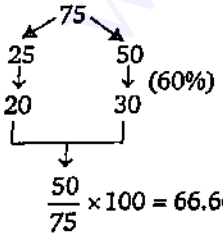
Now, $0.95 \times 800 + 1000 = 0.96 \times 1000 + 800$
Hence the presumed option is correct.

25.

H	E		U
$0.6x$	$0.24x$	→	$0.16x$
	↓		
	3600	⟶	2400

$(\because E : U = 24 : 16 = 3 : 2)$

26. Go through option



$$\frac{50}{75} \times 100 = 66.66\%$$

Hence, the presumed option is correct.

27.

Science	Commerce	Arts	Engineering
$\frac{x}{6}$	+ $\frac{x}{8}$	+ $\frac{x}{15}$	K

No. of Engineering student $= 1 - \left(\frac{x}{6} + \frac{x}{8} + \frac{x}{15}\right) = k = \frac{77x}{120}$

When $x = 120$ (the least possible number)
then the no of Engineering students = 77

28.

Copper		Aluminium	
7	:	4	
21 kg		12 kg	→ 33 kg

Required total alloy $= \frac{33}{0.88} = 37.5$ kg

29. $$1.08x = 1404$$
$$x = 1300$$

Therefore reduction in price = 1404 − 1300 = Rs. 104

$\left[\text{Since } 1300 + 1300 \times \frac{8}{100} = 1404\right]$

So you can solve it by using options.

30. Very fundamental question.

31. $$8x + 9x + 10x = 810$$
$\Rightarrow$ $$x = 30$$

Total marks in QA ⟶ 240
DI ⟶ 270
VA/RC ⟶ 300

Now her score in QA ⟶ $\frac{240}{1.2} = 200$

Her score in DI $= \frac{270}{1.08} = 250$

Her score in $\frac{VA}{RC} = \frac{300}{1.0714} = 280$

Her total score = 200 + 250 + 280 = 730

32. Commission up to 10000 $= 10000 \times \frac{9}{100} = 900$

Again after 10000,

Commission	:	Bonus
9	:	3
$3x$	:	x

$\Rightarrow$
$\therefore$ Bonus $= (1380 - 900) \times \frac{1}{4} =$ Rs. 120

33.

Men	Women
$600x$	$400x$

Total Engineer $= 480x$

Male Engineer $= 480x \times 0.66 = 320x$

$\therefore$ Women who are Engineers $= 160x$

$\therefore$ Women who are not Engineers $= 400x - 160x = 240x$

Required percentage $= \frac{240}{400} \times 100 = 60\%$

34.

	Veer	Zara
In the last month	160	100
After the first change	200	75
After final change	120	120

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Time × Rate = Total charges

35. $1 \times 1 = 1$

$$x \times 1.25 = 1.1$$

$$x = \frac{1.1}{1.25} \times 100 = 88\%$$

Thus decrease in time = 12%

36.

	Ambani	Sahara
Average earning	$4x$	$5x$
No. of family members	k	l
Total earning	$6y$	$5y$

Average earning × Number of family members = Total earning

$$l = \frac{5y}{5x} = \frac{y}{x}$$

$$k = \frac{6y}{4x}$$

$$\text{Required percentage} = \frac{l}{k} \times 100 = \frac{\frac{y}{x}}{\frac{6y}{4x}} \times 100$$

$$= \frac{4}{6} \times 100 = 66.66\%$$

Alternatively : After some steps you can use the options.

37.

Year	Value
2000	100
2001	110
2002	121
2003	133.1
2004	119.79
2005	107.811
2006	97.0299

HINT $(100) \times 1.1 \times 1.1 \times 1.1 \times 0.9 \times 0.9 \times 0.9 = 97.0299$

Now, $\frac{100 - 97.0299}{100} \times 100 \approx 3\%$

38. $30 \times 25 \times 35 = x \times 30 \times 28$

$\Rightarrow$ $x = 31.25$

It means 32 pages.

So, the percentages increase in the no. of pages

$$= \frac{2}{30} \times 100 = 6.66\%$$

39. Rate of increase of the price

= (rate of inflation + 2)% = 8 + 2 = 10%

Jan 2004	Jan 2005	Jan 2006	
20	20	22	24.2
	+2	+2.2	

Alternatively: Expected price after 2 years

$$= 20 \times 1.1 \times 1.1 = 24.2$$

40. Let the total number of people = x

then the amount donated by $0.6x$ people

$$= 600 \times 0.6x = 360x$$

Now since Rs. $360x$ is equal to 75% of the required amount. Hence we need only 25% more amount from the rest of the people *i.e.*, from $0.4x$ people

$$\text{Hence average requirement} = \frac{120x}{0.4x} = 300$$

Alternatively: $(600 \times 0.6 + k \times 0.4)\frac{3}{4} = 600 \times 0.6$

$\Rightarrow$ $k = 300$

41. By mistake $= \frac{x}{10}$

Actual value $= x \times 10$

$$\% \text{ change} = \frac{10x - \frac{x}{10}}{10x} \times 100 = \frac{99}{100} \times 100$$

= 99% (negative)

Since actual value is greater than the wrong value

Alternatively :

Actual result $= 10 \times 10 = 100$ (suppose $x = 10$)

wrong result $= \frac{10}{10} = 1$

$$\% \text{ change} = \frac{100 - 1}{100} \times 100 = 99\%$$

NOTE Percentage error is always calculated on the basis of actual (*i.e.*, correct) value.

42. Use some different values for x then verify.

Let $x = 150$ then % error $= \frac{200 - 100}{100} \times 100 = 100\%$

Again if $x = 100$ then % error $= \frac{150 - 50}{50} \times 100 = 200\%$

Hence, we cannot determine.

43. Number of visitors × Rate = Revenue collected

$$\underset{-20\%}{1} \quad \times 1 \quad = \underset{+50\%}{1}$$

$$0.8 \quad \times 1.5 \quad = 1.2$$

Therefore percentage change in the revenue

$$= \frac{1.2 - 1}{1} \times 100 = 20\%$$

44.

	Bata	Woodland
M.P	100	100
S.P.	51	51

(Since the marked prices are same)

NOTE M.P. → Marked price, S.P. → Selling price
decreased by 49% = reduced to 51%

45. Selling price = cost price + profit

Shirt = $7y + x$

Coat = $12y + x$

Since, the profit is same, so the selling price of shirt will certainly be less than Rs. 2000 (which is half of the total value) as it is clear that cost price of shirt is less than the cost price of coat.

NOTE If selling price of shirt is equal to or greater than the selling price of coat then the C.P. of coat will be equal to or less than the C.P. of shirt which is wrong. Hence the only possible choice is (d).

46. Since we don't have sufficient data. Further any value is possible as the required income tax.

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47. $2004 \longrightarrow 10000 \times x\% = k$

$2005 \longrightarrow 16000 \times \frac{5}{8} x\% = k$

but $x - \frac{5}{8}x = 9 \Rightarrow x = 24\%$

So, the income tax = 2400.

48. Only (c) is correct since it is divisible by 4.

Let the original number of element be x then the new no. of elements will be

$$\frac{4x}{5} = K$$

So K must be divisible by 4

Since, $x = \frac{K \times 5}{4}$

49. Go throught option

Boys	Girls	Total
14	11	25
56	44	100

(12)

Since out of 100 students no. of boys are greater than the no. of girls by 12 *i.e.*, 12% Hence correct.

Alternatively :

Let the no. of boys and girls be x and y respectively

then $(x - y) = \frac{12 \times (x + y)}{100}$

$\Rightarrow \frac{x}{y} = \frac{14}{11}$

50. Population after 2 years $= 5000\left(1 + \frac{2}{100}\right)^2$

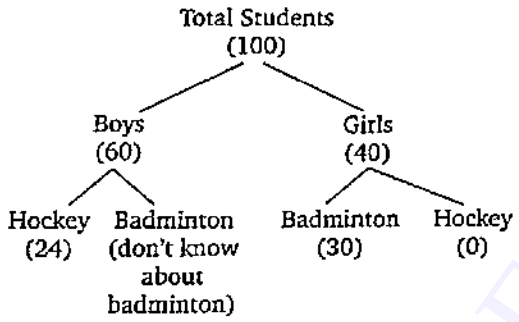
$$5000 \times \frac{51}{50} \times \frac{51}{50} = 5202$$

51. Daily supply $= (100 - z)\%$ of $y = \frac{(100 - z)y}{100}$

$\therefore$ Required no. of days $= \frac{x \times 100}{(100 - z)y}$

LEVEL 2

1.



Since we do not have information that whether the rest of the boys playing badminton or not. So we can not determine the total no. of students who are not playing any of the two games.

2. Go through option. Let us assume option (c)

$$10 = 2 \times 5 = 5 \times 2 = 1 \times 10 \times 10 \times 1$$

Consider the proper fraction $= \frac{2}{5}$

[Since the given percentage values are 25% and 20% that's why we have picked up option (c)].

$$\frac{2}{5} \rightarrow \frac{4}{25} \rightarrow \frac{5}{20}$$

To verify : $\frac{2}{5} \times \frac{5}{8} = \frac{1}{4} = \frac{5}{20}$

Hence presumed option is correct

Alternatively: $\frac{x}{y} \rightarrow \frac{x^2}{y^2} \rightarrow \frac{1.25x^2}{0.8y^2} = \frac{25}{16}\frac{x^2}{y^2}$

Now since $\frac{25x^2}{16y^2} = \frac{5}{8}\left(\frac{x}{y}\right)$

$\Rightarrow \frac{x}{y} = \frac{2}{5}$

3. Income = Expenditure + Savings

$$\begin{pmatrix} 8x = \\ 10x = \end{pmatrix} \begin{pmatrix} 5x + 3x \\ 8x + 2x \end{pmatrix} - x$$

Now the deficit $= (3x - 2x) = x = 3500$

and $\therefore$ the new salary $= 10x = 35{,}000$

Alternatively : Go through options.

4. Go through options

$2457 - 2143 = 314$ (True)

Again $(2457 + 2143) + 41 = 4641$

Now $\frac{4641}{0.85} \rightarrow 5460$

Again $\frac{5460 \times 45}{100} = 2457$ (True)

Hence the presumed option is correct.

Alternatively: Let there be total x eligible voters, and the no. of votes goes to loser is k then

$0.85x - 41 = 2k + 314$ ...(1)

$k + 314 = 0.45x$ ...(2)

Therefore $x = 5460$

then, $5460 \times 0.85 = 4641$

Again $4641 - 41 = 4600$

Again $k + (k + 314) = 4600$

$\Rightarrow k = 2143$ (loser)

and $k + 314 = 2457$ (winner)

5.

Income →	4	4.4	4.8	5.2]	18.4	lakh
Saving →	2	1.76	1.44	1.04]	6.24	lakh
Exp. →	2	2.64	3.36	4.16]	12.16	lakh

So, $\frac{6.24}{12.16} \times 100 = 51\frac{6}{19}\%$

6. Let there be x voters and k votes goes to loser then

$0.8x - 120 = k + (k + 200)$

$k + 200 = 0.41x$

$\Rightarrow k = 1440$

and $(k + 200) = 1640$

Therefore $\frac{1440}{3200} \times 100 = 45\%$

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Solutions for 7 – 9 :

$$P + R = 30{,}000 \quad ...(1)$$

$$N = R - 8000 \quad ...(2)$$

$$(R + N) = 233.3\,(P)$$

$$\Rightarrow \quad 3(R + N) = 7P$$

$$\Rightarrow \quad 6R - 7P = 24{,}000 \quad ...(3)$$

$$\therefore \quad R = 18{,}000 \quad ...(4)$$

and $P = 12{,}000$

and $N = 10{,}000$

7. $\dfrac{P + R + N}{3} = \dfrac{40{,}000}{3} = 1333.33$

8. Can't be determined

9. $\dfrac{8}{10} \times 100 = 80\%$

10. (Bonus) Commission $= \dfrac{20 \times 10{,}00{,}000}{100} = 2$ lakh

but total profit = net profit + $\dfrac{10}{100}$ × net profit

$\Rightarrow$ 1.32 lakh = 1.1 × net profit

$\Rightarrow$ net profit = 1.2 lakh = 1,20,000

$\therefore$ commission = total profit – net profit

= 1,32,000 – 1,20,000 = 12,000

hence total earning = 2,00,000 + 12,000 = 212,000

11. Let Mr. Scindia has x shares of 5.5%

$$x \times 92 = 32{,}200$$

$\Rightarrow$ $x = 350$ shares

Income = 350 × 5.5 = 1925

Now, after investment his income is

$$\left(\frac{1}{3} \times \frac{32200}{92} \times 4.5\right) + \left(\frac{2}{5} \times \frac{32200}{115} \times 5\right) + \left(\frac{4}{15} \times \frac{32{,}200}{56} \times 6\right)$$

$$= 525 + 560 + 920 = 2005$$

Profit = 2005 – 1925 = Rs. 80

12. The surface area of a cube = $6a^2 = 6 \times (\text{side})^2$

New surface area = $6 \times 1.44\,a^2$

$\therefore$ $\dfrac{0.44\,a^2}{a^2} \times 100 = 44\%$

Solution for 13 and 14 :

Pati → *Pt*, *Patni* → *Pn*, *Woh* → *W*

$$(Pt + Pn) = 2W \quad ...(i)$$

$$(Pn + W) = 4\,Pt \quad ...(ii)$$

Solving equation (i) and (ii) we get

$$\frac{Pn}{W} = \frac{7}{5} \quad \text{and} \quad \frac{Pt}{W} = \frac{3}{5}$$

$\Rightarrow$ $Pt : Pn : W = 3 : 7 : 5$

Again $(Pt + Pn) = 2W \quad ...(iii)$

and $(Pn + W) \times 7 = 8 \times Pt \quad ...(iv)$

Therefore $\dfrac{Pn}{W} = \dfrac{3}{5}, \quad \dfrac{Pt}{W} = \dfrac{7}{5}$

$\Rightarrow$ $Pt : Pn : W = 7 : 3 : 5$

Gain of Pati = $7x - 3x = 4x = 800$

$\Rightarrow$ $x = 200$

	Pati	*Patni*	*Woh*
Amount at the begining of Game	= 600	1400	1000
Amount at the end of the game	= 1400	600	1000

13. Only Patni has suffered the loss

14. $\dfrac{1400 - 600}{1400} \times 100 = 57.1428\%$

15. RM + MC = Total cost

Total cost + Profit = Sale price

$70 + 30 = 100 \quad 100 + 10 = 110$

$84 + 42 = 126 \quad 126 + 72 = 198$ + 80%

Therefore profit % = $\dfrac{72}{126} \times 100 = 57.14\%$

16. $A + B + C + D = 56$

$B + C + D = 4.6A$

$\Rightarrow$ $A + B + C + D = 5.6A$ (adding A in both sides)

56 lakh = $5.6A$

$\Rightarrow$ $A = 10$ lakh

Similarly $A + C + D = \dfrac{11}{3} B$

$\Rightarrow$ $A + B + C + D = \dfrac{14}{3} B$

$\Rightarrow$ $B = 12$ lakh

Similarly $4(A + B + D) = C$

$\Rightarrow$ $A + B + D = 2.5C$

$\Rightarrow$ $A + B + C + D = 3.5C$

$\Rightarrow$ $C = 16$ lakh

Therefore $D = (A + B + C + D) - (A + B + C) = 18$ lakh

17. Losing candidate = $0.3\,x$

$\therefore$ Other two candidates = $0.7\,x$

The share of winning candidate = $0.36\,x$

and the second ranker = $0.34\,x$

$\therefore$ Margin (min. possible) = $0.02\,x$

$\Rightarrow$ 2% of x

Let the minimum possible voters be 50 then

$$\frac{2 \times 50}{100} = 1$$

Hence the minimum possible margin of votes = 1

18.

Day	Initial amount	Sales	Remaining over night	Rotten	Stock for next day
I	x	$0.5x$	$0.5x$	$0.05x$	$0.45x$
II	$0.45x$	$0.225x$	$0.225x$	$0.0225x$	$0.2025x$
III	$0.2025x$	$0.10125x$	$0.10125x$	$0.010125x$	

$\therefore$ Total rotten amount = $0.082625x = 1983$

$\Rightarrow$ $x = 24{,}000$

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19. Check through option

Alternatively : Let the initial amount be x (with gambler), then

$$\left(\left((x+100)\frac{1}{2}+100\right)\frac{1}{2}+100\right)\frac{1}{2}=\frac{x}{2}$$

$$\Rightarrow \quad x=\frac{700}{3}$$

20. Non-defective products

$$\frac{25\times 0.98+35\times 0.96+40\times 0.95}{100}\times 100=96.1$$

21.

No. of Machines	Output	Manf. cost	Est. cost	Total cost	Profit
12	48,000	24,000	10,000	34,000	14,000
11	44,000	22,000	10,000	32,000	12,000

Profit = out put − Total cost

= 44,000 − 32,000 = 12,000

Initial value of share holders $= 14{,}000\times\frac{10}{100}=1400$

Changed value of share holders $= 12{,}000\times\frac{10}{100}=1200$

$$\%\text{ decrease}=\frac{200}{1400}\times 100=14.28\%$$

22.

Rice	Wheat
25	9
$\times x$	$\times 5x$
$25x$	$45x$

$$70x=350$$

$$\Rightarrow \quad x=5$$

Hence the price of Rice = Rs. 5 per kg

Price of wheat = Rs. 25 per kg

Now, the price of wheat = Rs. 30 per kg

Let the new amount of Rice be M kg, then

$$M\times 5+9\times 30=350$$

$$M=16$$

Hence decrease (in%) of amount of rice

$$=\frac{25-16}{25}\times 100=36\%$$

23.

Year	Rate of Commission	Commission in values
1	20%	$0.2\times 20{,}000=4000$
	25% (bonus)	$0.25\times 4000=1000$
2	16%	$0.16\times 20{,}000=3200$
3	12%	$0.12\times 20{,}000=2400$
4	10%	$0.1\times 20{,}000=2000$
5–10	4%	$6\times 0.04\times 20{,}000=4800$

Total commission

= (4000 + 3200 + 2400 + 2000 + 48 00) + (1000) = 17,400

Solution for 24-25

	OHIO	TEXAS
		(M : F)
Last year :	600 ← (2 : 1)	(5 : 6) → 1100
This year :	750 ← (2 : 1)	(unknown) → 1200

24. Since we don't know the number of female employees in the Texas office this year so we can't determine

25. 1100 + 600 = 1700

26. There is no need to use the no. of goats *i.e.*, (34, 398) let initially there be 1000 goats then

$$1000\rightarrow 1400\rightarrow 980\rightarrow 1274\rightarrow 1146.6$$

Thus the % increase $=\frac{1146.6-1000}{1000}\times 100=14.66\%$

27. In 2002 (980 goats) as per the flow chart

Optional	Science	Commerce	Engineering	Total
	5000	3000	8000	16,000
Finance	1000	1200	680	2880
HR	1600	720	1040	3360
Marketing	2400	1080	6280	9760

28. 6280 Students of Engineering opted marketing

29. $\frac{720}{16{,}000}\times 100=4.5\%$

30. Marketing, since maximum students have opted marketing.

31. Consider some values and then verify the option.

32. Go through option :

700 ↓ 91% = (100 − 9)% ↓ 637 ↓ 80% ↓ 509.6 ≈ 510 persons ↓ 70% (completely cured out) ↓ 357 ↓ (Partially cured) 153 = (510 − 357)

Hence, the presumed option is corret

33. Total expenditure per kg

= 3.2 + 1.8 + 2 + 3 = 10 = cost price

Selling price = Rs. 18 (per kg)

Gross profit = Rs. 8 per kg = (18 − 10)

Net profit $= 8\times\frac{80}{100}$ (since 20% is tax) = Rs. 6.4

Hence the net profit of the factory = 6.4 × 50,00,000

= Rs. 3,20,00,000 = Rs. 3.2 crore

34. Let the percentage marks in QA = $(10a+b)\%$

Let the percentage marks in DI = $(10b+a)\%$

Let the percentage marks in VA = $x\%$

then $\frac{(10a+b)+x+(10b+a)}{3}=x$

$$\Rightarrow \quad 11a+11b+x=3x$$

$$\Rightarrow \quad x=\frac{11}{2}(a+b)$$

Thus the percentage of the VA section is a multiple of 11.

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35. $P_1 = k\frac{T}{V}$

$$P_2 = k\frac{1.4T}{0.8} = K\frac{7T}{4V}$$

$$\frac{P_2 - P_1}{P_1} = \frac{\frac{7}{4}\frac{T}{V} - \frac{T}{V}}{\frac{T}{V}} = \frac{\frac{3}{4}\frac{T}{V}}{\frac{T}{V}} = \frac{3}{4}$$

Hence, the new pressure will be increased by 75%.

36. $20 \times 0.92 \Rightarrow$ 10 minutes.

$$\frac{23 \times 40 \times 0.90}{20 \times 0.92} = 45$$

Thus the required time is 45 times than the previous time

Hence, 450 minutes $= 7\frac{1}{2}$ hrs.

37. Original volume $= 16 \times 12 \times 5 = 960\ (\text{inch})^3$

Required capacity $= 1120\ (\text{inch})^3$

Increase in area $= \frac{1120}{5} - 16 \times 12$

$= 224 - 192 = 32\ (\text{inch})^2$

$$\% \text{ increase} = \frac{32}{192} \times 100 = 16.66\%$$

38. The total passengers in each compartment $= 25 \times \frac{7}{5} = 35$

Total no. of seats $= (35)^2 = 1225$

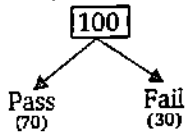
Maximum available capacity $= 1225 \times \frac{80}{100} = 980$ seats

39.

	Tata	Reliance
Prepaid	100	81
Postpaid	90	72

Thus the % decrease in talk time $= \frac{90 - 72}{90} \times 100 = 20\%$

40. **Half yearly exam**



Annual exam

$$\underbrace{\frac{70 \times 0.6}{42}} + \underbrace{\frac{30 \times 0.8}{24}}$$

$\therefore$ Total pass in annual exam $= 42 + 24 = 66$

41. Percentage of passed students $= 68\%$ [100 – 32]%

Number of girls passed the exam $= 408$

Number of boys passed the exam $= 476$

Total passed students $= 884$

Therefore total no. of students $= \frac{884}{68} \times 100 = 1300$

Solution for 42- 45 :

Name	Horse	Chariot	Land	Total (in Rs.)
Ram	2 lakh (10)	80,000 (10)	20 acre = 1 lakh	3.8 lakh
Sita	1.6 lakh (8)	80,000 (10)	8 acre = 40,000	2.8 lakh
Laxman	1 lakh (5)	2 lakh (25)	20 acre = 1 lakh	4 lakh
Urmila	1.4 lakh (7)	40,000 (5)	16 acre = 80,000	2.6 lakh

1. $R + S = L + U$ *and* $R > S$ *and* $L > U$

2. Horses $\rightarrow (R + S) : (L + U) = 3x : 2x = 18x : 12x$

Again Ram have $\frac{1}{3}$ *rd horses*

Therefore $30x \times \frac{1}{3} = 10x$

Therefore the horse of Sita $= 18x - 10x = 8x$

$\Rightarrow \quad x = 1$

Therefore the horse of Ram $= 10$ *and Laxman* $= 5$

No. of chariots of Sita = No. of chariots of Ram $= \frac{K}{5}$

and *no. of chariots of Laxman* $= \frac{K}{2}$

Hence the no. of chariots of Urmila $= K - \left(\frac{K}{5} + \frac{K}{5} + \frac{K}{2}\right) = \frac{K}{10}$

Again $\frac{k}{2} - \frac{k}{10} = 20 \Rightarrow k = 50$ *chariots*

Now the 50% property of Laxman $= 25$ *chariots* $= 2{,}00{,}000$

Hence the total property of Laxman $= 4{,}00{,}000$

Thus the area of Land of Laxman $= \frac{2{,}00{,}000 - 5 \times 20{,}000}{5000}$

$= 20$ acre $= (1$ lakh$)$

Total property of Urmila

$= 1{,}40{,}000 + 40{,}000 + 80{,}000 = 2{,}60{,}000$

Thus the total property of Laxman and Urmila $= 6.6$ *lakh*

42. $3.8 - 2.6 = 1.2$ lakh

43. Value of chariots of Laxman = 2 lakh

Now since only Ram has the horses of worth Rs. 2 lakh. So only Ram can exchange with Laxman.

45. $\frac{7.2 - 6.0}{6.0} \times 100 = 20\%$

46. Total cubes $160 + 56 = 216$

Therefore the side of cube = 6 unit

No. of cubes without any exposure $= (6 - 2)^3 = 64$

Thus 64 cubes will be inside of the big cube

Now rest of the cubes $= 160 - 64 = 96$

Again the no. of cubes with one face outside

$= 6 \times (4 \times 4) = 96$

Hence the required percentage $= \frac{96}{216} \times 100 = 44.44\%$

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47.

(Qualified CAT)
↓
x boys and x girls
↓
(Qualified G.D)
↓
$0.4x$ boys
↓
(Interview)

IIM-A	IIM-B	
[$0.2x$	$0.2x$]	appeared
↓	↓	
Qualified	Qualified	
↓	↓	
$0.8 \times 0.2x$	$0.4 \times 0.2x$	
$= 0.16x$	$= 0.08x$	

Total boys qualified the final stage = 0.24%

Thus $0.24x = 24$

$\Rightarrow$ $x = 100$

48. Go through option and consider some appropriate values

Alternatively : $\frac{p}{100+p} = \frac{q}{100}$

$\Rightarrow$ $100(p-q) = pq$

$\Rightarrow$ $(p-q) = \frac{pq}{100}$

49. Let the original price be P, then the decrease in value of P after one cycle

$$= p\left(\frac{x}{100}\right)^2 = 21025 \quad \text{...(i)}$$

Again the final value after second cycle

$$\Rightarrow P\left(1+\frac{x}{100}\right)\left(1-\frac{x}{100}\right)\left(1+\frac{x}{100}\right)\left(1-\frac{x}{100}\right) = 484416$$

$$\Rightarrow P\left[1-\left(\frac{x}{100}\right)^2\right]^2 = 484416 \quad \text{...(ii)}$$

Dividing equation (ii) by equation (i)

$$\frac{\left[1-\left(\frac{x}{100}\right)^2\right]^2}{\left(\frac{x}{100}\right)^2} = \frac{484416}{21025} = \frac{2304}{100}$$

$$\Rightarrow \frac{1-\left(\frac{x}{100}\right)^2}{\left(\frac{x}{100}\right)} = \sqrt{\frac{2304}{100}} = \frac{48}{10}$$

Let $\frac{x}{100} = k$, then

$$\frac{1-k^2}{k} = \frac{48}{10}$$

$\Rightarrow$ $10k^2 + 48k - 10 = 0$

$\Rightarrow$ $5k^2 - 24x - 5 = 0$

$\Rightarrow$ $k = 5$ or $k = -\frac{1}{5}$ (inadmissible value)

So $x = 20\%$

Hence, $P\left(\frac{x}{100}\right)^2 = 21025$

$\Rightarrow$ $P = 525625$

50.

Men	×	Time	=	Work
100	×	1	=	100 unit
150	×	1	=	150 unit

Extra man power required = 50

but since new workers are $\frac{5}{4}$ times as efficient as existing workers.

$\therefore$ Actual no. of workers $= \frac{50}{5/4} = 40$ men

Hence, required percentage $= \frac{40}{100} \times 100 = 40\%$

Final Round

1. Growth rate of Finance

$$= \frac{125-75}{75} \times 100 = \frac{2}{3} \times 100 = 66.66\%$$

Growth rate of Marketing $= \frac{250-100}{100} \times 100 = 150\%$

2. Total no. of students in 2003-04 is

$= (250 + 125 + 100 + 75) = 550$

No. of student in 2004-05

$$= 550\left(1+\frac{1}{11}\right) = 600 \text{ students}$$

3. $\frac{50}{(150+100+75+50)} \times 100 = \frac{50}{375} \times 100 = 13.33\%$

4. $\frac{4800-4000}{4000} \times 100 = 20\%$

5. $\frac{450}{4350} \times 100 = 10.34\%$, which is lowest in comparison to others

Thus the growth rate of logical reasoning book is lowest.

6. DI $\rightarrow \frac{5000-3750}{3750} \times 100 = 33.33$

and VA $\rightarrow \frac{5500-4140}{4140} \times 100 = 32.85\%$

7. Only LR has consistent growth, others have been fluctuated (QA – 2003, DI – 2003, VA – 2002)

8. $\frac{\left[\frac{8.5-2.97}{2.97} \times 100\right]}{5} = 37.23\%$

9. Growth rate of Hero = 37.23%

Growth rate of Atlas = 28.1%

Avon is least which is clear from the data.

Total bicycle production is highest in 95 hence false

Hero cycle's share $= \frac{2.97 + 4.22 + \ldots 8.5}{79.84} = 43\%$

Hence, (d) since none of (a), (b) and (c) is correct.

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10. $\frac{1.75+2.48+\ldots 4.21}{79.84} = 23.4\%$

11. $\frac{1.64}{64.8} \times 100 = 2.53$

12. Amul has max. unutilised capacity

HINT $\left(\frac{1.54}{59.35} \times 100\right) - 1.54 = 1.05$

Similarly for others can also be find out

13. $\frac{11.6}{61.3} \times 100 \approx 18.9$

14. $\frac{42.20}{132.8} \times 100 = 31.81\%$

15. $\frac{105}{360} \times 100 = \frac{7}{24} \times 100 = 29.166\%$

16. $\frac{30}{120} \times 100 = 25\%$

17. $\frac{15}{120} \times 100 = 12.5\%$

18. $105 - 45 = 60° = 4$ hours

19. $15° = 1$ hour (Maths)
$30° = 2$ hour (other subjects)

20. Sony TV → 19%
Sony Refrigerator → 15%
difference = 4% = 14.4° (Since 1% = 3.6°)

21.
12% → Samsung TV = 31 crore
18% → Electrolux TV = 46.5 crore
100% → Total market = 258 crore

11% → Samsung Refrg. = 9 crore
13% → Electrolux Refrg. = 10.6 crore
100% → Total market = 81.8 crore

$$\text{Market share} = \frac{46.5 + 10.6}{258 + 81.8} \approx 16.7\%$$

22. 18% = Rs. 42 crores, so 27% (= 16 + 11) = Rs. 63 crore
13% = Rs. 6 crore, so 28% (= 12 + 16) = Rs. 13 crore
Total Rs. 76 crores
∴ Annual approx. turnover = 4 × 76 = Rs. 304 crores

23. (a) (13 – 12) = 1% = 3.6
(b) We don't know the turnover of TV and refrigerator market for each brand.
(c) 6% total refrigerator market = 100 crore
⇒ Total refrigerator market ≅ 1667 crore
∴ Difference = 2% of 1667 = 33.34 crore
Thus, (b) can't be inffered.

24. $\frac{52,000 - 5,000}{5000} \times 100 = 9.4 \times 100 = 940\%$

25. Medical college in $1980 = \frac{11}{100} \times 32,000 = 3520$

$$1990 = \frac{9}{100} \times 52,000 = 4680$$

$$\therefore \quad \frac{4680 - 3520}{3520} \times 100 = 32.95$$

26. We don't have the information about the proportion (share) of Engineering colleges in the given years.

27. $\frac{32,000 - 12,000}{12,000} \times 100 = 166.66\%$

28. Number of medical colleges in 1990

$$= \frac{52,000 \times 9}{100} = 4680$$

Increase in the total no. of colleges

$$= 60,000 - 52000 = 8000$$

Increase in the no. of medical colleges $= \frac{8000}{4} = 2000$

Therefore, percentage of medical colleges in 2000

$$= \frac{4680 + 2000}{60,000} \times 100 = 11\%$$

29. If the amount of water consumed of sector 1 is the same then we can directly compare the tariffs to the two years.

	Tariff 2003–04	% change	Tariff 2000–01
Region 1	1000	+ 25%	800
Region 2	400	+ 14.28%	350
Region 3	250	–16.66%	300
Region 4	625	+ 4.166%	600
Region 5	720	+ 20%	600
	2995		**2650**

$$\frac{2995 - 2650}{2650} \times 100 = \frac{345}{2650} \times 100$$

$$= 13.01\%$$

30.

	Tariff 2003–04	% change over 2000-01	Tariff 2000–01
Sector 1	400	+ 14.28%	350
Sector 2	375	+ 7.14%	350
Sector 3	525	–12.5%	600
Sector 4	800	–20%	1000
	2100		**2300**

$$\text{Average tariff} = \frac{2300}{4} = 575$$

31. In 2003-04 the water consumed by various secters out of 20,000 kilo-litres can be given as follow:

Category	Percentage	Consumption in 2003–04
Domestic	40	8,000
Urban	25	5,000
Rural	15	3,000
Industrial	20	4,000
		20,000

Since there was a 20% decrease in the domestic consumption in 2003-04 the domestic consumption in

2000-01 $= \frac{8,000}{0.8} = 10,000$

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But this constitutes 40% of total water consumed in 2000-01 and the industrial consumption constitutes 20% of total water in 2000-01. Hence in 2000-01 the industrial consumption $= 10{,}000 \times \frac{20}{40} = 5{,}000$ kilo-liters

32. We do not know the category-wise break up of tariffs ie the rates of Urban sector is unknown.

33. The average of Region 2 $= \frac{400 + 375 + 525 + 800}{4} = 525$

Average tariff in region 4 $= \frac{62.5 + 750 + 240 + 360}{4} = 487.5$

Average tariff in region 5 $= \frac{220 + 360 + 320 + 400}{4} = 450$

Statements (c) can not be determined

34. Total number of people = 10,000

Business men	10%	1000
Govt. Servant	13%	1300
Professionals	30%	3000
Students	45%	4500
Housewives	2%	200

LML	1200
SUZUKI	1200
BAJAJ	3000
HERO HONDA	6600

The total number of Hero Honda bikes = 6600

Total numbers of Government servants housewives and students = 6000

Total no. of Businessmen and professional = 4000

∴ Percentage of remaining (*i.e.*, Businessmen and prof.) driving Hero Honda $= \frac{600}{4000} \times 100 = 15\%$

35. Let the no. of people who drive one two and three bikes be $15k$, $3k$ and k respectively.

Number of bikes which are being driven

$$= 15k + 2(3k) + 3(k) = 24k$$

Since LML and Suzuki can not be driven by same person and a person can drive maximum 3 bikes.

Total bikes which are being used to drive = 12,000

$$\therefore \quad 24k = 12{,}000 \Rightarrow k = 500$$

Total number of people driving the bikes

$$= 15k + 3k + k = 19k = 9500$$

∴ Number of people who do not drive any bike

$$= 10{,}000 - 9500 = 500$$

36. From the previous solution, number of people who drive more than 1 bike *i.e.*, 2 bikes and 3 bikes are 1500 and 500 respectively.

These people have total 4500 bikes

$$(= 1500 \times 2 + 500 \times 3)$$

Hence, the remaining Hero Honda bikes

$$= 6600 - 4500 = 2100$$

Thus, the number of persons who drive single Hero Honda

$$= 2100$$

37. Since 20% drive other bikes 80% drive only bajaj bike number of people who drive only Bajaj bike

$$= 0.8 \times 3000 = 2400$$

38. $\frac{11000 + 12000}{21{,}000} \times 100 = 109.52\%$

39. $\frac{27{,}000 - 23{,}000}{23{,}000} \times 100 = 17.4\%$

40. Demand = Domestic Production + Imports

Average demand $= \frac{44 + 54.5 + 57}{3} \approx 52$ million tonnes

41. (c) $\frac{30{,}000 - 21{,}000}{21{,}000} \times 100 \approx 43\%$

42. Offshore production in 2004 $= 16{,}000 \times 0.875$

$= 14{,}000$ thousand tonnes

Onshore production = 11,000 thousand tonnes

Demand in 2004 $= 57{,}000 \times 1.02 = 5814$ thousand tonnes

imports = 33.14 million tonnes ($\because$ 1 million $= 10^6$)

43. $201924 \times \frac{5}{6} = 33654 \times 5 = 168270$

44. $127890 \times \frac{11}{10} = 140679$

45. $126400 + 975264 + 237852 = 1339516$

46. $\frac{100956}{2445480} \times 100 = 4.128\%$

Alternatively: $\frac{1}{24} \times 100 \approx 4.16\%$

47. 1969 – 70 → 134790
1979 – 80 → 284620] 149830

48. 1929 – 30 → 9,75,264
1949 – 50 → 1,40,679

% decrease $= \frac{975264 - 140679}{975264} \times 100 = 85.57\%$

49. 1959 – 60 ⟶ 1960 – 61

$168020 \xrightarrow{+20\%} 201924$

Therefore $\frac{168020}{201924} \times 100 = 83.33\%$

Alternatively: From percentage charge graphic

Increase	Decrease
20%	16.66%
1/5	1/6

∴ The required value = 83.33% (100 – 16.66%)

Alternatively: $x \times \frac{6}{5} = k$

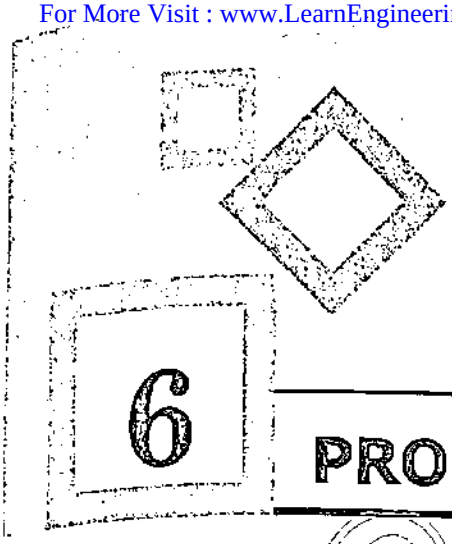
$$\Rightarrow \quad x = \frac{5k}{6}$$

$$(x)\% = \frac{5k}{6} \times 100 = 83.33\% \text{ of } k$$

50. Only 7,900 tonnes increase over the 1919 – 20.

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6 PROFIT, LOSS AND DISCOUNT

This chapter is as much important as other chapters in the arithmetic section, since it is a chapter of applications by which we calculate profit, loss, discount in day-to-day business. So in the entrance exams of MBA it plays a crucial role in determining the selection of an aspirant. Particularly in CAT the questions asked in QA are very logical in nature, apart from the simple application in Data interpretation (D.I.) and Data sufficiency (D.S.). Entrance exams like FMS, IIFT, NIFT and MAT ask plethora of questions. Infact a normal question paper seems to be incomplete without having the problems based on profit, loss and discount.

THEORY AND CONCEPTS

In day-to-day life we sell and purchase the things as per our requirement. A customer can get things in the following manner :

Manufacturer (or producer) → Whole-saler (dealer) → (Shopkeeper) Retailer (or sales person) → Customer

TERMINOLOGY

Cost price (CP) : The money paid by the shopkeeper to the manufacturer or whole-saler to buy the goods is called the cost price (CP) of the goods purchased by the shopkeeper.

NOTE If an article is purchased for some amount and there are some additional expenses on transportation labour, commission etc., these are to be added in the cost price. Such expenses are called overhead expenses or overheads.

Selling Price (SP) : The price at which the shopkeeper sells the goods is called the selling price (SP) of the goods sold by the shopkeeper.

Profit : If the selling price of an article is more than its cost price, then the dealer (or shopkeeper) makes a profit (or gain)

i.e., Profit = SP – CP; SP > CP

Loss : If the selling price of an article is less than its cost price, then the dealer suffers a loss.

i.e., loss = CP – SP; CP > SP

Important Formulae

(i) Profit = SP – CP

(ii) Loss = CP – SP

(iii) $\text{Profit percentage} = \frac{\text{profit}}{\text{cost price}} \times 100$

(iv) $\text{Loss percentage} = \frac{\text{loss}}{\text{cost price}} \times 100$

(v) $SP = \left(\frac{100 + \text{gain}\%}{100} \times CP\right) = \left(\frac{100 - \text{loss}\%}{100} \times CP\right)$

(vi) $CP = \left(\frac{100}{100 + \text{gain}\%} \times SP\right) = \left(\frac{100}{100 - \text{loss}\%} \times SP\right)$

(vii) SP = (100 + k)% of CP; when profit = k% of CP

(viii) SP = (100 – k)% of CP; when loss = k% of CP

NOTE Profit or loss is always calculated on the basis of cost price unless otherwise mentioned in the problem.

EXAMPLE 1 *A fruit seller buys 300 oranges at 5 for Rs. 8 and sold at 2 for Rs. 5. Find :*

(i) the cost price of each orange

(ii) the selling price of each orange

(iii) profit or loss on selling one orange

(iv) his total profit or loss on selling all the oranges.

SOLUTION (i) Since the cost price of 5 oranges = Rs. 8

∴ Cost price of one (or each) orange = Rs. $\frac{8}{5}$ = Rs. 1.60

(ii) Since the selling price of 2 oranges = Rs. 5

∴ the selling price of one orange = Rs. $\frac{5}{2}$ = Rs. 2.50

(iii) Since SP is more than CP, there is a profit.

So profit on selling one orange = SP – CP

= 2.5 – 1.6 = Re. 0.90

(iv) Profit on selling all the oranges

= Rs. (0.90 × 300) = Rs. 270

Hence the total profit on selling all oranges = Rs. 270

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EXAMPLE 2 *A shopkeeper buys 100 eggs at Rs. 1.20 per piece. Unfortunately 4 eggs got spoiled during transportation. The shopkeeper sells the remaining eggs at Rs. 15 a dozen. Find his profit or loss.*

SOLUTION Cost price of all eggs = Rs. 100×1.2 = Rs. 120

Selling price of one egg = $\frac{15}{12}$ = Rs. 1.25

$\therefore$ Selling price of 96 eggs = $96 \times \frac{15}{12}$ = Rs. 120

HINT After spoiling 4 eggs, only 96 eggs are left.

Now, since the total selling price and total cost price is same, the shopkeeper neither makes a profit nor suffers a loss.

EXAMPLE 3 *Aviral purchased a computer for Rs. 47,000. He had to sell it for Rs. 45,800. Find his profit or loss per cent.*

SOLUTION Since SP < CP, there will be loss

$$\text{loss (\%)} = \frac{\text{loss}}{\text{CP}} \times 100 = \frac{47000 - 45800}{47000} \times 100$$

$$= \frac{1200}{47000} \times 100 = 2\frac{26}{47}\%$$

EXAMPLE 4 *A dealer sold 600 quintals of sugar at a profit of 7%. If a quintal of sugar cost him Rs. 1600, find his total profit and the selling price.*

SOLUTION CP = 1600×600 = Rs. 9,60,000

Rate of Profit = 7%

$$\text{Profit} = 9{,}60{,}000 \times \frac{7}{100} = \text{Rs. } 67200$$

$\therefore$ SP = CP + Profit = Rs. 9,60,000 + Rs. 67,200

= Rs. 1027200

Thus profit = Rs. 67200 and selling price = Rs. 10,27,200

EXAMPLE 5 *A dealer buys 200 quintals of wheat at Rs. 1200 a quintal. He spends Rs. 10,000 on transportation and storage. Then he sells the wheat at Rs. 13 per kg. Find his profit or loss. Also calculate it as a percentage.*

SOLUTION CP = 1200×200 = Rs. 2,40,000

Transportation and storage cost = Rs. 10,000

Total CP = 2,40,000 + 10,000

= Rs. 2,50,000

Total SP = $13 \times 200 \times 100$

= Rs. 2,60,000

$\therefore$ Now, since SP > CP, hence there will be profit.

Profit = SP − CP = Rs. (2,60,000 − 2,50,000) = Rs. 10,000

$$\text{Profit (\%)} = \frac{\text{Profit}}{\text{CP}} \times 100$$

$$= \frac{10{,}000}{2{,}50{,}000} \times 100 = 4\%$$

EXAMPLE 6 *Find the cost price of an article which is sold for Rs. 220 at a loss of 12%.*

SOLUTION SP = Rs. 220, Loss = 12%

Let CP = Rs. x

then SP = 88% of CP

$$220 = \frac{88}{100} \times x$$

$$\Rightarrow \quad x = 250$$

Therefore cost price = Rs. 250

EXAMPLE 7 *By selling a colour TV for Rs. 23520, a dealer suffers a loss of 4%. What is the cost price of the colour TV? At what price should he sell it to gain 8%?*

SOLUTION SP = 96% of CP $\because$ (loss is 4%)

$$\therefore \quad 23520 = \frac{96}{100} \times \text{CP}$$

$$\text{CP} = 24500$$

Now gain % = 8%

$\therefore$ New selling price = (100 + 8)% of CP

$$= \frac{108}{100} \times 24500 = \text{Rs. } 26460$$

INTRODUCTORY EXERCISE-6.1

1. A towel is sold for Rs. 198 at a gain of 10%. What is the cost price of the towel? At what price must it be sold to gain 25%?
2. A man sold a watch at Rs. 6000, at a loss of $33\frac{1}{3}\%$. Find the cost price?
3. By selling a shirt for Rs. 285 a shopkeeper loses 5%. At what price should he sell the shirt so as to gain 15%?
4. Sufyan bought 1200 eggs at Rs. 16 a dozen. At what price per hundred must he sell the eggs so as to earn a profit of 15%?
5. Ram Singh purchased two camels for Rs. 18,000 and Rs. 15,000 respectively. He sold them at a loss of 15% and a gain of 19% respectively. Find the selling price of each of the camels. Also find the overall gain or loss per cent in the transaction.

MARK UP AND DISCOUNT

Marked price: Basically to avoid loss due to bargaining by the customer and to get the profit over the cost price trader increases the cost price by a certain value, this increase in value over cost price is known as markup and the increased price (*i.e.*, CP + Markup) is called the marked price or printed price or list price of the goods.

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Marked price = CP + Markup
Marked price = CP + (% markup on CP)

Generally goods are sold at marked price, if there is no further discount, then in this case selling price equals to marked price.

Discount : Discount means reduction of marked price to sell at a lower rate or literally discount means concession. Basically it is calculated on the basis of marked price.

∴ Selling price = Marked price – Discount
or Selling price = Marked price (MP) – (% discount on MP)

Since marked price = CP + % markup on CP

Remember **markup** is calculated on the basis of CP while **discount** is calculated on the basis of MP.

In general, CP < SP < MP at profit
CP = SP < MP at no profit no loss
SP < CP < MP at loss

Also 

and 

NOTE You will find wide application of percentage calculation, percentage change graphic etc, so to move further you must know the concepts of percentages.

EXAMPLE 8 *If the cost price of an article is Rs. 300 and the per cent markup is 20%. What is the marked price?*

SOLUTION MP = CP + % markup on CP

$= 300 + 300 \times \frac{20}{100}$

MP = Rs. 360

Alternatively : SP = 300 × 1.2 = 360

EXAMPLE 9 *If the marked price of an article is Rs. 450 and markup percentage is 12.5%, what is the cost price?*

SOLUTION MP = 112.5 of CP

$\therefore \quad 450 = \frac{112.5}{100} \times CP$

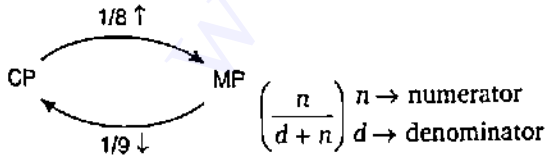
$450 = \frac{9}{8} \times CP$

$\Rightarrow \quad CP = Rs.\ 400$

Alternatively : Change your outlook to visulize the problem differently for smarter calculation.

You can see that MP is 12.5% (*i.e.*, $\frac{1}{8}$ times) greater than CP.

So CP will be $\frac{1}{9}$ times less than MP. This percentage (or fraction) change rule has been throughly discussed through different illustrations in percentage chapter. In my opinion if you change your observation *i.e.*, you mould your thinking in a CAT oriented approach you will not change only your attitude and aptitude but the whole scenario will be changed. So look this problem like me :



$\left(\frac{n}{d+n}\right)$

So, the CP will be $\frac{1}{9}$ times less than MP, which is rather too much easier to calculate than the traditional method given at the beginning of the solution.

EXAMPLE 10 *If the marked price of an article is Rs. 660 and the discount is 10%, then what is the selling price of the article?*

SOLUTION **General Solution :**

SP = MP – Discount

$SP = 660 - 660 \times \frac{10}{100}$

SP = 594

Alternatively: SP = 90% of MP

SP = 0.9 × 660

SP = 594

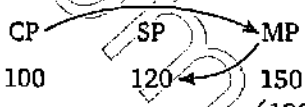
Alternatively: We can see that when SP is 10% (*i.e.*, $\frac{1}{10}$) less than MP, it means MP is $\frac{1}{9}$ times greater than SP.

Therefore $MP = \frac{10}{9} SP \quad \left(1 + \frac{1}{9} = \frac{10}{9}\right)$

Ultimately I would like to say that you should see the core of the problem and solution both, instead of taking help from any formula. In other words formulae make you logically blind particularly in context of CAT where your intelligence is tested rather than your memory and theory of the maths.

EXAMPLE 11 *If the markup percentage of an article is 50% and discount percentage is also 20%, then the profit percentage will be :*
(a) 10% (b) 0% (c) 30% (d) 20%

SOLUTION See the smartest calculation technique and follow me to be smarter.



Therefore profit = 20% $\left(\frac{120-100}{100} \times 100 = 20\%\right)$

Explanation: MP = 150% of CP (∵ markup is 50%)
SP = 80% of MP (∵ discount is 20%)

Once again I am reiterating on the gist of this chapter. What is that gist? The gist is that this chapter test nothing but your grip on the concepts of percentage change. I think CP, SP, MP etc., are just the words, the crux is that how CP, SP and MP etc., change with respect to each other. So those students who have grasped and internalised the most important concept of percentage change, they will not find any difficulty in understanding the concept of the profit-loss problems. Believe me when your visualisation will be improved, you will be master on this chapter. So try to have a different approach.

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EXAMPLE 12 *A trader markup the goods by 10% and then gives a discount of 10%. What is the profit or loss percentage?*

SOLUTION

CP → SP → MP

100 → 99 → 110

So there is a loss of 1%.

Explanation: MP = 110% of CP (10% markup)

SP = 90% of MP (10% discount)

EXAMPLE 13 *Successive discount of 10% and 5% is equivalent to :*

(a) 16.5% *(b) 15%*

(c) 15.5% *(d) 14.5%*

SOLUTION

100 ↓ (–10%) 90 ↓ (–5%) 85.5 (14.5)

NOTE There is no difference in two different cases *i.e.*, either you decrease 100 at first by 10% and then by 5% or decrease 100 at first by 5% and then by 10%.

100 ↓ (–5%) 95 ↓ (–10%) 85.5 (14.5)

Explanation : $100 \times 0.9 \times 0.95 = 85.5$

∴ Single discount equivalent to 10% and 5% is 14.5%

(100 – 85.5)%

or $(100 \times 0.95 \times 0.9 = 85.5)$

Some Important Concepts

(1) When two articles are sold at same price but one of them at a profit and another at a loss and the percentage profit is the same as the percentage loss. In this case there is always a loss.

$$\text{loss (\%)} = \left(\frac{\text{Common gain or loss}}{10}\right)^2$$

EXAMPLE 14 *A man sells two wrist watches one at a profit of 10% and another at a loss of 10%, but the selling price of each watch is Rs. 200. Find the :*

(1) percentage profit or loss

(2) net amount of profit or loss

SOLUTION (1) Since there is always loss.

(Logic: To get the logic you should refer the percentage chapter)

Now, $\text{loss \%} = \left(\frac{\text{Common gain or loss}}{10}\right)^2 \%$

$$\text{loss \%} = \left(\frac{10}{10}\right)^2 \% = 1\%$$

(2) Amount of loss.

Total SP = Rs. 400 (200 + 200)

again, total SP = 99% of CP (1% loss)

$$400 = \frac{99}{100} \times \text{CP}$$

$$\Rightarrow \quad \text{CP} = \text{Rs. } 404.04$$

$$\therefore \quad \text{loss} = \text{CP} - \text{SP} = \text{Rs. } 4.04 \qquad (= 404.04 - 400)$$

EXAMPLE 15 *There were two articles and the sum of cost prices of these articles is Rs. 500. One of them was sold at a profit of 20% and another at a loss of 20%. Besides if the selling prices of both the articles were same. Find the amount of overall loss.*

SOLUTION

$$x \times \frac{120}{100} = (500 - x)\frac{80}{100}$$

$$\Rightarrow \quad x = 200$$

So, the CP of profit yielding article = Rs. 200

and the CP of loss giving article = Rs. 300 (500 – 200)

and the common SP = Rs. 240 = (200 × 1.2 = 300 × 0.8)

So the loss = CP – SP

$$= 500 - 2 \times 240 = \text{Rs. } 20$$

(2) When the selling price of 'm' articles is same as the cost price of 'n' similar articles.

EXAMPLE 16 *If the cost price of 15 apples is same as the selling price of 20 apples. What is the gain or loss per cent?*

SOLUTION CP of 15 apples = SP of 20 apples

$$\Rightarrow \quad \text{CP} \times 15 = \text{SP} \times 20$$

$$\Rightarrow \quad \frac{\text{CP}}{\text{SP}} = \frac{4}{3}$$

So you can see that CP > SP, therefore, there will be loss.

Now consider CP = 4, then SP = 3

$$\therefore \quad \text{loss} = 1$$

$$\text{loss (\%)} = \frac{\text{loss}}{\text{CP}} \times 100$$

$$= \frac{1}{4} \times 100 = 25\%$$

$$\therefore \quad \text{loss} = 25\%$$

EXAMPLE 17 *If the selling price of 10 CDs is the same as the cost price of 12 CDs. What is the profit or loss per cent?*

SOLUTION SP of 10 CDs = CP of 12 CDs

$$\Rightarrow \quad \text{SP} \times 10 = \text{CP} \times 12$$

$$\Rightarrow \quad \frac{\text{SP}}{\text{CP}} = \frac{12}{10} = \frac{6}{5}$$

⇒ SP > CP, therefore there will be a profit

$$\text{Profit (\%)} = \frac{(\text{SP} - \text{CP})}{\text{CP}} \times 100$$

$$= \frac{(6-5)}{5} \times 100 = 20\% \text{ (profit)}$$

(3) When a person recovers the cost price of 'm' articles by selling 'n' articles ($n < m$), then

$$\text{Profit (\%)} = \frac{\text{goods left}}{\text{goods sold}} \times 100$$

$$= \frac{m-n}{n} \times 100$$

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In this case money is equated in terms of no. of (or amount) articles.

For your convenience always assume the CP of an article as Re 1. (or Rs. 100)

EXAMPLE 18. *A dealer by selling 12 oranges gets the cost price of 15 oranges. What is the percentage profit?*

SOLUTION $\text{Profit (\%)} = \frac{\text{goods left}}{\text{goods sold}} \times 100 = \frac{15-12}{12} \times 100 = 25\%$

Alternatively : Suppose CP of one orange = Re. 1

then CP of 12 oranges = Rs. 12

and SP of 12 oranges = CP of 15 oranges = Rs. 15

$\therefore$ $\text{profit} = \frac{15-12}{12} \times 100 = 25\%$

EXAMPLE 19. *By selling 8 bananas, a fruit seller gains the selling price of 1 banana. Calculate his gain per cent.*

SOLUTION Let the SP of one banana = Re. 1

then SP of 8 bananas = Rs. 8

and profit = Re. 1

$\therefore$ CP = 8 – 1 = Rs. 7

$\therefore$ $\text{profit \%} = \frac{1}{7} \times 100 = 14\frac{2}{7}\%$

EXAMPLE 20. *By selling 18 chocolates, a vendor loses the selling price of 2 chocolates. Find his loss per cent.*

SOLUTION Let the SP of 1 chocolate = Re. 1

$\therefore$ SP of 18 chocolate = Rs. 18

and loss = Rs. 2

$\therefore$ CP = SP + loss

= 18 + 2 = Rs. 20

$\therefore$ $\text{Percentage loss} = \frac{\text{loss}}{\text{CP}} \times 100$

$= \frac{2}{20} \times 100 = 10\%$

EXAMPLE 21. *A trader sell all his articles at the cost price but gives 10% less amount as he should give. What is his percentage profit?*

SOLUTION $\text{Profit (\%)} = \frac{\text{goods left}}{\text{goods sold}} \times 100$

$= \frac{10}{90} \times 100 = 11\frac{1}{9}\%$

Since if we assume that the CP of 1 article is Re. 1.

Now since he gives only 90% article instead of 100% and save 10% article. So his profit will be the equal to the remaining articles (over the sold articles).

It means when he sells the articles (actually) worth Rs. 90 then he gains by articles worth Rs. 10.

Hence profit % $= \frac{10}{90} \times 100 = 11\frac{1}{9}\%$

(The selling price $= \frac{100}{90} = \frac{10}{9}$ = Rs. 1.11

EXAMPLE 22. *A trader by means of his false balance defrauds to the extent of 10% in buying goods and also defrauds to 10% in selling. Find his gain per cent.*

SOLUTION Let the actual CP of an article be Re. 1, then the effective CP $= \frac{100}{110} = \frac{10}{11}$

(Since he purchases 110 articles by paying Rs. 100)

$$\text{again SP} = \frac{100}{90} = \frac{10}{9}$$

(Since he sells only 90 articles charging the CP of 100 articles)

$$\therefore \text{ Gain \%} = \frac{\text{SP} - \text{CP}}{\text{CP}} \times 100 = \frac{\frac{10}{9} - \frac{10}{11}}{\frac{10}{11}} \times 100$$

$$= \frac{20}{99} \times \frac{11}{10} \times 100$$

$$= \frac{200}{9} = 22\frac{2}{9}\%$$

INTRODUCTORY EXERCISE-6.2

1. The cost price of a scooter is Rs. 20,000 and the profit percent is 12%. What is the selling price?
   (a) 2400 (b) 22040
   (c) 2600 (d) 22400
2. The SP of an article is Rs. 3200 and the profit per cent is $33\frac{1}{3}\%$. Find the cost price.
   (a) Rs. 20000 (b) Rs. 2000
   (c) Rs. 2400 (d) Rs. 3000
3. The CP of an article is 5/6th of the SP. What is the percentage profit or loss?
   (a) 20% loss (b) 16.66% profit
   (c) 16.66% loss (d) 20% profit.
4. The MP of a camera is 3/2 of the CP and SP is 9/10 of MP. Find the percentage profit or loss.
   (a) 25% profit (b) 35% profit
   (c) 33.33% loss (d) none of these
5. The MP of an article is 30% higher than its CP and 20% discount is allowed on this article then the profit percentage :
   (a) 10% (b) 14%
   (c) 4% (d) 26%
6. An article is sold for Rs. 1980 at 10% profit. What is the cost price?
   (a) Rs. 198 (b) Rs. 1800
   (c) Rs. 1900 (d) Rs. 1600

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7. On selling an article for Rs. 576 a trader loses 4%. In order to gain $4\frac{1}{6}$%, he must sell that article for :
(a) Rs. 636 (b) Rs. 676
(c) Rs. 625 (d) can't be determined

8. The per cent profit made when an article is sold for Rs. 56 is thrice as when it is sold for Rs. 42. The cost price of the article is :
(a) Rs. 48 (b) Rs. 49
(c) Rs. 50 (d) Rs. 35

9. A shopkeeper uses a weight of 460 g instead of 500 g and sells the articles at the cost price. What is the profit percentage?
(a) 40% (b) 23%
(c) $8\frac{16}{23}$% (d) 20%

10. A trader uses a weight of 920 g instead of 1 kg and sells the articles at the marked price which is 15% above the cost price. Find the profit percentage.
(a) 20% (b) 23%
(c) 25% (d) can't be determined

11. If a gift pack is sold at a gain of 6% instead of at a loss of 6%, then the seller gets Rs. 6 more. The cost price of the article is :
(a) Rs. 60 (b) Rs. 66
(c) Rs. 50 (d) Rs. 36

12. A man sells a bicycle at a gain of 10%. If he had bought it at 10% less and sold it for Rs. 132 less, he would have still gained 10%. The cost price of the article is :
(a) Rs. 1000 (b) Rs. 1200
(c) Rs. 1500 (d) Rs. 1320

13. An item costing Rs. 600 is being sold at 20% loss. If the price is further reduced by 12.5%, the selling price will be :
(a) Rs. 400 (b) Rs. 380
(c) Rs. 420 (d) Rs. 525

14. While selling an electric fan, a dealer gives a discount of 5%. If he gives a discount of 8%, he earns Rs. 36 less as profit. The marked price of the fan is :
(a) Rs. 1000 (b) Rs. 1200
(c) Rs. 800 (d) none of these

15. If a commission of 10% is given on the marked price of a book, the publisher gains 20%. If the commission is increased to 15%, the gain of publisher is :
(a) 13.33% (b) 15%
(c) 18% (d) data insufficient

16. A retailer buys a cellphone at a discount of 15% and sells it for Rs. 5865. Thus, he makes a profit of 15%. The discount is :
(a) Rs. 200 (b) Rs. 850
(c) Rs. 750 (d) Rs. 900

17. At what per cent above the cost price must an articles be marked so as to gain 17% after allowing a discount of 10%?
(a) 34% (b) 70%
(c) 30% (d) 27%

18. A merchant marks his goods at Rs. 900 and allows a discount of 25%. If he still gain 12.5%, then the cost price of the article is :
(a) Rs. 500 (b) Rs. 600
(c) Rs. 720 (d) can't be determined

19. A vendor buys oranges @ Rs. 2 for 3 oranges and sells them at a rupee each. To make a profit of Rs. 10, he must sell :
(a) 10 oranges (b) 20 oranges
(c) 30 oranges (d) 40 oranges

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EXERCISE

LEVEL 1

1. Abhishek and Bhanu both are dealers of KML scooters. The price of a KML Scooter is Rs. 28,000. Abhishek gives a discount of 10% on whole, while Bhanu gives a discount of 12% on the first Rs. 20,000 and 8% on the rest Rs. 8000. What is the difference between their selling prices?
(a) Rs. 240 (b) Rs. 420
(c) Rs. 640 (d) none of these

2. A trader sells two articles, one at a loss of 10% and another at a profit of 15% but finally there is no loss or gain. If the total sale price of these two articles is Rs. 30,000, find the difference between their cost prices :
(a) Rs. 5000 (b) Rs. 6000
(c) Rs. 7500 (d) none of these

3. A milkman purchases the milk at Rs. x per litre and sells it at Rs. $2x$ per litre still he mixes 2 litres water with every 6 litres of pure milk. What is the profit percentage?
(a) 116% (b) 166.66%
(c) 60% (d) 100%

4. 60% goods are sold at 5% loss while rest are sold at 10% profit. If there is a total profit of Rs. 100, then the worth of goods sold is :
(a) Rs. 6000 (b) Rs. 5000
(c) Rs. 10000 (d) none of these

5. A retailer bought 20 kg tea at a discount of 10%. Besides 1 kg tea was freely offered to him by the wholesaler at the purchase of 20 kg tea. Now he sells all the tea at the marked price to a customer. What is profit percentage of retailer?
(a) 30% (b) 12%
(c) 16.66% (d) none of these

6. Two articles are sold at the same price. One at a profit of 75% and another one at a loss of 30%. What is the overall profit or loss?
(a) 22.5% profit (b) 57.5 profit
(c) $13\frac{2}{7}$% loss (d) none of these

7. What is percentage profit in selling an article at a discount of 20% which was earlier being sold at a 40% profit?
(a) 20% (b) 14%
(c) 28% (d) 12%

8. A man bought 18 oranges for a rupee and sold them at 12 oranges for a rupee. What is the profit percentage?
(a) 33.33% (b) 50%
(c) 66.66% (d) none of these

9. A dealer buys a product at Rs. 1920, He sells at a discount of 20% still he gets the profit of 20%. What is the selling price of that product?
(a) Rs. 2304 (b) Rs. 1536
(c) Rs. 2200 (d) it is not possible

10. Tinkawala purchased the articles for Rs. 123684. He sold 60% of those at a profit of 16.66% and rest at a loss. Find the loss percentage on the remaining if the overall loss is 14%?
(a) 20% (b) 30%
(c) 60% (d) 66.66%

11. What should be the minimum markup percentage such that after giving a discount of $66\frac{2}{3}$%, there will not be a loss?
(a) 200% (b) 133.33%
(c) 100% (d) 150%

12. The ratio of cost price and marked price of an article is 2 : 3 and ratio of percentage profit and percentage discount is 3 : 2 What is the discount percentage?
(a) 16.66% (b) 20%
(c) 25% (d) 33.33%

13. A dealer gives as much discount (in per cent) as the markup (in per cent) above the cost price. What is the profit or loss per cent?
(a) 10% (b) 1%
(c) 4% (d) can't be determined

14. In the above problem, what is the markup percentage when there is a loss in percentage be half of the percentage markup?
(a) 20% (b) 40%
(c) can't be determined (d) none of these

15. A shopkeeper sold 12 cameras at a profit of 20% and 8 cameras at a profit of 10%. If he had sold all the 20 cameras at a profit of 15%, then his profit would have been reduced by Rs. 36. What is the cost price of each camera?
(a) 100 (b) 150
(c) 180 (d) 220

16. Mr. Mittal purchased a car for Rs. 3,00,000 and a bike for his son for Rs. 1,00,000. He sold the car at a profit of 10% and bike at a loss of 20%. What is the net gain or loss?
(a) 2% gain (b) 1.5% loss
(c) 2.5% loss (d) 2.5% gain

17. A trader sells 20 kg of sugar at Rs. 400. A customer asks 20% discount and he agrees to it but instead of 1 kg he gives 4% less sugar. What is the effective discount that the customer gets?
(a) 16% (b) 16.66%
(c) 15.5% (d) 19.6%

18. The profit percentage on the three articles A, B and C is 10%, 20% and 25% and the ratio of the cost price is 1 : 2 : 4. Also the ratio of number of articles sold of A, B and C is 2 : 5 : 2, then the overall profit percentage is :
(a) 18.5% (b) 21%
(c) 75% (d) none of these

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19. The marked price of an article is increased by 25% and the selling price is increased by 16.66%, then the amount of profit doubles. If the original marked price be Rs. 400 which is greater than the corresponding cost price by 33.33%, what is the increased selling price?
(a) 240 (b) 360
(c) 420 (d) 600

20. A shopkeeper calculated his profit per cent on the selling price which comes out to be 30%. If it had been calculated as usual on the cost price then what is the required percentage profit?
(a) $42\frac{6}{7}\%$ (b) $4\frac{7}{27}\%$
(c) $7\frac{6}{42}\%$ (d) none of these

21. An item was sold after giving two successive discount of 20% and 10% respectively. If the item was sold for Rs. 468. The marked price of that item is :
(a) Rs. 600 (b) Rs. 500
(c) Rs. 575 (d) Rs. 650

22. The cost price of an article '*A*' is Rs. 160 and selling price of another article '*B*' is Rs. 240. If the selling price of *A* will be equal to the cost price of *B*, then the profit after selling *A* is 20%. What is the profit on '*B*'?
(a) 16.66% (b) 50%
(c) 25% (d) none of these

23. A single discount equivalent to three successive discounts of 5%, 10%, 20% is :
(a) 68.4% (b) 35%
(c) 31.6% (d) 32%

24. Ragini purchases oranges at Rs. 10 per dozen and sells them at Rs. 12 for every 10 oranges. What is the profit percentage?
(a) 40% (b) 44%
(c) 60% (d) 48%

25. A trader uses a weighing balance that shows 1250 g for a kilogram. He further marks-up his cost price by 20%. What is the profit percentage?
(a) 5% (b) 45%
(c) 50% (d) 30%

26. On selling an article for Rs. 240, a trader loses 4%. In order to gain 10% he must sell that article for :
(a) Rs. 275 (b) Rs. 340
(c) Rs. 320 (d) Rs. 264

27. A merchant marks his goods at Rs. 300 and allows a discount of 25%. If he still gains 12.5%, then the cost price of article is :
(a) Rs. 220 (b) Rs. 200
(c) Rs. 240 (d) Rs. 260

28. An item costing Rs. 200 is being sold at 10% loss. If the price is further reduces by 5%, the selling price will be :
(a) Rs. 170 (b) Rs. 171
(c) Rs. 180 (d) Rs. 181

29. A person sold two cows each for Rs. 9900. If he gained 10% on one and lost 20% on the other, then which of the following is true?
(a) He gained Rs. 200
(b) He lost Rs. 200
(c) He neither gained nor lost
(d) None of the above

30. Two third of a consignment was sold at a profit of 5% and the remainder at a loss of 2%. If the total profit was Rs. 400, the value of the consignment (in rupees) was :
(a) 15000 (b) 20000
(c) 10000 (d) 12000

31. A fruit seller declares that he sells fruits at the cost price. However, he uses a weight of 450 g instead of 500 g. His percentage profit is :
(a) 10% (b) $11\frac{1}{9}\%$
(c) 12% (d) $12\frac{2}{9}\%$

32. A person loses Rs. 20 by selling some bananas at the rate of Rs. 3 per banana and gains Rs. 30, if he sells them at Rs. 3.25 per banana. The number of bananas sold by him :
(a) 100 (b) 200
(c) 120 (d) 2400

33. Due to an increase of 30% in the price of eggs, 3 eggs less are available for Rs. 9.10. The present rate per egg is :
(a) 91 paise (b) 78 paise
(c) 48 paise (d) 84 paise

34. By selling 12 apples for a rupee, a man loses 20%. How many for a rupee should he sell to gain 20%?
(a) 8 (b) 10
(c) 15 (d) 16

35. A dealer buys a washing machine, listed at Rs. 10000 and gets 10% and 20% successive discounts. He spends 10% of his CP on transport. At what price (in rupees) should he sell the washing machine to earn a profit of 10%?
(a) 8722 (b) 7892
(c) 8712 (d) 8840

36. 6% more is gained by selling a coat for Rs. 1425 than by selling it for Rs. 1353. The cost price of the coat is :
(a) Rs. 1000 (b) Rs. 1250
(c) Rs. 1500 (d) Rs. 1200

37. By selling a wrist watch at Rs. 405 the shopkeeper incurs a loss of 10%. What is the gain or loss percentage if he sells the same watch at Rs. 465?
(a) profit of 10% (b) loss of 6%
(c) profit of 3.33% (d) no profit no loss

38. Titan sells a wrist watch to a wholesaler making a profit of 10%. The wholeseler, in turn, sells it to the retailer making a profit of 10%. A customer purchases it by paying Rs. 990. Thus the profit of retailer is $2\frac{3}{11}\%$. What is the cost incurred by the Titan to produce it?
(a) 768 (b) 750
(c) 800 (d) 820

39. Pepsi and Coke, there are two companies, selling the packs of cold-drinks. For the same selling price Pepsi gives two successive discounts of 10% and 25%. While Coke sells it by giving two successive discounts of 15% and 20%. What is the ratio of their marked price?
(a) 143 : 144 (b) 19 : 11
(c) 136 : 135 (d) 73 : 77

40. When a shopkeeper reduces the selling price from 1080 to 1026 its loss increases by 4 percentage point. What is the selling price of this same article when it fetches a profit of 4%?

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(a) Rs. 1392 (b) Rs. 1404
(c) Rs. 1450 (d) Rs. 1350

41. The difference between CP and SP of a table fan is Rs. 175 when it gives the profit of 14%. What is the selling price of that fan?
(a) 1225 (b) 1450
(c) 1425 (d) 1275

42. A company instead of raising the mark-up by 20% discounted the cost price by 20% while stiching the price tag on its product. Further the company offers a discount of 6.25% to its customer. In this process company incurs a loss of Rs. 37.5 on a single article. What is the selling price of that article?
(a) 417.5 (b) 112.5
(c) 365.5 (d) none of these

43. When an article is sold for Rs. 703 loss incurred is 25% less than the profit earned on selling it at Rs. 836. What is the selling price of the article when it earns a profit of 20%?
(a) 912 (b) 1576
(c) 1532 (d) 1092

44. Arun bought toffees at 6 for a rupee. How many for a rupee he should sell to gain 20%?
(a) 3 (b) 4
(c) 5 (d) can't be determined

45. A scientific calculator is available at Universal Shoppe in Hazratganz at 20% discount and the same is available at only 15% discount at Universal Shoppe Bhootnath Market. Ms. Agrawal has just sufficient amount of Rs. 800 to purchase it at Universal Shoppe Hazratganz. What is the amount that Ms. Agrawal has less than the required amount to purchase it at Universal Shoppe Bhootnath?
(a) Rs. 70 (b) Rs. 50
(c) Rs. 100 (d) data insufficient

46. A balance of a trader weighs 10% less than it should be. Still the trader marks-up his goods to get the overall profit of 20%. What is the markup on the cost price?
(a) 40% (b) 8%
(c) 25% (d) 16.66%

47. ITC sells one product at a profit of 20% another at a loss of 20% at the same selling price. What is the loss incurred by ITC?
(a) 1% (b) 2%
(c) 4% (d) 0%

48. In the previous question, if SP of each article be Rs. 200, what is the amount of loss?
(a) Rs. 10 (b) Rs. 16
(c) Rs. 16.66 (d) none of these

49. The cost price of 19 articles is same as the selling price of 29 articles. What is the loss %?
(a) 35% (b) 34.48%
(c) 52.63% (d) none of these

50. The selling price of 13 articles is same as the cost price of 23 articles. What is the profit percentage?
(a) 43.47% (b) 74.83%
(c) 78% (d) 76.92%

51. A trader can procure 34 pencils by selling 28 pencils. What is the ratio of cost price to the selling price of a pencil?
(a) 2 : 3 (b) 14 : 17
(c) 9 : 7 (d) 4 : 7

52. At style cloth emporium the shopkeeper measures 20% less for every metre of cloth also he marks-up goods by 20%. What is the profit percentage?
(a) 50% (b) 80%
(c) 75% (d) none of these

53. A bookseller procures 40 books for Rs. 3200 and sells them at a profit equal to the selling price of 8 books. What is the selling price of one dozen books, if the price of each book is same?
(a) 720 (b) 960
(c) 1200 (d) 1440

54. The profit percentage of A and B is same on selling the articles at Rs. 1800 each but A calculates his profit on the selling price while B calculates it correctly on the cost price which is equal to 20%. What is the difference in their profits?
(a) Rs. 360 (b) Rs. 60
(c) Rs. 540 (d) Rs. 450

55. Each of A and B sold their article at Rs. 1818 but A incurred a loss of 10% while B gained by 1%. What is the ratio of cost price of the articles of A to that of B?
(a) 101 : 90 (b) 85 : 89
(c) 81 : 75 (d) none of these

56. A trader sold an article at a loss of 5% but when he increased the selling price by Rs. 65 he gained 3.33% on the cost price. If he sells the same article at Rs. 936, what is the profit percentage?
(a) 15% (b) 16.66%
(c) 20% (d) data insufficient

57. Even after a discount of q% on marked price a trader gains by p%. What is the markup percentage over the cost price?
(a) $\frac{p+q}{(q-p)} \times 100$ (b) $\frac{p+q}{(100-p)} \times 100$
(c) $\frac{p+q}{(100-q)} \times 100$ (d) not possible

58. A milkman mixes 10% water in pure milk but he is not content with it so he again mixes 10% more water in the previous mixture. What is the profit percentage of milkman if he sells it at cost price :
(a) 11.11% (b) 20%
(c) 21% (d) 12.1%

59. A person sold an electronic watch at Rs. 96 in such a way that his percentage profit is same as the cost price of the watch. If he sells it at twice the percentage profit of its previous percentage profit then the new selling price will be :
(a) Rs. 132 (b) Rs. 150
(c) Rs. 192 (d) Rs. 180

60. A trader mixes 25% kerosene to his petrol and then he sells the whole mixture at the price of petrol. If the cost price of kerosene be 50% of the cost price of petrol, what is the net profit percentage?
(a) $11\frac{1}{9}\%$ (b) $12\frac{1}{9}\%$
(c) $9\frac{1}{11}\%$ (d) 20%

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61. A retailer cheats both to his whole-seller and his customer by 10% by his faulty balance *i.e.*, he actually weighs 10% more while purchasing from wholesaler and weighs 10% less while selling to his customer. What is his net profit percentage, when he sells at CP?
(a) $22\frac{2}{11}\%$ (b) $22\frac{2}{9}\%$
(c) 20% (d) 21%

62. A trader procures his goods from a wholesaler, whose balance reads 1200 g for 1000 g. The trader sells all the procured goods to a customer after marking up the goods at 20% above the cost price. What is his overall percentage profit or loss in the whole transaction?
(a) 38% profit
(b) 50% profit
(c) no profit no loss
(d) none of the above

63. A person wants to reduce the trade tax so he calculates his profit on the sale price instead of on the cost price. In this way by selling a article for Rs. 280 he calculates his profit as $14\frac{2}{7}\%$. What is his actual profit percentage?
(a) 20% (b) 16.66%
(c) 25% (d) data insufficient

64. A vendor sells his articles at a certain profit percentage. If he sells his articles at 1/3 of his actual selling price, then he incurs a loss of 40%. What is his actual profit percentage?
(a) 72% (b) 120%
(c) 80% (d) none of these

65. A retailer increases the selling price by 25% due to which his profit percentage increases from 20% to 25%. What is the percentage increase in cost price?
(a) 20% (b) 30%
(c) 25% (d) 50%

66. Abhinav saves Rs. 25 by getting 6.66% discount on a textbook. What is the amount of money (in Rs.) paid by him?
(a) 450 (b) 350
(c) 225 (d) 375

67. At kul-kul petrol pump the operator gives 5% less petrol but he sells it at the cost price. What is his profit in this way?
(a) 5% (b) 5.6%
(c) 5.26% (d) 4.78%

68. Due to reduction of 25% in price of oranges a customer can purchase 4 oranges more for Rs. 16. What is original price of an orange?
(a) Re 1 (b) Rs. 1.33
(c) Rs. 1.5 (d) Rs. 1.6

69. A reduction of 20% in the price of sugar enables a housewife to purchase 6 kg more for Rs. 240. What is the original price per kg of sugar?
(a) Rs. 10 per kg (b) Rs. 8 perkg
(c) Rs. 6 per kg (d) Rs. 5 per kg

70. A wholesaler sells toys at a profit of 20% to a retailer and retailer sells these toys to its customer at a profit of 25%. What is the profit percentage of the retailer?
(a) 5% (b) 80%
(c) 20% (d) 25%

LEVEL 2

1. An automobile agency launched a scheme that if a customer purchases two Jabaaj Discover bikes, one extra Jabaaj Discover will be free and if he purchases 3 Jabaaj Pulser he will get one extra Jabaaj Pulser free. If the cost price of 3 Jabaaj Discover and 4 Jabaaj Pulser be Rs. 67500 and Rs. 232500 respectively. If a customer purchases 2 bikes of Jabaaj Discover and 3 bikes of Jabaaj Pulser as per scheme he availed 1 bike free of each category, then at what price these bikes should be sold so that the agency can get overall profit of 17.5% :
(a) 235250 (b) 352500
(c) 368000 (d) 268000

2. Rahul went to purchase a Nokia mobile handset, the shopkeeper told him to pay 20% tax if he asked the bill. Rahul manages to get the discount of 5% on the actual saleprice of the mobile and he paid the shopkeeper Rs. 3325 without tax. Besides he manages to avoid to pay 20% tax on the already discounted price, what is the amount of discount that he has gottten?
(a) 750 (b) 375
(c) 875 (d) 525

3. When a bicycle manufacturer reduced its selling price by 50%, the number of bicycles sold radically increased by 600%. Initially the manufacturer was getting only 140% profit. What is the percentage increase of his profit?
(a) 10% (b) 14%
(c) 0% (d) can't be determined

4. A trader marks his goods such that he can make 32% profit after giving 12% discount. However a customer availed 20% discount instead of 12%. What is the new profit percentage of trader?
(a) 20% (b) 44%
(c) 30% (d) 28.8%

5. A retailer bought 3850 Linc pens and 1848 Cello pens at the same price. He sells Linc pens in such a way that he can buy 650 Linc pens with the sale price of 481 Linc pens. Again he can buy only 408 Cello pens with the sale price of 629 pens. What is the overall percentage of profit of the retailer?
(a) 4.8% (b) 9.6%
(c) 13% (d) none of these

6. The ratio of selling price of 3 articles *A*, *B* and *C* is 8 : 9 : 5and the ratio of percentage profit is 8 : 7 : 14 respectively. If the profit percentage of *A* is 14.28% and the cost price of *B* is Rs. 400, what is the overall percentage gain?
(a) 14.28% (b) 14.87%
(c) 16.66% (d) none of these

7. Anna sold his car to Boney at a profit of 20% and Boney sold it to Chakori at a profit of 10%. Chakori sold it to mechanic at a loss of 9.09%. Mechanic spent 10% of his purchasing price and then sold it at a profit of 8.33% to Anna once again. What is the loss of Anna?
(a) 23% (b) 29%
(c) 50% (d) 40%

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8. In an office the number of employees reduces in the ratio of 3 : 2 and the wages increases in the ratio of 20 : 27. What is the profit percentage of employees over the previous wages?
(a) 10% (b) 9.09%
(c) 11.11% (d) none of these

9. I asked the shopkeeper the price of a wristwatch. I found that I had just the required sum of money. When the shopkeeper allowed me a discount of 25%, I could bought another watch worth Rs. 940 for my younger sister. What is the price which I have paid for my own watch?
(a) Rs. 2700 (b) Rs. 1800
(c) Rs. 2820 (d) Rs. 3760

10. *A* and *B* are two partners and they have invested Rs. 54,000 and Rs. 90,000 in a business. After one year *A* recieved Rs. 1800 as his share of profit out of total profit of Rs. 3600 including his certain commission on total profit since he is a working partner and rest profit is received by *B*. What is the commission of *A* as a percentage of the total profit?
(a) 20% (b) 10%
(c) 5% (d) 25%

11. A trader sells goods to a customer at a profit of *k*% over the cost price, besides it he cheats his customer by giving 880 g only instead of 1 kg. Thus his overall profit percentage is 25%. Find the value of *k*?
(a) 8.33% (b) 8.25%
(c) 10% (d) 12.5%

12. A trader sells two brands of petrol; one is Extra Premium (EP) and other one is 'Speed' (SP). He mixes 12 litres of EP with 3 litres of speed and by selling this mixture at the price of EP he gets the profit of 9.09%. If the price of Extra Premium be Rs. 48 per litre, then the price of Speed (SP) is :
(a) Rs. 38 per litre (b) Rs. 42 per litre
(c) Rs. 28 per litre (d) none of these

13. *A*, *B* and *C* invest in the ratio of 3 : 4 : 5. The percentage of return on their investments are in the ratio of 6 : 5 : 4. Find the total earnings, if *B* earns Rs. 250 more than *A* :
(a) Rs. 6000 (b) Rs. 7250
(c) Rs. 5000 (d) none of these

14. Ajay bought a motor cycle for Rs. 50,000. 2 years later he sold it to Bijoy at 10% less of the cost price. Bijoy spend 5% of the purchasing price on its maintenance. Later Bijoy displayed the sale price of his motorcycle Rs. 50,000. Chetan wanted to purchase it at 15% discount but Bijoy gave him two successive discounts of 10% and 5% instead of 15% in one time. What is the actual discount availed by Chetan?
(a) 15% (b) 15.5%
(c) 14.5% (d) none of these

15. Kamal bought a house in Sushant city, whose sale price was Rs. 8 lakh. He availed 20% discount as an early bird offer and then 10% discount due to cash payment. After that he spent 10% of the cost price in interior decoration and lawn of the house. At what price should he sell the house to earn a profit of 25%?
(a) Rs. 9 lakh (b) Rs. 7.99 lakh
(c) Rs. 7.92 lakh (d) none of these

16. I wanted to purchase 10 chairs for the class room whose cost was Rs. 200 each. The trader offered me a discount if I were to purchase a set of 12 chairs. So I calculated that if I assume the normal price of 10 chairs then we can purchase 2 extra chairs which cost me only Rs. 80 each of two chairs at the cost price of 12 chairs after discount. What is the percentage discount?
(a) 6% (b) 8%
(c) 12% (d) 10%

17. The cost of servicing of a Maruti car at Maruti care Pvt. Ltd. is Rs. 400. Manager of service centre told me that for the second service within a year a customer can avail a 10% discount and further for third and fourth servicing he can avail 10% discount of the previous amount paid, within a year. Further if a customer gets more than 4 services within a year he has to pay just 60% of the servicing charges on these services. A customer availed 5 services from the same servicing station, what is the total percentage discount fetched by the customer?
(a) 19.42% (b) 18.5%
(c) 17.6% (d) 26%

18. The cost price of an article is *C* and the selling price of the same article is *S*, where *Z* is the profit or loss percentage. If the cost price and selling price both are increased by same amount then which of the following is true :
(a) *Z* increases (b) *Z* decreases
(c) remains constant (d) none of these

19. Cost price of 12 oranges is equal to the selling price of 9 oranges and the discount on 10 oranges is equal to the profit on 5 oranges. What is the percentage point difference between the profit percentage and discount percentage?
(a) 20 (b) 22.22
(c) 16.66 (d) 15

20. A car mechanic purchased four old cars for Rs. 1 lakh. He spent total 2 lakh in the maintenance and repairing of these four cars. What is the average sale price of the rest three cars to get 50% total profit if he has already sold one of the four cars at Rs. 1.2 lakh?
(a) 1.5 lakh (b) 1.1 lakh
(c) 1.2 lakh (d) 1.65 lakh

21. The cost of setting up a magazine is Rs. 2800. The cost of paper and ink etc is Rs. 80 per 100 copies and printing cost is Rs. 160 per 100 copies. In the last month 2000 copies were printed but only 1500 copies could be sold at Rs. 5 each. Total 25% profit on the sale price was realized. There is one more resource of income from the magazine which is advertising. What sum of money was obtained from the advertising in magazine?
(a) Rs. 1750 (b) Rs. 2350
(c) Rs. 1150 (d) Rs. 1975

22. DSNL charges a fixed rental of Rs. 350 per month. It allows 200 calls free per month. Each call is charged at Rs. 1.4 when the number of calls exceeds 200 per month and it charges Rs. 1.6 when the number of calls exceeds 400 per month and so on. A customer made 150 calls in February and 250 calls in March. By how much per cent the each call is cheaper in March than each call in February?
(a) 28% (b) 25%
(c) 18.5% (d) none of these

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23. In the Bargain Bazar everyone purchases with a fair bargaining, so the traders markup the prices too much. A trader marked up an article at Rs. M expected huge profit if it is sold on the marked price. But a customer purchased it at $M/2$ with his fine bargaining skills, so the expected profit of the trader diminished by 66.66%. What is the percentage discount fetched by the customer through bargaining?
(a) 33.33% (b) 50%
(c) 66.66% (d) none of these

24. Tika Chand has a weighing balance in which there is a technical fault. The right pan of his balance measures always 200 g more than its left pan. Tika Chand as usual misutilise this balance in his business. While purchasing the articles he puts goods in the left pan and weight in the right pan while selling he reverse the order *i.e.*, goods in the right pan and weight in the left pan. He uses only 2 kg weight for the measurement and to measure $2n$ kg weight he measures n times by 2-2 kg but he sells goods at cost price. What is his profit percentage?
(a) 20% (b) $22\frac{2}{9}\%$
(c) $18\frac{2}{11}\%$ (d) none of these

25. Akram Miya has two types of grapes. One is the fresh grapes containing 80% water and dry grapes containing 25% water. He sells 20 kg dry grapes, by adding water to the dry grapes, at cost price. What is the total profit percentage when after adding water the weight of 20 kg dry grapes increased in the proportion of water in fresh grapes?
(a) 275% (b) 200%
(c) 80% (d) 125%

26. Pankaj and Sushil invested some amount of money in the ratio of 3 : 5 for the same period in a business. They decided that at the end of year 20% profit was to be given to AIDS Control Society of India as a donation. Out of the remaining, 75% was to be reinvested and the rest of the profit was to be divided as interest on their capitals. If the difference in their shares is Rs. 1200. Find the total profit?
(a) Rs. 18000 (b) Rs. 24000
(c) Rs. 20000 (d) none of these

27. Jagran group launched a new magazine in January 2004. The group printed 10000 copies initially for Rs. 50000. It distributed 20% of its stock freely as specimen copy and 25% of the rest magazines are sold at 25% discount and rest at 16.66% discount whose printing price was Rs. 12 per copy. What is the overall gain or loss in the first month's issue of magazine, if the magazine could not realize the income from advertisements or other resources?
(a) 56% profit (b) 27% loss
(c) 16.66% profit (d) 38% profit

28. Teenagers shoe company sells the shoes whose prices *i.e.*, cost prices and selling prices are the multiples of either 13, 14, 15, 16, 17, 18 or 19, starting from Rs. 399 to Rs. 699 (i.e., $399 \le CP/SP \le 699$). What can be the maximum profit of the company?
(a) Rs. 292 (b) RS. 398
(c) Rs. 298 (d) Rs. 300

29. Jhun Jhunwala makes 1000 toys and incurs a cost of Rs. 1,2 for each toy. He marks-up the price in such a way that if he sells only 70% of the manufactured toys he will realize 16.66% overall profit. He sells only 750 articles at the marked price since rest of the toys are found to be defective so can't be sold. What is the net profit or loss of Jhun Jhunwala?
(a) 14.44% loss (b) 25% profit
(c) 33.33% profit (d) none of these

30. Anupam sells a painting to Bhargava at 4/5th the rate of profit at which Bhargava sells it to Chaudhary. Further Chaudhary sells it to Dara Singh at half the rate of profit at which Anupam sold it to Bhargava. If Chaudhary earns a profit of 10% by selling it to Dara Singh for Rs. 2805. What is the cost price of painting for Bhargava?
(a) 1896 (b) 2040
(c) 1680 (d) 2000

31. A dishonest retailer cheats his wholesaler and customer both. He purchases 19% more from the wholesaler and sells 15% less while selling to its customer. What is profit percentage by selling the goods at cost price?
(a) 36.78% (b) 34%
(c) 40% (d) 36.85%

32. Rotomac produces very fine quality of writing pens. Company knows that on an average 10% of the produced pens are always defective so are rejected before packing. Company promises to deliver 7200 pens to its wholesaler at Rs. 10 each. It estimates the overall profit on all the manufactured pens to be 25%. What is the manufacturing cost of each pen?
(a) Rs. 6 (b) Rs. 7.2
(c) Rs. 5.6 (d) Rs. 8

33. Pratibha printers prepares diaries expecting to earn a profit of 40% by selling on the marked price. But during transportation 8% diaries were got spoiled due to at random rain and 32% could be sold only at 75% of the cost price. Thus the remaining 60% diaries could be sold at the expected price. What is the net profit or loss in the whole consignment?
(a) 6% (b) 10%
(c) 8% (d) can't be determined

34. Radhey Lal markup the prices of sweets by 40% and he sold only 40% of those at this price. He sells half of the rest at $14\frac{2}{7}\%$ discount and rest at 25% discount. What is the net profit of Radhey Lal?
(a) 26.5% (b) 23.5%
(c) 30% (d) 28.6%

35. The price of an article reduces to 576 after two successive discounts. The markup is 80% above the cost price of Rs. 500. What is the new profit percentage if instead of two successive discounts the markup price was further increased successively two times by the same percentage?
(a) 259.2% (b) 59.2%
(c) 159.2% (d) can't be determined

36. A trader marks-up his goods by 80% and gives discount of 25%. Besides it he weighs 10% less amount while selling his goods. What is the net profit of trader?
(a) 50% (b) 35%
(c) 45% (d) 55%

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37. A dishonest trader marks up his goods by 80% and gives discount of 25%. Besides he gets 20% more amount per kg from wholesaler and sells 10% less per kg to customer. What is the overall profit percentage?
(a) 80% (b) 60%
(c) 70% (d) none of these

38. A dishonest dealer purchases goods at 20% discount of the cost price of Rs x and also cheats his wholesaler by getting 20% extra through false weighing, per kg. Then he marks up his goods by 80% of x, but he gives a discount of 25% besides he cheats his customer by weighing 10% less than the required. What is his overall profit percentage?
(a) 125% (b) 100%
(c) 98.66% (d) 120%

39. Anjuli, Bhoomika and Chawla went to market to purchase the rings whose costs were same. But each ring was available with two successive discounts. Anjuli availed two successive discounts of 5% and 20%. Bhoomika availed two successive discounts 10% and 15% while Chawla availed two successive discounts of 12% and 13%. Who gets the maximum possible discount?
(a) Anjuli (b) Bhoomika
(c) Chawla (d) all of these

40. An egg seller sells his eggs only in the packs of 3 eggs, 6 eggs, 9 eggs, 12 eggs etc., but the rate is not necessarily uniform. One day Raju (which is not the same egg seller) purchased at the rate of 3 eggs for a rupee and the next hour he purchased equal number of eggs at the rate of 6 eggs for a rupee. Next day he sold all the eggs at the rate of 9 eggs for Rs. 2. What is his percentage profit or loss?
(a) 10% loss (b) 11.11% loss
(c) 3% loss (d) 2.5% profit

41. Virendra and Gurindra purchased one camera each at the same prices. Later on Amrendra purchased both cameras at equal prices from Virendra and Gurindra. But the profit percentage of Virendra was P while the same of Gurindra was Q since Gurindra calculated his profit on the selling price. Thus $Q = 41\frac{2}{3}\%$ of P. If Amrendra sells one of the camera to Dholakiya at $P\%$ profit then what is the cost price for Dholakiya, while Amrendra purchased each of the camera at Rs. 240?
(a) Rs. 676 (b) Rs. 500
(c) Rs. 576 (d) none of these

42. A merchant earns 25% profit in general. Once his 25% consignment was abducted forever by some goondas. Trying to compensate his loss he sold the rest amount by increasing his selling price by 20%. What is the new percentage profit or loss?
(a) 10% loss (b) 12.5% loss
(c) 12.5% profit (d) 11.11% loss

43. A milkman purchases 10 litres of milk at Rs. 7 per litre and forms a mixture by adding freely available water which constitutes 16.66% of the mixture. Later on he replaced the mixture by some freely available water and thus the ratio of milk is to water is 2 : 1. He then sold the new mixture at cost price of milk and replaced amount of mixture at twice the cost of milk then what is the profit percentage?
(a) 68% (b) 34%
(c) 40% (d) none of these

44. Profit on selling 10 candles equals selling price of 3 bulbs. While loss on selling 10 bulbs equals selling price of 4 candles. Also profit percentage equals to the loss percentage and cost of a candle is half of the cost of a bulb. What is the ratio of selling price of candle to the selling price of a bulb?
(a) 5 : 4 (b) 3 : 2
(c) 4 : 5 (d) 3 : 4

45. Cost price of two motorcycles is same. One is sold at a profit of 15% and the other for Rs. 4800 more than the first. If the net profit is 20%. Find the cost price of each motorcycle :
(a) Rs. 48000 (b) Rs. 52000
(c) Rs. 36000 (d) Rs. 42500

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Answers

INTRODUCTORY EXERCISE-6.1

1. Rs. 180 and Rs. 225 2. Rs. 9000 3. Rs. 345 4. Rs. $153\frac{1}{3}$ per hundred 5. Rs. 15300, Rs. 17850, gain = $\frac{5}{11}$ %

INTRODUCTORY EXERCISE-6.2

1 (d)	2. (c)	3. (d)	4. (b)	5. (c)	6. (b)	7. (c)	8. (d)	9. (c)	10. (c)
11. (c)	12. (b)	13. (c)	14. (b)	15. (a)	16. (d)	17. (c)	18. (b)	19. (c)	

LEVEL-1

1 (a)	2. (b)	3. (b)	4. (c)	5. (c)	6. (d)	7. (d)	8. (b)	9. (a)	10. (c)
11. (a)	12. (a)	13. (d)	14. (d)	15. (c)	16. (d)	17. (b)	18. (b)	19. (c)	20. (a)
21. (d)	22. (c)	23. (c)	24. (b)	25. (c)	26. (a)	27. (b)	28. (b)	29. (d)	30. (a)
31. (b)	32. (b)	33. (a)	34. (a)	35. (c)	36. (d)	37. (c)	38. (c)	39. (c)	40. (b)
41. (c)	42. (b)	43. (a)	44. (c)	45. (b)	46. (b)	47. (c)	48. (c)	49. (b)	50. (d)
51. (b)	52. (a)	53. (c)	54. (b)	55. (a)	56. (c)	57. (c)	58. (c)	59. (a)	60. (a)
61. (b)	62. (c)	63. (b)	64. (c)	65. (a)	66. (b)	67. (c)	68. (b)	69. (a)	70. (d)

LEVEL-2

1 (b)	2. (c)	3. (c)	4. (a)	5. (d)	6. (d)	7. (a)	8. (a)	9. (c)	10. (a)
11. (c)	12. (c)	13. (b)	14. (c)	15. (c)	16. (d)	17. (a)	18. (b)	19. (b)	20. (b)
21. (d)	22. (a)	23. (b)	24. (b)	25. (a)	26. (b)	27. (a)	28. (c)	29. (b)	30. (b)
31. (c)	32. (b)	33. (c)	34. (b)	35. (c)	36. (a)	37. (a)	38. (a)	39. (a)	40. (b)
41. (c)	42. (c)	43. (a)	44. (b)	45. (a)					

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Hints & Solutions

LEVEL 1

1. Abhishek Bhanu

Discount → 2800 2400 + 640 = 3040

The difference in selling price is same as difference in discount which Rs. 240 (= 3040 – 2800)

2. ∵ 10 % x = 15% of y, where $x + y = 30000$

$$\Rightarrow \quad \frac{x}{y} = \frac{3k}{2k}$$

Hence, the difference = k = 6000

3. Let the cost price of 1 litre pure milk be Re. 1, then

$$\left.\begin{matrix}6 \text{ litres (milk)} \rightarrow \text{CP} = \text{Rs. } 6 \\ 2 \text{ litres (water)} \rightarrow \text{CP} = \text{Rs. } 0\end{matrix}\right\} \rightarrow \text{CP} = \text{Rs. 6 only.}$$

and 8 litre mixture → SP → 8 × 2 = Rs. 16

$$\text{Profit} = \frac{16-6}{6} \times 100 = \frac{1000}{6} = 166.66\%$$

4. $\left.\begin{matrix}\text{SP of 60\% goods} = 0.6x \times 0.95 = 0.57x \\ \text{SP of 40\% goods} = 0.4x \times 1.1 = 0.44x\end{matrix}\right\}$ Total SP = $1.01x$

$$\text{Profit} = 0.01x = 100$$

$$\Rightarrow \quad x = 10000$$

Alternatively: From option (c)

10000 → 6000, 4000

loss = 300 gain = 400

net gain = 400 – 300 = Rs. 100,

Hence, option (c) is correct.

5. Let the MP of 1 kg tea be Re. 1, then CP of 20 kg with discount = 20 × 0.9 = Rs. 18

also 1 kg tea is free. So the retailer gets tea worth Rs. 21 by paying Rs. 18 only.

$$\text{Profit \%} = \frac{\text{goods left}}{\text{goods sold}} \times 100$$

$$= \frac{21-18}{18} \times 100 = 16.66\%$$

(Since the retailer earns Rs. 3 on each Rs. 18)

6. Let the CP of profit yielding article be Rs. 100 then

250 CP → 30% loss

75% profit: SP 175 ↔ 175 SP; CP 100

Total CP = 350, Total SP = 350

So there is no profit no gain.

Alternatively: $1.75x = 0.7y$

$$\Rightarrow \quad \frac{x}{y} = \frac{2}{5}$$

again $[2 \times 1.75 = 3.5]$

and $[5 \times 0.7 = 3.5]$

Total cost price = 2 + 5 = Rs. 7

Total selling price = 3.5 + 3.5 = Rs. 7

Hence, no loss no gain.

7.

	CP	SP	MP	
Initially	: 100	140	×	(since profit = 40%)
New prices :	100	112	140	

profit = 12%, discount = 20%

8. $\frac{CP}{SP} = \frac{2}{3}$

$$\therefore \quad \text{profit (\%)} = \frac{1}{2} \times 100 = 50\%$$

Alternatively: $\text{CP} = \frac{100}{18} = 5.55$ paise

$$\text{SP} = \frac{100}{12} = 8.33 \text{ paise}$$

$$\text{Profit} = \frac{8.33 - 5.55}{5.55} \times 100 = 50\%$$

9.

CP	SP	MP
100	120	150
	20% profit	20% discount

But it can be directly solved as SP = 1.2 CP = Rs. 2304

NOTE There is no role of discount.

10. Note: There is no role of cost price of article (Rs. 123684)

(Consider) CP → 100

60 40

16.66% of profit + 10 −24 60% loss

70 16

(100 –14) = SP 86

Since, the overall % loss = 14%.

Thus option (c) is correct.

Alternatively: $0.6x + k \times 0.4x = 0.86x$

$$\Rightarrow \quad 0.4kx = 0.16x$$

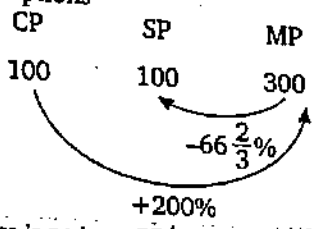
$$\Rightarrow \quad k = 0.4$$

Therefore loss = 1 – 0.4 = 0.6 *i.e.*, 60%

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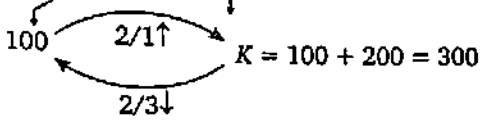
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11. Go through options



NOTE When there is no loss, CP = SP.

Alternatively: CP = SP, MP



From percentage change graphic :

Decrease	Increase
$\frac{2}{3}\downarrow$	$\frac{2}{3-2}=\frac{2}{1}\uparrow$
= 66.66%	= 200%

So, when CP = 100, markup % = 200, MP = 300

$$\text{Discount} = 66\frac{2}{3}\% = 200,\ \text{SP} = 100 = \text{CP}$$

Alternatively : $(CP = SP) = \frac{1}{3}MP$

$\Rightarrow$ $(CP = SP) : MP = 1 : 3$

12. $CP : MP = 2x : 3x$

$\Rightarrow$ $\text{Profit} = x$

(%) profit : (%) discount = 3 : 2

Let $CP = 200,\ SP = 300$

But $\frac{3x}{100}\times 200 + \frac{2x}{100}\times 300 = 100$

$\Rightarrow$ $x = 8.33\%$

$\text{Discount } 2x = 16.66\%$

13. It is dependent upon the markup (%) or discount (%)

Since,



loss = 4%

Again



loss = 25%

Hence we can't determine.

14. CP → Profit (%) → MP

$100 \to x\% \to 100 + \frac{100x}{100} = (100 + x)$

Again, MP → discount → SP

$(100 + x) \to x\% \to (100 + x) - (100 + x)\frac{x}{100}$

$= (100 + x)\left(1 - \frac{x}{100}\right)$

Now $\frac{(100 + x)}{(100 + x)\left(1 - \frac{x}{100}\right)} = \frac{2}{1}$

$\Rightarrow$ $\left(1 - \frac{x}{100}\right) = \frac{1}{2} \Rightarrow x = 50\%$

This can be observed in the solution of the previous problem.

15. Go through option

$180 \times 12 \times 1.2 + 180 \times 8 \times 1.1 = 180\,[14.4 + 8.8]$

$= 180\,(23.2) = 4176$

and $180 \times 20 \times 1.15 = 4140$

Therefore loss = 4176 − 4140 = 36

Hence option (c) is correct.

16. Gain = 30000

loss = 20000

net gain = 10000 over Rs. 4 lakh

Hence, $\text{profit} = \frac{10000}{400000}\times 100 = 2.5\%$

17. Let the MP = Re 1 per kg then

Weight	MP	Rate
100	100	1
96	80	$\frac{80}{96}$
	SP	

$\text{Effective discount} = 1 - \frac{80}{96} = \frac{16}{96}$

$\%\ \text{discount} = \frac{16}{96}\times 100 = 16.66\%$

18. CP of $A + B + C = 2x \times y \div 5x \times 2y + 2x \times 4y = 20xy$

Profit of $A = 0.2xy$

Profit of $B = 2xy$

Profit of $C = 2xy$

Total profit $= 4.2xy$

$\%\ \text{profit} = \frac{4.2xy}{20xy}\times 100 = 21\%$

19. Initially

CP	Profit	SP	MP
100	x	$(100 + x)$	133.33

After change 100 $2x$ $(100 + x)\frac{7}{6}$

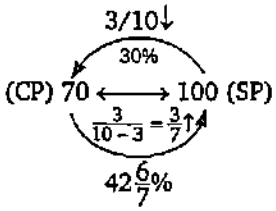
Now, since $(100 + x)\frac{7}{6} - 100 = 2x$

$\Rightarrow$ $x = 20\%$

	CP	Profit	SP	MP
	100	20	120	133.33
So,	300	60	360	400
Again	300	120	420	

So the increased selling price = Rs. 420

20. From percentage change graphic



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$$\text{Decrease} = \frac{3}{10}\downarrow = 30\%$$

$$\text{Increase} = \frac{3}{(10-3)} = \frac{3}{7}\uparrow = 42\frac{6}{7}\%$$

21. Let the price be x, then

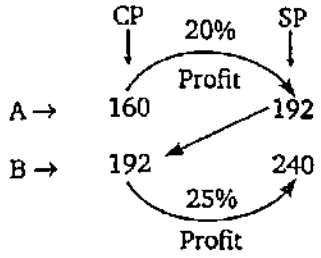
$$x \times 0.8 \times 0.9 = 468$$

$\Rightarrow x = 650$, therefore marked price = Rs. 650

Alternatively: $650 \times 0.8 \times 0.2 = 468$

Hence option (d) is correct.

22.



23. Reduced price = $100 \times 0.95 \times 0.90 \times 0.80 = 68.40$

$\therefore$ Single discount = $(100 - 68.4)\% = 31.6\%$

24. $CP = \frac{10}{12}$, $SP = \frac{12}{10}$

$$\text{Profit (\%)} = \frac{\frac{12}{10} - \frac{10}{12}}{\frac{10}{12}} \times 100 = 44\%$$

25. Let the cost price of one gram be Re. 1, then the markup price be Rs. 1.2 per g.

Now he sells 1000 g which seems to be 1250 g so he charges to a customer 1250×1.2 = Rs. 1500 for 1000 g (or Rs. 1000)

Thus his profit % = $\frac{1500 - 1000}{1000} \times 100 = 50\%$

26. Let the CP be Rs. x, then SP be $0.96x$

$\therefore$ $0.96x = 240 \Rightarrow x = 250$

Now the new SP = $250 \times 1.1 = 275$

27.

CP	SP	MP
x	$112.5x = 225$	300
		(−25%)

$\therefore x = 200$, hence the cost price be Rs. 200.

NOTE It can also be solved by using option.
First of all find the SP by decreasing MP by 25% then this SP will be equal to 112.5% (12.5% is the profit) of the cost price so the CP can be find as given above.

28.

CP	$\xrightarrow{(-10\%)}$	$(SP)_1$	$\xrightarrow{(-5\%)}$	$(SP)_2$
200		180		171

29. The CP of profitable cow = $\frac{9900}{1.1} = 9000$

and profit = Rs. 900

The CP of loss yielding cow = $\frac{9900}{0.8} = 12375$

and loss = Rs. 2475

So, the net loss = $2475 - 900 = 1575$

30. The best way is to go through option. Consider option (a)

15000	
$\left(\frac{2}{3}\right)$	$\left(\frac{1}{3}\right)$
10000	5000
(5% profit) ↓	↓ 2% loss
500	100

Net gain = Rs. 400 (= 500 − 100)

Hence, option (a) is correct.

Alternatively: $\left[\left(1.05 \times \frac{2x}{3} + 0.98\,\frac{x}{3}\right) - x\right] = 400$

$\Rightarrow$ $x = 15000$

31. Profit (%) = $\frac{\text{goods left}}{\text{goods sold}} \times 100 = \frac{50}{450} \times 100 = 11\frac{1}{9}\%$

Alternatively: Suppose CP of 1 g is Re. 1 then he sells goods worth Rs. 450 only and charges Rs. 500. So the profit is Rs. 50 over the sale price of Rs. 450.

32. Go through options. Consider option (b)

$200 \times 3.25 - 200 \times 3$ = Rs. 50 (= 30 + 20)

Hence, option (b) is correct.

Alternatively: Rs. 0.25 is the difference on 1 banana

$\therefore$ Re 1 is the difference on $\frac{1}{0.25} = 4$ bananas

$\therefore$ Rs. 50 is the difference on $4 \times 50 = 200$ bananas

33. By percentage change (or fraction change) graphic

increase decrease

$30\% = \frac{3}{10}\uparrow$ $\frac{3}{13}\downarrow$ [Since product *i.e.*, price is constant]

It means now $\frac{3}{13}$ times less eggs are available which is equal to 3 eggs.

i.e., $\frac{3x}{13} = 3$

$\Rightarrow$ $x = 13$ eggs (initially)

Now available eggs on the same price = 13 − 3 = 10

Thus, the new price = $\frac{910}{10} = 91$ paise

Alternatively : Go through options.

New or changed	No. of eggs	Rate		= total prices
			(+30%)	= 910 paise
Initial →	13	× 70		= 910 paise

Therefore option (a) is correct.

34. $SP = \frac{100}{12}$ paise

$CP = x$, $SP = 0.8x = \frac{100}{12} \Rightarrow x = \frac{100}{9.6}$ paise

Therefore, $CP = \frac{100}{9.6}$ paise

Thus, the new SP (with 20% profit) = $\frac{100}{9.6} \times 1.2 = \frac{100}{8}$ paise

Hence, 8 apples can be purchased for Re. 1 to gain 20%.

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35. $10000 \xrightarrow{-10\%} 9000 \xrightarrow{-20\%} 7200 \xrightarrow{+10\%} 7920 \xrightarrow{+10\%} 8712$

36. $1425 - 1353 = 72 = 6\%$ of CP

$\Rightarrow$ CP = 1200

37. $CP = \frac{405}{0.9} = 450$

New SP = Rs. 465

Profit (%) $= \frac{465 - 450}{450} \times 100 = 3.33\%$

38. $\left[((x \times 1.1) \times 1.1) \times \frac{1125}{1100}\right] = 990$

$\Rightarrow$ $x = 800$

Alternatively: Go through options or by the reverse process.

39. Best way is to go through option

Alternatively : $P \times 0.9 \times 0.75 = C \times 0.85 \times 0.80$

$\Rightarrow$ $\frac{P}{C} = \frac{136}{135}$

40. Let the CP be x then

$0.04x = 54 (= 1080 - 1026)$

$\Rightarrow$ $x = \frac{54}{0.04} = 1350$ (CP)

CP = 1350

SP = $1350 \times 1.04 = 1404$

41. SP − CP = 175 = 14% of CP

$\Rightarrow$ CP = 1250

$\therefore$ SP = 1250 + 175 = 1425

42. CP = 100, then

Tag price = 80

SP, after discount = 75

Total loss = 25

$\because$ Rs. 25 are being less at CP of Rs. 100

$\therefore$ Rs. 37.5 are being less at CP of Rs. $\frac{100}{25} \times 37.5 = 150$

Hence, SP = 150 − 37.5 = 112.5

Alternatively: 112.5 + 37.5 = 150 = CP

Again $150 \times 0.8 \times .9375 = 112.5 = SP$

Hence, option (b) is correct.

43. $CP + 4k = SP_1$ (gives profit)

$CP - 3k = SP_2$ (gives loss)

since loss ($3k$) is 25% less than profit ($4k$)

$\therefore$ $SP_1 - SP_2 = 7k = 836 - 703 = 133$

$\Rightarrow$ $k = 19$

$\therefore$ $CP = SP_1 - 4k = SP_2 + 3k = 760$

Therefore required SP = $760 \times 1.2 = 912$

44. You can go through options.

$CP = \frac{100}{6} = 16.66$ paise

$SP = \frac{100}{5} = 20$ paise

profit (%) $= \frac{20 - 16.66}{16.66} \times 100 = \frac{3.33}{16.66} \times 100$

$= \frac{1}{5} \times 100 = 20\%$

Hence, option (c) is correct.

Alternatively: $CP = \frac{100}{6}$

$SP = \frac{100}{6} \times 1.2 = \frac{100}{5}$

Hence, he should sell 5 toffees for Re. 1 (= 100 paise)

45. CP MP

$800 = 0.8x$ x

20% discount

$\Rightarrow$ x = Rs. 1000 = MP

SP at Hazratganz = Rs. 800

SP at Bhootnath = $1000 \times 0.85 = 850$

So, she requires Rs. 50 more to purchase the same calculator at Bhootnath.

46. Let the CP be Re. 1 per g

but he weighs 900 g for every 1000 g

$\therefore$ Value of goods sold = 900

Now, let the markup be x%

$\therefore$ $MP = 1000 + \frac{1000x}{100} = (1000 + 10x)$

but since MP = SP, $\therefore$ $SP = (1000 + 10x)$

Hence, profit (%) $= \frac{(1000 + 10x) - 900}{900} \times 100 = 20$

$\Rightarrow$ $x = 8$

Thus the markup = 8%.

47. Loss % $= \left(\frac{\text{common gain or loss}}{10}\right)^2 \%$

$= \left(\frac{20}{10}\right)^2 \% = 4\%$

48. Since loss = 4%

Now assume total CP of both articles be x, then

$SP = 0.96x = 400$

$\Rightarrow$ $x = \frac{400}{0.96} = CP$

$\therefore$ loss = 4% of CP

$\frac{4}{100} \times \frac{400}{0.96} =$ Rs. 16.66

49. $\frac{CP}{SP} = \frac{29x}{19x}$

loss % $= \frac{29x - 19x}{29x} \times 100 = 34.48\%$

(To understand the concept assume CP of each article Rs. 29 and SP of each article = Rs. 19)

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50. $\frac{CP}{SP} = \frac{13}{23}$

$\text{profit (\%)} = \frac{23-13}{13} \times 100 = \frac{10}{13} \times 100 = 76.92\%$

51. $\frac{CP}{SP} = \frac{28}{34} = \frac{14}{17}$

52. MP = Rs. 1.2, if CP = Re. 1
But he gives the cloth worth Rs. 0.8 instead of Re. 1.

$\text{Therefore profit \%} = \frac{1.2-0.8}{0.8} = \frac{0.4}{0.8} \times 100 = 50\%$

53. CP = Rs. 80 $\left(= \frac{3200}{40}\right)$

Now SP of 40 books = CP of 40 books + SP of 8 books

$\Rightarrow$ SP of 32 books = 3200

$\Rightarrow$ SP of 1 book = Rs. 100

$\therefore$ Required SP of 1 dozen books = Rs. 1200

54. Profit of $A = 1800 \times \frac{20}{100}$ = Rs. 360

CP of $B = \frac{1800}{1.2} = 1500$

$\therefore$ Profit of B = Rs. 300 (= 1800 – 1500)

$\therefore$ Difference in profit of A and B = Rs. 60

55. CP of $A = \frac{1818}{0.9} = 2020$

CP of $B = \frac{1818}{1.01} = 1800$

$\frac{\text{CP of } A}{\text{CP of } B} = \frac{2020}{1800} = \frac{101}{90}$

56. 103.33 CP – 0.95 CP = 65

$\Rightarrow$ CP = Rs. 780

$\text{Profit (\%)} = \frac{936-780}{780} \times 100 = 20\%$

57. Best way is to assume some proper values then verify it.

58. Let us assume initially he has 100 l of milk
So, after 10% of water addition new amount of
mixture = 110 l
Again adding 10% more then the final amount of
mixture = 121 l

$\text{Profit \%} = \frac{21}{100} \times 100 = 21\%$

59. $SP = x + \frac{x \times x}{100} = 96\%$

$\Rightarrow x = 60$

New, $SP = 60 + \left(\frac{60 \times 120}{100}\right)$ = Rs. 132

60. $K \rightarrow 1$ litre, price of $K \rightarrow 1$
$P \rightarrow 4$ litre, price of $P \rightarrow 2$
Price of total (new) mixture $= 1 \times 1 + 4 \times 2$ = Rs. 9
Price of pure petrol of same quantity = 5×2 = Rs. 10
Percentage profit $= \frac{1}{9} \times 100 = 11\frac{1}{9}\%$

61. Amount purchased = 1100 g
Amount sold = 900 g

$\text{Profit \%} = \frac{200}{900} \times 100 = 22\frac{2}{9}\%$

62. If CP be Re. 1 per g, then he pays Rs. 1200 for 1000 g
Again he obtains Rs. 1200 for 1000 g (by selling at 20% profit). Thus there is no loss no gain.

63.

CP	SP
240	280

$\left(-14\frac{2}{7}\%\right)$

$\text{Actual profit (\%)} = \frac{40}{240} \times 100 = 16.66\%$

NOTE Here you can apply base change concept.

increase 16.66% $\left(\frac{1}{6}\uparrow\right)$ decrease $14\frac{2}{7}\%$ $\left(\frac{1}{7}\downarrow\right)$

64. Go through options :

$100 \xrightarrow{80\%} 180$

$\text{New SP} = \frac{180}{3} = 60$

Percentage loss = 40% (100 – 60)
Hence, (c) is correct choice.

65.

CP	Profit	SP	
100	20	120	+25%
x	25	150	

$x \times 1.25 = 150 \Rightarrow x = 120$

$\text{\% change} = \frac{120-100}{100} \times 100 = 20\%$

66. 6.66% of MP = 25

$\Rightarrow$ MP = 375

$\therefore$ SP = MP – 25 = 350

67. $\text{Profit \%} = \frac{5}{95} \times 100 = 5.26\%$

68. Recall it is based on inverse proportion or product constancy concept.

Reduction in price	increase in amount
$25\%\left(\frac{1}{4}\right)\downarrow$	$\uparrow\left(\frac{1}{3}\right)33.33\% = 4$ oranges

It means original number of oranges = $4 \times 3 = 12$

$\therefore$ Original price of oranges $= \frac{16}{12}$ = Rs. 1.33

69.

Reduction in price	increase in amount
$20\%\left(\frac{1}{5}\right)\downarrow$	$\frac{1}{4}\uparrow(25\%) = 6$ kg

It means original amount of sugar needed = $6 \times 4 = 24$ kg

$\therefore$ Original price of the sugar $= \frac{240}{24}$ = Rs. 10 per kg.

70. Just a sitter. Don't be bogged down. Its simple i.e., 25%, nothing else, which is very obvious.

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LEVEL (2)

1. Just a sitter but a logical problem.

CP of 5 bikes = 67500 + 232500 = 300000

Now, since we require 17.5% profit, so SP = $300000 \times \frac{117.5}{100}$

= Rs. 352500

2. CP = 100, SP (with tax) = 120

New SP = 100 – 5 = 95

∴ Effective discount = 120 – 95 = 25

So, at SP of 95 → discount = 25

and at SP of 3325 → discount = $\frac{25}{95} \times 3325 = 875$

3. Let the CP of a bicycle = Rs. 100

Now, since profit 140%

∴ SP = 240

Now, 7 bicycles are being sold instead if 1 bicycle, but the sale price of new bicycle = Rs. 120

Therefore total sale price of new sale of bicycles

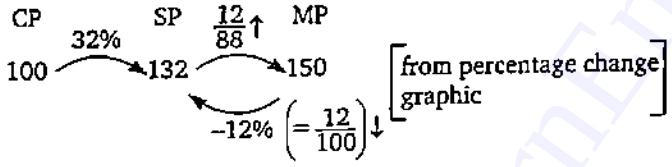
$= 7 \times 120 = 840$

and the CP = $7 \times 100 = 700$

So, the new profit = 840 – 700 = Rs. 140

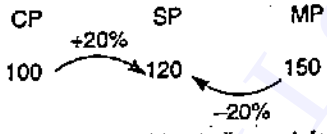
Since the initial profit is same as the new so there is no increase in percentage.

4.



NOTE In this case first of all find the SP, after adding profit percentage to CP then find the MP through SP.

Now,



Here first of all we subtract discount from the MP then the resultant value will be SP.

5.

Linc pens	Cello pens
CP : SP	CP : SP
37 : 50	37 : 24

$$\text{Profit \%} = \frac{13}{37} \times 100$$

and $$\text{Loss\%} = \frac{13}{37} \times 100$$

Since Profit = loss

Hence option (d) is correct.

6.

	A		B		C
SP	$\frac{1}{7}\binom{8}{7}\frac{1}{8}$	:	$\frac{1}{8}\binom{9}{8}\frac{1}{9}$	:	$\frac{1}{4}\binom{5}{4}\frac{1}{5}$

Since $14.28\% = \frac{1}{7}$

So, the ratio of profit percentage of

A		B		C	
8	:	7	:	14	(Given)
↓		↓		↓	
$\frac{1}{7}$		$\frac{1}{8}$		$\frac{1}{4}$	

Thus the ratio of CP of $A : B : C$

$7 : 8 : 4$

Therefore % profit = $\frac{(8+9+5)-(7+8+4)}{(7+8+4)} \times 100$

$= \frac{3}{19} \times 100 = 15.78\%$

7.

	A	B	C	M
CP →	100	120	132	(120 + 12) = 132
SP →	120	132	120	143
CP →	143	←		

Loss of $A = 143 - 120 = 23$

% loss of $A = \frac{23}{100} \times 100 = 23\%$

8. Total wages = no. of employees × wage per employee

$60xy = 3x \times 20y$

$54xy = 2x \times 27y$

$$\text{Profit (\%)} = \frac{60-54}{60} \times 100 = 10\%$$

9. If I had Rs. 100

Discount = 25 = cost of my sister's watch

then cost of my own watch = 75

Thus the ratio of cost of my own watch to that of my sister's watch = 3 : 1

10. Ratio of profit of $A : B$ (excluding commission of A)

$= 3 : 5$ (∵ 54 : 90 = 3 : 5)

Now the share of profit of B = 3600 – 1800 = Rs. 1800

So the share of profit of A (excluding commission) = Rs. 1080

So the commission of A = 1800 – 1080 = 720

Therefore the required percentage = $\frac{720}{3600} \times 100 = 20\%$

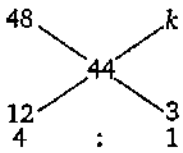
11. $\text{Profit \%} = \frac{25}{100} = \frac{120 + k\ (\text{Profit})}{880\ (\text{Sale})} \Rightarrow k = 100$

Therefore, net profit % = $\frac{100}{1000} \times 100 = 10\%$

12. $\text{SP} = \frac{12}{11}$ of CP

$48 = \frac{12}{11}$ of CP ⇒ CP = 44

Now, by alligation



∴ $k = 28$

Thus the price of speed brand is Rs. 28/litre.

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13.

	A	B	C
Investment	$3x$	$4x$	$5x$
Rate of return	$6y\%$	$5y\%$	$4y\%$
Return	$\frac{18xy}{100}$	$\frac{20xy}{100}$	$\frac{20xy}{100}$

$$\text{Total} = (18 + 20 + 20) = \frac{58xy}{100}$$

$$B\text{'s earnings} - A\text{'s earnings} = \frac{2xy}{100} = 250$$

$$\text{Total earning} = \frac{58xy}{100} = 7250$$

14. MP 100 → after first discount 90 → after second discount 85.5

So, the net discount = 100 − 85.5 = 14.5%

15. MP → CP → Total CP

100 → 80 → 72 → 79.2

(−20%, −10%, +10%)

SP = 125% of CP

SP = 1.25 × 79.2

SP = 99

So, initially marked price = 100 ⇒ 8,00,000

Final sale price = 99 ⇒ 7,92,000

16. Price of 10 chairs = 10 × 200 = 2000

Price of 12 chairs (without discount) = 12 × 200 = 2400

Price of 12 chairs with discount = 10 × 200 + 2 × 80 = 2160

therefore discount = 2400 − 2160 = 240

Hence discount % = $\frac{240}{2400} \times 100 = 10\%$

17. Amount paid in Ist service = 100 (suppose)

Amount paid in IInd service = 90

Amount paid in IIIrd service = 81

Amount paid in IVth service = 72.9

Amount paid in Vth service = 60

Total amount paid 403.9

Discount = 500 − 403.9 = 96.1

Discount % = $\frac{96.1}{500} \times 100 = 19.42\%$

18. Consider some proper values and then check out.

19.

CP	:	SP
3	:	4

Profit on 3 apples = Re 1 (consider CP = Re 1)

Profit = 33.33%

and discount = 11.11%

Since CP (3) — SP (4) — MP (4.5)

(1) (0.5)

Profit is double that of discount

So, the percentage point difference = 33.33% − 11.11%

= 22.22% point

20. Total cost of 4 cars = 1 + 2 = 3 lakh

Total SP of 4 cars = 3 × 1.5 = 4.5 lakh

SP of 1 car = 1.2 lakh

SP of rest 3 cars = 4.5 − 1.2 = 3.3 lakh

Average SP of all the 3 cars = 1.1 lakh

21. Setup cost = Rs. 2800

Paper etc. = Rs. 1600

Printing cost = Rs. 3200

Total cost = Rs. 7600

Total sale price = 1500 × 5 = 7500

Let the amount obtained from advertising is x then

(7500 + x) − (7600) = 25% of 7500

x = 1975

22. Charge of 1 call in February = $\frac{350}{150} = \frac{7}{3}$

Charge of 1 call in March = $\frac{350 + 50 \times 1.4}{250}$

$= \frac{420}{250} = \frac{42}{25}$

% cheapness of a call in March = $\frac{\frac{7}{3} - \frac{42}{25}}{\frac{7}{3}} \times 100 = 28\%$

23. Let the CP be 100 and % markup be k% then

MP = 100 + k

100 —k%→ (100 + k) MP (also expected SP)

but actual SP = $\frac{100 + k}{2}$

$\therefore \quad \frac{\left[\left(\frac{100 + k}{2}\right)\right]}{k} = \frac{200}{3 \times 100}$ (= 66.66%)

$\Rightarrow \quad k = 300$

Therefore CP MP (initially)

100 400

Finally SP = $\frac{400}{2} = 200$

$\therefore$ Discount = $\frac{200}{400} \times 100 = 50\%$

Alternatively: MP → M

SP → $\frac{M}{2}$

Discount (%) = $\frac{M/2}{M} \times 100 = 50\%$

24. Let the CP and SP of 1 g = Re 1, then

He spends Rs. 2000 and purchase 2200 g

and he charges Rs. 2000 and sells 1800 g

$$\text{profit (\%)} = \frac{\text{goods left}}{\text{goods sold}} \times 100$$

$$= \frac{400}{1800} \times 100 = 22\frac{2}{9}\%$$

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25. Fresh grapes

Water		Pulp
80%		20%
4	:	1

Dry grapes

Water		Pulp
25%		75%
1	:	3

5 kg — 15 kg] out of 20 kg dry grapes

+ (55 kg) → 60 kg

80%	20%	Required proportion of water and pulp
4	1	
60 kg	15 kg	

Thus to make dry grapes similar to the fresh grapes, Akram requires 55 kg water with 20 kg of dry grapes.

So, the profit (%) $= \frac{55}{20} \times 100 = 275\%$

26. Let the total profit be 100

Amount left after donation = 80

Amount left after reinvestment = 20

Now, $\frac{5x}{8} - \frac{3x}{8} = 1200$

where x is the amount left after reinvestment

$$\frac{2x}{8} = 1200 \Rightarrow x = 4800$$

Therefore, total profit $= 4800 \times 5 = 24000$

Alternatively: Go through options.

27. Total cost = Rs. 50,000

Total sale price (or revenue) $= 2000 \times 9 + 6000 \times 10 = 78{,}000$

$$\text{profit (\%)} = \frac{28000}{50000} \times 100 = 56\%$$

28. The maximum possible profit

= maximum possible difference in SP and CP.

It means SP be maximum and CP be minimum

$CP_{(min)}$ = Rs. 399 $\quad 19m = 399$

where m is an integer.

Again $SP_{(max)}$ = Rs. 697, which is very close to 699

Here $697 = 17k$; k is a positive integer.

So, the maximum profit = 697 − 399 = Rs. 298

29. Total cost price $= 1000 \times 1.2 =$ Rs. 1200

Expected selling price $= 700 \times x = 1200 \times 1.1666 = 1400$

$\Rightarrow$ $x =$ Rs. 2 per toy

Now the real selling price $= 750 \times 2 =$ Rs. 1500

$\therefore$ Profit = Rs. 300 (= 1500 − 1200)

$\therefore$ Profit % $= \frac{300}{1200} \times 100 = 25\%$

30. Chaudhary's profit = 10%

Anupam's profit = 20%

Bhargava's profit = 25%

i.e., $A \xrightarrow{20\%} B \xrightarrow{25\%} C \xrightarrow{10\%} D$

100 → 120 → 150 → 165

Now

B	D	
120	165	
17 times		17 times
2040	2805	

31. From the statements it is clear that he purchases 119 g instead of 100 g and he sells 85 g instead of 100 g. Therefore in this whole transaction he saves 19 + 15 = 34 g

Thus the profit $= \frac{34}{85} \times 100 = \left(\frac{\text{Goods left}}{\text{Goods sold}} \times 100\right) = 40\%$

NOTE In this type of question always equate either money or weight for simple solution.

32. You must know that the company is able to deliver only 90% of the manufactured pens. So let K be the manufacturing price of a pen, then

total income (including 25% profit) $= (8000 \times K) \times 1.25$

also this same income is obtained by selling 90% manufactured pens at Rs. 10 which is equal to 7200×10

Thus $(8000 \times K)\,1.25 = 7200 \times 10$

$\Rightarrow$ $K =$ Rs. 7.2 (90% of 8000 = 7200)

33. Let the number of diaries (produced) be 100 and the cost price of a diary be Re 1 then

Total cost incurred $= 100 \times 1 = 100$

Total sale price $= 32 \times 0.75 + 60 \times 1.4 = 108$

Therefore profit = Rs. 8

Thus there is 8% profit

NOTE Marked price (*i.e.*, expected) = 40% above the cost price.

34. Let the number of sweets be 100 and the cost price of one piece of sweet = Re. 1

then, total cost price $= 100 \times 1 =$ Rs. 100

Total sale price $= 40 \times 1.4 + 30 \times 1.2 + 30 \times 1.05 = 123.5$

$\therefore$ profit (%) = 23.5% (= 123.5 − 100)

35.

CP	SP	MP
500	576	900

Again $SP = MP\left[\left(1 - \frac{r}{100}\right)^2\right]$ [$r \rightarrow$ rate of discount in %]

$\Rightarrow$ $576 = 900\left(1 - \frac{r}{100}\right)^2$

$\Rightarrow$ $\frac{24}{30} = \left(1 - \frac{r}{100}\right) \Rightarrow r = 20\%$

Again, new $SP = MP\left(1 + \frac{r}{100}\right)^2 = 900\left(1 + \frac{20}{100}\right)^2 = 1296$

New, profit percentage $= \frac{SP - CP}{CP} \times 100$

$= \frac{1296 - 500}{500} \times 100 = 159.2\%$

36. Consider actual price of 1 g goods = Re. 1 then he sells the product equals to Rs. 90 only (10% less weighing)

Again MP = Rs. 1.8 and SP = 1.35 for 1 g.

Thus he gives the goods worth Rs. 90 and charges Rs. 135 after 25% discount. Thus the profit %

$$= \frac{135 - 90}{90} \times 100 = 50\%$$

37. $CP = \frac{100}{120} = \frac{10}{12}$

(since he purchases 120 g and pays Rs. 100, by assumption actual CP of 1 g = Re. 1)

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$$SP = \frac{135}{90} = \frac{3}{2} = \frac{18}{12}$$

(Since actual MP = 180, actual SP = 135, with 25% discount and he sells only 90 g instead of 100 g)

$$\text{profit (\%)} = \frac{\frac{18}{12} - \frac{10}{12}}{\frac{10}{12}} \times 100 = 80\%$$

38. Let the actual cost price of an article be Re 1 (in place of x)
Now he purchases goods worth Rs. 120 and pays Rs. 80, since 20% discount is allowed.
So the $CP = \frac{80}{120} = \frac{2}{3}$

Again MP = 180, SP = 135 (since 25% discount)
Thus the trader sells goods worth Rs. 90 instead of 100 g and charges Rs. 135. Therefore the effective $SP = \frac{135}{90} = \frac{3}{2}$

$$\therefore \quad \text{profit (\%)} = \frac{\frac{3}{2} - \frac{2}{3}}{2/3} \times 100 = 125\%$$

39.

Anjuli		Bhoomika		Chawla
100		100		100
↓ – 20%		↓ – 15%		↓ – 12%
24 ← 80		23.5 ← 85		23.44 ← 88
↓ – 5%		↓ – 10%		↓ – 13%
76		76.5		76.56

Thus it is clear from the graphical solution that the maximum discount is availed by Anjuli.

NOTE It does not matter that we first decrease by 20% and then by 5% or *vice-versa*. This concept has been already illustrated in percentage chapter. Try to do it for your concept clarification.

40. CP of one egg (in first case) $= \frac{1}{3} = 33.33$ paise

CP of one egg (in second case) $= \frac{1}{6} = 16.66$ paise

Average CP of one egg $= \frac{(33.33 + 16.66)}{2} = 25$ paise

SP of one egg $= \frac{200}{9}$ (Re. 1 = 100 paise)

$$\text{profit-loss} = \frac{25 - \frac{200}{9}}{25} \times 100 \qquad \left(\text{loss\%} = \frac{CP - SP}{CP} \times 100\right)$$

$= 11.11\%$ loss

41. This question is based on fundamental concept of percentage change.

$$\text{Virendra} = CP_V \xrightarrow{\rightarrow P\%} SP_V$$

$$\text{Gurindra} = CP_G \xrightarrow{Q\% \leftarrow} SP_G$$

Here, $CP_V = CP_G$ and $SP_V = SP_G$
But $P \neq Q$
also, $P\%$ of $CP_V = Q\%$ of SP_G
also $Q = 41\frac{2}{3}\%$ of $P = \frac{125}{3} \times \frac{P}{100}$

and $Q = \frac{P}{100 + P} \times 100$

(From the concept of percentage change)

$$\therefore \quad \frac{P}{100 + P} \times 100 = \frac{125}{3} \times \frac{P}{100} \qquad \left(41\frac{2}{3} = \frac{125}{3}\right)$$

$$\Rightarrow \quad P = 140$$

$\therefore$ CP = 100 when SP = 240

Where $CP = CP_V = CP_G$ and $SP = SP_V = SP_G$

Again SP for Amrindra = 240 + 140% of 240 = 576

42. Let the CP of one article be Rs. 1
then the SP be Re. 1.25
Again the new SP be $(1.25) \times 1.2 = 1.5$
Now, if he sell initially 100 articles, then

CP = 100 × 1 = Rs. 100

SP = 100 × 1.25 = Rs. 125

New SP = 75 × 1.5 = 112.5 (since 25% articles were abducted)

$\therefore$ New profit percentage = 12.5%

43. **Note :** First of all the price of milk does not matter. You can assume any convenient price. Besides it instead of 10 l of milk you can consider 100 l of milk to avoid calculations in decimal.
Now, since water is 16.66% in the mixture of milk, therefore with 100 l pure milk 20 l water is added. Again note that in replacement method the quantity of mixture does not increase except to the variation in ratio of contents.
Again by replacement formula

$$\frac{80}{120} = \frac{100}{120}\left(1 - \frac{K}{120}\right) \quad \Rightarrow \quad K = 24\,l$$

Thus he replaces 24 l of mixture with water.
(Note the required ratio of milk is to water is 2 : 1. It means in 3 l of new mixture, there will be 2 l of water)
Thus if the price of new mixture be Re 1, then the price of replaced mixture be Rs. 2.
Therefore, total SP = 120 × 1 + 24 × 2 = 168
and CP = 100 × 1 = 100
$\therefore$ Profit % = 68%

44.

	Candle	Bulb
CP	a	c
SP	b	d

and $c = 2a$

Profit = 10 $(b - a) = 3d$

and Loss = 10 $(c - d) = 4b$

$$\text{Profit (\%)} = \frac{3d}{10a} \times 100$$

and $$\text{Loss (\%)} = \frac{4b}{10c} \times 100$$

Again $$\frac{3d}{10a} \times 100 = \frac{4b}{10c} \times 100$$

$$\Rightarrow \quad \frac{3d}{a} = \frac{4b}{c} \Rightarrow \frac{3d}{a} = \frac{4b}{2a} \qquad (\because c = 2a)$$

$$\Rightarrow \quad \frac{b}{d} = \frac{3}{2}$$

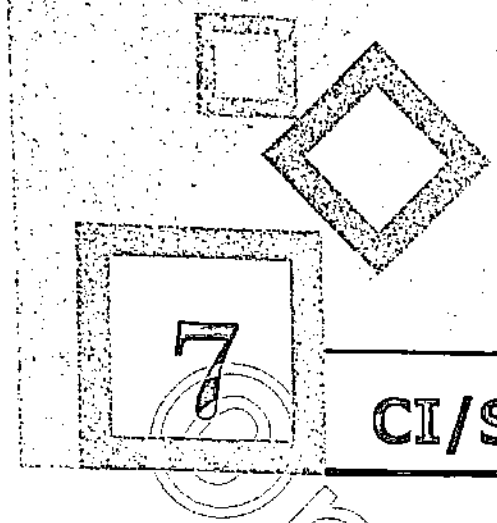
45. Let the CP of each motorcycle be Rs. x, then

$$2(1.15x) + 4800 = 2(1.2x)$$

$$\Rightarrow \quad 0.x = 4800 \Rightarrow x = 48000$$

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7 CI/SI/INSTALMENTS

The concept of simple interest and compound interest is one of those concepts which are widely used in Business and Banking etc. Questions are rarely asked in CAT from this chapter, but it is important for the purpose of Data interpretation. Since the questions based on the concept of 'interest' are rather easy so it is common that MAT, FMS, CET etc. asked enough questions from this chapter. The questions of this chapter are so easy that 95% questions can be solved orally or using unitary methods *i.e.*, sometimes you don't need to apply the given formula and I think that is essential.

Simple interest is nothing but the fix percentage of the principal (invested/borrowed amount of money)

Some key words used in the concept of interest.

Principal :(P) It is the sum of money deposited/loaned etc. also known as **capital.**

Interest : It is the money paid by borrower, calculated on the basis of **Principal.**

Time (T/n) : This is the duration for which money is lent/borrowed.

Rate of Interest (r/R) : It is the rate at which the interest is charged on principal.

Amount (A) = Principal + Interest

Simple Interest : When the interest is calculated uniformly only on the principal for the given time period.

Compound Interest : In this case for every next period of time the interest is charged on the total previous amount (which is the sum of principal and interest charged on it so far.) *i.e.*, every time we calculate successive increase in the previous amount.

IMPORTANT FORMULAE

Simple Interest (SI)

$$SI = \frac{P \times r \times t}{100}$$

P = Principal

r = rate of interest (in %)

t = time period (yearly, half yearly etc.)

$$\text{Amount } (A) = P + \frac{Prt}{100} = P\left(1 + \frac{rt}{100}\right)$$

NOTE Out of the five variables A, SI, P, r, t we can find any one of these, if we have the requisite information.

CONVERSION OF TIME PERIOD-RATE OF INTEREST

Given (r%)	Given (t)	Required (r%)	Required (t)
r% annual	t years	$\frac{r}{2}$(%) half-yearly	$2t$
r% annual	t years	$\frac{r}{4}$% quarterly	$4t$
r% annual	t years	$\frac{r}{12}$% monthly	$12t$

Compound Interest (CI)

1. $CI = A - P$ [A is the amount including interest and principal (P) both]

2. $$\text{Amount } (A) = P\left(1 + \frac{r}{100}\right)^t$$

3. When rate of interest is half-yearly

$$A = P\left(1 + \frac{(r/2)}{100}\right)^{2t}$$

4. When the rate of interest is quarterly

$$A = P\left(1 + \frac{r/4}{100}\right)^{4t}$$

Difference between CI and SI for two years

$$\text{(also in second year)} = P\left(\frac{r}{100}\right)^2$$

Difference between CI and SI for three years

$$= P\left(\frac{r}{100}\right)^2\left(\frac{r}{100} + 3\right)$$

Depreciation : It is known that the prices of some articles depriciates in their values over a time period. When the value in terms of currency decreases, we say that the value of the article is depreciating.

$$V_f = V_i\left(1 - \frac{r}{100}\right)^t$$

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V_i = initial value of the article
V_f = final (depreciated) value of article
r, is the rate of interest by which the price of article decreases over the time period 't'.

Population : It has been observed that the population of a particular locality/nation etc. increases or decreases successively over its previous value *i.e.*, it increases or decreases as compound interest for the money. Thus we use similar formulae for the calculation of population.

(Total population) $P = P_0\left(1+\frac{r}{100}\right)^n$,

when population increases

(Total population) $P = P_0\left(1-\frac{r}{100}\right)^n$,

when population decreases.

Instalments : When a borrower paid the sum in parts (*i.e.*, not in a single amount) then we say that he/she is paying in instalments. For example A borrowed Rs. 100 from B and he pays back it to B in several parts *i.e.*, Rs. 20 in 5 times or Rs. 50 in 2 times etc. But the important point is that borrower has to also pay the interest for using the borrowed sum/or purchased article. In general the value of each instalment is kept constant even when the interest charged on each instalment vary for each instalment.

For Simple Interest

$$A = \left[x+\left(x+\frac{x\times r\times 1}{100}\right)+\left(x+\frac{x\times r\times 2}{100}\right)+\left(x+\frac{x\times r\times 3}{100}\right)+\ldots\right]$$

A = Total amount paid
x = Value of each instalment

Also, $\boxed{A = P+\frac{P\times n\times r}{100}}$

P is the principal
n is the number of instalments
r is the rate of interest

For Compound Interest

(loan amount)

$$P = \left[\frac{x}{\left(1+\frac{r}{100}\right)}+\frac{x}{\left(1+\frac{r}{100}\right)^2}+\frac{x}{\left(1+\frac{r}{100}\right)^3}+\ldots+\frac{x}{\left(1+\frac{r}{100}\right)^n}\right]$$

$x \to x$ is the value of each instalment.

Total amount paid in instalments $(A) = P\left(1+\frac{r}{100}\right)^n$

n is the number of instalments.

Difference between CI and SI for n^{th} year

$$= \frac{Pr}{100}\left[\left(1+\frac{r}{100}\right)^{n-1}-1\right]$$

For Compound Interest

$$\frac{\text{increase in amount in } n^{th} \text{ year}}{\text{increase in amount in } (n+1)^{th} \text{ year}} = \frac{100}{(100+r)}$$

Similarly,

$$\frac{\text{decrease in amount in } n^{th} \text{ year}}{\text{decrease in amount in } (n+1)^{th} \text{ year}} = \frac{100}{(100-r)}$$

where r is rate of interest.

EXAMPLE 1. *Find the simple interest on Rs. 1000 at 12% per annum for 5 years.*

SOLUTION $SI = \frac{Prt}{100} = \frac{1000\times 12\times 5}{100} = \text{Rs. } 600$

{Total amount = P + SI = 1000 + 600 = Rs. 1600}

EXAMPLE 2. *Find the simple interest on Rs. 800 at 7% per annum and on Rs. 700 at 16% p.a. and on Rs. 500 at 4% p.a. for 2 years.*

SOLUTION $SI = \frac{P_1 r_1 t_1}{100}+\frac{P_2 r_2 t_2}{100}+\frac{P_3 r_3 t_3}{100}$

$= \frac{800\times 7\times 2}{100}+\frac{700\times 16\times 2}{100}+\frac{500\times 4\times 2}{100}$

$= 112 + 224 + 40$

$= \text{Rs. } 376$

EXAMPLE 3. *A sum of money (P) doubles in 10 years. In how many years it will be treble at the same rate of simple interest?*

SOLUTION $A = 2P$

$\therefore$ $SI = P$ $\quad (SI = 2P - P)$

$P = \frac{P\times r\times 10}{100}$

$\Rightarrow$ $r = 10\%$

So, the new amount = $3P$

But the new SI = $2P = (3P - P)$

$2P = \frac{P\times 10\times t}{100}$ $\quad (r = 10\%)$

$t = 20$ years

EXAMPLE 4. *A sum of money in 3 years becomes 1344 and in 7 years it becomes Rs. 1536. What is the principal sum where simple rate of interest is to be charged ?*

(a) 4000 (b) 1500 (c) 1200 (d) 2800

SOLUTION It would be very time saving if we do it by unitary method.

$1536 - 1344 = \text{Rs. } 192$

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Rs. 192 is the interest for 4 years

So, Rs. 48 is the interest for 1 year

It means Rs. 144 is the interest for 3 years at actual principal.

$\therefore$ Principal = 1344 – 144 = Rs. 1200

EXAMPLE 5. *A certain sum of money amounts to Rs. 15900 at simple rate of interest at 6% p.a. in 1 year. What is the value of principal sum ?*

(a) 12000 *(b) 18000*
(c) 15000 *(d) 14000*

SOLUTION Best way is to go through option as $\frac{15000 \times 6 \times 1}{100} = 900$

So the total amount = 15900, hence (c) is correct.

EXAMPLE 6. *A sum of money becomes 3 times in 12 years. In how many years it will become 5 times at the same rate of simple interest?*

(a) 20 years *(b) 16 years*
(c) 24 years *(d) 30 years*

SOLUTION In this type of question consider only SI not the amount. So,

(in first case) SI = 2P for principal P.

(in second case) SI = 4P for principal P.

Now since SI is directly proportional to the time at fixed rate of interest.

Therefore to make SI two times it will require 12 years

So to make SI four times it will require 24 years

Thus to make an amount 5 times of the principal requires 24 years at the given rate of interest as per question.

EXAMPLE 7. *What will be the amount when Rs. 10000 is deposited in a bank at 10% per annum compounded annually for 3 years?*

SOLUTION $A = P\left(1 + \frac{r}{100}\right)^n = 10000\left(1 + \frac{10}{100}\right)^3$

$= 10000 \times (1.1)^3$

$=$ Rs. 13310

EXAMPLE 8. *A sum of amount at r% compound interest doubles in 3 years. In 9 years it will be k times of the original principal. What is the value of k?*

(a) 10 *(b) 9* *(c) 6* *(d) 8*

SOLUTION $A = P\left(1 + \frac{r}{100}\right)^n$

1st case $2P = P\left(1 + \frac{r}{100}\right)^3$

$\Rightarrow$ $2 = \left(1 + \frac{r}{100}\right)^3$

2nd case $(2)^3 = \left[\left(1 + \frac{r}{100}\right)^3\right]^3 = \left(1 + \frac{r}{100}\right)^9$

$\Rightarrow$ $k = (2)^3 = 8$ times.

EXAMPLE 9. *Rs. 20000 is being compounded at 20% p.a. If the rate of interest is charged half yearly. What will be the amount after 2 years ?*

(a) 28292 *(b) 27292*
(c) 29282 *(d) 22358*

SOLUTION $20000\left(1 + \frac{20/2}{100}\right)^{2\times 2} = 20000\left(1 + \frac{10}{100}\right)^4$

$= 20000\,(1.1)^4$

$= 20000 \times 1.4641$

$=$ Rs. 29282

EXAMPLE 10. *If the rate of interest is 10% p.a. and Rs. 12000 lent at the compound interest, half yearly. What is the equivalent simple rate of interest for first year?*

SOLUTION $A = P\left(1 + \frac{r}{100}\right)^n = 12000\,(1.05)^2$

$= 12000 \times 1.1025$

$=$ Rs. 13230

Again SI = 13230 – 12000 = 1230

$\therefore$ $1230 = \frac{12000 \times r \times 2}{2 \times 100}$

$\Rightarrow$ $r = 10.25\%$

EXAMPLE 11. *Rs. 1000 is being charged at 50% per annum. What is the interest for 3rd year at compound interest ?*

(a) 1122 *(b) 1025* *(c) 1125* *(d) 625*

SOLUTION Total amount for two years = $1000\,(1.5)^2 = 2250$

Now, interest for third year = $\frac{2250 \times 50 \times 1}{100} = 1125$

Hence (c) is correct.

EXAMPLE 12. *CI and SI for a certain sum at certain rate of interest are Rs. 220 and Rs. 200. What is the principal for these interests?*

(a) Rs. 2200 *(b) Rs. 200*
(c) Rs. 500 *(d) Rs. 2000*

SOLUTION 220 – 200 = 20

Rs. 20 is the interest on the interest of the first year, which is Rs. $100 = \left(\frac{200}{2}\right)$

Thus the rate of interest = 20% $\left(20 = \frac{100 \times r \times 1}{100}\right)$

So the actual principal = P = 500

$\left(\text{since } 100 = \frac{P \times 20 \times 1}{100} \Rightarrow P = 500\right)$

Hence (c) is correct.

EXAMPLE 13. *The population of UP increases very rapidly @ 12% p.a. and 3 years ago its population was 1 million. What is present population?*

SOLUTION $A = 1000000\,(1.12)^3 = 1404928$

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EXAMPLE 14. *Jadeja purchased a maruti car 3 years ago for Rs. 2 lakh. Its value depreciated each year @ 25% p.a. What is the present value of the car ?*

SOLUTION $200000\left(1-\frac{25}{100}\right)^3 = 200000\times\left(\frac{3}{4}\right)^3$

$= 84{,}375$

EXAMPLE 15. *The difference between CI and SI for 3 years @ 20% p.a. is Rs. 152. What is the principal lent in each case ?*

(a) 1200 (b) 1155
(c) 1187.5 (d) none of these

SOLUTION Difference between CI and SI for 3 years = Rs. 152

$$P\left(\frac{r}{100}\right)^2\left(\frac{r}{100}+3\right)=152$$

$$P\left(\frac{1}{25}\right)\left(\frac{16}{5}\right)=152$$

$$P=\frac{152\times25\times5}{16}$$

$$P = 9.5\times25\times5 = 1187.5$$

EXAMPLE 16. *A sum is being lent at 20% p.a. compound interest. What is the ratio of increase in amount of 4th year to 5th year?*

(a) 4 : 5 (b) 5 : 4
(c) 5 : 6 (d) can't be determined

SOLUTION $$\frac{P\left(1+\frac{r}{100}\right)^4}{P\left(1+\frac{r}{100}\right)^5}=\frac{1}{\left(1+\frac{r}{100}\right)}$$

$$=\frac{100}{100+r}=\frac{100}{120}=\frac{5}{6}$$

EXAMPLE 17. *Rs. 12000 amounts to Rs. 20736 in 3 years at r% p.a. of compound interest. What is the value of r?*

(a) 10% (b) 25% (c) 12% (d) 20%

SOLUTION $$A = P\left(1+\frac{r}{100}\right)^3$$

$$20736 = 12000\left(1+\frac{r}{100}\right)^3$$

$$\Rightarrow \quad \frac{20736}{12000}=\left(1+\frac{r}{100}\right)^3$$

$$\Rightarrow \quad \frac{1728}{1000}=\left(1+\frac{r}{100}\right)^3$$

$$\Rightarrow \quad \left(\frac{12}{10}\right)^3=\left(1+\frac{r}{100}\right)^3$$

$$\Rightarrow \quad \left(1+\frac{2}{10}\right)^3=\left(1+\frac{r}{100}\right)^3$$

$$\Rightarrow \quad r = 20\%$$

Alternatively : The best way for this problem is to go through options.

$12000\times1.2 \Rightarrow 14400\times1.2 \Rightarrow 17280\times1.2 \Rightarrow 20736$

EXAMPLE 18. *A certain sum amounts to Rs. 14641 in 4 years @ 10% p.a. compounded annualy. What is the value of principal?*

(a) Rs. 6000 (b) Rs. 12000
(c) Rs. 10000 (d) data insufficient

SOLUTION $$14641 = P\left(1+\frac{10}{100}\right)^4$$

$$14641 = P\left(\frac{11}{10}\right)^4$$

$$P = 14641\times\left(\frac{10}{11}\right)^4$$

$$= 10000$$

EXAMPLE 19. *A sum of Rs. 10000 is borrowed at 8% p.a. compounded annualy which is paid back in 3 equal annual instalments. What is the amount of each instalments?*

SOLUTION $$10000 = x\left[\frac{25}{27}+\left(\frac{25}{27}\right)^2+\left(\frac{25}{27}\right)^3\right]$$

$$= x\times\frac{25}{27}\left[1+\frac{25}{27}+\frac{625}{729}\right]$$

$$=\frac{25x}{27}\left[\frac{2029}{729}\right]$$

$$\Rightarrow \quad x = \text{Rs. } 3880.335$$

Alternatively :

$$10000\,(1.08)^3 = x\,[1+(1.08)+(1.08)^2]$$

$$\Rightarrow \quad x = 3880.335$$

EXAMPLE 20. *A scooty is sold by an automobile agency for Rs. 19200 cash or for Rs. 4800 cash down payment together with five equal monthly instalments. If the rate of interest charged by the company is 12% per annum find each instalment.*

SOLUTION Balance of the price to be paid through instalments

= Rs. 14400

Rate of interest (r) = 12% p.a.

$$\therefore \left(14400+\frac{14400\times12\times5}{100\times12}\right)=\left[x+\left(x+\frac{12x}{1200}\right)+\left(x+\frac{12x\times2}{12\times100}\right)+\ldots+\left(x+\frac{12x\times4}{12\times100}\right)\right]$$

$\Rightarrow x = \text{Rs. } 2964.70$, where x is the value of each instalment.

NOTE In the left hand side and right hand side given amounts are equal. Each amount is equal to the total amount payable after 5 months.

Important : For quick and smooth calculation to the problems of this chapter you can remember the following values :

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Rate of Interest	For Rs. 100, amount				
	after one year	after 2 years	after 3 years	after 4 years	after 5 years
5%	105	110.25	115.7625	121.550625	127.62815625
6%	106	112.36	119.1016	126.247696	133.82255776
8%	108	116.64	125.9712	136.048896	146.93280768
10%	110	121.00	133.1000	146.4100	161.051
12%	112	125.44	140.4928	157.351936	176.23416832
15%	115	132.25	152.0875	174.900625	201.13571875
20%	120	144.00	172.80	207.36	248.832
25%	125	156.25	195.3125	244.140625	305.17578125
30%	130	169.00	219.70	285.61	371.293
50%	150	225.00	337.50	506.25	759.375

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EXERCISE

LEVEL 1

1. A man borrows Rs. 4000 and pays back after 5 years at 15% simple interest. The amount paid by the man is :
   (a) Rs. 1800 (b) Rs. 4800
   (c) Rs. 7500 (d) Rs. 7000

2. What is the time period for which Rs. 8000 amounts to Rs. 12000 at 20% p.a. of simple interest?
   (a) 4 years (b) 2.5 years
   (c) 3.25 years (d) 6 years

3. What is the rate of simple interest at which Rs. 14,000 gives interest of Rs. 1960 in two years?
   (a) 4% (b) 5%
   (c) 7% (d) 10%

4. What is the sum of amount which gives Rs. 6300 as interest @ 7% per annum of simple interest in $7\frac{1}{2}$ years?
   (a) 36000 (b) 24000
   (c) 63000 (d) 12000

5. If the rate of simple interest is 12% per annum, the amount that would fetch interest of Rs. 6000 per annum is :
   (a) Rs. 7200 (b) Rs. 48000
   (c) Rs. 50000 (d) Rs. 72000

6. A sum was put at simple interest at a certain rate for 2 years. Had it been put at 4% higher rate, it would have fetched Rs. 112 more. The sum is :
   (a) 1120 (b) 1400
   (c) 1200 (d) 8000

7. In what time will a sum of money double itself @ 20% per annum (p.a.) simple interest?
   (a) 10 years (b) 5 years
   (c) 2 years (d) 4 years

8. At r% per annum a sum doubles after 20 years. The rate of interest per annum is :
   (a) 4% (b) 5%
   (c) 8% (d) 10%

9. A sum of money trebles (*i.e.*, 3 times) in 15 years at r% of simple interest per annum. What is the value of r?
   (a) 12% (b) $\frac{40}{3}$%
   (c) $\frac{50}{3}$% (d) can't be determined

10. A sum of money doubles itself in 12 years. In how many years would it treble itself?
    (a) 36 years (b) 18 years
    (c) 24 years (d) 15 years

11. Out of a sum of Rs. 625, a part was lent at 5% SI and the other at 10% SI. If the interest on the first part after 2 years is equal to the interest on the second part after 4 years, then the second sum (in Rs.) is :
    (a) 250 (b) 300
    (c) 125 (d) 275

12. A sum of Rs. 2500 is lent out in two parts; one at 12% p.a. and another at 12.5% p.a. for one year. If the total annual income is Rs. 306, the money lent at 12% is :
    (a) 1000 (b) 1200
    (c) 1500 (d) 1300

13. Akul lent Rs. 6000 to Bakul for 2 years and Rs. 1500 to camlin for 4 years and received altogether from both Rs. 900 as simple interest. The rate of interest is :
    (a) 4% (b) 8%
    (c) 10% (d) 5%

14. A person takes a loan of Rs. 200 at 5% simple interest. He returns Rs. 100 at the end of one year. In order to clear his dues at the end of 2 years, he would pay :
    (a) 125.50 (b) 110
    (c) 115.50 (d) none of these

15. A lends a sum of money for 10 years at 5% SI. B lends double the amount for 5 years at the same rate of interest. Which of the following statements is true in this regard?
    (a) A will get twice the amount of interest that B would get
    (b) B will get twice the amount of interest that A would get
    (c) A will get the same amount of interest that B would get
    (d) B will get four times the amount of interest that A would get

16. Consider the following statements.
    If a money is loaned at simple interest then the :
    (i) money gets doubled in 5 years if the rate of interest is $16\frac{2}{3}$%.
    (ii) money gets doubled in 5 years if the rate of interest is 20%.
    (iii) money becomes four times in 10 years if it gets doubled in 5 years.
    of these statements :
    (a) (i) and (iii) are correct (b) (iii) alone is correct
    (c) (ii) alone is correct (d) (ii) and (iii) are correct

17. Pratibha invests an amount of Rs. 15,860 in the names of her three daughters A, B and C in such a way that they get the same interest after 2, 3 and 4 years respectively. If the rate of simple interest is 5% p.a., then the ratio of the amounts invested among A, B and C will be :
    (a) 5 : 10 : 12 (b) $\frac{1}{10} : \frac{1}{15} : \frac{1}{20}$
    (c) 6 : 7 : 8 (d) 6 : 5 : 4

18. What annual payment will discharge a debt of Rs. 580 in 5 years, the rate being 8% p.a.?
    (a) 120 (b) 100
    (c) 80 (d) 78

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19. Find the amount of Rs. 2500 invested at 12% during the period from 4th February, 2005 to 18 April 2005 :
(a) Rs. 3000 (b) Rs. 3800
(c) Rs. 2560 (d) Rs. 2600

20. Find the amount of Rs. 1700 invested at 16% half yearly at simple interest for one year :
(a) 2100 (b) 2244
(c) 2200 (d) 2500

21. The compound interest on Rs. 1000 at 10% p.a. in 3 years is :
(a) 331 (b) 1331
(c) 133 (d) 300

22. The compound interest on Rs. 10000 at 20% p.a. in 4 years :
(a) 10736 (b) 736
(c) 20736 (d) 7280

23. The compound interest on Rs. 4000 at 25% p.a. in 3 years :
(a) 1235 (b) 5625
(c) 3812.5 (d) 3750.5

24. The compound interest on Rs. 5000 at 30% per annum for 4 years :
(a) 4280.5 (b) 6700
(c) 9280.5 (d) 3857.5

25. A sum of Rs. 400 would become Rs. 441 after 2 years at r% compound interest, find the value of 'r' :
(a) 10% (b) 5%
(c) 15% (d) 20%

26. At compound interest, if a certain sum of money doubles in n years, then the amount will be four fold in :
(a) n^2 years (b) $2n^2$ years
(c) $2n$ years (d) $4n$ years

27. Rs. 6000 amounts to Rs. 7986 in 3 years at CI. The rate of interest is :
(a) 20% (b) 10%
(c) 6% (d) 7.5%

28. The least number of complete years in which a sum of money put at 20% CI will be more than doubled is :
(a) 4 (b) 5
(c) 6 (d) 8

29. The CI on Rs. 5000 for 3 years at 8% for first year, 10% for second year and 12% for third year will be :
(a) Rs. 1750 (b) Rs. 1652.80
(c) Rs. 1575 (d) Rs. 1685.20

30. A sum of Rs. 2400 deposited at CI, doubled after 5 years. After 20 years it will become :
(a) Rs. 24000 (b) Rs. 38400
(c) Rs. 19200 (d) can't be determine

31. A sum of Rs. 550 was taken as a loan. This is to be paid back in two equal annual instalments. If the rate of interest be 20% compounded annually, then the value of each instalments is :
(a) 300 (b) 360
(c) 250 (d) none of these

32. The difference between CI and SI on a sum of money lent for 2 years at 10% is Rs. 40. The sum is :
(a) 1600 (b)
(c) 4000 (d) none of these

33. The difference between simple and compound interest on Rs. 6000 for 1 year at 20% per annum reckoned half yearly is :
(a) 120 (b) 60
(c) 180 (d) 72

34. A certain sum amounts to Rs. 8988.8 in two years and to Rs. 9528.128 in three years, at compound interest per annum. What is the principal and rate of interest?
(a) Rs. 12,000, 5% (b) Rs. 6,000, 8%
(c) Rs. 8,000, 6% (d) Rs. 10,000, 8.5%

35. The compound interest and the simple interest for two years on a certain sum of money at a certain rate of interest are Rs. 2257.58, Rs. 2100 respectively. Find the principal and rate percent :
(a) 6000, 7% (b) 7500, 8%
(c) 14000, 10% (d) 7000, 15%

36. The compound interest on a certain sum at a certain rate of interest for the second year and third year is Rs. 21780 and Rs. 23958 respectively. What is the rate of interest?
(a) 6% (b) 12%
(c) 10% (d) 15%

37. Amit borrowed Rs. 800 at 10% rate of interest. He repaid Rs. 400 at the end of first year. What is the amount required to pay at the end of second year to discharge his loan which was calculated at compound interest?
(a) 420 (b) 440
(c) 450 (d) 528

38. A sonata watch is sold for Rs. 440 cash or for Rs. 200 cash down payment together with Rs. 244 to be paid after one month. Find the rate of interest charged in the instalment scheme :
(a) 10% (b) 15%
(c) 20% (d) 25%

39. A cellphone is available for Rs. 600 or for 300 cash down payment together with Rs. 360 to be paid after two months. Find the rate of interest charged under this scheme :
(a) 20% (b) 50%
(c) 120% (d) none of these

40. Abhinav purchases a track suit for Rs. 2400 cash or for Rs. 1000 cash down payments and two monthly instalments of Rs. 800 each. Find the rate of interest :
(a) 75% (b) 120%
(c) 50% (d) none of these

41. Indicom cell-phone is available for Rs. 2500 cash or Rs. 520 cash down payments followed by 4 equal monthly instalments. If the rate of interest charged is 25% per annum, calculate the monthly instalment :
(a) 520 (b) 480
(c) 550 (d) none of these

42. An article is sold for Rs. 500 cash or for Rs. 150 cash down payments followed by 5 equal monthly instalments. If the rate of interest charged is 18% p.a., compute the monthly instalment :
(a) 63.07% (b) 37.06%
(c) 75.0% (d) 73.06%

43. A sum of Rs. 390200 is to be paid back in three equal annual instalments. How much is each instalment, if the rate of interest charged is 4% per annum compounded annually?
(a) Rs. 140608 (b) Rs. 120560
(c) Rs. 10000 (d) Rs. 18000

44. Purnima borrowed a sum of money and returned it in three equal quarterly instalments of Rs. 17576 each. Find the sum borrowed, if the rate of interest charged was 16% per annum compounded quarterly. Find also the total interest charged :
(a) 46900 and 4700 (b) 48775 and 3953
(c) 68320 and 1200 (d) none of these

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45. Sunidhi borrowed Rs. 10815, which is to be paid back in 3 equal half yearly instalments. If the interest is compounded half yearly at $\frac{40}{3}$% per annum, how much is each instalment?
(a) 2048 (b) 3150
(c) 4096 (d) 5052

46. Sapna borrowed some money on compound interest and returned it in three years in equal annual instalments. If the rate of interest is 15% per annum and annual instalment is Rs. 456680, find the sum borrowed :
(a) 1112220 (b) 1111200
(c) 1122000 (d) none of these

47. P and Q invest some amount under SI and CI repectively but for the same period at 6% per annum. Each gets a total amount of Rs. 65,000 at the end of 6 years. Which of the following is definitely true?
(i) Q's initial principal is less than that of P
(ii) Q's initial principal is equal to that of P
(iii) P's percentage earning is less than that of Q
(a) (i) only (b) (ii) only
(c) (iii) only (d) (i) and (iii) only

48. In the above (i.e., previous) problem, what is the ratio of P's final amount to that of Q, if P and Q invest the same amounts?
(a) $(1.06)^6$ (b) $\frac{136}{100} \times \left(\frac{100}{106}\right)^6$
(c) $\frac{100}{136} \times \left(\frac{106}{100}\right)^6$ (d) none of these

49. Mr. Lala Ram has lent some money to Aaju at 6% p.a. and the Baaju at 8% p.a. At the end of the year he has gain the overall interest at 7% per annum. In what ratio has he lent the money to Aaju and Baaju?
(a) 2 : 3 (b) 1 : 1
(c) 5 : 6 (d) 4 : 3

50. The difference between simple and compound interest for the fourth year is Rs. 7280 at 20% p.a. What is the principal sum?
(a) 10000 (b) 50000
(c) 1 lakh (d) 40000

51. The simple interest on certain sum at 5% for 9 month is Rs. 10 greater than the simple interest on the same sum @ 3% for 14 months. What is the sum of interest in both the cases (i.e., total sum of interest)?
(a) Rs. 130 (b) Rs. 290
(c) Rs. 120 (d) Rs. 330

52. Mr. Bajaj invested $\frac{1}{7}$ of his total investment at 4% and $\frac{1}{2}$ at 5% and rest at 6% for the one year and received total interest of Rs. 730. What is the total sum invested?
(a) Rs. 70000 (b) Rs. 14000
(c) Rs. 24000 (d) Rs. 38000

53. The rate of interest in two banks DNB and HBI are in the ratio of 7 : 8. If a person invested some amount in both the banks and received equal amounts from both the banks in two year. The ratio of amount invested in DNB and HBI respectively is :
(a) 15 : 1 (b) 8 : 7
(c) 7 : 8 (d) 108 : 107

54. The ratio of CI for 3 years and SI for 1 year for a fixed amount at a rate of r% is 3.64. What is the value of r?
(a) 10% (b) 15%
(c) 20% (d) none of these

55. A sum of money becomes 13/5 times of itself in 32 years at r% of SI. What is the value of r?
(a) 6% (b) 7%
(c) 5% (d) 18%

56. The difference between interest received by A and B is Rs. 18 on Rs. 1500 for 3 year. What is the difference in rate of interest?
(a) 1% (b) 2.5%
(c) 0.5% (d) 0.4%

LEVEL 2

1. The compound interest on a certain sum for 2 years is Rs. 756 and SI (simple interest) is Rs. 720. If the sum is invested such that the SI is Rs. 900 and the number of years is equal to the rate per cent per annum, find the rate per cent :
(a) 4 (b) 5
(c) 6 (d) 1.0

2. Jalela and Dalela have to clear their respective loans by paying 3 equal annual instalments of Rs. 30000 each. Jalela pays @ 10% per annum of simple interest while Dalela pays 10% per annum compound interest. What is the difference in their payments?
(a) Rs. 300 (b) Rs. 425
(c) Rs. 245 (d) Rs. 333.33

3. Akbar lends twice the interest received from Birbal to Chanakya at the half of the interest at which he lent to Birbal. If Akbar lent Rs. P @ r% per annum for 1 year to Birbal then the interest received by Akbar from Chanakya is :
(a) $\frac{Pr^2}{100}$ (b) $\left(\frac{Pr}{10}\right)^2$
(c) $P\left(\frac{r}{10}\right)^2$ (d) $P\left(\frac{r}{100}\right)^2$

4. Equal amounts of each Rs. 43892 is lend to two persons for 3 years. One @ 30% SI and second @ 30% CI annually. By how much per cent the CI is greater than the simple interest received in this 3 years duration?
(a) 23% (b) 33%
(c) 33.33% (d) none of these

5. Rs. 3500 was lent partly @ 4% and partly @ 6% SI. The total interest received after 3 years is 498. What is the amount lent @ 4% SI?
(a) Rs. 1300 (b) Rs. 1800
(c) Rs. 200 (d) Rs. 2200

6. The population of vultures in a particular locality is decreases by a certain rate of interest (compounded annually). If the current population of vultures be 29160 and the ratio of decrease in population for second year and 3rd year be 10 : 9. What was the population of vultures 3 years ago?
(a) 30000 (b) 35000
(c) 40000 (d) 50000

7. The ratio of the amount for two years under CI annually and for one year under SI is 6 : 5. When the rate of interest is same, then the value of rate of interest is :

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(a) 12.5% (b) 18%
(c) 20% (d) 16.66%

8. A bicycle can be purchased on cash payment of Rs. 1500. The same bicycle can also be purchased at the down payment (initial payment, at the time of purchasing) of Rs. 350 and rest can be paid in 3 equal installments of Rs. 400 for next 3 months. The rate of SI per annum charged by the dealer is :
(a) $23\frac{9}{17}\%$ (b) $17\frac{9}{23}\%$
(c) $13\frac{9}{17}\%$ (d) none of these

9. Data Ram lends equal sum of money at the same rate of interest to *A* and *B*. The money lends to *A* becomes twice of the original amount in just four years at simple interest. While Data Ram lends to *B* for the first two years at compound interest and for the rest two years at simple interest. If the difference between the amount of *A* and *B* after 4 years is Rs. 2750. What is the amount of money that Data Ram lends to each one?
(a) Rs. 40000 (b) Rs. 6000
(c) Rs. 8000 (d) Rs. 80000

10. Akram Ali left an amount of Rs. 340000 to be divided between his two sons aged 10 years and 12 years such that both of them would get an equal amount when each attain 18 years age. What is the share of elder brother if the whole amount was invested at 10% simple interest :
(a) 12000 (b) 16000
(c) 160000 (d) 180000

11. Satyam took loan from IDIDI Bank for his 2 years course of MBA at IMD. He took the loan of Rs. 6 lakh such that he would be charged at 8% per annum at CI during his course and at 10% CI after the completion of course. He returned half of the amount which he had to be paid on the completion of his studies and remaining after 2 years. What is the total amount returned by Satyam ?
(a) Rs. 7.73323 lakh (b) Rs. 7.58 lakh
(c) Rs. 7.336 lakh (d) none of these

12. We had 1000 goats at the beginning of year 2001 and the no. of goats each year increases by 10% by giving birth (compounded annually). At the end of each year we double the no. of goats by purchasing the same no. of goats as there is the no. of goats with us at the time. What is the no. of goats at the beginning of 2004?
(a) 10600 (b) 10648
(c) 8848 (d) 8226

13. Rs. 100000 was invested by Mohan in a fixed deposit @ 10% per annum at CI. However every year he has to pay 20% tax on the compound interest. How much money does Mohan has after 3 year?
(a) 128414 (b) 108000
(c) 126079.2 (d) none of these

14. A property dealer bought a rectangular plot (of land) in Noida 5 years ago at the rate of Rs. 1000 per m^2. The cost of plot is increases by 5% in every 6 years and the worth of a rupee falls down at a rate of 2% in every 5 years. What is the approximate value of the land per meter^2 25 years hence?
(a) Rs. 995 (b) Rs. 1134
(c) Rs. 1500 (d) Rs. 1495

15. *A* and *B* run a joint venture in which the profit earned by *A* and *B* are in the ratio 28 : 15. *A* invest his share at the start of the year and *B* joins in after 9 months of the same year. What is the ratio of their initial investment respectively?
(a) 7 : 15 (b) 8 : 13
(c) 5 : 17 (d) 15 : 7

16. In the previous problem if *A* gets a profit of Rs. 4200 the amount invested by *B* is :
(a) 2250 (b) 2600
(c) 1350 (d) can't be determined

17. Arvind and Govind each invested Rs. 15000 for 3 years at the same rate of interest but Arvind's investment is compounded annually while Govind's investment is charged on simple interest. What amount did Arvind receive more than Govind?
(a) Rs. 680 (b) Rs. 3450
(c) data insufficient (d) none of these

18. Shyam Lal takes a loan of Rs. 10500 at 10% p.a. compounded annually which is to be repaid in two equal annual instalments. One at the end of one year and the other at the end of the second year. The value of each instalments is :
(a) 5987 (b) 6050
(c) 6352.5 (d) 5678.5

19. Hari Lal and Hari Prasad have equal amounts. Hari Lal invested all his amount at 10% compounded annually for 2 years and Hari Prasad invested 1/4 at 10% compound interest (annually) and rest at *r*% per annum at simple interest for the same 2 years period. The amount received by both at the end of 2 year is same. What is the value of *r*?
(a) 14% (b) 12.5%
(c) 10.5% (d) 11%

20. The annual sales of a company in the year 2000 was Rs. 1000 and in the year 2005 was Rs. 2490. Find the compounded annual growth rate (CAGR) of sales in the given period of the same company :
(a) 14.289% (b) 10%
(c) 15% (d) 20%

21. HDFC lends 1 million to HUDCO at 10% simple interest p.a. for 2 years and HUDCO lends the same amount to SAHARA STATES HOUSING corporation at 10% p.a. of compound interest for 2 years. What is the earning of HUDCO in this way?
(a) Rs. 133100 (b) Rs. 33100
(c) Rs. 131000 (d) no profit no loss

22. ICICI lent Rs. 1 lakh to captain Ram Singh @ 6% per annum of simple interest for 10 years period. Meanwhile ICICI offered a discount in rate of interest for armed forces. Thus the rate of interest ICICI decreased to 4%. In this way Ram Singh had to pay total amount 1.48 lakh.

After how many year Ram Singh got the discount in rate of interest?
(a) 3 years (b) 4 years
(c) 6 years (d) 5 years

23. Sanjay purchased a hotel worth Rs. 10 lakh and barkha purchased a car worth Rs. 16 lakh. The value of hotel every year increases by 20% of the previous value and the value of car every year depreciates by 25%. What is the difference between the price of hotel and car after 3 years?
(a) Rs. 925000 (b) Rs. 10,53,000
(c) remains constant (d) can't be determined

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LEVEL-1

1 (d)	2. (b)	3. (c)	4. (d)	5. (c)	6. (b)	7. (b)	8. (b)	9. (b)	10. (c)
1. (c)	12. (d)	13. (d)	14. (c)	15. (c)	16. (c)	17. (b)	18. (b)	19. (c)	20. (b)
1. (a)	22. (a)	23. (c)	24. (c)	25. (b)	26. (c)	27. (b)	28. (a)	29. (b)	30. (b)
1. (b)	32. (c)	33. (b)	34. (c)	35. (d)	36. (c)	37. (d)	38. (c)	39. (c)	40. (b)
1. (a)	42. (d)	43. (a)	44. (b)	45. (c)	46. (b)	47. (d)	48. (b)	49. (b)	50. (b)
1. (b)	52. (b)	53. (b)	54. (c)	55. (c)	56. (d)				

LEVEL-2

1 (b)	2. (a)	3. (d)	4. (b)	5. (d)	6. (c)	7. (c)	8. (d)	9. (c)	10. (d)
11. (a)	12. (b)	13. (d)	14. (b)	15. (a)	16. (d)	17. (c)	18. (b)	19. (c)	20. (a)
21. (c)	22. (b)	23. (b)							

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Hints & Solutions

LEVEL 1

1. $4000 + (4000 \times 5 \times 0.15) = \text{Rs. } 7000$

2. $(12000 - 8000) = \dfrac{8000 \times t \times 20}{100}$

$\Rightarrow \quad t = \dfrac{5}{2}$ years

3. $1960 = \dfrac{14000 \times r \times 2}{100} \Rightarrow r = 7\%$

4. $6300 = \dfrac{P \times 7 \times 15}{100 \times 2}$

$\Rightarrow \quad P = \text{Rs. } 12000$

5. $6000 = \dfrac{P \times 12 \times 1}{100}$

$\Rightarrow \quad P = 50000$

6. In my opinion this question should be solved by unitary method instead of making complex solution

Years	Rate of Interest	Interest
2	4%	112
2	1%	28
1	1%	(14)

It means the principal sum is Rs. 1400

Alternatively: $\dfrac{P}{100} \times 2[(r+4) - r] = 112$

$\Rightarrow \quad P = 1400$

7. $SI = 2P - P = P$ (Interest = Amount – Principal)

$P = \dfrac{P \times 20 \times t}{100} \Rightarrow t = 5$ years

8. $P = \dfrac{P \times r \times 20}{100}$ (Interest = Amount – Principal)

$\Rightarrow \quad r = 5\%$ p.a.

9. $2P = \dfrac{P \times 15 \times r}{100}$ $(2P = 3P - P)$

$r = \dfrac{40}{3}\%$ p.a.

10. $P = \dfrac{P \times r \times 12}{100} \Rightarrow r = \dfrac{100}{12}\%$ p.a.

Now, $2P = \dfrac{P \times 100 \times t}{12 \times 100} = 24$ years

Shortcut (for SI) Since SI in second case is double the SI in first case, so the time period will also be double since SI is directly proportional to the time period, provided that rate of interest be same and principal as well.

11. $\dfrac{P_1 \times 5 \times 2}{100} = \dfrac{P_2 \times 10 \times 4}{100}$

$\Rightarrow \quad P_1 : P_2 = 4 : 1$

Therefore second principal is Rs. $125 \left(= 625 \times \dfrac{1}{5}\right)$

Alternative: Go through options.

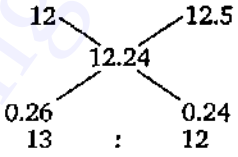
12. Best way is to go through option

So, $1300 \times 0.12 + 1200 \times 0.125 = 306$

Hence, (d) is the correct option.

Alternatively: Solve through alligation

Therefore the ratio of first principal to the second principal is 13 : 12.



$306 = \dfrac{2500 \times r \times 1}{100}$

$r = \dfrac{306}{25} = 12.24$

13. $\left(\dfrac{6000 \times 2 + 1500 \times 4}{100}\right) r = 900$

$\Rightarrow \quad r = 5\%$

14. Amount to be paid in first year $= \dfrac{200 \times 5 \times 1}{100} + 200 = 210$

Amount left as a principal for the second year

$= 210 - 100 = 110$

$\therefore$ Amount paid in second year $= 110 + \dfrac{110 \times 5 \times 1}{100} = 115.5$

15.

A	B
$P_A = P$	$P_B = 2P$
$r_A = 5\%$	$r_B = 5\%$
$t = 10$ years	$t = 5$ years
Interest of $A = \dfrac{P \times 5 \times 10}{100} = \dfrac{P}{2}$	Interest of $B = \dfrac{2P \times 5 \times 5}{100} = \dfrac{P}{2}$

Hence (c) is correct

16. **Statement (I)** $\dfrac{P \times 5 \times 50}{100 \times 3} \neq P$, hence wrong

Statement (II) $SI = \dfrac{P \times 5 \times 20}{100} = P$, hence correct

Statement (III) $P = \dfrac{P \times 5 \times r}{100} \Rightarrow r = 20\%$

Again $SI = \dfrac{P \times 10 \times 20}{100} = 2P$ hence wrong

17. $\dfrac{P_1 \times 2 \times 5}{100} = \dfrac{P_2 \times 3 \times 5}{100} = \dfrac{P_3 \times 4 \times 5}{100}$

$\Rightarrow \quad 10P_1 = 15P_2 = 20P_3$

$\Rightarrow \quad P_1 : P_2 : P_3 = 30 : 20 : 15 = \dfrac{1}{10} : \dfrac{1}{15} : \dfrac{1}{20}$

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18. $580 = \left[(x) + \left(x + \frac{x \times 1 \times 8}{100}\right) + \left(x + \frac{x \times 2 \times 8}{100}\right) + \left(x + \frac{x \times 3 \times 8}{100}\right) + \left(x + \frac{x \times 4 \times 8}{100}\right)\right]$

$$580 = 5x + \frac{8x}{10} = \frac{58x}{10}$$

$$\Rightarrow \quad x = 100$$

19. Time period = 24 (Feb) + 31 (March) + 18 (April)

$$= 73 \text{ days} = \frac{1}{5} \text{ year}$$

$$\therefore \quad 2500 + \frac{2500 \times 12 \times 1}{100 \times 5} = \text{Rs. } 2560$$

20. $1700 + \frac{1700 \times 16 \times 2}{100} = \text{Rs. } 2244$

21. $A = P\left(1 + \frac{r}{100}\right)^n$

$$A = 1000\left(1 + \frac{10}{100}\right)^3$$

$$A = 1000 \times (1.1)^3$$

$$A = 1331$$

$$\therefore \quad CI = A - P = 1331 - 1000 = 331$$

22. $A = 10000 \times (1.2)^4$

$$A = 10000 \times 2.0736$$

$$A = 20736$$

$$CI = A - P$$

$$CI = 20736 - 10000 = 10736$$

23. $CI = [4000 \times (1.25)^3] - (4000)$

$$= 4000 \times 1.953125 - 4000$$

$$= 4000 \times 0.953125$$

$$= 3812.5$$

24. $CI = 5000 \times (1.3)^4 - 5000$

$$= 5000 \times 2.8561 - 5000$$

$$= 5000\ (1.8561)$$

$$= 9280.5$$

25. $441 = 400\left(1 + \frac{r}{100}\right)^2$

$$\Rightarrow \quad \frac{21}{20} = 1 + \frac{r}{100}$$

$$\Rightarrow \quad r = 5\%$$

26. Since, $A = 2P$, then

$$2P = P\left(1 + \frac{r}{100}\right)^n$$

$$\Rightarrow \quad 2 = \left(1 + \frac{r}{100}\right)^n$$

$$\Rightarrow \quad 4 = (2)^2 = \left[\left(1 + \frac{r}{100}\right)^n\right]^2 = \left[\left(1 + \frac{r}{100}\right)^{2n}\right]$$

$$\Rightarrow \quad \text{time period} = 2n \qquad \{\because (a^m)^n = a^{m \cdot n}\}$$

(This question can be easily done by considering some appropriate values)

27. $7986 = 6000\left(1 + \frac{r}{100}\right)^3$

$$\frac{7986}{6000} = \left(1 + \frac{r}{100}\right)^3$$

$$\Rightarrow \quad \frac{1331}{1000} = \left(1 + \frac{r}{100}\right)^3$$

$$\Rightarrow \quad \frac{11}{10} = \left(1 + \frac{r}{100}\right)$$

$$\Rightarrow \quad r = 10\%$$

Alternatively: Go through options.

$$6000 \xrightarrow{+10\%} 6600 \xrightarrow{+10\%} 7260 \xrightarrow{+10\%} 7986$$

Hence, assumed option is correct.

28. Go through option

$$1 \times (1.2) = 1.2$$

$$1 \times (1.2)^2 = 1.44$$

$$1 \times (1.2)^3 = 1.728$$

$$1 \times (1.2)^4 = 2.0736$$

Hence minimum 4 years are required to double the sum.

29. $A = P\left(1 + \frac{R_1}{100}\right)\left(1 + \frac{R_2}{100}\right)\left(1 + \frac{R_3}{100}\right) \ldots$

$$A = 5000\ (1.08)\ (1.1)\ (1.12)$$

$$A = 6652.8$$

$$CI = 1652.8 = (6652.8 - 5000)$$

30. $2P = P\left(1 + \frac{r}{100}\right)^5$

$$\Rightarrow \quad 2 = \left(1 + \frac{r}{100}\right)^5$$

Now, $(2)^4 = \left(\left(1 + \frac{r}{100}\right)^5\right)^4 = \left(1 + \frac{r}{100}\right)^{20}$

$\Rightarrow$ The amount becomes 16 $(= 2^4)$ times

$\therefore$ Hence (b) is correct, $\because (2400 \times 16 = 38400)$

31. $550 = x\left[\left(\frac{1}{1.2}\right) + \left(\frac{1}{1.2}\right)^2\right]$

$$\Rightarrow \quad 550 = x\left[\frac{2.2}{1.44}\right]$$

$$\Rightarrow \quad x = \text{Rs. } 360$$

32. Difference between CI and SI for 2 years $= P\left(\frac{r}{100}\right)^2$

$$40 = P\left(\frac{10}{100}\right)^2 \quad \Rightarrow \quad P = 4000$$

Alternatively: Go through options

	First year	Second year	
SI	400	400	40
CI	400	440	

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Alternatively: Go back in the reverse process.
Rs. 40 at 10% implies that the principal for this interest was Rs. 400. Again by the same logic Rs. 400 as interest obtained at the principal of Rs. 4000 at 10%.

33. $6000\left(\frac{10}{100}\right)^2 = 60$

34. $$\frac{P\left(1+\frac{r}{100}\right)^3}{P\left(1+\frac{r}{100}\right)^2} = \left(1+\frac{r}{100}\right)$$

$$\therefore \quad \frac{9528.128}{8988.8} = \left(1+\frac{r}{100}\right)$$

$$\Rightarrow \quad \left(1+\frac{84270}{1404500}\right) = \left(1+\frac{r}{100}\right)$$

$$\Rightarrow \quad \left(1+\frac{6}{100}\right) = \left(1+\frac{r}{100}\right)$$

$$\Rightarrow \quad r = 6\%$$

So, $$8988.8 = P\left(1+\frac{6}{100}\right)^3$$

$$\Rightarrow \quad P = 8000$$

Alternatively: Best way is to go through options.

35. Difference between CI and SI for 2 years will be equal to the interest on SI for first year.

Hence, $6000 \times 0.07 \Rightarrow 420 \times 0.07 = 29.4 \neq 157.5$
and $7500 \times 0.08 \Rightarrow 600 \times 0.08 = 48 \neq 157.5$
and $14000 \times 0.1 \Rightarrow 1400 \times 0.1 = 140 \neq 157.5$
and $7000 \times 0.15 \Rightarrow 1050 \times 0.15 = 157.5 = 157.5$

Thus option (d) is correct.

36. $$\frac{P\left(1+\frac{r}{100}\right)^3}{P\left(1+\frac{r}{100}\right)^2} = \left(1+\frac{r}{100}\right)$$

$$\frac{23958}{21780} = \left(1+\frac{r}{100}\right)$$

$$1+\frac{2178}{21780} = 1+\frac{r}{100}$$

$$\boxed{r = 10\%}$$

Alternatively: Remember the difference between compound interest of any two consecutive years will be same as the interest on the amount of total previous years.

So, $23958 - 21780 = 2178$

Now $$r = \frac{2178}{21780} \times 100$$

$$r = 10\%$$

37. Amount to be paid at the end of 2 year

$$= \frac{800 \times 10 \times 2}{100} + 800 = 880$$

Amount left as principal for the second year

$$= 480 = (880 - 400)$$

Amount to be paid after 2nd year $= 480 + \frac{480 \times 10}{100}$

$$= \text{Rs. } 528$$

38. Principal for next month = 440 − 200 = 240

Amount paid after next month = 244

Therefore interest charged at Rs. 240 = 4

$$\therefore \quad 4 = \frac{240 \times r \times 1}{12 \times 100}$$

$$r = 20\% \text{ per annum}$$

39. Amount as a principal for first and second month = 600 − 300 = Rs. 300

Now, interest = 360 − 300 = Rs. 60

$$\therefore \quad 60 = \frac{300}{100} \times \frac{2}{12} \times r$$

$$r = 120\%$$

40. Amount as a principal for 2 month = 2400 − 1000 = 1400
At the rate of r% per annum after 2 months, Rs. 1400 will amount to

$$\text{Rs.}\left(1400 + \frac{1400 \times r \times 2}{100 \times 12}\right) \quad \text{...(i)}$$

Again total amount for the 2 instalments at the end of second month will be

$$\text{Rs}\left[800 + \left(800 + \frac{800 \times r \times 1}{100 \times 12}\right)\right] \quad \text{...(ii)}$$

from (i) and (ii), we get

$$1400 + \frac{2800r}{1200} = 1600 + \frac{800r}{1200}$$

$$\frac{2000r}{1200} = 200$$

$$\Rightarrow \quad r = 120\%$$

41. Balance price to be paid in instalments = 1980
At the rate of r% per annum after 4 months, Rs. 1980 will amount to Rs. $\left(1980 + \frac{1980 \times 4 \times 25}{12 \times 100}\right) = \text{Rs. } 2145$ ...(i)

Now, the total amount for the 4 instalments at the end of fourth month will be

$$\left[x + \left(x + \frac{25x \times 1}{12 \times 100}\right) + \left(x + \frac{25x \times 2}{12 \times 100}\right) + \left(x + \frac{25 \times 3x}{12 \times 100}\right)\right]$$

$$= 4x + \frac{25x}{1200}(1 + 2 + 3)$$

$$= \frac{33x}{8} \quad \text{...(ii)}$$

from (i) and (ii) $\frac{33x}{8} = 2145$

$$\boxed{x = 520}$$

42. Balance price to be paid in instalments = 350
At the rate of r% per annum after 5 months, Rs. 350 will amount to

$$\text{Rs.}\left(350 + \frac{350 \times 18 \times 5}{12 \times 100}\right) = \left(350 + \frac{1750 \times 18}{1200}\right) \quad \text{...(i)}$$

Again, total amount for the 5 instalments at the end of 5th month will be

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CI/SI/Instalments

$$\text{Rs.}\left[x+\left(x+\frac{18x\times1}{1200}\right)+\left(x+\frac{18x\times2}{12\times100}\right)+\left(x+\frac{18x\times3}{12\times100}\right)+\left(x+\frac{18x\times4}{12\times100}\right)\right]$$

$$=\left[5x+\frac{18x}{1200}(1+2+3+4)\right]$$

$$=5x+\frac{180x}{1200}=\left(\frac{6180x}{1200}\right) \quad \ldots(ii)$$

from (i) and (ii), we get

$$\left(350+\frac{1750\times18}{1200}\right)=\frac{6180x}{1200}$$

$$\Rightarrow \quad x=73.058$$

43. Let each instalments be Rs. x
The amount to be paid instalments = 390200
The total value of all the three instalments is

$$\text{Rs.}\left[x\left(\frac{25}{26}\right)+x\left(\frac{25}{26}\right)^2+x\left(\frac{25}{26}\right)^3\right]$$

and this must be equal to Rs. 390200

Hence, $x\times\frac{25}{26}\left[1+\frac{25}{26}+\frac{625}{676}\right]=390200$

$$x=\text{Rs. } 140608$$

44. Rate of interest = 16% annum
Actual rate of interest = 4% per quarter
Principal of all three instalments

$$=\left[17576\left(\left(\frac{25}{26}\right)+\left(\frac{25}{26}\right)^2+\left(\frac{25}{26}\right)^3\right)\right]$$

$$=\frac{17576\times25\times1951}{26\times676}$$

$$=48775$$

Total amount paid = Rs. $17576\times3=52728$

Interest charged = $52728-48775=3953$

45. Rate of interest $=\frac{40/3}{2}\%=\frac{20}{3}\%$ half yearly

$$\therefore \quad 10815=x\left[\frac{15}{16}+\left(\frac{15}{16}\right)^2+\left(\frac{15}{16}\right)^3\right]$$

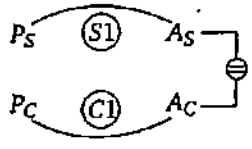
$$\Rightarrow \quad x=4096$$

46. $$P=486680\left[\frac{20}{23}+\left(\frac{20}{23}\right)^2+\left(\frac{20}{23}\right)^3\right]$$

$$\Rightarrow \quad P=\left[486680\times\frac{20}{23}\left(1+\frac{20}{23}+\frac{400}{529}\right)\right]$$

$$P=1111200$$

47.



Obviously $P_S > P_C$, therefore percentage gain of P_C is greater than P_S.

48. $$A_S=P+\frac{P\times6\times6}{100}=P\left(\frac{136}{100}\right)$$

$$A_C=P\left(1+\frac{6}{100}\right)^6=P\left(\frac{106}{100}\right)^6$$

$$\therefore \quad \frac{A_S}{A_C}=\frac{136}{100}\times\left(\frac{100}{106}\right)^6$$

49.



50. Difference between CI and SI for n^{th} year

$$=\frac{Pr}{100}\left[\left(1+\frac{r}{100}\right)^{n-1}-1\right]$$

$$7280=\frac{P\times20}{100}[(1.2)^3-1]$$

$$\Rightarrow \quad P=50000$$

Alternatively:

	Initially	Ist year	IInd year	IIIrd year	IVth year
SI	10000	12000	14000	16000	18000
CI	10000	12000	14400	17280	20736

CI for 4th year = 20736 − 17280 = 3456

SI for 4th year = 2000

Difference between CI and SI = 1456 for Rs. 10000
So, the difference of Rs. 7280 is for Rs. 50000

51. $$\frac{P\times5\times9}{100\times12}-\frac{P\times14\times3}{100\times12}=10$$

$$\Rightarrow \quad P=\text{Rs. } 4000$$

Now, $\frac{4000}{100\times12}[5\times9+14\times3]=\text{Rs. } 290$

52. Let the principal be x, then

$$\frac{\left(\frac{x}{7}\times4+\frac{x}{2}\times5+\frac{5x}{14}\times6\right)}{100}=730$$

$$\Rightarrow \quad x=14000$$

Alternatively: Go through suitable options.
Choose any middlemost option so that if the chosen option is not correct, then you can determine that whether you have to increase or decrease the value of the choices given.

53. $$\frac{P_1\times2\times7x}{100}=\frac{P_2\times2\times8x}{100}$$

$$\Rightarrow \quad \frac{P_1}{P_2}=\frac{8}{7}$$

54. $$\frac{P\left[\left(1+\frac{r}{100}\right)^3-1\right]}{\frac{pr}{100}}=\frac{\left(1+\frac{r}{100}\right)^3-1}{\frac{r}{100}}=3.64$$

Now go through options and verify

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$$\frac{\left(1+\frac{20}{100}\right)^3-1}{\frac{20}{100}}=\frac{(1.2)^3-1}{0.2}$$

$$=\frac{0.728}{0.2}=3.64$$

Hence (c) is correct.

55. $\frac{13p}{5}-p=\frac{8p}{5}$

Now, $\frac{8}{5}P=\frac{P\times r\times 32}{100}$

$\Rightarrow$ $r=5\%$

56. $\frac{1500\times 3}{100}(r_1-r_2)=18$

$\Rightarrow$ $r_1-r_2=0.4$

LEVEL 2

1. CI for 2 years = Rs. 756
SI for 2 years = Rs. 720
It means the interest on the interest of the first year = Rs. 36 (= 756 – 720)
This implies that the rate of interest is 10%
as $\frac{36}{360}\times 100=10\%$
It means the principal for first year was Rs. 3600
$\because$ $\frac{P\times 10\times 1}{100}=360$
$\Rightarrow$ $P=3600$
Now, $\frac{P\times k\times k}{100}=SI$, where $r=t=k$

$$\frac{3600\times k^2}{100}=900$$

$\Rightarrow$ $k=5$

2. $30000\,(1+1.1+(1.1)^2)-30000\,(1+1.1+1.2)$
$\Rightarrow$ Rs. 300

$$30{,}000\left(1+\frac{10}{100}\right)^2-\frac{30{,}000\times 10\times 2}{100}$$

$\Rightarrow$ Rs. 300

3. Interest received from Birbal = $\frac{Pr}{100}$

Interest received from Chanakya = $\frac{\left(2\frac{Pr}{100}\right)\times\frac{r}{2}}{100}$

$$=P\left(\frac{r}{100}\right)^2$$

4. $100\,(1.3)^3=219.7$
$\Rightarrow$ CI = 119.7
and $SI=\frac{100\times 3\times 30}{100}=90$
$\therefore$ CI is greater than SI by Rs. 29.7 (119.7 – 90)
$\therefore$ % increase = $\frac{29.7}{90}\times 100=33.0\%$

5. The best way is to go through options

$$\frac{2200\times 4\times 3}{100}+\frac{1300\times 6\times 3}{100}=\text{Rs. }498$$

Hence the presumed option is correct.

NOTE When we consider Rs. 2200 for 4%, then the rest amount *i.e.*, Rs. 1300 = (3500 – 2200) will be considered automatically for 6%.

Alternatively: Average % rate = $\frac{166}{35}\%$

$$\left[\because 498=\frac{3500\times r\times 3}{100}\right]\Rightarrow r=\frac{166}{35}\%$$

$\therefore$ $\frac{35}{35}\times 4$ — $\frac{166}{35}$ — $6\times\frac{35}{35}$

$\frac{44}{35}$: $\frac{26}{35}$

22 : 13

Thus the ratio of principal at 4% and 6% will be in the ratio of 22 : 13 respectively.

6. $\frac{\text{Decreases in second year}}{\text{Decreases in third year}}=\frac{100}{100-r}=\frac{10}{9}$

$\Rightarrow$ $r=10\%$
Let the population of vultures 3 years ago be P, then

$$P\left(1-\frac{10}{100}\right)^3=29160$$

$\Rightarrow$ $P=40000$

7. On the second year (in terms of CI) is

$$\frac{P\left(1+\frac{r}{100}\right)^2}{\left(P+\frac{Pr}{100}\right)}=\frac{6}{5}\quad\Rightarrow\quad\left(1+\frac{r}{100}\right)=\frac{6}{5}$$

$\Rightarrow$ $r=20\%$

8. Balance price to be paid in instalments = Rs. 1150 ...(i)
$\because$ $(1500-350)=1150$
Now, the total amount for the next 3 instalments at the end of 3rd month will be

$$\left(1150+\frac{1150\times r\times 3}{12\times 100}\right)=$$

$$\left[400+\left(400+\frac{400\times r\times 1}{100\times 12}\right)+\left(400+\frac{400\times r\times 2}{100\times 12}\right)\right]$$

$$\left(\frac{46000+115r}{40}\right)=\left(1200+\frac{400\times 3r}{1200}\right)$$

$\Rightarrow$ $r=\frac{80}{3}=26.66\%$ ...(ii)

9. A: $P=\frac{P\times 4\times r}{100}$
$\Rightarrow$ $r=25\%$
B: $P\left(1+\frac{25}{100}\right)^2=\frac{25P}{16}$

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Again $\frac{25P}{16} \times \frac{2 \times 25}{100} = \frac{25P}{32}$

Therefore total amount of A after 4 years $= 2P$

and total amount of B after 4 years $= \frac{25P}{16} + \frac{25P}{32} = \frac{75P}{32}$

Therefore difference in amount $= \frac{75P}{32} - 2P = \frac{11P}{32} = 2750$

$\Rightarrow \quad P = 8000$

10. Go through options

$$1.8 + \frac{1.8 \times 6 \times 10}{100} = 1.6 + \frac{1.6 \times 8 \times 10}{100},$$

Hence (d) is correct.

Alternatively: $P_1 + \frac{P_1 \times 6 \times 10}{100} = P_2 + \frac{P_2 \times 8 \times 10}{100}$

$\Rightarrow \quad \frac{P_1}{P_2} = \frac{9}{8}$

11. Amount which is to be returned on completion of studies

$= 600000 \times (1.08)^2$

$= 699840$

But only half of 699840 is returned which is equal to Rs. 349920

∴ Amount which is returned after two year of completion of studies

$$= 349920\left(1 + \frac{10}{100}\right)^2$$

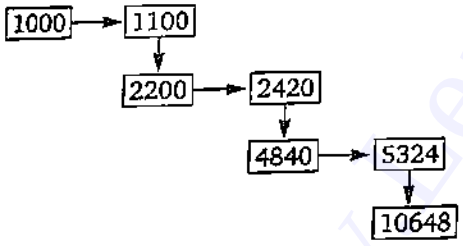
$= 423403.2$

Total amount returned

$= 349920 + 423403.2 = 773323.2$

$=$ Rs. 7.73323 lakh

12.



13. Note that, ultimately 8% interest is charged.
So the net value after 3 years = 125971.2

14. Total time = 25 + 5 = 30 years

Again no. of time periods for cost increment $= \frac{30}{6} = 5$

and no. of time periods for rupee depreciation $= \frac{30}{5} = 6$

Now, the net value of the plot $= 1000 \times (1.05)^5 \times (0.98)^6$

$\approx$ Rs. 1130

15. $\frac{A}{B} = \frac{12 \times x}{3 \times y} = \frac{28}{15}$

$\Rightarrow \quad \frac{A}{B} = \frac{7}{15}$

16. We can find the profit of B but not investment.

17. We don't know the rate of interest.

18. $10500 = x\left[\frac{10}{11} + \left(\frac{10}{11}\right)^2\right]$

$\Rightarrow \quad x = 6050$

19. Let the amount of investment with each one be Rs. 400, then

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$$[400\ (1.1)^2] = [100\ (1.1)^2] + \left[300 + \frac{300 \times r \times 2}{100}\right]$$

$\Rightarrow \quad r = 10.5\%$

20. Best way is to go through options

$1000 \times (1.2)^2 = 2488.32 \approx 2490$

21. Amount earned by HDFC $= 1000000 + \frac{1000000 \times 10 \times 2}{100}$

$= 1200000$

Amount earned by HUDCO $= 1000000\ (1.1)^3 = 1331000$

Net earning of HUDCO $= 1331000 - 1200000 = 131000$

22. Interest paid by Ram Singh = Rs. 48000
Now go through option

$$48000 = \frac{100000}{100}[6 \times 4 + 4 \times 6]$$

$48000 = 48000$

Hence proved that option (b) is correct. It means Ram Singh availed the discount after 4 years of loaning.

23. Worth of hotel after 3 years $= 1000000\ (1.2)^3$

$= 1728000$

Worth of car after 3 years $= 1600000\left(\frac{3}{4}\right)^3$

$=$ Rs. 6,75,000

So, the difference in their worth (pertaining to hotel and car) is

$= 1728000 - 675000 = 10,53,000$

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8. TIME AND WORK

This chapter is one of the easiest chapter for the students. Even an average student can perform better than in other chapters. There is basically one concept is involved in this chapter *i.e.*, concept of efficiency. So most of the problems are very similar in their basic characteristic. Almost every aptitude exam ask the problems from this chapter. On an average 2-3 problems from this chapter have been asked in past years in CAT.

As it is very clear to all of us that the work is directly related with time. As one can say if a particular person or machine works for more time then more work will be done and if it devotes less time then it yields less work *i.e.*, output of a machine or person is directly proportional to time, provided that he/she maintains his/her efficiency during the work.

CONCEPT OF EFFICIENCY

Suppose a person can complete a particular work in 2 days then we can say that each day he does half of the work or 50% work each day. Thus it is clear that his efficiency is 50% per day. Efficiency is generally considered with respect to the time. The time can be calculated either in days, hours, minutes or months etc. So if a person completes his work in 4 days, then his efficiency (per day) is 25%. Since each day he works 1/4th of the total work (*i.e.*, 25% of the total work).

I would like to mention that the calculation of percentage and conversion of ratios and fractions into percentage and vice versa is the prerequisite for this chapter.

Now, if a person can complete a work in n days then his one day's work $= \frac{1}{n}$

and this one day's work in terms of percentage is called his efficiency.

Also if a person can complete $\frac{1}{n}$ work in one day, then he can complete the whole work in n days.

Relation between work of 1 unit of time and percentage efficiency.

A person can complete his work in n days, then his one day's work $= \frac{1}{n}$, his percentage efficiency $= \frac{1}{n} \times 100$

No. of days/hours etc. required to complete the whole work	Work of 1 day/hour	Percentage efficiency
n	$1/n$	$100/n$
1	1/1	100%
2	1/2	50%
3	1/3	$33.33\% = 33\frac{1}{3}\%$
4	1/4	25%
5	1/5	20%
6	1/6	$16.66\% = 16\frac{2}{3}\%$
7	1/7	$14.28\% = 14\frac{2}{7}\%$
8	1/8	12.5%
9	1/9	$11.11\% = 11\frac{1}{9}\%$
10	1/10	10%

This table is very similar to the percentage fraction table given in the chapter of percentage. This table just manifests as a model for efficiency conversion.

Basically for faster and smarter calculation you have to have your percentage calculation very smart.

All the problems of this chapter can be solved through two methods :

1. Unitary method
2. Percentage efficiency

Unitary method is generally obsolete in respect to high level aptitude exam of CAT since it involves typical calculation of LCM each and every time. But when the problems are solved through percentage efficiency it becomes inevitable to save the time which in turn helps to do some more problems within the stipulated time. I admit that initially it might be difficult to solve for those students who are not so good and confident in percentage and fraction calculation, but a little bit of extra Practice will yield an unexpected result in quicker calculations. Now I have some good examples to show you both the methods of solving the same problems.

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EXAMPLE 1 *A can do a job in 12 days and B can do the same job in 6 days, in how many days working together they can complete the job?*

SOLUTION A's 1 day's work $= \frac{1}{12}$

B's 1 day's work $= \frac{1}{6}$

$\therefore$ $(A + B)$'s 1 day's work $= \frac{1}{12} + \frac{1}{6} = \frac{3}{12} = \frac{1}{4}$

$\therefore$ Time taken by both to finish the whole work $= \frac{1}{1/4} = 4$ days

Alternatively :

efficiency of $A = \frac{100}{12} = 8.33\%$

and efficiency of $B = \frac{100}{6} = 16.66\%$

Combined efficiency of A and B both $= 8.33 + 16.66 = 25\%$

$\therefore$ Time taken by both to finish the work (working together) $= \frac{100}{25} = 4$ days

NOTE As per my experience, I have found that in this chapter only selected (numerals) numbers are always used and thus there are almost 20 – 25 numbers are frequently used. So one can very easily remember (and calculate) the percentage efficiency but with different combinations to calculate the LCM becomes a very tedious job. Still you can choose your own method, which is comfortable to you.

RELATION BETWEEN EFFICIENCY AND TIME

Efficiency is inversely proportional to the time (*i.e.* number of days, hours, minutes) etc.

EXAMPLE 1 *A takes 16 days to finish a job alone, while B takes 8 days to finish the same job. What is the ratio of their efficiency and who is less efficient.*

SOLUTION Since A takes more time than B to finish the same job hence A is less efficient or

efficiency of $A = \frac{100}{16} = 6.25\%$

and efficiency of $B = \frac{100}{8} = 12.5\%$

ratio of efficiency of $A : B = \frac{1}{16} : \frac{1}{8} = 1 : 2$

Hence, B is twice efficient as A.

EXAMPLE 2 *A is thrice efficient as B and A takes 20 days to do a job, then in how many days B can finish the same job?*

SOLUTION Ratio of efficiency of $A : B = 3 : 1$

$\therefore$ Ratio of required days of $A : B = \frac{1}{3} : \frac{1}{1} = 1 : 3$

Now since A takes 20 days. So B will take 60 days to finish the same job.

NOTE Ratio of number of days is equal to the ratio of the reciprocals of efficiency and *vice-versa*.

EXAMPLE 2 *A can do a job in 10 days, B can do the same job in 12 days and C can do the same job in 15 days. In how many days they will finish the work together?*

SOLUTION A's 1 day's work $= \frac{1}{10}$

B's 1 day's work $= \frac{1}{12}$

C's 1 day's work $= \frac{1}{15}$

$(A + B + C)$'s one day's work $= \frac{1}{10} + \frac{1}{12} + \frac{1}{15} = \frac{15}{60} = \frac{1}{4}$

Since they can complete $\frac{1}{4}$ work in 1 day. So they will finish the whole work in $\frac{1}{1/4} = 4$ days.

Alternatively : A's efficiency $= 10\%$

B's efficiency $= 8.33\%$

C's efficiency $= 6.66\%$

Combined efficiency of A, B and $C = 10 + 8.33 + 6.66 = 25\%$

Hence, they will take $\frac{100}{25} = 4$ days to finish the job working together

(Since in one day they complete 25% work)

For example if A is twice efficient as B, it means, A takes half the time to finish the same job as B requires working alone.

EXAMPLE 3 *P is thrice as efficient as Q and is therefore able to finish a piece of work in 60 days less than Q. Find the time in which P and Q can complete the work individually.*

SOLUTION Efficiency of $P : Q = 3 : 1$

Required number of days of $P : Q = 1 : 3$

i.e., if P requires x days then Q requires $3x$ days

but $3x - x = 60$

$\Rightarrow$ $2x = 60$

$\Rightarrow$ $x = 30$ and $3x = 90$

Thus P can finish the work in 30 days and Q can finish the work in 90 days.

EXAMPLE 4 *A is twice as good a workman as B and is therefore able to finish a piece of work in 30 days less than B. In how many days they can complete the whole work; working together?*

SOLUTION Ratio of efficiency $= 2 : 1$ $(A : B)$

Ratio of required time $= 1 : 2$ $(A : B)$ $\Rightarrow$ $x : 2x$

but $2x - x = 30$

$\Rightarrow$ $x = 30$ and $2x = 60$

now efficiency of $A = 3.33\%$

and efficiency of $B = 1.66\%$

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Combined efficiency of A and B together = 5%

$\therefore$ time required by A and B working together to finish the work

$$= \frac{100}{5} = 20 \text{ days.}$$

NOTE

$$\text{Efficiency} \propto \frac{1}{\text{number of time units}}$$

$\therefore$ Efficiency × time = constant work

Hence $$\text{Required time} = \frac{\text{work}}{\text{efficiency}}$$

Whole work is always considered as 1, in terms of fraction and 100%, in terms of percentage.

In general, number of day's or hours $= \frac{100}{\text{efficiency}}$

EXAMPLE 5 *A can do a works in x days while B can do the same work in y days then in how many days will they complete the work, working together?*

SOLUTION A's one day's work $= \frac{1}{x}$

B's one day's work $= \frac{1}{y}$

So, both A and B completes $\frac{1}{x} + \frac{1}{y} = \frac{x+y}{xy}$ work in one day.

Now, **by Unitary method**

$\frac{x+y}{xy}$ work can be completed in 1 day

$\therefore$ 1 (means complete) work will be finished in $\frac{1}{\frac{(x+y)}{xy}}$ days

$= \frac{xy}{x+y}$ days.

$\left(\because \text{Required time to complete the work} = \frac{1}{\text{one day's work}}\right)$

CONCEPT OF NEGATIVE WORK

In this case one person works but another destroyes it or cancells it. For example Sonu can write 20 pages per hour but his younger sister Rimjhim erases 10 pages per hour which Sonu writes.

It means finally Sonu writes 10 pages per hour since each hour his sister erases 10 pages out of 20 written pages. Take an other example :

EXAMPLE 1 *A tub can be filled in 20 minutes but there is a leakage in it which can empty the full tub in 60 minutes. In how many minutes it can be filled?*

SOLUTION Filling efficiency = 5% $\left(\because 5 = \frac{100}{20}\right)$

emptying efficiency = 1.66% $\left(\because 1.66 = \frac{100}{60}\right)$

Net efficiency = 5 − 1.66 = 3.33%

$\therefore$ Required time to fill the tub $= \frac{100}{3.33} = 30$ minutes

Alternatively : In 1 minute tub is filling $= \frac{1}{20}$

In 1 minute tub is emptying $= \frac{1}{60}$

$\therefore$ In 1 minute, effective filling of tub $= \frac{1}{20} - \frac{1}{60}$

$= \frac{2}{60} = \frac{1}{30}$

Since $\frac{1}{30}$ part of tub is filled in 1 minute.

Therefore complete tub will be filled in $\frac{1}{1/30} = 30$ minutes

APPLICATION OF INVERSE PROPORTION (OR PRODUCT CONSTANCY)

As I have already discussed thoroughly the concept of inverse proportion and product constancy in ratio-proportion and it has been widely used in profit loss chapter also.

Since the efficiency or rate of work done in one unit of time (mentioned) is inversely proportional to the time *i.e.*, if the rate of work done is greater then the time required will be less and if the rate of work done is less then the time required for the same amount of work will be more.

This product constancy method is limited to the constant work, if the amount of work gets changed, then it does not work, then we have to take help from the unitary method.

When more than one man/machiine work on a particular work/project, the rate of work is calculated as the strength of workers working in a particular time. So the amount of work is defined in terms of man-days or man-hours or man-days-hours.

EXAMPLE 1 *If 20 persons can do a piece of work in 7 days then calculate the number of persons required to complete the work in 28 days*

SOLUTION Number of persons × days = work

$20 \times 7 = 140$ man-days

Now, $x \times 28 = 140$ man-days

$\Rightarrow$ $x = 5$

Therefore in second case the required number of person is 5.

Second Method : Since work is constant, therefore

$M_1 \times D_1 = M_2 \times D_2 =$ work done

$20 \times 7 = M_2 \times 28$

$\Rightarrow$ $M_2 = 5$

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Third Method : Men × days = work, which is constant

$$\downarrow\frac{3}{4}\begin{pmatrix}20 & 7\\ 5 & 28\end{pmatrix}3\uparrow$$

Since number of days is increased by 3 times (*i.e.*, 300%). So the number of men will be decreased by $\frac{3}{4}$ times (*i.e.*, 75%)

(remember percentage change graphic)

EXAMPLE 2 *If 25 men can do a piece of work in 36 days working 10 hours a day, then how many men are required to complete the work working 6 hours a day in 20 days?*

SOLUTION
$$M_1 \times D_1 \times H_1 = M_2 \times D_2 \times H_2$$
$$25 \times 36 \times 10 = M_2 \times 20 \times 6$$
$$\Rightarrow \quad M_2 = 75 \text{ persons}$$

Alternatively :

$$\begin{matrix}\text{Men} & \times & \text{Time}\\ 2\uparrow\begin{pmatrix}25 & & 360\\ 75 & & 120\end{pmatrix}\frac{2}{3}\downarrow & & \end{matrix}$$

[Since product (work) is constant]

By percentage change graphic, when time is decreased by $\frac{2}{3}$ (*i.e.*, 66.66%), number of men is increased by 2 times (*i.e.*, 200%)

EXAMPLE 3 *A contractor employed 30 men to complete the project in 100 days. But later on he realised that just after 25 days only 20% of the work had been completed.*

(a) How many extra days, than the scheduled time are required?

(b) To complete the work on the scheduled time how many men he has to increase?

(c) If the amount of work is also increased by 20% of the actual work, then how many extra days are required (in comparison with scheduled time) but the number of men remained constant.

(d) How many men should be increased so that the work will be completed in 25 days less than the scheduled time.

SOLUTION (a) Men × days = work done

$$30 \times 25 = 750 = 20\% \text{ of the actual work}$$

Now, the work to be done is 4 times than the work done but the number of days is only 3 times. So he is required 4 times the number of days, thus he has to work for extra 25 days.

(b)
$$M_1D_1 = M_2D_2$$
$$4\,(30 \times 25) = M_2 \times 75$$
$$\Rightarrow \quad M_2 = 40$$

(Since, the work to be done is 4 times of the work done. Hence he requires 4 times man-days.)

Alternatively : Since the new product (*i.e.*, work) is 4 times of the original product (*i.e.*, work). But the new product is being multiplied by 3. Thus to make it 4 times we have to multiply it by 4/3. Thus without changing number of days we get the new value of number of men which is 40 (being multiplied by 4/3)

Therefore he has to increase 10 more men.

(c) New work = 3750 man-days

and the available number of men = 30

$$\therefore \quad \text{number of days required} = \frac{3750}{30} = 125$$

So, he has to work for extra 50 days where

$$50 = (125 - 75)$$

(d) Work = 3000 unit (man days)

number of available days = 50 (75 – 25)

$$\therefore \quad \text{number of men required} = \frac{3000}{50} = 60$$

Thus he has to increase (60 – 30) = 30 more men.

EXAMPLE 4 *16 workers working 6 hours a day can build a wall of length 150 metres, breadth 20 m and height 12 m in 25 days. In how many days 12 workers, working 8 hours a day can build a wall of length 800 m, breadth 15 m and height 6 m.*

SOLUTION Here work is the volume of the wall. So the work force should be increased/decreased in the ratio of volume of the work. Therefore

$$\frac{L_1B_1H_1}{L_2B_2H_2} = \frac{M_1D_1T_1}{M_2D_2T_2}$$

where L, B, H are length, breadth and height of the wall respectively and M, D, T are men, days and time in hours per day, respectively. 1 indicates the first case, while 2 indicates second case. So the ratio of work force remains constant as the ratio of volume and work.

$$\therefore \quad \frac{150 \times 20 \times 12}{800 \times 15 \times 6} = \frac{16 \times 6 \times 25}{12 \times 8 \times D_2}$$
$$\Rightarrow \quad D_2 = 50$$

hence the required number of days = 50

Please notice that the volume of work becomes twice (in the second case) so the work force will also be twice to the previous work force.

RELATION BETWEEN EFFICIENCIES

In this case the efficiencies of different persons are different but when they work in a group, so the efficiency of the group is required to know the time taken. For example 3 men can do a work in 4 days while 12 boys can do the same work in 3 days. It means we need $3 \times 4 = 12$ man-days *i.e.*, 12 men can finish the job in 1 day.

Similarly we need $12 \times 3 = 36$ boys days *i.e.*, 36 boys can finish the same job in 1 day.

Here we can see that to finish the work in only 1 day 12 men are needed while 36 boys are needed. Thus we can conclude that work of 12 men is equal to the work of 36 boys. Therefore efficiency of 12 men is equal to 36 boys *i.e.*, we can say the efficiency of 1 man is equal to 3 boys. Thus a man is thrice efficient as a boy or we can say that a man is two times more efficient than a boy.

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EXAMPLE 1 *6 boys and 8 women finish a job in 6 days and 14 boys and 10 women finish the same job in 4 days. In how many days working together 1 boy and 1 woman can finish the work?*

SOLUTION In this kind of questions we find the work force required to complete the work in 1 day (or given unit of time) then we equate the work force to find the relationship between the efficiencies (or work rate) between the different workers.

Therefore $6B + 8W = 6$ days

$\Rightarrow$ $6(6B + 8W) = 1$ day (inversely proportional)

$\Rightarrow$ $36B + 48W = 1$ (by unitary method)

Again $14B + 10W = 4$ days

$\Rightarrow$ $56B + 40W = 1$

So, here it is clear that either we employ $36B$ and $48W$ to finish the work in 1 day or $56B$ and $40W$ to finish the same job in 1 day. Thus, we can say

$\Rightarrow$ $36B + 48W = 56B + 40W$

$\Rightarrow$ $20B = 8W$

$\Rightarrow$ $W = 2.5B$

Thus a woman is 2.5 times as efficient as a boy.

Now, since $36B + 48W = 1$

$\Rightarrow$ $36B + 48 \times (2.5B) = 1$

$\Rightarrow$ $156B = 1$

i.e., to finish the job in 1 day 156 boys are required or the amount of work is 156 boys-days.

Again $1W + 1B = 2.5B + 1B = 3.5B$

Now, since 156 boys can finish the job in 1 day

So 1 boy can finish the job in 1×156 days

$\therefore$ 3.5 boys can finish the job in $\frac{1 \times 156}{3.5} = 44\frac{4}{7}$ days

NOTE There is a great difference between 'and' & 'or'
For example 4 men and 8 women can do a piece of work in 10 days : means it is unknown that who is faster or slower (*i.e.*, we don't know the relation between efficiencies of a man and a woman)
Again, 4 men or 8 women can do a piece of work in 10 days : means 4 men are equal to 8 women. Hence a man is twice efficient as a woman.

EXAMPLE 2 *6 men and 8 women can do a job in 10 days. In how many days can 5 men and 9 women do the same job?*

SOLUTION Since we don't know the relation between work rate (or efficiency) between a man and that of a woman so we can't find the required number of days.

EXAMPLE 3 *6 men and 8 women can do a job in 10 days. In how many days can 3 men and 4 women finish the same job working together?*

SOLUTION Notice here we don't know the relation of efficiencies but we can solve the problem due to clear relation between the work force.

Since $6M + 8W = 10$ days

$2(3M + 4W) = 10$ days

$\Rightarrow$ $(3M + 4W) = 20$ days

Since, the work force has become half of the original force so number of days must be double.

Thus required number of days = 20

EXAMPLE 4 *A can complete a work in 12 days, B in 15 days. Find the time taken by them :*

(a) when A and B worked together.

(b) when A and B worked alternatively started by A.

(c) when A and B worked alternatively started by B.

(d) if A started two days later, in comparison to B.

(e) if B started two days later, in comparison to A.

(f) if A leaves two days before the actual completion of the work.

(g) if B leaves two days before the actual completion of the work.

(h) if A leaves two days before the scheduled completion of the work.

(i) if B leaves two days before the scheduled completion of the work.

(j) if B does negative work with his same work rate.

SOLUTION (a) A's efficiency = 8.33%

B's efficiency = 6.66%

Combined efficiency = 15%

So, the required time $= \frac{100}{15} = 6\frac{2}{3}$ days.

(b) In every two days A and B work 15%. So in 12 days they will complete 90% work. Now on the 13th day, A will finish 8.33% of the remaining (*i.e.*, 10% work) and the rest 1.66% will be finished by B on 14th day by taking time $= \frac{1.66}{6.66} = \frac{1}{4}$ day.

Thus, total required time $= 12 + 1 + \frac{1}{4} = 13\frac{1}{4}$ days.

See the chart below :

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Work	8.33	6.66	8.33	6.66	8.33	6.66	8.33	6.66	8.33	6.66	8.33	6.66	8.33	1.66
	15		15		15		15		15		15			
	$15 \times 6 = 90\%$												10%	

(c) Here the difference is in the last day/days only.

See the chart below :

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Work	6.66	8.33	6.66	8.33	6.66	8.33	6.66	8.33	6.66	8.33	6.66	8.33	6.66	3.33
	15		15		15		15		15		15		10	
	$15 \times 6 = 90\%$												10%	

So, after the completion of 90% work in 12 days, on the 13th day 6.66% work will be finished by B and then 3.33% (the last part of the work) will be finished by A by taking time $= \frac{3.33}{8.33} = \frac{2}{5}$ day.

Thus, the required time $= 12 + 1 + \frac{2}{5} = 13\frac{2}{5}$ days.

(d) It means B worked for 2 days more than A has worked for. Now, since B completes $2 \times 6.66 = 13.33\%$ work in two days. So the remaining work will be done by A and B together.

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Therefore the time taken by A and B, working together

$$= \frac{86.66}{15} \frac{\text{(remaining work)}}{\text{(combined efficiency)}} = 5\frac{7}{9} \text{ days}$$

Thus the total required time to finish the whole work

$$= 2 + 5\frac{7}{9} = 7\frac{7}{9} \text{ days.}$$

(e) It means A alone worked for 2 days which is equal to

$$8.33 \times 2 = 16.66\%$$

So, the remaining (83.33%) work will be done by A and B together.

$$\therefore \quad \text{Time taken together} = \frac{83.33}{15} = 5\frac{5}{9} \text{ days}$$

Thus, the total required time $= 2 + 5\frac{5}{9} = 7\frac{5}{9}$ days.

(f) 'A leaves two days before the actual completion of the work' means in the last two days B has worked alone which is $6.66 \times 2 = 13.33\%$

Therefore the remaining work *i.e.*, 86.66% work was only done by A and B together in $\frac{86.66}{15} = 5\frac{7}{9}$ days

Thus, the required time $= 2 + 5\frac{7}{9} = 7\frac{7}{9}$ days.

NOTE This case is exactly same as part (d) of the same example. The difference is only in order of days. In part (d) B worked alone initially for 2 days and in part (f) B worked alone in the last 2 days.

(g) Applying the same concept as above we can calculate the required number of days which is same as in case (e)

Work	$A + B$	A
	$15 \times 5\frac{2}{5}$	8.33×2
	83.33%	16.66%

(h) 'A leaves two days before the scheduled completion of the work' means A works with B only for $4\frac{2}{3}$ days since the scheduled time is $6\frac{2}{3}$ [see the case (a)]

Scheduled time is the time in which work could be finished by working together without any change in efficiency or work force.

Now since both A and B have worked together for $4\frac{2}{3}$ days it means the last 2 days work (scheduled) which is equal to 30% (since if A and B working together can finish 30% work in 2 days) will be done by B only.

$\therefore$ required time to complete 30% work by B alone

$$= \frac{30}{6.66} = 4.5 \text{ days}$$

Thus the total required time to finish the whole work

$$= 4\frac{2}{3} + 4\frac{1}{2} = 9\frac{1}{6} \text{ days.}$$

(i) Very similar to the case (h) A and B worked only for 4 days (2 days less than the scheduled time $6\frac{2}{3}$ days)

(It is obvious that in $4\frac{2}{3}$ days A and B, working together completes 70% work, but we can directly calculate that they can complete 30% if they work together, in 2 days)

Thus 30% work is done by A alone

$$\therefore \quad \text{time taken} = \frac{30}{8.33} = 3\frac{3}{5} \text{ days}$$

Thus, the total time $= 4\frac{2}{3} + 3\frac{3}{5} = 8\frac{4}{15}$ days.

(j) Combined efficiency of A and $B = (8.33) + (-6.66)$

$= 1.66\%$

$\therefore$ Time required by A and B working together

$$= \frac{100}{1.66} = 60 \text{ days}$$

(Negative work means, B works against A)

EXAMPLE 5 *A can complete a work in 10 days, B in 12 days and C in 15 days. All of them began the work together, but A had to leave the work after 2 days of the start and B 3 days before the completion of the work. How long did the work last?*

SOLUTION See the diagram and then interpret the language of the question.

Initial 2 days		Last 3 days
$A + B + C$	$B + C$	C
$25 \times 2 = 50\%$	$15\% \times 2$ $[100 - (50 + 20)]$	$6.66 \times 3 = 20\%$

Since initially for 2 days all of them A, B and C work together so they complete the 50% work

and for the last 3 days only C works which is equal to 20% work.

Thus, the remaining work $= 30\%[100 - (50 + 20)]$

This 30% work was done by B and C in 2 days $= \left(\frac{30}{15}\right)$.

NOTE

Efficiency of $A = 10\%$

Efficiency of $B = 8.33\%$

Efficiency of $C = 6.66\%$

So, the total number of required days $= 2 + 2 + 3 = 7$ days.

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EXERCISE

LEVEL 1

1. *A* can finish a piece of work in 12 days while *B* can do it in 15 days. If both work at it together, what time will they take to do the work?
(a) 6 days (b) 8 days
(c) $6\frac{2}{3}$ days (d) $10\frac{1}{5}$ days

2. *A* can do a piece of work in 10 days, *B* can do it in 12 days and *C* can do it in 15 days. In how many days they can complete the work, working together?
(a) 6 (b) 4
(c) 5 (d) 10

3. *A* can do a piece of work in 5 days, *B* can do it in 10 days. With the help of *C*, they finish the work in 2 days. In how many days *C* alone can do the whole work?
(a) 3 (b) 4
(c) 5 (d) 6

4. *A* can do a piece of work in 8 days, *B* can do it in 16 days, while *C* can do it in 80 days. In how many days they can complete the whole work, working together?
(a) 5 (b) 6
(c) $8\frac{2}{3}$ (d) $20\frac{2}{5}$

5. *A* can do a piece of work in 10 days, *B* can do it in 15 days working together they can finish the work in :
(a) 9 (b) 8
(c) 10 (d) 6

6. *A* can do a piece of work in 24 days, while *B* can do it in 30 days. With the help of *C* they can finish the whole work in 12 days. How much time is required for *C* to complete the work, alone?
(a) 100 days (b) 120 days
(c) 125 days (d) 72 days

7. *A* can do a piece of work in 10 days. *B* can do it in 24 days. If *C* also works with them then it takes only 6 days to complete the whole work. In how many days *C* alone can complete the whole work?
(a) 25 (b) 40
(c) 50 (d) 75

8. *A* can do a piece of work in 14 days while *B* can do it in 21 days. In how many days, working together they will complete the whole work?
(a) 10.5 (b) 8
(c) 8.4 (d) 9

9. *A* can do a piece of work in 24 days. If *B* is 60% more efficient than *A*, then the number of days required by *B* to do the same piece of work is :
(a) 10 (b) 15
(c) 12 (d) 9.6

10. *A* is twice as good a workman as *B* and together they finish a piece of work in 14 days. In how many days can *A* alone finish the work?
(a) 18 (b) 20
(c) 21 (d) 24

11. *A* is twice as good a workman as *B* and together they finish a piece of work in 7 days. In how many days can *A* alone finish the work?
(a) 10 (b) 10.5
(c) 12 (d) 11

12. *A* is thrice as good a workman as *B* and takes 10 days less to do a piece of work than *B* takes. The number of days taken by *B* to finish the work is :
(a) 9 (b) 18
(c) 15 (d) 12

13. *A* is twice as good a workman as *B* and therefore *A* takes 6 days less than *B* to finish the work individually. If *A* and *B* working together complete the work in 4 days, then how many days are required by *B* to complete the work alone?
(a) 12 (b) 18
(c) 8 (d) 6

14. *A* is thrice efficient as *B* and *C* is twice as efficient as *B*. What is the ratio of number of days taken by *A*, *B* and *C*, when they work individually?
(a) 2 : 6 : 3 (b) 2 : 3 : 6
(c) 1 : 2 : 3 (d) 3 : 1 : 2

15. Ajit is 3 times as efficient as Bablu, then the ratio of number of days required by each to work alone, completely?
(a) 3 : 1 (b) 1 : 3
(c) 6 : 3 (d) 3 : 6

16. *X* is 4 times more efficient as *Y*, then the number of days taken by *X* and *Y* are in the ratio :
(a) 8 : 1 (b) 1 : 5
(c) 6 : 8 (d) 5 : 1

17. If *A* takes 4/5 days as *B* takes and working together they require $\frac{20}{3}$ days to complete the whole work. What is the efficiency of *B* ?
(a) $6\frac{2}{3}\%$ (b) 16%
(c) 5.55% (d) $8\frac{1}{3}\%$

18. *A* is thrice as efficient as *B*. Working together they complete the work in 3 days. If *B* takes 8 days more than *A*, what is the number of days taken by *A* to finish the whole work, alone?
(a) 4 (b) 2
(c) 12 (d) 16

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19. A and B can do a piece of work in 8 days, B and C can do the same work in 12 days and A and C complete it in 8 days. In how many days A, B and C can complete the whole work, working together?
(a) 4 (b) 6
(c) 12 (d) 9

20. A and B can do a piece of work in 12 days, B and C in 15 days C and A in 20 days. In how many days can C alone do it?
(a) 60 (b) 50
(c) 25 (d) 24

21. Ganga, Jamuna and Saraswati can do a piece of work, working together, in 1 day. Ganga is thrice efficient as Jamuna and Jamuna takes the twice the number of days as Saraswati takes to do it alone. What is the difference between the number of days taken by Ganga and Saraswati?
(a) 1 (b) 2
(c) 3 (d) 4

22. A can finish a work in 12 days and B can do it in 15 days. After A had worked for 3 days, B also joined A to finish the remaining work. In how many days, the remaining work will be finished?
(a) 3 (b) 4
(c) 5 (d) 6

23. Raja can do a piece of work in 14 days, while Rani can do the same work in 21 days. They started the work together but 3 days before the completion of the work, Raja left the work. The total number of days to complete the work is :
(a) 7 (b) 8.5
(c) 5 (d) $10\frac{1}{5}$

24. A and B can complete a task in 30 days when working together after A and B have been working together for 11 days, B is called away and A, all by himself completes the task in the next 28 days. Had A been working alone, the number of days taken by him to complete the task would have been :
(a) $33\frac{3}{19}$ (b) $19\frac{6}{25}$
(c) $44\frac{4}{19}$ (d) none of these

25. Sonu can do a piece of work in 20 days. He started the work and left after some days, when 25% work was done. After it Abhijeet joined and completed it working for 10 days. In how many days Sonu and Abhijeet can do the complete work, working together?
(a) 6 (b) 8
(c) 10 (d) 12

26. Efficiency of Asha is 25% more than Usha and Usha takes 25 days to complete a piece of work. Asha started a work alone and then usha joined her 5 days before actual completion of the work. For how many days Asha worked alone?
(a) 9 (b) 11
(c) 10 (d) 15

27. Krishna can do a work in 10 days while Mohan can do the same work in 20 days. They started work together. After 3 days Krishna left the work and Mohan completed it. For how many days Mohan worked alone more than the number of days required when both worked together?
(a) $4\frac{1}{3}$ (b) $3\frac{1}{4}$
(c) $2\frac{3}{5}$ (d) $3\frac{2}{3}$

28. Kareena can do a piece of work in 9 days and Karishma can do the same work in 18 days. They started the work. After 3 days Shahid joined them, who can complete alone the same whole work in 3 days. What is the total number of days in which they had completed the work?
(a) 12 (b) 8
(c) 4 (d) 6

29. Kavita, Babita and Samita started a work. 5 days later Samita left the work and Babita left the work after working 8 days. In how many more days Kavita would have completed the rest work if they take 20, 60 and 30 days individually to finish a work?
(a) 4 (b) 5
(c) 6 (d) 8

30. The ratio of efficiency of A is to C is 5 : 3. The ratio of number of days taken by B is to C is 2 : 3. A takes 6 days less than C, when A and C completes the work individually. B and C started the work and left after 2 days. The number of days taken by A to finish the remaining work is :
(a) 4.5 (b) 5
(c) 6 (d) $9\frac{1}{3}$

31. Anand can do a piece of work in 45 days, but Bahuguna can do the same work in 5 days less, than Anand, when working alone. Anand and Bahuguna both started the work together but Bahuguna left after some days and Anand finished the remaining work in 56 days with half of his efficiency but he did the work with Bahuguna with his complete efficiency. For how many days they had worked together?
(a) 6 (b) 8
(c) 9 (d) 12

32. Chandni and Divakar can do a piece of work in 9 days and 12 days respectively. If they work for a day alternatively, Chandni beginning, in how many days, the work will be completed?
(a) $10\frac{1}{4}$ (b) $9\frac{1}{5}$
(c) 11.11 (d) 10

33. Fatima and Zahira can do a piece of work in 12 days and 15 days respectively. If they work for alternate day and Fatima starts the work first, then in how many days the work will be completed?
(a) $12\frac{1}{5}$ (b) $13\frac{1}{4}$
(c) $13\frac{1}{5}$ (d) 15

34. In the previous question (number 33) if Zahira starts first then in how many days, the work will be completed?
(a) $14\frac{1}{5}$ (b) 14
(c) $13\frac{1}{5}$ (d) $13\frac{2}{5}$

35. The number of days required by A, B and C to work individually is 6, 12 and 8 respectively. They started a work doing it alternatively. If A has started then followed by B and so on, how many days are needed to complete the whole work?
(a) 8 (b) 7.5
(c) 8.5 (d) $9\frac{1}{2}$

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36. In the previous question if the order of working days be as *B*, *C*, *A*, *B*, *C*, *A* ... (starting with *B* and followed by *C* and *A* respectively), then in how many days the work will be completed?

(a) $7\frac{3}{4}$ (b) 8

(c) $8\frac{1}{4}$ (d) 9

37. *A* takes 6 days less than *B* to do a certain job and 2 days more than *C*. *A* and *B* together can do the work in the same time as *C*. In how many days *B* alone can do the complete work?

(a) 10 (b) 14

(c) 12 (d) 16

38. *A* and *B* undertook a work for Rs. 350. *A* got Rs. 150 more than that of *B*, when they worked together. *B* takes 9 days more than *A*, when they work individually. In how many days *A* and *B* working together can do the whole work :

(a) 5 (b) $4\frac{2}{7}$

(c) $4\frac{5}{7}$ (d) $5\frac{4}{7}$

39. Alen and Border can do a work individually in 21 and 42 days respectively. In how many days they can complete the work, working alternatively?

(a) 14 (b) 28

(c) 42 (d) 35

40. *C* takes twice the number of days to do a piece of work than *A* takes. *A* and *B* together can do it in 6 days while *B* and *C* can do it in 10 days. In how many days *A* alone can do the work?

(a) 60 (b) 30

(c) 6 (d) 7.5

41. When *A*, *B* and *C* are deployed for a task, *A* and *B* together do 70% of the work and *B* and *C* together do 50% of the work. Who is most efficient?

(a) *A* (b) *B*

(c) *C* (d) can't be determined

42. Colonel, Major and General started a work together for Rs. 816. Colonel and Major did $\frac{8}{17}$ of the total work, while Major and General together did $\frac{12}{17}$ of the whole work. What is the amount of the least efficient person?

(a) Rs. 256 (b) Rs. 144

(c) Rs. 85 (d) can't be determined

43. Sharma is 20% less efficient than Kelkar. If Kelkar can do a piece of work in 24 days. The number of days required by Sharma to complete the same work alone?

(a) 20 (b) 30

(c) 28.8 (d) can't be determined

44. 30 persons can do a piece of work in 24 days. How many more people are required to complete the work in 20 days?

(a) 4 (b) 5

(c) 6 (d) none of these

45. 12 women can do a piece of work in 20 days. If the 4 women deny to work, then how many more days are required?

(a) 6 (b) 10

(c) 15 (d) none of these

46. 35 boys can do a piece of work in 15 days. The work w completed in 25 days. How many boys did not turn up for t job?

(a) 14 (b) 20

(c) 6 (d) 7

47. 24 men can complete a job in 40 days. The number of m required to complete the job in 32 days is :

(a) 30 (b) 40

(c) 25 (d) 50

48. 16 men finished one-third work in 6 days. The number additional men are required to complete the job in next days :

(a) 10 (b) 8

(c) 16 (d) 32

49. If 10 persons can do a job in 20 days, then 20 person wit twice the efficiency can do the same job in :

(a) 5 days (b) 40 days

(c) 10 days (d) 20 days

50. A certain job was assigned to a group of men to do in 20 days But 12 men did not turn up for the job and the remaining men did the job in 32 days. The original number of men in the group was :

(a) 32 (b) 36

(c) 42 (d) 40

51. 30 workers can finish a work in 20 days. After how many days should 9 workers leave the job so that the work is completed in total 26 days :

(a) 12 (b) 10

(c) 6 (d) none of these

52. 25 men can complete a job in 30 days. After how many days should the strength of work force be increased by 50 men so that the work will be completed in $\frac{2}{3}$rd of the actual time :

(a) 15 (b) 10

(c) 18 (d) 5

53. A group of workers can complete a job in 120 days. If there were 4 more such workers then the work could be finished in 12 days less. What was the actual strength of workers?

(a) 30 workers (b) 40 workers

(c) 42 workers (d) 36 workers

54. Mr. Modi can copy 40 pages in 10 minutes, Mr Xerox and Mr. Modi both working together can copy 250 in 25 minutes. In how many minutes Mr. Xerox can copy 36 pages?

(a) 5 minutes (b) 6 minutes

(c) 3 minutes (d) 12 minutes

55. 20 persons completed $\frac{1}{3}$rd of the work in 12 days. How many more person are required to finish the rest work in next 12 days?

(a) 20 (b) 12

(c) 18 (d) 40

56. A contractor undertook a work to complete in 60 days. But just after 20 days he observed that only $\frac{1}{5}$th of the project work had been completed. To complete the work in time (i.e.,

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in rest days) minimum how many workers he had to increase, if there were initially 75 workers were deployed for the task?
(a) 25 (b) 50
(c) 75 (d) can't be determined

57. 6 men or 10 women can reap a field in 15 days, then the number of days that 12 men and 5 women will take to reap the same field is :
(a) 5 (b) 6
(c) 8 (d) 12

58. If 2 men or 3 women or 4 boys can do a piece of work in 52 days, then the same piece of work will be done by 1 man, 1 woman and 1 boy in :
(a) 48 days (b) 36 days
(c) 45 days (d) none of these

59. 2 men or 5 women or 7 boys can finish a work in 469 days, then the number of days taken by 7 men, 5 women and 2 boys to finish the work is :
(a) 134 (b) 106
(c) 100 (d) 98

60. 6 children and 2 men complete a certain piece of work in 6 days. Each child takes twice the time taken by a man to finish the work. In how many days will 5 men finish the same work?
(a) 6 (b) 8
(c) 9 (d) 15

61. 2 men and 3 women finish 25% of the work in 4 days, while 6 men and 14 women can finish the whole work in 5 days. In how many days will 20 women finish it?
(a) 20 (b) 25
(c) 24 (d) 88

Directions : For question number 62 and 63:
A can do a work in 15 days and B can do it in 18 days. With the help of C, all of them complete the work in 6 days.

62. How long will it take C to finish the work alone?
(a) 30 days (b) 22 days
(c) $\frac{45}{2}$ days (d) 25 days

63. A, B and C received total Rs. 27,000 for the whole work. What is the share of C, if the money is distributed in the ratio of amount of work done, individually?
(a) 2700 (b) 7200
(c) 14400 (d) 6300

Directions : Solve the following questions individually.

64. 314 weavers weaves 6594 shawls in $\frac{1}{6}$ hours. What is the number of shawls weaved per hour by an average weaver?
(a) 42 (b) 21
(c) 102 (d) 126

65. Three men and two women can do a piece of work in 4 days, while two men and three women can do the same work in 5 days. Rs 44 is given to a woman for her contribution towards work, per day. What is the amount received by a man per day?
(a) Rs. 88 (b) Rs. 144
(c) Rs. 154 (d) can't be determined

66. 30 girls can finish a work in 40 days. After how many days should 10 girls leave the work so that it may be finished in 46 days, if all the 30 girls started the work?
(a) 18 days (b) 28 days
(c) 22 days (d) 30 days

67. If 8 women collect 200 kg of tea leaves in 10 hours. How many more (in kg) of tea leaves will 12 women collect in 8 hours?
(a) 24 kg (b) 40 kg
(c) 50 kg (d) 100 kg

68. 450 man-days of work can be completed by a certain number of men in some days. If the number of people (men) are increased by 27, then the number of day required to complete the same work is decreased by 15. The number of days are required to complete the three times work (than the previous/actual work) by 27 men?
(a) 50 (b) 60
(c) 54 (d) 45

69. 4 boys and 5 girls can do a piece of work in 10 days. 6 boys and 6 girls can do the same work in 7 days. In how many days can 2 boys and 7 girls complete the same work, working together?
(a) 15 days (b) 14 days
(c) 21 days (d) 18 days

70. If 20 engineers and 20 workers can together construct a 20 km road in 20 days. 40 engineers and 40 workers together construct 40 km road in how many days?
(a) 10 (b) 20
(c) 40 (d) can't be determined

71. $(x-2)$ men can do a piece of work in x days and $(x+7)$ men can do 75% of the same work in $(x-10)$ days. Then in how many days can $(x+10)$ men finish the work?
(a) 27 days (b) 12 days
(c) 25 days (d) 18 days

72. A man, a woman and a girl worked for a contractor for the same period. A man is twice efficient as a woman and a woman is thrice efficient as a girl Rs. 10000 were given to all of them. What is the sum of money received by a woman and a girl together?
(a) Rs. 5500 (b) Rs. 4500
(c) Rs. 4000 (d) Rs. 6000

73. 33 men can do a job in 30 days. If 44 men started the job together and after every day of the work, one person leaves. What is the minimum number of days required to complete the whole work?
(a) 21 (b) 42
(c) 45 (d) none of these

74. Abhishek can do a piece of work in 40 days. He alone worked at it for 8 days and then Bacchhan completed alone the rest work in 24 days. In how many days they will complete the whole work, working together?
(a) $17\frac{1}{7}$ days (b) $18\frac{1}{7}$ days
(c) $9\frac{1}{6}$ days (d) 14 days

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75. C is twice efficient as A. B takes thrice as many days as C. A takes 12 days to finish the work alone. If they work in pairs (i.e., AB, BC, CA) starting with AB on the first day then BC on the second day and AC on the third day and so on, then how many days are required to finish the work?

(a) $6\frac{1}{5}$ days (b) 4.5 days

(c) $5\frac{1}{9}$ days (d) 8 days

76. Ahluwalia and Bimal together take 6 days to finish the work. Bimal and Jalan together take 10 days to finish the work. What is the difference between number of days taken by Ahluwalia and Jalan when they worked alone to complete the whole work?

(a) 12 days (b) 16 days

(c) 15 days (d) can't be determined

77. B is twice efficient as A and A can do a piece of work in 15 days. A started the work and after a few days B joined him. They completed the work in 11 days, from the starting. For how many days they worked together?

(a) 1 day (b) 2 days

(c) 6 days (d) 5 days

78. A, B and C can complete a piece of work in 15, 30 and 40 days respectively. They started the work together and A left 2 days before the completion of the work and B left 4 days before the completion of the work. In how many days was the work completed?

(a) $7\frac{3}{10}$ (b) $10\frac{2}{15}$

(c) $10\frac{7}{30}$ (d) none of these

79. There was a leakage in the container of the refined oil. If 11 kg oil is leaked out per day then it would have lasted for 50 days, if the leakage was 15 kg per day, then it would have lasted for only 45 days. For how many days would the oil have lasted, if there was no leakage and it was completely used for eating purpose?

(a) 80 days (b) 72 days

(c) 100 days (d) 120 days

80. A contractor undertook to complete the work in 40 days and he deployed 20 men for his work. 8 days before the scheduled time he realised that $\frac{1}{3}$rd of the work was still to be done. How many more men were required to complete the work in stipulated time?

(a) 16 (b) 15

(c) 20 (d) 25

81. 7 Indian and 4 Chinese finish a job in 5 days. 7 Japanese and 3 Chinese finish the same job in 7 days. Given that the efficiency of each person of a particular nationality is same but different from others. One Indian, one Chinese and one Japanese will complete the work in :

(a) $18\frac{3}{13}$ days (b) $20\frac{5}{12}$ days

(c) $21\frac{6}{14}$ days (d) $20\frac{7}{12}$ days

82. A, B and C are three book binders. A takes 8 minutes, B takes 12 minutes and C takes 16 minutes to bind a book. If they work each day for 12 hours, then on an average, how many books each one bind per day?

(a) 65 (b) 52

(c) 48 (d) 70

83. A piece of work can be completed by 10 men and 6 women in 18 days. Men works 9 hours per day while women works 7.5 hours per day. Per hour efficiency of a woman is $\frac{2}{3}$rd of a man's efficiency. In how many days 10 men and 9 women complete the work?

(a) 16 days (b) 20 days

(c) 30 days (d) 25 days

84. B and C are equally efficient, but the efficiency of A is half of each B and C. A and B started a work and 3 days later C joined them. If A alone can do the work in 14 days, then in how many more days the work will be completed?

(a) 1 (b) 2

(c) 3 (d) 4.5

85. A can do a piece of work in 10 days, B in 15 days. They work together for 5 days, the rest of the work is finished by C in two more days. If they get Rs. 3000 as wages for the whole work, what are the daily wages of A, B and C respectively (in Rs.):

(a) 200,250,300 (b) 300,200,250

(c) 200,300,400 (d) none of these

86. A can do a piece of work in 2 hours, B can do thrice the work in 8 hours and 'C' can do the same work as A in 8 hours. If all of them work together, how long it would take them to complete the work :

(a) 1 hour (b) 2 hours

(c) 3 hours (d) 4 hours

87. A is twice efficient as B and together they do the same work in as much time as C and D together. If C and D can complete the work in 20 and 30 days respectively, working alone, then in how many days A can complete the work individually :

(a) 12 days (b) 18 days

(c) 24 days (d) 30 days

88. 4 men and 2 boys can finish a piece of work in 5 days. 3 women and 4 boys can finish the same work in 5 days. Also 2 men and 3 women can finish the same work in 5 days. In how many days 1 man, 1 woman and one boy can finish the work, at their double efficiency?

(a) $4\frac{8}{13}$ (b) $4\frac{7}{13}$

(c) $3\frac{7}{13}$ (d) none of these

89. If m men can do a work in r days, then the number of days taken by $(m + n)$ men to do it is :

(a) $\frac{m+n}{mn}$ (b) $\frac{m+n}{mr}$

(c) $\frac{mr}{(m+n)}$ (d) $\frac{(m+n)r}{mn}$

90. Pipe A can fill a tank in 36 minutes and pipe B can fill it in 45 minutes. If both the pipes are opened to fill an empty tank, in how many minutes will it be full?

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(a) 15 (b) 18
(c) 20 (d) 25

91. Tap *A* can fill the empty tank in 12 hours, but due to a leak in the bottom it is filled in 15 hours. If the tank is full and then tap *A* is closed then in how many hours the leak can empty it?
(a) 45 hours (b) 48 hours
(c) 52 hours (d) 60 hours

92. Pipe *A* and *B* can fill a cistern in 10 hours and 15 hours respectively. When a third pipe *C* which works as an outlet pipe is also open then the cistern can be filled in 18 hours. The outlet pipe can empty a full cistern in :
(a) 12 hours (b) 8 hours
(c) 9 hours (d) 14 hours

93. A cistern has a leak which would empty it in 6 hours. A tap is turned on which fills the cistern @ 10 liters per hour and then it is emptied in 15 hours. What is the capacity of the cistern?
(a) 100 litres (b) 166.66 litres
(c) 60.66 litres (d) none of these

94. Tap *a* fills a tank in 10 hours and *B* can fill it in 15 hours. Both are opened simultaneously. Sometimes later tap *B* was closed, then it takes total 8 hours to fill uo the whole tank. After how many hours *B* was closed?
(a) 2 (b) 3
(c) 4 (d) 5

95. Tap *A* can fill a tank in 20 hours, *B* in 25 hours but tap *C* can empty a full tank in 30 hours. Starting with *A*, followed by *B* and *C* each tap opens alternatively for one hour period till the tank gets filled up completely. In how many hour the tank will be filled up completely?
(a) $51\frac{11}{15}$ (b) $52\frac{2}{3}$
(c) $24\frac{4}{11}$ (d) none of these

96. If one pipe *A* can fill a tank in 20 minutes, then 5 pipes, each of 20% efficiency of *A*, can fill the tank in :
(a) 80 min (b) 100 min
(c) 20 min (d) 25 min

97. Pipe *A* basically used as inlet pipe and pipe *B* is used as outlet pipe. Pipes *A* and *B* both are opened simultaneously, all the time. When pipe *A* fills the tank and *B* empty the tank, it will take double the time than when both the pipe fill the tank. When pipe *B* is used for filling the tank, its efficiency remains constant. What is the ratio of efficiency of pipe *A* and pipe *B* respectively?
(a) 3 : 1 (b) 5 : 2
(c) 1 : 3 (d) 3 : 2

98. Pipe *A* can fill the tank in 4 hours, while pipe *B* can fill it in 6 hours working separately. Pipe *C* can empty whole the tank in 4 hours. He opened the pipe *A* and *B* simultaneously to fill the empty tank. He wanted to adjust his alarm so that he could open the pipe *C* when it was half-filled, but he mistakenely adjusted his alarm at a time when his tank would be 3/4th filled. What is the time difference between both the cases, to fill the tank fully :
(a) 48 min. (b) 54 min.
(c) 30 min. (d) none of these

99. Two pipes *A* and *B* can fill a cistern in 15 hours and 10 hours respectively. *A* tap *C* can empty the full cistern in 30 hours. All the three taps were open for 2 hours, when it was remembered that the emptying tap had been left open. It was then closed. How many hours more would it take for the cistern to be filled?
(a) 30 min. (b) 1.2 hours
(c) 24 min. (d) 35 min.

100. Pipe *A* can fill an empty tank in 30 hours while *B* can fill it in 45 hours. Pipe *A* and *B* are opened and closed alternatively *i.e.*, first pipe *A* is opened, then *B*, again *A* and then *B* and so on for 1 hour each time without any time lapse. In how many hours the tank will be filled when it was empty, initially?
(a) 36 (b) 54
(c) 48 (d) 60

LEVEL (2)

1. *A*, *B* and *C* three weavers have to supply an order of 100 shawls. *A* can weave a shawl in 2 hours, *B* in 3 hours and *C* in 4 hours respectively. It is known that even being a joint contract each one weaves his own shawl completely *i.e.*, no other weaver help to the rest weavers. In how many hours they will complete the order irrespective of day or night?
(a) 93 hours (b) 100 hours
(c) $92\frac{4}{13}$ hours (d) 94 hours

2. Arun and Satyam can complete a work individually in 12 working days and 15 working days respectively with their full efficiencies. Arun does work only on Monday, Wednesday and Friday while Satyam does the work on Tuesday, Thursday and Saturday. Sunday is always off. But Arun and Satyam both works with half of their efficiencies on Friday and Saturday respectively. If Arun started the work on Ist January which falls on Monday followed by Satyam on the next day and so on (*i.e.*, they work collectively in alternate days), then on which day work will be completed?
(a) Tuesday (b) Wednesday
(c) Thursday (d) Friday

3. Kaushalya can do a work in 20 days, while Kaikeyi can do the same work in 25 days. They started the work jointly. Few days later Sumitra also joined them and thus all of them completed the whole work in 10 days. All of them were paid total Rs. 700. What is the share of Sumitra?
(a) Rs. 130 (b) Rs. 185
(c) Rs. 70 (d) can't be determined

4. *A* and *B* can complete the work individually in 24 days and 30 days respectively, working 10 hours a day. Work is to be done in two shift. Morning shift lasts for 6 hours and evening shift lasts for 4 hours. On the first day *A* works in the morning shift while *B* works in the evening shift. Next day *A* works in the evening shift while *B* works in the morning shift and so on. It means they work alternatively with respect to their shifts. Thus they work on this pattern till the work is completed. On which day the work got completed?

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(a) 26th day (b) 27th day
(c) 28th day (d) 30th day

5. In the previous question what per cent of the time of the morning shift was utilised?
(a) 33.33% (b) 60%
(c) 100% (d) none of these

6. Working together B and C take 50% more number of days than A, B and C together take and A and B working together, take $\frac{8}{3}$ more number of days than A, B and C take together. If A, B and C all have worked together till the completion of the work and B has received Rs. 120 out of the total earning of Rs. 450, then in how many days did A, B and C together complete the whole work?
(a) 10 (b) 6
(c) 4 (d) 2

7. At Technosys Pvt Ltd. there are some engineering students employed as trainee engineers, belong to two eminent institutions of India. One group belong to MIT and another to NIT. Each student of MIT works for 10 hours a day till 60 days and each student of NIT works for 8 hours till 80 days on the two same projects. The ratio of number of students of MIT and that of NIT is 4 : 5 respectively. Students of which institution is slower in work and by how much?
(a) Each student of MIT is 20% less efficient than that of NIT
(b) Each student of NIT is 33.33% less efficient than that of MIT
(c) Each student of NIT is 25% less efficient than that of MIT
(d) Each student of MIT is 33.33% less efficient than that of NIT.

Directions for questions 8, 9 and 10 : At Arihant Prakashan every book goes through 3 phases (or stages) typing, composing and binding. There are 16 typists, 10 composers and 15 binders. A typist can type 8 books in each hour, a composer can compose 12 books in each hour and a binder can bind 12 books in each hour. All of the people at Arihant Prakashan works for 10 hours a day and each person is trained to do only the job of 1 category.

8. How many books can be prepared in each day?
(a) 1500 (b) 1200
(c) 1440 (d) 1380

9. If company has hired 12 more people, who can do any of the three jobs, then maximum how many books can be prepared in each day?
(a) 1500
(b) 1680
(c) 1800
(d) more than 2000

10. If company wanted to reduce the number of employees by 3, then from which category it reduces the number of employees without reducing the amount of production of books?
(a) It should reduce two binders and 1 typist
(b) It should reduce three binders only
(c) It should reduce 1 typist, 1 composer and 1 binder
(d) both (a) and (b) are possible

11. Bunty and Babli working together completed a job in 8 days. If Bunty worked twice efficiently as he actually did and Babli worked $\frac{1}{3}$ as efficiently as she actually did, the work would have been completed in 6 days. Find the time taken by Bunty to complete the job alone :
(a) 8 days (b) $\frac{38}{35}$ days
(c) $\frac{15}{2}$ days (d) $13\frac{1}{3}$ days

12. A single reservoir supplies the petrol to the whole city, while the reservoir is fed by a single pipeline filling the reservoir with the stream of uniform volume. When the reservoir is full and if 40,000 litres of petrol is used daily, the supply fails in 90 days. If 32,000 litres of petrol is used daily, it fails in 60 days. How much petrol can be used daily without the supply ever failing?
(a) 64000 litres (b) 56000 litres
(c) 78000 litres (d) 60000 litres

13. Railneer is packaged in a water bottling plant, with the help of two machines M_1 and M_2. M_1 and M_2 produces 400 and 600 bottles per minute. One day's production can be processed by M_1 operating alone for 9 hours, by M_2 operating alone for 6 hours or by both M_1 and M_2 operating simultaneously for 3 hours and 36 minutes. If one days production is processed by M_1 operating alone for $\frac{1}{3}$ of the time and M_1 and M_2 simultaneously operating for $\frac{2}{3}$ of the time, then in how many hours total production of one day will be completed?
(a) 2 (b) 3
(c) 4.5 (d) 4.8

Directions for questions 14 and 15 : A contractor employed a certain number of workers to finish constructing a road in a certain scheduled time. Sometime later, when a part of work had been completed, he realised that the work would get delayed by three-fourth of the scheduled time, so he at once doubled the number of workers and thus he managed to finish the road on the scheduled time.

14. How much work he had been completed, before increasing the number of workers?
(a) 10%
(b) $14\frac{2}{7}\%$
(c) 20%
(d) can't be determined

15. Some-time after the new workers were introduced, all of the newly introduced workers left the work due to heavy rain and the efficiency of the remaining workers reduced by 20% due to which the work finally got completed by delay of 60% of the scheduled time then how much work still remained incomplete by the end of the scheduled time?
(a) $17\frac{3}{5}\%$ (b) 21%
(c) $27\frac{5}{7}\%$ (d) 28%

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Directions for questions 16 to 20 : Mr. Stenley employed a certain number of typists for his project. 8 days later 20% of the typist left the job and it was found that it took as much time to complete the rest work from then as the entire work needed with all the employed typists. The average speed of a typist is 20 pages/hour.

16. How many typists left the work?
(a) 10 (b) 5
(c) 16 (d) can't be determined

17. Minimum how many typist could be employed?
(a) 10 (b) 5
(c) 15 (d) 4

18. What could be the number of typists remained at work when 20% of the employed typists left the job?
(a) 15 (b) 18
(c) 68 (d) 78

19. What is the actual number of days required, when it is done with actual work force, through out the completion :
(a) 32 days (b) 48 days
(c) 40 days (d) can't be determined

20. 16 days after the 20% typist left the job it was decided to complete the work on time by increasing the work-force again. By how much percentage increase in work-force is required?
(a) 100% (b) 50%
(c) 200% (d) none of these

21. Five tailors *A*, *B*, *C*, *D* and *E* stich 1800 shirts in 90 days working alternatively. Find the minimum possible number of shirts that can be stiched in a single day by working together :
(a) 100 (b) 20
(c) 50 (d) 4

Directions for question number 22, 23 : 8 men and 5 women working 6 hours a day can complete a work in 4 days. Also 4 men and 5 women working for 8 hours a day can complete the same job in 5 days. Similarly 5 boys working 8 hours a day can complete the same job in 30 days.

22. If 4 men, 3 women and 4 boys worked together everyday for 5 hours, then in how many days they have completed the work?
(a) 3 (b) 4
(c) 8 (d) 6

23. If women and children (boys) can't be employed, then minimum how many men are required to complete the job in 6 days if the working hours per day cannot exceed 9?
(a) 4 (b) 5
(c) 6 (d) 7

24. Eklavya can do the 6 times the actual work in 36 days while Faizal can do the one-fourth of the original work in 3 days. In how many days will both working together complete the 3 times of the original work?
(a) 6 (b) 10
(c) 12 (d) 15

25. Progressive Company Pvt. Ltd. hired some employees in a fix pattern. On the first day it hired one person, on the second day one more joined him. On the third, fourth etc (*i.e.*, every next day) one more person increased in this group. The capacity of each person was same. The whole work was completed on the 24th day then out of total Rs. 5000, maximum how much a person had earned?
(a) Rs. 500 (b) Rs. 400
(c) Rs. 200 (d) Rs. 250

26. Two persons having different productivity of labour, working together can reap a field in 2 days. If one-third of the field was reaped by the first man and rest by the other one working alternatively took 4 days. How long did it take for the faster person to reap the whole field working alone?
(a) 12 (b) 8
(c) 6 (d) 3

27. The total number of men, women and children working in a factory is 18. They earn Rs. 4000 in a day. If the sum of the wages of all men, all women and all children is in the ratio of 18 : 10 : 12 and if the wages of an individual man, woman and child is in the ratio 6 : 5 : 3, then how much a woman earn in a day?
(a) Rs. 400 (b) Rs. 250
(c) Rs. 150 (d) Rs. 120

28. A group of workers was put on a job. From the second day onwards, one worker was withdrawn each day. The job was finished when the last worker was withdrawn. Had no worker been withdrawn at any stage, the group would have finished the job in 55% of the time. How many workers were there in the group?
(a) 50 (b) 40
(c) 45 (d) 10

29. Two workers undertake to do a job. The second worker started working 2 hours after the first. Five hours after the second worker has begun working there is still $\frac{9}{20}$ of the work to be done. When the assignment is completed, it turns out that first worker has done 60% of the work, while second worker has done rest of the work. How many hours would it take each one to do the whole job individually?
(a) 10 hours and 12 hours
(b) 15 hours and 10 hours
(c) 20 hours and 25 hours
(d) 18 hours and 20 hours

30. A group of men decided to do a job in 4 days. But since 20 men dropped out every day, the job completed at the end of the 7th day. How many men were there at the beginning?
(a) 240 (b) 140
(c) 280 (d) 150

31. Brahma, Vishnu and Mahesh are three friends with different productivity. Brahma working alone needs as much time as Vishnu and Mahesh working together, while Vishnu himself needs 8 hours more working alone than when he works with Mahesh. Brahma working alone needs 8 hours less than Vishnu needs working alone. In how much time Brahma, Vishnu and Mahesh working together can complete the job?
(a) 4 hours (b) 5 hours
(c) 6 hours (d) 8 hours

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32. Milinda takes $8\frac{1}{3}$ hours more when she works alone in comparison of when she works with Bill. While Bill takes $5\frac{1}{3}$ hours more when he work alone in comparison to the time, when he works with Milinda. How long it will take Bill to complete the work alone?

(a) 10 hours (b) 15 hours
(c) 18 hours (d) 12 hours

33. Pascal and Rascal are two workers. Working together they can complete the whole work in 10 hours. If the Pascal worked for 2.5 hours and Rascal worked for 8.5 hours, still there was half of the work to be done. In how many hours Pascal working alone, can complete the whole work?

(a) 24 hours (b) $17\frac{1}{7}$ hours
(c) 40 hours (d) can't be determined

34. Boston, Churchill and David are three workers, employed by a contractor. They completed the whole work in 10 days. Initially all of them worked together, but the last 60% of the work was completed by only Churchill and David together. Boston worked with Churchill and David only for initial two days then he left the work due to his poor health. Also Churchill takes 20% less time to finish the work alone than that of David working alone. If they were paid Rs. 3000 for the entire work, then what is the share of least efficient person?

(a) Rs. 900 (b) Rs. 1200
(c) Rs. 1000 (d) none of these

35. There are three boats B_1, B_2 and B_3 working together they carry 60 people in each trip. One day an early morning B_1 carried 50 people in few trips alone. When it stopped carrying the passengers B_2 and B_3 started carrying the people together. It took a total of 10 trips to carry 300 people by B_1, B_2 and B_3. It is known that each day on an average 300 people cross the river using only one of the 3 boats B_1, B_2 and B_3. How many trips it would take to B_1 to carry 150 passengers alone?

(a) 15 (b) 30
(c) 25 (d) 10

36. Three men and 5 women together can finish a job in 3 days. Working on the same job 3 women take 5 days more than the time required by 2 men. What is the ratio of efficiency of a man to a woman?

(a) 2 : 1 (b) 3 : 2
(c) 5 : 2 (d) 4 : 1

37. Henry and Ford are two different persons, but when they worked together, they complete it in 10 days. Had Henry worked at half of his efficiency and Ford at five times of his efficiency it would have taken them to finish the job in 50% of the scheduled time. In how many days Ford can complete the job working alone?

(a) 12 (b) 24
(c) 15 (d) 30

38. Anne, Benne and Cenne are three friends. Anne and Benne are twins. Benne takes 2 days more than Cenne to complete the work. If Anne started a work and 3 days later Benne joins him, then the work gets completed in 3 more days. Working together Anne, Benne and Cenne can complete thrice the original work in 6 days. In how many days Benne can complete twice the original work with double the efficiency working alone?

(a) 2 (b) 3
(c) 4 (d) 6

39. Three typists A, B and C working together 8 hours per day can type 900 pages in 20 days. In a day B types as many pages more than A as C types as many pages more than B. The number of pages typed by A in 4 hours equal to the number of pages typed by C in 1 hour. How many pages C types in each hour?

(a) 1 (b) 2
(c) 3 (d) 4

Directions for question number 40 and 41: Four pipes A, B, C and D can fill a cistern in 20, 25, 40 and 50 hours respectively.

40. The first pipe A was opened at 6:00 am, B at 8:00 am, C at 9:00 am and D at 10:00 am. When will the cistern be full?

(a) 4:18 pm (b) 3:09 pm
(c) 12:15 pm (d) 11:09 am

41. If A and B are opened as inlet pipe into the cistern and C and D are opened as outlet pipes from the cistern and all the four pipes are opened simultaneously, how many hours will it take to fill the cistern completely?

(a) 20 hours (b) $11\frac{1}{9}$ hours
(c) $22\frac{2}{9}$ hours (d) 45 hours

42. A tank is connected with four pipes A, B, C and D of which two are filling the tank and other two are emptying it. The time taken by A, B, C and D to finish their jobs are 10 hours, 15 hours, 20 hours and 30 hours respectively. All four pipes are opened. When the tank was empty, it took 12 hours to fill it completely. Which two are the outlet pipes?

(a) A and B (b) C and D
(c) A and C (d) B and D

43. Two pipes A and B can fill a tank in 24 hours and $\frac{120}{7}$ hours respectively. Harihar opened the pipes A and B to fill an empty tank and some times later he closed the taps A and B, when the tank was supposed to be full. After that it was found that the tank was emptied in 2.5 hours because an outlet pipe 'C' connected to the tank was open from the beginning. If Harihar closed the pipe C instead of closing pipes A and B the remaining tank would have been filled in :

(a) 2 hours (b) 8 hours
(c) 6 hours (d) 4 hours

44. Pipe A can fill a tank in 12 hours and pipe B can fill it in 15 hours, separately. A third pipe C can empty it in 20 hours. Initially pipe A was opened, after one hour pipe B was opened and then after 1 hour when pipe B was opened pipe C was also opened. In how many hours the tank will be full?

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(a) $9\frac{2}{3}$ hours (b) $6\frac{2}{3}$ hours
(c) 10 hours (d) none of these

Directions for question number 45 and 46: A tank has an inlet and outlet pipe. The inlet pipe fills the tank completely in 2 hours when the outlet pipe is plugged. The outlet pipe empties the tank completely in 6 hours when the inlet pipe is plugged.

45. If both pipes are opened simultaneously at a time when the tank was one-third filled, when will the tank fill thereafter?
(a) $\frac{3}{2}$ hours (b) $\frac{2}{3}$ hour
(c) 2 hours (d) $1\frac{2}{3}$ hours

46. If there is a leakage also which is capable of draining out the liquid from the tank at half of the rate of outlet pipe, then what is the time taken to fill the empty tank when both the pipes are opened?
(a) 3 hours (b) $3\frac{2}{3}$ hours
(c) 4 hours (d) none of these

47. An inlet pipe can fill a tank in 5 hours and an outlet pipe can empty the same tank in 36 hours, working individually. How many additional number of outlet pipes of the same capacity are required to be opened, so that the tank never over flows?
(a) 3 (b) 6
(c) 8 (d) 7

Directions for question number 48 and 49: In a public bathroom there are n taps 1, 2, 3... n. Tap 1 and tap 2 take equal time to fill the tank while tap 3 takes half the time taken by tap 2 and tap 4 takes half the time taken by tap 3. Similarly each next number of tap takes half the time taken by previous number of tap i.e., K^{th} tap takes half the time taken by $(K-1)^{th}$ tap.

48. If the 10th tap takes 2 hours to fill the tank alone then what is the ratio of efficiency of 8th tap and 12th tap, respectively?
(a) 4 : 1 (b) 5 : 3
(c) 16 : 1 (d) 1 : 16

49. If the 8th tap takes 80 hours to fill the tank then the 10th and 12th taps working together take how many hours to fill the tank?
(a) 2 hours (b) 4 hours
(c) 6 hours (d) none of these

50. Pipe *A* takes 3/4 of the times required by pipe *B* to fill the empty tank individually. When an outlet pipe *C* is also opened simultaneously with pipe *A* and pipe *B*, it takes 3/4 more time to fill the empty tank than it takes, when only pipe *A* and pipe *B* are opened together. If it takes to fill 33 hours when all the three pipes are opened simultaneously, then in what time pipe *C* can empty the full tank operating alone?
(a) 66 hours (b) 50 hours
(c) 44 hours (d) can't be determined

Directions for question number 51, 52 and 53: A contractor undertook a project to complete it, in 20 days which needed 5 workers to work continuously for all the days estimated. But before the start of the work the client wanted to complete it earlier than the scheduled time, so the contractor calculated that he needed to increase 5 additional men every 2 days to complete the work in the time the client wanted it :

51. How many men were working on the day the project was completed as per the date of client wanted to complete it?
(a) 5 (b) 10
(c) 20 (d) none of these

52. Find the number of days in which client wanted to complete his work.
(a) 15 (b) 10
(c) 8 (d) can't be determined

53. If the work was further increased by 50% but the contractor continues to increase the 5 workers on every 2 days then how many more days are required over the initial time specified by the client?
(a) 1 day (b) 2 days
(c) 5 days (d) none of these

54. A tank is connected with 8 pipes. Some of them are inlet pipes and rest work as outlet pipes. Each of the inlet pipe can fill the tank in 8 hours, individually, while each of those that empty the tank *i.e.*, outlet pipe, can empty it in 6 hours individually. If all the pipes are kept open when the tank is full, it will take exactly 6 hours for the tank to empty. How many of these are inlet pipes?
(a) 2 (b) 4
(c) 5 (d) 6

55. A tank has two inlet pipes which can fill the empty tank in 12 hours and 15 hours working alone and one outlet pipe which can empty the full tank in 8 hours working alone. The inlet pipes are kept open for all the time but the outlet pipe was opened after 2 hours for one hour and then again closed for 2 hours then once again opened for one hour. This pattern of outlet pipe continued till the tank got completely filled. In how many hours the tank has been filled, working on the given pattern?
(a) 8 hours 24 minutes (b) 10 hours 15 minutes
(c) 9 hours 10 minutes (d) 9 hours 6 minutes

Final Round

Directions for question number 1 to 5 : In a nut-bolt factory 180 workers are working for 6 hours a day. Out of 180 workers there are some men, some women and rest are boys. All the workers can produce either nut or bolt or both of them. A man can produce 60 nuts and 80 bolts in each hour and a woman can produce 30 nuts and 60 bolts per hour. A man is thrice as efficient as a boy and 3/2 times as efficient as a woman. Given that all men, all women and all boys produce equal number of articles of one kind (i.e., either nut or bolt) per hour.

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1. Working 6 hours a day, how many nuts they can produce with 52500 bolt in each day?
(a) 17500 (b) 26250
(c) 50850 (d) can't be determined

2. In how many hours can 15 men, 12 women and 8 boys can produce 12000 nuts and 8200 bolts?
(a) 3 (b) 5
(c) 6 (d) none of these

3. If 30 women and 50 boys can produce 5400 bolts in one hour then to produce equal number of nuts in one hour how many men are required?
(a) 30 (b) 40
(c) 50 (d) none of these

4. If the efficiency of each boy is doubled then what is the increase in production per hour?
(a) 100% (b) 50%
(c) 33.33% (d) none of these

5. If the manager of factory wanted equal number of men, women and boys in his factory but the efficiency of a man, woman and a boy remains constant, then the change in production is :
(a) increased by 10% (b) decreased by 10%
(c) increased by 8.33% (d) none of these

6. Ram Lal is a renowned packager of fruits in Varanasi. He packs 70 mangoes or 56 guavas every day working 7 hours per day. His wife also helps him. She packs 30 mangoes or 24 guavas working 6 hours per day. Ram Lal has to pack 3300 mangoes and 2400 guavas with the help of his wife. They work alternatively, each day 10 hours. His wife started packaging on the first day and works on every alternate days. Similarly Ram Lal started his work on second day and worked alternatively till the completion of the work. In how many days the work will be finished?
(a) 85 days (b) $85\frac{2}{5}$ days
(c) 84 days (d) none of these

Directions for question number 7 to 11 : *A company produces three products. The products are processed on 3 different machines. The time required to manufacture one unit of each the three products and the daily capacity of three machines are given in the table below :*

Machine	Time per unit (in min.)			Machine capacity (min./day)
	Product 1	Product 2	Product 3	
M_1	2	2	3	450
M_2	2	5	–	410
M_3	3	–	4	480

7. How many units of product 1 can be produced in one day?
(a) 160 (b) 205
(c) 225 (d) 64

8. If minimum 20 units of P_1 and 30 units of P_2 have to be produced, then what is the maximum units of P_3 that can be produced in a day?
(a) 116 (b) 105
(c) 205 (d) 220

Directions for questions 9, 10 and 11 : *Read the following additional data for question number 9, 10 and 11. The profit per unit for product 1, 2 and 3 is Rs. 3, Rs. 4 and Rs. 5.*

9. What combination of P_1, P_2 and P_3 will yield maximum profit, under the manufacturing constraints?
1. $P_1 - 25$, $P_2 - 50$, $P_3 - 100$
2. $P_1 - 20$, $P_2 - 60$, $P_3 - 80$
3. $P_1 - 100$, $P_2 - 0$, $P_3 - 50$
4. $P_1 - 0$, $P_2 - 80$, $P_3 - 100$
(a) 4 (b) 2
(c) 3 (d) 1

10. Which of the machine if it breaks down will affect profitability the least?
(a) Machine 1 (b) Machine 2
(c) Machine 3 (d) Machine 1 or 3

11. If no production of product 2 is scheduled today and it is decided to only produce one type of product today, then what is the maximum profits that can be had today?
(a) Rs. 480 (b) Rs. 750
(c) Rs. 600 (d) none of these

Directions for question number 12-15 : *Ready Tailoring Services is very well known in its quality and time bound services. The company (Ready Tailoring Services) received a large order for stiching military uniforms. It has two different orders to prepare the shirts one for Officers and second for Jawans (non-officers). It has three cutters who will cut the fabric. Six tailors who will do the stiching and 3 assistant to stich the buttons and iron the shirts. Each of these 12 persons will work for exactly 8 hours a day. Each of the Officers uniform requires 20 minutes for cutting the fabric, 1 hour for stiching and 20 minutes for stiching buttons and ironing the shirts. Whereas the Jawan's uniform requires 15 minutes for cutting the fabric and 60 minutes for stiching and 10 minutes for buttons and ironing.*

12. If the company has to supply 40 officers uniforms only and no other on a particular day, how many man-hours are utilised on that day?
(a) $33\frac{1}{3}$ hours (b) $66\frac{2}{3}$ hours
(c) 40 hours (d) 60 hours

13. If the number of tailors will be increased by 50% then maximum how many uniforms for officers can be completed in one day?
(a) 48 (b) 50
(c) 60 (d) 72

14. If the company can increase maximum 3 employees of any category then for which category should it hire to get maximum increase in production capacity, assuming that it needs to stich only officer's uniform :
(a) cutter (b) tailor
(c) assistant (d) can't be determined

15. If the company has to produce the shirts for only one category then of which category it can produce maximum number of uniforms?
(a) Officers (b) Jawans
(c) either (a) or (b) (d) none of these

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Answers

LEVEL-1

1 (c)	2. (b)	3. (c)	4. (a)	5. (d)	6. (b)	7. (b)	8. (c)	9. (b)	10. (c)
11. (b)	12. (c)	13. (a)	14. (a)	15. (b)	16. (b)	17. (a)	18. (a)	19. (b)	20. (a)
21. (a)	22. (c)	23. (d)	24. (c)	25. (b)	26. (b)	27. (a)	28. (c)	29. (c)	30. (c)
31. (b)	32. (a)	33. (b)	34. (d)	35. (a)	36. (c)	37. (c)	38. (b)	39. (b)	40. (d)
41. (a)	42. (b)	43. (b)	44. (c)	45. (b)	46. (a)	47. (a)	48. (c)	49. (a)	50. (a)
51. (c)	52. (a)	53. (d)	54. (b)	55. (a)	56. (c)	57. (b)	58. (a)	59. (d)	60. (a)
61. (a)	62. (c)	63. (b)	64. (d)	65. (c)	66. (b)	67. (b)	68. (a)	69. (b)	70. (b)
71. (b)	72. (c)	73. (d)	74. (a)	75. (c)	76. (d)	77. (b)	78. (b)	79. (b)	80. (c)
81. (b)	82. (a)	83. (a)	84. (a)	85. (b)	86. (a)	87. (b)	88. (d)	89. (c)	90. (c)
91. (d)	92. (c)	93. (a)	94. (b)	95. (a)	96. (c)	97. (a)	98. (b)	99. (c)	100. (a)

LEVEL-2

1 (a)	2. (c)	3. (c)	4. (b)	5. (c)	6. (c)	7. (c)	8. (b)	9. (c)	10. (d)
11. (d)	12. (b)	13. (c)	14. (b)	15. (c)	16. (d)	17. (b)	18. (c)	19. (c)	20. (d)
21. (a)	22. (c)	23. (b)	24. (c)	25. (b)	26. (d)	27. (b)	28. (d)	29. (c)	30. (b)
31. (c)	32. (d)	33. (b)	34. (c)	35. (a)	36. (c)	37. (d)	38. (d)	39. (c)	40. (b)
41. (c)	42. (b)	43. (b)	44. (a)	45. (c)	46. (c)	47. (d)	48. (d)	49. (b)	50. (c)
51. (c)	52. (c)	53. (b)	54. (b)	55. (c)					

FINAL ROUND

1 (c)	2. (b)	3. (a)	4. (c)	5. (c)	6. (c)	7. (a)	8. (b)	9. (d)	10. (b)
11. (c)	12. (b)	13. (d)	14. (b)	15. (c)					

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Hints & Solutions

LEVEL 1

1. Efficiency of $A = \frac{100}{12} = 8.33\%$

Efficiency of $B = \frac{100}{15} = 6.66\%$

Combined efficiency of A and $B = 8.33 + 6.66 = 15\%$

Number of days taken by A and B, when worked together

$$= \frac{100}{15} = 6\frac{10}{15} = 6\frac{2}{3} \text{ days}$$

NOTE Efficiency × Time period = Fixed amount of work

Also, in terms of percentage total work to be done is considered as 100% (in fraction it is 1), unless otherwise stated.

Alternatively : It can be done through unitary method, also.

2. Efficiency of $A = \frac{100}{10} = 10\%$

Efficiency of $B = \frac{100}{12} = 8.33\%$

Efficiency of $C = \frac{100}{15} = 6.66\%$

Combined efficiency of A, B and $C = 10 + 8.33 + 6.66 = 25\%$

$\therefore$ Required number of days, when A, B and C worked together

$$= \frac{100}{25} = 4 \text{ days}$$

3. Efficiency of $A = 20\%\left(= \frac{100}{5}\right)$

Efficiency of $B = 10\%\left(= \frac{100}{10}\right)$

Efficiency of A, B and $C = 50\%\left(= \frac{100}{2}\right)$

$\therefore$ Efficiency of C = (Efficiency of A, B and C) − (Efficiency of A and B)

$= (50) - (20 + 10) = 20\%$

$\therefore$ Number of days required by C to work alone $= \frac{100}{20} = 5$ days

Alternatively : Go through options and satisfy the values.

	A	B	C	$(A + B + C)$
Days →	5	10	5	2
Efficiency →	20%	10%	20%	50%

Consider option (c)

4. Efficiency of $A = 12.5\%$

Efficiency of $B = 6.25\%$

Efficiency of $C = 1.25\%$

Efficiency of $(A + B + C) = 20\%\ (= 12.5 + 6.25 + 1.25)$

Required number of days $= \frac{100}{20} = 5$ days

5. Efficiency of $A = 10\%$

Efficiency of $B = 6.66\%$

$\therefore$ Required number of days $= \frac{100}{16.66} = 6$ days

6. Efficiency of $A = \frac{100}{24} = 4.16\%$

Efficiency of $B = \frac{100}{30} = 3.33\%$

Efficiency of $(A + B + C) = \frac{100}{12} = 8.33\%$

$\therefore$ Efficiency of $C = (8.33) - (4.16 + 3.33) = 0.83\%$

$\therefore$ Number of days required by C to complete the work alone

$$= \frac{100}{0.83} = \frac{100}{5/6} = 120 \text{ days}$$

NOTE $\frac{5}{6} = 0.833$

7. Efficiency of $A = 10\%$

Efficiency of $B = 4.16\%$

Efficiency of $(A + B + C) = 16.66\%$

$\left\{\begin{array}{l} \frac{100}{10} = 10 \\ \frac{100}{24} = 4.16 \\ \frac{100}{6} = 16.66 \end{array}\right.$

Efficiency of $C = (16.66) - (10 + 4.16) = 2.5\%$

Number of days required by C alone to finish the work

$$= \frac{100}{2.5} = 40 \text{ days}$$

8. Efficiency of $A = 7.14\%$

Efficiency of $B = 4.76\%$

Efficiency of $A + B = 11.9\%$

$\therefore$ Number of days required by A and B, working together

$$= \frac{100}{11.9} = 8.4 \text{ days}$$

HINT You can see that there is only one option between 8 and 9 which is 8.4 hence (c) is the correct choice.

Explanation : For 8 days denominator should be 12.5 and for 9 days denominator should be almost 11.

Alternatively : One day's work of A and B

$$= \frac{1}{14} + \frac{1}{21} = \frac{5}{42}$$

$\therefore$ Required number of days $= \frac{42}{5} = 8.4$ days

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9. Efficiency of $A = 4.16\%$

Efficiency of $B = 1.6 \times 4.16 = 6.66\%$

$\therefore$ Number of days required by $B = \dfrac{100}{6.66} = 15$ days

Alternatively :

		A	B
Efficiency	$\rightarrow$	x	$1.6x$
Days	$\rightarrow$	$1.6k$	k

(Since number of days are inversely proportional to efficiency)

Now $1.6k = 24 \Rightarrow k = 15$ day

Alternatively : A's one day's work $= \dfrac{1}{24}$

$\therefore$ B's one day's work $= 160\%$ of $\dfrac{1}{24} = \dfrac{1}{15}$

$\therefore$ Required number of days $= 15$

10. Efficiency of $A + B = \dfrac{100}{14} = 7.14\%$

Again the ratio of efficiency of A and $B = 2:1$

$\therefore$ Efficiency of $A = \dfrac{2}{3} \times 7.14 = 4.76\%$

$\therefore$ required number of days by $A = \dfrac{100}{4.76} = 21$ days

NOTE Only choice (c) lies between 20 and 24.

Explanation : For 20 days denominator should be 5 and for 24 days, denominator should be 4.16 also the only choice (c) gives a value which is multiple of 7 and very close to the answer (as appears).

Alternatively : One day's work of

$$A \text{ and } B = \frac{1}{x} + \frac{1}{2x} = \frac{1}{14}$$

$$\Rightarrow \quad x = 21$$

Since A is twice efficient as B so A will take half of the days taken by B.

11. Efficiency of $A + B = 14.28\% \left(= \dfrac{100}{7}\right)$

Now, since the ratio of efficiency of A and B is $2:1$

So, the efficiency of $A = \dfrac{2}{3} \times 14.28 = 9.52\%$

$\therefore$ Number of days required by $A = 10.5$ days

HINT Only choice (b) is correct since there is no other value (In option) lies between 10 and 11.

Explanation : For 10 days, denominator should be 10 and for 11 days, denominator should be 9.09.

Alternatively : Let A takes x days and then B takes $2x$ days

then 1 day's work of A and $B = \dfrac{1}{x} + \dfrac{1}{2x} = \dfrac{1}{7}$

$$\Rightarrow \quad x = 10.5$$

Thus A takes 10.5 days.

12.

		A		B
Efficiency	$\rightarrow$	3	:	1
Days	$\rightarrow$	1	:	3

(Since number of days are inversely proportional to the efficiency)

Now if A requires x days, so B requires $3x$ days

$\therefore$ Difference of required days $(= 3x - x) = 2x = 10$

$$\Rightarrow \quad x = 5$$

Hence the number of days required by $B = 3x = 3 \times 5 = 15$ days

13.

		A		B
Efficiency	$\rightarrow$	2	:	1
Days	$\rightarrow$	1	:	2

$$\left(\text{Days} \propto \frac{1}{\text{Efficiency}}\right)$$

Now, let A requires x days, then B requires $2x$ days

$\therefore$ difference in number of days $(= 2x - x) = x = 6$

$$\Rightarrow \quad x = 6$$

$\therefore$ B requires $2x = 2 \times 6 = 12$ days

Alternatively : If A takes x days, then B takes $x + 6$ days

Now, A's 1 day's work $= \dfrac{1}{x}$

B's 1 day's work $= \dfrac{1}{x+6}$

$$\therefore \quad \frac{1/x}{\dfrac{1}{x+6}} = \frac{2}{1}$$

(Since A does twice the work as B does)

$$\Rightarrow \quad x = 6$$

$\therefore$ B takes $2x = 12$ days.

Alternatively : $\dfrac{1}{x} + \dfrac{1}{(x+6)} = \dfrac{1}{4}$

$\Rightarrow$ $x = 6$ and $2x = 12$ days (required by B)

Alternatively : Go through option and satisfy all the conditions.

14.

	A	:	B	:	C
Ratio of efficiency	3	:	1	:	2
Ratio of number of days	$\left\{\frac{1}{3}\right.$	:	$\frac{1}{1}$	:	$\left.\frac{1}{2}\right\}$
or	2	:	6	:	3

Hence, (a) is correct $\left[\because \text{Time} \propto \dfrac{1}{\text{Efficiency}}\right]$

15.

	Ajit		Bablu
Efficiency	3	:	1
No. of days	1	:	3

16.

	X		Y
Efficiency	5	:	1
No. of days	1	:	5

17. Efficiency of $A + B = \dfrac{100}{\frac{20}{3}} = 15\%$

	A		B
No. of days	4	:	5
Efficiency	5	:	4

$\therefore$ Efficiency of $B = \dfrac{4}{9} \times 15 = \dfrac{20}{3} = 6\dfrac{2}{3}\%$

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18. Efficiency of $A + B = 33.33\% \left(= \frac{100}{3}\right)$

Ratio of efficiency of A and $B = 3 : 1$

$\therefore$ Efficiency of $A = \frac{3}{4} \times 33.33 = 25\%$

$\therefore$ Number of days taken by $A = 4 \left(= \frac{100}{25}\right)$

Alternatively :

	A		B
Efficiency	3	:	1
No. of days	1 (x)	:	3 (x)

$\therefore$ Difference in days $= 2x = 8$

$\Rightarrow$ $x = 4$ and $3x = 12$

Therefore number of days taken by A, working alone = 4 days

Alternatively : $\frac{1}{x} + \frac{1}{(x+8)} = \frac{1}{3}$

Now, you can use the options, or solve the equations to get the value of x which is equal to 4.

19. Efficiency of $(A + B) = 12.5\%$
Efficiency of $(B + C) = 8.33\%$
Efficiency of $(C + A) = 12.5\%$
$\therefore$ Efficiency of $[(A + B) + (B + C) + (C + A)] = 33.33\%$
$\therefore$ Efficiency of $(A + B + C) = 16.66\%$
$\therefore$ Number of days required by A, B and C together = 6 days

$\left(= \frac{100}{16.66}\right)$

20. Efficiency of $(A + B) = 8.33\%$
Efficiency of $(B + C) = 6.66\%$
Efficiency of $(C + A) = 5.00\%$
$\therefore$ Efficiency of $A + B + C = \frac{1}{2}(8.33 + 6.66 + 5) = 10\%$
$\therefore$ Efficiency of C = Efficiency of $[(A + B + C) - (A + B)]$
$= (10 - 8.33) = 1.66\%$
$\therefore$ Number of days required by C alone $= \frac{100}{1.66} = 60$ days

21. Remember $\text{Time} \propto \frac{1}{\text{Efficiency}}$

Now,

		Ganga	:	Jamuna	:	Saraswati
Efficiency	→	3	:	1	:	2
No. of days	→	2	:	6	:	3

Again, efficiency of Ganga, Jamuna and Saraswati

$= \frac{100}{1} = 100\%$

$\therefore$ Efficiency of Ganga $= \frac{3}{6} \times 100 = 50\%$

Efficiency of Saraswati $= \frac{2}{6} \times 100 = 33.33\%$

Now, number of days taken by Ganga $= \frac{100}{50} = 2$

Number of days taken by Saraswati $= \frac{100}{33.33} = 3$

$\therefore$ Difference in number of days taken by Ganga and Saraswati
$= 3 - 2 = 1$ day

22.

Day	1	2	3	4	5	6	7	8
	A	A	A	$(A+B)$	$(A+B)$	$(A+B)$	$(A+B)$	$(A+B)$

$3 \times \frac{1}{12} = \frac{3}{12} = 25\%$ work completed

Remaining work $= 75\% = (100 - 25)$

Now, efficiency of $A + B = 15\% = (8.33 + 6.66)$

$\therefore$ Number of days required by $A + B = \frac{75}{15} = 5$

to complete rest work (= 75%)

23. '3 days before the completion of the work Raja left the work' means in last 3 days only Rani has worked alone.

So, in last 3 days worked done by Rani $= 3 \times \frac{1}{21} = \frac{1}{7}$

So, the rest $\left(1 - \frac{1}{7}\right) = \frac{6}{7}$ work was done by Raja and Rani both.

Number of days in which Raja and Rani worked together

$= \frac{6/7}{5/42} = \frac{36}{5} = 7\frac{1}{5}$ days

NOTE Work done by Raja and Rani in one day $= \frac{1}{14} + \frac{1}{21} = \frac{5}{42}$

$\left(\text{Also, number of days} = \frac{\text{Total work to be done}}{\text{Work to be done in one day}}\right)$

24. Work done in 11 days $= \frac{11}{30}$

Rest work $= \frac{19}{30}$

1 day's work of $A = \frac{19/30}{28} = \frac{19}{30 \times 28} = \frac{19}{840}$

Total number of days required to complete the whole work alone

$= \frac{1}{19/840} = \frac{840}{19} = 44\frac{4}{19}$ days

25. Efficiency of Sonu $= 5\% = \left(\frac{100}{20}\right)$

Rest work = 75%

$\therefore$ Efficiency of Abhijeet $= \frac{75}{10} = 7.5\%$

$\therefore$ Combined efficiency of Sonu and Abhijeet
$= 12.5\% = (7.5 + 5)$

$\therefore$ Number of days required by Sonu and Abhijeet, to work together

$= \frac{100}{12.5} = 8$ days

26.

	Asha	Usha
Efficiency	5	4
No. of days	$4x$	$5x = 25$

$\therefore$ Number of days required by Asha to finish the work alone
$= 20$ $\quad (\because 4x = 4 \times 5)$

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(Alternatively, from percentage change graphic, number of days taken by Asha will be 20% less than Usha, if efficiency of Asha is 25% more than Usha)

Now, since Asha and Usha did work together for last 5 days

$= 5 \times 9 = 45\%$

(Since efficiency of Asha = 5% and Usha's efficiency = 4%)

It means Asha completed 55% work alone

∴ Number of days taken by Asha to complete 55% work

$= \frac{55}{5} = 11$ days

27. Krishna's efficiency = 10%

Mohan's efficiency = 5%

Work done by Krishna and Mohan together in 3 days

$= 15 \times 3 = 45\%$

Now, number of days in which B completed rest (55%) work alone $= \frac{55}{5} = 11$

Total number of days in which B worked = 3 + 11 = 14

Now number of days required by B, when A and B both worked together $= \frac{100}{15} = 6\frac{2}{3}$

∴ Required difference in number of days $= (11) - \left(6\frac{2}{3}\right)$

$= \frac{13}{3} = 4\frac{1}{3}$ days

28. Efficiency of Kareena and Karishma

$= 11.11 + 5.55 = 16.66\%$

Work done in 3 days $= 3 \times 16.66 = 50\%$

Rest work done by Kareena, Karishma and Shahid $= \frac{50}{50} = 1$ day

(Since efficiency of Shahid = 33.33%)

Thus in 4 (= 3 + 1) days they have completed the work.

29. Efficiency of Kavita = 5%

Efficiency of Babita = 1.66%

Efficiency of Samita = 3.33%

Work done in 5 days by $K + B + S = 5 \times (10) = 50\%$

Work done in 3 days by $K + B = 3 \times (6.66) = 20\%$

Remaining work (30%) done by Kavita alone $= \frac{30}{5} = 6$ days

30.

	A	:	B	:	C
Efficiency →	10	:	9	:	6
No. of days →	$9x$	:	$10x$	:	$15x$

Now, $15x - 9x = 6$

$\Rightarrow x = 1$

∴ Number of days taken by $A = 9$

Number of days taken by $B = 10$

Number of days taken by $C = 15$

Work done by B and C in initial 2 days $= \frac{2 \times 1}{6} = \frac{1}{3}$

rest work $= \frac{2}{3}$

Number of days required by A to finish $\frac{2}{3}$ work $= \frac{2/3}{1/9} = 6$ days

31.

	Anand	Bahuguna
No. of days	45	40
Efficiency	$2.22\%\left(=\frac{1}{45}\right)$	$2.5\%\left(=\frac{1}{40}\right)$

Anand did the work in 56 days $= 56 \times \frac{1}{45 \times 2} = \frac{28}{45}$

∴ Rest work $\left(\frac{17}{45}\right)$ was done by Anand and Bahuguna

$= \frac{17/45}{17/360} = 8$ days

(Since Anand and Bahuguna do the work in one day $= \frac{1}{45} + \frac{1}{40} = \frac{17}{360}$)

32. Efficiency of Chandni = 11.11%

Efficiency of Divakar = 8.33%

They do 19.44% work in 2 days

∴ They need 10 days to do 97.22% work,

Now the rest work (2.78) was done by Chandni in $\frac{2.78}{11.11} = \frac{1}{4}$ day

Therefore total number of days required $= 10 + \frac{1}{4} = 10\frac{1}{4}$ days

Alternatively : Chandni's one day's work $= \frac{1}{9}$

Divakar's one day's work $= \frac{1}{12}$

Chandni's and Divakar's (1 + 1) = 2 day's work $= \frac{1}{9} + \frac{1}{12} = \frac{7}{36}$

So, in 10 days they do $= \frac{7 \times 5}{36} = \frac{35}{36}$ work

So, the remaining $\frac{1}{36}\left(=1 - \frac{35}{36}\right)$ work will be done by Chandni $= \frac{1/36}{1/9} = \frac{1}{4}$ day

Thus total number of required days $= 10 + \frac{1}{4} = 10\frac{1}{4}$ days

HINT

Day	1	2	3	4	5	6	7	8	9	10	11
Turn	C	D	C	D	C	D	C	D	C	D	C
Work	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{36}$
	7/36		7/36		7/36		7/36		7/36		
	35/36										1/36

33.

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14
	F	Z	F	Z	F	Z	F	Z	F	Z	F	Z	F	Z
	$\frac{1}{12}$	$\frac{1}{15}$	$\frac{1}{12}$	$\frac{1}{15}$	$\frac{1}{12}$	$\frac{1}{15}$	$\frac{1}{12}$	$\frac{1}{15}$	$\frac{1}{12}$	$\frac{1}{15}$	$\frac{1}{12}$	$\frac{1}{15}$	$\frac{1}{12}$	$\frac{1}{60}$
	9/60		9/60		9/60		9/60		9/60		9/60			
	54/60												6/60	

In two days Fatima and Zahira do $= \frac{1}{12} + \frac{1}{15} = \frac{9}{60}$ work. In 12 days they do $= \frac{6 \times 9}{60} = \frac{54}{60}$ work

So, the remaining work $= \frac{6}{60} = \frac{1}{10}$

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Now, $\frac{1}{12}$ work will be done by Fatima in 13th day

So, the remaining work $= \frac{1}{10} - \frac{1}{12} = \frac{1}{60}$

This $\frac{1}{60}$ work will be done by Zahira $= \frac{1/60}{1/15} = \frac{1}{4}$ day

Thus the total number of days required

$$= 12 + 1 + \frac{1}{4} = 13\frac{1}{4} \text{ days}$$

Alternatively : A's efficiency = 8.33%, B's efficiency = 6.66%] 15%

Thus in 12 days 90% work will be done and in 13th day 8.33% more work will be done so the rest work 1.66% will be done by Zahira

$$= \frac{1.66}{6.66} = \frac{1}{4} \text{ day}$$

Thus, total number of required days $= 12 + 1 + \frac{1}{4} = 13\frac{1}{4}$

34. The difference will arise only on the last two days only. The work done on 13th day $= \frac{1}{15}$

$$\text{rest work} = \frac{1}{10} - \frac{1}{15} = \frac{1}{30}$$

This rest work $\left(\frac{1}{30}\right)$ will be done by Fatima $= \frac{1/30}{1/12} = \frac{2}{5}$ day

Thus total number of required days $= 12 + 1 + \frac{2}{5} = 13\frac{2}{5}$ days.

35.

1	2	3	4	5	6	7	8
A	B	C	A	B	C	A	B
1/6	1/12	1/8	1/6	1/12	1/8	1/6	1/12
3/8			3/8			1/4	

$$\frac{6}{8} = \frac{3}{4}$$

In 3 days A, B, C do $\frac{3}{8}$ work

In 6 days A, B, C do $\frac{3}{4}$ work

Rest work $= \frac{1}{4}$, which is less than $\frac{3}{8}$

On the 7th day, $\frac{1}{6}$ more work will be done by A

Now rest work $= \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$

Now, this rest work $\left(\frac{1}{12}\right)$ will be done by B in 1 complete day.

Thus, total number of days $= 6 + 1 + 1 = 8$ days

Alternatively : Efficiency of $A = 16.66\%$

Efficiency of $B = 8.33\%$

Efficiency of $C = 12.5\%$

Efficiency of $A + B = 25\%$

Efficiency of $A + B + C = 37.5\%$

In 3 days A, B, C completes 37.5% work

In 6 days A, B, C completes 75% work

Rest work = 25%

This 25% work will be completed by A and B in next 2 days

Thus total $6 + 2 = 8$ days are needed.

36. From the previous solution 75% work will be completed in 6 days. In the next two days (*i.e.*, on 7th and 8th day) B and C will complete 20.83% (12.5 + 8.33) more work.

So the remaining work = 4.16%

This 4.16% work will be completed by $A = \frac{4.16}{16.66} = \frac{1}{4}$ day

So, the total number of required days $= 6 + 2 + \frac{1}{4} = 8\frac{1}{4}$ days.

37. Let B takes x days to complete the work individually.

Then, the B's 1 day's work $= \frac{1}{x}$

A's 1 day's work $= \frac{1}{x-6}$

C's 1 day's work $= \frac{1}{x-8}$

$$\therefore \quad \frac{1}{x-6} + \frac{1}{x} = \frac{1}{x-8}$$

Now either solve the equation or satisfy the equation from the choices given in the question.

Thus option (c) is correct.

Alternatively : Select an appropriate option and then solve through % efficiency.

38. A's share = Rs. 250

B's share = Rs. 100

It means the ratio of efficiency of $A : B = 250 : 100 = 5 : 2$

$\therefore$ Ratio of days taken by A and $B = 2x : 5x$

Now, $5x - 2x = 9 \Rightarrow x = 3$

$\therefore$ Number of days taken by $A = 6$ (efficiency = 16.66%)

Number of days taken by $B = 15$ (efficiency = 6.66%)

Therefore number of days taken by A and B, working together

$$= \frac{100}{23.33} = \frac{300}{70} = 4\frac{2}{7} \text{ days}$$

39. Alen's one day's work $= \frac{1}{21}$

Border's one day's work $= \frac{1}{42}$

(working alternatively) Alen and Border's two days work

$$= \frac{1}{21} + \frac{1}{42} = \frac{1}{14}$$

So, Alen and Border do $\frac{1}{14}$ work in 2 days

So, they complete the work in $14 \times 2 = 28$ days.

40. Efficiency of A and $B = 16.66\%$

Efficiency of B and $C = 10\%$

But efficiency of A is twice that of C

$\because$ Therefore, $A = 2C$

Now $A + B = 16.66$ and $B + C = 10$

$\Rightarrow \quad 2C + B = 16.66 \quad \ldots(1)$

and $\quad C + B = 10 \quad \ldots(2)$

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$\therefore$ from eq. (1) and (2) $C = 6.66\%$

$A = 13.33\%$

$\therefore$

$\therefore$ Number of days taken by A to complete the work alone

$$= \frac{100}{13.33} = 7\frac{1}{2} \text{ days}$$

41. $A + B = 70\%$

$B + C = 50\%$ $\quad [\because A + B + B + C - (A + B + C) = B$

$70 + 50 - 100 = 20\%]$

$\Rightarrow \quad B = 20\%$

$A = 50\%$

and $\quad C = 30\%$

Hence, A is most efficient.

42. $$C + M = \frac{8}{17}$$

$$M + G = \frac{12}{17}$$

$$\Rightarrow \quad M = \frac{3}{17}$$

$[\because (C + M + M + G) - (C + M + G) = M]$

$$\left[\left(\frac{8}{17} + \frac{12}{17}\right) - 1 = \frac{3}{17}\right]$$

$$\therefore \quad C = \frac{5}{17}$$

$$G = \frac{9}{17}$$

So the whole amount will be distributed in the ratio of 5 : 3 : 9 among C, M and G respectively.

Now since M is least efficient so he get his own share

$$= \frac{3}{17} \times 816 = \text{Rs. } 144$$

43.

	Sharma	Kelkar
Efficiency	$0.8x$	x
Number of day	k	$0.8k = 24$

$\because \quad 0.8k = 24 \Rightarrow k = 30$

Thus Sharma requires 30 days, to complete the work, alone.

44.

	Man	Days	
$\frac{1}{5}\uparrow$	30	24	$\frac{1}{6}\downarrow$
	36	20	

Applying product constancy method.

For a constant work when days are reduced by $\frac{1}{6}$, then number of men is increased by $\frac{1}{5}$. Hence 6 men will increase.

Alternatively : $24 \times 30 = 20 \times x$

$\Rightarrow \quad x = 36$

Therefore 6 more men are required.

45.

	Women	Days	
$\frac{1}{3}\downarrow$	12	20	$\frac{1}{2}\uparrow$
	8	30	

Thus 10 more days are required.

Alternatively : $12 \times 20 = 8 \times x$

$\Rightarrow \quad x = 30$

$\therefore$ 10 more days are required.

46.

	Boys	Days	
$\frac{2}{5}\downarrow$	35	15	$\frac{2}{3}\uparrow$
	21	25	

Thus 14 boys did not turn up for the job.

Alternatively : $35 \times 15 = 25 \times x$

$\Rightarrow \quad x = 21$

$\therefore$ $35 - 21 = 14$ boys did not turn up for the job.

47. $x \times 32 = 24 \times 40$ $\quad (M_1D_1 = M_2D_2)$

48. $$M \times D = W$$

$$16 \times 6 = \frac{1}{3}W$$

$$\text{Rest work} = \frac{2}{3}W$$

For double work in same time we need double men. So 16 more men are required.

Alternatively : $2 \times (16 \times 6) = 6 \times M$

$\Rightarrow \quad M = 32$

$\therefore$ 16 more men are required.

49. $M \times D = 10 \times 20 = 200$ Man-days

New Man-days $= (20 \times 2) \times x$

$$200 = 20 \times 2 \times x$$

$$x = 5 \text{ days}$$

or $\quad M_1D_1 = M_2D_2$

$$10 \times 20 = (20 \times 2) \times x$$

$\Rightarrow \quad x = 5$

50. $$M_1D_1 = M_2D_2$$

$$M_1 \times 20 = (M_1 - 12) \times 32$$

$\Rightarrow \quad M_1 = 32$

Also, using the above concept you can go through options.

51. Go through options. Consider option (c)

$$30 \times 20 = 30 \times 6 + 21 \times 20$$

$600 = 600$, hence presumed option is correct.

Alternatively : $30 \times 20 = 30 \times x + 21 \times (26 - x)$

$\Rightarrow \quad x = 6$

52. Go through options. Consider choice (a)

$25 \times 30 = 25 \times 15 + 75 \times 5$

$750 = 750$, hence choice (a) is correct.

Alternatively : $25 \times 30 = 25 \times x + 75 \times (20 - x)$

$\left[\because 30 \times \frac{2}{3} = 20\right]$

53. Go through options. Consider choice (d)

$$36 \times 120 = 40 \times 108$$

$$4320 = 4320$$

Hence, choice (d) is correct.

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Alternatively :

	Men	Days	
$\frac{1}{9}\uparrow$	x	120	$\left(\frac{1}{10}\right)\downarrow$
	$\downarrow$	$\downarrow$	
	$\left(x\times\frac{10}{9}\right)$	(12)	

From percentage change (product constancy) graphic when number of days are decreased by $\frac{1}{10}$ then the number of men are increased by $\frac{1}{9}$ and $\frac{1}{9}$ is equivalent to 4 men so the actual number of men are $9\times4=36$.

Alternatively : $M_1\ D_1 = M_2\ D_2$

$$x\times120=(x+4)\times108$$

$$\Rightarrow \quad x=36$$

54. Efficiency (per minute) of Modi = 4 copies/min

Efficiency of Modi and Xerox together = 10 pages/min

∴ Efficiency of Xerox alone = 10 − 4 = 6 pages/min

∴ Mr. Xerox needs 6 minutes to copy 36 pages.

55. $$\text{Work done}=\frac{1}{3}$$

$$\text{Remaining work}=\frac{2}{3}$$

$$2\times(20\times12)=12\times x$$

$$\Rightarrow \quad x=40$$

So, 20 men will be increased.

56. $$\text{Work done}=\frac{1}{5}$$

$$\text{remaining work}=\frac{4}{5}$$

$$\therefore \quad 4\,(20\times75)=40\times x$$

$$x=150$$

Therefore 75 men should be increased.

57. Work done by 6 men = Work done by 10 women.

$\Rightarrow$ Work done by 1 man = work done by $\frac{10}{6}=\frac{5}{3}$ women

$$\therefore \quad 12\text{ men}+5\text{ women}=12\times\frac{5}{3}+5=25\text{ women}$$

$$\therefore \quad W_1\times D_1=W_2\times D_2,\quad W=\text{women},\ D=\text{days}$$

$$10\times15=25\times D_2$$

$$D_2=6$$

58. Work done by 2 men = 3 women = 4 boys

$$\therefore \quad 1\text{ man}=2\text{ boys}$$

$$1\text{ woman}=\frac{4}{3}\text{ boys}$$

$$\therefore \quad \text{boys}\times\text{days}=4\times52\ \text{(boys-days)}$$

Again 1 man + 1 woman + 1 boy $=2+\frac{4}{3}+1=\frac{13}{3}$ boys

$$B_1\times D_1=B_2\times D_2,\quad B=\text{boys},\ D=\text{days}$$

$$4\times52=\frac{13}{3}\times D_2$$

$$D_2=48\text{ days}$$

59. 2 men = 7 boys $\Rightarrow$ 1 man $=\frac{7}{2}$ days

5 women = 7 boys $\Rightarrow$ 1 woman $=\frac{7}{5}$ boys

7 men + 5 women + 2 boys $=7\times\frac{7}{2}+5\times\frac{7}{5}+2=\frac{67}{2}$ boys

Now, $$B_1\times D_1=B_2\times D_2$$

$$7\times469=\frac{67}{2}\times D_2$$

$$\Rightarrow \quad D_2=98\text{ days}$$

60. $$6C+2M=6\text{ days}$$

$$\Rightarrow \quad 36C+12M=1\text{ days}$$

Again $$1M=2C$$

$$\therefore \quad 36+12\times2=1\text{ day}$$

60 children can do the work in 1 day

Now, 5 men = 10 children

∴ 10 children can do the work in 6 days.

61. $$8M+12W=4\text{ days (whole work)}$$

$$\Rightarrow \quad 32M+48W=1\text{ day} \qquad \ldots(1)$$

Again $$6M+14W=5\text{ days}$$

$$\Rightarrow \quad 30M+70W=1\text{ day} \qquad \ldots(2)$$

From eq. (1) and (2)

$$32M+48W=30M+70W$$

$$\Rightarrow \quad 2M=22W$$

$$\Rightarrow \quad 1M=11W$$

Now, $$30M+70W=1\text{ day}$$

$$(30\times11+70)W=1\text{ day}$$

Therefore 400W requires 1 day to complete the whole work.

Thus 20W needs 20 days to complete the whole work.

62. Efficiency of A = 6.66%

Efficiency of B = 5.55%

Efficiency of $A+B+C$ = 16.66%

∴ Efficiency of C = 4.44%

Now, number of days required by

$$C=\frac{100}{4.44}=\frac{100}{(10\times4)/9}=\frac{45}{2}\text{ days}$$

63. Ratio of efficiencies of A, B and C = 6 : 5 : 4

∴ Share of $C=\frac{4}{15}\times27000=7200$

64. In 1 hour 314 weavers weave = 6594 × 6 shawls

In 1 hour 1 weaver weaves $=\frac{6594\times6}{314}$ shawls = 126 shawls

65. $$3M+2W=4\text{ days}$$

$$\Rightarrow \quad 12M+8W=1\text{ day} \qquad \ldots(1)$$

Again, $$2M+3W=5D$$

$$\Rightarrow \quad 10M+15W=1D \qquad \ldots(2)$$

From eq. (1) and (2)

$$12M+8W=10M+15W$$

$$\Rightarrow \quad 2M=7W$$

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$$\Rightarrow \quad \frac{M}{W} = \frac{7}{2}$$

Since the ratio of efficiency of women : men = 2 : 7
So, the amount of a man per day = Rs. 154

$$\left[\left(\frac{7}{2} \times 44\right) = 154\right]$$

66. Go through options.
Alternatively :

$$30 \times 40 = 30 \times x + 20 \times (46 - x)$$

$$\Rightarrow \quad x = 28 \text{ days}$$

67. Women × hours = 8 × 10 = 80 women hours
Now, women × hours = 12 × 8 = 96 women hours
Since new work force is 20% greater than previous work force.
So, the new quantity of tea leaves will be increased by 20% which is equal to $200 \times \frac{20}{100} = 40$ kg, hence (b)

68. New work = 3 × 450 man-day

$$\therefore \quad 3 \times 450 = 27 \times x$$

$$x = 50 \text{ days}$$

69.

$$4B + 5G = 10$$

$$\Rightarrow \quad 40B + 50G = 1 \qquad \ldots(1)$$

Again,

$$6B + 6G = 7$$

$$\Rightarrow \quad 42B + 42G = 1 \qquad \ldots(2)$$

Comparing eq. (1) and (2)

$$40B + 50G = 42B + 42G$$

$$\Rightarrow \quad 2B = 8G \quad \Rightarrow \quad 1B = 4G$$

Now, $(42 \times 4 + 42)$ girls = 1 day
210 girls can do a work in 1 day
Again $2B + 7G = 15$ girls
So, 15 girls require $\frac{210}{15} = 14$ days

70. Equate the man-days
For 20 km road, 20 × 20 = 400 man-days are required
∴ For 40 km road 800 man-days are required
So,

$$800 = 40 \times x$$

$$\Rightarrow \quad x = 20 \text{ days}$$

71.

$$\frac{3}{4} \times (x - 2)\, x = (x + 7)(x - 10)$$

$$\Rightarrow \quad x^2 - 6x - 280 = 0$$

$$\Rightarrow \quad x = 20 \quad \text{and} \quad x = -14$$

So, the acceptable values is $x = 20$
∴ Total work = $(x - 2) \times x = 18 \times 20 = 360$ unit
Now

$$360 = 30 \times k \qquad \because (30 = 20 + 10)$$

$$\Rightarrow \quad k = 12 \text{ days}$$

72. Efficiency of a man : woman : girl = 6 : 3 : 1

$$\therefore \text{ Share of a woman and girl} = \frac{(3+1)}{(6+3+1)} \times 10{,}000$$

$$= \frac{4}{10} \times 10000 = \text{Rs. } 4000$$

73. Total work = 33 × 30 = 990 man-days

$$\text{First day's work} = \frac{1}{990} \times 44$$

$$\text{Second day's work} = \frac{1}{990} \times 43$$

$$\text{Third day's work} = \frac{1}{990} \times 42 \text{ and so on}$$

$$\text{So, the total work in 44 days} = \frac{1}{990}(44 + 43 + 42 + \ldots)$$

$$= \frac{1}{990} \times \frac{44 \times 45}{2} = 1$$

Hence in 44 days total work will be completed.

74. Efficiency of Abhishek = 2.5% = $\left(\frac{100}{40}\right)$

Work done in 8 days = 20%
Rest work = 80%

$$\text{Efficiency of Bacchhan} = \frac{80}{24} = 3.33\%$$

$$\therefore \text{ Required number of days} = \frac{100}{(2.5 + 3.33)}$$

$$= \frac{100}{5.83} = \frac{100 \times 6}{35} = 17\frac{1}{7}$$

HINT $0.83 = \frac{5}{6}$, so, $5.83 = 5 + \frac{5}{6} = \frac{35}{6}$

75.

	A		B		C
Efficiency	3	:	2	:	6
Number of days	2	:	3	:	1

∴ Number of days taken by A = 12
Number of days taken by B = 18
Number of days taken by C = 6

$$1 \text{ day's work of } (A + B) = \frac{5}{36}$$

$$1 \text{ day's work of } (B + C) = \frac{8}{36}$$

$$1 \text{ day's work of } (C + A) = \frac{9}{36}$$

Day →	1	2	3	4	5	6
Work →	5/36	8/36	9/36	5/36	8/36	
	35/36					1/36

$$\text{In 5 days total work done} = \frac{35}{36}$$

Now, the rest work (1/36), which is done by AC.
∴ Number of days taken by AC for the rest work

$$= \frac{1/36}{9/36} = \frac{1}{9}$$

Therefore, total time $= 5 + \frac{1}{9} = 5\frac{1}{9}$ days

76. Efficiency of A + B = 16.66
Efficiency of B + J = 10
We have no further relevent informations, so we cannot determine.

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77.

	A	B
Efficiency	x	$2x$
Number of days	15	15/2

$$A\text{'s one day's work} = \frac{1}{15}$$

$$B\text{'s one day's work} = \frac{2}{15}$$

$$(A+B)\text{'s one day's work} = \frac{3}{15}$$

Now, let us assume B joined A after $(11-x)$ days, then

$$\frac{(11-x)}{15} + \frac{x \times 3}{15} = 1$$

$$\Rightarrow \quad (11-x) + 3x = 15$$

$$\Rightarrow \quad x = 2$$

It means they worked together for 2 days.

Alternatively: Go through options and check easily with the percentage efficiency.

$$9 \times 6.66 + 2 \times 20 = 100\%$$

78. Efficiency of A = 6.66%

Efficiency of B = 3.33%

Efficiency of C = 2.5%

Work done in last two days (only C do it) $= 2 \times 2.5 = 5\%$

Work done in the 3rd and 4th day from the last day (only A and C do it)

$$= 2 \times 9.16 = 18.33\%$$

Remaining work $= 100 - (5 + 18.33) = 76.66\%$

This 76.66% work was done by all of A, B and C.

$$\therefore \text{ Number of days taken by them} = \frac{76.66}{12.5} = \frac{460}{75} = 6\frac{2}{15}$$

$$\therefore \text{ Total time required} = 6\frac{2}{15} + 2 + 2 = 10\frac{2}{15} \text{ days}$$

79. Let x kg of oil is used for eating purpose, daily, then

$$(x + 11) \times 50 = (x + 15) \times 45$$

$$x = 25$$

$$\therefore \quad \text{Total quantity of oil} = (25 + 11) \times 50 = 1800$$

$$\therefore \quad \text{Required number of days} = \frac{1800}{25} = 72 \text{ days}$$

80. Work done $= \frac{2}{3}$

Remaining work $= \frac{1}{3}$, which is half of $\frac{2}{3}$

$$\therefore \quad \frac{1}{2} \times (20 \times 32) = 8 \times x$$

$$\Rightarrow \quad x = 40 \text{ men}$$

Therefore, 20 more men were required.

81. One day's work of 7 Indian with 4 Chinese $= \frac{1}{5}$

and one day's work of 7 Japanese and 3 Chinese $= \frac{1}{7}$

Therefore, one day's work of 7 Indian, 7 Chinese and 7 Japanese

$$= \frac{1}{5} + \frac{1}{7} = \frac{12}{35}$$

Therefore, one day's work of 1 Indian, 1 Chinese and 1 Japanese

$$= \frac{12}{35} \times \frac{1}{7} = \frac{12}{35 \times 7}$$

Therefore, number of days required by 1 Indian, 1 Chinese and 1 Japanese

$$= \frac{1}{12/(35 \times 7)} = \frac{35 \times 7}{12} = 20\frac{5}{12} \text{ days}$$

82. In 1 minute A, B and C bind $= \frac{1}{8} + \frac{1}{12} + \frac{1}{16} = \frac{13}{48}$ book

In 12 hours A, B and C bind $= 12 \times 60 \times \frac{13}{48} = 195$ books

$\therefore$ Average number of books bind by each $= \frac{195}{3} = 65$ books

83. Work of a man for 1 hour $= \frac{3}{2}$ women's work for 1 hour

Again, work of a man for 1 day

$$= \left(\frac{3}{2} \times \frac{9}{7.5}\right) \text{ women's work for 1 hour}$$

$\Rightarrow$ Work of a man for 1 day $= \frac{9}{5}$ women's work for 1 day

$$\Rightarrow \quad 1 \text{ man} = \frac{9}{5} \text{ women}$$

$$\therefore \quad 10 \text{ men} + 6 \text{ women} = 10 \times \frac{9}{5} + 6 = 24 \text{ women}$$

$$\therefore \quad 10 \text{ men} + 9 \text{ women} = 10 \times \frac{9}{5} + 9 = 27 \text{ women}$$

Now, $\quad D_1 \times W_1 = D_2 \times W_2$

$$18 \times 24 = D_2 \times 27$$

$$\Rightarrow \quad D_2 = 16 \text{ days}$$

84. Number of days taken by A to complete work alone = 14 days

Number of days taken by B to complete work alone = 7 days

Number of days taken by C to complete work alone = 7 days

$$\text{One day's work of } A \text{ and } B = \frac{1}{14} + \frac{1}{7} = \frac{3}{14}$$

$$\text{and one day's work of } A, B \text{ and } C = \frac{1}{14} + \frac{1}{7} + \frac{1}{7} = \frac{5}{14}$$

$$3 \text{ day's work of } A \text{ and } B = 3 \times \frac{3}{14} = \frac{9}{14}$$

$$\text{remaining work} = \frac{5}{14} \qquad \left(1 - \frac{9}{14}\right)$$

This remaining work will be done by A, B and C

$$= \frac{5/14}{5/14} = 1 \text{ day}$$

85. A's 5 days work = 50%

B's 5 days work = 33.33%

C's 2 days work = 16.66% $\qquad [100 - (50 + 33.33)]$

Ratio of contribution of work of A, B and $C = 50 : 33\frac{1}{3} : 16\frac{2}{3}$

$$= 3 : 2 : 1$$

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A's total share = Rs. 1500

B's total share = Rs. 1000

C's total share = Rs. 500

A's one day's earning = Rs. 300

B's one day's earning = Rs. 200

and C's one day's earning = Rs. 250

86. Efficiency of A = 50% $\left(\frac{100}{2}\right)$

Efficiency of B = 37.5% $\left(\frac{100}{8/3}\right)$

Efficiency of C = 12.5% $\left(\frac{100}{8}\right)$

Combined efficiency of A, B and C = 100%

So, they complete the work in 1 hour.

87.

	A	+	B	=	C	+	D
	↓		↓		↓		↓
Ratio of efficiency	$10x$	+	$5x$		$9x$	+	$6x$
	$15x$				$15x$		

Therefore, ratio of efficiency of $A : C = 10 : 9$

Therefore, ratio of days taken by $A : C = 9 : 10$

Therefore, number of days taken by A = 18 days

88. Efficiency of 4 men and 2 boys = 20%

Efficiency of 3 women and 4 boys = 20%

Efficiency of 2 men and 3 women = 20%

∴ Efficiency of 6 men, 6 women and 6 boys = 60%

∴ Efficiency of 1 man, 1 woman and 1 boy = 10%

Now, since they will work at double their efficiency

∴ Efficiency of 1 man, 1 woman and 1 boy = 20%

∴ Required number of days = 5

89. $M_1 \times D_1 = M_2 \times D_2$

$$m \times r = (m + n) \times D_2$$

$$D_2 = \frac{mr}{(m + n)}$$

90. Efficiency of A = 2.77%

Efficiency of B = 2.22%

∴ Combined efficiency of A and B = 5%

$= (2.77 + 2.22)$

Thus, it will take total of 20 minutes $\left(\because 20 = \frac{100}{5}\right)$

91. Efficiency of A = 8.33%

Effective efficiency = 6.66%, when there is leakage

∴ Efficiency of leakage = 1.66% $= (8.33 - 6.66)$

It means due to leakage a full tank will be empty in 60 hours.

92. Efficiency of $A + B = 10 + 6.66 = 16.66\%$

Efficiency of $A + B + C$ = 5.55%

∴ Efficiency of C (outlet pipe) = 16.66 − 5.55

= 11.11%

It means outlet pipe C can empty in 9 hours.

93. Efficiency of only leakage = 16.66%

Effective efficiency of leakage = 6.66%

It means the capacity of filling pipe = 10%

Therefore, the inlet pipe can fill the tank in 10 hours hence the capacity of tank = 100 l.

94. Efficiency of tap A and B = 16.66% = (10 + 6.66)

$$\therefore \quad 16.66x + 10 \times (8 - x) = 100\%$$

$$\Rightarrow \quad x = 3$$

95. Efficiency of A = 5%

Efficiency of B = 4%

Efficiency of C = − 3.33%

It means in every 3 consecutive hours tops A, B and C can fill 5.66% (= 5 + 4 − 3.33)

Therefore in 51 hours (= 3 × 17) tops A, B and C can fill 96.33% (= 5.66 × 17)

∴ the remaining part *i.e.*, 3.66% (= 100 − 96.33) can be filled up by A in $\frac{11}{15}$ hours $\left(= \frac{3.66}{5}\right)$, since it is now A's turn.

Hence, the total time required = $51 + \frac{11}{15} = 51\frac{11}{15}$

97. Efficiency when both pipes used to fill = $A + B$

and efficiency when pipe A is used to fill and pipe B is used to empty the tank = $A - B$

$$\therefore \quad \frac{A + B}{A - B} = \frac{2}{1}$$

$$\Rightarrow \quad \frac{A}{B} = \frac{3}{1} \qquad \text{(By componendo and dividendo)}$$

Thus, the ratio of efficiency of pipe A and B = 3 : 1.

98. **In ideal case:**

Time taken to fill the half tank by A and $B = \frac{50}{41.66} = \frac{6}{5}$ hours

Time taken by A, B and C to fill rest half of the tank

$$= \frac{50}{16.66} = 3 \text{ hours}$$

Total time = $\frac{6}{5} + 3 = 4$ hours 12 minutes

In second case:

Time taken to fill $\frac{3}{4}$ tank by A and $B = \frac{75}{41.66} = \frac{9}{5}$ hours

Time taken by A, B and C to fill rest $\frac{1}{4}$ tank $= \frac{25}{16.66} = \frac{3}{2}$ hours

Total time = $\frac{9}{5} + \frac{3}{2} = 3$ hours 18 minutes

Therefore, difference in time = 54 minutes

99. Time taken by pipes A and B to fill the whole tank = $\frac{100}{16.66}$

= 6 hours

Capacity filled in 2 hours by pipes A, B and C = 2 × 13.33

= 26.66%

Remaining capacity = 73.33%

This remaining capacity can be filled by A and B

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$$= \frac{73.33}{16.66} = 4\frac{2}{5} \text{ hours}$$

So, the total time required $= 2 + 4\frac{2}{5} = 6$ hours 24 minutes

Thus, in this case 24 minutes extra are required.

100. In one hour pipe A can fill $= \frac{1}{30}$ part of the tank

In one hour pipe B can fill $= \frac{1}{45}$ part of the tank

In two hour pipes A and B can fill $= \frac{1}{18}$ part of the tank

Therefore in 36 hours the tank will be completely filled.

Alternatively: Efficiency of pipe $A = 3.33\%$

Efficiency of pipe $B = 2.22\%$

and Combined efficiency $= 5.55\%$

Therefore in 2 hours pipe A and B fill 5.55%.

Thus to fill 100% tank, these pipe will take 36 hours.

LEVEL (2)

1. LCM of 2, 3 and 4 = 12

In 12 hours A will make 6 shawls
" " " B will make 4 shawls
" " " C will make 3 shawls
} 13 shawls

i.e., in 12 hours they will weave 13 shawls

so, in 84 hours they will weave 91 shawls

Now, in 9 hours A will make 4 shawls
in 9 hours B will make 3 shawls
in 9 hours C will make 2 shawls
} 9 shawls

So, they will complete 100 shawls in 93 hours.

NOTE Since, they cannot share each-others work so B will take completely 9 hours to make 3 shawls, even when A and C stay idle for the last 1 hour till B completes his own work.

2.

	M	T	W	Th	F	S
	A	S	A	S	A	S
Work →	15%		15%		7.5%	

37.5%

{This pattern continued for total 2 weeks only till 75% work got completed.}

Thus in 2 weeks they will complete 75% work.

Now 15% of the remaining (25% of the work) will be done in the third week in Monday and Tuesday. Again 10% work remained undone. Out of this 8.33% work will be done by Arun on Wednesday and remaining 1.66% work will be completed on Thursday by Satyam.

Final week

M	T	W	Th
A	S	A	S
15%		8.33%	1.66%

25%

3. Efficiency of Kaushalya = 5%

Efficiency of Kaikeyi = 4%

Thus, in 10 days working together they will complete only 90% of the work.

$$[(5+4)\times 10] = 90$$

Hence, the remaining work will surely done by Sumitra, which is 10%.

Thus, Sumitra will get 10% of Rs. 700, which is Rs. 70.

✍ ***Solutions for question number 4 and 5:***

	1st day	*2nd day*
Morning shift)	*A*	*B*
Evening shift)	*B*	*A*

It is clear that in two days finally they work very similar to the alternate days i.e., finally A work for 10 hours and B also works for 10 hours.

Thus in every two days they will complete 7.5% work. So, in 26 days they will complete 97.5% of the total work.

Now, the remaining work = 2.5%

Now, this is the turn of A,

Since A does 4.16% work in 10 hours.

So, he will do 2.5% work in 6 hours.

4. Thus, the work will be finished on 27th day.

5. Since, A does 2.5% work in 6 hours, which is the actual duration of morning shift. So 100% time of the morning shift was utilised.

6. **From the first statement :**

	$(B+C)$	$(A+B+C)$
Efficiency	$2x$	$3x$
Days	$3y$	$2y$

Thus, we can say that efficiency of A is $\frac{1}{3}$ the efficiency of $(A+B+C)$.

From the last statement:

$$\text{Share of } B \text{ out of total amount} = \frac{120}{450} = \frac{4}{15}$$

From these two results we can conclude that:

		A	:	B	:	C
Ratio of efficiency		$\frac{5}{15}\left(=\frac{1}{3}\right)$	:	$\frac{4}{15}$	:	$\frac{6}{15}$
	$\Rightarrow$	5	:	4	:	6
Ratio of number of days	$\Rightarrow$	$\frac{1}{5}$	:	$\frac{1}{4}$	:	$\frac{1}{6}$
	$\Rightarrow$	$12x$	:	$15x$	:	$10x$

One day's work of A and $B = \frac{1}{12x} + \frac{1}{15x} = \frac{9}{60x}$

$\therefore$ A and B will take $\frac{60x}{9}$ days to complete the whole work,

Again one day's work of A, B and C

$$= \frac{1}{12x} + \frac{1}{15x} + \frac{1}{10x} = \frac{15}{60x}$$

$\therefore$ A, B and C working together complete the work in $\frac{60x}{15}$ days

$$\therefore \quad \frac{60x}{9} - \frac{60x}{15} = \frac{8}{3}$$

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(Since, A and B take $\frac{8}{3}$ days more than A, B and C)

$$\Rightarrow \quad x = 1$$

$\therefore$ Number of days required to complete the whole work by A, B and $C = 4x = 4 \times 1 = 4$ days

Alternatively: You can solve through option.

Alternatively: Ratio of efficiencies of A, B and $C = 5x : 4x : 6x$

$\therefore$ Number of days required by A and $B = \frac{100}{9x}$ ...(1)

and number of days required by A, B and $C = \frac{100}{15x}$ ...(2)

$$\therefore \quad \frac{100}{9x} - \frac{100}{15x} = \frac{8}{3} \Rightarrow x = \frac{5}{3}$$

From eq. (2) number of days required by A, B and C working together

$$= \frac{100}{15x} = \frac{100}{15 \times \frac{5}{3}} = \frac{100}{25} = 4 \text{ days}$$

7.

MIT	NIT
$4 \times 10 \times 60 \times E_1$	$= 5 \times 8 \times 80 \times E_2$

$$\Rightarrow \quad \frac{E_1}{E_2} = \frac{4}{3}$$

where E_1 and E_2 are the respective working efficiencies per hour.

$\therefore$ Each engineer from NIT is 25% less efficient than each engineer from MIT.

8.

T	C	B	
16	10	15	
8	12	12	
128	120	180	← in one hour
1280	1200	1800	← in 10 hour

Since, restriction is imposed by composers *i.e.*, since only 1200 books can be composed in 10 hours so not more than 1200 books can be finally prepared.

9. To maximise the production we locate 5 persons for composing and 7 persons for typing. Only then we can maximise our production which is 1800 books per day.

T	C	B
(16 + 7)	(10 + 5)	15
8	12	12
184	180	180
1840	1800	1800

10. **1st case**

T	C	B
15	10	13
8	12	12
120	120	156
1200	1200	1560

No change in critical value

2nd case

T	C	B
16	10	12
8	12	12
128	120	144

No change in critical value*

*Critical value means the minimum amount of job which creates the restriction.

NOTE In third case it will reduce the production below 1200 books per day (or 120 books per hour)

So, option (d) is correct.

11. Efficiency of Bunty and Babli (jointly) = 12.5%

Now, go through options and satisfy the conditions.

Consider option (d).

	Bunty		Babli
Efficiency	7.5%	→	5%
Days	$\frac{40}{3}$		20

Now, the new efficiency of Bunty = 15%

and the new efficiency of Babli = $\frac{5}{3}$%

$\therefore$ Combined efficiency = $\frac{50}{3}$%

$\therefore$ Number of days taken by them = $\frac{100}{50/3} = 6$ days

Hence, the presumed option (d) is correct.

NOTE Without solving the complete problem we can say that only option (d) is true since other 3 options gives the efficiency of Bunty equal to or more than 12.5% which is inadmissible *i.e.*, cannot be equal to or greater than the combined efficiency of both persons together.

12. Let x litre be the per day filling and v litre be the capacity of the reservoir, then

$$90x + v = 40000 \times 90 \quad \text{...(1)}$$

and $$60x + v = 32000 \times 60 \quad \text{...(2)}$$

Solving eq. (1) and (2), we get

$$x = 56000$$

Hence, 56000 litres per day can be used without the failure of supply.

13. One day's production = $400 \times 9 \times 60 = 2160000$ bottles per day

Ratio of time utilised by M_1 and $(M_1 + M_2) = 1 : 2$

Now, the production of bottles by M_1 in 1 minute = 400

and the production of bottles by M_1 and M_2 together in 2 minutes = 2000

Thus total 2400 bottles can be processed in 3 minutes

$\therefore$ 216000 bottles can be processed in

$$216000 \times \frac{3}{2400} = 4.5 \text{ hours}$$

14. Let he initially employed x workers which works for D days and he estimated 100 days for the whole work and then he doubled the worker for $(100 - D)$ days.

$$D \times x + (100 - D) \times 2x = 175x$$

$$\Rightarrow \quad D = 25 \text{ days}$$

NOTE $175 = 100 + \frac{3}{4} \times 100$, since required number of days are 75% $\left(i.e., \frac{3}{4}\right)$ more than the estimated number of days.

Now, the work done in 25 days = $25x$

Total work = $175x$

$\therefore$ Work done before increasing the number of workers

$$= \frac{25x}{175x} \times 100 = 14\frac{2}{7}\%$$

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15. **Note:** For easier calculation consider some convenient value of x (i.e., number of workers). Let initially there were 20 workers employed.

It means work done till 25 days

$$= 25 \times 20 = 500 \text{ man-days}$$

Now, since delay in works is 60%.

It means the work was completed in 160 days. Let the increased workers worked for k days then

$$40 \times k + 16 \times (135 - k) = 75 \times 40$$

$$\Rightarrow \quad k = 35 \text{ days}$$

$\therefore$ 40 workers work for only 35 days

Here 40 means twice the work force of 20

and 16 means 80% efficiency of the original work-force

and 135 means (160 – 25) days

and 75 means (100 – 25) days

Since, number of days are increased by 60 in which only 16 workers work.

$\therefore$ Remaining work after 100 days = $60 \times 16 = 960$

$$= \frac{960}{3500} \times 100 = 27\frac{5}{7}\%$$

Solutions for 16–20: *Let the actual number of typists be n required to work for D days, then*

$$nD = (n \times 8) + \left(\frac{4n}{5} \times D\right)$$

$$\Rightarrow \quad D = 40$$

16. Since, the above equation is independent of n (*i. e.*, number of typists) so cannot be calculated. It means there are many possible values.

17. Since 20% $\left(i.e., \frac{1}{5}\right)$ typists left the job. So, there can be any value which is multiple of 5 *i. e.*, whose 20% is always an integer. Hence, 5 is the least possible value.

18. The remaining value must be divisible by 4.

Since, $\frac{4x}{5} = k$ $\quad x \rightarrow$ actual number of typists

$$\Rightarrow \quad x = \frac{k \times 5}{4} \quad k \rightarrow \text{remaining number of typists}$$

So, k must be divisible by 4, since a person cannot be a fraction. Hence, option (c) could be the possible answer.

19. Since, $D = 40$

20. Work done in 8 days = $8x$

Work done in further 16 days = $16 \times \frac{4}{5}x = \frac{64x}{5}$

Remaining work = $40x - \left(8x + \frac{64x}{5}\right) = \frac{96x}{5}$

Remaining number of days = $40 - (8 + 16) = 16$

$\therefore$ New work-force = $\frac{96x/5}{16} = \frac{6x}{5}$

$\therefore$ Change (or increase) in work-force

$$= \frac{\left(\frac{6x}{5}\right) - x}{x} \times 100 = 20\%$$

21. Let A, B, C, D and E stiches a, b, c, d and e shirts per day then

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} + \frac{1}{e} = \frac{90}{1800} = \frac{1}{20}$$

when $a = b = c = d = e$, we get the minimum value

$$\therefore \quad 5 \times \frac{1}{a} = \frac{1}{20}$$

$$\Rightarrow \quad a = 100 = b = c = d = e$$

22. $$(8M + 5W) \times 6 \times 4 = (4M + 5W) \times 8 \times 5$$

$$\Rightarrow \quad 4M = 10W$$

$$\Rightarrow \quad 1M = 2.5W$$

Now, substituting the value M by W, we get total work-done

Work = $(8 \times 2.5 + 5) \times 6 \times 4 = 600$ women-days-hours

$= 240$ man-days-hours

Again work = $5 \times 8 \times 30 = 1200$ boy-days-hours

$$\therefore \quad 1M = 2.5W = 5B$$

$$4M + 3W + 4B = 30 \text{ boys}$$

$\therefore$ Required number of days = $\frac{1200}{30 \times 5} = 8$ days

23. $$9 \times 6 \times M = 240$$

$$M = \frac{240}{6 \times 9} = 4.4 \text{ men}$$

Therefore, minimum 5 men are required.

24. Efficiency of Eklaya = 16.66%

Efficiency of Faizal = 8.33%

Total efficiency of Eklaya and Faizal = 25%

So, they can do actual work in 4 days

$\therefore$ 3 times work requires 12 days.

25. $1 + 2 + 3 + 4 + 5 + \ldots + 24 = \frac{24 \times 25}{2} = 300$

Total work = 300 man-days

But, the person who started the work on the first day works for 24 days. Hence, his share will be maximum which is equal to $\frac{24}{300} = \frac{2}{25}$.

Thus, he will receive $5000 \times \frac{2}{25} =$ Rs. 400.

26. Total efficiency of two persons = 50%

Ratio of efficiencies of first person to the second person = 1 : 2

Therefore, efficiency of second person = 33.33%

Hence, he will take 3 days to complete the work alone.

27. Ratio of number of men, women and children

$$= \frac{18}{6} : \frac{10}{5} : \frac{12}{3} = 3x : 2x : 4x$$

$$\therefore \quad (3x + 2x + 4x) = 18$$

$$\therefore \quad x = 2$$

Therefore, number of women = 4

Share of all women = $\frac{10}{40} \times 4000 =$ Rs. 1000

$(\because 18 + 10 + 12 = 40)$

$\therefore$ Share of each woman = $\frac{1000}{4} =$ Rs. 250

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28. It can be solved easily through option.

$$(10 + 9 + 8 + \ldots + 1) = 10 \times \left(10 \times \frac{55}{100}\right)$$

$$55 = 55$$ Hence correct.

Alternatively: $\frac{n(n+1)}{2} = n \times \frac{55n}{100}$

$$\Rightarrow \quad n = 10$$

In both cases total work is 55 man-days.

29. Go through option. Consider choice (c).

Efficiency of first worker = 5%

Efficiency of second worker = 4%

$\therefore$ In 7 hours first worker completed 35% work

In 5 hours second worker completed 20% work

Thus, work completed = 55%

Remaining work = 45%

Hence, one condition is satisfied.

Again, they will take 5 more hours to complete 45% work

$$\left[\frac{45}{4+5} = 5\right]$$

Thus, first person completes $7 \times 5 + 5 \times 5 = 60\%$ work

and second person completes $5 \times 4 + 5 \times 4 = 40\%$ work

Hence, second condition is also satisfied. Hence, correct option is (c).

30. Go through option

$$140 \times 4 = (140 + 120 + 100 + \ldots + 20)$$

$$560 = 560$$

Alternatively: Let n be the initial number of worker then

$$n \times 4 = n + (n-20) + (n-40) + \ldots + (n-120)$$

$$4n = 7n - 420$$

$$\Rightarrow \quad 3n = 420$$

$$\Rightarrow \quad n = 140 \text{ workers}$$

31. **From the first statement**

	B		$(V + M)$
Number of days	x	=	x
Efficiency	1	:	1

From the second statement

	V	$(V + M)$
Number of days →	$(K + 8)$	K

From the third statement

	B	V
Number of days →	$(n - 8)$	n

Now, go through option and consider option (c).

Efficiency of $(B + V + M) = 16.66\%$ (number of days = 6)

$\therefore$ Efficiency of $B = 8.33\%$ (number of days = 12)

and Efficiency of $(V + M) = 8.33\%$ (number of days = 12)

Therefore, B will take 12 days.

Now, from third statement V will take 20 days.

Hence, the efficiency of V is 5%.

Therefore, V will take 20 days. Hence, second statement is also true. Thus, the presumed option (c) is correct.

32. Consider option (d)

Time taken by Bill = 12 hours

Efficiency of Bill = 8.33%

Therefore, time taken by Bill and Milinda working together

$$= 6\frac{2}{3} \text{ hours}$$

Hence, the efficiency of Milinda and Bill = 15%

Therefore efficiency of Milinda = 15 – 8.33 = 6.66%

Thus, the number of days taken by Milinda = 15, which is $8\frac{1}{3}$ days more than when both work together.

Hence, all the conditions satisfied.

		M	$B + M$	B
Efficiency	→	6.66%	15%	8.33%
Number of days	→	15	$6\frac{2}{3}$	12

33. Efficiency of Pascal and Rascal = 10%

Pascal worked for 2.5 hours and Rascal worked separately 8.5 hours. Which means it can be considered that Pascal and Rascal worked together for 2.5 hours and Rascal worked alone for 6 hours.

Thus, Pascal and Rascal in 2.5 hours can complete 25% work.

It means the remaining (50 – 25) = 25% of the work was done by Rascal in 6 hours.

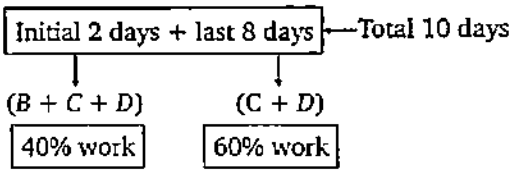
Therefore, Rascal can do 100% work in 24 hours. It means the efficiency of Rascal = 4.16%

Therefore, efficiency of Pascal = (10 – 4.16) = 5.83%

Thus, Pascal require $\frac{100}{5.83} = 17\frac{1}{7}$ hours to complete the work alone.

Alternatively: Go through option.

34.



From the above diagram it is clear that efficiency of C and D is 7.5%, since C and D complets 60% work in 8 days and efficiency of B, C and D is 20%. It means efficiency of B alone is 12.5% = (20 – 7.5).

Now

	C	:	D
Number of days	$4x$	:	$5x$
Efficiency	$5y$	:	$4y$

$\therefore$ Efficiency of $C = \frac{5}{9} \times 7.5 = 4.16\%$

and Efficiency of $D = \frac{4}{9} \times 7.5 = 3.33\%$

Thus, D is the least efficient person.

Now share of work done by David

$$(D) = 3.33\% \times 10 = 33.33\%$$

Hence, his share of amount = 33.33% of Rs. 3000 = Rs. 1000

35. Combined efficiency of all the three boats = 60 passenger/trip

Now, consider option (a).

15 trips and 150 passengers means efficiency of $B_1 = 10\frac{p}{t}$

which means in carrying 50 passengers B_1 must has taken 5 trips. So the rest trips equal to 5 (10 – 5 = 5) in which B_2 and B_3 together carried remaining 250 (300 – 50) passengers.

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Therefore the efficiency of B_2 and $B_3 = \frac{250}{5} = 50\frac{P}{t}$

Since, the combined efficiency of B_1, B_2 and B_3 is 60. Which is same as given in the first statement hence option (a) is correct.

Alternatively: It can be solved by framing quadratic equation.

36. Efficiency of 3 men + 5 women = 33.33%

Required number of days by 2 men = x

$\therefore$ Required number of days by 3 women = $x + 5$

Now, consider option (c).

Therefore, $3M + 5W = 3M + 2M = 5$ men

Therefore, efficiency of a man = 6.66%

Hence, a man needs 15 days to finish the job, working alone.

Again $3M + 5W = 7.5W + 5W = 12.5W$

Therefore, efficiency of a woman = 2.66%

Therefore, a woman needs 37.5 days.

Thus, 2 men needs 7.5 days to work alone $\because \left(7.5 = \frac{15}{2}\right)$

and 3 women needs 12.5 days to work alone $\because \left(12.5 = \frac{37.5}{3}\right)$

Hence, the difference in number of days = 5 which is same as given in the problem. Hence correct option is (c).

37. Efficiency of Henry and Ford (combined) = 10%

Consider option (d).

Efficiency of Ford = 3.33% (30 days)

Therefore, Efficiency of Henry = 6.66%

Now, the new efficiency of Ford = 16.66%

and the new efficiency of Henry = 3.33%

Therefore, newly combined efficiency of H and F = 20%

Therefore, required number of days by Henry and Ford working together = 5

Since 5 is half of 10, hence the option (d) is correct.

38. From the last statement:

Efficiency of Anne (A), Benne (B) and Cenne (C) = 50%

From the first statement: Number of days taken by B is 2 more than C.

From the second statement: Anne had worked for 6 days and Benne had worked for 3 days only. Now, consider option (d).

Number of days taken by B = 6,

Efficiency = 16.66%

It means Benne had completed $16.66 \times 3 = 50\%$ work in 3 days.

Therefore Anne had completed 50% work in 6 days.

Thus, the efficiency of Anne = $8.33\%\left(\frac{50}{6}\right)$

Hence, the efficiency of Cenne = $50 - (16.66 + 8.33) = 25\%$

Thus B takes 6 days ($\because$ efficiency = 16.66%)

and C takes 4 days ($\because$ efficiency = 25%)

which is true according to the first statement, hence option (d) is correct.

39. Number of pages typed by A, B and C together per day = 45

Now let the number of pages typed by B is x

then the number of pages typed by $A = x - d$

and the number of pages typed by $C = x + d$

$\Rightarrow (x - d) + (x) + (x + d) = 45$

$\Rightarrow x = 15$ pages per day.

Again let C types k pages per day then A types $\frac{k}{4}$ pages per day.

Therefore, the ratio of typing of pages per day of A and C = 1 : 4

$\therefore$ Number of pages typed by C in one day

$= \frac{4}{5} \times 30 = 24$ pages $(30 = 45 - 15)$

$\therefore$ Number of pages typed by C per hour $= \frac{24}{8} = 3$ pages/hour

40. Efficiency of P = 5%

Efficiency of Q = 4%

Efficiency of R = 2.5%

Efficiency of S = 2%

Till 10 am pipe P filled 20%
Till 10 am pipe Q filled 8%
Till 10 am pipe R filled 2.5%
} 30.5%

Thus, at 10 am pipe P, Q and R filled 30.5% of the cistern.

Now, the time taken by P, Q, R and S together to fill the remaining capacity of the cistern

$= \frac{69.5}{13.5} = \frac{139}{27} = 5$ hours and 9 minutes (approx)

Therefore, total time = 4 hours + 5 hours 9 minutes

= 9 hours and 9 minutes

It means cistern will be filled up at 3 : 09 pm

41. Efficiency of $P + Q$ = 9% (inlet pipes)

Efficiency of $R + S$ = 4.5% (outlet pipes)

Net efficiency = 4.5%

So, the time taken $= \frac{100}{4.5} = 22\frac{2}{9}$ hours

42. Efficiency of A = 10%

Efficiency of B = 6.66%

Efficiency of C = 5%

Efficiency of D = 3.33%

Efficiency of $A + B + C + D$ = 8.33 (time = 12 hours)

Now, go through options and consider A and B as inlet pipes and C and D as outlet pipes, then

$(10 + 6.66) - (5 + 3.33) = 8.33$

which is required hence it is certain that C and D are outlet pipes.

NOTE There is no any other such combination.

43. Efficiency of inlet pipe A = 4.16% $\left(\frac{100}{24}\right)$

Efficiency of inlet pipe B = 5.83% $\left(\frac{100}{120/7}\right)$

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$\therefore$ Efficiency of A and B together $= 10\%$ $\quad\therefore$ (time $= 10$ hours)

Now, if the efficiency of outlet pipe be $x\%$ then in 10 hours the capacity of tank which will be filled $= 10 \times (10 - x)$

Now, since this amount of water is being emptied by C at $x\%$ per hour, then

$$\frac{10 \times (10 - x)}{x} = 2.5 \text{ hours} \quad \Rightarrow \quad x = 8\%$$

Therefore in 10 hours 20% tank is filled only. Hence, the remaining 80% of the capacity will be filled by pipes A and B in $\frac{80}{10} = 8$ hours.

44. Efficiency of pipe $A = 8.33\%$

Efficiency of pipe $B = 6.67\%$

Efficiency of pipe $C = 5\%$

When tap C was opened pipe A filled 16.66% capacity

When tap C was opened pipe B filled 6.67% capacity

Therefore rest capacity of the tank to be filled

$$= 100 - 23.34 = 76.66\%$$

Now, the net efficiency of A, B and $C = 10\%$

Hence, pipes A, B and C will take $= \frac{76.66}{10} = 7.66 = 7\frac{2}{3}$ hours

$\therefore$ Total time $= 2$ hours $+ 7\frac{2}{3}$ hours $= 9\frac{2}{3}$ hours

45. Efficiency of inlet pipe $= 50\%$

Efficiency of outlet pipe $= 16.66\%$

Net efficiency of pipes A and $B = 33.33\%$

Capacity of tank to be filled up $= 66.66\%$

Hence, required time $= \frac{66.66}{33.33} = 2$ hours

46. Rate of leakage $= 8.33\%$ per hour

Net efficiency $= 50 - (16.66 + 8.33) = 25\%$

Time required $= \frac{100}{25} = 4$ hours

47. Since, an inlet pipe is 7.2 times efficient than an outlet pipe. Therefore, in order to tank never overflow we will need total 8 outlet pipes.

Thus we need only 7 more $(8 - 1 = 7)$ outlet pipes.

48. Time taken by 8th tap $= 2 \times 2 \times 2 = 8$ hours

and time taken by 12th tap $= 2 \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}$ hour

Ratio of time taken by 8th tap and 12th tap $= 8 : \frac{1}{2} = 16 : 1$

$\therefore$ Ratio of efficiencies of 8th tap and 12th tap $= 1 : 16$

49. Time taken by 10th tap $= 80 \times \frac{1}{2} \times \frac{1}{2} = 20$ hours

Time taken by 12th tap $= 80 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = 5$ hours

Thus 10th tap and 12th tap together will take 4 hours.

50. Let pipe A fill the tank in $3x$ hours then pipe B fill it in $4x$ hours.

Therefore, in 1 hour they will fill $= \frac{1}{3x} + \frac{1}{4x} = \frac{7}{12x}$

i.e., they will take $\frac{12x}{7}$ hours

When an inlet pipe C is also opened then it takes

$$= \frac{12x}{7} + \frac{12x}{7} \times \frac{3}{4}$$

$$= \frac{12x}{7}\left(\frac{7}{4}\right) = 3x \text{ hours}$$

Now, in one hour pipe A, B and C working together fill

$$= \frac{1}{3x} + \frac{1}{4x} - \frac{1}{C} = \frac{1}{3x}$$

$$\Rightarrow \quad \frac{7}{12x} - \frac{1}{C} = \frac{1}{3x}$$

$$\Rightarrow \quad \frac{1}{C} = \left(\frac{1}{12x} - \frac{1}{3x}\right)$$

$$\Rightarrow \quad C = 4x$$

Hence in $4x$ hours pipe C can empty the whole tank.

Now, since $\quad 3x = 33 \quad \Rightarrow \quad x = 11$

$\therefore \quad 4x = 4 \times 11 = 44$ hours

51. Total work $= 5 \times 20 = 100$ man-days.

Let the client needed to complete it in n days then

$$(5 \times 2) + (10 \times 2) + (15 \times 2) + (20 \times 2) = 100$$

Hence, in 8 days all the work will be completed as per the requirement of client.

And on the 8th day 20 men were working.

52. It requires total 8 days by adding the work force successively.

53. Total work $= 100 + 50 = 150$ man-days

In 8 days 100 man-days work has been completed. Now on 9th and 10th day there will be 25 workers. So in 2 days they will complete additional 50 man-days work. Thus the work requires 2 more days.

54. Let there be 'n' inlet pipes then there must be $(8 - n)$ outlet pipes.

Therefore $(8 - n)\frac{1}{6} - n \times \frac{1}{8} = \frac{1}{6} \quad \Rightarrow \quad n = 4$

Alternatively: $(8 - n)\,16.66 - n \times 12.5 = 16.66$

$$\Rightarrow \quad n = \frac{116.66}{29.16} = \frac{700}{175} = 4$$

Alternatively: It can be solve through options very easily.

55. Efficiency of two inlet pipes A and $B = 8.33 + 6.66 = 15\%$

Efficiency of two inlet pipes A and B along with an outlet pipe

$$C = 8.33 + 6.66 - 12.5 = 2.5\%$$

Hours	1	2	3	4	5	6	7	8	9	10
Efficiency →	15	15	2.5	15	15	2.5	15	15	2.5	
	32.5			32.5			32.5			
	97.5%									2.5%

In 9 hours 97.5% tank will be completely filled. On the 10th hour 2.5% (remaining capacity) will be fill by pipe A, B with its 15% efficiency.

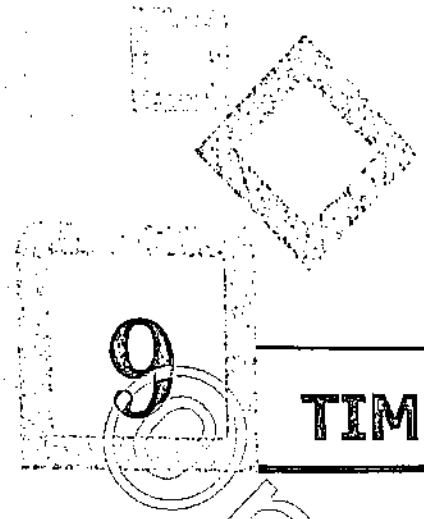
Thus, A will take $= \frac{2.5}{15} = \frac{1}{6}$ hour $= 10$ minutes

Therefore, total time $= 9$ hours and 10 minutes.



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9 TIME, SPEED AND DISTANCE

It is sure and certain that it is one of the most important chapter, which includes various fundamental and logical concepts and therefore most of the problems are complex in their nature. Every year 4-6 problems are generally asked in CAT. Besides CAT almost every aptitude exam contains the questions pertaining to the concepts of TSD.

This chapter includes the following :

(a) Motion in a straight line
(b) Circular motion and races
(c) Problems based on trains, boats, river and clocks etc.

CONCEPT OF MOTION

When a body moves from a point A to another point B at a distance of D, then it requires some time (T) to cover a distance (D) with a particular speed (S).

The relation between T, S and D is as follows:

$$T \times S = D$$

i.e., Time × Speed = Distance

Therefore, when D is constant,

$$T \propto \frac{1}{S}$$

and when T is constant,

$$D \propto S$$

and when S is constant,

$$D \propto T$$

NOTE This relation of proportionality is very important.

Formulae : Distance = Speed × Time

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

NOTE To solve the problem, all the units involved in the calculation must be uniform *i.e.*, either all of them be in metres and seconds or in kilometres and hours etc.

Conversion of unit:

$$1 \text{ km/h} = \frac{5}{18} \text{ m/s}$$

$$1 \text{ m/s} = \frac{18}{5} \text{ km/h}$$

[1 km = 1000 m, 1 h = 60 min, 1 min = 60 s]

HINT Try to find these relations by unitary method.

1 mile = 1609.30 m = 1.6093 km

and 1 km = 0.621 mile

and 1 yard = 0.9144 m

and 1 m = 1.0936 yards

and 1 m = 39.4 inches

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time taken}}$$

When distances are equal :

$$\text{Average speed} = \frac{2xy}{x+y} \quad \text{(speeds } x, y\text{)}$$

and $$\text{Average speed} = \frac{3xyz}{xy+yz+zx} \quad \text{(speeds } x, y, z\text{)}$$

EXAMPLE 1 *Abhishek drives his bike at the speed of 150 km/h. What is the distance covered by him in 3 hours.*

SOLUTION $D = S \times T = 150 \times 3 = 450$ km

EXAMPLE 2 *Udai travels half of his journey by train at the speed of 120 km/h and rest half by car at 80 km/h. What is the average speed?*

SOLUTION Let the total distance be $2D$ km, then

Total time = Time taken by train + Time taken by car

$$= \frac{D}{120} + \frac{D}{80}$$

$$\therefore \quad \text{Average speed} = \frac{2D}{\frac{D}{120} + \frac{D}{80}}$$

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$$\frac{2D}{D\left(\frac{1}{120}+\frac{1}{80}\right)} = \frac{2}{\left(\frac{2+3}{240}\right)} = 96 \text{ km/h}$$

Alternatively: Consider the distance in numerals (since it is independent of the distance) as the LCM of the speeds.

Then, the total distance = LCM of 120 and 80 = 240

Therefore, time taken by train $= \frac{120}{120} = 1$ h

and time taken by car $= \frac{120}{80} = \frac{3}{2}$ h

$\therefore$ Average speed $= \frac{\text{Total distance}}{\text{Total time taken}}$

$$= \frac{240}{1+\frac{3}{2}} = 96 \text{ km/h}$$

Alternatively: Average speed $= \frac{2(xy)}{(x+y)}$

where x and y are the speeds.

$\therefore$ Average speed $= \frac{2 \times 120 \times 80}{120+80} = 96$ km/h

NOTE This formula is applicable only when the distances travelled at the speed of x km/h and y km/h are same.

Alternatively: Solve through alligations.

EXAMPLE 3 *Speed of Karan is 40 km/h and speed of Arjun is 60 km/h. What is the ratio of time taken by each to cover the same distance?*

SOLUTION

	Karan		Arjun
Speed	40	:	60
$\Rightarrow$	2	:	3
$\therefore$ Time	3	:	2

Alternatively: Time taken $= \frac{1}{40} : \frac{1}{60} = 3 : 2$

NOTE For the constant distance covered, the ratio of time taken by each person is the ratio of the reciprocal of their speeds and *vice-versa*.

If the ratio of speeds of A, B, C, D be $a : b : c : d$, then the ratio of time taken by A, B, C and D to cover the same distance

$$= \frac{1}{a} : \frac{1}{b} : \frac{1}{c} : \frac{1}{d}$$

and if the time taken by A, B, C and D be $p : q : r : s$, then the ratio of their speeds

$$= \frac{1}{p} : \frac{1}{q} : \frac{1}{r} : \frac{1}{s}$$

EXAMPLE 4 *A car takes half of the time taken by truck to go from Lucknow to Bombay. A truck takes 20 hours to go for the same journey. What is the speed of truck, if the speed of car be 120 km/h.*

SOLUTION

	Car		Truck
Time $\rightarrow$	1	:	2
Speed $\rightarrow$	2	:	1

Therefore, speed of the truck = 60 km/h

EXAMPLE 5 *A cycle covers 75 km distance in 5 hours. What is the distance covered by the cycle in 6 hours?*

SOLUTION Speed $= \frac{\text{Distance}}{\text{Time}}$

$\therefore$ Speed $= \frac{75}{5} = 15$ km/h

Now, distance covered in 6 hours at the speed of 15 km/h

$= 15 \times 6 = 90$ km

EXAMPLE 6 *The distance of the college and the home of Rajeev is 80 km. One day he was late by 1 hour than the normal time to leave for the college, so he increased his speed by 4 km/h and thus he reached to college at the normal time. What is the changed (or increased) speed of Rajeev?*

(a) 28 km/h *(b) 30 km/h*
(c) 40 km/h *(d) 20 km/h*

SOLUTION Let the normal speed be x km/h, then

$$\frac{80}{x} - \frac{80}{(x+4)} = 1$$

$\Rightarrow$ $x^2 + 4x - 320 = 0$

$\Rightarrow$ $x(x+20) - 16(x+20) = 0$

$(x+20)(x-16) = 0$

$x = 16$ km/h

$\therefore$ $(x+4) = 20$ km/h

Therefore increased speed = 20 km/h

Alternatively: If t_1 and t_2 be the original and changed time and S_1 and S_2 be the original and changed speeds, then

$$S_1 \times S_2 = \frac{\text{Distance} \times (S_1 \sim S_2)}{(t_1 \sim t_2)}$$

Also, $$\boxed{t_1 \times t_2 = \frac{\text{Distance} \times (t_1 \sim t_2)}{(S_1 \sim S_2)}}$$

$\therefore$ $S_1 \times S_2 = \frac{80}{1} \times 4 = 320$

$\therefore$ $S_1 \times S_2 = 320$

or $S_1 \times (S_1 + 4) = 320$

$\Rightarrow$ $S_1 = 16$ and $S_2 = 20$

NOTE Here you need not to solve necessarily quadratic equation. You can just try and find two factors of 320 in such a way that (in this particular problem) one factor must be greater than the other factor by 4.

Alternatively: Go through options. Consider 20

$$t_2 = \frac{80}{20} = 4$$

$$t_1 = 4 + 1 = 5$$

$\therefore$ $S_1 = \frac{80}{5} = 16$

Since, S_1 is 4 km/h less than S_2. Hence, option (d) is correct.

Alternatively: If you find yourself comfortable then factorise 80 in two factors and check the correct combinations.

$$80 = \left[\begin{matrix} 1 \times 80 \\ 2 \times 40 \\ 4 \times 20 \\ 5 \times 16 \\ 8 \times 10 \end{matrix}\right] \begin{matrix} \\ \\ \text{(-1)} \\ \\ \\ \end{matrix} \text{+4}$$

So, the increased speed = 20 km/h.

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EXAMPLE 7 *Shweta when increases her speed from 24 km/h to 30 km/h she takes one hour less than the usual time to cover a certain distance. What is the distance usually covered by Shweta?*

(a) 160 km (b) 240 km (c) 120 km (d) 90 km

SOLUTION Let the original time be t hours, then

$$24t = 30 \times (t-1) = D \quad \text{(distance)}$$

then $t = 5$

$\therefore$ Distance $= 24 \times 5 = 120$ km

Alternatively: Go through options.

$\frac{120}{24} = 5$ h

$\frac{120}{30} = 4$ h $\Big\}$ 1 hour less

Hence, the option (c) is correct.

Alternatively: Since distance (D) is constant.

Therefore, $D = S_1 \times t_1 = S_2 \times t_2$

It means here we can apply product constancy

Speed	Time	
$\frac{1}{4}\uparrow$	$\frac{1}{5}\downarrow$	$= 1$ hour

$\therefore$ Original time taken $= 5 \times 1 = 5$ hours

Therefore, Distance = Original speed × Original time

$= 24 \times 5 = 120$ km

NOTE In the given exercise or in the whole chapter you have to solve maximum problems through product constancy concept described in the chapter of ratio proportion and variation. Solving through product constancy gives faster results.

Alternatively: Let the distance be D, then

$$\frac{D}{24} - \frac{D}{30} = 1$$

$\Rightarrow$ $D = 120$ km

EXAMPLE 8 *Kriplani goes to school at 20 km/h and reaches the school 4 minutes late. Next time, she goes at 25 km/h and reaches the school 2 minutes earlier than the scheduled time. What is the distance of her school?*

SOLUTION Increase in speed = 5 km/h

Decrease in time = 6 min (4 + 2)

By product constancy :

Speed	Time	
$\frac{1}{4}\uparrow$	$\frac{1}{5}\downarrow$	$= 6$ min

It means original time = 30 min $\left(\because \frac{x}{5} = 6 \Rightarrow x = 30\right)$

$\therefore$ Total distance = Original speed × Original time

$= 20 \times \frac{30}{60} = 10$ km

EXAMPLE 9 *Amit covers a certain distance with his own speed, but when he reduces his speed by 10 km/h his time duration for the journey increases by 40 hours, while if he increases his speed by 5 km/h from his original speed he takes 10 hours less than the original time taken. Find the distance covered by him.*

SOLUTION

$\left.\begin{matrix} S & & T \\ -10 & \times & +40 \end{matrix}\right\}$ $40S - 10T = |-400|$ ...(i)

$\left.\begin{matrix} S & & T \\ +5 & \times & -10 \end{matrix}\right\}$ $-10S + 5T = |-50|$ ...(ii)

Solving eq. (i) and (ii), we get

$S = 25$ and $T = 60$

$\therefore$ Distance $(D) = S \times T$

$= 25 \times 60 = 1500$ km

where $D \rightarrow$ Distance, $S \rightarrow$ Speed, $T \rightarrow$ Time

'+' means increase in value.

and '–' means decrease in value.

Alternatively: Let distance be x km and usual speed be y km/h.

$$\frac{x}{(y-10)} - \frac{x}{y} = 40$$

$$\Rightarrow \quad x\left[\frac{10}{y(y-10)}\right] = 40$$

$$\Rightarrow \quad x = 4y(y-10) \quad \text{...(i)}$$

and $$\frac{x}{y} - \frac{x}{(y+5)} = 10$$

$$\Rightarrow \quad x = 2y(y+5) \quad \text{...(ii)}$$

from eq. (i) and (ii)

$$4y(y-10) = 2y(y+5)$$

$$2y - 20 = y + 5$$

$$y = 25 \text{ km/h}$$

$$x = 1500 \text{ km}$$

EXAMPLE 10 *A train met with an accident 60 km away from Anantpur station. It completed the remaining journey at $\frac{5}{6}$th of the previous speed and reached the Baramula station 1 hour 12 min late. Had the accident taken place 60 km further, it would have been only 1 hour late.*

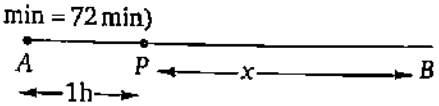
(a) What is the normal speed of the train?

(b) What is the distance between Anantpur and Baramula?

SOLUTION **Case I.** Since the speed is decreased by $\frac{1}{6}$. So, the time will be increased by 1/5, which is equal to 1 hour 12 minutes.

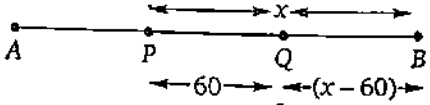
It means the normal time required for this remaining part (x) of the journey is 5×72 min $= 360$ min $= 6$ h.

($\because$ 1 h 12 min = 72 min)



P is the place of accident.

Case II. When accident is supposed to be happened at Q.



Since, the speed is decreased by $\frac{1}{6}$, hence, the time will be increased by $\frac{1}{5}$, which is equal to 1 hour, hence the normal time required for this remaining part $(x - 60)$ of journey $= 5 \times 1 = 5$ hours.

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Thus, it is clear that when the train runs 60 km of its normal speed it takes 1 hour less, which implies that in 1 hour the train can run 60 km with its normal speed. Thus, the normal speed of the train is 60 km/h.

(b) Since the train requires 6 hours at its normal speed of 60 km/h for the x km. Hence,

$$x = 6 \times 60 = 360 \text{ km}$$

Thus, the total distance = Distance travelled before accident + Distance travelled after accident

$$= 60 \times 1 + 60 \times 6 = 420 \text{ km}$$

EXAMPLE 11 *A and B started simultaneously towards each other from P and Q respectively. The distance between P and Q is 600 km and the ratio of speeds of A is to B is 5 : 7. If they meet at a point M:*

(i) Find the ratio of PM : QM.

(ii) Find the distance PM.

SOLUTION (i) Since the time is constant so, the distance covered by A and B is directly proportional to the speeds of A and B.

Hence, $PM : QM = 5 : 7$

(ii) Since, the ratio of their speeds (or their distances covered) is 5 : 7. Hence, A will cover $\frac{5}{12}$ of the total length.

$$\therefore \quad PM = \frac{5}{12} \times 600 = 250 \text{ km}$$

(Similarly $QM = \frac{7}{12} \times 600 = 350$ km)

RELATIVE MOTION WITH TWO OR MORE BODIES

(i) When two bodies move in the same direction: If the speeds of the two bodies A and B be S_A and S_B, then their relative speed $= S_A - S_B$ or $S_B - S_A$ *i.e.*, in the same direction the relative speed or effective speed between two bodies is the difference of their speeds. (The difference is always considered as positive)

(ii) When two bodies move in the opposite direction: If the speeds of the two bodies A and B be S_A and S_B, then their relative speed $= S_A + S_B$.

i.e., in the opposite direction the relative speed or effective speed between two bodies is the sum of their speeds.

EXAMPLE 12 *The distance between two places P and Q is 700 km. Two persons A and B started towards Q and P from P and Q simultaneously. The speed of A is 30 km/h and speed of B is 40 km/h. They meet at a point M which lies on the way from P to Q.*

(i) How long will they take to meet each other at M?

(ii) What is the ratio of PM : MQ?

(iii) What is the distance MQ?

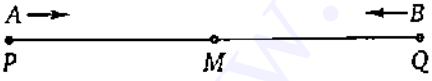
(iv) What is the extra time needed by A to reach at Q than to reach at P by B?

(v) What is the ratio of time taken by A and B to reach their respective destinations after meeting at M?

(vi) In how many hours will they be separated by only 560 km before meeting each other.

(vii) How long will it take to separate them by 280 km from each other when they cross M (time to be considered after their meeting)?

SOLUTION



(i) Since, they are coming towards each other from opposite ends, therefore the relative speed will be the sum of their speeds = 30 + 40 = 70 km/h.

Thus, the required time to meet at M

= Time required to cover 700 km (combined)

$$= \frac{700}{70} = 10 \text{ h}$$

Thus in 10 hours they will meet each other at M.

(ii) The ratio of their distances covered to meet at M

= Ratio of their speeds = 3 : 4

(Since, time is constant *i. e.*, same for each)

Thus $PM : MQ = 3 : 4$

(iii) $MQ = \frac{4}{7} \times 700 = 400$ km

(iv) Time required by A to reach at $Q = \frac{700}{30} = \frac{70}{3}$ h

Time required by B to reach at $P = \frac{700}{40} = \frac{70}{4}$ h

$\therefore$ Extra time required by $A = \frac{70}{3} - \frac{70}{4}$

$$= 70 \times \frac{1}{12} = 5 \text{ h } 50 \text{ min}$$

(v) Time required by A to cover $MQ = \frac{400}{30}$

and time required by B to cover $MP = \frac{300}{40}$

$\therefore$ Required ratio $= \frac{400/30}{300/40} = \frac{16}{9}$

Remember: If speed of A is S_A and speed of B is S_B and A takes t_A time to cover MQ and B takes t_B time to cover MP, then

$$\frac{S_A}{S_B} = \sqrt{\frac{t_B}{t_A}}$$

(vi) It means they have to cover (700 − 560) = 140 km. Thus, the required time to cover 140 km distance

$$= \frac{140}{70} = 2 \text{ h}$$

(vii) Since in each hour they separate by 70 km from each other. Hence, to separate by 280 km, time required

$$= \frac{280}{70} = 4 \text{ h}$$

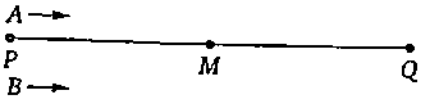
EXAMPLE 13 *Two places P and Q are 800 km apart from each other. Two persons start from P towards Q at an interval of 2 hours. Whereas A leaves P for Q before B. The speeds of A and B are 40 km/h and 60 km/h respectively. B overtakes (or catches or meets) A at M, which is on the way from P to Q.*

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(i) *How long will B take to overtake?*
(ii) *What is the distance from P, where B overtakes A (i.e., PM)?*
(iii) *What is the ratio of time taken by A and B to meet at M?*
(iv) *What is the extra time required by A to reach at Q?*
(v) *How many hours late A will reach at Q than that of B?*
(vi) *After how many hours A and B will be separated by 50 km before M, when both are moving?*
(vii) *How many hours does B require to advance himself, by 100 km, in comparison to A?*

SOLUTION



(i) Since A moves 2 hours earlier than B at the speed of 40 km/h, so in 2 hours A will cover 80 km. Thus A will be 80 km away from B.
Now, since they are moving in the same direction therefore their relative speed will be $(60 - 40) = 20$ km/h *i.e.*, with respect to A, B moves at 20 km/h. It means either they reduce (or create) the difference of 20 km in each hour between themselves.
Thus, the required time to overtake (or reduce the difference upto zero from 80 km)

$$= \frac{\text{Distance advanced}}{\text{Relative speed}} = \frac{80}{20} = 4 \text{ h}$$

(ii) The distance between P and M
= (Time required to overtake × Speed of the faster body)
$= 4 \times 60 = 240$ km
Since, B has to move for 4 hours at 60 km/h. Hence, distance covered = 240 km.

(iii) Time taken by B to reach at $M = 4$ h
Time taken by A to reach at $M = (4 + 2) = 6$ h
Thus, the ratio of time taken by A and $B = 6 : 4 = 3 : 2$
(Since, A has left 2 hours earlier)

(iv) Time taken by A to reach at $Q = \frac{800}{40} = 20$ h

Time taken by B to reach at $Q = \frac{800}{60} = 13$ h 20 min

Hence, A takes 6 hrs 40 min extra to reach at Q.

(v) Since, A leaves 2 hrs earlier, thus he will reach at Q only 4 hours 40 min late.
Since, A takes 6 hrs 40 min extra to reach at Q.

(vi) When B starts to move towards Q the difference between A and B is 80 km.
The required difference between A and $B = 50$ km
Hence, they have to reduce it by 30 km.

Thus, the required time $= \frac{(80 - 50)}{20} = 1.5$ h

$= 1$ h 30 min

Thus, after 3/2 hrs A will be only 50 metres ahead of B.

(vii) When B starts to follow A (towards Q) A was 80 km ahead of B. Also B wants to overtake A and further go ahead of A by 100 km. Thus, the net difference (required) = 180 km.

$\therefore$ Required time

$$= \frac{(\text{Distance advanced} + \text{Required difference})}{\text{Relative speed}}$$

$$= \frac{80 + 100}{20} = \frac{180}{20} = 9 \text{ h}$$

Thus, after 9 hrs (when B starts moving) B will be 100 km ahead of A *i.e.*, they will be separated by 100 km from each other after crossing M.

TO AND FRO MOTION IN A STRAIGHT LINE

This concept is just the extension of the previous concepts of relative motion between more than one dynamic (or moving) bodies.

(a) When two bodies start moving towards each other

(i) To meet each other they cover the distances in the ratio of their individual speeds.
(ii) If the initial distance (or gaping) between two bodies A and B is D, then A and B together have to cover D unit of distance for the first meeting.
(iii) For the next number of meeting (*e.g.*, second, third, fourth meeting and so on) both A and B together have to cover $2D$ distance more from the previous meeting *i.e.*, to meet for the fourth time they have to cover together $D + (3 \times 2D) = 7D$ unit of distances. Similarly for seventh meeting they have to cover together $D + (6 \times 2D) = 13D$ units of distance. Thus for each subsequent meeting they have to cover $2D$ distance extra from the previous one.

NOTE Individually they will cover the distances in the ratio of their speeds for any number of meeting.
Thus, the total distance covered for the nth meeting $= (2n - 1) D$.

(b) When two bodies start moving towards the same direction

(i) For the first meeting after they start to move they have to cover $2D$ distance, if the distance between two particular points (or places) be D unit.
Since, the faster body reaches the next (or opposite) end first than the slower body and the faster body starts returning before the slower body reaches the same opposite end and thus the two bodies meet somewhere between the two ends covering individually the distances in their respective speeds.
(ii) For every subsequent meeting they have to cover together $2D$ unit distance more from the previous meeting.
Thus, for n^{th} meeting they have to cover together $(n \times 2D)$ unit of distance.
(iii) At any point of time the distances covered by the bodies will be equal to the ratio of their speeds.

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EXAMPLE 14 *The distance between P and Q is 100 m and the speeds of A and B are 20 m/s and 30 m/s respectively. Initially A and B are at P. They move between P and Q. Calculate:*

(i) the distance covered by A in first meeting.
(ii) the time required for the second meeting.
(iii) the distance covered by B to meet for the third time.
(iv) the ratio of distances covered by A and B till the fourth meeting.
(v) the distance between P and the place of fifth meeting.
(vi) the distance between Q and the point of third meeting.

SOLUTION

Meeting	Total distance covered by A and B together	Distance covered by A	Distance covered by B	Point of meeting from P	Point of meeting from Q	Time (in second)
1st	200 m	80 m	120 m	80 m	20 m	4
2nd	400 m	160 m	240 m	40 m	60 m	8
3rd	600 m	240 m	360 m	40 m	60 m	12
4th	800 m	320 m	480 m	80 m	20 m	16
5th	1000 m	400 m	600 m	0 m (at P)	100 m	20

(i) The distance covered by A in the first meeting = 80 m

(ii) Time required for the second meeting = $\frac{400}{50} = 8$ s

(iii) The distance covered by B for the third time meeting = $12 \times 30 = 360$ m

(iv) It is always (for any moment, after the starting of movement) will be in the ratio of their respective speeds. So, the required ratio of distances covered by A and $B = 20 : 30 = 2 : 3$ *i. e.*,

$$A : B = 20 : 30 = 2 : 3$$

(v) Since for the fifth meeting they have to cover $50 \times 20 = 1000$ m.

(vi) 60 m.

EXAMPLE 15 *The distance between two points P and Q is 100 m. A is initially at P and B is at Q. The speeds of A and B is 20 m/s and 30 m/s. They move between P and Q to and fro :*

(i) Find the time required for the first meeting.
(ii) Distance covered by A till the third meeting.
(iii) Distance covered by B till the fifth meeting.
(iv) The distance between P and the place of fourth meeting.
(v) The distance between Q and the place of fifth meeting.
(vi) The ratio of distances covered by each one till the third meeting.

SOLUTION

No. of meeting	Distance covered by A and B together	Time (in second)	Distance covered by A	Distance covered by B	Distance between P and point of meeting	Distance between Q and point of meeting
1st	100 m	2	40 m	60 m	40 m	60 m
2nd	300 m	6	120 m	180 m	80 m	20 m
3rd	500 m	10	200 m	300 m	at P	at P
4th	700 m	14	280 m	420 m	80 m	20 m
5th	900 m	18	360 m	540 m	40 m	60 m

(i) 2 second $\left(\text{Time} = \frac{100}{50} = 2\right)$

(ii) 200 m (Distance = Speed × Time = $20 \times 10 = 200$ m)

(iii) 540 m ($D = 18 \times 30 = 540$ m)

(iv) 80 m

(v) 60 m

(vi) 2 : 3 always equal to the ratio of respective speeds.

EXAMPLE 16 *A and B are two friends. A lives at a place P and B lives at another place Q. Everyday A goes to Q to meet B at 120 km/h. Thus, he takes 3 hours. On a particular day B started to meet A so he moved towards P. On that day A took only 2 hours to meet B on the way instead at Q.*

(i) What is the ratio of speeds of A is to B?

(ii) What is the speed of B?

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SOLUTION Distance between P and $Q = 120 \times 3 = 360$ km

Let the speed of B be S_B, then

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$2 = \frac{360}{(120 + S_B)}$$

$$\Rightarrow \quad S_B = 60 \text{ km/h}$$

Here, A and B are moving towards each other. So, the relative speed will be the sum of the speeds of A and B both.

Therefore, ratio of speeds of $A : B = 2 : 1$.

(i) 2 : 1 (ii) 60 km/h

NOTE In this case ratio of speeds of

$$A : B = \frac{\text{Actual time required when } B \text{ is also moving}}{\text{Time difference when } B \text{ is also moving}}$$

$$= \frac{(3 - 1)}{1} = \frac{2}{1}$$

EXAMPLE 17 *A lives at P and B lives at Q. A usually goes to meet B at Q. He covers the distance in 3 hours at 150 km/h. On a particular day B started moving away from A. While A was moving towards Q thus A took total 5 hours to meet B.*

(i) What is the speed of B?

(ii) What is the ratio of speeds of A : B ?

SOLUTION Distance $= 3 \times 150 = 450$ km

$$\text{Time} = \frac{\text{Distance}}{\text{Relative speed}}$$

$$5 = \frac{450}{(150 - S_B)} \qquad S_B \rightarrow \text{Speed of } B$$

$$\Rightarrow \quad S_B = 60 \text{ km/h}$$

Ratio of speeds of $A : B = \frac{150}{60} = \frac{5}{2} = 5 : 2$

(i) 60 km/h (ii) 5 : 2

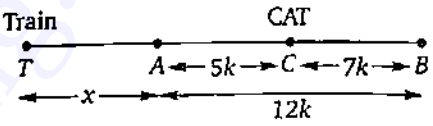
In this case ratio of speeds of

$$A : B = \frac{\text{Actual time required when } B \text{ is also moving}}{\text{Time difference}}$$

$$= \frac{5}{2}$$

EXAMPLE 18 *A train approaches a tunnel AB. Inside the tunnel a cat located at a point i. e., $\frac{5}{12}$ of the distance AB measured from the entrance A. When the train whistles, the cat runs. If the cat moves to the entrance of the tunnel A, the train catches the cat exactly at the entrance. If the cat moves to the exit B, the train catches the cat at exactly the exit. The speed of the train is greater than the speed of the cat by what order?*

SOLUTION Train



Let the speed of train be u and the speed of cat be v and train whistles at a point T, x km away from A, then

$$\frac{u}{v} = \frac{x}{5k} = \frac{x + 12k}{7k}$$

$$\Rightarrow \quad 7x = 5(x + 12k)$$

$$\Rightarrow \quad \frac{x}{k} = \frac{30}{1}$$

$$\therefore \quad \frac{u}{v} = \frac{30}{5 \times 1} = \frac{6}{1}$$

Alternatively: $\frac{u}{v} = \frac{7k + 5k}{7k - 5k} = \frac{6}{1}$

NOTE Since time is constant, therefore distances covered by train and cat will be in the ratio of their respective speeds.

CONCEPT BASED ON TRAINS

(i) When two trains (or bodies) are moving in opposite direction, their relative speed will be equal to the sum of their individual speeds.

(ii) When two trains are moving in the same direction their relative speed will be equal to the difference of their speeds.

(iii) Distance to be covered to cross each other is always equal to the sum of their individual lengths.

(iv) Distance to be covered such as bridge, platform etc., is always equal to the sum of the length of train and the length of the particular object such as bridge, platform etc.

(v) Distance to be covered such as pole, man, tree etc, is always equal to the length of the train only.

(vi) If a man is travelling in a train, then this man has to cover the distance to cross another train is equal to the length of the train which is passing or crossing him. In this case the relative speed of both the trains will be considered.

EXAMPLE 19 *A train crosses a tree in 10 seconds. If the length of the train be 150 m, then find the speed of the train.*

SOLUTION Distance = Length of train

= Speed of train × Time

$150 = \text{Speed} \times 10$

$\Rightarrow$ Speed = 15 m/s

$\text{Speed} = 15 \times \frac{18}{5} = 54$ km/h

NOTE A train starts to cross a stationary thin object (of inconsiderable thickness) when the engine of the train meets the object and it completes the crossing when the last wagon (or backend) of the train just crosses the object.

EXAMPLE 20 *A train crosses a man coming from the opposite direction in 7.5 seconds. If the speed of man be 10 m/s and speed of train is 20 m/s, find the length of the train.*

SOLUTION Length of train = Time × Relative speed

$= 7.5 \times (10 + 20) = 7.5 \times 30 = 225$ m

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EXAMPLE 21 *A train of length 250 m crosses a bridge of length 150 m in 20 seconds. What is the speed of train?*

SOLUTION (Length of train + Length of bridge).

= Speed of train × Time

$(250 + 150) = 20 \times \text{Speed}$

$$\text{Speed} = \frac{400}{20} = 20 \text{ m/s} = 72 \text{ km/h}$$

EXAMPLE 22 *Two trains coming from the opposite sides crosses each other in 10 seconds if the lengths of first train and second train be 125 m and 175 m respectively, also the speed of first train be 36 km/h, find the speed of second train.*

SOLUTION Speed of first train = 36 km/h = 10 m/s

Now, $\text{Time} = \dfrac{\text{Sum of length of the two trains}}{\text{Sum of their speeds}}$

$$10 = \frac{125 + 175}{(10 + x)}$$

$$\Rightarrow \quad x = 20 \text{ m/s} = 72 \text{ km/h}$$

EXAMPLE 23 *A fast moving superfast express crosses another passenger train in 20 seconds. The speed of faster train is 72 km/hr and speeds of slower train is 27 km/h. Also the length of faster train is 100 m, then find the length of the slower train if they are moving in the same direction.*

SOLUTION $\text{Time} = \dfrac{\text{Sum of length of the two train}}{\text{Difference in speeds}}$

$$20 = \frac{(100 + x)}{25/2}$$

$$\Rightarrow \quad x = 150 \text{ m}$$

NOTE Relative speed = (72 – 27) = 45 km/h

$$= 45 \times \frac{5}{18} = \frac{25}{2} \text{ m/s}$$

CONCEPT BASED ON BOATS AND RIVERS (OR STREAMS)

(i) When the boat and stream (or current) of river move in the same direction, then the relative speed of the boat is the sum of the individual speeds of boat and river. It is known as **downstream speed**.

(ii) When the boat moves against the current of the river (*i.e.*, in opposite direction), then the relative speed of the boat is the difference of the speeds of the boat and stream (of the river). It is known as **upstream speed**.

Let the speed of boat in still water be B and speed of current of river be C then,

$$\begin{cases} \text{Downstream speed} = (B + C) \\ \text{Upstream speed} = (B - C) \end{cases} ; \qquad B > C$$

$$\begin{cases} \text{Speed of the boat in still water} = \dfrac{(D + U)}{2} \\ \text{Speed of current (or stream)} = \dfrac{(D - U)}{2} \end{cases}$$

where $D \rightarrow$ downstream speed of the boat

and $U \rightarrow$ upstream speed of the boat

When the distance covered by boat in downstream (*i.e.*, with the flow of water) is same as the distance covered by boat in upstream (against the flow of the water) then,

$$\frac{\text{Time taken by boat in DS}}{\text{Time taken by boat in US}} = \frac{\text{Upstream speed}}{\text{Downstream speed}}$$

DS → Downstream, US → Upstream

EXAMPLE 24 *A boat can move at 5 km/h in still water (i.e., when water is not flowing). The speed of stream of the river is 1 km/h. A boat takes 80 minutes to go from a point A to another point B and return to the same point.*

(i) What is the distance between the two points?

(ii) What is the ratio of downstream speed and upstream speed?

(iii) What is the ratio of time taken in downstream speed to the upstream speed?

SOLUTION Downstream speed of boat = (5 + 1) = 6 km/h

Upstream speed of boat = (5 – 1) = 4 km/h

Therefore, $\dfrac{\text{Downstream speed}}{\text{Upstream speed}} = \dfrac{\text{Upstream time}}{\text{Downstream time}}$

$$\frac{6}{4} = \frac{3}{2} = \frac{\text{Time taken in upstream direction}}{\text{Time taken in downstream direction}}$$

$\therefore$ Time taken in downstream $= \dfrac{2}{5} \times 80 = 32$ min

and time taken in upstream direction $= \dfrac{3}{5} \times 80 = 48$ min

$\therefore$ Distance between two points = DS speed × DS time

= US speed × US time

where DS → Downstream and US → Upstream

$$D = \frac{6 \times 32}{60} = 3.2 \text{ km}$$

or $$D = 4 \times \frac{48}{60} = 3.2 \text{ km}$$

(i) 3.2 km (ii) 3 : 2 (iii) 2 : 3

EXAMPLE 25 *A man can row 9 km/h in still water. It takes him twice as long as to row up as to row down. Find the rate of stream of the river.*

SOLUTION $\dfrac{\text{Time taken in upstream}}{\text{Time taken in downstream}} = \dfrac{2}{1}$

$\therefore$ $\dfrac{\text{Downstream speed}}{\text{Upstream speed}} = \dfrac{2}{1}$ where $\dfrac{B + R}{B - R} = \dfrac{2}{1}$

$B \rightarrow$ Speed of boat in still water

$R \rightarrow$ Speed of current

$\Rightarrow$ $\dfrac{B}{R} = \dfrac{3}{1}$ (By componendo and dividendo)

$\Rightarrow$ $\dfrac{9}{R} = \dfrac{3}{1} \Rightarrow R = 3$ km/h

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RACES

Terminology

(i) **Startup or headstart:** When a runner allows to another runner to stay ahead in the same race, then it is said that there is a startup in the race.
For example if A allows B to go ahead before starting the race, then it is said that A gives startup to B and B has the startup. If before starting the race B goes ahead of x metre, then we can say A gives x metre startup to B or B has startup (or headstart) of x metre.

(ii) **Dead heat:** When the runners reach the finishing line (or the final post) then it is said that these runners finish (or end) the race in dead heat.

SOME MORE USEFUL CONCEPTS

(i) When it is said that A can give B a start of x metre in y metre race, then it means in y metre race B runs x metre less than A in the same time.

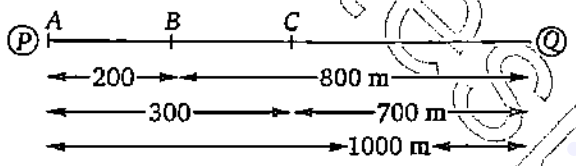
NOTE There is a great difference between 'can' and 'gives'.

(ii) When A beats B by t seconds in a race of y metre then it means B is the loser and A is the winner and when A reaches the finishing line B is still some distance back to A. Thus B takes t seconds to cover the remaining distance. Hence, we can calculate the speed of loser (B).

(iii) Throughout the race there is always a certain relationship among runners *i.e.*, they always maintain the ratio of speeds. I think this is the nub of all the problems pertaining to the races. Go ahead...

EXAMPLE 1 *A can give B a 200 m startup and C a 300 m startup in a race of 1 km. How many metres startup can B gives to C in a 1 km race.*

SOLUTION



Ratio of speeds of $A : B = 1000 : 800 = 5 : 4$

Ratio of speeds of $A : C = 1000 : 700 = 10 : 7$

Ratio of speeds of $B : C = 800 : 700 = 8 : 7$

Since, when B moves 8 m, C moves 7 metre. Therefore, when B moves 1000 m, C moves 875 metre. Thus, B can give C a start of $1000 - 875 = 125$ m.

Alternatively: Since C is 12.5% slower than B. So, C will cover 12.5% less distance than B in the same time.

(Since when time is constant, they cover the distances in the ratio of their speeds.)

Thus, in 1000 m (or 1 km) when B runs 1000 m, C will run 125 m less than B.

Hence, B can give a start of 125 m to C in a 1 km race.

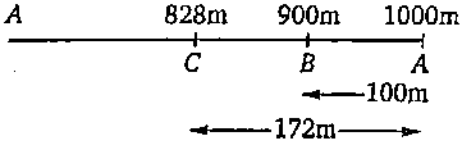
EXAMPLE 2 *In a one km race A gives B a start of 100 m and in a one km race B gives a start of 80 m to C. In a 1 km race who will win and by how much distance from the worst performer between two losers?*

SOLUTION Ratio of speeds of $A : B = 1000 : 900 = 100 : 90$

Ratio of speeds of $B : C = 1000 : 920 = 100 : 92$

Therefore, when A moves 1000 m, B moves 900 m and when B moves 900 m, C moves 828 m.

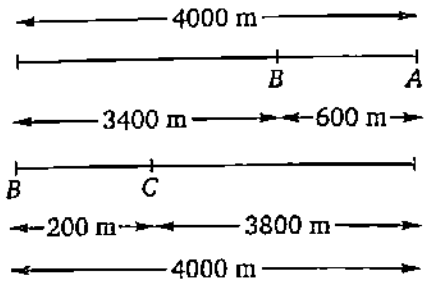
Thus,



Since, C moves 8% less than B in the same time. Thus, C is the worst performer and A will win by him by 172 m.

EXAMPLE 3 *In a 4 km race A wins by 600 m over B. B can give start of 200 m to C in a 4 km race. By how much distance C gets start up so that the race between A and C ends in dead heat in the same race of 4 km?*

SOLUTION



Ratio of speeds of $A : B = 20 : 17$

and Ratio of speeds of $B : C = 20 : 19$

$\therefore$ Ratio of speeds of $A : B : C = 400 : 340 : 323$

Therefore, in 4000 m race A run 4000 m, B run 3400 m and C run 3230 m. Thus C can get 770 m start up from A.

EXAMPLE 4 *In a 1500 m race A wins over B by 350 m and in 1500 m race C can give a startup of 250 m to B. By how much distance A give start up to C, so that A beats C by 50 metres?*

SOLUTION Ratio of speeds of $A : B = 30 : 23$

Ratio of speeds of $B : C = 5 : 6$

$\therefore$ Ratio of speeds of $A : B : C = 150 : 115 : 138$

So, when A moves 1500 m, B moves 1150 m and C moves 1380 m. Thus C moves 120 m less than A. To win A just by 50 m over C, A should give $120 - 50 = 70$ m startup to C.

EXAMPLE 5 *In a race of 2500 m, A beats B by 500 m and in a race of 2000 m, B beats C by 800 m. By what distance A gives startup to C so that they will end up with dead heat in 3 km race. Also find that by what distance A will win over C in a 1 km race?*

SOLUTION Ratio of speeds of $A : B = 5 : 4$

Ratio of speeds of $B : C = 5 : 3$

$\therefore$ Ratio of speeds of $A : B : C = 25 : 20 : 12$

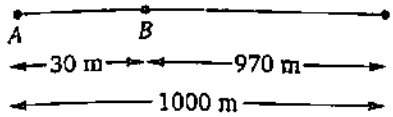
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In 3 km race A run 3000 m, B run 2400 m, C run 1440 m, so to end up the race in dead heat A should give C the startup of 1560 m and therefore in 1 km the same will be 520 m.

EXAMPLE 6 *A gives B, a start of 30 m or 10 seconds and end up the race of 1 km in dead heat. What is the ratio of speeds of A and B?*

SOLUTION



Since, either B has the startup of 30 m or 10 seconds. It means B runs 30 m in 10 seconds. Hence, the speed of B is 3 m/s.

NOTE Don't be confused that A's speed = 3 m/s.

Also, it is very simple when A moves 1000 m, B moves 970 m. Since, the ratio of speeds is equal to the ratio of distances covered by A and B in the same time. Thus, the speed of

$$A : B = 1000 : 970 = 100 : 97$$

EXAMPLE 7 *In a 1 km race A wins over B by 80 m or 20 seconds. B can give a start of 100 m to C in 1 km race. Find out that by how much time A will win over C? Also, find the ratio of speeds of B and C.*

SOLUTION Ratio of speeds of $A : B = 100 : 92$

Ratio of speeds of $B : C = 10 : 9$

$\therefore$ Ratio of speeds of $A : B : C = 1000 : 920 : 828$

Also, Speed of $B = \frac{80}{20} = 4$ m/s

Therefore, Speed of $C = 3.6$ m/s

Now, C has to cover 172 m distance in extra time. So, the time taken by C to cover the remaining distance $= \frac{172}{3.6} = 47.77$ s.

Ratio of speeds of $B : C$ = Ratio of distances covered by $B : C$

$$= 1000 : 900 = 10 : 9$$

$$\therefore \quad \frac{\text{Speed of } B}{\text{Speed of } C} = \frac{10}{9}$$

EXAMPLE 8 *A can win B by 250 m in a 2 km race. What should be the change in distance of startup? So, that B must cover 20% less distance than that by A in the same time.*

SOLUTION

$$\frac{\text{Distance covered by } A}{\text{Distance covered by } B} = \frac{5}{4}$$

$$\therefore \quad \frac{\text{Speed of } A}{\text{Speed of } B} = \frac{5}{4}$$

So, when A moves 2000 m, B should move 1600 m. But since initially B moves 1750 m. Therefore the new startup will be increased by 150 m.

EXAMPLE 9 *The ratio of speeds of A and B is 4 : 7 and A loses the race by 270 m, then what is the length of the race course?*

SOLUTION When B moves 7 m, A moves only 4 m. Hence, A loses the race by 3 m.

Now, since B loses by 3 m in the race of 7 m.

$\therefore$ B will lose 270 m in the race of 630 m.

EXAMPLE 10 *The ratio of time taken to run a certain distance by Pythagorus and Hawkins is 4 : 3 and thus Hawkins wins the race by 360 m. What is the distance of race course?*

SOLUTION

$$\frac{\text{Time taken by Pythagorus}}{\text{Time taken by Hawkins}} = \frac{4}{3}$$

$$\frac{\text{Speed of Pythagorus}}{\text{Speed of Hawkins}} = \frac{3}{4}$$

$$= \frac{\text{Distance covered by P}}{\text{Distance covered by H}}$$

Now, when Hawkins runs 4 m, Pythagorus runs 3 m and thus Hawkins wins by 1 m.

So, when Hawkins wins the race by 1 m, race course is 4 m. when Hawkins wins the race by 360 m, race course is

$$360 \times 4 = 1440 \text{ m}$$

Alternatively : 360 = 25% of the total length of race.

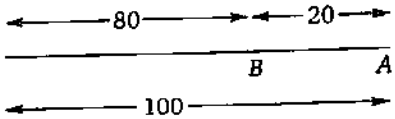
$\therefore$ length of race course = 1440 m.

EXAMPLE 11 *In a race of D km, A wins over B by 0.2 D and in the same length of race B wins over C by 0.25 D. What should be the head- start to C, So that A and C finish the race at the same time.*

SOLUTION **Note:** Instead of solving the problem using 'D' as a distance convert it into 100 i. e , suppose $D = 100$.

(It is just for your convenience.)

Now,



Now, Speed of $A : B = 100 : 80 = 5 : 4$

Speed of $B : C = 4 : 3$

$\therefore$ Speed of $A : B : C = 20 : 16 : 12$

$$= \underbrace{100 : 80 : 60}_{\{40\}}$$

So, A can give 'C' a start of 40 m in 100 m race. Therefore, A can give 0.4 D. Start to 'C' in the race of D unit.

EXAMPLE 12 *In a 3 km race the speeds of A and B are in the ratio of 6 : 5 and A wins by 10 seconds. What is the time taken by B to finish the race. Also, to end the race in dead heat what per cent of total distance A should give the startup to B?*

SOLUTION Ratio of distances covered by A and B when A just reaches the finishing line = 6 : 5.

Thus, B has to cover 500 m distance in extra time and this 500 m distance is covered by B in 10 seconds. So, the speed of B = 50 m/s.

Thus, the total time required by B to complete the 3 km race

$$= \frac{3000}{50} = 60 \text{ s} = 1 \text{ min}$$

Since, B coveres 16.66% distance less than A covers in the same time, so A should give 16.66% of the total distance as a startup to B.

EXAMPLE 13 *In a 1 km race A gives B a startup of 5 seconds and still wins over B by 15 seconds. The ratio of speeds of A and B is 2 : 1. Find the time taken by A to finish 2.5 km race.*

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SOLUTION

$$\frac{\text{Speed of } A}{\text{Speed of } B} = \frac{2}{1}$$

$$\frac{\text{Time taken by } A}{\text{Time taken by } B} = \frac{1}{2}$$

$$\frac{t}{(t+20)} = \frac{1}{2}$$

$$\Rightarrow \quad t = 20 \text{ s}$$

Thus, A needs 20 seconds to cover 1 km. Thus, to cover 2.5 km race he needs $20 \times 2.5 = 50$ seconds.

EXAMPLE 14 *X can beat Y by 200 m in a race of 2000 m. Y can beat Z by 100 m in a race of 2500 m. By how many metres can X beat Z in a race of 1000 m.*

SOLUTION Ratio of speeds of X and $Y = 10 : 9$ (2000 : 1800)

Ratio of speeds of Y and $Z = 25 : 24$ (2500 : 2400)

Ratio of speeds of X, Y and $Z = 250 : 225 : 216$

Since in a race of 250 m, X beats Z by 34 m.

So, in a race of 1000 m, X will beat Z by 136 m.

EXAMPLE 15 *In a 6 km race B has 250 m headstart and C has 500 m headstart by A, still A beats C and B by 235 m and 350 m respectively. How many metres startup can B give to C so as to end up the race at the same time with C in the race of 6 km. Also find the ratio of speeds of $A : B : C$.*

SOLUTION When A runs 6000 m, B runs 5400 m only and C runs 5265 m only.

So, when B runs 6000 m, C will run 5850 m. So, B can give 150 m startup to C.

Ratio of speeds of $A : B : C = 6000 : 5400 : 5265$

$= 1200 : 1080 : 1053$

$= 400 : 360 : 351$

EXAMPLE 16 *A can run 1 km in 2 min 20 second and B can run the same distance in 3 min. What is the distance travelled by B in the same time as A travels, when they start simultaneously in the race of 4.5 km.*

SOLUTION

$$\frac{\text{Time taken by } A}{\text{Time taken by } B} = \frac{140 \text{ s}}{180 \text{ s}} = \frac{7}{9}$$

$$\therefore \quad \frac{\text{Speed of } A}{\text{Speed of } B} = \frac{\text{Distance travelled by } A}{\text{Distance travelled by } B} = \frac{9}{7}$$

Therefore, B travels $\frac{7}{9} \times 4.5 = 3.5$ km.

EXAMPLE 17 *Shahrukh takes 4 min to cover the same distance for which Urmila takes 6 min 30 sec. What is the ratio of distances covered by Shahrukh and Urmila in the race of 2.6 km and by what distance Shahrukh wins over Urmila?*

SOLUTION

$$\frac{\text{Time taken by Shahrukh}}{\text{Time taken by Urmila}} = \frac{240}{390} = \frac{8}{13}$$

$$\frac{\text{Distance covered by Shahrukh}}{\text{Distance covered by Urmila}} = \frac{13}{8}$$

Hence, Shahrukh will win the race by 1 km.

EXAMPLE 18 *A runs at the speed of $\frac{7}{4}$ times the speed of B. By calculation B finds that she has to run 300 m after A reaches to the winning post. What is the total distance of race?*

SOLUTION

$$\frac{\text{Speed of } A}{\text{Speed of } B} = \frac{7}{4}$$

$$\frac{\text{Distance of } A}{\text{Distance of } B} = \frac{7}{4}$$

So, the total distance of race = 700 m.

$(\because 7x - 4x = 300 \Rightarrow 7x = 700)$

EXAMPLE 19 *Time taken by A is 5/7 of B's time for the same length of race. The speed of A is 84 m/s and A beats B by 240 m. What is the length of race course?*

SOLUTION

$$\frac{\text{Speed of } A}{\text{Speed of } B} = \frac{\text{Time taken by } B}{\text{Time taken by } A} = \frac{7}{5}$$

Now since, $7x - 5x = 240$

$$\Rightarrow \quad 7x = 840 \text{ m}$$

EXAMPLE 20 *In a race of 3000 m, Michal beats Nicholas by 600 m and Nicholas in a race of 2000 m beats Oscar by 600 m. In a 1 km race by how much distance Michal beats Oscar?*

SOLUTION Ratio of speeds of $M : N : O = 100 : 80 : 56$

So, Michal (M) beats Oscar (O) by 440 m in 1 km race.

EXAMPLE 21 *In a race of 500 m, President runs at 5 m/s. If president gives chairman a start of 20 m and still beats him by 20 seconds, what is the chairman's speed?*

SOLUTION Chairman has to cover only 480 m. He takes total

$$100 + 20 = 120 \text{ s}$$

Therefore, speed of chairman $= \frac{480}{120} = 4$ m/s

EXAMPLE 22 *In a race of 800 m Dholakiya gives Preetam a start of 200 m and then loses the race by 20 seconds. What is the speed of Preetam and Dholakiya respectively? If the ratio of respective speeds be 3 : 2.*

SOLUTION Let the speed of Dholakiya be S_D and speed of Preetam be S_P and let the time taken by Preetam be t second then,

$$\frac{S_D}{S_P} = \frac{\frac{800}{(t+20)}}{\frac{600}{t}} = \frac{2}{3}$$

$$\frac{800}{600} \times \frac{t}{(t+20)} = \frac{2}{3}$$

$$t = 20 \text{ s}$$

$$\therefore \quad \text{Speed of Preetam} = \frac{600}{20} = 30 \text{ m/s}$$

and speed of Dholakiya $= \frac{800}{40} = 20$ m/s

EXAMPLE 23 *In race of 1 km Sahara gives Birla a startup of 250 m still Sahara wins by 150 m. What is the ratio of speeds of Sahara and Birla?*

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SOLUTION When Sahara covers 1000 m, Birla covers only 600 m.
So, the ratio of speeds of Sahara and Birla
$= 1000 : 600 = 5 : 3$

EXAMPLE 24 *Priyambda wins the race over Kokilaben by 150 m in a race of 1 km but when she gives a startup of 5 seconds to Kokilaben she wins by 65 m. Find the speed of Kokilaben.*

SOLUTION In 5 seconds Kokilaben runs 85 m.
So, the speed of Kokilaben = 17 m/s.

CIRCULAR MOTION

(i) When the bodies are moving in the opposite direction, their relative speed becomes equal to the sum of their individual speed.

(ii) When two bodies are moving in the same direction their relative speed becomes equal to the difference of the individual speeds.

First Meeting

(i) Let A and B be two runners. Time taken by them to meet for the first time

$$= \frac{\text{Length of the circular track}}{\text{Relative speed}}$$

(ii) When there are more than two runners, then suppose A is the fastest runner and A meets B for the first time in t_{AB} seconds/hours and A meets C for the first time in t_{AC} seconds/hours and A meets D for the first time in t_{AD} seconds/hours and so on. Then time taken by all of them to meet for the first time is the LCM of t_{AB}, t_{AC}, t_{AD} etc.

First Meeting at the Starting Point

Let A takes t_A time to complete one round and B takes t_B time and C takes t_C time and so on, then the time taken to meet for the first time at the starting point = LCM of t_A, t_B, t_C etc.

EXAMPLE 1 *Arjun and Bhishma are running on a circular track of length 600 m (i.e., circumference of the track). Speed of Arjun is 30 m/s and that of Bhishma is 20 m/s. They start from the same point at the same time in the same direction. When will they meet again for the first time?*

SOLUTION
$$\text{Time} = \frac{\text{Circumference}}{\text{Relative speed}}$$
$$= \frac{600}{10} = 60 \text{ s}$$

Actually Arjun (faster one) has to make a lead of 600 m, because when Arjun will be 600 m ahead (or extra distance) of Bhishma, they will be together again as a person when completes the total length (or circumference) it starts retracing the same path and thus Arjun and Bhishma can be together again.

Since, they make a difference (or Arjun makes a lead) of 10 m in 1 second. So, he will create 600 m difference in 60 second.

EXAMPLE 2 *In the previous problem if they move in opposite direction, then what is the time taken by them to meet again for the first time?*

SOLUTION
$$\text{Time} = \frac{\text{Circumference}}{\text{Relative speed}}$$
$$= \frac{600}{50} = 12 \text{ s} \qquad (30 + 20 = 50)$$

EXAMPLE 3 *Arjun and Bhishma run at the speed 30 m/s and 20 m/s respectively on the circular track of 600 m, as its circumference, when would the Arjun and Bhishma meet for the first time at the starting point if they start simultaneously from the same point?*

SOLUTION Time taken by Arjun to complete one round
$$= \frac{600}{30} = 20 \text{ s}$$
Time taken by Bhishma to complete one round $= \frac{600}{20} = 30$ s

Hence, after every 20 second, Arjun would be at the starting point and after every 30 second. Bhishma would be at the starting point. Thus the time taken by both to be at the starting point again for the first time
$$= \text{LCM of } 20 \text{ and } 30 = 60 \text{ s}$$
Thus, every 60 seconds they would be together at the starting point.

NOTE The required time for the bodies to meet for the first time at the starting point is immaterial to the direction of bodies *i.e.*, whether they move in the same direction or in opposite direction.

EXAMPLE 4 *Arjun, Bhishma and Nakul run on the circular path at the speed of 20 m/s, 30 m/s and 50 m/s respectivley in the same direction. The circumference of the track (or path) is 600 m.*
(i) When will they be together again for the first time?
(ii) When will they be together again for the first time at the starting point?

SOLUTION (i) Nakul meets Arjun after every $= \frac{600}{(50 - 20)} = 20$ s

Nakul meets Bhishma after every $= \frac{600}{(50 - 30)} = 30$ s

Therefore, all of the three would meet after every 60 seconds. (60 = LCM of 20 and 30).
Hence, they would all meet for the first time after 60 seconds.

(ii) Arjun takes $\frac{600}{20} = 30$ s to complete one round

Bhisma takes $\frac{600}{30} = 20$ s to complete one round

and Nakul takes $\frac{600}{50} = 12$ s to complete one round

Hence, they would meet for the first time at the starting point after 60 seconds.

HINT 60 = LCM of 30, 20 and 12.

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EXAMPLE 5 *A tortoise can complete a full round on the circular track in 8 min, while the rabbit can do the same on the same circular track in 5 min. A, B, C and D are the four consecutive points on the circular track which are equidistant from each other. A is opposite to C and B is opposite to D.*

(i) After how many minutes will they meet together for the first time, when both have started simultaneously from the same point in the same direction?

(ii) After how many minutes will they meet together for the first time at the starting point, when both have started simultaneously from the same point in the same direction?

(iii) If they start from a point A, simultaneously in the same direction then after how many minutes will they meet at C which is just opposite to A?

(iv) If tortoise has a lead of 5 min, when would they meet for the first time, where they have started from the same point in the same direction?

SOLUTION (i) Ratio of time of rabbit and tortoise = 5 : 8

$\therefore$ Ratio of the speed of rabbit and tortoise = 8 : 5

Let the length of the circular track be 40 m, then the speed of rabbit = 8 m/min and speed of tortoise = 5 m/min.

(40 is the LCM of 8 and 5, taken just for convenience in calculation)

Now, the time when they will be together for the first time

$$= \frac{\text{Circumference}}{\text{Relative speed}} = \frac{40}{3} \text{ min}$$

(ii) Time taken by rabbit to complete one round = 5 min

Time taken by tortoise to complete one round = 8 min

$\therefore$ They will meet at the starting point, for the first time

= LCM of 5 and 8 = 40 min

(iii) Rabbit reaches point C in $\frac{20}{8} = 2.5$ min.

Rabbit reaches the same point C in 7.5, 12.5, 17.5, 22.5 min etc. Tortoise reaches after 4 min, 12 min, 20 min, ... etc. They will never meet at C since time cannot coincide as time taken by rabbit is the decimal number and time taken by tortoise is a natural number.

(iv) In 5 min tortoise travels 25 m. Now, rabbit starts chasing tortoise @ 8 m/min.

$\therefore$ Time taken by rabbit to overtake tortoise

$$= \frac{25}{(8-5)} = \frac{25}{3} \text{ min}$$

EXAMPLE 6 *Alfred and Bernard run on the circular track of 600 m. Speeds of Alfred and Bernard are 30 m/s and 20 m/s respectively. Initially they are diametrically opposite to each other.*

(i) When will they meet for the first time if both move in the same direction?

(ii) If both of them move in opposite directions, when will they meet for the second time?

SOLUTION (i) Relative speed of Alfred = 30 – 20 = 10 m/s

Bernard is 300 m ahead in the race.

So, time taken by Alfred to catch Bernard $= \frac{300}{10} = 30$ s

(ii) To meet for the second time they have to cover

= 300 + 600 = 900 m

there relative speed = 30 + 20 = 50 m/s

(since direction is opposite)

$\therefore$ Time taken to meet for the second time $= \frac{900}{50} = 18$ s

EXAMPLE 7 *A, B and C run on a circular track of 800 m. Speeds of A, B and C are 20 m/s, 26 m/s and 33 m/s.*

(i) When will they meet for the first time?

(ii) What is the ratio of distances covered by each one to meet for the first time.

SOLUTION (i) Time taken by C to meet A for the first time

$$= \frac{800}{33-20} = \frac{800}{13} \text{ s}$$

Time taken by C to meet B for the first time

$$= \frac{800}{33-26} = \frac{800}{7} \text{ s}$$

LCM of $\frac{800}{13}$ and $\frac{800}{7} = 800$ s.

Therefore, all of them (i.e., A, B and C) meet after 800 seconds = 13 min 20 sec.

(ii) The ratio of distances = ratio of speeds

= 20 : 26 : 33

INTRODUCTORY EXERCISE-9.1

1. P and Q starts running on the circular track in opposite directions. They start from a point A and meet for the first, second and third time at B, C and A respectively. What is the ratio of speeds of P and Q?
   (a) 2 : 1 (b) 3 : 1
   (c) 4 : 1 (d) none of these
2. P and Q run on a circular track of 100 m. Speed of P is 250 m/s and that of Q is 400 m/s. In how many points on a circular track will they meet if they start from the same point, simultaneously in the same direction?
   (a) 1 (b) 2
   (c) 3 (d) 4
3. A, B and C start to walk around a circular track. A completes one round in 91/6 seconds, B completes in 65/4 seconds and C completes in 56/3 seconds respectively. After how many seconds will they meet to each other at the starting point again?
   (a) 3460 s (b) 1820 s
   (c) 3640 s (d) none of these

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CLOCKS

Actually the movement of hour-hand and minute-hand follows the relative motion. The dial of the clock behaves like a circular track and where minute-hand is a faster runner and hour-hand is a slower one.

For better understanding with the clocks, assume 60 minutes shown on the dial as 60 points. Here we give an arbitrary new unit of distance as 'point'.

So, minute-hand (MH) runs on the circular track of 60 points at 60 points per hour and hour-hand (HH) runs at 5 points per hour. Now we become familiar with the relative motion of two hands of a clock.

Here, 1 point $= 6°$ and 60 point $= 360°$

also 1 point $= 6° = 1$ min

EXAMPLE 1 *How many times minute-hand coincides with hour-hand in 12 hours?*

SOLUTION Time taken by minute-hand to meet hour-hand for the first time $= \frac{60}{55} = \frac{12}{11}$ hours.

(Assume initially both hands are at 12 *i. e.*, 12 O' clock is shown by them.)

Therefore, after every 12/11 hours-minute hand and hour-hand meet each other (or coincide or overtake).

Now, since in $\frac{12}{11}$ hours they coincide 1 time

So, in 12 hours they will coincide

$$= \frac{12}{12/11} = \frac{12}{12} \times 11 = 11 \text{ times}$$

Remember: Between 11 O'clock and 1 O'clock, two hands coincide only one time, that's why they coincide with each other only 11 times in 12 hours.

NOTE In every $\frac{12}{11}$ hours (or 65 $\frac{5}{11}$ min) two hands of a clock coincide.

EXAMPLE 2 *In 12 hours how many times the two hands of clock will be just opposite to each other i.e., they make a straight line having the difference of 180° between them?*

SOLUTION $180° = 30$ points

For the first time minute-hand and hour-hand will be separated in $\frac{30}{55}$ h $\left(\text{Time} = \frac{\text{Required distance}}{\text{Relative speed}}\right)$ and for every next time they will take $\frac{60}{55}$ minutes more to occur as opposite to each other.

(Since in 60/55 hours they complete one round of clock's dial *i.e.*, 60 points undergoing the relative motion, we have total 12 hours.)

Thus, $\frac{30}{55} + \frac{60}{55} + \frac{60}{55} + \frac{60}{55} + \ldots = 12$

$\Rightarrow \quad \frac{30}{55} + \frac{60}{55} + \frac{60}{55} + \ldots = \frac{12 \times 55}{55}$

$\Rightarrow \quad 6 + 12 + 12 + \ldots = 12 \times 11 = 132$

$\Rightarrow \quad 6 + 12n = 132$

$\Rightarrow \quad 12n = 126 \Rightarrow n = 10$

only integral value is admissible.

Thus, total $10 + 1 = 11$ times both hands of a clock will be opposite to each other.

Remember: Between 5 O'clock and 7 O'clock the two hands make 180° angle only one time, that's why they make 180° angle only 11 times in 12 hours, *i.e.*, at exactly 6 O'clock they are 180° apart.

EXAMPLE 3 *In 12 hours how many times a minute-hand and hour-hand of a clock makes 90° between them or becomes perpendicular to each other?*

SOLUTION $90° = 15$ points

This problem can be solved in two parts.

(i) When minute-hand goes ahead of hour-hand.

(ii) When hour-hand goes ahead of minute-hand.

(i) For the first time minute-hand and hour-hand will make 90° (or 15 points) difference in $\frac{15}{55}$ hours.

$\left(\text{time} = \frac{\text{distance}}{\text{relative speed}}\right)$

Now, after every $\frac{60}{55}$ hours they will occur at 90°. Since, in every $\frac{60}{55}$ hours they create a difference of 360° or 60 points (as the circumference of dial).

Now, we have 12 hours,

$\therefore \quad \frac{15}{55} + \frac{60}{55} + \frac{60}{55} + \ldots = 12\text{ h} = 12 \times \frac{55}{55}$

$\Rightarrow \quad \frac{3}{11} + \frac{12}{11} + \frac{12}{11} + \ldots = \frac{12 \times 11}{11}$

$\Rightarrow \quad \frac{3}{11} + \left(\frac{12}{11} n\right) = \frac{12 \times 11}{11}$

$\Rightarrow \quad 3 + 12n = 132$

$\Rightarrow \quad n = \frac{129}{12} = 10$

(only integral value of n is acceptable)

Therefore, $10 + 1 = 11$ times in 12 hours minute-hand makes 90° angle between the two hands, but when minute hand is ahead of hour-hand.

(ii) For the first time minute-hand and hour-hand will make 90° (or 15 points) difference in $\frac{60 - 15}{55} = \frac{45}{55}$ hours.

Since, in this case minute-hand goes till it appears to be 15 points behind of hour-hand (consider initially they are showing 12 O'clock) then you will see that at 12 : 49 : 05 two hand are making 90° angle between them, while it appears to be hour-hand is ahead of minute-hand.

Now, for every next time after $\frac{60}{55}$ hours they will show the same situation.

we have 12 hours.

So, $\frac{45}{55} + \frac{60}{55} + \frac{60}{55} + \frac{60}{55} + \ldots = 12\text{ h} = 12 \times \frac{55}{55}$

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$$\Rightarrow \quad \frac{3}{11}+\frac{4}{11}+\frac{4}{11}\ldots=\frac{4\times 11}{11}$$

$$\Rightarrow \quad 3+4+4+\ldots=44$$

$$\Rightarrow \quad 3+4n=144$$

$\Rightarrow n=\frac{41}{4}=10$, consider only integral value.

Thus, total $10+1=11$ times they will make 90° angle.

Hence, in 12 hours both hands make 90° angle $(11+11)=22$ times in different positions.

Remember: At 3 O'clock and 9 O'clock they are at right angled. Since, except between 2–4 O'clock and 8–10 O'clock in each hour both hands make 90° angle 4 times while in the 2–4 O'clock and 8–10 O'clock two hands makes three-three times in every two hour.

EXAMPLE 4 *Between 2 O'clock and 3 O'clock when two hands of a clock overlap each other?*

SOLUTION To overlap or overtake minute-hand has to reduce the gap of 10 points. Since at 2 O'clock two hands are 10 point apart.

$$\therefore \quad \text{Time}=\frac{\text{Distance advanced}}{\text{Relative speed}}$$

$$=\frac{10}{55}\text{h}=\frac{10}{55}\times 60\text{ min}$$

$$=10\text{ min }54\text{ s}$$

Thus, at 2 : 10 : 54 both hands of a clock coincide.

EXAMPLE 5 *Between 6 am and 7 am when the two hands of a clock coincide.*

SOLUTION $\text{Time}=\frac{30}{55}\text{h}=\frac{30}{55}\times 60\text{ min}=32\text{ min }43\text{ s}$

(Distance advanced $=6\times 5=30$ points)

Thus at 6 : 32 : 43 two hands of a clock coincide.

EXAMPLE 6 *Between 11 O'clock and 12 O'clock when will they coincide.*

SOLUTION Distance advanced at 11 O'clock = 55 points

Relative speed = 55 point/h

$$\therefore \quad \text{Time}=\frac{55}{55}=1\text{ h}$$

Hence, they will coincide at $(11+1)=12$ O'clock.

EXAMPLE 7 *Between 3 O'clock and 4 O'clock when will the two hands make 36° angle between them:*

(i) when hour-hand is ahead of minute-hand.

(ii) when minute-hand is ahead of hour-hand.

SOLUTION (i) 36° = 6 points

Now, at 3 O'clock two hands are separated by exactly 15 points to which we have to reduce upto 6 points. Thus, we have to reduce $15-6=9$ points distance, with the relative speed of 55 point/h.

$$\therefore \quad \text{Time required}=\frac{9}{55}\text{h}=\frac{9}{55}\times 60$$

$$=9\text{ min }49\text{ s}$$

Thus at 3 : 09 : 49 they are 36° apart from each other.

(ii) At 3 O'clock both hands are 15 points apart so to make them 6 points apart minute-hand has to move for $(15+6)=21$ points, since minute-hand has to go 6 points ahead of hour hand, after crossing the hour-hand.

$$\therefore \quad \text{Time}=\frac{21}{55}\text{h}=\frac{21}{55}\times 60\text{ min}=22\text{ min }54\text{ s}$$

Thust at 3 : 22 : 54, both hands will be 6 points (or 36°) apart from each other.

Did you notice something?

The same angle can be formed in two situations, one when hour-hand is ahead of minute-hand and when minute-hand is ahead of hour-hand.

Thus, you can find the required time by dividing the required difference of points (which you have to either create or reduce) by the relative speed.

EXAMPLE 8 *What is the angle between the two hands at 3 : 10 am?*

SOLUTION Assume 60th point (*i.e.*, when it is 12 O'clock) as the origin.

Step 1. Find the distance of minute-hand from the origin.

Step 2. Find the distance of hour-hand from the origin.

Step 3. Take the difference between two values obtained in step 1 and step 2.

∴ **Step 1.** 10 point = 60°

Step 2. 90° + 5° = 95° (In 10 min hour-hand moves 5°)

Step 3. 95° – 60° = 35°

Thus, at 3 : 10 am two hands are 35° apart.

EXAMPLE 9 *What is the angle between two hands of a clock at 7 : 35?*

SOLUTION **Step 1.** At 7 : 35, minute-hand is $35\times 6=210°$ away from origin.

Step 2. At 7 : 35 hour-hand is $7\times 30+35\times\frac{1}{2}=210+17.5$ = 227.5° away from the origin.

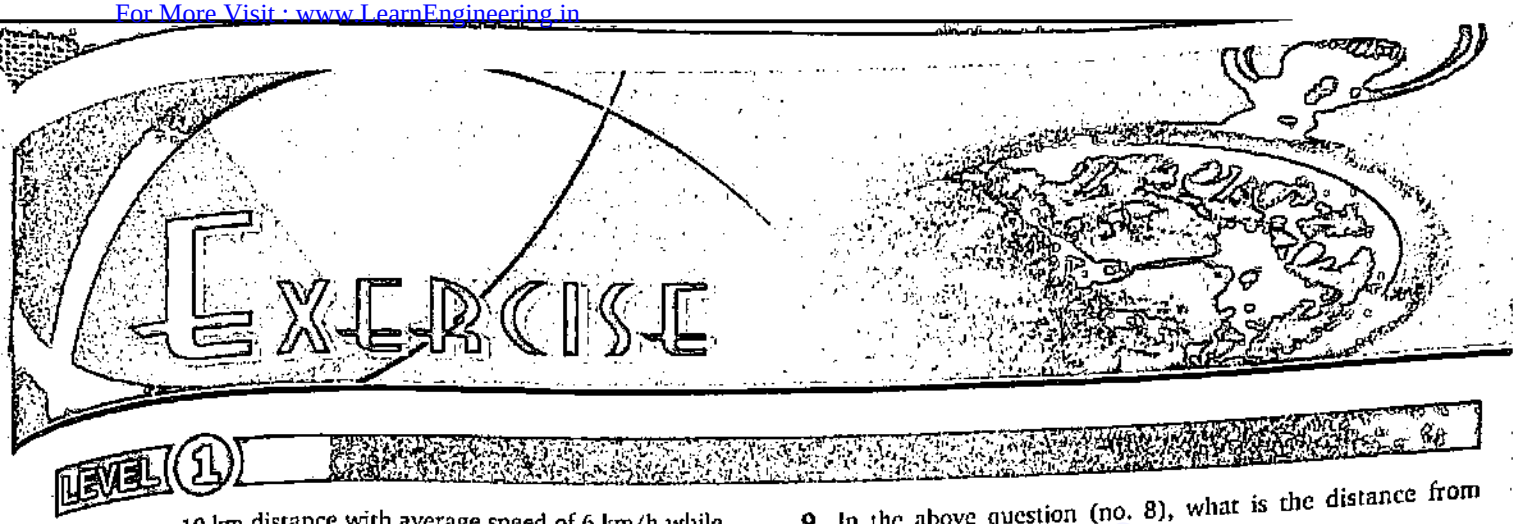
Step 3. 227.5 – 210 = 17.5°

Thus at 7 : 35, both hands make 17.5° angle between them.

NOTE A minute-hand moves 6° in one minute while a hour-hand moves $\left(\frac{1}{2}\right)°$ in one minute.

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LEVEL 1

1. 'A' goes 10 km distance with average speed of 6 km/h while rest 20 km he travels with an average speed of 15 km/h. What is the average speed of 'A' **during** the whole journey?
(a) 10 km/h (b) 12 km/h
(c) 13 km/h (d) 14.5 km/h

2. A covers half of his distance with 20 km/h and rest with 30 km/h. What is the average speed during the whole journey?
(a) 20 km/h (b) 24 km/h
(c) 25 km/h (d) 26 km/h

3. A covers 1/3rd of his journey at the speed of 10 km/h and half of the rest at the speed of 20 km/h and rest at the speed of 30 km/h. What is the average speed of A?
(a) $6\frac{2}{11}$ km/h (b) $16\frac{4}{11}$ km/h
(c) $16\frac{4}{11}$ m/s (d) none of these

4. A covers 1/4th of his journey at 20 km/h and 1/3rd of the rest at 25 km/h and half of the rest at 30 km/h and rest at the speed of 40 km/h. What is the average speed of A?
(a) $13\frac{78}{89}$ km/h (b) 12 km/h
(c) $26\frac{86}{89}$ km/h (d) 28 km/h

5. A covered half of his journey at 20 km/h and rest at x km/h, then his average speed is 24 km/h. What is the value of x?
(a) 30 (b) 32
(c) 36 (d) 40

6. A man covered half of the distance at $3x$ km/h and rest at $5x$ km/h. What is the average speed of the man?
(a) $4x$ km/h (b) $3.5x$ km/h
(c) $3.75x$ km/h (d) none of these

7. A person goes to his office at 1/3rd of the speed at which he returns from his office. If the average speed during the whole trip (i.e., one round) is 12 km/h. What is the speed of the person while he was going to his office?
(a) 10 (b) 6
(c) 8 (d) can't be determined

8. A person X starts from Lucknow and another persons Y starts from Kanpur to meet each other. Speed of X is 25 km/h, while speed of Y is 35 km/h. If the distance between Lucknow and Kanpur be 120 km and both X and Y start their journey at the same time, when will they meet?
(a) 1 h later (b) 2 h later
(c) $\frac{1}{2}$ h later (d) 3 h later

9. In the above question (no. 8), what is the distance from Lucknow where they meet?
(a) 50 km (b) 60 km
(c) 100 km (d) 80 km

10. Two persons A and B started from two different places towards each other. If the ratio of their speeds be 3 : 5, then what is the ratio of distance covered by A and B respectively till the point of meeting?
(a) 1 : 2 (b) 3 : 4
(c) 3 : 5 (d) 5 : 3

Directions for question number 11 to 16: A person P is at X and another person Q is at Y. The distance between X and Y is 100 km. The speed of P is 20 km/h. While the speed of Q is 60 km/h?

11. If they first time meet at point Z somewhere between X and Y then the distance between X and Z is :
(a) 20 km (b) 40 km
(c) 25 km (d) 30 km

12. If they continue to move to and fro between X and Y then what is the distance covered by P when they meet second time?
(a) 105 km (b) 100 km
(c) 80 km (d) 75 km

13. If they continue to move to and fro between X and Y then what is the distance travelled by Q, when they meet each other for the third time?
(a) 375 km (b) 225 km
(c) 350 km (d) 445 km

14. If P and Q continue to move between X and Y in the given manner and if they meet for the fourth time at a place M somewhere between X and Y, then the distance between X and M is :
(a) 10 km (b) 90 km
(c) 75 km (d) 25 km

15. If P and Q continue to move between X and Y, then the ratio of distances covered by P and Q when they meet for the 5th time?
(a) 1 : 4 (b) 1 : 3
(c) 2 : 3 (d) 3 : 4

16. If P and Q continue to move between X and Y, then the distance covered by P and Q together between any two consecutive meeting?
(a) 100 (b) 300
(c) 200 (d) can't be determined

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Directions for questions number 17, 18 and 19: *A persons P starts his journey from A and another person Q starts his journey from B, towards each other. The speeds of P and Q are 16 km/h and 25 km/h respectively and they meet at point M somewhere between A and B when they start their journey simultaneously.*

17. What is the ratio of time taken by P and Q to reach at M?
(a) 1 : 4 (b) 1 : 1
(c) 4 : 5 (d) 16 : 25

18. What is the ratio of time taken by P and Q to reach B and A respectively?
(a) 16 : 25 (b) 1 : 1
(c) 25 : 16 (d) 4 : 5

19. What is the ratio of time taken by P and Q after meeting each other at M to reach B and A respectively?
(a) 25 : 16 (b) 625 : 256
(c) 16 : 25 (d) 4 : 5

20. The speeds of Vimal and Kamal are 30 km/h and 40 km/h. Initially Kamal is at a place L and Vimal is at a place M. The distance between L and M is 650 km. Vimal started his journey 3 hours earlier than Kamal to meet each other. If they meet each other at a place P somewhere between L and M, then the distance between P and M is :
(a) 220 km (b) 250 km
(c) 330 km (d) 320 km

21. In the above question (no. 20) what is the distance between L and P ?
(a) 220 km (b) 320 km
(c) 330 km (d) none of these

Directions for questions number 22 to 27: *There are two places X and Y, 200 km apart from each other. Initially two persons P and Q both are at 'X'. The speed of P is 20 km/h and speed of Q is 30 km/h. Later on they starts to move to and fro between X and Y.*

22. If they starts to move between X and Y, then for the first time when they will meet each other?
(a) after 12 hours (b) after 24 hours
(c) after 30 hours (d) after 8 hours

23. If they meet first time at a point M somewhere between X and Y, then what is the distance travelled by P?
(a) 160 km (b) 150 km
(c) 200 km (d) 210 km

24. If they meet second time each other at a point N somewhere between X and Y, then the distance travelled by Q is :
(a) 240 km (b) 480 km
(c) 360 km (d) none of these

25. If they meet third time each other at a point C, somewhere between X and Y, then the ratio of distances CX and CY is :
(a) 3 : 2 (b) 1 : 3
(c) 2 : 3 (d) 2 : 5

26. If they meet fourth time each other at a point D somewhere between X and Y, then what is the distance between D and X?
(a) 75 (b) 80
(c) 150 (d) 160

27. After starting their race, they meet each other for the nth time at point X, then what is the minimum possible value of n?
(a) 1 (b) 2
(c) 3 (d) 5

Directions for questions number 28 to 33: *A person X started 3 hours earlier at 40 km/h from a place P, then another person Y followed him at 60 km/h, started his journey at 3 O'clock, afternoon.*

28. At what time will they meet to each other (or at what time Y will overtake X)?
(a) 4:30 pm (b) 5 pm
(c) 6 pm (d) 9 pm

29. At what time the difference between X and Y was 30 km, but before Y overtakes X?
(a) 6:30 pm (b) 7:30 pm
(c) 8:75 pm (d) none of these

30. At what time Y will be 30 km ahead of X, after overtaking it?
(a) 6:45 pm (b) 7:30 pm
(c) 10:30 pm (d) 8 pm

31. What is the distance travelled by Y to overtake X?
(a) 180 km (b) 420 km
(c) 320 km (d) 360 km

32. What distance should Y cover so that he may reach 360km ahead of X?
(a) 1440 km (b) 1200 km
(c) 920 km (d) 750 km

33. What is difference in time when X was 30 km ahead of Y and when Y was 30 km ahead of X?
(a) 2 (b) 3
(c) 3.5 (d) 4.25

34. A postman goes with a speed of 36 km/h what is the speed of postman in m/s?
(a) 4.5 m/s (b) 6 m/s
(c) 10 m/s (d) can't be determined

35. In the above question (no. 34) what is the speed in m/min?
(a) 325 (b) 432
(c) 360 (d) 600

36. In the above question (no. 34) what is the speed of postman in mile/h?
(a) 22.37 (b) 30.08
(c) 28.30 (d) 38.12

37. A train goes with a speed of 20 m/s. What is the speed of train in km/h?
(a) 57 km/h (b) 72 km/h
(c) 80 km/h (d) 120 km/h

38. As per the question (no. 37) what is the speed of train in km/min?
(a) 1.2 (b) 12
(c) 1200 (d) 120

39. A is twice fast as B and B is thrice as fast as C. The journey covered by C in 78 minutes will be covered by A in :
(a) 12 min (b) 13 min
(c) 15.5 min (d) none of these

40. The ratio of speeds of A is to B is 2 : 3 and therefore A takes 20 minutes less time than B takes. What is the ratio of time taken by A and B?
(a) 2 : 3 (b) 2 : 5
(c) 3 : 2 (d) 3 : 5

41. What is the time taken by A (in the above question)?
(a) 1 h (b) 1.2 h
(c) 0.6 h (d) 30 min

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Time, Speed and Distance

42. A certain distance is covered at a certain speed. If half of this distance is covered in double the time, the ratio of the two speeds is :
(a) 1 : 16 (b) 4 : 1
(c) 2 : 1 (d) 2 : 8

43. Two runner start runing together for a certain distance, one at 5 km/h and another at 3 km/h. The former arrives one and half an hour before the latter. The distance (in km) is :
(a) 12 (b) 20
(c) 25 (d) 36

44. The ratio between the rates of walking of *A* and *B* is 2 : 3. If the time taken by *B* to cover a certain distance is 48 minutes, the time taken (in minutes) by *A* to cover the distance is :
(a) 52 min (b) 68 min
(c) 72 min (d) 32 min

45. Two trains starting at the same time from two stations 300 km apart and going in opposite directions, cross each other at a distance of 160 km from one of them. The ratio of their speeds is :
(a) 7 : 9 (b) 16 : 20
(c) 8 : 7 (d) 8 : 12

46. *A* and *B* travel the same distance at 9 km/h and 10 km/h respectively. If *A* takes 20 minutes longer than *B*, the distance travelled by each is :
(a) 16 (b) 20
(c) 30 (d) none of these

47. Abhinav leaves Mumbai at 6 am and reaches Bangalore at 10 am Praveen leaves Bangalore at 8 am and reaches Mumbai at 11:30 am. At what time do they cross each other?
(a) 10 am (b) 8:32 am
(c) 8:56 am (d) 9:20 am

48. Two persons, Alif and Laila start at the same time from Allahabad and Lucknow and proceed towards each other at 45 km/h and 54 km/h respectively. When they meet, it is found that one of them has travelled 72 km more than the other. The distance between the places (in km) is :
(a) 729 (b) 792
(c) 540 (d) none of these

49. Walking at 4/5 of his normal speed, Dewang is 15 minutes late in reaching his club. What is the usual time taken by him to cover the distance?
(a) 1 h (b) 1 h 20 min
(c) 45 min (d) none of these

50. Walking at 3/4 of her normal speed Malleshwari takes 2 hours more than the normal time. What is the normal time?
(a) 4 h (b) 5 h
(c) 6 h (d) 8 h

51. Walking at 3/2 of his normal speed Shekhawat takes 40 minutes less than the usual time? What is the changed (new) time taken by Shekhawat?
(a) 1 h (b) 1.2 h
(c) 3 h (d) 2 h

52. A man reduces his speed from 20 km/h to 18 km/h. So, he takes 10 minutes more than the normal time. What is the distance travelled by him?
(a) 30 km (b) 25 km
(c) 50 km (d) 36 km

53. Osaka walks from his house at 5 km/h and reaches his office 10 minutes late. If this speed had been 6 km/h he would have reached 15 minutes early. The distance of his office from his house is :
(a) 15 km (b) 12.5 km
(c) 10.5 km (d) 18 km

54. Sachin and Mongiya starting from the same place walk at the rate of 7.5 km/h and 9 km/h respectively. What time will they take to be 7.5 km apart, if they walk in the same direction?
(a) 3 h (b) 4 h
(c) 5 h (d) none of these

55. Two aeroplanes start from the same place in opposite directions. One goes towards east at 320 km/h and the other goes towards west at 400 km/h what time will they take to be 720 km apart?
(a) 4 h (b) 3 h
(c) 1 h (d) 1.5 h

56. A man covers a certain distance by his own car. Had he moved 6 km/h faster he would have taken 4 hours less. If he had moved 4 km/h slower, he would have taken 4 hours more. The distance (in km) is :
(a) 240 km (b) 640 km
(c) 480 km (d) none of these

57. If Deepesh had walked 20 km/h faster he would have saved 1 hour in the distance of 600 km. What is the usual speed of Deepesh?
(a) 100 (b) 120
(c) 150 (d) none of these

58. Harsha takes 3 hours more than Ashok, who drives his car 5 km/h faster than Harsha drives, to cover 180 km distance. What is the speed of Harsha?
(a) 12 km/h (b) 15 km/h
(c) 30 km/h (d) 40 km/h

59. A minibus takes 6 hours less to cover 1680 km distance, if its speed is increased by 14 km/h? What is the usual time taken by minibus?
(a) 15 h (b) 24 h
(c) 25 h (d) 30 h

60. The driver of an ambulance sees a school bus 40 m ahead of him. After 20 second, the school bus is 60 metre behind. If the speed of the ambulance is 30 km/h, what is the speed of the school bus?
(a) 10 km/h (b) 12 km/h
(c) 15 km/h (d) 22 km/h

61. A postman riding a bicycle at 15 km/h can reach a village in 4 hours. If he is delayed by 1 hour at the start, then in order to reach his destination in time, he should ride with a speed of:
(a) 20 km/h (b) 16 km/h
(c) 14 km/h (d) 12 km/h

62. What is the time required by a train of length of 350 m to cross an electric pole with a speed of 70 m/s?
(a) 3 s (b) 3.5 s
(c) 5 s (d) 10 s

63. A train 270 m long is running over a bridge of length of 130 m with a speed of 40 m/s. What is the time taken by this train to cross the bridge?

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(a) 6 s (b) 16 s
(c) 10 s (d) none of these

64. If a train 225 m long passes a telegraphic pole in 9 seconds, then the time taken (in seconds) by it to cross a tunnel 450 m long is :
(a) 8 s (b) 10 s
(c) 27 s (d) none of these

65. A train 350 m long is running at the speed of 36 km/h. If it crosses a tunnel in 1 minute, then the length of the tunnel (in metres) is :
(a) 200 m (b) 250 m
(c) 150 m (d) none of these

66. If a 250 m long train crosses a platform of the same length as that of the train in 25 seconds, then the speed of the train is :
(a) 150 m/s (b) 200 m/s
(c) 20 km/h (d) 72 km/h

67. Sabarmati express takes 18 seconds to pass completely through a station 162 m long and 15 seconds through another station 120 m long. The length of the Sabarmati express is :
(a) 132 m (b) 100 m
(c) 80 m (d) 90 m

68. A train 200 m long travels at the speed of 72 km/h. A man is running at 3.6 km/h in the same direction in which the train is going. The train will pass the man in :
(a) 10 s (b) $12\frac{3}{19}$ s
(c) $10\frac{10}{19}$ s (d) none of these

69. A train 350 m long is moving at the speed of 20 km/h. It will cross a man coming from the opposite direction at the speed of 1 km/h in :
(a) 27 s (b) 35 s
(c) 45 s (d) 60 s

70. The length of Lucknow mail is 120 m and that of Punjab mail is 80 m. These two trains are running in the same direction with velocities of 40 km/h and 50 km/h respectively. The time taken by them to cross each other is :
(a) 8 s (b) 72 s
(c) 11.5 s (d) 12.5 s

71. In the above question if the trains are running in opposite directions. The time taken by them to cross each other is :
(a) 8 s (b) 72 s
(c) 12.5 s (d) none of these

72. A train passes an electric pole in 10 seconds and a platform 120 m long in 18 seconds. Its length in metres is :
(a) 150 m (b) 130 m
(c) 240 m (d) 180 m

73. A 175 m long train crosses a man walking at a speed of 9 km/h in the opposite direction in 10 sec. The speed of the train (in km/h) is :
(a) 45 (b) 54
(c) 72 (d) 68

74. A train of length 100 m takes 1/6 minute to pass over another train 150 m long coming from the opposite direction. If the speed of first train is 60 km/h, the speed of the second train is :
(a) 45 km/h (b) 28 km/h
(c) 30 km/h (d) none of these

75. A train overtakes two girls who are walking in the opposite direction in which the train is going at the rate of 3 km/h and 6 km/h and passes them completely in 36 seconds and 30 seconds respectively. The length of the train (in metres) is :
(a) 120 m (b) 150 m
(c) 125 m (d) none of these

76. A coolie standing on a railway platform observes that a train going in one direction takes 4 seconds to pass him. Another train of same length going in opposite direction takes 5 seconds to pass him. The time taken (in seconds) by the two trains to cross each other will be :
(a) 35 (b) 36.5
(c) $\frac{40}{9}$ (d) none of these

77. Pushpak express leaves Lucknow at 6 am and two hours later another train Bhopal express leaves Lucknow. Both trains arrive Bhopal at 4 pm on the same day. If the difference between their speeds be 10 km/h, what is the average speeds of both the trains over entire route :
(a) 40 km/h (b) $44\frac{4}{9}$ km/h
(c) $42\frac{3}{5}$ km/h (d) none of these

Directions for question number 78, 79: Two trains leave Meerut at the difference of 4 hours. The first train leaves at 8 am at 40 km/h and the faster train leaves later at 60 km/h in the same direction.

78. When the faster train will overtake the slower train?
(a) 4 pm (b) 2 pm
(c) 8 pm (d) 6:30 pm

79. What is the distance from Meerut, where one train overtakes another train?
(a) 480 km (b) 420 km
(c) 360 km (d) 250 km

80. The distance between Lucknow and Delhi is 700 km. Rajdhani express starts from Delhi for the Lucknow at 60 km/h. 50 minutes later Lucknow express leaves Lucknow for Delhi on the parallel tracks at 70 km/h. How far from Lucknow will they cross each other?
(a) 250 km (b) 360 km
(c) 350 km (d) 475 km

81. Patna express travels first 560 km in 7 hours and rest 360 km in 9 hours. What is the average speed of the train?
(a) 39 km/h (b) 43 km/h
(c) 63 km/h (d) 57.5 km/h

82. Jammutavi express leaves Jammu for Kanya Kumari at 120 km/h and returns to Jammu at 80 km/h. What is the average speed of the train during the whole journey?
(a) 47.5 km/h (b) 96 km/h
(c) 38 km/h (d) 57.5 km/h

83. In the above question if the total time taken by the train is 25 hours, what is the distance between these two places?
(a) 1365.5 km (b) 1369 km
(c) 1200 km (d) can't be determined

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84. Roorkee express normally reaches its destination at 50 km/h in 30 hours. Find the speed at which it travels to reduce the time by 10 hours?
(a) 38 km/h (b) 76 km/h
(c) 75 km/h (d) 60 km/h

85. Two trains *A* and *B* start simultaneously in the opposite direction from two points P and Q and arrive at their destinations 16 and 9 hours respectively after their meeting each other. At what speed does the second train *B* travel if the first train travels at 120 km/h per hour :
(a) 90 km/h (b) 160 km/h
(c) 67.5 km/h (d) none of these

86. There are two trains running on two parallel tracks. Length of each train is 120 m. When they are running in opposite directions, they cross each other in 4 seconds and when they are running in the same direction they cross in 12 seconds. What is the speed of the faster train?
(a) 80 km/h (b) 72 km/h
(c) 120 km/h (d) 144 km/h

87. Two trains are travelling in the same direction at 22.5 km/h and 7.5 km/h respectively. The faster train crosses a man in the slower train in 18 seconds. What is length of the faster train?
(a) 87.5 m (b) 75 m
(c) 122.5 m (d) none of these

88. A train covers a certain distance moving at a speed of 60 km/h. However if it were to halt for a fixed time every hour, its average speed comes out to be 50 km/h. For how much time does the train halt for every hour?
(a) 6 min (b) 10 min
(c) 12 min (d) none of these

89. Two horses start trotting towards each other, one from *A* to *B* and another from *B* to *A*. They cross each other after one hour and the first horse reaches *B*, 5/6 hour before the second horse reaches *A*. If the distance between *A* and *B* is 50 km. What is the speed of the slower horse?
(a) 30 km/h (b) 15 km/h
(c) 25 km/h (d) 20 km/h

90. Pankaj walked at 5 km/h for certain part of the journey and then he took an auto for the remaining part of the journey travelling at 25 km/h. If he took 10 hours for the entire journey. What part of journey did he travelled by auto if the average speeed of the entire journey be 17 km/h :
(a) 750 km (b) 100 km
(c) 150 km (d) 200 km

91. A car travelled first 36 km at 6 km/h faster than the usual speed, but it returned the same distance at 6 km/h slower than the usual speed. If the total time taken by car is 8 hours, for how many hours does it travelled at the faster speed?
(a) 4 (b) 3
(c) 2 (d) 1

92. A dog starts chasing to a cat 2 hours later. It takes 2 hours to dog to catch the cat. If the speed of the dog is 30 km/h, what is the speed of cat?
(a) 10 km/h (b) 15 km/h
(c) 20 km/h (d) can't be determined

93. Prachi starts from Barabanki at 6 am at constant speed of 60 km/h. She halts at Lucknow for half an hour and then drives at 40 km/h. If she reaches Kanpur at 9 : 30 am, which is 160 km from Barabanki, how far is Barabanki from Lucknow?
(a) 75 km (b) 80 km
(c) 100 km (d) 120 km

94. Two trains whose respective lengths are 200 m and 250 m cross each other in 18 s, when they are travelling in opposite direction and in 1 min, when they are travelling in the same direction. What is the speed of the faster train (in km/h)?
(a) 38.5 (b) 48.5
(c) 54 (d) 58.5

95. Abhinav started for the station half a km from his home walking at 1 km/h to catch the train in time. After 3 minutes he realised that he had forgotten a document at home and returned with increased, but constant speed to get it succeded in catching the train. Find his latter speed in km/h :
(a) 1.25 (b) 1.1
(c) $\frac{11}{9}$ (d) 2

Directions for question number 96-97: The ratio of speeds at which Anil and Mukesh walk is 3 : 4. Anil takes 30 minutes more than the time taken by Mukesh in reaching the destination.

96. If Anil drives the car at twice the speed of his walking then the time required to reach his destination by car is :
(a) 45 min (b) 60 min
(c) 1.5 h (d) 1 h 20 min

97. What is the total distance travelled by each of them, if the average of speeds of Anil and Mukesh is 28 km/h?
(a) 48 (b) 60
(c) 17 km (d) 70

98. Train *X* starts from point *A* for point *B* at the same time that train *Y* starts from *B* to *A*. Point *A* and *B* are 300 km apart. The trains are moving at a constant speed atleast at 25 km/h. The trains meet eachother 3 hours after they start. If the faster train takes atleast 2 more hours to reach the destination. By which time will the slower train have definitely reached its destination? (Ignoring the length of trains in crossing)
(a) 4 hours after the start
(b) 7.5 hours after the start
(c) 6 hours after the start
(d) none of the above

99. In reaching the Purnagiri a man took half as long again to climb the second third as he did to climb the first third and a quarter as long again for the last third as for the second third. He took altogether 5 hs 50 minutes. Find the time he spent on the first third of the journey?
(a) 72 min (b) 80 min
(c) 81 min (d) 88 min

100. Walking at four fifth of his usual speed Vijay Malya reaches his office 15 minutes late on a particular day. The next day, he walked at 5/4 of his usual speed. How early would he be to the office when compared to the previous day?
(a) 27 min (b) 32 min
(c) 30 min (d) none of these

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101. Abdul starts in a car from Ahmedabad towards Bangalore. After sometime he realises that he will cover only 75% of the distance in the scheduled time and he therefore doubles his speed immediately and thus manages to reach Bangalore exactly on time. Find the time after which Abdul changed his speed, given that he could have been late by 3 hours if he had not changed his speed :
(a) 3 h (b) 4 h
(c) 5 h (d) 6 h

102. A man travels the first part of his journey at 20 km/h and the next at 70 km/h, covering the entire journey at an average speed of 50 km/h. What is the ratio of the distance that he covered at 20 km/h to that he covered at 70 km/h?
(a) 4 : 21 (b) 3 : 22
(c) 1 : 4 (d) 3 : 5

103. Anjali fires two bullets from the same place at an interval of 6 minutes but Bhagwat sitting in a car approaching the place of firing hears the second fire 5 minute 32 seconds after the first firing. What is the speed of car, if the speed of sound is 332 m/s?
(a) 56 m/s (b) 102 m/s
(c) 28 m/s (d) 32 m/s

104. A car crosses a man walking at 6 km/h. The man can see the things upto 450 m only in one direction due to fog. He sees the car which was going in the same direction for 4.5 minutes. What is the speed of the car?
(a) 9 km/h (b) 12 km/h
(c) 12.5 km/h (d) 15 km/h

105. A man takes 4 h 20 minutes in walking to a certain place and riding back. If he walk on both sides he loses 1 h. The time he would take by riding both ways is :
(a) 2 h 20 min (b) 3 h 20 min
(c) 2 h (d) 4 h 40 min

106. A train met with an accident 120 km from station *A*. It completed the remaining journey at 5/6 of its previous speed and reached 2 hours late at station *B*. Had the accident taken place 300 km further, it would have been only 1 hour late? What is the speed of the train?
(a) 100 km/h (b) 120 km/h
(c) 60 km/h (d) 50 km/h

107. For the above question what is the total distance between *A* and *B* ?
(a) 480 km (b) 520 km
(c) 600 km (d) 720 km

108. The wheel of an engine of 300 cm in circumference makes 10 revolutions in 6 seconds. What is the speed of the wheel (in km/h)?
(a) 18 (b) 20
(c) 27 (d) 36

109. A man can row downstream at 12 km/h and upstream at 8 km/h. What is the speed of man in still water?
(a) 12 km/h (b) 10 km/h
(c) 8 km/h (d) 9 km/h

110. A man can row upstream at 15 km/h and downstream at 21 km/h. The speed of water current of the river is :
(a) 8 km/h (b) 6 km/h
(c) 3 km/h (d) 5 km/h

111. A boat moves downstream at 1 km in 5 minutes and upstream at 1 km in 12 minutes. What is the speed of current?
(a) 4.5 km/h (b) 3.5 km/h
(c) 2 km/h (d) 2.5 km/h

112. A man rows downstream 60 km and upstream 36 km, taking 4 hours each time. The speed of the man is :
(a) 15 km/h (b) 16 km/h
(c) 8 km/h (d) 12 km/h

113. A man can row 5 km/h in still water. If the rate of current is 1 km/h, it takes $\frac{5}{4}$ hours to row to a place and back. How far is the place?
(a) 2 km (b) 2.5 km
(c) 3 km (d) 4 km

114. A man can swim 5 km/h in still water. If the speed of current be 3 km/h, the time taken by him to swim to a place 16 km upstream and back is :
(a) 8 h (b) 7.5 h
(c) 6.66 h (d) 10 h

115. A boat covers 48 km upstream and 72 km downstream in 12 hours, while it covers 72 km upstream and 48 km downstream in 13 hours. The speed of stream is :
(a) 2 km/h (b) 2.2 km/h
(c) 2.5 km/h (d) 4 km/h

116. A motor boat takes 2 hours to travel a distance of 9 km downstream and it takes 6 hours to travel the same distance against the current. The speed of the boat in still water and that of the current (in km/h) respectively are :
(a) 6, 5 (b) 3, 1.5
(c) 8, 5 (d) 9, 3

117. A man can row 15 km/h in still water and he finds that it takes him twice as much time to row up than as to row down the same distance in the river. The speed of the current (in km/h) is :
(a) 6 km/h (b) 6.5 km/h
(c) 4.5 km/h (d) 5 km/h

118. A motor boat takes 12 hours to go downstream and it takes 24 hours to return the same distance. What is the time taken by boat in still water?
(a) 15 h (b) 16 h
(c) 8 h (d) 20 h

119. The speed of a boat in upstream is 2/3 that of downstream. Find the ratio of speed of boat in still water and to the average speed of boat in downstream and upstream?
(a) $\frac{24}{25}$ (b) $\frac{25}{24}$
(c) $\frac{5}{12}$ (d) none of these

120. The difference between downstream speed and upstream speed is 3 km/h and the total time taken during upstream and downstream is 3 hours. What is the downstream speed, if the downstream and upstream distance are 3 km each?
(a) 2.5 km/h (b) 4.33 km/h
(c) 4 km/h (d) 3.3 km/h

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121. A boat which sails at 10 km/h in still water starts chasing, from 10 km behind, another one which sails at 4 km/h in the upstream direction. After how long will it catchup if the stream is flowing at 2 km/h :
(a) 4 h (b) 2.5 h
(c) 2 h (d) 3.5 h

Directions for question number 122 and 123 : *A motor boat went downstream for 120 km and immediately returned. It took the boat 15 hours to complete the round trip. If the speed of the river were twice as high, the trip downstream and back would take 24 hours.*

122. What is the speed of the boat in still water?
(a) 20 km/h (b) 18 km/h
(c) 15 km/h (d) 16 km/h

123. What is the speed of the stream?
(a) 3.5 km/h (b) 4 km/h
(c) 6 km/h (d) 8 km/h

124. A boat sails 15 km of a river towards upstream in 5 hours. How long will it take to cover the same distance downstream, if the speed of current is one-fourth the speed of the boat in still water :
(a) 1.8 h (b) 3 h
(c) 4 h (d) 5 h

125. A boat takes 5 hours more while going back in upstream than in downstream. If the distance between two places is 24 km and the speed of boat in still water be 5.5 km/h. What must be the speed of boat in still water so that it can row donwstream, 24 km, in 4 hours?
(a) 1.5 km/h (b) 3.5 km/h
(c) 4.5 km/h (d) 3 km/h

126. A boat takes 7 hours to go from P to R, through a midpoint Q, but it takes 8 hours to go from P to Q and then return from Q to P. How long it would take to go from R to P?
(a) 7 h (b) 8 h
(c) 9 h (d) none of these

127. Mallah can row 40 km upstream and 55 km downstream in 13 h and 30 km upstream and 44 km downstream in 10 hours. What is the speed of Mallah in still water?
(a) 6 km/h (b) 12 km/h
(c) 3 km/h (d) 8 km/h

Directions for question number 128-129 : *In a kilometre race, A can give B a start of 20 m and also in a half kilometre race C beats A by 50 m.*

128. If A, B and C run a 2 km race, what is the difference between the distances covered by the two losers, when the winner finishes the race?
(a) 64 m (b) 36 m
(c) 32 m (d) 58 m

129. B and C run a half km race, who should give a start to the slower runner and of how many metres so that they both finish the race at the same time?
(a) C, 59 m (b) B, 34 m
(c) C, 48 m (d) B, 56 m

130. In a 1600 m race, A beats B by 80 m and C by 60 m. If they run at the same time then by what distance will C beat B in a 400 m race?
(a) $5\frac{15}{77}$ m (b) $5\frac{20}{76}$ m
(c) $15\frac{5}{77}$ m (d) none of these

131. Aman can run a distance in 190 seconds and Shakti can run the same distance in 200 seconds. If they start together, by what distance Aman can beat Shakti in 1 km race?
(a) 48 m (b) 25 m
(c) 24 m (d) 50 m

132. A runs $\frac{7}{4}$ times as fast as B. If A gives B a start of 300 m, how far must the winning post be if both A and B have to end the race at same time?
(a) 1400 m (b) 700 m
(c) 350 m (d) 210 m

133. A beats B by 100 m in a race of 1200 m and B beats C by 200 m in a race of 1600 m. Approximately by how many metres can A beat C in a race of 9600 m?
(a) 1600 m (b) 1800 m
(c) 1900 m (d) 2400 m

134. In a 1000 m race Ameesha gives a headstart of 100 m to Bipasha and beats her by 200 m. In the same race Ameesha gives a headstart of 100 m to Celina and beats her by 300 m. By how many metres would Bipasha beat Celina in a 50 m race?
(a) 6.66 m (b) 7.143 m
(c) 8 m (d) none of these

135. In a 1000 metres race Ravi gives Vinod a start of 40 m and beats him by 19 seconds. If Ravi gives a start of 30 seconds then Vinod beats Ravi by 40 m. What is the ratio of speed of Ravi to that of Vinod?
(a) 4 : 5 (b) 6 : 5
(c) 3 : 8 (d) 5 : 4

136. In a race, the man who came two places ahead of the last man finished one place ahead of the man who came three places behind the man just ahead of the one who stood second. How many men finished the race?
(a) 6 (b) 5
(c) 4 (d) 8

137. Vinay and Versha run a race with their speeds in the ratio of 5 : 3. They prefer to run on a circular track of circumference 1.5 km. What is the distance covered by Vinay when he passes Versha for the seventh time?
(a) 25.25 km (b) 26.25 km
(c) 132 m (d) none of these

138. A gives both B and C a start of 60 m in a 1500 m race. However, while B finishes with him, C is 15 m behind them when A and B cross the finishing line. How much start can B give C for the 1500 m race course?
(a) $7\frac{6}{23}$ m (b) $15\frac{5}{8}$ m
(c) $7\frac{11}{16}$ m (d) $5\frac{5}{24}$ m

139. In a 600 m race Prabhat has a start of 200 m and the ratio of speeds of Prabhat and Nishith is 4 : 5, then the distance by which Prabhat wins by :
(a) 100 m (b) 80 m
(c) 120 m (d) none of these

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140. In the game of billiards, *A* can give *B*, 20 points in 80 and *B* can give *C*, 16 points in 80. How many points can *A* give *C* in a game of 200?
(a) 64 (b) 72
(c) 80 (d) none of these

141. In a day how many times the minute and hour-hands coincide (or overlap or come together)?
(a) 12 (b) 20
(c) 22 (d) 24

142. In a day how many times the minute-hand and hour-hand make right angle between them?
(a) 12 (b) 20
(c) 22 (d) 44

143. In a day how many times the minute-hand and hour-hand are opposite to each other?
(a) 12 (b) 24
(c) 11 (d) 22

144. The relative speed of minute-hand with respect to hour-hand is :
(a) $\left(5\frac{1}{2}\right)^\circ$ per minute
(b) $\left(\frac{11}{12}\right)$ minute per minute
(c) 6° per minute
(d) both (a) and (b)

145. When do the hands of a clock coincide between 5 and 6?
(a) 5 : 30 (b) 5 : 27 : 16
(c) 5 : 32 : 16 (d) 5 : 28 : 56

146. What is the angle between hour-hand and minute-hand at 2 : 25 pm?
(a) 77.5° (b) 68°
(c) 67.5° (d) none of these

147. When do the two hands of a clock of just after 3 pm make 30° angle between them?
(a) 3 : 15 : 00 (b) 3 : 10 : 54
(c) 3 : 01 : 59 (d) none of these

148. At what time are the hands of clock together between 7 pm and 8 pm?
(a) 7 : 45 : 54 (b) 7 : 36 : 27
(c) 7 : 37 : 49 (d) 7 : 38 : 11

149. What is the angle between the hands of a clock when the time is 15 minutes past 6?
(a) 97.5° (b) 87.5°
(c) 77° (d) none of these

150. What are the possible times when a clock shows 35° angle between two hands between 3 pm and 4 pm?
(a) 20 min 25 sec (b) 25 min 20 sec
(c) 22 min 43 sec (d) none of these

LEVEL 2

Directions for question number 1-4 : Aishwarya is going to cover a distance of 360 km from Ambala to Khandala. The first one-third of the distance she covers on a cycle. The second one-third she covers by an auto-rickshaw and the remaining distance she travels by car. The average speed of the journey by a car is 5 times the average speed by cycle and 20 km/h more than the average speed by auto-rickshaw, but she took 1 hour more by auto-rickshaw than by car.

1. What is the average speed of the whole journey?
(a) 15 km/h (b) 24 km/h
(c) 20 km/h (d) none of these

2. What is the time taken in the whole journey?
(a) 10 h (b) 12 h
(c) 15 h (d) none of these

3. What is the distance covered by her in last five hours of her journey?
(a) 250 km (b) 240 km
(c) 200 km (d) can't be determined

4. Instead of travelling the first one-third by cycle if she travels by same auto-rickshaw with the same average speed, then what is the percentage decrease/increase in time taken during the entire journey?
(a) 46.66% (b) 33.33%
(c) 50% (d) 25%

5. Bipasha and Mallika leave towns Kolkata and Ambala at 6 am and travel towards Ambala and Kolkata respectively. Speed of Bipasha is 60 km/h and speed of Mallika is 120 km/h. Rani leaves Kolkata for Ambala sometime later and travels at a speed of 90 km/h. If the distance between Kolkata and Ambala is 1080 km and all three meet at the same point on the way, at same time, then at what time did Rani leave Kolkata?
(a) 7 am (b) 8 am
(c) 7:30 am (d) 10 am

6. A passenger sitting in a train of length *l* m, which is running with speed of 60 km/h passing through two bridges, notices that he crosses the first bridge and the second bridge in time intervals which are in the ratio of 7 : 4 respectively. If the length of first bridge be 280 m, then the length of second bridge is :
(a) 490 m (b) 220 m
(c) 160 m (d) can't be determined

7. Pathik and Rahi started from two places Andheri and Bhavnagar towards Bhavnagar and Andheri respectively at 8 : 20 am. The speeds of Pathik and Rahi are in the ratio of 4 : 5. They meet at Chandni Chowk, somewhere between Andheri and Bhavnagar, spent some-time together enjoyed coffee and burger and then both started towards their destination at 9 : 27 am. If Pathik reaches Bhavnagar at 10 : 32 am, how much time did they spend together?
(a) 8 min (b) 12 min
(c) 15 min (d) can't be determined

8. A train with 120 wagons crosses Arjun who is going in the same direction, in 36 seconds. It travels for half an hour from the time it starts overtaking the Arjun (he is riding on the horse) before it starts overtaking Srikrishna (who is also riding on his horse) coming from the opposite direction in 24 seconds. In how much time (in seconds) after the train has crossed the Srikrishna do the Arjun meets to Srikrishna?
(a) 3560 sec (b) 3600 sec
(c) 3576 sec (d) can't be determined

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9. Kareena and Shahid start from Kurla and Worli towards Worli and Kurla respectively, at the same time. After they meet at Shantakruz on the way from Kurla to Worli, Kareena reduces her speed by 33.33% and returns back to Kurla and Shahid increases his speed by 33.33% and returns back to Worli. If Kareena takes 2 hours for the entire journey, what is the time taken by Shahid for the entire journey?
(a) 96 min (b) 84 min
(c) 168 min (d) can't be determined

10. Due to the technical snag in the signal system two trains start approaching each other on the same rail track from two different stations, 240 km away from each other. When the train starts a bird also starts moving to and fro between the two trains at 60 km/h touching each time each train. The bird is initially sitting on the top of the engine of one of the trains and it moves so till these trains collide. If these trains collide one and a half hour after the start, then how many kilometres bird travells till the time of collision of trains?
(a) 90 km (b) 130 km
(c) 120 km (d) none of these

11. Einstein walks on an escalator at a rate of 5 steps per second and reaches the other end in 10 seconds. While coming back, walking at the same speed he reaches the starting point in 40 seconds. What is the number of steps on the escalator?
(a) 40 (b) 60
(c) 120 (d) 80

12. A girl while walking diametrically across a semicircular playground, takes 3 minutes less than if she had kept walking round the circular path from A to B. If she walks 60 metres a minute, what is the diameter of the play ground :
(a) 60 m (b) 48 m
(c) 84 m (d) 315 m

13. Two trains start simultaneously from two stations Howrah and Bandra, respectively towards each other on the same track. The distance between the two stations is 560 km and the speed of trains are 30 and 40 km/h.
Simultaneously with the trains, a sparrow sitting on the top of one of the train starts towards the other and reverses its direction on reaching the other train and so on. If the speed of sparrow is 80 km/h then the distance that the sparrow flies before being crushed between the train is :
(a) 70 km (b) 560 km
(c) 640 km (d) 650 km

14. A surveillance plane is moving between two fixed places Pukhwara and Kargil at 120 km/h. The distance between two places is 600 km. After 18 hour what will be the distance between the Kargil and its position if it is starts moving from Pukhwara?
(a) 360 km (b) 300 km
(c) 240 km (d) none of these

15. The speed of a car during the second hour of its journey is thrice that in the first hour. Also its third hours speed is the average speed of the first two hours. Had the car travelled at the second hours speed during all the first three hours, then it would have travelled 150 km more. Find the percentage reduction in time in the second case for the first three hours :
(a) $33\frac{1}{3}\%$ (b) 40%
(c) 25% (d) 50%

16. There are three runners Tom, Dick and Harry with their respective speeds of 10 km/h, 20 km/h and 30 km/h. They are initially at P and they have to run between the two points P and Q which are 10 km apart from each other. They start their race at 6 am and end at 6 pm on the same day. If they run between P and Q without any break, then how many times they will be together either at P and Q during the given time period?
(a) 5 (b) 7
(c) 4 (d) 12

Directions for question number 17 and 18 : Arjun and Srikrishna go by chariot from Mathura to Kurukshetra which is on the way to Hastinapur. Abhimanyu goes from Hastinapur to Kurukshetra. The distance between Mathura to Hastinapur is 700 km and the distance between Hastinapur and Kurukshetra is 300 km. Speed of Arjun and Srikrishna's chariot is 25 km/h and speed of Abhimanyu is 10 km/h. All the three persons start their journey at 10 am. After travelling some miles Srikrishna sees Duryodhan going (by riding on his horse) at 20 km/h to Kurukshetra. Arjun and Srikrishna go ahead meet Abhimanyu and pick him up. Then they return immediately to Kurukshetra and thus all the four reach at the same time.

17. What is the total distance travelled by Arjun?
(a) 400 (b) 500
(c) 600 (d) can't be determined

18. What is the total time taken to reach Kurukshetra?
(a) 10 h (b) 15 h
(c) 18 h (d) 24 h

19. Priyanka, Akshay and Salman started out on a journey to watch the newly released movie "Mujhse Shaadi Karogi", which was being shown at wave cine-multiplex. The multiplex was 120 km away from their starting point of journey. Priyanka and Salman went by car at the speed of 50 km/h, while Akshay travelled by Tonga (horse cart) at 10 km/h. After a certain distance Salman got off and travelled the rest distance by another Tonga at 10 km/h, while Priyanka went back for Akshay and reached the destination at the same time that Salman arrived. The number of hours required for the trip was :
(a) 4 h (b) 5 h
(c) 4.8 h (d) can't be determined

Directions for question number 20 and 21 : Ajai and Kajol start towards each other at the same time from Barabanki and Fatehpur for their destinations Fatehpur and Barabanki respectively which are 300 km apart. They meet each other 120 km away from Barabanki.

20. Shahrukh starts from Barabanki to Fatehpur, 1 hour after Ajai starts. Shahrukh meets Kajol 1.5 hours after Shahrukh starts. If the speed of Shahrukh is atleast 20 km/h faster than the speed of Kajol.
Which of the following statements is true?

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(a) The minimum possible speed of Ajai is 45 km/h
(b) The maximum possible speed of Ajai is 45 km/h
(c) The minimum possible speed of Kajol is 60 km/h
(d) The maximum possible speed of Kajol is 60 km/h

21. What is the minimum speed of Shahrukh to overtake Ajai, before he meets Kajol? (Use the data from previous question, if necessary)
(a) 30 (b) 40
(c) 60 (d) none of these

Directions for question number 22-24 : *Raghupati goes at a speed of 60 km/h. Raghav goes at a speed of 36 km/h. Raja Ram can go from Azamgarh to Bareilley in 2 hours. The distance between Azamgarh to Bareilley is equal to the distance between Azamgarh to Chandoli. Raghav takes the same time travelling from Bareilley to Azamgarh as from Bareilley to Chandoli at his regular speed which is twice the speed of Raja Ram.*

22. What is the distance between Azamgarh and Chandoli?
(a) 60 km (b) 27 km
(c) 36 km (d) 18 km

23. How much time will Raghupati take to complete a round trip of the three cities?
(a) 1 h 12 min (b) 1 h 48 min
(c) 1 h 30 min (d) 1 h 36 min

24. If Raghupati and Raja Ram travel towards each other from Bareilley and Chandoli respectively, how far from Bareilley will they meet each other?
(a) $\frac{60}{13}$ (b) $27\frac{9}{13}$ km
(c) $37\frac{9}{13}$ (d) $\frac{360}{9}$

Directions for question number 25-26 : *Mohan, Namit and Pranav travel from Shantipur to Hulchulpur. They have a two seeter bike which can be driven by only Mohan. It is known that due to very stringent traffic rules only two persons can ride at a time. Hulchulpur is 180 km away from Shantipur. All of them can walk at 6 km/h, but reach to Hulchulpur simultaneously also they started their journey simultaneously.*

25. If the speed of the bike is 36 km/h, then what is the total distance that the bike travels?
(a) 400 km (b) 380 km
(c) 200 km (d) 320 km

26. If the speed of the bike is 42 km/h, then what is the shortest possible time in which all three of them can complete the journey?
(a) $7\frac{1}{3}$ h (b) $9\frac{4}{7}$ h
(c) $9\frac{3}{7}$ h (d) can't be determined

27. While walking down on the pavements of New York city. I notice that every 20 minute there is a city bus coming in the opposite direction and every 30 minute there is a city bus overtaking me from behind. What is the time gap between one city bus passing a stationary point known as Local Bus Stop beside the route and the immediately next city bus in the same direction passing the same stationary point?
(a) 27 min (b) 24 min
(c) 25 min (d) can't determined

28. Abhinav and Brijesh start from Allahabad and Barabanki respectively with uniform velocities. Abhinav is headed towards Barabanki and Brijesh towards Allahabad and both cities are 600 km apart. Abhinav rests whenever Brijesh is on the move and Brijesh rest whenever Abhinav is on the move. Abhinav's speed is 25 km/h and Brijesh's speed is 30 km/h. If Abhinav starts first and reaches Barabanki in 36 hours, then find the least time that Brijesh would take to reach his destination after Abhinav makes a start :
(a) 20 h (b) 36 h
(c) 44 h (d) none of these

29. A man can cross a downstream river by steamer in 40 minutes and same by boat in 1 hour. If the time of crossing the river in upstream direction by steamer is 50% more than downstream time by the steamer and the time required by boat to cross the same river by boat in upstream is 50% more than the time required in downstream by boat. What is the time taken for the man to cross the river downstream by steamer and then return to same place by boat half the way and by steamer the rest of the way?
(a) 85 min (b) 115 min
(c) 120 min (d) 125 min

Directions for question number 30 and 31 : *Awadh express and Bokaro express start simultaneously from Lucknow and Jamshedpur towards each other and continuously shuttle between these two places. Every time these trains meet each other, they turn back after exchanging their respective speeds, the initial ratio of their speeds is 2 : 1.*

30. What is the number of distinct places at which they will meet?
(a) 1 (b) 2
(c) 5 (d) none of these

31. Let these two trains first time meet at Patna, then what is the ratio of distances covered by Awadh express and Bokaro express till they meet for the third time at the same place Patna :
(a) 1 : 1 (b) 14 : 13
(c) 10 : 11 (d) none of these

32. Mahindra starts a journey for his office, which is in the north east of his home. An hour after starting meets with a minor accident. He takes one hour in resuming his journey. After that he proceeds at 5/6th of his former speed and arrives at the office 1 hour 36 minutes late than the scheduled time. Had the accident occurred 80 kms further from the actual place of accident, he would have arrived 1 hour 20 minutes beyond the scheduled time. What is the distance between his office and his home?
(a) 180 km (b) 240 km
(c) 250 km (d) 300 km

33. A soldier fired two bullets at an interval of 335 seconds moving at a uniform speed v_1. A terrorist who was running ahead of the soldier in the same direction, hears the two shots at an interval of 330 seconds? If the speed of sound is 1188 km/h, then who is the faster and by how much?

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(a) Soldier, 22 km/h (b) Terrorist, 25 km/h
(c) Soldier, 18 km/h (d) Terrorist, 20 km/h

34. A hunter fired two shots from the branch of a tree at an interval of 76 seconds. A tiger separating too fast hears the two shots at an interval of 83 seconds. If the velocity of the sound is 1195.2 km/h, then find the speed of tiger?
(a) 112.8 km/h (b) 100.8 km/h
(c) 80.16 km/h (d) none of these

35. A man goes to the fair in Funcity with his son and faithful dog. Unfortunately man misses his son which he realises 20 minutes later. The son comes back towards his home at the speed of 20 m/min and man follows him at 40 m/min. The dog runs to the son (child) and comes back to the man (father) to show him the direction of his son. It keeps moving to and fro at 60 m/min between son and father, till the man meets the son. What is the distance travelled by the dog in the direction of the son?
(a) 800 m (b) 1675 m
(c) 848 m (d) 1000 m

36. Amarnath express left Amritsar for Gorakhpur. Two hours later Gorakhnath express left from Amritsar to Gorakhpur. Both trains reached Gorakhpur simultaneously. If Amarnath express had started from Amritsar and Gorakhnath express had started from Gorakhpur at the same time and travelled towards each other they would meet in 1 h 20 min. Find the time taken by Amarnath express to travel from Amritsar to Gorakhpur (in hours) :
(a) 2 (b) 4
(c) 5 (d) 6

37. Akbar and Birbal setout at the same time to walk towards eachother respectively from Agra and Banaras 144 km apart. Akbar walks at the constant speed of 8 km/h, while Birbal walks 4 km in the first hour, 5 km in the second hour, 6 km in the third hour and so on. Then the Akbar and Birbal will meet :
(a) in 6 h
(b) in 8 h
(c) midway between Agra and Banaras
(d) 80 km away from Banaras

38. A tiger is 50 of its own leaps behind a deer. The tiger takes 5 leaps per minute to the deer's 4. If the tiger and the deer cover 8 m and 5 m per leap respectively, what distance will the tiger have to run before it catches the deer?
(a) 600 m (b) 700 m
(c) 800 m (d) 1000 m

39. Soniya and Priyanka started from Amethi and Bellari for Bellari and Amethi, which are 645 km apart. They meet after 15 hours. After their meeting, Sonia increased her speed by 3 km/h and Priyanka reduced her speed by 3 km/h, they arrived at Bellari and Amethi respectively at the same time. What is their initial speeds?
(a) 24 km/h and 30 km/h (b) 25 km/h and 18 km/h
(c) 18 km/h and 21 km/h (d) 20 km/h and 23 km/h

40. Den Bosch and Eastbourne are two famous cities 300 km apart. Maradona starts from Den Bosch at 8 : 24 am. An hour later Pele starts from Den Bosch. After travelling for 1 hour, Pele reaches Nottingham that Maradona had passed 40 minutes earlier. Nottingham falls on the way from Den Bosch to Eastbourne. If Pele and Maradona just reaches Eastbourne at the same time, what are the speeds of the Maradona and Pele respectively?
(a) 100 km/h, 125 km/h (b) 60 km/h, 80 km/h
(c) 60 km/h, 75 km/h (d) 75 km/h, 100 km/h

41. A thief sees a jeep at a distance of 250 m, coming towards him at 36 km/h. Thief takes 5 seconds to realise that there is nothing but the police is approaching him by the jeep and start running away from police at 54 km/h. But police realise after 10 seconds, when the thief starts running away, that he is actually a thief and gives chase at 72 km/h. How long after thief saw police did police catchup with him and what is the distance police had to travel to do so?
(a) 50 s, 1000 m (b) 65 s, 1150 m
(c) 65 s, 1300 m (d) 45 s, 1050 m

42. Inspired by the 'Golden quadrilateral project' UP Government recently accomplished a diamond triangular project. Under this project the State Government laid down 6 lane roads connecting three cities Ayodhya, Banaras and Chitrakoot, which are equally separated from each other *i. e.*, in terms of geometry they form an equilateral triangle. Angad and Bajrang start simultaneously from Ayodhya and Banaras respectively, towards Chitrakoot. When Angad covers 100 kms, Bajrang covers such a distance that the distance betweeen Angad and Bajrang makes 90° angle with the road joining Banaras and Chitrakoot. When Bajrang reaches Chitrakoot, Angad is still 150 km away from Chitrakoot. What is the distance between Ayodhya and Banaras?
(a) 250 km (b) 450 km
(c) 300 km (d) none of these

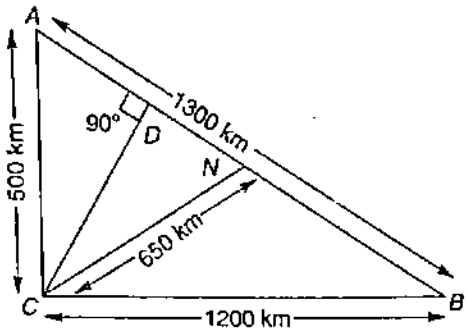
43. Two trains Ajanta express and Barouni express simultaneously started on two parallet tracks from Meerut to Nagpur, which are 390 km apart. The ratio of the speed of Ajanta express and Barouni express is 6 : 7. After how long (in kms) travelling, Barouni express exchanges the speed with Ajanta express so that both the trains reach at their destination simultaneously :
(a) 150 km (b) 190 km
(c) 210 km (d) can't be determined

44. In a circus there was a leopard and a tiger walking in the two different rings of same radii. There I observed that when leopard moved 3 steps, tiger moved 5 steps in the same time, but the distance traversed by leopard in 5 steps is equal to the distance traversed by tiger in 4 steps. What is the number of rounds that a leopard made when tiger completed 100 rounds?
(a) 120 (b) 48
(c) 75 (d) none of these

Directions for question number 45-48 : *In the following figure the route is shown which is followed by Professor Jai and Professor Jaya, who are visting faculty at IIM-A and IIM-B respectively. A, B denote IIM-A and IIM-B respectively and C denotes the residence of Prof. Jai and Prof. Jaya. They leave home for classes at the same time and their driving speeds are* $\frac{500}{13}$ *km/h and* $\frac{1200}{13}$ *km/h respectively. Also they finish the classes at the same time to reach home.*

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The path adopted by Jai and Jaya is CADC and CBDC respectively. Prof. Jai and Prof. Jaya are husband and wife respectively.

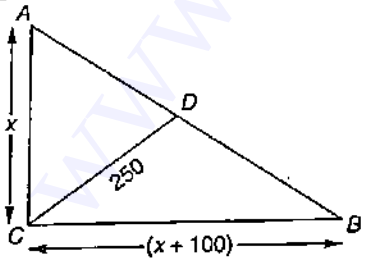
45. If both of them start and finish the classes at the same time, then who returned home earlier than other, if no one of them halts for anywhere in the route and they just leave the institution as soon as they finish the lectures?
(a) Prof. Jai
(b) Prof. Jaya
(c) Return at the same time
(d) Can't be determined

46. In the shown figure *N* and *D* denotes Noida and Delhi respectively, who returned home late and by how much time, if Jaya turned from Noida instead of Delhi :
(a) Jai, 9 h 10 min (b) Jaya, 9 h 50 min
(c) Jai, 2 h 55 min (d) Jai, 16 h 10 min

47. In the above question how many per cent time Jaya saved in going via Noida of the total time taken previously :
(a) 10% (b) 25%
(c) 50% (d) 17%

48. If Mrs. Jaya wants to watch the premier show of a movie at Wave Cinema in Noida while returning from institute through BNC. When will she return home given that she spends total time 3 hours at wave cinema?
(a) at the same time as normal
(b) 5 min late than her husband
(c) at the same time when her husband returns
(d) can't be determined

49. Preetam and Devi start running a race on the given track as shown in figure.



Where *AC* and *BC* are mutually perpendicular and *CD* is the median of triangular paths *ABC*. *BC* is 100 km longer than that of *AC*, again *CD* is 250 km. The speeds of Preetam and Devi are 30 km/h and 40 km/h, initially and their respective paths of running are *CADC* and *CBDC*. After how much time they reverse their speeds so that they return *C* at the same time?
(a) $\frac{50}{7}$ h (b) $\frac{120}{7}$ h
(c) $\frac{80}{11}$ h (d) none of these

Directions for question number 50 ... markets $M_1, M_2, M_3, \ldots$ etc of Vyapar ... circular paths $P_1, P_2, P_3, \ldots$ etc. All the ... are concentric at centre 'O' and their distances are 1 km, 2 km, 3 km, ... etc from the centre 'O' respectively. At the centre 'O' there is a "Khoob Khao" tiffin agency which supplies the tiffins to all the markets $M_1, M_2, M_3, \ldots$ etc. A tiffin carrier starts from 'O' goes directly east of the shop and then on reaching the circular path it moved 1 km in counter clock direction on it. After completing its 1 km distance on P_1 the carrier moves to P_2 in the radial direction. Then it goes 2 km on P_2. Similarly 3 km on P_3 and 4 km on P_4 etc in counter clock direction moving radially from P_2 to P_3 and P_3 to P_4 etc and motion of the carrier continued in this manner till it reaches exactly in the east direction.

50. After reaching east of the shop it can't move on further than the given distance on the current path. For how many markets can it supply its tiffins directly?
(a) 4 (b) 5
(c) 7 (d) can't be determined

51. The total distance covered by the carriers in providing the tiffins from centre 'O' to the last point in one way only is :
(a) 30 km (b) 28 km
(c) 35 km (d) none of these

52. The ratio of distances covered on the circular path P_2 to that on the last path, where the carrier reaches directly eastward of its shop is :
(a) 1 : 1 (b) 2 : 7
(c) 2 : π (d) none of these

Directions for question number 53-57 : A train enters into a tunnel AB at A and exits at B. A jackal is sitting at O in another by passing tunnel AOB, which is connected to AB at A and B, where OA is perpendicular to OB. A cat is sitting at P inside the tunnel AB making the shortest possible distance between O and P, such that AO : PB = 30 : 32. When a train before entering into the tunnel AB makes a whistle (or siren) somewhere before A, the jackal and cat run towards A, they meet with accident (with the train) at the entrance A. Further if the cat moves towards B instead of A it again meets with accident at the exit of the tunnel by the same train coming from the same direction.

53. What is the ratio of speeds of jackal and cat?
(a) 4 : 3 (b) 5 : 3
(c) 1 : 1 (d) can't be determined

54. The ratio of speeds of jackal is to train is :
(a) 5 : 1 (b) 3 : 5
(c) 1 : 5 (d) can't be determined

55. If jackal moves towards *OPA*, it will meet with train at M_1 then AM_1 is :
(a) 20 km (b) 16 km
(c) 10 km (d) can't be determined

56. If jackal moves towards *OPB* and cat moves towards *POA* who will not meet with accident with the train?
(a) Jackal (b) Cat
(c) Both (a) and (b) (d) Can't be determined

57. The ratio of time taken by cat and jackal in moving *OAPO* and *PBOP* respectively given that they do not meet with accident :
(a) 1 : 1 (b) 3 : 4
(c) 5 : 4 (d) none of these

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58. A candle of 6 cm long burns at the rate of 5 cm in 5 h and another candle of 8 cm long burns at the rate of 6 cm in 4 h. What is the time required by each candle to remain of equal lengths after burning for some hours, when they start to burn simultaneously with uniform rate of burning?
(a) 1 h (b) 1.5 h
(c) 2 h (d) none of these

59. Two boats start at the same instant to cross a river W metre wide. The faster boat reaches the other bank and returns back immediately. What are the distances travelled by them when they meet, where the speeds of these boats are b_1 & b_2?
(a) $\frac{2W}{(b_1+b_2)}, \frac{2W}{(b_1-b_2)}$
(b) $\frac{2W}{(b_1+b_2)} b_1$ and $\frac{2W}{(b_1+b_2)} b_2$
(c) $\frac{W}{(b_1+b_2)} b_1, \frac{W}{(b_1+b_2)} b_2$
(d) data insufficient

60. Mariya was travelling in her boat when the wind blew her hat off and the hat started floating back downstream. The boat continued to travel upstream for twelve more minutes before Mariya realized that her hat had fallen off and turned back downstream. She caught up with that as soon as it reached the starting point. Find the speed of river if Mariya's hat flew off exactly 3 km from where she started :
(a) 5 km/h (b) 6 km/h
(c) 7.5 km/h (d) can't be determined

61. Akbar, Birbal and Chanakya run around a circular track of length 500 m. Akbar and Birbal run with the speeds of 15 m/s and 20 m/s in the same direction respectively and Chanakya being very intelligent run in the opposite direction with a speed of 25 m/s. If all three of them start at the same time, then :
(a) Akbar meets Chanakya more frequently than Birbal does
(b) Akbar and Chanakya meets as frequently as Birbal and Chanakya
(c) Akbar meets Birbal least frequently
(d) Nothing can be concluded

62. Arun and Barun run with the speeds of 30 m/s and 20 m/s around a circular track of 600 m. They participate in a 3000 m race. What is the distance covered by Arun when he passes Barun for the 5th time?
(a) 2200 m (b) 2250 m
(c) 2850 m (d) none of these

63. Akkal and Bakkal are running on a circular track of radius 175 metres. Akkal can complete a round in 100 seconds and the speed of Bakkal is twice the speed of Akkal. They started simultaneously towards each other from two points 350 metres diametrically opposite on the circular path. If they first meet at a point they called it love point, which is between the two points P and Q from where they have started their race, after how much time from the start do they meet at love point for the third time?
(a) $218\frac{2}{5}$ s (b) $216\frac{2}{3}$ s
(c) 221 s (d) none of these

64. Arti and Barkha start swimming towards each other from the deep end and shallow end respectively of a swimming pool in Funcity. They start their swimming simultaneously in the length of 300 m pool. The ratio of their speeds is 1 : 2 respectively. Each swimmer rests for 6 seconds once she reaches the other end and starts swimming back. Where will they meet for the second time in the still water of swimming pool?
(a) 30 m from the shallow end
(b) at the shallow end
(c) at the deepend
(d) can't be determined

65. A and B runs around a circular track. A beats B by one round or 10 minutes. In this race, they had completed 4 rounds. If the race was only of one round, find the A's time over the course :
(a) 8 min (b) 7.5 min
(c) 12.5 min (d) 12 min .

66. A, B and C participated in a race. A covers the same distance in 49 steps, as B covers in 50 steps and C in 51 steps. A takes 10 steps in the same time as B takes 9 steps and C takes 8 steps. Who is the winner of the race?
(a) A (b) B
(c) C (d) can't be determined

67. Shambhu drives his car very fast at 360 m/s. Moving ahead for some hours he finds some problem in headlights of the car. So he takes 20 seconds in changing the bulb of the headlight by stopping the car. Mean while he notices that another car which was 400 m back is now 200 m ahead of his car. What is the speed of this car?
(a) 100 km/h (b) 92 km/h
(c) 108 km/h (d) 300 km/h

68. Two persons start from the opposite ends of a 90 km straight track and run to and fro between the two ends. The speed of first person is 30 m/s and the speed of other is 125/6 m/s. They continue their motion for 10 hours. How many times they pass each other?
(a) 10 (b) 9
(c) 12 (d) none of these

69. At what time after 3 : 10 am, the acute angle made by the minute and hour-hand is double to that of at 3 : 10 am, for the first time?
(a) 4 h 43 min (b) 3 h 48 min
(c) $3\text{ h }\frac{320}{11}$ min (d) none of these

70. If the two incorrect watches are set at 12 : 00 noon at correct time, when will both the watches show the correct time for the first time given that the first watch gains 1 min in 1 hour and second watch loses 4 min in 2 hours :
(a) 6 pm, 25 days later
(b) 12 : 00 noon, 30 days later
(c) 12 noon, 15 days later
(d) 6 am 45 days later

71. Rajeev and Sanjeev are too close friends Rajeev's watch gains 1 minute in an hour and Sanjeev's watch loses 2 minutes in an hour. Once they set both the watches at 12 : 00 noon, with my correct watch. When will the two incorrect watches of Rajeev and Sanjeev show the same time together?
(a) 8 days later (b) 10 days later
(c) 6 days later (d) can't be determined

72. At a railway station a 24 hour watch loses 3 minutes in 4 hours. If it is set correctly on Sunday noon when will the watch show the correct time?
(a) 6 pm after 40 days (b) 12 noon after 75 days
(c) 12 pm after 100 days (d) 12 noon after 80 days

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73. A swiss watch is being shown in a museum which has a very peculiar property. It gains as much in the day as it loses during night between 8 pm to 8 am. In a week how many times will the clock show the correct time?
(a) 6 times (b) 14 times
(c) 7 times (d) 8 times

74. A wrist watch which is running 12 minutes late on a Sunday noon is 16 minutes ahead of the correct time at 12 noon on the next Sunday. When is the clock 8 minutes ahead of time?
(a) Thursday 10 am (b) Friday noon
(c) Friday 8 pm (d) Tuesday noon

75. A clock loses 2 minutes in an hour and another clock gains 2 minutes in every 2 hours. Both these clocks are set correctly at a certain time on Sunday and both the clocks stop simultaneously on the next day with the time shown being 9 am and 10 : 06 am. What is the correct time at which they stopped?
(a) 9 : 54 am (b) 9 : 44 pm
(c) 9 : 46 am (d) 9 : 44 am

76. David sets his watch at 6 : 10 am on Sunday, which gains 12 minutes in a day. On Wednesday if this watch is showing 2 : 50 pm. What is the correct time?
(a) 1 : 50 pm (b) 2 : 10 pm
(c) 2 : 30 pm (d) 3 : 30 pm

77. Ramu purchased a second hand Swiss watch which is very costly. In this watch the minute-hand and hour hand coincide after every $65\frac{3}{11}$ minutes. How much time does the watch lose or gain per day?
(a) 4 min (b) 5 min
(c) 4 min, 20 sec (d) none of these

78. My watch was 8 minutes behind at 8 pm on Sunday but within a week at 8 pm on Wednesday it was 7 minutes ahead of time. During this period at which time this watch has shown the correct time :
(a) Tuesday 10 : 24 am
(b) Wednesday 9 : 16 pm
(c) It cannot show the correct time during this period
(d) None of the above

79. Out of the following four choices which does not show the coinciding of the hour hand and minute-hand :
(a) 3 : 16 : 2 (b) 6 : 32 : 43
(c) 9 : 59 : 05 (d) 5 : 27 : 16

80. Kumbhakarna starts sleeping between 1 am and 2 am and he wakes up when his watch shows such a time that the two hands (i.e., hour-hand and minute-hand) interchange the respective places. He wakes up between 2 am and 3 am on the same night. How long does he sleep?
(a) $55\frac{5}{13}$ min (b) $110\frac{10}{13}$ min
(c) $54\frac{6}{13}$ min (d) none of these

SPEED TEST (TSD)

Directions for question number 1 to 3: Arun took part in a triathlon, an athletic event. He had to swim, run and bicycle to 10 km, 24 km and 30 km, respectively and return the same way. Arun's average speed for the triathelon is 4 km/h. He took a total of 4 min for swimming and 20 min for bicycling in the triathelon.

1. His speed of running is:
(a) 5 km/min (b) 6 km/min
(c) 7.5 km/min (d) $\frac{8}{3}$ km/min

2. He finishes race in:
(a) 16 min (b) 23 min
(c) 32 min (d) 35 min

3. How much time he took while returning for the bicycle if the time taken for the return of each phase (i. e., each event) is 50% greater than that of taken initially:
(a) 12 min (b) 11 min
(c) $11\frac{1}{9}$ min (d) none of these

Directions for question number 4 to 5: A, B and C start running a race from the same point P in the same direction. A runs around a path which is an equilateral triangle PQX and B runs around a square path PQYZ and C runs on the regular hexagonal path PQRSTU. Where the one side PQ of each path is common.

4. If all of them complete one equal round at the same time then which of the following is true?
(a) Speed of *C* is twice that of *B*
(b) Speed of *A* is half that of *C*
(c) Speed of *B* is 50% more than that of *A*
(d) none of the above

5. If each of them run at a same constant speed, what is the maximum number of rounds anyone of them would have completed when they meet for the first time?
(a) 1 (b) 2 (c) 3 (d) 4

Directions for the question number 6 to 9: City X and City Y are connected to a straight road. A and B start moving simultaneously towards each other. After travelling some distance, B takes a 60° turn to his left. Two hours later after B turns, A takes a 60° turn to his right. A travels 60 km after turning, before he meets B. A and B meet 10 hours after they start their journey. A and B together travel 200 km before turning and they arrive at the meeting point simultaneously.

6. How many hours after turning does *A* meet *B* ?
(a) 1 (b) 3 (c) 4 (d) 5

7. What distance does *A* travel before turning 60 degree to his right?
(a) 140 km (b) 150 km
(c) 160 km (d) 170 km

8. If *A* and *B* had not turned, after how many hours would they have met?
(a) $7\frac{1}{8}$ h (b) $9\frac{1}{8}$ h (c) $6\frac{1}{8}$ h (d) $8\frac{1}{8}$ h

9. What is *B*'s speed?
(a) 10 km/h (b) 12.5 km/h
(c) 12 km/h (d) 15 km/h

10. Chetak and Ashwa, two horses, start galloping from Patna to Ranchi which are 80 km apart. The speed of Chetak is 160 km/h and that of Ashwa is 150 km/h. They start galloping simultaneously, from Patna to Ranchi. Chetak reached to Ranchi and returned to Patna but Ashwa returned from Jamshedpur (which is somewhere between Patna and Ranchi) to Patna at the same time. What is the ratio of distances between Patna and Jamshedpur and Jamshedpur and Ranchi?
(a) 15 : 1 (b) 3 : 25
(c) 15 : 2 (d) none of these

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Answers

INTRODUCTORY EXERCISE-9.1

1 (a)	2. (c)	3. (c)							

LEVEL-1

1 (a)	2. (b)	3. (b)	4. (c)	5. (a)	6. (c)	7. (c)	8. (b)	9. (a)	10. (c)
11. (c)	12. (d)	13. (a)	14. (d)	15. (b)	16. (c)	17. (b)	18. (c)	19. (b)	20. (c)
21. (b)	22. (d)	23. (a)	24. (b)	25. (c)	26. (d)	27. (d)	28. (d)	29. (b)	30. (c)
31. (d)	32. (a)	33. (b)	34. (c)	35. (d)	36. (a)	37. (b)	38. (a)	39. (b)	40. (c)
41. (a)	42. (b)	43. (b)	44. (c)	45. (c)	46. (c)	47. (c)	48. (b)	49. (a)	50. (b)
51. (b)	52. (a)	53. (b)	54. (c)	55. (c)	56. (c)	57. (a)	58. (b)	59. (d)	60. (b)
61. (a)	62. (c)	63. (c)	64. (c)	65. (b)	66. (d)	67. (d)	68. (c)	69. (d)	70. (a)
71. (a)	72. (a)	73. (b)	74. (c)	75. (b)	76. (c)	77. (b)	78. (c)	79. (a)	80. (c)
81. (d)	82. (b)	83. (c)	84. (c)	85. (b)	86. (d)	87. (b)	88. (b)	89. (d)	90. (c)
91. (c)	92. (b)	93. (d)	94. (d)	95. (c)	96. (b)	97. (a)	98. (b)	99. (b)	100. (a)
101. (d)	102. (a)	103. (c)	104. (b)	105. (b)	106. (c)	107. (d)	108. (a)	109. (b)	110. (c)
111. (b)	112. (d)	113. (c)	114. (d)	115. (a)	116. (b)	117. (d)	118. (b)	119. (b)	120. (b)
121. (b)	122. (b)	123. (c)	124. (b)	125. (b)	126. (c)	127. (d)	128. (b)	129. (a)	130. (a)
131. (d)	132. (b)	133. (c)	134. (b)	135. (b)	136. (b)	137. (b)	138. (b)	139. (a)	140. (c)
141. (c)	142. (d)	143. (d)	144. (d)	145. (b)	146. (a)	147. (b)	148. (d)	149. (a)	150. (c)

LEVEL-2

1 (b)	2. (c)	3. (b)	4. (a)	5. (b)	6. (c)	7. (c)	8. (c)	9. (b)	10. (a)
11. (d)	12. (d)	13. (c)	14. (c)	15. (a)	16. (b)	17. (c)	18. (d)	19. (c)	20. (b)
21. (d)	22. (c)	23. (b)	24. (b)	25. (b)	26. (c)	27. (b)	28. (c)	29. (b)	30. (b)
31. (c)	32. (b)	33. (c)	34. (b)	35. (d)	36. (b)	37. (c)	38. (c)	39. (d)	40. (d)
41. (b)	42. (c)	43. (c)	44. (b)	45. (c)	46. (c)	47. (d)	48. (b)	49. (b)	50. (c)
51. (a)	52. (a)	53. (b)	54. (c)	55. (c)	56. (b)	57. (c)	58. (d)	59. (b)	60. (c)
61. (c)	62. (d)	63. (b)	64. (b)	65. (b)	66. (a)	67. (c)	68. (c)	69. (c)	70. (b)
71. (b)	72. (d)	73. (d)	74. (b)	75. (d)	76. (b)	77. (a)	78. (a)	79. (c)	80. (a)

SPEED TEST-1

1 (b)	2. (c)	3. (b)	4. (b)	5. (d)	6. (b)	7. (a)	8. (d)	9. (c)	10. (a)

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Hints & Solutions

LEVEL 1

1. Average speed $= \dfrac{\text{Total distance}}{\text{Total time}} = \dfrac{10+20}{\frac{10}{6}+\frac{20}{15}}$

$= \dfrac{30}{\left(\frac{90}{30}\right)} = \dfrac{30}{3} = 10$ km/h

2. Average speed (when distances are same) $= \dfrac{2uv}{u+v}$

$= \dfrac{2\times 20\times 30}{(20+30)} = 24$ km/h

where u, v are the different speeds.
(or use the general formula–Total distance/Total time)

3. Average speed $= \dfrac{3uvw}{uv+vw+uw}$

$= \dfrac{3\times 10\times 20\times 30}{200+600+300} = \dfrac{18000}{1100}$

$= 16\dfrac{4}{11}$ km/h

Alternatively: Suppose the total distance 3 times the LCM of the given speeds, then solve by general formula.

Total distance = 180 km (say)

then Total time $= \dfrac{60}{10}+\dfrac{60}{20}+\dfrac{60}{30}$

$= 6+3+2 = 11$ h

$\therefore$ Average speed $= \dfrac{180}{11} = 16\dfrac{4}{11}$ km/h

Alternatively: Suppose the total distance equals to $3x$ km then solve as above.

4. Suppose the total distance equals to 4 times the LCM of the speeds.

$\therefore$ Total distance = 2400 km

$\therefore$ Total time $= \dfrac{600}{20}+\dfrac{600}{25}+\dfrac{600}{30}+\dfrac{600}{40}$

$= 30+24+20+15 = 89$ h

$\therefore$ Average speed $= \dfrac{2400}{89} = 26\dfrac{86}{89}$ km/h

5. $\dfrac{2\times 20\times x}{(20+x)} = 24$

$\Rightarrow$ $x = 30$ km/h

Alternatively: Go through options.

6. Average speed $= \dfrac{2uv}{u+v}$

$= \dfrac{2\times 3x\times 5x}{(3x+5x)} = 3.75x$ km/h

7. $u = k$, $v = 3k$

$\therefore$ $\dfrac{2uv}{u+v} \Rightarrow \dfrac{2\times k\times 3k}{(k+3k)} = 12$

$\Rightarrow$ $1.5k = 12$

$\Rightarrow$ $k = 8$ km/h

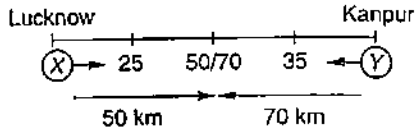
Alternatively: Go through options.
Alternatively: Solve through alligations.

8. Effective speed = 25 + 35 = 60 km/h

Total distance to be covered = 120 km

$\therefore$ Time taken $= \dfrac{120}{60} = 2$ h

HINT Since in each hour X and Y together covers 60 km.

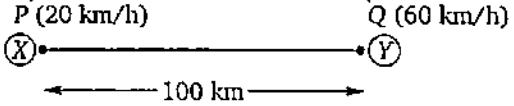


9. Since they take 2 hours to meet each other. Thus in 2 h X is 50 km away from Lucknow.
Alternatively: Both X and Y will cover the respective distances in the ratio of their speeds.

So, distance from Lucknow $= \dfrac{5}{12}\times 120 = 50$ km

10. When time is constant the distances covered by A and B will be in the ratio of their speeds, respectively.

11.



To meet each other they have to cover 100 km distance together and the ratio of distances covered by each one is directly proportional to the ratio of their speeds, respectively. Since the time taken by each one is same (i. e., constant).

Hence, $\dfrac{XZ}{YZ} = \dfrac{20}{60} = \dfrac{1}{3}$

$\therefore$ $XZ = \dfrac{1}{4}\times 100 = 25$ km

12. To meet the second time they have to cover 300 km distance together [for n^{th} time distance $= (2n-1)d$]
Time taken by them to meet each other, for the second time

$= \dfrac{300}{80} = 3\dfrac{3}{4}$ h

Distance covered by $P = 20\times 3\dfrac{3}{4} = 75$ km

Alternatively: The ratio of distance covered = Ratio of their speeds

$\therefore$ Distance covered by $P = \dfrac{1}{4}\times 300 = 75$ km

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13. The ratio of distances covered by each of them
= Ratio of their respective speeds

$\therefore$ Distance covered by Q for the third meeting $= \frac{3}{4} \times 500$

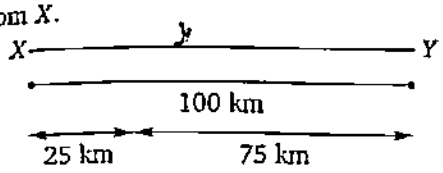
$= 375$ km

14. Consider the distance travelled by any one of them, then find the distance between X and M, where they meet.

$\therefore$ The distance travelled by P for the fourth meeting

$$= \frac{1}{4} \times 700 = 175 \text{ km}$$

Therefore P will be 75 km from Y. It means P will be 25 km away from X.



15. This is constant for any number of meeting and is equal to the ratio of their speeds. Hence, 1 : 3.

16. It is always twice the length of race course. Hence, between any two consecutive meeting they have to cover total 200 km distance to meet each other for the next meeting.

HINT For the first meeting they will cover 100 km.
For the second meeting they will cover 300 km.
For the third meeting they will cover 500 km.
For the fourth meeting they will cover 700 km and so on.

17. To meet each other they will take equal time since they start their journey simultaneously.

18. To reach their destination the time taken by A and B is equal to the ratio of reciprocal to their speeds. Since when distance is constant time is inversely proportional to the respective speeds.

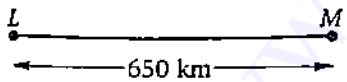
Hence, time taken by A and $B = \frac{1}{16} : \frac{1}{25} = 25 : 16$

19. $\frac{A\text{'s speed}}{B\text{'s speed}} = \sqrt{\frac{\text{Time taken by } B\ (t_B)}{\text{Time taken by } A\ (t_A)}}$

$$\frac{16}{25} = \sqrt{\frac{t_B}{t_A}}$$

$$\Rightarrow \quad \frac{t_B}{t_A} = \frac{256}{625} \Rightarrow \frac{t_A}{t_B} = \frac{625}{256}$$

20.



(Kamal) 40 km/h (Vimal) 30 km/h

In the first 3 hours Vimal covers 90 km.

So, the rest distance = 560 km

Now, Kamal and Vimal both travels together, towards each other.

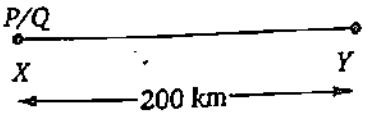
So, the time $= \frac{\text{Distance}}{\text{Speed}} = \frac{560}{70} = 8$ h

Thus, Vimal travels total = 3 + 8 = 11 h

Thus, the distance travelled by Vimal = 11 × 30 = 330 km

21. The distance covered by Kamal = 8 × 40 = 320 km

22.



To meet the first time they have to cover 200 × 2 = 400 km

Now, the time taken to meet each other $= \frac{400}{50} = 8$ h

23. Distance travelled by $P = 20 \times 8 = 160$ km

24. Distance travelled by $Q = 30 \times \frac{800}{50} = 480$ km

25. Distance travelled by $P = 20 \times \frac{1200}{50} = 480$ km

So, P will be 80 km away from X. Therefore ratio of distances

$$= \frac{80}{120} = \frac{2}{3}$$

26. The distance travelled by $P = \frac{1600}{50} \times 20 = 640$ km

Thus, P will be 160 km away from X.
[Since 640 = (200 × 2) + 240]
$\therefore$ Distance between D and X is 160 km.

27. This is possible only when the distance covered by P in n round be the multiple of 400.

28. X started at 12 : 00 noon
Y started at 3 pm

$$\text{Required time} = \frac{\text{Distance advanced in 3 h}}{\text{Relative speed}}$$

$$= \frac{40 \times 3}{20} = 6 \text{ h} \qquad (20 = 60 - 40)$$

Hence, Y will overtake X at 9 pm. $(3 + 6 = 9)$

29. $$\text{Required time} = \frac{\text{Distance advance} - \text{Required difference}}{\text{Relative speed}}$$

$$= \frac{120 - 30}{20} = \frac{90}{20} = 4.5 \text{ h}$$

Thus, at 7 : 30 pm X and Y will be 30 km apart.

30. $$\text{Required time} = \frac{\text{Distance advanced} + \text{Required difference}}{\text{Relative speed}}$$

$$= \frac{120 + 30}{20} = \frac{150}{20} = 7.5 \text{ h}$$

Thus at 10 : 30 pm X and Y will be 30 km apart.

31. Distance travelled by Y to overtake X

= Time taken × Speed of Y

= 6 × 60 = 360 km

Thus, Y will overtake X at a distance of 360 km from P.

32. Required distance

$$= \left(\frac{\text{Distance advanced} + \text{Required difference}}{\text{Relative speed}}\right) \times (\text{Speed of } Y)$$

$$= \frac{120 + 360}{20} \times 60 = 1440 \text{ km}$$

33. Time (when X was 30 km ahead of Y) $= \frac{120 - 30}{20} = 4.5$ h

Time (when Y was 30 km ahead of X) $= \frac{120 + 30}{20} = 7.5$ h

Thus, required difference in time = 3 h

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34. Speed = 36 km/h

$$= 36 \times \frac{5}{18} = 10 \text{ m/s}$$

35. Since in 60 minutes postman goes 36000 metre.

So in 1 minute postman goes $\frac{36000}{60} = 600$ metre

Thus, his required speed = 600 m/min

36. 1 mile = 1609.3 m = 1.6093 km

$\therefore$ 1 km $= \frac{1}{1.6093}$ mile = 0.6213882 mile

$\therefore$ Required speed = 36×0.6213882 mile/h

= 22.37 mile/h

37. Speed = 20 m/s $= 20 \times \frac{18}{5} = 72$ km/h

38. In one hour train goes 72 km. So in one minute train will go $\frac{72}{60}$.

$\therefore$ Speed $= \frac{6}{5} = 1.2$ km/min

39. The ratio of speeds of $A, B, C = 6 : 3 : 1$

$\therefore$ The ratio of time taken by $A, B, C = 1 : 2 : 6$

$\therefore$ Time taken by $A = 13$ min

40.

	A	:	B
Speed	2	:	3
Time	$3x$	:	$2x$

41. $\because$ $3x - 2x = 20 \Rightarrow x = 20$

$\therefore$ $3x = 60$ min = 1 h

42. Let the original speed be S_1 and time t_1 and distance be D

Now, $\frac{D/2}{2t_1} = S_2$

$\therefore$ $S_2 = \frac{D}{4t_1}$ and $S_1 = \frac{D}{t_1}$

$\therefore$ $\frac{S_1}{S_2} = \frac{D/t_1}{D/4t_1} = \frac{4}{1}$

43. You can go through options to check the required difference.

Alternatively: Required distance

$$= \frac{S_1 S_2}{(S_1 \sim S_2)} \times \text{Time difference}$$

$$= \frac{5 \times 8}{3} \times \frac{3}{2} = 20 \text{ km}$$

Alternatively: Take the LCM of distances then solve by unitary method.

$\therefore$ LCM of 5, 8 = 40

Now, consider 40 km as a distance, then there is a 3 hours difference in 40 km. So, 3/2 hours difference will be in 20 km.

Alternatively: Let x be the distance, then

$$\frac{x}{5} - \frac{x}{8} = \frac{3}{2}$$

$\Rightarrow$ $x = 20$ km

44.

	A	:	B
Speed	2	:	3
Time	3	:	2

$\because$ B takes 48 minutes so A will take 72 minutes

= 1 h 12 min

45. The ratio of distances = 160 : 140 = 8 : 7

$\therefore$ The ratio of speeds = 8 : 7

46. $\frac{x}{9} - \frac{x}{10} = \frac{20}{60}$

$x = 30$ km

47. Time taken by Abhinav = 4 h

Time taken by Praveen = 3.5 h

For your convenience take the product of times taken by both as a distance.

Then the distance = 14 km

Since, Abhinav covers half of the distance in 2 hours (*i.e.*, at 8 am)

Now, the rest half (*i.e.*, 7 km) will be covered by both Praveen and Abhinav.

Time taken by them $= \frac{7}{7.5} = 56$ min

Thus, they will cross each other at 8 : 56 am.

48. 9 km difference arises in the 99 km distance.

$\therefore$ 72 km difference will arise in the 792 km distance.

49. Apply the product constancy concept

Speed	Time	
$\frac{1}{5}\downarrow$	$\frac{1}{4}\uparrow$	= 15 min

Since, $\frac{x}{4} = 15$ min $\Rightarrow$ $x = 60$ min = 1 h

So, the original (or usual) time = 60 min = 1 h

50.

Speed	Time	
$\frac{1}{4}\downarrow$	$\frac{1}{3}\uparrow$	= 2 h

$\Rightarrow$ Usual time = $2 \times 3 = 6$ h

51.

Speed	Time	
$\frac{1}{2}\uparrow$	$\frac{1}{3}\downarrow$	= 40 min

$\Rightarrow$ Usual time = $3 \times 40 = 120$ min = 2 h

52.

Speed	Time	
$\frac{2}{20} = \frac{1}{10}\downarrow$	$\frac{1}{9}\uparrow$	= 10 min

$\Rightarrow$ Usual time = $9 \times 10 = 90$ min $= \frac{3}{2}$ h

$\therefore$ Distance travelled = Speed $\times$ Time

$= 20 \times \frac{3}{2} = 30$ km

53. Increase in speed = 1 km

Change in time = 25 min

Speed	Time	
$\frac{1}{5}\uparrow$	$\frac{1}{6}\downarrow$	= 25

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$\Rightarrow$ Usual (normal) time $= 6 \times 25 = 150 \text{ min} = \frac{5}{2}\text{h}$

$\therefore$ Distance = Normal speed × Normal time

$= 5 \times \frac{5}{2} = 12.5 \text{ km}$

54. Since, they are 1.5 km apart in each hour.
$\therefore$ 7.5 km apart they will be in 5 hours.
(Since, in the same direction speeds are subtracted)

55. In each hour they will be 720 km apart.
(Since in opposite direction speeds are added)

56. Let the distance = x km and usual rate = y km/h

$$\therefore \quad \frac{x}{y} - \frac{x}{(y+6)} = 4 \text{ h} \qquad \ldots(i)$$

$$\frac{x}{(y-4)} - \frac{x}{y} = 4 \qquad \ldots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{x}{y} - \frac{x}{(y+6)} = \frac{x}{(y-4)} - \frac{x}{y}$$

$$\Rightarrow \quad y = 24$$

Now putting the value of $y = 24$ in Eq (i), we get

$$x = 480$$

Alternatively:

S T / $+6$ -4 $\Rightarrow$ $-4S + 6T = |-4 \times 6|$

S T / -4 $+4$ $\Rightarrow$ $4S - 4T = |-4 \times 4|$

$$\therefore \quad -4S + 6T = 24 \qquad \ldots(i)$$

$$4S - 4T = 16 \qquad \ldots(ii)$$

$\therefore$ Solving these two equations, we get

$$T = 20 \quad \text{and} \quad S = 24$$

$\therefore$ Distance = Speed × Time

$= 24 \times 20 = 480 \text{ km}$

57. Let the original speed be x km/h, then

$$\frac{600}{x} - \frac{600}{(x+20)} = 1$$

$$\Rightarrow \quad 600\left(\frac{x+20-x}{x(x+20)}\right) = 1$$

$$\Rightarrow \quad x^2 + 120x - 12000 = 0$$

$$\Rightarrow \quad (x-120)(x-100) = 0$$

$$\Rightarrow \quad x = 100 \quad \text{and} \quad x = 120$$

$\therefore$ Original speed = 100 km/h

Alternatively:

$$S_1 \times S_2 = \text{Distance} \times \frac{\text{Difference in speed}}{\text{Difference in time}}$$

$$= 600 \times \frac{20}{1} = 600 \times 20 = 12000$$

$$\Rightarrow \quad S_1 \times S_2 = 12000$$

$$\therefore \quad S_1 = 100 \quad \text{and} \quad S_2 = 120$$

(Factorise 12000 in such a way that the difference be 20) S_1 is the original speed and S_2 is the changed speed.

58. Very similar to question number 57.

$$180 \times \frac{5}{3} = 60 \times 5$$

$$S_1 \times S_2 = 300 = 15 \times 20$$

$$S_1 = 15 \text{ km/h}$$

and $\quad S_2 = 20 \text{ km/h}$

Alternatively: $\quad S_1 \times (S_1 + 5) = 300$

$$\Rightarrow \quad S_1 = 15 \text{ km/h}$$

59. Very similar to previous problems

$$t_1 \times t_2 = \text{Distance} \times \frac{\text{Difference in time}}{\text{Difference in speed}}$$

$$\Rightarrow \quad t_1 \times t_2 = 1680 \times \frac{6}{14}$$

$$t_1 \times t_2 = 120 \times 6 = 720 = 30 \times 24$$

$$\Rightarrow \quad t_1 = 30 \quad \text{and} \quad t_2 = 24$$

[since $t_2 = (t_1 - 6) \Rightarrow t_1 \times (t_1 - 6) = 720 \Rightarrow t_1 = 30$]

Therefore, usual time = 30 h

60. Relative speecd $= \frac{\text{Total distance}}{\text{Total time}}$

$$= \frac{60 + 40}{20} = 5 \text{ m/s}$$

$$\therefore \quad 5 \text{ m/s} = 5 \times \frac{18}{5} = 18 \text{ km/h}$$

Now, Relative speed = 18 km/h

= Speed of ambulance − Speed of school bus

18 = 30 − speed of school bus

Speed of school bus = 12 km/h

61.

Time	Speed
$\frac{1}{4}\downarrow$	$\frac{1}{3}\uparrow$

Since he has to increase his speed by $\frac{1}{3}$rd of the original speed.

So, the new speed = 15 + 5 = 20 km/h

Alternatively: $\quad 15 \times 4 = 3 \times x$

$$\Rightarrow \quad x = 20 \text{ km/h}$$

62. Time (required) $= \frac{\text{Length of train}}{\text{Speed of train}}$ $\quad \left(T = \frac{D}{S}\right)$

$$= \frac{350}{70} = 5 \text{ s}$$

63. Time taken to cross the bridge $= \frac{\text{Total length}}{\text{Speed of train}}$

$$= \frac{270 + 130}{40} = \frac{400}{40}$$

$$= 10 \text{ s}$$

64. Speed of train $= \frac{225}{9} = 25 \text{ s}$

Now, Time taken to cross the tunnel $= \frac{225 + 450}{25} = 27 \text{ s}$

65. Let the length of the tunnel be x m, then

$$\text{Time} = \frac{\text{Length of train + Length of tunnel}}{\text{Speed}}$$

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$$60 = \frac{350 + x}{10} \qquad \left(\text{Speed} = 36 \times \frac{5}{18} = 10 \text{ m/s}\right)$$

$\Rightarrow \quad x = 250 \text{ m}$

66. Speed of train $= \dfrac{\text{Length of train + Length of platform}}{\text{Time}}$

$$= \frac{250 + 250}{25} = 20 \text{ m/s}$$

$$= 72 \text{ km/h}$$

67. Let the length of the Sabarmati express is x metre then,

$$\frac{x + 162}{18} = \frac{x + 120}{15} \text{ Speed of train}$$

$\Rightarrow \quad x = 90 \text{ m}$

68. Time $= \dfrac{\text{Length of train}}{\text{Relative speed}}$

$$= \frac{200}{19} = 10\frac{10}{19} \text{ s}$$

(Relative speed = 72 – 3.6 = 68.4 km/h = 19 m/s)

69. Time $= \dfrac{\text{Length of train}}{\text{Relative speed}}$

$$= \frac{350}{35} \times 6 = 60 \text{ s}$$

(Relative speed = 20 + 1 = 21 km/h $= 21 \times \dfrac{5}{18} = \dfrac{35}{6}$ m/s)

70. Relative speed = 50 – 40 = 10 km/h $= \dfrac{50}{18}$ m/s

$\therefore$ Time taken $= \dfrac{\text{Sum of length of the trains}}{\text{Relative speed}}$

$$= \frac{200}{50} \times 18 = 72 \text{ s}$$

71. Relative speed = 50 + 40 = 90 km/h

$$= 90 \times \frac{5}{18} = 25 \text{ m/s}$$

$\therefore$ Time taken $= \dfrac{\text{Sum of lengths of train}}{\text{Relative speed}} = \dfrac{200}{25} = 8 \text{ s}$

72. Let the length of train be x m, then

$$\frac{x}{10} = \frac{120 + x}{18}$$

$\Rightarrow \quad x = 150 \text{ m}$

73. Let the speed of train be x km/h, then

$$\frac{175}{10} = (9 + x) \times \frac{5}{18}$$

$\Rightarrow \quad x = 54 \text{ km/h}$

74. Relative speed = Sum of speeds of two trains

$$= (60 + x)$$

$$\text{Time} = \frac{\text{Sum of length of two trains}}{\text{Relative speed}}$$

$$10 = \frac{250}{(60 + x) \times 5} \times 18$$

$$(60 + x) = 90$$

$$x = 30 \text{ km/h}$$

75. Let the length of the train be x metre, and let the speed of train be y km/h, then

$$x = (y + 3)\frac{5}{18} \times 36 \quad \text{...(i)}$$

and $$x = (y + 6) \times \frac{5}{18} \times 30 \quad \text{...(ii)}$$

From Eq. (i) and (ii), we get

$$(y + 3) \times 36 = (y + 6) \times 30$$

$$y = 12 \text{ km/h}$$

$\therefore \quad x = (y + 3) \times \dfrac{5}{18} \times 36$

or $\quad x = 15 \times \dfrac{5}{18} \times 36$

or $\quad x = 150 \text{ m}$

76. Let the length of each train be x m, then

$S_1 = \dfrac{x}{4}$ and $S_2 = \dfrac{x}{5}$; S_1 and S_2 are speeds

Now, required time $= \dfrac{2x}{\frac{x}{4} + \frac{x}{5}} = \dfrac{40}{9}$ s

77.

	Pushpak		Bhopal
Time →	10	:	8
Speed →	8	:	10
$\Rightarrow$	$4x$	:	$5x$

Since, $\quad 5x - 4x = 10$

$\Rightarrow \quad 5x = 50$ and $4x = 40$

$\therefore$ Average speed $= \dfrac{2 \times 40 \times 50}{40 + 50} = 44\dfrac{4}{9}$ km/h

78. Required time $= \dfrac{\text{Distance advanced}}{\text{Relative speed}} = \dfrac{4 \times 40}{20} = 8 \text{ h}$

Thus, the faster train will overtake at 8 pm.

79. Required distance = Time taken in overtaking × Faster's speed

$= 8 \times 60 = 480$ km from Meerut

80. In 50 minute Rajdhani express can cover 50 km. So, the rest distance = 650 km, which will be jointly covered by both trains.

$\therefore \quad$ Time taken $= \dfrac{650}{(60 + 70)} = 5 \text{ h}$

Distance from Lucknow $= 5 \times 70 = 350$ km

81. Average speed $= \dfrac{\text{Total distance}}{\text{Total time}}$

$$= \frac{560 + 360}{16} = 57\frac{1}{2} \text{ h}$$

82. Average speed $= \dfrac{2 \times 80 \times 120}{200} = 96$ km/h

83. Distance = Average speed × Average time

or $\quad \text{Distance} = 96 \times \dfrac{25}{2}$

or $\quad \text{Distance} = 1200$ km

84. $50 \times 30 = x \times 20$

$\Rightarrow \quad x = 75 \text{ km/h}$

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85. $$\frac{S_1}{S_2} = \sqrt{\frac{t_2}{t_1}}$$

$$\frac{120}{S_2} = \sqrt{\frac{9}{16}} = \frac{3}{4}$$

$$\Rightarrow \quad S_2 = 160 \text{ km/h}$$

86. $$S_1 + S_2 = \frac{240}{4} = 60$$

$$S_1 - S_2 = \frac{240}{12} = 20$$

$$\therefore \quad S_1 = 40 \text{ m/s} \quad \text{and} \quad S_2 = 20 \text{ m/s}$$

$$\therefore \quad S_1 = 40 \times \frac{18}{5} = 144 \text{ km/h}$$

87. Time taken to cross the man $= \dfrac{\text{Length of the faster train}}{\text{Relative speed}}$

$$18 = \frac{x}{15 \times \frac{5}{18}} \quad \Rightarrow \quad x = 75 \text{ m}$$

88. Suppose the total distance be 300 km (LCM of 50 and 60) then in the first case it takes only 5 hours and in the second case it takes 6 hours.
Thus, in 6 hours trains halts for 1 hour.
Therefore in 1 hour train halts for 1/6 hour = 10 m
Alternatively: Difference in speeds = 10 km/h
Faster speed = 60 km/h
$\therefore$ Required time per hour $= \frac{10}{60} = \frac{1}{6}$ h = 10 min

89. If the speed of faster horse be f_S and that of slower horse be s_S, then

$$f_S + s_S = \frac{50}{1} = 50$$

and $$\frac{50}{s_S} - \frac{50}{f_S} = \frac{5}{6}$$

Now, you can go through options.
The speed of slower horse is 20 km/h.
Since, $20 + 30 = 50$
and $$\frac{50}{20} - \frac{50}{30} = \frac{5}{6}$$

90. Let he walked for x hours, then

$$5x + 25(10 - x) = 17 \times 10$$

$$\Rightarrow \quad x = 4$$

$$\therefore \quad 10 - x = 6 \text{ h}$$

Hence, distance travelled by auto $= 25 \times 6 = 150$ km.

91. Let the original speed be x km/h then,

$$\frac{36}{(x-6)} + \frac{36}{(x+6)} = 8$$

Now, you can go though options or solve it as follows

$$\frac{(x+6+x-6)}{(x^2-36)} = \frac{8}{36}$$

$$\Rightarrow \quad x = 12 \quad \text{and} \quad x = -3$$

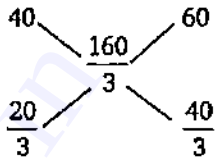
Thus, the possible value of $x = 12$
$\therefore$ Time taken by faster speed = 2 h

92. $$\text{Time} = \frac{\text{Distance advanced}}{\text{Relative speed}}$$

$$2 = \frac{2 \times x}{(30 - x)}$$

$$\Rightarrow \quad x = 15 \text{ km/h}$$

93. Prachi travels only for 3 hours, since half an hour she halts at Lucknow.
Now, the average speed (except the halt) $= \frac{160}{3}$ km/h
Therefore, by alligation



$$\Rightarrow \quad 1 : 2$$

Therefore, the ratio of time taken at 40 km/h and at 60 km/h is 1 : 2.
Thus, the distance between Lucknow and Kanpur
$= 2 \times 60 = 120$ km

94. Let the speed of the faster train be f_S and slower train be s_S, then

$$f_S + s_S = \frac{200 + 250}{18} = 25 \text{ m/s}$$

and $$f_s - s_S = \frac{200 + 250}{60} = \frac{450}{60} = 7.5 \text{ m/s}$$

$$\therefore \quad f_S = 16.25 \text{ m/s}$$

$$= 16.25 \times \frac{18}{5} = 58.5 \text{ km/h}$$

95. Distance covered in 3 minutes $= 3 \times \frac{1000}{60} = 50$ m

Now he has to cover (500 + 50) metres in (30 – 3) minutes

$$\therefore \quad \text{New speed} = \frac{550/1000}{27/60} = \frac{11}{9} \text{ km/h}$$

96.

	Anil		Mukesh
Speed →	3	:	4
Time →	4	:	3

But $$4x - 3x = \frac{1}{2} \text{ h}$$

$$\Rightarrow \quad 4x = 2 \text{ h} \quad \text{and} \quad 3x = 1.5 \text{ h}$$

Now, since Anil doubles the speed so time will be half of the actual time. Hence, new time will be 1 hour.

97. Average speed of Anil and Mukesh $= \frac{3x + 4x}{2} = 28$

$$\Rightarrow \quad x = 8$$

$$\therefore \quad \text{Speed of Sameer} = 3 \times 8 = 24 \text{ km/h}$$

$$\therefore \quad \text{Distance travelled} = 2 \times 24 = 48 \text{ km}$$

98. Let the speed of X and Y be the x km/h and y km/h respectively. Since they meet after 3 hours, so $x + y = 100$.
Since, the faster train takes atleast 3 + 2 = 5 hours to complete the 300 km journey. Hence, minimum possible speed for the slower train = 40 km/h at which speed it will take 7.5 h to complete the journey. $\left(7.5 = \frac{300}{40}\right)$

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99. Let the time taken in first third part of the journey be x minutes, then the time required in second third part of the journey is $\frac{3x}{2}$ and in the last third part of the journey time required is $\frac{15x}{8}$.

Therefore, $x + \frac{3x}{2} + \frac{15x}{8} = 350$ min

$\Rightarrow$ $x = 80$ min

100.

Speed	Time	
$\frac{1}{5}\downarrow$	$\frac{1}{4}\uparrow$	$= 15$ min

Therefore usual time $= 4 \times 15 = 60$ min

Now,

Speed	Time	
$\frac{1}{4}\uparrow$	$\frac{1}{5}\downarrow$	$= 12$ min

(Since original time = 60 min)

Therefore he will be 15 + 12 = 27 minutes early in comparison to the previous day.

101. Let the original speed be s km/h and scheduled time $= t$ hours

and total distance $= D$ km

then $s \times t = \frac{3}{4}D$ ...(i)

and $s \times (t + 3) = D$ ...(ii)

From Eq. (i) and (ii), we get

$st = \frac{3}{4}[s(t + 3)] \Rightarrow t = 9$ h

and let $s = 1$ km/h, then $D = 12$ km

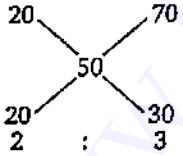
Again, since he doubles his speed after k hours then,

$s_1t_1 + s_2t_2 = D$

$1 \times k + 2 \times (9 - k) = 12$

$\Rightarrow$ $k = 6$ h

102. By alligation rule



Ratio of time = 2 : 3

$\therefore$ Ratio of distances $= 2 \times 20 : 3 \times 70 = 4 : 21$

Alternatively: $\frac{x}{20} + \frac{y}{70} = \frac{x + y}{50}$

$\Rightarrow$ $\frac{76x + 20y}{1400} = \frac{x + y}{50}$

$\Rightarrow$ $42x = 8y \Rightarrow \frac{x}{y} = \frac{4}{21}$

103. $\frac{\text{(Speed of wind)}}{\text{(Speed of car)}} = \frac{\text{(Time utilised)}}{\text{(Time saved)}}$

$\frac{332}{x} = \frac{332}{28}$

$\Rightarrow$ $x = 28$ m/s

104. Time $= \frac{\text{Total distance}}{\text{Relative speed}}$

$\frac{4.5}{60} = \frac{450/1000}{x} \Rightarrow x = 6$ km/h

Relative speed = Speed of car − Speed of man

$6 = x - 6$

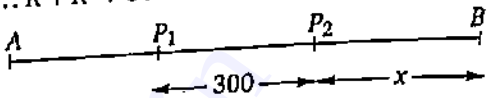
$\Rightarrow$ $x = 12$ km/h

105. $W + R \rightarrow 4$ h 20 min

$W + W \rightarrow 5$ h 20 min

$\therefore R + R \rightarrow 3$ h 20 min

106.



$P_1 \rightarrow$ Place of accident

$P_2 \rightarrow$ Imaginary place of accident

For the distance x

Speed	Time	
$\frac{1}{6}\downarrow$	$\frac{1}{5}\uparrow$	$= 1$ h

Thus, the usual time required for the distance x km is $5 \times 1 = 5$ hours.

For the distance $(x + 300)$

Speed	Time	
$\frac{1}{6}\downarrow$	$\frac{1}{5}\uparrow$	$= 2$ h

Thus, the usual time required for the distance $(x + 300)$ km is $5 \times 2 = 10$ h.

It means he covers 300 km distance in 5 h

$\therefore$ Speed $= \frac{300}{5} = 60$ km/h (normal speed)

107. Since, he can cover x km at 60 km/h in 5 h it means $x = 300$ km.

Therefore, total distance = (120 + 300 + 300) = 720 km.

108. Circumference means one revolution.

Therefore, distance covered in 10 revolutions $= 300 \times 10 = 30$ m

i.e., 30 metre in 6 seconds.

$\therefore$ Speed of wheel $= \frac{30}{6}$ m/s $= 5$ m/s

$\therefore$ 5 m/s $= 5 \times \frac{18}{5} = 18$ km/h

109. Speed of man in still water $= \frac{S_D + S_U}{2}$

$= \frac{12 + 8}{2} = 10$ km/h

$S_D \rightarrow$ Speed in downstream (= Boat + River)

$S_U \rightarrow$ Speed in upstream (=Boat − River)

110. Speed of water current $= \frac{S_D - S_U}{2} = \frac{21 - 15}{2} = 3$ km/h

111. $S_D = 12$ km/h

$S_U = 5$ km/h

Speed of current $= \frac{12 - 5}{2} = 3.5$ km/h

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112. $S_2 \rightarrow 15$ km/h
$S_2 \rightarrow 9$ km/h

$$\text{Speed of man} = \frac{15+9}{2} = 12 \text{ km/h}$$

113. Let the required distance be D km, then

$$\frac{D}{6} + \frac{D}{4} = \frac{5}{4}$$

$$D\left(\frac{10}{24}\right) = \frac{5}{4}$$

$$\Rightarrow \quad D = 3 \text{ km}$$

114. $\frac{16}{S} + \frac{16}{2} = 10$ h

115. Let the downstream speed be D and upstream speed be U, then

$$\frac{48}{U} + \frac{72}{D} = 12$$

and $$\frac{48}{D} + \frac{72}{U} = 13$$

$$\therefore \quad 48m + 72n = 12 \quad \ldots(i)$$

and $$48n + 72m = 13 \quad \ldots(ii)$$

Solving Eq. (i) and (ii), we get

$$m + n = \frac{5}{24} \quad \text{and} \quad m - n = \frac{1}{24}$$

$$\therefore \quad D = 12 \text{ km/h} \quad \text{and} \quad U = 8 \text{ km/h}$$

$$\therefore \quad \text{Speed of current} = 2 \text{ km/h}$$

116.

$$D_S = \frac{9}{2} = 4.5 \text{ km/h}$$

$$U_S = \frac{9}{6} = 1.5 \text{ km/h}$$

$$\therefore \text{ Speed of boat in still water} = \frac{4.5+1.5}{2} = 3 \text{ km/h}$$

$$\text{and Speed of river in still water} = \frac{4.5-1.5}{2} = 1.5 \text{ km/h}$$

117. $\frac{D_T}{U_T} = \frac{1}{2} \Rightarrow \frac{D_S}{U_S} = \frac{2}{1} \Rightarrow \frac{B+S}{B-S} = \frac{2}{1} \Rightarrow \frac{B}{S} = \frac{3}{1}$

HINT D_T and U_T are the downstream and upstream times and D_S and U_S are the downstream and upstream speeds. Here we can use componendo and dividendo.

$$\because \quad \frac{\text{Speed of boat}}{\text{Speed of stream}} = \frac{3}{1} = \frac{15}{x}$$

$$\therefore \quad \text{Speed of stream} = 5 \text{ km/h}$$

118. If t_1 and t_2 are the upstream and downstream times. Then time taken in still water is given by

$$\frac{2 \times t_1 \times t_2}{t_1 + t_2} = \frac{2 \times 12 \times 24}{36} = 16 \text{ h}$$

Alternatively: $D = (B + S) \times 12$

and $D = (B - S) \times 24$

where $(B + S)$ is downstream speed
$(B - S)$ is upstream speed

$$\Rightarrow \quad \frac{B+S}{B-S} = \frac{2}{1}$$

$$\Rightarrow \quad \frac{B}{S} = \frac{3}{1} \quad \text{(By componendo and dividendo)}$$

Now $D = 4S \times 12 = 48S$

$D = 48S = 16B$ (Distance = Time × Speed)

$\therefore$ Required time = 16 h

119. $\frac{D_S}{U_S} = \frac{B+S}{B-S} = \frac{3}{2}$

where $B \rightarrow$ Speed of boat in still water
$S \rightarrow$ Speed of current/stream

$$\Rightarrow \quad \frac{2B}{2S} = \frac{5}{1} \quad \text{(By componendo and dividendo)}$$

$$\Rightarrow \quad \frac{B}{S} = \frac{5/2}{1/2} \Rightarrow B = \frac{5}{2}$$

$$\text{Average speed of downstream and upstream} = \frac{2 \times 3 \times 2}{3+2} = \frac{12}{5}$$

$$\therefore \quad \text{Required ratio} = \frac{5/2}{12/5} = \frac{25}{24}$$

120. Let x be the upstream speed, then the downstream speed will be $(x + 3)$.

$$\therefore \quad \frac{3}{x} + \frac{3}{x+3} = 3$$

$$\Rightarrow \quad x^2 + x - 3 = 0$$

$$\Rightarrow \quad x = \frac{-1 + \sqrt{13}}{2}$$

$$= \frac{-1 + 3.6}{2} = 1.3 \text{ km/h}$$

$$\therefore \quad (x + 3) = 4.3 \text{ km/h}$$

121. Upstream speed of first boat = 8 km/h

Upstream speed of second boat = 4 km/h

$$\therefore \quad \text{Relative speed} = 4 \text{ km/h} \quad (= 8 - 4)$$

$$\therefore \quad \text{Required time} = \frac{10}{4} = 2.5 \text{ h}$$

122. Downstream speed = $B + S$ $B \rightarrow$ Speed of boat

Upstream speed = $B - S$ $S \rightarrow$ Speed of stream

$$\therefore \quad \frac{120}{B+S} + \frac{120}{B-S} = 15$$

$$\Rightarrow \quad \frac{B}{B^2 - S^2} = \frac{1}{16} \quad \ldots(i)$$

Again $$\frac{120}{B+2S} + \frac{120}{B-2S} = 24$$

$$\Rightarrow \quad \frac{B}{B^2 - 4S^2} = \frac{1}{10} \quad \ldots(ii)$$

From Eq. (i) and (ii), we get

$$B = (B^2 - S^2)16$$

and $$B = (B^2 - 4S^2)10$$

$$\Rightarrow \quad 10B^2 - 40S^2 = 16B^2 - 16S^2$$

$$\Rightarrow \quad \frac{B^2}{S^2} = 9 \Rightarrow \frac{B}{S} = 3$$

i.e., $$B : S = 3 : 1$$

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Now, you can go through options or solve by equations. Since now you know the ratio of speeds of boat and stream. The correct choice is (b).

124. Upstream speed $= B - S$

Downstream speed $= B + S$

$$B - S = \frac{15}{5} = 3\,\text{km/h}$$

Again $B = 4S$

$\therefore$ $B - S = 3 = 3S$

$\Rightarrow$ $S = 1$ and $B = 4$ (km/h)

$\therefore$ $B + S = 5\,\text{km/h}$

$\therefore$ Time during downstream $= \frac{15}{5} = 3\,\text{h}$

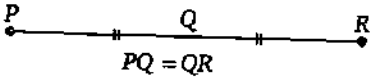
125. $\frac{24}{(5.5 - R)} - \frac{24}{(5.5 + R)} = 5$

$\Rightarrow$ $R = 2.5\,\text{km/h}$; $R \rightarrow$ Speed of river/current

Again $(B_2 + R) = \frac{24}{4} = 6$

$\Rightarrow$ $(B_2 + 2.5) = 6 \Rightarrow B_2 = 3.5\,\text{km/h}$

126.



$PQ = QR$

$P \rightarrow Q \rightarrow R$ (7 h)

It means $P \rightarrow Q$ (3.5 h)

Again $\{P \rightarrow Q$ and $Q \rightarrow P\}$ (8 h)

It means $Q \rightarrow P$ (4.5 h)

Therefore $R \rightarrow Q$ (4.5 h)

Thus, from R to P boat will take 9 hours.

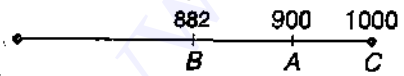
HINT $P \rightarrow R$ (Downstream)
$R \rightarrow P$ (Upstream)

127. $\frac{40}{(B - S)} + \frac{55}{(B + S)} = 13$

$\frac{30}{(B - S)} + \frac{44}{(B + S)} = 10$

HINT Go though options for quicker answer and prefer the value which can help in dividing 44 and 55.

Solutions for question number 128 and 129: B is 2% slow than A and A is 10% slow than C. Therefore, in 1 km race.

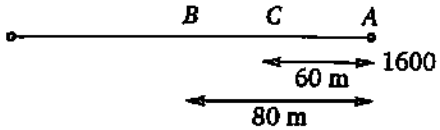


128. Since in 1000 m (1 km) race difference between A and B is 18 m.

So in 2 km race it will be 36 m.

129. In a 1 km race C can give B a start of 118 m. Therefore in a half km race C can give B a start of 59 m.

130. In a 1600 m race



$C \rightarrow 1540$, $B \rightarrow 1520$

The ratio of speeds of $C : B$ is 77 : 76. It means in 77 m race C beats B by 1 m. So, in 400 m race C will beat B by

$$400 \times \frac{1}{77}\,\text{m} = 5\frac{15}{77}\,\text{m}$$

131.

	Aman	Shakti
Time	190	200
Speed	$20x$	$19x$

$\because$ $20x = 1000$ m

$\Rightarrow$ $x = 50$ m

Again $20x - 19x = x = 50$ m

So, Aman can beat Shakti by 50 m.

132.

	A		B
Speed	7	:	4
Time	4	:	7
Distance	4	:	7

(Since distance $\propto$ time)

Now, $7x - 4x = 300$

$3x = 300$

$\Rightarrow$ $x = 100$

$\therefore$ $7x = 700$

Thus, the winning post be 700 m away from the starting point.

133. Ratio of speeds of $A : B = 12 : 11$

and ratio of speeds of $B : C = 8 : 7$

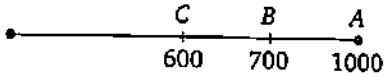
Therefore ratio of speeds of $A : B : C = 96 : 88 : 77$

So in 9600 m race A will beat C by 1900 m.

134. When Ameesha runs 1000 m, then Bipasha runs only 700 m [1000 – (100 + 200)]

when Ameesha runs 1000 m, then Celina runs only 600 m [1000 – (100 + 300)]

Therefore,



Now, since in 700 m race Bipasha beats Celina by 100 m. So, in 50 m race Bipasha will beat Celina by

$$50 \times \frac{100}{700} = 7.14\,\text{m}$$

Alternatively: Bipasha beats Celina by 14.28% of the distance then in 50 m race Bipasha will beat Celina by 7.14 m.

135.

		Distance	Time
Case I:	Ravi	1000	t_1
	Vinod	960	$t_1 + 19$
Case II:	Ravi	960	t_2
	Vinod	1000	$t_2 + 30$

Therefore, $\frac{1000}{t_1} = \frac{960}{t_2}$ = Speed of Ravi

$\Rightarrow$ $t_1 = \frac{25}{24} t_2$

Also $\frac{960}{t_1 + 19} = \frac{1000}{t_2 + 30}$ = Speed of Vinod

$\therefore$ $(t_2 + 30)\,24 = 25\left[\frac{25}{24} t_2 + 19\right]$

$\Rightarrow$ $49 t_2 = 5880$

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$\Rightarrow$ $t_2 = 120$ s

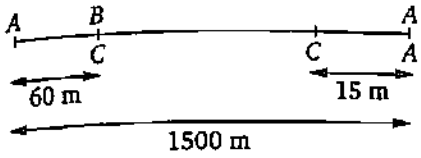
$\therefore$ Required ratio $= \dfrac{960/t_2}{1000/t_2 + 30} = \dfrac{6}{5}$

137. Since, the speeds of Vinay and Versha are in the ratio of 5 : 3 *i.e.*, when Vinay covers 5 rounds, then Versha covers 3 rounds, but first time Vinay and Versha meet when Vinay completes $2\frac{1}{2}$ round and Versha completes $1\frac{1}{2}$ round.

For Vinay to pass Versha 7th time, Vinay would have completed $7 \times 2\frac{1}{2}$ rounds. Since, each round is $1\frac{1}{2}$ km, the distance covered by Vinay is

$$7 \times 2\frac{1}{2} \times 1\frac{1}{2} = 7 \times \frac{5}{2} \times \frac{3}{2} = 26\frac{1}{4} \text{ km}$$

138.



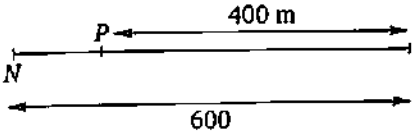
In the same time, when A covers 1500 m, B covers 1440 m and C covers 1425 m.

So, in 1440 m race B can give a start of 15 m.

$\therefore$ In 1500 m race B will give a start of

$$\frac{15}{1440} \times 1500 = 15\frac{5}{8} \text{ m}$$

139. In 600 m race Prabhat can have only start of 120 m. Now since he has more than 200 m start up so he will win the race.



Now, when Prabhat will cover 400 m distance, then in the same time Nishith will cover only 500 m. So, Prabhat will win by 100 m.

140. B is 25% slower than A and C is 20% slower than B.

Therefore in a game of 200 points B can have 150 points and C can have 120 points.

Thus, A can give C 80 points.

141. Since between 11 am and 1 pm and 11 pm and 1 am two hands of a clock coincide only once, each time.

142. Since between 2 am and 3 am (2 pm and 3 pm) and 8 am and 10 am (8 pm and 10 pm) two hands of a clock make 90° angle only 3 times in rest of the each hour two hands make 90° angle 2 times.

143. Since between 5 am and 7 am (5 pm and 7 pm) this happens only once. In rest each of the hours it happens one time.

144. Both (a) and (b) are correct.

Relative speed = Speed of minute-hand − Speed of hour-hand

$$= 6° - \frac{1}{2}° = 5\frac{1}{2}°$$

and $\quad 1 \text{ min} - \dfrac{1}{12} \text{ min} = \dfrac{11}{12} \text{ min}$

145. $\dfrac{5 \times 30}{11/2} = \dfrac{300}{11} = 27\dfrac{3}{11}$ min = 27 min 16 s

Therefore, required time = 5 : 27 : 16

146. The angle made by minute-hand = $5 \times 30 = 150°$

The angle made by hour-hand = $2 \times 30 + 25 \times \dfrac{1}{2} = 72.5°$

Hence, required angle = $150 - 72.5 = 77.5°$

147. $\dfrac{90° - 30°}{5.5} = \dfrac{60}{11} \times 2 = \dfrac{120}{11} = 10\dfrac{10}{11}$ minute

$= 10$ min 54 s

$\therefore$ Required time = 3 : 10 : 54

148. $\dfrac{210}{5.5} = \dfrac{210}{11} \times 2 = \dfrac{420}{11} = 38\dfrac{2}{11}$ min = 38 min 11 s

Therefore, required time = 7 : 38 : 11

149. The angle made by minute-hand = 90°

The angle made by hour-hand = $6 \times 30 + 15 \times \dfrac{1}{2} = 187.5°$

$\therefore$ Required difference = 97.5°

150. $\dfrac{90 - 35}{5.5} = \dfrac{55}{11} \times 2 = 10$ min

So, the required time = 3 : 10 : 00

Again $\dfrac{90 + 35}{5.5} = \dfrac{125}{11} \times 2 = \dfrac{250}{11} = 22\dfrac{8}{11}$ min

$= 22$ min 43 s

LEVEL (2)

1.

	Cycle	Auto	Car	
Speed	x	$(5x - 20)$	$5x$	
Time		$(t + 1)$	t	
Distance	120	120	120	(in km)

$\therefore$ $\dfrac{120}{(5x - 20)} - \dfrac{120}{5x} = 1$

$\Rightarrow$ $x^2 - 4x - 96 = 0$

$\Rightarrow$ $x = 12$

$\therefore$ Average speed $= \dfrac{360}{(10 + 3 + 2)} = 24$ km/h

2. Time taken by cycle $= \dfrac{120}{12} = 10$ h

Time taken by auto $= \dfrac{120}{40} = 3$ h

Time taken by car $= \dfrac{120}{60} = 2$ h

Total time = 15 h

3. In last 5 hours she covers 240 km (120 + 120)

4. New time = 3 + 3 + 2 = 8 h

Hence, decrease in time = 7 h (15 − 8

$\therefore$ Percentage change $= \dfrac{7}{15} \times 100 = 46.66\%$

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5. Time taken to meet Bipasha and Mallika $= \frac{1080}{(60+120)} = 6$ h

So, in 6 hours Bipasha covers 360 km and this 360 km distance Rani covers in $\frac{360}{90} = 4$ h.

Hence, Rani leaves Kolkata 2 hours later than Bipasha *i. e.*, at 8 am. Rani leaves Kolkata.

NOTE The distance 360 covered by Bipasha to meet Mallika can also be calculated by the ratio of their speeds.

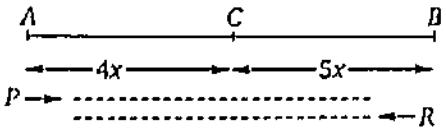
6. Note here the length of the train in which passenger is travelling is not considered since we are concerned with the passenger instead of train. So, the length of the bridge will be directly proportional to the time taken by the passenger respectively.

Therefore, $\frac{t_1}{t_2} = \frac{l_1}{l_2}$ $t \rightarrow$ Time, $l \rightarrow$ Length of bridge

$$\frac{7}{4} = \frac{280}{x}$$

$$\Rightarrow \quad x = 160 \text{ m}$$

7.



Note that the distances covered by them to meet at C are in the direct ratio of their speeds. Therefore

$$AC : BC = 4x : 5x$$

Now, for any particular person (say Pathik) the time required to cover different distances is directly proportional to the different distances. So, time taken by Pathik to cover AC and BC are the ratio of 4 : 5 (excluding staying or halt time at Chandni Chowk).

Thus time required to cover AC is 52 minutes only since he covers BC in 65 minutes.

But since he leaves Chandni Chowk for Bhavnagar at 9 : 27 am *i. e.*, 67 minutes later, when he left Andheri. It means he must have stayed at C for (67 – 52) = 15 minutes.

8. Let the length of the train be L metres and speeds of the train Arjun and Srikrishna be R, A and K respectively, then

$$\frac{L}{R-A} = 36 \qquad \ldots(i)$$

and $$\frac{L}{(R+K)} = 24 \qquad \ldots(ii)$$

From eq. (i) and (ii)

$$3(R-A) = 2(R+K)$$

$$\Rightarrow \quad R = 3A + 2K$$

In 30 minutes (*i. e.*, 1800 seconds), the train covers $1800R$ (distance) but the Arjun also covers $1800A$ (distance) in the same time.

Therefore distance between Arjun and Srikrishna, when the train has just crossed Srikrishna

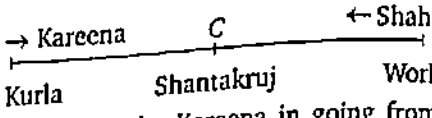
$$= 1800(R-A) - 24(A+K)$$

$$\therefore \quad \text{Time required} = \frac{1800(R-A) - 24(A+K)}{(A+K)}$$

$$= (3600 - 24) = 3576 \text{ s}$$

(Since $R = 3A + 2K$)

9.



Let the time taken by Kareena in going from K to S is x minutes and the time taken by Shahid in going from Worli to Shantakruj be y min.

Since, the new speed of Kareena is $\frac{2}{3}$, therefore time taken in returning $= \frac{3}{2}x$.

$$\therefore \quad x + \frac{3}{2}x = 120$$

$$\Rightarrow \quad x = 48 \text{ min}$$

but $$x = y$$

Again since the new speed of Shahid is $\frac{4}{3}$, therefore the time taken in returning $= \frac{3}{4}y$.

$$\therefore \quad \text{Total time} = y + \frac{3}{4}y$$

$$= 48 + 36 = 84 \text{ min}$$

10. Time taken to collide the two trains $= \frac{3}{2}$ h

So, in $\frac{3}{2}$ h bird travels $\frac{3}{2} \times 60 = 90$ km

11. Let there be l steps in the escalator and x be the speed (in steps/second) of escalator, then

$$\frac{l}{(5+x)} = 10 \quad \text{and} \quad \frac{l}{(5-x)} = 40$$

then $$\frac{5+x}{(5-x)} = \frac{40}{10} \quad \Rightarrow \quad x = 3$$

$\therefore$ Number of steps in the escalator $= l = 8 \times 10 = 80$

12. Let the radius be r, then difference in the distance

$$= (\pi r - 2r) = r(\pi - 2)$$

$$= r\left(\frac{22}{7} - 2\right) = 60 \times 3$$



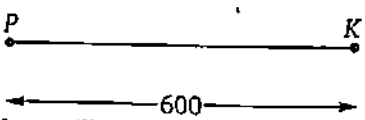
$$\Rightarrow \quad 2r = 315 \text{ m}$$

[$\pi r \rightarrow$ semiperimeter and $2r \rightarrow$ diameter]

13. Time taken by trains to collide $= \frac{560}{70} = 8$ h

In 8 h sparrow will cover $8 \times 80 = 640$ km

14.



In 18 h plane will cover $18 \times 120 = 2160$ km

Now, $2160 = (600 \times 2) + 600 + 360$

So, the plane will be 360 km away from Kargil it means it will be 240 km (600 – 360) away from Pukhwara.

15.

	First hour	Second hour	Third hour	Total
Initial speed	x	$3x$	$2x$	$6x$
New speed	$3x$	$3x$	$3x$	$9x$

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$\therefore$ Percentage increase in speed $= \frac{3x}{6x} \times 100 = 50\%$

Since speed is increased by (50%) $\frac{1}{2}$.

Therefore, time will reduce by (33.33%) $\frac{1}{3}$.

16. P ——— Q

They will be together at every two hours. Therefore in 12 h they will be $(6 + 1) = 7$ times together at P and they will never meet altogether at Q.

17.

Mathura — Kurukshetra — Hastinapur

400 km, 300 km

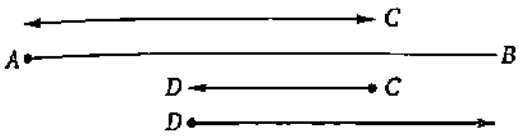
700 km

Consider only one person either Arjun or Srikrishna since their speed is same and move together.

Now, the distance covered by Arjun and Abhimanyu is in the ratio of their speeds. So, Arjun will cover 500 km to meet Abhimanyu and thus Arjun has to return back 100 km for Kurukshetra. Therefore, Arjun will cover total 600 km distance.

18. Total time $= \frac{600}{25} = 24$ h

19.



A is the starting point of journey.

B is the destination.

$C \rightarrow$ where Salman has got off.

$D \rightarrow$ where Priyanka picks up Akshay

Let $AD = l$ and $BC = k$ and $CD = x$

then $$\frac{CD + DB}{BC} = \frac{50}{10}$$

$$\frac{2x + k}{k} = \frac{5}{1}$$

$$\Rightarrow \quad \frac{x}{k} = \frac{2}{1}$$

Again $$\frac{AC + CD}{AD} = \frac{50}{10}$$

$$\frac{2x + l}{l} = \frac{5}{1}$$

$$\Rightarrow \quad \frac{x}{l} = \frac{2}{1}$$

$$\Rightarrow \quad x = 2k = 2l \quad \text{or} \quad k = l = \frac{x}{2}$$

$$\therefore \quad k + x + l = 120$$

$\Rightarrow \quad k = 30$ km, $\quad x = 60$ km and $\quad l = 30$ km

Total distance travelled $= AC + CD + DB$

$$= l + x + x + x + k = 240 \text{ km}$$

$$\therefore \quad \text{Time (required)} = \frac{240}{50} = 4.8 \text{ h}$$

20.

Barabanki ——— Fatehpur

300

120, 180

Let the speeds of Ajai, Kajol and Shahrukh be x, y and z respectively, then

$$\frac{y}{x} = \frac{180}{120} \Rightarrow x = \frac{2y}{3}$$

Note Kajol is faster since she covers 180 km while Ajai covers only 120 km in the same time.

Shahrukh meets Kajol 1.5 hours after Shahrukh himself starts and 2.5 hours after Kajol starts.

Hence, $$2.5y + 1.5z = 300$$

$$\Rightarrow \quad z = \frac{600 - 5y}{3}$$

Since $z \geq (y + 20) \Rightarrow \frac{600 - 5y}{3} \geq (y + 20)$

$$\Rightarrow \quad y \leq 67.5$$

or $$x \leq 45 \text{ km/h}$$

21. Let t be the time after Kajol starts, when she meets Ajai, then

$$t = \frac{300}{(x + y)}$$

This should be less than 2.5 or $(x + y) > 120$

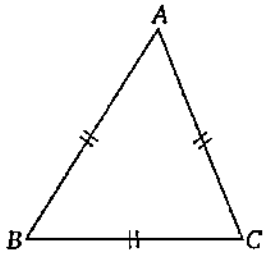
Since $$y = \frac{3x}{2} \Rightarrow y > 72$$

This $(y > 72)$ is greater than 67.5 km/h and hence Shahrukh will always overtake Ajai before he meets Kajol.

22. Speed of Raghupati $(R_P) = 60$ km/h

Speed of Raghav $(R_V) = 36$ km/h

Speed of Raja Ram $(RR) = 18$ km/h



$AB = AC = BC$

Time taken to cover AB by (RR) is 2 hours

$\therefore$ Time taken to cover AB by Raghav is 1 hour

$\therefore$ Time taken to cover AB by Raghupati = 36 min

$$\left(t_{RP} : t_{RV} : t_{RR} = \frac{1}{S_{RP}} : \frac{1}{S_{RV}} : \frac{1}{S_{RR}}\right)$$

$t \rightarrow$ Time, $S \rightarrow$ Speed

$$AB = 2 \times 18 = 36 \text{ km}$$

23. Time $= \frac{3 \times 36}{60} = \frac{9}{5}$ h $= 1$ h 48 min

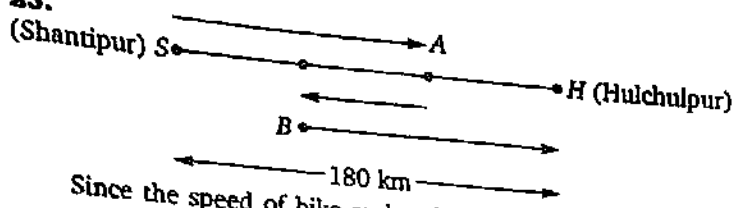
24. Distance from Bareilley $= \frac{60}{(60 + 18)} \times 36$

$$= \frac{360}{13} = 27\frac{9}{13} \text{ km}$$

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25.



Since the speed of bike and walking are different. So, two people partially travelled by bike and rest by walking since all the three persons take equal time to reach the destination. It means initially Mohan will carry either Namit or Pranav to a point A, then this person reach to H by walking and Mohan return to B where he will pick up the third person and reach at H at the same time as the second person.

Let $SB = k$, $AB = x$ and $AH = l$

Now, $$\frac{SA + AB}{SB} = \frac{36}{6}$$

$$\frac{2x + k}{k} = \frac{6}{1}$$

$$\Rightarrow \quad \frac{x}{k} = \frac{5}{2}$$

and $$\frac{AB + BH}{AH} = \frac{36}{6}$$

$$\Rightarrow \quad \frac{2x + l}{l} = \frac{6}{1}$$

$$\Rightarrow \quad \frac{x}{l} = \frac{5}{2}$$

$$\therefore \quad x : k : l = 5 : 2 : 2$$

$$\Rightarrow \quad x + k + l = 180$$

$$\Rightarrow \quad x = 100, \quad k = 40 \quad \text{and} \quad l = 40 \text{ km}$$

Total distance travelled by bike $= SA + AB + BH$

$$= k + 3x + l = 380 \text{ km}$$

26. $$\frac{2x + k}{k} = \frac{42}{6} = \frac{7}{1}$$

$$\Rightarrow \quad \frac{x}{k} = \frac{3}{1}$$

Similarly $$\frac{x}{l} = \frac{3}{1}$$

$$\therefore \quad x : k : l = 3 : 1 : 1$$

$$\therefore \quad x = 108, \quad k = 36, \quad l = 36 \text{ km}$$

Total distance travelled $= k + 3x + l = 396$ km

$$\therefore \quad \text{Required time} = \frac{396}{42} = 9\frac{3}{7} \text{ h}$$

27. Let the buses leave from both the stations at time intervals of T, then the distance between any two consecutive buses coming opposite to me = the distance betwen any two consecutive buses coming in the same direction as me = VT.
(where V is the velocity of the buses)
Let the speed of walking be W, then

$$\frac{VT}{V + W} = 20 \quad \text{and} \quad \frac{VT}{V - W} = 30$$

$$\therefore \quad \frac{V + W}{V - W} = \frac{30}{20} = \frac{3}{2}$$

$$\Rightarrow \quad \frac{V}{W} = \frac{5}{1}$$

$$\therefore \quad \frac{VT}{V + W} = 20$$

$$\Rightarrow \quad \frac{5}{6} \times T = 20$$

$$\Rightarrow \quad T = 24 \text{ min}$$

28. Time taken by Abhinav = 36 h

Ideal time required by Abhinav $= \frac{600}{25} = 24$ h

It means Abhinav rests for $(36 - 24) = 12$ h

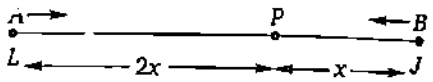
Now, the required time for Brijesh $= \frac{600}{30} = 20$ h

But Brijesh utilised those 12 hours in which Abhinav rests, so he needs only $(20 - 12) = 8$ hours extra.

Thus, the total time taken by Brijesh $= 36 + 8 = 44$ h

29. Downstream (Steamer) = 40 min
Downstream (Boat) = 60 min
Upstream (Steamer) = 60 min
Upstream (Boat) = 90 min
Required time $= 40 + 30 + 45 = 115$ min

30.



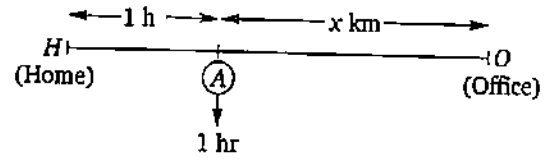
These two trains meet only at P and L i. e., there are only two points.

31. For the first meeting they have to cover only $2x + x = 3x$ distance and for the further meeting for each next meeting they have to cover $6x$ distance together.

Distance covered by A	$2x$	$2x$	$4x$	$2x$
Distance covered by B	x	$4x$	$2x$	$4x$
Point of meeting	P	L	P	P
Total distance travelled	$3x$	$6x$	$6x$	$6x$

when A and B meet at P for the third time A goes $10x$ and B goes $11x$.
Thus, the required ratio = 10 : 11

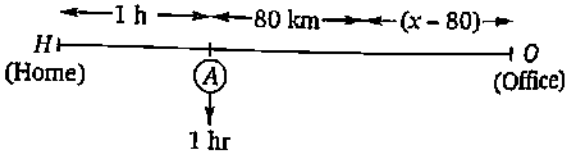
32.



1 hr

Speed	Time
$\frac{1}{6}\downarrow$	$\frac{1}{5}\uparrow = 36$ min

$\Rightarrow$ actual time required to cover x km is $5 \times 36 = 180$ min



1 hr

Speed	Time
$\frac{1}{6}\downarrow$	$\frac{1}{5}\uparrow = 20$

$\Rightarrow$ actual time required for $(x - 80)$ km $= 5 \times 20 = 100$ min
It means he can move $= x - (x - 80) = 80$ km in

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$$(180 - 80) = 80 \text{ min}$$

It means his actual speed = 60 km/h

Thus, the total distance from his home to his office

$$= 60 \times 1 + 60 \times 3 = 240 \text{ km}$$

NOTE Since 1 hour he lost at the place of accident, so the actual delay due to reduced speed is always 1 hour less than it is found to be in both the cases.
It means due to reduced speed he becomes late only 36 minutes and 20 minutes in respective cases.

33. $\frac{\text{Speed of wind (Sound)}}{\text{Relative speed of soldier and terrorist}} = \frac{\text{Time utilised}}{\text{Difference in time}}$

$$\frac{1188}{x} = \frac{330}{5}$$

$$\Rightarrow \quad x = 18 \text{ km/h}$$

34. In case of increasing gap between two objects.

$$\frac{\text{Speed of sound}}{\text{Speed of tiger}} = \frac{\text{Time utilised}}{\text{Difference in time}}$$

$$\frac{1195.2}{x} = \frac{83}{7}$$

$$\Rightarrow \quad x = 100.8 \text{ km/h}$$

35. In 20 minutes the difference between man and his son

$$= 20 \times 20 = 400 \text{ m}$$

Distance travelled by dog when he goes towards son

$$= \frac{400}{40} \times 60$$

= 600 m and time required is 10 minutes

In 10 minutes the remaining difference between man and son.

$$400 - (20 \times 10) = 200 \text{ m}$$

NOTE Relative speed of dog with child is 40 km/h and the same with man is 100 km/h.

Time taken by dog to meet the man $= \frac{200}{100} = 2$ min

In 2 min the remaining distance between child and man

$$200 - (2 \times 20) = 160 \text{ m}$$

Now, the time taken by dog to meet the child again

$$= \frac{160}{40} = 4 \text{ min}$$

In 4 minutes he covers $4 \times 60 = 240$ m distance while going towards the son.

In 4 minute the remaining distance between man and child

$$= 160 - (4 \times 20) = 80 \text{ m}$$

Time required by dog to meet man once again

$$= \frac{80}{100} = 0.8 \text{ min}$$

In 0.8 min remaining distance between man and child

$$= 80 - (0.8 \times 20) = 64 \text{ m}$$

Now, time taken by dog to meet the child again

$$= \frac{64}{40} \times \frac{8}{5} \text{ min}$$

$$\therefore \quad \text{Distance travelled by dog} = \frac{8}{5} \times 60 = 96 \text{ m}$$

Thus, we can observe that every next time dog just go 2/5th of the previous distance to meet the child in the direction of child.

So, we can calculate the total distance covered by dog in the direction of child with the help of GP formula.

Here, first term (a) = 600 and common ratio (r) $= \frac{2}{5}$

$\because$ Sum of the infinite GP $= \frac{a}{1-r}$

$$= \frac{600}{\left(1 - \frac{2}{5}\right)} = \frac{600}{3/5} = 1000 \text{ m}$$

36. Let Amarnath express takes x hours, then Gorakhnath express takes $(x - 2)$ hours.

$$\therefore \quad \frac{1}{x} + \frac{1}{(x-2)} = \frac{60}{80}$$

$$\Rightarrow \quad x = 4 \text{ h}$$

37. Distance travelled by them in first hour = 12 km

Distance travelled by them in second hour = 13 km

Distance travelled by them in third hour = 14 km and so on

Thus, in 9 hours they will cover exactly 144 km and in 9 h each will cover half-half the total distance.

$(8 \times 9 = 72$ and $4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 + 12 = 72)$

38. Speed of tiger = 40 m/min

Speed of deer = 20 m/min

Relative speed = 40 – 20 = 20 m/min

Difference in distances = $50 \times 8 = 400$ m

$\therefore$ Time taken in overtaking (or catching) $= \frac{400}{20} = 20$ min

$\therefore$ Distance travelled in 20 min $= 20 \times 40 = 800$ m

39. The sum of their speeds $= \frac{615}{15} = 43$ km/h

Notice that they are actually exchanging their speeds. Only then they can arrive at the same time at their respective destinations. It means the difference in speeds is 3 km/h.

Thus, $\quad x + (x + 3) = 43$

$\Rightarrow \quad x = 20$ and $x + 3 = 23$

The concept is very similar to the case when after meeting each other they returned to their own places of departure. It can be solved through option also.

40. Let Pele covers x km in 1 hour. So Maradona takes (2 h – 40 min) = 1 h 20 min to cover x km. Let speed of Maradona and Pele be M and P respectively than

$$x = M \times \frac{4}{3} \quad \text{and} \quad x = P \times 1$$

$$\Rightarrow \quad \frac{M}{P} = \frac{3}{4}$$

Again $\quad \frac{300}{M} - \frac{300}{P} = 1$

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$\Rightarrow \quad \frac{300}{3k} - \frac{300}{4k} = 1$

$\Rightarrow \quad k = 25$

$\Rightarrow \quad M = 3k = 75 \text{ km/h}$

and $\quad P = 4k = 100 \text{ km/h}$

(Through option it is very easy to solve.)

41. Initial speed of police = 10 m/s
Increased speed of police = 20 m/s
Speed of thief = 15 m/s
Initial difference between thief and police = 250 m
After 5 seconds difference between thief and police

$$= 250 - (5 \times 10) = 200 \text{ m}$$

After 10 seconds more the difference between thief and police
$= 200 + (5 \times 10) = 250$ m.
Now, the time required by police to catch the thief

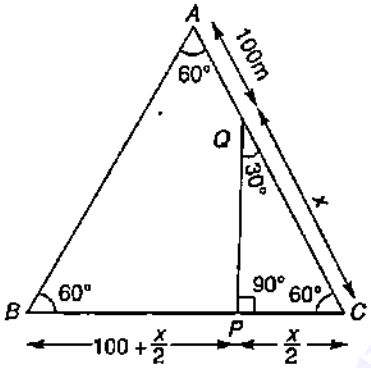
$$= \frac{250}{5} = 50 \text{ s}$$

$$\text{Distance travelled} = 50 \times 20 = 1000 \text{ m}$$

$$\text{Total time} = 50 + 15 = 65 \text{ s}$$

$$\text{Total distance} = 1000 + (15 \times 10) = 1150 \text{ m}$$

42.



$$\frac{100 + \frac{x}{2}}{100} = \frac{100 + x}{(100 + x - 150)}$$

$$= \frac{\text{Speed of Bajrang}}{\text{Speed of Angad}}$$

$$\frac{200 + x}{200} = \frac{(100 + x)}{(x - 50)}$$

$\Rightarrow \quad (200 + x)(x - 50) = 200(100 + x)$

$\Rightarrow \quad x^2 + 150x - 10000 = 20000 + 200x$

$\Rightarrow \quad x^2 - 50x - 30000 = 0$

$\Rightarrow \quad (x - 200)(x + 150) = 0$

$\Rightarrow \quad x = 200 \text{ km}$

Therefore distance between Ayodhya and Banaras is 300 km since $AB = BC = AC$.
(With the help of trigonometry we can find the value of PC in terms of x i. e., $\cos 60° = \frac{PC}{QC} = \frac{1}{2}$. Hence $PC = \frac{x}{2}$)

43. Basically they will exchange their speeds just after half of the time required for the whole journey. It means after covering 210 km distance they will exchange their speeds.
Check it out graphically for more clarification.

44. The ratio of speeds
= The ratio of distances, when time is constant
∴ The ratio of distances covered by leopard to the tiger = 12 : 25
Again, ratio of rounds made by leopard to the tiger = 12 : 25
Hence, leopard makes 48 rounds, when tiger makes 100 rounds.

45. Length of $DC = \frac{6000}{13}$ (for this, refer geometry section)

Total distance covered in the returning by Jai

$$= AD + CD$$

$$= \frac{2500}{13} + \frac{6000}{13} = \frac{8500}{13} \text{ km}$$

$$\text{Required time} = \frac{8500/13}{500/13} = 17 \text{ h}$$

Total distance covered by Jaya while returning

$$= BD + DC$$

$$= \frac{14400}{13} + \frac{6000}{13}$$

$$\therefore \quad \text{Required time} = \frac{20400/13}{1200/13} = 17$$

Hence, both will reach at the same time.
Alternatively: Since the ratio of speeds is same as that of distances. So, they will take same time to reach the home.

46. The distance of route $ADC = \frac{8500}{13}$

and the distance of route $BNC = 1300$

and the time taken by Jai is $\frac{8500/13}{500/13} = 17$ h

and the time taken by Jaya is $\frac{1300}{1200/13} = \frac{169}{12} \text{ h} = 14\frac{1}{12} \text{ h}$

$= 14 \text{ h } 05 \text{ min}$

Hence, option (c) is correct.

47. Time saved in percentage $= \frac{175}{1020} \times 100 = 17.15\%$

48. Husband takes 17 hours and she takes 14 h 05 min + 3h
= 17 h 05 min
So, she becomes late by 05 min than her husband.

49. $x^2 + (x + 100)^2 = (500)^2$ (Using Pythagorus theorem)

$\Rightarrow \quad x = 300 \text{ km}$

Now, let they change their speeds after t_1 hours and then the rest time is t_2 then

$$30t_1 + 40t_2 = 800 \quad \ldots(i)$$

$$40t_1 + 30t_2 = 900 \quad \ldots(ii)$$

Solving Eq. (i) and (ii), we get

$$t_1 = \frac{120}{7} \quad \text{and} \quad t_2 = \frac{50}{7}$$

50. Since it moves only one radian on every path and it has to move 2π radian to reach directly eastward. Hence, it has to run on more than 6 paths i. e., the last path is 7th one (or P_7)
($\because n \times 1$ radian $\geq 2\pi$ radian)

$\Rightarrow \quad n \geq 2\pi$

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or $n = 7$, for interger values

Hence, option (c) is correct.

51. Since it stops directly eastward of the shop so the total distance covered so for

$$= 7 + (1 + 2 + 3 + 4 + 5 + 6 + 2) = 30 \text{ km}$$

NOTE Total radial movement = 7 km
Again on the last path it will move only 2 km.

Actually it has to cover total 2π radian distance but on 6 paths it covers only 6 radian hence the remaining distance which will be covered on the 7th path i. e., $2\pi - 6$

$$= 2 \times \frac{22}{7} - 6 = \frac{2}{7} \text{ radian}$$

But, the radius of the last path (i. e., P_7) = 7 km

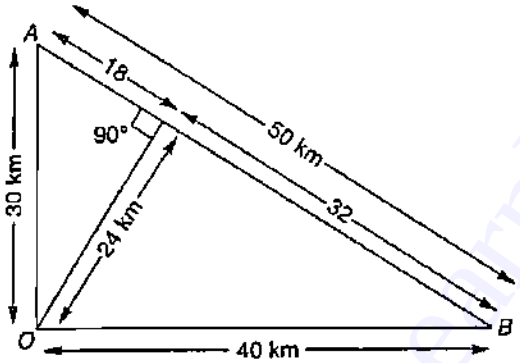
Hence, the distance covered in km $= \frac{2}{7} \times 7 = 2$ km

$$\left(\theta = \frac{\text{Arc}}{\text{Radius}}\right)$$

Thus, on the last path it moves only 2 km. Hence, (a) is the correct choice.

52. The ratio of distance covered on P_2 and $P_7 = \frac{2}{2} = \frac{1}{1}$

53. Since it is clear from the statement itself that ΔAOB is a right angle triangle and further OP must be perpendicular to AB then we can find that $AO = 30$ km and $BO = 40$ km by using Pythogorus theorem and its corrollaries.



HINT

$$OP^2 = OA^2 - AP^2$$

$$\therefore \quad OP^2 = 900 - x^2$$

and $$OB^2 = OP^2 + BP^2$$

$$\therefore \quad OB^2 = 900 - x^2 + 1024$$

and $$AB^2 = OA^2 + OB^2$$

$$\therefore \quad (x + 32)^2 = 900 + 900 - x^2 + 1024$$

$$\Rightarrow \quad x = 18 \text{ km}$$

Hence, $AP = 18$, $OA = 30$ and $OB = 40$ and $OP = 24$ km

Now, since jackal and cat reaches A at the same time, so the ratio of speeds = ratio of distances covered by them.

$$\therefore \quad \frac{\text{Speed of Jackal}}{\text{Speed of cat}} = \frac{30}{18} = \frac{5}{3}$$

54. Again, since jackal and train both arrive at A at the same time and let the train was x km away from A, before entering into the tunnels, i.e., when it makes a whistle then the ratio of distances covered by train and jackal.

$$= \frac{x}{30} = \frac{x + 50}{40}$$

$$\Rightarrow \quad x = 150 \text{ km}$$

Thus, the ratio of speeds of Jackal is to train is 1 : 5.

55. Since, when the train arrive at A, the jackal can move 30 km. So, at the time when train is at A the jackal will cover 6 km from P on PA in addition to 24 km at OP. Now, the rest distance at AP is 12 km this remaining distance will be covered by train and jackal according to their respective speeds.

So, distance covered by train $= 12 \times \frac{5}{6} = 10$ km

and distance covered by jackal $= 12 \times \frac{1}{6} = 2$ km

Hence, jackal will meet with train at M_1 which is 10 km away from A (inside AB).

NOTE It can be solved using options in lesser time.

56. It is obvious from the path of cat that if cat moves in the POA direction it will never meet with accident and now jackal follows the path OPB. Again when the train is at A then jackal will cover 30 km (i. e., 24 (OP) + 6 km on PB).
So, the ratio of distances covered by jackal is to train = ratio of their respective speeds.
Now let the jackal and train meet each other at AB, $(6 + x)$ km away from P towards B, then

$$\frac{x}{24 + x} = \frac{1}{5}$$

$$\Rightarrow \quad 4x = 24 \quad \Rightarrow \quad x = 6$$

Hence, train meets with jackal at $(18 + 6 + 6) = 30$ km away from A.

Alternatively: $\frac{150 + 18 + 6 + x}{30 + x} = \frac{5}{1} \Rightarrow x = 6$

Hence, $18 + 6 + 6 = 30$ km.
Thus, option (b) is correct.

57. The ratio of time taken by cat and jackal $= \frac{72/3}{96/5} = \frac{5}{4}$

Hence, option (c) is correct.

58.

$$(6 - x) = (8 - 1.5x)$$

$$\Rightarrow \quad x = 4 \text{ cm}$$

So, it will take 4 hours to burn in such a way that they remain equal in length.

59. Total distance covered by them when they meet = $2W$

and $$\text{Total time} = \frac{2W}{b_1 + b_2}$$

$$\therefore \quad d_1 = \frac{2W}{(b_1 + b_2)} b_1 \quad \text{and} \quad d_2 = \frac{2W}{(b_1 + b_2)} b_2$$

60. Let the speed of boat be B and that of river be R. In 12 minutes the distance between boat and hat

$$= 12(B - R) + 12R = 12B$$

Now time taken by boat to reach to the hat

$$= \frac{12B}{(B + R) - R} = 12 \text{ min}$$

Total time = 24 min

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In 24 minutes had flown off = 3 km

$\therefore \quad \frac{24}{60} \times R = 3$

$\Rightarrow \quad R = 7.5$ km/h

61. Akbar meets Birbal once $\frac{500}{20-15} = 100$ s

Birbal meets Chanakya once $= \frac{500}{20+25} = 11\frac{1}{9}$ s

Akbar meets Chanakya once $= \frac{500}{15+25} = 12.5$ s

62. Time taken by them to meet $= \frac{600}{30-20} = 60$ s

Time taken to meet 5th time $= 5 \times 60 = 300$ s

Total duration of race $= \frac{3000}{30} = 100$ s

So, they will not meet 5th time in the race of 3000 metre.

63. Length of the track $= 2 \times \frac{22}{7} \times 175 = 1100$ m

Distance to be covered for the first meeting = 550 m

Speed of Akkal $= \frac{1100}{100} = 11$ m/s

Speed of Bakkal $= \frac{1100}{50} = 22$ m/s

Time taken from the start of the first meeting

$= \frac{550}{(11+22)} = \frac{50}{3}$ s

Time taken for Akkal and Bakkal to meet again at Love point = LCM of times taken by them to go around the track once.

= LCM of $\frac{1100}{11}$ and $\frac{1100}{22}$

= LCM of 100 and 50

= 100 s

So, the total required time $= \frac{50}{3} + 100 + 100$

$= \frac{650}{3} = 216\frac{2}{3}$ s

64. Since both rest for 6 seconds so when B is just about to start the journey A reaches there at the shallow end so they meet at they shallow end.

65. B runs around the track in 10 min.

i. e., Speed of B = 10 min per round

$\therefore$ A beats B by 1 round

Time taken by A to complete 4 rounds

= Time taken by B to complete 3 rounds

= 30 min

$\therefore$ A's speed $= \frac{30}{4}$ min per round

= 7.5 min per round

Hence, if the race is only of one round A's time over the course = 7 min 30 sec.

66. The ratio of speeds of $A, B, C = \frac{10}{49} : \frac{9}{50} : \frac{8}{51}$

Hence, A is the fastest.

67. Speed of this car $= \frac{400+200}{20} \times \frac{18}{5}$ km/h

= 108 km/h

68. The speeds of two persons is 108 km/h and 75 km/h. The first person covers 1080 km in 10 hours and thus he makes 12 rounds. Thus, he will pass over another person 12 times in any one of the direction.

69. Angle between two hands at 3 : 10 am

$= (90 + 5) - 60 = 35°$

So, the required angle = 70°, after 3 : 10 am

Total time required to make 70° angle when minute-hand is ahead of hour-hand.

$= \frac{90+70}{11/2} = \frac{320}{11}$ min

So, at 3 h $\frac{320}{11}$ min the required angle will be formed.

Alternatively: Check through options.

70. For the first watch: When a watch creates the difference of 12 hours, it shows correct time.

So to create the difference of 12 h required time

$= \frac{60 \times 12}{24} = 30$ days

For the second watch: To create the difference of 12 h required time

$= \frac{30 \times 12}{24} = 15$ days

So, after 30 days at the same time both watches show the correct time.

HINT Take the LCM of 30 and 15.

71. To show the same time together the difference between two watches must be 12 h.

Now, since they create 3 min differnce in 1 h

So they will create 12 h differnce in $\frac{1}{3} \times \frac{12 \times 60}{24}$

= 10 days later

72. To show the correct time again, watch must create 24 h difference. (Since in one round hour-hand covers 24 h.)

So, the required time $= \frac{4}{3} \times \frac{60 \times 24}{24} = 80$ day

73. $(n + 1)$ times in $n$days

74. Actually the watch gains (12 + 16)

= 28 min in $7 \times 24 \times 60$ min.

Thus, it gains 1 min in 360 minutes.

Therefore, it will gain (12 + 8) min in $\frac{20 \times 360}{60 \times 24} = 5$ day

Hence, (b) is the correct choice.

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75. Actually they create a difference of 3 min per hour and the two watches are showing a difference of 66 minutes. Thus, they must have been corrected 22 hours earlier.
Now, the correct time can be found by comparing any one of the watch.
Since, second watch gains 1 min in 1 hour so it will must show 22 min extra than the correct time in 22 hours.
Hence, the correct time can be found by subtracting 22 min from 10 : 06.
Hence, (d) is the correct answer.

NOTE For quick answer go through options.

76. Incorrect watch covers 1452 min in 1440 min

So, it will cover 1 min in $\frac{1440}{1452}$ min

Therefore it will cover 4840 min in $\frac{1440}{1452} \times 4840$

$= 4800$ min

$= 80$ h

Therefore 80 h = 3 days and 8 h.

77. You must know that a correct watch coincide just after $65\frac{5}{11}$ min.

Therefore in every $65\frac{5}{11}$ hours the watch gains $\frac{2}{11}$.

Hence, in 24 hours it will gain $\frac{2}{11} \times \frac{11}{720} \times 24 \times 60 = 4$ min

78. In 72 hours my watch gains $(8 + 7) = 15$ min. To show the correct time watch must gain 8 minutes.
Since the watch gains 15 min in 72×60 min.
Therefore, the watch will gain 8 min in $\frac{72 \times 60 \times 8}{15}$ min

$$= \frac{72 \times 60 \times 8}{15} = 38 \text{ h } 24 \text{ min}$$

Hence, (a) is the correct choice.

80. To exchange the position both hands to cover 360° together.
In one minute, hour-hand moves $\frac{1°}{2}$ and in one minute, minute-hand moves 6°. Let the required time be t min, then

$$6t + \frac{1}{2}t = 360$$

$$\Rightarrow \quad t = \frac{360}{13} \times 2 = \frac{720}{13} = 55\frac{5}{13} \text{ min}$$

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10 MENSURATION

It is one of the easiest chapters which contributes almost 6-8% problems in Quantitative Aptitude Section of CAT. Besides there are several other aptitude tests which includes plethora of questions from this topic itself. Therefore it is advised that those students who are not so good in other sections such as algebra or sort of logical questions they must emphasise on this chapter. Even the questions asked from this chapter are not as much complex as in Geometry.

Definition : Mensuration is a science of measurement of the lengths of lines, areas of surfaces and volumes of solids.

Planes : Planes are two dimensional *i.e.*, these two dimensions are namely length and breadth. These occupy surface.

Solids : Solids are three dimensional, namely length, breadth and height. These occupy space.

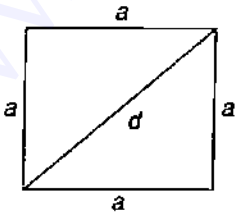
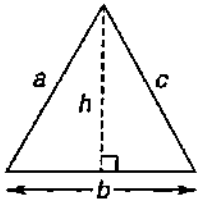
CONVERSION OF SOME IMPORTANT UNITS

1 km = 10 hectometre = 100 decametre
= 1000 metre = 10,000 decimetre
= 1,00,000 centimetre = 10,00,000 millimetre
1 hectare = 10,000 square metre
1 are = 100 square metre
1 square hectometre = 100 square decametre
1 square decametre = 100 square metre
1 square metre = 100 square decimetre
1 square decimetre = 100 square centimetre
1 square centimetre = 100 square millimetre

$\sqrt{2} = 1.414$, $\sqrt{3} = 1.732$,
$\sqrt{5} = 2.236$, $\sqrt{6} = 2.45$

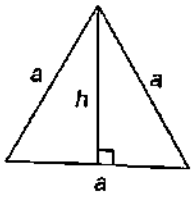
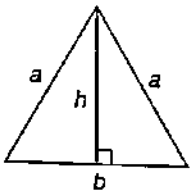
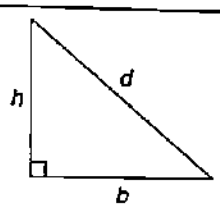
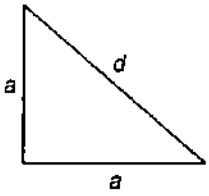
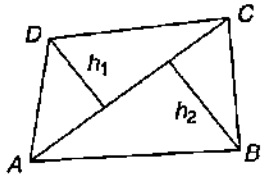
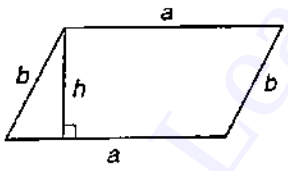
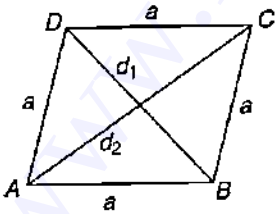
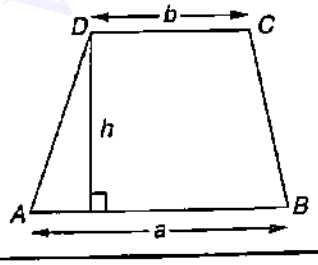
Weight = Volume × density

2-D Figures (Plane figures)

S. No.	Name	Figure	Nomenclature	Area	Perimeter
1.	Rectangle		$l \rightarrow$ length $b \rightarrow$ breadth	$l \times b = lb$	$2l + 2b = 2(l + b)$
2.	Square		$a \rightarrow$ side $d \rightarrow$ diagonal $d = a\sqrt{2}$	(i) $a \times a = a^2$ (ii) $\frac{d^2}{2}$	$a + a + a + a = 4a$
3.	Triangle (Scalene)		a, b and c are three sides of triangle and s the semiperimeter, where $s = \left(\frac{a+b+c}{2}\right)$ b is the base and h is the altitude of triangle	(i) $\frac{1}{2} \times b \times h$ (ii) $\sqrt{s(s-a)(s-b)(s-c)}$ (Hero's formula)	$a + b + c = 2s$

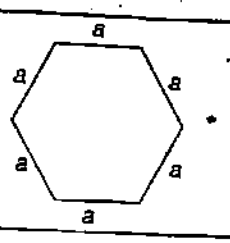
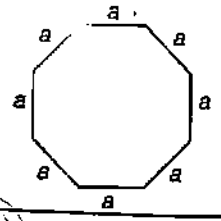
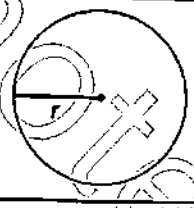
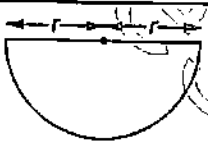
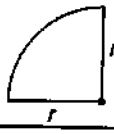
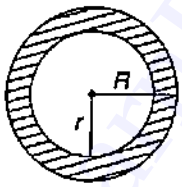
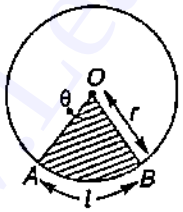
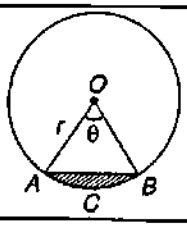
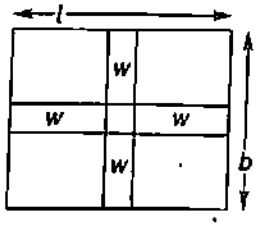
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S. No.	Name	Figure	Nomenclature	Area	Perimeter
4.	Equilateral triangle		$a \rightarrow$ side $h \rightarrow$ height or altitude $h = \frac{\sqrt{3}}{2}a$	(i) $\frac{1}{2} \times a \times h$ (ii) $\frac{\sqrt{3}}{4}a^2$	$3a$
5.	Isosceles triangle		$a \rightarrow$ equal sides $b \rightarrow$ base $h \rightarrow$ height or altitude $h = \frac{\sqrt{4a^2 - b^2}}{2}$	(i) $\frac{1}{2} \times b \times h$ (ii) $\frac{1}{4} \times b \times \sqrt{4a^2 - b^2}$	$2a + b$
6.	Right angled triangle		$b \rightarrow$ base $h \rightarrow$ altitude/height $d \rightarrow$ diagonal $d = \sqrt{b^2 + h^2}$	$\frac{1}{2} \times b \times h$	$b + h + d$
7.	Isosceles right angled triangle		$a \rightarrow$ equal sides $d \rightarrow$ diagonal $d = a\sqrt{2}$	$\frac{1}{2}a^2$	$2a + d$
8.	Quadrilateral		AC is the diagonal and h_1, h_2 are the altitudes on AC from the vertices D and B respectively	$\frac{1}{2} \times AC \times (h_1 + h_2)$	$AB + BC + CD + AD$
9.	Parallelogram		a and b are sides adjacent to each other. $h \rightarrow$ distance between the parallel sides	$a \times h$	$2(a + b)$
10.	Rhombus		$a \rightarrow$ each equal side of rhombus d_1 and d_2 are the diagonals $d_1 \rightarrow BD$ $d_2 \rightarrow AC$	$\frac{1}{2} \times d_1 \times d_2$	$4a$
11.	Trapezium		a and b are parallel sides to each other and h is the perpendicular distance between parallel sides	$\left(\frac{a+b}{2}\right) \times h$	$AB + BC + CD + AD$

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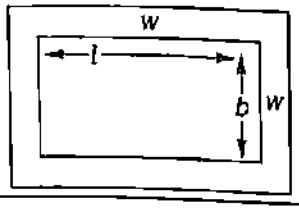
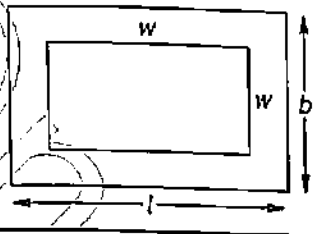
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S. No.	Name	Figure	Nomenclature	Area	Perimeter
12.	Regular hexagon		$a \rightarrow$ each of the equal side	$\frac{3\sqrt{3}}{2}a^2$	$6a$
13.	Regular octagon		$a \rightarrow$ each of equal side	$2a^2(1+\sqrt{2})$	$8a$
14.	Circle		$r \rightarrow$ radius of the circle $\pi = \frac{22}{7} = 3.1416$ (approx)	πr^2	$2\pi r$ (called as circumference)
15.	Semicircle		$r \rightarrow$ radius of the circle	$\frac{1}{2}\pi r^2$	$\pi r + 2r$
16.	Quadrant		$r \rightarrow$ radius	$\frac{1}{4}\pi r^2$	$\frac{1}{2}\pi r + 2r$
17.	Ring or circular path (shaded region)		$R \rightarrow$ outer radius $r \rightarrow$ inner radius	$\pi(R^2 - r^2)$	(outer) $\rightarrow 2\pi R$ (inner) $\rightarrow 2\pi r$
18.	Sector of a circle		$O \rightarrow$ centre of the circle $r \rightarrow$ radius $l \rightarrow$ length of the arc $\theta \rightarrow$ angle of the sector $l = 2\pi r\left(\frac{\theta}{360°}\right)$	(i) $\pi r^2\left(\frac{\theta}{360°}\right)$ (ii) $\frac{1}{2} r \times l$	$l + 2r$
19.	Segment of a circle		$\theta \rightarrow$ angle of the sector $r \rightarrow$ radius $AB \rightarrow$ chord $ACB \rightarrow$ arc of the circle	Area of segment ACB (minor segment) $= r^2\left(\frac{\pi\theta}{360°} - \frac{\sin\theta}{2}\right)$	$2r\left[\frac{\pi\theta}{360°} + \sin\left(\frac{\theta}{2}\right)\right]$
20.	Pathways running across the middle of a rectangle		$l \rightarrow$ length $b \rightarrow$ breadth $w \rightarrow$ width of the path (road)	$(l + b - w)\,w$	$2(l+b) - 4w$ $= 2[l + b - 2w]$

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S. No.	Name	Figure	Nomenclature	Area	Perimeter
21.	Outer pathways		$l \rightarrow$ length $b \rightarrow$ breadth $w \rightarrow$ widthness of the path	$(l+b+2w)\,2w$	(inner) $\rightarrow 2(l+b)$ (outer) $\rightarrow 2(l+b+4w)$
22.	Inner path		$l \rightarrow$ length $b \rightarrow$ breadth $w \rightarrow$ widthness of the path	$(l+b-2w)\,2w$	(outer) $\rightarrow 2(l+b)$ (inner) $\rightarrow 2(l+b-4w)$

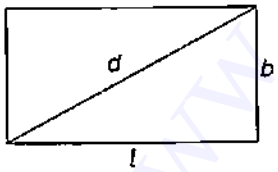
RECTANGLES AND SQUARES

1. Area of a rectangle = Length × breadth = $l \times b$
2. Area of a square = $(\text{side})^2 = \frac{1}{2}(\text{diagonal})^2$
$= \frac{1}{2}d^2 = a^2$
3. Diagonal of a rectangle
$= \sqrt{(\text{length})^2 + (\text{breadth})^2} = \sqrt{l^2 + b^2}$
4. Diagonal of a square
$= \sqrt{\text{side}^2 + \text{side}^2} = \text{side}\sqrt{2} = a\sqrt{2}$
5. Perimeter of a rectangle
$= 2(\text{length} + \text{breadth}) = 2(l+b)$
6. Perimeter of a square $= 4 \times \text{side} = 4a$
7. Area of four walls of a room $= 2(l+b) \times h$

EXAMPLE 1 *The length and breadth of a rectangular room are 15 m and 8 m respectively :*
(a) Find the perimeter of room.
(b) Find the area of the floor of room.
(c) Find the maximum possible length of the rod that can be put on the floor.

SOLUTION (a) Perimeter $= 2(15+8) = 46$ m
(b) Area of floor $= 15 \times 8 = 120$ m^2
(c) Length of diagonal $= \sqrt{(15)^2 + (8)^2} = 17$ m

NOTE The maximum possible length of any rod that can be placed on a rectangular floor is equal to the diagonal of the floor of the room.



Here $d > l$

EXAMPLE 2 *The side of a square shaped garden is 20 m. Find the :*
(a) area of the garden
(b) perimeter (or boundary) of the garden
(c) maximum possible distance between any two corners of the garden.

SOLUTION (a) Area $= (\text{side})^2 = (20)^2 = 400$ m^2 (square metre)
(b) Perimeter $= 4 \times \text{side} = 4 \times 20 = 80$ m
(c) Diagonal $= \text{side}\sqrt{2} = 20 \times \sqrt{2}$
$= 20 \times 1.414 = 28.28$ m

EXAMPLE 3 *One side of a rectangular lawn is 12 m and its diagonal is 13 m. Find the area of the field.*

SOLUTION $d = \sqrt{l^2 + b^2} \Rightarrow 13 = \sqrt{12^2 + b^2}$
$\Rightarrow b = 5$ m
$\therefore$ Area $= l \times b = 12 \times 5 = 60$ m^2

EXAMPLE 4 *The length of a rectangle is 1 cm more than its breadth. The diagonal is 29 cm. Find the area of the rectangle.*
(a) 481 cm^2 (b) 841 cm^2
(c) 420 cm^2 (d) 870 m^2

SOLUTION $l = (b+1)$
and $d = \sqrt{l^2 + b^2} = \sqrt{(b+1)^2 + b^2} = 29$
$\Rightarrow b^2 + b = 420$
$\Rightarrow b = 20$
$\therefore l = 21$
$\therefore$ Area $= l \times b = 420$ cm^2

EXAMPLE 5 *The length of a wall is 5/4 times of its height. If the area of the wall be 180 m^2. What is the sum of the length and height of the wall?*

SOLUTION Let the length be $5x$ and height be $4x$
then $l \times h = 180 = 5x \times 4x = 20x^2$
$\Rightarrow x = 3$
$\therefore l + h = 15 + 12 = 27$
Alternatively : $1.25\,h^2 = 180$

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$\Rightarrow$ $h = 12$ m

$\therefore$ $l = 15$ m

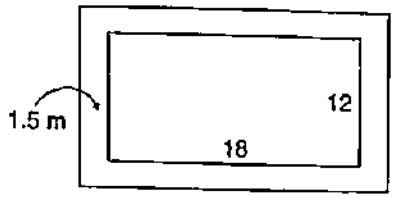
$l + h = 27$ m

EXAMPLE 6 *A rectangular grassy lawn is 18 m by 12 m. It has a gravel path 1.5 m wide all around it on the outside. What is the area of the path.*

SOLUTION Area of path (outside the lawn) $= (l + b + 2w)\,2w$

$= (18 + 12 + 3)\,3$

$= 99$ m^2



Alternatively :

Area of lawn $= 18 \times 12 = 216$ m^2

Total area (lawn + path) $= (18 + 3) \times (12 + 3)$

$= 21 \times 15 = 315$ m^2

$\therefore$ Area of path (only) $= 315 - 216 = 99$ m^2

EXAMPLE 7 *Find the cost of carpeting a room 17 m long and 9 m wide with a carpet 60 cm broad at 40 paise per metre.*

SOLUTION Area of carpet = Area of room

$l \times 0.6 = 17 \times 9$ (60 cm = 0.6 m)

$\Rightarrow$ $l = 255$ m

$\therefore$ Cost of carpeting the floor = rate $\times$ length of carpet

$= 0.4 \times 255 =$ Rs. 102.00

EXAMPLE 8 *The dimensions of a lawn are in the ratio 4 : 1 and its area is 1/4 hectares. What is the length of the lawn?*

SOLUTION $l \times b = 4x \times x = \frac{1}{4} \times 10{,}000$

$\Rightarrow$ $x = 25$

$\Rightarrow$ length $= 4x = 100$ m

EXAMPLE 9 *A room is 16 m long, 7 m broad and 8 m high. Find the cost of white washing the four walls of room at Rs. 7.5 per m^2, white washing is not to be done on the doors and windows, which occupy 65 m^2.*

SOLUTION Area of 4 walls of a room $= 2(16 + 7) \times 8 = 368$ m^2

Net area of 4 walls $= 368 - 65 = 303$ m^2

$\therefore$ Cost of white washing $= 303 \times 7.5 =$ Rs. 2272.5

EXAMPLE 10 *The ratio between the sides of a room is 5 : 3. The cost of white washing the ceiling of the room at 50 P per square m is Rs. 270 and the cost of papering the walls at 10 P per square metre is Rs. 48. The height of the room is :*

(a) 6 m (b) 8 m (c) 5 m (d) 10 m

SOLUTION Area of ceiling $= \frac{\text{Total cost}}{\text{cost of 1 sq. unit}}$

$= \frac{270}{0.5} = 540$ sq. m

Now since $l : b = 5x : 3x$

$\Rightarrow$ $l \times b = 15x^2 = 540$ m^2

$\Rightarrow$ $l = 30$ and $b = 18$ m

Now, Area of the 4 walls $= \frac{\text{Total cost}}{\text{Cost of 1 sq. unit}} = \frac{48}{0.1} = 480$ m^2

$\therefore$ Height $= \frac{480}{2(30 + 18)} = 5$ m

INTRODUCTORY EXERCISE-10.1

1. A rectangular field has its length and breadth in the ratio of 16 : 9. If its perimeter is 750 cm. What is its area?
   (a) 7500 cm^2 (b) 32400 cm^2
   (c) 14400 cm^2 (d) 14000 cm^2
2. A rectangular field costs Rs. 110 for levelling at 50 paise per square metre. If the ratio of length : breadth is 11 : 5. Find the length of the field :
   (a) 16 m (b) 21 m
   (c) 22 m (d) none of these
3. A room is half as long again as it is wide. The cost of carpeting it at 62 paise per square metre is Rs. 2916.48. Find the cost of white washing the ceiling at 30 paise per metre :
   (a) Rs. 2211.5 (b) Rs. 1114.2
   (c) Rs. 1411.2 (d) can't be determined
4. The length of a rectangular plot of ground is four times its breadth and its area is 4 hectares. How long will it take to a dog to walk round it at the rate of 3 km/hr?
   (a) 12 min (b) 20 min
   (c) 21 min (d) 18.5 min
5. Find the length of the wire required to fence a square field 6 times having its area 5 hectares and 76 ares :
   (a) 5760 m (b) 6760 m
   (c) 52500 m (d) 11760 m
6. A room is 19 m long and 3.50 m broad. What will be the cost of covering its floor with a carpet of 70 cm wide at 95 paise per metre?
   (a) Rs. 90.25 (b) Rs. 99.25
   (c) Rs. 90.75 (d) none of these
7. Find the cost of paving a courtyard 316.8 m $\times$ 65 m with stones measuring 1.3 m $\times$ 1.1 m at Rs. 0.5 per stone :
   (a) Rs. 1440 (b) Rs. 7200
   (c) Rs. 72,000 (d) none of these
8. A rectangular garden 63 m long and 54 m broad has a path 3 m wide inside it. Find the cost of paving the path at Rs. 37/2 per square metre :

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(a) Rs. 12321 (b) Rs. 11100
(c) Rs. 74000 (d) none of these

9. If the length of a rectangular field is doubled and its breadth is halved (*i.e.*, reduced by 50%). What is percentage change in its area?
(a) 0% (b) 10%
(c) 25% (d) 33.33%

10. A path of uniform width runs all around the inside of rectangular field 116 m by 68 m and occupies 720 sq. m. Find the width of the path :
(a) 1 m (b) 1.5 m
(c) 2 m (d) 4 m

11. A drawing room is 7.5 m long 6.5 m broad and 6 m high. Find the length of paper 2.5 dm wide to cover its walls allowing 8 sq. m for doors :
(a) 368 m (b) 640 m
(c) 625 m (d) 888 m

12. A square field of 2 sq. km is to be divided into two equal parts by a wall which coincides with a diagonal. Find the length of the wall :
(a) $\sqrt{2}$ km (b) 1 km
(c) 4.2 km (d) 2 km

13. There are two square fields. Of the two square fields one contains 1 hectare area while the other is broader by 11 per cent. Find the difference in area expressed in sq. m :
(a) 2321 sq. m (b) 1210 sq. m
(c) 2121 sq. m (d) 7700 sq. m

14. The expenses of carpeting a half of the floor were Rs. 759, but if the length had been 6 m less than it was, the expenses would have been Rs. 561. What is the length?
(a) 21 m (b) 23 m
(c) 45 m (d) 27 m

15. If a roll of paper 1 km long has area 1/25 hectare, how wide is the paper?
(a) 4 m (b) 40 cm
(c) 40 dm (d) 25 cm

16. The diagonal and one side of a rectangular plot are 289 m and 240 m respectively. Find the other side :
(a) 237 m (b) 181 m
(c) 161 m (d) 159 m

17. How many tiles 20 cm by 40 cm will be required to pave the floor of a prayer hall of a room 16 m long and 9 m wide :
(a) 18000 (b) 2700
(c) 1800 (d) 14400

18. If the area of a square be 22050 sq. cm. Find the length of diagonal :
(a) 201 cm (b) 220 cm
(c) 211 cm (d) 210 cm

19. If requires 90 g paint for painting a door 12 cm × 9 cm, how much paint is required for painting a similar door 4 cm × 3 cm?
(a) 30 g (b) 27 g
(c) 10 g (d) 45 g

20. The area of a rectangular foot ball field is 24200 sq. m. It is half as broad as it is long. What is the approx minimum distance a man will cover if he wishes to go from one corner to the opposite one?
(a) 283 m (b) 246 m
(c) 576 m (d) 289 m

21. The area of the four walls of a room is 2640 sq. m and the length is twice the breadth and the height is given as 11 m. What is the area of the ceiling?
(a) 2800 m^2 (b) 3200 m^2
(c) 320 m^2 (d) none of these

22. If the perimeter of a square and a rectangle are the same, then the areas *A* and *B* enclosed by them would satisfy the inequality :
(a) $A > B$ (b) $A \geq B$
(c) $A < B$ (d) $A \leq B$

23. If the perimeter of a rectangle and a square each is equal to 80 cm and the difference of their areas is 100 sq. cm, the sides of the rectangle are :
(a) 25 cm, 15 cm (b) 28 cm, 12 cm
(c) 30 cm, 10 cm (d) 35 cm, 15 cm

24. The number of square shaped tin sheets of side 25 cm that can be cut off from a square tin sheet of side 1 m, is :
(a) 4 (b) 40
(c) 16 (d) 400

25. The length and breadth of a rectangular field are 120 m and 80 m respectively. Inside the field, a park of 12 m width is made around the field. The area of the park is :
(a) 2358 m^2 (b) 7344 m^2
(c) 4224 m^2 (d) 3224 m^2

26. A 5m wide lawn is cultivated all along the outside of a rectangular plot measuring 90m × 40 m. The total area of the lawn is :
(a) 1441 m^2 (b) 1400 m^2
(c) 2600 m^2 (d) 420 m^2

27. The length of a rectangle is 2 cm more than its breadth. The perimeter is 48 cm. The area of the rectangle (in cm^2) is :
(a) 96 (b) 128
(c) 143 (d) 144

28. The area of a rectangular field is 52000 m^2. This rectangular area has been drawn on a map to the scale 1 cm to 100 m. The length is shown as 3.25 cm on the map. The breadth of the rectangular field is :
(a) 210 m (b) 150 m
(c) 160 m (d) 123 m

29. If the length of diagonal *BD* of a square *ABCD* is 4.8 cm, the area of the square *ABCD* is :
(a) 9.6 cm^2 (b) 11.52 cm^2
(c) 12.52 cm^2 (d) 5.76 cm^2

30. If the side of square is increased by 20%, then how much per cent does its area get increased :
(a) 40% (b) 20%
(c) 44% (d) 24%

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31. The ratio of the area of a square to that of the square drawn on its diagonal is :
(a) 1 : 1 (b) 1 : 2
(c) 1 : 3 (d) 1 : 4

32. The length and breadth of a square are increased by 60% and 40% respectively. The area of the resulting rectangle exceeds the area of the square by :
(a) 224% (b) 24%
(c) 124% (d) 100%

TRIANGLES

(1) Area of a triangle $= \frac{1}{2} \times$ base $\times$ height (General formula)

(2) Area of an scalene triangle

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

where s is the semiperimeter of the triangle and a, b and c are the three sides of the triangle, $\left(s = \frac{a+b+c}{2}\right)$

(**Scalene triangle :** A triangle whose all sides are different in length)

(3) Area of a right angle triangle $= \frac{1}{2} \times$ base $\times$ height

(**Right angle triangle :** A triangle in which two sides are perpendicular)

Also, Hypotenuse $= \sqrt{(\text{base})^2 + (\text{height})^2}$

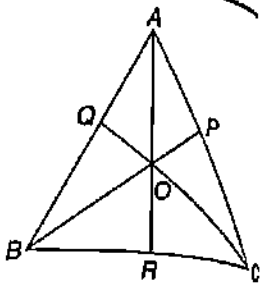
(4) Area of an equilateral triangle

$$= \frac{1}{2} \times \text{side} \times \left(\frac{\sqrt{3}}{2}\text{ side}\right) = \frac{\sqrt{3}}{4} \times (\text{side})^2$$

(**Equilateral triangle :** A triangle in which all the three sides are equal also all the three internal angles are equal)

Height of an equilateral triangle $(h) = \frac{\sqrt{3}}{2} \times$ side

An important property : In an equilateral Δ perpendiculars drawn from all the three vertices intersect each other in the ratio of 2 : 1 from the vertex to the base.

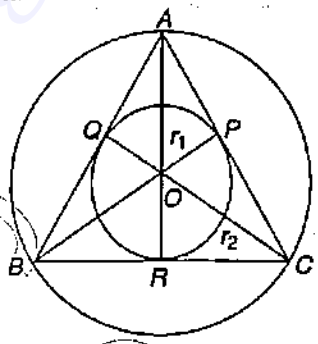


$$\therefore \quad \frac{AO}{OR} = \frac{BO}{OP} = \frac{CO}{OQ} = \frac{2}{1} = \frac{r_2}{r_1}$$

$OP = OQ = OR$, all are the inradii

$OA = OB = OC$, all are the circumradii

NOTE Radii means more than one radius.



O is the centre of two circles. Circle PQR is called as incircle (touching the sides) and circle ABC is called as circumcircle (touching the vertices)

$$\therefore \quad \text{Inradius} = \frac{\text{side}}{2\sqrt{3}}$$

and $$\text{Circumradius} = \frac{\text{side}}{\sqrt{3}}$$

EXAMPLE 1 *The base of a right angled triangle is 8 cm and hypotenuse is 17 cm. Find its area.*

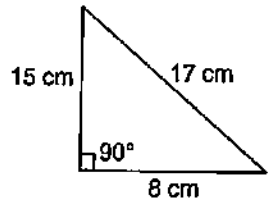
SOLUTION Hypotenuse $= \sqrt{(\text{base})^2 + (\text{altitude})^2}$

$$(17) = \sqrt{64 + (x)^2}$$

$$\Rightarrow \quad 289 = 64 + x^2$$

$$\Rightarrow \quad x^2 = 225$$

$$\Rightarrow \quad x = 15\text{cm}$$



$$\therefore \quad \text{Area} = \frac{1}{2} \times \text{base} \times \text{altitude}$$

(height is called altitude also)

$$= \frac{1}{2} \times 8 \times 15$$

$$= 60 \text{ cm}^2$$

EXAMPLE 2 *Find the area of an equilateral triangle whose side is* $4\sqrt{3}$ *cm.*

SOLUTION Area of an equilateral triangle $= \frac{\sqrt{3}}{4} \times (\text{side})^2$

$$= \frac{\sqrt{3}}{4} \times (4\sqrt{3})^2$$

$$= \frac{\sqrt{3}}{4} \times 48 = 12\sqrt{3} \text{ cm}^2$$

EXAMPLE 3 *The area of a right angled triangle is 24 cm² and the length of its hypotenuse is 10 cm. The length of the shorter leg is :*
(a) 5 cm (b) 4 cm (c) 6 cm (d) 3 cm

SOLUTION $H^2 = A^2 + B^2 = 100 \text{ cm}^2$

$$\text{Area} = \frac{1}{2} \times A \times B$$

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$\Rightarrow \quad \frac{1}{2} A \times B = 24 \text{ cm}^2$

$\Rightarrow \quad A \times B = 48 \text{ cm}^2$

$\Rightarrow \quad (A+B)^2 = A^2 + B^2 + 2AB$

$\therefore \quad (A+B)^2 = 196$

$A + B = 14 \quad \ldots(1)$

$\Rightarrow$

Again $\quad (A-B)^2 = A^2 + B^2 - 2AB$

$\Rightarrow \quad (A-B)^2 = 4$

$\Rightarrow \quad A - B = 2 \quad \ldots(2)$

Therefore by solving equation (1) and (2) we get

$A = 8 \quad$ and $\quad B = 6$

Therefore the shorter leg is 6 cm.

Alternatively : Go through options.

$\because \quad \frac{1}{2} \times A \times B = 24$

$\Rightarrow \quad A \times B = 48$

Let us assume $B = 6$, then $A = \frac{48}{6} = 8$

Now, $\quad A^2 + B^2 = 100$

$(8)^2 + (6)^2 = 100$

$100 = 100$

Hence choice (c) is correct.

EXAMPLE 4 *Find the area and perimeter of a triangle whose sides are 17 cm, 8 cm and 15 cm long.*

SOLUTION Since $\quad 8^2 + 15^2 = 17^2$

$64 + 225 = 289$

$289 = 289$

Hence it is a right angled triangle.

$\therefore \quad$ Area $= \frac{1}{2} \times b \times h$

$= \frac{1}{2} \times 15 \times 8$

$= 60 \text{ cm}^2$

and $\quad$ perimeter $= 8 + 15 + 17 = 40$ cm

EXAMPLE 5 *Find the area and perimeter of an isosceles triangle whose equal sides are 5 cm each and base is 6 cm.*

SOLUTION Area of an isosceles triangle

$= \frac{b}{4} \sqrt{4a^2 - b^2}$

$= \frac{6}{4} \sqrt{4 \times 25 - 36}$

$= \frac{3}{2} \times \sqrt{64} = 12 \text{ cm}^2$

Alternatively : We know that the altitude CD bisects the base AB in the isosceles triangle ABC.

$\therefore \quad AD = BD = 3$ cm

Using Pythagorus theorem in ΔADC we have,

$CD^2 = AC^2 - AD^2$

$\therefore \quad CD = 4$ cm

$\therefore \quad$ Area of triangle $= \frac{1}{2} \times$ base $\times$ height

$= \frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2$

Also, the perimeter of triangle $= 5 + 5 + 6 = 16$ cm

INTRODUCTORY EXERCISE-10.2

1. What is the area of the triangle whose sides are 84 m, 80 m and 52 m?
   (a) 1620 sq. m (b) 2016 sq. m
   (c) 1818 sq. m (d) none of these
2. Two poles 15 m and 30 m high stand upright in a play ground. If their feet be 36 m apart, find the distance between their tops.
   (a) 41 m (b) 36 m
   (c) 39 m (d) none of these
3. The sides of a triangle are 25 m, 39 m and 56 m respectively. Find the perpendicular distance from the vertex opposite to the side 56 m.
   (a) 15 m (b) 16.5 m
   (c) 18.6 m (d) 21 m
4. ABC is a triangle and D, E, F are the mid-points of the sides BC, CA, AB respectively. The ratio of the areas of ΔABC and ΔDEF is :
   (a) 4 : 1 (b) 5 : 1
   (c) 3 : 1 (d) can't be determined
5. The integral base of an isosceles triangle can be whose area is 60 cm^2 and the length of one of the equal sides is 13 cm :
   (a) 20 cm (b) 10 cm
   (c) 16 cm (d) data insufficient
6. A ladder is resting with one end in contact with the top of a wall of height 60 m and the other end on the ground is at a distance of 11 m from the wall. The length of the ladder is :
   (a) 61 m (b) 71 m
   (c) 87 m (d) none of these
7. The base of a triangular field is three times its height. If the cost of cultivating the field is Rs. 36.72 per hectare is Rs. 495.72, find the height and base of the triangular field :
   (a) 480 m, 1120 m (b) 400 m, 1200 m
   (c) 300 m, 900 m (d) 250 m, 650 m
8. If every side of a triangle is doubled, then increase in area of the triangle is :
   (a) 200% (b) 300%
   (c) 400% (d) none of these

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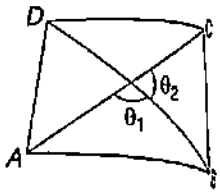
9. If the altitude of an equilateral triangle is $2\sqrt{3}$, then its area is :
(a) $4\sqrt{3}$ cm^2 (b) $12\sqrt{3}$ cm^2
(c) $\frac{8}{\sqrt{3}}$ cm^2 (d) none of these

10. If the perimeter of an equilateral triangle and a square are same and the area of equilateral triangle is P and the area of square is Q, then :
(a) $P < Q$ (b) $P \le Q$
(c) $P > Q$ (d) $P \ge Q$

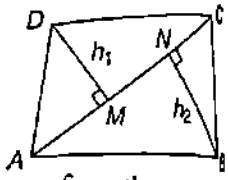
PARALLELOGRAM, RHOMBUS AND TRAPEZIUM

1. (a) Area of parallelogram = base (b) × height (h)
(b) Area of parallelogram
= product of any two adjacent side × sine of the included angle
2. Perimeter = 2 (sum of any two adjacent sides)
3. (a) Area of rhombus $= \frac{1}{2} \times$ (product of diagonals)
$= \frac{1}{2} \times d_1 d_2$
(b) Area of rhombus = product of adjacent sides × sine of the angle included by them
4. Perimeter of rhombus = 4 × side
5. Area of a trapezium $= \frac{1}{2} \times$ sum of parallel sides × height
height → distance between the two parallel sides
6. Perimeter of trapezium = sum of all the four sides

7. (a) Area of quadrilateral $= \frac{1}{2} \times$ product of diagonal × sine of the angle between them ($AC = d_1$ and $BD = d_2$)



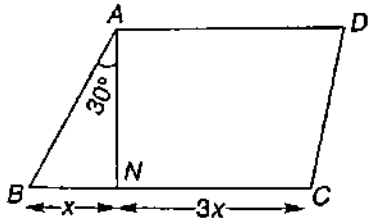
$\therefore$ Area $= \frac{1}{2} d_1 d_2 \sin\theta_1$
$= \frac{1}{2} d_1 d_2 \sin\theta_2$



(b) Area $= \frac{1}{2} \times$ diagonal × sum of the perpendiculars drawn from the opposite vertices on it.
$= \frac{1}{2} d \times (h_1 + h_2)$
where, $MD = h_1$ and $BN = h_2$
$AC = d$

INTRODUCTORY EXERCISE-10.3

1. The adjacent sides of a parallelogram are 6 cm and 8 cm and the angle between them is 30°. What is the area of the parallelogram?
(a) 24 cm^2 (b) 12 cm^2
(c) 40 cm^2 (d) $24\sqrt{3}$ cm^2
2. A parallelogram has sides 30 cm and 20 cm and one of its diagonal is 40 cm long. Then its area is :
(a) $75\sqrt{5}$ cm^2 (b) 245 cm^2
(c) $150\sqrt{15}$ cm^2 (d) 300 cm^2
3. The distance of a 24 cm long side of a parallelogram from the opposite side is 22 cm. The area of the parallelogram is :
(a) 264 cm^2 (b) 246 cm^2
(c) 460 cm^2 (d) 528 cm^2
4. The two adjacent sides of a parallelogram are 25 cm and 40 cm respectively. The altitude drawn on the longer side is 18 cm, then the area of the parallelogram is :
(a) 450 cm^2 (b) 720 cm^2
(c) 500 cm^2 (d) none of these
5. In the adjoining figure, the ratio of the areas of the parallelogram $ABCD$ and that of triangle ABN is :



(a) 6 : 1 (b) 5 : 1
(c) 4 : 1 (d) 8 : 1
6. If the perimeter of a rhombus is $4p$ and lengths of its diagonals are a and b, then its area is :
(a) $\frac{a}{b}$ (b) $\frac{ab}{2}$
(c) ab/p (d) $p(a^2 + b^2)$
7. The ratio of the lengths of the diagonal of a rhombus is 2 : 5. Then, the ratio of the area of the rhombus to the square of the shorter diagonal :
(a) 5 : 4 (b) 5 : 2
(c) 2 : 5 (d) none of these
8. Area of a rhombus is 256 cm^2. One of the diagonal is half of the other diagonal. The sum of the diagonals is :
(a) 38 cm (b) 48 cm
(c) 28 cm (d) 56 cm

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9. The lengths of two parallel sides of a trapezium are 30 cm and 50 cm and its height is 16 cm. Its area is :
(a) 460 cm^2 (b) 750 cm^2
(c) 320 cm^2 (d) 640 cm^2

10. $ABCD$ is a trapezium in which $AB \parallel CD$ and $AB = 2CD$. If its diagonals intersect each other at O, then ratio of areas of triangles AOB and COD is :
(a) 1 : 4 (b) 1 : 2
(c) 4 : 1 (d) 2 : 1

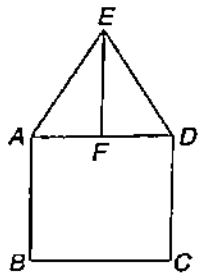
11. The area of a trapezium is 441 cm^2 and the ratio of parallel sides is 5 : 9. Also the perpendicular distance between them is 21 cm. The longer of parallel sides is :
(a) 36 cm (b) 27 cm
(c) 18 cm (d) 28 cm

12. The cross-section of a canal is in the shape of a trapezium and the area of cross-section is 360 m^2. If the canal is 12 m wide at the top and 8 m wide at the bottom the depth of the canal is :
(a) 36 m (b) 180 m
(c) 45 m (d) none of these

13. The area of a hexagon whose one side is 4 m, is :
(a) $6\sqrt{3}$ m^2 (b) $24\sqrt{3}$ m^2
(c) $42\sqrt{3}$ m^2 (d) 24 m^2

14. $ABCD$ is a quadrilateral $AC = 19$ cm. The lengths of perpendiculars from B and D on AC are 5 cm and 7 cm respectively. Then, the area of $ABCD$ (in cm^2) is :
(a) 162 (b) 144
(c) 228 (d) 114

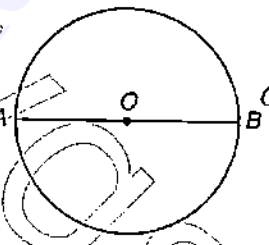
15. $ABCD$ is a square, $AC = BD = 4\sqrt{2}$ cm, $AE = DE = 2.5$ cm. Find the area of the adjoining figure $ABCDE$:
(a) 19 cm^2
(b) 22 cm^2
(c) 17 cm^2
(d) none of the above



CIRCLES

1. Area of a circle $= \pi R^2$ ($R \rightarrow$ radius of the circle)
2. Circumference of the circle $= 2\pi R$
3. Length of an arc $= 2\pi R\left(\frac{\theta}{360°}\right)$
4. Area of a sector $= \pi R^2\left(\frac{\theta}{360°}\right) = \frac{1}{2}(\text{arc} \times R)$
5. Area of segment $= \pi R^2\left(\frac{\theta}{360°}\right) - \frac{R^2}{2}\sin\theta$

Diameter $= 2 \times$ radius



$OB = OA \rightarrow$ radius
$AB \rightarrow$ diameter

EXAMPLE 1 *The radius of a circular wheel is $1\frac{3}{4}$ m. How many revolutions will it make in travelling 11 km?*

SOLUTION Total distance (travelled) = 11 km = 11000 m

Distance travelled in one revolution

= circumference of the wheel

$= 2 \times \pi \times r = 2 \times \frac{22}{7} \times \frac{7}{4} = 11$ m

$\therefore$ number of revolutions in 11 km $= \frac{11000}{11}$

= 1000 revolution

EXAMPLE 2 *It takes 13.5 mL to paint the surface of the circular sheet of radius 17 cm. How much paint is required to paint a similar circular sheet with double the radius?*

SOLUTION Ratio of radii = 1 : 2

$\therefore$ Ratio of areas = 1 : 4

$$\left(\text{Since } \frac{\text{Area of } C_1}{\text{Area of } C_2} = \frac{\pi r_1^2}{\pi r_2^2} = \frac{1 \times 1}{2 \times 2} = \frac{1}{4}\right)$$

$\therefore$ Quantity of required paint is 4 times

Thus we need $4 \times 13.5 = 54$ mL paint.

EXAMPLE 3 *A circular road runs round a circular garden. If the difference between the circumference of the outer circle and the inner circle is 44 m. Find the width of the road.*

SOLUTION Let R and r be radii of outer circle and inner circle respectively

$\therefore$ Width of the road $= R - r$

$\therefore$ $2\pi R - 2\pi r = 44$ m

$\Rightarrow$ $2\pi(R - r) = 44$ m

$\Rightarrow$ $(R - r) = 7$ m $\quad (\because \pi = \frac{22}{7})$

EXAMPLE 4 *The radius of a circle is 5 m. What is the radius of another circle whose area is 25 times that of the first?*

SOLUTION Ratio of areas = (ratio of radii)^2

$$\frac{25}{1} = (\text{ratio of radii})^2$$

$\Rightarrow$ ratio of radii $= \frac{5}{1}$

Therefore radius of another circle is 5 times

Hence the required radius = 25 m

EXAMPLE 5 *What is the radius of a circle whose area is equal to the sum of the areas of two circles whose radii are 20 cm and 21 cm.*

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SOLUTION

$$\pi R^2 = \pi r_1^2 + \pi r_2^2$$
$$\pi R^2 = \pi (r_1^2 + r_2^2)$$
$$R^2 = (400 + 441)$$
$$R^2 = 841$$
$$\Rightarrow \quad R = 29 \text{ cm}$$

EXAMPLE 6 *In a circle of radius 28 cm, an arc subtends an angle of 108° at the centre.*

(a) Find the area of the sector.

(b) Find the length of the arc.

SOLUTION (a) Area of the sector $= \pi r^2 \left(\frac{\theta}{360^\circ}\right)$

$$= \frac{22}{7} \times 28 \times 28 \times \frac{108^\circ}{360^\circ}$$
$$= 22 \times 4 \times 28 \times \frac{3}{10}$$
$$= 739.2 \text{ cm}$$

(b) Length of the arc $= 2\pi r \left(\frac{\theta}{360^\circ}\right)$

$$= 2 \times \frac{22}{7} \times 28 \times \frac{108}{360^\circ} = 52.8 \text{ cm}$$

INTRODUCTORY EXERCISE-10.4

1. If the circumference of a circle is 704 cm, then its area is :
(a) 49324 m^2 (b) 39424 m^2
(c) 3672 cm^2 (d) 39424 cm^2

2. If the circumference of a circle is 4.4 m, then the area of the circle (in m^2) is :
(a) $49/\pi$ (b) 49π
(c) 4.9π (d) none of these

3. A circular wire of radius 4.2 m is cut and bent in the form of a rectangle whose longer side is 20% more than its shorter side. The longer side of the rectangle is :
(a) 7.2 m (b) 72 cm
(c) 8 m (d) none of these

4. The inner circumference of a circular path around a circular lawn is 440 m. What is the radius of the outer circumference of the path, if the path is 14 m wide?
(a) 96 m (b) 84 m
(c) 70 m (d) 88 m

5. The sum of the radius and the circumference of a circle is 51 cm. The area of the circle is :
(a) 151 cm (b) 152 cm
(c) 154 cm (d) data insufficient

6. The difference between the circumference and the diameter of the circle is 15 m. What is the area of the circle?
(a) 225 m^2 (b) 165 m^2
(c) 156 m^2 (d) none of these

7. The radius of a circle is increased by 2 cm from 5 cm to 7 cm. What is the percentage change in area of the circle?
(a) 96% (b) 35%
(c) 70% (d) 74%

8. If the circumference of a circle is increased by 20%, then its area will be increased by :
(a) 44% (b) 32%
(c) 40% (d) none of these

9. The area of a circular field is 124.74 hectares. The cost of fencing it at the rate of 80 paise per metre is :
(a) Rs. 3168 (b) Rs. 1584
(c) Rs. 1729 (d) none of these

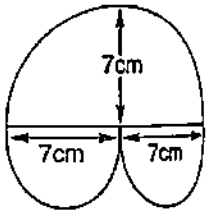
10. Eldeco Housing Pvt. Ltd purchased a circular plot of land for Rs. 158400 at the rate of 1400 per sq. metre. The radius of the plot is :
(a) 5 m (b) 6 m
(c) 7 m (d) 14 m

11. A figure consists of a square of side 'a' m with semicircles drawn on the outside of the square. The area (in m^2) of the figure so formed will be :
(a) $a^2(\pi + 1)$ (b) $a^2\left(\pi + \frac{1}{4}\right)$
(c) $a^2 + \frac{\pi a^2}{2}$ (d) none of these

12. The length of a rope by which a buffalo must be tethered so that she may be able to graze a grassy area of 2464 sq. m is :
(a) 35 m (b) 27 m
(c) 24 m (d) 28 m

13. A circle of radius 'a' is divided into 6 equal sectors. An equilateral triangle is drawn on the chord of each sector to lie outside the circle. Area of the resulting figure is:
(a) $3a^2(\pi + \sqrt{3})$ (b) $3\sqrt{3}\,a^2$
(c) $3(a^2\sqrt{3} + \pi)$ (d) $\frac{3\sqrt{3}\pi a^2}{2}$

14. In the following figure, the area in (cm^2) is :
(a) 115.5
(b) 228.5
(c) 154
(d) none of the above



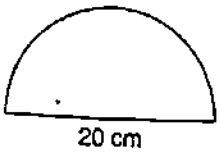
15. If a piece of wire 25 cm long is bent into an arc of a circle subtending an angle of 75° at the centre, then the radius of the circle (in cm) is :
(a) $\frac{\pi}{120}$ (b) $\frac{60}{\pi}$
(c) 60π (d) none of these

16. Four horses are tethered at four corners of a square plot of 42 m so that they just cannot reach one another. The area left ungrazed is :
(a) 378 m^2 (b) 438 m^2
(c) 786 m^2 (d) none of these

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17. The circumference of the following figure is :



(a) $(20 + 10\pi)$ (b) 20π
(c) 10π (d) 30π

18. The area of a minor sector subtending the central angle at the centre 40° is 8.25 cm^2. What is the area of the remaining part (*i.e.*, major sector) of the circle?
(a) 82.5 cm^2 (b) 74.25 cm^2
(c) 66 cm^2 (d) none of these

19. The area of a sector of a circle of radius 8 cm, formed by an arc of length 5.6 cm, is :
(a) 22.4 cm^2 (b) 2.24 cm^2
(c) 56 cm^2 (d) none of these

20. How long will a man take to go, walking at 13.2 km/h, round a circular garden of 700 m radius?
(a) 12 minute (b) 30 minute
(c) 20 minute (d) none of these

21. What is the radius of circular field whose area is equal to the sum of the areas of three smaller circular fields of radii 8 m, 9 m and 12 m respectively?
(a) 17 m (b) 20 m
(c) 21 m (d) 29 m

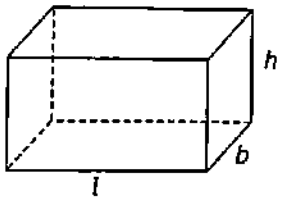
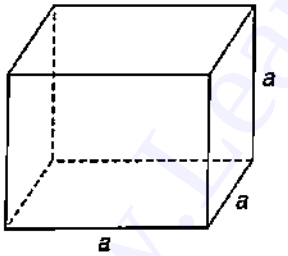
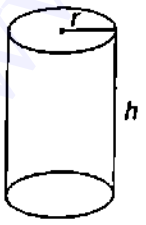
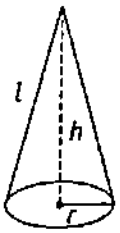
22. A rope by which a calf is tied is decreased from 23 m to 12 m. What is the decrease in area to be grazed by it?
(a) 1110 m^2 (b) 1210 m^2
(c) 1120 m^2 (d) 1221 m^2

23. A wire is bent in the form of a square of side 66 m. It is cut and again bent in the form of a circle. The diameter of this circle is :
(a) 42 m (b) 84 m
(c) 21 m (d) none of these

24. A wire is in the form of a circle of radius 42 m is cut and again bent in the form of a square. What is the diagonal of the square?
(a) 66 m (b) $66\sqrt{3}$ m
(c) $66\sqrt{2}$ m (d) none of these

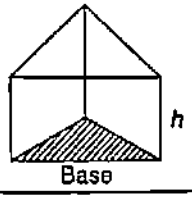
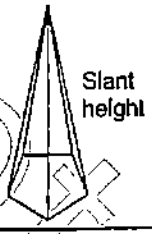
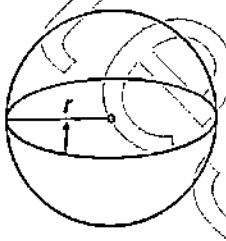
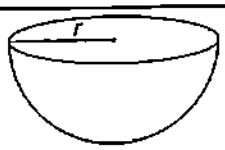
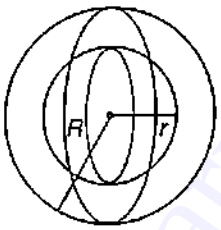
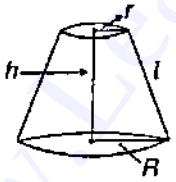
25. If the driving wheel of a bicycle makes 560 revolutions in travelling 1.1 km. Find the diameter of the wheel :
(a) 31·5 cm (b) 30.5 cm
(c) 62.5 cm (d) none of these

3-D FIGURES (SOLIDS)

S. No.	Name	Figure	Nomenclature	Volume	Curved/Lateral surface area	Total surface area
1.	Cuboid		$l \rightarrow$ length $b \rightarrow$ breadth $h \rightarrow$ height	lbh	$2(l+b)h$	$2(lb+bh+hl)$
2.	Cube		$a \rightarrow$ edge/side	a^3	$4a^2$	$6a^2$
3.	Right circular cylinder		$r \rightarrow$ radius of base $h \rightarrow$ height of the cylinder	$\pi r^2 h$	$2\pi rh$	$2\pi r(h+r)$
4.	Right circular cone		$r \rightarrow$ radius $h \rightarrow$ height $l \rightarrow$ slant height $l = \sqrt{r^2 + h^2}$	$\frac{1}{3}\pi r^2 h$	πrl	$\pi r(l+r)$

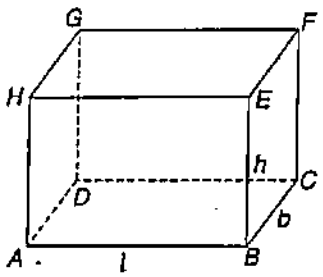
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S. No.	Name	Figure	Nomenclature	Volume	Curved/Lateral surface area	Total surface area
5.	Right triangular prism		—	area of base × height	perimeter of base × height	lateral surface area + 2 (area of base)
6.	Right pyramid		—	$\frac{1}{3}$ × area of the base × height	$\frac{1}{2}$ × perimeter of the base × slant height	lateral surface area + area of the base
7.	Sphere		$r \rightarrow$ radius	$\frac{4}{3}\pi r^3$	—	$4\pi r^2$
8.	Hemisphere		$r \rightarrow$ radius	$\frac{2}{3}\pi r^3$	$2\pi r^2$	$3\pi r^2$
9.	Spherical shell		$r \rightarrow$ inner radius $R \rightarrow$ outer radius	$\frac{4}{3}\pi[R^3 - r^3]$	—	$4\pi[R^2 + r^2]$
10.	Frustum of a cone		—	$\frac{\pi}{3}h(r^2 + Rr + R^2)$	$\pi(r + R)\,l$	lateral surface area + $\pi[R^2 + r^2]$

CUBOID AND CUBE

Cuboid : A cuboid has 6 faces, 12 edges, 8 vertices and 4 diagonals.



Faces : *ABCD, EFGH, ABEH, CFGD, BCFE, ADGH*

Edges : *AB, BC, CD, AD, EF, FG, GH, EH, BE, CF, DG, AH*

Vertices : *A, B, C, D, E, F, G, H*

Diagonals : *AF, BG, CH, DE*

Formulae : Volume $= l \times b \times h$

($l \rightarrow$ length, $b \rightarrow$ breadth, $h \rightarrow$ height)

Total surface area $= 2(lb + bh + hl)$

Diagonal $(d) = \sqrt{l^2 + b^2 + h^2}$

Cube : A cube has 6 equal faces, 12 equal edges, 8 vertices and 4 equal diagonals

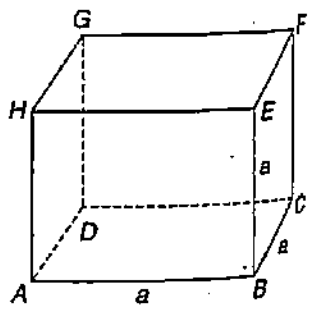
Formulae :

Volume : $(a)^3$

Total surface area $= 6(a)^2$

($a \rightarrow$ edge of the cube)

Diagonal $(d) = a\sqrt{3}$



Euler's Theorem $\rightarrow (V + F) = (E + 2)$; where $V \rightarrow$ no. of vertices, $F \rightarrow$ no. of faces, $E \rightarrow$ no. of edges

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EXAMPLE 1 *The dimensions of a cuboid are 16 cm, 18 cm and 24 cm. Find :*
(a) volume (b) surface area (c) diagonal

SOLUTION (a) Volume $= l \times b \times h = 16 \times 18 \times 24 = 6912 \text{ cm}^3$

(b) Surface area $= 2(lb + bh + hl)$
$= 2(16 \times 18 + 18 \times 24 + 24 \times 16)$
$= 2208 \text{ cm}^2$

(c) Diagonal $= \sqrt{l^2 + b^2 + h^2}$
$= \sqrt{16^2 + 18^2 + 24^2} = \sqrt{1156} = 34 \text{ cm}$

EXAMPLE 2 *Edge of a cube is 5 cm. Find :*
(a) volume (b) surface area (c) diagonal

SOLUTION Volume $= a^3 = (5)^3 = 125 \text{ cm}^3$
Surface area $= 6a^2 = 6 \times (5)^2 = 150 \text{ cm}^3$
Diagonal $= a\sqrt{3} = 5\sqrt{3} = 8.660 = 8.66 \text{ cm}$

EXAMPLE 3 *Three cubes of volumes, 1 cm³, 216 cm³ and 512 cm³ are melted to form a new cube. What is the diagonal of the new cube?*

SOLUTION Volume of new cube $= 1 + 216 + 512 = 729 \text{ cm}^3$
$\therefore$ Edge of new cube $= \sqrt[3]{729} = 9 \text{ cm}$
$\therefore$ Surface area $= 6a^2 = 6 \times (9)^2 = 486 \text{ cm}^2$
$\therefore$ Diagonal of the new cube $= a\sqrt{3} = 9\sqrt{3}$
$= 15.6 \text{ cm (approx)}$

EXAMPLE 4 *The surface area of a cube is 864 cm². Find its volume.*

SOLUTION $6a^2 = 864 \Rightarrow a^2 = 144 \Rightarrow a = 12 \text{ cm}$
$\therefore a^3 = (12)^3 = 1728 \text{ cm}^3$

EXAMPLE 5 *Find the length of the longest pole that can be placed in a room 30 m long, 24 m broad and 18 m high.*

SOLUTION $d = \sqrt{l^2 + b^2 + h^2}$
$d = \sqrt{900 + 576 + 324}$
$d = 30\sqrt{2} \text{ m}$

EXAMPLE 6 *A brick measures 20 cm × 10 cm × 7.5 cm. How many bricks will be required for a wall 20 m × 2 m × 0.75 m?*

SOLUTION Number of bricks $= \dfrac{\text{total volume of a wall}}{\text{volume of one brick}}$
$= \dfrac{20 \times 2 \times 0.75 \times 100 \times 100 \times 100}{20 \times 10 \times 7.5}$
$= 20{,}000$

INTRODUCTORY EXERCISE-10.5

1. A cube of metal, each edge of which measures 4 cm, weighs 400 kgs. What is the length of each edge of a cube of the same metal which weighs 3200 kg?
(a) 64 cm (b) 8 cm
(c) 2 cm (d) none of these

2. The length of a tank is thrice that of breadth, which is 256 cm deep and holds 3000 L water. What is the base area of the tank? (1000 L = 1 cubic metre)
(a) 111775 m² (b) 1171.875 m²
(c) 1.171875 m² (d) none of these

3. The three co-terminus edges of a rectangular solid are 36 cm, 75 cm and 80 cm respectively. Find the edge of a cube which will be of the same capacity :
(a) 60 cm (b) 52 cm
(c) 46 cm (d) none of these

4. A tank 10 m long and 4 m wide is filled with water. How many litres of water must be drawn off to make the surface sink by 1 m. (1000 L = 1 cubic metre)
(a) 20 kilolitre (b) 40 kilolitre
(c) 50 kilolitre (d) none of these

5. How many cubes each of surface area 24 sq. dm can be made out of a metre cube, without any wastage :
(a) 75 (b) 250
(c) 125 (d) 62

6. Three cubes of metal, whose edges are 3 cm, 4 cm and 5 cm respectively are melted to form a new cube. What is the surface area of the new cube?
(a) 216 cm² (b) 56 cm²
(c) 36 cm² (d) none of these

7. A lid of rectangular box of sides 39.5 cm by 9.35 cm is sealed all around with tape such that there is an overlapping of 3.75 cm of the tape. What is the length of the tape used?
(a) 111.54 cm (b) 101.45 cm
(c) 110.45 cm (d) none of these

8. A cistern from inside is 12.5 m long, 8.5 m broad and 4 m high and is open at top. Find the cost of cementing the inside of a cistern at Rs. 24 per sq. m :
(a) Rs. 6582 (b) Rs. 8256
(c) Rs. 7752 (d) Rs. 8752

9. 250 men took a dip in a water tank at a time, which is 80 m × 50 m. What is the rise in the water level if the average displacement of 1 man is 4 m³?
(a) 22 cm (b) 25 cm
(c) 18 cm (d) 30 cm

10. The edge of a cube is increased by 100%, the surface area of the cube is increased by :
(a) 100% (b) 200%
(c) 300% (d) 400%

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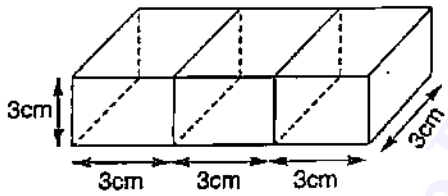
11. The edge of a cube is doubled. What will be the new volume?
(a) 2 times (b) 3 times
(c) 4 times (d) 8 times

12. The external dimensions of a wooden box closed at both ends are 24 cm, 16 cm and 10 cm respectively and thickness of the wood is 5 mm. If the empty box weighs 7.35 kg, find the weight of 1 cubic cm of wood :
(a) 10 g (b) 12.5 g
(c) 27 g (d) 15 g

13. The internal dimensions of a tank are 12 dm, 8 dm and 5 dm. How many cubes each of edge 7 cm can be placed in the tank with faces parallel to the sides of the tank. Find also, how much space is left unoccupied?
(a) 35; 113 dm^3 (b) 1313; 31.13 dm^3
(c) 1309; 31.013 dm^3 (d) 1309; 13.31 dm^3

14. The length, breadth and height of box are 2 m, 1.5 m and 80 cm respectively. What would be the cost of canvas to cover it up fully, if one square metre of canvas costs Rs. 25.00?
(a) Rs. 260 (b) Rs. 290
(c) Rs. 285 (d) none of these

15. Three cubes each of edge 3 cm long are placed together as shown in the adjoining figure. Find the surface area of the cuboid so formed :



(a) 182 sq. cm (b) 162 sq. cm
(c) 126 sq. cm (d) none of these

16. A room is 36 m long, 12 m wide and 10 m high. It has 6 windows, each 3 m × 2.5 m; one door 9.5 m × 6 m and one fire chimney 4 m × 4.5 m. Find the expenditure of papering its walls at the rate of 70 paise per metre, if the width of the paper is 1.2 m :
(a) Rs. 490 (b) Rs. 690
(c) Rs. 1000 (d) none of these

17. A school hall has the dimensions 30 m, 12 m by 6 m. Find the number of children who can be accommodated, if each child should get 8 m^3 of space :
(a) 240 (b) 270
(c) 250 (d) 150

18. When each side of a cube is increased by 2 cm, the volume is increased by 1016 cm^3. Find the side of the cube. If each side of it is decreased by 2 cm, by how much will the volume decrease?
(a) 12 cm, 729 cm^3 (b) 8 cm, 512 cm^3
(c) 9 cm, 729 cm^3 (d) 12 cm, 728 cm^3

19. Three equal cubes are placed adjacently in a row. Find the ratio of the total surface area of the resulting cuboid to that of the sum of the total surface areas of the three cubes :
(a) 5 : 7 (b) 7 : 9
(c) 9 : 7 (d) none of these

20. A hollow square shaped tube open at both ends is made of iron. The internal square is of 5 cm side and the length of the tube is 8 cm. There are 192 cm^3 of iron in the tube. Find its thickness :
(a) 2 cm (b) 0.5 cm
(c) 1 cm (d) can't be determined

21. A cube of 11 cm edge is immersed completely in a rectangular vessel containing water. If the dimensions of base are 15 cm and 12 cm. Find the rise in water level in the vessel :
(a) 6.85 cm (b) 7 cm
(c) 7.31 cm (d) 7.39 cm

22. A rectangular tank 25 cm long and 20 cm wide contains water to a depth of 5 cm. A metal cube of side 10 cm is placed in the tank so that one face of the cube rests on the bottom of the tank. Find how many litres of water must be poured into the tank so as to just cover the cube?
(a) 1 L (b) 1.5 L
(c) 2 L (d) 2.5 L

23. A rectangular block has length 10 cm, breadth 8 cm and height 2 cm. From this block, a cubical hole of side 2 cm is drilled out. Find the volume and the surface area of the remaining solid :
(a) 152 cm^3, 512 cm^2 (b) 125 cm^3, 215 cm^2
(c) 152 cm^3, 240 cm^2 (d) 125 cm^3, 512 cm^2

24. A rectangular tank of dimensions 24 m × 12 m × 8 m is dug inside a rectangular field 600 m long and 200 m broad. The earth taken out is evenly spread over the field. By how much will the level of the field rise?
(a) 1.925 cm (b) 0.02 m
(c) 0.2 cm (d) none of these

25. How many bricks (number near to next hundred) will be required to build a wall 30 m long, 30 cm thick and 5 m high with a provision of 2 doors, each 4 m × 2.5 m and each brick being 20 cm × 16 cm × 8 cm when one-ninth of the wall is filled with lime?
(a) 13500 bricks (b) 13600 bricks
(c) 20050 bricks (d) 18500 bricks

26. A rectangular water reservoir is 15 m by 12 m at the base. Water flows into it through a pipe whose cross-section is 5 cm by 3 cm at the rate of 16 m per second. Find the height to which the water will rise in the reservoir in 25 minutes :
(a) 0.2 m (b) 2 cm
(c) 0.5 m (d) none of these

27. Which of the following pairs is not correctly matched :

Geometrical objects	Number of vertices
(A) Tetrahedron	4
(B) Pyramid with rectangular base	5
(C) Cube	6
(D) Triangle	3

(a) (A) (b) (B)
(c) (C) (d) (D)

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28. If the length of diagonal of a cube is $6\sqrt{3}$ cm, then the length of its edge is :
(a) 2 cm (b) 3 cm
(c) 6 cm (d) $\frac{36}{\sqrt{3}}$ cm

29. The length of longest pole that can be placed on the floor of a room is 12 m and the length of longest pole that can be placed in the room is 15 m. The height of the room is :
(a) 3 m (b) 6 m
(c) 9 m (d) none of these

30. The sum of length, breadth and depth of a cuboid is 12 cm and its diagonal is $5\sqrt{2}$ cm. Its surface area is :
(a) 152 cm^2 (b) 94 cm^2
(c) 108 cm^2 (d) $60\sqrt{2}$ cm^2

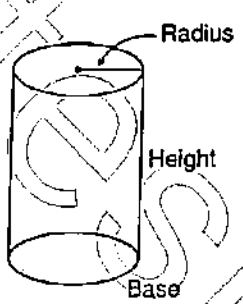
31. The volume of a wall, 3 times as high as it is broad and 8 times as long as it is high, is 36.864 m^3. The height of the wall is :
(a) 1.8 m (b) 2.4 m
(c) 4.2 m (d) none of these

32. If the areas of 3 adjacent sides of a cuboid are x, y, z respectively, then the volume of the cuboid is :
(a) xyz (b) $\sqrt{xyz}$
(c) $3xyz$ (d) none of these

CYLINDER AND CONE

Cylinder :

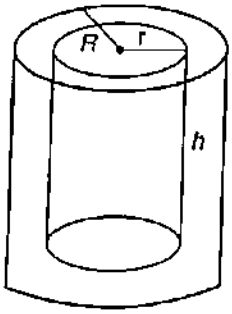
Volume = base area × height
$= \pi r^2 h$



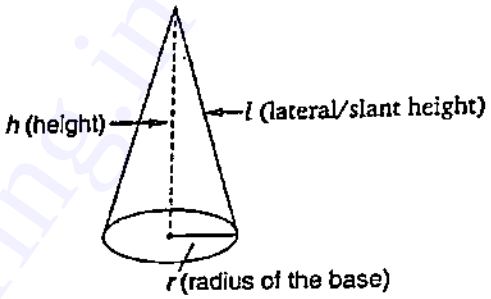
Curved surface area
= perimeter × height
$= 2\pi rh$

Total surface of the cylinder
= curved surface area + 2 (base area)
$= 2\pi rh + 2\pi r^2 = 2\pi r(h + r)$

Volume of a hollow cylinder = $\pi h (R^2 - r^2)$



Cone :



Volume (V) = $\frac{1}{3}$ × base area × height = $\frac{1}{3}\pi r^2 h$

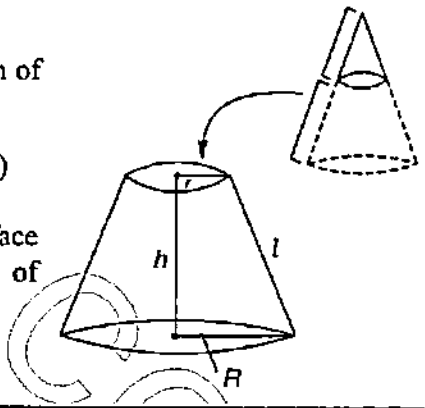
Curved surface area = πrl

Total surface area = $\pi rl + \pi r^2 = \pi r (l + r)$

Frustum of a cone :

Volume of the frustum of a cone
$= \frac{\pi}{3} h (r^2 + Rr + R^2)$

Curved/Lateral surface area of the frustum of cone
$= \pi l (r + R)$



EXAMPLE 1 *The base radius of a cylinder is 14 cm and its height is 30 cm. Find :*
(a) volume *(b) curved surface area*
(c) total surface area

SOLUTION (a) Volume of cylinder = $\pi r^2 \times h = \frac{22}{7} \times 14 \times 14 \times 30$
$= 18480 \text{ cm}^3$

(b) Curved surface area = $2\pi rh = 2 \times \frac{22}{7} \times 14 \times 30$
$= 2640 \text{ cm}^2$

(c) Total surface area = $2\pi r (h + r)$
$= 2 \times \frac{22}{7} \times 14 (30 + 14)$
$= 2 \times \frac{22}{7} \times 14 \times 44 = 3872 \text{ cm}^2$

EXAMPLE 2 *How many cubic metres of earth must be dug to make a well 14 m deep and 4 m in diameter?*

SOLUTION Earth to be dugout from the well
= volume of the cylindrical well
$= \pi r^2 h = \frac{22}{7} \times 2 \times 2 \times 14$
$= 176 \text{ m}^3$

EXAMPLE 3 *A hollow cylindrical tube open at both ends is made of iron 2 cm thick. If the external diameter be 50 cm and the length of the tube is 210 cm, find the number of cubic cm of iron in it.*

SOLUTION External radius (R) = 25 cm
Internal radius (r) = (25 – 2) = 23 cm
Volume of iron = $\pi h (R^2 - r^2)$

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$$= \frac{22}{7} \times 210 \times (25^2 - 23^2)$$

$$= 63360 \text{ cm}^3$$

EXAMPLE 4 *A well with 14 m inside diameter is dugout 15 m deep. The earth taken out of it has been evenly spread all around it to a width of 21 m to form an embankment. What is the height of the embankment?*

SOLUTION Area of embankment × height of embankment = volume of earth dugout

$$\pi (R^2 - r^2) \times h = \pi \times 7 \times 7 \times 15$$

$$\Rightarrow \quad (28^2 - 7^2) h = 7 \times 7 \times 15$$

$$\Rightarrow \quad (35 \times 21) \times h = 7 \times 7 \times 15$$

$$\Rightarrow \quad h = 1 \text{ m}$$

EXAMPLE 5 *A cylindrical cistern whose diameter is 21 cm is partly filled with water. If a rectangular block of iron 14 cm in length, 10.5 cm in breadth and 11 cm in thickness is wholly immersed in water, by how many centimetres will the water level rise?*

SOLUTION Volume of the block $= 14 \times 10.5 \times 11 \text{ cm}^3$

Radius of the tank $= \frac{21}{2} = 10.5$ cm

Volume of the cylinder $= \pi r^2 h = \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times h$

$$\therefore \quad \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times h = 14 \times 10.5 \times 11$$

$$h = \frac{14}{3} = 4\frac{2}{3} \text{ cm}$$

EXAMPLE 6 *If the radius of cylinder is doubled, but height is reduced by 50%. What is the percentage change in volume?*

SOLUTION $\frac{r_1}{r_2} = \frac{r}{2r}$ and $\frac{h_1}{h_2} = \frac{h}{h/2}$

$\therefore$ Actual volume $= \pi r^2 h$

New volume $= \pi(2r)^2 \times \frac{h}{2} = 2\pi r^2 h$

Therefore new volume is the twice of the original volume.

Hence the change in volume $= \frac{2-1}{1} \times 100 = 100\%$

EXAMPLE 7 *The radius of the base of a right cone is 35 cm and its height is 84 cm. Find :*

(a) slant height *(b) curved surface area*
(c) total surface area *(d) volume*

SOLUTION (a) Slant height $(l) = \sqrt{r^2 + h^2}$

($r \rightarrow$ radius of the circular base)

$= \sqrt{35^2 + 84^2}$ ($h \rightarrow$ height of the cone)

$= \sqrt{1225 + 7056}$

$= \sqrt{8281} = 91$ cm

(b) Curved surface area $= \pi r l = \frac{22}{7} \times 35 \times 91$

$= 10010 \text{ cm}^2$

(c) Total surface area = lateral surface area + base area

$= \pi r l + \pi r^2 = \pi r (l + r)$

$= \frac{22}{7} \times 35 (91 + 84)$

$= 110 \times 175 = 19250 \text{ cm}^2$

(d) Volume $= \frac{1}{3} \pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 35 \times 35 \times 84$

$= 107800 \text{ cm}^3$

EXAMPLE 8 *Find the area of the iron sheet required to prepare a cone 20 cm high with base radius 21 cm.*

SOLUTION $r = 21$ cm, $h = 20$ cm

$\therefore \quad l = \sqrt{r^2 + h^2} = 29$ cm

$\therefore$ Area of the sheet = total surface area of the cone

$= \pi r l + \pi r^2$

$= \pi r (l + r) = \frac{22}{7} \times 21 [29 + 21]$

$= 3300 \text{ cm}^2$

EXAMPLE 9 *A solid metallic cylinder of base radius 3 cm and height 5 cm is melted to make n solid cones of height 1 cm and base radius 1 mm. Find the value of n.*

SOLUTION $n = \frac{\text{volume of cylinder}}{\text{volume of one cone}}$

$= \frac{\pi \times 3 \times 3 \times 5}{\frac{1}{3} \pi \times \frac{1}{10} \times \frac{1}{10} \times 1} = 13500$

INTRODUCTORY EXERCISE-10.6

1. How many cubic metres of water fill a pipe which is 3500 m long and 0.08 m in diameter?
   (a) 17.5 m^3 (b) 17.6 m^3
   (c) 21 m^3 (d) 35 m^3
2. A cube of metal, whose edge is 10 cm, is wholly immersed in water contained in cylindrical tube whose diameter is 20 cm. By how much will the water level rise in the tube?
   (a) 3.3 cm (b) $6\frac{3}{11}$ cm
   (c) $3\frac{2}{11}$ cm (d) none of these

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3. Find the height of the cylinder whose volume is 511 m^3 and the area of the base is 36.5 m^2 :
(a) 7 m (b) 10.5 m
(c) 14 m (d) none of these

4. The lateral surface area of a cylinder is 1056 cm^2 and its height is 16 cm. What is its volume?
(a) 5566 cm^3 (b) 4455 cm^3
(c) 5544 cm^3 (d) none of these

5. There is a cubical block of wood of side 2 cm. If the cylinder of the largest possible volume is carved out from it. Find the volume of the remaining wood :
(a) $\frac{7}{12}$ cm^2 (b) $\frac{12}{7}$ cm^3
(c) $5\frac{5}{7}$ cm^3 (d) none of these

6. The amount of concrete required to build a cylindrical pillar whose base has a perimeter of 8.8 m and whose curved surface area is 17.6 m^2 :
(a) 12.32 m^3 (b) 12.23 m^3
(c) 9.235 m^3 (d) 8.88 m^3

7. If the diameter of the base of a closed right circular cylinder be equal to its height 'h', then its whole surface area is :
(a) $\frac{2}{3}\pi h^2$ (b) $\frac{3}{2}\pi h^3$
(c) $\frac{3}{2}\pi h^2$ (d) πh^3

8. A right circular cylindrical tunnel of diameter 4 m and length 10 m is to be constructed from a sheet of iron. The area of the iron sheet required :
(a) $\frac{280}{\pi}$ (b) 40 π
(c) 80 π (d) none of these

9. The ratio between curved surface area and total surface area is 2 : 3. If the total surface area be 924 cm^2, find the volume of the cylinder :
(a) 2156 cm^3 (b) 1256 cm^3
(c) 1265 cm^3 (d) none of these

10. If the volume and curved surface area of a cylinder are 269.5 cm^3 and 154 cm^2 respectively, what is the height of the cylinder?
(a) 6 (b) 3.5
(c) 7 (d) can't be determined

11. If the curved surface area of a cylinder is 1320 cm^2 and its base radius is 21 cm, then its total surface area is :
(a) 4092 cm^2 (b) 2409 cm^2
(c) 4920 cm^2 (d) none of these

12. The ratio between the radius of the base and the height of a cylindrical pillar is 3 : 4. If its volume is 4851 m^3, the curved surface area of the pillar is :
(a) 924 m^2 (b) 1617 m^2
(c) 425 m^2 (d) none of these

13. If the ratio of total surface area to the curved surface area of a cylinder be 4 : 1, what is the ratio of radius to the height?
(a) 4 : 1 (b) 2 : 3
(c) 3 : 2 (d) 3 : 1

14. The circumference of the base of a right cylinder is 33 cm and height is 330 cm. What is the volume of this cylinder?
(a) 28586.25 cm^3 (b) 3344 cm^3
(c) 4433 cm^3 (d) 3456 cm^3

15. The radius of an iron rod decreased to one-fourth. If its volume remains constant, the length will become :
(a) 2 times (b) 8 times
(c) 4 times (d) 16 times

16. The total surface area of the cylinder is 2640 m^2 and the sum of height and radius of base of cylinder is 30 m. What is the ratio of height and radius of the cylinder?
(a) 7 : 9 (b) 9 : 7
(c) 8 : 7 (d) 3 : 7

17. The radii of two cylinders are in the ratio of 3 : 5 and their heights are in the ratio 4 : 3. The ratio of their volumes is :
(a) 12 : 25 (b) 13 : 25
(c) 4 : 5 (d) 5 : 4

18. The heights of two cylinders are in the ratio of 3 : 1. If the volumes of two cylinders be same, the ratio of their respective radii are :
(a) $\sqrt{3}$: 1 (b) 1 : $\sqrt{3}$
(c) 1 : 9 (d) none of these

19. The ratio of heights of two cylinders is 3 : 2 and the ratio of their radii is 6 : 7. What is the ratio of their curved surface areas?
(a) 9 : 7 (b) 1 : 1
(c) 7 : 9 (d) 7 : 4

20. A hollow garden roller 42 cm wide with a girth of 132 cm is made of iron 3 cm thick. The volume of the iron of the roller is :
(a) 15544 cm^3 (b) 15444 cm^3
(c) 15545 cm^3 (d) none of these

21. A conical vessel has a capacity of 15 L of milk. Its height is 50 cm and base radius is 25 cm. How much milk can be contained in a vessel in cylindrical form having the same dimensions as that of the cone?
(a) 15 L (b) 30 L
(c) 45 L (d) none of these

22. The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base. If its volume be $\frac{1}{27}$ of the volume of the given cone, at what height above the base is the section made?
(a) 20 cm (b) 18 cm
(c) 27 cm (d) 15 cm

23. A tent is in the form of right circular cone 10.5 m high, the diameter of the base being 13 m. If 8 men are in the tent, find the average number of cubic metres of air space per man :
(a) $32\frac{3}{58}$ (b) 59.75
(c) $36\frac{9}{13}$ (d) $58\frac{3}{32}$

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24. The radius and height of a right circular cone are in the ratio of 5 : 12. If its volume is $314\frac{2}{7}$ m^3, its Slant height is :
(a) 26 m (b) 19.5 m
(c) 13 m (d) none of them

25. The volume and height of a right circular cone are 1232 cm^3 and 24 cm respectively, the area of its curved surface (in cm^2) is :
(a) 1100 (b) 225
(c) 616 (d) 550

26. The circumference of the base of a right circular cone is 220 cm^3 and its height 84 cm. The curved surface area of the cone is :
(a) 20020 cm^2 (b) 2020 cm^2
(c) 2200 cm^2 (d) 10010 cm^2

27. How many metres of cloth 10 m wide will be required to make a conical tent with base radius of 14 m and height is 48 m?
(a) 110 m (b) 55 m
(c) 77 m (d) 220 m

28. A cone of height 2.8 cm has a lateral surface area 23.10 m^2. The radius of the base is :
(a) 3.5 cm (b) 2 cm
(c) 2.1 cm (d) 4 cm

29. The radii of two cones are equal and their slant heights are in the ratio 3 : 2. If the curved surface area of the smaller cone is 300 cm^2, then the curved surface area of the bigger cone (in cm^2) is :
(a) 250 (b) 450
(c) 150 (d) 200

30. The ratio of the volume of a right circular cylinder and a right circular cone of the same base and height will be :
(a) 2 : 3 (b) 1 : 3
(c) 3 : 1 (d) 9 : 1

31. If the diameter of the base of right circular cone is equal to 8 cm and its slant height is 5 cm, then the area of its axial section is :
(a) 9 cm^2 (b) 12 cm^2
(c) 20 cm^2 (d) 40 cm^2

32. If the base radius and the height of a right circular cone are increased by 40% then the percentage increase in volume (approx) is :
(a) 175% (b) 120%
(c) 64% (d) 540%

33. From a circular sheet of paper of radius 25 cm, a sector area 4% is removed. If the remaining part is used to make a conical surface, then the ratio of the radius and height of the cone is :
(a) 16 : 25 (b) 9 : 25
(c) 7 : 12 (d) 24 : 7

34. If the radius of the base is doubled, keeping the height constant, what is the ratio of the volume of the larger cone to the smaller cone?
(a) 2 : 1 (b) 3 : 1
(c) 4 : 1 (d) 4 : 3

35. A largest posible cone is cut out from a cube of volume 1000 cm^3. The volume of the cone is :
(a) 280 cm^3 (b) 261.9 cm^3
(c) 269.1 cm^3 (d) 296.1 cm^3

36. If the height and the radius of a cone are doubled, the volume of the cone becomes :
(a) 2 times (b) 8 times
(c) 16 times (d) 4 times

37. A conical tent has 60° angle at the vertex. The ratio of its radius and slant height is :
(a) 3 : 2 (b) 1 : 2
(c) 1 : 3 (d) can't be determined

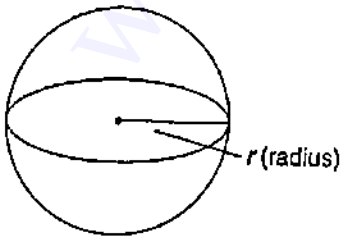
38. Water flows at the rate of 5 m per min from a cylindrical pipe 8 mm in diameter. How long will it take to fill up a conical vessel whose radius is 12 cm and depth 35 cm?
(a) 315 s (b) 365 s
(c) 5 min (d) none of these

39. A reservoir is in the shape of a frustum of a right circular cone. It is 8 m across at the top and 4 m across at the bottom. It is 6 m deep its capacity is :
(a) 224 m^3 (b) 176 m^3
(c) 225 m^3 (d) none of these

40. A conical vessel whose internal radius is 10 cm and height 72 cm is full of water. If this water is poured into a cylindrical vessel with internal radius 30 cm, the height of the water level rises in it is :
(a) $2\frac{2}{3}$ cm (b) $3\frac{2}{3}$ cm
(c) $5\frac{2}{3}$ cm (d) none of these

SPHERE, PRISM AND PYRAMID

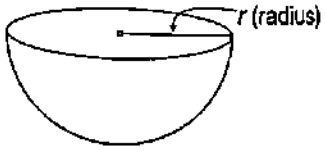
Sphere :



$$\text{Volume} = \frac{4}{3}\pi r^3$$

$$\text{Surface area} = 4\pi r^2$$

Hemisphere :



$$\text{Volume} = \frac{2}{3}\pi r^3$$

$$\text{Curved surface area} = 2\pi r^2$$

$$\text{Total surface area} = 3\pi r^2 \quad (2\pi r^2 + \pi r^2)$$

Spherical shell :

$$\text{Volume} = \frac{4}{3}\pi(R^3 - r^3)$$

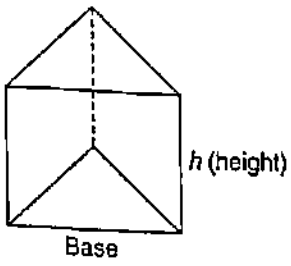
$$\text{Total surface area} = 4\pi(R^2 + r^2)$$

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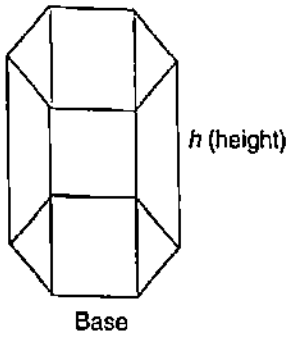
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Prism :

Volume = Base area × height

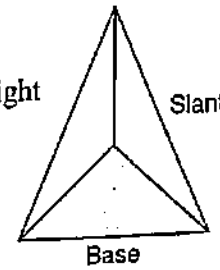


Lateral surface area
= perimeter of the base × height



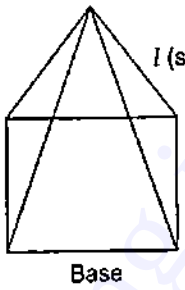
Pyramid :

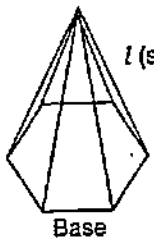
Volume $= \frac{1}{3} \times$ base area × height



Lateral surface area
$= \frac{1}{2} \times$ perimeter of the base × slant height

Total surface area = lateral surface area + base area





EXAMPLE 1 *Find the volume and surface area of a sphere of radius 3.5 cm.*

SOLUTION Volume $= \frac{4}{3}\pi r^3$

$= \frac{4}{3} \times \frac{22}{7} \times 3.5 \times 3.5 \times 3.5 = 179.66 \text{ cm}^3$

Surface area $= 4\pi r^2$

$= 4 \times \frac{22}{7} \times 3.5 \times 3.5 = 154 \text{ cm}^2$

EXAMPLE 2 *Find the volume, curved surface area and total surface area of a hemisphere of diameter 7 cm.*

SOLUTION Volume $= \frac{2}{3}\pi r^3$

$= \frac{2}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2} = 89.833 \text{ cm}^3$

Curved surface area $= 2\pi r^2$

$= 2 \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} = 77 \text{ cm}^2$

Total surface area $= 3\pi r^2$

$= 3 \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} = 115.5 \text{ cm}^2$

EXAMPLE 3 *How many bullets can be made from a sphere of 8 cm radius. The radius of each bullet must be 0.2 cm.*

SOLUTION Number of bullets $= \frac{\text{volume of sphere}}{\text{volume of 1 bullet}}$

$= \frac{\frac{4}{3}\pi \times 8 \times 8 \times 8}{\frac{4}{3}\pi \times 0.2 \times 0.2 \times 0.2} = 64000$

EXAMPLE 4 *A sphere has the same curved surface as a cone of height 12 cm and base radius 5 cm. Find the radius to the nearest cm.*

SOLUTION

$4\pi r^2 = \pi \times 5 \times 13$

$\Rightarrow \quad r^2 = \frac{65}{4}$

$\Rightarrow \quad r = 4 \text{ cm}$ (approx)

INTRODUCTORY EXERCISE-10.7

1. A spherical ball of lead 6 cm in radius is melted and recast into three spherical balls. The radii of two of these balls are 3 cm and 4 cm. What is the radius of the third sphere?
   (a) 4.5 cm (b) 5 cm
   (c) 6 cm (d) 7 cm
2. The radius of a copper sphere is 12 cm. The sphere is melted and drawn into a long wire of uniform circular cross-section. If the length of the wire is 144 cm, the radius of wire is :
   (a) 1 cm (b) 2 cm
   (c) 4 cm (d) none of these
3. A hemispherical bowl of internal radius 6 cm contains alcohol. This alcohol is to be filled into cylindrical shaped small bottles of diameter 6 cm and height 1 cm. How many bottles will be needed to empty the bowl?
   (a) 36 (b) 27
   (c) 16 (d) 4

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4. A hemisphere of lead of diameter 14 cm is cast into a right circular cone of height 14 cm. The radius of the base of the cone is :
(a) 7 cm (b) 14 cm
(c) 21 cm (d) none of these

5. The volume of a spherical shell whose external and internal diameters are 14 cm and 10 cm respectively :
(a) 42π cm^3 (b) $\frac{872}{3}\pi$ cm^3
(c) 118π cm^3 (d) 86π cm^3

6. A solid metal ball of diameter 16 cm is melted and cast into smaller balls, each of radius 1 cm. The number of such balls is :
(a) 256 (b) 2048
(c) 512 (d) 4096

7. If a hemispherical dome has an inner radius 21 cm then its volume (in m^3) is :
(a) 4910 m^3 (b) 18354 m^3
(c) 19404 m^3 (d) none of these

8. A sphere of radius 9 cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 12 cm. If the sphere is submerged completely, then the surface of the water rises by :
(a) 27.5 cm (b) 27 cm
(c) 12 cm (d) 6.75 cm

9. If the height of a cone is half the radius of a sphere then the radius of the base of the cone, which has the same volume as a sphere of radius 7 cm is :
(a) 14 m (b) $\frac{14}{\sqrt{2}}$ cm
(c) $14\sqrt{2}$ cm (d) none of these

10. From a solid sphere of radius 15 cm, a right circular cylindrical hole of radius 9 cm whose axis passing through the centre is removed. The total surface area of the remaining solid is :
(a) 1188π cm^2 (b) 108π cm^2
(c) 1170π cm^2 (d) 144π cm^2

11. The volume of a pyramid of base area 25 cm^2 and height 12 cm is :
(a) 200 cm^3 (b) 100 cm^3
(c) 400 cm^3 (d) 800 cm^3

12. If the base of right rectangular prism remains constant and the measures of the lateral edges are halved, then its volume will be reduced by :
(a) 50% (b) 33.33%
(c) 66.66% (d) none of these

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EXERCISE

MISCELLANEOUS

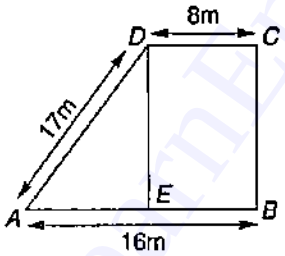
1. If the side of an equilateral triangle is r, then the area of the triangle varies directly as :
   (a) $\sqrt{r}$ (b) r
   (c) r^2 (d) r^3

2. The length of the minute hand of the clock is 6 cm. The area swept by the minute hand in 30 minutes is :
   (a) $\frac{1}{36\pi}$ cm^2 (b) $\frac{1}{18\pi}$ cm^2
   (c) 18π cm^2 (d) 36π cm^2

3. If the diagonals of a rhombus are 18 cm and 24 cm respectively, then its perimeter is :
   (a) 15 cm (b) 42 cm
   (c) 60 cm (d) 70 cm

4. If the ratio of diagonals of two squares is 3 : 2 then the ratio of the areas of two squares is :
   (a) 4 : 5 (b) 6 : 5
   (c) 9 : 4 (d) $\sqrt{3} : \sqrt{2}$

5. In the given figure, $ABCD$ is a trapezium in which the parallel sides AB, CD are both perpendicular to BC. Find the area of the trapezium :
   (a) 140 m^2
   (b) 168 m^2
   (c) 180 m^2
   (d) 156.4 m^2



6. One cubic metre piece of copper is melted and recast into a square cross-section bar, 36 m long. An exact cube is cut off from this bar. If cubic metre of copper cost Rs. 108, then the cost of this cube is :
   (a) 50 paisa (b) 75 paisa
   (c) one rupee (d) 1.50 rupee

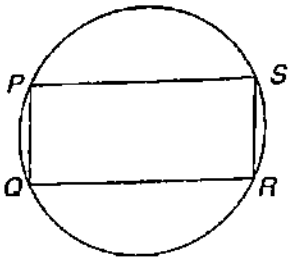
7. If 'h' be the height of a pyramid standing on a base which is an equilateral triangle of side 'a' units, then the slant height is :
   (a) $\sqrt{h^2 + a^2/4}$ (b) $\sqrt{h^2 + a^2/8}$
   (c) $\sqrt{h^2 + a^2/3}$ (d) $\sqrt{h^2 + a^2}$

8. The area of the square base of a right pyramid is 64 cm^2. If the area of each triangle forming the slant surface is 20 cm^2, then the volume of the pyramid is :
   (a) 64 cm^3 (b) $\frac{128}{3}$ cm^3
   (c) $\frac{64}{3}\sqrt{3}$ cm^3 (d) $64\sqrt{2}$ cm^3

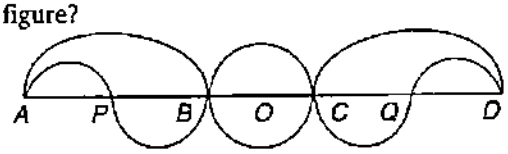
9. If the surface areas of two spheres are in the ratio 4 : 9, then the ratio of their volumes is :
   (a) 8 : 25 (b) 8 : 26
   (c) 8 : 27 (d) 8 : 28

10. The side of a rhombus are 10 cm and one of its diagonal is 16 cm. The area of the rhombus is :
   (a) 96 cm^2 (b) 95 cm^2
   (c) 94 cm^2 (d) 93 cm^2

11. In the adjoining figure $PQRS$ is a rectangle 8 cm × 6 cm, inscribed in the circle. The area of the shaded portion will be :
   (a) 48 cm^2
   (b) 42.50 cm^2
   (c) 32.50 cm^2
   (d) 30.5 cm^2



12. In the adjoining figure $AB = CD = 2BC = 2BP = 2CQ$. In the middle, a circle with radius 1 cm is drawn. In the rest figure all are the semicircular arcs. What is the perimeter of the whole figure?



   (a) 4π (b) 8π
   (c) 10π (d) none of these

13. In a shower 10 cm of rain fall the volume of water that falls on 1.5 hectares of ground is :
   (a) 1500 m^3 (b) 1400 m^3
   (c) 1200 m^3 (d) 1000 m^3

14. The base of a prism is a right angle triangle and the two sides containing the right angle are 8 cm and 15 cm. If its height is 20 cm, then the volume of the prism is :
   (a) 1600 cc (b) 300 cc
   (c) 1200 cc (d) 600 cc

15. A conical circus tent is to be made of canvas. The height of the tent is 35 m and the radius of the base is 84 m. If $\pi = \frac{22}{7}$, then the canvas required is :
   (a) 24000 m^2 (b) 24004 m^2
   (c) 24014 m^2 (d) 24024 m^2

16. The radius of base and the volume of a right circular cone are doubled. The ratio of the length of the larger cone to that of the smaller cone is :
   (a) 1 : 4 (b) 1 : 2
   (c) 2 : 1 (d) 4 : 1

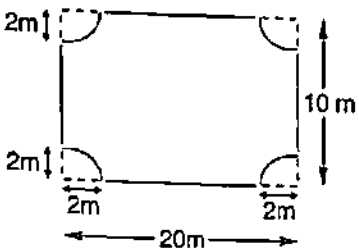
17. A cone and a hemisphere have equal base diameter and equal volumes. The ratio of their heights is :
   (a) 3 : 1 (b) 2 : 1
   (c) 1 : 2 (d) 1 : 3

18. A hollow sphere of internal and external diameters 4 cm and 8 cm respectively is melted to form a solid cylinder of base diameter 8 cm. The height of the cylinder is approximately :
   (a) 4.5 cm (b) 4.57 cm
   (c) 4.67 cm (d) 4.7 cm

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19. The perimeter of the figure given below correct to one decimal place is :



(a) 56 m
(b) 56.6 m
(c) 57.2 m
(d) 57.9 m

20. The sum of the radii of the two circle is 140 cm and the difference between their circumference is 88 cm. The radius of the larger circle is :
(a) 60 cm (b) 70 cm
(c) 63 cm (d) 77 cm

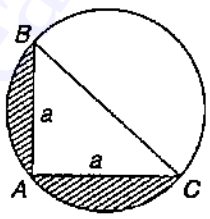
21. If the lateral surface of a right circular cone is 2 times its base, then the semi-vertical angle of the cone must be :
(a) 15° (b) 30°
(c) 45° (d) 60°

22. There is a pyramid on a base which is a regular hexagon of side $2a$. If every slant edge of this pyramid is of length $\frac{(5a)}{2}$ then the volume of this pyramid must be :
(a) $3a^3$ (b) $3a^3\sqrt{2}$
(c) $3a^3\sqrt{3}$ (d) $6a^3$

23. The slant height of a conical tent made of canvas is $\frac{14}{3}$ m. The radius of tent is 2.5 m. The width of the canvas is 1.25 m. If the rate of canvas per metre is Rs. 33, then the total cost of the canvas required for the tent (in Rs.) is :
(a) 726 (b) 950
(c) 960 (d) 968

24. A hemispherical basin 150 cm in diameter holds water one hundred and twenty times as much a cylindrical tube. If the height of the tube is 15 cm, then the diameter of the tube (in cm) is :
(a) 23 (b) 24
(c) 25 (d) 26

25. If BC passes through centre of the circle, then the area of the shaded region in the given figure is :



(a) $\frac{a^2}{2}(3-\pi)$
(b) $a^2\left(\frac{\pi}{2}-1\right)$
(c) $2a^2(\pi-1)$
(d) $\frac{a^2}{2}\left(\frac{\pi}{2}-1\right)$

26. A river 3 m deep and 60 m wide is flowing at the rate of 2.4 km/h. The amount of water running into the sea per minute is :
(a) 6000 m^3 (b) 6400 m^3
(c) 6800 m^3 (d) 7200 m^3

27. A cone whose height is 15 cm and radius of base is 6 cm is trimmed sufficiently to reduce it to a pyramid whose base is an equilateral triangle. The volume of the portion removed is :
(a) 330 cm^3 (b) 328 cm^3
(c) 325 cm^3 (d) 331 cm^3

28. If a solid right circular cylinder is made of iron is heated to increase its radius and height by 1% each, then the volume of the solid is increased by :
(a) 1.01% (b) 3.03%
(c) 2.02% (d) 1.2%

29. The base of a prism is a regular hexagon. If every edge of the prism measures 1 m, then the volume of the prism is :
(a) $\frac{3\sqrt{2}}{2}$ m^3 (b) $\frac{3\sqrt{3}}{2}$ m^3
(c) $\frac{6\sqrt{2}}{5}$ m^3 (d) $\frac{5\sqrt{3}}{2}$ m^3

30. If the side of a square is 24 cm, then the circumference of its circumscribed circle (in cm) is :
(a) $24\sqrt{3}\pi$ (b) $24\sqrt{2}\pi$
(c) $12\sqrt{2}\pi$ (d) 24π

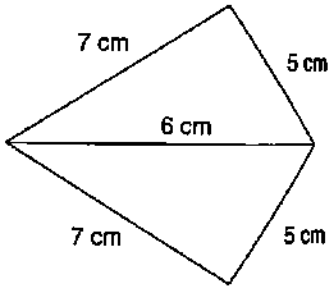
31. An isosceles right triangle has area 112.5 m^2. The length of its hypotenuse (in cm) is :
(a) 21.213 (b) 21.013
(c) 21.113 (d) 21.313

32. Two circles of unit radii, are so drawn that the centre of each lies on the circumference of the other. The area of the region, common to both the circles, is :
(a) $\frac{(4\pi-3\sqrt{3})}{12}$ (b) $\frac{(4\pi-6\sqrt{3})}{12}$
(c) $\frac{(4\pi-3\sqrt{3})}{6}$ (d) $\frac{(4\pi-6\sqrt{3})}{6}$

33. If the right circular cone is separated into three solids of volumes V_1, V_2 and V_3 by two planes which are parallel to the base and trisects the altitude, then $V_1 : V_2 : V_3$ is :
(a) 1 : 2 : 3 (b) 1 : 4 : 6
(c) 1 : 6 : 9 (d) 1 : 7 : 19

34. Water flows at the rate of 10 m per minute from a cylindrical pipe 5 mm in diameter. A conical vessel whose diameter is 40 cm and depth 24 cm is filled. The time taken to fill the conical vessel is :
(a) 50 min (b) 50 min. 12 sec.
(c) 51 min. 12 sec (d) 51 min. 15 sec

35. The length of four sides and a diagonal of the given quadrilateral are indicated in the diagram. If A denotes the area and l the length of the other diagonal, then A and l are respectively :



(a) $12\sqrt{6}, 4\sqrt{6}$
(b) $12\sqrt{6}, 5\sqrt{6}$
(c) $6\sqrt{6}, 4\sqrt{6}$
(d) $6\sqrt{6}, 5\sqrt{6}$

36. If a regular square pyramid has a base of side 8 cm and height of 30 cm, then its volume is :
(a) 120 cc (b) 240 cc
(c) 640 cc (d) 900 cc

37. A cylinder circumscribes a sphere. The ratio of their volumes is :
(a) 1 : 2 (b) 3 : 2
(c) 4 : 3 (d) 5 : 6

38. In triangle ABC, $BC = 8$ cm, $AC = 15$ cm and $AB = 17$ cm. The length of the altitude drawn from B on AC is :
(a) 6 cm (b) 7 cm
(c) 8 cm (d) 10 cm

39. The area of the largest possible square inscribed in a circle of unit radius (in square unit) is :
(a) 3 (b) 4
(c) $2\sqrt{3}\pi$ (d) 2

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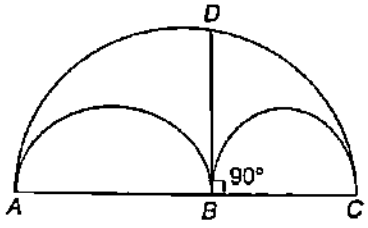
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40. The area of the largest triangle that can be inscribed in a semicircle of radius r is :
(a) r^2 cm^2 (b) $\left(\frac{r}{3}\right)^2$ cm^2
(c) $r\sqrt{2}$ cm^2 (d) $3\sqrt{3}r$ cm^2

41. If a regular hexagon is inscribed in a circle of radius r, then its perimeter is :
(a) $6\sqrt{3}r$ (b) $6r$
(c) $3r$ (d) $12r$

42. If a regular hexagon circumscribes a circle of radius r, then its perimeter is :
(a) $4\sqrt{3}r$ (b) $6\sqrt{3}r$
(c) $6r$ (d) $12\sqrt{3}r$

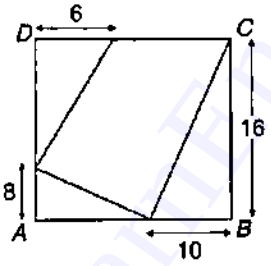
43. In the adjoining figure there are three semicircles in which $BC = 6$ cm and $BD = 6\sqrt{3}$ cm. What is the area of the shaded region (in cm) :



(a) 12π (b) 9π
(c) 27π (d) 28π

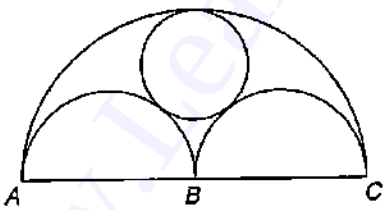
44. Area of a rhombus is 144 cm^2 and the ratio of length of two diagonals is 1 : 2. The sum of lengths of its diagonals are :
(a) 72 cm (b) 40 cm
(c) 36 cm (d) $18\sqrt{2}$ cm

45. Find the area of the shaded region in the given figure of square $ABCD$:



(a) 128 cm^2
(b) 192 cm^2
(c) 148 cm^2
(d) 168 cm^2

46. In the following figure $AB = BC$ and $AC = 84$ cm. The radius of the inscribed circle is 14 cm. B is the centre of the largest semi- circle. What is the area of the shaded region?



(a) 335 cm^2 (b) 770 cm^2
(c) 840 cm^2 (d) 650 cm^2

47. A tank 4 m long and 2.5 m wide and 6 m deep is dug in a field 10 m long and 9 m wide. If the earth dugout is evenly spread over the field, the rise in level of the field will be :
(a) 80 cm (b) 75 cm
(c) 60 cm (d) 30 cm

48. An open box is made of wood 2 cm thick. Its internal length is 86 cm, breadth 46 cm and height is 38 cm. The cost of painting the outer surface of the box at Rs. 10 per m^2 is :
(a) Rs. 18.5 (b) Rs. 8.65
(c) Rs. 11.65 (d) Rs. 17.50

49. A rectangular tin sheet is 22 m long and 8 m broad. It is rolled along its length to form a cylinder by making the opposite edges just to touch each other. The volume of the cylinder (in m^3) is :
(a) 385 (b) 204
(c) 280π (d) 308

50. The lateral surface of a cylinder is developed into a square whose diagonal is $2\sqrt{2}$ cm. The area of the base of the cylinder (in cm^2) is :
(a) 3π (b) $1/\pi$
(c) π (d) 6π

51. If from a circular sheet of paper of radius 15 cm, a sector of 144° is removed and the remaining is used to make a conical surface, then the angle at the vertex will be :
(a) $\sin^{-1}\left(\frac{3}{10}\right)$ (b) $\sin^{-1}\left(\frac{6}{5}\right)$
(c) $2\sin^{-1}\left(\frac{3}{5}\right)$ (d) $2\sin^{-1}\left(\frac{4}{5}\right)$

52. A right circular cone of radius 4 cm and slant height 5 cm is carved out from a cylindrical piece of wood of same radius and height 5 cm. The surface area of the remaining wood is :
(a) 84π (b) 70π
(c) 76π (d) 50π

53. If h, s, V be the height, curved surface area and volume of a cone respectively, then $(3\pi Vh^3 + 9V^2 - s^2h^2)$ is equal to :
(a) 0 (b) π
(c) $\frac{V}{sh}$ (d) $\frac{36}{V}$

54. If a cone is cut into two parts by a horizontal plane passing through the mid point of its axis, the ratio of the volumes of the upper part and the frustum is :
(a) 1 : 1 (b) 1 : 2
(c) 1 : 3 (d) 1 : 7

55. A cone, a hemisphere and a cylinder stand on equal bases of radius R and have equal heights H. Their whole surfaces are in the ratio :
(a) $(\sqrt{3}+1):3:4$ (b) $(\sqrt{2}+1):7:8$
(c) $(\sqrt{2}+1):3:4$ (d) none of these

56. If a sphere is placed inside a right circular cylinder so as to touch the top, base and the lateral surface of the cylinder. if the radius of the sphere is R, the volume of the cylinder is :
(a) $2\pi R^3$ (b) $8\pi R^3$
(c) $\frac{4}{3}\pi R^3$ (d) none of these

57. A cylinder is circumscribed about a hemisphere and a cone is inscribed in the cylinder so as to have its vertex at the centre of one end and the other end as its base. The volumes of the cylinder, hemisphere and the cone are respectively in the ratio of :
(a) $3:\sqrt{3}:2$ (b) 3 : 2 : 1
(c) 1 : 2 : 3 (d) 2 : 3 : 1

58. The base of a pyramid is a rectangle 40 m long and 20 m wide. The slant height of the pyramid from the mid-point of the shorter side of the base to the apex is 29 m. What is the volume of pyramid?
(a) 5600 m^3 (b) 400 m^3
(c) 6500 m^3 (d) $1753\sqrt{110}$ m^3

59. A copper wire when bent in the form of a square, encloses an area of 121 m^2. If the same wire is bent to form a circle, the area enclosed by it would be :
(a) 122 m^2 (b) 112 m^2
(c) 154 m^2 (d) 308 m^2

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LEVEL 1

1. The perimeter of a parallelogram with one internal angles 150° is 64 cm. Find the length of its sides when its area is maximum :
(a) 16, 16 (b) 15, 17
(c) 14, 18 (d) can't be determined

2. A sphere of 30 cm radius is dropped into a cylindrical vessel of 80 cm diameter, which is partly filled with water, then its level rises by x cm. Find x :
(a) 27.5 cm (b) 22.5 cm
(c) 18.5 cm (d) none of these

3. Amit walked 12 m toward east, then he turned to his right and walked 18 m. He then turned to his right and walked 12 m. He again turned to his right and walked 28 m then he again turned to his right and walked 24 m. At what distance is he from the starting point and in which direction?
(a) 23 m north-east (b) 26 m north-east
(c) 26 m west (d) 34 m north-east

4. Find the inradius of triangle if its area is 30 cm^2 and hypotenuse is 13 cm :
(a) 1 cm (b) 2 cm
(c) 2.5 cm (d) $2\sqrt{2}$ cm

5. Which of the following figure will have maximum area if the perimeter of all figures is same?
(a) Square (b) Octagon
(c) Circle (d) Hexagon

6. $ABCD$ is a trapezium with $\angle A = 90°$ and AB parallel to CD. Then $\angle B$ is :
(a) 90° (b) $90° - \angle C$
(c) $360° - \angle C$ (d) $180° - \angle C$

7. A square $ABCD$ has an equilateral triangle drawn on the side AB (interior of the square). The triangle has vertex at G. What is the measure of the angle CGB ?
(a) 60° (b) 80°
(c) 75° (d) 90°

8. There are two concentric circles whose areas are in the ratio of 9 : 16 and the difference between their diameters is 4 cm. What is the area of the outer circle?
(a) 32 cm^2 (b) 64π cm^2
(c) 36 cm^2 (d) 48 cm^2

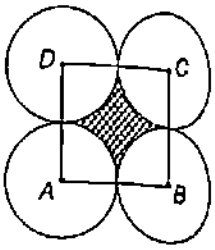
9. A square and rhombus have the same base. If the rhombus is inclined at 60°, find the ratio of area of square to the area of the rhombus :
(a) $2\sqrt{3} : 3$ (b) $1 : \sqrt{3}$
(c) $\sqrt{3} : 2$ (d) none of these

10. Four isosceles triangles are cut off from the corners of a square of area 400 m^2. Find the area of new smaller square (in m^2) :
(a) $200\sqrt{2}$ (b) $\frac{200}{\sqrt{2}}$
(c) 200 (d) $100\sqrt{2}$

11. Altitude and base of a right angle triangle are $(x + 2)$ and $(2x + 3)$ (in cm). If the area of the triangle be 60 cm^2, the length of the hypotenuse is :
(a) 21 cm (b) 13 cm
(c) 17 cm (d) 15 cm

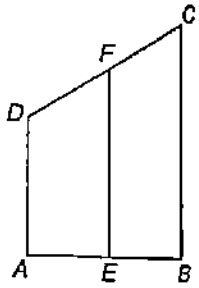
12. Find the area of a regular octagon with each side 'a' cm :
(a) $2a^2(1 + \sqrt{2})$ (b) $\sqrt{2}a(1 + \pi)$
(c) $a^2(\sqrt{2} + 2)$ (d) none of these

13. $ABCD$ is a square, 4 equal circles are just touching each other whose centres are the vertices A, B, C, D of the square. What is the ratio of the shaded to the unshaded area within square?



(a) $\frac{8}{11}$ (b) $\frac{3}{11}$
(c) $\frac{5}{11}$ (d) $\frac{6}{11}$

14. $ABCD$ is a trapezium, in which $AD \parallel BC$, E and F are the mid-points of AB and CD respectively, then EF is :

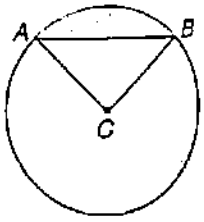


(a) $\frac{(AD + BC)}{2}$
(b) $\frac{(AB + CD)}{2}$
(c) $\frac{DF \times CF}{AE \times BE}$
(d) $\frac{AD + EF + BC}{2}$

15. A right circular cone resting on its base is cut at $\frac{4}{5}$th its height along a plane parallel to the circular base. The height of original cone is 75 cm and base diameter is 42 cm. What is the base radius of cut out (top portion) cone?
(a) 4.2 cm (b) 2.8 cm
(c) 3.5 cm (d) 8.4 cm

16. l, b are the length and breadth of a rectangle respectively. If the perimeter of this rectangle is numerically equal to the area of the rectangle. What is the value of $l - b$ (where $l > b$)?
(a) 1 (b) 2
(c) 3 (d) can't be determined

17. In the adjoining figure ABC is an equilateral triangle and C is the centre of the circle, A and B lie on the circle. What is the area of the shaded region, if the diameter of the circle is 28 cm?



(a) $\left(102\frac{2}{3} - 49\sqrt{3}\right)$ cm^2
(b) $\left(103\frac{2}{3} - 98\sqrt{3}\right)$ cm^2
(c) $(109 - 38\sqrt{3})$ cm
(d) none of the above

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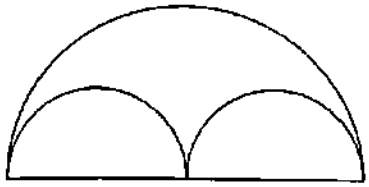
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18. l_1, b_1 and l_2, b_2 are the lengths and breadths of the two rectangles respectively, but the areas of the rectangles are same. l_1 is increased by 25% and b_1 is decreased by 25%. Similarly l_2 is decreased by 25% and b_2 is increased by 25%. If A_1 and A_2 is the new areas of the two rectangles respectively, then :
(a) $A_1 > A_2$ (b) $A_1 < A_2$
(c) $A_1 = A_2$ (d) can't be determine

19. An acute angle made by a side of parallelogram with other pair of parallel sides is 60°. If the distance between these parallel sides is $6\sqrt{3}$, the other side is :
(a) 12 cm (b) $12\sqrt{3}$ cm
(c) $15\sqrt{3}$ cm (d) none of these

20. A solid sphere is melted and recast into a right circular cone with a base radius equal to the radius of the sphere. What is the ratio of the height and radius of the cone so formed?
(a) 4 : 3 (b) 2 : 3
(c) 3 : 4 (d) none of these

21. In the given figure there are 3 semicircles, the radii of each smaller circle is equal. If the radius of the larger circle be 22 cm, then the area of the shaded region is :

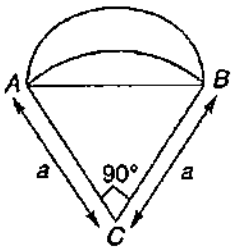


(a) $363\frac{\pi}{4}$ (b) $363\frac{\pi}{3}$
(c) 236.5π (d) 363π

22. A rectangular lawn 60 m × 40 m has two roads each 5 m wide running in the middle of it, one parallel to length and the other parallel to breadth. The cost of gravelling the roads at 80 paise per sq. m is :
(a) Rs. 380 (b) Rs. 385
(c) Rs. 400 (d) none of these

23. There are two rectangular fields of same area. The length of first rectangular field is x% less than the length of the second field and breadth of the first field is $(5x)$% greater than the breadth of the second field. What is the value of x?
(a) 15 (b) 25
(c) 50 (d) 80

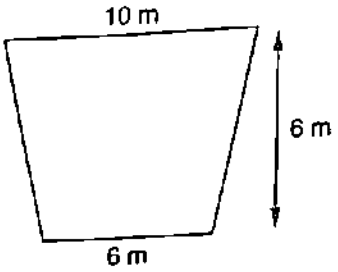
24. In the adjoining figure ACB is a quadrant with radius 'a'. A semicircle is drawn outside the quadrant taking AB as a diameter. Find the area of shaded region :



(a) $\frac{1}{4}(\pi - 2a^2)$
(b) $\left(\frac{1}{4}\right)(\pi a^2 - a^2)$
(c) $\frac{a^2}{2}$
(d) can't be determined

25. Ravi made an error of 5% in excess while measuring the length of rectangle and an error of 8% deficit was made while measuring the breadth. What is the percentage error in the area?
(a) – 3% (b) – 40%
(c) – 3.4% (d) can't be determined

26. In the adjoining figure the cross-section of a swimming pool is shown. If the length of the swimming pool is 120 m then the amount of water it can hold is :

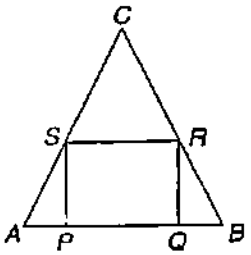


(a) 5760 m^3
(b) 9600 m^3
(c) 7200 m^3
(d) none of these

27. Around a circular garden a circular road is to be repair which costs Rs. 22176 at the rate of Rs. 1 per sq. m. If the inner radius is 112 m, find the width of the circular road :
(a) 18 m (b) 28 m
(c) 14 m (d) none of these

28. An equilateral triangle is cut from its three vertices to form a regular hexagon. What is the percentage of area wasted?
(a) 20% (b) 50%
(c) 33.33% (d) 66.66%

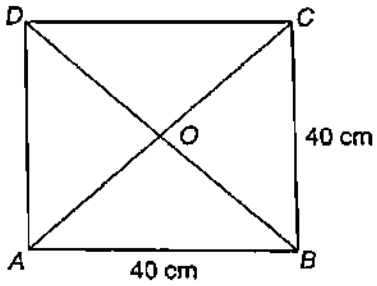
29. ABC is an equilateral triangle and $PQRS$ is a square inscribed in the triangle in such a way that P and Q lie on AB and R, S lie on BC and AC respectively. What is the value of $RC : RB$?



(a) $1 : \sqrt{2}$
(b) $1 : \sqrt{3}$
(c) $\sqrt{3} : 2$
(d) 1 : 2

30. The area of a square and circle is same and the perimeter of square and equilateral triangle is same, then the ratio between the area of circle and the area of equilateral triangle is :
(a) $\pi : 3$ (b) $9 : 4\sqrt{3}$
(c) $4 : 9\sqrt{3}$ (d) none of these

31. Adjoining figure shows a square $ABCD$ in which O is the point of intersection of diagonals AC and BD. Four squares of maximum possible area are formed inside each four triangles AOB, BOC, COD and AOD. What is the total area of these 4 squares?



(a) 400 cm^2 (b) 100 cm^2
(c) 80 cm^2 (d) none of these

32. What is the ratio of the area of circumcircle of equilateral triangle to the area of square with the same side length as the equilateral triangle?
(a) $\pi : 3$ (b) $\pi : \sqrt{3}$
(c) $\sqrt{3} : 2$ (d) none of these

33. It is required to construct a big rectangular hall that can accommodate 400 people with 25 m^3 space for each person. The height of the wall has been fixed at 10 m and the total inner surface area of the walls must be 1300 m^2. What is the length and breadth of the hall (in metres)?

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(a) 30, 20 (b) 45, 20
(c) 40, 25 (d) 35, 30

34. The perimeter of a rectangle and an equilateral triangle are same. Also, one of the sides of the rectangle is equal to the side of the triangle. The ratio of the areas of the rectangle and the triangle is :
(a) $\sqrt{3}:1$ (b) $1:\sqrt{3}$
(c) $2:\sqrt{3}$ (d) $4:\sqrt{3}$

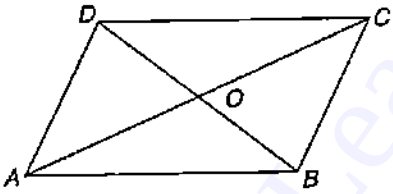
35. If l, b, p be the length, breadth and perimeter of a rectangle and b, l, p are in GP (in order), then l/b is :
(a) 2 : 1 (b) $(\sqrt{3}-1):1$
(c) $(\sqrt{3}+1):1$ (d) $2:\sqrt{3}$

36. A spherical steel ball was silver polished then it was cut into 4 similar pieces. What is ratio of the polished area to the non polished area :
(a) 1 : 1 (b) 1 : 2
(c) 2 : 1 (d) can't be determined

37. What is the total surface area of the identical cubes of largest possible size that are cut from a cuboid of size 85 cm × 17 cm × 5.1 cm?
(a) 26010 cm^2 (b) 21600 cm^2
(c) 26100 cm^2 (d) none of these

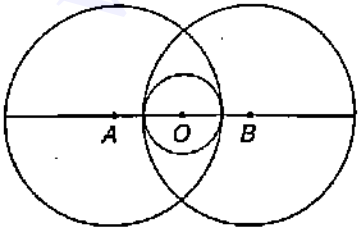
38. 125 identical cubes are cut from a big cube and all the smaller cubes are arranged in a row to form a long cuboid. What is the percentage increase in the total surface area of the cuboid over the total surface area of the cube?
(a) $234\frac{2}{3}\%$ (b) $235\frac{1}{3}\%$
(c) $134\frac{2}{3}\%$ (d) none of these

39. In the adjoining figure a parallelogram $ABCD$ is shown. $AB = 24$ cm and $AO = BO = 13$ cm. Find BC :



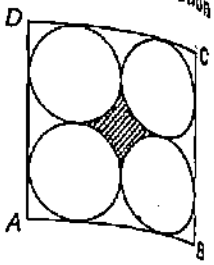
(a) 8 cm (b) 10 cm
(c) 11 cm (d) none of these

40. There are two circles intersecting each other. Another smaller circle with centre O, is lying between the common region of two larger circles. Centres of the circle (*i.e.*, A, O and B) are lying on a straight line. $AB = 16$ cm and the radii of the larger circles are 10 cm each. What is the area of the smaller circle?



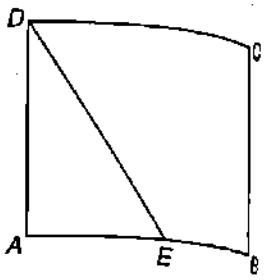
(a) 4π cm^2 (b) 2π cm^2
(c) $\frac{4}{\pi}$ cm^2 (d) $\frac{\pi}{4}$ cm^2

41. $ABCD$ is a square, inside which 4 circles with radius 1 cm, each are touching each other. What is the area of the shaded region?



(a) $(2\pi - 3)$ cm^2
(b) $(4 - \pi)$ cm^2
(c) $(16 - 4\pi)$ cm^2
(d) none of these

42. $ABCD$ is a square, E is a point on AB such that $BE = 17$ cm. The area of triangle ADE is 84 cm^2. What is the area of square?

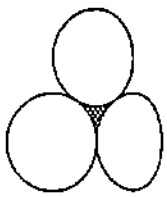


(a) 400 cm^2
(b) 625 cm^2
(c) 729 cm^2
(d) 576 cm^2

43. If the volume of a sphere, a cube, a tetrahedron and a octahedron be same then which of the following has maximum surface area?
(a) Sphere (b) Cube
(c) Octahedron (d) Tetrahedron

44. In a rectangle the ratio of the length is to breadth is same as that of the sum of the length and breadth to the length. If l and b be the length and breadth of the rectangle then which of the following is true?
(i) $\frac{l}{b} = \frac{l^2}{b^2} + 1$ (ii) $\frac{b}{l-b} = \frac{l+b}{l}$
(iii) $lb = (l+b)(l-b)$
(a) only (i) is true
(b) only (ii) is true
(c) only (ii) and (iii) are true
(d) only (i) and (ii) are true

45. Three circle of equal radii touch each other as shown in figure. The radius of each circle is 1 cm. What is the area of shaded region?



(a) $\left(\frac{2\sqrt{3}-\pi}{2}\right)$ cm^2
(b) $\left(\frac{3\sqrt{2}-\pi}{3}\right)$ cm^2
(c) $\frac{2\sqrt{3}}{\pi}$ cm^2
(d) none of the above

46. How many spheres of radius 1.5 cm can be cut out of a wooden cube of edge 9 cm?
(a) 216 (b) 81
(c) 27 (d) can't be determined

47. Kaurav and Pandav have a rectangular field of area 20,000 sq. m. They decided to divide it into two equal parts by dividing it with a single straight line. Kaurav wanted to fence their land immediately, so they incurred total expenses for the fencing all the four sides alone at Rs. 2 per metre. What is the minimum cost that Kaurav had to incur?
(a) Rs. 800 (b) Rs. 1600
(c) Rs. 1200 (d) Rs. 600

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48. There is a cone of height 12 cm, out of which a smaller cone (which is the top portion of the original cone) with the same vertex and vertical axis is cut out. What is the ratio of the volume of the larger (actual) cone to the remaining part (frustum) of the cone, if the height of the smaller cone is 9 cm?
(a) 3 : 1 (b) 9 : 1
(c) 64 : 37 (d) 16 : 7

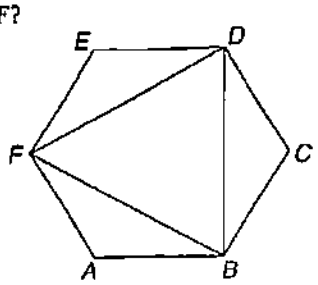
49. Radhey can walk along the boundary of a rectangular field and also along the diagonals of the field. His speed is 53 km/h. The length of the field is 45 km. Radhey started from one corner and reached to the diagonally opposite corner in 1 hour. What is the area of the field?
(a) 860 km^2 (b) 1260 km^2
(c) 1060 km^2 (d) can't be determined

50. Charles has a right circular cylinder which he inserted completely into a right circular cone of height 30 cm. The vertical angle of the cone is 60° and the diameter of the cylinder is $8\sqrt{3}$ cm. What is the volume of the cone?
(a) $\frac{3000}{7}\pi$ cm^3 (b) 3000π cm^3
(c) 4860π cm^3 (d) can't be determind

51. There are six faces in a cube. Rajeev fix one cube on each of the faces. The dimensions of all the cubes are same. What is the ratio of total surface area of the newly formed solid to the area of a single cube?
(a) 7 : 1 (b) 6 : 1
(c) 5 : 1 (d) 41 : 9

52. If the ratio of diagonals of two cubes is 3 : 2 then the ratio of the surface areas of the two cubes respectively is :
(a) 5 : 4 (b) 9 : 5
(c) 9 : 4 (d) can't be determind

53. *ABCDEF* is a regular hexagon of side 6 cm. What is the area of triangle *BDF*?



(a) $32\sqrt{3}$ cm^2 (b) $27\sqrt{3}$ cm^2
(c) 24 cm^2 (d) none of these

Directions for question number 54 and 55 : King Dashratha of Ayodhya had a rectangular plot of area 9792 m^2. He divided it into 4 square shaped plots by fencing parallel fences to the breadth of the rectangular plot. All the four sons got each square shaped plot. However, some area of plot was still left which could not be formed as a square shaped. So, four more square shaped plots were formed by fencing parallel to the longer side of the original plot. The king gave one smaller square shaped plot to each of his wives and one of the smaller square shaped plot retained with himself and then nothing left to divide.

54. What is the ratio of the area of larger square shaped plot to the area of the smaller square shaped plot?
(a) 17 : 1 (b) 25 : 9
(c) 16 : 1 (d) can't be determined

55. What are dimensions of the original plot?
(a) 288 m, 34 m (b) 102 m, 96 m
(c) 306 m, 32 m (d) 204 m, 48 m

LEVEL 2

Directions for question number 1, 2 and 3 : Each edge of an equilateral triangle is 'a' cm. A cone is formed by joining any two sides of the triangle.

1. What is the radius and slant height of the cone?
(a) $a, \frac{a}{2\pi}$ (b) $\frac{a}{\pi}, \frac{a}{2}$
(c) $\frac{a}{2\pi}, a$ (d) $2a, \frac{a}{\pi}$

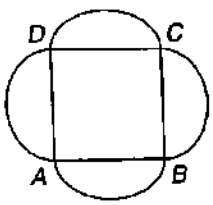
2. What is the volume of the cone?
(a) $\frac{a^2}{24\pi^3}\sqrt{4-\pi^2}$ (b) $\frac{a^3}{24\pi^2}\sqrt{4\pi^2-1}$
(c) $\frac{a^3}{8\pi^2}\sqrt{1-4\pi^2}$ (d) $\frac{a}{\sqrt{3}}\pi^2\left(1-\frac{2}{\pi}\right)$

3. If the cone is cut along its axis from the middle, the new shape we obtain after opening the paper is :
(a) isosceles triangle (b) equilateral triangle
(c) right angle triangle (d) none of these

4. If the sum of the radius and the height of a closed cylinder is 35 cm and the total surface area of the cylinder is 1540 cm^2, then the circumference of the base of the cylinder is :
(a) 66 cm (b) 44 cm
(c) 56 cm (d) can't be determined

5. An iron pillar has some part in the form of a right circular cylinder and remaining in the form of a right circular cone. The radius of base of cone, as well as cylinder is 21 cm. The cylindrical part is 80 cm high and conical part is 16 cm high. Find the weight of the pillar, if 1 cm^3 of iron weighs 8.45 g :
(a) 999.39 kg (b) 111 kg
(c) 1001 kg (d) 989 kg

6. *ABCD* is a square of side *a* cm. *AB*, *BC*, *CD* and *AD* all are the chords of circles with equal radii each. If the chords subtends an angle of 120° at their respective centres, find the total area of the given figure, where arcs are part of the circles :



(a) $\left[a^2+4\left(\frac{\pi a^2}{9}-\frac{a^2}{3\sqrt{2}}\right)\right]$ (b) $\left[a^2+4\left(\frac{\pi a^2}{9}-\frac{a^2}{4\sqrt{3}}\right)\right]$
(c) $[9a^2-4\pi+3\sqrt{3}a^2]$ (d) none of these

7. A rectangle has a perimeter of 26. How many combinations of integral valued length are possible?
(a) 4 (b) 8
(c) 6 (d) 12

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8. A hollow sphere of outer diameter 24 cm is cut into two equal hemisphere. The total surface area of one of the hemisphere is $1436\frac{2}{7}$ cm^2. Each one of the hemisphere is filled with water. What is the volume of water that can be filled in each of the hemisphere?

(a) $3358\frac{2}{3}$ cm^3 (b) $3528\frac{2}{3}$ cm^3

(c) $2359\frac{2}{3}$ cm^3 (d) $9335\frac{2}{3}$ cm^3

9. A big cube of side 8 cm is formed by rearranging together 64 small but identical cubes each of side 2 cm. Further, if the corner cubes in the topmost layer of the big cube are removed, what is the change in total surface area of the big cube?

(a) 16 cm^2, decreases
(b) 48 cm^2, decreases
(c) 32 cm^2, decreases
(d) remains the same as previously

10. A large solid sphere of diameter 15 m is melted and recast into several small spheres of diameter 3 m. What is the percentage increase in the surface area of the smaller spheres over that of the large sphere?

(a) 200% (b) 400%
(c) 500% (d) can't be determined

11. A cone is made of a sector with a radius of 14 cm and an angle of 60°. What is total surface area of the cone?

(a) 119.78 cm^2 (b) 191.87 cm^2
(c) 196.5 cm^2 (d) none of these

12. Kishan Chand is a very labourious farmer, he erected a fence around his paddy field in a square shape. He used 26 poles in each side, each at a distance of 4 m. What is the area of field?

(a) 1.6 hectare (b) 2.6 hectare
(c) 5.76 hectare (d) 1 hectare

13. A rectangular lawn is surrounded by a path of width 2 m on all sides. Now if the length of the lawn is reduced by 2 m the lawn becomes a square lawn and the area of path becomes 13/11 times, what is the length of the original lawn?

(a) 8 m (b) 9 m
(c) 10 m (d) 12 m

Directions for question number 14, 15 : A cylinder with height and radius 2 : 1 is filled with soft drink and then it is tilted so as to allow some soft drink to flow off to an extent where the level of soft drink just touches the lowest point of the upper mouth.

14. If the 2.1 L soft drink is retained in the cylinder, what is the capacity of the cylinder?

(a) 3.6 L (b) 4 L
(c) 1.2 L (d) 4.2 L

15. If the quantity of soft drink left is poured into a conical flask whose height and base radius are same as that of the cylinder so as to fill the conical flask completely, the quantity of soft drink left in the cylinder as a fraction of its total capacity is :

(a) $\frac{1}{3}$ (b) $\frac{1}{6}$

(c) $\frac{1}{9}$ (d) $\frac{1}{10}$

16. An elephant of length 4 m is at one corner of a rectangular cage 16 m × 30 m and facing towards the diagonally opposite corner. If the elephant starts moving towards the diagonally opposite corner it takes 15 seconds to reach the opposite corner. Find the speed of the elephant :

(a) 1 m/s (b) 2 m/s
(c) 1.87 m/s (d) can't be determined

17. What is the height of the cone which is formed by joining the two ends of a sector of circle with radius r and angle 60°?

(a) $\frac{\sqrt{35}}{6}r$ (b) $\frac{\sqrt{25}}{6}r$

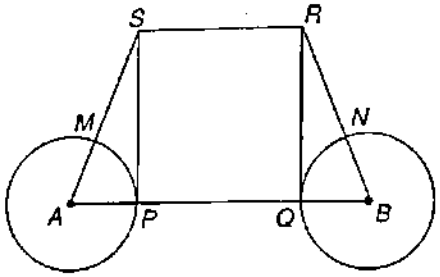
(c) $\frac{r^2}{\sqrt{3}}$ (d) none of these

18. If a cube of maximum possible volume is cut off from a solid sphere of diameter d, then the volume of the remaining (waste) material of the sphere would be equal to :

(a) $\frac{d^3}{3}\left(\pi - \frac{d}{2}\right)$ (b) $\frac{d^3}{3}\left(\frac{\pi}{2} - \frac{1}{\sqrt{3}}\right)$

(c) $\frac{d^2}{4}(\sqrt{2} - \pi)$ (d) none of these

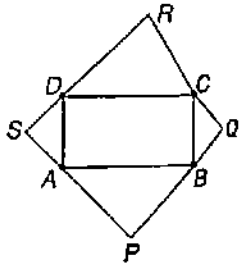
19. In the adjoining figure $PQRS$ is a square and $MS = RN$ and A, P, Q and B lie on the same line. Find the ratio of the area of two circles to the area of the square. Given that $AP = MS$:



(a) $\frac{\pi}{3}$ (b) $\frac{2\pi}{3}$

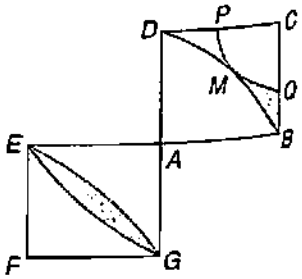
(c) $\frac{3\pi}{2}$ (d) $\frac{6}{\pi}$

20. $ABCD$ is a rectangle and there are four equilateral triangles. Area of ΔASD equals to area of ΔBQC and area of ΔDRC equals to area of ΔAPB. The perimeter of the rectangle is 12 cm. Also the sum of the areas of the four triangles is $10\sqrt{3}$ cm^2 then the total area of the figure thus formed :



(a) $2(4 + 5\sqrt{3})$ cm^2 (b) $5(4 + 2\sqrt{3})$ cm^2
(c) $42\sqrt{3}$ cm^2 (d) none of these

21. $ABCD$ and $EFGA$ are the squares of side 4 cm, each. In square $ABCD$, DMB and PMQ are the arcs of circles with centres at A and C respectively. In square $AEFG$, the shaded region is enclosed by two arcs of circles with centres at A and F, respectively. What is the ratio of the shaded regions of the squares $ABCD$ and $AEFG$ respectively :



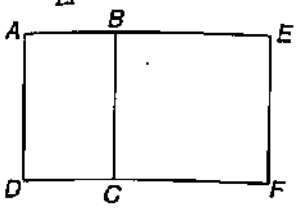
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(a) $\frac{2+\pi(\sqrt{2}-2)}{(\pi-2)}$ (b) $\frac{(\pi-2)}{2(\sqrt{2}+1-\pi)}$

(c) $\frac{4}{3}$ (d) none of these

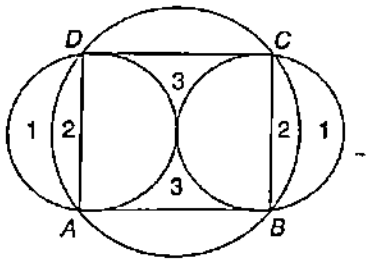
22. In the adjoining figure $\frac{AB}{BC}=\frac{AD}{DF}$, where *EBCF* is a square. Find the ratio of $\frac{AE}{EF}$:



(a) $\frac{(1\pm\sqrt{7})}{3}$ (b) $\frac{(1-\sqrt{7})}{2}$

(c) $\frac{(1+\sqrt{5})}{2}$ (d) $\frac{(1\pm\sqrt{5})}{2}$

Direction for question number 23–25 : In the adjoining figure ABCD is a square. A circle ABCD is passing through all the four vertices of the square. There are two more circles on the sides AD and BC touching each other inside the square, AD and BC are the respective diameters of the two smaller circles. Area of the square is 16 cm².



23. What is the area of region 1?

(a) 2.4 cm² (b) $\left(2-\frac{\pi}{4}\right)$ cm²

(c) 8 cm² (d) $(4\pi-2)$ cm²

24. What is the area of region 2?

(a) $3(\pi-2)$ cm² (b) $(\pi-3)$ cm²

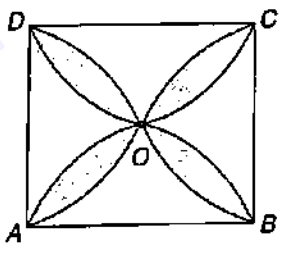
(c) $(2\pi-3)$ cm² (d) $4(\pi-2)$ cm²

25. What is the area of region 3?

(a) $(4-4\pi)$ cm² (b) $4(4-\pi)$ cm²

(c) $(4\pi-2)$ cm² (d) $(3\pi+2)$ cm²

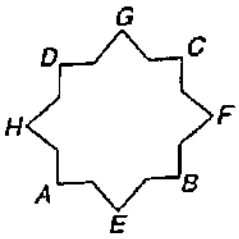
26. In the adjoining figure *ABCD* is a square. Four equal semicircles are drawn in such a way that they meet each other at 'O'. Sides *AB*, *BC*, *CD* and *AD* are the respective diameters of the four semicircles. Each of the side of the square is 8 cm. Find the area of the shaded region :



(a) $32(\pi-2)$ cm² (b) $16(\pi-2)$ cm²

(c) $(2\pi-8)$ cm² (d) $\left(\frac{3}{4}\pi-4\right)$ cm²

27. *ABCD* is a square. Another square *EFGH* with the same area is placed on the square *ABCD* such that the point of intersection of diagonals of square *ABCD* and square *EFGH* coincide and the sides of square *EFGH* are parallel to the diagonals of square *ABCD*. Thus a new figure is formed as shown in the figure. What is the area enclosed by the given figure if each side of the square is 4 cm :



(a) $32(2-\sqrt{2})$ (b) $16\left(\frac{3+\sqrt{2}}{2+\sqrt{2}}\right)$

(c) $32\left(\frac{2+\sqrt{2}}{3+\sqrt{2}}\right)$ (d) none of these

28. A piece of paper is in the form of a right angle triangle in which the ratio of base and perpendicular is 3 : 4 and hypotenuse is 20 cm. What is the volume of the biggest cone that can be formed by taking right angle vertex of the paper as the vertex of the cone?

(a) 45.8 cm³ (b) 56.1 cm³

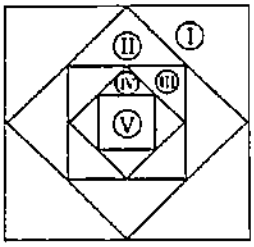
(c) 61.5 cm³ (d) 48 cm³

29. In a particular country the value of diamond is directly proportional to the surface area (exposed) of the diamond. Four thieves steel a cubical diamond piece and then divide equally in four parts. What is the maximum percentage increase in the value of diamond after cutting it?

(a) 50% (b) 66.66%

(c) 100% (d) none of these

Directions for question number 30 and 31 : In the figure shown square II is formed by joining the mid-points of square I, square III is formed by joining the mid-points of square II



and so on. In this way total five squares are drawn. The side of the square I is 'a' cm.

30. What is the perimeter of all the five squares?

(a) $\frac{(4\sqrt{2}+1)a}{(\sqrt{2}+1)}$ (b) $\frac{(4\sqrt{2}-1)a}{(\sqrt{2}+1)}$

(c) $\frac{5}{6}a$ (d) $(7+3\sqrt{2})a$

31. What is the total area of all the five squares?

(a) $\frac{(4\sqrt{2}-1)a^2}{(4\sqrt{2}-1)}$ (b) $\frac{(4\sqrt{2}-1)a^2}{4(\sqrt{2}-1)}$

(c) $\frac{31}{16}a^2$ (d) none of these

Directions for question number 32-35 : Each edge of a cube is equally divided into n parts, thus there are total n^3 smaller cubes. Let,

N_0 → Number of smaller cubes with no exposed surfaces

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$N_1 \rightarrow$ *Number of smaller cubes with one exposed surfaces*
$N_2 \rightarrow$ *Number of smaller cubes with two exposed surfaces*
$N_3 \rightarrow$ *Number of smaller cubes with three exposed surfaces*

32. What is the number of unexposed smaller cubes (N_0)?
(a) $(n-2)^3$ (b) n^3
(c) $n!$ (d) 8

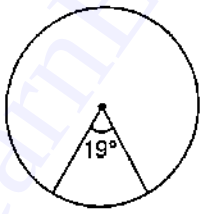
33. What is the number of smaller cubes with one exposed surface (N_1)?
(a) $4(n-3)^3$ (b) $6(n-2)^2$
(c) $(n-3)^2$ (d) $(n+1)^2$

34. What is the value of (N_2)?
(a) $8(n-2)^2$ (b) $6(n-2)$
(c) $12(n-2)$ (d) $3(n-3)^2$

35. What is the value of N_3?
(a) $(n-1)!$ (b) $(n-2)^2$
(c) $\frac{n(n+1)}{2}$ (d) 8

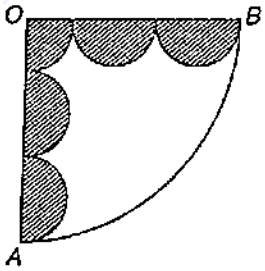
36. In a bullet the gun powder is to be filled up inside the metallic enclosure. The metallic enclosure is made up of a cylindrical base and conical top with the base of radius 5 cm. The ratio of height of cylinder and cone is 3 : 2. A cylindrical hole is drilled through the metal solid with height two-third the height of metal solid. What should be the radius of the hole, so that the volume of the hole (in which gun powder is to be filled up) is one-third the volume of metal solid after drilling?
(a) $\sqrt{\frac{88}{5}}$ cm (b) $\sqrt{\frac{55}{8}}$ cm
(c) $\frac{55}{8}$ cm (d) 33π cm

37. A sector of the circle measures 19° (see the figure). Using only a scale, a compass and a pencil, is it possible to split the circle into 360 equal sectors of 1° central angle?



(a) Yes
(b) No
(c) Yes, only if radius is known
(d) can't be determined

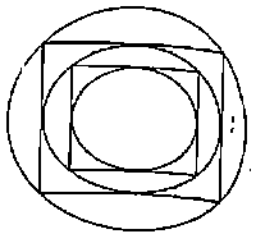
38. A circular paper is folded along its diameter, then again it is folded to form a quadrant. Then it is cut as shown in the figure, after it the paper was reopened in the original circular shape. Find the ratio of the original paper to that of the remaining paper? (The shaded portion is cut off from the quadrant. The radius of quadrant *OAB* is 5 cm and radius of each semicircle is 1 cm) :



(a) 25 : 16 (b) 25 : 9
(c) 20 : 9 (d) none of these

39. A cubical cake is cut into several smaller cubes by dividing each edge in 7 equal parts. The cake is cut from the top along the two diagonals forming four prisms. Some of them get cut and rest remained in the cubical shape. A complete cubical (smaller) cake was given to adults and the cut off part of a smaller cake is given to a child (which is not an adult). If all the cakes were given equally each piece to a person, total how many people could get the cake?
(a) 343 (b) 448
(c) 367 (d) 456

Directions for question number 40-42 : *A square is inscribed in a circle then another circle is inscribed in the square. Another square is then inscribed in the circle. Finally a circle is inscribed in the innermost square. Thus there are 3 circles and 2 squares as shown in the figure. The radius of the outer-most circle is R.*

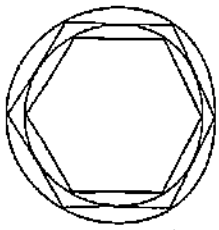


40. What is the radius of the inner-most circle?
(a) $\frac{R}{2}$ (b) $\frac{R}{\sqrt{2}}$
(c) $\sqrt{2}R$ (d) none of these

41. What is the sum of areas of all the squares shown in the figure?
(a) $3R^2$ (b) $3\sqrt{2}R^2$
(c) $\frac{3}{\sqrt{2}}R^2$ (d) none of these

42. What is the ratio of sum of circumferences of all the circles to the sum of perimeters of all the squares?
(a) $(2+\sqrt{3})\pi R$ (b) $(3+\sqrt{2})\pi R$
(c) $3\sqrt{3}\pi R$ (d) none of these

Directions for question number 43-45 : *A regular hexagon is inscribed in a circle of radius R. Another circle is inscribed in the hexagon. Now another hexagon is inscribed in the second (smaller) circle.*

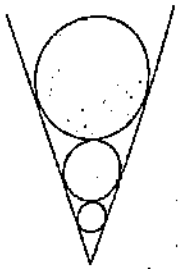


43. What is the sum of perimeters of both the hexagons?
(a) $(2+\sqrt{3})R$ (b) $3(2+\sqrt{3})R$
(c) $3(3+\sqrt{2})R$ (d) none of these

44. What is the ratio of area of inner circle to the outer circle?
(a) 3 : 4 (b) 9 : 16
(c) 3 : 8 (d) none of these

45. If there are some more circles and hexagons inscribed in the similar way as given above, then the ratio of each side of outermost hexagon (largest one) to that of the fourth (smaller one) hexagon is (fourth hexagon means the hexagon which is inside the third hexagon from the outside):
(a) $9 : 3\sqrt{2}$ (b) 16 : 9
(c) $8 : 3\sqrt{3}$ (d) none of these

Directions for question number 46-47 : *Five spheres are kept in a cone in such a way that each sphere touch each other and also touch the lateral surface of the cone, this is due to increasing radius of the spheres starting from the vertex of the cone. The radius of the smallest sphere is 16 cm.*



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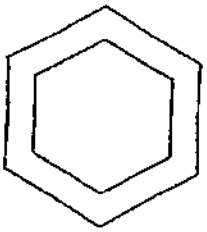
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46. If the radius of the fifth (i.e., largest) sphere be 81 cm, then find the radius of the third (i.e., middlemost) sphere :
(a) 25 cm (b) $25\sqrt{3}$ cm
(c) 36 cm (d) data insufficient

47. What is the least distance between the smallest sphere and the vertex of the cone?
(a) 64 cm (b) 60 cm
(c) 28 cm (d) none of these

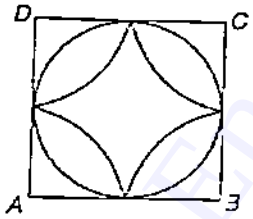
48. Saumya has a pencil box of volume 60 cm^3. What can be the maximum length of a pencil that can be accommodated in the box. Given that all the sides are integral (in cm) and different from each other?
(a) $7\sqrt{2}$ cm (b) $\sqrt{905}$ cm
(c) $\sqrt{170}$ cm (d) $\sqrt{3602}$ cm

49. There are two concentric hexagons. Each of the side of both the hexagons are parallel. Each side of an internal regular hexagon is 8 cm. What is the area of the shaded region, if the distance between corresponding parallel sides is $2\sqrt{3}$ cm :



(a) $120\sqrt{3}$ cm^2
(b) $148\sqrt{3}$ cm^2
(c) 126 cm^2
(d) none of the above

50. $ABCD$ is a square. A circle is inscribed in the square. Also taking A, B, C, D (the vertices of square) as the centres of four quadrants, drawn inside the circle, which are touching each other on the mid-points of the sides of square. Area of square is 4 cm^2. What is the area of the shaded region?



(a) $\left(4-\frac{3\pi}{2}\right)$ cm^2 (b) $(2\pi - 4)$ cm^2
(c) $(4-2\pi)$ cm^2 (d) none of these

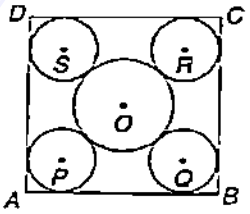
51. In a factory there are two identical solid blocks of iron. When the first block is melted and recast into spheres of equal radii 'r', then 14 cc of iron was left, but when the second block was melted and recast into sphere each of equal radii '$2r$', then 36 cc of iron was left. The volumes of the solid blocks and all the spheres are in integers. What is the volume (in cm^3) of each of the larger spheres of radius '$2r$'?
(a) 176 (b) 12π
(c) 192 (d) data insufficient

52. There is a vast grassy farm in which there is a rectangular building of the farm-house whose length and breadth is 50 m and 40 m respectively. A horse is tethered at a corner of the house with a tether of 80 m long. What is the maximum area that the horse can graze?
(a) 5425π (b) 5245π
(c) 254π (d) none of these

53. A cube of side 6 cm is painted on all its 6 faces with red colour. It is then broken up into 216 smaller identical cubes. What is the ratio of $N_0 : N_1 : N_2$.
Where, $N_0 \rightarrow$ number of smaller cubes with no coloured surface.
$N_1 \rightarrow$ number of smaller cubes with 1 red face.
$N_2 \rightarrow$ number of smaller cubes with 2 red faces :
(a) 3 : 4 : 6 (b) 3 : 4 : 5
(c) 4 : 6 : 3 (d) can't be determined

54. Assume that a mango and its seed, both are spherical, now if the radius of seed is 2/5 of the thickness of the pulp. The seed lies exactly at the centre of the fruit. What per cent of the total volume of the mango is its pulp?
(a) $63\frac{3}{5}\%$ (b) 97.67%
(c) $68\frac{2}{3}\%$ (d) none of these

55. In the adjoining diagram $ABCD$ is a square with side 'a' cm. In the diagram the area of the larger circle with centre 'O' is equal to the sum of the areas of all the rest four circles with equal radii, whose centres are P, Q, R and S. What is the ratio between the side of square and radius of a smaller circle?



(a) $(2\sqrt{2}+3)$ (b) $(2+3\sqrt{2})$
(c) $(4+3\sqrt{2})$ (d) can't be determined

56. Initially the diameter of a balloon is 28 cm. It can explode when the diameter becomes 5/2 times of the initial diameter. Air is blown at 156 cc/s. It is known that the shape of balloon always remains spherical. In how many seconds the balloon will explode?
(a) 1078 s (b) 1368 s
(c) 1087 s (d) none of these

57. The radius of a cone is $\sqrt{2}$ times the height of the cone. A cube of maximum possible volume is cut from the same cone. What is the ratio of the volume of the cone to the volume of the cube?
(a) 3.18π (b) 2.25π
(c) 2.35 (d) can't be determined

58. Raju has 64 small cubes of 1 cm^3. He wants to arrange all of them in a cuboidal shape, such that the surface area will be minimum. What is the diagonal of this larger cuboid?
(a) $8\sqrt{2}$ cm (b) $\sqrt{273}$
(c) $4\sqrt{3}$ cm (d) $\sqrt{129}$ cm

59. The volume of a cylinder is 48.125 cm^3, which is formed by rolling a rectangular paper sheet along the length of the paper. If a cuboidal box (without any lid i.e., open at the top) is made from the same sheet of paper by cutting out the quare of side 0.5 cm from each of the four corners of the paper sheet, then what is the volume of this box?
(a) 20 cm^3 (b) 38 cm^3
(c) 19 cm^3 (d) none of these

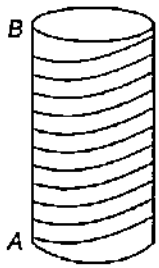
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Directions for question number 60-62 : *Consider a cylinder of height h cm and radius* $r = \frac{4}{\pi}$ *cm as shown in the figure. A string of certain length when wound on its cylindrical surface, starting at point A, gives a maximum of n turns.*

60. What is the vertical spacing (in cm) between two consecutive turns?



(a) $\frac{h}{n}$

(b) $\frac{\sqrt{h}}{n}$

(c) $\frac{h^2}{n}$

(d) can't be determined

61. If there is no spacing between any two courecutive turns and the width of string be x cm, then the required length of the string is :

(a) $\frac{8x}{h}$ cm (b) $\frac{8h}{x}$ cm

(c) $8hx$ cm (d) $2\sqrt{2}\frac{h}{x}$ cm

62. If the string is wound on the exterior four walls of a cube of side a cm starting at point C and ending at point D exactly above C, making equally spaced 4 turns. The side of the cube is :

(a) $a = \frac{2n}{\sqrt{255}}$ (b) $a = \frac{(n)^2}{16}$

(c) $a = \frac{8n}{\sqrt{257}}$ (d) $a = 2\sqrt{15}n$

63. A blacksmith has a rectangular iron sheet 10 ft long. He has to cut out 7 circular discs from this sheet. What is the minimum possible width of the iron sheet if the radius of each disc is 1 ft?

(a) $2\sqrt{3}$ ft (b) $(2+\sqrt{3})$ ft

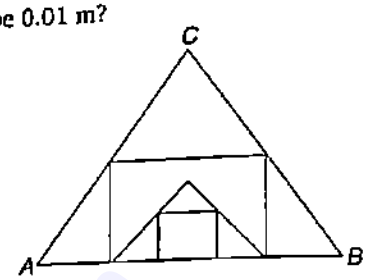
(c) $(3+\sqrt{2})$ ft (d) $(2+2\sqrt{3})$ ft

64. The perimeter of a square, a rhombus and a hexagon are same. The area of square, rhombus and hexagon be s, r, h, respectively then which of the following is correct :

(a) $r > s > h$ (b) $s > h > r$

(c) $h > s > r$ (d) data insufficient

65. In the adjoining figure ABC is an equilateral triangle inscribing a square of maximum possible area. Again in this square there is an equilateral triangle whose side is same as that of the square. Further the smaller equilateral triangle inscribes a square of maximum possible area. What is the area of the innermost square if the each side of the outermost triangle be 0.01 m?



(a) $(873 - 504\sqrt{3})$ cm^2 (b) $(738 - 504\sqrt{3})$ cm^2

(c) $(873 - 405\sqrt{2})$ cm^2 (d) none of these

66. A blacksmith has a rectangular sheet of iron. He has to make a cylindrical vessel both circular ends are closed. When he minimises the wastage of the sheet of iron, then what is the ratio of the wastage to the utilised area of sheet?

(a) $\frac{1}{11}$ (b) $\frac{2}{17}$

(c) $\frac{3}{22}$ (d) none of these

67. Barun needs an open box of capacity 864 m^3. Actually where he lives, the rates of paints are soaring high so he wants to minimize the surface area of the box keeping the capacity of the box same as required. What is the base area and height of such a box?

(a) 36 m^2, 24 m (b) 216 m^2, 4 m

(c) 144 m^2, 6 m (d) none of these

68. There are two cylindrical containers of equal capacity and equal dimensions. If the radius of one of the container is increased by 12 ft and the height of another container is increased by 12 ft, then the capacity of both the container is equally increased by K cubic ft. If the actual heights of each of the container be 4 ft, then find the increased volume of each of the container :

(a) 1680 π cu ft (b) 2304 π cu ft

(c) 1480 π cu ft (d) can't be determined

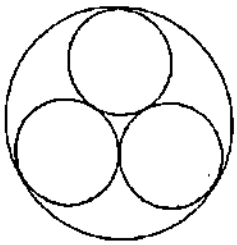
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Test of Your Learning

1. Three equal circles each of radius 1 cm are circumscribed by a larger circle. Find the perimeter of the circumscribing circle :



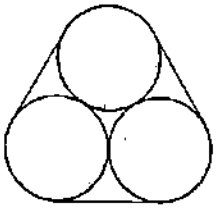
(a) $\frac{\sqrt{3}}{2}(2-\sqrt{3})\pi$ cm

(b) $\left(\frac{2+\sqrt{3}}{\sqrt{3}}\right)$ cm

(c) $\frac{2}{\sqrt{3}}(2+\sqrt{3})\pi$ cm

(d) none of the above

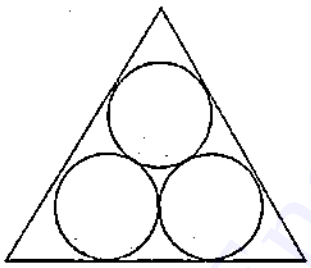
2. Three circular rings of equal radii of 1 cm each are touching each other. A string runs all around the set of rings very tightly. What is the minimum length of string required to bind all the three rings in the given manner?



(a) $\frac{6}{\pi}$ cm (b) $2(3+\pi)$ cm

(c) $\frac{\pi}{6}$ cm (d) can't be determined

3. An equilateral triangle circumscribes all the three circles each of radius 1 cm. What is the perimeter of the equilateral triangle?



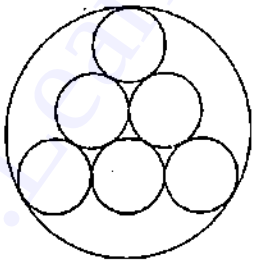
(a) $6(\sqrt{3}+1)$ cm

(b) $\sqrt{3}(8+\sqrt{2})$ cm

(c) $15(\sqrt{3}-1)$ cm

(d) none of the above

4. Six circles each of unit radius are being circumscribed by another larger circle. All the smaller circles touch each other. What is the circumference of the larger circle?



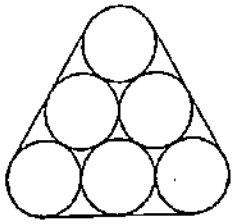
(a) $\left(\frac{\sqrt{3}+4}{\sqrt{2}}\right)\pi$ cm

(b) $4\sqrt{3}\pi$ cm

(c) $2\left(\frac{4+\sqrt{3}}{\sqrt{3}}\right)\pi$ cm

(d) can't be determined

5. There are six circular rings of iron, kept close to each other. A string binds them as tightly as possible. If the radius of each circular iron ring is 1 cm. What is the minimum possible length of string required to bind them?



(a) $2(6+3\sqrt{3}+\pi)$ cm

(b) $6(2+\sqrt{3})\pi$ cm

(c) $2(6+\pi)$ cm

(d) none of the above

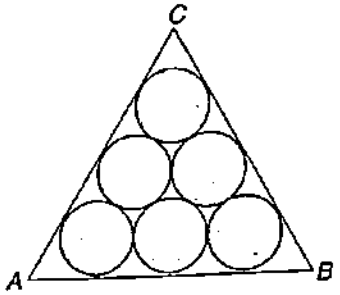
6. An equilateral triangle circumscribes all the six circles, each with radius 1 cm. What is the perimeter of the equilateral triangle?

(a) $6(2+\sqrt{3})$ cm

(b) $3(\sqrt{3}+2)$ cm

(c) $12(\sqrt{3}+4)$ cm

(d) none of the above



7. A cube of maximum possible volume is cut from the sphere of diameter $3\sqrt{3}$ cm. What is the ratio of volume of the sphere to that of cube?

(a) $\frac{4\sqrt{3}}{\pi}$ (b) $\frac{\sqrt{3}}{2}\pi$

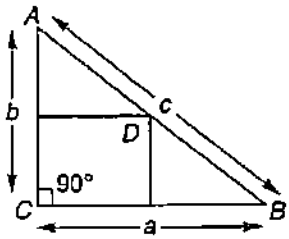
(c) $\frac{4}{\sqrt{3}}\pi$ (d) none of these

8. A cube of maximum possible volume is cut from the solid right circular cylinder. What is the ratio of volume of cube to that of cylinder if the edge of a cube is equal to the height of the cylinder?

(a) $\frac{11}{7}$ (b) $\sqrt{2}\frac{\pi}{7}$

(c) $\frac{7}{11}$ (d) none of these

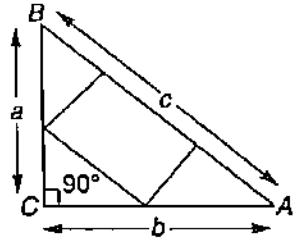
9. In a right angle triangle ABC, what is the maximum possible area of a square that can be inscribed when one of its vertices coincide with the vertex of right angle of the triangle?



(a) $\frac{a}{b}$ (b) $\frac{ab}{a+b}$

(c) $\frac{a+b}{ab}$ (d) $\left(\frac{ab}{a+b}\right)^2$

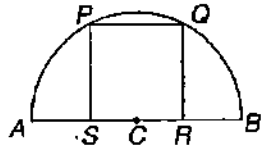
10. In the adjoining figure a square of maximum possible area is circumscribed by the right angle triangle ABC in such a way that one of its side just lies on the hypotenuse of the triangle. What is the area of the square?



(a) $\left(\frac{abc}{a^2+b^2+ab}\right)^2$ (b) $\frac{a^2+b^2+c^2}{abc}$

(c) $\frac{abc}{a^2+b^2+c^2}$ (d) none of these

11. In the adjoining figure $PQRS$ is a square of maximum possible area which is circumscribed by the semicircle. Points R and S lie on the diameter AB. What is the area of the square if the radius of the circle is 'r'?



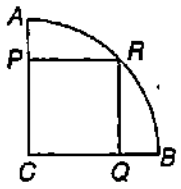
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(a) $\frac{\sqrt{3}}{4}r^2$ (b) $\frac{4}{5}r^2$

(c) $\frac{3}{5}r^2$ (d) $\frac{\sqrt{5}}{4}r^2$

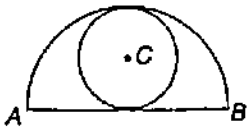
12. In the adjoining figure a quadrant (of circle) inscribes a square of maximum possible area. If the radius of the circle be 'r' then what is the area of the square?



(a) $\frac{r^2}{2}$ (b) $\frac{3r^2}{5}$

(c) $\frac{r^2}{\sqrt{3}}$ (d) $2\sqrt{6}r$

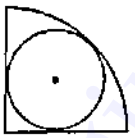
13. In the adjoining figure, AB is the diameter of a semicircle which inscribes a circle of maximum possible area. If the radius of the larger circle (i. e., semicircle) is r, the area of the inscribed circle is :



(a) $\frac{5r^2}{\pi}$ (b) $\frac{2\pi}{\sqrt{3}}r^2$

(c) $\frac{\pi r^2}{4}$ (d) none of these

14. In a quadrant (of a circle) a circle of maximum possible area is given. If the radius of the circumscribing quadrant be r, then what is the area of the inscribed circle?



(a) $(2+3\sqrt{2})r^2$ (b) $\frac{\pi r^2}{(3+2\sqrt{2})}$

(c) $\left(\frac{3+2\sqrt{2}}{r^2}\right)\pi$ (d) none of these

15. A cylindrical chocobar has its radius r unit and height 'h' unit. If we wish to increase the volume by same unit either by increasing its radius alone or its height alone, then how many unit we have to increase the radius or height?

(a) $\frac{r^2+2r}{h}$ (b) $\frac{r^2-2rh}{h}$

(c) $\frac{2r^2-rh}{h^2}$ (d) $\frac{\pi r^2}{2h}$

16. A 12 cm long wire is bent to form a triangle with one of its angle as 60°. Find the sides of the triangle (in cm) when its area is largest?

(a) 3, 4, 5 (b) 2, 4, 6

(c) 4, 4, 4 (d) 2.66, 3.66, 4.66

17. Let $S_1, S_2, \ldots, S_n$ be the squares such that for each $n > 1$, the length of a side of S_n equals the length of the diagonal of $S_{(n+1)}$. If the length of a circle of S_1 is 10 cm, then for which of the following values of n the area of S_n is less than 1 square cm?

(a) 4 (b) 8

(c) 6 (d) 7

18. Area of a regular hexagon and a regular octagon is same. Which one of the two has larger perimeter?

(a) Hexagon (b) Octagon

(c) can't be determined (d) none of these

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Answers

INTRODUCTORY EXERCISE-10.1

1 (b)	2. (c)	3. (c)	4. (b)	5. (a)	6. (a)	7. (b)	8. (a)	9. (a)	10. (c)
11. (b)	12. (d)	13. (a)	14. (b)	15. (b)	16. (c)	17. (c)	18. (d)	19. (c)	20. (b)
21. (b)	22. (a)	23. (c)	24. (c)	25. (c)	26. (b)	27. (c)	28. (c)	29. (b)	30. (c)
31. (b)	32. (c)								

INTRODUCTORY EXERCISE-10.2

1 (b)	2. (c)	3. (a)	4. (a)	5. (d)	6. (a)	7. (c)	8. (b)	9. (a)	10. (a)

INTRODUCTORY EXERCISE-10.3

1 (a)	2. (c)	3. (d)	4. (b)	5. (d)	6. (b)	7. (a)	8. (b)	9. (d)	10. (c)
11. (b)	12. (a)	13. (b)	14. (d)	15. (a)					

INTRODUCTORY EXERCISE-10.4

1 (d)	2. (d)	3. (a)	4. (b)	5. (c)	6. (d)	7. (a)	8. (a)	9. (a)	10. (b)
11. (c)	12. (d)	13. (b)	14. (a)	15. (b)	16. (a)	17. (a)	18. (c)	19. (a)	20. (c)
21. (a)	22. (b)	23. (b)	24. (c)	25. (c)					

INTRODUCTORY EXERCISE-10.5

1 (b)	2. (c)	3. (a)	4. (b)	5. (c)	6. (a)	7. (b)	8. (a)	9. (b)	10. (c)
11. (d)	12. (a)	13. (c)	14. (b)	15. (c)	16. (a)	17. (b)	18. (d)	19. (b)	20. (c)
21. (d)	22. (b)	23. (c)	24. (a)	25. (b)	26. (a)	27. (c)	28. (c)	29. (c)	30. (b)
31. (b)	32. (b)								

INTRODUCTORY EXERCISE-10.6

1 (b)	2. (c)	3. (c)	4. (c)	5. (b)	6. (a)	7. (c)	8. (b)	9. (a)	10. (c)
11. (a)	12. (a)	13. (d)	14. (a)	15. (d)	16. (c)	17. (a)	18. (b)	19. (a)	20. (b)
21. (c)	22. (a)	23. (d)	24. (c)	25. (d)	26. (d)	27. (d)	28. (c)	29. (b)	30. (c)
31. (b)	32. (a)	33. (d)	34. (c)	35. (c)	36. (b)	37. (b)	38. (a)	39. (b)	40. (a)

INTRODUCTORY EXERCISE-10.7

1 (b)	2. (c)	3. (c)	4. (a)	5. (b)	6. (c)	7. (c)	8. (d)	9. (c)	10. (c)
11. (b)	12. (a)								

MISCELLANEOUS

1 (c)	2. (c)	3. (c)	4. (c)	5. (c)	6. (a)	7. (c)	8. (a)	9. (c)	10. (a)
11. (d)	12. (c)	13. (a)	14. (c)	15. (d)	16. (b)	17. (b)	18. (c)	19. (b)	20. (d)
21. (b)	22. (c)	23. (d)	24. (c)	25. (d)	26. (d)	27. (d)	28. (b)	29. (b)	30. (b)
31. (a)	32. (c)	33. (d)	34. (c)	35. (a)	36. (c)	37. (b)	38. (c)	39. (d)	40. (a)
41. (b)	42. (a)	43. (c)	44. (c)	45. (a)	46. (b)	47. (b)	48. (c)	49. (d)	50. (b)
51. (c)	52. (c)	53. (a)	54. (d)	55. (c)	56. (a)	57. (b)	58. (a)	59. (c)	

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